nRF52840

Product Specification

v1.10

NORDIC® SEMICONDUCTOR

Feature list

Features:

- *Bluetooth*® 5, IEEE 802.15.4-2006, 2.4 GHz transceiver
  - -95 dBm sensitivity in 1 Mbps Bluetooth low energy mode
  - -103 dBm sensitivity in 125 kbps Bluetooth low energy mode (long range)
  - -20 to +8 dBm TX power, configurable in 4 dB steps
  - On-air compatible with nRF52, nRF51, nRF24L, and nRF24AP Series devices
  - Supported data rates:
    - Bluetooth 5 – 2 Mbps, 1 Mbps, 500 kbps, and 125 kbps
    - IEEE 802.15.4-2006 – 250 kbps
    - Proprietary 2.4 GHz – 2 Mbps, 1 Mbps
  - Single-ended antenna output (on-chip balun)
  - 128-bit AES/ECB/CCM/AAR co-processor (on-the-fly packet encryption)
  - 4.8 mA peak current in TX (0 dBm)
  - 4.6 mA peak current in RX
  - RSSI (1 dB resolution)
- Arm® Cortex®-M4 32-bit processor with FPU, 64 MHz
  - 212 EEMBC CoreMark® score running from flash memory
  - 52 µA/MHz running CoreMark from flash memory
  - Watchpoint and trace debug modules (DWT, ETM, and ITM)
  - Serial wire debug (SWD)
- Rich set of security features
  - Arm TrustZone® CryptoCell™ 310 security subsystem
    - NIST SP800-90A and SP800-90B compliant random number generator
    - AES-128 – ECB, CBC, CMAC/CBC-MAC, CTR, CCM/CCM*
    - Chacha20/Poly1305 AEAD supporting 128- and 256-bit key size
    - SHA-1 and SHA-2 up to 256 bit
    - Keyed-hash message authentication code (HMAC)
    - RSA up to 2048-bit key size
    - SRP up to 3072-bit key size
    - ECC support for most used curves, including P-256 (secp256r1) and Ed25519/Curve25519
    - Application key management using derived key model
  - Secure boot ready
    - Flash access control list (ACL)
    - Root-of-trust (RoT)
    - Debug control and configuration
    - Access port protection (CTRL-AP)
  - Secure erase
- Flexible power management
  - 1.7 V to 5.5 V supply voltage range
  - On-chip DC/DC and LDO regulators with automated low current modes
  - 1.8 V to 3.3 V regulated supply for external components
  - Automated peripheral power management
  - Fast wake-up using 64 MHz internal oscillator
  - 0.4 µA at 3 V in System OFF mode, no RAM retention
  - 1.5 µA at 3 V in System ON mode, no RAM retention, wake on RTC
- 1 MB flash and 256 kB RAM
- Advanced on-chip interfaces
  - USB 2.0 full speed (12 Mbps) controller
  - QSPI 32 MHz interface
  - High-speed 32 MHz SPI
  - Type 2 near field communication (NFC-A) tag with wake-on field
    - Touch-to-pair support
  - Programmable peripheral interconnect (PPI)
  - 48 general purpose I/O pins
  - EasyDMA automated data transfer between memory and peripherals
- Nordic SoftDevice ready with support for concurrent multiprotocol
- 12-bit, 200 ksps ADC – 8 configurable channels with programmable gain
- 64 level comparator
- 15 level low-power comparator with wake-up from System OFF mode
- Temperature sensor
- Four 4 channel pulse width modulator (PWM) units with EasyDMA
- Audio peripherals – I^2S, digital microphone interface (PDM)
- Five 32-bit timers with counter mode
- Up to four SPI masters/three SPI slaves with EasyDMA
- Up to two I^2C compatible two-wire master/slave
- Two UART (CTS/RTS) with EasyDMA
- Quadrature decoder (QDEC)
- Three real-time counters (RTC)
- Single crystal operation
- Package variants
  - aQFN73™ package, 7 x 7 mm
  - QFN48 package, 6 x 6 mm
  - WLCSP package, 3.544 x 3.607 mm

Applications:

- Advanced computer peripherals and I/O devices
  - Mouse
  - Keyboard
  - Multi-touch trackpad
- Advanced wearable devices
  - Health/fitness sensors and monitoring devices
  - Wireless payment enabled devices
- Internet of things (IoT)
  - Smart home sensors and controllers
  - Industrial IoT sensors and controllers
- Interactive entertainment devices
  - Remote controls
  - Gaming controllers

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1 Revision history

Date	Version	Description
July 2024	1.10	The following content has been added or updated: • Added content in the following chapters: • UICR – Unused registers • Instantiation – Added CRYPTOCELL components • Updated content in the following chapters: • PWM – Wave counter section • CRYPTOCELL – Updated description
February 2024	1.9	The following content has been added or updated: • Added content in the following chapters: • Pin assignments – Recommended usage for pin P0.18 as QSPI CSN for WLCSP package • RADIO – Parameters $P_{ACP,R,\ IEEE\ 802.15.4}$ and $P_{ACP,A,\ IEEE\ 802.15.4}$ • SAADC – Sections Shared Resources, and Operation Modes • SPI, TWI, and UART – RXD register description • Updated content in the following chapters: • FICR – INFO.VARIANTS and INFO.PART • NVMC – Removed deprecated registers ERASEPCR1 and ERASEPCR0 • Ordering information – MOQ for the nRF52840-QFAA-F-R7 device • PWM – Example and images in Wave Counter section • SAADC – Removed t_{ACQ} parameters • SPIS – Electrical parameters $t_{SPIS,VSO}$, $t_{SPIS,HSO}$, $t_{SPIS,SUSI}$, and $t_{SPIS,HSI}$. Removed deprecated registers • About this document – Permissions table • Reference circuitry – Circuit configuration no. 1 for QFAA QFN48 package • Editorial changes
December 2023	1.8	The following content has been added: • Reference circuitry – PMIC support section
November 2021	1.7	The following content has been added or updated: • Absolute maximum ratings – Updated aQFN73 ESD CDM maximum value according to PCN162
November 2021	1.6	The following content has been added or updated:

Date	Version	Description
		• Ordering information – Build codes Dxx not recommended for new designs • Editorial changes
September 2021	1.5	The following content has been added or updated: • Mechanical specifications – Updated aQFN73 mechanical specification according to PCN148
June 2021	1.4	The following content has been added or updated: • Added information for the QFN48 package variant in • Absolute maximum ratings • FICR • Ordering information • Package variation • Reference circuitry • Mechanical specifications • Pin assignments • POWER • Package thermal characteristics – Added QFN48 and WLCSP thermal resistance and updated aQFN73 with JEDEC PCB numbers
April 2021	1.3	The following content has been added or updated: • UICR – Added value HwDisabled to APPROTECT register. • Debug and trace – Added description of APPROTECT functionality for devices where APPROTECT is controlled by hardware and software. Added peripheral APPROTECT with necessary registers to control APPROTECT for devices where APPROTECT is controlled by hardware and software. • GPIO – Added missing NFC parameters C_{PAD_NFC} and I_{NFC_LEAK}. • Pin assignments – Added note that DEC5 is not connected for aQFN73, and WLCSP build codes Fxx and later. • Mechanical specifications – Corrected min/max values of WLCSP D and E dimensions. • Reference circuitry – Updated aQFN73 and WLCSP reference circuitry with note on DEC5. • Ordering information – Updated box labels. Added new product options.
January 2021	1.2	The following content has been added or updated: • Updated minimum valid value for EasyDMA MAXCNT and AMOUNT registers in SPIM, SPIS, TWIM, TWIS, and UARTE. • FICR – Added nRF52820 value to INFO.PART register. Added size of packages to package description. Updated

Date	Version	Description
		INFO.VARIANT device variants. Removed reset value for register TRNG90B.STARTUP and all TEMP module calibration registers. • UICR – Updated reference to the VEXTDIF parameter in the REGOUT0 register. • NVMC – Updated reset value of register READYNEXT. Updated registers ERASEPAGE, ERASEALL, ERASEUICR, ERASEPARTIAL, IHIT, and IMISS. • POWER – Added Wake from System OFF reset source for the WDT. Added parameter $R_{SOURCE,VBUSVDDH}$. Renamed parameters to clarify REG0 in the Electrical Specification. Updated value of parameter $R_{SOURCE,VBUS}$ to 6 Ω. • CLOCK – Added parameter $V_{AMP,IN,XO,LOW}$. • Current consumption – Added USBD current measurements. Updated values in Electrical Specification - Sleep. • ACL – Added clarification for the maximum protected region size being limited to half the flash. • LPCOMP – Added clarification about not disabling LPCOMP in the same write as COMP is enabled. • QSPI – Added clarification about XIP region being read only. • RADIO – Added EVENTS_SYNC. • SPIS – Updated parameter $t_{SPIS,HCSN}$. • SPIM – Updated parameter $t_{SPIM,CSK}$. • TWIM – Updated t_{TWIM,HD_STA} parameter. Clarified use of SUSPEND and RESUME tasks. • USBD – Updated the ENABLE register functionality description. Moved current measurements from the electrical specification section to CURRENT. • Mechanical specifications – Added tolerances for D and E dimensions for WLCSP and aQFN73 packages. • Reference circuitry – Corrected links. Corrected tables in WLCSP package Configurations 1 to 6. Added a new reference design with 4 component RF-match for the QIAA aQFN73 package. Updated recommended value for USB serial resistor. • Package thermal characteristics – Added aQFN73 thermal resistance. • Ordering information – Corrected WLCSP ball count. • Absolute maximum ratings – Added footnote on HTOL supply. Increased Flash memory retention to 10 years at 85°C. • Recommended operating conditions – Added parameter T_J and WLCSP package light sensitivity section. • Editorial changes.
February 2019	1.1	The following content has been added or updated:

Date	Version	Description
		• Added information for the WLCSP package variant in Pin assignments, Mechanical specifications, Reference circuitry, FICR, Absolute maximum ratings, and Ordering information. • Reference circuitry – Updated RF-Match in aQFN73 reference circuitry for all configurations. Added optional 4.7 Ω resistor to USB supply. • UICR – Removed NRFFW[13] and NRFFW[14] registers. • CPU on page 20 – Corrected value of parameter $CM_{FLASH/mA}$. • POWER – Clarified range of voltages in both Normal and High voltage modes. • CLOCK – Corrected value of parameter P_{D_LFXO} to a less restrictive value. • EasyDMA – Added section about EasyDMA error handling. Corrected example code in section EasyDMA array list. • NVMC – Added note about the necessity to halt the CPU before issuing NVMC commands from the debugger. • ACL – Corrected register access to ReadWriteOnce (RWO) for some registers. • I_2S – Removed invalid values from register MCKFREQ, see parameter f_{MCK}. Fixed figure for Memory mapping for 8-bit stereo. • SAADC – Corrected description of functionality of SAMPLE task. • SPIS – Exposed the LIST register. Corrected SPI modes table. • TWIS – Exposed the LIST register. • UART – Added STOP bit configuration description. • RADIO – Added equations to convert from HW RSSI to 802.15.4 range and dBm. Clarified RSSI timing. Clarified that TX ramp up time is affected by RU field in MODECNF0. Added IEEE 802.15.4 radio timing parameters to the electrical specifications. Added sensitivity parameter for 2 Mbit NRF mode. • USBD – Pointed that isochronous transfers have to be finished before the next SOF event, or the result of the transfer is undefined. • Legal notices on page 979 – Updated text and image.
March 2018	1.0	First release

2 About this document

This document is organized into chapters that are based on the modules and peripherals available in the IC.

2.1 Document status

The document status reflects the level of maturity of the document.

Document name	Description
Objective Product Specification (OPS)	Applies to document versions up to 1.0. This document contains target specifications for product development.
Product Specification (PS)	Applies to document versions 1.0 and higher. This document contains final product specifications. Nordic Semiconductor ASA reserves the right to make changes at any time without notice in order to improve design and supply the best possible product.

Table 1: Defined document names

2.2 Peripheral chapters

The chapters describing peripherals include the following information:

- A detailed functional description of the peripheral.
- The register configuration for the peripheral.
- The electrical specification tables, containing performance data which apply for the operating conditions described in Recommended operating conditions on page 971.

2.2.1 Peripheral naming conventions

Every peripheral has a unique capitalized name or an abbreviation of its name, e.g. TIMER, used for identification and reference.

This name is used in chapter headings and references, and it will appear in the Arm Cortex Microcontroller Software Interface Standard (CMSIS) hardware abstraction layer to identify the peripheral.

The peripheral instance name, which is different from the peripheral name, is constructed using the peripheral name followed by a numbered postfix, starting with 0, for example, TIMER0. A postfix is normally only used if a peripheral can be instantiated more than once. The peripheral instance name is also used in the CMSIS to identify the peripheral instance.

2.3 Register tables

Individual registers are described using register tables. These tables are built up of two sections. The first three colored rows describe the position and size of the different fields in the register. The following rows describe the fields in more detail.

2.3.1 Fields and values

The **Id (Field Id)** row specifies the bits that belong to the different fields in the register. If a field has enumerated values, then every value will be identified with a unique value id in the **Value ID** column.

A blank space means that the field is reserved and read as undefined, and it also must be written as `0` to secure forward compatibility. If a register is divided into more than one field, a unique field name is specified for each field in the **Field** column. The **Value ID** may be omitted in the single-bit bit fields when values can be substituted with a Boolean type enumerator range, e.g. true/false, disable(d)/enable(d), on/off, and so on.

Values are usually provided as decimal or hexadecimal. Hexadecimal values have a `0x` prefix, decimal values have no prefix.

The **Value** column can be populated in the following ways:

- Individual enumerated values, for example 1, 3, 9.
- Range of values, e.g. [0..4], indicating all values from and including 0 and 4.
- Implicit values. If no values are indicated in the **Value** column, all bit combinations are supported, or alternatively the field's translation and limitations are described in the text instead.

If two or more fields are closely related, the **Value ID**, **Value**, and **Description** may be omitted for all but the first field. Subsequent fields will indicate inheritance with '..'.

A feature marked **Deprecated** should not be used for new designs.

2.3.2 Permissions

Different fields in a register might have different access permissions enforced by hardware.

The access permission for each register field is documented in the **Access** column in the following ways:

Access	Description	Hardware behavior
R	Read-only	Field can only be read. A write will be ignored.
W	Write-only	Field can only be written. A read will return an undefined value.
RW	Read-write	Field can be read and written multiple times.
W1	Write-once	Field can only be written once per reset. Any subsequent write will be ignored. A read will return an undefined value.
RW1	Read-write-once	Field can be read multiple times, but only written once per reset. Any subsequent write will be ignored.
W0C	Write 0 to clear	Field can be read multiple times. A zero clears (set to zero) the corresponding bit in the register. Bits set to one are ignored.
W1C	Write 1 to clear	Field can be read multiple times. A one clears (set to zero) the corresponding bit in the register. Bits set to zero are ignored.
W1S	Write 1 to set	Field can be read multiple times. A one sets the corresponding bit in the register. Bits set to zero are ignored.
RME	Read Modify External	When read, a side effect occurs.

Table 2: Register field permission schemes

2.4 Registers

Register overview

Register	Offset	Description
DUMMY	0x514	Example of a register controlling a dummy feature

2.4.1 DUMMY

Address offset: 0x514

Example of a register controlling a dummy feature

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID									I	H	G	F	E	D	D	D	C	C	C								B							A	A
Reset 0x00050002				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	FIELD0																																	
			Disabled	0																															
			NormalMode	1																															
			ExtendedMode	2																															
B	RW	FIELD1																																	
				This field is deprecated.																															
			Disabled	0																															
			Enabled	1																															
C	RW	FIELD2																																	
			ValidRange	[2..7]																															
D	RW	FIELD3																																	
E	R	FIELD4																																	
F	W	FIELD5																																	
G	RW W1C	FIELD6																																	
H	RW W0C	FIELD7																																	
I	RW RME	FIELD8																																	

3 Block diagram

This block diagram illustrates the overall system. Arrows with white heads indicate signals that share physical pins with other signals.

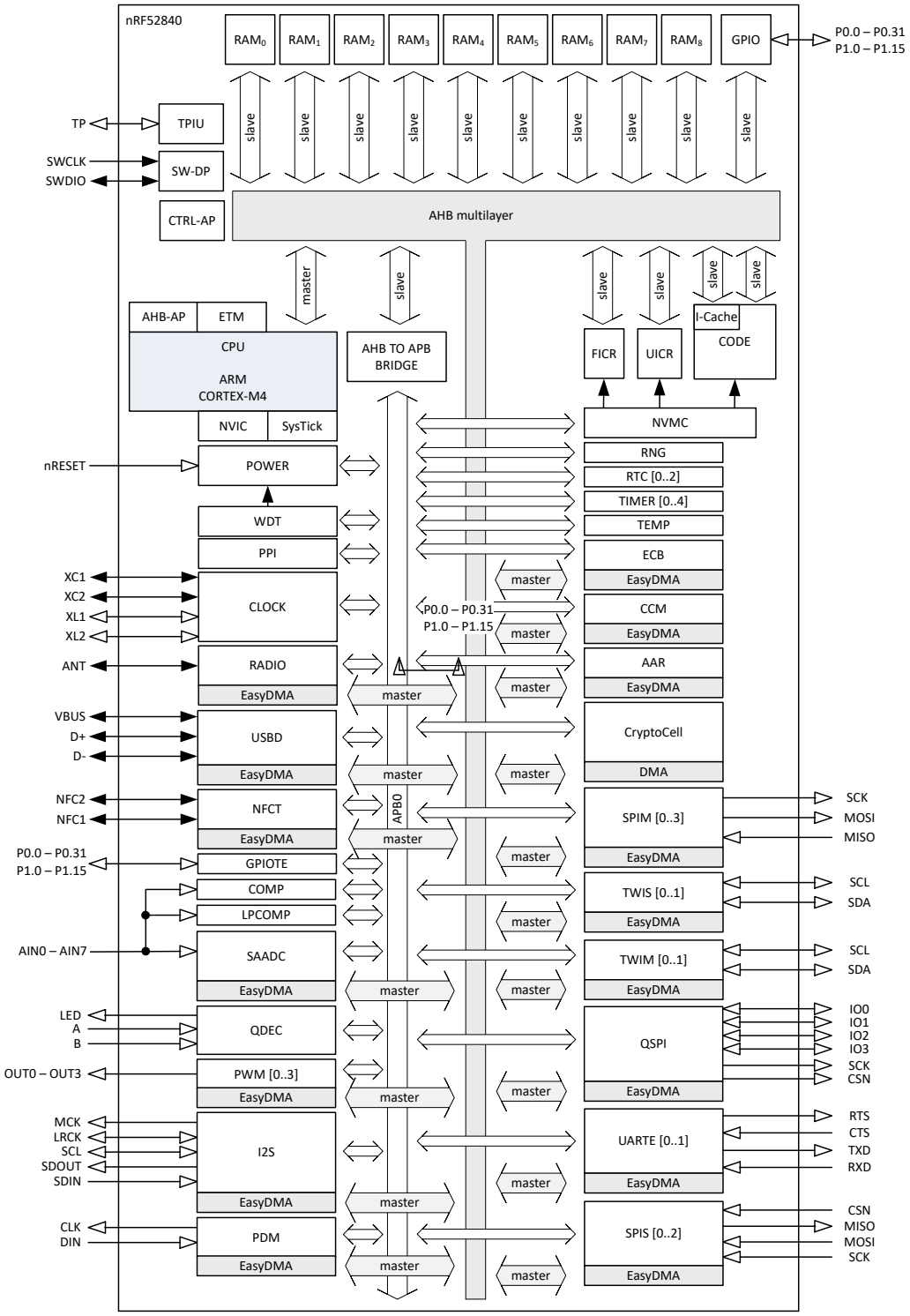


Figure 1: Block diagram

4 Core components

4.1 CPU

The Arm Cortex-M4 processor with floating-point unit (FPU) has a 32-bit instruction set (Thumb®-2 technology) that implements a superset of 16- and 32-bit instructions to maximize code density and performance.

This processor implements the following features that enable energy-efficient arithmetic and high-performance signal processing.

- Digital signal processing (DSP) instructions
- Single-cycle multiply and accumulate (MAC) instructions
- Hardware divide
- 8- and 16-bit single instruction multiple data (SIMD) instructions
- Single-precision floating-point unit (FPU)

The Arm Cortex Microcontroller Software Interface Standard (CMSIS) hardware abstraction layer for the Arm Cortex processor series is implemented and available for the M4 CPU.

Real-time execution is highly deterministic in thread mode, to and from sleep modes, and when handling events at configurable priority levels via the nested vectored interrupt controller (NVIC).

Executing code from flash memory will have a wait state penalty on the nRF52 Series. An instruction cache can be enabled to minimize flash wait states when fetching instructions. For more information on cache, see Cache on page 27. The Electrical specification on page 21 shows CPU performance parameters including wait states in different modes, CPU current and efficiency, and processing power and efficiency based on the CoreMark benchmark.

The Arm system timer (SysTick) is present on nRF52840. The SysTick's clock will only tick when the CPU is running or when the system is in debug interface mode.

4.1.1 Floating point interrupt

The floating point unit (FPU) may generate exceptions when used due to e.g. overflow or underflow, which in turn will trigger the FPU interrupt.

See Instantiation on page 24 for more information about the exceptions triggering the FPU interrupt.

To clear the IRQ (interrupt request) line when an exception has occurred, the relevant exception bit within the floating-point status and control register (FPSCR) needs to be cleared. For more information about the FPSCR or other FPU registers, see *Cortex-M4 Devices Generic User Guide*.

4.1.2 CPU and support module configuration

The Arm Cortex-M4 processor has a number of CPU options and support modules implemented on the IC.

Option / Module	Description	Implemented
Core options		
NVIC	Nested vector interrupt controller	48 vectors
PRIORITIES	Priority bits	3
WIC	Wakeup interrupt controller	NO
Endianness	Memory system endianness	Little endian
Bit-banding	Bit banded memory	NO
DWT	Data watchpoint and trace	YES
SysTick	System tick timer	YES
Modules		
MPU	Memory protection unit	YES
FPU	Floating-point unit	YES
DAP	Debug access port	YES
ETM	Embedded trace macrocell	YES
ITM	Instrumentation trace macrocell	YES
TPIU	Trace port interface unit	YES
ETB	Embedded trace buffer	NO
FPB	Flash patch and breakpoint unit	YES
HTM	AMBA® AHB trace macrocell	NO

4.1.3 Electrical specification

4.1.3.1 CPU performance

The CPU clock speed is 64 MHz. Current and efficiency data is taken when in System ON and the CPU is executing the CoreMark benchmark. It includes power regulator and clock base currents. All other blocks are IDLE.

Symbol	Description	Min.	Typ.	Max.	Units
W_{FLASH}	CPU wait states, running CoreMark from flash, cache disabled			2	
$W_{FLASHCACHE}$	CPU wait states, running CoreMark from flash, cache enabled			3	
W_{RAM}	CPU wait states, running CoreMark from RAM			0	
CM_{FLASH}	CoreMark, running CoreMark from flash, cache enabled		212		CoreMark
$CM_{FLASH/MHz}$	CoreMark per MHz, running CoreMark from flash, cache enabled		3.3		CoreMark/MHz
$CM_{FLASH/mA}$	CoreMark per mA, running CoreMark from flash, cache enabled, DCDC 3V		64		CoreMark/mA

4.2 Memory

The nRF52840 contains 1024 kB of flash memory and 256 kB of RAM that can be used for code and data storage.

The CPU and peripherals with EasyDMA can access memory via the AHB multilayer interconnect. In additon, peripherals are accessed by the CPU via the AHB multilayer interconnect, as shown in the following figure.

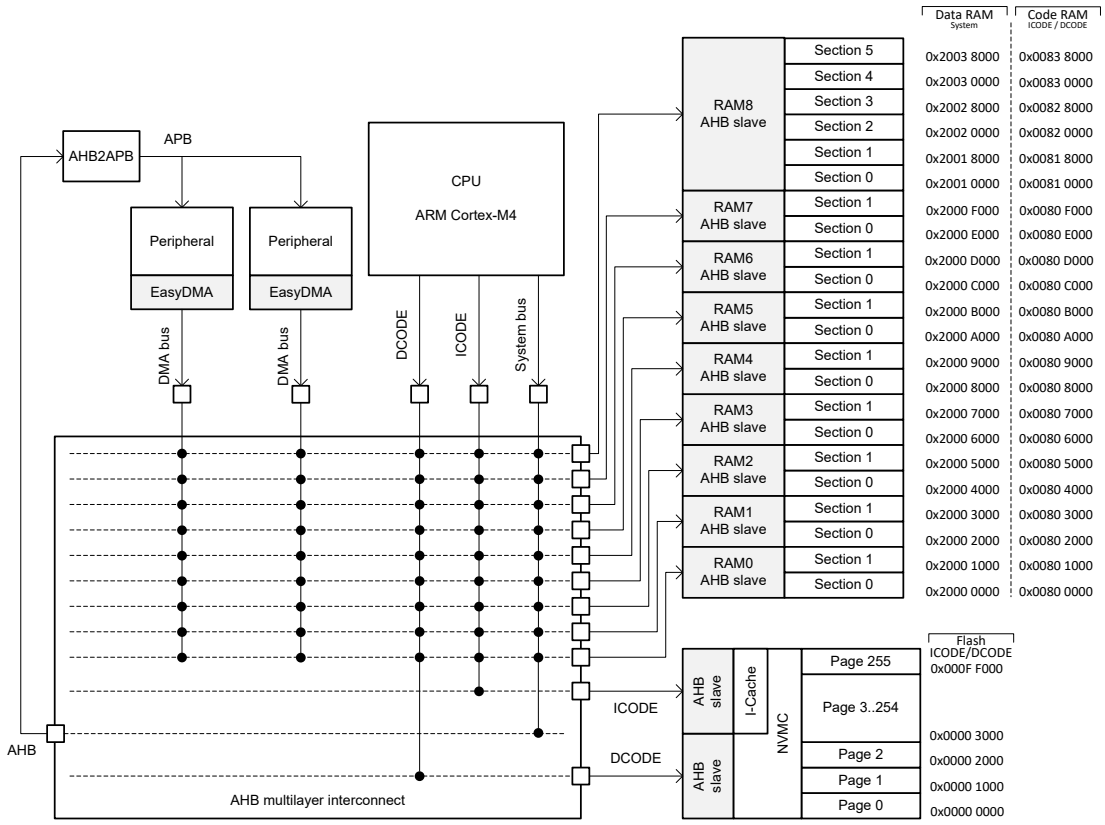


Figure 2: Memory layout

See AHB multilayer on page 66 and EasyDMA on page 63 for more information about the AHB multilayer interconnect and EasyDMA.

The same physical RAM is mapped to both the Data RAM region and the Code RAM region. It is up to the application to partition the RAM within these regions so that one does not corrupt the other.

4.2.1 RAM - Random access memory

The RAM interface is divided into nine RAM AHB slaves.

RAM AHB slaves 0 to 7 are connected to two 4 kB RAM sections each, while RAM AHB slave 8 is connected to six 32 kB sections, as shown in Memory layout on page 22.

Each RAM section has separate power control for System ON and System OFF mode operation, which is configured via RAM register (see the POWER — Power supply on page 81).

4.2.2 Flash - Non-volatile memory

The CPU can read from flash memory an unlimited number of times, but is restricted in how it writes to flash and the number of writes and erases it can perform.

Writing to flash memory is managed by the non-volatile memory controller (NVMC), see NVMC — Non-volatile memory controller on page 25.

Flash memory is divided into 256 pages of 4 kB each that can be accessed by the CPU via the ICODE and DCODE buses as shown in Memory layout on page 22.

4.2.3 Memory map

The complete memory map for the nRF52840 is shown in the following figure. As described in Memory on page 21, Code RAM and Data RAM are the same physical RAM.

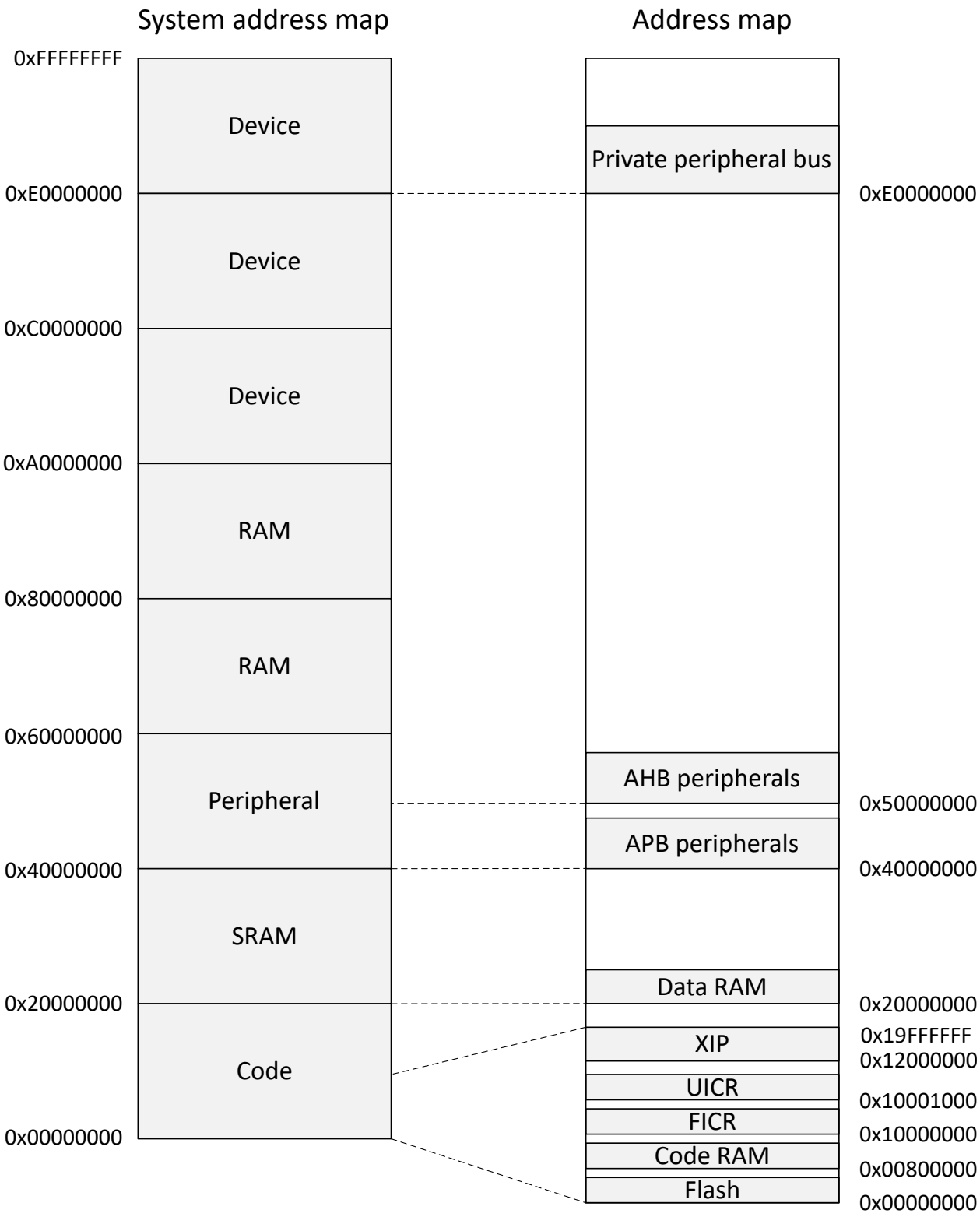


Figure 3: Memory map

4.2.4 Instantiation

ID	Base address	Instance	Description
0	0x40000000	APPROTECT	APPROTECT control
0	0x40000000	CLOCK	Clock control
0	0x40000000	POWER	Power control
0	0x50000000	GPIO	General purpose input and output This instance is deprecated.
0	0x50000000	P0	General purpose input and output, port 0
0	0x50000300	P1	General purpose input and output, port 1
1	0x40001000	RADIO	2.4 GHz radio
2	0x40002000	UART0	Universal asynchronous receiver/transmitter This instance is deprecated.
2	0x40002000	UARTE0	Universal asynchronous receiver/transmitter with EasyDMA, unit 0
3	0x40003000	SPI0	SPI master 0 This instance is deprecated.
3	0x40003000	SPIM0	SPI master 0
3	0x40003000	SPIS0	SPI slave 0
3	0x40003000	TWI0	Two-wire interface master 0 This instance is deprecated.
3	0x40003000	TWIM0	Two-wire interface master 0
3	0x40003000	TWIS0	Two-wire interface slave 0
4	0x40004000	SPI1	SPI master 1 This instance is deprecated.
4	0x40004000	SPIM1	SPI master 1
4	0x40004000	SPIS1	SPI slave 1
4	0x40004000	TWI1	Two-wire interface master 1 This instance is deprecated.
4	0x40004000	TWIM1	Two-wire interface master 1
4	0x40004000	TWIS1	Two-wire interface slave 1
5	0x40005000	NFCT	Near field communication tag
6	0x40006000	GPIOTE	GPIO tasks and events
7	0x40007000	SAADC	Analog to digital converter
8	0x40008000	TIMER0	Timer 0
9	0x40009000	TIMER1	Timer 1
10	0x4000A000	TIMER2	Timer 2
11	0x4000B000	RTC0	Real-time counter 0
12	0x4000C000	TEMP	Temperature sensor
13	0x4000D000	RNG	Random number generator
14	0x4000E000	ECB	AES electronic code book (ECB) mode block encryption
15	0x4000F000	AAR	Accelerated address resolver
15	0x4000F000	CCM	AES counter with CBC-MAC (CCM) mode block encryption
16	0x40010000	WDT	Watchdog timer
17	0x40011000	RTC1	Real-time counter 1
18	0x40012000	QDEC	Quadrature decoder
19	0x40013000	COMP	General purpose comparator
19	0x40013000	LPCOMP	Low power comparator
20	0x40014000	EGU0	Event generator unit 0
20	0x40014000	SWI0	Software interrupt 0
21	0x40015000	EGU1	Event generator unit 1
21	0x40015000	SWI1	Software interrupt 1

ID	Base address	Instance	Description
22	0x40016000	EGU2	Event generator unit 2
22	0x40016000	SWI2	Software interrupt 2
23	0x40017000	EGU3	Event generator unit 3
23	0x40017000	SWI3	Software interrupt 3
24	0x40018000	EGU4	Event generator unit 4
24	0x40018000	SWI4	Software interrupt 4
25	0x40019000	EGU5	Event generator unit 5
25	0x40019000	SWI5	Software interrupt 5
26	0x4001A000	TIMER3	Timer 3
27	0x4001B000	TIMER4	Timer 4
28	0x4001C000	PWM0	Pulse width modulation unit 0
29	0x4001D000	PDM	Pulse Density modulation (digital microphone) interface
30	0x4001E000	ACL	Access control lists
30	0x4001E000	NVMC	Non-volatile memory controller
31	0x4001F000	PPI	Programmable peripheral interconnect
32	0x40020000	MWU	Memory watch unit
33	0x40021000	PWM1	Pulse width modulation unit 1
34	0x40022000	PWM2	Pulse width modulation unit 2
35	0x40023000	SPI2	SPI master 2 This instance is deprecated.
35	0x40023000	SPIM2	SPI master 2
35	0x40023000	SPIS2	SPI slave 2
36	0x40024000	RTC2	Real-time counter 2
37	0x40025000	I2S	Inter-IC sound interface
38	0x40026000	FPU	FPU interrupt
39	0x40027000	USBD	Universal serial bus device
40	0x40028000	UARTE1	Universal asynchronous receiver/transmitter with EasyDMA, unit 1
41	0x40029000	QSPI	External memory interface
42	0x5002A000	CRYPTOCELL	CRYPTOCELL 310 security subsystem
43	0x5002B000	CC_AES	CRYPTOCELL AES engine
43	0x5002B000	CC_CHACHA	CRYPTOCELL CHACHA engine
43	0x5002B000	CC_CTL	CRYPTOCELL CTL interface
43	0x5002B000	CC_DIN	CRYPTOCELL DIN DMA engine
43	0x5002B000	CC_DOUT	CRYPTOCELL DOUT DMA engine
43	0x5002B000	CC_HASH	CRYPTOCELL HASH engine
43	0x5002B000	CC_HOST_RGF	CRYPTOCELL HOST register interface
43	0x5002B000	CC_MISC	CRYPTOCELL MISC interface
43	0x5002B000	CC_PKA	CRYPTOCELL PKA engine
43	0x5002B000	CC_RNG	CRYPTOCELL RNG engine
43	0x5002B000	CC_RNG_SRAM	CRYPTOCELL RNG SRAM interface
45	0x4002D000	PWM3	Pulse width modulation unit 3
47	0x4002F000	SPIM3	SPI master 3
N/A	0x10000000	FICR	Factory information configuration
N/A	0x10001000	UICR	User information configuration

Table 3: Instantiation table

4.3 NVMC — Non-volatile memory controller

The non-volatile memory controller (NVMC) is used for writing and erasing of the internal flash memory and the UICR (user information configuration registers).

The CONFIG on page 28 is used to enable the NVMC for writing (CONFIG.WEN = Wen) and erasing (CONFIG.WEN = Een).

The CPU must be halted before initiating a NVMC operation from the debug system.

4.3.1 Writing to flash

When write is enabled, full 32-bit words can be written to word-aligned addresses in flash memory.

As illustrated in Memory on page 21, the flash is divided into multiple pages. The same 32-bit word in flash memory can only be written n_{WRITE} number of times before a page erase must be performed.

The NVMC is only able to write `0` to bits in flash memory that are erased (set to `1`). It cannot rewrite a bit back to `1`. Only full 32-bit words can be written to flash memory using the NVMC interface. To write less than 32 bits, write the data as a full 32-bit word and set all the bits that should remain unchanged in the word to `1`. The restriction on the number of writes (n_{WRITE}) still applies in this case.

Only word-aligned writes are allowed. Byte or half-word-aligned writes will result in a hard fault.

The time it takes to write a word to flash is specified by t_{WRITE}. The CPU is halted if the CPU executes code from the flash while the NVMC is writing to the flash.

NVM writing time can be reduced by using READYNEXT. If this status bit is set to `1`, code can perform the next data write to the flash. This write will be buffered and will be taken into account as soon as the ongoing write operation is completed.

4.3.2 Erasing a page in flash

When erase is enabled, the flash memory can be erased page by page using the ERASEPAGE on page 29.

After erasing a flash page, all bits in the page are set to `1`. The time it takes to erase a page is specified by $t_{ERASEPAGE}$. The CPU is halted if the CPU executes code from the flash while the NVMC is writing to the flash.

See Partial erase of a page in flash on page 27 for information on dividing the page erase time into shorter chunks.

4.3.3 Writing to user information configuration registers (UICR)

User information configuration registers (UICR) are written in the same way as flash. After UICR has been written, the new UICR configuration will only take effect after a reset.

UICR can only be written n_{WRITE} number of times before an erase must be performed using ERASEUICR on page 29 or ERASEALL on page 29. The time it takes to write a word to UICR is specified by t_{WRITE}.
The CPU is halted if the CPU executes code from the flash while the NVMC is writing to the UICR.

4.3.4 Erasing user information configuration registers (UICR)

When erase is enabled, UICR can be erased using the ERASEUICR on page 29.

After erasing UICR, all bits in UICR are set to `1`. The time it takes to erase UICR is specified by $t_{ERASEPAGE}$.
The CPU is halted if the CPU executes code from the flash while the NVMC performs the erase operation.

4.3.5 Erase all

When erase is enabled, flash and UICR can be erased completely in one operation by using the ERASEALL on page 29. This operation will not erase the factory information configuration registers (FICR).

The time it takes to perform an ERASEALL command is specified by $t_{ERASEALL}$. The CPU is halted if the CPU executes code from the flash while the NVMC performs the erase operation.

4.3.6 Access port protection behavior

When access port protection is enabled, parts of the NVMC functionality will be blocked in order to prevent intentional or unintentional erase of UICR.

	CTRL-AP ERASEALL	NVMC ERASEPAGE	NVMC ERASEPAGE PARTIAL	NVMC ERASEALL	NVMC ERASEUICR
APPROTECT					
Disabled	Allowed	Allowed	Allowed	Allowed	Allowed
Enabled	Allowed	Allowed	Allowed	Allowed	Blocked

Table 4: NVMC Protection

4.3.7 Partial erase of a page in flash

Partial erase is a feature in the NVMC to split a page erase time into shorter chunks to prevent longer CPU stalls in time-critical applications. Partial erase is only applicable to the code area in flash memory and does not work with UICR.

When erase is enabled, the partial erase of a flash page can be started by writing to ERASEPAGEPARTIAL on page 30. The duration of a partial erase can be configured in ERASEPAGEPARTIALCFG on page 30. A flash page is erased when its erase time reaches $t_{ERASEPAGE}$. Use ERASEPAGEPARTIAL N number of times so that N * ERASEPAGEPARTIALCFG ≥ $t_{ERASEPAGE}$, where N * ERASEPAGEPARTIALCFG gives the cumulative (total) erase time. Every time the cumulative erase time reaches $t_{ERASEPAGE}$, it counts as one erase cycle.

After the erase is complete, all bits in the page are set to `1`. The CPU is halted if the CPU executes code from the flash while the NVMC performs the partial erase operation.

The bits in the page are undefined if the flash page erase is incomplete, i.e. if a partial erase has started but the total erase time is less than $t_{ERASEPAGE}$.

4.3.8 Cache

An instruction cache (I-Cache) can be enabled for the ICODE bus in the NVMC.

A cache hit is an instruction fetch from the cache, and it has a 0 wait-state delay. The number of wait-states for a cache miss, where the instruction is not available in the cache and needs to be fetched from flash, is shown in CPU on page 20.

Enabling the cache can increase CPU performance and reduce power consumption by reducing the number of wait cycles and the number of flash accesses. This will depend on the cache hit rate. Cache will use some current when enabled. If the reduction in average current due to reduced flash accesses is larger than the cache power requirement, the average current to execute the program code will decrease.

When disabled, the cache does not use current and does not retain its content.

It is possible to enable cache profiling to analyze the performance of the cache for your program using the ICACHECNF register. When profiling is enabled, the IHIT and IMISS registers are incremented for every instruction cache hit or miss, respectively. The hit and miss profiling registers do not wrap around after reaching the maximum value. If the maximum value is reached, consider profiling for a shorter duration to get correct numbers.

4.3.9 Registers

Instances

Instance	Base address	Description
NVMC	0x4001E000	Non-volatile memory controller

Register overview

Register	Offset	Description
READY	0x400	Ready flag
READYNEXT	0x408	Ready flag
CONFIG	0x504	Configuration register
ERASEPAGE	0x508	Register for erasing a page in code area
ERASEALL	0x50C	Register for erasing all non-volatile user memory
ERASEUICR	0x514	Register for erasing user information configuration registers
ERASEPAGEPARTIAL	0x518	Register for partial erase of a page in code area
ERASEPAGEPARTIALCFG	0x51C	Register for partial erase configuration
ICACHECNF	0x540	I-code cache configuration register
IHIT	0x548	I-code cache hit counter
IMISS	0x54C	I-code cache miss counter

4.3.9.1 READY

Address offset: 0x400

Ready flag

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000001				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
ID	R/W	Field	Value ID	Value																															
A	R	READY																																	
			Busy	0																															
			Ready	1																															

4.3.9.2 READYNEXT

Address offset: 0x408

Ready flag

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000001				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
ID	R/W	Field	Value ID	Value																															
A	R	READYNEXT																																	
			Busy	0																															
			Ready	1																															

4.3.9.3 CONFIG

Address offset: 0x504

Configuration register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																		A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	WEN																																	
			Ren	0																															
			Wen	1																															
			Een	2																															

Program memory access mode. It is strongly recommended to only activate erase and write modes when they are actively used.

Enabling write or erase will invalidate the cache and keep it invalidated.

Read only access

Write enabled

Erase enabled

4.3.9.4 ERASEPAGE

Address offset: 0x508

Register for erasing a page in code area

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	ERASEPAGE																																	

Register for starting erase of a page in code area

The value is the address to the page to be erased. (Addresses of first word in page). The erase must be enabled using CONFIG.WEN before the page can be erased. Attempts to erase pages that are outside the code area may result in undesirable behavior, e.g. the wrong page may be erased.

4.3.9.5 ERASEALL

Address offset: 0x50C

Register for erasing all non-volatile user memory

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	ERASEALL																																	
			NoOperation	0																															
			Erase	1																															

Erase all non-volatile memory including UICR registers. The erase must be enabled using CONFIG.WEN before the non-volatile memory can be erased.

No operation

Start chip erase

4.3.9.6 ERASEUICR

Address offset: 0x514

Register for erasing user information configuration registers

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	ERASEUICR			Register starting erase of all user information configuration registers. The erase must be enabled using CONFIG.WEN before the UICR can be erased.
			NoOperation	0	No operation
			Erase	1	Start erase of UICR

4.3.9.7 ERASEPAGEPARTIAL

Address offset: 0x518

Register for partial erase of a page in code area

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	ERASEPAGEPARTIAL			Register for starting partial erase of a page in code area The value is the address to the page to be partially erased (address of the first word in page). The erase must be enabled using CONFIG.WEN before every erase page partial and disabled using CONFIG.WEN after every erase page partial. Attempts to erase pages that are outside the code area may result in undesirable behavior, e.g. the wrong page may be erased.

4.3.9.8 ERASEPAGEPARTIALCFG

Address offset: 0x51C

Register for partial erase configuration

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A
Reset 0x0000000A				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DURATION			Duration of the partial erase in milliseconds The user must ensure that the total erase time is long enough for a complete erase of the flash page.

4.3.9.9 ICACHECNF

Address offset: 0x540

I-code cache configuration register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	CACHEEN			Cache enable
			Disabled	0	Disable cache. Invalidates all cache entries.
			Enabled	1	Enable cache
B	RW	CACHEPROFEN			Cache profiling enable
			Disabled	0	Disable cache profiling
			Enabled	1	Enable cache profiling

4.3.9.10 IHIT

Address offset: 0x548

I-code cache hit counter

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	HITS			Number of cache hits. Register is writable, but only to '0'.

4.3.9.11 IMISS

Address offset: 0x54C

I-code cache miss counter

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	MISSES			Number of cache misses. Register is writable, but only to '0'.

4.3.10 Electrical specification

4.3.10.1 Flash programming

Symbol	Description	Min.	Typ.	Max.	Units
n_{WRITE}	Number of times a 32-bit word can be written before erase			2	
$n_{ENDURANCE}$	Erase cycles per page	10000			
t_{WRITE}	Time to write one 32-bit word			41[1]	µs
$t_{ERASEPAGE}$	Time to erase one page			85[1]	ms
$t_{ERASEALL}$	Time to erase all flash			169[1]	ms
$t_{ERASEPAGEPARTIAL,acc}$	Accuracy of the partial page erase duration. Total execution time for one partial page erase is defined as ERASEPAGEPARTIALCFG * $t_{ERASEPAGEPARTIAL,acc}$.			1.05[1]	

[1] Applies when HFXO is used. Timing varies according to HFINT accuracy when HFINT is used.

4.3.10.2 Cache size

Symbol	Description	Min.	Typ.	Max.	Units
$Size_{ICODE}$	I-Code cache size		2048		Bytes

4.4 FICR — Factory information configuration registers

Factory information configuration registers (FICR) are pre-programmed in factory and cannot be erased by the user. These registers contain chip-specific information and configuration.

4.4.1 Registers

Instances

Instance	Base address	Description
FICR	0x10000000	Factory information configuration

Register overview

Register	Offset	Description
CODEPAGESIZE	0x010	Code memory page size
CODESIZE	0x014	Code memory size
DEVICEID[0]	0x060	Device identifier
DEVICEID[1]	0x064	Device identifier
ER[0]	0x080	Encryption root, word 0
ER[1]	0x084	Encryption root, word 1
ER[2]	0x088	Encryption root, word 2
ER[3]	0x08C	Encryption root, word 3
IR[0]	0x090	Identity Root, word 0
IR[1]	0x094	Identity Root, word 1
IR[2]	0x098	Identity Root, word 2
IR[3]	0x09C	Identity Root, word 3
DEVICEADDRTYPE	0x0A0	Device address type
DEVICEADDR[0]	0x0A4	Device address 0
DEVICEADDR[1]	0x0A8	Device address 1
INFO.PART	0x100	Part code
INFO.VARIANT	0x104	Build code, last two letters of Package Variant and first two characters of Build Code, encoded in ASCII.
INFO.PACKAGE	0x108	Package option
INFO.RAM	0x10C	RAM variant
INFO.FLASH	0x110	Flash variant
PRODTEST[0]	0x350	Production test signature 0
PRODTEST[1]	0x354	Production test signature 1
PRODTEST[2]	0x358	Production test signature 2
TEMP.A0	0x404	Slope definition A0
TEMP.A1	0x408	Slope definition A1
TEMP.A2	0x40C	Slope definition A2
TEMP.A3	0x410	Slope definition A3
TEMP.A4	0x414	Slope definition A4
TEMP.A5	0x418	Slope definition A5

Register	Offset	Description
TEMP.B0	0x41C	Y-intercept B0
TEMP.B1	0x420	Y-intercept B1
TEMP.B2	0x424	Y-intercept B2
TEMP.B3	0x428	Y-intercept B3
TEMP.B4	0x42C	Y-intercept B4
TEMP.B5	0x430	Y-intercept B5
TEMP.T0	0x434	Segment end T0
TEMP.T1	0x438	Segment end T1
TEMP.T2	0x43C	Segment end T2
TEMP.T3	0x440	Segment end T3
TEMP.T4	0x444	Segment end T4
NFC.TAGHEADER0	0x450	Default header for NFC tag. Software can read these values to populate NFCID1_3RD_LAST, NFCID1_2ND_LAST, and NFCID1_LAST.
NFC.TAGHEADER1	0x454	Default header for NFC tag. Software can read these values to populate NFCID1_3RD_LAST, NFCID1_2ND_LAST, and NFCID1_LAST.
NFC.TAGHEADER2	0x458	Default header for NFC tag. Software can read these values to populate NFCID1_3RD_LAST, NFCID1_2ND_LAST, and NFCID1_LAST.
NFC.TAGHEADER3	0x45C	Default header for NFC tag. Software can read these values to populate NFCID1_3RD_LAST, NFCID1_2ND_LAST, and NFCID1_LAST.
TRNG90B.BYTES	0xC00	Amount of bytes for the required entropy bits
TRNG90B.RCCUTOFF	0xC04	Repetition counter cutoff
TRNG90B.APCUTOFF	0xC08	Adaptive proportion cutoff
TRNG90B.STARTUP	0xC0C	Amount of bytes for the startup tests
TRNG90B.ROSC1	0xC10	Sample count for ring oscillator 1
TRNG90B.ROSC2	0xC14	Sample count for ring oscillator 2
TRNG90B.ROSC3	0xC18	Sample count for ring oscillator 3
TRNG90B.ROSC4	0xC1C	Sample count for ring oscillator 4

4.4.1.1 CODEPAGESIZE

Address offset: 0x010

Code memory page size

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	R	CODEPAGESIZE			Code memory page size

4.4.1.2 CODESIZE

Address offset: 0x014

Code memory size

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	R	CODESIZE			Code memory size in number of pages Total code space is: CODEPAGESIZE * CODESIZE

4.4.1.3 DEVICEID[0]

Address offset: 0x060

Device identifier

Bit number																															
ID																															
Reset 0xFFFFFFFF																															
ID	R/W	Field	Value ID	Value	Description																										
A	R	DEVICEID			64 bit unique device identifier DEVICEID[0] contains the least significant bits of the device identifier. DEVICEID[1] contains the most significant bits of the device identifier.																										

4.4.1.4 DEVICEID[1]

Address offset: 0x064

Device identifier

Bit number																															
ID																															
Reset 0xFFFFFFFF																															
ID	R/W	Field	Value ID	Value	Description																										
A	R	DEVICEID			64 bit unique device identifier DEVICEID[0] contains the least significant bits of the device identifier. DEVICEID[1] contains the most significant bits of the device identifier.																										

4.4.1.5 ER[0]

Address offset: 0x080

Encryption root, word 0

Bit number																															
ID																															
Reset 0xFFFFFFFF																															
ID	R/W	Field	Value ID	Value	Description																										
A	R	ER			Encryption root, word 0																										

4.4.1.6 ER[1]

Address offset: 0x084

Encryption root, word 1

Bit number																															
ID																															
Reset 0xFFFFFFFF																															
ID	R/W	Field	Value ID	Value	Description																										
A	R	ER			Encryption root, word 1																										

4.4.1.7 ER[2]

Address offset: 0x088

Encryption root, word 2

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF					**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID		Value																															
A	R	ER																																		
					Encryption root, word 2																																

4.4.1.8 ER[3]

Address offset: 0x08C

Encryption root, word 3

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF					**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID		Value																															
A	R	ER																																		
					Encryption root, word 3																																

4.4.1.9 IR[0]

Address offset: 0x090

Identity Root, word 0

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF					**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID		Value																															
A	R	IR																																		
					Identity Root, word 0																																

4.4.1.10 IR[1]

Address offset: 0x094

Identity Root, word 1

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF					**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID		Value																															
A	R	IR																																		
					Identity Root, word 1																																

4.4.1.11 IR[2]

Address offset: 0x098

Identity Root, word 2

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	R	IR			Identity Root, word 2

4.4.1.12 IR[3]

Address offset: 0x09C

Identity Root, word 3

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	R	IR			Identity Root, word 3

4.4.1.13 DEVICEADDRTYPE

Address offset: 0x0A0

Device address type

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	R	DEVICEADDRTYPE			Device address type
			Public	0	Public address
			Random	1	Random address

4.4.1.14 DEVICEADDR[0]

Address offset: 0x0A4

Device address 0

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	R	DEVICEADDR			48 bit device address DEVICEADDR[0] contains the least significant bits of the device address. DEVICEADDR[1] contains the most significant bits of the device address. Only bits [15:0] of DEVICEADDR[1] are used.

4.4.1.15 DEVICEADDR[1]

Address offset: 0x0A8

Device address 1

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	R	DEVICEADDR			48 bit device address DEVICEADDR[0] contains the least significant bits of the device address. DEVICEADDR[1] contains the most significant bits of the device address. Only bits [15:0] of DEVICEADDR[1] are used.

4.4.1.16 INFO

Device info

4.4.1.16.1 INFO.PART

Address offset: 0x100

Part code

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00052840				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 1 0 0 1 0 1 0 0 0 0 1 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	PART			Part code
			N52840	0x52840	nRF52840
			Unspecified	0xFFFFFFFF	Unspecified

4.4.1.16.2 INFO.VARIANT

Address offset: 0x104

Build code, last two letters of Package Variant and first two characters of Build Code, encoded in ASCII.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	R	VARIANT			For valid values see SoC revisions and variants.

4.4.1.16.3 INFO.PACKAGE

Address offset: 0x108

Package option

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	R	PACKAGE			Package option
			QI	0x2004	QIxx - 7x7 73-pin aQFN
			QF	0x2000	QFxx - 6x6 48-pin QFN
			CK	0x2005	CKxx - 3.544 x 3.607 WLCSP
			Unspecified	0xFFFFFFFF	Unspecified

4.4.1.16.4 INFO.RAM

Address offset: 0x10C

RAM variant

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	R	RAM																																	
			K16	0x10																															
			K32	0x20																															
			K64	0x40																															
			K128	0x80																															
			K256	0x100																															
			Unspecified	0xFFFFFFFF																															

4.4.1.16.5 INFO.FLASH

Address offset: 0x110

Flash variant

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	R	FLASH																																	
			K128	0x80																															
			K256	0x100																															
			K512	0x200																															
			K1024	0x400																															
			K2048	0x800																															
			Unspecified	0xFFFFFFFF																															

4.4.1.17 PRODTEST[0]

Address offset: 0x350

Production test signature 0

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	R	PRODTEST																																	
			Done	0xBB42319F																															
			NotDone	0xFFFFFFFF																															

4.4.1.18 PRODTEST[1]

Address offset: 0x354

Production test signature 1

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	R	PRODTEST			Production test signature 1
			Done	0xBB42319F	Production tests done
			NotDone	0xFFFFFFFF	Production tests not done

4.4.1.19 PRODTEST[2]

Address offset: 0x358

Production test signature 2

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	R	PRODTEST			Production test signature 2
			Done	0xBB42319F	Production tests done
			NotDone	0xFFFFFFFF	Production tests not done

4.4.1.20 TEMP

Registers storing factory TEMP module linearization coefficients

4.4.1.20.1 TEMP.A0

Address offset: 0x404

Slope definition A0

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	R	A			A (slope definition) register.

4.4.1.20.2 TEMP.A1

Address offset: 0x408

Slope definition A1

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	R	A			A (slope definition) register.

4.4.1.20.3 TEMP.A2

Address offset: 0x40C

Slope definition A2

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																					A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	R	A			A (slope definition) register.

4.4.1.20.4 TEMP.A3

Address offset: 0x410

Slope definition A3

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																					A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	R	A			A (slope definition) register.

4.4.1.20.5 TEMP.A4

Address offset: 0x414

Slope definition A4

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																					A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	R	A			A (slope definition) register.

4.4.1.20.6 TEMP.A5

Address offset: 0x418

Slope definition A5

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																					A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	R	A			A (slope definition) register.

4.4.1.20.7 TEMP.B0

Address offset: 0x41C

Y-intercept B0

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																			A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	R	B			B (y-intercept)

4.4.1.20.8 TEMP.B1

Address offset: 0x420

Y-intercept B1

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																			A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	R	B			B (y-intercept)

4.4.1.20.9 TEMP.B2

Address offset: 0x424

Y-intercept B2

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																			A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	R	B			B (y-intercept)

4.4.1.20.10 TEMP.B3

Address offset: 0x428

Y-intercept B3

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																			A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	R	B			B (y-intercept)

4.4.1.20.11 TEMP.B4

Address offset: 0x42C

Y-intercept B4

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																			A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	R	B			B (y-intercept)

4.4.1.20.12 TEMP.B5

Address offset: 0x430

Y-intercept B5

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A A A A A A
Reset 0xFFFFFFFF	**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**

ID	R/W	Field	Value ID	Value	Description
A	R	B			B (y-intercept)

4.4.1.20.13 TEMP.T0

Address offset: 0x434

Segment end T0

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A
Reset 0xFFFFFFFF	**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**

ID	R/W	Field	Value ID	Value	Description
A	R	T			T (segment end) register

4.4.1.20.14 TEMP.T1

Address offset: 0x438

Segment end T1

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A
Reset 0xFFFFFFFF	**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**

ID	R/W	Field	Value ID	Value	Description
A	R	T			T (segment end) register

4.4.1.20.15 TEMP.T2

Address offset: 0x43C

Segment end T2

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A
Reset 0xFFFFFFFF	**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**

ID	R/W	Field	Value ID	Value	Description
A	R	T			T (segment end) register

4.4.1.20.16 TEMP.T3

Address offset: 0x440

Segment end T3

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A
Reset 0xFFFFFFFF	**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**

ID	R/W	Field	Value ID	Value	Description
A	R	T			T (segment end) register

4.4.1.20.17 TEMP.T4

Address offset: 0x444

Segment end T4

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																												A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	R	T																																	
				T (segment end) register																															

4.4.1.21 NFC.TAGHEADER0

Address offset: 0x450

Default header for NFC tag. Software can read these values to populate NFCID1_3RD_LAST, NFCID1_2ND_LAST, and NFCID1_LAST.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				D	D	D	D	D	D	D	D	C	C	C	C	C	C	C	C	B	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A
Reset 0xFFFFFF5F				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**0**	**1**	**0**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	R	MFGID																																	
				Default Manufacturer ID: Nordic Semiconductor ASA has ICM 0x5F																																	
B	R	UD1																																	
				Unique identifier byte 1																																	
C	R	UD2																																	
				Unique identifier byte 2																																	
D	R	UD3																																	
				Unique identifier byte 3																																	

4.4.1.22 NFC.TAGHEADER1

Address offset: 0x454

Default header for NFC tag. Software can read these values to populate NFCID1_3RD_LAST, NFCID1_2ND_LAST, and NFCID1_LAST.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				D	D	D	D	D	D	D	D	C	C	C	C	C	C	C	C	B	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	R	UD4																																	
				Unique identifier byte 4																																	
B	R	UD5																																	
				Unique identifier byte 5																																	
C	R	UD6																																	
				Unique identifier byte 6																																	
D	R	UD7																																	
				Unique identifier byte 7																																	

4.4.1.23 NFC.TAGHEADER2

Address offset: 0x458

Default header for NFC tag. Software can read these values to populate NFCID1_3RD_LAST, NFCID1_2ND_LAST, and NFCID1_LAST.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				D	D	D	D	D	D	D	D	C	C	C	C	C	C	C	C	B	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	R	UD8																																	
				Unique identifier byte 8																															
B	R	UD9																																	
				Unique identifier byte 9																															
C	R	UD10																																	
				Unique identifier byte 10																															
D	R	UD11																																	
				Unique identifier byte 11																															

4.4.1.24 NFC.TAGHEADER3

Address offset: 0x45C

Default header for NFC tag. Software can read these values to populate NFCID1_3RD_LAST, NFCID1_2ND_LAST, and NFCID1_LAST.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				D	D	D	D	D	D	D	D	C	C	C	C	C	C	C	C	B	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	R	UD12																																	
				Unique identifier byte 12																															
B	R	UD13																																	
				Unique identifier byte 13																															
C	R	UD14																																	
				Unique identifier byte 14																															
D	R	UD15																																	
				Unique identifier byte 15																															

4.4.1.25 TRNG90B

NIST800-90B RNG calibration data

4.4.1.25.1 TRNG90B.BYTES

Address offset: 0xC00

Amount of bytes for the required entropy bits

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	R	BYTES																																	
				Amount of bytes for the required entropy bits																															

4.4.1.25.2 TRNG90B.RCCUTOFF

Address offset: 0xC04

Repetition counter cutoff

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	R	RCCUTOFF																																	
				Repetition counter cutoff																															

4.4.1.25.3 TRNG90B.APCUTOFF

Address offset: 0xC08

Adaptive proportion cutoff

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	R	APCUTOFF			Adaptive proportion cutoff

4.4.1.25.4 TRNG90B.STARTUP

Address offset: 0xC0C

Amount of bytes for the startup tests

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	R	STARTUP			Amount of bytes for the startup tests

4.4.1.25.5 TRNG90B.ROSC1

Address offset: 0xC10

Sample count for ring oscillator 1

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	R	ROSC1			Sample count for ring oscillator 1

4.4.1.25.6 TRNG90B.ROSC2

Address offset: 0xC14

Sample count for ring oscillator 2

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	R	ROSC2			Sample count for ring oscillator 2

4.4.1.25.7 TRNG90B.ROSC3

Address offset: 0xC18

Sample count for ring oscillator 3

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	R	ROSC3			Sample count for ring oscillator 3

4.4.1.25.8 TRNG90B.ROSC4

Address offset: 0xC1C

Sample count for ring oscillator 4

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	R	ROSC4			Sample count for ring oscillator 4

4.5 UICR — User information configuration registers

The user information configuration registers (UICRs) are non-volatile memory (NVM) registers for configuring user-specific settings.

For information on writing UICR registers, see the NVMC — Non-volatile memory controller on page 25 and Memory on page 21 chapters.

4.5.1 Registers

Instances

Instance	Base address	Description
UICR	0x10001000	User information configuration

Register overview

Register	Offset	Description
UNUSED0	0x000	This address is reserved
UNUSED1	0x004	This address is reserved
UNUSED2	0x008	This address is reserved
UNUSED3	0x010	This address is reserved
NRFFW[0]	0x014	Reserved for Nordic firmware design
NRFFW[1]	0x018	Reserved for Nordic firmware design
NRFFW[2]	0x01C	Reserved for Nordic firmware design
NRFFW[3]	0x020	Reserved for Nordic firmware design
NRFFW[4]	0x024	Reserved for Nordic firmware design
NRFFW[5]	0x028	Reserved for Nordic firmware design
NRFFW[6]	0x02C	Reserved for Nordic firmware design
NRFFW[7]	0x030	Reserved for Nordic firmware design
NRFFW[8]	0x034	Reserved for Nordic firmware design
NRFFW[9]	0x038	Reserved for Nordic firmware design
NRFFW[10]	0x03C	Reserved for Nordic firmware design
NRFFW[11]	0x040	Reserved for Nordic firmware design
NRFFW[12]	0x044	Reserved for Nordic firmware design
NRFHW[0]	0x050	Reserved for Nordic hardware design
NRFHW[1]	0x054	Reserved for Nordic hardware design
NRFHW[2]	0x058	Reserved for Nordic hardware design
NRFHW[3]	0x05C	Reserved for Nordic hardware design
NRFHW[4]	0x060	Reserved for Nordic hardware design

Register	Offset	Description
NRFHW[5]	0x064	Reserved for Nordic hardware design
NRFHW[6]	0x068	Reserved for Nordic hardware design
NRFHW[7]	0x06C	Reserved for Nordic hardware design
NRFHW[8]	0x070	Reserved for Nordic hardware design
NRFHW[9]	0x074	Reserved for Nordic hardware design
NRFHW[10]	0x078	Reserved for Nordic hardware design
NRFHW[11]	0x07C	Reserved for Nordic hardware design
CUSTOMER[0]	0x080	Reserved for customer
CUSTOMER[1]	0x084	Reserved for customer
CUSTOMER[2]	0x088	Reserved for customer
CUSTOMER[3]	0x08C	Reserved for customer
CUSTOMER[4]	0x090	Reserved for customer
CUSTOMER[5]	0x094	Reserved for customer
CUSTOMER[6]	0x098	Reserved for customer
CUSTOMER[7]	0x09C	Reserved for customer
CUSTOMER[8]	0x0A0	Reserved for customer
CUSTOMER[9]	0x0A4	Reserved for customer
CUSTOMER[10]	0x0A8	Reserved for customer
CUSTOMER[11]	0x0AC	Reserved for customer
CUSTOMER[12]	0x0B0	Reserved for customer
CUSTOMER[13]	0x0B4	Reserved for customer
CUSTOMER[14]	0x0B8	Reserved for customer
CUSTOMER[15]	0x0BC	Reserved for customer
CUSTOMER[16]	0x0C0	Reserved for customer
CUSTOMER[17]	0x0C4	Reserved for customer
CUSTOMER[18]	0x0C8	Reserved for customer
CUSTOMER[19]	0x0CC	Reserved for customer
CUSTOMER[20]	0x0D0	Reserved for customer
CUSTOMER[21]	0x0D4	Reserved for customer
CUSTOMER[22]	0x0D8	Reserved for customer
CUSTOMER[23]	0x0DC	Reserved for customer
CUSTOMER[24]	0x0E0	Reserved for customer
CUSTOMER[25]	0x0E4	Reserved for customer
CUSTOMER[26]	0x0E8	Reserved for customer
CUSTOMER[27]	0x0EC	Reserved for customer
CUSTOMER[28]	0x0F0	Reserved for customer
CUSTOMER[29]	0x0F4	Reserved for customer
CUSTOMER[30]	0x0F8	Reserved for customer
CUSTOMER[31]	0x0FC	Reserved for customer
PSELRESET[0]	0x200	Mapping of the nRESET function (see POWER chapter for details)
PSELRESET[1]	0x204	Mapping of the nRESET function (see POWER chapter for details)
APPROTECT	0x208	Access port protection
NFCPINS	0x20C	Setting of pins dedicated to NFC functionality: NFC antenna or GPIO
DEBUGCTRL	0x210	Processor debug control
REGOUT0	0x304	Output voltage from REG0 regulator stage. The maximum output voltage from this stage is given as VDDH - V_VDDH-VDD.

4.5.1.1 UNUSED0

Address offset: 0x000

This address is reserved

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description

4.5.1.2 UNUSED1

Address offset: 0x004

This address is reserved

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description

4.5.1.3 UNUSED2

Address offset: 0x008

This address is reserved

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description

4.5.1.4 UNUSED3

Address offset: 0x010

This address is reserved

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description

4.5.1.5 NRFFW[0]

Address offset: 0x014

Reserved for Nordic firmware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

4.5.1.6 NRFFW[1]

Address offset: 0x018

Reserved for Nordic firmware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

4.5.1.7 NRFFW[2]

Address offset: 0x01C

Reserved for Nordic firmware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

4.5.1.8 NRFFW[3]

Address offset: 0x020

Reserved for Nordic firmware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

4.5.1.9 NRFFW[4]

Address offset: 0x024

Reserved for Nordic firmware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

4.5.1.10 NRFFW[5]

Address offset: 0x028

Reserved for Nordic firmware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

4.5.1.11 NRFFW[6]

Address offset: 0x02C

Reserved for Nordic firmware design

Bit number																																
ID																																
Reset 0xFFFFFFFF																																
ID	R/W	Field																														
A	RW	NRFFW																														

Value ID	Value	Description
		Reserved for Nordic firmware design

4.5.1.12 NRFFW[7]

Address offset: 0x030

Reserved for Nordic firmware design

Bit number																																
ID																																
Reset 0xFFFFFFFF																																
ID	R/W	Field																														
A	RW	NRFFW																														

Value ID	Value	Description
		Reserved for Nordic firmware design

4.5.1.13 NRFFW[8]

Address offset: 0x034

Reserved for Nordic firmware design

Bit number																																
ID																																
Reset 0xFFFFFFFF																																
ID	R/W	Field																														
A	RW	NRFFW																														

Value ID	Value	Description
		Reserved for Nordic firmware design

4.5.1.14 NRFFW[9]

Address offset: 0x038

Reserved for Nordic firmware design

Bit number																																
ID																																
Reset 0xFFFFFFFF																																
ID	R/W	Field																														
A	RW	NRFFW																														

Value ID	Value	Description
		Reserved for Nordic firmware design

4.5.1.15 NRFFW[10]

Address offset: 0x03C

Reserved for Nordic firmware design

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

4.5.1.16 NRFFW[11]

Address offset: 0x040

Reserved for Nordic firmware design

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

4.5.1.17 NRFFW[12]

Address offset: 0x044

Reserved for Nordic firmware design

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	NRFFW			Reserved for Nordic firmware design

4.5.1.18 NRFHW[0]

Address offset: 0x050

Reserved for Nordic hardware design

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	NRFHW			Reserved for Nordic hardware design

4.5.1.19 NRFHW[1]

Address offset: 0x054

Reserved for Nordic hardware design

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	NRFHW			Reserved for Nordic hardware design

4.5.1.20 NRFHW[2]

Address offset: 0x058

Reserved for Nordic hardware design

Bit number																																
ID																																
Reset 0xFFFFFFFF																																
ID	R/W	Field																														
A	RW	NRFHW																														

Value ID	Value	Description
		Reserved for Nordic hardware design

4.5.1.21 NRFHW[3]

Address offset: 0x05C

Reserved for Nordic hardware design

Bit number																																
ID																																
Reset 0xFFFFFFFF																																
ID	R/W	Field																														
A	RW	NRFHW																														

Value ID	Value	Description
		Reserved for Nordic hardware design

4.5.1.22 NRFHW[4]

Address offset: 0x060

Reserved for Nordic hardware design

Bit number																																
ID																																
Reset 0xFFFFFFFF																																
ID	R/W	Field																														
A	RW	NRFHW																														

Value ID	Value	Description
		Reserved for Nordic hardware design

4.5.1.23 NRFHW[5]

Address offset: 0x064

Reserved for Nordic hardware design

Bit number																																
ID																																
Reset 0xFFFFFFFF																																
ID	R/W	Field																														
A	RW	NRFHW																														

Value ID	Value	Description
		Reserved for Nordic hardware design

4.5.1.24 NRFHW[6]

Address offset: 0x068

Reserved for Nordic hardware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	NRFHW			Reserved for Nordic hardware design

4.5.1.25 NRFHW[7]

Address offset: 0x06C

Reserved for Nordic hardware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	NRFHW			Reserved for Nordic hardware design

4.5.1.26 NRFHW[8]

Address offset: 0x070

Reserved for Nordic hardware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	NRFHW			Reserved for Nordic hardware design

4.5.1.27 NRFHW[9]

Address offset: 0x074

Reserved for Nordic hardware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	NRFHW			Reserved for Nordic hardware design

4.5.1.28 NRFHW[10]

Address offset: 0x078

Reserved for Nordic hardware design

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	NRFHW			Reserved for Nordic hardware design

4.5.1.29 NRFHW[11]

Address offset: 0x07C

Reserved for Nordic hardware design

Bit number																																
ID																																
Reset 0xFFFFFFFF																																
ID	R/W	Field																														
A	RW	NRFHW																														

Value ID	Value	Description
		Reserved for Nordic hardware design

4.5.1.30 CUSTOMER[0]

Address offset: 0x080

Reserved for customer

Bit number																																
ID																																
Reset 0xFFFFFFFF																																
ID	R/W	Field																														
A	RW	CUSTOMER																														

Value ID	Value	Description
		Reserved for customer

4.5.1.31 CUSTOMER[1]

Address offset: 0x084

Reserved for customer

Bit number																																
ID																																
Reset 0xFFFFFFFF																																
ID	R/W	Field																														
A	RW	CUSTOMER																														

Value ID	Value	Description
		Reserved for customer

4.5.1.32 CUSTOMER[2]

Address offset: 0x088

Reserved for customer

Bit number																																
ID																																
Reset 0xFFFFFFFF																																
ID	R/W	Field																														
A	RW	CUSTOMER																														

Value ID	Value	Description
		Reserved for customer

4.5.1.33 CUSTOMER[3]

Address offset: 0x08C

Reserved for customer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.34 CUSTOMER[4]

Address offset: 0x090

Reserved for customer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.35 CUSTOMER[5]

Address offset: 0x094

Reserved for customer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.36 CUSTOMER[6]

Address offset: 0x098

Reserved for customer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.37 CUSTOMER[7]

Address offset: 0x09C

Reserved for customer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.38 CUSTOMER[8]

Address offset: 0x0A0

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.39 CUSTOMER[9]

Address offset: 0x0A4

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.40 CUSTOMER[10]

Address offset: 0x0A8

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.41 CUSTOMER[11]

Address offset: 0x0AC

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.42 CUSTOMER[12]

Address offset: 0x0B0

Reserved for customer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW	CUSTOMER																																	
				Reserved for customer																															

4.5.1.43 CUSTOMER[13]

Address offset: 0x0B4

Reserved for customer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW	CUSTOMER																																	
				Reserved for customer																															

4.5.1.44 CUSTOMER[14]

Address offset: 0x0B8

Reserved for customer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW	CUSTOMER																																	
				Reserved for customer																															

4.5.1.45 CUSTOMER[15]

Address offset: 0x0BC

Reserved for customer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW	CUSTOMER																																	
				Reserved for customer																															

4.5.1.46 CUSTOMER[16]

Address offset: 0x0C0

Reserved for customer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW	CUSTOMER																																	
				Reserved for customer																															

4.5.1.47 CUSTOMER[17]

Address offset: 0x0C4

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.48 CUSTOMER[18]

Address offset: 0x0C8

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.49 CUSTOMER[19]

Address offset: 0x0CC

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.50 CUSTOMER[20]

Address offset: 0x0D0

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.51 CUSTOMER[21]

Address offset: 0x0D4

Reserved for customer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.52 CUSTOMER[22]

Address offset: 0x0D8

Reserved for customer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.53 CUSTOMER[23]

Address offset: 0x0DC

Reserved for customer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.54 CUSTOMER[24]

Address offset: 0x0E0

Reserved for customer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.55 CUSTOMER[25]

Address offset: 0x0E4

Reserved for customer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.56 CUSTOMER[26]

Address offset: 0x0E8

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.57 CUSTOMER[27]

Address offset: 0x0EC

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.58 CUSTOMER[28]

Address offset: 0x0F0

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.59 CUSTOMER[29]

Address offset: 0x0F4

Reserved for customer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.60 CUSTOMER[30]

Address offset: 0x0F8

Reserved for customer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.61 CUSTOMER[31]

Address offset: 0x0FC

Reserved for customer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	CUSTOMER			Reserved for customer

4.5.1.62 PSELRESET[0]

Address offset: 0x200

Mapping of the nRESET function (see POWER chapter for details)

Note: All PSELRESET registers have to contain the same value for a pin mapping to be valid. If values are not the same, there will be no nRESET function exposed on a GPIO. As a result, the device will always start independently of the levels present on any of the GPIOs.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		18	GPIO pin number onto which nRESET is exposed
B	RW	PORT		0	Port number onto which nRESET is exposed
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

4.5.1.63 PSELRESET[1]

Address offset: 0x204

Mapping of the nRESET function (see POWER chapter for details)

Note: All PSELRESET registers have to contain the same value for a pin mapping to be valid. If values are not the same, there will be no nRESET function exposed on a GPIO. As a result, the device will always start independently of the levels present on any of the GPIOs.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		18	GPIO pin number onto which nRESET is exposed
B	RW	PORT		0	Port number onto which nRESET is exposed
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

4.5.1.64 APPROTECT

Address offset: 0x208

Access port protection

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PALL			Enable or disable access port protection. See Debug and trace on page 67 for more information.
			Disabled	0xFF	Hardware disable of access port protection for devices where access port protection is controlled by hardware
			HwDisabled	0x5A	Hardware disable of access port protection for devices where access port protection is controlled by hardware and software
			Enabled	0x00	Enable

4.5.1.65 NFCPINS

Address offset: 0x20C

Setting of pins dedicated to NFC functionality: NFC antenna or GPIO

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PROTECT			Setting of pins dedicated to NFC functionality
			Disabled	0	Operation as GPIO pins. Same protection as normal GPIO pins.
			NFC	1	Operation as NFC antenna pins. Configures the protection for NFC operation.

4.5.1.66 DEBUGCTRL

Address offset: 0x210

Processor debug control

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B B B B B B B B A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	CPUNIDEN			Configure CPU non-intrusive debug features
			Enabled	0xFF	Enable CPU ITM and ETM functionality (default behavior)
			Disabled	0x00	Disable CPU ITM and ETM functionality
B	RW	CPUFPBEN			Configure CPU flash patch and breakpoint (FPB) unit behavior
			Enabled	0xFF	Enable CPU FPB unit (default behavior)
			Disabled	0x00	Disable CPU FPB unit. Writes into the FPB registers will be ignored.

4.5.1.67 REGOUT0

Address offset: 0x304

Output voltage from REG0 regulator stage. The maximum output voltage from this stage is given as VDDH - V_VDDH-VDD.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	VOUT			Output voltage from REG0 regulator stage.
			1V8	0	1.8 V
			2V1	1	2.1 V
			2V4	2	2.4 V
			2V7	3	2.7 V
			3V0	4	3.0 V
			3V3	5	3.3 V
			DEFAULT	7	Default voltage: 1.8 V

4.6 EasyDMA

EasyDMA is a module implemented by some peripherals to gain direct access to Data RAM.

EasyDMA is an AHB bus master similar to CPU and is connected to the AHB multilayer interconnect for direct access to Data RAM. EasyDMA is not able to access flash.

A peripheral can implement multiple EasyDMA instances to provide dedicated channels. For example, for reading and writing of data between the peripheral and RAM. This concept is illustrated in EasyDMA example on page 64.

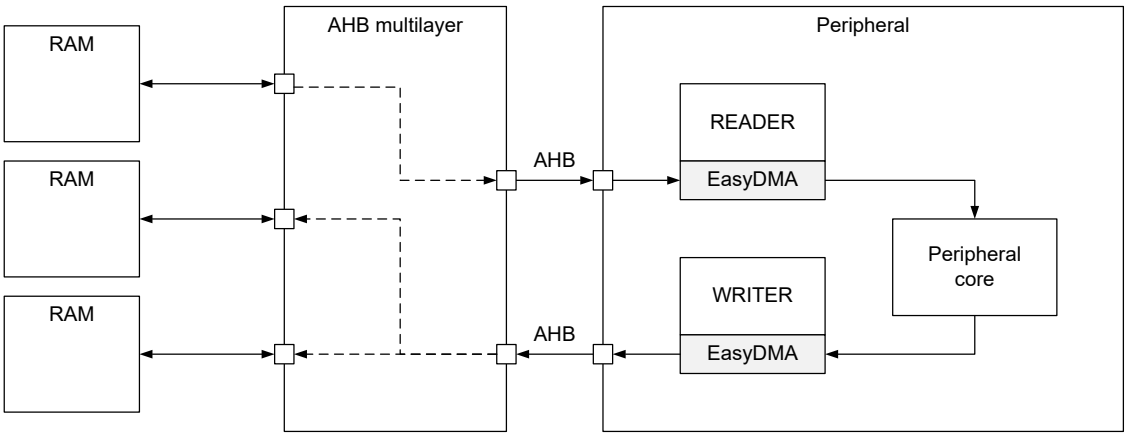


Figure 4: EasyDMA example

An EasyDMA channel is implemented in the following way, but some variations may occur:

```
READERBUFFER_SIZE 5
WRITERBUFFER_SIZE 6

uint8_t readerBuffer[READERBUFFER_SIZE]  __at__ 0x20000000;
uint8_t writerBuffer[WRITERBUFFER_SIZE]  __at__ 0x20000005;

// Configuring the READER channel
MYPERIPHERAL->READER.MAXCNT = READERBUFFER_SIZE;
MYPERIPHERAL->READER.PTR = &readerBuffer;

// Configure the WRITER channel
MYPERIPHERAL->WRITER.MAXCNT = WRITEERBUFFER_SIZE;
MYPERIPHERAL->WRITER.PTR = &writerBuffer;
```

This example shows a peripheral called MYPERIPHERAL that implements two EasyDMA channels - one for reading called READER, and one for writing called WRITER. When the peripheral is started, it is assumed that the peripheral will perform the following tasks:

- Read 5 bytes from the readerBuffer located in RAM at address 0x20000000
- Process the data
- Write no more than 6 bytes back to the writerBuffer located in RAM at address 0x20000005

The memory layout of these buffers is illustrated in EasyDMA memory layout on page 64.

0x20000000	readerBuffer[0]	readerBuffer[1]	readerBuffer[2]	readerBuffer[3]
0x20000004	readerBuffer[4]	writerBuffer[0]	writerBuffer[1]	writerBuffer[2]
0x20000008	writerBuffer[3]	writerBuffer[4]	writerBuffer[5]	

Figure 5: EasyDMA memory layout

The WRITER.MAXCNT register should not be specified larger than the actual size of the buffer (writerBuffer). Otherwise, the channel would overflow the writerBuffer.

Once an EasyDMA transfer is completed, the AMOUNT register can be read by the CPU to see how many bytes were transferred. For example, CPU can read MYPERIPHERAL->WRITER.AMOUNT register to see how many bytes WRITER wrote to RAM.

Note: The PTR register of a READER or WRITER must point to a valid memory region before use. The reset value of a PTR register is not guaranteed to point to valid memory. See Memory on page 21 for more information about the different memory regions and EasyDMA connectivity.

4.6.1 EasyDMA error handling

Some errors may occur during DMA handling.

If READER.PTR or WRITER.PTR is not pointing to a valid memory region, an EasyDMA transfer may result in a HardFault or RAM corruption. See Memory on page 21 for more information about the different memory regions.

If several AHB bus masters try to access the same AHB slave at the same time, AHB bus congestion might occur. An EasyDMA channel is an AHB master. Depending on the peripheral, the peripheral may either stall and wait for access to be granted, or lose data.

4.6.2 EasyDMA array list

EasyDMA is able to operate in Array List mode.

The Array List mode is implemented in channels where the LIST register is available.

The array list does not provide a mechanism to explicitly specify where the next item in the list is located. Instead, it assumes that the list is organized as a linear array where items are located one after the other in RAM.

The EasyDMA Array List can be implemented by using the data structure ArrayList_type as illustrated in the code example below using a READER EasyDMA channel as an example:

```
#define BUFFER_SIZE  4

typedef struct ArrayList
{
  uint8_t buffer[BUFFER_SIZE];
} ArrayList_type;

ArrayList_type ReaderList[3]  __at__ 0x20000000;

MYPERIPHERAL->READER.MAXCNT = BUFFER_SIZE;
MYPERIPHERAL->READER.PTR = &ReaderList;
MYPERIPHERAL->READER.LIST = MYPERIPHERAL_READER_LIST_ArrayList;
```

The data structure only includes a buffer with size equal to the size of READER.MAXCNT register. EasyDMA uses the READER.MAXCNT register to determine when the buffer is full.

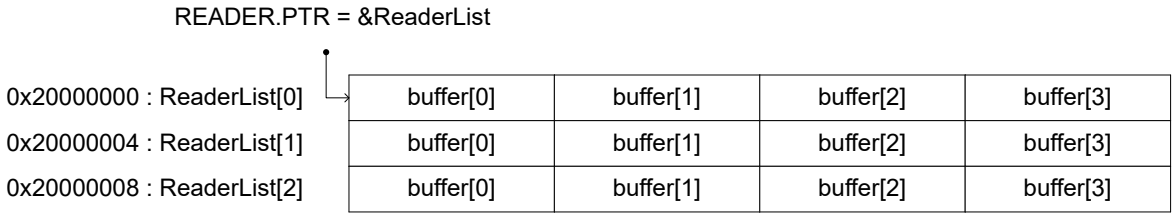


Figure 6: EasyDMA array list

4.7 AHB multilayer

AHB multilayer enables parallel access paths between multiple masters and slaves in a system. Access is resolved using priorities.

Each bus master is connected to all the slave devices using an interconnection matrix. The bus masters are assigned priorities, which are used to resolve access when two (or more) bus masters request access to the same slave device. When that occurs, the following rules apply:

- If two (or more) bus masters request access to the same slave device, the master with the highest priority is granted the access first.
- Bus masters with lower priority are stalled until the higher priority master has completed its transaction.
- If the higher priority master pauses at any point during its transaction, the lower priority master in queue is temporarily granted access to the slave device until the higher priority master resumes its activity.
- Bus masters that have the same priority are mutually exclusive, thus cannot be used concurrently.

Some peripherals, such as RADIO, do not have a safe stalling mechanism (no internal data buffering, or opportunity to pause incoming data). Being a low priority bus master might cause loss of data for such peripherals upon bus contention. To avoid AHB bus contention when using multiple bus masters, follow these guidelines:

- Avoid situations where more than one bus master is accessing the same slave.
- If more than one bus master is accessing the same slave, make sure that the bus bandwidth is not exhausted.

Below is a list of bus masters in the system and their priorities.

Bus master name	Description
CPU	
CTRL-AP	
USB	
CRYPTOCELL	
SPIM1/SPIS1/TWIM1/TWIS1	Same priority and mutually exclusive
RADIO	
CCM/ECB/AAR	Same priority and mutually exclusive
SAADC	
UARTE0	
SPIM0/SPIS0/TWIM0/TWIS0	Same priority and mutually exclusive
SPIM2/SPIS2	Same priority and mutually exclusive
NFCT	
I2S	
PDM	
PWM0	
PWM1	
PWM2	
QSPI	
PWM3	
UARTE1	
SPIM3	

Table 5: AHB bus masters (listed from highest to lowest priority)

Defined bus masters are the CPU and peripherals with implemented EasyDMA. The available slaves are RAM AHB slaves. How the bus masters and slaves are connected using the interconnection matrix is illustrated in Memory on page 21.

4.8 Debug and trace

The debug and trace system offers a flexible and powerful mechanism for non-intrusive debugging.

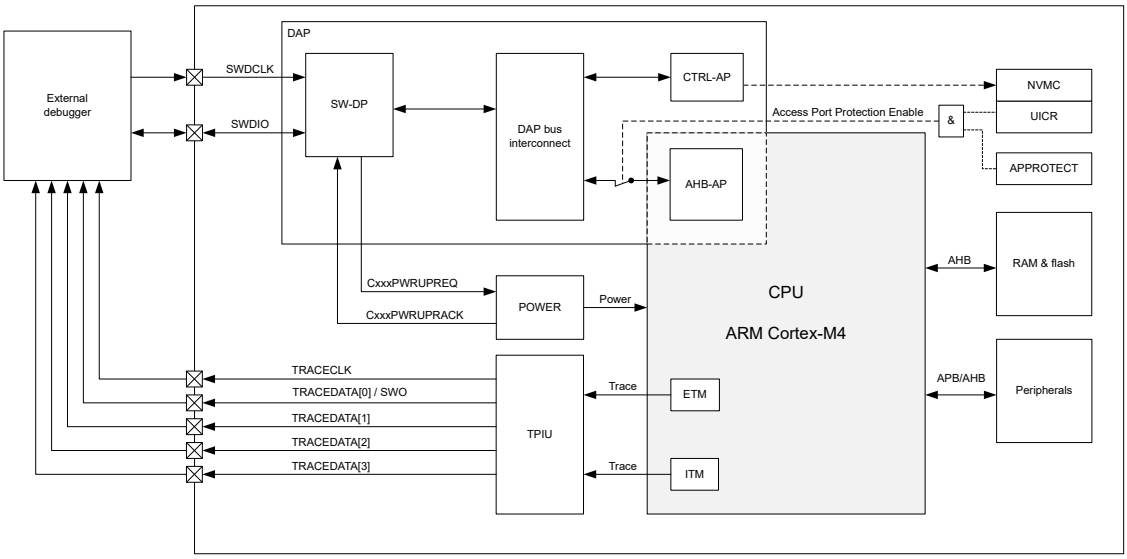


Figure 7: Debug and trace overview

The main features of the debug and trace system are the following:

- Two-pin serial wire debug (SWD) interface

- Flash patch and breakpoint (FPB) unit that supports the following comparators:
  - Two literal comparators
  - Six instruction comparators
- Data watchpoint and trace (DWT) unit with four comparators
- Instrumentation trace macrocell (ITM)
- Embedded trace macrocell (ETM)
- Trace port interface unit (TPIU)
  - 4-bit parallel trace of ITM and ETM trace data
  - Serial wire output (SWO) trace of ITM data

4.8.1 DAP - Debug access port

An external debugger can access the device via the DAP.

The debug access port (DAP) implements a standard ARM CoreSight serial wire debug port (SW-DP), which implements the serial wire debug protocol (SWD). SWD is a two-pin serial interface, see SWDCLK and SWDIO in Debug and trace overview on page 67.

In addition to the default access port in CPU (AHB-AP), the DAP includes a custom control access port (CTRL-AP). The CTRL-AP is described in more detail in CTRL-AP - Control access port on page 71.

Note:

- The SWDIO line has an internal pull-up resistor.
- The SWDCLK line has an internal pull-down resistor.

4.8.2 Access port protection

Access port protection blocks the debugger from read and write access to all CPU registers and memory-mapped addresses when enabled.

Access port protection is enabled and disabled differently depending on the build code of the device.

Access port protection controlled by hardware

This information refers to build codes Dxx and earlier.

By default, access port protection is disabled.

Access port protection is enabled by writing UICR.APPROTECT to `Enabled` and performing any reset. See Reset on page 89 for more information.

Access port protection is disabled by issuing an ERASEALL command via CTRL-AP. This command will erase the flash, UICR, and RAM, including UICR.APPROTECT. Erasing UICR will set UICR.APPROTECT value to `Disabled`. CTRL-AP is described in more detail in CTRL-AP - Control access port on page 71.

Access port protection controlled by hardware and software

This information refers to build codes Fxx and later.

By default, access port protection is enabled.

Access port protection is disabled by issuing an ERASEALL command via CTRL-AP. Read CTRL-AP.APPROTECTSTATUS to ensure that access port protection is disabled, and repeat the ERASEALL command if needed. This command will erase the flash, UICR, and RAM. CTRL-AP is described in more detail in CTRL-AP - Control access port on page 71. Access port protection will remain disabled until one of the following occurs:

- Pin reset
- Power or brownout reset
- Watchdog reset if not in Debug Interface Mode, see Debug Interface mode on page 73
- Wake from System OFF if not in Emulated System OFF

To keep access port protection disabled, the following actions must be performed:

- Program UICR.APPROTECT to `HwDisabled`. This disables the hardware part of the access port protection scheme after the first reset of any type. The hardware part of the access port protection will stay disabled as long as UICR.APPROTECT is not overwritten.
- Firmware must write APPROTECT.DISABLE to `SwDisable`. This disables the software part of the access port protection scheme.

Note: Register APPROTECT.DISABLE is reset after pin reset, power or brownout reset, watchdog reset, or wake from System OFF as mentioned above.

The following figure is an example on how a device with access port protection enabled can be erased, programmed, and configured to allow debugging. Operations sent from debugger as well as registers written by firmware will affect the access port state.

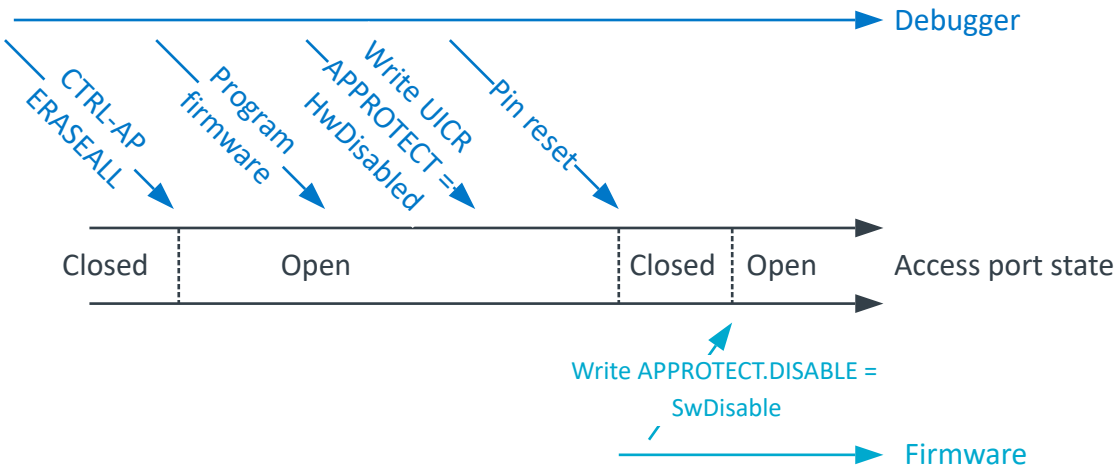


Figure 8: Access port unlocking

Access port protection is enabled when the disabling conditions are not present. For additional security, it is recommended to write `Enabled` to UICR.APPROTECT, and have firmware write `Force` to APPROTECT.FORCEPROTECT. This is illustrated in the following figure.

Note: Register APPROTECT.FORCEPROTECT is reset after any reset.

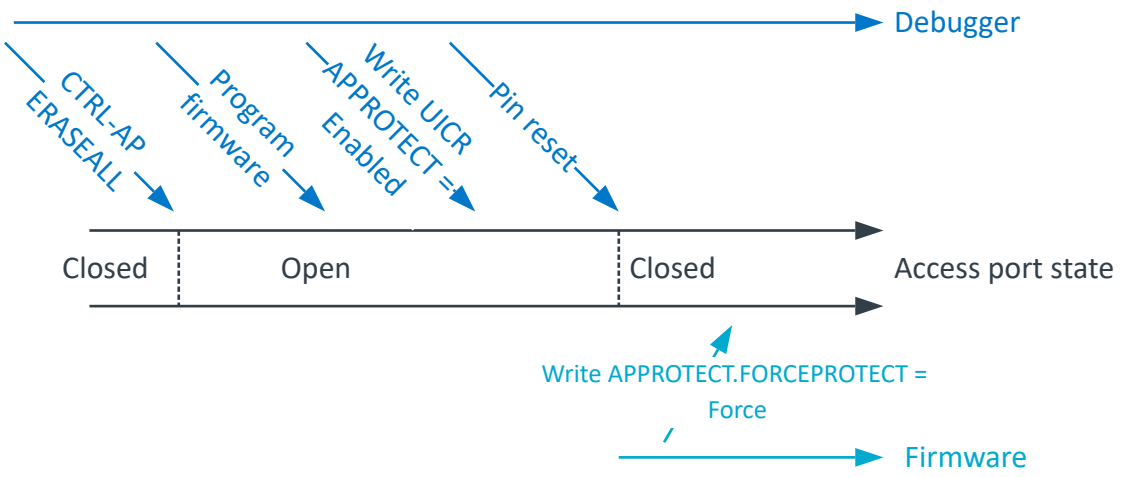


Figure 9: Force access port protection

4.8.2.1 Registers

Instances

Instance	Base address	Description
APPROTECT	0x40000000	APPROTECT control

Register overview

Register	Offset	Description
FORCEPROTECT	0x550	Software force enable APPROTECT mechanism until next reset.
DISABLE	0x558	Software disable APPROTECT mechanism

4.8.2.1.1 FORCEPROTECT

Address offset: 0x550

Software force enable APPROTECT mechanism until next reset.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																												A	A	A	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW1	FORCEPROTECT																																	
			Force	0x0																															

ID	R/W	Field	Value ID	Value	Description
A	RW1	FORCEPROTECT			Write 0x0 to force enable APPROTECT mechanism
			Force	0x0	Software force enable APPROTECT mechanism

4.8.2.1.2 DISABLE

Address offset: 0x558

Software disable APPROTECT mechanism

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DISABLE			Software disable APPROTECT mechanism
			SwDisable	0x5A	Software disable APPROTECT mechanism

4.8.3 CTRL-AP - Control access port

The control access port (CTRL-AP) is a custom access port that enables control of the device when other access ports in the DAP are disabled by the access port protection.

Access port protection is described in more detail in Access port protection on page 68.

Control access port has the following features:

- Soft reset - see Reset on page 89 for more information
- Disabling of access port protection - device control is allowed through CTRL-AP even when all other access ports in DAP are disabled by access port protection

4.8.3.1 Registers

Register overview

Register	Offset	Description
RESET	0x000	Soft reset triggered through CTRL-AP
ERASEALL	0x004	Erase all
ERASEALLSTATUS	0x008	Status register for the ERASEALL operation
APPROTECTSTATUS	0x00C	Status register for access port protection
IDR	0x0FC	CTRL-AP identification register, IDR

4.8.3.1.1 RESET

Address offset: 0x000

Soft reset triggered through CTRL-AP

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RESET			Soft reset triggered through CTRL-AP. See Reset behavior in POWER chapter for more details.
			NoReset	0	Reset is not active
			Reset	1	Reset is active. Device is held in reset.

4.8.3.1.2 ERASEALL

Address offset: 0x004

Erase all

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	W	ERASEALL			Erase all flash and RAM
			NoOperation	0	No operation
			Erase	1	Erase all flash and RAM

4.8.3.1.3 ERASEALLSTATUS

Address offset: 0x008

Status register for the ERASEALL operation

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	ERASEALLSTATUS			Status register for the ERASEALL operation
			Ready	0	ERASEALL is ready
			Busy	1	ERASEALL is busy (on-going)

4.8.3.1.4 APPROTECTSTATUS

Address offset: 0x00C

Status register for access port protection

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	APPROTECTSTATUS			Status register for access port protection
			Enabled	0	Access port protection enabled
			Disabled	1	Access port protection not enabled

4.8.3.1.5 IDR

Address offset: 0x0FC

CTRL-AP identification register, IDR

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E E E D D D D C C C C C C C B B B B A A A A A A A A
Reset 0x02880000				**0 0 0 0 0 0 1 0 1 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	APID			AP identification
B	R	CLASS			Access port (AP) class
			NotDefined	0x0	No defined class
			MEMAP	0x8	Memory access port
C	R	JEP106ID			JEDEC JEP106 identity code
D	R	JEP106CONT			JEDEC JEP106 continuation code
E	R	REVISION			Revision

4.8.3.2 Electrical specification

4.8.3.2.1 Control access port

Symbol	Description	Min.	Typ.	Max.	Units
R_{pull}	Internal SWDIO and SWDCLK pull up/down resistance		13		kΩ
f_{SWDCLK}	SWDCLK frequency	0.125		8	MHz

4.8.4 Debug Interface mode

Before an external debugger can access either CPU's access port (AHB-AP) or the control access port (CTRL-AP), the debugger must first request the device to power up via CxxxPWRUPREQ in the SWJ-DP.

If the device is in System OFF when power is requested via CxxxPWRUPREQ, the system will wake up and the DIF flag in RESETREAS on page 95 will be set. The device is in the Debug Interface mode as long as the debugger is requesting power via CxxxPWRUPREQ. Once the debugger stops requesting power via CxxxPWRUPREQ, the device is back in normal mode. Some peripherals behave differently in Debug Interface mode compared to normal mode. These differences are described in more detail in the chapters of the peripherals that are affected.

When a debug session is over, the external debugger must make sure to put the device back into normal mode since the overall power consumption is higher in Debug Interface mode than in normal mode.

For details on how to use the debug capabilities, read the debug documentation of your IDE.

4.8.5 Real-time debug

The nRF52840 supports real-time debugging.

Real-time debugging allows interrupts to execute to completion in real time when breakpoints are set in Thread mode or lower priority interrupts. This enables developers to set breakpoints and single-step through the code without the risk of real-time event-driven threads running at higher priority failing. For example, this enables the device to continue to service the high-priority interrupts of an external controller or sensor without failure or loss of state synchronization while the developer steps through code in a low-priority thread.

4.8.6 Trace

The device supports ETM and ITM trace.

Trace data from the ETM and the ITM is sent to an external debugger via a 4-bit wide parallel trace port interface unit (TPIU), see TRACEDATA[0] through TRACEDATA[3] and TRACECLK in Debug and trace overview on page 67.

In addition to parallel trace, the TPIU supports serial trace via the serial wire output (SWO) trace protocol. Parallel and serial trace cannot be used at the same time. ETM trace is only supported in Parallel Trace mode, while ITM trace is supported in both Parallel and Serial Trace modes.

For details on how to use the trace capabilities, read the debug documentation of your IDE.

TPIU's trace pins are multiplexed with GPIOs. SWO and TRACEDATA[0] use the same GPIO. See Pin assignments on page 926 for more information.

Trace speed is configured in register TRACECONFIG on page 170. The speed of the trace pins depends on the DRIVE setting of the GPIOs that the trace pins are multiplexed with. Only S0S1 and H0H1 drives are suitable for debugging. S0S1 is the default DRIVE setting at reset. If parallel or serial trace port signals are not fast enough with the default settings, all GPIOs in use for tracing should be set to high drive (H0H1). The DRIVE setting for these GPIOs should not be overwritten by firmware during the debugging session.

4.8.6.1 Registers

4.8.6.2 Electrical specification

4.8.6.2.1 Trace port

Symbol	Description	Min.	Typ.	Max.	Units
T_{cyc}	Clock period as defined by Arm in the Timing specifications for Trace Port Physical Interface of the Embedded Trace Macrocell Architecture Specification	62.5		500	ns

5 Power and clock management

5.1 Power management unit (PMU)

Power and clock management in nRF52840 is designed to automatically ensure maximum power efficiency.

The core of the power and clock management system is the power management unit (PMU) illustrated in the following figure.

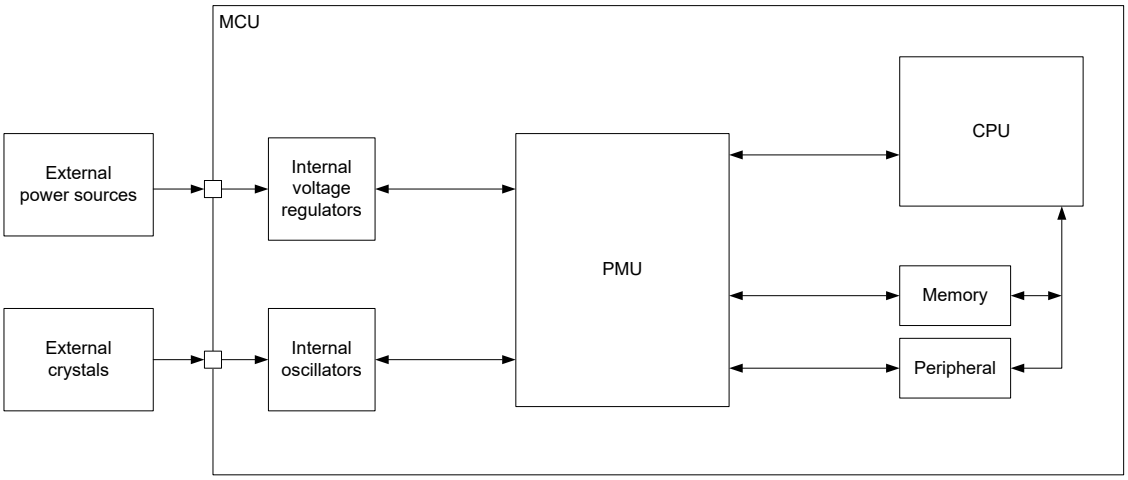


Figure 10: Power management unit

The PMU automatically detects which power and clock resources are required by the different system components at any given time. The PMU will then automatically start/stop and choose operation modes in supply regulators and clock sources, to achieve the lowest power consumption possible.

5.2 Current consumption

Because the system is continually being tuned by the Power management unit (PMU) on page 75, estimating an application's current consumption can be challenging when measurements cannot be directly performed on the hardware. To facilitate the estimation process, a set of current consumption scenarios are provided to show the typical current drawn from the VDD supply.

Each scenario specifies a set of operations and conditions applying to the given scenario. The following table shows a set of common conditions used in all scenarios, unless otherwise stated in the description of a given scenario. All scenarios are listed in Electrical specification on page 76.

Condition	Value
Supply	3 V on VDD/VDDH (Normal voltage mode)
Temperature	25°C
CPU	WFI (wait for interrupt)/WFE (wait for event) sleep
Peripherals	All idle
Clock	Not running
Regulator	LDO
RAM	In System ON, full 256 kB powered. In System OFF, full 256 kB retention.
Compiler	GCC v4.9.3 20150529 (arm-none-eabi-gcc). • Compiler flags: `-O0 -falign-functions=16 -fno-strict-aliasing -mcpu=cortex-m4 -mfloat-abi=soft -msoft-float -mthumb`.
Cache enabled[2]	Yes
32 MHz crystal[3]	SMD 2520, 32 MHz, 10 pF +/- 10 ppm

Table 6: Current consumption scenarios, common conditions

5.2.1 Electrical specification

5.2.1.1 Sleep

Symbol	Description	Min.	Typ.	Max.	Units
$I_{ON_RAMOFF_EVENT}$	System ON, no RAM retention, wake on any event		0.97		µA
$I_{ON_RAMON_EVENT}$	System ON, full 256 kB RAM retention, wake on any event		2.35		µA
$I_{ON_RAMON_POF}$	System ON, full 256 kB RAM retention, wake on any event, power-fail comparator enabled		2.35		µA
$I_{ON_RAMON_GPIOTE}$	System ON, full 256 kB RAM retention, wake on GPIOTE input (event mode)		17.37		µA
$I_{ON_RAMON_GPIOTEPORT}$	System ON, full 256 kB RAM retention, wake on GPIOTE PORT event		2.36		µA
$I_{ON_RAMOFF_RTC}$	System ON, no RAM retention, wake on RTC (running from LFRC clock)		1.50		µA
$I_{ON_RAMON_RTC}$	System ON, full 256 kB RAM retention, wake on RTC (running from LFRC clock)		3.16		µA
$I_{OFF_RAMOFF_RESET}$	System OFF, no RAM retention, wake on reset		0.40		µA
$I_{OFF_RAMOFF_LPCOMP}$	System OFF, no RAM retention, wake on LPCOMP		0.86		µA
$I_{OFF_RAMON_RESET}$	System OFF, full 256 kB RAM retention, wake on reset		1.86		µA
$I_{ON_RAMOFF_EVENT_5V}$	System ON, no RAM retention, wake on any event, 5 V supply on VDDH, REG0 output = 3.3 V		1.29		µA
$I_{OFF_RAMOFF_RESET_5V}$	System OFF, no RAM retention, wake on reset, 5 V supply on VDDH, REG0 output = 3.3 V		0.95		µA

[2] Applies only when CPU is running from flash memory
[3] Applies only when HFXO is running

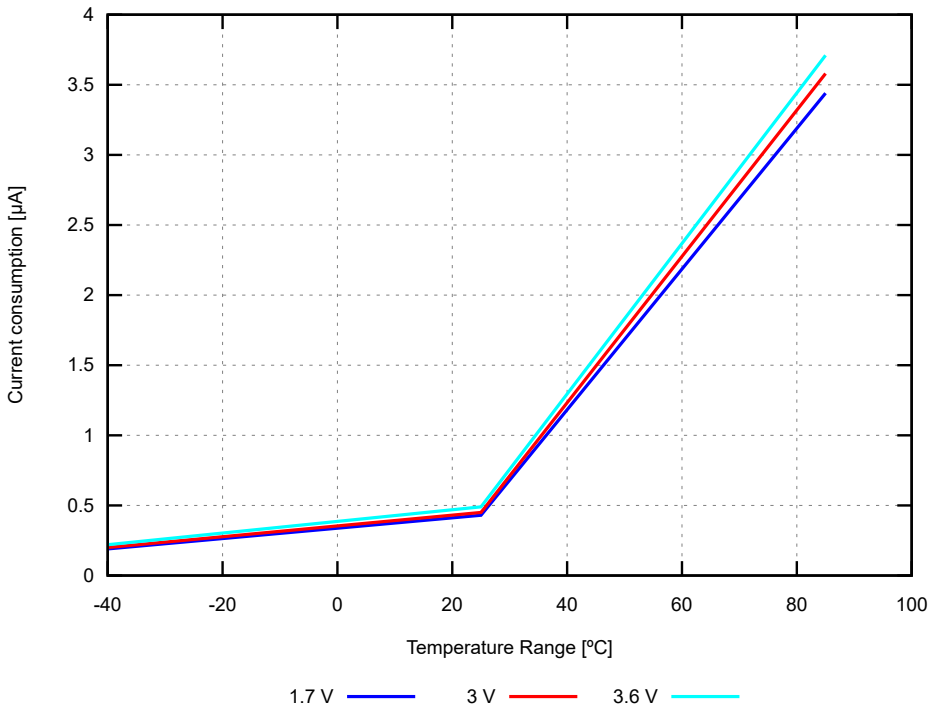


Figure 11: System OFF, no RAM retention, wake on reset (typical values)

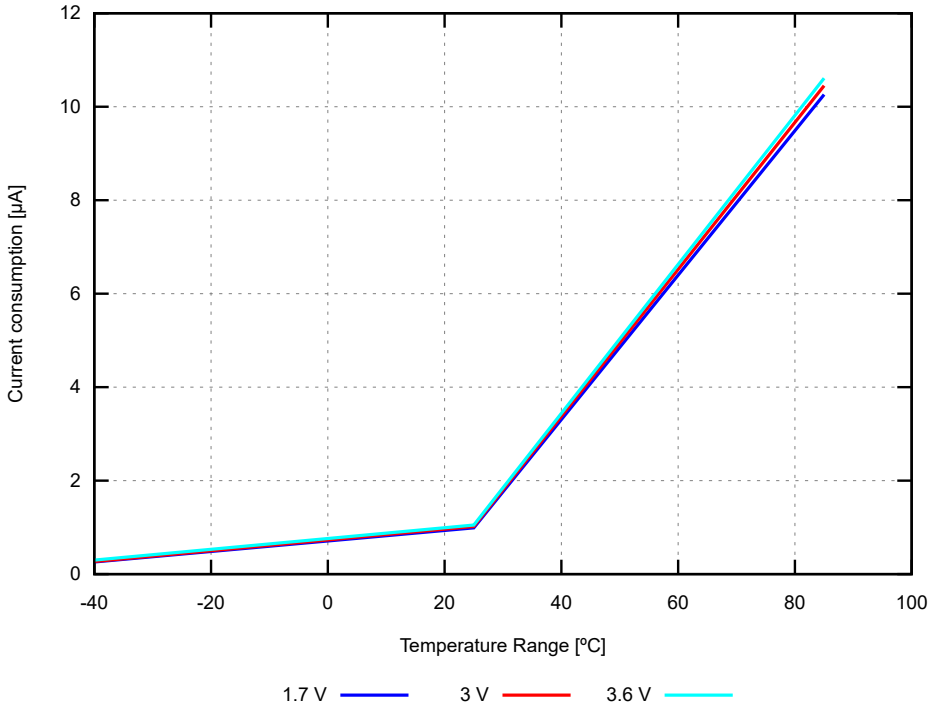


Figure 12: System ON, no RAM retention, wake on any event (typical values)

5.2.1.2 COMP active

Symbol	Description	Min.	Typ.	Max.	Units
$I_{COMP,LP}$	COMP enabled, low power mode		30.1		µA
$I_{COMP,NORM}$	COMP enabled, normal mode		31.8		µA
$I_{COMP,HS}$	COMP enabled, high-speed mode		35.1		µA

5.2.1.3 CPU running

Symbol	Description	Min.	Typ.	Max.	Units
I_{CPU0}	CPU running CoreMark @64 MHz from flash, Clock = HFXO, Regulator = DC/DC		3.3		mA
I_{CPU1}	CPU running CoreMark @64 MHz from flash, Clock = HFXO		6.3		mA
I_{CPU2}	CPU running CoreMark @64 MHz from RAM, Clock = HFXO, Regulator = DC/DC		2.8		mA
I_{CPU3}	CPU running CoreMark @64 MHz from RAM, Clock = HFXO		5.2		mA
I_{CPU4}	CPU running CoreMark @64 MHz from flash, Clock = HFINT, Regulator = DC/DC		3.1		mA

5.2.1.4 NFCT active

Symbol	Description	Min.	Typ.	Max.	Units
I_{sense}	Current in SENSE STATE[4]		100		nA
$I_{activated}$	Current in ACTIVATED STATE		400		µA

5.2.1.5 Radio transmitting/receiving

Symbol	Description	Min.	Typ.	Max.	Units
I_{RADIO_TX0}	Radio transmitting @ 8 dBm output power, 1 Mbps *Bluetooth* Low Energy (BLE) mode, Clock = HFXO, Regulator = DC/DC		16.40		mA
I_{RADIO_TX1}	Radio transmitting @ 0 dBm output power, 1 Mbps BLE mode, Clock = HFXO, Regulator = DC/DC		6.40		mA
I_{RADIO_TX2}	Radio transmitting @ -40 dBm output power, 1 Mbps BLE mode, Clock = HFXO, Regulator = DC/DC		3.83		mA
I_{RADIO_TX3}	Radio transmitting @ 0 dBm output power, 1 Mbps BLE mode, Clock = HFXO		10.80		mA
I_{RADIO_TX4}	Radio transmitting @ -40 dBm output power, 1 Mbps BLE mode, Clock = HFXO		4.82		mA
I_{RADIO_TX5}	Radio transmitting @ 0 dBm output power, 250 kbit/s IEE 802.15.4-2006 mode, Clock = HFXO, Regulator = DC/DC		6.40		mA
I_{RADIO_RX0}	Radio receiving @ 1 Mbps BLE mode, Clock = HFXO, Regulator = DC/DC		6.26		mA
I_{RADIO_RX1}	Radio receiving @ 1 Mbps BLE mode, Clock = HFXO		10.10		mA
I_{RADIO_RX2}	Radio receiving @ 250 kbit/s IEE 802.15.4-2006 mode, Clock = HFXO, Regulator = DC/DC		6.53		mA

[4] This current does not apply when in NFC field

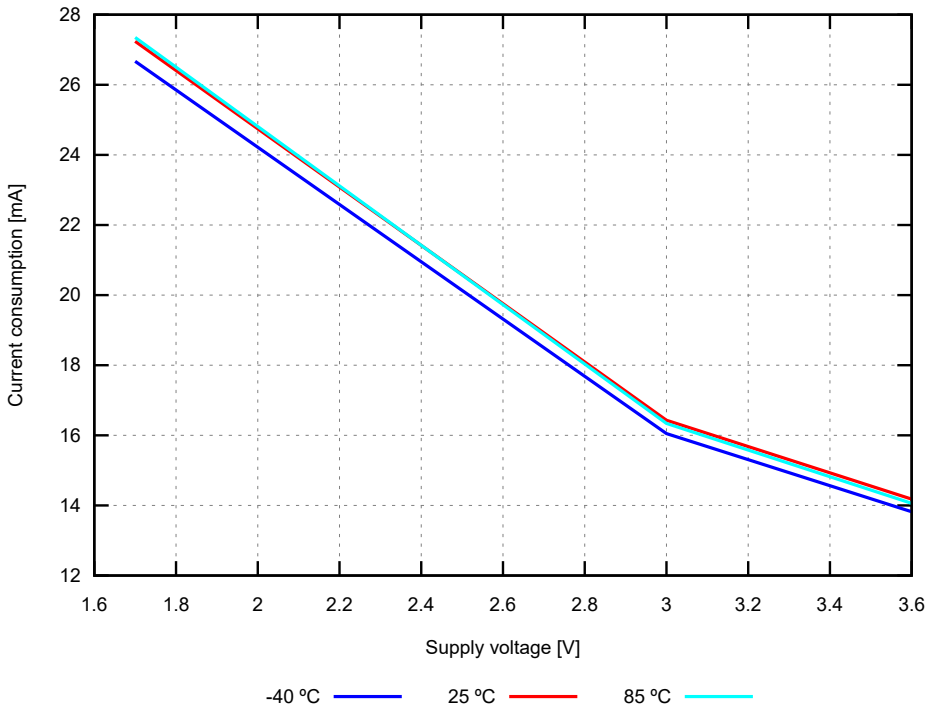


Figure 13: Radio transmitting @ 8 dBm output power, 1 Mbps BLE mode, Clock = HFXO, Regulator = DC/DC (typical values)

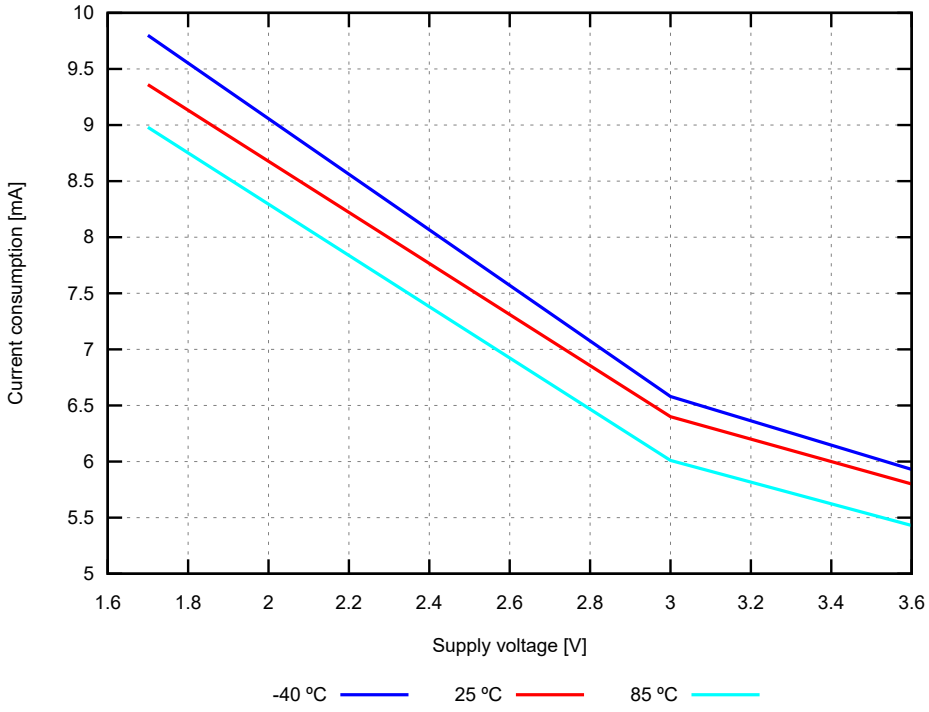


Figure 14: Radio transmitting @ 0 dBm output power, 1 Mbps BLE mode, Clock = HFXO, Regulator = DC/DC (typical values)

5.2.1.6 RNG active

Symbol	Description	Min.	Typ.	Max.	Units
I_{RNG0}	RNG running		635		µA

5.2.1.7 SAADC active

Symbol	Description	Min.	Typ.	Max.	Units
$I_{SAADC,RUN}$	SAADC sampling @ 16 ksps, Acquisition time = 20 µs, Clock = HFXO, Regulator = DC/DC		1.24		mA

5.2.1.8 TEMP active

Symbol	Description	Min.	Typ.	Max.	Units
I_{TEMP0}	TEMP started		1.05		mA

5.2.1.9 TIMER running

Symbol	Description	Min.	Typ.	Max.	Units
I_{TIMER0}	One TIMER instance running @ 1 MHz, Clock = HFINT		418		µA
I_{TIMER1}	Two TIMER instances running @ 1 MHz, Clock = HFINT		418		µA
I_{TIMER2}	One TIMER instance running @ 1 MHz, Clock = HFXO		646		µA
I_{TIMER3}	One TIMER instance running @ 16 MHz, Clock = HFINT		595		µA
I_{TIMER4}	One TIMER instance running @ 16 MHz, Clock = HFXO		823		µA

5.2.1.10 USBD running

Symbol	Description	Min.	Typ.	Max.	Units
$I_{USB,ACTIVE,VBUS}$	Current from VBUS supply, USB active		2.4		mA
$I_{USB,SUSPEND,VBUS}$	Current from VBUS supply, USB suspended, CPU sleeping		262		µA
$I_{USB,ACTIVE,VDD}$	Current from VDD supply (normal voltage mode), all RAM retained, regulator=LDO, CPU running, USB active		7.73		mA
$I_{USB,SUSPEND,VDD}$	Current from VDD supply (normal voltage mode), all RAM retained, regulator=LDO, CPU sleeping, USB suspended		173		µA
$I_{USB,ACTIVE,VDDH}$	Current from VDDH supply (high voltage mode), VDD=3 V (REG0 output), all RAM retained, regulator=LDO, CPU running, USB active		7.46		mA
$I_{USB,SUSPEND,VDDH}$	Current from VDDH supply (high voltage mode), VDD=3 V (REG0 output), all RAM retained, regulator=LDO, CPU sleeping, USB suspended		178		µA
$I_{USB,DISABLED,VDD}$	Current from VDD supply, USB disabled, VBUS supply connected, all RAM retained, regulator=LDO, CPU sleeping		7		µA

5.2.1.11 WDT active

Symbol	Description	Min.	Typ.	Max.	Units
$I_{WDT,STARTED}$	WDT started		3.1		µA

5.2.1.12 Compounded

Symbol	Description	Min.	Typ.	Max.	Units
I_{S0}	CPU running CoreMark from flash, Radio transmitting @ 0 dBm output power, 1 Mbps *Bluetooth* Low Energy (BLE) mode, Clock = HFXO, Regulator = DC/DC		8.1		mA
I_{S1}	CPU running CoreMark from flash, Radio receiving @ 1 Mbps BLE mode, Clock = HFXO, Regulator = DC/DC		8.6		mA
I_{S2}	CPU running CoreMark from flash, Radio transmitting @ 0 dBm output power, 1 Mbps BLE mode, Clock = HFXO		15.4		mA
I_{S3}	CPU running CoreMark from flash, Radio receiving @ 1 Mbps BLE mode, Clock = HFXO		16.2		mA
I_{S4}	CPU running CoreMark from flash, Radio transmitting @ 0 dBm output power, 1 Mbps BLE mode, Clock = HFXO, Regulator = DC/DC, 5 V supply on VDDH, REG0 output = 3.3 V		11.9		mA
I_{S5}	CPU running CoreMark from flash, Radio receiving @ 1 Mbps BLE mode, Clock = HFXO, Regulator = DC/DC, 5 V supply on VDDH, REG0 output = 3.3 V		12.7		mA

5.3 POWER — Power supply

The power supply consists of a number of LDO and DC/DC regulators that are utilized to maximize the system's power efficiency.

This device has the following power supply features:

- On-chip LDO and DC/DC regulators
- Global System ON/OFF modes
- Individual RAM section power control for all system modes
- Analog or digital pin wakeup from System OFF
- Supervisor hardware to manage power-on reset, brownout, and power failure
- Auto-controlled refresh modes for LDO and DC/DC regulators to maximize efficiency
- External circuitry supply
- Separate USB supply

5.3.1 Main supply

The main supply voltage is connected to the VDD/VDDH pins. The system will enter one of two supply voltage modes, Normal or High Voltage mode, depending on how the supply voltage is connected to these pins.

> **Note:** VDD and VDDH are shortcircuited inside the QFN48 package. Therefore the QFN48 device is only usable in Normal Voltage supply mode, and not High Voltage supply mode.

The system enters Normal Voltage mode when the supply voltage is connected to both the VDD and VDDH pins (pin VDD shorted to pin VDDH). For the supply voltage range to connect to both VDD and VDDH pins, see parameter V_{DD}.

The system enters High Voltage mode when the supply voltage is only connected to the VDDH pin and the VDD pin is not connected to any voltage supply. For the supply voltage range to connect to the VDDH pin, see parameter V_{DDH}.

The register MAINREGSTATUS on page 99 can be used to read the current supply voltage mode.

5.3.1.1 Main voltage regulators

The system contains two main supply regulator stages, REG0 and REG1.

Each regulator stage has the following regulator type options:

- Low-dropout regulator (LDO)
- Buck regulator (DC/DC)

In Normal Voltage mode, only the REG1 regulator stage is used, and the REG0 stage is automatically disabled. In High Voltage mode, both regulator stages (REG0 and REG1) are used. The output voltage of REG0 can be configured in register REGOUT0 on page 63. This output voltage is connected to VDD and is the input voltage to REG1.

Note: In High Voltage mode, the configured output voltage for REG0 (REGOUT0 on page 63) must not be greater than REG0 input voltage minus the voltage drop in REG0 (VDDH - $V_{VDDH-VDD}$).

By default, the LDO regulators are enabled and the DC/DC regulators are disabled. Registers DCDCEN0 on page 99 and DCDCEN on page 98 are used to enable the DC/DC regulators for REG0 and REG1 stages respectively.

When a DC/DC converter is enabled, the corresponding LDO regulator is disabled. External LC filters must be connected for each of the DC/DC regulators if they are being used. The advantage of using a DC/DC regulator is that the overall power consumption is normally reduced as the efficiency of such a regulator is higher than that of a LDO. The efficiency gained by using a DC/DC regulator is best seen when the regulator voltage drop (difference between input and output voltage) is high. The efficiency of internal regulators vary with the supply voltage and the current drawn from the regulators.

Note: Do not enable the DC/DC regulator without an external LC filter being connected as this will inhibit device operation, including debug access, until an LC filter is connected.

5.3.1.2 GPIO levels

The GPIO high reference voltage is equal to the level on the VDD pin.

In Normal Voltage mode, the GPIO high level equals the voltage supplied to the VDD pin. In High Voltage mode, it equals the level specified in register REGOUT0 on page 63.

5.3.1.3 External circuitry supply

In High Voltage mode, the output from REG0 can be used to supply external circuitry from the VDD pin.

The VDD output voltage is configured in the register REGOUT0 on page 63.

The supported output voltage range depends on the supply voltage provided by the VDDH pin. Minimum difference between voltage supplied on the VDDH pin and the voltage output on the VDD pin is defined by the $V_{REG0,DROP}$ parameter in Regulator specifications, REG0 stage on page 155.

Supplying external circuitry is allowed in both System OFF and System ON mode.

Note: The maximum allowed current drawn by external circuitry is dependent on the total internal current draw. The maximum current that can be drawn externally from REG0 is defined in Regulator specifications, REG0 stage on page 155).

5.3.1.4 Regulator configuration examples

The voltage regulators can be configured in several ways, depending on the selected supply voltage mode (Normal/High) and the regulator type option (LDO or DC/DC).

Four configuration examples are illustrated in the following figures.

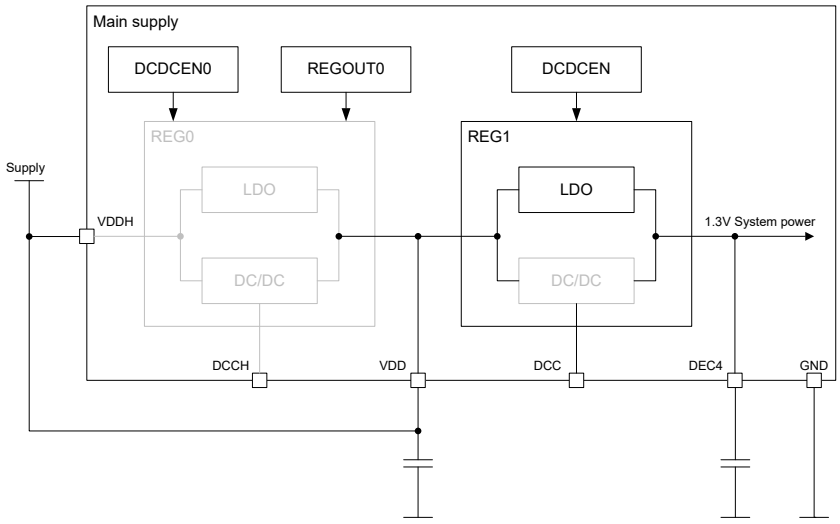


Figure 15: Normal Voltage mode, LDO only

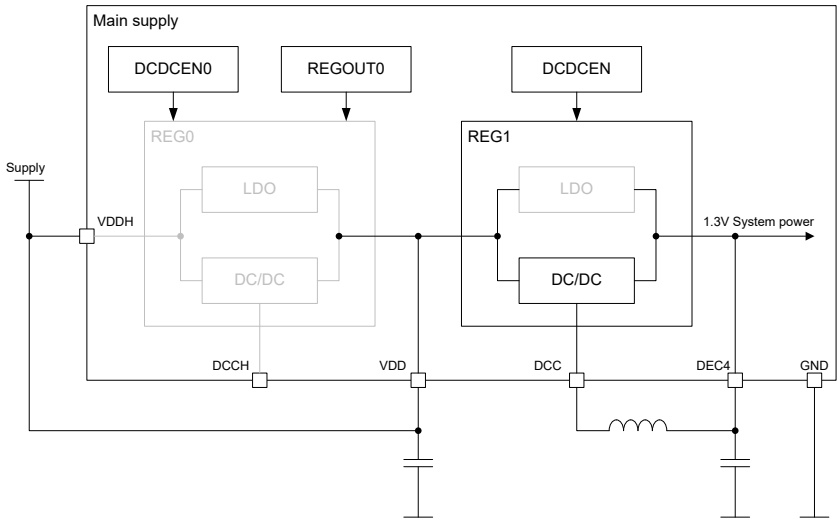


Figure 16: Normal Voltage mode, DC/DC REG1 enabled

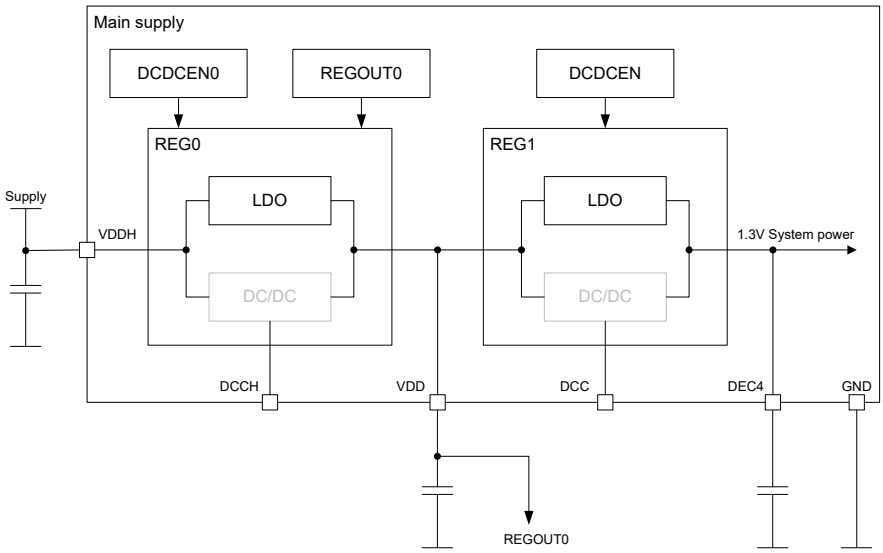


Figure 17: High Voltage mode, LDO only

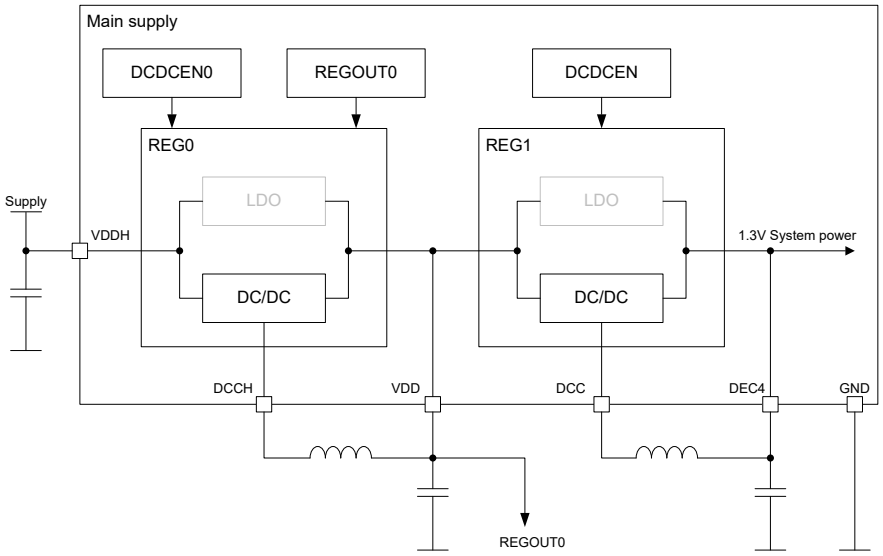


Figure 18: High Voltage mode, DC/DC for REG0 and REG1 enabled

5.3.1.5 Power supply supervisor

The power supply supervisor enables monitoring of the connected power supply.

The power supply supervisor provides the following functionality:

- Power-on reset - signals the circuit when a supply is connected
- An optional power-fail comparator (POF) - signals the application when the supply voltages drop below a configured threshold
- A fixed brownout reset detector - holds the system in reset when the voltage is too low for safe operation

The power supply supervisor is illustrated in the following figure.

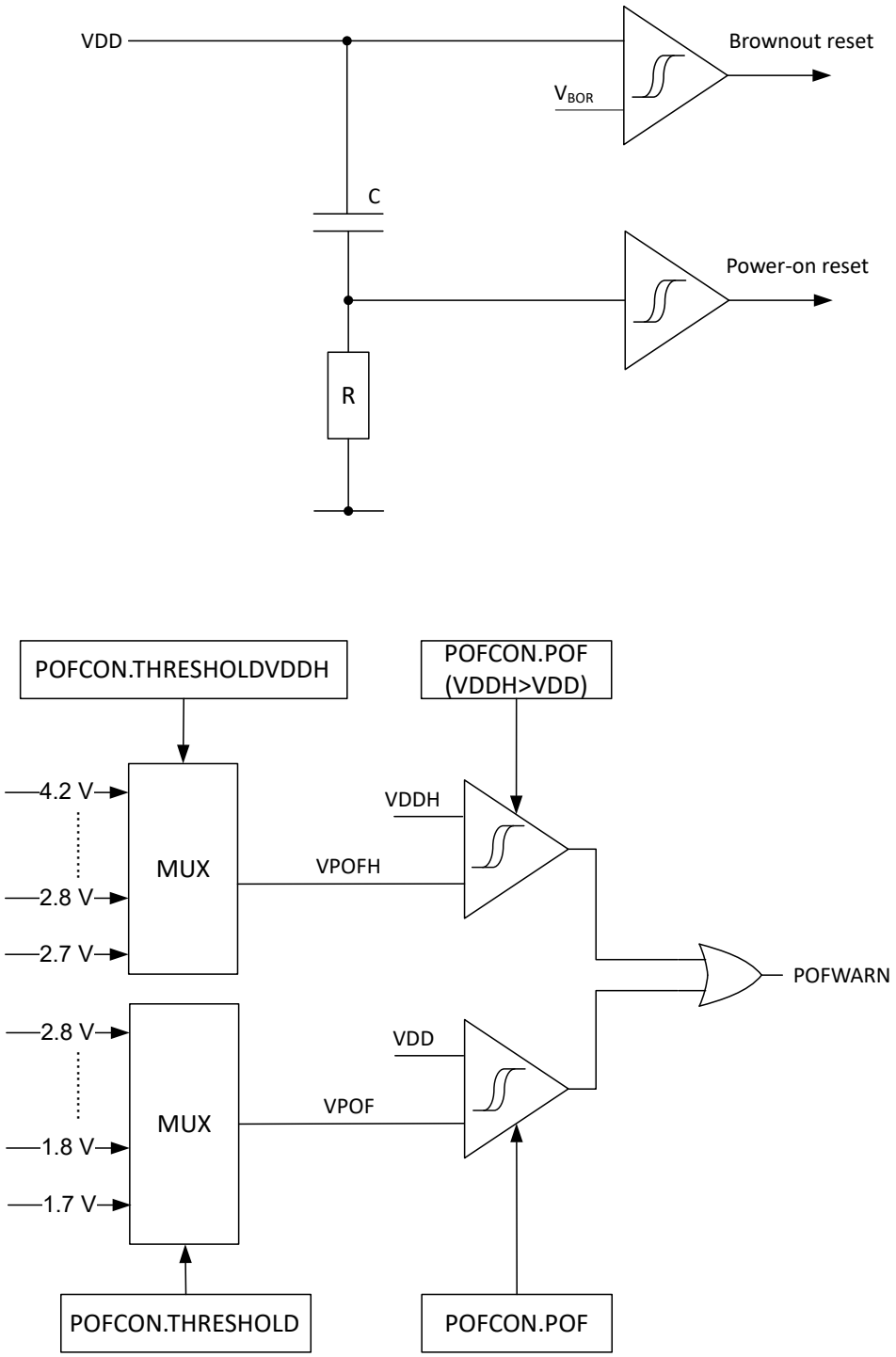


Figure 19: Power supply supervisor

5.3.1.6 Power-fail comparator

Using the power-fail comparator (POF) is optional. When enabled, it can provide an early warning to the CPU of an impending power supply failure.

To enable and configure the power-fail comparator, see the register POFCON on page 97.

When the supply voltage falls below the defined threshold, the power-fail comparator generates an event (POFWARN) that can be used by an application to prepare for power failure. This event is also generated when the supply voltage is already below the threshold at the time the power-fail comparator is enabled, or if the threshold is re-configured to a level above the supply voltage.

If the power failure warning is enabled, and the supply voltage is below the threshold, the power-fail comparator will prevent the NVMC from performing write operations to the flash.

The comparator features a hysteresis of V_{HYST}, as illustrated in the following figure.

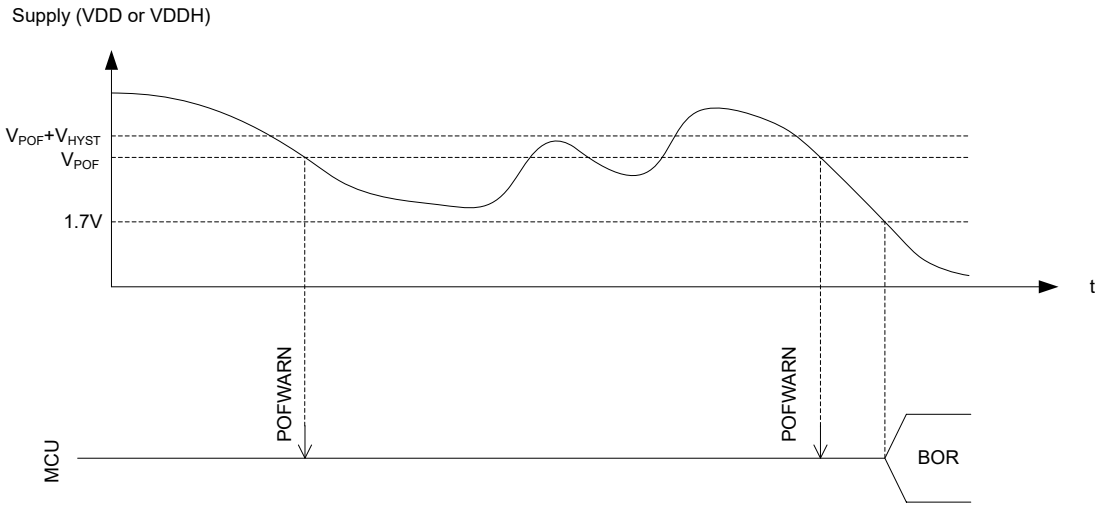


Figure 20: Power-fail comparator (BOR = brownout reset)

To save power, the power-fail comparator is not active in System OFF or System ON when HFCLK is not running.

5.3.2 USB supply

When using the USB peripheral, a 5 V USB supply needs to be provided to the VBUS pin.

The USB peripheral has a dedicated internal voltage regulator for converting the VBUS supply to 3.3 V used by the USB signalling interface (D+ and D- lines, and pull-up on D+). The remainder of the USB peripheral (USBD) is supplied through the main supply like other on-chip features. As a consequence, VBUS and either VDDH or VDD supplies are required for USB peripheral operation.

When VBUS rises into its valid range, the software is notified through a USBDETECTED event. A USBREMOVED event is sent when VBUS goes below its valid range. Use these events to implement the USBD start-up sequence described in the USBD chapter.

When VBUS rises into its valid range while the device is in System OFF, the device resets and transitions to System ON mode. The RESETREAS register will have the VBUS bit set to indicate the source of the wake-up.

See VBUS detection specifications on page 157 for the levels at which the events are sent ($V_{BUS,DETECT}$ and $V_{BUS,REMOVE}$) or at which the system is woken up from System OFF ($V_{BUS,DETECT}$).

When the USBD peripheral is enabled through the ENABLE register, and VBUS is detected, the regulator is turned on. A USBPWRRDY event is sent when the regulator's worst case settling time has elapsed, indicating to the software that it can enable the USB pull-up to signal a USB connection to the host.

The software can read the state of the VBUS detection and regulator output readiness at any time through the USBREGSTATUS register.

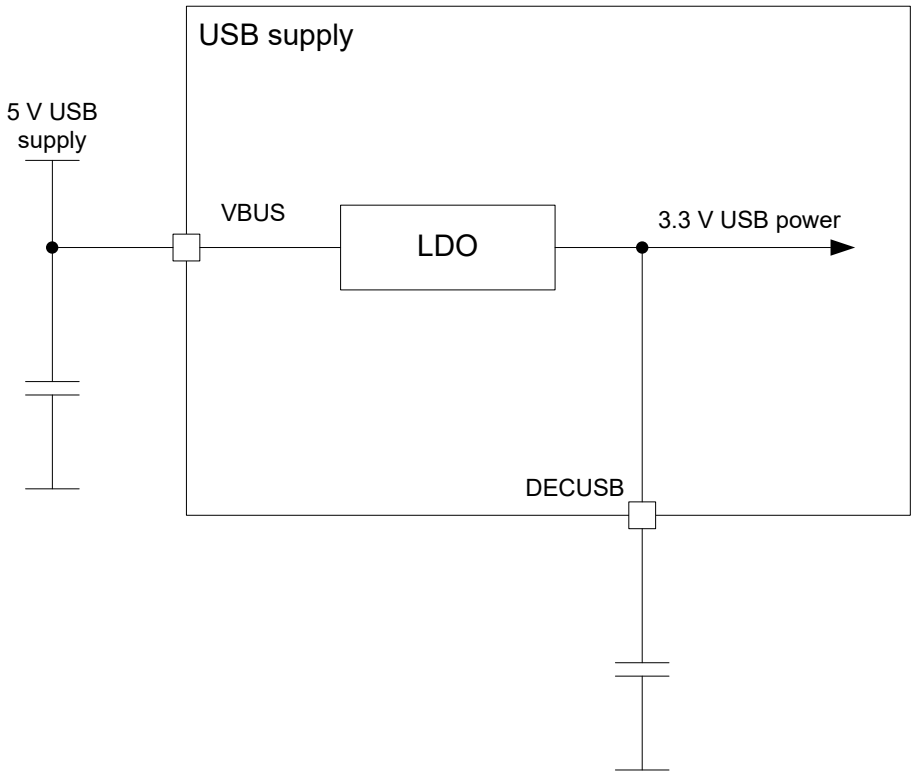


Figure 21: USB voltage regulator

To ensure stability, the input and output of the USB regulator need to be decoupled with a suitable decoupling capacitor. See Reference circuitry on page 937 for the recommended values.

5.3.3 System OFF mode

System OFF is the deepest power saving mode the system can enter. In this mode, the system's core functionality is powered down and all ongoing tasks are terminated.

The device can be put into System OFF mode using the register SYSTEMOFF on page 97. When in System OFF mode, the device can be woken up through one of the following signals:

- The DETECT signal, optionally generated by the GPIO peripheral.
- The ANADETECT signal, optionally generated by the LPCOMP module.
- The SENSE signal, optionally generated by the NFC module to wake-on-field.
- Detecting a valid USB voltage on the VBUS pin ($V_{BUS,DETECT}$).
- A reset.

The system is reset when it wakes up from System OFF mode.

One or more RAM sections can be retained in System OFF mode, depending on the settings in the RAM[n].POWER registers. RAM[n].POWER are retained registers. These registers are usually overwritten by the start-up code provided with the nRF application examples.

Before entering System OFF mode, all on-going EasyDMA transactions need to have completed. See peripheral specific chapters for more information about how to acquire the status of EasyDMA transactions.

5.3.3.1 Emulated System OFF mode

If the device is in Debug Interface mode, System OFF will be emulated to secure that all required resources needed for debugging are available during System OFF.

Required resources needed for debugging include the following key components:

- Debug and trace on page 67
- CLOCK — Clock control on page 157
- POWER — Power supply on page 81
- NVMC — Non-volatile memory controller on page 25
- CPU on page 20
- Flash memory
- RAM

See Debug and trace on page 67 for more information.

Because the CPU is kept on in an emulated System OFF mode, it is recommended to add an infinite loop directly after entering System OFF, to prevent the CPU from executing code that normally should not be executed.

5.3.4 System ON mode

System ON is the default state after power-on reset. In System ON mode, all functional blocks such as the CPU or peripherals can be in IDLE or RUN mode, depending on the configuration set by the software and the state of the application executing.

Register RESETREAS on page 95 provides information about the source causing the wakeup or reset.

The system can switch the appropriate internal power sources on and off, depending on the amount of power needed at any given time. The power requirement of a peripheral is directly related to its activity level, and the activity level of a peripheral fluctuates when specific tasks are triggered or events are generated.

5.3.4.1 Sub-power modes

In System ON mode, when the CPU and all peripherals are in IDLE mode, the system can reside in one of the two sub-power modes.

The sub-power modes are:

- Constant Latency
- Low-power

In Constant Latency mode, the CPU wakeup latency and the PPI task response are constant and kept at a minimum. This is secured by forcing a set of basic resources to be turned on while in sleep. The cost of constant and predictable latency is increased power consumption. Constant Latency mode is selected by triggering the CONSTLAT task.

In Low-power mode, the automatic power management system described in System ON mode on page 88 ensures that the most efficient supply option is chosen to save power. The cost of having the lowest possible power consumption is a varying CPU wakeup latency and PPI task response. Low-power mode is selected by triggering the LOWPWR task.

When the system enters System ON mode, it is by default in the sub-power mode Low-power.

5.3.5 RAM power control

The RAM power control registers are used for configuring the following:

- The RAM sections to be retained during System OFF
- The RAM sections to be retained and accessible during System ON

In System OFF, retention of a RAM section is configured in the RETENTION field of the corresponding register RAM[n].POWER.

In System ON, retention and accessibility of a RAM section is configured in the RETENTION and POWER fields of the corresponding register RAM[n].POWER.

The following table summarizes the behavior of these registers.

Configuration			RAM section status	
System on/off	RAM[n].POWER.POWER	RAM[n].POWER.RETENTION	Accessible	Retained
Off	x	Off	No	No
Off	x	On	No	Yes
On	Off	Off	No	No
On	Off[5]	On	No	Yes
On	On	x	Yes	Yes

Table 7: RAM section configuration

The advantage of not retaining RAM contents is that the overall current consumption is reduced.

See Memory on page 21 for more information on RAM sections.

5.3.6 Reset

Several sources may trigger a reset.

After a reset has occurred, register RESETREAS can be read to determine which source triggered the reset.

5.3.6.1 Power-on reset

The power-on reset generator initializes the system at power-on.

The system is held in reset state until the supply has reached the minimum operating voltage and the internal voltage regulators have started.

5.3.6.2 Pin reset

A pin reset is generated when the physical reset pin on the device is asserted.

Pin reset is configured via both registers PSELRESET[n].

5.3.6.3 Wakeup from System OFF mode reset

The device is reset when it wakes up from System OFF mode.

The debug access port (DAP) is not reset following a wake up from System OFF mode if the device is in Debug Interface mode. See chapter Debug and trace on page 67 for more information.

5.3.6.4 Soft reset

A soft reset is generated when the SYSRESETREQ bit of the application interrupt and reset control register (AIRCR) in the ARM® core is set.

See ARM documentation for more details.

A soft reset can also be generated via the register RESET on page 71 in the CTRL-AP.

5.3.6.5 Watchdog reset

A Watchdog reset is generated when the watchdog times out.

See chapter WDT — Watchdog timer on page 918 for more information.

[5] Not useful. RAM section power off gives negligible reduction in current consumption when retention is on.

5.3.6.6 Brownout reset

The brownout reset generator puts the system in a reset state if VDD drops below the brownout reset (BOR) threshold.

See section Power fail comparator on page 156 for more information.

5.3.6.7 Retained registers

A retained register is one that will retain its value in System OFF mode and through a reset, depending on the reset source. See the individual peripheral chapters for information on which of their registers are retained.

5.3.6.8 Reset behavior

The various reset sources and their targets are summarized in the table below.

Reset source	Reset target								
	CPU	Peripherals	GPIO	Debug[6]	SWJ-DP	RAM	WDT	Retained registers	RESETREAS
CPU lockup [7]	x	x	x						
Soft reset	x	x	x						
Wakeup from System OFF mode reset	x	x		x [8]		x [9]	x		
Watchdog reset [10]	x	x	x	x		x	x	x	
Pin reset	x	x	x	x		x	x	x	
Brownout reset	x	x	x	x	x	x	x	x	x
Power-on reset	x	x	x	x	x	x	x	x	x

Note: The RAM is never reset, but depending on a reset source the content of RAM may be corrupted.

5.3.7 Registers

Instances

Instance	Base address	Description
POWER	0x40000000	Power control

Register overview

Register	Offset	Description
TASKS_CONSTLAT	0x78	Enable Constant Latency mode

6 All debug components excluding SWJ-DP. See Debug and trace on page 67 for more information about the different debug components.

7 Reset from CPU lockup is disabled if the device is in Debug Interface mode. CPU lockup is not possible in System OFF.

8 The debug components will not be reset if the device is in Debug Interface mode.

9 RAM is not reset on wakeup from System OFF mode. RAM, or certain parts of RAM, may not be retained after the device has entered System OFF mode, depending on the settings in the RAM registers.

10 Watchdog reset is not available in System OFF.

Register	Offset	Description
TASKS_LOWPWR	0x7C	Enable Low-power mode (variable latency)
EVENTS_POFWARN	0x108	Power failure warning
EVENTS_SLEEPENTER	0x114	CPU entered WFI/WFE sleep
EVENTS_SLEEPEXIT	0x118	CPU exited WFI/WFE sleep
EVENTS_USBDETECTED	0x11C	Voltage supply detected on VBUS
EVENTS_USBREMOVED	0x120	Voltage supply removed from VBUS
EVENTS_USBPWRRDY	0x124	USB 3.3 V supply ready
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
RESETREAS	0x400	Reset reason
RAMSTATUS	0x428	RAM status register This register is deprecated.
USBREGSTATUS	0x438	USB supply status
SYSTEMOFF	0x500	System OFF register
POFCON	0x510	Power-fail comparator configuration
GPREGRET	0x51C	General purpose retention register
GPREGRET2	0x520	General purpose retention register
DCDCEN	0x578	Enable DC/DC converter for REG1 stage
DCDCEN0	0x580	Enable DC/DC converter for REG0 stage
MAINREGSTATUS	0x640	Main supply status
RAM[0].POWER	0x900	RAM0 power control register
RAM[0].POWERSET	0x904	RAM0 power control set register
RAM[0].POWERCLR	0x908	RAM0 power control clear register
RAM[1].POWER	0x910	RAM1 power control register
RAM[1].POWERSET	0x914	RAM1 power control set register
RAM[1].POWERCLR	0x918	RAM1 power control clear register
RAM[2].POWER	0x920	RAM2 power control register
RAM[2].POWERSET	0x924	RAM2 power control set register
RAM[2].POWERCLR	0x928	RAM2 power control clear register
RAM[3].POWER	0x930	RAM3 power control register
RAM[3].POWERSET	0x934	RAM3 power control set register
RAM[3].POWERCLR	0x938	RAM3 power control clear register
RAM[4].POWER	0x940	RAM4 power control register
RAM[4].POWERSET	0x944	RAM4 power control set register
RAM[4].POWERCLR	0x948	RAM4 power control clear register
RAM[5].POWER	0x950	RAM5 power control register
RAM[5].POWERSET	0x954	RAM5 power control set register
RAM[5].POWERCLR	0x958	RAM5 power control clear register
RAM[6].POWER	0x960	RAM6 power control register
RAM[6].POWERSET	0x964	RAM6 power control set register
RAM[6].POWERCLR	0x968	RAM6 power control clear register
RAM[7].POWER	0x970	RAM7 power control register
RAM[7].POWERSET	0x974	RAM7 power control set register
RAM[7].POWERCLR	0x978	RAM7 power control clear register
RAM[8].POWER	0x980	RAM8 power control register
RAM[8].POWERSET	0x984	RAM8 power control set register
RAM[8].POWERCLR	0x988	RAM8 power control clear register

5.3.7.1 TASKS_CONSTLAT

Address offset: 0x78

Enable Constant Latency mode

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CONSTLAT			Enable Constant Latency mode
			Trigger	1	Trigger task

5.3.7.2 TASKS_LOWPWR

Address offset: 0x7C

Enable Low-power mode (variable latency)

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_LOWPWR			Enable Low-power mode (variable latency)
			Trigger	1	Trigger task

5.3.7.3 EVENTS_POFWARN

Address offset: 0x108

Power failure warning

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_POFWARN			Power failure warning
			NotGenerated	0	Event not generated
			Generated	1	Event generated

5.3.7.4 EVENTS_SLEEPENTER

Address offset: 0x114

CPU entered WFI/WFE sleep

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_SLEEPENTER			CPU entered WFI/WFE sleep
			NotGenerated	0	Event not generated
			Generated	1	Event generated

5.3.7.5 EVENTS_SLEEPEXIT

Address offset: 0x118

CPU exited WFI/WFE sleep

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_SLEEPEXIT			CPU exited WFI/WFE sleep
			NotGenerated	0	Event not generated
			Generated	1	Event generated

5.3.7.6 EVENTS_USBDETECTED

Address offset: 0x11C

Voltage supply detected on VBUS

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_USBDETECTED			Voltage supply detected on VBUS
			NotGenerated	0	Event not generated
			Generated	1	Event generated

5.3.7.7 EVENTS_USBREMOVED

Address offset: 0x120

Voltage supply removed from VBUS

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_USBREMOVED			Voltage supply removed from VBUS
			NotGenerated	0	Event not generated
			Generated	1	Event generated

5.3.7.8 EVENTS_USBPWRRDY

Address offset: 0x124

USB 3.3 V supply ready

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_USBPWRRDY			USB 3.3 V supply ready
			NotGenerated	0	Event not generated
			Generated	1	Event generated

5.3.7.9 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	POFWARN			Write '1' to enable interrupt for event POFWARN
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	SLEEPENTER			Write '1' to enable interrupt for event SLEEPENTER
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	SLEEPEXIT			Write '1' to enable interrupt for event SLEEPEXIT
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	USBDETECTED			Write '1' to enable interrupt for event USBDETECTED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	USBREMOVED			Write '1' to enable interrupt for event USBREMOVED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	USBPWRRDY			Write '1' to enable interrupt for event USBPWRRDY
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

5.3.7.10 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	POFWARN			Write '1' to disable interrupt for event POFWARN
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	SLEEPENTER			Write '1' to disable interrupt for event SLEEPENTER
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	SLEEPEXIT			Write '1' to disable interrupt for event SLEEPEXIT
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	USBDETECTED			Write '1' to disable interrupt for event USBDETECTED
			Clear	1	Disable
			Disabled	0	Read: Disabled

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
			Enabled	1	Read: Enabled
E	RW	USBREMOVED			Write '1' to disable interrupt for event USBREMOVED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	USBPWRRDY			Write '1' to disable interrupt for event USBPWRRDY
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

5.3.7.11 RESETREAS

Address offset: 0x400

Reset reason

Note: Unless cleared, the RESETREAS register will be cumulative. A field is cleared by writing '1' to it. If none of the reset sources are flagged, this indicates that the chip was reset from the on-chip reset generator, which will indicate a power-on-reset or a brownout reset.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	RESETPIN			Reset from pin-reset detected
			NotDetected	0	Not detected
			Detected	1	Detected
B	RW	DOG			Reset from watchdog detected
			NotDetected	0	Not detected
			Detected	1	Detected
C	RW	SREQ			Reset from soft reset detected
			NotDetected	0	Not detected
			Detected	1	Detected
D	RW	LOCKUP			Reset from CPU lock-up detected
			NotDetected	0	Not detected
			Detected	1	Detected
E	RW	OFF			Reset due to wake up from System OFF mode when wakeup is triggered from DETECT signal from GPIO
			NotDetected	0	Not detected
			Detected	1	Detected
F	RW	LPCOMP			Reset due to wake up from System OFF mode when wakeup is triggered from ANADETECT signal from LPCOMP
			NotDetected	0	Not detected
			Detected	1	Detected
G	RW	DIF			Reset due to wake up from System OFF mode when wakeup is triggered from entering into debug interface mode
			NotDetected	0	Not detected
			Detected	1	Detected
H	RW	NFC			Reset due to wake up from System OFF mode by NFC field detect
			NotDetected	0	Not detected

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
			Detected	1	Detected
I	RW	VBUS			Reset due to wake up from System OFF mode by VBUS rising into valid range
			NotDetected	0	Not detected
			Detected	1	Detected

5.3.7.12 RAMSTATUS (Deprecated)

Address offset: 0x428

RAM status register

This register is deprecated.

Note: Since this register is deprecated the following substitutions have been made: RAM block 0 is equivalent to a block comprising RAM0.S0 and RAM1.S0. RAM block 1 is equivalent to a block comprising RAM2.S0 and RAM3.S0. RAM block 2 is equivalent to a block comprising RAM4.S0 and RAM5.S0. RAM block 3 is equivalent to a block comprising RAM6.S0 and RAM7.S0. A RAM block field will indicate ON as long as any of the RAM sections associated with a block are on.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	RAMBLOCK0			RAM block 0 is on or off/powering up
			Off	0	Off
			On	1	On
B	R	RAMBLOCK1			RAM block 1 is on or off/powering up
			Off	0	Off
			On	1	On
C	R	RAMBLOCK2			RAM block 2 is on or off/powering up
			Off	0	Off
			On	1	On
D	R	RAMBLOCK3			RAM block 3 is on or off/powering up
			Off	0	Off
			On	1	On

5.3.7.13 USBREGSTATUS

Address offset: 0x438

USB supply status

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	VBUSDETECT			VBUS input detection status (USBDETECTED and USBREMOVED events are derived from this information)
			NoVbus	0	VBUS voltage below valid threshold
			VbusPresent	1	VBUS voltage above valid threshold
B	R	OUTPUTRDY			USB supply output settling time elapsed
			NotReady	0	USBREG output settling time not elapsed
			Ready	1	USBREG output settling time elapsed (same information as USBPWRRDY event)

5.3.7.14 SYSTEMOFF

Address offset: 0x500

System OFF register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	W	SYSTEMOFF			Enable System OFF mode
			Enter	1	Enable System OFF mode

5.3.7.15 POFCON

Address offset: 0x510

Power-fail comparator configuration

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C C C C B B B B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	POF			Enable or disable power failure warning
			Disabled	0	Disable
			Enabled	1	Enable
B	RW	THRESHOLD			Power-fail comparator threshold setting. This setting applies both for normal voltage mode (supply connected to both VDD and VDDH) and high voltage mode (supply connected to VDDH only). Values 0-3 set threshold below 1.7 V and should not be used as brown out detection will be activated before power failure warning on such low voltages.
			V17	4	Set threshold to 1.7 V
			V18	5	Set threshold to 1.8 V
			V19	6	Set threshold to 1.9 V
			V20	7	Set threshold to 2.0 V
			V21	8	Set threshold to 2.1 V
			V22	9	Set threshold to 2.2 V
			V23	10	Set threshold to 2.3 V
			V24	11	Set threshold to 2.4 V
			V25	12	Set threshold to 2.5 V
			V26	13	Set threshold to 2.6 V
			V27	14	Set threshold to 2.7 V

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C C C C B B B B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			V28	15	Set threshold to 2.8 V
C	RW	THRESHOLDVDDH			Power-fail comparator threshold setting for high voltage mode (supply connected to VDDH only). This setting does not apply for normal voltage mode (supply connected to both VDD and VDDH).
			V27	0	Set threshold to 2.7 V
			V28	1	Set threshold to 2.8 V
			V29	2	Set threshold to 2.9 V
			V30	3	Set threshold to 3.0 V
			V31	4	Set threshold to 3.1 V
			V32	5	Set threshold to 3.2 V
			V33	6	Set threshold to 3.3 V
			V34	7	Set threshold to 3.4 V
			V35	8	Set threshold to 3.5 V
			V36	9	Set threshold to 3.6 V
			V37	10	Set threshold to 3.7 V
			V38	11	Set threshold to 3.8 V
			V39	12	Set threshold to 3.9 V
			V40	13	Set threshold to 4.0 V
			V41	14	Set threshold to 4.1 V
			V42	15	Set threshold to 4.2 V

5.3.7.16 GPREGRET

Address offset: 0x51C

General purpose retention register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	GPREGRET			General purpose retention register This register is a retained register

5.3.7.17 GPREGRET2

Address offset: 0x520

General purpose retention register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	GPREGRET			General purpose retention register This register is a retained register

5.3.7.18 DCDCEN

Address offset: 0x578

Enable DC/DC converter for REG1 stage

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DCDCEN			Enable DC/DC converter for REG1 stage.
			Disabled	0	Disable
			Enabled	1	Enable

5.3.7.19 DCDCEN0

Address offset: 0x580

Enable DC/DC converter for REG0 stage

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DCDCEN			Enable DC/DC converter for REG0 stage.
			Disabled	0	Disable
			Enabled	1	Enable

5.3.7.20 MAINREGSTATUS

Address offset: 0x640

Main supply status

Note:

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	MAINREGSTATUS			Main supply status
			Normal	0	Normal voltage mode. Voltage supplied on VDD.
			High	1	High voltage mode. Voltage supplied on VDDH.

5.3.7.21 RAM[0].POWER

Address offset: 0x900

RAM0 power control register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW	S0POWER																																	
			Off	0																															
			On	1																															
B	RW	S1POWER																																	
			Off	0																															
			On	1																															
C	RW	S2POWER																																	
			Off	0																															
			On	1																															
D	RW	S3POWER																																	
			Off	0																															
			On	1																															
E	RW	S4POWER																																	
			Off	0																															
			On	1																															
F	RW	S5POWER																																	
			Off	0																															
			On	1																															
G	RW	S6POWER																																	
			Off	0																															
			On	1																															
H	RW	S7POWER																																	
			Off	0																															
			On	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
I	RW	S8POWER																																	
			Off	0																															
			On	1																															
J	RW	S9POWER																																	
			Off	0																															
			On	1																															
K	RW	S10POWER																																	
			Off	0																															
			On	1																															
L	RW	S11POWER																																	
			Off	0																															
			On	1																															
M	RW	S12POWER																																	
			Off	0																															
			On	1																															
N	RW	S13POWER																																	
			Off	0																															
			On	1																															
O	RW	S14POWER																																	
			Off	0																															
			On	1																															
P	RW	S15POWER																																	
			Off	0																															
			On	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
Q	RW	S0RETENTION																																	
			Off	0																															
			On	1																															
R	RW	S1RETENTION																																	
			Off	0																															
			On	1																															
S	RW	S2RETENTION																																	
			Off	0																															
			On	1																															
T	RW	S3RETENTION																																	
			Off	0																															
			On	1																															
U	RW	S4RETENTION																																	
			Off	0																															
			On	1																															
V	RW	S5RETENTION																																	
			Off	0																															
			On	1																															
W	RW	S6RETENTION																																	
			Off	0																															
			On	1																															
X	RW	S7RETENTION																																	
			Off	0																															
			On	1																															
Y	RW	S8RETENTION																																	
			Off	0																															
			On	1																															
Z	RW	S9RETENTION																																	
			Off	0																															
			On	1																															
a	RW	S10RETENTION																																	
			Off	0																															
			On	1																															
b	RW	S11RETENTION																																	
			Off	0																															
			On	1																															
c	RW	S12RETENTION																																	
			Off	0																															
			On	1																															
d	RW	S13RETENTION																																	
			Off	0																															
			On	1																															
e	RW	S14RETENTION																																	
			Off	0																															
			On	1																															
f	RW	S15RETENTION																																	
			Off	0																															
			On	1																															

5.3.7.22 RAM[0].POWERSET

Address offset: 0x904

RAM0 power control set register

When read, this register will return the value of the POWER register.

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value	Description																															
A	W	S0POWER			Keep RAM section S0 of RAM0 on or off in System ON mode																																	
			On	1	On																															
B	W	S1POWER			Keep RAM section S1 of RAM0 on or off in System ON mode																																	
			On	1	On																															
C	W	S2POWER			Keep RAM section S2 of RAM0 on or off in System ON mode																																	
			On	1	On																															
D	W	S3POWER			Keep RAM section S3 of RAM0 on or off in System ON mode																																	
			On	1	On																															
E	W	S4POWER			Keep RAM section S4 of RAM0 on or off in System ON mode																																	
			On	1	On																															
F	W	S5POWER			Keep RAM section S5 of RAM0 on or off in System ON mode																																	
			On	1	On																															
G	W	S6POWER			Keep RAM section S6 of RAM0 on or off in System ON mode																																	
			On	1	On																															
H	W	S7POWER			Keep RAM section S7 of RAM0 on or off in System ON mode																																	
			On	1	On																															
I	W	S8POWER			Keep RAM section S8 of RAM0 on or off in System ON mode																																	
			On	1	On																															
J	W	S9POWER			Keep RAM section S9 of RAM0 on or off in System ON mode																																	
			On	1	On																															
K	W	S10POWER			Keep RAM section S10 of RAM0 on or off in System ON mode																																	
			On	1	On																															
L	W	S11POWER			Keep RAM section S11 of RAM0 on or off in System ON mode																																	
			On	1	On																															
M	W	S12POWER			Keep RAM section S12 of RAM0 on or off in System ON mode																																	
			On	1	On																															
N	W	S13POWER			Keep RAM section S13 of RAM0 on or off in System ON mode																																	
			On	1	On																															
O	W	S14POWER			Keep RAM section S14 of RAM0 on or off in System ON mode																																	
			On	1	On																															
P	W	S15POWER			Keep RAM section S15 of RAM0 on or off in System ON mode																																	
			On	1	On																															
Q	W	S0RETENTION			Keep retention on RAM section S0 when RAM section is switched off																																	
			On	1	On																															
R	W	S1RETENTION			Keep retention on RAM section S1 when RAM section is switched off																																	
			On	1	On																															
S	W	S2RETENTION			Keep retention on RAM section S2 when RAM section is switched off																																	
			On	1	On																															
T	W	S3RETENTION			Keep retention on RAM section S3 when RAM section is switched off																																	
			On	1	On																															
U	W	S4RETENTION			Keep retention on RAM section S4 when RAM section is switched off																																	
			On	1	On																															
V	W	S5RETENTION			Keep retention on RAM section S5 when RAM section is switched off																																	

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
			On	1	On
W	W	S6RETENTION			Keep retention on RAM section S6 when RAM section is switched off
			On	1	On
X	W	S7RETENTION			Keep retention on RAM section S7 when RAM section is switched off
			On	1	On
Y	W	S8RETENTION			Keep retention on RAM section S8 when RAM section is switched off
			On	1	On
Z	W	S9RETENTION			Keep retention on RAM section S9 when RAM section is switched off
			On	1	On
a	W	S10RETENTION			Keep retention on RAM section S10 when RAM section is switched off
			On	1	On
b	W	S11RETENTION			Keep retention on RAM section S11 when RAM section is switched off
			On	1	On
c	W	S12RETENTION			Keep retention on RAM section S12 when RAM section is switched off
			On	1	On
d	W	S13RETENTION			Keep retention on RAM section S13 when RAM section is switched off
			On	1	On
e	W	S14RETENTION			Keep retention on RAM section S14 when RAM section is switched off
			On	1	On
f	W	S15RETENTION			Keep retention on RAM section S15 when RAM section is switched off
			On	1	On

5.3.7.23 RAM[0].POWERCLR

Address offset: 0x908

RAM0 power control clear register

When read, this register will return the value of the POWER register.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	W	S0POWER			Keep RAM section S0 of RAM0 on or off in System ON mode
			Off	1	Off
B	W	S1POWER			Keep RAM section S1 of RAM0 on or off in System ON mode
			Off	1	Off
C	W	S2POWER			Keep RAM section S2 of RAM0 on or off in System ON mode
			Off	1	Off
D	W	S3POWER			Keep RAM section S3 of RAM0 on or off in System ON mode
			Off	1	Off
E	W	S4POWER			Keep RAM section S4 of RAM0 on or off in System ON mode
			Off	1	Off
F	W	S5POWER			Keep RAM section S5 of RAM0 on or off in System ON mode
			Off	1	Off
G	W	S6POWER			Keep RAM section S6 of RAM0 on or off in System ON mode
			Off	1	Off
H	W	S7POWER			Keep RAM section S7 of RAM0 on or off in System ON mode
			Off	1	Off
I	W	S8POWER			Keep RAM section S8 of RAM0 on or off in System ON mode

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
			Off	1																															
J	W	S9POWER																																	
			Off	1																															
K	W	S10POWER																																	
			Off	1																															
L	W	S11POWER																																	
			Off	1																															
M	W	S12POWER																																	
			Off	1																															
N	W	S13POWER																																	
			Off	1																															
O	W	S14POWER																																	
			Off	1																															
P	W	S15POWER																																	
			Off	1																															
Q	W	S0RETENTION																																	
			Off	1																															
R	W	S1RETENTION																																	
			Off	1																															
S	W	S2RETENTION																																	
			Off	1																															
T	W	S3RETENTION																																	
			Off	1																															
U	W	S4RETENTION																																	
			Off	1																															
V	W	S5RETENTION																																	
			Off	1																															
W	W	S6RETENTION																																	
			Off	1																															
X	W	S7RETENTION																																	
			Off	1																															
Y	W	S8RETENTION																																	
			Off	1																															
Z	W	S9RETENTION																																	
			Off	1																															
a	W	S10RETENTION																																	
			Off	1																															
b	W	S11RETENTION																																	
			Off	1																															
c	W	S12RETENTION																																	
			Off	1																															
d	W	S13RETENTION																																	
			Off	1																															
e	W	S14RETENTION																																	
			Off	1																															
f	W	S15RETENTION																																	
			Off	1																															

5.3.7.24 RAM[1].POWER

Address offset: 0x910

RAM1 power control register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	S0POWER			Keep RAM section S0 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S0RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
B	RW	S1POWER			Keep RAM section S1 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S1RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
C	RW	S2POWER			Keep RAM section S2 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S2RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
D	RW	S3POWER			Keep RAM section S3 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S3RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
E	RW	S4POWER			Keep RAM section S4 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S4RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
F	RW	S5POWER			Keep RAM section S5 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S5RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
G	RW	S6POWER			Keep RAM section S6 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S6RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
H	RW	S7POWER																																	
			Off	0																															
			On	1																															
I	RW	S8POWER																																	
			Off	0																															
			On	1																															
J	RW	S9POWER																																	
			Off	0																															
			On	1																															
K	RW	S10POWER																																	
			Off	0																															
			On	1																															
L	RW	S11POWER																																	
			Off	0																															
			On	1																															
M	RW	S12POWER																																	
			Off	0																															
			On	1																															
N	RW	S13POWER																																	
			Off	0																															
			On	1																															
O	RW	S14POWER																																	
			Off	0																															
			On	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
P	RW	S15POWER			Keep RAM section S15 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S15RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
Q	RW	S0RETENTION			Keep retention on RAM section S0 when RAM section is off
			Off	0	Off
			On	1	On
R	RW	S1RETENTION			Keep retention on RAM section S1 when RAM section is off
			Off	0	Off
			On	1	On
S	RW	S2RETENTION			Keep retention on RAM section S2 when RAM section is off
			Off	0	Off
			On	1	On
T	RW	S3RETENTION			Keep retention on RAM section S3 when RAM section is off
			Off	0	Off
			On	1	On
U	RW	S4RETENTION			Keep retention on RAM section S4 when RAM section is off
			Off	0	Off
			On	1	On
V	RW	S5RETENTION			Keep retention on RAM section S5 when RAM section is off
			Off	0	Off
			On	1	On
W	RW	S6RETENTION			Keep retention on RAM section S6 when RAM section is off
			Off	0	Off
			On	1	On
X	RW	S7RETENTION			Keep retention on RAM section S7 when RAM section is off
			Off	0	Off
			On	1	On
Y	RW	S8RETENTION			Keep retention on RAM section S8 when RAM section is off
			Off	0	Off
			On	1	On
Z	RW	S9RETENTION			Keep retention on RAM section S9 when RAM section is off
			Off	0	Off
			On	1	On
a	RW	S10RETENTION			Keep retention on RAM section S10 when RAM section is off
			Off	0	Off
			On	1	On
b	RW	S11RETENTION			Keep retention on RAM section S11 when RAM section is off
			Off	0	Off
			On	1	On
c	RW	S12RETENTION			Keep retention on RAM section S12 when RAM section is off
			Off	0	Off
			On	1	On
d	RW	S13RETENTION			Keep retention on RAM section S13 when RAM section is off
			Off	0	Off
			On	1	On
e	RW	S14RETENTION			Keep retention on RAM section S14 when RAM section is off
			Off	0	Off

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
			On	1	On
f	RW	S15RETENTION			Keep retention on RAM section S15 when RAM section is off
			Off	0	Off
			On	1	On

5.3.7.25 RAM[1].POWERSET

Address offset: 0x914

RAM1 power control set register

When read, this register will return the value of the POWER register.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	W	S0POWER			Keep RAM section S0 of RAM1 on or off in System ON mode
			On	1	On
B	W	S1POWER			Keep RAM section S1 of RAM1 on or off in System ON mode
			On	1	On
C	W	S2POWER			Keep RAM section S2 of RAM1 on or off in System ON mode
			On	1	On
D	W	S3POWER			Keep RAM section S3 of RAM1 on or off in System ON mode
			On	1	On
E	W	S4POWER			Keep RAM section S4 of RAM1 on or off in System ON mode
			On	1	On
F	W	S5POWER			Keep RAM section S5 of RAM1 on or off in System ON mode
			On	1	On
G	W	S6POWER			Keep RAM section S6 of RAM1 on or off in System ON mode
			On	1	On
H	W	S7POWER			Keep RAM section S7 of RAM1 on or off in System ON mode
			On	1	On
I	W	S8POWER			Keep RAM section S8 of RAM1 on or off in System ON mode
			On	1	On
J	W	S9POWER			Keep RAM section S9 of RAM1 on or off in System ON mode
			On	1	On
K	W	S10POWER			Keep RAM section S10 of RAM1 on or off in System ON mode
			On	1	On
L	W	S11POWER			Keep RAM section S11 of RAM1 on or off in System ON mode
			On	1	On
M	W	S12POWER			Keep RAM section S12 of RAM1 on or off in System ON mode
			On	1	On
N	W	S13POWER			Keep RAM section S13 of RAM1 on or off in System ON mode
			On	1	On
O	W	S14POWER			Keep RAM section S14 of RAM1 on or off in System ON mode
			On	1	On
P	W	S15POWER			Keep RAM section S15 of RAM1 on or off in System ON mode
			On	1	On
Q	W	S0RETENTION			Keep retention on RAM section S0 when RAM section is switched off
			On	1	On

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
R	W	S1RETENTION			Keep retention on RAM section S1 when RAM section is switched off
			On	1	On
S	W	S2RETENTION			Keep retention on RAM section S2 when RAM section is switched off
			On	1	On
T	W	S3RETENTION			Keep retention on RAM section S3 when RAM section is switched off
			On	1	On
U	W	S4RETENTION			Keep retention on RAM section S4 when RAM section is switched off
			On	1	On
V	W	S5RETENTION			Keep retention on RAM section S5 when RAM section is switched off
			On	1	On
W	W	S6RETENTION			Keep retention on RAM section S6 when RAM section is switched off
			On	1	On
X	W	S7RETENTION			Keep retention on RAM section S7 when RAM section is switched off
			On	1	On
Y	W	S8RETENTION			Keep retention on RAM section S8 when RAM section is switched off
			On	1	On
Z	W	S9RETENTION			Keep retention on RAM section S9 when RAM section is switched off
			On	1	On
a	W	S10RETENTION			Keep retention on RAM section S10 when RAM section is switched off
			On	1	On
b	W	S11RETENTION			Keep retention on RAM section S11 when RAM section is switched off
			On	1	On
c	W	S12RETENTION			Keep retention on RAM section S12 when RAM section is switched off
			On	1	On
d	W	S13RETENTION			Keep retention on RAM section S13 when RAM section is switched off
			On	1	On
e	W	S14RETENTION			Keep retention on RAM section S14 when RAM section is switched off
			On	1	On
f	W	S15RETENTION			Keep retention on RAM section S15 when RAM section is switched off
			On	1	On

5.3.7.26 RAM[1].POWERCLR

Address offset: 0x918

RAM1 power control clear register

When read, this register will return the value of the POWER register.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	W	S0POWER			Keep RAM section S0 of RAM1 on or off in System ON mode
			Off	1	Off
B	W	S1POWER			Keep RAM section S1 of RAM1 on or off in System ON mode
			Off	1	Off
C	W	S2POWER			Keep RAM section S2 of RAM1 on or off in System ON mode
			Off	1	Off
D	W	S3POWER			Keep RAM section S3 of RAM1 on or off in System ON mode
			Off	1	Off

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
E	W	S4POWER			Keep RAM section S4 of RAM1 on or off in System ON mode
			Off	1	Off
F	W	S5POWER			Keep RAM section S5 of RAM1 on or off in System ON mode
			Off	1	Off
G	W	S6POWER			Keep RAM section S6 of RAM1 on or off in System ON mode
			Off	1	Off
H	W	S7POWER			Keep RAM section S7 of RAM1 on or off in System ON mode
			Off	1	Off
I	W	S8POWER			Keep RAM section S8 of RAM1 on or off in System ON mode
			Off	1	Off
J	W	S9POWER			Keep RAM section S9 of RAM1 on or off in System ON mode
			Off	1	Off
K	W	S10POWER			Keep RAM section S10 of RAM1 on or off in System ON mode
			Off	1	Off
L	W	S11POWER			Keep RAM section S11 of RAM1 on or off in System ON mode
			Off	1	Off
M	W	S12POWER			Keep RAM section S12 of RAM1 on or off in System ON mode
			Off	1	Off
N	W	S13POWER			Keep RAM section S13 of RAM1 on or off in System ON mode
			Off	1	Off
O	W	S14POWER			Keep RAM section S14 of RAM1 on or off in System ON mode
			Off	1	Off
P	W	S15POWER			Keep RAM section S15 of RAM1 on or off in System ON mode
			Off	1	Off
Q	W	S0RETENTION			Keep retention on RAM section S0 when RAM section is switched off
			Off	1	Off
R	W	S1RETENTION			Keep retention on RAM section S1 when RAM section is switched off
			Off	1	Off
S	W	S2RETENTION			Keep retention on RAM section S2 when RAM section is switched off
			Off	1	Off
T	W	S3RETENTION			Keep retention on RAM section S3 when RAM section is switched off
			Off	1	Off
U	W	S4RETENTION			Keep retention on RAM section S4 when RAM section is switched off
			Off	1	Off
V	W	S5RETENTION			Keep retention on RAM section S5 when RAM section is switched off
			Off	1	Off
W	W	S6RETENTION			Keep retention on RAM section S6 when RAM section is switched off
			Off	1	Off
X	W	S7RETENTION			Keep retention on RAM section S7 when RAM section is switched off
			Off	1	Off
Y	W	S8RETENTION			Keep retention on RAM section S8 when RAM section is switched off
			Off	1	Off
Z	W	S9RETENTION			Keep retention on RAM section S9 when RAM section is switched off
			Off	1	Off
a	W	S10RETENTION			Keep retention on RAM section S10 when RAM section is switched off
			Off	1	Off
b	W	S11RETENTION			Keep retention on RAM section S11 when RAM section is switched off
			Off	1	Off
c	W	S12RETENTION			Keep retention on RAM section S12 when RAM section is switched off
			Off	1	Off

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
d	W	S13RETENTION			Keep retention on RAM section S13 when RAM section is switched off
			Off	1	Off
e	W	S14RETENTION			Keep retention on RAM section S14 when RAM section is switched off
			Off	1	Off
f	W	S15RETENTION			Keep retention on RAM section S15 when RAM section is switched off
			Off	1	Off

5.3.7.27 RAM[2].POWER

Address offset: 0x920

RAM2 power control register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	S0POWER			Keep RAM section S0 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S0RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
B	RW	S1POWER			Keep RAM section S1 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S1RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
C	RW	S2POWER			Keep RAM section S2 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S2RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
D	RW	S3POWER			Keep RAM section S3 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S3RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
E	RW	S4POWER			Keep RAM section S4 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S4RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
F	RW	S5POWER			Keep RAM section S5 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S5RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
G	RW	S6POWER			Keep RAM section S6 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S6RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
H	RW	S7POWER			Keep RAM section S7 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S7RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
I	RW	S8POWER			Keep RAM section S8 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S8RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
J	RW	S9POWER			Keep RAM section S9 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S9RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
K	RW	S10POWER			Keep RAM section S10 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S10RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
L	RW	S11POWER			Keep RAM section S11 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S11RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
M	RW	S12POWER			Keep RAM section S12 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S12RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
N	RW	S13POWER			Keep RAM section S13 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S13RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
O	RW	S14POWER			Keep RAM section S14 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S14RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
P	RW	S15POWER			Keep RAM section S15 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S15RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
Q	RW	S0RETENTION			Keep retention on RAM section S0 when RAM section is off
			Off	0	Off
			On	1	On
R	RW	S1RETENTION			Keep retention on RAM section S1 when RAM section is off
			Off	0	Off
			On	1	On
S	RW	S2RETENTION			Keep retention on RAM section S2 when RAM section is off
			Off	0	Off
			On	1	On
T	RW	S3RETENTION			Keep retention on RAM section S3 when RAM section is off
			Off	0	Off
			On	1	On
U	RW	S4RETENTION			Keep retention on RAM section S4 when RAM section is off
			Off	0	Off
			On	1	On
V	RW	S5RETENTION			Keep retention on RAM section S5 when RAM section is off
			Off	0	Off
			On	1	On
W	RW	S6RETENTION			Keep retention on RAM section S6 when RAM section is off
			Off	0	Off
			On	1	On
X	RW	S7RETENTION			Keep retention on RAM section S7 when RAM section is off
			Off	0	Off
			On	1	On
Y	RW	S8RETENTION			Keep retention on RAM section S8 when RAM section is off
			Off	0	Off
			On	1	On
Z	RW	S9RETENTION			Keep retention on RAM section S9 when RAM section is off
			Off	0	Off
			On	1	On
a	RW	S10RETENTION			Keep retention on RAM section S10 when RAM section is off

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
			Off	0	Off
			On	1	On
b	RW	S11RETENTION			Keep retention on RAM section S11 when RAM section is off
			Off	0	Off
			On	1	On
c	RW	S12RETENTION			Keep retention on RAM section S12 when RAM section is off
			Off	0	Off
			On	1	On
d	RW	S13RETENTION			Keep retention on RAM section S13 when RAM section is off
			Off	0	Off
			On	1	On
e	RW	S14RETENTION			Keep retention on RAM section S14 when RAM section is off
			Off	0	Off
			On	1	On
f	RW	S15RETENTION			Keep retention on RAM section S15 when RAM section is off
			Off	0	Off
			On	1	On

5.3.7.28 RAM[2].POWERSET

Address offset: 0x924

RAM2 power control set register

When read, this register will return the value of the POWER register.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	W	S0POWER			Keep RAM section S0 of RAM2 on or off in System ON mode
			On	1	On
B	W	S1POWER			Keep RAM section S1 of RAM2 on or off in System ON mode
			On	1	On
C	W	S2POWER			Keep RAM section S2 of RAM2 on or off in System ON mode
			On	1	On
D	W	S3POWER			Keep RAM section S3 of RAM2 on or off in System ON mode
			On	1	On
E	W	S4POWER			Keep RAM section S4 of RAM2 on or off in System ON mode
			On	1	On
F	W	S5POWER			Keep RAM section S5 of RAM2 on or off in System ON mode
			On	1	On
G	W	S6POWER			Keep RAM section S6 of RAM2 on or off in System ON mode
			On	1	On
H	W	S7POWER			Keep RAM section S7 of RAM2 on or off in System ON mode
			On	1	On
I	W	S8POWER			Keep RAM section S8 of RAM2 on or off in System ON mode
			On	1	On
J	W	S9POWER			Keep RAM section S9 of RAM2 on or off in System ON mode
			On	1	On
K	W	S10POWER			Keep RAM section S10 of RAM2 on or off in System ON mode

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
			On	1																															
L	W	S11POWER																																	
			On	1																															
M	W	S12POWER																																	
			On	1																															
N	W	S13POWER																																	
			On	1																															
O	W	S14POWER																																	
			On	1																															
P	W	S15POWER																																	
			On	1																															
Q	W	S0RETENTION																																	
			On	1																															
R	W	S1RETENTION																																	
			On	1																															
S	W	S2RETENTION																																	
			On	1																															
T	W	S3RETENTION																																	
			On	1																															
U	W	S4RETENTION																																	
			On	1																															
V	W	S5RETENTION																																	
			On	1																															
W	W	S6RETENTION																																	
			On	1																															
X	W	S7RETENTION																																	
			On	1																															
Y	W	S8RETENTION																																	
			On	1																															
Z	W	S9RETENTION																																	
			On	1																															
a	W	S10RETENTION																																	
			On	1																															
b	W	S11RETENTION																																	
			On	1																															
c	W	S12RETENTION																																	
			On	1																															
d	W	S13RETENTION																																	
			On	1																															
e	W	S14RETENTION																																	
			On	1																															
f	W	S15RETENTION																																	
			On	1																															

5.3.7.29 RAM[2].POWERCLR

Address offset: 0x928

RAM2 power control clear register

When read, this register will return the value of the POWER register.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	W	S0POWER																																	
			Off	1																															
B	W	S1POWER																																	
			Off	1																															
C	W	S2POWER																																	
			Off	1																															
D	W	S3POWER																																	
			Off	1																															
E	W	S4POWER																																	
			Off	1																															
F	W	S5POWER																																	
			Off	1																															
G	W	S6POWER																																	
			Off	1																															
H	W	S7POWER																																	
			Off	1																															
I	W	S8POWER																																	
			Off	1																															
J	W	S9POWER																																	
			Off	1																															
K	W	S10POWER																																	
			Off	1																															
L	W	S11POWER																																	
			Off	1																															
M	W	S12POWER																																	
			Off	1																															
N	W	S13POWER																																	
			Off	1																															
O	W	S14POWER																																	
			Off	1																															
P	W	S15POWER																																	
			Off	1																															
Q	W	S0RETENTION																																	
			Off	1																															
R	W	S1RETENTION																																	
			Off	1																															
S	W	S2RETENTION																																	
			Off	1																															
T	W	S3RETENTION																																	
			Off	1																															
U	W	S4RETENTION																																	
			Off	1																															
V	W	S5RETENTION																																	
			Off	1																															
W	W	S6RETENTION																																	
			Off	1																															
X	W	S7RETENTION																																	
			Off	1																															
Y	W	S8RETENTION																																	
			Off	1																															

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
Z	W	S9RETENTION			Keep retention on RAM section S9 when RAM section is switched off
			Off	1	Off
a	W	S10RETENTION			Keep retention on RAM section S10 when RAM section is switched off
			Off	1	Off
b	W	S11RETENTION			Keep retention on RAM section S11 when RAM section is switched off
			Off	1	Off
c	W	S12RETENTION			Keep retention on RAM section S12 when RAM section is switched off
			Off	1	Off
d	W	S13RETENTION			Keep retention on RAM section S13 when RAM section is switched off
			Off	1	Off
e	W	S14RETENTION			Keep retention on RAM section S14 when RAM section is switched off
			Off	1	Off
f	W	S15RETENTION			Keep retention on RAM section S15 when RAM section is switched off
			Off	1	Off

5.3.7.30 RAM[3].POWER

Address offset: 0x930

RAM3 power control register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	S0POWER			Keep RAM section S0 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S0RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
B	RW	S1POWER			Keep RAM section S1 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S1RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
C	RW	S2POWER			Keep RAM section S2 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S2RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
D	RW	S3POWER			Keep RAM section S3 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S3RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
E	RW	S4POWER			Keep RAM section S4 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S4RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
F	RW	S5POWER			Keep RAM section S5 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S5RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
G	RW	S6POWER			Keep RAM section S6 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S6RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
H	RW	S7POWER			Keep RAM section S7 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S7RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
I	RW	S8POWER			Keep RAM section S8 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S8RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
J	RW	S9POWER			Keep RAM section S9 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S9RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
K	RW	S10POWER			Keep RAM section S10 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S10RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
L	RW	S11POWER			Keep RAM section S11 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S11RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
M	RW	S12POWER			Keep RAM section S12 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S12RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
N	RW	S13POWER			Keep RAM section S13 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S13RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
O	RW	S14POWER			Keep RAM section S14 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S14RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
P	RW	S15POWER			Keep RAM section S15 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S15RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
Q	RW	S0RETENTION			Keep retention on RAM section S0 when RAM section is off
			Off	0	Off
			On	1	On
R	RW	S1RETENTION			Keep retention on RAM section S1 when RAM section is off
			Off	0	Off
			On	1	On
S	RW	S2RETENTION			Keep retention on RAM section S2 when RAM section is off
			Off	0	Off
			On	1	On
T	RW	S3RETENTION			Keep retention on RAM section S3 when RAM section is off
			Off	0	Off
			On	1	On
U	RW	S4RETENTION			Keep retention on RAM section S4 when RAM section is off
			Off	0	Off
			On	1	On
V	RW	S5RETENTION			Keep retention on RAM section S5 when RAM section is off
			Off	0	Off
			On	1	On
W	RW	S6RETENTION			Keep retention on RAM section S6 when RAM section is off
			Off	0	Off
			On	1	On
X	RW	S7RETENTION			Keep retention on RAM section S7 when RAM section is off
			Off	0	Off
			On	1	On
Y	RW	S8RETENTION			Keep retention on RAM section S8 when RAM section is off

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
			Off	0	Off
			On	1	On
Z	RW	S9RETENTION			Keep retention on RAM section S9 when RAM section is off
			Off	0	Off
			On	1	On
a	RW	S10RETENTION			Keep retention on RAM section S10 when RAM section is off
			Off	0	Off
			On	1	On
b	RW	S11RETENTION			Keep retention on RAM section S11 when RAM section is off
			Off	0	Off
			On	1	On
c	RW	S12RETENTION			Keep retention on RAM section S12 when RAM section is off
			Off	0	Off
			On	1	On
d	RW	S13RETENTION			Keep retention on RAM section S13 when RAM section is off
			Off	0	Off
			On	1	On
e	RW	S14RETENTION			Keep retention on RAM section S14 when RAM section is off
			Off	0	Off
			On	1	On
f	RW	S15RETENTION			Keep retention on RAM section S15 when RAM section is off
			Off	0	Off
			On	1	On

5.3.7.31 RAM[3].POWERSET

Address offset: 0x934

RAM3 power control set register

When read, this register will return the value of the POWER register.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	W	S0POWER			Keep RAM section S0 of RAM3 on or off in System ON mode
			On	1	On
B	W	S1POWER			Keep RAM section S1 of RAM3 on or off in System ON mode
			On	1	On
C	W	S2POWER			Keep RAM section S2 of RAM3 on or off in System ON mode
			On	1	On
D	W	S3POWER			Keep RAM section S3 of RAM3 on or off in System ON mode
			On	1	On
E	W	S4POWER			Keep RAM section S4 of RAM3 on or off in System ON mode
			On	1	On
F	W	S5POWER			Keep RAM section S5 of RAM3 on or off in System ON mode
			On	1	On
G	W	S6POWER			Keep RAM section S6 of RAM3 on or off in System ON mode
			On	1	On
H	W	S7POWER			Keep RAM section S7 of RAM3 on or off in System ON mode

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
			On	1																															
I	W	S8POWER																																	
			On	1																															
J	W	S9POWER																																	
			On	1																															
K	W	S10POWER																																	
			On	1																															
L	W	S11POWER																																	
			On	1																															
M	W	S12POWER																																	
			On	1																															
N	W	S13POWER																																	
			On	1																															
O	W	S14POWER																																	
			On	1																															
P	W	S15POWER																																	
			On	1																															
Q	W	S0RETENTION																																	
			On	1																															
R	W	S1RETENTION																																	
			On	1																															
S	W	S2RETENTION																																	
			On	1																															
T	W	S3RETENTION																																	
			On	1																															
U	W	S4RETENTION																																	
			On	1																															
V	W	S5RETENTION																																	
			On	1																															
W	W	S6RETENTION																																	
			On	1																															
X	W	S7RETENTION																																	
			On	1																															
Y	W	S8RETENTION																																	
			On	1																															
Z	W	S9RETENTION																																	
			On	1																															
a	W	S10RETENTION																																	
			On	1																															
b	W	S11RETENTION																																	
			On	1																															
c	W	S12RETENTION																																	
			On	1																															
d	W	S13RETENTION																																	
			On	1																															
e	W	S14RETENTION																																	
			On	1																															
f	W	S15RETENTION																																	
			On	1																															

5.3.7.32 RAM[3].POWERCLR

Address offset: 0x938

RAM3 power control clear register

When read, this register will return the value of the POWER register.

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value	Description																															
A	W	S0POWER			Keep RAM section S0 of RAM3 on or off in System ON mode																																	
			Off	1	Off																															
B	W	S1POWER			Keep RAM section S1 of RAM3 on or off in System ON mode																																	
			Off	1	Off																															
C	W	S2POWER			Keep RAM section S2 of RAM3 on or off in System ON mode																																	
			Off	1	Off																															
D	W	S3POWER			Keep RAM section S3 of RAM3 on or off in System ON mode																																	
			Off	1	Off																															
E	W	S4POWER			Keep RAM section S4 of RAM3 on or off in System ON mode																																	
			Off	1	Off																															
F	W	S5POWER			Keep RAM section S5 of RAM3 on or off in System ON mode																																	
			Off	1	Off																															
G	W	S6POWER			Keep RAM section S6 of RAM3 on or off in System ON mode																																	
			Off	1	Off																															
H	W	S7POWER			Keep RAM section S7 of RAM3 on or off in System ON mode																																	
			Off	1	Off																															
I	W	S8POWER			Keep RAM section S8 of RAM3 on or off in System ON mode																																	
			Off	1	Off																															
J	W	S9POWER			Keep RAM section S9 of RAM3 on or off in System ON mode																																	
			Off	1	Off																															
K	W	S10POWER			Keep RAM section S10 of RAM3 on or off in System ON mode																																	
			Off	1	Off																															
L	W	S11POWER			Keep RAM section S11 of RAM3 on or off in System ON mode																																	
			Off	1	Off																															
M	W	S12POWER			Keep RAM section S12 of RAM3 on or off in System ON mode																																	
			Off	1	Off																															
N	W	S13POWER			Keep RAM section S13 of RAM3 on or off in System ON mode																																	
			Off	1	Off																															
O	W	S14POWER			Keep RAM section S14 of RAM3 on or off in System ON mode																																	
			Off	1	Off																															
P	W	S15POWER			Keep RAM section S15 of RAM3 on or off in System ON mode																																	
			Off	1	Off																															
Q	W	S0RETENTION			Keep retention on RAM section S0 when RAM section is switched off																																	
			Off	1	Off																															
R	W	S1RETENTION			Keep retention on RAM section S1 when RAM section is switched off																																	
			Off	1	Off																															
S	W	S2RETENTION			Keep retention on RAM section S2 when RAM section is switched off																																	
			Off	1	Off																															
T	W	S3RETENTION			Keep retention on RAM section S3 when RAM section is switched off																																	
			Off	1	Off																															
U	W	S4RETENTION			Keep retention on RAM section S4 when RAM section is switched off																																	
			Off	1	Off																															
V	W	S5RETENTION			Keep retention on RAM section S5 when RAM section is switched off																																	

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
			Off	1	Off
W	W	S6RETENTION			Keep retention on RAM section S6 when RAM section is switched off
			Off	1	Off
X	W	S7RETENTION			Keep retention on RAM section S7 when RAM section is switched off
			Off	1	Off
Y	W	S8RETENTION			Keep retention on RAM section S8 when RAM section is switched off
			Off	1	Off
Z	W	S9RETENTION			Keep retention on RAM section S9 when RAM section is switched off
			Off	1	Off
a	W	S10RETENTION			Keep retention on RAM section S10 when RAM section is switched off
			Off	1	Off
b	W	S11RETENTION			Keep retention on RAM section S11 when RAM section is switched off
			Off	1	Off
c	W	S12RETENTION			Keep retention on RAM section S12 when RAM section is switched off
			Off	1	Off
d	W	S13RETENTION			Keep retention on RAM section S13 when RAM section is switched off
			Off	1	Off
e	W	S14RETENTION			Keep retention on RAM section S14 when RAM section is switched off
			Off	1	Off
f	W	S15RETENTION			Keep retention on RAM section S15 when RAM section is switched off
			Off	1	Off

5.3.7.33 RAM[4].POWER

Address offset: 0x940

RAM4 power control register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	S0POWER			Keep RAM section S0 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S0RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
B	RW	S1POWER			Keep RAM section S1 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S1RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
C	RW	S2POWER			Keep RAM section S2 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S2RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
			On	1																															
D	RW	S3POWER																																	
			Off	0																															
			On	1																															
E	RW	S4POWER																																	
			Off	0																															
			On	1																															
F	RW	S5POWER																																	
			Off	0																															
			On	1																															
G	RW	S6POWER																																	
			Off	0																															
			On	1																															
H	RW	S7POWER																																	
			Off	0																															
			On	1																															
I	RW	S8POWER																																	
			Off	0																															
			On	1																															
J	RW	S9POWER																																	
			Off	0																															
			On	1																															
K	RW	S10POWER																																	
			Off	0																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
			On	1	On
L	RW	S11POWER			Keep RAM section S11 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S11RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
M	RW	S12POWER			Keep RAM section S12 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S12RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
N	RW	S13POWER			Keep RAM section S13 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S13RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
O	RW	S14POWER			Keep RAM section S14 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S14RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
P	RW	S15POWER			Keep RAM section S15 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S15RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
Q	RW	S0RETENTION			Keep retention on RAM section S0 when RAM section is off
			Off	0	Off
			On	1	On
R	RW	S1RETENTION			Keep retention on RAM section S1 when RAM section is off
			Off	0	Off
			On	1	On
S	RW	S2RETENTION			Keep retention on RAM section S2 when RAM section is off
			Off	0	Off
			On	1	On
T	RW	S3RETENTION			Keep retention on RAM section S3 when RAM section is off
			Off	0	Off
			On	1	On
U	RW	S4RETENTION			Keep retention on RAM section S4 when RAM section is off
			Off	0	Off
			On	1	On
V	RW	S5RETENTION			Keep retention on RAM section S5 when RAM section is off
			Off	0	Off
			On	1	On

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
W	RW	S6RETENTION			Keep retention on RAM section S6 when RAM section is off
			Off	0	Off
			On	1	On
X	RW	S7RETENTION			Keep retention on RAM section S7 when RAM section is off
			Off	0	Off
			On	1	On
Y	RW	S8RETENTION			Keep retention on RAM section S8 when RAM section is off
			Off	0	Off
			On	1	On
Z	RW	S9RETENTION			Keep retention on RAM section S9 when RAM section is off
			Off	0	Off
			On	1	On
a	RW	S10RETENTION			Keep retention on RAM section S10 when RAM section is off
			Off	0	Off
			On	1	On
b	RW	S11RETENTION			Keep retention on RAM section S11 when RAM section is off
			Off	0	Off
			On	1	On
c	RW	S12RETENTION			Keep retention on RAM section S12 when RAM section is off
			Off	0	Off
			On	1	On
d	RW	S13RETENTION			Keep retention on RAM section S13 when RAM section is off
			Off	0	Off
			On	1	On
e	RW	S14RETENTION			Keep retention on RAM section S14 when RAM section is off
			Off	0	Off
			On	1	On
f	RW	S15RETENTION			Keep retention on RAM section S15 when RAM section is off
			Off	0	Off
			On	1	On

5.3.7.34 RAM[4].POWERSET

Address offset: 0x944

RAM4 power control set register

When read, this register will return the value of the POWER register.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	W	S0POWER			Keep RAM section S0 of RAM4 on or off in System ON mode
			On	1	On
B	W	S1POWER			Keep RAM section S1 of RAM4 on or off in System ON mode
			On	1	On
C	W	S2POWER			Keep RAM section S2 of RAM4 on or off in System ON mode
			On	1	On
D	W	S3POWER			Keep RAM section S3 of RAM4 on or off in System ON mode
			On	1	On

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
E	W	S4POWER			Keep RAM section S4 of RAM4 on or off in System ON mode
			On	1	On
F	W	S5POWER			Keep RAM section S5 of RAM4 on or off in System ON mode
			On	1	On
G	W	S6POWER			Keep RAM section S6 of RAM4 on or off in System ON mode
			On	1	On
H	W	S7POWER			Keep RAM section S7 of RAM4 on or off in System ON mode
			On	1	On
I	W	S8POWER			Keep RAM section S8 of RAM4 on or off in System ON mode
			On	1	On
J	W	S9POWER			Keep RAM section S9 of RAM4 on or off in System ON mode
			On	1	On
K	W	S10POWER			Keep RAM section S10 of RAM4 on or off in System ON mode
			On	1	On
L	W	S11POWER			Keep RAM section S11 of RAM4 on or off in System ON mode
			On	1	On
M	W	S12POWER			Keep RAM section S12 of RAM4 on or off in System ON mode
			On	1	On
N	W	S13POWER			Keep RAM section S13 of RAM4 on or off in System ON mode
			On	1	On
O	W	S14POWER			Keep RAM section S14 of RAM4 on or off in System ON mode
			On	1	On
P	W	S15POWER			Keep RAM section S15 of RAM4 on or off in System ON mode
			On	1	On
Q	W	S0RETENTION			Keep retention on RAM section S0 when RAM section is switched off
			On	1	On
R	W	S1RETENTION			Keep retention on RAM section S1 when RAM section is switched off
			On	1	On
S	W	S2RETENTION			Keep retention on RAM section S2 when RAM section is switched off
			On	1	On
T	W	S3RETENTION			Keep retention on RAM section S3 when RAM section is switched off
			On	1	On
U	W	S4RETENTION			Keep retention on RAM section S4 when RAM section is switched off
			On	1	On
V	W	S5RETENTION			Keep retention on RAM section S5 when RAM section is switched off
			On	1	On
W	W	S6RETENTION			Keep retention on RAM section S6 when RAM section is switched off
			On	1	On
X	W	S7RETENTION			Keep retention on RAM section S7 when RAM section is switched off
			On	1	On
Y	W	S8RETENTION			Keep retention on RAM section S8 when RAM section is switched off
			On	1	On
Z	W	S9RETENTION			Keep retention on RAM section S9 when RAM section is switched off
			On	1	On
a	W	S10RETENTION			Keep retention on RAM section S10 when RAM section is switched off
			On	1	On
b	W	S11RETENTION			Keep retention on RAM section S11 when RAM section is switched off
			On	1	On
c	W	S12RETENTION			Keep retention on RAM section S12 when RAM section is switched off
			On	1	On

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
d	W	S13RETENTION			Keep retention on RAM section S13 when RAM section is switched off
			On	1	On
e	W	S14RETENTION			Keep retention on RAM section S14 when RAM section is switched off
			On	1	On
f	W	S15RETENTION			Keep retention on RAM section S15 when RAM section is switched off
			On	1	On

5.3.7.35 RAM[4].POWERCLR

Address offset: 0x948

RAM4 power control clear register

When read, this register will return the value of the POWER register.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	W	S0POWER			Keep RAM section S0 of RAM4 on or off in System ON mode
			Off	1	Off
B	W	S1POWER			Keep RAM section S1 of RAM4 on or off in System ON mode
			Off	1	Off
C	W	S2POWER			Keep RAM section S2 of RAM4 on or off in System ON mode
			Off	1	Off
D	W	S3POWER			Keep RAM section S3 of RAM4 on or off in System ON mode
			Off	1	Off
E	W	S4POWER			Keep RAM section S4 of RAM4 on or off in System ON mode
			Off	1	Off
F	W	S5POWER			Keep RAM section S5 of RAM4 on or off in System ON mode
			Off	1	Off
G	W	S6POWER			Keep RAM section S6 of RAM4 on or off in System ON mode
			Off	1	Off
H	W	S7POWER			Keep RAM section S7 of RAM4 on or off in System ON mode
			Off	1	Off
I	W	S8POWER			Keep RAM section S8 of RAM4 on or off in System ON mode
			Off	1	Off
J	W	S9POWER			Keep RAM section S9 of RAM4 on or off in System ON mode
			Off	1	Off
K	W	S10POWER			Keep RAM section S10 of RAM4 on or off in System ON mode
			Off	1	Off
L	W	S11POWER			Keep RAM section S11 of RAM4 on or off in System ON mode
			Off	1	Off
M	W	S12POWER			Keep RAM section S12 of RAM4 on or off in System ON mode
			Off	1	Off
N	W	S13POWER			Keep RAM section S13 of RAM4 on or off in System ON mode
			Off	1	Off
O	W	S14POWER			Keep RAM section S14 of RAM4 on or off in System ON mode
			Off	1	Off
P	W	S15POWER			Keep RAM section S15 of RAM4 on or off in System ON mode
			Off	1	Off

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
Q	W	S0RETENTION			Keep retention on RAM section S0 when RAM section is switched off
			Off	1	Off
R	W	S1RETENTION			Keep retention on RAM section S1 when RAM section is switched off
			Off	1	Off
S	W	S2RETENTION			Keep retention on RAM section S2 when RAM section is switched off
			Off	1	Off
T	W	S3RETENTION			Keep retention on RAM section S3 when RAM section is switched off
			Off	1	Off
U	W	S4RETENTION			Keep retention on RAM section S4 when RAM section is switched off
			Off	1	Off
V	W	S5RETENTION			Keep retention on RAM section S5 when RAM section is switched off
			Off	1	Off
W	W	S6RETENTION			Keep retention on RAM section S6 when RAM section is switched off
			Off	1	Off
X	W	S7RETENTION			Keep retention on RAM section S7 when RAM section is switched off
			Off	1	Off
Y	W	S8RETENTION			Keep retention on RAM section S8 when RAM section is switched off
			Off	1	Off
Z	W	S9RETENTION			Keep retention on RAM section S9 when RAM section is switched off
			Off	1	Off
a	W	S10RETENTION			Keep retention on RAM section S10 when RAM section is switched off
			Off	1	Off
b	W	S11RETENTION			Keep retention on RAM section S11 when RAM section is switched off
			Off	1	Off
c	W	S12RETENTION			Keep retention on RAM section S12 when RAM section is switched off
			Off	1	Off
d	W	S13RETENTION			Keep retention on RAM section S13 when RAM section is switched off
			Off	1	Off
e	W	S14RETENTION			Keep retention on RAM section S14 when RAM section is switched off
			Off	1	Off
f	W	S15RETENTION			Keep retention on RAM section S15 when RAM section is switched off
			Off	1	Off

5.3.7.36 RAM[5].POWER

Address offset: 0x950

RAM5 power control register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	S0POWER			Keep RAM section S0 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S0RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
B	RW	S1POWER																																	
			Off	0																															
			On	1																															
C	RW	S2POWER																																	
			Off	0																															
			On	1																															
D	RW	S3POWER																																	
			Off	0																															
			On	1																															
E	RW	S4POWER																																	
			Off	0																															
			On	1																															
F	RW	S5POWER																																	
			Off	0																															
			On	1																															
G	RW	S6POWER																																	
			Off	0																															
			On	1																															
H	RW	S7POWER																																	
			Off	0																															
			On	1																															
I	RW	S8POWER																																	
			Off	0																															
			On	1																															

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
J	RW	S9POWER			Keep RAM section S9 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S9RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
K	RW	S10POWER			Keep RAM section S10 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S10RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
L	RW	S11POWER			Keep RAM section S11 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S11RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
M	RW	S12POWER			Keep RAM section S12 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S12RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
N	RW	S13POWER			Keep RAM section S13 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S13RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
O	RW	S14POWER			Keep RAM section S14 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S14RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
P	RW	S15POWER			Keep RAM section S15 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S15RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
Q	RW	S0RETENTION			Keep retention on RAM section S0 when RAM section is off
			Off	0	Off
			On	1	On
R	RW	S1RETENTION			Keep retention on RAM section S1 when RAM section is off
			Off	0	Off
			On	1	On

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
S	RW	S2RETENTION																																	
			Off	0																															
			On	1																															
T	RW	S3RETENTION																																	
			Off	0																															
			On	1																															
U	RW	S4RETENTION																																	
			Off	0																															
			On	1																															
V	RW	S5RETENTION																																	
			Off	0																															
			On	1																															
W	RW	S6RETENTION																																	
			Off	0																															
			On	1																															
X	RW	S7RETENTION																																	
			Off	0																															
			On	1																															
Y	RW	S8RETENTION																																	
			Off	0																															
			On	1																															
Z	RW	S9RETENTION																																	
			Off	0																															
			On	1																															
a	RW	S10RETENTION																																	
			Off	0																															
			On	1																															
b	RW	S11RETENTION																																	
			Off	0																															
			On	1																															
c	RW	S12RETENTION																																	
			Off	0																															
			On	1																															
d	RW	S13RETENTION																																	
			Off	0																															
			On	1																															
e	RW	S14RETENTION																																	
			Off	0																															
			On	1																															
f	RW	S15RETENTION																																	
			Off	0																															
			On	1																															

5.3.7.37 RAM[5].POWERSET

Address offset: 0x954

RAM5 power control set register

When read, this register will return the value of the POWER register.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	W	S0POWER																																	
			On	1																															
B	W	S1POWER																																	
			On	1																															
C	W	S2POWER																																	
			On	1																															
D	W	S3POWER																																	
			On	1																															
E	W	S4POWER																																	
			On	1																															
F	W	S5POWER																																	
			On	1																															
G	W	S6POWER																																	
			On	1																															
H	W	S7POWER																																	
			On	1																															
I	W	S8POWER																																	
			On	1																															
J	W	S9POWER																																	
			On	1																															
K	W	S10POWER																																	
			On	1																															
L	W	S11POWER																																	
			On	1																															
M	W	S12POWER																																	
			On	1																															
N	W	S13POWER																																	
			On	1																															
O	W	S14POWER																																	
			On	1																															
P	W	S15POWER																																	
			On	1																															
Q	W	S0RETENTION																																	
			On	1																															
R	W	S1RETENTION																																	
			On	1																															
S	W	S2RETENTION																																	
			On	1																															
T	W	S3RETENTION																																	
			On	1																															
U	W	S4RETENTION																																	
			On	1																															
V	W	S5RETENTION																																	
			On	1																															
W	W	S6RETENTION																																	
			On	1																															
X	W	S7RETENTION																																	
			On	1																															
Y	W	S8RETENTION																																	
			On	1																															

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
Z	W	S9RETENTION			Keep retention on RAM section S9 when RAM section is switched off
			On	1	On
a	W	S10RETENTION			Keep retention on RAM section S10 when RAM section is switched off
			On	1	On
b	W	S11RETENTION			Keep retention on RAM section S11 when RAM section is switched off
			On	1	On
c	W	S12RETENTION			Keep retention on RAM section S12 when RAM section is switched off
			On	1	On
d	W	S13RETENTION			Keep retention on RAM section S13 when RAM section is switched off
			On	1	On
e	W	S14RETENTION			Keep retention on RAM section S14 when RAM section is switched off
			On	1	On
f	W	S15RETENTION			Keep retention on RAM section S15 when RAM section is switched off
			On	1	On

5.3.7.38 RAM[5].POWERCLR

Address offset: 0x958

RAM5 power control clear register

When read, this register will return the value of the POWER register.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	W	S0POWER			Keep RAM section S0 of RAM5 on or off in System ON mode
			Off	1	Off
B	W	S1POWER			Keep RAM section S1 of RAM5 on or off in System ON mode
			Off	1	Off
C	W	S2POWER			Keep RAM section S2 of RAM5 on or off in System ON mode
			Off	1	Off
D	W	S3POWER			Keep RAM section S3 of RAM5 on or off in System ON mode
			Off	1	Off
E	W	S4POWER			Keep RAM section S4 of RAM5 on or off in System ON mode
			Off	1	Off
F	W	S5POWER			Keep RAM section S5 of RAM5 on or off in System ON mode
			Off	1	Off
G	W	S6POWER			Keep RAM section S6 of RAM5 on or off in System ON mode
			Off	1	Off
H	W	S7POWER			Keep RAM section S7 of RAM5 on or off in System ON mode
			Off	1	Off
I	W	S8POWER			Keep RAM section S8 of RAM5 on or off in System ON mode
			Off	1	Off
J	W	S9POWER			Keep RAM section S9 of RAM5 on or off in System ON mode
			Off	1	Off
K	W	S10POWER			Keep RAM section S10 of RAM5 on or off in System ON mode
			Off	1	Off
L	W	S11POWER			Keep RAM section S11 of RAM5 on or off in System ON mode
			Off	1	Off

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
M	W	S12POWER																																	
			Off	1																															
N	W	S13POWER																																	
			Off	1																															
O	W	S14POWER																																	
			Off	1																															
P	W	S15POWER																																	
			Off	1																															
Q	W	S0RETENTION																																	
			Off	1																															
R	W	S1RETENTION																																	
			Off	1																															
S	W	S2RETENTION																																	
			Off	1																															
T	W	S3RETENTION																																	
			Off	1																															
U	W	S4RETENTION																																	
			Off	1																															
V	W	S5RETENTION																																	
			Off	1																															
W	W	S6RETENTION																																	
			Off	1																															
X	W	S7RETENTION																																	
			Off	1																															
Y	W	S8RETENTION																																	
			Off	1																															
Z	W	S9RETENTION																																	
			Off	1																															
a	W	S10RETENTION																																	
			Off	1																															
b	W	S11RETENTION																																	
			Off	1																															
c	W	S12RETENTION																																	
			Off	1																															
d	W	S13RETENTION																																	
			Off	1																															
e	W	S14RETENTION																																	
			Off	1																															
f	W	S15RETENTION																																	
			Off	1																															

5.3.7.39 RAM[6].POWER

Address offset: 0x960

RAM6 power control register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW	S0POWER																																	
			Off	0																															
			On	1																															
B	RW	S1POWER																																	
			Off	0																															
			On	1																															
C	RW	S2POWER																																	
			Off	0																															
			On	1																															
D	RW	S3POWER																																	
			Off	0																															
			On	1																															
E	RW	S4POWER																																	
			Off	0																															
			On	1																															
F	RW	S5POWER																																	
			Off	0																															
			On	1																															
G	RW	S6POWER																																	
			Off	0																															
			On	1																															
H	RW	S7POWER																																	
			Off	0																															
			On	1																															

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
I	RW	S8POWER			Keep RAM section S8 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S8RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
J	RW	S9POWER			Keep RAM section S9 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S9RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
K	RW	S10POWER			Keep RAM section S10 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S10RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
L	RW	S11POWER			Keep RAM section S11 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S11RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
M	RW	S12POWER			Keep RAM section S12 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S12RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
N	RW	S13POWER			Keep RAM section S13 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S13RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
O	RW	S14POWER			Keep RAM section S14 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S14RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
P	RW	S15POWER			Keep RAM section S15 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S15RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
Q	RW	S0RETENTION																																	
			Off	0																															
			On	1																															
R	RW	S1RETENTION																																	
			Off	0																															
			On	1																															
S	RW	S2RETENTION																																	
			Off	0																															
			On	1																															
T	RW	S3RETENTION																																	
			Off	0																															
			On	1																															
U	RW	S4RETENTION																																	
			Off	0																															
			On	1																															
V	RW	S5RETENTION																																	
			Off	0																															
			On	1																															
W	RW	S6RETENTION																																	
			Off	0																															
			On	1																															
X	RW	S7RETENTION																																	
			Off	0																															
			On	1																															
Y	RW	S8RETENTION																																	
			Off	0																															
			On	1																															
Z	RW	S9RETENTION																																	
			Off	0																															
			On	1																															
a	RW	S10RETENTION																																	
			Off	0																															
			On	1																															
b	RW	S11RETENTION																																	
			Off	0																															
			On	1																															
c	RW	S12RETENTION																																	
			Off	0																															
			On	1																															
d	RW	S13RETENTION																																	
			Off	0																															
			On	1																															
e	RW	S14RETENTION																																	
			Off	0																															
			On	1																															
f	RW	S15RETENTION																																	
			Off	0																															
			On	1																															

5.3.7.40 RAM[6].POWERSET

Address offset: 0x964

RAM6 power control set register

When read, this register will return the value of the POWER register.

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value	Description																															
A	W	S0POWER			Keep RAM section S0 of RAM6 on or off in System ON mode																																	
			On	1	On																															
B	W	S1POWER			Keep RAM section S1 of RAM6 on or off in System ON mode																																	
			On	1	On																															
C	W	S2POWER			Keep RAM section S2 of RAM6 on or off in System ON mode																																	
			On	1	On																															
D	W	S3POWER			Keep RAM section S3 of RAM6 on or off in System ON mode																																	
			On	1	On																															
E	W	S4POWER			Keep RAM section S4 of RAM6 on or off in System ON mode																																	
			On	1	On																															
F	W	S5POWER			Keep RAM section S5 of RAM6 on or off in System ON mode																																	
			On	1	On																															
G	W	S6POWER			Keep RAM section S6 of RAM6 on or off in System ON mode																																	
			On	1	On																															
H	W	S7POWER			Keep RAM section S7 of RAM6 on or off in System ON mode																																	
			On	1	On																															
I	W	S8POWER			Keep RAM section S8 of RAM6 on or off in System ON mode																																	
			On	1	On																															
J	W	S9POWER			Keep RAM section S9 of RAM6 on or off in System ON mode																																	
			On	1	On																															
K	W	S10POWER			Keep RAM section S10 of RAM6 on or off in System ON mode																																	
			On	1	On																															
L	W	S11POWER			Keep RAM section S11 of RAM6 on or off in System ON mode																																	
			On	1	On																															
M	W	S12POWER			Keep RAM section S12 of RAM6 on or off in System ON mode																																	
			On	1	On																															
N	W	S13POWER			Keep RAM section S13 of RAM6 on or off in System ON mode																																	
			On	1	On																															
O	W	S14POWER			Keep RAM section S14 of RAM6 on or off in System ON mode																																	
			On	1	On																															
P	W	S15POWER			Keep RAM section S15 of RAM6 on or off in System ON mode																																	
			On	1	On																															
Q	W	S0RETENTION			Keep retention on RAM section S0 when RAM section is switched off																																	
			On	1	On																															
R	W	S1RETENTION			Keep retention on RAM section S1 when RAM section is switched off																																	
			On	1	On																															
S	W	S2RETENTION			Keep retention on RAM section S2 when RAM section is switched off																																	
			On	1	On																															
T	W	S3RETENTION			Keep retention on RAM section S3 when RAM section is switched off																																	
			On	1	On																															
U	W	S4RETENTION			Keep retention on RAM section S4 when RAM section is switched off																																	
			On	1	On																															
V	W	S5RETENTION			Keep retention on RAM section S5 when RAM section is switched off																																	

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
			On	1	On
W	W	S6RETENTION			Keep retention on RAM section S6 when RAM section is switched off
			On	1	On
X	W	S7RETENTION			Keep retention on RAM section S7 when RAM section is switched off
			On	1	On
Y	W	S8RETENTION			Keep retention on RAM section S8 when RAM section is switched off
			On	1	On
Z	W	S9RETENTION			Keep retention on RAM section S9 when RAM section is switched off
			On	1	On
a	W	S10RETENTION			Keep retention on RAM section S10 when RAM section is switched off
			On	1	On
b	W	S11RETENTION			Keep retention on RAM section S11 when RAM section is switched off
			On	1	On
c	W	S12RETENTION			Keep retention on RAM section S12 when RAM section is switched off
			On	1	On
d	W	S13RETENTION			Keep retention on RAM section S13 when RAM section is switched off
			On	1	On
e	W	S14RETENTION			Keep retention on RAM section S14 when RAM section is switched off
			On	1	On
f	W	S15RETENTION			Keep retention on RAM section S15 when RAM section is switched off
			On	1	On

5.3.7.41 RAM[6].POWERCLR

Address offset: 0x968

RAM6 power control clear register

When read, this register will return the value of the POWER register.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	W	S0POWER			Keep RAM section S0 of RAM6 on or off in System ON mode
			Off	1	Off
B	W	S1POWER			Keep RAM section S1 of RAM6 on or off in System ON mode
			Off	1	Off
C	W	S2POWER			Keep RAM section S2 of RAM6 on or off in System ON mode
			Off	1	Off
D	W	S3POWER			Keep RAM section S3 of RAM6 on or off in System ON mode
			Off	1	Off
E	W	S4POWER			Keep RAM section S4 of RAM6 on or off in System ON mode
			Off	1	Off
F	W	S5POWER			Keep RAM section S5 of RAM6 on or off in System ON mode
			Off	1	Off
G	W	S6POWER			Keep RAM section S6 of RAM6 on or off in System ON mode
			Off	1	Off
H	W	S7POWER			Keep RAM section S7 of RAM6 on or off in System ON mode
			Off	1	Off
I	W	S8POWER			Keep RAM section S8 of RAM6 on or off in System ON mode

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
			Off	1																															
J	W	S9POWER																																	
			Off	1																															
K	W	S10POWER																																	
			Off	1																															
L	W	S11POWER																																	
			Off	1																															
M	W	S12POWER																																	
			Off	1																															
N	W	S13POWER																																	
			Off	1																															
O	W	S14POWER																																	
			Off	1																															
P	W	S15POWER																																	
			Off	1																															
Q	W	S0RETENTION																																	
			Off	1																															
R	W	S1RETENTION																																	
			Off	1																															
S	W	S2RETENTION																																	
			Off	1																															
T	W	S3RETENTION																																	
			Off	1																															
U	W	S4RETENTION																																	
			Off	1																															
V	W	S5RETENTION																																	
			Off	1																															
W	W	S6RETENTION																																	
			Off	1																															
X	W	S7RETENTION																																	
			Off	1																															
Y	W	S8RETENTION																																	
			Off	1																															
Z	W	S9RETENTION																																	
			Off	1																															
a	W	S10RETENTION																																	
			Off	1																															
b	W	S11RETENTION																																	
			Off	1																															
c	W	S12RETENTION																																	
			Off	1																															
d	W	S13RETENTION																																	
			Off	1																															
e	W	S14RETENTION																																	
			Off	1																															
f	W	S15RETENTION																																	
			Off	1																															

5.3.7.42 RAM[7].POWER

Address offset: 0x970

RAM7 power control register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW	S0POWER																																	
			Off	0																															
			On	1																															
B	RW	S1POWER																																	
			Off	0																															
			On	1																															
C	RW	S2POWER																																	
			Off	0																															
			On	1																															
D	RW	S3POWER																																	
			Off	0																															
			On	1																															
E	RW	S4POWER																																	
			Off	0																															
			On	1																															
F	RW	S5POWER																																	
			Off	0																															
			On	1																															
G	RW	S6POWER																																	
			Off	0																															
			On	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
H	RW	S7POWER																																	
			Off	0																															
			On	1																															
I	RW	S8POWER																																	
			Off	0																															
			On	1																															
J	RW	S9POWER																																	
			Off	0																															
			On	1																															
K	RW	S10POWER																																	
			Off	0																															
			On	1																															
L	RW	S11POWER																																	
			Off	0																															
			On	1																															
M	RW	S12POWER																																	
			Off	0																															
			On	1																															
N	RW	S13POWER																																	
			Off	0																															
			On	1																															
O	RW	S14POWER																																	
			Off	0																															
			On	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
P	RW	S15POWER																																	
			Off	0																															
			On	1																															
Q	RW	S0RETENTION																																	
			Off	0																															
			On	1																															
R	RW	S1RETENTION																																	
			Off	0																															
			On	1																															
S	RW	S2RETENTION																																	
			Off	0																															
			On	1																															
T	RW	S3RETENTION																																	
			Off	0																															
			On	1																															
U	RW	S4RETENTION																																	
			Off	0																															
			On	1																															
V	RW	S5RETENTION																																	
			Off	0																															
			On	1																															
W	RW	S6RETENTION																																	
			Off	0																															
			On	1																															
X	RW	S7RETENTION																																	
			Off	0																															
			On	1																															
Y	RW	S8RETENTION																																	
			Off	0																															
			On	1																															
Z	RW	S9RETENTION																																	
			Off	0																															
			On	1																															
a	RW	S10RETENTION																																	
			Off	0																															
			On	1																															
b	RW	S11RETENTION																																	
			Off	0																															
			On	1																															
c	RW	S12RETENTION																																	
			Off	0																															
			On	1																															
d	RW	S13RETENTION																																	
			Off	0																															
			On	1																															
e	RW	S14RETENTION																																	
			Off	0																															

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
			On	1	On
f	RW	S15RETENTION			Keep retention on RAM section S15 when RAM section is off
			Off	0	Off
			On	1	On

5.3.7.43 RAM[7].POWERSET

Address offset: 0x974

RAM7 power control set register

When read, this register will return the value of the POWER register.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	W	S0POWER			Keep RAM section S0 of RAM7 on or off in System ON mode
			On	1	On
B	W	S1POWER			Keep RAM section S1 of RAM7 on or off in System ON mode
			On	1	On
C	W	S2POWER			Keep RAM section S2 of RAM7 on or off in System ON mode
			On	1	On
D	W	S3POWER			Keep RAM section S3 of RAM7 on or off in System ON mode
			On	1	On
E	W	S4POWER			Keep RAM section S4 of RAM7 on or off in System ON mode
			On	1	On
F	W	S5POWER			Keep RAM section S5 of RAM7 on or off in System ON mode
			On	1	On
G	W	S6POWER			Keep RAM section S6 of RAM7 on or off in System ON mode
			On	1	On
H	W	S7POWER			Keep RAM section S7 of RAM7 on or off in System ON mode
			On	1	On
I	W	S8POWER			Keep RAM section S8 of RAM7 on or off in System ON mode
			On	1	On
J	W	S9POWER			Keep RAM section S9 of RAM7 on or off in System ON mode
			On	1	On
K	W	S10POWER			Keep RAM section S10 of RAM7 on or off in System ON mode
			On	1	On
L	W	S11POWER			Keep RAM section S11 of RAM7 on or off in System ON mode
			On	1	On
M	W	S12POWER			Keep RAM section S12 of RAM7 on or off in System ON mode
			On	1	On
N	W	S13POWER			Keep RAM section S13 of RAM7 on or off in System ON mode
			On	1	On
O	W	S14POWER			Keep RAM section S14 of RAM7 on or off in System ON mode
			On	1	On
P	W	S15POWER			Keep RAM section S15 of RAM7 on or off in System ON mode
			On	1	On
Q	W	S0RETENTION			Keep retention on RAM section S0 when RAM section is switched off
			On	1	On

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
R	W	S1RETENTION			Keep retention on RAM section S1 when RAM section is switched off
			On	1	On
S	W	S2RETENTION			Keep retention on RAM section S2 when RAM section is switched off
			On	1	On
T	W	S3RETENTION			Keep retention on RAM section S3 when RAM section is switched off
			On	1	On
U	W	S4RETENTION			Keep retention on RAM section S4 when RAM section is switched off
			On	1	On
V	W	S5RETENTION			Keep retention on RAM section S5 when RAM section is switched off
			On	1	On
W	W	S6RETENTION			Keep retention on RAM section S6 when RAM section is switched off
			On	1	On
X	W	S7RETENTION			Keep retention on RAM section S7 when RAM section is switched off
			On	1	On
Y	W	S8RETENTION			Keep retention on RAM section S8 when RAM section is switched off
			On	1	On
Z	W	S9RETENTION			Keep retention on RAM section S9 when RAM section is switched off
			On	1	On
a	W	S10RETENTION			Keep retention on RAM section S10 when RAM section is switched off
			On	1	On
b	W	S11RETENTION			Keep retention on RAM section S11 when RAM section is switched off
			On	1	On
c	W	S12RETENTION			Keep retention on RAM section S12 when RAM section is switched off
			On	1	On
d	W	S13RETENTION			Keep retention on RAM section S13 when RAM section is switched off
			On	1	On
e	W	S14RETENTION			Keep retention on RAM section S14 when RAM section is switched off
			On	1	On
f	W	S15RETENTION			Keep retention on RAM section S15 when RAM section is switched off
			On	1	On

5.3.7.44 RAM[7].POWERCLR

Address offset: 0x978

RAM7 power control clear register

When read, this register will return the value of the POWER register.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	W	S0POWER			Keep RAM section S0 of RAM7 on or off in System ON mode
			Off	1	Off
B	W	S1POWER			Keep RAM section S1 of RAM7 on or off in System ON mode
			Off	1	Off
C	W	S2POWER			Keep RAM section S2 of RAM7 on or off in System ON mode
			Off	1	Off
D	W	S3POWER			Keep RAM section S3 of RAM7 on or off in System ON mode
			Off	1	Off

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
E	W	S4POWER			Keep RAM section S4 of RAM7 on or off in System ON mode
			Off	1	Off
F	W	S5POWER			Keep RAM section S5 of RAM7 on or off in System ON mode
			Off	1	Off
G	W	S6POWER			Keep RAM section S6 of RAM7 on or off in System ON mode
			Off	1	Off
H	W	S7POWER			Keep RAM section S7 of RAM7 on or off in System ON mode
			Off	1	Off
I	W	S8POWER			Keep RAM section S8 of RAM7 on or off in System ON mode
			Off	1	Off
J	W	S9POWER			Keep RAM section S9 of RAM7 on or off in System ON mode
			Off	1	Off
K	W	S10POWER			Keep RAM section S10 of RAM7 on or off in System ON mode
			Off	1	Off
L	W	S11POWER			Keep RAM section S11 of RAM7 on or off in System ON mode
			Off	1	Off
M	W	S12POWER			Keep RAM section S12 of RAM7 on or off in System ON mode
			Off	1	Off
N	W	S13POWER			Keep RAM section S13 of RAM7 on or off in System ON mode
			Off	1	Off
O	W	S14POWER			Keep RAM section S14 of RAM7 on or off in System ON mode
			Off	1	Off
P	W	S15POWER			Keep RAM section S15 of RAM7 on or off in System ON mode
			Off	1	Off
Q	W	S0RETENTION			Keep retention on RAM section S0 when RAM section is switched off
			Off	1	Off
R	W	S1RETENTION			Keep retention on RAM section S1 when RAM section is switched off
			Off	1	Off
S	W	S2RETENTION			Keep retention on RAM section S2 when RAM section is switched off
			Off	1	Off
T	W	S3RETENTION			Keep retention on RAM section S3 when RAM section is switched off
			Off	1	Off
U	W	S4RETENTION			Keep retention on RAM section S4 when RAM section is switched off
			Off	1	Off
V	W	S5RETENTION			Keep retention on RAM section S5 when RAM section is switched off
			Off	1	Off
W	W	S6RETENTION			Keep retention on RAM section S6 when RAM section is switched off
			Off	1	Off
X	W	S7RETENTION			Keep retention on RAM section S7 when RAM section is switched off
			Off	1	Off
Y	W	S8RETENTION			Keep retention on RAM section S8 when RAM section is switched off
			Off	1	Off
Z	W	S9RETENTION			Keep retention on RAM section S9 when RAM section is switched off
			Off	1	Off
a	W	S10RETENTION			Keep retention on RAM section S10 when RAM section is switched off
			Off	1	Off
b	W	S11RETENTION			Keep retention on RAM section S11 when RAM section is switched off
			Off	1	Off
c	W	S12RETENTION			Keep retention on RAM section S12 when RAM section is switched off
			Off	1	Off

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
d	W	S13RETENTION			Keep retention on RAM section S13 when RAM section is switched off
			Off	1	Off
e	W	S14RETENTION			Keep retention on RAM section S14 when RAM section is switched off
			Off	1	Off
f	W	S15RETENTION			Keep retention on RAM section S15 when RAM section is switched off
			Off	1	Off

5.3.7.45 RAM[8].POWER

Address offset: 0x980

RAM8 power control register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	S0POWER			Keep RAM section S0 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S0RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
B	RW	S1POWER			Keep RAM section S1 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S1RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
C	RW	S2POWER			Keep RAM section S2 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S2RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
D	RW	S3POWER			Keep RAM section S3 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S3RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
E	RW	S4POWER			Keep RAM section S4 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S4RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
F	RW	S5POWER																																	
			Off	0																															
			On	1																															
G	RW	S6POWER																																	
			Off	0																															
			On	1																															
H	RW	S7POWER																																	
			Off	0																															
			On	1																															
I	RW	S8POWER																																	
			Off	0																															
			On	1																															
J	RW	S9POWER																																	
			Off	0																															
			On	1																															
K	RW	S10POWER																																	
			Off	0																															
			On	1																															
L	RW	S11POWER																																	
			Off	0																															
			On	1																															
M	RW	S12POWER																																	
			Off	0																															
			On	1																															

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
N	RW	S13POWER			Keep RAM section S13 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S13RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
O	RW	S14POWER			Keep RAM section S14 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S14RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
P	RW	S15POWER			Keep RAM section S15 on or off in System ON mode. RAM sections are always retained when on, but can also be retained when off depending on the settings in S15RETENTION. All RAM sections will be off in System OFF mode.
			Off	0	Off
			On	1	On
Q	RW	S0RETENTION			Keep retention on RAM section S0 when RAM section is off
			Off	0	Off
			On	1	On
R	RW	S1RETENTION			Keep retention on RAM section S1 when RAM section is off
			Off	0	Off
			On	1	On
S	RW	S2RETENTION			Keep retention on RAM section S2 when RAM section is off
			Off	0	Off
			On	1	On
T	RW	S3RETENTION			Keep retention on RAM section S3 when RAM section is off
			Off	0	Off
			On	1	On
U	RW	S4RETENTION			Keep retention on RAM section S4 when RAM section is off
			Off	0	Off
			On	1	On
V	RW	S5RETENTION			Keep retention on RAM section S5 when RAM section is off
			Off	0	Off
			On	1	On
W	RW	S6RETENTION			Keep retention on RAM section S6 when RAM section is off
			Off	0	Off
			On	1	On
X	RW	S7RETENTION			Keep retention on RAM section S7 when RAM section is off
			Off	0	Off
			On	1	On
Y	RW	S8RETENTION			Keep retention on RAM section S8 when RAM section is off
			Off	0	Off
			On	1	On
Z	RW	S9RETENTION			Keep retention on RAM section S9 when RAM section is off
			Off	0	Off
			On	1	On
a	RW	S10RETENTION			Keep retention on RAM section S10 when RAM section is off

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
			Off	0	Off
			On	1	On
b	RW	S11RETENTION			Keep retention on RAM section S11 when RAM section is off
			Off	0	Off
			On	1	On
c	RW	S12RETENTION			Keep retention on RAM section S12 when RAM section is off
			Off	0	Off
			On	1	On
d	RW	S13RETENTION			Keep retention on RAM section S13 when RAM section is off
			Off	0	Off
			On	1	On
e	RW	S14RETENTION			Keep retention on RAM section S14 when RAM section is off
			Off	0	Off
			On	1	On
f	RW	S15RETENTION			Keep retention on RAM section S15 when RAM section is off
			Off	0	Off
			On	1	On

5.3.7.46 RAM[8].POWERSET

Address offset: 0x984

RAM8 power control set register

When read, this register will return the value of the POWER register.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x0000FFFF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	W	S0POWER			Keep RAM section S0 of RAM8 on or off in System ON mode
			On	1	On
B	W	S1POWER			Keep RAM section S1 of RAM8 on or off in System ON mode
			On	1	On
C	W	S2POWER			Keep RAM section S2 of RAM8 on or off in System ON mode
			On	1	On
D	W	S3POWER			Keep RAM section S3 of RAM8 on or off in System ON mode
			On	1	On
E	W	S4POWER			Keep RAM section S4 of RAM8 on or off in System ON mode
			On	1	On
F	W	S5POWER			Keep RAM section S5 of RAM8 on or off in System ON mode
			On	1	On
G	W	S6POWER			Keep RAM section S6 of RAM8 on or off in System ON mode
			On	1	On
H	W	S7POWER			Keep RAM section S7 of RAM8 on or off in System ON mode
			On	1	On
I	W	S8POWER			Keep RAM section S8 of RAM8 on or off in System ON mode
			On	1	On
J	W	S9POWER			Keep RAM section S9 of RAM8 on or off in System ON mode
			On	1	On
K	W	S10POWER			Keep RAM section S10 of RAM8 on or off in System ON mode

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
			On	1																															
L	W	S11POWER																																	
			On	1																															
M	W	S12POWER																																	
			On	1																															
N	W	S13POWER																																	
			On	1																															
O	W	S14POWER																																	
			On	1																															
P	W	S15POWER																																	
			On	1																															
Q	W	S0RETENTION																																	
			On	1																															
R	W	S1RETENTION																																	
			On	1																															
S	W	S2RETENTION																																	
			On	1																															
T	W	S3RETENTION																																	
			On	1																															
U	W	S4RETENTION																																	
			On	1																															
V	W	S5RETENTION																																	
			On	1																															
W	W	S6RETENTION																																	
			On	1																															
X	W	S7RETENTION																																	
			On	1																															
Y	W	S8RETENTION																																	
			On	1																															
Z	W	S9RETENTION																																	
			On	1																															
a	W	S10RETENTION																																	
			On	1																															
b	W	S11RETENTION																																	
			On	1																															
c	W	S12RETENTION																																	
			On	1																															
d	W	S13RETENTION																																	
			On	1																															
e	W	S14RETENTION																																	
			On	1																															
f	W	S15RETENTION																																	
			On	1																															

5.3.7.47 RAM[8].POWERCLR

Address offset: 0x988

RAM8 power control clear register

When read, this register will return the value of the POWER register.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	W	S0POWER																																	
			Off	1																															
B	W	S1POWER																																	
			Off	1																															
C	W	S2POWER																																	
			Off	1																															
D	W	S3POWER																																	
			Off	1																															
E	W	S4POWER																																	
			Off	1																															
F	W	S5POWER																																	
			Off	1																															
G	W	S6POWER																																	
			Off	1																															
H	W	S7POWER																																	
			Off	1																															
I	W	S8POWER																																	
			Off	1																															
J	W	S9POWER																																	
			Off	1																															
K	W	S10POWER																																	
			Off	1																															
L	W	S11POWER																																	
			Off	1																															
M	W	S12POWER																																	
			Off	1																															
N	W	S13POWER																																	
			Off	1																															
O	W	S14POWER																																	
			Off	1																															
P	W	S15POWER																																	
			Off	1																															
Q	W	S0RETENTION																																	
			Off	1																															
R	W	S1RETENTION																																	
			Off	1																															
S	W	S2RETENTION																																	
			Off	1																															
T	W	S3RETENTION																																	
			Off	1																															
U	W	S4RETENTION																																	
			Off	1																															
V	W	S5RETENTION																																	
			Off	1																															
W	W	S6RETENTION																																	
			Off	1																															
X	W	S7RETENTION																																	
			Off	1																															
Y	W	S8RETENTION																																	
			Off	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
Z	W	S9RETENTION																																	
			Off	1																															
a	W	S10RETENTION																																	
			Off	1																															
b	W	S11RETENTION																																	
			Off	1																															
c	W	S12RETENTION																																	
			Off	1																															
d	W	S13RETENTION																																	
			Off	1																															
e	W	S14RETENTION																																	
			Off	1																															
f	W	S15RETENTION																																	
			Off	1																															

5.3.8 Electrical specification

5.3.8.1 Regulator operating conditions

Symbol	Description	Min.	Typ.	Max.	Units
$V_{DD,POR}$	VDD supply voltage needed during power-on reset	1.75			V
V_{DD}	Normal voltage mode operating voltage	1.7	3.0	3.6	V
V_{DDH}	High voltage mode operating voltage	2.5	3.7	5.5	V
C_{VDD}	Effective decoupling capacitance on the VDD pin	2.7	4.7	5.5	µF
C_{DEC4}	Effective decoupling capacitance on the DEC4 pin	0.7	1	1.3	µF

5.3.8.2 Regulator specifications, REG0 stage

Symbol	Description	Min.	Typ.	Max.	Units
$V_{REG0OUT}$	REG0 output voltage	1.8		3.3	V
$V_{REG0OUT,ERR}$	REG0 output voltage error (deviation from setting in REGOUT0 on page 63)	-10		5	%
$V_{VDDH-VDD}$	Required difference between input voltage (VDDH) and output voltage (VDD, configured in REGOUT0 on page 63), VDDH > VDD	0.3			V
$I_{EXT,OFF}$	External current draw[11] allowed in High voltage mode (supply on VDDH) during System OFF.			1	mA
$I_{EXT,LOW}$	External current draw[11] allowed in High voltage mode (supply on VDDH) when radio output power is higher than 4 dBm.			5	mA
$I_{EXT,HIGH}$	External current draw[11] allowed in High voltage mode (supply on VDDH) when radio output power is lower than or equal to 4 dBm.			25	mA

[11] External current draw is defined as the sum of all GPIO currents and the current being drawn from VDD.

5.3.8.3 Device startup times

Symbol	Description	Min.	Typ.	Max.	Units
t_{POR}	Time in power-on reset after supply reaches minimum operating voltage, depending on supply rise time				
$t_{POR,10µs}$	VDD rise time 10 µs[12]		1	10	ms
$t_{POR,10ms}$	VDD rise time 10 ms[12]		9		ms
$t_{POR,60ms}$	VDD rise time 60 ms[12]		23	110	ms
$t_{RISE,REG0OUT}$	REG0 output (VDD) rise time after VDDH reaches minimum VDDH supply voltage[12]				
$t_{RISE,REG0OUT,10µs}$	VDDH rise time 10 µs[12]		0.22	1.55	ms
$t_{RISE,REG0OUT,10ms}$	VDDH rise time 10 ms[12]		5		ms
$t_{RISE,REG0OUT,100ms}$	VDDH rise time 100 ms[12]	30	50	80	ms
t_{PINR}	Reset time when using pin reset, depending on pin capacitance				
$t_{PINR,500nF}$	500 nF capacitance at reset pin			32.5	ms
$t_{PINR,10µF}$	10 µF capacitance at reset pin			650	ms
t_{R2ON}	Time from power-on reset to System ON				
$t_{R2ON,NOTCONF}$	If reset pin not configured	tPOR			ms
$t_{R2ON,CONF}$	If reset pin configured	tPOR + tPINR			ms
t_{OFF2ON}	Time from OFF to CPU execute		16.5		µs
$t_{IDLE2CPU}$	Time from IDLE to CPU execute		3.0		µs
$t_{EVTSET,CL1}$	Time from HW event to PPI event in Constant Latency System ON mode		0.0625		µs
$t_{EVTSET,CL0}$	Time from HW event to PPI event in Low Power System ON mode		0.0625		µs

5.3.8.4 Power fail comparator

Symbol	Description	Min.	Typ.	Max.	Units
$V_{POF,NV}$	Nominal power level warning thresholds (falling supply voltage) in Normal voltage mode (supply on VDD). Levels are configurable between Min. and Max. in 100 mV increments	1.7		2.8	V
$V_{POF,HV}$	Nominal power level warning thresholds (falling supply voltage) in High voltage mode (supply on VDDH). Levels are configurable in 100 mV increments	2.7		4.2	V
V_{POFTOL}	Threshold voltage tolerance (applies in both Normal voltage mode and High voltage mode)	-5		5	%
$V_{POFHYST}$	Threshold voltage hysteresis (applies in both Normal voltage mode and High voltage mode)	40	50	60	mV
$V_{BOR,OFF}$	Brownout reset voltage range System OFF mode. Brownout only applies to the voltage on VDD	1.2		1.62	V
$V_{BOR,ON}$	Brownout reset voltage range System ON mode. Brownout only applies to the voltage on VDD	1.57	1.6	1.63	V

5.3.8.5 USB operating conditions

Symbol	Description	Min.	Typ.	Max.	Units
V_{BUS}	Supply voltage on VBUS pin	4.35	5	5.5	V
V_{DPDM}	Voltage on D+ and D- lines	VSS - 0.3		VUSB33 + 0.3	V

[12] See Recommended operating conditions on page 971 for more information.

5.3.8.6 USB regulator specifications

Symbol	Description	Min.	Typ.	Max.	Units
$I_{USB,QUIES}$	USB regulator quiescent current drawn from VBUS (USBD enabled)		170		µA
$t_{USBPWRRDY}$	Time from USB enabled to USBPWRRDY event triggered, V_{BUS} supply provided		1		ms
V_{USB33}	On voltage at the USB regulator output (DECUSB pin)	3.0	3.3	3.6	V
$R_{SOURCE,VBUS}$	Maximum source resistance on VBUS, including cable, when VDDH is not connected to VBUS			6	Ω
$R_{SOURCE,VBUSVDDH}$	Maximum source resistance on VBUS, including cable, when VDDH is connected to VBUS			3.8	Ω
C_{DECUSB}	Decoupling capacitor on the DECUSB pin	2.35	4.7	5.5	µF

5.3.8.7 VBUS detection specifications

Symbol	Description	Min.	Typ.	Max.	Units
$V_{BUS,DETECT}$	Voltage at which rising VBUS gets reported by USBDETECTED	3.4	4.0	4.3	V
$V_{BUS,REMOVE}$	Voltage at which decreasing VBUS gets reported by USBREMOVED	3.0	3.6	3.9	V

5.4 CLOCK — Clock control

The clock control system can source the system clocks from a range of internal or external high and low frequency oscillators and distribute them to modules based upon a module's individual requirements. Clock distribution is automated and grouped independently by module to limit current consumption in unused branches of the clock tree.

Listed here are the main features for CLOCK:

- 64 MHz on-chip oscillator
- 64 MHz crystal oscillator, using external 32 MHz crystal
- 32.768 kHz +/-500 ppm RC oscillator
- 32.768 kHz crystal oscillator, using external 32.768 kHz crystal
- 32.768 kHz oscillator synthesized from 64 MHz oscillator
- Firmware (FW) override control of crystal oscillator activity for low latency start up
- Automatic internal oscillator and clock control, and distribution for ultra-low power

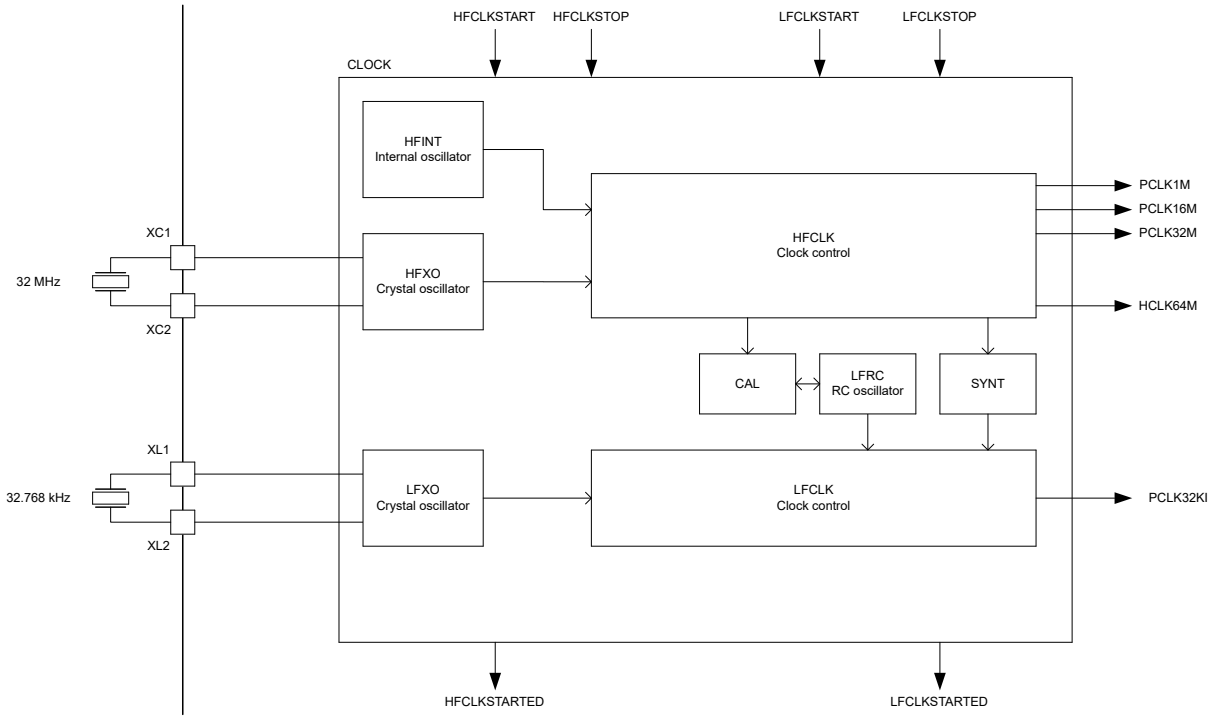


Figure 22: Clock control

5.4.1 HFCLK controller

The HFCLK controller provides several clock signals in the system.

These are as follows:

- HCLK64M: 64 MHz CPU clock
- PCLK1M: 1 MHz peripheral clock
- PCLK16M: 16 MHz peripheral clock
- PCLK32M: 32 MHz peripheral clock

The HFCLK controller uses the following high frequency clock (HFCLK) sources:

- 64 MHz internal oscillator (HFINT)
- 64 MHz crystal oscillator (HFXO)

For illustration, see Clock control on page 158.

The HFCLK controller will automatically provide the clock(s) requested by the system. If the system does not request any clocks from the HFCLK controller, the controller will enter a power saving mode.

The HFINT source will be used when HFCLK is requested and HFXO has not been started.

The HFXO is started by triggering the HFCLKSTART task and stopped by triggering the HFCLKSTOP task. When the HFCLKSTART task is triggered, the HFCLKSTARTED event is generated once the HFXO startup time has elapsed. The HFXO startup time is given as the sum of the following:

- HFXO power-up time, as specified in 64 MHz crystal oscillator (HFXO) on page 171.
- HFXO debounce time, as specified in register HFXODEBOUNCE on page 169.

The HFXO must be running to use the RADIO or the calibration mechanism associated with the 32.768 kHz RC oscillator.

5.4.1.1 64 MHz crystal oscillator (HFXO)

The 64 MHz crystal oscillator (HFXO) is controlled by a 32 MHz external crystal.

The crystal oscillator is designed for use with an AT-cut quartz crystal in parallel resonant mode. To achieve correct oscillation frequency, the load capacitance must match the specification in the crystal data sheet.

Circuit diagram of the 64 MHz crystal oscillator on page 159 shows how the 32 MHz crystal is connected to the 64 MHz crystal oscillator.

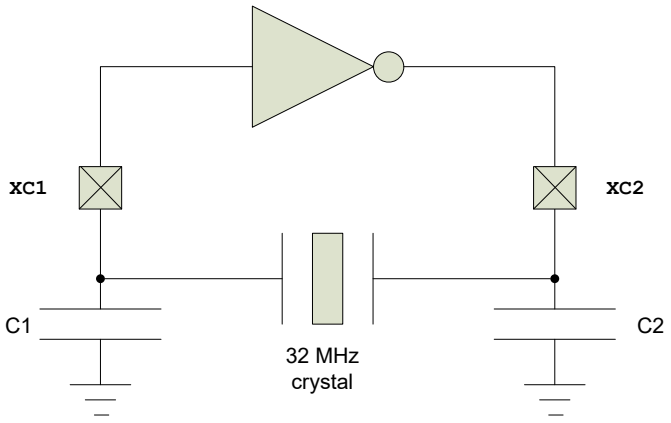


Figure 23: Circuit diagram of the 64 MHz crystal oscillator

The load capacitance (CL) is the total capacitance seen by the crystal across its terminals and is given by:

$$CL = \frac{(C1' \cdot C2')}{(C1' + C2')}$$

$$C1' = C1 + C_{pcb1} + C_{pin}$$
$$C2' = C2 + C_{pcb2} + C_{pin}$$

C1 and C2 are ceramic SMD capacitors connected between each crystal terminal and ground. For more information, see Reference circuitry on page 937. C_{pcb1} and C_{pcb2} are stray capacitances on the PCB. C_{pin} is the pin input capacitance on the `XC1` and `XC2` pins. See table 64 MHz crystal oscillator (HFXO) on page 171. The load capacitors C1 and C2 should have the same value.

For reliable operation, the crystal load capacitance, shunt capacitance, equivalent series resistance, and drive level must comply with the specifications in table 64 MHz crystal oscillator (HFXO) on page 171. It is recommended to use a crystal with lower than maximum load capacitance and/or shunt capacitance. A low load capacitance will reduce both start up time and current consumption.

5.4.2 LFCLK controller

The system supports several low frequency clock sources.

As illustrated in Clock control on page 158, the system supports the following low frequency clock sources:

- 32.768 kHz RC oscillator (LFRC)
- 32.768 kHz crystal oscillator (LFXO)
- 32.768 kHz synthesized from HFCLK (LFSYNT)

The LFCLK controller and all of the LFCLK clock sources are always switched off when in System OFF mode.

The LFCLK clock is started by first selecting the preferred clock source in register LFCLKSRC on page 169 and then triggering the LFCLKSTART task. If the LFXO is selected as the clock source, the LFCLK will initially start running from the 32.768 kHz LFRC while the LFXO is starting up and automatically switch to using the LFXO once this oscillator is running. The LFCLKSTARTED event will be generated when the LFXO has been started.

The LFCLK clock is stopped by triggering the LFCLKSTOP task.

Register LFCLKSRC on page 169 controls the clock source, and its allowed swing. The truth table for various situations is as follows:

SRC	EXTERNAL	BYPASS	Comment
0	0	0	Normal operation, LFRC is source
0	0	1	DO NOT USE
0	1	X	DO NOT USE
1	0	0	Normal XTAL operation
1	1	0	Apply external low swing signal to XL1, ground XL2
1	1	1	Apply external full swing signal to XL1, leave XL2 grounded or unconnected
1	0	1	DO NOT USE
2	0	0	Normal operation, LFSYNT is source
2	0	1	DO NOT USE
2	1	X	DO NOT USE

Table 8: LFCLKSRC configuration depending on clock source

It is not allowed to write to register LFCLKSRC on page 169 when the LFCLK is running.

A LFCLKSTOP task will stop the LFCLK oscillator. However, the LFCLKSTOP task can only be triggered after the STATE field in register LFCLKSTAT on page 168 indicates LFCLK running state.

The synthesized 32.768 kHz clock depends on the HFCLK to run. If high accuracy is required for the LFCLK running off the synthesized 32.768 kHz clock, the HFCLK must running from the HFXO source.

5.4.2.1 32.768 kHz RC oscillator (LFRC)

The default source of the low frequency clock (LFCLK) is the 32.768 kHz RC oscillator (LFRC).

The LFRC oscillator does not require additional external components.

The LFRC oscillator has two modes of operation, normal and ultra-low power (ULP) mode, enabling the user to trade power consumption against accuracy of the clock. The LFRC mode is configured in register LFRCMODE. The LFRC oscillator has to be stopped before changing the mode of the oscillator.

The LFRC frequency will be affected by variation in temperature. The LFRC oscillator can be calibrated to improve accuracy by using the HFXO as a reference oscillator during calibration.

5.4.2.2 Calibrating the 32.768 kHz RC oscillator

After the LFRC oscillator is started and running, it can be calibrated by triggering the CAL task.

The LFRC oscillator will then temporarily request the HFCLK to be used as a reference for the calibration. A DONE event will be generated when calibration has finished. The HFCLK crystal oscillator has to be started (by triggering the HFCLKSTART task) in order for the calibration mechanism to work.

It is not allowed to stop the LFRC or write to LFRCMODE during an ongoing calibration.

5.4.2.3 Calibration timer

The calibration timer can be used to time the calibration interval of the 32.768 kHz RC oscillator.

The calibration timer is started by triggering the CTSTART task and stopped by triggering the CTSTOP task. The calibration timer will always start counting down from the value specified in CTIV (Retained) on page

169 and generate a CTTO event when it reaches 0. The calibration timer will automatically stop when it reaches 0.

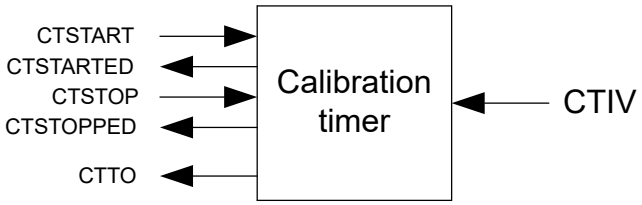


Figure 24: Calibration timer

After a CTSTART task has been triggered, the calibration timer will ignore further tasks until it has returned the CTSTARTED event. Likewise, after a CTSTOP task has been triggered, the calibration timer will ignore further tasks until it has returned a CTSTOPPED event. Triggering CTSTART while the calibration timer is running will immediately return a CTSTARTED event. Triggering CTSTOP when the calibration timer is stopped will immediately return a CTSTOPPED event.

5.4.2.4 32.768 kHz crystal oscillator (LFXO)

For higher LFCLK accuracy (when better than +/- 500 ppm accuracy is required), the low frequency crystal oscillator (LFXO) must be used.

The following external clock sources are supported:

- Low swing clock signal applied to the *XL1* pin. The *XL2* pin shall then be grounded.
- Rail-to-rail clock signal applied to the *XL1* pin. The *XL2* pin shall then be grounded or left unconnected.

To achieve correct oscillation frequency, the load capacitance must match the specification in the crystal data sheet. Circuit diagram of the 32.768 kHz crystal oscillator on page 161 shows the LFXO circuitry.

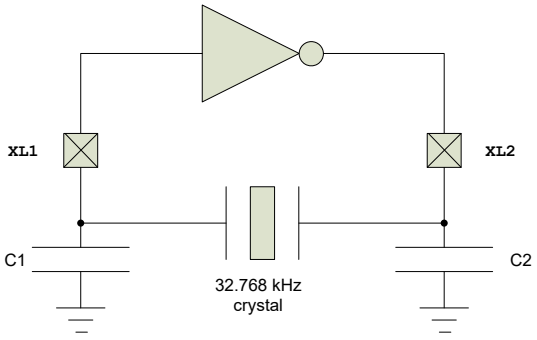


Figure 25: Circuit diagram of the 32.768 kHz crystal oscillator

The load capacitance (CL) is the total capacitance seen by the crystal across its terminals and is given by:

$$CL = \frac{(C1' \cdot C2')}{(C1' + C2')}$$

$$C1' = C1 + C_{pcb1} + C_{pin}$$
$$C2' = C2 + C_{pcb2} + C_{pin}$$

C1 and C2 are ceramic SMD capacitors connected between each crystal terminal and ground. C_{pcb1} and C_{pcb2} are stray capacitances on the PCB. C_{pin} is the pin input capacitance on the XC1 and XC2 pins (see Low frequency crystal oscillator (LFXO) on page 171). The load capacitors C1 and C2 should have the same value.

For more information, see Reference circuitry on page 937.

5.4.2.5 32.768 kHz synthesized from HFCLK (LFSYNT)

LFCLK can also be synthesized from the HFCLK clock source. The accuracy of LFCLK will then be the accuracy of the HFCLK.

Using the LFSYNT clock avoids the requirement for a 32.768 kHz crystal, but increases average power consumption as the HFCLK will need to be requested in the system.

5.4.3 Registers

Instances

Instance	Base address	Description
CLOCK	0x40000000	Clock control

Register overview

Register	Offset	Description
TASKS_HFCLKSTART	0x000	Start HFXO crystal oscillator
TASKS_HFCLKSTOP	0x004	Stop HFXO crystal oscillator
TASKS_LFCLKSTART	0x008	Start LFCLK
TASKS_LFCLKSTOP	0x00C	Stop LFCLK
TASKS_CAL	0x010	Start calibration of LFRC
TASKS_CTSTART	0x014	Start calibration timer
TASKS_CTSTOP	0x018	Stop calibration timer
EVENTS_HFCLKSTARTED	0x100	HFXO crystal oscillator started
EVENTS_LFCLKSTARTED	0x104	LFCLK started
EVENTS_DONE	0x10C	Calibration of LFRC completed
EVENTS_CTTO	0x110	Calibration timer timeout
EVENTS_CTSTARTED	0x128	Calibration timer has been started and is ready to process new tasks
EVENTS_CTSTOPPED	0x12C	Calibration timer has been stopped and is ready to process new tasks
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
HFCLKRUN	0x408	Status indicating that HFCLKSTART task has been triggered
HFCLKSTAT	0x40C	HFCLK status
LFCLKRUN	0x414	Status indicating that LFCLKSTART task has been triggered
LFCLKSTAT	0x418	LFCLK status
LFCLKSRCCOPY	0x41C	Copy of LFCLKSRC register, set when LFCLKSTART task was triggered
LFCLKSRC	0x518	Clock source for the LFCLK
HFXODEBOUNCE	0x528	HFXO debounce time. The HFXO is started by triggering the TASKS_HFCLKSTART task.
CTIV	0x538	Calibration timer interval This register is retained.
TRACECONFIG	0x55C	Clocking options for the trace port debug interface
LFRCMODE	0x5B4	LFRC mode configuration

5.4.3.1 TASKS_HFCLKSTART

Address offset: 0x000

Start HFXO crystal oscillator

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_HFCLKSTART			Start HFXO crystal oscillator
			Trigger	1	Trigger task

5.4.3.2 TASKS_HFCLKSTOP

Address offset: 0x004

Stop HFXO crystal oscillator

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_HFCLKSTOP			Stop HFXO crystal oscillator
			Trigger	1	Trigger task

5.4.3.3 TASKS_LFCLKSTART

Address offset: 0x008

Start LFCLK

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_LFCLKSTART			Start LFCLK
			Trigger	1	Trigger task

5.4.3.4 TASKS_LFCLKSTOP

Address offset: 0x00C

Stop LFCLK

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_LFCLKSTOP			Stop LFCLK
			Trigger	1	Trigger task

5.4.3.5 TASKS_CAL

Address offset: 0x010

Start calibration of LFRC

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CAL			Start calibration of LFRC
			Trigger	1	Trigger task

5.4.3.6 TASKS_CTSTART

Address offset: 0x014

Start calibration timer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CTSTART			Start calibration timer
			Trigger	1	Trigger task

5.4.3.7 TASKS_CTSTOP

Address offset: 0x018

Stop calibration timer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CTSTOP			Stop calibration timer
			Trigger	1	Trigger task

5.4.3.8 EVENTS_HFCLKSTARTED

Address offset: 0x100

HFXO crystal oscillator started

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_HFCLKSTARTED			HFXO crystal oscillator started
			NotGenerated	0	Event not generated
			Generated	1	Event generated

5.4.3.9 EVENTS_LFCLKSTARTED

Address offset: 0x104

LFCLK started

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_LFCLKSTARTED			LFCLK started
			NotGenerated	0	Event not generated
			Generated	1	Event generated

5.4.3.10 EVENTS_DONE

Address offset: 0x10C

Calibration of LFRC completed

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_DONE			Calibration of LFRC completed
			NotGenerated	0	Event not generated
			Generated	1	Event generated

5.4.3.11 EVENTS_CTTO

Address offset: 0x110

Calibration timer timeout

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_CTTO			Calibration timer timeout
			NotGenerated	0	Event not generated
			Generated	1	Event generated

5.4.3.12 EVENTS_CTSTARTED

Address offset: 0x128

Calibration timer has been started and is ready to process new tasks

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_CTSTARTED			Calibration timer has been started and is ready to process new tasks
			NotGenerated	0	Event not generated
			Generated	1	Event generated

5.4.3.13 EVENTS_CTSTOPPED

Address offset: 0x12C

Calibration timer has been stopped and is ready to process new tasks

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_CTSTOPPED			Calibration timer has been stopped and is ready to process new tasks
			NotGenerated	0	Event not generated
			Generated	1	Event generated

5.4.3.14 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	HFCLKSTARTED			Write '1' to enable interrupt for event HFCLKSTARTED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	LFCLKSTARTED			Write '1' to enable interrupt for event LFCLKSTARTED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	DONE			Write '1' to enable interrupt for event DONE
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	CTTO			Write '1' to enable interrupt for event CTTO
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	CTSTARTED			Write '1' to enable interrupt for event CTSTARTED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	CTSTOPPED			Write '1' to enable interrupt for event CTSTOPPED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

5.4.3.15 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	HFCLKSTARTED			Write '1' to disable interrupt for event HFCLKSTARTED
			Clear	1	Disable

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	LFCLKSTARTED			Write '1' to disable interrupt for event LFCLKSTARTED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	DONE			Write '1' to disable interrupt for event DONE
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	CTTO			Write '1' to disable interrupt for event CTTO
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	CTSTARTED			Write '1' to disable interrupt for event CTSTARTED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	CTSTOPPED			Write '1' to disable interrupt for event CTSTOPPED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

5.4.3.16 HFCLKRUN

Address offset: 0x408

Status indicating that HFCLKSTART task has been triggered

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	STATUS			HFCLKSTART task triggered or not
			NotTriggered	0	Task not triggered
			Triggered	1	Task triggered

5.4.3.17 HFCLKSTAT

Address offset: 0x40C

HFCLK status

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	SRC			Source of HFCLK
			RC	0	64 MHz internal oscillator (HFINT)
			Xtal	1	64 MHz crystal oscillator (HFXO)
B	R	STATE			HFCLK state
			NotRunning	0	HFCLK not running
			Running	1	HFCLK running

5.4.3.18 LFCLKRUN

Address offset: 0x414

Status indicating that LFCLKSTART task has been triggered

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	STATUS			LFCLKSTART task triggered or not
			NotTriggered	0	Task not triggered
			Triggered	1	Task triggered

5.4.3.19 LFCLKSTAT

Address offset: 0x418

LFCLK status

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	SRC			Source of LFCLK
			RC	0	32.768 kHz RC oscillator (LFRC)
			Xtal	1	32.768 kHz crystal oscillator (LFXO)
			Synth	2	32.768 kHz synthesized from HFCLK (LFSYNT)
B	R	STATE			LFCLK state
			NotRunning	0	LFCLK not running
			Running	1	LFCLK running

5.4.3.20 LFCLKSRCCOPY

Address offset: 0x41C

Copy of LFCLKSRC register, set when LFCLKSTART task was triggered

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	SRC			Clock source
			RC	0	32.768 kHz RC oscillator (LFRC)
			Xtal	1	32.768 kHz crystal oscillator (LFXO)
			Synth	2	32.768 kHz synthesized from HFCLK (LFSYNT)

5.4.3.21 LFCLKSRC

Address offset: 0x518

Clock source for the LFCLK

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	SRC			Clock source
			RC	0	32.768 kHz RC oscillator (LFRC)
			Xtal	1	32.768 kHz crystal oscillator (LFXO)
			Synth	2	32.768 kHz synthesized from HFCLK (LFSYNT)
B	RW	BYPASS			Enable or disable bypass of LFCLK crystal oscillator with external clock source
			Disabled	0	Disable (use with Xtal or low-swing external source)
			Enabled	1	Enable (use with rail-to-rail external source)
C	RW	EXTERNAL			Enable or disable external source for LFCLK
			Disabled	0	Disable external source (use with Xtal)
			Enabled	1	Enable use of external source instead of Xtal (SRC needs to be set to Xtal)

5.4.3.22 HFXODEBOUNCE

Address offset: 0x528

HFXO debounce time. The HFXO is started by triggering the TASKS_HFCLKSTART task.

The EVENTS_HFCLKSTARTED event is generated after the HFXO power up time + the HFXO debounce time has elapsed. It is not allowed to change the value of this register while the HFXO is starting.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A
Reset 0x00000010				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	HFXODEBOUNCE		0x01..0xFF	HFXO debounce time. Debounce time = HFXODEBOUNCE * 16 µs.
			Db256us	0x10	256 µs debounce time. Recommended for TSX-3225, FA-20H and FA-128 crystals.
			Db1024us	0x40	1024 µs debounce time. Recommended for NX1612AA and NX1210AB crystals.

5.4.3.23 CTIV (Retained)

Address offset: 0x538

Calibration timer interval

This register is retained.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	CTIV			Calibration timer interval in multiple of 0.25 seconds. Range: 0.25 seconds to 31.75 seconds.

5.4.3.24 TRACECONFIG

Address offset: 0x55C

Clocking options for the trace port debug interface

This register is a retained register. Reset behavior is the same as debug components.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B B A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	TRACEPORTSPEED			Speed of trace port clock. Note that the TRACECLK pin will output this clock divided by two.
			32MHz	0	32 MHz trace port clock (TRACECLK = 16 MHz)
			16MHz	1	16 MHz trace port clock (TRACECLK = 8 MHz)
			8MHz	2	8 MHz trace port clock (TRACECLK = 4 MHz)
			4MHz	3	4 MHz trace port clock (TRACECLK = 2 MHz)
B	RW	TRACEMUX			Pin multiplexing of trace signals. See pin assignment chapter for more details.
			GPIO	0	No trace signals routed to pins. All pins can be used as regular GPIOs.
			Serial	1	SWO trace signal routed to pin. Remaining pins can be used as regular GPIOs.
			Parallel	2	All trace signals (TRACECLK and TRACEDATA[n]) routed to pins.

5.4.3.25 LFRCMODE

Address offset: 0x5B4

LFRC mode configuration

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	MODE			Set LFRC mode
			Normal	0	Normal mode
			ULP	1	Ultra-low power mode (ULP)
B	RW	STATUS			Active LFRC mode. This field is read only.
			Normal	0	Normal mode
			ULP	1	Ultra-low power mode (ULP)

5.4.4 Electrical specification

5.4.4.1 64 MHz internal oscillator (HFINT)

Symbol	Description	Min.	Typ.	Max.	Units
f_{NOM_HFINT}	Nominal output frequency		64		MHz
f_{TOL_HFINT}	Frequency tolerance		±1.5	±8	%

5.4.4.2 64 MHz crystal oscillator (HFXO)

Symbol	Description	Min.	Typ.	Max.	Units
f_{NOM_HFXO}	Nominal output frequency		64		MHz
f_{XTAL_HFXO}	External crystal frequency		32		MHz
f_{TOL_HFXO}	Frequency tolerance requirement for 2.4 GHz proprietary radio applications			±60	ppm
$f_{TOL_HFXO_BLE}$	Frequency tolerance requirement, Bluetooth low energy applications, packet length ≤ 200 bytes			±40	ppm
$f_{TOL_HFXO_BLE_LP}$	Frequency tolerance requirement, Bluetooth low energy applications, packet length > 200 bytes			±30	ppm
C_{L_HFXO}	Load capacitance			12	pF
C_{0_HFXO}	Shunt capacitance			7	pF
$R_{S_HFXO_7PF}$	Equivalent series resistance 3 pF < C0 ≤ 7 pF			60	Ω
$R_{S_HFXO_3PF}$	Equivalent series resistance C0 ≤ 3 pF			100	Ω
P_{D_HFXO}	Drive level			100	µW
C_{PIN_HFXO}	Input capacitance XC1 and XC2		3		pF
I_{STBY_X32M}	Core standby current for various crystals				
$I_{STBY_X32M_X0}$	Epson TSX-3225		80		µA
$I_{STBY_X32M_X1}$	Epson FA-20H		72		µA
$I_{STBY_X32M_X2}$	Epson FA-128		70		µA
$I_{STBY_X32M_X3}$	NDK NX1612AA		136		µA
$I_{STBY_X32M_X4}$	NDK NX1210AB		143		µA
I_{START_X32M}	Average startup current for various crystals, first 1 ms				
$I_{START_X32M_X0}$	Epson TSX-3225		328		µA
$I_{START_X32M_X1}$	Epson FA-20H		363		µA
$I_{START_X32M_X2}$	Epson FA-128		396		µA
$I_{START_X32M_X3}$	NDK NX1612AA		783		µA
$I_{START_X32M_X4}$	NDK NX1210AB		833		µA
t_{POWER_X32M}	Power-up time for various crystals				
$t_{POWER_X32M_X0}$	Epson TSX-3225		50		µs
$t_{POWER_X32M_X1}$	Epson FA-20H		60		µs
$t_{POWER_X32M_X2}$	Epson FA-128		75		µs
$t_{POWER_X32M_X3}$	NDK NX1612AA		195		µs
$t_{POWER_X32M_X4}$	NDK NX1210AB		210		µs

5.4.4.3 Low frequency crystal oscillator (LFXO)

Symbol	Description	Min.	Typ.	Max.	Units
f_{NOM_LFXO}	Crystal frequency		32.768		kHz
$f_{TOL_LFXO_BLE}$	Frequency tolerance requirement for BLE stack			±500	ppm
$f_{TOL_LFXO_ANT}$	Frequency tolerance requirement for ANT stack			±50	ppm
C_{L_LFXO}	Load capacitance			12.5	pF
C_{0_LFXO}	Shunt capacitance			2	pF
R_{S_LFXO}	Equivalent series resistance			100	kΩ
P_{D_LFXO}	Drive level			0.5	µW
C_{pin}	Input capacitance on XL1 and XL2 pads		4		pF
I_{LFXO}	Run current for 32.768 kHz crystal oscillator		0.23		µA
t_{START_LFXO}	Startup time for 32.768 kHz crystal oscillator		0.25		s

Symbol	Description	Min.	Typ.	Max.	Units
$V_{AMP,IN,XO,LOW}$	Peak to peak amplitude for external low swing clock. Input signal must not swing outside supply rails.	200		1000	mV

5.4.4.4 Low frequency RC oscillator (LFRC)

Symbol	Description	Min.	Typ.	Max.	Units
f_{NOM_LFRC}	Nominal frequency		32.768		kHz
f_{TOL_LFRC}	Frequency tolerance, uncalibrated			±5	%
$f_{TOL_CAL_LFRC}$	Frequency tolerance after calibration[13]			±500	ppm
I_{LFRC}	Run current		0.7		µA
t_{START_LFRC}	Startup time		1000		µs

5.4.4.5 Low frequency RC oscillator (LFRC), Ultra-low power mode (ULP)

Symbol	Description	Min.	Typ.	Max.	Units
f_{NOM_LFULP}	Nominal frequency		32.768		kHz
$f_{TOL_UNCAL_LFULP}$	Frequency tolerance, uncalibrated			±7	%
$f_{TOL_CAL_LFULP}$	Frequency tolerance after calibration[14]			±2000	ppm
I_{LFULP}	Run current		0.3		µA
t_{START_LFULP}	Startup time		1500		µs

5.4.4.6 Synthesized low frequency clock (LFSYNT)

Symbol	Description	Min.	Typ.	Max.	Units
f_{NOM_LFSYNT}	Nominal frequency		32.768		kHz

[13] Constant temperature within ±0.5 °C, calibration performed at least every 8 seconds, averaging interval > 7.5 ms, defined as 3 sigma

[14] Constant temperature within ±0.5 °C, calibration performed at least every 8 seconds, averaging interval > 125 ms, defined as 3 sigma

6 Peripherals

6.1 Peripheral interface

Peripherals are controlled by the CPU by writing to configuration registers and task registers. Peripheral events are indicated to the CPU by event registers and interrupts if they are configured for a given event.

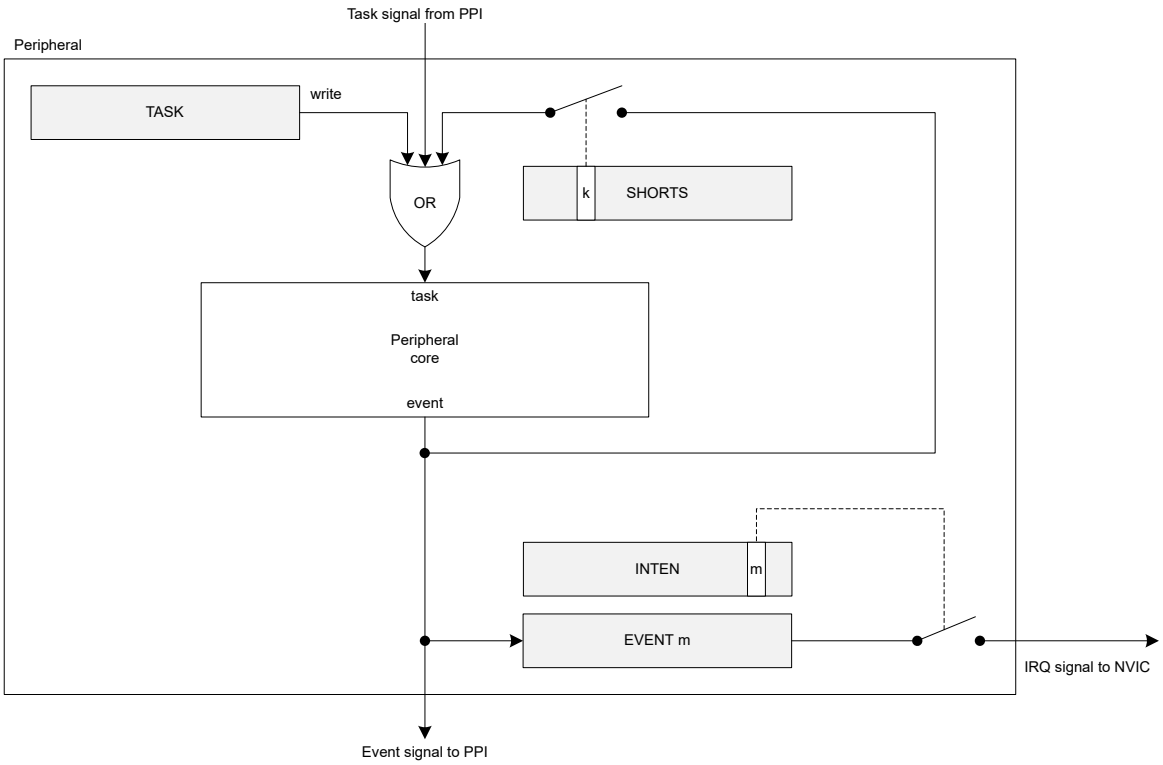


Figure 26: Tasks, events, shortcuts, and interrupts

6.1.1 Peripheral ID

Every peripheral is assigned a fixed block of 0x1000 bytes of address space, which is equal to 1024 x 32 bit registers.

See Instantiation on page 24 for more information about which peripherals are available and where they are located in the address map.

There is a direct relationship between peripheral ID and base address. For example, a peripheral with base address 0x40000000 is assigned ID=0, a peripheral with base address 0x40001000 is assigned ID=1, and a peripheral with base address 0x4001F000 is assigned ID=31.

Peripherals may share the same ID, which may impose one or more of the following limitations:

- Some peripherals share some registers or other common resources.
- Operation is mutually exclusive. Only one of the peripherals can be used at a time.
- Switching from one peripheral to another must follow a specific pattern (disable the first, then enable the second peripheral).

6.1.2 Peripherals with shared ID

In general (with the exception of ID 0), peripherals sharing an ID and base address may not be used simultaneously. The user can only enable one peripheral at the time on this specific ID.

When switching between two peripherals sharing an ID, the user should do the following to prevent unwanted behavior:

1. Disable the previously used peripheral.
2. Remove any programmable peripheral interconnect (PPI) connections set up for the peripheral that is being disabled.
3. Clear all bits in the INTEN register, i.e. INTENCLR = 0xFFFFFFFF.
4. Explicitly configure the peripheral that you are about to enable and do not rely on configuration values that may be inherited from the peripheral that was disabled.
5. Enable the now configured peripheral.

See which peripherals are sharing ID in Instantiation on page 24.

6.1.3 Peripheral registers

Most peripherals feature an ENABLE register. Unless otherwise specified in the relevant chapter, the peripheral registers (in particular the PSEL registers) must be configured before enabling the peripheral.

The peripheral must be enabled before tasks and events can be used.

6.1.4 Bit set and clear

Registers with multiple single-bit bit fields may implement the set-and-clear pattern. This pattern enables firmware to set and clear individual bits in a register without having to perform a read-modify-write operation on the main register.

This pattern is implemented using three consecutive addresses in the register map, where the main register is followed by dedicated SET and CLR registers (in that exact order).

The SET register is used to set individual bits in the main register while the CLR register is used to clear individual bits in the main register. Writing `1` to a bit in SET or CLR register will set or clear the same bit in the main register respectively. Writing `0` to a bit in SET or CLR register has no effect. Reading the SET or CLR register returns the value of the main register.

Note: The main register may not be visible and hence not directly accessible in all cases.

6.1.5 Tasks

Tasks are used to trigger actions in a peripheral, for example to start a particular behavior. A peripheral can implement multiple tasks with each task having a separate register in that peripheral's task register group.

A task is triggered when firmware writes `1` to the task register, or when the peripheral itself or another peripheral toggles the corresponding task signal. See Tasks, events, shortcuts, and interrupts on page 173.

6.1.6 Events

Events are used to notify peripherals and the CPU about events that have happened, for example a state change in a peripheral. A peripheral may generate multiple events with each event having a separate register in that peripheral's event register group.

An event is generated when the peripheral itself toggles the corresponding event signal, and the event register is updated to reflect that the event has been generated. See Tasks, events, shortcuts, and interrupts on page 173. An event register is only cleared when firmware writes `0` to it.

Events can be generated by the peripheral even when the event register is set to `1`.

6.1.7 Shortcuts

A shortcut is a direct connection between an event and a task within the same peripheral. If a shortcut is enabled, the associated task is automatically triggered when its associated event is generated.

Using a shortcut is the equivalent to making the same connection outside the peripheral and through the PPI. However, the propagation delay through the shortcut is usually shorter than the propagation delay through the PPI.

Shortcuts are predefined, which means their connections cannot be configured by firmware. Each shortcut can be individually enabled or disabled through the shortcut register, one bit per shortcut, giving a maximum of 32 shortcuts for each peripheral.

6.1.8 Interrupts

All peripherals support interrupts. Interrupts are generated by events.

A peripheral only occupies one interrupt, and the interrupt number follows the peripheral ID. For example, the peripheral with ID=4 is connected to interrupt number 4 in the nested vectored interrupt controller (NVIC).

Using the INTEN, INTENSET, and INTENCLR registers, every event generated by a peripheral can be configured to generate that peripheral's interrupt. Multiple events can be enabled to generate interrupts simultaneously. To resolve the correct interrupt source, the event registers in the event group of peripheral registers will indicate the source.

Some peripherals implement only INTENSET and INTENCLR registers, and the INTEN register is not available on those peripherals. See the individual peripheral chapters for details. In all cases, reading back the INTENSET or INTENCLR register returns the same information as in INTEN.

Each event implemented in the peripheral is associated with a specific bit position in the INTEN, INTENSET, and INTENCLR registers.

The relationship between tasks, events, shortcuts, and interrupts is shown in Tasks, events, shortcuts, and interrupts on page 173.

Interrupt clearing

Clearing an interrupt by writing `0` to an event register, or disabling an interrupt using the INTENCLR register, can take up to four CPU clock cycles to take effect. This means that an interrupt may reoccur immediately, even if a new event has not come, if the program exits an interrupt handler after the interrupt is cleared or disabled but before four clock cycles have passed.

> **Note:** To avoid an interrupt reoccurring before a new event has come, the program should perform a read from one of the peripheral registers. For example, the event register that has been cleared, or the INTENCLR register that has been used to disable the interrupt. This will cause a one to three-cycle delay and ensure the interrupt is cleared before exiting the interrupt handler.

Care should be taken to ensure the compiler does not remove the read operation as an optimization. If the program can guarantee a four-cycle delay after an event is cleared or an interrupt is disabled, then a read of a register is not required.

6.2 AAR — Accelerated address resolver

Accelerated address resolver is a cryptographic support function for implementing the Resolvable Private Address Resolution procedure described in *Bluetooth Core Specification* v4.0. Resolvable Private Address generation should be achieved using ECB and is not supported by AAR.

The procedure allows two devices that share a secret key to generate and resolve a hash based on their device address. The AAR block enables real-time address resolution on incoming packets when configured as described in this chapter. This allows real-time packet filtering (whitelisting) using a list of known shared keys (Identity Resolving Keys (IRK) in *Bluetooth*).

6.2.1 EasyDMA

AAR implements EasyDMA for reading and writing to RAM. EasyDMA will have finished accessing RAM when the END, RESOLVED, and NOTRESOLVED events are generated.

If the IRKPTR on page 181, ADDRPTR on page 181, and the SCRATCHPTR on page 181 is not pointing to the Data RAM region, an EasyDMA transfer may result in a HardFault or RAM corruption. See Memory on page 21 for more information about the different memory regions.

6.2.2 Resolving a resolvable address

A private resolvable address is composed of six bytes according to the *Bluetooth Core Specification*.

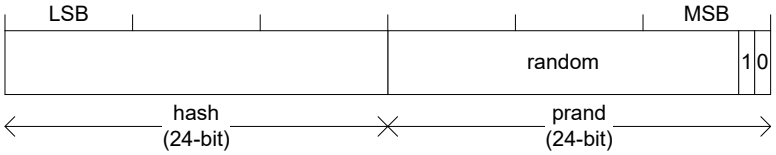


Figure 27: Resolvable address

To resolve an address, the register ADDRPTR on page 181 must point to the start of the packet. The resolver is started by triggering the START task. A RESOLVED event is generated when AAR manages to resolve the address using one of the Identity Resolving Keys (IRK) found in the IRK data structure. AAR will use the IRK specified in the register IRK0 to IRK15 starting from IRK0. The register NIRK on page 181 specifies how many IRKs should be used. The AAR module will generate a NOTRESOLVED event if it is not able to resolve the address using the specified list of IRKs.

AAR will go through the list of available IRKs in the IRK data structure and for each IRK try to resolve the address according to the Resolvable Private Address Resolution Procedure described in the *Bluetooth Core specification* v4.0 [Vol 3] chapter 10.8.2.3. The time it takes to resolve an address varies due to the location in the list of the resolvable address. The resolution time will also be affected by RAM accesses performed by other peripherals and the CPU. See the Electrical specifications for more information about resolution time.

AAR only compares the received address to those programmed in the module without checking the address type.

AAR will stop as soon as it has managed to resolve the address, or after trying to resolve the address using NIRK number of IRKs from the IRK data structure. AAR will generate an END event after it has stopped.

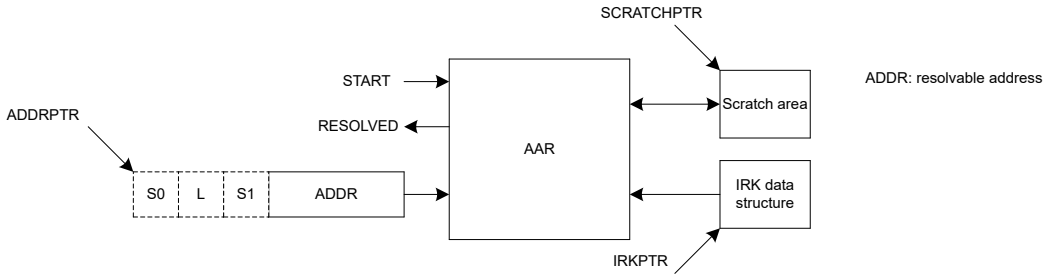


Figure 28: Address resolution with packet preloaded into RAM

6.2.3 Example

The following example shows how to chain RADIO packet reception with address resolution using AAR.

AAR may be started as soon as the 6 bytes required by AAR have been received by RADIO and stored in RAM. The ADDRPTR pointer must point to the start of packet.

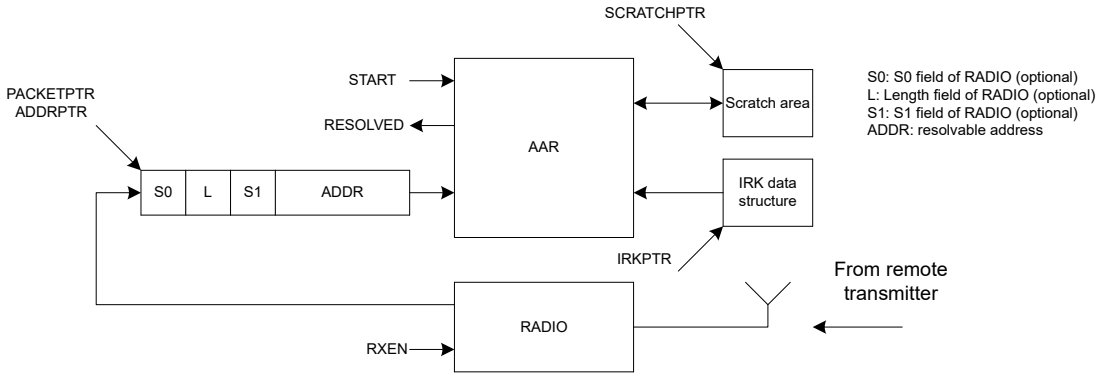


Figure 29: Address resolution with packet loaded into RAM by RADIO

6.2.4 IRK data structure

The IRK data structure is located in RAM at the memory location specified by the IRKPTR register.

Property	Address offset	Description
IRK0	0	IRK number 0 (16 bytes)
IRK1	16	IRK number 1 (16 bytes)
..	..	..
IRK15	240	IRK number 15 (16 bytes)

Table 9: IRK data structure overview

6.2.5 Registers

Instances

Instance	Base address	Description
AAR	0x4000F000	Accelerated address resolver

Register overview

Register	Offset	Description
TASKS_START	0x000	Start resolving addresses based on IRKs specified in the IRK data structure
TASKS_STOP	0x008	Stop resolving addresses
EVENTS_END	0x100	Address resolution procedure complete
EVENTS_RESOLVED	0x104	Address resolved
EVENTS_NOTRESOLVED	0x108	Address not resolved
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
STATUS	0x400	Resolution status
ENABLE	0x500	Enable AAR
NIRK	0x504	Number of IRKs
IRKPTR	0x508	Pointer to IRK data structure
ADDRPTR	0x510	Pointer to the resolvable address
SCRATCHPTR	0x514	Pointer to data area used for temporary storage

6.2.5.1 TASKS_START

Address offset: 0x000

Start resolving addresses based on IRKs specified in the IRK data structure

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_START			Start resolving addresses based on IRKs specified in the IRK data structure
			Trigger	1	Trigger task

6.2.5.2 TASKS_STOP

Address offset: 0x008

Stop resolving addresses

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STOP			Stop resolving addresses
			Trigger	1	Trigger task

6.2.5.3 EVENTS_END

Address offset: 0x100

Address resolution procedure complete

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_END			Address resolution procedure complete
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.2.5.4 EVENTS_RESOLVED

Address offset: 0x104

Address resolved

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_RESOLVED			Address resolved
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.2.5.5 EVENTS_NOTRESOLVED

Address offset: 0x108

Address not resolved

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_NOTRESOLVED			Address not resolved
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.2.5.6 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	END			Write '1' to enable interrupt for event END
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	RESOLVED			Write '1' to enable interrupt for event RESOLVED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	NOTRESOLVED			Write '1' to enable interrupt for event NOTRESOLVED

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.2.5.7 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	END			Write '1' to disable interrupt for event END
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	RESOLVED			Write '1' to disable interrupt for event RESOLVED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	NOTRESOLVED			Write '1' to disable interrupt for event NOTRESOLVED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.2.5.8 STATUS

Address offset: 0x400

Resolution status

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	STATUS		[0..15]	The IRK that was used last time an address was resolved

6.2.5.9 ENABLE

Address offset: 0x500

Enable AAR

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ENABLE			Enable or disable AAR
			Disabled	0	Disable
			Enabled	3	Enable

6.2.5.10 NIRK

Address offset: 0x504

Number of IRKs

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	NIRK		[1..16]	Number of Identity Root Keys available in the IRK data structure

6.2.5.11 IRKPTR

Address offset: 0x508

Pointer to IRK data structure

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	IRKPTR			Pointer to the IRK data structure

6.2.5.12 ADDRPTR

Address offset: 0x510

Pointer to the resolvable address

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ADDRPTR			Pointer to the resolvable address (6-bytes)

6.2.5.13 SCRATCHPTR

Address offset: 0x514

Pointer to data area used for temporary storage

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	SCRATCHPTR			Pointer to a scratch data area used for temporary storage during resolution. A space of minimum 3 bytes must be reserved.

6.2.6 Electrical specification

6.2.6.1 AAR Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
t_{AAR}	Address resolution time per IRK. Total time for several IRKs is given as (1 µs + n * t_AAR), where n is the number of IRKs. (Given priority to the actual destination RAM block).			6	µs
$t_{AAR,8}$	Time for address resolution of 8 IRKs. (Given priority to the actual destination RAM block).			49	µs

6.3 ACL — Access control lists

The Access control lists (ACL) peripheral is designed to assign and enforce access permission schemes for different regions of the on-chip flash memory map.

Flash memory regions can be assigned individual ACL permission schemes. The following registers are involved:

- PERM register - configures permission schemes
- ADDR register - defines the flash page start address (word-aligned)
- SIZE register - determines the size of the region where the permission schemes are applied

Note: The size of the region is restricted to a multiple of the flash page size, measured in bytes. The maximum region is limited to half the flash size. See Memory on page 21 for more information.

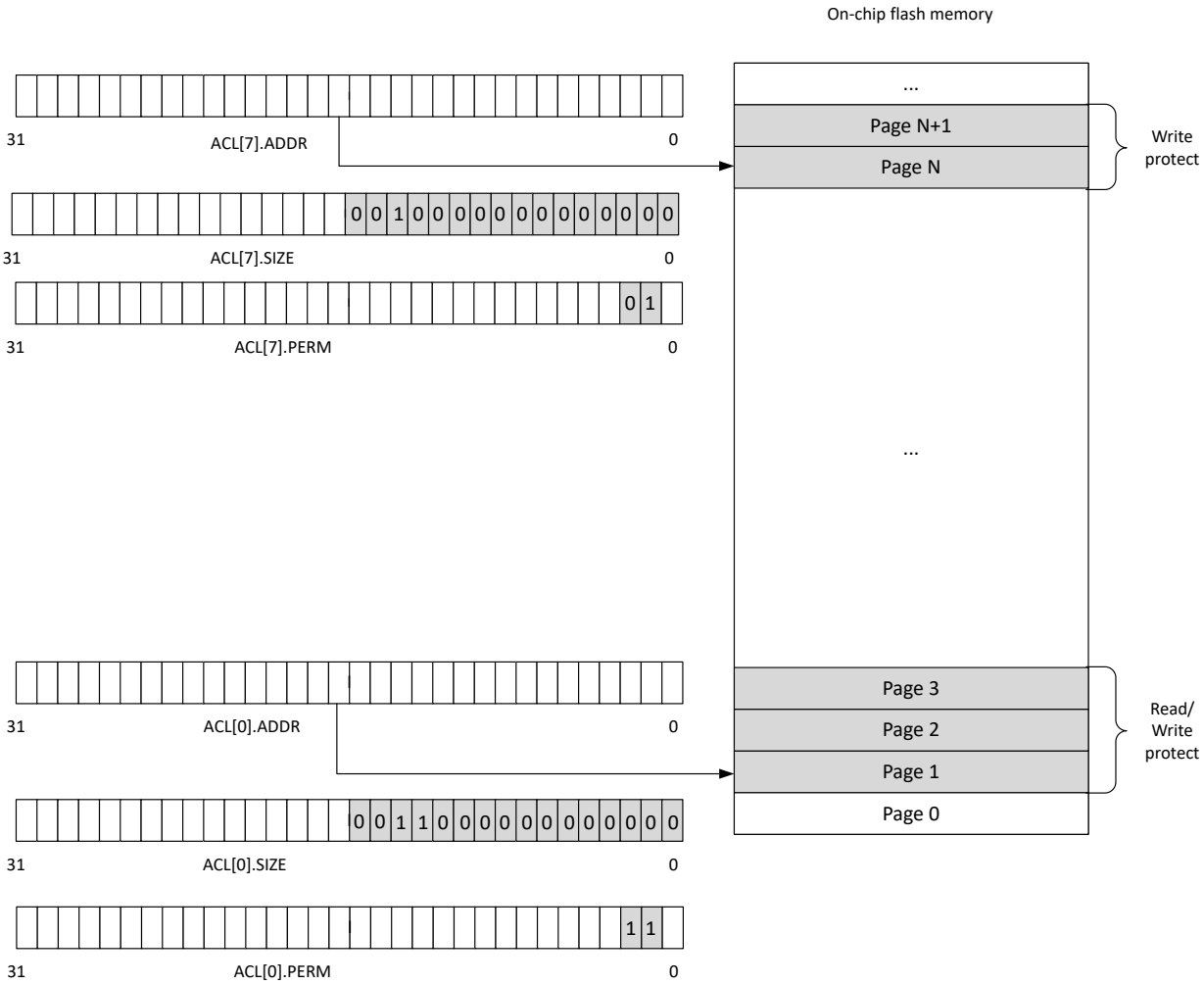


Figure 30: On-chip flash memory protected regions

There are four defined ACL permission schemes, each with different combinations of read/write permissions, as shown in the following table.

Read	Write	Protection description
0	0	No protection. Entire region can be executed, read, written to, or erased.
0	1	Region can be executed and read, but not written to or erased.
1	0	Region can be written to and erased, but not executed or read.
1	1	Region is locked for all access until next reset.

Table 10: ACL permission schemes

Note: If a permission violation to a protected region is detected by the ACL peripheral, the request is blocked and a Bus Fault exception is triggered.

Access control to a configured region is enforced by the hardware. This goes into effect two CPU clock cycles after the ADDR, SIZE, and PERM registers for an ACL instance are written successfully. There are two dependencies for protection to be enforced. First, a valid start address for the flash page boundary must be written to the ADDR register. Second, the SIZE and PERM registers cannot be zero.

The ADDR, SIZE, and PERM registers can only be written once. All ACL configuration registers are cleared on reset by resetting the device from a reset source. This is the only way of clearing the configuration registers. To ensure that the ACL peripheral always enforces the desired permission schemes, the device boot sequence must perform the necessary configuration.

Debugger read access to a read-protected region will be Read-As-Zero (RAZ), while debugger write access to a write-protected region will be Write-Ignored (WI).

6.3.1 Registers

Instances

Instance	Base address	Description
ACL	0x4001E000	Access control lists

Register overview

Register	Offset	Description
ACL[0].ADDR	0x800	Start address of region to protect. The start address must be word-aligned.
ACL[0].SIZE	0x804	Size of region to protect counting from address ACL[0].ADDR. Writing a '0' has no effect.
ACL[0].PERM	0x808	Access permissions for region 0 as defined by start address ACL[0].ADDR and size ACL[0].SIZE
ACL[1].ADDR	0x810	Start address of region to protect. The start address must be word-aligned.
ACL[1].SIZE	0x814	Size of region to protect counting from address ACL[1].ADDR. Writing a '0' has no effect.
ACL[1].PERM	0x818	Access permissions for region 1 as defined by start address ACL[1].ADDR and size ACL[1].SIZE
ACL[2].ADDR	0x820	Start address of region to protect. The start address must be word-aligned.
ACL[2].SIZE	0x824	Size of region to protect counting from address ACL[2].ADDR. Writing a '0' has no effect.
ACL[2].PERM	0x828	Access permissions for region 2 as defined by start address ACL[2].ADDR and size ACL[2].SIZE
ACL[3].ADDR	0x830	Start address of region to protect. The start address must be word-aligned.
ACL[3].SIZE	0x834	Size of region to protect counting from address ACL[3].ADDR. Writing a '0' has no effect.
ACL[3].PERM	0x838	Access permissions for region 3 as defined by start address ACL[3].ADDR and size ACL[3].SIZE
ACL[4].ADDR	0x840	Start address of region to protect. The start address must be word-aligned.
ACL[4].SIZE	0x844	Size of region to protect counting from address ACL[4].ADDR. Writing a '0' has no effect.
ACL[4].PERM	0x848	Access permissions for region 4 as defined by start address ACL[4].ADDR and size ACL[4].SIZE
ACL[5].ADDR	0x850	Start address of region to protect. The start address must be word-aligned.
ACL[5].SIZE	0x854	Size of region to protect counting from address ACL[5].ADDR. Writing a '0' has no effect.
ACL[5].PERM	0x858	Access permissions for region 5 as defined by start address ACL[5].ADDR and size ACL[5].SIZE
ACL[6].ADDR	0x860	Start address of region to protect. The start address must be word-aligned.
ACL[6].SIZE	0x864	Size of region to protect counting from address ACL[6].ADDR. Writing a '0' has no effect.
ACL[6].PERM	0x868	Access permissions for region 6 as defined by start address ACL[6].ADDR and size ACL[6].SIZE
ACL[7].ADDR	0x870	Start address of region to protect. The start address must be word-aligned.
ACL[7].SIZE	0x874	Size of region to protect counting from address ACL[7].ADDR. Writing a '0' has no effect.
ACL[7].PERM	0x878	Access permissions for region 7 as defined by start address ACL[7].ADDR and size ACL[7].SIZE

6.3.1.1 ACL[0].ADDR

Address offset: 0x800

Start address of region to protect. The start address must be word-aligned.

Note: This register can only be written once.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW1	ADDR			Start address of flash region 0. The start address must point to a flash page boundary.

6.3.1.2 ACL[0].SIZE

Address offset: 0x804

Size of region to protect counting from address ACL[0].ADDR. Writing a '0' has no effect.

Note: This register can only be written once.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW1	SIZE			Size of flash region 0 in bytes. Must be a multiple of the flash page size.

6.3.1.3 ACL[0].PERM

Address offset: 0x808

Access permissions for region 0 as defined by start address ACL[0].ADDR and size ACL[0].SIZE

Note: This register can only be written once.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW1	WRITE			Configure write and erase permissions for region 0. Writing a '0' has no effect.
			Enable	0	Allow write and erase instructions to region 0.
			Disable	1	Block write and erase instructions to region 0.
B	RW1	READ			Configure read permissions for region 0. Writing a '0' has no effect.
			Enable	0	Allow read instructions to region 0.
			Disable	1	Block read instructions to region 0.

6.3.1.4 ACL[1].ADDR

Address offset: 0x810

Start address of region to protect. The start address must be word-aligned.

Note: This register can only be written once.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW1	ADDR			Start address of flash region 1. The start address must point to a flash page boundary.

6.3.1.5 ACL[1].SIZE

Address offset: 0x814

Size of region to protect counting from address ACL[1].ADDR. Writing a '0' has no effect.

Note: This register can only be written once.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW1	SIZE			Size of flash region 1 in bytes. Must be a multiple of the flash page size.

6.3.1.6 ACL[1].PERM

Address offset: 0x818

Access permissions for region 1 as defined by start address ACL[1].ADDR and size ACL[1].SIZE

Note: This register can only be written once.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW1	WRITE			Configure write and erase permissions for region 1. Writing a '0' has no effect.
			Enable	0	Allow write and erase instructions to region 1.
			Disable	1	Block write and erase instructions to region 1.
B	RW1	READ			Configure read permissions for region 1. Writing a '0' has no effect.
			Enable	0	Allow read instructions to region 1.
			Disable	1	Block read instructions to region 1.

6.3.1.7 ACL[2].ADDR

Address offset: 0x820

Start address of region to protect. The start address must be word-aligned.

Note: This register can only be written once.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW1	ADDR			Start address of flash region 2. The start address must point to a flash page boundary.

6.3.1.8 ACL[2].SIZE

Address offset: 0x824

Size of region to protect counting from address ACL[2].ADDR. Writing a '0' has no effect.

Note: This register can only be written once.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW1	SIZE			Size of flash region 2 in bytes. Must be a multiple of the flash page size.

6.3.1.9 ACL[2].PERM

Address offset: 0x828

Access permissions for region 2 as defined by start address ACL[2].ADDR and size ACL[2].SIZE

Note: This register can only be written once.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW1	WRITE			Configure write and erase permissions for region 2. Writing a '0' has no effect.
			Enable	0	Allow write and erase instructions to region 2.
			Disable	1	Block write and erase instructions to region 2.
B	RW1	READ			Configure read permissions for region 2. Writing a '0' has no effect.
			Enable	0	Allow read instructions to region 2.
			Disable	1	Block read instructions to region 2.

6.3.1.10 ACL[3].ADDR

Address offset: 0x830

Start address of region to protect. The start address must be word-aligned.

Note: This register can only be written once.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW1	ADDR			Start address of flash region 3. The start address must point to a flash page boundary.

6.3.1.11 ACL[3].SIZE

Address offset: 0x834

Size of region to protect counting from address ACL[3].ADDR. Writing a '0' has no effect.

Note: This register can only be written once.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW1	SIZE			Size of flash region 3 in bytes. Must be a multiple of the flash page size.

6.3.1.12 ACL[3].PERM

Address offset: 0x838

Access permissions for region 3 as defined by start address ACL[3].ADDR and size ACL[3].SIZE

Note: This register can only be written once.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW1	WRITE			Configure write and erase permissions for region 3. Writing a '0' has no effect.
			Enable	0	Allow write and erase instructions to region 3.
			Disable	1	Block write and erase instructions to region 3.
B	RW1	READ			Configure read permissions for region 3. Writing a '0' has no effect.
			Enable	0	Allow read instructions to region 3.
			Disable	1	Block read instructions to region 3.

6.3.1.13 ACL[4].ADDR

Address offset: 0x840

Start address of region to protect. The start address must be word-aligned.

Note: This register can only be written once.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW1	ADDR			Start address of flash region 4. The start address must point to a flash page boundary.

6.3.1.14 ACL[4].SIZE

Address offset: 0x844

Size of region to protect counting from address ACL[4].ADDR. Writing a '0' has no effect.

Note: This register can only be written once.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW1	SIZE			Size of flash region 4 in bytes. Must be a multiple of the flash page size.

6.3.1.15 ACL[4].PERM

Address offset: 0x848

Access permissions for region 4 as defined by start address ACL[4].ADDR and size ACL[4].SIZE

Note: This register can only be written once.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW1	WRITE			Configure write and erase permissions for region 4. Writing a '0' has no effect.
			Enable	0	Allow write and erase instructions to region 4.
			Disable	1	Block write and erase instructions to region 4.
B	RW1	READ			Configure read permissions for region 4. Writing a '0' has no effect.
			Enable	0	Allow read instructions to region 4.
			Disable	1	Block read instructions to region 4.

6.3.1.16 ACL[5].ADDR

Address offset: 0x850

Start address of region to protect. The start address must be word-aligned.

Note: This register can only be written once.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW1	ADDR			Start address of flash region 5. The start address must point to a flash page boundary.

6.3.1.17 ACL[5].SIZE

Address offset: 0x854

Size of region to protect counting from address ACL[5].ADDR. Writing a '0' has no effect.

Note: This register can only be written once.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW1	SIZE			Size of flash region 5 in bytes. Must be a multiple of the flash page size.

6.3.1.18 ACL[5].PERM

Address offset: 0x858

Access permissions for region 5 as defined by start address ACL[5].ADDR and size ACL[5].SIZE

Note: This register can only be written once.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW1	WRITE			Configure write and erase permissions for region 5. Writing a '0' has no effect.
			Enable	0	Allow write and erase instructions to region 5.
			Disable	1	Block write and erase instructions to region 5.
B	RW1	READ			Configure read permissions for region 5. Writing a '0' has no effect.
			Enable	0	Allow read instructions to region 5.
			Disable	1	Block read instructions to region 5.

6.3.1.19 ACL[6].ADDR

Address offset: 0x860

Start address of region to protect. The start address must be word-aligned.

Note: This register can only be written once.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW1	ADDR																																	

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW1	ADDR			Start address of flash region 7. The start address must point to a flash page boundary.

6.3.1.23 ACL[7].SIZE

Address offset: 0x874

Size of region to protect counting from address ACL[7].ADDR. Writing a '0' has no effect.

Note: This register can only be written once.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW1	SIZE			Size of flash region 7 in bytes. Must be a multiple of the flash page size.

6.3.1.24 ACL[7].PERM

Address offset: 0x878

Access permissions for region 7 as defined by start address ACL[7].ADDR and size ACL[7].SIZE

Note: This register can only be written once.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW1	WRITE			Configure write and erase permissions for region 7. Writing a '0' has no effect.
			Enable	0	Allow write and erase instructions to region 7.
			Disable	1	Block write and erase instructions to region 7.
B	RW1	READ			Configure read permissions for region 7. Writing a '0' has no effect.
			Enable	0	Allow read instructions to region 7.
			Disable	1	Block read instructions to region 7.

6.4 CCM — AES CCM mode encryption

Cipher block chaining - message authentication code (CCM) mode is an authenticated encryption algorithm designed to provide both authentication and confidentiality during data transfer. CCM combines counter mode encryption and CBC-MAC authentication. The CCM terminology "Message authentication code (MAC)" is called the "Message integrity check (MIC)" in *Bluetooth* terminology and also in this document.

The CCM block generates an encrypted keystream that is applied to input data using the XOR operation and generates the four byte MIC field in one operation. CCM and RADIO can be configured to work synchronously. CCM will encrypt in time for transmission and decrypt after receiving bytes into memory from the radio. All operations can complete within the packet RX or TX time. CCM on this device is

implemented according to *Bluetooth* requirements and the algorithm as defined in IETF RFC3610, and depends on the AES-128 block cipher. A description of the CCM algorithm can also be found in NIST Special Publication 800-38C. The *Bluetooth* specification describes the configuration of counter mode blocks and encryption blocks to implement compliant encryption for Bluetooth Low Energy.

The CCM block uses EasyDMA to load key counter mode blocks (including the nonce required), and to read/write plain text and cipher text.

The AES CCM peripheral supports three operations: keystream generation, packet encryption, and packet decryption. These operations are performed in compliance with the *Bluetooth* AES CCM 128 bit block encryption, see *Bluetooth Core specification Version 4.0*.

The following figure illustrates keystream generation followed by encryption or decryption. The shortcut is optional.

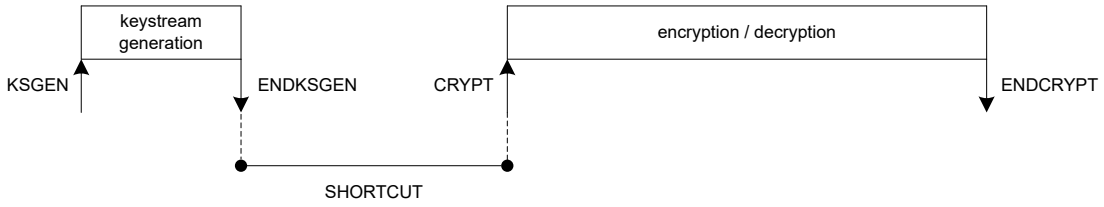


Figure 31: Keystream generation

6.4.1 Keystream generation

A new keystream needs to be generated before a new packet encryption or packet decryption operation can start.

A keystream is generated by triggering the KSGEN task. An ENDKSGEN event is generated after the keystream has been generated.

Keystream generation, packet encryption, and packet decryption operations utilize the configuration specified in the data structure pointed to by CNFPTR on page 203. It is necessary to configure this pointer and its underlying data structure, and register MODE on page 202 before the KSGEN task is triggered.

The keystream will be stored in the AES CCM peripheral's temporary memory area, specified by the SCRATCHPTR on page 204, where it will be used in subsequent encryption and decryption operations.

For default length packets (MODE.LENGTH = Default), the size of the generated keystream is 27 bytes. When using extended length packets (MODE.LENGTH = Extended), register MAXPACKETSIZE on page 204 specifies the length of the keystream to be generated. The length of the generated keystream must be greater or equal to the length of the subsequent packet payload to be encrypted or decrypted. The maximum length of the keystream in extended mode is 251 bytes, which means that the maximum packet payload size is 251.

If a shortcut is used between the ENDKSGEN event and CRYPT task, pointer INPTR on page 203 and the pointers OUTPTR on page 203 must also be configured before the KSGEN task is triggered.

6.4.2 Encryption

The AES CCM periheral is able to read an unencrypted packet, encrypt it, and append a four byte MIC field to the packet.

During packet encryption, the AES CCM peripheral performs the following:

- Reads the unencrypted packet located in RAM address specified in the INPTR pointer
- Encrypts the packet
- Appends a four byte long Message Integrity Check (MIC) field to the packet

Encryption is started by triggering the CRYPT task with register MODE on page 202 set to ENCRYPTION. An ENDCRYPT event is generated when packet encryption is completed.

The AES CCM peripheral will also modify the length field of the packet to adjust for the appended MIC field. It adds four bytes to the length and stores the resulting packet in RAM at the address specified in pointer OUTPTR on page 203, see Encryption on page 194.

Empty packets (length field is set to 0) will not be encrypted but instead moved unmodified through the AES CCM peripheral.

AES CCM supports different widths of the LENGTH field in the data structure for encrypted packets. This is configured in register MODE on page 202.

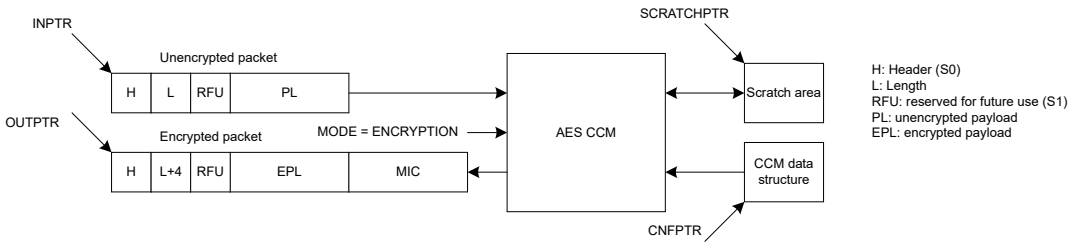


Figure 32: Encryption

6.4.3 Decryption

The AES CCM peripheral is able to read an encrypted packet, decrypt it, authenticate the MIC field, and generate an appropriate MIC status.

During packet decryption, the AES CCM peripheral performs the following:

- Reads the encrypted packet located in RAM at the address specified in the INPTR pointer
- Decrypts the packet
- Authenticates the packet's MIC field
- Generates the appropriate MIC status

The packet header (S0) and payload are included in the MIC authentication.

Decryption is started by triggering the CRYPT task with register MODE on page 202 set to DECRYPTION. An ENDCRYPT event is generated when packet decryption is completed.

The AES CCM peripheral modifies the length field of the packet to adjust for the MIC field. It subtracts four bytes from the length and stores the decrypted packet in RAM at the address specified in the pointer OUTPTR, see Decryption on page 195.

CCM is only able to decrypt packet payloads that are at least five bytes long (one byte or more encrypted payload (EPL) and four bytes of MIC). CCM will therefore generate a MIC error for packets where the length field is set to 1, 2, 3, or 4.

Empty packets (length field is set to 0) will not be decrypted but instead moved unmodified through the AES CCM peripheral. These packets will always pass the MIC check.

CCM supports different widths of the LENGTH field in the data structure for decrypted packets. This is configured in register MODE on page 202.

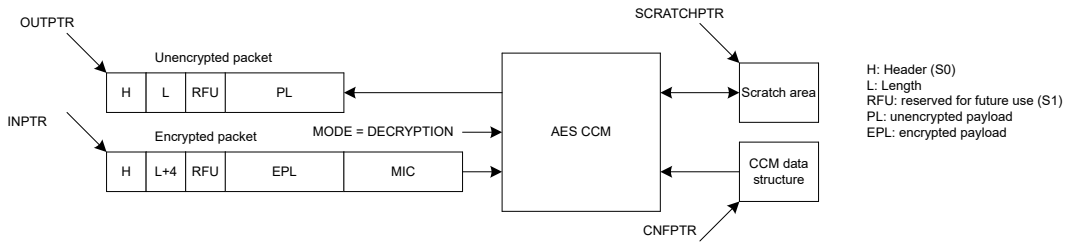


Figure 33: Decryption

6.4.4 AES CCM and RADIO concurrent operation

The CCM peripheral is able to encrypt/decrypt data synchronously to data being transmitted or received on the radio.

In order for CCM to run synchronously with the radio, the data rate setting in register MODE on page 202 needs to match the radio data rate. The settings in this register apply whenever either the KSGEN or CRYPT tasks are triggered.

The data rate setting of register MODE on page 202 can also be overridden on-the-fly during an ongoing encrypt/decrypt operation by the contents of register RATEOVERRIDE on page 204. The data rate setting in this register applies whenever the RATEOVERRIDE task is triggered. This feature can be useful in cases where the radio data rate is changed during an ongoing packet transaction.

6.4.5 Encrypting packets on-the-fly in radio transmit mode

When the AES CCM peripheral encrypts a packet on-the-fly while RADIO is transmitting it, RADIO must read the encrypted packet from the same memory location that the AES CCM peripheral is writing to.

The OUTPTR on page 203 pointer in the AES CCM must point to the same memory location as the PACKETPTR pointer in the radio, see Configuration of on-the-fly encryption on page 195.

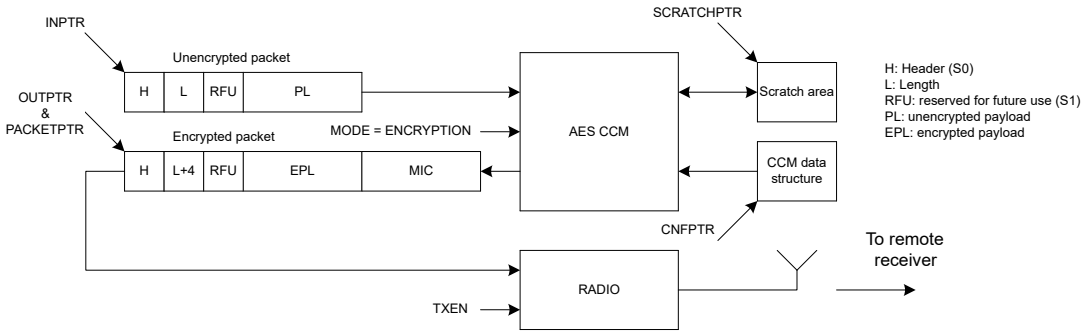


Figure 34: Configuration of on-the-fly encryption

In order to match RADIO's timing, the KSGEN task must be triggered early enough to allow the keystream generation to complete before packet encryption begins.

For short packets (MODE.LENGTH = Default), the KSGEN task must be triggered before or at the same time as the START task in RADIO is triggered. In addition, the shortcut between the ENDKSGEN event and the CRYPT task must be enabled. This use-case is illustrated in On-the-fly encryption of short packets (MODE.LENGTH = Default) using a PPI connection on page 196. It uses a PPI connection between the READY event in RADIO and the KSGEN task in the AES CCM peripheral.

For long packets (MODE.LENGTH = Extended), the keystream generation needs to start earlier, such as when the TXEN task in RADIO is triggered.

Refer to Timing specification on page 205 for information about the time needed for generating a keystream.

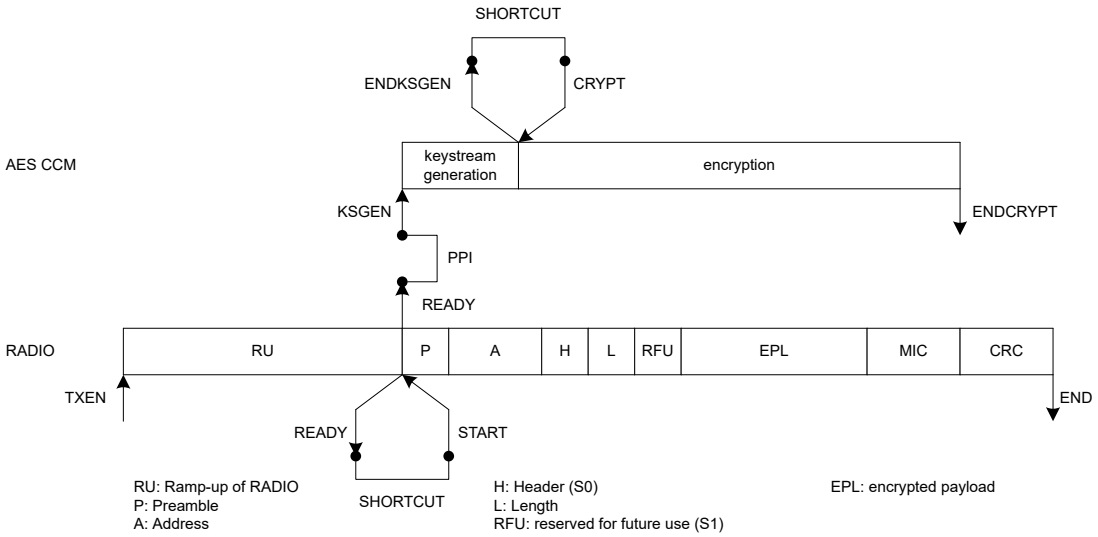


Figure 35: On-the-fly encryption of short packets (MODE.LENGTH = Default) using a PPI connection

6.4.6 Decrypting packets on-the-fly in RADIO receive mode

When the AES CCM peripheral decrypts a packet on-the-fly while RADIO is receiving it, the AES CCM peripheral must read the encrypted packet from the same memory location that RADIO is writing to.

The INPTR on page 203 pointer in the AES CCM must point to the same memory location as the PACKETPTR pointer in RADIO, see Configuration of on-the-fly decryption on page 196.

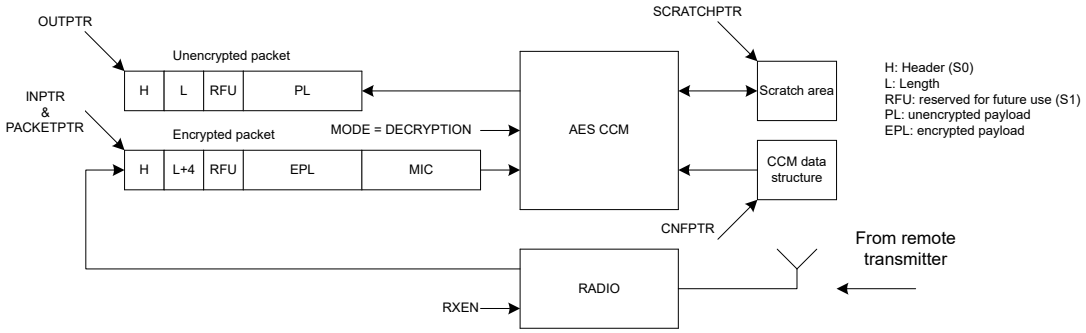


Figure 36: Configuration of on-the-fly decryption

In order to match RADIO's timing, the KSGEN task must be triggered early enough to allow the keystream generation to complete before the decryption of the packet shall start.

For short packets (MODE.LENGTH = Default) the KSGEN task must be triggered no later than when the START task in RADIO is triggered. In addition, the CRYPT task must be triggered no earlier than when the ADDRESS event is generated by RADIO.

If the CRYPT task is triggered exactly at the same time as the ADDRESS event is generated by RADIO, the AES CCM peripheral will guarantee that the decryption is completed no later than when the END event in RADIO is generated.

This use-case is illustrated in On-the-fly decryption of short packets (MODE.LENGTH = Default) using a PPI connection on page 197 using a PPI connection between the ADDRESS event in RADIO and the CRYPT task in the AES CCM peripheral. The KSGEN task is triggered from the READY event in RADIO through a PPI connection.

For long packets (MODE.LENGTH = Extended) the keystream generation will need to start even earlier, such as when the RXEN task in RADIO is triggered.

Refer to Timing specification on page 205 for information about the time needed for generating a keystream.

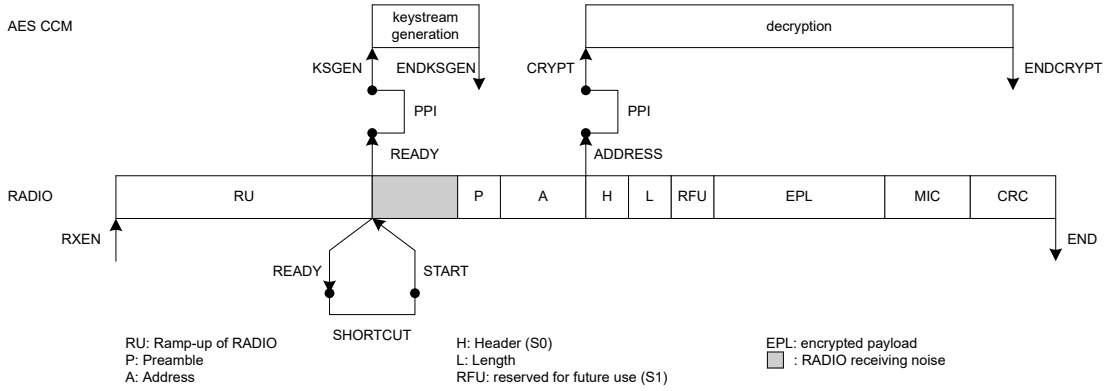


Figure 37: On-the-fly decryption of short packets (MODE.LENGTH = Default) using a PPI connection

6.4.7 CCM data structure

The CCM data structure is located in Data RAM at the memory location specified by the CNFPTR pointer register.

Property	Address offset	Description
KEY	0	16 byte AES key
PKTCTR	16	Octet0 (LSO) of packet counter
	17	Octet1 of packet counter
	18	Octet2 of packet counter
	19	Octet3 of packet counter
	20	Bit 6 – Bit 0: Octet4 (7 most significant bits of packet counter, with Bit 6 being the most significant bit) Bit7: Ignored
	21	Ignored
	22	Ignored
	23	Ignored
	24	Bit 0: Direction bit Bit 7 – Bit 1: Zero padded
IV	25	8 byte initialization vector (IV) Octet0 (LSO) of IV, Octet1 of IV, … , Octet7 (MSO) of IV

Table 11: CCM data structure overview

The NONCE vector (as specified by the *Bluetooth* Core Specification) will be generated by hardware based on the information specified in the CCM data structure from CCM data structure overview on page 197 .

Property	Address offset	Description
HEADER	0	Packet Header
LENGTH	1	Number of bytes in unencrypted payload
RFU	2	Reserved Future Use
PAYLOAD	3	Unencrypted payload

Table 12: Data structure for unencrypted packet

Property	Address offset	Description
HEADER	0	Packet Header
LENGTH	1	Number of bytes in encrypted payload including length of MIC LENGTH will be 0 for empty packets since the MIC is not added to empty packets
RFU	2	Reserved Future Use
PAYLOAD	3	Encrypted payload
MIC	3 + payload length	ENCRYPT: 4 bytes encrypted MIC MIC is not added to empty packets

Table 13: Data structure for encrypted packet

6.4.8 EasyDMA and ERROR event

CCM implements an EasyDMA mechanism for reading and writing to RAM.

When the CPU and EasyDMA enabled peripherals access the same RAM block at the same time, increased bus collisions might disrupt on-the-fly encryption. This will generate an ERROR event.

EasyDMA stops accessing RAM when the ENDKSGEN and ENDCRYPT events are generated.

If the CNFPTR, SCRATCHPTR, INPTR, and the OUTPTR are not pointing to the Data RAM region, an EasyDMA transfer may result in a HardFault or RAM corruption. See Memory on page 21 for more information about the different memory regions.

6.4.9 Registers

Instances

Instance	Base address	Description
CCM	0x4000F000	AES counter with CBC-MAC (CCM) mode block encryption

Register overview

Register	Offset	Description
TASKS_KSGEN	0x000	Start generation of keystream. This operation will stop by itself when completed.
TASKS_CRYPT	0x004	Start encryption/decryption. This operation will stop by itself when completed.
TASKS_STOP	0x008	Stop encryption/decryption
TASKS_RATEOVERRIDE	0x00C	Override DATARATE setting in MODE register with the contents of the RATEOVERRIDE register for any ongoing encryption/decryption
EVENTS_ENDKSGEN	0x100	Keystream generation complete
EVENTS_ENDCRYPT	0x104	Encrypt/decrypt complete
EVENTS_ERROR	0x108	CCM error event This register is deprecated.
SHORTS	0x200	Shortcuts between local events and tasks
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
MICSTATUS	0x400	MIC check result
ENABLE	0x500	Enable
MODE	0x504	Operation mode
CNFPTR	0x508	Pointer to data structure holding the AES key and the NONCE vector
INPTR	0x50C	Input pointer
OUTPTR	0x510	Output pointer
SCRATCHPTR	0x514	Pointer to data area used for temporary storage

Register	Offset	Description
MAXPACKETSIZE	0x518	Length of keystream generated when MODE.LENGTH = Extended
RATEOVERRIDE	0x51C	Data rate override setting.

6.4.9.1 TASKS_KSGEN

Address offset: 0x000

Start generation of keystream. This operation will stop by itself when completed.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_KSGEN			Start generation of keystream. This operation will stop by itself when completed.
			Trigger	1	Trigger task

6.4.9.2 TASKS_CRYPT

Address offset: 0x004

Start encryption/decryption. This operation will stop by itself when completed.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CRYPT			Start encryption/decryption. This operation will stop by itself when completed.
			Trigger	1	Trigger task

6.4.9.3 TASKS_STOP

Address offset: 0x008

Stop encryption/decryption

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STOP			Stop encryption/decryption
			Trigger	1	Trigger task

6.4.9.4 TASKS_RATEOVERRIDE

Address offset: 0x00C

Override DATARATE setting in MODE register with the contents of the RATEOVERRIDE register for any ongoing encryption/decryption

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_RATEOVERRIDE			Override DATARATE setting in MODE register with the contents of the RATEOVERRIDE register for any ongoing encryption/decryption
			Trigger	1	Trigger task

6.4.9.5 EVENTS_ENDKSGEN

Address offset: 0x100

Keystream generation complete

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDKSGEN			Keystream generation complete
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.4.9.6 EVENTS_ENDCRYPT

Address offset: 0x104

Encrypt/decrypt complete

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDCRYPT			Encrypt/decrypt complete
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.4.9.7 EVENTS_ERROR (Deprecated)

Address offset: 0x108

CCM error event

This register is deprecated.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ERROR			CCM error event This field is deprecated.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.4.9.8 SHORTS

Address offset: 0x200

Shortcuts between local events and tasks

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ENDKSGEN_CRYPT			Shortcut between event ENDKSGEN and task CRYPT
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

6.4.9.9 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ENDKSGEN			Write '1' to enable interrupt for event ENDKSGEN
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	ENDCRYPT			Write '1' to enable interrupt for event ENDCRYPT
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	ERROR			Write '1' to enable interrupt for event ERROR This register is deprecated.
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.4.9.10 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ENDKSGEN			Write '1' to disable interrupt for event ENDKSGEN
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	ENDCRYPT			Write '1' to disable interrupt for event ENDCRYPT
			Clear	1	Disable

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	ERROR			Write '1' to disable interrupt for event ERROR This register is deprecated.
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.4.9.11 MICSTATUS

Address offset: 0x400

MIC check result

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	MICSTATUS			The result of the MIC check performed during the previous decryption operation
			CheckFailed	0	MIC check failed
			CheckPassed	1	MIC check passed

6.4.9.12 ENABLE

Address offset: 0x500

Enable

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ENABLE			Enable or disable CCM
			Disabled	0	Disable
			Enabled	2	Enable

6.4.9.13 MODE

Address offset: 0x504

Operation mode

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B B A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**

ID	R/W	Field	Value ID	Value	Description
A-	RW	MODE			The mode of operation to be used. Settings in this register apply whenever either the KSGEN task or the CRYPT task is triggered.
			Encryption	0	AES CCM packet encryption mode

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B B A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
			Decryption	1	AES CCM packet decryption mode
B	RW	DATARATE			Radio data rate that the CCM shall run synchronous with
			1Mbit	0	1 Mbps
			2Mbit	1	2 Mbps
			125Kbps	2	125 kbps
			500Kbps	3	500 kbps
C	RW	LENGTH			Packet length configuration
			Default	0	Default length. Effective length of LENGTH field in encrypted/decrypted packet is 5 bits. A keystream for packet payloads up to 27 bytes will be generated.
			Extended	1	Extended length. Effective length of LENGTH field in encrypted/decrypted packet is 8 bits. A keystream for packet payloads up to MAXPACKETSIZE bytes will be generated.

6.4.9.14 CNFPTR

Address offset: 0x508

Pointer to data structure holding the AES key and the NONCE vector

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	CNFPTR			Pointer to the data structure holding the AES key and the CCM NONCE vector (see table CCM data structure overview)

6.4.9.15 INPTR

Address offset: 0x50C

Input pointer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	INPTR			Input pointer

6.4.9.16 OUTPTR

Address offset: 0x510

Output pointer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	OUTPTR			Output pointer

6.4.9.17 SCRATCHPTR

Address offset: 0x514

Pointer to data area used for temporary storage

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	SCRATCHPTR			Pointer to a scratch data area used for temporary storage during keystream generation, MIC generation and encryption/decryption. The scratch area is used for temporary storage of data during keystream generation and encryption. When MODE.LENGTH = Default, a space of 43 bytes is required for this temporary storage. When MODE.LENGTH = Extended, a space of (16 + MAXPACKETSIZE) bytes is required.																												

6.4.9.18 MAXPACKETSIZE

Address offset: 0x518

Length of keystream generated when MODE.LENGTH = Extended

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																												A	A	A	A	A	A	A	A
Reset 0x000000FB				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**0**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW	MAXPACKETSIZE		[0x001B..0x00FB]	Length of keystream generated when MODE.LENGTH = Extended. This value must be greater than or equal to the subsequent packet payload to be encrypted/decrypted.																																

6.4.9.19 RATEOVERRIDE

Address offset: 0x51C

Data rate override setting.

Override value to be used instead of the setting of MODE.DATARATE. This override value applies when the RATEOVERRIDE task is triggered.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																		A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	RATEOVERRIDE			Data rate override setting																																
			1Mbit	0	1 Mbps																																
			2Mbit	1	2 Mbps																																
			125Kbps	2	125 kbps																																
			500Kbps	3	500 kbps																																

6.4.10 Electrical specification

6.4.10.1 Timing specification

Symbol	Description	Min.	Typ.	Max.	Units
t_{kgen}	Time needed for keystream generation (given priority access to destination RAM block)			50	µs

6.5 COMP — Comparator

The comparator (COMP) compares an input voltage (VIN+) against a second input voltage (VIN-). VIN+ can be derived from an analog input pin (AIN0-AIN7). VIN- can be derived from multiple sources depending on the operation mode of the comparator.

The main features of COMP are the following:

- Input range from 0 V to VDD
- Single-ended mode
  - Fully flexible hysteresis using a 64-level reference ladder
- Differential mode
  - Configurable hysteresis
- Reference inputs (VREF):
  - VDD
  - External reference from AIN0 to AIN7 (between 0 V and VDD)
  - Internal references 1.2 V, 1.8 V, and 2.4 V
- Three speed/power consumption modes:
  - Low-power
  - Normal
  - High-speed
- Event generation on output changes
  - UP event on VIN- > VIN+
  - DOWN event on VIN- < VIN+
  - CROSS event on VIN+ and VIN- crossing
  - READY event on core and internal reference (if used) ready

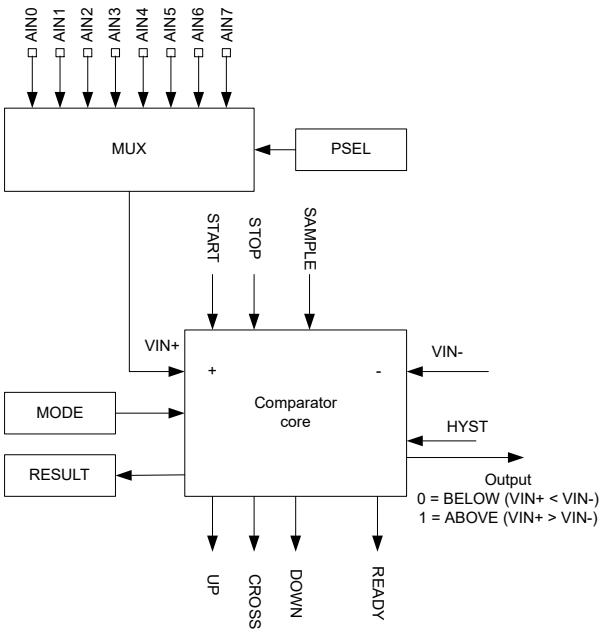


Figure 38: Comparator overview

Once enabled (using the ENABLE register), the comparator is started by triggering the START task and stopped by triggering the STOP task. The comparator will generate a READY event to indicate when it is ready for use and the output is correct. The delay between START and READY is $t_{INT_REF,START}$ if an internal reference is selected, or $t_{COMP,START}$ if an external reference is used. When the COMP module is started, events will be generated every time VIN+ crosses VIN-.

Operation modes

The comparator can be configured to operate in two main operation modes: differential mode and single-ended mode. See the MODE register for more information. In both operation modes, the comparator can operate in different speed and power consumption modes (low-power, normal and high-speed). High-speed mode will consume more power compared to low-power mode, and low-power mode will result in slower response time compared to high-speed mode.

Use the PSEL register to select any of the AIN0-AIN7 pins as VIN+ input, regardless of the operation mode selected for the comparator. The source of VIN- depends on which of the following operation mode are used:

- Differential mode - Derived directly from AIN0 to AIN7
- Single-ended mode - Derived from VREF. VREF can be derived from VDD, AIN0-AIN7 or internal 1.2 V, 1.8 V and 2.4 V references.

The selected analog pins will be acquired by the comparator once it is enabled.

An optional hysteresis on VIN+ and VIN- can be enabled when the module is used in differential mode through the HYST register. In single-ended mode, VUP and VDOWN thresholds can be set to implement a hysteresis using the reference ladder (see Comparator in single-ended mode on page 208). This hysteresis is in the order of magnitude of $V_{DIFFHYST}$, and shall prevent noise on the signal to create unwanted events. See Hysteresis example where VIN+ starts below VUP on page 209 for an illustration of the effect of an active hysteresis on a noisy input signal.

An upward crossing will generate an UP event and a downward crossing will generate a DOWN event. The CROSS event will be generated every time there is a crossing, independent of direction.

The immediate value of the comparator can be sampled to RESULT register by triggering the SAMPLE task.

6.5.1 Differential mode

In differential mode, the reference input VIN- is derived directly from one of the AINx pins.

Before enabling the comparator via the ENABLE register, the following registers must be configured for the differential mode:

- PSEL
- MODE
- EXTREFSEL

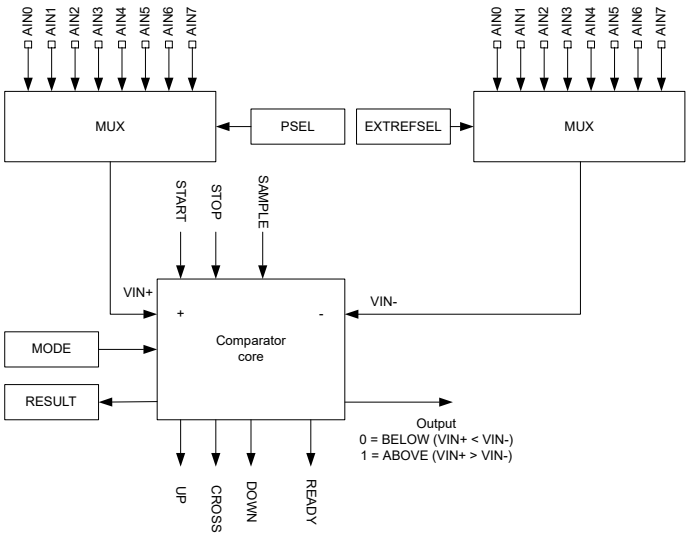


Figure 39: Comparator in differential mode

Note: Depending on the device, not all the analog inputs may be available for each MUX. See definitions for PSEL and EXTREFSEL for more information about which analog pins are available on a particular device.

When the HYST register is turned on during this mode, the output of the comparator and associated events do the following:

- Change from ABOVE to BELOW when VIN+ drops below VIN- - ($V_{DIFFHYST}$/2)
- Change from BELOW to ABOVE when VIN+ raises above VIN- + ($V_{DIFFHYST}$/2)

This behavior is illustrated in the following figure.

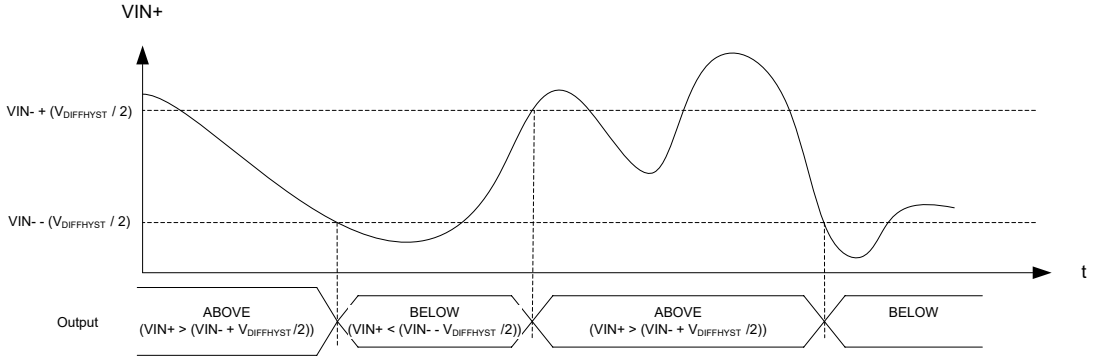


Figure 40: Hysteresis enabled in differential mode

6.5.2 Single-ended mode

In single-ended mode, VIN- is derived from the reference ladder.

Before enabling the comparator via the ENABLE register, the following registers must be configured for the single-ended mode:

- PSEL
- MODE
- REFSEL
- EXTREFSEL
- TH

The reference ladder uses the reference voltage (VREF) to derive two new voltage references, VUP and VDOWN. VUP and VDOWN are configured using THUP and THDOWN respectively in the TH register. VREF can be derived from any of the available reference sources, configured using the EXTREFSEL and REFSEL registers as shown in the following figure. When AREF is selected in the REFSEL register, the EXTREFSEL register is used to select one of the AIN0-AIN7 analog input pins as reference input. The selected analog pins will be acquired by the comparator once it is enabled.

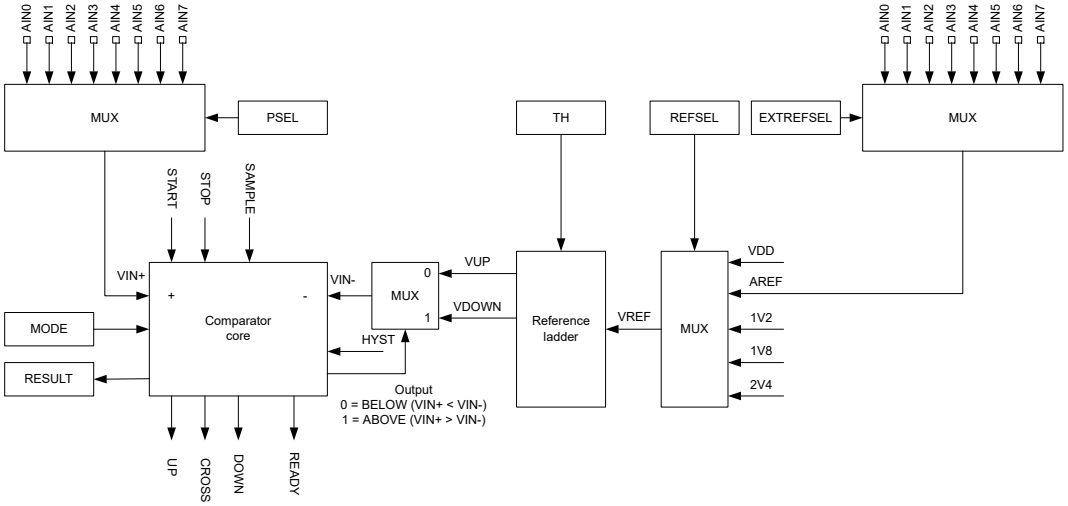


Figure 41: Comparator in single-ended mode

Note: Depending on the device, not all the analog inputs may be available for each MUX. See definitions for PSEL and EXTREFSEL for more information about which analog pins are available on a particular device.

When the comparator core detects that VIN+ > VIN-, i.e. ABOVE as per the RESULT register, VIN- will switch to VDOWN. When VIN+ falls below VIN- again, VIN- will be switched back to VUP. By specifying VUP larger than VDOWN, a hysteresis can be generated as illustrated in the following figures.

Writing to HYST has no effect in single-ended mode, and the content of this register is ignored.

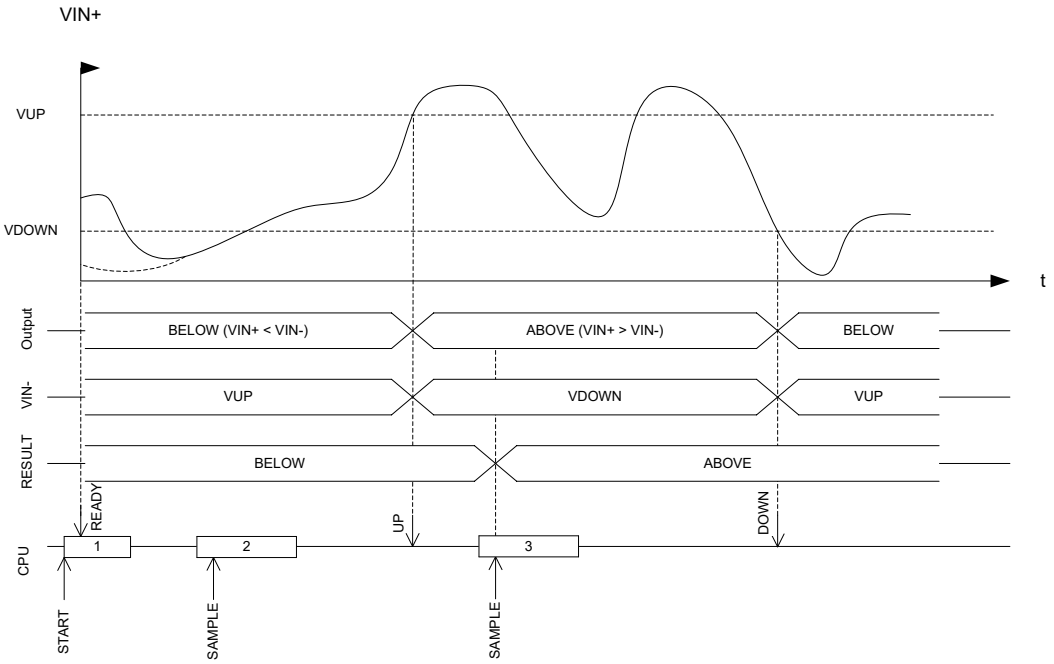


Figure 42: Hysteresis example where VIN+ starts below VUP

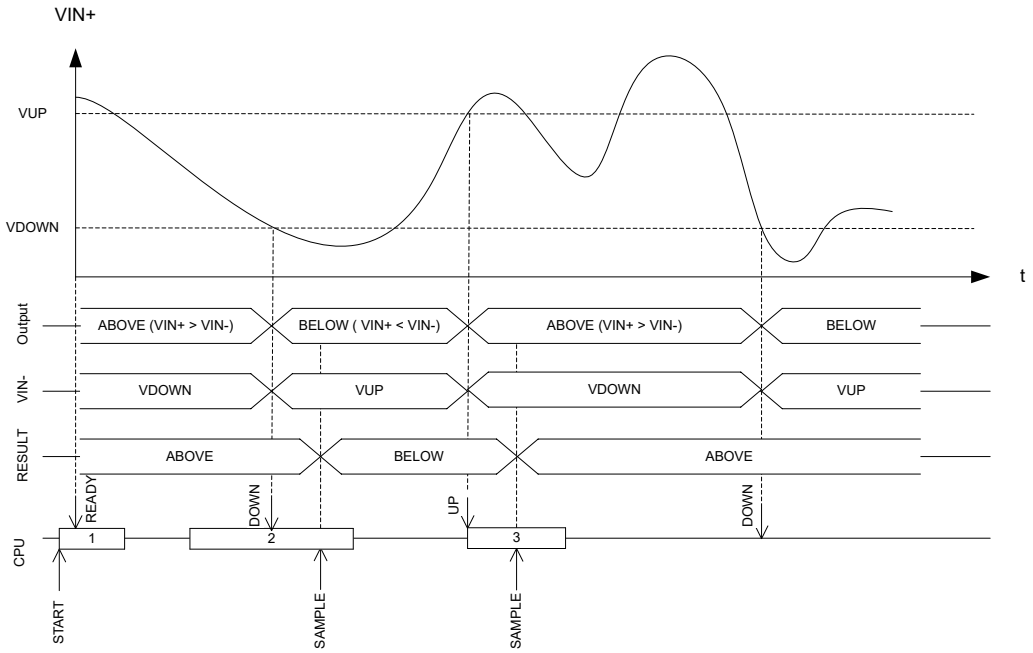


Figure 43: Hysteresis example where VIN+ starts above VUP

6.5.3 Registers

Instances

Instance	Base address	Description
COMP	0x40013000	General purpose comparator

Register overview

Register	Offset	Description
TASKS_START	0x000	Start comparator
TASKS_STOP	0x004	Stop comparator
TASKS_SAMPLE	0x008	Sample comparator value
EVENTS_READY	0x100	COMP is ready and output is valid
EVENTS_DOWN	0x104	Downward crossing
EVENTS_UP	0x108	Upward crossing
EVENTS_CROSS	0x10C	Downward or upward crossing
SHORTS	0x200	Shortcuts between local events and tasks
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
RESULT	0x400	Compare result
ENABLE	0x500	COMP enable
PSEL	0x504	Pin select
REFSEL	0x508	Reference source select for single-ended mode
EXTREFSEL	0x50C	External reference select
TH	0x530	Threshold configuration for hysteresis unit
MODE	0x534	Mode configuration
HYST	0x538	Comparator hysteresis enable

6.5.3.1 TASKS_START

Address offset: 0x000

Start comparator

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_START			Start comparator
			Trigger	1	Trigger task

6.5.3.2 TASKS_STOP

Address offset: 0x004

Stop comparator

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STOP			Stop comparator
			Trigger	1	Trigger task

6.5.3.3 TASKS_SAMPLE

Address offset: 0x008

Sample comparator value

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_SAMPLE			Sample comparator value
			Trigger	1	Trigger task

6.5.3.4 EVENTS_READY

Address offset: 0x100

COMP is ready and output is valid

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_READY			COMP is ready and output is valid
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.5.3.5 EVENTS_DOWN

Address offset: 0x104

Downward crossing

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_DOWN			Downward crossing
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.5.3.6 EVENTS_UP

Address offset: 0x108

Upward crossing

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_UP			Upward crossing
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.5.3.7 EVENTS_CROSS

Address offset: 0x10C

Downward or upward crossing

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_CROSS			Downward or upward crossing
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.5.3.8 SHORTS

Address offset: 0x200

Shortcuts between local events and tasks

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	READY_SAMPLE			Shortcut between event READY and task SAMPLE
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
B	RW	READY_STOP			Shortcut between event READY and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
C	RW	DOWN_STOP			Shortcut between event DOWN and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
D	RW	UP_STOP			Shortcut between event UP and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
E	RW	CROSS_STOP			Shortcut between event CROSS and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

6.5.3.9 INTEN

Address offset: 0x300

Enable or disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	READY			Enable or disable interrupt for event READY
			Disabled	0	Disable
			Enabled	1	Enable
B	RW	DOWN			Enable or disable interrupt for event DOWN
			Disabled	0	Disable
			Enabled	1	Enable
C	RW	UP			Enable or disable interrupt for event UP
			Disabled	0	Disable
			Enabled	1	Enable
D	RW	CROSS			Enable or disable interrupt for event CROSS
			Disabled	0	Disable

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Enabled	1	Enable

6.5.3.10 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	READY			Write '1' to enable interrupt for event READY
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	DOWN			Write '1' to enable interrupt for event DOWN
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	UP			Write '1' to enable interrupt for event UP
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	CROSS			Write '1' to enable interrupt for event CROSS
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.5.3.11 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	READY			Write '1' to disable interrupt for event READY
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	DOWN			Write '1' to disable interrupt for event DOWN
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	UP			Write '1' to disable interrupt for event UP
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
D	RW	CROSS			Write '1' to disable interrupt for event CROSS
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.5.3.12 RESULT

Address offset: 0x400

Compare result

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	RESULT			Result of last compare. Decision point SAMPLE task.
			Below	0	Input voltage is below the threshold (VIN+ < VIN-)
			Above	1	Input voltage is above the threshold (VIN+ > VIN-)

6.5.3.13 ENABLE

Address offset: 0x500

COMP enable

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ENABLE			Enable or disable COMP
			Disabled	0	Disable
			Enabled	2	Enable

6.5.3.14 PSEL

Address offset: 0x504

Pin select

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PSEL			Analog pin select
			AnalogInput0	0	AIN0 selected as analog input
			AnalogInput1	1	AIN1 selected as analog input
			AnalogInput2	2	AIN2 selected as analog input
			AnalogInput3	3	AIN3 selected as analog input
			AnalogInput4	4	AIN4 selected as analog input
			AnalogInput5	5	AIN5 selected as analog input
			AnalogInput6	6	AIN6 selected as analog input

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			AnalogInput7	7	AIN7 selected as analog input

6.5.3.15 REFSEL

Address offset: 0x508

Reference source select for single-ended mode

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A
Reset 0x00000004				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	REFSEL			Reference select
			Int1V2	0	VREF = internal 1.2 V reference (VDD >= 1.7 V)
			Int1V8	1	VREF = internal 1.8 V reference (VDD >= VREF + 0.2 V)
			Int2V4	2	VREF = internal 2.4 V reference (VDD >= VREF + 0.2 V)
			VDD	4	VREF = VDD
			ARef	5	VREF = AREF

6.5.3.16 EXTREFSEL

Address offset: 0x50C

External reference select

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EXTREFSEL			External analog reference select
			AnalogReference0	0	Use AIN0 as external analog reference
			AnalogReference1	1	Use AIN1 as external analog reference
			AnalogReference2	2	Use AIN2 as external analog reference
			AnalogReference3	3	Use AIN3 as external analog reference
			AnalogReference4	4	Use AIN4 as external analog reference
			AnalogReference5	5	Use AIN5 as external analog reference
			AnalogReference6	6	Use AIN6 as external analog reference
			AnalogReference7	7	Use AIN7 as external analog reference

6.5.3.17 TH

Address offset: 0x530

Threshold configuration for hysteresis unit

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B B B B B B A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	THDOWN		[63:0]	VDOWN = (THDOWN+1)/64*VREF
B	RW	THUP		[63:0]	VUP = (THUP+1)/64*VREF

6.5.3.18 MODE

Address offset: 0x534

Mode configuration

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	SP			Speed and power modes
			Low	0	Low-power mode
			Normal	1	Normal mode
			High	2	High-speed mode
B	RW	MAIN			Main operation modes
			SE	0	Single-ended mode
			Diff	1	Differential mode

6.5.3.19 HYST

Address offset: 0x538

Comparator hysteresis enable

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	HYST			Comparator hysteresis
			NoHyst	0	Comparator hysteresis disabled
			Hyst50mV	1	Comparator hysteresis enabled

6.5.4 Electrical specification

6.5.4.1 COMP Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
$t_{PROPDLY,LP}$	Propagation delay, Low-power mode[15]		0.6		µs
$t_{PROPDLY,N}$	Propagation delay, Normal mode[15]		0.2		µs
$t_{PROPDLY,HS}$	Propagation delay, High-speed mode[15]		0.1		µs
$V_{DIFFHYST}$	Optional hysteresis applied to differential input	20	30	80	mV
$V_{VDD-VREF}$	Required difference between VDD and a selected VREF, VDD > VREF	0.3			V
$t_{INT_REF,START}$	Startup time for the internal bandgap reference		50	80	µs
E_{INT_REF}	Internal bandgap reference error	-3		3	%
$V_{INPUTOFFSET}$	Input offset	-10		10	mV
$t_{COMP,START}$	Startup time for the comparator core		3		µs

[15] Propagation delay is with 10 mV overdrive.

6.6 CRYPTOCELL — Arm TrustZone CryptoCell 310

Arm TrustZone CryptoCell 310 (CRYPTOCELL) is a security subsystem providing root of trust (RoT) and cryptographic services for a device.

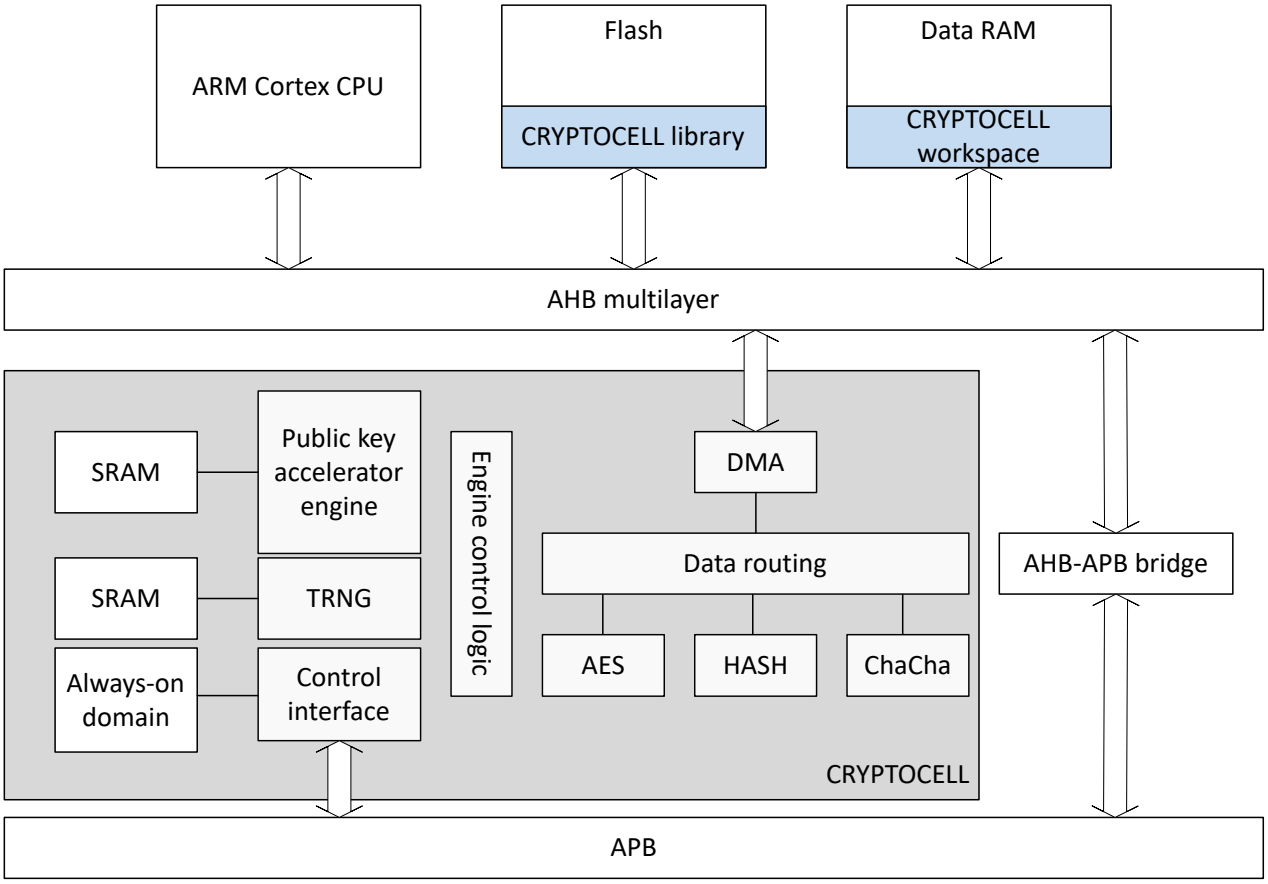


Figure 44: CRYPTOCELL block diagram

The following cryptographic features are among the functionality that can be supported:

- True random number generator (TRNG) compliant with FIPS 140-2, BSI AIS-31, and NIST 800-90B.
- Pseudorandom number generator (PRNG) using underlying AES engine compliant with NIST 800-90A
- RSA public key cryptography
  - Signature verification up to key sizes of 2048 bits
  - Key generation up to key sizes of 2048 bits
  - PKCS#1 v2.1/v1.5
- Elliptic curve cryptography (ECC)
  - NIST FIPS 186-4 recommended curves using pseudorandom parameters, up to 521 bits:
    - Prime field: P-192, P-224, P-256, P-384, P-521
  - SEC 2 recommended curves using pseudorandom parameters, up to 521 bits:
    - Prime field: secp160r1, secp192r1, secp224r1, secp256r1, secp384r1, secp521r1
  - Koblitz curves using fixed parameters, up to 256 bits:
    - Prime field: secp160k1, secp192k1, secp224k1, secp256k1
  - Brainpool curves:
    - Prime field: BrainpoolP256r1
  - Edwards/Montgomery curves:

    - Ed25519, Curve25519
  - ECDH/ECDSA support
- Secure remote password protocol (SRP), up to 3072 bits operations
- Hashing functions
  - SHA-1, SHA-2 up to 256 bits
  - Keyed-hash message authentication code (HMAC)
- AES symmetric encryption
  - General purpose AES engine (encrypt/decrypt, sign/verify)
  - 128 bits key size
  - Supported encryption modes: ECB, CBC, CMAC/CBC-MAC, CTR, CCM/CCM*
- ChaCha20/Poly1305 symmetric encryption
  - 128 and 256 bits key size
  - Authenticated encryption with associated data (AEAD) mode

6.6.1 Disclaimer

This section contains an important disclaimer about the CRYPTOCELL subsystem documentation.

The CRYPTOCELL subsystem is recommended for use with the libraries in the Nordic Semiconductor ASA SDK. These libraries are tested and verified to work with the CRYPTOCELL subsystem hardware. The CRYPTOCELL subsystem documentation and register descriptions are for reference only and can be used for modifying the Nordic supplied SDK libraries or implementing new features.

Nordic Semiconductor ASA reserves the right to change the CRYPTOCELL documentation and register descriptions without further notice. Changes will not trigger erratas and will not be seen as changing form/fit/function of the device.

Please note that Nordic cannot support questions directly related to the register interface or modification of the source code implementation. Nordic provide support for the top-level API in the software library distributed as part of the device SDK.

6.6.2 Usage

The CRYPTOCELL subsystem is a hardware and software solution where software is delivered as libraries in Nordic device SDKs. Recommended usage of the CRYPTOCELL subsystem is to use the SDK library implementation available for the device. The CRYPTOCELL subsystem is documented for reference purpose only, please see section Disclaimer on page 218 for more information.

To enable CRYPTOCELL, use register ENABLE on page 224. The device will not enter the System ON IDLE mode until CRYPTOCELL has been disabled, see POWER — Power supply on page 81 for more information. The Nordic SDK software library automatically controls enabling and disabling of the CRYPTOCELL subsystem as a part of its function calls.

6.6.3 Security configuration

CRYPTOCELL has internal storage for its security configuration, which is preserved even if CRYPTOCELL is disabled.

The following configuration settings are retained:

- Device life cycle state (LCS)
- Various lock bits
- 128 bits device root key, K_{DR}, see Device root key on page 220

Any reset source will erase the CRYPTOCELL internal storage, see Reset on page 89 for more information.

6.6.3.1 Lifecycle state (LCS)

Lifecycle refers to the multiple states a device goes through during its lifetime. `DebugEnable` and `Secure` are the two CRYPTOCELL lifecycle states available to the device.

The CRYPTOCELL lifecycle state (LCS) is controlled through register HOST_IOT_LCS on page 301. The LCS is configured by writing either `DebugEnable` or `Secure` to the LCS field of this register. To validate that the register is configured correctly, read back the read-only field LCS_IS_VALID from the register HOST_IOT_LCS on page 301. The LCS_IS_VALID field will change from `Invalid` to `Valid` once a valid LCS value is written.

The following debug override functionality is available if LCS is configured as `DebugEnable`:

- Registers HOST_IOT_KDR0 through HOST_IOT_KDR3 can be written multiple times.
- The TRNG output can be overridden. This is done by writing the desired value to register EHR_DATA[0] through EHR_DATA[5] in RNG engine. If LCS is configured as `Secure`, registers EHR_DATA are read-only and its content is randomly generated by the RNG engine.

LCS field value	LCS_IS_VALID field value	Description
Secure	Invalid	Default reset value indicating that LCS has not been configured.
Secure	Valid	LCS set to `Secure` mode, and LCS is valid. Registers HOST_IOT_KDR0 through HOST_IOT_KDR3 can only be written once. Any additional writes are ignored.
DebugEnable	Valid	LCS set to `DebugEnable` mode, and LCS is valid. Registers HOST_IOT_KDR0 through HOST_IOT_KDR3 can be written multiple times.

Table 14: Lifecycle states

6.6.4 Cryptographic flow

The following section describe a typical cryptographic flow for the CRYPTOCELL subsystem.

1. Enable CRYPTOCELL subsystem as described in Usage on page 218.
2. Perform clock control for the desired cryptographic engine(s) as described in Power and clock on page 221.
3. Configure the desired cryptographic mode as described in CTL interface on page 286.
4. Depending on the selected cryptographic mode the active engine(s) must be configured, including which cryptographic key to use as described in Cryptographic key selection on page 219.
5. Optionally configure DMA engines as described in Direct memory access (DMA) on page 220.
6. Initiate the operation, and wait for an event as described in Interrupt handling on page 221.
7. Check status register(s) for the active engine(s).

6.6.5 Cryptographic key selection

The CRYPTOCELL subsystem can operate on different cryptographic keys.

6.6.5.1 Hardware unique keys

The AES engine can be instructed to use different key input sources.

The cryptographic key input for the AES engine on page 224 can either be a hard-coded RTL key referred to as K_{PRTL}, a device root key referred to as K_{DR} which is typically programmed into CRYPTOCELL during boot by an immutable bootloader, or a session key provided runtime by the application .

Register HOST_CRYPTOKEY_SEL on page 299 selects one of the following keys for the AES cryptographic operations:

- RTL key K_{PRTL}
- Device root key K_{DR}
- Session key

6.6.5.1.1 RTL key

CRYPTOCELL contains one hard-coded RTL key referred to as K_{PRTL}. This key is set to the same value for all devices with the same part code and cannot be changed.

CRYPTOCELL can perform cryptographic operations using the K_{PRTL} key without a bootloader or application having access to the key value itself. Usage of K_{PRTL} can be disabled until next reset by writing to register HOST_IOT_KPRTL_LOCK on page 299. If a locked K_{PRTL} key is requested, a zero vector key will be used by the AES engine instead.

6.6.5.1.2 Device root key

The device root key, K_{DR}, is a 128 bits AES key typically programmed by an immutable bootloader as part of the CRYPTOCELL initialization process during device boot sequence. It is kept in the CRYPTOCELL internal storage until the next reset.

To configure the K_{DR} key, write the key value into registers HOST_IOT_KDR0 through HOST_IOT_KDR3. These registers are write-only when LCS is set to `DebugEnable` mode, and write-once when LCS is set to `Secure` mode. The K_{DR} key value is kept when the read-back value of register HOST_IOT_KDR0 is `Retained`. Once configured, CRYPTOCELL can perform cryptographic operations using the K_{DR} key without an updatable bootloader or application having access to the key value itself.

The K_{DR} key should be protected by the immutable bootloader using ACL — Access control lists on page 182 .

6.6.5.2 Session keys

Session keys are supported by the AES and CHACHA engine.

Before starting a cryptographic operation using a session key, the desired key value must be written in clear-text by the CPU into the write-only key registers of the corresponding engine. One session key can be overwritten by another as long as the write order of the write-only key registers are respected. Please refer to the corresponding chapter of each cryptographic engine for more information about write order.

The AES engine on page 224 supports 128 bits session keys, and CHACHA engine on page 234 supports 128/256 bits session keys.

The last written session key for each engine is retained until CRYPTOCELL is disabled, the engine is reset, or the device is reset.

6.6.5.3 Asymmetric keys

Asymmetric cryptographic keys are supported by the PKA engine.

Before starting a cryptographic operation using an asymmetric key, the desired key value must be written into the PKA SRAM together with the payload.

See PKA engine on page 253 for more information.

6.6.6 Internal memories

CRYPTOCELL contains two dedicated memory blocks; one 4 kB SRAM block for the PKA engine calculations, and one 2 kB SRAM block for the RNG engine entropy collector.

See PKA SRAM on page 257 and RNG SRAM on page 273 for more information about these dedicated memory blocks.

6.6.7 Direct memory access (DMA)

CRYPTOCELL support direct memory access (DMA) to allow cryptographic operations on memory mapped regions without involving the CPU.

The following table indicates which memory is accessible by CRYPTOCELL DMA engines.

Memory type	Read	Write
SRAM	Yes	Yes
Flash	No	No
External flash (QSPI)	No	No

Table 15: DMA transaction types

Data stored in a memory type not accessible by CRYPTOCELL DMA engines must be copied to an accessible memory type before it can be processed by the CRYPTOCELL subsystem. Maximum DMA transaction size is limited to 2^{16}-1 bytes.

The CRYPTOCELL DMA engine can also run in Bypass mode, meaning data is read and written without being piped through a cryptographic engine. Thus CRYPTOCELL can act as a general purpose DMA engine for moving data.

Operating the DMA engines in Bypass mode involve the following steps:

1. Enable DMA engines clock using register DMA_CLK on page 303.
2. Configure cryptographic control for `Bypass` mode using register CRYPTO_CTL on page 286.
3. Set the the output destination address and size of the receiving buffer.
4. Start the DMA transaction by configuring the input source address and the number of bytes to transfer.
5. Status of the DMA transaction can be monitored by either polling register DOUT_DMA_MEM_BUSY on page 293, or by unmasking the interrupt for field `DOUT_TO_MEM_MASK` in register IMR on page 296.

See DIN DMA engine on page 288 and DOUT DMA engine on page 292 for more information.

6.6.8 Power and clock

Power and clock management of the CRYPTOCELL subsystem is handled automatically in hardware, as long as the neccessary conditions are fulfilled by software.

Clock gating

CRYPTOCELL implements separate clock domains for each cryptographic engine. Internal clock gating control is handled through the MISC interface on page 302, as well as register RNG_CLK on page 284. The registers of a cryptographic engine are only accessible when its clock is enabled.

Power gating

CRYPTOCELL must be disabled to ensure lowest possible power consumption when the subsystem is not needed.

The CRYPTOCELL subsystem power is controlled through register ENABLE on page 224. Even though external clock input is gated away automatically by hardware, the CRYPTOCELL subsystem power will still be enabled. To initiate a full power-down sequence software must perform the following steps:

1. Make sure there are no pending tasks
2. Clear all pending interrupts in register RNG_ICR on page 278 and register ICR on page 297.
3. Disable CRYPTOCELL subsystem using register ENABLE on page 224.

6.6.9 Interrupt handling

CRYPTOCELL triggers interrupt once processing is complete.

See register IRR on page 296 for more information on which CRYPTOCELL subsystem components are able to trigger an interrupt request.

To clear the IRQ line when an interrupt has occurred, the relevant interrupt bit in register ICR on page 297 must be cleared. Interrupt sources can be masked using register IMR on page 296. If an interrupt source is masked, no interrupt request will be triggered.

In addition if field `RNG_INT` in register IRR on page 296 is asserted, the relevant RNG engine interrupt bit in register RNG_ICR on page 278 must be cleared *before* clearing that interrupt bit in register ICR on page 297 as described above.

The figure below shows how the CRYPTOCELL subsystem interrupt handling is designed and how it is connected to the NVIC module in the CPU.

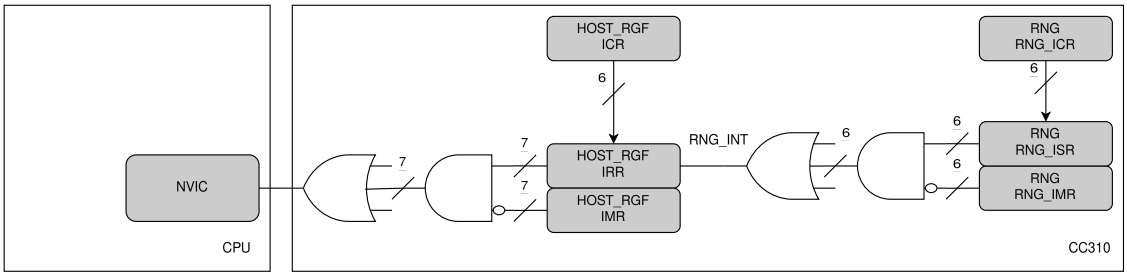


Figure 45: CRYPTOCELL interrupt handling

6.6.10 Standards

Arm TrustZone CryptoCell 310 (CRYPTOCELL) is compliant with the protocol specifications and standards shown in the following table.

Algorithm family	Identification code	Document title
TRNG	NIST SP 800-90B	*Recommendation for the Entropy Sources Used for Random Bit Generation*
	BSI AIS-31	*Functionality Classes and Evaluation Methodology for True Random Number Generators*
	FIPS 140-2	*Security Requirements for Cryptographic Modules*
PRNG	NIST SP 800-90A	*Recommendation for Random Number Generation Using Deterministic Random Bit Generators*
Stream cipher	Chacha	*ChaCha, a variant of Salsa20*, Daniel J. Bernstein, January 28th 2008
MAC	Poly1305	*The Poly1305-AES message-authentication code*, Daniel J. Bernstein *Cryptography in NaCl*, Daniel J. Bernstein
Key agreement	SRP	*The Secure Remote Password Protocol*, Thomas Wu, November 11th 1997
Key derivation	NIST SP 800-108	*Recommendation for Key Derivation Using Pseudorandom Functions.*
AES	FIPS-197	*Advanced Encryption Standard (AES)*. Compliant with 128 bits key size only
	NIST SP 800-38A	*Recommendation for Block Cipher Modes of Operation - Methods and Techniques*
	NIST SP 800-38B	*Recommendation for Block Cipher Modes of Operation: The CMAC Mode for Authentication*
	NIST SP 800-38C	*Recommendation for Block Cipher Modes of Operation: The CCM Mode for Authentication and Confidentiality*
	ISO/IEC 9797-1	AES CBC-MAC per ISO/IEC 9797-1 MAC algorithm 1
	IEEE 802.15.4-2011	*IEEE Standard for Local and metropolitan area networks - Part 15.4: Low-Rate Wireless Personal Area Networks (LR-WPANs)*, Annex B.4: *Specification of generic CCM* mode of operation*
Hash	FIPS 180-4	Secure Hash Standard (SHA1, SHA-224, SHA-256)
	RFC2104	*HMAC: Keyed-Hashing for Message Authentication*
RSA	PKCS#1	*Public-Key Cryptography Standards (PKCS) #1: RSA Cryptography Specifications* v1.5/2.1. RSA signature verification supported up to key sizes of 2048 bits. RSA key generation supported up to key sizes of 2048 bits.
Diffie-Hellman	ANSI X9.42	*Public Key Cryptography for the Financial Services Industry: Agreement of Symmetric Keys Using Discrete Logarithm Cryptography*
	PKCS#3	*Diffie-Hellman Key-Agreement Standard*
ECC	ANSI X9.63	*Public Key Cryptography for the Financial Services Industry - Key Agreement and Key Transport Using Elliptic Curve Cryptography*
	IEEE 1363	*Standard Specifications for Public-Key Cryptography*
	ANSI X9.62	*Public Key Cryptography For The Financial Services Industry: The Elliptic Curve Digital Signature Algorithm (ECDSA)*
	Ed25519	Edwards-curve, *Ed25519: high-speed high-security signatures*, Daniel J. Bernstein, Niels Duif, Tanja Lange, Peter Schwabe, and Bo-Yin Yang
	Curve25519	Montgomery curve, *Curve25519: new Diffie-Hellman speed records*, Daniel J. Bernstein
	FIPS 186-4	*Digital Signature Standard (DSS)*
	SEC 2	*Recommended Elliptic Curve Domain Parameters*, Certicom Research
	NIST SP 800-56A rev. 2	*Recommendation for Pair-Wise Key Establishment Schemes Using Discrete Logarithm Cryptography*

Table 16: CRYPTOCELL cryptography standards

6.6.11 Registers

Instances

Instance	Base address	Description
CRYPTOCELL	0x5002A000	CRYPTOCELL 310 security subsystem

Register overview

Register	Offset	Description
ENABLE	0x500	Enable CRYPTOCELL subsystem.

6.6.11.1 ENABLE

Address offset: 0x500

Enable CRYPTOCELL subsystem.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field		Value ID			Value					Description																							
A	RW	ENABLE										Enable or disable the CRYPTOCELL subsystem.																							
				Disabled			0					CRYPTOCELL subsystem disabled.																							
				Enabled			1					CRYPTOCELL subsystem enabled.																							

6.6.12 Accelerators

This chapter contains register interfaces for each of the hardware accelerator engines.

6.6.12.1 AES engine

The Advanced Encryption Standard (AES) hardware engine is designed according to FIPS197 for use in encrypt/decrypt and sign/verify operations for 128 bits key sizes.

The following cipher modes are supported:

- ECB
- CBC
- CBC-MAC
- CMAC
- CTR
- CCM
- CCM*

Note:

To ensure proper operation when writing 128 bits AES keys, the write-only key registers of the AES engine must be written in ascending order, starting with:

- AES_KEY_0[0]
- AES_KEY_0[1]
- AES_KEY_0[2]
- AES_KEY_0[3]

6.6.12.1.1 Cryptographic flow

The following section describe a simple cryptographic flow for this engine.

```
uint8_t buf_dst[16] = { 0 };
uint8_t buf_src[16] = { 0x81, 0x02, 0xF2, 0x40, 0xD5, 0xB9, 0x44, 0x59,
                        0xA2, 0xEB, 0x6F, 0xF2, 0x49, 0xF5, 0xEB, 0x94 };

/* Enable CRYPTOCELL subsystem */
NRF_CRYPTOCELL->ENABLE =  CRYPTOCELL_ENABLE_ENABLE_Enabled;

/* Enable engine and DMA clock */
NRF_CC_MISC->AES_CLK = CC_MISC_AES_CLK_ENABLE_Enable;
NRF_CC_MISC->DMA_CLK = CC_MISC_DMA_CLK_ENABLE_Enable;

/* Wait until crypto engine is Idle  */
while (NRF_CC_CTL->CRYPTO_BUSY == CC_CTL_CRYPTO_BUSY_STATUS_Busy) { }

/* Configure AES as cryptographic flow */
NRF_CC_CTL->CRYPTO_CTL = CC_CTL_CRYPTO_CTL_MODE_AESActive;

/* Configure AES engine control for decryption using ECB mode (default) */
NRF_CC_AES->AES_CONTROL = CC_AES_AES_CONTROL_DEC_KEY0_Decrypt;

/* Load the AES key value into the engine */
NRF_CC_AES->AES_KEY_0[0] = 0x51515151;
NRF_CC_AES->AES_KEY_0[1] = 0x52525252;
NRF_CC_AES->AES_KEY_0[2] = 0x53535353;
NRF_CC_AES->AES_KEY_0[3] = 0x54545454;

/* Configure default init vector */
NRF_CC_AES->AES_IV_0[0] = 0x0;
NRF_CC_AES->AES_IV_0[1] = 0x0;
NRF_CC_AES->AES_IV_0[2] = 0x0;
NRF_CC_AES->AES_IV_0[3] = 0x0;

/* Configure DMA output destination address */
NRF_CC_DOUT->DST_MEM_ADDR = (uint32_t) buf_dst;
NRF_CC_DOUT->DST_MEM_SIZE = (uint32_t) sizeof(buf_dst);

/* Configure DMA input source address to start the cryptographic operation */
NRF_CC_DIN->SRC_MEM_ADDR = (uint32_t) buf_src;
NRF_CC_DIN->SRC_MEM_SIZE = (uint32_t) sizeof(buf_src);

/* Wait on DOUT DMA interrupt */
while(!(NRF_CC_HOST_RGF->IRR & CC_HOST_RGF_IRR_DOUT_TO_MEM_INT_Msk)) {}
```

6.6.12.1.2 Registers

Instances

Instance	Base address	Description
CC_AES	0x5002B000	CRYPTOCELL AES engine

Register overview

Register	Offset	Description
AES_KEY_0[0]	0x400	AES key value to use. The initial AES_KEY_0[0] register holds the least significant bits [31:0] of the key value.
AES_KEY_0[1]	0x404	AES key value to use. The initial AES_KEY_0[0] register holds the least significant bits [31:0] of the key value.
AES_KEY_0[2]	0x408	AES key value to use. The initial AES_KEY_0[0] register holds the least significant bits [31:0] of the key value.
AES_KEY_0[3]	0x40C	AES key value to use. The initial AES_KEY_0[0] register holds the least significant bits [31:0] of the key value.
AES_KEY_0[4]	0x410	AES key value to use. The initial AES_KEY_0[0] register holds the least significant bits [31:0] of the key value.
AES_KEY_0[5]	0x414	AES key value to use. The initial AES_KEY_0[0] register holds the least significant bits [31:0] of the key value.
AES_KEY_0[6]	0x418	AES key value to use. The initial AES_KEY_0[0] register holds the least significant bits [31:0] of the key value.
AES_KEY_0[7]	0x41C	AES key value to use. The initial AES_KEY_0[0] register holds the least significant bits [31:0] of the key value.
AES_IV_0[0]	0x440	AES Initialization Vector (IV) to use. The initial AES_IV_0[0] register holds the least significant bits [31:0] of the IV.
AES_IV_0[1]	0x444	AES Initialization Vector (IV) to use. The initial AES_IV_0[0] register holds the least significant bits [31:0] of the IV.
AES_IV_0[2]	0x448	AES Initialization Vector (IV) to use. The initial AES_IV_0[0] register holds the least significant bits [31:0] of the IV.
AES_IV_0[3]	0x44C	AES Initialization Vector (IV) to use. The initial AES_IV_0[0] register holds the least significant bits [31:0] of the IV.
AES_CTR[0]	0x460	AES counter (CTR) to use. The initial AES_CTR[0] register holds the least significant bits [31:0] of the CTR.
AES_CTR[1]	0x464	AES counter (CTR) to use. The initial AES_CTR[0] register holds the least significant bits [31:0] of the CTR.
AES_CTR[2]	0x468	AES counter (CTR) to use. The initial AES_CTR[0] register holds the least significant bits [31:0] of the CTR.
AES_CTR[3]	0x46C	AES counter (CTR) to use. The initial AES_CTR[0] register holds the least significant bits [31:0] of the CTR.
AES_BUSY	0x470	Status register for AES engine activity.
AES_SK	0x478	Writing to this address trigger sampling of the HW key to the AES_KEY_0 register
AES_CMAC_INIT	0x47C	Writing to this address triggers the AES engine to generate K1 and K2 for AES-CMAC operations.
AES_REMAINING_BYTES	0x4BC	This register should be set with the amount of remaining bytes until the end of the current AES operation.
AES_CONTROL	0x4C0	Control the AES engine behavior.
AES_HW_FLAGS	0x4C8	Hardware configuration of the AES engine. Reset value holds the supported features.
AES_CTR_NO_INCREMENT	0x4D8	This register enables the AES CTR no increment mode in which the counter mode is not incremented between two blocks
AES_SW_RESET	0x4F4	Reset the AES engine.

Register	Offset	Description
AES_CMAC_SIZE0_KICK	0x524	Writing to this address triggers the AES engine to perform a CMAC operation with size 0. The CMAC result can be read from the AES_IV_0 register.

6.6.12.1.2.1 AES_KEY_0[0]

Address offset: 0x400

AES key value to use. The initial AES_KEY_0[0] register holds the least significant bits [31:0] of the key value.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	VALUE			AES key value.

6.6.12.1.2.2 AES_KEY_0[1]

Address offset: 0x404

AES key value to use. The initial AES_KEY_0[0] register holds the least significant bits [31:0] of the key value.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	VALUE			AES key value.

6.6.12.1.2.3 AES_KEY_0[2]

Address offset: 0x408

AES key value to use. The initial AES_KEY_0[0] register holds the least significant bits [31:0] of the key value.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	VALUE			AES key value.

6.6.12.1.2.4 AES_KEY_0[3]

Address offset: 0x40C

AES key value to use. The initial AES_KEY_0[0] register holds the least significant bits [31:0] of the key value.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	VALUE			AES key value.

6.6.12.1.2.5 AES_KEY_0[4]

Address offset: 0x410

AES key value to use. The initial AES_KEY_0[0] register holds the least significant bits [31:0] of the key value.

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value	Description																															
A	W	VALUE			AES key value.																															

6.6.12.1.2.6 AES_KEY_0[5]

Address offset: 0x414

AES key value to use. The initial AES_KEY_0[0] register holds the least significant bits [31:0] of the key value.

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value	Description																															
A	W	VALUE			AES key value.																															

6.6.12.1.2.7 AES_KEY_0[6]

Address offset: 0x418

AES key value to use. The initial AES_KEY_0[0] register holds the least significant bits [31:0] of the key value.

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value	Description																															
A	W	VALUE			AES key value.																															

6.6.12.1.2.8 AES_KEY_0[7]

Address offset: 0x41C

AES key value to use. The initial AES_KEY_0[0] register holds the least significant bits [31:0] of the key value.

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value	Description																															
A	W	VALUE			AES key value.																															

6.6.12.1.2.9 AES_IV_0[0]

Address offset: 0x440

AES Initialization Vector (IV) to use. The initial AES_IV_0[0] register holds the least significant bits [31:0] of the IV.

AES_IV_0 must be configured according to the selected AES mode:

- AES CBC/CBC-MAC : Loaded with the IV.

This register is a 'R/W change' register, as the written register values changes during processing.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	VALUE																																	
				AES non-tunneling or first tunnel stage IV value.																																	

6.6.12.1.2.10 AES_IV_0[1]

Address offset: 0x444

AES Initialization Vector (IV) to use. The initial AES_IV_0[0] register holds the least significant bits [31:0] of the IV.

AES_IV_0 must be configured according to the selected AES mode:

- AES CBC/CBC-MAC : Loaded with the IV.

This register is a 'R/W change' register, as the written register values changes during processing.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	VALUE																																	
				AES non-tunneling or first tunnel stage IV value.																																	

6.6.12.1.2.11 AES_IV_0[2]

Address offset: 0x448

AES Initialization Vector (IV) to use. The initial AES_IV_0[0] register holds the least significant bits [31:0] of the IV.

AES_IV_0 must be configured according to the selected AES mode:

- AES CBC/CBC-MAC : Loaded with the IV.

This register is a 'R/W change' register, as the written register values changes during processing.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	VALUE																																	
				AES non-tunneling or first tunnel stage IV value.																																	

6.6.12.1.2.12 AES_IV_0[3]

Address offset: 0x44C

AES Initialization Vector (IV) to use. The initial AES_IV_0[0] register holds the least significant bits [31:0] of the IV.

AES_IV_0 must be configured according to the selected AES mode:

- AES CBC/CBC-MAC : Loaded with the IV.

This register is a 'R/W change' register, as the written register values changes during processing.

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value	Description																															
A	RW	VALUE			AES non-tunneling or first tunnel stage IV value.																																	

6.6.12.1.2.13 AES_CTR[0]

Address offset: 0x460

AES counter (CTR) to use. The initial AES_CTR[0] register holds the least significant bits [31:0] of the CTR.

AES_CTR must be configured according to the selected AES mode:

- AES CTR : Loaded with the counter value.

This register is a 'R/W change' register, as the written register values changes during processing.

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value	Description																															
A	RW	VALUE			AES CTR value.																																	

6.6.12.1.2.14 AES_CTR[1]

Address offset: 0x464

AES counter (CTR) to use. The initial AES_CTR[0] register holds the least significant bits [31:0] of the CTR.

AES_CTR must be configured according to the selected AES mode:

- AES CTR : Loaded with the counter value.

This register is a 'R/W change' register, as the written register values changes during processing.

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value	Description																															
A	RW	VALUE			AES CTR value.																																	

6.6.12.1.2.15 AES_CTR[2]

Address offset: 0x468

AES counter (CTR) to use. The initial AES_CTR[0] register holds the least significant bits [31:0] of the CTR.

AES_CTR must be configured according to the selected AES mode:

- AES CTR : Loaded with the counter value.

This register is a 'R/W change' register, as the written register values changes during processing.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	VALUE			AES CTR value.

6.6.12.1.2.16 AES_CTR[3]

Address offset: 0x46C

AES counter (CTR) to use. The initial AES_CTR[0] register holds the least significant bits [31:0] of the CTR.

AES_CTR must be configured according to the selected AES mode:

- AES CTR : Loaded with the counter value.

This register is a 'R/W change' register, as the written register values changes during processing.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	VALUE			AES CTR value.

6.6.12.1.2.17 AES_BUSY

Address offset: 0x470

Status register for AES engine activity.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	STATUS			AES engine status.
			Idle	0	AES engine is idle
			Busy	1	AES engine is busy

6.6.12.1.2.18 AES_SK

Address offset: 0x478

Writing to this address trigger sampling of the HW key to the AES_KEY_0 register

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	AES_SK			Sample HW key to AES_KEY_0 registers.

6.6.12.1.2.19 AES_CMAC_INIT

Address offset: 0x47C

Writing to this address triggers the AES engine to generate K1 and K2 for AES-CMAC operations.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	ENABLE			Generate K1 and K2 for the AES-CMAC operations.
			Enable	1	Initialize AES-CMAC operations.

6.6.12.1.2.20 AES_REMAINING_BYTES

Address offset: 0x4BC

This register should be set with the amount of remaining bytes until the end of the current AES operation.

The AES engine counts down from this value to determine the last block or the block before the last blocks in mode AES CMAC and mode AES CCM.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	VALUE			Remaining bytes util the end of the current AES operation.

6.6.12.1.2.21 AES_CONTROL

Address offset: 0x4C0

Control the AES engine behavior.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D C C B B B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DEC_KEY0			Set AES encrypt or decrypt mode in non-tunneling operations.
			Encrypt	0	Perform AES encryption
			Decrypt	1	Perform AES decryption
B	RW	MODE_KEY0			Set the AES mode.
			ECB	0x0	Electronic codebook mode
			CBC	0x1	Cipher block chaining mode
			CTR	0x2	Counter mode
			CBC_MAC	0x3	Cipher Block Chaining Message Authentication Code
			CMAC	0x7	Cipher-based Message Authentication Code
C	RW	NK_KEY0			Set the AES key length.
			128Bits	0x0	128 bits key length
D	RW	AES_XOR_CRYPTOKEY			This field determines the value that is written to AES_KEY_0, when AES_SK is kicked.
			Disable	0	The value that is written to AES_KEY_0 is the value of the HW cryptokey as is.
			Enable	1	The value that is written to AES_KEY_0 is the value of the HW cryptokey XOR with the current value of AES_KEY_0.
E	RW	DIRECT_ACCESS			Using direct access and not the DIN-DOUT DMA interface
			Disable	0	Access using the DIN-DOUT DMA interface
			Enable	1	Access using direct access

6.6.12.1.2.22 AES_HW_FLAGS

Address offset: 0x4C8

Hardware configuration of the AES engine. Reset value holds the supported features.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				K J I H G F E D C B A
Reset 0x00000108				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 1 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	SUPPORT_256_192_KEY			If this flag is set, the engine support 192 bits and 256 bits key size.
B	R	AES_LARGE_RKEK			If this flag is set, the engine support AES_LARGE_RKEK.
C	R	DPA_CNTRMSR_EXIST			If this flag is set, the engine support DPA countermeasures.
D	R	CTR_EXIST			If this flag is set, the engine support AES CTR mode.
E	R	ONLY_ENCRYPT			If this flag is set, the engine only support encrypt operations.
F	R	USE_SBOX_TABLE			If this flag is set, the engine uses SBOX tables.
G	R	USE_5_SBOXES			If this flag is set, the engine uses 5 SBOX where each AES round takes 4 cycles.
H	R	AES_SUPPORT_PREV_IV			If this flag is set, the engine contains the PREV_IV register for faster AES XCBC MAC calculation.
I	R	AES_TUNNEL_EXIST			If this flag is set, the engine support tunneling operations.
J	R	SECOND_REGS_SET_EXIST			If this flag is set, the engine support a second register set for tunneling operations.
K	R	DFA_CNTRMSR_EXIST			If this flag is set, the engine support DFA countermeasures.

6.6.12.1.2.23 AES_CTR_NO_INCREMENT

Address offset: 0x4D8

This register enables the AES CTR no increment mode in which the counter mode is not incremented between two blocks

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ENABLE			This field enables the AES CTR no increment mode in which the counter mode is not incremented between two blocks
			Disable	0	Counter always incremented between blocks
			Enable	1	Do not increment counter between blocks

6.6.12.1.2.24 AES_SW_RESET

Address offset: 0x4F4

Reset the AES engine.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	RESET			Writing any value to this address resets the AES engine. The reset takes 4 CPU clock cycles to complete.
			Enable	1	Reset AES engine.

6.6.12.1.2.25 AES_CMAC_SIZE0_KICK

Address offset: 0x524

Writing to this address triggers the AES engine to perform a CMAC operation with size 0. The CMAC result can be read from the AES_IV_0 register.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	ENABLE																																	
			Disable	0																															
			Enable	1																															

ID	R/W	Field	Value ID	Value	Description
A	W	ENABLE			Force AES CMAC operation with size 0.
			Disable	0	Normal AES CMAC operation
			Enable	1	Force CMAC operation with size 0

6.6.12.2 CHACHA engine

The ChaCha algorithm is a family of stream ciphers.

The ChaCha family of stream ciphers can be used as both a stand-alone algorithm, and in combination with the Poly1305 authenticator to form an Authenticated Encryption with Associated Data (AEAD) algorithm as defined in RFC7539 for IETF protocols.

The CHACHA engine provide acceleration for the stream encryption, while the PKA engine is used for acceleration of the Poly1305 authenticator. The core of the ChaCha algorithm is a hash function which is based on rotation operations. In the default configuration the hash function consist of 20 rounds of rotation permutations. The implementation support ChaCha stream ciphers using key sizes up to 256 bits in 8, 12 and 20 rounds. The ChaCha20/Poly1305 combination is perfectly suited for embedded environments, and can achieve much higher throughput than AES using similar power consumption and execution time.

Note: To ensure proper operation when writing 128 bits CHACHA keys, the write-only key registers of the CHACHA engine must be written in ascending order, starting with:

- CHACHA_KEY[0]
- CHACHA_KEY[1]
- CHACHA_KEY[2]
- CHACHA_KEY[3]

For 256 bits CHACHA keys, this must be followed by:

- CHACHA_KEY[4]
- CHACHA_KEY[5]
- CHACHA_KEY[6]
- CHACHA_KEY[7]

6.6.12.2.1 Cryptographic flow

The following section describe a simple cryptographic flow for this engine.

```
uint8_t buf_dst[16] = { 0 };
uint8_t buf_src[16] = { 0x18, 0x35, 0x9B, 0x75, 0x18, 0x6F, 0x33, 0xBE,
                        0x22, 0x0A, 0x3D, 0xB7, 0x66, 0xFD, 0x98, 0x35 };

/* Enable CRYPTOCELL subsystem */
NRF_CRYPTOCELL->ENABLE = CRYPTOCELL_ENABLE_ENABLE_Enabled;

/* Enable engine and DMA clock */
NRF_CC_MISC->CHACHA_CLK = CC_MISC_CHACHA_CLK_ENABLE_Enable;
NRF_CC_MISC->DMA_CLK = CC_MISC_DMA_CLK_ENABLE_Enable;

/* Wait until crypto engine is Idle  */
while (NRF_CC_CTL->CRYPTO_BUSY == CC_CTL_CRYPTO_BUSY_STATUS_Busy) { }

/* Configure CHACHA as cryptographic flow */
NRF_CC_CTL->CRYPTO_CTL = CC_CTL_CRYPTO_CTL_MODE_ChaChaActive;

/* Configure testing NONCE */
NRF_CC_CHACHA->CHACHA_IV[0] = 0xBBBBAAAA;
NRF_CC_CHACHA->CHACHA_IV[1] = 0x22221111;

/* Load the CHACHA test key value into the engine */
NRF_CC_CHACHA->CHACHA_KEY[0] = 0x51515151;
NRF_CC_CHACHA->CHACHA_KEY[1] = 0x52525252;
NRF_CC_CHACHA->CHACHA_KEY[2] = 0x53535353;
NRF_CC_CHACHA->CHACHA_KEY[3] = 0x54545454;
NRF_CC_CHACHA->CHACHA_KEY[4] = 0x51515151;
NRF_CC_CHACHA->CHACHA_KEY[5] = 0x52525252;
NRF_CC_CHACHA->CHACHA_KEY[6] = 0x53535353;
NRF_CC_CHACHA->CHACHA_KEY[7] = 0x54545454;

/* Configure CHACHA mode - using default (0x0), adding new message init */
NRF_CC_CHACHA->CHACHA_CONTROL =
  (CC_CHACHA_CHACHA_CONTROL_INIT_Enable <<
   CC_CHACHA_CHACHA_CONTROL_INIT_Pos);

/* Configure DMA output destination address */
NRF_CC_DOUT->DST_MEM_ADDR = (uint32_t) buf_dst;
NRF_CC_DOUT->DST_MEM_SIZE = (uint32_t) sizeof(buf_dst);

/* Configure DMA input source address to start the cryptographic operation */
NRF_CC_DIN->SRC_MEM_ADDR = (uint32_t) buf_src;
NRF_CC_DIN->SRC_MEM_SIZE = (uint32_t) sizeof(buf_src);

/* Wait on DOUT DMA interrupt */
while(!(NRF_CC_HOST_RGF->IRR & CC_HOST_RGF_IRR_DOUT_TO_MEM_INT_Msk)) {}
```

6.6.12.2.2 Registers

Instances

Instance	Base address	Description
CC_CHACHA	0x5002B000	CRYPTOCELL CHACHA engine

Register overview

Register	Offset	Description
CHACHA_CONTROL	0x380	Control the CHACHA engine behavior.
CHACHA_VERSION	0x384	CHACHA engine HW version
CHACHA_KEY[0]	0x388	CHACHA key value to use. The initial CHACHA_KEY[0] register holds the least significant bits [31:0] of the key value.
CHACHA_KEY[1]	0x38C	CHACHA key value to use. The initial CHACHA_KEY[0] register holds the least significant bits [31:0] of the key value.
CHACHA_KEY[2]	0x390	CHACHA key value to use. The initial CHACHA_KEY[0] register holds the least significant bits [31:0] of the key value.
CHACHA_KEY[3]	0x394	CHACHA key value to use. The initial CHACHA_KEY[0] register holds the least significant bits [31:0] of the key value.
CHACHA_KEY[4]	0x398	CHACHA key value to use. The initial CHACHA_KEY[0] register holds the least significant bits [31:0] of the key value.
CHACHA_KEY[5]	0x39C	CHACHA key value to use. The initial CHACHA_KEY[0] register holds the least significant bits [31:0] of the key value.
CHACHA_KEY[6]	0x3A0	CHACHA key value to use. The initial CHACHA_KEY[0] register holds the least significant bits [31:0] of the key value.
CHACHA_KEY[7]	0x3A4	CHACHA key value to use. The initial CHACHA_KEY[0] register holds the least significant bits [31:0] of the key value.
CHACHA_IV[0]	0x3A8	CHACHA Initialization Vector (IV) to use. The IV is also known as the nonce.
CHACHA_IV[1]	0x3AC	CHACHA Initialization Vector (IV) to use. The IV is also known as the nonce.
CHACHA_BUSY	0x3B0	Status register for CHACHA engine activity.
CHACHA_HW_FLAGS	0x3B4	Hardware configuration of the CHACHA engine. Reset value holds the supported features.
CHACHA_BLOCK_CNT_LSB	0x3B8	Store the LSB value of the block counter, in order to support suspend/resume of operation
CHACHA_BLOCK_CNT_MSB	0x3BC	Store the MSB value of the block counter, in order to support suspend/resume of operation
CHACHA_SW_RESET	0x3C0	Reset the CHACHA engine.
CHACHA_POLY1305_KEY[0]	0x3C4	The auto-generated key to use in Poly1305 MAC calculation. The initial CHACHA_POLY1305_KEY[0] register holds the least significant bits [31:0] of the key value.
CHACHA_POLY1305_KEY[1]	0x3C8	The auto-generated key to use in Poly1305 MAC calculation. The initial CHACHA_POLY1305_KEY[0] register holds the least significant bits [31:0] of the key value.
CHACHA_POLY1305_KEY[2]	0x3CC	The auto-generated key to use in Poly1305 MAC calculation. The initial CHACHA_POLY1305_KEY[0] register holds the least significant bits [31:0] of the key value.
CHACHA_POLY1305_KEY[3]	0x3D0	The auto-generated key to use in Poly1305 MAC calculation. The initial CHACHA_POLY1305_KEY[0] register holds the least significant bits [31:0] of the key value.
CHACHA_POLY1305_KEY[4]	0x3D4	The auto-generated key to use in Poly1305 MAC calculation. The initial CHACHA_POLY1305_KEY[0] register holds the least significant bits [31:0] of the key value.
CHACHA_POLY1305_KEY[5]	0x3D8	The auto-generated key to use in Poly1305 MAC calculation. The initial CHACHA_POLY1305_KEY[0] register holds the least significant bits [31:0] of the key value.
CHACHA_POLY1305_KEY[6]	0x3DC	The auto-generated key to use in Poly1305 MAC calculation. The initial CHACHA_POLY1305_KEY[0] register holds the least significant bits [31:0] of the key value.

Register	Offset	Description
CHACHA_POLY1305_KEY[7]	0x3E0	The auto-generated key to use in Poly1305 MAC calculation. The initial CHACHA_POLY1305_KEY[0] register holds the least significant bits [31:0] of the key value.
CHACHA_ENDIANNESS	0x3E4	CHACHA engine data order configuration.
CHACHA_DEBUG	0x3E8	Debug register for the CHACHA engine

6.6.12.2.2.1 CHACHA_CONTROL

Address offset: 0x380

Control the CHACHA engine behavior.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																									G	F				E	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	CHACHA_OR_SALSA																																	
			ChaCha	0																															
			Salsa	1																															
B	RW	INIT																																	
			Disable	0																															
			Enable	1																															
C	RW	GEN_KEY_POLY1305																																	
			Disable	0																															
			Enable	1																															
D	RW	KEY_LEN																																	
			256Bits	0																															
			128Bits	1																															
E	RW	NUM_OF_ROUNDS																																	
			Default	0																															
			12Rounds	1																															
			8Rounds	2																															
F	RW	RESET_BLOCK_CNT																																	
			Disable	0																															
			Enable	1																															
G	RW	USE_IV_96BIT																																	
			Disable	0																															
			Enable	1																															

6.6.12.2.2.2 CHACHA_VERSION

Address offset: 0x384

CHACHA engine HW version

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000001				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
ID	R/W	Field	Value ID	Value																															
A	R	CHACHA_VERSION																																	

6.6.12.2.2.3 CHACHA_KEY[0]

Address offset: 0x388

CHACHA key value to use. The initial CHACHA_KEY[0] register holds the least significant bits [31:0] of the key value.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	VALUE			CHACHA key value.

6.6.12.2.2.4 CHACHA_KEY[1]

Address offset: 0x38C

CHACHA key value to use. The initial CHACHA_KEY[0] register holds the least significant bits [31:0] of the key value.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	VALUE			CHACHA key value.

6.6.12.2.2.5 CHACHA_KEY[2]

Address offset: 0x390

CHACHA key value to use. The initial CHACHA_KEY[0] register holds the least significant bits [31:0] of the key value.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	VALUE			CHACHA key value.

6.6.12.2.2.6 CHACHA_KEY[3]

Address offset: 0x394

CHACHA key value to use. The initial CHACHA_KEY[0] register holds the least significant bits [31:0] of the key value.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	VALUE			CHACHA key value.

6.6.12.2.2.7 CHACHA_KEY[4]

Address offset: 0x398

CHACHA key value to use. The initial CHACHA_KEY[0] register holds the least significant bits [31:0] of the key value.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	VALUE			CHACHA key value.

6.6.12.2.2.8 CHACHA_KEY[5]

Address offset: 0x39C

CHACHA key value to use. The initial CHACHA_KEY[0] register holds the least significant bits [31:0] of the key value.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	VALUE			CHACHA key value.

6.6.12.2.2.9 CHACHA_KEY[6]

Address offset: 0x3A0

CHACHA key value to use. The initial CHACHA_KEY[0] register holds the least significant bits [31:0] of the key value.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	VALUE			CHACHA key value.

6.6.12.2.2.10 CHACHA_KEY[7]

Address offset: 0x3A4

CHACHA key value to use. The initial CHACHA_KEY[0] register holds the least significant bits [31:0] of the key value.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	VALUE			CHACHA key value.

6.6.12.2.2.11 CHACHA_IV[0]

Address offset: 0x3A8

CHACHA Initialization Vector (IV) to use. The IV is also known as the nonce.

The size of the nonce is controlled from register CHACHA_CONTROL on page 237.

For 64 bits IV size the nonce value must be encoded using:

- CHACHA_IV[0] : Bits [31:0] of the nonce
- CHACHA_IV[1] : Bits [63:32] of the nonce

For 96 bits IV size the nonce value must be encoded using:

- CHACHA_BLOCK_CNT_MSB on page 241 : Bits [31:0] of the nonce
- CHACHA_IV[0] : Bits [63:32] of the nonce
- CHACHA_IV[1] : Bits [95:64] of the nonce

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	VALUE																																	

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
A	R	CHACHA_EXISTS			If this flag is set, the engine include ChaCha support
B	R	SALSA_EXISTS			If this flag is set, the engine include Salsa support
C	R	FAST_CHACHA			If this flag is set, the next matrix calculated when the current one is written to data output path.

6.6.12.2.2.15 CHACHA_BLOCK_CNT_LSB

Address offset: 0x3B8

Store the LSB value of the block counter, in order to support suspend/resume of operation

The two first words (n) in the last row of the cipher matrix are the block counter. At the end of each block (512b), the block counter for the next block is written by HW to register CHACHA_BLOCK_CNT_LSB on page 241 and register CHACHA_BLOCK_CNT_MSB on page 241. If starting a new message the block counter must also be reset.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	VALUE			This register holds the ChaCha block counter bits [31:0] and must be read and written during respectively suspend and resume operations.

6.6.12.2.2.16 CHACHA_BLOCK_CNT_MSB

Address offset: 0x3BC

Store the MSB value of the block counter, in order to support suspend/resume of operation

For the description of register CHACHA_BLOCK_CNT_MSB on page 241, see register CHACHA_BLOCK_CNT_LSB on page 241.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	VALUE			This register holds the ChaCha block counter bits [63:32] and must be read and written during respectively suspend and resume operations.

6.6.12.2.2.17 CHACHA_SW_RESET

Address offset: 0x3C0

Reset the CHACHA engine.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	RESET			Writing any value to this address resets the CHACHA engine. The reset takes 4 CPU clock cycles to complete.
			Enable	1	Reset CHACHA engine.

6.6.12.2.2.18 CHACHA_POLY1305_KEY[0]

Address offset: 0x3C4

The auto-generated key to use in Poly1305 MAC calculation.

The initial CHACHA_POLY1305_KEY[0] register holds the least significant bits [31:0] of the key value.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	VALUE			Poly1305 key value.

6.6.12.2.2.19 CHACHA_POLY1305_KEY[1]

Address offset: 0x3C8

The auto-generated key to use in Poly1305 MAC calculation.

The initial CHACHA_POLY1305_KEY[0] register holds the least significant bits [31:0] of the key value.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	VALUE			Poly1305 key value.

6.6.12.2.2.20 CHACHA_POLY1305_KEY[2]

Address offset: 0x3CC

The auto-generated key to use in Poly1305 MAC calculation.

The initial CHACHA_POLY1305_KEY[0] register holds the least significant bits [31:0] of the key value.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	VALUE			Poly1305 key value.

6.6.12.2.2.21 CHACHA_POLY1305_KEY[3]

Address offset: 0x3D0

The auto-generated key to use in Poly1305 MAC calculation.

The initial CHACHA_POLY1305_KEY[0] register holds the least significant bits [31:0] of the key value.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	VALUE			Poly1305 key value.

6.6.12.2.2.22 CHACHA_POLY1305_KEY[4]

Address offset: 0x3D4

The auto-generated key to use in Poly1305 MAC calculation.

The initial CHACHA_POLY1305_KEY[0] register holds the least significant bits [31:0] of the key value.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	VALUE			Poly1305 key value.

6.6.12.2.2.23 CHACHA_POLY1305_KEY[5]

Address offset: 0x3D8

The auto-generated key to use in Poly1305 MAC calculation.

The initial CHACHA_POLY1305_KEY[0] register holds the least significant bits [31:0] of the key value.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	VALUE			Poly1305 key value.

6.6.12.2.2.24 CHACHA_POLY1305_KEY[6]

Address offset: 0x3DC

The auto-generated key to use in Poly1305 MAC calculation.

The initial CHACHA_POLY1305_KEY[0] register holds the least significant bits [31:0] of the key value.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	VALUE			Poly1305 key value.

6.6.12.2.2.25 CHACHA_POLY1305_KEY[7]

Address offset: 0x3E0

The auto-generated key to use in Poly1305 MAC calculation.

The initial CHACHA_POLY1305_KEY[0] register holds the least significant bits [31:0] of the key value.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	VALUE			Poly1305 key value.

6.6.12.2.2.26 CHACHA_ENDIANNESS

Address offset: 0x3E4

CHACHA engine data order configuration.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	CHACHA_DIN_WORD_ORDER			Change the word order of the input data.
			Default	0	Use default word order for 128-bits input, where words are ordered as follows: w0, w1, w2, w3.
			Reverse	1	Reverses the word order for 128-bits input, where words are re-ordered as follows: w3, w2, w1, w0.
B	RW	CHACHA_DIN_BYTE_ORDER			Change the byte order of the input data.
			Default	0	Use default byte order within each input word, where bytes are ordered as follows: B0, B1, B2, B3.
			Reverse	1	Reverse the byte order within each input word, where bytes are re-ordered as follows: B3, B2, B1, B0.
C	RW	CHACHA_CORE_MATRIX_LBE_ORDER			Change the quarter of a matrix order in the engine.
			Default	0	Use default quarter of matrix order, where quarters are ordered as follows: q0, q1, q2, q3. Each quarter represents a 128-bits section of the matrix.
			Reverse	1	Reverse the order of matrix quarters, where quarters are re-ordered as follows: q3, q2, q1, q0. Each quarter represents a 128-bits section of the matrix.
D	RW	CHACHA_DOUT_WORD_ORDER			Change the word order of the output data.
			Default	0	Uses default word order for 128-bits output, where words are ordered as follows: w0, w1, w2, w3.
			Reverse	1	Reverse the word order for 128-bits output, where words are re-ordered as follows: w3, w2, w1, w0.
E	RW	CHACHA_DOUT_BYTE_ORDER			Change the byte order of the output data.
			Default	0	Use default byte order within each output word, where bytes are ordered as follows: B0, B1, B2, B3.
			Reverse	1	Reverse the byte order within each output word, where bytes are re-ordered as follows: B3, B2, B1, B0.

6.6.12.2.2.27 CHACHA_DEBUG

Address offset: 0x3E8

Debug register for the CHACHA engine

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	FSM_STATE			Reflects the debug state of the CHACHA FSM.
			IDLE_STATE	0	CHACHA FSM is in idle state
			INIT_STATE	1	CHACHA FSM is in init state
			ROUNDS_STATE	2	CHACHA FSM is in rounds state
			FINAL_STATE	3	CHACHA FSM is in final state

6.6.12.3 HASH engine

The HASH engine is designed according to FIPS 180-4, and support both the SHA1 and SHA2 family of digest algorithms up to 256 bits.

The following SHA modes are supported:

- SHA-1
- SHA-224

- SHA-256

Note:

To ensure proper operation, the FIPS 180-4 defined initial hash values written to the registers of the HASH engine must be written in descending order, starting with:

- HASH_H[7] for SHA-256, and SHA-224.
- HASH_H[6] for SHA-256, and SHA-224.
- HASH_H[5] for SHA-256, and SHA-224.
- HASH_H[4] for SHA-256, SHA-224, and SHA-1.
- HASH_H[3] for SHA-256, SHA-224, and SHA-1.
- HASH_H[2] for SHA-256, SHA-224, and SHA-1.
- HASH_H[1] for SHA-256, SHA-224, and SHA-1.
- HASH_H[0] for SHA-256, SHA-224, and SHA-1.

6.6.12.3.1 Cryptographic flow

The following section describe a simple cryptographic flow for this engine.

```
uint8_t buf_src[32] =  {
      0xFA,0xFA,0xFA,0xFA,0xFA,0xFA,0xFA,0xFA,
      0xFA,0xFA,0xFA,0xFA,0xFA,0xFA,0xFA,0xFA,
      0xFA,0xFA,0xFA,0xFA,0xFA,0xFA,0xFA,0xFA,
      0xFA,0xFA,0xFA,0xFA,0xFA,0xFA,0xFA,0xFA };

/* Enable CRYPTOCELL subsystem */
NRF_CRYPTOCELL->ENABLE = CRYPTOCELL_ENABLE_ENABLE_Enabled;

/* Enable engine and DMA clock */
NRF_CC_MISC->HASH_CLK = CC_MISC_HASH_CLK_ENABLE_Enable;
NRF_CC_MISC->DMA_CLK = CC_MISC_DMA_CLK_ENABLE_Enable;

/* Wait until hash engine is Idle  */
while (NRF_CC_CTL->HASH_BUSY == CC_CTL_HASH_BUSY_STATUS_Busy) {}

/* Clear all interrupts */
NRF_CC_HOST_RGF->ICR = 0xFFFFFFFF;

/* Configure HASH as cryptographic flow */
NRF_CC_CTL->CRYPTO_CTL = CC_CTL_CRYPTO_CTL_MODE_HashActive;

/* Configure engine for SHA256 */
NRF_CC_HASH->HASH_CONTROL = CC_HASH_HASH_CONTROL_MODE_SHA256;

/* Configure initial SHA256 values */
NRF_CC_HASH->HASH_H[7] = 0x5BE0CD19;
NRF_CC_HASH->HASH_H[6] = 0x1F83D9AB;
NRF_CC_HASH->HASH_H[5] = 0x9B05688C;
NRF_CC_HASH->HASH_H[4] = 0x510E527F;
NRF_CC_HASH->HASH_H[3] = 0xA54FF53A;
NRF_CC_HASH->HASH_H[2] = 0x3C6EF372;
NRF_CC_HASH->HASH_H[1] = 0xBB67AE85;
NRF_CC_HASH->HASH_H[0] = 0x6A09E667;

/* Configure DMA input source address to start the cryptographic operation */
NRF_CC_DIN->SRC_MEM_ADDR = (uint32_t) buf_src;
NRF_CC_DIN->SRC_MEM_SIZE = (uint32_t) sizeof(buf_src);

/* Wait on DIN DMA interrupt indicating data has been fetched */
while(!(NRF_CC_HOST_RGF->IRR & CC_HOST_RGF_IRR_MEM_TO_DIN_INT_Msk)) {}

/* Wait until hash engine is Idle */
while (NRF_CC_CTL->HASH_BUSY == CC_CTL_HASH_BUSY_STATUS_Busy) {}

/* Calculated SHA256 digest now available in
NRF_CC_HASH->HASH_H[0] to NRF_CC_HASH->HASH_H[7]  */
```

6.6.12.3.2 Registers

Instances

Instance	Base address	Description
CC_HASH	0x5002B000	CRYPTOCELL HASH engine

Register overview

Register	Offset	Description
HASH_H[0]	0x640	HASH_H value registers. The initial HASH_H[0] register holds the least significant bits [31:0] of the value.
HASH_H[1]	0x644	HASH_H value registers. The initial HASH_H[0] register holds the least significant bits [31:0] of the value.
HASH_H[2]	0x648	HASH_H value registers. The initial HASH_H[0] register holds the least significant bits [31:0] of the value.
HASH_H[3]	0x64C	HASH_H value registers. The initial HASH_H[0] register holds the least significant bits [31:0] of the value.
HASH_H[4]	0x650	HASH_H value registers. The initial HASH_H[0] register holds the least significant bits [31:0] of the value.
HASH_H[5]	0x654	HASH_H value registers. The initial HASH_H[0] register holds the least significant bits [31:0] of the value.
HASH_H[6]	0x658	HASH_H value registers. The initial HASH_H[0] register holds the least significant bits [31:0] of the value.
HASH_H[7]	0x65C	HASH_H value registers. The initial HASH_H[0] register holds the least significant bits [31:0] of the value.
HASH_PAD_AUTO	0x684	Configure the HASH engine to automatically pad data at the end of the DMA transfer to complete the digest operation.
HASH_INIT_STATE	0x694	Configure HASH engine initial state registers.
HASH_VERSION	0x7B0	HASH engine HW version
HASH_CONTROL	0x7C0	Control the HASH engine behavior.
HASH_PAD	0x7C4	Enable the hardware padding feature of the HASH engine.
HASH_PAD_FORCE	0x7C8	Force the hardware padding operation to trigger if the input data length is zero bytes.
HASH_CUR_LEN_0	0x7CC	Bits [31:0] of the number of bytes that have been digested so far.
HASH_CUR_LEN_1	0x7D0	Bits [63:32] of the number of bytes that have been digested so far.
HASH_HW_FLAGS	0x7DC	Hardware configuration of the HASH engine. Reset value holds the supported features.
HASH_SW_RESET	0x7E4	Reset the HASH engine.
HASH_ENDIANNESS	0x7E8	Configure the endianness of HASH data and padding generation.

6.6.12.3.2.1 HASH_H[0]

Address offset: 0x640

HASH_H value registers. The initial HASH_H[0] register holds the least significant bits [31:0] of the value.

This register is a 'R/W change' register, as the written register values changes during processing.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	VALUE			Write the initial hash value before start of digest operation, and read the final hash value result after the digest operation has been completed.

6.6.12.3.2.2 HASH_H[1]

Address offset: 0x644

HASH_H value registers. The initial HASH_H[0] register holds the least significant bits [31:0] of the value.

This register is a 'R/W change' register, as the written register values changes during processing.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	VALUE			Write the initial hash value before start of digest operation, and read the final hash value result after the digest operation has been completed.

6.6.12.3.2.3 HASH_H[2]

Address offset: 0x648

HASH_H value registers. The initial HASH_H[0] register holds the least significant bits [31:0] of the value.

This register is a 'R/W change' register, as the written register values changes during processing.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	VALUE			Write the initial hash value before start of digest operation, and read the final hash value result after the digest operation has been completed.

6.6.12.3.2.4 HASH_H[3]

Address offset: 0x64C

HASH_H value registers. The initial HASH_H[0] register holds the least significant bits [31:0] of the value.

This register is a 'R/W change' register, as the written register values changes during processing.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	VALUE			Write the initial hash value before start of digest operation, and read the final hash value result after the digest operation has been completed.

6.6.12.3.2.5 HASH_H[4]

Address offset: 0x650

HASH_H value registers. The initial HASH_H[0] register holds the least significant bits [31:0] of the value.

This register is a 'R/W change' register, as the written register values changes during processing.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	VALUE			Write the initial hash value before start of digest operation, and read the final hash value result after the digest operation has been completed.

6.6.12.3.2.6 HASH_H[5]

Address offset: 0x654

HASH_H value registers. The initial HASH_H[0] register holds the least significant bits [31:0] of the value.

This register is a 'R/W change' register, as the written register values changes during processing.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	VALUE			Write the initial hash value before start of digest operation, and read the final hash value result after the digest operation has been completed.

6.6.12.3.2.7 HASH_H[6]

Address offset: 0x658

HASH_H value registers. The initial HASH_H[0] register holds the least significant bits [31:0] of the value.

This register is a 'R/W change' register, as the written register values changes during processing.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	VALUE			Write the initial hash value before start of digest operation, and read the final hash value result after the digest operation has been completed.

6.6.12.3.2.8 HASH_H[7]

Address offset: 0x65C

HASH_H value registers. The initial HASH_H[0] register holds the least significant bits [31:0] of the value.

This register is a 'R/W change' register, as the written register values changes during processing.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	VALUE			Write the initial hash value before start of digest operation, and read the final hash value result after the digest operation has been completed.

6.6.12.3.2.9 HASH_PAD_AUTO

Address offset: 0x684

Configure the HASH engine to automatically pad data at the end of the DMA transfer to complete the digest operation.

This feature can only be used if HASH_PAD on page 251 is enabled, and must be disabled after a digest operation is completed. In the event of zero bytes input data length the hardware padding must be manually triggered using register HASH_PAD_FORCE on page 251.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	HWPAD			Enable automatic padding in hardware. Disable this register when the digest operation is completed.
			Disable	0	Do not enable automatic hardware padding.
			Enable	1	Enable automatic hardware padding.

6.6.12.3.2.10 HASH_INIT_STATE

Address offset: 0x694

Configure HASH engine initial state registers.

Data fetched using the DIN DMA engine will be loaded into initial hash value registers HASH_H[0] on page 247 or used as IV for AES MAC.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	LOAD			Enable loading of data to initial state registers. Digest/IV for HASH/AES_MAC. Disable this register when loading of data using DIN DMA is done.
			Disable	0	Disable loading of data to initial state registers.
			Enable	1	Enable loading of data to initial state registers.

6.6.12.3.2.11 HASH_VERSION

Address offset: 0x7B0

HASH engine HW version

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C C C C B B B A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	PATCH			
B	R	MINOR_VERSION_NUMBER			Minor version number
C	R	MAJOR_VERSION_NUMBER			Major version number

6.6.12.3.2.12 HASH_CONTROL

Address offset: 0x7C0

Control the HASH engine behavior.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	MODE			Select HASH mode to execute
			SHA1	1	Select SHA1 mode
			SHA256	2	Select SHA256 mode
			SHA224	10	Select SHA224 mode

6.6.12.3.2.13 HASH_PAD

Address offset: 0x7C4

Enable the hardware padding feature of the HASH engine.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	ENABLE			Configure hardware padding feature.
			Disable	0	Disable hardware padding feature.
			Enable	1	Enable hardware padding feature.

6.6.12.3.2.14 HASH_PAD_FORCE

Address offset: 0x7C8

Force the hardware padding operation to trigger if the input data length is zero bytes.

This feature can only be used if HASH_PAD on page 251 is enabled, and must be disabled after a digest operation is completed.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ENABLE			Trigger hardware padding operation. Disable this register when the digest operation is completed.
			Disable	0	Do not force hardware padding to trigger.
			Enable	1	Force hardware padding to trigger.

6.6.12.3.2.15 HASH_CUR_LEN_0

Address offset: 0x7CC

Bits [31:0] of the number of bytes that have been digested so far.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	VALUE			Bits [31:0] of current length of digested data in bytes.

6.6.12.3.2.16 HASH_CUR_LEN_1

Address offset: 0x7D0

Bits [63:32] of the number of bytes that have been digested so far.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	VALUE																																	
				Bits [63:32] of current length of digested data in bytes.																																	

6.6.12.3.2.17 HASH_HW_FLAGS

Address offset: 0x7DC

Hardware configuration of the HASH engine. Reset value holds the supported features.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																	J	I	H	G	F	E	D	C	C	C	B	B	B	B	A	A	A	A
Reset 0x00012001				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
ID	R/W	Field	Value ID	Value																															
A	R	CW																																	
			One	1																															
			Two	2																															
B	R	CH																																	
			One	0																															
			All	1																															
C	R	DW																																	
			32Bits	0																															
			64Bits	1																															
D	R	SHA_512_EXISTS																																	
E	R	PAD_EXISTS																																	
F	R	MD5_EXISTS																																	
G	R	HMAC_EXISTS																																	
H	R	SHA_256_EXISTS																																	
I	R	HASH_COMPARE_EXISTS																																	
J	R	DUMP_HASH_TO_DOUT_EXISTS																																	

6.6.12.3.2.18 HASH_SW_RESET

Address offset: 0x7E4

Reset the HASH engine.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	RESET																																	
			Enable	1																															

6.6.12.3.2.19 HASH_ENDIANNESS

Address offset: 0x7E8

Configure the endianness of HASH data and padding generation.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	ENDIAN			Endianness of HASH data and padding generation. The default value is little-endian.
			LittleEndian	0	Use little-endian format for data and padding
			BigEndian	1	Use big-endian format for data and padding

6.6.12.4 PKA engine

The Public Key Accelerator (PKA) engine is designed to accelerate asymmetric cryptographic algorithms.

The PKA design is a general purpose bignum modular ALU capable of supporting operand sizes between 128-3136 bits in the following operations:

- Modular exponentiation/inversion
- Modular/regular addition/subtraction
- Modular/regular increment/decrement
- Modular/regular multiplication/division
- Logical operations (AND, OR, XOR, SHIFT)

The PKA engine can be used to hardware accelerate various arithmentic regular and modular mathematical operations involving very large numbers which are used in both RSA and Elliptic Curve Cryptographic (ECC) public-key cryptosystems.

6.6.12.4.1 Virtual memory mapping

The PKA engine uses virtual register mapping to facilitate flexible data management across a variety of cryptographic algorithms.

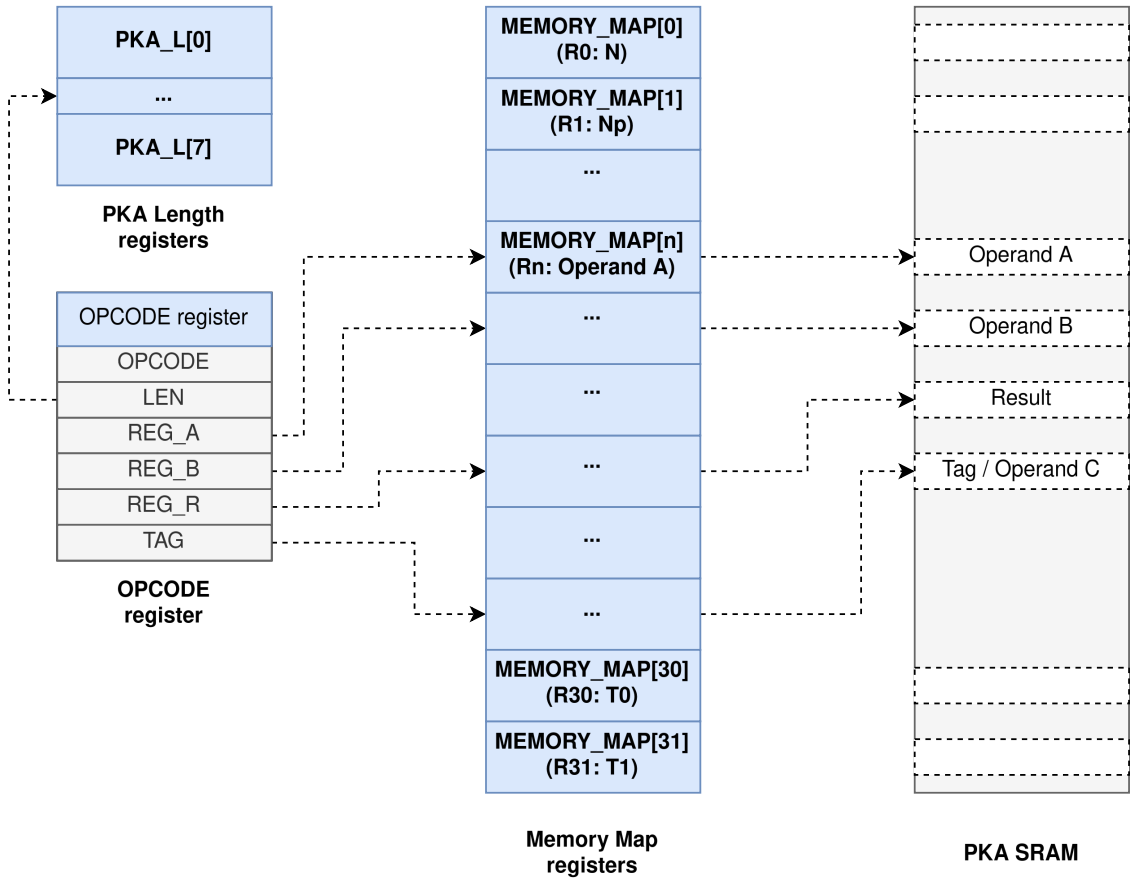


Figure 46: Virtual register mapping

All virtual registers must be defined and configured in the dedicated PKA SRAM on page 257 before they can be accessed by the PKA engine during processing. This SRAM acts as a private memory pool for the PKA engine, where all other access is blocked during processing. The virtual registers are used as input for the PKA calculation and as a placeholder for intermediate and final results.

The PKA engine can be configured to perform multiple operations on virtual operand registers and store the result of the operation in a virtual result or intermediate result register. During the next iteration the PKA engine can then use an intermediate result register from a previous operation as a virtual operand register for further calculations. This re-mapping strategy enables the PKA engine to efficiently handle complex cryptographic operations.

In total there are 32 virtual registers that can be mapped to different PKA SRAM regions using register MEMORY_MAP[0] on page 260, denoted as virtual register R0 - R31. Four of these 32 virtual registers are special registers, and their register index mapping can be changed using register N_NP_T0_T1_ADDR on page 268:

- N - holds the modulus number, by default mapped to virtual register R0. This register is used by the PKA engine for modular operations, and its modulus N value does not change during processing.
- Np - holds the inverse modulus number, by default mapped to virtual register R1. This register is used by the PKA engine for the Barrett reduction algorithm, and its inverse modulus Np value does not change during processing.
- T0 - temporary register, by default mapped to virtual register R30. This register is for internal use by the PKA engine.
- T1 - temporary register, by default mapped to virtual register R31. This register is for internal use by the PKA engine.

All virtual registers must be 64 bits word size aligned, and the size of the virtual registers must be at least the size of the largest operand plus an extra 64 bits for internal PKA calculations. These extra 64 bits must be initialized to zero. This is applicable for all virtual registers R0 - R31. The configured virtual register size does not define the size of the operation, it only limits the largest operand size that can be used with the corresponding virtual register.

The memory map configuration can be altered dynamically by the PKA engine, depending on the operation. Not all virtual registers need to be configured for each operation. It is recommended to re-write the memory map configuration after a reset.

6.6.12.4.2 Engine operations

The PKA engine can perform multiple operations on operands stored in virtual registers.

PKA processing is triggered by writing to register OPCODE on page 267. This register contains both the PKA operation to perform, and which virtual register indexes to use as operand inputs, tag, and intermediate or final result output of the operation. Register PKA_DONE on page 271 will indicate `Processing` until the PKA operation is done, after which the result can be read from the result register in PKA SRAM.

The following OPCODE virtual register indexes must be configured prior to starting the PKA engine:

- Field `REG_R` configure which virtual register to use for storing an intermediate or final result.
- Field `REG_A` and `REG_B` configure which virtual registers to use as operand input. The operand input fields can be interpreted by the PKA engine as constants instead of virtual register indexes by setting fields `CONST_A` and `CONST_B` for certain operations, as documented in the table below.
- The size of the operands are set in field `LEN`, which must point to one of the pre-configured operand sizes in bits configured in register PKA_L[0] on page 269.

6.6.12.4.2.1 OPCODE overview

Supported PKA operation codes and the corresponding required virtual register configurations.

OPCODE	Operation
Terminate	Terminate ongoing PKA operation
AddInc	Add or Increment • ADD: `REG_R = REG_A + REG_B` • INC: `REG_R = REG_A + 0x1`, when `REG_B` and `CONST_B` are 0x1
SubDecNeg	Subtract, Decrement, or Negate • SUB: `REG_R = REG_A - REG_B` • DEC: `REG_R = REG_A - 0x1`, when `REG_B` and `CONST_B` are 0x1 • NEG: `REG_R = 0x0 - REG_B`, when `REG_A` is 0x0 and `CONST_A` is 0x1
ModAddInc	Modular Add or Modular Increment • ModADD: `REG_R = (REG_A + REG_B) % REG_N` • ModINC: `REG_R = (REG_A + 0x1) % REG_N`, when `REG_B` and `CONST_B` are 0x1
ModSubDecNeg	Modular Subtract, Modular Decrement, or Modular Negate • ModSUB: `REG_R = (REG_A - REG_B) % REG_N` • ModDEC: `REG_R = (REG_A - 0x1) % REG_N`, when `REG_B` and `CONST_B` is 0x1 • ModNEG: `REG_R = (0x0 - REG_B) % REG_N`, when `REG_A` is 0x0
ANDTST0CLR0	And, Test bit 0, or Clear • AND: `REG_R = REG_A & REG_B` • TST0: `REG_R = REG_A & 0x1`, when `REG_B` is 0x1, and `CONST_B` is 0x1 • CLR: `REG_R = 0x0`, when `REG_B` is 0x0 and `CONST_B` is 0x1. `REG_A` is ignored.
ORCOPYSET0	Or, Copy, or Set bit 0 • OR: `REG_R = REG_A | REG_B` • COPY: `REG_R = REG_A`, when `REG_B` is 0x0 and `CONST_B` is 0x1. • SET0: `REG_R = REG_A | 0x1`, when `REG_B` and `CONST_B` is 0x1.
XORFLP0INVCMP	XOR, Flip bit 0, Invert, or Compare • XOR: `REG_R = REG_A XOR REG_B` • FLP0: `REG_R = REG_A XOR 0x1`, when `REG_B` and `CONST_B` is 0x1. • INV: `REG_R = REG_A XOR 0xFFFFFFFF`, when `REG_B` is 0x1F and `CONST_B` is 0x1. • CMP: `REG_A XOR REG_B`, when `DISCARD_R` is 0x1, result of comparison is provided by the `ALU_OUT_ZERO` flag in PKA_STATUS register.
SHR0	Shift right 0. This operation performs a logical right shift on the contents of `REG_A` by a specified number of bit positions and stores the result in `REG_R`. The leftmost bits of `REG_R` that are vacated by the shift operation are filled with zeros. `REG_R = REG_A >> s`, `CONST_B` must be set to 0x1. To perform `s` shifts, `REG_B` should be set to `s - 1` (where `1 <= s <= 31`).
SHR1	Shift right 1. This operation performs a logical right shift on the contents of `REG_A` by a specified number of bit positions and stores the result in `REG_R`. The leftmost bits of `REG_R` that are vacated by the shift operation are filled with ones. `REG_R = REG_A >> s`, `CONST_B` must be set to 0x1. To perform `s` shifts, `REG_B` should be set to `s - 1` (where `1 <= s <= 31`).
SHL0	Shift left 0. This operation performs a logical left shift on the contents of `REG_A` by a specified number of bit positions and stores the result in `REG_R`. The leftmost bits of `REG_R` that are vacated by the shift operation are filled with zeros. `REG_R = REG_A << s`, `CONST_B` must be set to 0x1. To perform `s` shifts, `REG_B` should be set to `s - 1` (where `1 <= s <= 31`).
SHL1	Shift left 1. This operation performs a logical left shift on the contents of `REG_A` by a specified number of bit positions and stores the result in `REG_R`. The leftmost bits of `REG_R` that are vacated by the shift operation are filled with ones. `REG_R = REG_A << s`, `CONST_B` must be set to 0x1. To perform `s` shifts, `REG_B` should be set to `s - 1` (where `1 <= s <= 31`).
MulLow	Multiply Low. This operation performs a multiplication of the values in `REG_A` and `REG_B` and stores the result in the destination register `REG_R`. Any bits of the product that exceed the operand size are discarded, effectively keeping only the least significant bits (LSBs) that fit within the operand size.

OPCODE	Operation
	`REG_R = (REG_A * REG_B) & operand size mask`
ModMul	Modular Multiply. `REG_R = (REG_A * REG_B) % REG_N`
ModMulN	The output of this operation is a number that is potentially larger than the modulus `N`, but guaranteed to be smaller than `2N`. Assuming `REG_A` and `REG_B` are already reduced modulo `N` or are less than `N`, the operation is simply `REG_R = (REG_A * REG_B)`.
ModExp	Modular Exponentiation. `REG_R = (REG_A ^ REG_B) % REG_N`
Division	Integer Division. This operation performs integer division of the value in `REG_A` by the value in `REG_B`. The quotient of the division is stored in `REG_R`, and the remainder is stored back in `REG_A`. • `REG_R = REG_A / REG_B` • `REG_A = REG_A % REG_B` If `REG_B` is zero (0x0), the operation is invalid, and the divide by zero bit in the status register is set to indicate a division error.
ModInv	Modular Inversion. `REG_R = 1/REG_B % REG_N`
ModDiv	Modular division is done by calculating the modular inverse of the divisor, check that the inverse value exists by examining the GCD, and then use modular multiplication to multiply the inverse result by the divided. `REG_A = (REG_A * REG_B^(-1)) % REG_N`
MulHigh	Multiply High. This operation multiplies `REG_A` by `REG_B` and captures the high-order bits of the result that exceed the operand size. It places these significant bits, along with an additional `PKA_WORD` number of bits, into the destination register `REG_R`. `REG_R = (REG_A * REG_B) >> operand size`
ModMLAC	Modular Multiplication Acceleration. Performs a modular multiplication and addition. `REG_C` is defined using the operation tag. `REG_R = ((REG_A * REG_B) + REG_C) % REG_N`
ModMLACNR	Modular Multiplication Acceleration No Reduction. Same as ModMLAC, but this omits the final reduction of the result.
Reduction	Reduction. This operation performs a modular reduction, where the result `REG_R` is the remainder of `REG_A` divided by `REG_N`. The length of the operation is flexible and can be chosen based on the specific requirements of the use case. `REG_R = REG_A % REG_N`

Table 17: PKA OPCODE descriptions

6.6.12.4.3 Pipeline configuration

The following section describe how the PKA engine is used to accelerate asymmetric cryptographic algorithms.

The PKA engine supports pipelined operations; the pipeline depth is one opcode, thus the next operation can be set up while the previous operation is executing. Register PKA_PIPE on page 270 will indicate if the pipeline is ready for a new opcode and register PKA_DONE on page 271 will indicate when the PKA operation has been completed and no operation is waiting in the pipeline.

1. Enable CRYPTOCELL subsystem as described in Cryptographic flow on page 219.
2. Initialize the PKA engine to accommodate the maximum bit size of all intended operations
   a. Configure registers PKA_L[0] on page 269 for all required operand bit sizes. The desired operand length is selected using field `LEN` in register OPCODE on page 267.
   b. Define the PKA SRAM memory map partitioning using register MEMORY_MAP[0] on page 260 for register `N`, `Np`, `T0`, and `T1`, as well as any other virtual registers intended to be used in the operations. The PKA SRAM memory map partitioning must allow for the max operand bit size plus an additional 64 bits reserved for PKA engine internal calculations.
3. For all operations

   a. Load the PKA SRAM virtual registers `N` and `Np` as required
   b. Load the remaining PKA SRAM virtual registers as required
   c. Execute the operation by writing register OPCODE on page 267
   d. Prepare the next opcode once register PKA_PIPE on page 270 is ready.
   e. Handle any status bits in register PKA_STATUS on page 268
   f. Re-use intermediate results of the previous operation as needed.
4. Wait for the operation to complete by either polling register PKA_DONE on page 271, or by unmasking the interrupt for field `PKA_MASK` in register IMR on page 296
5. Read the result from the result register.

6.6.12.4.4 PKA SRAM

The 4 kB PKA SRAM memory connected to the PKA engine is used exclusively by the engine during cryptographic operations. All access to this memory is blocked while the PKA engine is processing.

The PKA SRAM memory is not directly mapped to the device memory map. Instead, any read or write operation to this memory region must be done using the PKA engine on page 253.

Writing data to the PKA SRAM involves the following steps:

1. **Set the Address Offset**: Specify the starting byte address for writing by setting register PKA_SRAM_WADDR on page 271. An offset value of `0x0` points to the first 32-bits word in the PKA SRAM memory. An offset value of `0x10` points to the fourth 32-bits word in the PKA SRAM memory.
2. **Write Data**: After setting the address offset, data is written to register PKA_SRAM_WDATA on page 271. The address will automatically increment after each write, allowing writes to the next word without needing to set the offset again.

Reading data from the PKA SRAM involves the following steps:

1. **Set the Read Address**: Specify the starting byte address for reading by setting register PKA_SRAM_RADDR on page 272
2. **Read Data**: Retrieve the data from register PKA_SRAM_RDATA on page 272. Similar to the write address, the read address will auto-increment with each read, setting it to the next word.

Note: Before switching from writing to reading operations (or vice versa), the PKA SRAM write buffer must be cleared. This is done using register PKA_SRAM_WCLEAR on page 272. Clearing the buffer ensures that the next operation starts cleanly without any leftover data from the previous operation.

6.6.12.4.5 Cryptographic flow

The following section describe a simple cryptographic flow for this engine.

```
/* Enable CRYPTOCELL and its PKA engine */
NRF_CRYPTOCELL->ENABLE = CRYPTOCELL_ENABLE_ENABLE_Enabled;
NRF_CC_MISC->PKA_CLK = CC_MISC_PKA_CLK_ENABLE_Enable;

/* Define the operand bit size as 2048 */
NRF_CC_PKA->PKA_L[1] = 0x800;

/* Define the 32-bits PKA SRAM address of the selected R4 and R5 */
NRF_CC_PKA->MEMORY_MAP[4] = 0x108;
NRF_CC_PKA->MEMORY_MAP[5] = 0x14A;

/* Initialize the SRAM registers with one word of data */
NRF_CC_PKA->PKA_SRAM_WADDR = NRF_CC_PKA->MEMORY_MAP[4];
NRF_CC_PKA->PKA_SRAM_WDATA = 0x5;
NRF_CC_PKA->PKA_SRAM_WADDR = NRF_CC_PKA->MEMORY_MAP[5];
NRF_CC_PKA->PKA_SRAM_WDATA = 0x2;

/* Execute subtract, OPCODE SubDecNeg: R4 = R4 - R5 */
NRF_CC_PKA->OPCODE =
    (4 << CC_PKA_OPCODE_REG_R_Pos) |
    (5 << CC_PKA_OPCODE_REG_B_Pos) |
    (4 << CC_PKA_OPCODE_REG_A_Pos) |
    (1 << CC_PKA_OPCODE_LEN_Pos)   |
    (CC_PKA_OPCODE_OPCODE_SubDecNeg << CC_PKA_OPCODE_OPCODE_Pos);

/* Wait for operation to complete, result will be in R4 */
while (!NRF_CC_PKA->PKA_DONE) { }
```

This cryptographic flow example perform a subtract operation with the following assumptions:

- All PKA SRAM registers, including the special virtual registers `N`, `Np`, `T0`, and `T1`, have been cleared before the operation is run.
- The operation is using index 1 in register PKA_L[0] on page 269, which is set to accommodate an operand size of 2048 bits.
- Register `R4` and `R5` have been selected to run this operation. Register `R4` is used both as the operand A register and the result register.
- The memory map is configured to allow operands of 2048 bits plus an additional 64 bits for the internal PKA engine calculations. The configuration of the MEMORY_MAP[0] on page 260 for virtual register `N`, `Np`, `T0`, and `T1` is not included in the example. The memory map is thus configured with 66 words per register, leading to the following:

Virtual register	Memory map register	PKA SRAM address
N (R0)	MEMORY_MAP[0]	0x0
Np (R1)	MEMORY_MAP[1]	0x42
...	...	...
R4	MEMORY_MAP[4]	0x108
R5	MEMORY_MAP[5]	0x14A
...	...	...

6.6.12.4.6 Registers

Instances

Instance	Base address	Description
CC_PKA	0x5002B000	CRYPTOCELL PKA engine

Register overview

Register	Offset	Description
MEMORY_MAP[0]	0x0	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[1]	0x4	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[2]	0x8	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[3]	0xC	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[4]	0x10	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[5]	0x14	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[6]	0x18	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[7]	0x1C	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[8]	0x20	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[9]	0x24	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[10]	0x28	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[11]	0x2C	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[12]	0x30	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[13]	0x34	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[14]	0x38	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[15]	0x3C	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[16]	0x40	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[17]	0x44	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[18]	0x48	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[19]	0x4C	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[20]	0x50	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[21]	0x54	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[22]	0x58	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[23]	0x5C	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[24]	0x60	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[25]	0x64	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[26]	0x68	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[27]	0x6C	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[28]	0x70	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[29]	0x74	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
MEMORY_MAP[30]	0x78	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Register	Offset	Description
MEMORY_MAP[31]	0x7C	Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.
OPCODE	0x80	Operation code to be executed by the PKA engine. Writing to this register triggers the PKA operation.
N_NP_T0_T1_ADDR	0x84	This register defines the N, Np, T0, and T1 virtual register index.
PKA_STATUS	0x88	This register holds the status for the PKA pipeline.
PKA_SW_RESET	0x8C	Reset the PKA engine.
PKA_L[0]	0x90	This register holds the operands bit size.
PKA_L[1]	0x94	This register holds the operands bit size.
PKA_L[2]	0x98	This register holds the operands bit size.
PKA_L[3]	0x9C	This register holds the operands bit size.
PKA_L[4]	0xA0	This register holds the operands bit size.
PKA_L[5]	0xA4	This register holds the operands bit size.
PKA_L[6]	0xA8	This register holds the operands bit size.
PKA_L[7]	0xAC	This register holds the operands bit size.
PKA_PIPE	0xB0	Status register indicating if the PKA pipeline is ready to receive a new OPCODE.
PKA_DONE	0xB4	Status register indicating if the PKA operation has been completed.
PKA_VERSION	0xC4	PKA engine HW version. Reset value holds the version.
PKA_SRAM_WADDR	0xD4	Start address in PKA SRAM for subsequent write transactions.
PKA_SRAM_WDATA	0xD8	Write data to PKA SRAM. Writing to this register triggers a DMA transaction writing data into PKA SRAM. The DMA address offset is automatically incremented during write.
PKA_SRAM_RDATA	0xDC	Read data from PKA SRAM. Reading from this register triggers a DMA transaction read data from PKA SRAM. The DMA address offset is automatically incremented during read.
PKA_SRAM_WCLEAR	0xE0	Register for clearing PKA SRAM write buffer.
PKA_SRAM_RADDR	0xE4	Start address in PKA SRAM for subsequent read transactions.

6.6.12.4.6.1 MEMORY_MAP[0]

Address offset: 0x0

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																							A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.2 MEMORY_MAP[1]

Address offset: 0x4

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																							A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.3 MEMORY_MAP[2]

Address offset: 0x8

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.4 MEMORY_MAP[3]

Address offset: 0xC

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.5 MEMORY_MAP[4]

Address offset: 0x10

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.6 MEMORY_MAP[5]

Address offset: 0x14

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.7 MEMORY_MAP[6]

Address offset: 0x18

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.8 MEMORY_MAP[7]

Address offset: 0x1C

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																							A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.9 MEMORY_MAP[8]

Address offset: 0x20

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																							A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.10 MEMORY_MAP[9]

Address offset: 0x24

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																							A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.11 MEMORY_MAP[10]

Address offset: 0x28

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																							A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.12 MEMORY_MAP[11]

Address offset: 0x2C

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																							A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.13 MEMORY_MAP[12]

Address offset: 0x30

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																							A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.14 MEMORY_MAP[13]

Address offset: 0x34

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																							A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.15 MEMORY_MAP[14]

Address offset: 0x38

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																							A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.16 MEMORY_MAP[15]

Address offset: 0x3C

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																							A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.17 MEMORY_MAP[16]

Address offset: 0x40

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.18 MEMORY_MAP[17]

Address offset: 0x44

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.19 MEMORY_MAP[18]

Address offset: 0x48

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.20 MEMORY_MAP[19]

Address offset: 0x4C

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.21 MEMORY_MAP[20]

Address offset: 0x50

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.22 MEMORY_MAP[21]

Address offset: 0x54

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.23 MEMORY_MAP[22]

Address offset: 0x58

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.24 MEMORY_MAP[23]

Address offset: 0x5C

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.25 MEMORY_MAP[24]

Address offset: 0x60

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.26 MEMORY_MAP[25]

Address offset: 0x64

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.27 MEMORY_MAP[26]

Address offset: 0x68

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																							A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.28 MEMORY_MAP[27]

Address offset: 0x6C

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																							A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.29 MEMORY_MAP[28]

Address offset: 0x70

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																							A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.30 MEMORY_MAP[29]

Address offset: 0x74

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																							A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.31 MEMORY_MAP[30]

Address offset: 0x78

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.32 MEMORY_MAP[31]

Address offset: 0x7C

Register for mapping the virtual register R[n] to a physical address in the PKA SRAM.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			The physical word address used for the virtual register.

6.6.12.4.6.33 OPCODE

Address offset: 0x80

Operation code to be executed by the PKA engine.

Writing to this register triggers the PKA operation.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				I I I I I H H H G F F F F F E D D D D D C B B B B A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	TAG			Holds the operation tag or the operand C virtual register index.
B	RW	REG_R			Result register virtual register index.
C	RW	DISCARD_R			This field controls the interpretation of REG_R.
			Register	0x0	REG_R is intepreted as a register index.
			Discard	0x1	Result is discarded.
D	RW	REG_B			Operand B virtual register index.
E	RW	CONST_B			This field controls the interpretation of REG_B.
			Register	0x0	REG_B is intepreted as a register index.
			Constant	0x1	REG_B is intepreted as a constant.
F	RW	REG_A			Operand A virtual register index.
G	RW	CONST_A			This field controls the interpretation of REG_A.
			Register	0x0	REG_A is intepreted as a register index.
			Constant	0x1	REG_A is intepreted as a constant.
H	RW	LEN			The length of the operands. This value serves as an PKA length register index. E.g.: if LEN field value is set to 0, PKA_L[0] holds the size of the operands.
I	RW	OPCODE			Operation code to be executed by the PKA engine
			Terminate	0x0	Terminate operation
			AddInc	0x4	Add or Increment
			SubDecNeg	0x5	Subtract, Decrement, or Negate
			ModAddInc	0x6	Modular Add or Modular Increment
			ModSubDecNeg	0x7	Modular Subtract, Modular Decrement, or Modular Negate
			ANDTST0CLR0	0x8	Perform AND, test, or clear
			ORCOPYSET0	0x9	Perform OR, copy, or set bits
			XORFLP0INVCMP	0xA	Perform XOR, flip bits, invert, or compare
			SHR0	0xC	Shift right 0 operation

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				I I I I I H H H G F F F F F E D D D D D C B B B B B A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			SHR1	0xD	Shift right 1 operation
			SHL0	0xE	Shift left 0 operation
			SHL1	0xF	Shift left 1 operation
			MulLow	0x10	Multiply low operation
			ModMul	0x11	Modular multiply operation
			ModMulN	0x12	Modular multiply N operation
			ModExp	0x13	Modular exponentiation operation
			Division	0x14	Division operation
			ModInv	0x15	Modular inversion operation
			ModDiv	0x16	Modular division operation
			MulHigh	0x17	Multiply high operation
			ModMLAC	0x18	Modular multiplication acceleration
			ModMLACNR	0x19	Modular multiplication acceleration where final reduction is omitted
			Reduction	0x1B	Reduction operation

6.6.12.4.6.34 N_NP_T0_T1_ADDR

Address offset: 0x84

This register defines the N, Np, T0, and T1 virtual register index.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				D D D D D C C C C C B B B B B A A A A A
Reset 0x000FF820				**0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 0 0 0 0 0 1 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	N_VIRTUAL_ADDR			Register N virtual register index. Default is R0.
B	RW	NP_VIRTUAL_ADDR			Register Np virtual register index. Default is R1.
C	RW	T0_VIRTUAL_ADDR			Temporary register 0 virtual register index. Default is R30.
D	RW	T1_VIRTUAL_ADDR			Temporary register 1 virtual register index. Default is R31.

6.6.12.4.6.35 PKA_STATUS

Address offset: 0x88

This register holds the status for the PKA pipeline.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				K K K K K J I H G F E D C B B B B A A A A
Reset 0x00001000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	ALU_MSB_4BITS			The most significant 4-bits of the operand updated in shift operation.
B	R	ALU_LSB_4BITS			The least significant 4-bits of the operand updated in shift operation.
C	R	ALU_SIGN_OUT			Indicates the MSB sign of the last operation.
D	R	ALU_CARRY			Holds the carry of the last ALU operation.
E	R	ALU_CARRY_MOD			Holds the carry of the last modular operation.
F	R	ALU_SUB_IS_ZERO			Indicates the last subtraction operation sign.
G	R	ALU_OUT_ZERO			Indicates if the result of ALU OUT is zero.
H	R	ALU_MODOVRFLW			Modular overflow flag.
I	R	DIV_BY_ZERO			Indication if the division is done by zero.
J	R	MODINV_OF_ZERO			Indicates the modular inverse of zero.
K	R	OPCODE			Opcode of the last operation

6.6.12.4.6.36 PKA_SW_RESET

Address offset: 0x8C

Reset the PKA engine.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	RESET																																	
			Enable	1																															

6.6.12.4.6.37 PKA_L[0]

Address offset: 0x90

This register holds the operands bit size.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																							A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	OpSize																																	

6.6.12.4.6.38 PKA_L[1]

Address offset: 0x94

This register holds the operands bit size.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																							A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	OpSize																																	

6.6.12.4.6.39 PKA_L[2]

Address offset: 0x98

This register holds the operands bit size.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																							A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	OpSize																																	

6.6.12.4.6.40 PKA_L[3]

Address offset: 0x9C

This register holds the operands bit size.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																				A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	OpSize			Operand bit size.

6.6.12.4.6.41 PKA_L[4]

Address offset: 0xA0

This register holds the operands bit size.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																				A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	OpSize			Operand bit size.

6.6.12.4.6.42 PKA_L[5]

Address offset: 0xA4

This register holds the operands bit size.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																				A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	OpSize			Operand bit size.

6.6.12.4.6.43 PKA_L[6]

Address offset: 0xA8

This register holds the operands bit size.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																				A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	OpSize			Operand bit size.

6.6.12.4.6.44 PKA_L[7]

Address offset: 0xAC

This register holds the operands bit size.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																				A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	OpSize			Operand bit size.

6.6.12.4.6.45 PKA_PIPE

Address offset: 0xB0

Status register indicating if the PKA pipeline is ready to receive a new OPCODE.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
A	R	STATUS			PKA pipeline status.
			NotReady	0	PKA pipeline is not ready for a new OPCODE
			Ready	1	PKA pipeline is ready for a new OPCODE

6.6.12.4.6.46 PKA_DONE

Address offset: 0xB4

Status register indicating if the PKA operation has been completed.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
A	R	STATUS			PKA operation status.
			Processing	0	PKA operation is processing
			Completed	1	PKA operation is completed and pipeline is empty

6.6.12.4.6.47 PKA_VERSION

Address offset: 0xC4

PKA engine HW version. Reset value holds the version.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x16110215				**0 0 0 1 0 1 1 0 0 0 0 1 0 0 0 1 0 0 0 0 0 0 1 0 0 0 0 1 0 1 0 1**
ID	R/W	Field	Value ID	Value	Description
A	R	PKA_VERSION			

6.6.12.4.6.48 PKA_SRAM_WADDR

Address offset: 0xD4

Start address in PKA SRAM for subsequent write transactions.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	ADDR			PKA SRAM start address for write transaction

6.6.12.4.6.49 PKA_SRAM_WDATA

Address offset: 0xD8

Write data to PKA SRAM. Writing to this register triggers a DMA transaction writing data into PKA SRAM. The DMA address offset is automatically incremented during write.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	DATA																																	
				Data to write to PKA SRAM.																															

6.6.12.4.6.50 PKA_SRAM_RDATA

Address offset: 0xDC

Read data from PKA SRAM. Reading from this register triggers a DMA transaction read data from PKA SRAM. The DMA address offset is automatically incremented during read.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	R	DATA																																	
				Data to read from PKA SRAM																															

6.6.12.4.6.51 PKA_SRAM_WCLEAR

Address offset: 0xE0

Register for clearing PKA SRAM write buffer.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	CLEAR																																	
				Clear the PKA SRAM write buffer.																															

6.6.12.4.6.52 PKA_SRAM_RADDR

Address offset: 0xE4

Start address in PKA SRAM for subsequent read transactions.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	ADDR																																	
				PKA SRAM start address for read transaction																															

6.6.12.5 RNG engine

CRYPTOCELL implements a Random Number Generator (RNG) engine which uses a True Random Number Generator (TRNG) for its entropy collection.

The TRNG is a full entropy design compliant with:

- FIPS 140-2: *Security requirements for Cryptographic Modules*
- BSI AIS-31: *Functionality Classes and Evaluation Methodology for True Random Number Generators*
- NIST SP 800-90B: *Recommendation for the Entropy Sources Used for Random Bit Generation*

where a ring-oscillator is used as the noise source.

The entropy collected using the RNG engine can in turn be used for seeding a Pseudo Random Number Generator (PRNG) as defined in NIST SP 800-90A: *Recommendation for Random Number Generation Using Deterministic Random Bit Generators.*

NIST SP 800-90A define three Deterministic Random Bit Generator (DRBG) that are considered cryptographically secure pseudorandom number generators for use in cryptography: Hash DRBG, HMAC DRBG, and CTR DRBG.

The CRYPTOCELL DRBG implementation is a combination of hardware and software, where CTR DRBG is implemented using the the AES engine running AES encryption in counter (CTR) mode as the underlying cipher. This DRBG instance is seeded with random entropy from the RNG engine.

6.6.12.5.1 Ring oscillator length configuration

The RNG engine must be configured with specific parameters to ensure correct operation in order to output random bits with sufficient entropy.

The noise source used for collecting entropy is based on inverter timing jitter that is collected from a dedicated on-chip free-running ring oscillator. The ring oscillator length, i.e. the number of inverters in the chain, can be configured using register TRNG_CONFIG on page 279.

In total there are four different ring oscillator lengths that can be selected, referred to as ROSC1 through ROSC4. For each of these four configurable lengths a corresponding sample count value is provided in register TRNG90B.ROSC1 on page 45 through register TRNG90B.ROSC4 on page 46.

The sampling frequency is configured using register SAMPLE_CNT on page 281, and the programmed value defines the number of CPU clock cycles between two consecutive ring oscillator samples. The configured sample count value is the minimum number of clock cycles that is enough to get independent outputs from the ring oscillator and must match that of the configured ROSC length.

The following steps describe how to set the RNG engine parameters described above:

1. Enable RNG engine clock using register RNG_CLK on page 284.
2. Reset the RNG engine using register RNG_SW_RESET on page 282.
3. Re-enable RNG engine clock and select a device-specific sample count from registers TRNG90B.ROSC1 on page 45 through TRNG90B.ROSC4 on page 46 starting with the smallest one, and program the value into register SAMPLE_CNT on page 281.
4. Perform a readback of the selected sample count value.
5. Set the corresponding ROSC length in register TRNG_CONFIG on page 279 to match the selected sample count selection.
6. Enable the noise source using register NOISE_SOURCE on page 281.
7. Wait until event `EHR_VALID_INT` in register RNG_ISR on page 278 trigger to indicate successful collection of 192 bits of random data. The result can be read from registers EHR_DATA[0] on page 279.
8. If events `AUTOCORR_ERR_INT`, `CRNGT_ERR_INT`, or `VNC_ERR_INT` in register RNG_ISR on page 278 trigger, the RNG engine must be re-configured starting from step 2 above. Increase the ROSC length by a factor of one, and pick the corresponding sample count value from FICR. This step must be repeated until the collection of 192 bits of random data can be collected without an error event being triggered.

It is recommended to always try the shortest ROSC length first, allowing the RNG engine to complete the entropy collection in a shorter time and keep the ring oscillator turned off for longer periods in order to save power.

6.6.12.5.2 RNG SRAM

The 2 kB SRAM memory connected to the RNG engine can be used for storing a large pool of random entropy.

The RNG SRAM memory is not directly mapped to the device memory map. Instead, any read or write operation using word granularity to this memory region must be done using RNG SRAM interface on page 301. Larger payloads than word granularity can be processed using the DIN DMA engine on page 288 and DOUT DMA engine on page 292.

Before any RNG SRAM read or write transaction can be performed, the CRYPTOCELL must be enabled.

Writing data to the RNG SRAM involves the following steps:

1. **Set the Address Offset**: Specify the starting byte address for writing by setting register SRAM_ADDR on page 302. An offset value of `0x0` points to the first 32-bits word in the RNG SRAM memory. An offset value of `0x10` points to the fourth 32-bits word in the RNG SRAM memory.
2. **Write Data**: When register SRAM_DATA_READY on page 302 indicates DMA engine is idle, data is written to register SRAM_DATA on page 301. The address will automatically increment after each write, allowing writes to the next word without needing to set the offset again.

Reading data from the RNG SRAM involves the following steps:

1. **Set the Read Address**: Specify the starting byte address for reading by setting register SRAM_ADDR on page 302
2. **Discard first read**: Read and discard the first value from register SRAM_DATA on page 301, as it will contain the previous value pointed to by register SRAM_ADDR on page 302.
3. **Read Data**: When register SRAM_DATA_READY on page 302 indicates DMA engine is idle, retrieve the data from register SRAM_DATA on page 301. Similar to the write address, the read address will auto-increment with each read, setting it to the next word.

Note: Once the address register reaches the last RNG SRAM address, the automatic address incrementation halts. Any subsequent read or write transaction will cause the DMA engine to continue operating on the last 32-bits word in the RNG SRAM memory.

6.6.12.5.3 TRNG hardware tests

The RNG engine has a number of built-in hardware tests for making sure the collected entropy from the TRNG is of sufficient quality.

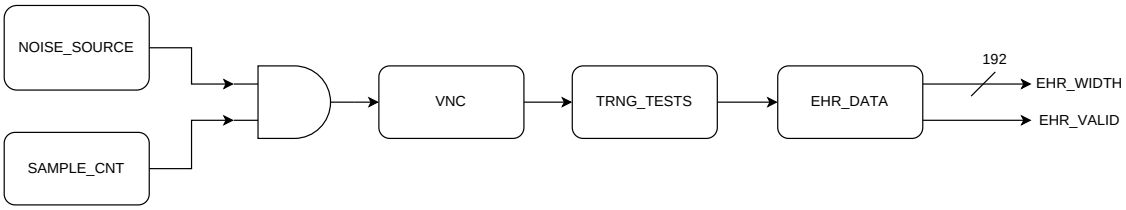


Figure 47: CRYPTOCELL True Random Number Generator

The TRNG collects random bits from the noise source according to the programmed sample counter value in register SAMPLE_CNT on page 281. The sampled bits are post-processed in a von Neumann corrector (VNC) before being subjected to a continuous random number generation test (CRNGT) and autocorrelation test.

192 bits of random data can be read from the entropy holding registers EHR_DATA[0] on page 279 once interrupt `EHR_VALID_INT` in register RNG_ISR on page 278 trigger. If this interrupt is masked away in register RNG_IMR on page 277, the status register TRNG_VALID on page 279 contains field `EHR_DATA` which can be polled when the random data is valid. Reading the most significant word from EHR_DATA registers will reset register TRNG_VALID and a new 192 bits collection period will start.

Note:

To ensure proper operation when reading 192 bits of random data from the EHR_DATA registers of the RNG engine the data must be read in ascending order, starting with:

- EHR_DATA[0]
- EHR_DATA[1]
- EHR_DATA[2]
- EHR_DATA[3]
- EHR_DATA[4]
- EHR_DATA[5]

6.6.12.5.3.1 von Neumann Corrector

The von Neumann Corrector (VNC) is designed to balance the succession of '1' and '0' bits being output by the TRNG noise source.

The input bits to the VNC is tested for bit equality, meaning a sequence of 32 consequtive bits with the same bit value will trigger event `VNC_ERR_INT` in register RNG_ISR on page 278.

If no error event is triggered, the input bits will be balanced using the VNC as shown in the figure below, and the resulting output bits will be subjected to additional TRNG tests. The VNC produce output only if the noise source is active, see register NOISE_SOURCE on page 281.

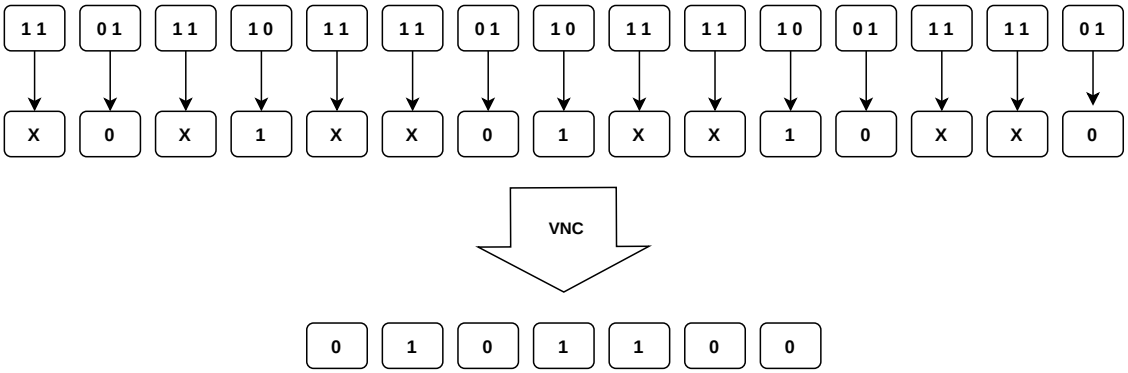


Figure 48: CRYPTOCELL von Neumann corrector

The VNC statistically output one bit for each 4 input bits sampled, meaning the average output rate of the TRNG is `1/(SAMPLE_CNT * 4)` bits per CPU clock cycle.

6.6.12.5.3.2 Continuous random number generation test

The Continuous random number generation test (CRNGT) process the balanced output of random data from the von Neumann corrector.

In the event that two consecutive blocks of 16 collected bits are equal, the CRNGT will trigger event `CRNGT_ERR_INT` in register RNG_ISR on page 278.

6.6.12.5.3.3 Autocorrelation test

The Autocorrelation test determine if there over time is a bias in the random bit sequences towards certain values or patterns, or if the bits in the sequence are truly independent.

If a bias in the collected bit stream is detected, the output will be discarded and the error flagged in register AUTOCORR_STATISTIC on page 282. If a bias is detected four consecutive times in a row, the autocorrelation test will trigger event `AUTOCORR_ERR_INT` in register RNG_ISR on page 278. In this situation the TRNG will cease to function until manually reset using register RNG_SW_RESET on page 282.

6.6.12.5.4 Cryptographic flow

The following section describe a simple cryptographic flow for this engine.

```
/* Enable CRYPTOCELL subsystem */
NRF_CRYPTOCELL->ENABLE = CRYPTOCELL_ENABLE_ENABLE_Enabled;

/* Enable engine clock */
NRF_CC_RNG->RNG_CLK = CC_RNG_RNG_CLK_ENABLE_Enable;

/* Reset engine */
NRF_CC_RNG->RNG_SW_RESET = CC_RNG_RNG_SW_RESET_RESET_Enable;

/* Configure sampling rate between consecutive bits */
do {
    NRF_CC_RNG->RNG_CLK = CC_RNG_RNG_CLK_ENABLE_Enable;
    NRF_CC_RNG->SAMPLE_CNT = NRF_FICR->TRNG90B.ROSC1;
} while ( NRF_CC_RNG->SAMPLE_CNT != NRF_FICR->TRNG90B.ROSC1 );

/* Configure ROSC length */
NRF_CC_RNG->TRNG_CONFIG = CC_RNG_TRNG_CONFIG_ROSC_LEN_ROSC1;

/* Enable noise source */
NRF_CC_RNG->NOISE_SOURCE = CC_RNG_NOISE_SOURCE_ENABLE_Enabled;

/* Wait for random data to be sampled */
while ((NRF_CC_RNG->RNG_ISR & CC_RNG_RNG_ISR_EHR_VALID_INT_Msk) == 0) {}

/* 192 bits of random data now available in
   NRF_CC_RNG->EHR_DATA[0] to NRF_CC_RNG->EHR_DATA[5] */
```

6.6.12.5.5 Registers

Instances

Instance	Base address	Description
CC_RNG	0x5002B000	CRYPTOCELL RNG engine

Register overview

Register	Offset	Description
RNG_IMR	0x100	Interrupt mask register. Each bit of this register holds the mask of a single interrupt source.
RNG_ISR	0x104	Interrupt status register. Each bit of this register holds the interrupt status of a single interrupt source. If corresponding RNG_IMR bit is unmasked, an interrupt is generated.
RNG_ICR	0x108	Interrupt clear register. Writing a 1 bit into a field in this register will clear the corresponding bit in RNG_ISR.
TRNG_CONFIG	0x10C	TRNG ring oscillator length configuration
TRNG_VALID	0x110	This register indicates if TRNG entropy collection is valid.

Register	Offset	Description
EHR_DATA[0]	0x114	The entropy holding registers (EHR) hold 192-bits random data collected by the TRNG. The initial EHR_DATA[0] register holds the least significant bits [31:0] of the random data value.
EHR_DATA[1]	0x118	The entropy holding registers (EHR) hold 192-bits random data collected by the TRNG. The initial EHR_DATA[0] register holds the least significant bits [31:0] of the random data value.
EHR_DATA[2]	0x11C	The entropy holding registers (EHR) hold 192-bits random data collected by the TRNG. The initial EHR_DATA[0] register holds the least significant bits [31:0] of the random data value.
EHR_DATA[3]	0x120	The entropy holding registers (EHR) hold 192-bits random data collected by the TRNG. The initial EHR_DATA[0] register holds the least significant bits [31:0] of the random data value.
EHR_DATA[4]	0x124	The entropy holding registers (EHR) hold 192-bits random data collected by the TRNG. The initial EHR_DATA[0] register holds the least significant bits [31:0] of the random data value.
EHR_DATA[5]	0x128	The entropy holding registers (EHR) hold 192-bits random data collected by the TRNG. The initial EHR_DATA[0] register holds the least significant bits [31:0] of the random data value.
NOISE_SOURCE	0x12C	This register controls the ring oscillator circuit used as a noise source.
SAMPLE_CNT	0x130	Sample count defining the number of CPU clock cycles between two consecutive noise source samples.
AUTOCORR_STATISTIC	0x134	Statistics counter for autocorrelation test activations. Statistics collection is stopped if one of the counters reach its limit of all ones.
TRNG_DEBUG	0x138	Debug register for the TRNG. This register is used to bypass TRNG tests in hardware.
RNG_SW_RESET	0x140	Reset the RNG engine.
RNG_BUSY	0x1B8	Status register for RNG engine activity.
TRNG_RESET	0x1BC	Reset the TRNG, including internal counter of collected bits and registers EHR_DATA and TRNG_VALID.
RNG_HW_FLAGS	0x1C0	Hardware configuration of RNG engine. Reset value holds the supported features.
RNG_CLK	0x1C4	Control clock for the RNG engine.
RNG_DMA	0x1C8	Writing to this register enables the RNG DMA engine.
RNG_DMA_ROSC_LEN	0x1CC	This register defines which ring oscillator length configuration should be used when using the RNG DMA engine.
RNG_DMA_SRAM_ADDR	0x1D0	This register defines the start address in TRNG SRAM for the TRNG data to be collected by the RNG DMA engine.
RNG_DMA_SAMPLES_NUM	0x1D4	This register defines the number of 192-bits samples that the RNG DMA engine collects per run.
RNG_WATCHDOG_VAL	0x1D8	This register defines the maximum number of CPU clock cycles per TRNG collection of 192-bits samples. If the number of cycles for a collection exceeds this threshold the WATCHDOG interrupt is triggered.
RNG_DMA_BUSY	0x1DC	Status register for RNG DMA engine activity.

6.6.12.5.5.1 RNG_IMR

Address offset: 0x100

Interrupt mask register. Each bit of this register holds the mask of a single interrupt source.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																														F	E	D	C	B	A
Reset 0x0000003F				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field		Value ID				Value																				Description							
A	RW	EHR_VALID_MASK																										See RNG_ISR for explanation on this interrupt.							
				IRQEnable				0																				Do not mask EHR interrupt i.e. interrupt is generated							
				IRQDisable				1																				Mask EHR interrupt i.e. no interrupt is generated							
B	RW	AUTOCORR_ERR_MASK																										See RNG_ISR for explanation on this interrupt.							
				IRQEnable				0																				Do not mask autocorrelation interrupt i.e. interrupt is generated							
				IRQDisable				1																				Mask autocorrelation interrupt i.e. no interrupt is generated							
C	RW	CRNGT_ERR_MASK																										See RNG_ISR for explanation on this interrupt.							

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F E D C B A
Reset 0x0000003F				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
			IRQEnable	0	Do not mask the CRNGT error interrupt i.e. interrupt is generated
			IRQDisable	1	Mask the CRNGT error interrupt i.e. no interrupt is generated
D	RW	VNC_ERR_MASK			See RNG_ISR for explanation on this interrupt.
			IRQEnable	0	Do not mask the von Neumann corrector error interrupt i.e. interrupt is generated
			IRQDisable	1	Mask the von Neumann corrector error interrupt i.e. no interrupt is generated
E	RW	WATCHDOG_MASK			See RNG_ISR for explanation on this interrupt.
			IRQEnable	0	Do not mask the watchdog interrupt i.e. interrupt is generated
			IRQDisable	1	Mask the watchdog interrupt i.e. no interrupt is generated
F	RW	DMA_DONE_MASK			See RNG_ISR for explanation on this interrupt.
			IRQEnable	0	Do not mask the RNG DMA completion interrupt i.e. interrupt is generated
			IRQDisable	1	Mask the RNG DMA completion interrupt i.e. no interrupt is generated

6.6.12.5.5.2 RNG_ISR

Address offset: 0x104

Interrupt status register. Each bit of this register holds the interrupt status of a single interrupt source. If corresponding RNG_IMR bit is unmasked, an interrupt is generated.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	EHR_VALID_INT			192-bits have been collected and are ready to be read.
B	R	AUTOCORR_ERR_INT			Autocorrelation error. Failure occurs when autocorrelation test has failed four times in a row. Once set, the TRNG ceases to function until next reset.
C	R	CRNGT_ERR_INT			Continuous random number generator test error. Failure occurs when two consecutive blocks of 16 collected bits are equal.
D	R	VNC_ERR_INT			von Neumann corrector error. Failure occurs if 32 consecutive collected bits are identical, ZERO, or ONE.
E	R	WATCHDOG_INT			Maximum number of CPU clock cycles per sample have been exceeded. See RNG_WATCHDOG_VAL for more information.
F	R	DMA_DONE_INT			RNG DMA to SRAM is completed.

6.6.12.5.5.3 RNG_ICR

Address offset: 0x108

Interrupt clear register. Writing a 1 bit into a field in this register will clear the corresponding bit in RNG_ISR.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	EHR_VALID_CLEAR			Writing value '1' clears corresponding bit in RNG_ISR
B	W	AUTOCORR_ERR_CLEAR			Cannot be cleared by software! Only RNG reset clears this bit.
C	W	CRNGT_ERR_CLEAR			Writing value '1' clears corresponding bit in RNG_ISR
D	W	VNC_ERR_CLEAR			Writing value '1' clears corresponding bit in RNG_ISR
E	W	WATCHDOG_CLEAR			Writing value '1' clears corresponding bit in RNG_ISR
F	W	DMA_DONE_CLEAR			Writing value '1' clears corresponding bit in RNG_ISR

6.6.12.5.5.4 TRNG_CONFIG

Address offset: 0x10C

TRNG ring oscillator length configuration

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ROSC_LEN			Set the length of the oscillator ring (= the number of inverters) out of four possible configurations.
			ROSC1	0	Use shortest ROSC1 ring oscillator configuration.
			ROSC2	1	Use ROSC2 ring oscillator configuration.
			ROSC3	2	Use ROSC3 ring oscillator configuration.
			ROSC4	3	Use longest ROSC4 ring oscillator configuration.

6.6.12.5.5.5 TRNG_VALID

Address offset: 0x110

This register indicates if TRNG entropy collection is valid.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	EHR_DATA			A value of 1 indicates that collection of bits in the TRNG is completed, and data can be read from EHR_DATA registers.
			NotValid	0	Collection of bits not valid.
			Valid	1	Collection of bits valid.

6.6.12.5.5.6 EHR_DATA[0]

Address offset: 0x114

The entropy holding registers (EHR) hold 192-bits random data collected by the TRNG.

The initial EHR_DATA[0] register holds the least significant bits [31:0] of the random data value.

These registers are readable if register TRNG_VALID on page 279 is `Valid`. Reading register EHR_DATA[5] will clear the content, reset TRNG_VALID, and start a new 192 bits collection period.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	VALUE			Random data value.

6.6.12.5.5.7 EHR_DATA[1]

Address offset: 0x118

The entropy holding registers (EHR) hold 192-bits random data collected by the TRNG.

The initial EHR_DATA[0] register holds the least significant bits [31:0] of the random data value.

These registers are readable if register TRNG_VALID on page 279 is `Valid`. Reading register EHR_DATA[5] will clear the content, reset TRNG_VALID, and start a new 192 bits collection period.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	VALUE			Random data value.

6.6.12.5.5.8 EHR_DATA[2]

Address offset: 0x11C

The entropy holding registers (EHR) hold 192-bits random data collected by the TRNG.

The initial EHR_DATA[0] register holds the least significant bits [31:0] of the random data value.

These registers are readable if register TRNG_VALID on page 279 is `Valid`. Reading register EHR_DATA[5] will clear the content, reset TRNG_VALID, and start a new 192 bits collection period.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	VALUE			Random data value.

6.6.12.5.5.9 EHR_DATA[3]

Address offset: 0x120

The entropy holding registers (EHR) hold 192-bits random data collected by the TRNG.

The initial EHR_DATA[0] register holds the least significant bits [31:0] of the random data value.

These registers are readable if register TRNG_VALID on page 279 is `Valid`. Reading register EHR_DATA[5] will clear the content, reset TRNG_VALID, and start a new 192 bits collection period.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	VALUE			Random data value.

6.6.12.5.5.10 EHR_DATA[4]

Address offset: 0x124

The entropy holding registers (EHR) hold 192-bits random data collected by the TRNG.

The initial EHR_DATA[0] register holds the least significant bits [31:0] of the random data value.

These registers are readable if register TRNG_VALID on page 279 is `Valid`. Reading register EHR_DATA[5] will clear the content, reset TRNG_VALID, and start a new 192 bits collection period.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	R	VALUE																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x0000FFFF				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW	VALUE																																	

Number of CPU clock cycles between two consecutive noise source samples.

If the von Neumann corrector is bypassed, the minimum value set in this register must not be smaller than decimal 17.

6.6.12.5.5.14 AUTOCORR_STATISTIC

Address offset: 0x134

Statistics counter for autocorrelation test activations. Statistics collection is stopped if one of the counters reach its limit of all ones.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID														B	B	B	B	B	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	AUTOCORR_TRYS																																	

Count each time an autocorrelation test starts. Any write to the field resets the counter.

B	RW	AUTOCORR_FAILS			Count each time an autocorrelation test fails. Any write to the field resets the counter.

6.6.12.5.5.15 TRNG_DEBUG

Address offset: 0x138

Debug register for the TRNG. This register is used to bypass TRNG tests in hardware.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																C	B	A	
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	VNC_BYPASS																																	

Bypass the von Neumann corrector post-processing test, including the 32 consecutive bits test.

			Disabled	0	von Neumann corrector post-processing is active
			Enabled	1	Bypass the von Neumann corrector
B	RW	CRNGT_BYPASS			Bypass the Continuous Random Number Generator Test (CRNGT).
			Disabled	0	CRNGT is active
			Enabled	1	Bypass CRNGT
C	RW	AUTOCORR_BYPASS			Bypass the autocorrelation test.
			Disabled	0	Autocorrelation test is active
			Enabled	1	Bypass the autocorrelation test

6.6.12.5.5.16 RNG_SW_RESET

Address offset: 0x140

Reset the RNG engine.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	W	RESET			Writing any value to this address resets the RNG engine. The reset takes 4 CPU clock cycles to complete.
			Enable	1	Reset RNG engine.

6.6.12.5.5.17 RNG_BUSY

Address offset: 0x1B8

Status register for RNG engine activity.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	STATUS			RNG engine status.
			Idle	0	RNG engine is idle
			Busy	1	RNG engine is busy
B	R	TRNG_STATUS			TRNG status.
			Idle	0	TRNG is idle
			Busy	1	TRNG is busy

6.6.12.5.5.18 TRNG_RESET

Address offset: 0x1BC

Reset the TRNG, including internal counter of collected bits and registers EHR_DATA and TRNG_VALID.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	W	RESET			Writing any value to this address resets the internal bits counter and registers EHR_DATA and TRNG_VALID. Register NOISE_SOURCE must be disabled in order for the reset to take place.
			Enable	1	Reset TRNG.

6.6.12.5.5.19 RNG_HW_FLAGS

Address offset: 0x1C0

Hardware configuration of RNG engine. Reset value holds the supported features.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				H G F E D C B A
Reset 0x0000000F				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1**

ID	R/W	Field	Value ID	Value	Description
A	R	EHR_WIDTH			Data width supported by the entropy collector
			128Bits	0	128 bits EHR width
			192Bits	1	192 bits EHR width
B	R	CRNGT_EXISTS			If this flag is set, the engine include support for continuous random number generator test.
C	R	AUTOCORR_EXISTS			If this flag is set, the engine include support for autocorrelation test.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				H G F E D C B A
Reset 0x0000000F				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
D	R	BYPASS_EXISTS			If this flag is set, the engine include support for bypassing TRNG tests.
E	R	PRNG_EXISTS			If this flag is set, the engine include a pseudo-random number generator.
F	R	KAT_EXISTS			If this flag is set, the engine include support for known answer tests.
G	R	RESEEDING_EXISTS			If this flag is set, the engine include support for automatic reseeding.
H	R	RNG_USE_5_SBOXES			
			Disable	0	20 SBOX AES
			Enable	1	5 SBOX AES

6.6.12.5.5.20 RNG_CLK

Address offset: 0x1C4

Control clock for the RNG engine.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	ENABLE			Enables clock for the RNG engine.
			Disable	0	Disable clock for RNG engine.
			Enable	1	Enable clock for RNG engine.

6.6.12.5.5.21 RNG_DMA

Address offset: 0x1C8

Writing to this register enables the RNG DMA engine.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ENABLE			
			Disable	0	Disable RNG DMA engine
			Enable	1	Enable RNG DMA engine This value is cleared when the RNG DMA engine completes its operation.

6.6.12.5.5.22 RNG_DMA_ROSC_LEN

Address offset: 0x1CC

This register defines which ring oscillator length configuration should be used when using the RNG DMA engine.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ROSC1			Use shortest ROSC1 ring oscillator configuration.
			Disable	0	Disable ROSC1
			Enable	1	Enable ROSC1

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
B	RW	ROSC2			Use ROSC2 ring oscillator configuration.
			Disable	0	Disable ROSC2
			Enable	1	Enable ROSC2
C	RW	ROSC3			Use ROSC3 ring oscillator configuration.
			Disable	0	Disable ROSC3
			Enable	1	Enable ROSC3
D	RW	ROSC4			Use longest ROSC4 ring oscillator configuration.
			Disable	0	Disable ROSC4
			Enable	1	Enable ROSC4

6.6.12.5.5.23 RNG_DMA_SRAM_ADDR

Address offset: 0x1D0

This register defines the start address in TRNG SRAM for the TRNG data to be collected by the RNG DMA engine.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RNG_SRAM_DMA_ADDR			Start address of the TRNG data in TRNG SRAM.

6.6.12.5.5.24 RNG_DMA_SAMPLES_NUM

Address offset: 0x1D4

This register defines the number of 192-bits samples that the RNG DMA engine collects per run.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RNG_SAMPLES_NUM			Defines the number of 192-bits samples that the DMA engine collects per run.

6.6.12.5.5.25 RNG_WATCHDOG_VAL

Address offset: 0x1D8

This register defines the maximum number of CPU clock cycles per TRNG collection of 192-bits samples. If the number of cycles for a collection exceeds this threshold the WATCHDOG interrupt is triggered.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RNG_WATCHDOG_VAL			Defines the maximum number of CPU clock cycles per TRNG collection of 192-bits samples. If the number of cycles for a collection exceeds this threshold the WATCHDOG interrupt is triggered.

6.6.12.5.5.26 RNG_DMA_BUSY

Address offset: 0x1DC

Status register for RNG DMA engine activity.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																									C	C	C	C	C	C	C	C	B	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	R	STATUS																																	
			Idle	0																															
			Busy	1																															
B	R	ROSC_LEN																																	
			ROSC1	0																															
			ROSC2	1																															
			ROSC3	2																															
			ROSC4	3																															
C	R	NUM_OF_SAMPLES																																	

ID	R/W	Field	Value ID	Value	Description
A	R	STATUS			RNG DMA engine status.
			Idle	0	RNG DMA engine is idle
			Busy	1	RNG DMA engine is busy
B	R	ROSC_LEN			The active ring oscillator length configuration used by the RNG DMA engine.
			ROSC1	0	Shortest ROSC1 ring oscillator configuration used.
			ROSC2	1	ROSC2 ring oscillator configuration used.
			ROSC3	2	ROSC3 ring oscillator configuration used.
			ROSC4	3	Longest ROSC4 ring oscillator configuration used.
C	R	NUM_OF_SAMPLES			Number of samples already collected using the current ring oscillator configuration.

6.6.13 Host integration

This chapter describes host registers used to control CRYPTOCELL behavior.

6.6.13.1 CTL interface

The CTL interface controls the cryptographic flow and provide busy status for individual components in the CRYPTOCELL subsystem.

6.6.13.1.1 Registers

Instances

Instance	Base address	Description
CC_CTL	0x5002B000	CRYPTOCELL CTL interface

Register overview

Register	Offset	Description
CRYPTO_CTL	0x900	Defines the cryptographic flow.
CRYPTO_BUSY	0x910	Status register for cryptographic cores engine activity.
HASH_BUSY	0x91C	Status register for HASH engine activity.
CONTEXT_ID	0x930	A general-purpose read/write register.

6.6.13.1.1.1 CRYPTO_CTL

Address offset: 0x900

Defines the cryptographic flow.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	W	MODE			Configure the cryptographic engine mode.
			Bypass	0	Bypass cryptographic engine
			AESActive	1	Use AES engine
			AESToHashActive	2	Pipe AES engine output to HASH engine input
			AESAndHashActive	3	Process input using both AES and HASH engine in parallell
			HashActive	7	Use HASH engine
			AESMACAndBypassActive	9	Calculate AES MAC and bypass
			AESToHashAndDOUTActive	10	Pipe AES engine output to HASH engine input. The resulting digest output is piped to DOUT buffer.
			ChaChaActive	16	Use CHACHA engine

6.6.13.1.1.2 CRYPTO_BUSY

Address offset: 0x910

Status register for cryptographic cores engine activity.

This register will be asserted whenever register AES_BUSY on page 231 or register HASH_BUSY on page 287 is asserted or when register DIN_FIFO_EMPTY on page 291 indicate that the DIN FIFO is not empty.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	STATUS			Cryptographic core engines status.
			Idle	0	Cryptographic core engines are idle
			Busy	1	Cryptographic core engines are busy

6.6.13.1.1.3 HASH_BUSY

Address offset: 0x91C

Status register for HASH engine activity.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	STATUS			Hash engine status.
			Idle	0	HASH engine is idle
			Busy	1	HASH engine is busy

6.6.13.1.1.4 CONTEXT_ID

Address offset: 0x930

A general-purpose read/write register.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																									A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	CONTEXT_ID			Context ID

6.6.13.2 DIN DMA engine

The Data IN (DIN) DMA engine transfers data into the CRYPTOCELL subsystem and its various cryptographic engines.

The DIN DMA engine provides a comprehensive interface for to facilitate the transfer of data from the CPU or memory to the cryptographic engines. It includes a variety of registers that control direct data buffering, DMA operations, and data flow management.

Maximum DMA transaction size is limited to 2^{16}-1 bytes. If a DMA transaction is configured with a payload size above the maximum DMA transaction size limit, the DMA engine must be reset before being functional again using register DIN_SW_RESET on page 290.

The flow demonstrated in Cryptographic flow on page 225 shows how the DIN DMA engine is configured to provide data to the AES engine using registers SRC_MEM_ADDR on page 289 and SRC_MEM_SIZE on page 289 to define the input source address and number of input bytes, respectively.

6.6.13.2.1 Registers

Instances

Instance	Base address	Description
CC_DIN	0x5002B000	CRYPTOCELL DIN DMA engine

Register overview

Register	Offset	Description
DIN_BUFFER	0xC00	Used by CPU to write data directly to the DIN buffer, which is then sent to the cryptographic engines for processing.
DIN_DMA_MEM_BUSY	0xC20	Status register for DIN DMA engine activity when accessing memory.
SRC_MEM_ADDR	0xC28	Data source address in memory.
SRC_MEM_SIZE	0xC2C	The number of bytes to be read from memory. Writing to this register triggers the DMA operation.
SRC_SRAM_ADDR	0xC30	Data source address in RNG SRAM.
SRC_SRAM_SIZE	0xC34	The number of bytes to be read from RNG SRAM. Writing to this register triggers the DMA operation.
DIN_DMA_SRAM_BUSY	0xC38	Status register for DIN DMA engine activity when accessing RNG SRAM.
DIN_DMA_SRAM_ENDIANNESS	0xC3C	Configure the endianness of DIN DMA transactions towards RNG SRAM.
DIN_SW_RESET	0xC44	Reset the DIN DMA engine.
DIN_CPU_DATA	0xC48	Specifies the number of bytes the CPU will write to the DIN_BUFFER, ensuring the cryptographic engine processes the correct amount of data.
DIN_WRITE_ALIGN	0xC4C	Indicates that the next CPU write to the DIN_BUFFER is the last in the sequence. This is needed only when the data size is NOT modulo 4 (e.g. HASH padding).
DIN_FIFO_EMPTY	0xC50	Register indicating if DIN FIFO is empty and if more data can be accepted.
DIN_FIFO_RESET	0xC58	Reset the DIN FIFO, effectively clearing the FIFO for new data.

6.6.13.2.1.1 DIN_BUFFER

Address offset: 0xC00

Used by CPU to write data directly to the DIN buffer, which is then sent to the cryptographic engines for processing.

The number of bytes to write is defined in DIN_CPU_DATA on page 291.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	DATA																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	R	STATUS																																	
			Idle	0																															
			Busy	1																															

6.6.13.2.1.2 DIN_DMA_MEM_BUSY

Address offset: 0xC20

Status register for DIN DMA engine activity when accessing memory.

6.6.13.2.1.3 SRC_MEM_ADDR

Address offset: 0xC28

Data source address in memory.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	ADDR																																	

6.6.13.2.1.4 SRC_MEM_SIZE

Address offset: 0xC2C

The number of bytes to be read from memory. Writing to this register triggers the DMA operation.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				C	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	SIZE																																	
B	W	FIRST																																	
C	W	LAST																																	

6.6.13.2.1.5 SRC_SRAM_ADDR

Address offset: 0xC30

Data source address in RNG SRAM.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR			Source address in RNG SRAM.

6.6.13.2.1.6 SRC_SRAM_SIZE

Address offset: 0xC34

The number of bytes to be read from RNG SRAM. Writing to this register triggers the DMA operation.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	SIZE			Total number of bytes to read from RNG SRAM.

6.6.13.2.1.7 DIN_DMA_SRAM_BUSY

Address offset: 0xC38

Status register for DIN DMA engine activity when accessing RNG SRAM.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	STATUS			DIN RNG SRAM DMA engine status.
			Idle	0	DIN RNG SRAM DMA engine is idle
			Busy	1	DIN RNG SRAM DMA engine is busy

6.6.13.2.1.8 DIN_DMA_SRAM_ENDIANNESS

Address offset: 0xC3C

Configure the endianness of DIN DMA transactions towards RNG SRAM.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ENDIAN			Endianness of DIN DMA transactions towards RNG SRAM. The default value is little-endian.
			LittleEndian	0	Use little-endian format for RNG SRAM DMA transactions
			BigEndian	1	Use big-endian format for RNG SRAM DMA transactions

6.6.13.2.1.9 DIN_SW_RESET

Address offset: 0xC44

Reset the DIN DMA engine.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	RESET			Writing any value to this address resets the DIN DMA engine. The reset takes 4 CPU clock cycles to complete.
			Enable	1	Reset DIN DMA engine.

6.6.13.2.1.10 DIN_CPU_DATA

Address offset: 0xC48

Specifies the number of bytes the CPU will write to the DIN_BUFFER, ensuring the cryptographic engine processes the correct amount of data.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	SIZE			When using CPU direct write to the DIN_BUFFER, the size of input data in bytes should be written to this register.

6.6.13.2.1.11 DIN_WRITE_ALIGN

Address offset: 0xC4C

Indicates that the next CPU write to the DIN_BUFFER is the last in the sequence. This is needed only when the data size is NOT modulo 4 (e.g. HASH padding).

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	LAST			Next CPU write to the DIN_BUFFER is the last word.
			Confirm	1	The next CPU write is the last in the sequence.

6.6.13.2.1.12 DIN_FIFO_EMPTY

Address offset: 0xC50

Register indicating if DIN FIFO is empty and if more data can be accepted.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
A	R	STATUS			DIN FIFO status
			NotEmpty	0	DIN FIFO is not empty
			Empty	1	DIN FIFO is empty, and more data can be accepted

6.6.13.2.1.13 DIN_FIFO_RESET

Address offset: 0xC58

Reset the DIN FIFO, effectively clearing the FIFO for new data.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	RESET																																	
			Enable	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	RESET																																	
			Enable	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	RESET																																	
			Enable	1																															

6.6.13.3 DOUT DMA engine

The Data OUT (DOUT) DMA engine transfers data from the CRYPTOCELL subsystem and its various cryptographic engines.

The DOUT DMA engine provides a comprehensive interface for to facilitate the transfer of data to the CPU or memory from the cryptographic engines. It includes a variety of registers that control direct data buffering, DMA operations, and data flow management.

Maximum DMA transaction size is limited to 2^{16}-1 bytes. If a DMA transaction is configured with a payload size above the maximum DMA transaction size limit, the DMA engine must be reset before being functional again using register DOUT_SW_RESET on page 295.

The flow demonstrated in Cryptographic flow on page 225 shows how the DOUT DMA engine is configured to output data from the AES engine using registers DST_MEM_ADDR on page 293 and DST_MEM_SIZE on page 293 to define the output source address and number of output bytes, respectively.

6.6.13.3.1 Registers

Instances

Instance	Base address	Description
CC_DOUT	0x5002B000	CRYPTOCELL DOUT DMA engine

Register overview

Register	Offset	Description
DOUT_BUFFER	0xC00	Cryptographic results directly accessible by the CPU.
DOUT_DMA_MEM_BUSY	0xD20	Status register for DOUT DMA engine activity when accessing memory.
DST_MEM_ADDR	0xD28	Data destination address in memory.
DST_MEM_SIZE	0xD2C	The number of bytes to be written to memory.
DST_SRAM_ADDR	0xD30	Data destination address in RNG SRAM.
DST_SRAM_SIZE	0xD34	The number of bytes to be written to RNG SRAM.
DOUT_DMA_SRAM_BUSY	0xD38	Status register for DOUT DMA engine activity when accessing RNG SRAM.
DOUT_DMA_SRAM_ENDIANNESS	0xD3C	Configure the endianness of DOUT DMA transactions towards RNG SRAM.
DOUT_READ_ALIGN	0xD44	Indication that the next CPU read from the DOUT_BUFFER is the last in the sequence. This is needed only when the data size is NOT modulo 4 (e.g. HASH padding).
DOUT_FIFO_EMPTY	0xD50	Register indicating if DOUT FIFO is empty or if more data will come.
DOUT_SW_RESET	0xD58	Reset the DOUT DMA engine.

6.6.13.3.1.1 DOUT_BUFFER

Address offset: 0xC00

Cryptographic results directly accessible by the CPU.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	R	DATA																																	

This address can be used by the CPU to read data directly from the DOUT buffer.

6.6.13.3.1.2 DOUT_DMA_MEM_BUSY

Address offset: 0xD20

Status register for DOUT DMA engine activity when accessing memory.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	R	STATUS																																	
			Idle	0																															
			Busy	1																															

DOUT memory DMA engine status.

DOUT memory DMA engine is idle

DOUT memory DMA engine is busy

6.6.13.3.1.3 DST_MEM_ADDR

Address offset: 0xD28

Data destination address in memory.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	ADDR																																	

Destination address in memory.

6.6.13.3.1.4 DST_MEM_SIZE

Address offset: 0xD2C

The number of bytes to be written to memory.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				C	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	SIZE																																	
B	W	FIRST																																	
C	W	LAST																																	

Total number of bytes to write to memory.

This field is reserved

This field is reserved

6.6.13.3.1.5 DST_SRAM_ADDR

Address offset: 0xD30

Data destination address in RNG SRAM.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	ADDR																																	

Destination address in RNG SRAM.

6.6.13.3.1.6 DST_SRAM_SIZE

Address offset: 0xD34

The number of bytes to be written to RNG SRAM.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	SIZE																																	

Total number of bytes to write to RNG SRAM.

6.6.13.3.1.7 DOUT_DMA_SRAM_BUSY

Address offset: 0xD38

Status register for DOUT DMA engine activity when accessing RNG SRAM.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	R	STATUS																																	
			Idle	0																															
			Busy	1																															

DOUT RNG SRAM DMA engine status.

DOUT RNG SRAM DMA engine is idle

DOUT RNG SRAM DMA engine is busy

6.6.13.3.1.8 DOUT_DMA_SRAM_ENDIANNESS

Address offset: 0xD3C

Configure the endianness of DOUT DMA transactions towards RNG SRAM.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	ENDIAN																																	
			LittleEndian	0																															
			BigEndian	1																															

Endianness of DOUT DMA transactions towards RNG SRAM. The default value is little-endian.

Use little-endian format for RNG SRAM DMA transactions

Use big-endian format for RNG SRAM DMA transactions

6.6.13.3.1.9 DOUT_READ_ALIGN

Address offset: 0xD44

Indication that the next CPU read from the DOUT_BUFFER is the last in the sequence. This is needed only when the data size is NOT modulo 4 (e.g. HASH padding).

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	LAST			Next CPU read from the DOUT_BUFFER is the last word, and the remaining read aligned content can be flushed.
			Flush	1	Flush the remaining read aligned content.

6.6.13.3.1.10 DOUT_FIFO_EMPTY

Address offset: 0xD50

Register indicating if DOUT FIFO is empty or if more data will come.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
A	R	STATUS			DOUT FIFO status
			NotEmpty	0	DOUT FIFO is not empty, and more data will come
			Empty	1	DOUT FIFO is empty

6.6.13.3.1.11 DOUT_SW_RESET

Address offset: 0xD58

Reset the DOUT DMA engine.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	RESET			Writing any value to this address resets the DOUT DMA engine. The reset takes 4 CPU clock cycles to complete.
			Enable	1	Reset DOUT DMA engine.

6.6.13.4 HOST register interface

The HOST_RGF interface contains registers for CRYPTOCELL interrupt handling, configuring CRYPTOCELL lifecycle state and CRYPTOCELL key management where different cryptographic key inputs can be connected to the AES engine.

Use of the CRYPTOCELL K_{PRTL} key or the device root key K_{DR} is selected using this interface. Availability and configuration of these two key types are typically controlled from an immutable bootloader. Once CRYPTOCELL has been correctly configured it will be possible for an application to either use session keys directly or perform cryptographic operations with the device root key key K_{DR} without having access to the key value.

6.6.13.4.1 Registers

Instances

Instance	Base address	Description
CC_HOST_RGF	0x5002B000	CRYPTOCELL HOST register interface

Register overview

Register	Offset	Description
IRR	0xA00	Interrupt request register. Each bit of this register holds the interrupt status of a single interrupt source. If corresponding IMR bit is unmasked, an interrupt is generated.
IMR	0xA04	Interrupt mask register. Each bit of this register holds the mask of a single interrupt source.
ICR	0xA08	Interrupt clear register. Writing a 1 bit into a field in this register will clear the corresponding bit in IRR.
ENDIANNESS	0xA0C	This register defines the endianness of the Host-accessible registers, and can only be written once.
HOST_SIGNATURE	0xA24	This register holds the CRYPTOCELL subsystem signature. See reset value.
HOST_BOOT	0xA28	Hardware configuration of the CRYPTOCELL subsystem. Reset value holds the supported features.
HOST_CRYPTOKEY_SEL	0xA38	AES hardware key select.
HOST_IOT_KPRTL_LOCK	0xA4C	This write-once register is the K_PRTL lock register. When this register is set, K_PRTL cannot be used and a zeroed key will be used instead. The value of this register is saved in the CRYPTOCELL AO power domain.
HOST_IOT_KDR0	0xA50	This register holds bits 31:0 of K_DR. The value of this register is saved in the CRYPTOCELL AO power domain. Reading from this address returns the K_DR valid status indicating if K_DR is successfully retained.
HOST_IOT_KDR1	0xA54	This register holds bits 63:32 of K_DR. The value of this register is saved in the CRYPTOCELL AO power domain.
HOST_IOT_KDR2	0xA58	This register holds bits 95:64 of K_DR. The value of this register is saved in the CRYPTOCELL AO power domain.
HOST_IOT_KDR3	0xA5C	This register holds bits 127:96 of K_DR. The value of this register is saved in the CRYPTOCELL AO power domain.
HOST_IOT_LCS	0xA60	Controls life-cycle state (LCS) for CRYPTOCELL subsystem

6.6.13.4.1.1 IRR

Address offset: 0xA00

Interrupt request register. Each bit of this register holds the interrupt status of a single interrupt source. If corresponding IMR bit is unmasked, an interrupt is generated.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																						G	F	E	D	C	B	A				
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	SRAM_TO_DIN_INT			The RNG SRAM to DIN DMA done interrupt status. This interrupt is asserted when all data was delivered from RNG SRAM to DIN buffer.
B	R	DOUT_TO_SRAM_INT			The DOUT to RNG SRAM DMA done interrupt status. This interrupt is asserted when all data was delivered from DOUT buffer to RNG SRAM.
C	R	MEM_TO_DIN_INT			The memory to DIN DMA done interrupt status. This interrupt is asserted when all data was delivered from memory to DIN buffer.
D	R	DOUT_TO_MEM_INT			The DOUT to memory DMA done interrupt status. This interrupt is asserted when all data was delivered from DOUT buffer to memory.
E	R	AHB_ERR_INT			The AHB error interrupt status.
F	R	PKA_INT			The PKA end of operation interrupt status.
G	R	RNG_INT			The RNG interrupt status.

6.6.13.4.1.2 IMR

Address offset: 0xA04

Interrupt mask register. Each bit of this register holds the mask of a single interrupt source.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				G F E D C B A
Reset 0x01FFFFFF				**0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	SRAM_TO_DIN_MASK			The RNG SRAM to DIN DMA done interrupt mask.
			IRQEnable	0	Do not mask RNG SRAM to DIN DMA done interrupt i.e. interrupt is generated
			IRQDisable	1	Mask RNG SRAM to DIN DMA done interrupt i.e. no interrupt is generated
B	RW	DOUT_TO_SRAM_MASK			The DOUT to RNG SRAM DMA done interrupt mask.
			IRQEnable	0	Do not mask DOUT to RNG SRAM DMA done interrupt i.e. interrupt is generated
			IRQDisable	1	Mask DOUT to RNG SRAM DMA done interrupt i.e. no interrupt is generated
C	RW	MEM_TO_DIN_MASK			The memory to DIN DMA done interrupt mask.
			IRQEnable	0	Do not mask memory to DIN DMA done interrupt i.e. interrupt is generated
			IRQDisable	1	Mask memory to DIN DMA done interrupt i.e. no interrupt is generated
D	RW	DOUT_TO_MEM_MASK			The DOUT to memory DMA done interrupt mask.
			IRQEnable	0	Do not mask DOUT to memory DMA done interrupt i.e. interrupt is generated
			IRQDisable	1	Mask DOUT to memory DMA done interrupt i.e. no interrupt is generated
E	RW	AHB_ERR_MASK			The AHB error interrupt mask.
			IRQEnable	0	Do not mask AHB error interrupt i.e. interrupt is generated
			IRQDisable	1	Mask AHB error interrupt i.e. no interrupt is generated
F	RW	PKA_MASK			The PKA end of operation interrupt mask.
			IRQEnable	0	Do not mask PKA end of operation interrupt i.e. interrupt is generated
			IRQDisable	1	Mask PKA end of operation interrupt i.e. no interrupt is generated
G	RW	RNG_MASK			The RNG interrupt mask.
			IRQEnable	0	Do not mask RNG interrupt i.e. interrupt is generated
			IRQDisable	1	Mask RNG interrupt i.e. no interrupt is generated

6.6.13.4.1.3 ICR

Address offset: 0xA08

Interrupt clear register. Writing a 1 bit into a field in this register will clear the corresponding bit in IRR.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	SRAM_TO_DIN_CLEAR			The RNG SRAM to DIN DMA done interrupt clear.
B	W	DOUT_TO_SRAM_CLEAR			The DOUT to RNG SRAM DMA done interrupt clear.
C	W	MEM_TO_DIN_CLEAR			The memory to DIN DMA done interrupt clear.
D	W	DOUT_TO_MEM_CLEAR			The DOUT to memory DMA done interrupt clear.
E	W	AHB_ERR_CLEAR			The AHB error interrupt clear.
F	W	PKA_CLEAR			The PKA end of operation interrupt clear.
G	W	RNG_CLEAR			The RNG interrupt clear. Register RNG_ISR in the RNG engine must be cleared before this interrupt can be cleared.

6.6.13.4.1.4 ENDIANNESS

Address offset: 0xA0C

This register defines the endianness of the Host-accessible registers, and can only be written once.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DOUT_WR_BG			DOUT write endianness.
			LittleEndian	0	Configure DOUT write as little-endian
			BigEndian	1	Configure DOUT write as big-endian
B	RW	DIN_RD_BG			DIN read endianness.
			LittleEndian	0	Configure DIN read as little-endian
			BigEndian	1	Configure DIN read as big-endian
C	RW	DOUT_WR_WBG			DOUT write word endianness.
			LittleEndian	0	Configure DOUT write word as little-endian
			BigEndian	1	Configure DOUT write word as big-endian
D	RW	DIN_RD_WBG			DIN read word endianness.
			LittleEndian	0	Configure DIN read word as little-endian
			BigEndian	1	Configure DIN read word as big-endian

6.6.13.4.1.5 HOST_SIGNATURE

Address offset: 0xA24

This register holds the CRYPTOCELL subsystem signature. See reset value.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x20E00000				**0 0 1 0 0 0 0 0 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	VALUE			Fixed-value identification signature used by host driver to verify CRYPTOCELL presence at this address.

6.6.13.4.1.6 HOST_BOOT

Address offset: 0xA28

Hardware configuration of the CRYPTOCELL subsystem. Reset value holds the supported features.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				b a Z Y X W V U T S R Q P O N M L K J I H G F F F E D C B A
Reset 0x4622982C				**0 1 0 0 0 1 1 0 0 0 1 0 0 0 1 0 1 0 0 1 1 0 0 0 0 0 1 0 1 1 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	POWER_GATING_EXISTS_LOCAL			If this flag is set, full power gating is implemented
B	R	LARGE_RKEK_LOCAL			If this flag is set, large RKEK is supported
C	R	HASH_IN_FUSES_LOCAL			If this flag is set, HASH in fuses is supported
D	R	EXT_MEM_SECURED_LOCAL			If this flag is set, external secure memory is supported
E	R	RKEK_ECC_EXISTS_LOCAL_N			If this flag is set, RKEK ECC is supported
F	R	SRAM_SIZE_LOCAL			SRAM size
G	R	DSCRPTR_EXISTS_LOCAL			If this flag is set, Descriptors are supported
H	R	PAU_EXISTS_LOCAL			If this flag is set, PAU is supported
I	R	RNG_EXISTS_LOCAL			If this flag is set, the RNG engine is present
J	R	PKA_EXISTS_LOCAL			If this flag is set, the PKA engine is present
K	R	RC4_EXISTS_LOCAL			If this flag is set, the RC4 engine is present
L	R	SHA_512_PRSNT_LOCAL			If this flag is set, the HASH engine supports SHA512
M	R	SHA_256_PRSNT_LOCAL			If this flag is set, the HASH engine supports SHA256
N	R	MD5_PRSNT_LOCAL			If this flag is set, the HASH engine supports MD5
O	R	HASH_EXISTS_LOCAL			If this flag is set, the HASH engine is present

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID		b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	F	F	E		D	C	B	A
Reset 0x4622982C	**0**	**1**	**0**	**0**	**0**	**1**	**1**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**1**	**0**	**1**	**0**	**0**	**1**	**1**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**1**	**1**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
P	R	C2_EXISTS_LOCAL			If this flag is set, the C2 engine is present
Q	R	DES_EXISTS_LOCAL			If this flag is set, the DES engine is present
R	R	AES_XCBC_MAC_EXISTS_LOCAL			If this flag is set, AES XCBC-MAC mode is supported
S	R	AES_CMAC_EXISTS_LOCAL			If this flag is set, AES CMAC mode is supported
T	R	AES_CCM_EXISTS_LOCAL			If this flag is set, AES CCM mode is supported
U	R	AES_XEX_HW_T_CALC_LOCAL			If this flag is set, AES XEX mode T-value calculation in HW is supported
V	R	AES_XEX_EXISTS_LOCAL			If this flag is set, AES XEX mode is supported
W	R	CTR_EXISTS_LOCAL			If this flag is set, AES CTR mode is supported
X	R	AES_DIN_BYTE_RESOLUTION_LOCAL			If this flag is set, the AES engine data input support byte size resolution
Y	R	TUNNELING_ENB_LOCAL			If this flag is set, the AES engine supports tunneling operations
Z	R	SUPPORT_256_192_KEY_LOCAL			If this flag is set, the AES engine supports 192/256 bits key sizes
a	R	ONLY_ENCRYPT_LOCAL			If this flag is set, the AES engine only support encryption
b	R	AES_EXISTS_LOCAL			If this flag is set, the AES engine is present

6.6.13.4.1.7 HOST_CRYPTOKEY_SEL

Address offset: 0xA38

AES hardware key select.

If the HOST_IOT_KPRTL_LOCK register is set, and the HOST_CRYPTOKEY_SEL register set to 1, then the HW key that is connected to the AES engine is zero

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																															A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	HOST_CRYPTOKEY_SEL			Select the source of the HW key that is used by the AES engine
			K_DR	0	Use device root key K_DR from CRYPTOCELL AO power domain
			K_PRTL	1	Use hard-coded RTL key K_PRTL
			Session	2	Use provided session key

6.6.13.4.1.8 HOST_IOT_KPRTL_LOCK

Address offset: 0xA4C

This write-once register is the K_PRTL lock register. When this register is set, K_PRTL cannot be used and a zeroed key will be used instead. The value of this register is saved in the CRYPTOCELL AO power domain.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW1	HOST_IOT_KPRTL_LOCK			This register is the K_PRTL lock register. When this register is set, K_PRTL cannot be used and a zeroed key will be used instead. The value of this register is saved in the CRYPTOCELL AO power domain.
			Disabled	0	K_PRTL can be selected for use from register HOST_CRYPTOKEY_SEL
			Enabled	1	K_PRTL has been locked until next power-on reset (POR). If K_PRTL is selected anyway, a zeroed key will be used instead.

6.6.13.4.1.9 HOST_IOT_KDR0

Address offset: 0xA50

This register holds bits 31:0 of K_DR. The value of this register is saved in the CRYPTOCELL AO power domain. Reading from this address returns the K_DR valid status indicating if K_DR is successfully retained.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW1	HOST_IOT_KDR0																																	
			NotRetained	0																															
			Retained	1																															

ID	R/W	Field	Value ID	Value	Description
A	RW1	HOST_IOT_KDR0			This register holds bits 31:0 of K_DR. The value of this register is saved in the CRYPTOCELL AO power domain. Reading from this address returns the K_DR valid status indicating if K_DR is successfully retained. Write: K_DR bits 31:0.
			NotRetained	0	Read: 128 bits K_DR key value is not yet retained in the CRYPTOCELL AO power domain.
			Retained	1	Read: 128 bits K_DR key value is successfully retained in the CRYPTOCELL AO power domain.

6.6.13.4.1.10 HOST_IOT_KDR1

Address offset: 0xA54

This register holds bits 63:32 of K_DR. The value of this register is saved in the CRYPTOCELL AO power domain.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W1	HOST_IOT_KDR1																																	

ID	R/W	Field	Value ID	Value	Description
A	W1	HOST_IOT_KDR1			K_DR bits 63:32

6.6.13.4.1.11 HOST_IOT_KDR2

Address offset: 0xA58

This register holds bits 95:64 of K_DR. The value of this register is saved in the CRYPTOCELL AO power domain.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W1	HOST_IOT_KDR2																																	

ID	R/W	Field	Value ID	Value	Description
A	W1	HOST_IOT_KDR2			K_DR bits 95:64

6.6.13.4.1.12 HOST_IOT_KDR3

Address offset: 0xA5C

This register holds bits 127:96 of K_DR. The value of this register is saved in the CRYPTOCELL AO power domain.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W1	HOST_IOT_KDR3			K_DR bits 127:96

6.6.13.4.1.13 HOST_IOT_LCS

Address offset: 0xA60

Controls life-cycle state (LCS) for CRYPTOCELL subsystem

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A A A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW1	LCS			Life-cycle state value. This field is write-once per reset.
			DebugEnable	0	CC310 operates in debug mode
			Secure	2	CC310 operates in secure mode
B	R	LCS_IS_VALID			Read-only field. Indicates if CRYPTOCELL LCS has been successfully configured since last reset.
			Invalid	0	Valid LCS not yet retained in the CRYPTOCELL AO power domain
			Valid	1	Valid LCS successfully retained in the CRYPTOCELL AO power domain

6.6.13.5 RNG SRAM interface

The RNG_SRAM interface enable reading and writing data to RNG SRAM.

6.6.13.5.1 Registers

Instances

Instance	Base address	Description
CC_RNG_SRAM	0x5002B000	CRYPTOCELL RNG SRAM interface

Register overview

Register	Offset	Description
SRAM_DATA	0xF00	Read/Write data from RNG SRAM
SRAM_ADDR	0xF04	First address given to RNG SRAM DMA for read/write transactions from/to RNG SRAM.
SRAM_DATA_READY	0xF08	RNG SRAM DMA engine is ready to read/write from/to RNG SRAM.

6.6.13.5.1.1 SRAM_DATA

Address offset: 0xF00

Read/Write data from RNG SRAM

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	SRAM_DATA			32 bits DMA read/write from/to RNG SRAM. A 'read' or 'write' operation to this register will trigger the DMA address to be automatically incremented.

6.6.13.5.1.2 SRAM_ADDR

Address offset: 0xF04

First address given to RNG SRAM DMA for read/write transactions from/to RNG SRAM.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	SRAM_ADDR			RNG SRAM starting address

6.6.13.5.1.3 SRAM_DATA_READY

Address offset: 0xF08

RNG SRAM DMA engine is ready to read/write from/to RNG SRAM.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
A	R	SRAM_READY			RNG SRAM DMA status.
			Busy	0	DMA is busy
			Idle	1	DMA is idle

6.6.13.6 MISC interface

The MISC interface controls clocks for the individual engines within the CRYPTOCELL subsystem.

Each cryptographic engine has an individual register for performing clock gating. Engine clock status is displayed in register CLK_STATUS on page 304.

> **Note:** Clock control for the RNG engine on page 272 is handled by register RNG_CLK on page 284 and not through the MISC interface.

6.6.13.6.1 Registers

Instances

Instance	Base address	Description
CC_MISC	0x5002B000	CRYPTOCELL MISC interface

Register overview

Register	Offset	Description
AES_CLK	0x810	Clock control for the AES engine.
HASH_CLK	0x818	Clock control for the HASH engine.
PKA_CLK	0x81C	Clock control for the PKA engine.
DMA_CLK	0x820	Clock control for the DMA engines.
CLK_STATUS	0x824	CRYPTOCELL clocks status register.
CHACHA_CLK	0x858	Clock control for the CHACHA engine.

6.6.13.6.1.1 AES_CLK

Address offset: 0x810

Clock control for the AES engine.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	ENABLE			Enables clock for the AES engine.
			Disable	0	Disable clock for the AES engine.
			Enable	1	Enable clock for the AES engine.

6.6.13.6.1.2 HASH_CLK

Address offset: 0x818

Clock control for the HASH engine.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	ENABLE			Enables clock for the HASH engine.
			Disable	0	Disable clock for the HASH engine.
			Enable	1	Enable clock for the HASH engine.

6.6.13.6.1.3 PKA_CLK

Address offset: 0x81C

Clock control for the PKA engine.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	ENABLE			Enables clock for the PKA engine.
			Disable	0	Disable clock for the PKA engine.
			Enable	1	Enable clock for the PKA engine.

6.6.13.6.1.4 DMA_CLK

Address offset: 0x820

Clock control for the DMA engines.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	ENABLE			Enables clock for the DMA engines.
			Disable	0	Disable clock for the DMA engines.
			Enable	1	Enable clock for the DMA engines.

6.6.13.6.1.5 CLK_STATUS

Address offset: 0x824

CRYPTOCELL clocks status register.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																											E	D				C	B		A
Reset 0x00000100				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	R	AES_CLK																																	
			Disabled	0																															
			Enabled	1																															
B	R	HASH_CLK																																	
			Disabled	0																															
			Enabled	1																															
C	R	PKA_CLK																																	
			Disabled	0																															
			Enabled	1																															
D	R	CHACHA_CLK																																	
			Disabled	0																															
			Enabled	1																															
E	R	DMA_CLK																																	
			Disabled	0																															
			Enabled	1																															

6.6.13.6.1.6 CHACHA_CLK

Address offset: 0x858

Clock control for the CHACHA engine.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	ENABLE																																	
			Disable	0																															
			Enable	1																															

6.7 ECB — AES electronic codebook mode encryption

The AES electronic codebook mode encryption (ECB) can be used for a range of cryptographic functions like hash generation, digital signatures, and keystream generation for data encryption/decryption. The ECB encryption block supports 128 bit AES encryption (encryption only, not decryption).

AES ECB operates with EasyDMA access to system Data RAM for in-place operations on cleartext and ciphertext during encryption. ECB uses the same AES core as the CCM and AAR blocks, and is an asynchronous operation which may not complete if the AES core is busy.

AES ECB features:

- 128 bit AES encryption
- Supports standard AES ECB block encryption
- Memory pointer support

- DMA data transfer

AES ECB performs a 128 bit AES block encrypt. At the STARTECB task, data and key is loaded into the algorithm by EasyDMA. When output data has been written back to memory, the ENDECB event is triggered.

AES ECB can be stopped by triggering the STOPECB task.

6.7.1 Shared resources

The ECB, CCM, and AAR share the same AES module. The ECB will always have lowest priority, and if there is a sharing conflict during encryption, the ECB operation will be aborted and an ERRORECB event will be generated.

6.7.2 EasyDMA

The ECB implements an EasyDMA mechanism for reading and writing to the Data RAM. This DMA cannot access the program memory or any other parts of the memory area except RAM.

If the ECBDATAPTR is not pointing to the Data RAM region, an EasyDMA transfer may result in a HardFault or RAM corruption. See Memory on page 21 for more information about the different memory regions.

The EasyDMA will have finished accessing the Data RAM when the ENDECB or ERRORECB is generated.

6.7.3 ECB data structure

Block encrypt input and output is stored in the same data structure. ECBDATAPTR should point to this data structure before STARTECB is initiated.

Property	Address offset	Description
KEY	0	16 byte AES key
CLEARTEXT	16	16 byte AES cleartext input block
CIPHERTEXT	32	16 byte AES ciphertext output block

Table 18: ECB data structure overview

6.7.4 Registers

Instances

Instance	Base address	Description
ECB	0x4000E000	AES electronic code book (ECB) mode block encryption

Register overview

Register	Offset	Description
TASKS_STARTECB	0x000	Start ECB block encrypt
TASKS_STOPECB	0x004	Abort a possible executing ECB operation
EVENTS_ENDECB	0x100	ECB block encrypt complete
EVENTS_ERRORECB	0x104	ECB block encrypt aborted because of a STOPECB task or due to an error
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
ECBDATAPTR	0x504	ECB block encrypt memory pointers

6.7.4.1 TASKS_STARTECB

Address offset: 0x000

Start ECB block encrypt

If a crypto operation is already running in the AES core, the STARTECB task will not start a new encryption and an ERRORECB event will be triggered.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTECB			Start ECB block encrypt If a crypto operation is already running in the AES core, the STARTECB task will not start a new encryption and an ERRORECB event will be triggered.
			Trigger	1	Trigger task

6.7.4.2 TASKS_STOPECB

Address offset: 0x004

Abort a possible executing ECB operation

If a running ECB operation is aborted by STOPECB, the ERRORECB event is triggered.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STOPECB			Abort a possible executing ECB operation If a running ECB operation is aborted by STOPECB, the ERRORECB event is triggered.
			Trigger	1	Trigger task

6.7.4.3 EVENTS_ENDECB

Address offset: 0x100

ECB block encrypt complete

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDECB			ECB block encrypt complete
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.7.4.4 EVENTS_ERRORECB

Address offset: 0x104

ECB block encrypt aborted because of a STOPECB task or due to an error

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ERRORECB			ECB block encrypt aborted because of a STOPECB task or due to an error
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.7.4.5 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ENDECB			Write '1' to enable interrupt for event ENDECB
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	ERRORECB			Write '1' to enable interrupt for event ERRORECB
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.7.4.6 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ENDECB			Write '1' to disable interrupt for event ENDECB
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	ERRORECB			Write '1' to disable interrupt for event ERRORECB
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.7.4.7 ECBDATAPTR

Address offset: 0x504

ECB block encrypt memory pointers

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ECBDATAPTR			Pointer to the ECB data structure (see Table 1 ECB data structure overview)

6.7.5 Electrical specification

6.7.5.1 ECB Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
t_{ECB}	Run time per 16 byte block in all modes			7.2	µs

6.8 EGU — Event generator unit

Event generator unit (EGU) provides support for interlayer signaling. This means providing support for atomic triggering of both CPU execution and hardware tasks, from both firmware (by CPU) and hardware (by PPI). This feature can be used for triggering CPU execution at a lower priority execution from a higher priority execution, or to handle a peripheral's interrupt service routine (ISR) execution at a lower priority for some of its events. However, triggering any priority from any priority is possible.

Listed here are the main EGU features:

- Software-enabled interrupt triggering
- Separate interrupt vectors for every EGU instance
- Up to 16 separate event flags per interrupt for multiplexing

EGU implements a set of tasks which can individually be triggered to generate the corresponding event. For example, the corresponding event for TASKS_TRIGGER[n] is EVENTS_TRIGGERED[n].

6.8.1 Registers

Instances

Instance	Base address	Description
EGU0	0x40014000	Event generator unit 0
EGU1	0x40015000	Event generator unit 1
EGU2	0x40016000	Event generator unit 2
EGU3	0x40017000	Event generator unit 3
EGU4	0x40018000	Event generator unit 4
EGU5	0x40019000	Event generator unit 5

Register overview

Register	Offset	Description
TASKS_TRIGGER[0]	0x000	Trigger 0 for triggering the corresponding TRIGGERED[0] event
TASKS_TRIGGER[1]	0x004	Trigger 1 for triggering the corresponding TRIGGERED[1] event
TASKS_TRIGGER[2]	0x008	Trigger 2 for triggering the corresponding TRIGGERED[2] event
TASKS_TRIGGER[3]	0x00C	Trigger 3 for triggering the corresponding TRIGGERED[3] event
TASKS_TRIGGER[4]	0x010	Trigger 4 for triggering the corresponding TRIGGERED[4] event
TASKS_TRIGGER[5]	0x014	Trigger 5 for triggering the corresponding TRIGGERED[5] event

Register	Offset	Description
TASKS_TRIGGER[6]	0x018	Trigger 6 for triggering the corresponding TRIGGERED[6] event
TASKS_TRIGGER[7]	0x01C	Trigger 7 for triggering the corresponding TRIGGERED[7] event
TASKS_TRIGGER[8]	0x020	Trigger 8 for triggering the corresponding TRIGGERED[8] event
TASKS_TRIGGER[9]	0x024	Trigger 9 for triggering the corresponding TRIGGERED[9] event
TASKS_TRIGGER[10]	0x028	Trigger 10 for triggering the corresponding TRIGGERED[10] event
TASKS_TRIGGER[11]	0x02C	Trigger 11 for triggering the corresponding TRIGGERED[11] event
TASKS_TRIGGER[12]	0x030	Trigger 12 for triggering the corresponding TRIGGERED[12] event
TASKS_TRIGGER[13]	0x034	Trigger 13 for triggering the corresponding TRIGGERED[13] event
TASKS_TRIGGER[14]	0x038	Trigger 14 for triggering the corresponding TRIGGERED[14] event
TASKS_TRIGGER[15]	0x03C	Trigger 15 for triggering the corresponding TRIGGERED[15] event
EVENTS_TRIGGERED[0]	0x100	Event number 0 generated by triggering the corresponding TRIGGER[0] task
EVENTS_TRIGGERED[1]	0x104	Event number 1 generated by triggering the corresponding TRIGGER[1] task
EVENTS_TRIGGERED[2]	0x108	Event number 2 generated by triggering the corresponding TRIGGER[2] task
EVENTS_TRIGGERED[3]	0x10C	Event number 3 generated by triggering the corresponding TRIGGER[3] task
EVENTS_TRIGGERED[4]	0x110	Event number 4 generated by triggering the corresponding TRIGGER[4] task
EVENTS_TRIGGERED[5]	0x114	Event number 5 generated by triggering the corresponding TRIGGER[5] task
EVENTS_TRIGGERED[6]	0x118	Event number 6 generated by triggering the corresponding TRIGGER[6] task
EVENTS_TRIGGERED[7]	0x11C	Event number 7 generated by triggering the corresponding TRIGGER[7] task
EVENTS_TRIGGERED[8]	0x120	Event number 8 generated by triggering the corresponding TRIGGER[8] task
EVENTS_TRIGGERED[9]	0x124	Event number 9 generated by triggering the corresponding TRIGGER[9] task
EVENTS_TRIGGERED[10]	0x128	Event number 10 generated by triggering the corresponding TRIGGER[10] task
EVENTS_TRIGGERED[11]	0x12C	Event number 11 generated by triggering the corresponding TRIGGER[11] task
EVENTS_TRIGGERED[12]	0x130	Event number 12 generated by triggering the corresponding TRIGGER[12] task
EVENTS_TRIGGERED[13]	0x134	Event number 13 generated by triggering the corresponding TRIGGER[13] task
EVENTS_TRIGGERED[14]	0x138	Event number 14 generated by triggering the corresponding TRIGGER[14] task
EVENTS_TRIGGERED[15]	0x13C	Event number 15 generated by triggering the corresponding TRIGGER[15] task
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt

6.8.1.1 TASKS_TRIGGER[0]

Address offset: 0x000

Trigger 0 for triggering the corresponding TRIGGERED[0] event

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_TRIGGER			Trigger 0 for triggering the corresponding TRIGGERED[0] event
			Trigger	1	Trigger task

6.8.1.2 TASKS_TRIGGER[1]

Address offset: 0x004

Trigger 1 for triggering the corresponding TRIGGERED[1] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_TRIGGER			Trigger 1 for triggering the corresponding TRIGGERED[1] event
			Trigger	1	Trigger task

6.8.1.3 TASKS_TRIGGER[2]

Address offset: 0x008

Trigger 2 for triggering the corresponding TRIGGERED[2] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_TRIGGER			Trigger 2 for triggering the corresponding TRIGGERED[2] event
			Trigger	1	Trigger task

6.8.1.4 TASKS_TRIGGER[3]

Address offset: 0x00C

Trigger 3 for triggering the corresponding TRIGGERED[3] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_TRIGGER			Trigger 3 for triggering the corresponding TRIGGERED[3] event
			Trigger	1	Trigger task

6.8.1.5 TASKS_TRIGGER[4]

Address offset: 0x010

Trigger 4 for triggering the corresponding TRIGGERED[4] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_TRIGGER			Trigger 4 for triggering the corresponding TRIGGERED[4] event
			Trigger	1	Trigger task

6.8.1.6 TASKS_TRIGGER[5]

Address offset: 0x014

Trigger 5 for triggering the corresponding TRIGGERED[5] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_TRIGGER			Trigger 5 for triggering the corresponding TRIGGERED[5] event
			Trigger	1	Trigger task

6.8.1.7 TASKS_TRIGGER[6]

Address offset: 0x018

Trigger 6 for triggering the corresponding TRIGGERED[6] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_TRIGGER			Trigger 6 for triggering the corresponding TRIGGERED[6] event
			Trigger	1	Trigger task

6.8.1.8 TASKS_TRIGGER[7]

Address offset: 0x01C

Trigger 7 for triggering the corresponding TRIGGERED[7] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_TRIGGER			Trigger 7 for triggering the corresponding TRIGGERED[7] event
			Trigger	1	Trigger task

6.8.1.9 TASKS_TRIGGER[8]

Address offset: 0x020

Trigger 8 for triggering the corresponding TRIGGERED[8] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_TRIGGER			Trigger 8 for triggering the corresponding TRIGGERED[8] event
			Trigger	1	Trigger task

6.8.1.10 TASKS_TRIGGER[9]

Address offset: 0x024

Trigger 9 for triggering the corresponding TRIGGERED[9] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_TRIGGER			Trigger 9 for triggering the corresponding TRIGGERED[9] event
			Trigger	1	Trigger task

6.8.1.11 TASKS_TRIGGER[10]

Address offset: 0x028

Trigger 10 for triggering the corresponding TRIGGERED[10] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_TRIGGER			Trigger 10 for triggering the corresponding TRIGGERED[10] event
			Trigger	1	Trigger task

6.8.1.12 TASKS_TRIGGER[11]

Address offset: 0x02C

Trigger 11 for triggering the corresponding TRIGGERED[11] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_TRIGGER			Trigger 11 for triggering the corresponding TRIGGERED[11] event
			Trigger	1	Trigger task

6.8.1.13 TASKS_TRIGGER[12]

Address offset: 0x030

Trigger 12 for triggering the corresponding TRIGGERED[12] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_TRIGGER			Trigger 12 for triggering the corresponding TRIGGERED[12] event
			Trigger	1	Trigger task

6.8.1.14 TASKS_TRIGGER[13]

Address offset: 0x034

Trigger 13 for triggering the corresponding TRIGGERED[13] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_TRIGGER			Trigger 13 for triggering the corresponding TRIGGERED[13] event
			Trigger	1	Trigger task

6.8.1.15 TASKS_TRIGGER[14]

Address offset: 0x038

Trigger 14 for triggering the corresponding TRIGGERED[14] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_TRIGGER			Trigger 14 for triggering the corresponding TRIGGERED[14] event
			Trigger	1	Trigger task

6.8.1.16 TASKS_TRIGGER[15]

Address offset: 0x03C

Trigger 15 for triggering the corresponding TRIGGERED[15] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_TRIGGER			Trigger 15 for triggering the corresponding TRIGGERED[15] event
			Trigger	1	Trigger task

6.8.1.17 EVENTS_TRIGGERED[0]

Address offset: 0x100

Event number 0 generated by triggering the corresponding TRIGGER[0] task

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TRIGGERED			Event number 0 generated by triggering the corresponding TRIGGER[0] task
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.8.1.18 EVENTS_TRIGGERED[1]

Address offset: 0x104

Event number 1 generated by triggering the corresponding TRIGGER[1] task

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TRIGGERED			Event number 1 generated by triggering the corresponding TRIGGER[1] task
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.8.1.19 EVENTS_TRIGGERED[2]

Address offset: 0x108

Event number 2 generated by triggering the corresponding TRIGGER[2] task

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TRIGGERED			Event number 2 generated by triggering the corresponding TRIGGER[2] task
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.8.1.20 EVENTS_TRIGGERED[3]

Address offset: 0x10C

Event number 3 generated by triggering the corresponding TRIGGER[3] task

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TRIGGERED			Event number 3 generated by triggering the corresponding TRIGGER[3] task
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.8.1.21 EVENTS_TRIGGERED[4]

Address offset: 0x110

Event number 4 generated by triggering the corresponding TRIGGER[4] task

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TRIGGERED			Event number 4 generated by triggering the corresponding TRIGGER[4] task
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.8.1.22 EVENTS_TRIGGERED[5]

Address offset: 0x114

Event number 5 generated by triggering the corresponding TRIGGER[5] task

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TRIGGERED			Event number 5 generated by triggering the corresponding TRIGGER[5] task
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.8.1.23 EVENTS_TRIGGERED[6]

Address offset: 0x118

Event number 6 generated by triggering the corresponding TRIGGER[6] task

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TRIGGERED			Event number 6 generated by triggering the corresponding TRIGGER[6] task
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.8.1.24 EVENTS_TRIGGERED[7]

Address offset: 0x11C

Event number 7 generated by triggering the corresponding TRIGGER[7] task

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TRIGGERED			Event number 7 generated by triggering the corresponding TRIGGER[7] task
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.8.1.25 EVENTS_TRIGGERED[8]

Address offset: 0x120

Event number 8 generated by triggering the corresponding TRIGGER[8] task

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TRIGGERED			Event number 8 generated by triggering the corresponding TRIGGER[8] task
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.8.1.26 EVENTS_TRIGGERED[9]

Address offset: 0x124

Event number 9 generated by triggering the corresponding TRIGGER[9] task

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TRIGGERED			Event number 9 generated by triggering the corresponding TRIGGER[9] task
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.8.1.27 EVENTS_TRIGGERED[10]

Address offset: 0x128

Event number 10 generated by triggering the corresponding TRIGGER[10] task

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TRIGGERED			Event number 10 generated by triggering the corresponding TRIGGER[10] task
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.8.1.28 EVENTS_TRIGGERED[11]

Address offset: 0x12C

Event number 11 generated by triggering the corresponding TRIGGER[11] task

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TRIGGERED			Event number 11 generated by triggering the corresponding TRIGGER[11] task
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.8.1.29 EVENTS_TRIGGERED[12]

Address offset: 0x130

Event number 12 generated by triggering the corresponding TRIGGER[12] task

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TRIGGERED			Event number 12 generated by triggering the corresponding TRIGGER[12] task
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.8.1.30 EVENTS_TRIGGERED[13]

Address offset: 0x134

Event number 13 generated by triggering the corresponding TRIGGER[13] task

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TRIGGERED			Event number 13 generated by triggering the corresponding TRIGGER[13] task
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.8.1.31 EVENTS_TRIGGERED[14]

Address offset: 0x138

Event number 14 generated by triggering the corresponding TRIGGER[14] task

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TRIGGERED			Event number 14 generated by triggering the corresponding TRIGGER[14] task
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.8.1.32 EVENTS_TRIGGERED[15]

Address offset: 0x13C

Event number 15 generated by triggering the corresponding TRIGGER[15] task

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TRIGGERED			Event number 15 generated by triggering the corresponding TRIGGER[15] task
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.8.1.33 INTEN

Address offset: 0x300

Enable or disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TRIGGERED[0]			Enable or disable interrupt for event TRIGGERED[0]
			Disabled	0	Disable
			Enabled	1	Enable
B	RW	TRIGGERED[1]			Enable or disable interrupt for event TRIGGERED[1]
			Disabled	0	Disable
			Enabled	1	Enable
C	RW	TRIGGERED[2]			Enable or disable interrupt for event TRIGGERED[2]
			Disabled	0	Disable
			Enabled	1	Enable
D	RW	TRIGGERED[3]			Enable or disable interrupt for event TRIGGERED[3]
			Disabled	0	Disable
			Enabled	1	Enable
E	RW	TRIGGERED[4]			Enable or disable interrupt for event TRIGGERED[4]
			Disabled	0	Disable
			Enabled	1	Enable
F	RW	TRIGGERED[5]			Enable or disable interrupt for event TRIGGERED[5]
			Disabled	0	Disable
			Enabled	1	Enable
G	RW	TRIGGERED[6]			Enable or disable interrupt for event TRIGGERED[6]
			Disabled	0	Disable
			Enabled	1	Enable
H	RW	TRIGGERED[7]			Enable or disable interrupt for event TRIGGERED[7]
			Disabled	0	Disable
			Enabled	1	Enable
I	RW	TRIGGERED[8]			Enable or disable interrupt for event TRIGGERED[8]
			Disabled	0	Disable
			Enabled	1	Enable
J	RW	TRIGGERED[9]			Enable or disable interrupt for event TRIGGERED[9]
			Disabled	0	Disable
			Enabled	1	Enable
K	RW	TRIGGERED[10]			Enable or disable interrupt for event TRIGGERED[10]
			Disabled	0	Disable
			Enabled	1	Enable
L	RW	TRIGGERED[11]			Enable or disable interrupt for event TRIGGERED[11]
			Disabled	0	Disable
			Enabled	1	Enable
M	RW	TRIGGERED[12]			Enable or disable interrupt for event TRIGGERED[12]
			Disabled	0	Disable
			Enabled	1	Enable
N	RW	TRIGGERED[13]			Enable or disable interrupt for event TRIGGERED[13]
			Disabled	0	Disable
			Enabled	1	Enable
O	RW	TRIGGERED[14]			Enable or disable interrupt for event TRIGGERED[14]
			Disabled	0	Disable
			Enabled	1	Enable
P	RW	TRIGGERED[15]			Enable or disable interrupt for event TRIGGERED[15]
			Disabled	0	Disable
			Enabled	1	Enable

6.8.1.34 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																				P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	TRIGGERED[0]																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	TRIGGERED[1]																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	TRIGGERED[2]																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
D	RW	TRIGGERED[3]																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
E	RW	TRIGGERED[4]																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
F	RW	TRIGGERED[5]																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
G	RW	TRIGGERED[6]																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
H	RW	TRIGGERED[7]																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
I	RW	TRIGGERED[8]																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
J	RW	TRIGGERED[9]																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
K	RW	TRIGGERED[10]																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
L	RW	TRIGGERED[11]			Write '1' to enable interrupt for event TRIGGERED[11]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
M	RW	TRIGGERED[12]			Write '1' to enable interrupt for event TRIGGERED[12]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
N	RW	TRIGGERED[13]			Write '1' to enable interrupt for event TRIGGERED[13]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
O	RW	TRIGGERED[14]			Write '1' to enable interrupt for event TRIGGERED[14]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
P	RW	TRIGGERED[15]			Write '1' to enable interrupt for event TRIGGERED[15]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.8.1.35 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	TRIGGERED[0]			Write '1' to disable interrupt for event TRIGGERED[0]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	TRIGGERED[1]			Write '1' to disable interrupt for event TRIGGERED[1]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	TRIGGERED[2]			Write '1' to disable interrupt for event TRIGGERED[2]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	TRIGGERED[3]			Write '1' to disable interrupt for event TRIGGERED[3]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	TRIGGERED[4]			Write '1' to disable interrupt for event TRIGGERED[4]
			Clear	1	Disable
			Disabled	0	Read: Disabled

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
			Enabled	1	Read: Enabled
F	RW	TRIGGERED[5]			Write '1' to disable interrupt for event TRIGGERED[5]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
G	RW	TRIGGERED[6]			Write '1' to disable interrupt for event TRIGGERED[6]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
H	RW	TRIGGERED[7]			Write '1' to disable interrupt for event TRIGGERED[7]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
I	RW	TRIGGERED[8]			Write '1' to disable interrupt for event TRIGGERED[8]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
J	RW	TRIGGERED[9]			Write '1' to disable interrupt for event TRIGGERED[9]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
K	RW	TRIGGERED[10]			Write '1' to disable interrupt for event TRIGGERED[10]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
L	RW	TRIGGERED[11]			Write '1' to disable interrupt for event TRIGGERED[11]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
M	RW	TRIGGERED[12]			Write '1' to disable interrupt for event TRIGGERED[12]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
N	RW	TRIGGERED[13]			Write '1' to disable interrupt for event TRIGGERED[13]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
O	RW	TRIGGERED[14]			Write '1' to disable interrupt for event TRIGGERED[14]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
P	RW	TRIGGERED[15]			Write '1' to disable interrupt for event TRIGGERED[15]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.8.2 Electrical specification

6.8.2.1 EGU Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
$t_{EGU,EVT}$	Latency between setting an EGU event flag and the system setting an interrupt		1		cycles

6.9 GPIO — General purpose input/output

The general purpose input/output pins (GPIOs) are grouped as one or more ports, with each port having up to 32 GPIOs.

The number of ports and GPIOs per port varies with product variant and package. Refer to Registers on page 324 and Pin assignments on page 926 for more information about the number of GPIOs that are supported.

GPIO has the following user-configurable features:

- Up to 32 GPIO pins per GPIO port
- Output drive strength
- Internal pull-up and pull-down resistors
- Wake-up from high or low level triggers on all pins
- Trigger interrupt on state changes on any pin
- All pins can be used by the PPI task/event system
- One or more GPIO outputs can be controlled through the PPI and GPIOTE channels
- Any pin can be mapped to a peripheral for layout flexibility
- GPIO state changes captured on the SENSE signal can be stored by the LATCH register

The GPIO port peripheral implements up to 32 pins, `PIN0` through `PIN31`. Each of these pins can be individually configured in the PIN_CNF[n] registers (n=0..31).

The following parameters can be configured through these registers:

- Direction
- Drive strength
- Enabling of pull-up and pull-down resistors
- Pin sensing
- Input buffer disconnect
- Analog input (for selected pins)

The PIN_CNF registers are retained registers. See POWER — Power supply on page 81 for more information about retained registers.

6.9.1 Pin configuration

Pins can be individually configured through the SENSE field in the PIN_CNF[n] register to detect either a high or low level input.

When the correct level is detected on a configured pin, the sense mechanism will set the DETECT signal high. Each pin has a separate DETECT signal. Default behavior, defined by the DETECTMODE register, combines all DETECT signals from the pins in the GPIO port into one common DETECT signal and routes it through the system to be utilized by other peripherals. This mechanism is functional in both System ON and System OFF mode. See GPIO port and the GPIO pin details on page 323.

The following figure illustrates the GPIO port containing 32 individual pins, where PIN0 is shown in more detail for reference. All signals on the left side of the illustration are used by other peripherals in the system and therefore not directly available to the CPU.

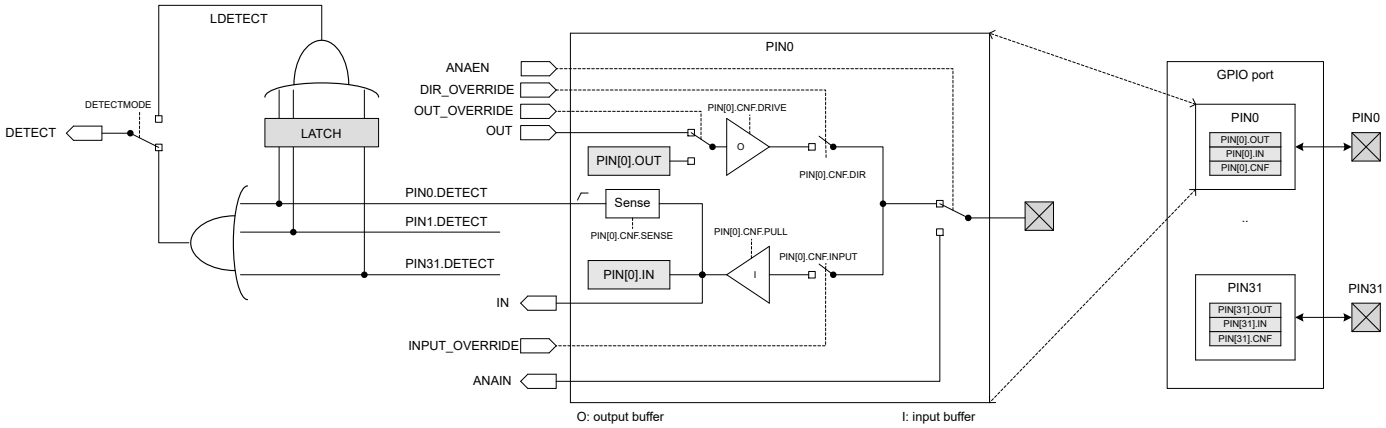


Figure 49: GPIO port and the GPIO pin details

Pins should be in a level that cannot trigger the sense mechanism before being enabled. If the SENSE condition configured in the PIN_CNF registers is met when the sense mechanism is enabled, the DETECT signal will immediately go high. A PORT event is triggered if the DETECT signal was low before enabling the sense mechanism. See GPIOTE — GPIO tasks and events on page 371.

See the following peripherals for more information about how the DETECT signal is used:

- POWER — Power supply on page 81 - uses the DETECT signal to exit from System OFF mode.
- GPIOTE — GPIO tasks and events on page 371 - uses the DETECT signal to generate the PORT event.

When a pin's PINx.DETECT signal goes high, a flag is set in the LATCH register. For example, when the PIN0.DETECT signal goes high, bit 0 in the LATCH register is set to 1. If the CPU performs a clear operation on a bit in the LATCH register when the associated PINx.DETECT signal is high, the bit in the LATCH register will not be cleared. The LATCH register will only be cleared if the CPU explicitly clears it by writing a 1 to the bit that shall be cleared, i.e. the LATCH register will not be affected by a PINx.DETECT signal being set low.

The LDETECT signal will be set high when one or more bits in the LATCH register are 1. The LDETECT signal will be set low when all bits in the LATCH register are successfully cleared to 0.

If one or more bits in the LATCH register are 1 after the CPU has performed a clear operation on the LATCH register, a rising edge will be generated on the LDETECT signal. This is illustrated in DETECT signal behavior on page 324.

Note: The CPU can read the LATCH register at any time to check if a SENSE condition has been met on any of the GPIO pins. This is still valid if that condition is no longer met at the time the CPU queries the LATCH register. This mechanism will work even if the LDETECT signal is not used as the DETECT signal.

The LDETECT signal is by default not connected to the GPIO port's DETECT signal, but via the DETECTMODE register. It is possible to change from default behavior to the DETECT signal that is derived directly from the LDETECT signal. See GPIO port and the GPIO pin details on page 323. The following figure illustrates the DETECT signal behavior for these two alternatives.

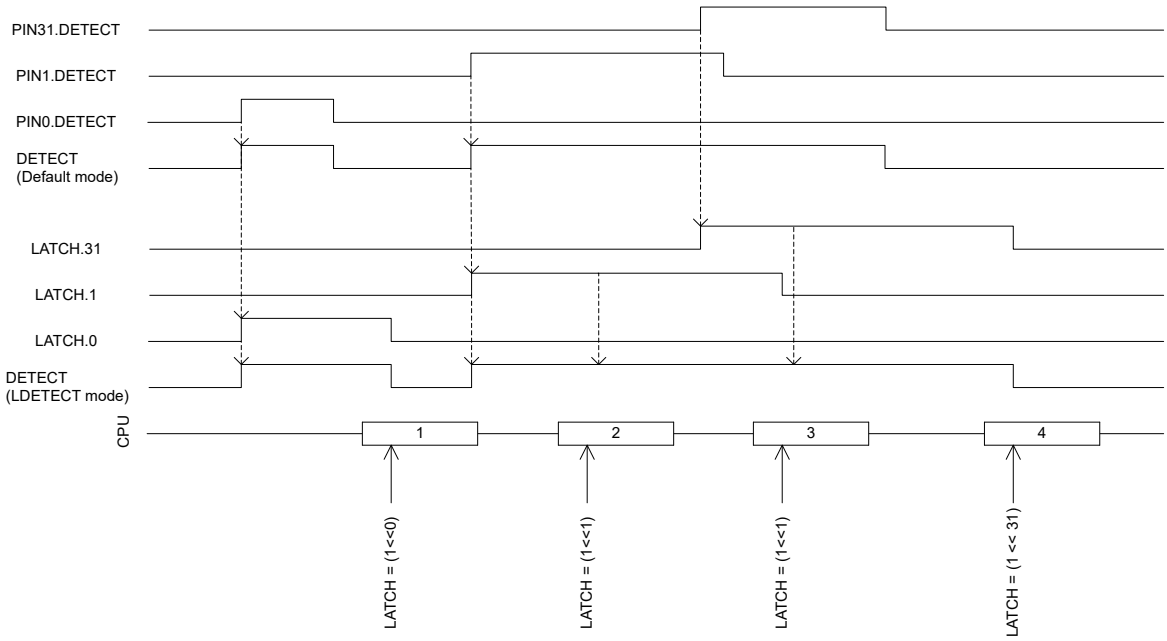


Figure 50: DETECT signal behavior

A GPIO pin input buffer can be disconnected from the pin to enable power savings when the pin is not used as an input, see GPIO port and the GPIO pin details on page 323. Input buffers must be connected to get a valid input value in the IN register, and for the sense mechanism to get access to the pin.

Other peripherals in the system can connect to GPIO pins and override their output value and configuration, or read their analog or digital input value. See GPIO port and the GPIO pin details on page 323.

Selected pins also support analog input signals, see ANAIN in GPIO port and the GPIO pin details on page 323. The assignment of the analog pins can be found in Pin assignments on page 926.

Note: When a pin is configured as digital input, increased current consumption occurs when the input voltage is between V_{IL} and V_{IH}. It is good practice to ensure that the external circuitry does not drive that pin to levels between V_{IL} and V_{IH} for a long period of time.

6.9.2 Registers

Instances

Instance	Base address	Description
GPIO	0x50000000	General purpose input and output This instance is deprecated.
P0	0x50000000	General purpose input and output, port 0
P1	0x50000300	General purpose input and output, port 1

Configuration

Instance	Configuration
GPIO	
P0	P0.00 to P0.31 implemented
P1	P1.00 to P1.15 implemented

Register overview

Register	Offset	Description
OUT	0x504	Write GPIO port
OUTSET	0x508	Set individual bits in GPIO port
OUTCLR	0x50C	Clear individual bits in GPIO port
IN	0x510	Read GPIO port
DIR	0x514	Direction of GPIO pins
DIRSET	0x518	DIR set register
DIRCLR	0x51C	DIR clear register
LATCH	0x520	Latch register indicating what GPIO pins that have met the criteria set in the PIN_CNF[n].SENSE registers
DETECTMODE	0x524	Select between default DETECT signal behavior and LDETECT mode
PIN_CNF[0]	0x700	Configuration of GPIO pins
PIN_CNF[1]	0x704	Configuration of GPIO pins
PIN_CNF[2]	0x708	Configuration of GPIO pins
PIN_CNF[3]	0x70C	Configuration of GPIO pins
PIN_CNF[4]	0x710	Configuration of GPIO pins
PIN_CNF[5]	0x714	Configuration of GPIO pins
PIN_CNF[6]	0x718	Configuration of GPIO pins
PIN_CNF[7]	0x71C	Configuration of GPIO pins
PIN_CNF[8]	0x720	Configuration of GPIO pins
PIN_CNF[9]	0x724	Configuration of GPIO pins
PIN_CNF[10]	0x728	Configuration of GPIO pins
PIN_CNF[11]	0x72C	Configuration of GPIO pins
PIN_CNF[12]	0x730	Configuration of GPIO pins
PIN_CNF[13]	0x734	Configuration of GPIO pins
PIN_CNF[14]	0x738	Configuration of GPIO pins
PIN_CNF[15]	0x73C	Configuration of GPIO pins
PIN_CNF[16]	0x740	Configuration of GPIO pins
PIN_CNF[17]	0x744	Configuration of GPIO pins
PIN_CNF[18]	0x748	Configuration of GPIO pins
PIN_CNF[19]	0x74C	Configuration of GPIO pins
PIN_CNF[20]	0x750	Configuration of GPIO pins
PIN_CNF[21]	0x754	Configuration of GPIO pins
PIN_CNF[22]	0x758	Configuration of GPIO pins
PIN_CNF[23]	0x75C	Configuration of GPIO pins
PIN_CNF[24]	0x760	Configuration of GPIO pins
PIN_CNF[25]	0x764	Configuration of GPIO pins
PIN_CNF[26]	0x768	Configuration of GPIO pins
PIN_CNF[27]	0x76C	Configuration of GPIO pins
PIN_CNF[28]	0x770	Configuration of GPIO pins
PIN_CNF[29]	0x774	Configuration of GPIO pins
PIN_CNF[30]	0x778	Configuration of GPIO pins
PIN_CNF[31]	0x77C	Configuration of GPIO pins

6.9.2.1 OUT

Address offset: 0x504

Write GPIO port

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID		Value																															
A	RW	PIN0																																		
			Low		0																															
			High		1																															
B	RW	PIN1																																		
			Low		0																															
			High		1																															
C	RW	PIN2																																		
			Low		0																															
			High		1																															
D	RW	PIN3																																		
			Low		0																															
			High		1																															
E	RW	PIN4																																		
			Low		0																															
			High		1																															
F	RW	PIN5																																		
			Low		0																															
			High		1																															
G	RW	PIN6																																		
			Low		0																															
			High		1																															
H	RW	PIN7																																		
			Low		0																															
			High		1																															
I	RW	PIN8																																		
			Low		0																															
			High		1																															
J	RW	PIN9																																		
			Low		0																															
			High		1																															
K	RW	PIN10																																		
			Low		0																															
			High		1																															
L	RW	PIN11																																		
			Low		0																															
			High		1																															
M	RW	PIN12																																		
			Low		0																															
			High		1																															
N	RW	PIN13																																		
			Low		0																															
			High		1																															
O	RW	PIN14																																		
			Low		0																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
			High	1																															
P	RW	PIN15																																	
			Low	0																															
			High	1																															
Q	RW	PIN16																																	
			Low	0																															
			High	1																															
R	RW	PIN17																																	
			Low	0																															
			High	1																															
S	RW	PIN18																																	
			Low	0																															
			High	1																															
T	RW	PIN19																																	
			Low	0																															
			High	1																															
U	RW	PIN20																																	
			Low	0																															
			High	1																															
V	RW	PIN21																																	
			Low	0																															
			High	1																															
W	RW	PIN22																																	
			Low	0																															
			High	1																															
X	RW	PIN23																																	
			Low	0																															
			High	1																															
Y	RW	PIN24																																	
			Low	0																															
			High	1																															
Z	RW	PIN25																																	
			Low	0																															
			High	1																															
a	RW	PIN26																																	
			Low	0																															
			High	1																															
b	RW	PIN27																																	
			Low	0																															
			High	1																															
c	RW	PIN28																																	
			Low	0																															
			High	1																															
d	RW	PIN29																																	
			Low	0																															
			High	1																															
e	RW	PIN30																																	
			Low	0																															
			High	1																															
f	RW	PIN31																																	

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Low	0	Pin driver is low
			High	1	Pin driver is high

6.9.2.2 OUTSET

Address offset: 0x508

Set individual bits in GPIO port

Note: Read: reads value of OUT register.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW W1S	PIN0			Pin 0
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Set	1	Write: a '1' sets the pin high; a '0' has no effect
B	RW W1S	PIN1			Pin 1
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Set	1	Write: a '1' sets the pin high; a '0' has no effect
C	RW W1S	PIN2			Pin 2
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Set	1	Write: a '1' sets the pin high; a '0' has no effect
D	RW W1S	PIN3			Pin 3
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Set	1	Write: a '1' sets the pin high; a '0' has no effect
E	RW W1S	PIN4			Pin 4
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Set	1	Write: a '1' sets the pin high; a '0' has no effect
F	RW W1S	PIN5			Pin 5
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Set	1	Write: a '1' sets the pin high; a '0' has no effect
G	RW W1S	PIN6			Pin 6
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Set	1	Write: a '1' sets the pin high; a '0' has no effect

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
H	RW	PIN7																																	
	W1S																																		
			Low	0																															
			High	1																															
			Set	1																															
I	RW	PIN8																																	
	W1S																																		
			Low	0																															
			High	1																															
			Set	1																															
J	RW	PIN9																																	
	W1S																																		
			Low	0																															
			High	1																															
			Set	1																															
K	RW	PIN10																																	
	W1S																																		
			Low	0																															
			High	1																															
			Set	1																															
L	RW	PIN11																																	
	W1S																																		
			Low	0																															
			High	1																															
			Set	1																															
M	RW	PIN12																																	
	W1S																																		
			Low	0																															
			High	1																															
			Set	1																															
N	RW	PIN13																																	
	W1S																																		
			Low	0																															
			High	1																															
			Set	1																															
O	RW	PIN14																																	
	W1S																																		
			Low	0																															
			High	1																															
			Set	1																															
P	RW	PIN15																																	
	W1S																																		
			Low	0																															
			High	1																															
			Set	1																															
Q	RW	PIN16																																	
	W1S																																		
			Low	0																															
			High	1																															
			Set	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
				Description																															
R	RW W1S	PIN17																																	
				Pin 17																															
			Low	0																															
				Read: pin driver is low																															
			High	1																															
				Read: pin driver is high																															
			Set	1																															
				Write: a '1' sets the pin high; a '0' has no effect																															
S	RW W1S	PIN18																																	
				Pin 18																															
			Low	0																															
				Read: pin driver is low																															
			High	1																															
				Read: pin driver is high																															
			Set	1																															
				Write: a '1' sets the pin high; a '0' has no effect																															
T	RW W1S	PIN19																																	
				Pin 19																															
			Low	0																															
				Read: pin driver is low																															
			High	1																															
				Read: pin driver is high																															
			Set	1																															
				Write: a '1' sets the pin high; a '0' has no effect																															
U	RW W1S	PIN20																																	
				Pin 20																															
			Low	0																															
				Read: pin driver is low																															
			High	1																															
				Read: pin driver is high																															
			Set	1																															
				Write: a '1' sets the pin high; a '0' has no effect																															
V	RW W1S	PIN21																																	
				Pin 21																															
			Low	0																															
				Read: pin driver is low																															
			High	1																															
				Read: pin driver is high																															
			Set	1																															
				Write: a '1' sets the pin high; a '0' has no effect																															
W	RW W1S	PIN22																																	
				Pin 22																															
			Low	0																															
				Read: pin driver is low																															
			High	1																															
				Read: pin driver is high																															
			Set	1																															
				Write: a '1' sets the pin high; a '0' has no effect																															
X	RW W1S	PIN23																																	
				Pin 23																															
			Low	0																															
				Read: pin driver is low																															
			High	1																															
				Read: pin driver is high																															
			Set	1																															
				Write: a '1' sets the pin high; a '0' has no effect																															
Y	RW W1S	PIN24																																	
				Pin 24																															
			Low	0																															
				Read: pin driver is low																															
			High	1																															
				Read: pin driver is high																															
			Set	1																															
				Write: a '1' sets the pin high; a '0' has no effect																															
Z	RW W1S	PIN25																																	
				Pin 25																															
			Low	0																															
				Read: pin driver is low																															
			High	1																															
				Read: pin driver is high																															
			Set	1																															
				Write: a '1' sets the pin high; a '0' has no effect																															
a	RW W1S	PIN26																																	
				Pin 26																															
			Low	0																															
				Read: pin driver is low																															
			High	1																															
				Read: pin driver is high																															
			Set	1																															
				Write: a '1' sets the pin high; a '0' has no effect																															

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
b	RW W1S	PIN27			Pin 27
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Set	1	Write: a '1' sets the pin high; a '0' has no effect
c	RW W1S	PIN28			Pin 28
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Set	1	Write: a '1' sets the pin high; a '0' has no effect
d	RW W1S	PIN29			Pin 29
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Set	1	Write: a '1' sets the pin high; a '0' has no effect
e	RW W1S	PIN30			Pin 30
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Set	1	Write: a '1' sets the pin high; a '0' has no effect
f	RW W1S	PIN31			Pin 31
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Set	1	Write: a '1' sets the pin high; a '0' has no effect

6.9.2.3 OUTCLR

Address offset: 0x50C

Clear individual bits in GPIO port

Note: Read: reads value of OUT register.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW W1C	PIN0			Pin 0
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Clear	1	Write: a '1' sets the pin low; a '0' has no effect
B	RW W1C	PIN1			Pin 1
			Low	0	Read: pin driver is low
			High	1	Read: pin driver is high
			Clear	1	Write: a '1' sets the pin low; a '0' has no effect
C	RW W1C	PIN2			Pin 2

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field		Value ID				Value																		Description									
				Low				0																		Read: pin driver is low									
				High				1																		Read: pin driver is high									
				Clear				1																		Write: a '1' sets the pin low; a '0' has no effect									
D	RW W1C	PIN3																								Pin 3									
				Low				0																		Read: pin driver is low									
				High				1																		Read: pin driver is high									
				Clear				1																		Write: a '1' sets the pin low; a '0' has no effect									
E	RW W1C	PIN4																								Pin 4									
				Low				0																		Read: pin driver is low									
				High				1																		Read: pin driver is high									
				Clear				1																		Write: a '1' sets the pin low; a '0' has no effect									
F	RW W1C	PIN5																								Pin 5									
				Low				0																		Read: pin driver is low									
				High				1																		Read: pin driver is high									
				Clear				1																		Write: a '1' sets the pin low; a '0' has no effect									
G	RW W1C	PIN6																								Pin 6									
				Low				0																		Read: pin driver is low									
				High				1																		Read: pin driver is high									
				Clear				1																		Write: a '1' sets the pin low; a '0' has no effect									
H	RW W1C	PIN7																								Pin 7									
				Low				0																		Read: pin driver is low									
				High				1																		Read: pin driver is high									
				Clear				1																		Write: a '1' sets the pin low; a '0' has no effect									
I	RW W1C	PIN8																								Pin 8									
				Low				0																		Read: pin driver is low									
				High				1																		Read: pin driver is high									
				Clear				1																		Write: a '1' sets the pin low; a '0' has no effect									
J	RW W1C	PIN9																								Pin 9									
				Low				0																		Read: pin driver is low									
				High				1																		Read: pin driver is high									
				Clear				1																		Write: a '1' sets the pin low; a '0' has no effect									
K	RW W1C	PIN10																								Pin 10									
				Low				0																		Read: pin driver is low									
				High				1																		Read: pin driver is high									
				Clear				1																		Write: a '1' sets the pin low; a '0' has no effect									
L	RW W1C	PIN11																								Pin 11									
				Low				0																		Read: pin driver is low									
				High				1																		Read: pin driver is high									
				Clear				1																		Write: a '1' sets the pin low; a '0' has no effect									
M	RW W1C	PIN12																								Pin 12									

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
			Low	0																															
			High	1																															
			Clear	1																															
N	RW W1C	PIN13																																	
			Low	0																															
			High	1																															
			Clear	1																															
O	RW W1C	PIN14																																	
			Low	0																															
			High	1																															
			Clear	1																															
P	RW W1C	PIN15																																	
			Low	0																															
			High	1																															
			Clear	1																															
Q	RW W1C	PIN16																																	
			Low	0																															
			High	1																															
			Clear	1																															
R	RW W1C	PIN17																																	
			Low	0																															
			High	1																															
			Clear	1																															
S	RW W1C	PIN18																																	
			Low	0																															
			High	1																															
			Clear	1																															
T	RW W1C	PIN19																																	
			Low	0																															
			High	1																															
			Clear	1																															
U	RW W1C	PIN20																																	
			Low	0																															
			High	1																															
			Clear	1																															
V	RW W1C	PIN21																																	
			Low	0																															
			High	1																															
			Clear	1																															
W	RW W1C	PIN22																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
			Low	0																															
			High	1																															
			Clear	1																															
X	RW W1C	PIN23																																	
			Low	0																															
			High	1																															
			Clear	1																															
Y	RW W1C	PIN24																																	
			Low	0																															
			High	1																															
			Clear	1																															
Z	RW W1C	PIN25																																	
			Low	0																															
			High	1																															
			Clear	1																															
a	RW W1C	PIN26																																	
			Low	0																															
			High	1																															
			Clear	1																															
b	RW W1C	PIN27																																	
			Low	0																															
			High	1																															
			Clear	1																															
c	RW W1C	PIN28																																	
			Low	0																															
			High	1																															
			Clear	1																															
d	RW W1C	PIN29																																	
			Low	0																															
			High	1																															
			Clear	1																															
e	RW W1C	PIN30																																	
			Low	0																															
			High	1																															
			Clear	1																															
f	RW W1C	PIN31																																	
			Low	0																															
			High	1																															
			Clear	1																															

6.9.2.4 IN

Address offset: 0x510

Read GPIO port

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	R	PIN0																																	
			Low	0																															
			High	1																															
B	R	PIN1																																	
			Low	0																															
			High	1																															
C	R	PIN2																																	
			Low	0																															
			High	1																															
D	R	PIN3																																	
			Low	0																															
			High	1																															
E	R	PIN4																																	
			Low	0																															
			High	1																															
F	R	PIN5																																	
			Low	0																															
			High	1																															
G	R	PIN6																																	
			Low	0																															
			High	1																															
H	R	PIN7																																	
			Low	0																															
			High	1																															
I	R	PIN8																																	
			Low	0																															
			High	1																															
J	R	PIN9																																	
			Low	0																															
			High	1																															
K	R	PIN10																																	
			Low	0																															
			High	1																															
L	R	PIN11																																	
			Low	0																															
			High	1																															
M	R	PIN12																																	
			Low	0																															
			High	1																															
N	R	PIN13																																	
			Low	0																															
			High	1																															
O	R	PIN14																																	
			Low	0																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
			High	1																															
P	R	PIN15																																	
			Low	0																															
			High	1																															
Q	R	PIN16																																	
			Low	0																															
			High	1																															
R	R	PIN17																																	
			Low	0																															
			High	1																															
S	R	PIN18																																	
			Low	0																															
			High	1																															
T	R	PIN19																																	
			Low	0																															
			High	1																															
U	R	PIN20																																	
			Low	0																															
			High	1																															
V	R	PIN21																																	
			Low	0																															
			High	1																															
W	R	PIN22																																	
			Low	0																															
			High	1																															
X	R	PIN23																																	
			Low	0																															
			High	1																															
Y	R	PIN24																																	
			Low	0																															
			High	1																															
Z	R	PIN25																																	
			Low	0																															
			High	1																															
a	R	PIN26																																	
			Low	0																															
			High	1																															
b	R	PIN27																																	
			Low	0																															
			High	1																															
c	R	PIN28																																	
			Low	0																															
			High	1																															
d	R	PIN29																																	
			Low	0																															
			High	1																															
e	R	PIN30																																	
			Low	0																															
			High	1																															
f	R	PIN31																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
			Low	0																															
			High	1																															

6.9.2.5 DIR

Address offset: 0x514

Direction of GPIO pins

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	PIN0																																	
			Input	0																															
			Output	1																															
B	RW	PIN1																																	
			Input	0																															
			Output	1																															
C	RW	PIN2																																	
			Input	0																															
			Output	1																															
D	RW	PIN3																																	
			Input	0																															
			Output	1																															
E	RW	PIN4																																	
			Input	0																															
			Output	1																															
F	RW	PIN5																																	
			Input	0																															
			Output	1																															
G	RW	PIN6																																	
			Input	0																															
			Output	1																															
H	RW	PIN7																																	
			Input	0																															
			Output	1																															
I	RW	PIN8																																	
			Input	0																															
			Output	1																															
J	RW	PIN9																																	
			Input	0																															
			Output	1																															
K	RW	PIN10																																	
			Input	0																															
			Output	1																															
L	RW	PIN11																																	
			Input	0																															
			Output	1																															
M	RW	PIN12																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
			Input	0																															
			Output	1																															
N	RW	PIN13																																	
			Input	0																															
			Output	1																															
O	RW	PIN14																																	
			Input	0																															
			Output	1																															
P	RW	PIN15																																	
			Input	0																															
			Output	1																															
Q	RW	PIN16																																	
			Input	0																															
			Output	1																															
R	RW	PIN17																																	
			Input	0																															
			Output	1																															
S	RW	PIN18																																	
			Input	0																															
			Output	1																															
T	RW	PIN19																																	
			Input	0																															
			Output	1																															
U	RW	PIN20																																	
			Input	0																															
			Output	1																															
V	RW	PIN21																																	
			Input	0																															
			Output	1																															
W	RW	PIN22																																	
			Input	0																															
			Output	1																															
X	RW	PIN23																																	
			Input	0																															
			Output	1																															
Y	RW	PIN24																																	
			Input	0																															
			Output	1																															
Z	RW	PIN25																																	
			Input	0																															
			Output	1																															
a	RW	PIN26																																	
			Input	0																															
			Output	1																															
b	RW	PIN27																																	
			Input	0																															
			Output	1																															
c	RW	PIN28																																	
			Input	0																															
			Output	1																															

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
d	RW	PIN29			Pin 29
			Input	0	Pin set as input
			Output	1	Pin set as output
e	RW	PIN30			Pin 30
			Input	0	Pin set as input
			Output	1	Pin set as output
f	RW	PIN31			Pin 31
			Input	0	Pin set as input
			Output	1	Pin set as output

6.9.2.6 DIRSET

Address offset: 0x518

DIR set register

Note: Read: reads value of DIR register.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW W1S	PIN0			Set as output pin 0
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Set	1	Write: a '1' sets pin to output; a '0' has no effect
B	RW W1S	PIN1			Set as output pin 1
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Set	1	Write: a '1' sets pin to output; a '0' has no effect
C	RW W1S	PIN2			Set as output pin 2
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Set	1	Write: a '1' sets pin to output; a '0' has no effect
D	RW W1S	PIN3			Set as output pin 3
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Set	1	Write: a '1' sets pin to output; a '0' has no effect
E	RW W1S	PIN4			Set as output pin 4
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Set	1	Write: a '1' sets pin to output; a '0' has no effect
F	RW W1S	PIN5			Set as output pin 5
			Input	0	Read: pin set as input

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
			Output	1																															
			Set	1																															
G	RW W1S	PIN6																																	
			Input	0																															
			Output	1																															
			Set	1																															
H	RW W1S	PIN7																																	
			Input	0																															
			Output	1																															
			Set	1																															
I	RW W1S	PIN8																																	
			Input	0																															
			Output	1																															
			Set	1																															
J	RW W1S	PIN9																																	
			Input	0																															
			Output	1																															
			Set	1																															
K	RW W1S	PIN10																																	
			Input	0																															
			Output	1																															
			Set	1																															
L	RW W1S	PIN11																																	
			Input	0																															
			Output	1																															
			Set	1																															
M	RW W1S	PIN12																																	
			Input	0																															
			Output	1																															
			Set	1																															
N	RW W1S	PIN13																																	
			Input	0																															
			Output	1																															
			Set	1																															
O	RW W1S	PIN14																																	
			Input	0																															
			Output	1																															
			Set	1																															
P	RW W1S	PIN15																																	
			Input	0																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
			Output	1																															
			Set	1																															
Q	RW W1S	PIN16																																	
			Input	0																															
			Output	1																															
			Set	1																															
R	RW W1S	PIN17																																	
			Input	0																															
			Output	1																															
			Set	1																															
S	RW W1S	PIN18																																	
			Input	0																															
			Output	1																															
			Set	1																															
T	RW W1S	PIN19																																	
			Input	0																															
			Output	1																															
			Set	1																															
U	RW W1S	PIN20																																	
			Input	0																															
			Output	1																															
			Set	1																															
V	RW W1S	PIN21																																	
			Input	0																															
			Output	1																															
			Set	1																															
W	RW W1S	PIN22																																	
			Input	0																															
			Output	1																															
			Set	1																															
X	RW W1S	PIN23																																	
			Input	0																															
			Output	1																															
			Set	1																															
Y	RW W1S	PIN24																																	
			Input	0																															
			Output	1																															
			Set	1																															
Z	RW W1S	PIN25																																	
			Input	0																															

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Output	1	Read: pin set as output
			Set	1	Write: a '1' sets pin to output; a '0' has no effect
a	RW W1S	PIN26			Set as output pin 26
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Set	1	Write: a '1' sets pin to output; a '0' has no effect
b	RW W1S	PIN27			Set as output pin 27
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Set	1	Write: a '1' sets pin to output; a '0' has no effect
c	RW W1S	PIN28			Set as output pin 28
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Set	1	Write: a '1' sets pin to output; a '0' has no effect
d	RW W1S	PIN29			Set as output pin 29
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Set	1	Write: a '1' sets pin to output; a '0' has no effect
e	RW W1S	PIN30			Set as output pin 30
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Set	1	Write: a '1' sets pin to output; a '0' has no effect
f	RW W1S	PIN31			Set as output pin 31
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Set	1	Write: a '1' sets pin to output; a '0' has no effect

6.9.2.7 DIRCLR

Address offset: 0x51C

DIR clear register

Note: Read: reads value of DIR register.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW W1C	PIN0			Set as input pin 0
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Clear	1	Write: a '1' sets pin to input; a '0' has no effect

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
B	RW W1C	PIN1																																	Set as input pin 1
			Input	0																															Read: pin set as input
			Output	1																															Read: pin set as output
			Clear	1																															Write: a '1' sets pin to input; a '0' has no effect
C	RW W1C	PIN2																																	Set as input pin 2
			Input	0																															Read: pin set as input
			Output	1																															Read: pin set as output
			Clear	1																															Write: a '1' sets pin to input; a '0' has no effect
D	RW W1C	PIN3																																	Set as input pin 3
			Input	0																															Read: pin set as input
			Output	1																															Read: pin set as output
			Clear	1																															Write: a '1' sets pin to input; a '0' has no effect
E	RW W1C	PIN4																																	Set as input pin 4
			Input	0																															Read: pin set as input
			Output	1																															Read: pin set as output
			Clear	1																															Write: a '1' sets pin to input; a '0' has no effect
F	RW W1C	PIN5																																	Set as input pin 5
			Input	0																															Read: pin set as input
			Output	1																															Read: pin set as output
			Clear	1																															Write: a '1' sets pin to input; a '0' has no effect
G	RW W1C	PIN6																																	Set as input pin 6
			Input	0																															Read: pin set as input
			Output	1																															Read: pin set as output
			Clear	1																															Write: a '1' sets pin to input; a '0' has no effect
H	RW W1C	PIN7																																	Set as input pin 7
			Input	0																															Read: pin set as input
			Output	1																															Read: pin set as output
			Clear	1																															Write: a '1' sets pin to input; a '0' has no effect
I	RW W1C	PIN8																																	Set as input pin 8
			Input	0																															Read: pin set as input
			Output	1																															Read: pin set as output
			Clear	1																															Write: a '1' sets pin to input; a '0' has no effect
J	RW W1C	PIN9																																	Set as input pin 9
			Input	0																															Read: pin set as input
			Output	1																															Read: pin set as output
			Clear	1																															Write: a '1' sets pin to input; a '0' has no effect
K	RW W1C	PIN10																																	Set as input pin 10
			Input	0																															Read: pin set as input
			Output	1																															Read: pin set as output
			Clear	1																															Write: a '1' sets pin to input; a '0' has no effect

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
L	RW W1C	PIN11			Set as input pin 11
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Clear	1	Write: a '1' sets pin to input; a '0' has no effect
M	RW W1C	PIN12			Set as input pin 12
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Clear	1	Write: a '1' sets pin to input; a '0' has no effect
N	RW W1C	PIN13			Set as input pin 13
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Clear	1	Write: a '1' sets pin to input; a '0' has no effect
O	RW W1C	PIN14			Set as input pin 14
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Clear	1	Write: a '1' sets pin to input; a '0' has no effect
P	RW W1C	PIN15			Set as input pin 15
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Clear	1	Write: a '1' sets pin to input; a '0' has no effect
Q	RW W1C	PIN16			Set as input pin 16
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Clear	1	Write: a '1' sets pin to input; a '0' has no effect
R	RW W1C	PIN17			Set as input pin 17
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Clear	1	Write: a '1' sets pin to input; a '0' has no effect
S	RW W1C	PIN18			Set as input pin 18
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Clear	1	Write: a '1' sets pin to input; a '0' has no effect
T	RW W1C	PIN19			Set as input pin 19
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Clear	1	Write: a '1' sets pin to input; a '0' has no effect
U	RW W1C	PIN20			Set as input pin 20
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Clear	1	Write: a '1' sets pin to input; a '0' has no effect

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
V	RW W1C	PIN21																																	
			Input	0																															
			Output	1																															
			Clear	1																															
W	RW W1C	PIN22																																	
			Input	0																															
			Output	1																															
			Clear	1																															
X	RW W1C	PIN23																																	
			Input	0																															
			Output	1																															
			Clear	1																															
Y	RW W1C	PIN24																																	
			Input	0																															
			Output	1																															
			Clear	1																															
Z	RW W1C	PIN25																																	
			Input	0																															
			Output	1																															
			Clear	1																															
a	RW W1C	PIN26																																	
			Input	0																															
			Output	1																															
			Clear	1																															
b	RW W1C	PIN27																																	
			Input	0																															
			Output	1																															
			Clear	1																															
c	RW W1C	PIN28																																	
			Input	0																															
			Output	1																															
			Clear	1																															
d	RW W1C	PIN29																																	
			Input	0																															
			Output	1																															
			Clear	1																															
e	RW W1C	PIN30																																	
			Input	0																															
			Output	1																															
			Clear	1																															

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
f	RW W1C	PIN31			Set as input pin 31
			Input	0	Read: pin set as input
			Output	1	Read: pin set as output
			Clear	1	Write: a '1' sets pin to input; a '0' has no effect

6.9.2.8 LATCH

Address offset: 0x520

Latch register indicating what GPIO pins that have met the criteria set in the PIN_CNF[n].SENSE registers

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN0			Status on whether PIN0 has met criteria set in PIN_CNF0.SENSE register. Write '1' to clear.
			NotLatched	0	Criteria has not been met
			Latched	1	Criteria has been met
B	RW	PIN1			Status on whether PIN1 has met criteria set in PIN_CNF1.SENSE register. Write '1' to clear.
			NotLatched	0	Criteria has not been met
			Latched	1	Criteria has been met
C	RW	PIN2			Status on whether PIN2 has met criteria set in PIN_CNF2.SENSE register. Write '1' to clear.
			NotLatched	0	Criteria has not been met
			Latched	1	Criteria has been met
D	RW	PIN3			Status on whether PIN3 has met criteria set in PIN_CNF3.SENSE register. Write '1' to clear.
			NotLatched	0	Criteria has not been met
			Latched	1	Criteria has been met
E	RW	PIN4			Status on whether PIN4 has met criteria set in PIN_CNF4.SENSE register. Write '1' to clear.
			NotLatched	0	Criteria has not been met
			Latched	1	Criteria has been met
F	RW	PIN5			Status on whether PIN5 has met criteria set in PIN_CNF5.SENSE register. Write '1' to clear.
			NotLatched	0	Criteria has not been met
			Latched	1	Criteria has been met
G	RW	PIN6			Status on whether PIN6 has met criteria set in PIN_CNF6.SENSE register. Write '1' to clear.
			NotLatched	0	Criteria has not been met
			Latched	1	Criteria has been met
H	RW	PIN7			Status on whether PIN7 has met criteria set in PIN_CNF7.SENSE register. Write '1' to clear.
			NotLatched	0	Criteria has not been met
			Latched	1	Criteria has been met
I	RW	PIN8			Status on whether PIN8 has met criteria set in PIN_CNF8.SENSE register. Write '1' to clear.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			NotLatched	0	Criteria has not been met
			Latched	1	Criteria has been met
J	RW	PIN9			Status on whether PIN9 has met criteria set in PIN_CNF9.SENSE register.
					Write '1' to clear.
			NotLatched	0	Criteria has not been met
			Latched	1	Criteria has been met
K	RW	PIN10			Status on whether PIN10 has met criteria set in PIN_CNF10.SENSE register.
					Write '1' to clear.
			NotLatched	0	Criteria has not been met
			Latched	1	Criteria has been met
L	RW	PIN11			Status on whether PIN11 has met criteria set in PIN_CNF11.SENSE register.
					Write '1' to clear.
			NotLatched	0	Criteria has not been met
			Latched	1	Criteria has been met
M	RW	PIN12			Status on whether PIN12 has met criteria set in PIN_CNF12.SENSE register.
					Write '1' to clear.
			NotLatched	0	Criteria has not been met
			Latched	1	Criteria has been met
N	RW	PIN13			Status on whether PIN13 has met criteria set in PIN_CNF13.SENSE register.
					Write '1' to clear.
			NotLatched	0	Criteria has not been met
			Latched	1	Criteria has been met
O	RW	PIN14			Status on whether PIN14 has met criteria set in PIN_CNF14.SENSE register.
					Write '1' to clear.
			NotLatched	0	Criteria has not been met
			Latched	1	Criteria has been met
P	RW	PIN15			Status on whether PIN15 has met criteria set in PIN_CNF15.SENSE register.
					Write '1' to clear.
			NotLatched	0	Criteria has not been met
			Latched	1	Criteria has been met
Q	RW	PIN16			Status on whether PIN16 has met criteria set in PIN_CNF16.SENSE register.
					Write '1' to clear.
			NotLatched	0	Criteria has not been met
			Latched	1	Criteria has been met
R	RW	PIN17			Status on whether PIN17 has met criteria set in PIN_CNF17.SENSE register.
					Write '1' to clear.
			NotLatched	0	Criteria has not been met
			Latched	1	Criteria has been met
S	RW	PIN18			Status on whether PIN18 has met criteria set in PIN_CNF18.SENSE register.
					Write '1' to clear.
			NotLatched	0	Criteria has not been met
			Latched	1	Criteria has been met
T	RW	PIN19			Status on whether PIN19 has met criteria set in PIN_CNF19.SENSE register.
					Write '1' to clear.
			NotLatched	0	Criteria has not been met
			Latched	1	Criteria has been met
U	RW	PIN20			Status on whether PIN20 has met criteria set in PIN_CNF20.SENSE register.
					Write '1' to clear.
			NotLatched	0	Criteria has not been met
			Latched	1	Criteria has been met

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
V	RW	PIN21																																	
			NotLatched	0																															
			Latched	1																															
W	RW	PIN22																																	
			NotLatched	0																															
			Latched	1																															
X	RW	PIN23																																	
			NotLatched	0																															
			Latched	1																															
Y	RW	PIN24																																	
			NotLatched	0																															
			Latched	1																															
Z	RW	PIN25																																	
			NotLatched	0																															
			Latched	1																															
a	RW	PIN26																																	
			NotLatched	0																															
			Latched	1																															
b	RW	PIN27																																	
			NotLatched	0																															
			Latched	1																															
c	RW	PIN28																																	
			NotLatched	0																															
			Latched	1																															
d	RW	PIN29																																	
			NotLatched	0																															
			Latched	1																															
e	RW	PIN30																																	
			NotLatched	0																															
			Latched	1																															
f	RW	PIN31																																	
			NotLatched	0																															
			Latched	1																															

6.9.2.9 DETECTMODE

Address offset: 0x524

Select between default DETECT signal behavior and LDETECT mode

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DETECTMODE			Select between default DETECT signal behavior and LDETECT mode
			Default	0	DETECT directly connected to PIN DETECT signals
			LDETECT	1	Use the latched LDETECT behavior

6.9.2.10 PIN_CNF[0]

Address offset: 0x700

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.11 PIN_CNF[1]

Address offset: 0x704

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.12 PIN_CNF[2]

Address offset: 0x708

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.13 PIN_CNF[3]

Address offset: 0x70C

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.14 PIN_CNF[4]

Address offset: 0x710

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.15 PIN_CNF[5]

Address offset: 0x714

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.16 PIN_CNF[6]

Address offset: 0x718

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.17 PIN_CNF[7]

Address offset: 0x71C

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.18 PIN_CNF[8]

Address offset: 0x720

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.19 PIN_CNF[9]

Address offset: 0x724

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.20 PIN_CNF[10]

Address offset: 0x728

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.21 PIN_CNF[11]

Address offset: 0x72C

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.22 PIN_CNF[12]

Address offset: 0x730

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.23 PIN_CNF[13]

Address offset: 0x734

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.24 PIN_CNF[14]

Address offset: 0x738

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.25 PIN_CNF[15]

Address offset: 0x73C

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.26 PIN_CNF[16]

Address offset: 0x740

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.27 PIN_CNF[17]

Address offset: 0x744

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.28 PIN_CNF[18]

Address offset: 0x748

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.29 PIN_CNF[19]

Address offset: 0x74C

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.30 PIN_CNF[20]

Address offset: 0x750

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.31 PIN_CNF[21]

Address offset: 0x754

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.32 PIN_CNF[22]

Address offset: 0x758

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.33 PIN_CNF[23]

Address offset: 0x75C

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.34 PIN_CNF[24]

Address offset: 0x760

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.35 PIN_CNF[25]

Address offset: 0x764

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.36 PIN_CNF[26]

Address offset: 0x768

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.37 PIN_CNF[27]

Address offset: 0x76C

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.38 PIN_CNF[28]

Address offset: 0x770

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.39 PIN_CNF[29]

Address offset: 0x774

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.40 PIN_CNF[30]

Address offset: 0x778

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.2.41 PIN_CNF[31]

Address offset: 0x77C

Configuration of GPIO pins

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E D D D C C B A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DIR			Pin direction. Same physical register as DIR register
			Input	0	Configure pin as an input pin
			Output	1	Configure pin as an output pin
B	RW	INPUT			Connect or disconnect input buffer
			Connect	0	Connect input buffer
			Disconnect	1	Disconnect input buffer
C	RW	PULL			Pull configuration
			Disabled	0	No pull
			Pulldown	1	Pull down on pin
			Pullup	3	Pull up on pin
D	RW	DRIVE			Drive configuration
			S0S1	0	Standard '0', standard '1'
			H0S1	1	High drive '0', standard '1'
			S0H1	2	Standard '0', high drive '1'
			H0H1	3	High drive '0', high 'drive '1''
			D0S1	4	Disconnect '0' standard '1' (normally used for wired-or connections)
			D0H1	5	Disconnect '0', high drive '1' (normally used for wired-or connections)
			S0D1	6	Standard '0'. disconnect '1' (normally used for wired-and connections)
			H0D1	7	High drive '0', disconnect '1' (normally used for wired-and connections)
E	RW	SENSE			Pin sensing mechanism
			Disabled	0	Disabled
			High	2	Sense for high level
			Low	3	Sense for low level

6.9.3 Electrical specification

6.9.3.1 GPIO Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
V_{IH}	Input high voltage	0.7 x VDD		VDD	V
V_{IL}	Input low voltage	VSS		0.3 x VDD	V
$V_{OH,SD}$	Output high voltage, standard drive, 0.5 mA, VDD ≥ 1.7	VDD - 0.4		VDD	V
$V_{OH,HDH}$	Output high voltage, high drive, 5 mA, VDD ≥ 2.7 V	VDD - 0.4		VDD	V
$V_{OH,HDL}$	Output high voltage, high drive, 3 mA, VDD ≥ 1.7 V	VDD - 0.4		VDD	V
$V_{OL,SD}$	Output low voltage, standard drive, 0.5 mA, VDD ≥ 1.7	VSS		VSS + 0.4	V
$V_{OL,HDH}$	Output low voltage, high drive, 5 mA, VDD ≥ 2.7 V	VSS		VSS + 0.4	V
$V_{OL,HDL}$	Output low voltage, high drive, 3 mA, VDD ≥ 1.7 V	VSS		VSS + 0.4	V
$I_{OL,SD}$	Current at VSS+0.4 V, output set low, standard drive, VDD ≥ 1.7	1	2	4	mA
$I_{OL,HDH}$	Current at VSS+0.4 V, output set low, high drive, VDD ≥ 2.7 V	6	10	15	mA
$I_{OL,HDL}$	Current at VSS+0.4 V, output set low, high drive, VDD ≥ 1.7 V	3			mA
$I_{OH,SD}$	Current at VDD-0.4 V, output set high, standard drive, VDD ≥ 1.7	1	2	4	mA
$I_{OH,HDH}$	Current at VDD-0.4 V, output set high, high drive, VDD ≥ 2.7 V	6	9	14	mA
$I_{OH,HDL}$	Current at VDD-0.4 V, output set high, high drive, VDD ≥ 1.7 V	3			mA
$t_{RF,15pF}$	Rise/fall time, standard drive mode, 10-90%, 15 pF load[16]		9		ns
$t_{RF,25pF}$	Rise/fall time, standard drive mode, 10-90%, 25 pF load[16]		13		ns
$t_{RF,50pF}$	Rise/fall time, standard drive mode, 10-90%, 50 pF load[16]		25		ns
$t_{HRF,15pF}$	Rise/Fall time, high drive mode, 10-90%, 15 pF load[16]		4		ns

[16] Rise and fall times based on simulations

Symbol	Description	Min.	Typ.	Max.	Units
$t_{HRF,25pF}$	Rise/Fall time, high drive mode, 10-90%, 25 pF load[16]		5		ns
$t_{HRF,50pF}$	Rise/Fall time, high drive mode, 10-90%, 50 pF load[16]		8		ns
R_{PU}	Pull-up resistance	11	13	16	kΩ
R_{PD}	Pull-down resistance	11	13	16	kΩ
C_{PAD}	Pad capacitance		3		pF

6.9.3.2 NFC Pads Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
C_{PAD_NFC}	Pad capacitance on NFC pads		4		pF
I_{NFC_LEAK}	Leakage current between NFC pads when driven to different states		1	10	µA

6.10 GPIOTE — GPIO tasks and events

The GPIO tasks and events (GPIOTE) module provides functionality for accessing GPIO pins using tasks and events. Each GPIOTE channel can be assigned to one pin.

A GPIOTE block enables GPIOs to generate events on pin state change which can be used to carry out tasks through the PPI system. A GPIO can also be driven to change state on system events using the PPI system. Tasks and events are briefly introduced in Peripheral interface on page 173, and GPIO is described in more detail in GPIO — General purpose input/output on page 322.

Low power detection of pin state changes is possible when in System ON or System OFF.

Instance	Number of GPIOTE channels
GPIOTE	8

Table 19: GPIOTE properties

Up to three tasks can be used in each GPIOTE channel for performing write operations to a pin. Two tasks are fixed (SET and CLR), and one (OUT) is configurable to perform following operations:

- Set
- Clear
- Toggle

An event can be generated in each GPIOTE channel from one of the following input conditions:

- Rising edge
- Falling edge
- Any change

6.10.1 Pin events and tasks

The GPIOTE module has a number of tasks and events that can be configured to operate on individual GPIO pins.

The tasks SET[n], CLR[n], and OUT[n] can write to individual pins, and events IN[n] can be generated from input changes of individual pins.

The SET task will set the pin selected in GPIOTE.CONFIG[n].PSEL to high. The CLR task will set the pin low.

The effect of the OUT task on the pin is configurable in CONFIG[n].POLARITY. It can set the pin high, set it low, or toggle it.

Tasks and events are configured using the CONFIG[n] registers. One CONFIG[n] register is associated with a set of SET[n], CLR[n], and OUT[n] tasks and IN[n] events.

As long as a SET[n], CLR[n], and OUT[n] task or an IN[n] event is configured to control pin **n**, the pin's output value will only be updated by the GPIOTE module. The pin's output value, as specified in the GPIO, will be ignored as long as the pin is controlled by GPIOTE. Attempting to write to the pin as a normal GPIO pin will have no effect. When the GPIOTE is disconnected from a pin, the associated pin gets the output and configuration values specified in the GPIO module, see MODE field in CONFIG[n] register.

When conflicting tasks are triggered simultaneously (i.e. during the same clock cycle) in one channel, the priority of the tasks is as described in the following table.

Priority	Task
1	OUT
2	CLR
3	SET

Table 20: Task priorities

When setting the CONFIG[n] registers, MODE=Disabled does not have the same effect as MODE=Task and POLARITY=None. In the latter case, a CLR or SET task occurring at the exact same time as OUT will end up with no change on the pin, based on the priorities described in the table above.

When a GPIOTE channel is configured to operate on a pin as a task, the initial value of that pin is configured in the OUTINIT field of CONFIG[n].

6.10.2 Port event

PORT is an event that can be generated from multiple input pins using the GPIO DETECT signal.

The event will be generated on the rising edge of the DETECT signal. See GPIO — General purpose input/output on page 322 for more information about the DETECT signal.

The GPIO DETECT signal will not wake the system up again if the system is put into System ON IDLE while the DETECT signal is high. Clear all DETECT sources before entering sleep. If the LATCH register is used as a source, a new rising edge will be generated on DETECT if any bit in LATCH is still high after clearing all or part of the register. This could occur if one of the PINx.DETECT signals is still high, for example. See Pin configuration on page 322 for more information.

Setting the system to System OFF while DETECT is high will cause a wakeup from System OFF reset.

This feature is always enabled even if the peripheral itself appears to be IDLE, meaning no clocks or other power intensive infrastructure have to be requested to keep this feature enabled. This feature can therefore be used to wake up the CPU from a WFI or WFE type sleep in System ON when all peripherals and the CPU are idle, meaning the lowest power consumption in System ON mode.

In order to prevent spurious interrupts from the PORT event while configuring the sources, the following must be performed:

1. Disable interrupts on the PORT event (through INTENCLR.PORT).
2. Configure the sources (PIN_CNF[n].SENSE).
3. Clear any potential event that could have occurred during configuration (write '0' to EVENTS_PORT).
4. Enable interrupts (through INTENSET.PORT).

6.10.3 Tasks and events pin configuration

Each GPIOTE channel is associated with one physical GPIO pin through the CONFIG.PSEL field.

When Event mode is selected in CONFIG.MODE, the pin specified by CONFIG.PSEL will be configured as an input, overriding the DIR setting in GPIO. Similarly, when Task mode is selected in CONFIG.MODE,

the pin specified by CONFIG.PSEL will be configured as an output overriding the DIR setting and OUT value in GPIO. When Disabled is selected in CONFIG.MODE, the pin specified by CONFIG.PSEL will use its configuration from the PIN[n].CNF registers in GPIO.

Note: A pin can only be assigned to one GPIOTE channel at a time. Failing to do so may result in unpredictable behavior.

6.10.4 Registers

Instances

Instance	Base address	Description
GPIOTE	0x40006000	GPIO tasks and events

Register overview

Register	Offset	Description
TASKS_OUT[0]	0x000	Task for writing to pin specified in CONFIG[0].PSEL. Action on pin is configured in CONFIG[0].POLARITY.
TASKS_OUT[1]	0x004	Task for writing to pin specified in CONFIG[1].PSEL. Action on pin is configured in CONFIG[1].POLARITY.
TASKS_OUT[2]	0x008	Task for writing to pin specified in CONFIG[2].PSEL. Action on pin is configured in CONFIG[2].POLARITY.
TASKS_OUT[3]	0x00C	Task for writing to pin specified in CONFIG[3].PSEL. Action on pin is configured in CONFIG[3].POLARITY.
TASKS_OUT[4]	0x010	Task for writing to pin specified in CONFIG[4].PSEL. Action on pin is configured in CONFIG[4].POLARITY.
TASKS_OUT[5]	0x014	Task for writing to pin specified in CONFIG[5].PSEL. Action on pin is configured in CONFIG[5].POLARITY.
TASKS_OUT[6]	0x018	Task for writing to pin specified in CONFIG[6].PSEL. Action on pin is configured in CONFIG[6].POLARITY.
TASKS_OUT[7]	0x01C	Task for writing to pin specified in CONFIG[7].PSEL. Action on pin is configured in CONFIG[7].POLARITY.
TASKS_SET[0]	0x030	Task for writing to pin specified in CONFIG[0].PSEL. Action on pin is to set it high.
TASKS_SET[1]	0x034	Task for writing to pin specified in CONFIG[1].PSEL. Action on pin is to set it high.
TASKS_SET[2]	0x038	Task for writing to pin specified in CONFIG[2].PSEL. Action on pin is to set it high.
TASKS_SET[3]	0x03C	Task for writing to pin specified in CONFIG[3].PSEL. Action on pin is to set it high.
TASKS_SET[4]	0x040	Task for writing to pin specified in CONFIG[4].PSEL. Action on pin is to set it high.
TASKS_SET[5]	0x044	Task for writing to pin specified in CONFIG[5].PSEL. Action on pin is to set it high.
TASKS_SET[6]	0x048	Task for writing to pin specified in CONFIG[6].PSEL. Action on pin is to set it high.
TASKS_SET[7]	0x04C	Task for writing to pin specified in CONFIG[7].PSEL. Action on pin is to set it high.
TASKS_CLR[0]	0x060	Task for writing to pin specified in CONFIG[0].PSEL. Action on pin is to set it low.
TASKS_CLR[1]	0x064	Task for writing to pin specified in CONFIG[1].PSEL. Action on pin is to set it low.
TASKS_CLR[2]	0x068	Task for writing to pin specified in CONFIG[2].PSEL. Action on pin is to set it low.
TASKS_CLR[3]	0x06C	Task for writing to pin specified in CONFIG[3].PSEL. Action on pin is to set it low.
TASKS_CLR[4]	0x070	Task for writing to pin specified in CONFIG[4].PSEL. Action on pin is to set it low.
TASKS_CLR[5]	0x074	Task for writing to pin specified in CONFIG[5].PSEL. Action on pin is to set it low.
TASKS_CLR[6]	0x078	Task for writing to pin specified in CONFIG[6].PSEL. Action on pin is to set it low.
TASKS_CLR[7]	0x07C	Task for writing to pin specified in CONFIG[7].PSEL. Action on pin is to set it low.
EVENTS_IN[0]	0x100	Event generated from pin specified in CONFIG[0].PSEL
EVENTS_IN[1]	0x104	Event generated from pin specified in CONFIG[1].PSEL
EVENTS_IN[2]	0x108	Event generated from pin specified in CONFIG[2].PSEL

Register	Offset	Description
EVENTS_IN[3]	0x10C	Event generated from pin specified in CONFIG[3].PSEL
EVENTS_IN[4]	0x110	Event generated from pin specified in CONFIG[4].PSEL
EVENTS_IN[5]	0x114	Event generated from pin specified in CONFIG[5].PSEL
EVENTS_IN[6]	0x118	Event generated from pin specified in CONFIG[6].PSEL
EVENTS_IN[7]	0x11C	Event generated from pin specified in CONFIG[7].PSEL
EVENTS_PORT	0x17C	Event generated from multiple input GPIO pins with SENSE mechanism enabled
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
CONFIG[0]	0x510	Configuration for OUT[n], SET[n], and CLR[n] tasks and IN[n] event
CONFIG[1]	0x514	Configuration for OUT[n], SET[n], and CLR[n] tasks and IN[n] event
CONFIG[2]	0x518	Configuration for OUT[n], SET[n], and CLR[n] tasks and IN[n] event
CONFIG[3]	0x51C	Configuration for OUT[n], SET[n], and CLR[n] tasks and IN[n] event
CONFIG[4]	0x520	Configuration for OUT[n], SET[n], and CLR[n] tasks and IN[n] event
CONFIG[5]	0x524	Configuration for OUT[n], SET[n], and CLR[n] tasks and IN[n] event
CONFIG[6]	0x528	Configuration for OUT[n], SET[n], and CLR[n] tasks and IN[n] event
CONFIG[7]	0x52C	Configuration for OUT[n], SET[n], and CLR[n] tasks and IN[n] event

6.10.4.1 TASKS_OUT[0]

Address offset: 0x000

Task for writing to pin specified in CONFIG[0].PSEL. Action on pin is configured in CONFIG[0].POLARITY.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_OUT			Task for writing to pin specified in CONFIG[0].PSEL. Action on pin is configured in CONFIG[0].POLARITY.
			Trigger	1	Trigger task

6.10.4.2 TASKS_OUT[1]

Address offset: 0x004

Task for writing to pin specified in CONFIG[1].PSEL. Action on pin is configured in CONFIG[1].POLARITY.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_OUT			Task for writing to pin specified in CONFIG[1].PSEL. Action on pin is configured in CONFIG[1].POLARITY.
			Trigger	1	Trigger task

6.10.4.3 TASKS_OUT[2]

Address offset: 0x008

Task for writing to pin specified in CONFIG[2].PSEL. Action on pin is configured in CONFIG[2].POLARITY.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_OUT			Task for writing to pin specified in CONFIG[2].PSEL. Action on pin is configured in CONFIG[2].POLARITY.
			Trigger	1	Trigger task

6.10.4.4 TASKS_OUT[3]

Address offset: 0x00C

Task for writing to pin specified in CONFIG[3].PSEL. Action on pin is configured in CONFIG[3].POLARITY.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_OUT			Task for writing to pin specified in CONFIG[3].PSEL. Action on pin is configured in CONFIG[3].POLARITY.
			Trigger	1	Trigger task

6.10.4.5 TASKS_OUT[4]

Address offset: 0x010

Task for writing to pin specified in CONFIG[4].PSEL. Action on pin is configured in CONFIG[4].POLARITY.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_OUT			Task for writing to pin specified in CONFIG[4].PSEL. Action on pin is configured in CONFIG[4].POLARITY.
			Trigger	1	Trigger task

6.10.4.6 TASKS_OUT[5]

Address offset: 0x014

Task for writing to pin specified in CONFIG[5].PSEL. Action on pin is configured in CONFIG[5].POLARITY.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_OUT			Task for writing to pin specified in CONFIG[5].PSEL. Action on pin is configured in CONFIG[5].POLARITY.
			Trigger	1	Trigger task

6.10.4.7 TASKS_OUT[6]

Address offset: 0x018

Task for writing to pin specified in CONFIG[6].PSEL. Action on pin is configured in CONFIG[6].POLARITY.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_OUT			Task for writing to pin specified in CONFIG[6].PSEL. Action on pin is configured in CONFIG[6].POLARITY.
			Trigger	1	Trigger task

6.10.4.8 TASKS_OUT[7]

Address offset: 0x01C

Task for writing to pin specified in CONFIG[7].PSEL. Action on pin is configured in CONFIG[7].POLARITY.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_OUT			Task for writing to pin specified in CONFIG[7].PSEL. Action on pin is configured in CONFIG[7].POLARITY.
			Trigger	1	Trigger task

6.10.4.9 TASKS_SET[0]

Address offset: 0x030

Task for writing to pin specified in CONFIG[0].PSEL. Action on pin is to set it high.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_SET			Task for writing to pin specified in CONFIG[0].PSEL. Action on pin is to set it high.
			Trigger	1	Trigger task

6.10.4.10 TASKS_SET[1]

Address offset: 0x034

Task for writing to pin specified in CONFIG[1].PSEL. Action on pin is to set it high.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_SET			Task for writing to pin specified in CONFIG[1].PSEL. Action on pin is to set it high.
			Trigger	1	Trigger task

6.10.4.11 TASKS_SET[2]

Address offset: 0x038

Task for writing to pin specified in CONFIG[2].PSEL. Action on pin is to set it high.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_SET			Task for writing to pin specified in CONFIG[2].PSEL. Action on pin is to set it high.
			Trigger	1	Trigger task

6.10.4.12 TASKS_SET[3]

Address offset: 0x03C

Task for writing to pin specified in CONFIG[3].PSEL. Action on pin is to set it high.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_SET			Task for writing to pin specified in CONFIG[3].PSEL. Action on pin is to set it high.
			Trigger	1	Trigger task

6.10.4.13 TASKS_SET[4]

Address offset: 0x040

Task for writing to pin specified in CONFIG[4].PSEL. Action on pin is to set it high.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_SET			Task for writing to pin specified in CONFIG[4].PSEL. Action on pin is to set it high.
			Trigger	1	Trigger task

6.10.4.14 TASKS_SET[5]

Address offset: 0x044

Task for writing to pin specified in CONFIG[5].PSEL. Action on pin is to set it high.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_SET			Task for writing to pin specified in CONFIG[5].PSEL. Action on pin is to set it high.
			Trigger	1	Trigger task

6.10.4.15 TASKS_SET[6]

Address offset: 0x048

Task for writing to pin specified in CONFIG[6].PSEL. Action on pin is to set it high.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_SET			Task for writing to pin specified in CONFIG[6].PSEL. Action on pin is to set it high.
			Trigger	1	Trigger task

6.10.4.16 TASKS_SET[7]

Address offset: 0x04C

Task for writing to pin specified in CONFIG[7].PSEL. Action on pin is to set it high.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_SET			Task for writing to pin specified in CONFIG[7].PSEL. Action on pin is to set it high.
			Trigger	1	Trigger task

6.10.4.17 TASKS_CLR[0]

Address offset: 0x060

Task for writing to pin specified in CONFIG[0].PSEL. Action on pin is to set it low.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CLR			Task for writing to pin specified in CONFIG[0].PSEL. Action on pin is to set it low.
			Trigger	1	Trigger task

6.10.4.18 TASKS_CLR[1]

Address offset: 0x064

Task for writing to pin specified in CONFIG[1].PSEL. Action on pin is to set it low.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CLR			Task for writing to pin specified in CONFIG[1].PSEL. Action on pin is to set it low.
			Trigger	1	Trigger task

6.10.4.19 TASKS_CLR[2]

Address offset: 0x068

Task for writing to pin specified in CONFIG[2].PSEL. Action on pin is to set it low.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CLR			Task for writing to pin specified in CONFIG[2].PSEL. Action on pin is to set it low.
			Trigger	1	Trigger task

6.10.4.20 TASKS_CLR[3]

Address offset: 0x06C

Task for writing to pin specified in CONFIG[3].PSEL. Action on pin is to set it low.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CLR			Task for writing to pin specified in CONFIG[3].PSEL. Action on pin is to set it low.
			Trigger	1	Trigger task

6.10.4.21 TASKS_CLR[4]

Address offset: 0x070

Task for writing to pin specified in CONFIG[4].PSEL. Action on pin is to set it low.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CLR			Task for writing to pin specified in CONFIG[4].PSEL. Action on pin is to set it low.
			Trigger	1	Trigger task

6.10.4.22 TASKS_CLR[5]

Address offset: 0x074

Task for writing to pin specified in CONFIG[5].PSEL. Action on pin is to set it low.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CLR			Task for writing to pin specified in CONFIG[5].PSEL. Action on pin is to set it low.
			Trigger	1	Trigger task

6.10.4.23 TASKS_CLR[6]

Address offset: 0x078

Task for writing to pin specified in CONFIG[6].PSEL. Action on pin is to set it low.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CLR			Task for writing to pin specified in CONFIG[6].PSEL. Action on pin is to set it low.
			Trigger	1	Trigger task

6.10.4.24 TASKS_CLR[7]

Address offset: 0x07C

Task for writing to pin specified in CONFIG[7].PSEL. Action on pin is to set it low.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CLR			Task for writing to pin specified in CONFIG[7].PSEL. Action on pin is to set it low.
			Trigger	1	Trigger task

6.10.4.25 EVENTS_IN[0]

Address offset: 0x100

Event generated from pin specified in CONFIG[0].PSEL

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_IN			Event generated from pin specified in CONFIG[0].PSEL
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.10.4.26 EVENTS_IN[1]

Address offset: 0x104

Event generated from pin specified in CONFIG[1].PSEL

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_IN			Event generated from pin specified in CONFIG[1].PSEL
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.10.4.27 EVENTS_IN[2]

Address offset: 0x108

Event generated from pin specified in CONFIG[2].PSEL

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_IN			Event generated from pin specified in CONFIG[2].PSEL
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.10.4.28 EVENTS_IN[3]

Address offset: 0x10C

Event generated from pin specified in CONFIG[3].PSEL

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_IN			Event generated from pin specified in CONFIG[3].PSEL
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.10.4.29 EVENTS_IN[4]

Address offset: 0x110

Event generated from pin specified in CONFIG[4].PSEL

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_IN			Event generated from pin specified in CONFIG[4].PSEL
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.10.4.30 EVENTS_IN[5]

Address offset: 0x114

Event generated from pin specified in CONFIG[5].PSEL

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_IN			Event generated from pin specified in CONFIG[5].PSEL
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.10.4.31 EVENTS_IN[6]

Address offset: 0x118

Event generated from pin specified in CONFIG[6].PSEL

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_IN			Event generated from pin specified in CONFIG[6].PSEL
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.10.4.32 EVENTS_IN[7]

Address offset: 0x11C

Event generated from pin specified in CONFIG[7].PSEL

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_IN			Event generated from pin specified in CONFIG[7].PSEL
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.10.4.33 EVENTS_PORT

Address offset: 0x17C

Event generated from multiple input GPIO pins with SENSE mechanism enabled

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_PORT			Event generated from multiple input GPIO pins with SENSE mechanism enabled
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.10.4.34 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	IN[0]			Write '1' to enable interrupt for event IN[0]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	IN[1]			Write '1' to enable interrupt for event IN[1]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
C	RW	IN[2]			Write '1' to enable interrupt for event IN[2]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	IN[3]			Write '1' to enable interrupt for event IN[3]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	IN[4]			Write '1' to enable interrupt for event IN[4]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	IN[5]			Write '1' to enable interrupt for event IN[5]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
G	RW	IN[6]			Write '1' to enable interrupt for event IN[6]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
H	RW	IN[7]			Write '1' to enable interrupt for event IN[7]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
I	RW	PORT			Write '1' to enable interrupt for event PORT
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.10.4.35 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	IN[0]			Write '1' to disable interrupt for event IN[0]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	IN[1]			Write '1' to disable interrupt for event IN[1]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	IN[2]			Write '1' to disable interrupt for event IN[2]
			Clear	1	Disable
			Disabled	0	Read: Disabled

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Enabled	1	Read: Enabled
D	RW	IN[3]			Write '1' to disable interrupt for event IN[3]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	IN[4]			Write '1' to disable interrupt for event IN[4]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	IN[5]			Write '1' to disable interrupt for event IN[5]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
G	RW	IN[6]			Write '1' to disable interrupt for event IN[6]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
H	RW	IN[7]			Write '1' to disable interrupt for event IN[7]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
I	RW	PORT			Write '1' to disable interrupt for event PORT
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.10.4.36 CONFIG[0]

Address offset: 0x510

Configuration for OUT[n], SET[n], and CLR[n] tasks and IN[n] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D D C B B B B B A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	MODE			Mode
			Disabled	0	Disabled. Pin specified by PSEL will not be acquired by the GPIOTE module.
			Event	1	Event mode The pin specified by PSEL will be configured as an input and the IN[n] event will be generated if operation specified in POLARITY occurs on the pin.
			Task	3	Task mode The GPIO specified by PSEL will be configured as an output and triggering the SET[n], CLR[n] or OUT[n] task will perform the operation specified by POLARITY on the pin. When enabled as a task the GPIOTE module will acquire the pin and the pin can no longer be written as a regular output pin from the GPIO module.
B	RW	PSEL		[0..31]	GPIO number associated with SET[n], CLR[n], and OUT[n] tasks and IN[n] event
C	RW	PORT		[0..1]	Port number

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D D C B B B B B A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
D	RW	POLARITY			When In task mode: Operation to be performed on output when OUT[n] task is triggered. When In event mode: Operation on input that shall trigger IN[n] event.
			None	0	Task mode: No effect on pin from OUT[n] task. Event mode: no IN[n] event generated on pin activity.
			LoToHi	1	Task mode: Set pin from OUT[n] task. Event mode: Generate IN[n] event when rising edge on pin.
			HiToLo	2	Task mode: Clear pin from OUT[n] task. Event mode: Generate IN[n] event when falling edge on pin.
			Toggle	3	Task mode: Toggle pin from OUT[n]. Event mode: Generate IN[n] when any change on pin.
E	RW	OUTINIT			When in task mode: Initial value of the output when the GPIOTE channel is configured. When in event mode: No effect.
			Low	0	Task mode: Initial value of pin before task triggering is low
			High	1	Task mode: Initial value of pin before task triggering is high

6.10.4.37 CONFIG[1]

Address offset: 0x514

Configuration for OUT[n], SET[n], and CLR[n] tasks and IN[n] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D D C B B B B B A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	MODE			Mode
			Disabled	0	Disabled. Pin specified by PSEL will not be acquired by the GPIOTE module.
			Event	1	Event mode The pin specified by PSEL will be configured as an input and the IN[n] event will be generated if operation specified in POLARITY occurs on the pin.
			Task	3	Task mode The GPIO specified by PSEL will be configured as an output and triggering the SET[n], CLR[n] or OUT[n] task will perform the operation specified by POLARITY on the pin. When enabled as a task the GPIOTE module will acquire the pin and the pin can no longer be written as a regular output pin from the GPIO module.
B	RW	PSEL		[0..31]	GPIO number associated with SET[n], CLR[n], and OUT[n] tasks and IN[n] event
C	RW	PORT		[0..1]	Port number
D	RW	POLARITY			When In task mode: Operation to be performed on output when OUT[n] task is triggered. When In event mode: Operation on input that shall trigger IN[n] event.
			None	0	Task mode: No effect on pin from OUT[n] task. Event mode: no IN[n] event generated on pin activity.
			LoToHi	1	Task mode: Set pin from OUT[n] task. Event mode: Generate IN[n] event when rising edge on pin.
			HiToLo	2	Task mode: Clear pin from OUT[n] task. Event mode: Generate IN[n] event when falling edge on pin.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D D C B B B B B A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Toggle	3	Task mode: Toggle pin from OUT[n]. Event mode: Generate IN[n] when any change on pin.
E	RW	OUTINIT			When in task mode: Initial value of the output when the GPIOTE channel is configured. When in event mode: No effect.
			Low	0	Task mode: Initial value of pin before task triggering is low
			High	1	Task mode: Initial value of pin before task triggering is high

6.10.4.38 CONFIG[2]

Address offset: 0x518

Configuration for OUT[n], SET[n], and CLR[n] tasks and IN[n] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D D C B B B B B A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	MODE			Mode
			Disabled	0	Disabled. Pin specified by PSEL will not be acquired by the GPIOTE module.
			Event	1	Event mode The pin specified by PSEL will be configured as an input and the IN[n] event will be generated if operation specified in POLARITY occurs on the pin.
			Task	3	Task mode The GPIO specified by PSEL will be configured as an output and triggering the SET[n], CLR[n] or OUT[n] task will perform the operation specified by POLARITY on the pin. When enabled as a task the GPIOTE module will acquire the pin and the pin can no longer be written as a regular output pin from the GPIO module.
B	RW	PSEL		[0..31]	GPIO number associated with SET[n], CLR[n], and OUT[n] tasks and IN[n] event
C	RW	PORT		[0..1]	Port number
D	RW	POLARITY			When In task mode: Operation to be performed on output when OUT[n] task is triggered. When In event mode: Operation on input that shall trigger IN[n] event.
			None	0	Task mode: No effect on pin from OUT[n] task. Event mode: no IN[n] event generated on pin activity.
			LoToHi	1	Task mode: Set pin from OUT[n] task. Event mode: Generate IN[n] event when rising edge on pin.
			HiToLo	2	Task mode: Clear pin from OUT[n] task. Event mode: Generate IN[n] event when falling edge on pin.
			Toggle	3	Task mode: Toggle pin from OUT[n]. Event mode: Generate IN[n] when any change on pin.
E	RW	OUTINIT			When in task mode: Initial value of the output when the GPIOTE channel is configured. When in event mode: No effect.
			Low	0	Task mode: Initial value of pin before task triggering is low
			High	1	Task mode: Initial value of pin before task triggering is high

6.10.4.39 CONFIG[3]

Address offset: 0x51C

Configuration for OUT[n], SET[n], and CLR[n] tasks and IN[n] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D D C B B B B B A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	MODE			Mode
			Disabled	0	Disabled. Pin specified by PSEL will not be acquired by the GPIOTE module.
			Event	1	Event mode The pin specified by PSEL will be configured as an input and the IN[n] event will be generated if operation specified in POLARITY occurs on the pin.
			Task	3	Task mode The GPIO specified by PSEL will be configured as an output and triggering the SET[n], CLR[n] or OUT[n] task will perform the operation specified by POLARITY on the pin. When enabled as a task the GPIOTE module will acquire the pin and the pin can no longer be written as a regular output pin from the GPIO module.
B	RW	PSEL		[0..31]	GPIO number associated with SET[n], CLR[n], and OUT[n] tasks and IN[n] event
C	RW	PORT		[0..1]	Port number
D	RW	POLARITY			When In task mode: Operation to be performed on output when OUT[n] task is triggered. When In event mode: Operation on input that shall trigger IN[n] event.
			None	0	Task mode: No effect on pin from OUT[n] task. Event mode: no IN[n] event generated on pin activity.
			LoToHi	1	Task mode: Set pin from OUT[n] task. Event mode: Generate IN[n] event when rising edge on pin.
			HiToLo	2	Task mode: Clear pin from OUT[n] task. Event mode: Generate IN[n] event when falling edge on pin.
			Toggle	3	Task mode: Toggle pin from OUT[n]. Event mode: Generate IN[n] when any change on pin.
E	RW	OUTINIT			When in task mode: Initial value of the output when the GPIOTE channel is configured. When in event mode: No effect.
			Low	0	Task mode: Initial value of pin before task triggering is low
			High	1	Task mode: Initial value of pin before task triggering is high

6.10.4.40 CONFIG[4]

Address offset: 0x520

Configuration for OUT[n], SET[n], and CLR[n] tasks and IN[n] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D D C B B B B B A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	MODE			Mode
			Disabled	0	Disabled. Pin specified by PSEL will not be acquired by the GPIOTE module.
			Event	1	Event mode The pin specified by PSEL will be configured as an input and the IN[n] event will be generated if operation specified in POLARITY occurs on the pin.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D D C B B B B B A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
			Task	3	Task mode The GPIO specified by PSEL will be configured as an output and triggering the SET[n], CLR[n] or OUT[n] task will perform the operation specified by POLARITY on the pin. When enabled as a task the GPIOTE module will acquire the pin and the pin can no longer be written as a regular output pin from the GPIO module.
B	RW	PSEL		[0..31]	GPIO number associated with SET[n], CLR[n], and OUT[n] tasks and IN[n] event
C	RW	PORT		[0..1]	Port number
D	RW	POLARITY			When In task mode: Operation to be performed on output when OUT[n] task is triggered. When In event mode: Operation on input that shall trigger IN[n] event.
			None	0	Task mode: No effect on pin from OUT[n] task. Event mode: no IN[n] event generated on pin activity.
			LoToHi	1	Task mode: Set pin from OUT[n] task. Event mode: Generate IN[n] event when rising edge on pin.
			HiToLo	2	Task mode: Clear pin from OUT[n] task. Event mode: Generate IN[n] event when falling edge on pin.
			Toggle	3	Task mode: Toggle pin from OUT[n]. Event mode: Generate IN[n] when any change on pin.
E	RW	OUTINIT			When in task mode: Initial value of the output when the GPIOTE channel is configured. When in event mode: No effect.
			Low	0	Task mode: Initial value of pin before task triggering is low
			High	1	Task mode: Initial value of pin before task triggering is high

6.10.4.41 CONFIG[5]

Address offset: 0x524

Configuration for OUT[n], SET[n], and CLR[n] tasks and IN[n] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D D C B B B B B A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	MODE			Mode
			Disabled	0	Disabled. Pin specified by PSEL will not be acquired by the GPIOTE module.
			Event	1	Event mode The pin specified by PSEL will be configured as an input and the IN[n] event will be generated if operation specified in POLARITY occurs on the pin.
			Task	3	Task mode The GPIO specified by PSEL will be configured as an output and triggering the SET[n], CLR[n] or OUT[n] task will perform the operation specified by POLARITY on the pin. When enabled as a task the GPIOTE module will acquire the pin and the pin can no longer be written as a regular output pin from the GPIO module.
B	RW	PSEL		[0..31]	GPIO number associated with SET[n], CLR[n], and OUT[n] tasks and IN[n] event
C	RW	PORT		[0..1]	Port number

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D D C B B B B B A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
D	RW	POLARITY			When In task mode: Operation to be performed on output when OUT[n] task is triggered. When In event mode: Operation on input that shall trigger IN[n] event.
			None	0	Task mode: No effect on pin from OUT[n] task. Event mode: no IN[n] event generated on pin activity.
			LoToHi	1	Task mode: Set pin from OUT[n] task. Event mode: Generate IN[n] event when rising edge on pin.
			HiToLo	2	Task mode: Clear pin from OUT[n] task. Event mode: Generate IN[n] event when falling edge on pin.
			Toggle	3	Task mode: Toggle pin from OUT[n]. Event mode: Generate IN[n] when any change on pin.
E	RW	OUTINIT			When in task mode: Initial value of the output when the GPIOTE channel is configured. When in event mode: No effect.
			Low	0	Task mode: Initial value of pin before task triggering is low
			High	1	Task mode: Initial value of pin before task triggering is high

6.10.4.42 CONFIG[6]

Address offset: 0x528

Configuration for OUT[n], SET[n], and CLR[n] tasks and IN[n] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D D C B B B B B A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	MODE			Mode
			Disabled	0	Disabled. Pin specified by PSEL will not be acquired by the GPIOTE module.
			Event	1	Event mode The pin specified by PSEL will be configured as an input and the IN[n] event will be generated if operation specified in POLARITY occurs on the pin.
			Task	3	Task mode The GPIO specified by PSEL will be configured as an output and triggering the SET[n], CLR[n] or OUT[n] task will perform the operation specified by POLARITY on the pin. When enabled as a task the GPIOTE module will acquire the pin and the pin can no longer be written as a regular output pin from the GPIO module.
B	RW	PSEL		[0..31]	GPIO number associated with SET[n], CLR[n], and OUT[n] tasks and IN[n] event
C	RW	PORT		[0..1]	Port number
D	RW	POLARITY			When In task mode: Operation to be performed on output when OUT[n] task is triggered. When In event mode: Operation on input that shall trigger IN[n] event.
			None	0	Task mode: No effect on pin from OUT[n] task. Event mode: no IN[n] event generated on pin activity.
			LoToHi	1	Task mode: Set pin from OUT[n] task. Event mode: Generate IN[n] event when rising edge on pin.
			HiToLo	2	Task mode: Clear pin from OUT[n] task. Event mode: Generate IN[n] event when falling edge on pin.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D D C B B B B B A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Toggle	3	Task mode: Toggle pin from OUT[n]. Event mode: Generate IN[n] when any change on pin.
E	RW	OUTINIT			When in task mode: Initial value of the output when the GPIOTE channel is configured. When in event mode: No effect.
			Low	0	Task mode: Initial value of pin before task triggering is low
			High	1	Task mode: Initial value of pin before task triggering is high

6.10.4.43 CONFIG[7]

Address offset: 0x52C

Configuration for OUT[n], SET[n], and CLR[n] tasks and IN[n] event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D D C B B B B B A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	MODE			Mode
			Disabled	0	Disabled. Pin specified by PSEL will not be acquired by the GPIOTE module.
			Event	1	Event mode The pin specified by PSEL will be configured as an input and the IN[n] event will be generated if operation specified in POLARITY occurs on the pin.
			Task	3	Task mode The GPIO specified by PSEL will be configured as an output and triggering the SET[n], CLR[n] or OUT[n] task will perform the operation specified by POLARITY on the pin. When enabled as a task the GPIOTE module will acquire the pin and the pin can no longer be written as a regular output pin from the GPIO module.
B	RW	PSEL		[0..31]	GPIO number associated with SET[n], CLR[n], and OUT[n] tasks and IN[n] event
C	RW	PORT		[0..1]	Port number
D	RW	POLARITY			When In task mode: Operation to be performed on output when OUT[n] task is triggered. When In event mode: Operation on input that shall trigger IN[n] event.
			None	0	Task mode: No effect on pin from OUT[n] task. Event mode: no IN[n] event generated on pin activity.
			LoToHi	1	Task mode: Set pin from OUT[n] task. Event mode: Generate IN[n] event when rising edge on pin.
			HiToLo	2	Task mode: Clear pin from OUT[n] task. Event mode: Generate IN[n] event when falling edge on pin.
			Toggle	3	Task mode: Toggle pin from OUT[n]. Event mode: Generate IN[n] when any change on pin.
E	RW	OUTINIT			When in task mode: Initial value of the output when the GPIOTE channel is configured. When in event mode: No effect.
			Low	0	Task mode: Initial value of pin before task triggering is low
			High	1	Task mode: Initial value of pin before task triggering is high

6.11 I²S — Inter-IC sound interface

The I²S (Inter-IC Sound) module, supports the original two-channel I²S format, and left or right-aligned formats. It implements EasyDMA for sample transfer directly to and from RAM without CPU intervention.

The I²S peripheral has the following main features:

- Master and Slave mode
- Simultaneous bi-directional (TX and RX) audio streaming
- Original I²S and left- or right-aligned format
- 8, 16 and 24-bit sample width
- Low-jitter Master Clock generator
- Various sample rates

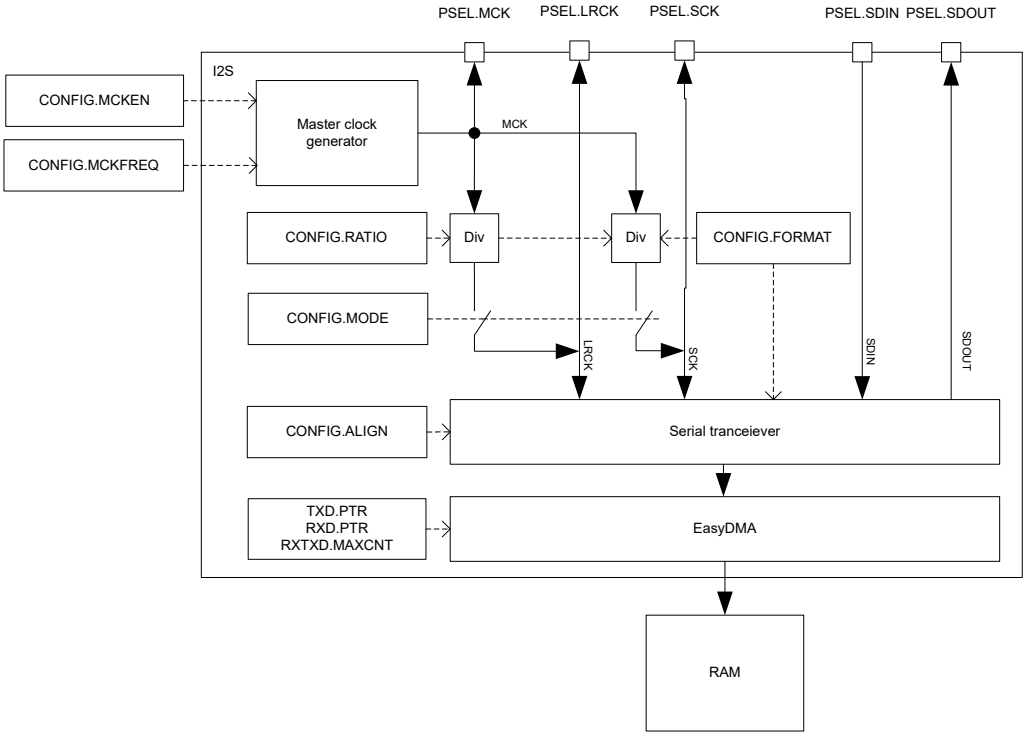


Figure 51: I²S master

6.11.1 Mode

The I²S protocol specification defines two modes of operation, Master and Slave.

The I²S mode decides which of the two sides (Master or Slave) shall provide the clock signals LRCK and SCK, and these signals are always supplied by the Master to the Slave.

6.11.2 Transmitting and receiving

The I²S module supports both transmission (TX) and reception (RX) of serial data. In both cases the serial data is shifted synchronously to the clock signals SCK and LRCK.

TX data is written to the SDOUT pin on the falling edge of SCK, and RX data is read from the SDIN pin on the rising edge of SCK. The most significant bit (MSB) is always transmitted first.

TX and RX are available in both Master and Slave modes and can be enabled/disabled independently in the CONFIG.TXEN on page 404 and CONFIG.RXEN on page 404.

Transmission and/or reception is started by triggering the START task. When started and transmission is enabled (in CONFIG.TXEN on page 404), the TXPTRUPD event will be generated for every RXTXD.MAXCNT on page 407 number of transmitted data words (containing one or more samples). Similarly, when started and reception is enabled (in CONFIG.RXEN on page 404), the RXPTRUPD event will be generated for every RXTXD.MAXCNT on page 407 received data words.

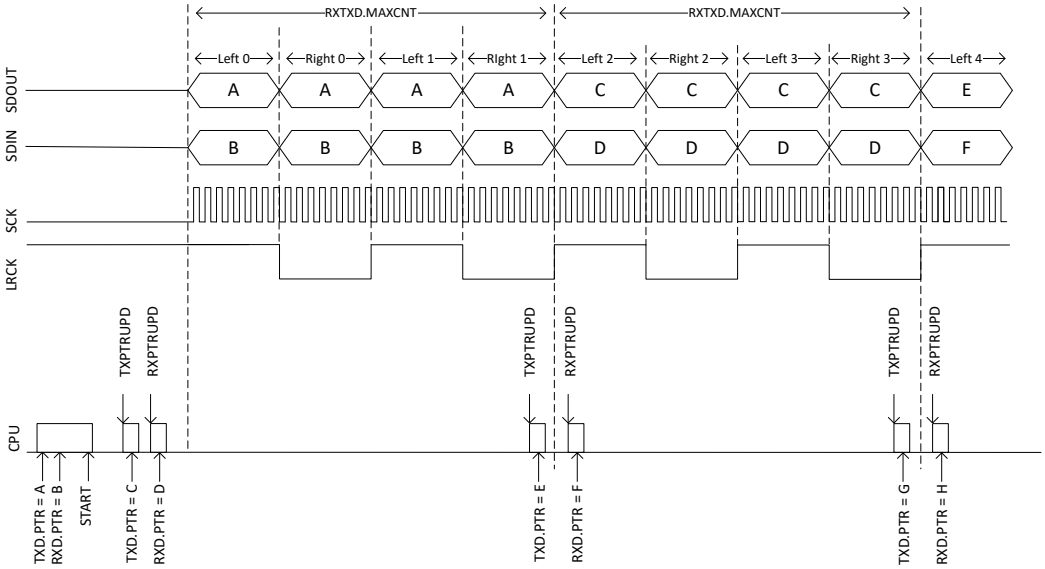


Figure 52: Transmitting and receiving. CONFIG.FORMAT = Aligned, CONFIG.SWIDTH = 8Bit, CONFIG.CHANNELS = Stereo, RXTXD.MAXCNT = 1.

6.11.3 Left right clock (LRCK)

The Left Right Clock (LRCK), often referred to as "word clock", "sample clock" or "word select" in I^2S context, is the clock defining the frames in the serial bit streams sent and received on SDOUT and SDIN, respectively.

In I2S mode, each frame contains one left and right sample pair, with the left sample being transferred during the low half period of LRCK followed by the right sample being transferred during the high period of LRCK.

In Aligned mode, each frame contains one left and right sample pair, with the left sample being transferred during the high half period of LRCK followed by the right sample being transferred during the low period of LRCK.

Consequently, the LRCK frequency is equivalent to the audio sample rate.

When operating in Master mode, the LRCK is generated from the MCK, and the frequency of LRCK is then given as:

```
LRCK = MCK / CONFIG.RATIO
```

LRCK always toggles around the falling edge of the serial clock SCK.

6.11.4 Serial clock (SCK)

The serial clock (SCK), often referred to as the serial bit clock, pulses once for each data bit being transferred on the serial data lines SDIN and SDOUT.

When operating in Master mode the SCK is generated from the MCK, and the frequency of SCK is then given as:

```
SCK = 2 * LRCK * CONFIG.SWIDTH
```

The falling edge of the SCK falls on the toggling edge of LRCK.

When operating in Slave mode SCK is provided by the external I^{2}S master.

6.11.5 Master clock (MCK)

The master clock (MCK) is the clock from which LRCK and SCK are derived when operating in Master mode.

The MCK is generated by an internal MCK generator. This generator always needs to be enabled when in Master mode, but the generator can also be enabled when in Slave mode. Enabling the generator when in slave mode can be useful in the case where the external Master is not able to generate its own master clock.

The MCK generator is enabled/disabled in the register CONFIG.MCKEN on page 404, and the generator is started or stopped by the START or STOP tasks.

In Master mode the LRCK and the SCK frequencies are closely related, as both are derived from MCK and set indirectly through CONFIG.RATIO on page 405 and CONFIG.SWIDTH on page 406.

When configuring these registers, the user is responsible for fulfilling the following requirements:

1. SCK frequency can never exceed the MCK frequency, which can be formulated as:

   ```
   CONFIG.RATIO >= 2 * CONFIG.SWIDTH
   ```

2. The MCK/LRCK ratio shall be a multiple of 2 * CONFIG.SWIDTH, which can be formulated as:

```
Integer = (CONFIG.RATIO / (2 * CONFIG.SWIDTH))
```

The MCK signal can be routed to an output pin (specified in PSEL.MCK) to supply external I^{2}S devices that require the MCK to be supplied from the outside.

When operating in Slave mode, the I^{2}S module does not use the MCK and the MCK generator does not need to be enabled.

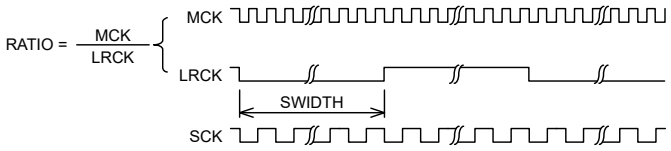


Figure 53: Relation between RATIO, MCK and LRCK.

Desired LRCK [Hz]	CONFIG.SWID	CONFIG.RATIO	CONFIG.MCKF	MCK [Hz]	LRCK [Hz]	LRCK error [%]
16000	16Bit	32X	32MDIV63	507936.5	15873.0	-0.8
16000	16Bit	64X	32MDIV31	1032258.1	16129.0	0.8
16000	16Bit	256X	32MDIV8	4000000.0	15625.0	-2.3
32000	16Bit	32X	32MDIV31	1032258.1	32258.1	0.8
32000	16Bit	64X	32MDIV16	2000000.0	31250.0	-2.3
44100	16Bit	32X	32MDIV23	1391304.3	43478.3	-1.4
44100	16Bit	64X	32MDIV11	2909090.9	45454.5	3.1

Table 21: Configuration examples

6.11.6 Width, alignment and format

The CONFIG.SWIDTH register primarily defines the sample width of the data written to memory. In master mode, it then also sets the amount of bits per frame. In Slave mode it controls padding/trimming if required. Left, right, transmitted, and received samples always have the same width. The CONFIG.FORMAT register specifies the position of the data frames with respect to the LRCK edges in both Master and Slave modes.

When using I^2S format, the first bit in a half-frame (containing one left or right sample) gets sampled on the second rising edge of the SCK after a LRCK edge. When using Aligned mode, the first bit in a half-frame gets sampled on the first rising edge of SCK following a LRCK edge.

For data being received on SDIN the sample value can be either right or left-aligned inside a half-frame, as specified in CONFIG.ALIGN on page 406. CONFIG.ALIGN on page 406 affects only the decoding of the incoming samples (SDIN), while the outgoing samples (SDOUT) are always left-aligned (or justified).

When using left-alignment, each half-frame starts with the MSB of the sample value (both for data being sent on SDOUT and received on SDIN).

When using right-alignment, each half-frame of data being received on SDIN ends with the LSB of the sample value, while each half-frame of data being sent on SDOUT starts with the MSB of the sample value (same as for left-alignment).

In Master mode, the size of a half-frame (in number of SCK periods) equals the sample width (in number of bits), and in this case the alignment setting does not care as each half-frame in any case will start with the MSB and end with the LSB of the sample value.

In slave mode, however, the sample width does not need to equal the frame size. This means you might have extra or fewer SCK pulses per half-frame than what the sample width specified in CONFIG.SWIDTH requires.

In the case where we use **left-alignment** and the number of SCK pulses per half-frame is **higher** than the sample width, the following will apply:

- For data received on SDIN, all bits after the LSB of the sample value will be discarded.
- For data sent on SDOUT, all bits after the LSB of the sample value will be 0.

In the case where we use **left-alignment** and the number of SCK pulses per frame is **lower** than the sample width, the following will apply:

- Data sent and received on SDOUT and SDIN will be truncated with the LSBs being removed first.

In the case where we use **right-alignment** and the number of SCK pulses per frame is **higher** than the sample width, the following will apply:

- For data received on SDIN, all bits before the MSB of the sample value will be discarded.
- For data sent on SDOUT, all bits after the LSB of the sample value will be 0 (same behavior as for left-alignment).

In the case where we use **right-alignment** and the number of SCK pulses per frame is **lower** than the sample width, the following will apply:

- Data received on SDIN will be sign-extended to "sample width" number of bits before being written to memory.
- Data sent on SDOUT will be truncated with the LSBs being removed first (same behavior as for left-alignment).

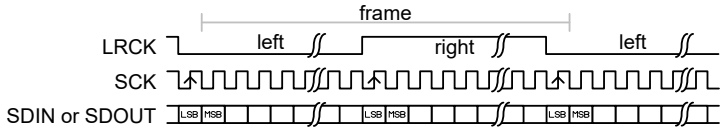


Figure 54: I²S format. CONFIG.SWIDTH equalling half-frame size.

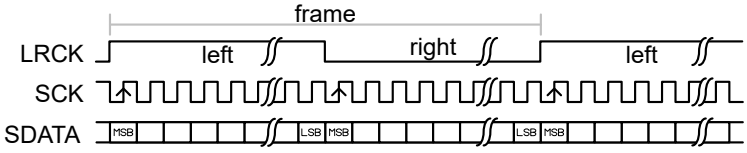


Figure 55: Aligned format. CONFIG.SWIDTH equalling half-frame size.

6.11.7 EasyDMA

The I²S module implements EasyDMA for accessing internal Data RAM without CPU intervention.

The source and destination pointers for the TX and RX data are configured in TXD.PTR on page 407 and RXD.PTR on page 407. The memory pointed to by these pointers will only be read or written when TX or RX are enabled in CONFIG.TXEN on page 404 and CONFIG.RXEN on page 404.

The addresses written to the pointer registers TXD.PTR on page 407 and RXD.PTR on page 407 are double-buffered in hardware, and these double buffers are updated for every RXTXD.MAXCNT on page 407 words (containing one or more samples) read/written from/to memory. The events TXPTRUPD and RXPTRUPD are generated whenever the TXD.PTR and RXD.PTR are transferred to these double buffers.

If TXD.PTR on page 407 is not pointing to the Data RAM region when transmission is enabled, or RXD.PTR on page 407 is not pointing to the Data RAM region when reception is enabled, an EasyDMA transfer may result in a HardFault and/or memory corruption. See Memory on page 21 for more information about the different memory regions.

Due to the nature of I²S, where the number of transmitted samples always equals the number of received samples (at least when both TX and RX are enabled), one common register RXTXD.MAXCNT on page 407 is used for specifying the sizes of these two memory buffers. The size of the buffers is specified in a number of 32-bit words. Such a 32-bit memory word can either contain four 8-bit samples, two 16-bit samples or one right-aligned 24-bit sample sign extended to 32 bit.

In stereo mode (CONFIG.CHANNELS=Stereo), the samples are stored as "left and right sample pairs" in memory. Figure Memory mapping for 8 bit stereo. CONFIG.SWIDTH = 8Bit, CONFIG.CHANNELS = Stereo. on page 396, Memory mapping for 16 bit stereo. CONFIG.SWIDTH = 16Bit, CONFIG.CHANNELS = Stereo. on page 396 and Memory mapping for 24 bit stereo. CONFIG.SWIDTH = 24Bit, CONFIG.CHANNELS = Stereo. on page 397 show how the samples are mapped to memory in this mode. The mapping is valid for both RX and TX.

In mono mode (CONFIG.CHANNELS=Left or Right), RX sample from only one channel in the frame is stored in memory, the other channel sample is ignored. Illustrations Memory mapping for 8 bit mono. CONFIG.SWIDTH = 8Bit, CONFIG.CHANNELS = Left. on page 396, Memory mapping for 16 bit mono, left

channel only. CONFIG.SWIDTH = 16Bit, CONFIG.CHANNELS = Left. on page 396 and Memory mapping for 24 bit mono, left channel only. CONFIG.SWIDTH = 24Bit, CONFIG.CHANNELS = Left. on page 397 show how RX samples are mapped to memory in this mode.

For TX, the same outgoing sample read from memory is transmitted on both left and right in a frame, resulting in a mono output stream.

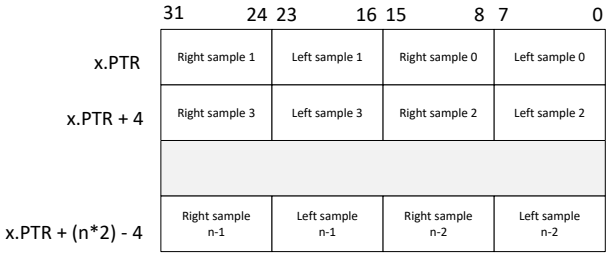


Figure 56: Memory mapping for 8 bit stereo. CONFIG.SWIDTH = 8Bit, CONFIG.CHANNELS = Stereo.

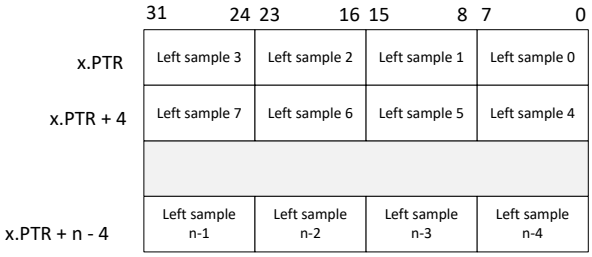


Figure 57: Memory mapping for 8 bit mono. CONFIG.SWIDTH = 8Bit, CONFIG.CHANNELS = Left.

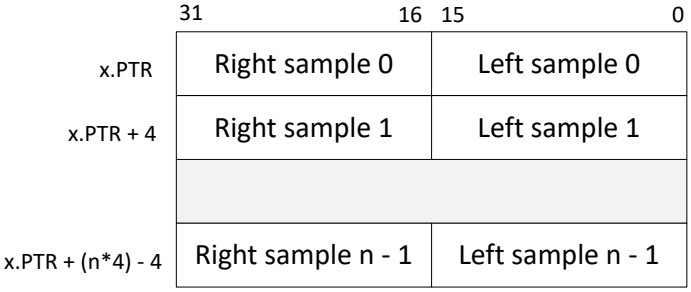


Figure 58: Memory mapping for 16 bit stereo. CONFIG.SWIDTH = 16Bit, CONFIG.CHANNELS = Stereo.

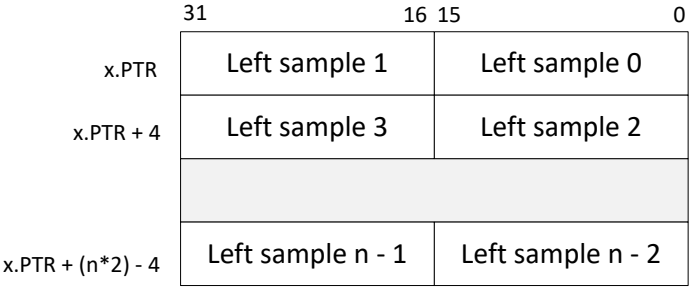


Figure 59: Memory mapping for 16 bit mono, left channel only. CONFIG.SWIDTH = 16Bit, CONFIG.CHANNELS = Left.

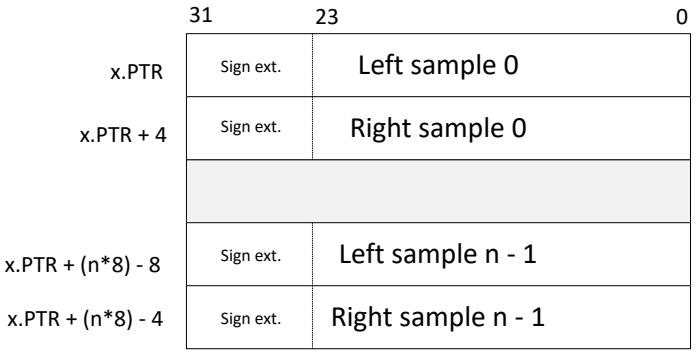


Figure 60: Memory mapping for 24 bit stereo. CONFIG.SWIDTH = 24Bit, CONFIG.CHANNELS = Stereo.

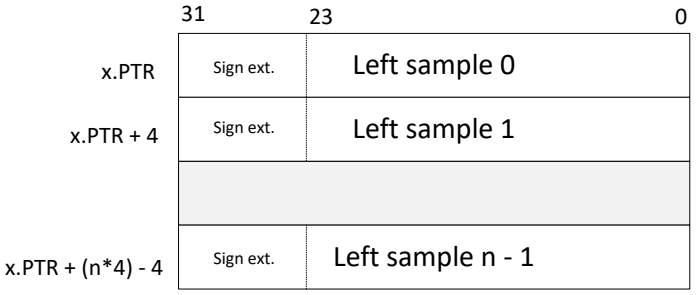


Figure 61: Memory mapping for 24 bit mono, left channel only. CONFIG.SWIDTH = 24Bit, CONFIG.CHANNELS = Left.

6.11.8 Module operation

Described here is a typical operating procedure for the I^2S module.

1. Configure the I²S module using the CONFIG registers

```
// Enable reception
NRF_I2S->CONFIG.RXEN = (I2S_CONFIG_RXEN_RXEN_Enabled <<
                                        I2S_CONFIG_RXEN_RXEN_Pos);
// Enable transmission
NRF_I2S->CONFIG.TXEN = (I2S_CONFIG_TXEN_TXEN_Enabled <<
                                        I2S_CONFIG_TXEN_TXEN_Pos);
// Enable MCK generator
NRF_I2S->CONFIG.MCKEN = (I2S_CONFIG_MCKEN_MCKEN_Enabled <<
                                        I2S_CONFIG_MCKEN_MCKEN_Pos);
// MCKFREQ = 4 MHz
NRF_I2S->CONFIG.MCKFREQ = I2S_CONFIG_MCKFREQ_MCKFREQ_32MDIV8 <<
                                        I2S_CONFIG_MCKFREQ_MCKFREQ_Pos;
// Ratio = 256
NRF_I2S->CONFIG.RATIO = I2S_CONFIG_RATIO_RATIO_256X <<
                                        I2S_CONFIG_RATIO_RATIO_Pos;
// MCKFREQ = 4 MHz and Ratio = 256 gives sample rate = 15.625 ks/s
// Sample width = 16 bit
NRF_I2S->CONFIG.SWIDTH = I2S_CONFIG_SWIDTH_SWIDTH_16Bit <<
                                        I2S_CONFIG_SWIDTH_SWIDTH_Pos;
// Alignment = Left
NRF_I2S->CONFIG.ALIGN = I2S_CONFIG_ALIGN_ALIGN_Left <<
                                        I2S_CONFIG_ALIGN_ALIGN_Pos;
// Format = I2S
NRF_I2S->CONFIG.FORMAT = I2S_CONFIG_FORMAT_FORMAT_I2S <<
                                        I2S_CONFIG_FORMAT_FORMAT_Pos;
// Use stereo
NRF_I2S->CONFIG.CHANNELS = I2S_CONFIG_CHANNELS_CHANNELS_Stereo <<
                                        I2S_CONFIG_CHANNELS_CHANNELS_Pos;
```

2. Map IO pins using the PINSEL registers

```
// MCK routed to pin 0
NRF_I2S->PSEL.MCK = (0 << I2S_PSEL_MCK_PIN_Pos) |
                    (I2S_PSEL_MCK_CONNECT_Connected <<
                                                I2S_PSEL_MCK_CONNECT_Pos);
// SCK routed to pin 1
NRF_I2S->PSEL.SCK = (1 << I2S_PSEL_SCK_PIN_Pos) |
                    (I2S_PSEL_SCK_CONNECT_Connected <<
                                                I2S_PSEL_SCK_CONNECT_Pos);
// LRCK routed to pin 2
NRF_I2S->PSEL.LRCK = (2 << I2S_PSEL_LRCK_PIN_Pos) |
                     (I2S_PSEL_LRCK_CONNECT_Connected <<
                                                I2S_PSEL_LRCK_CONNECT_Pos);
// SDOUT routed to pin 3
NRF_I2S->PSEL.SDOUT = (3 << I2S_PSEL_SDOUT_PIN_Pos) |
                      (I2S_PSEL_SDOUT_CONNECT_Connected <<
                                                I2S_PSEL_SDOUT_CONNECT_Pos);
// SDIN routed on pin 4
NRF_I2S->PSEL.SDIN = (4 << I2S_PSEL_SDIN_PIN_Pos) |
                     (I2S_PSEL_SDIN_CONNECT_Connected <<
                                                I2S_PSEL_SDIN_CONNECT_Pos);
```

3. Configure TX and RX data pointers using the TXD, RXD and RXTXD registers

```
NRF_I2S->TXD.PTR = my_tx_buf;
NRF_I2S->RXD.PTR = my_rx_buf;
NRF_I2S->TXD.MAXCNT = MY_BUF_SIZE;
```

4. Enable the I²S module using the ENABLE register

```
NRF_I2S->ENABLE = 1;
```

5. Start audio streaming using the START task

```
NRF_I2S->TASKS_START = 1;
```

6. Handle received and transmitted data when receiving the TXPTRUPD and RXPTRUPD events

```
if(NRF_I2S->EVENTS_TXPTRUPD  != 0)
{
    NRF_I2S->TXD.PTR = my_next_tx_buf;
    NRF_I2S->EVENTS_TXPTRUPD = 0;
}

if(NRF_I2S->EVENTS_RXPTRUPD != 0)
{
    NRF_I2S->RXD.PTR = my_next_rx_buf;
    NRF_I2S->EVENTS_RXPTRUPD = 0;
}
```

6.11.9 Pin configuration

The MCK, SCK, LRCK, SDIN and SDOUT signals associated with the I²S module are mapped to physical pins according to the pin numbers specified in the PSEL.x registers.

These pins are acquired whenever the I²S module is enabled through the register ENABLE on page 403.

When a pin is acquired by the I²S module, the direction of the pin (input or output) will be configured automatically, and any pin direction setting done in the GPIO module will be overridden. The directions for the various I²S pins are shown below in GPIO configuration before enabling peripheral (master mode) on page 399 and GPIO configuration before enabling peripheral (slave mode) on page 400.

To secure correct signal levels on the pins when the system is in OFF mode, and when the I²S module is disabled, these pins must be configured in the GPIO peripheral directly.

I²S signal	I²S pin	Direction	Output value	Comment
MCK	As specified in PSEL.MCK	Output	0	
LRCK	As specified in PSEL.LRCK	Output	0	
SCK	As specified in PSEL.SCK	Output	0	
SDIN	As specified in PSEL.SDIN	Input	Not applicable	
SDOUT	As specified in PSEL.SDOUT	Output	0	

Table 22: GPIO configuration before enabling peripheral (master mode)

I²S signal	I²S pin	Direction	Output value	Comment
MCK	As specified in PSEL.MCK	Output	0	
LRCK	As specified in PSEL.LRCK	Input	Not applicable	
SCK	As specified in PSEL.SCK	Input	Not applicable	
SDIN	As specified in PSEL.SDIN	Input	Not applicable	
SDOUT	As specified in PSEL.SDOUT	Output	0	

Table 23: GPIO configuration before enabling peripheral (slave mode)

6.11.10 Registers

Instances

Instance	Base address	Description
I2S	0x40025000	Inter-IC sound interface

Register overview

Register	Offset	Description
TASKS_START	0x000	Starts continuous I2S transfer. Also starts MCK generator when this is enabled.
TASKS_STOP	0x004	Stops I2S transfer. Also stops MCK generator. Triggering this task will cause the STOPPED event to be generated.
EVENTS_RXPTRUPD	0x104	The RXD.PTR register has been copied to internal double-buffers. When the I2S module is started and RX is enabled, this event will be generated for every RXTXD.MAXCNT words that are received on the SDIN pin.
EVENTS_STOPPED	0x108	I2S transfer stopped.
EVENTS_TXPTRUPD	0x114	The TDX.PTR register has been copied to internal double-buffers. When the I2S module is started and TX is enabled, this event will be generated for every RXTXD.MAXCNT words that are sent on the SDOUT pin.
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
ENABLE	0x500	Enable I2S module.
CONFIG.MODE	0x504	I2S mode.
CONFIG.RXEN	0x508	Reception (RX) enable.
CONFIG.TXEN	0x50C	Transmission (TX) enable.
CONFIG.MCKEN	0x510	Master clock generator enable.
CONFIG.MCKFREQ	0x514	Master clock generator frequency.
CONFIG.RATIO	0x518	MCK / LRCK ratio.
CONFIG.SWIDTH	0x51C	Sample width.
CONFIG.ALIGN	0x520	Alignment of sample within a frame.
CONFIG.FORMAT	0x524	Frame format.
CONFIG.CHANNELS	0x528	Enable channels.
RXD.PTR	0x538	Receive buffer RAM start address.
TXD.PTR	0x540	Transmit buffer RAM start address.
RXTXD.MAXCNT	0x550	Size of RXD and TXD buffers.
PSEL.MCK	0x560	Pin select for MCK signal.
PSEL.SCK	0x564	Pin select for SCK signal.
PSEL.LRCK	0x568	Pin select for LRCK signal.
PSEL.SDIN	0x56C	Pin select for SDIN signal.
PSEL.SDOUT	0x570	Pin select for SDOUT signal.

6.11.10.1 TASKS_START

Address offset: 0x000

Starts continuous I2S transfer. Also starts MCK generator when this is enabled.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_START			Starts continuous I2S transfer. Also starts MCK generator when this is enabled.
			Trigger	1	Trigger task

6.11.10.2 TASKS_STOP

Address offset: 0x004

Stops I2S transfer. Also stops MCK generator. Triggering this task will cause the STOPPED event to be generated.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STOP			Stops I2S transfer. Also stops MCK generator. Triggering this task will cause the STOPPED event to be generated.
			Trigger	1	Trigger task

6.11.10.3 EVENTS_RXPTRUPD

Address offset: 0x104

The RXD.PTR register has been copied to internal double-buffers. When the I2S module is started and RX is enabled, this event will be generated for every RXTXD.MAXCNT words that are received on the SDIN pin.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_RXPTRUPD			The RXD.PTR register has been copied to internal double-buffers. When the I2S module is started and RX is enabled, this event will be generated for every RXTXD.MAXCNT words that are received on the SDIN pin.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.11.10.4 EVENTS_STOPPED

Address offset: 0x108

I2S transfer stopped.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_STOPPED			I2S transfer stopped.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.11.10.5 EVENTS_TXPTRUPD

Address offset: 0x114

The TDX.PTR register has been copied to internal double-buffers. When the I2S module is started and TX is enabled, this event will be generated for every RXTXD.MAXCNT words that are sent on the SDOUT pin.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TXPTRUPD			The TDX.PTR register has been copied to internal double-buffers. When the I2S module is started and TX is enabled, this event will be generated for every RXTXD.MAXCNT words that are sent on the SDOUT pin.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.11.10.6 INTEN

Address offset: 0x300

Enable or disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F C B
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
B	RW	RXPTRUPD			Enable or disable interrupt for event RXPTRUPD
			Disabled	0	Disable
			Enabled	1	Enable
C	RW	STOPPED			Enable or disable interrupt for event STOPPED
			Disabled	0	Disable
			Enabled	1	Enable
F	RW	TXPTRUPD			Enable or disable interrupt for event TXPTRUPD
			Disabled	0	Disable
			Enabled	1	Enable

6.11.10.7 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F C B
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
B	RW	RXPTRUPD			Write '1' to enable interrupt for event RXPTRUPD
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	STOPPED			Write '1' to enable interrupt for event STOPPED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	TXPTRUPD			Write '1' to enable interrupt for event TXPTRUPD
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.11.10.8 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F C B
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
B	RW	RXPTRUPD			Write '1' to disable interrupt for event RXPTRUPD
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	STOPPED			Write '1' to disable interrupt for event STOPPED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	TXPTRUPD			Write '1' to disable interrupt for event TXPTRUPD
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.11.10.9 ENABLE

Address offset: 0x500

Enable I2S module.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ENABLE			Enable I2S module.
			Disabled	0	Disable
			Enabled	1	Enable

6.11.10.10 CONFIG.MODE

Address offset: 0x504

I2S mode.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	MODE																																	
			Master	0																															
			Slave	1																															

6.11.10.11 CONFIG.RXEN

Address offset: 0x508

Reception (RX) enable.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	RXEN																																	
			Disabled	0																															
			Enabled	1																															

6.11.10.12 CONFIG.TXEN

Address offset: 0x50C

Transmission (TX) enable.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000001				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW	TXEN																																	
			Disabled	0																															
			Enabled	1																															

6.11.10.13 CONFIG.MCKEN

Address offset: 0x510

Master clock generator enable.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	MCKEN			Master clock generator enable.
			Disabled	0	Master clock generator disabled and PSEL.MCK not connected(available as GPIO).
			Enabled	1	Master clock generator running and MCK output on PSEL.MCK.

6.11.10.14 CONFIG.MCKFREQ

Address offset: 0x514

Master clock generator frequency.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x20000000				**0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	MCKFREQ			Master clock generator frequency.
			32MDIV8	0x20000000	32 MHz / 8 = 4.0 MHz
			32MDIV10	0x18000000	32 MHz / 10 = 3.2 MHz
			32MDIV11	0x16000000	32 MHz / 11 = 2.9090909 MHz
			32MDIV15	0x11000000	32 MHz / 15 = 2.1333333 MHz
			32MDIV16	0x10000000	32 MHz / 16 = 2.0 MHz
			32MDIV21	0x0C000000	32 MHz / 21 = 1.5238095
			32MDIV23	0x0B000000	32 MHz / 23 = 1.3913043 MHz
			32MDIV30	0x08800000	32 MHz / 30 = 1.0666667 MHz
			32MDIV31	0x08400000	32 MHz / 31 = 1.0322581 MHz
			32MDIV32	0x08000000	32 MHz / 32 = 1.0 MHz
			32MDIV42	0x06000000	32 MHz / 42 = 0.7619048 MHz
			32MDIV63	0x04100000	32 MHz / 63 = 0.5079365 MHz
			32MDIV125	0x020C0000	32 MHz / 125 = 0.256 MHz

6.11.10.15 CONFIG.RATIO

Address offset: 0x518

MCK / LRCK ratio.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A
Reset 0x00000006				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RATIO			MCK / LRCK ratio.
			32X	0	LRCK = MCK / 32
			48X	1	LRCK = MCK / 48
			64X	2	LRCK = MCK / 64
			96X	3	LRCK = MCK / 96
			128X	4	LRCK = MCK / 128
			192X	5	LRCK = MCK / 192
			256X	6	LRCK = MCK / 256
			384X	7	LRCK = MCK / 384
			512X	8	LRCK = MCK / 512

6.11.10.16 CONFIG.SWIDTH

Address offset: 0x51C

Sample width.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																		A	A
Reset 0x00000001				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW	SWIDTH																																	
			8Bit	0																															
			16Bit	1																															
			24Bit	2																															

6.11.10.17 CONFIG.ALIGN

Address offset: 0x520

Alignment of sample within a frame.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	ALIGN																																	
			Left	0																															
			Right	1																															

6.11.10.18 CONFIG.FORMAT

Address offset: 0x524

Frame format.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	FORMAT																																	
			I2S	0																															
			Aligned	1																															

6.11.10.19 CONFIG.CHANNELS

Address offset: 0x528

Enable channels.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	CHANNELS			Enable channels.
			Stereo	0	Stereo.
			Left	1	Left only.
			Right	2	Right only.

6.11.10.20 RXD.PTR

Address offset: 0x538

Receive buffer RAM start address.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Receive buffer Data RAM start address. When receiving, words containing samples will be written to this address. This address is a word aligned Data RAM address.

6.11.10.21 TXD.PTR

Address offset: 0x540

Transmit buffer RAM start address.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Transmit buffer Data RAM start address. When transmitting, words containing samples will be fetched from this address. This address is a word aligned Data RAM address.

6.11.10.22 RXTXD.MAXCNT

Address offset: 0x550

Size of RXD and TXD buffers.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	MAXCNT			Size of RXD and TXD buffers in number of 32 bit words.

6.11.10.23 PSEL.MCK

Address offset: 0x560

Pin select for MCK signal.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				C																										B	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW	PIN		[0..31]																															
B	RW	PORT		[0..1]																															
C	RW	CONNECT																																	
			Disconnected	1																															
			Connected	0																															

6.11.10.24 PSEL.SCK

Address offset: 0x564

Pin select for SCK signal.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				C																										B	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW	PIN		[0..31]																															
B	RW	PORT		[0..1]																															
C	RW	CONNECT																																	
			Disconnected	1																															
			Connected	0																															

6.11.10.25 PSEL.LRCK

Address offset: 0x568

Pin select for LRCK signal.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				C																										B	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW	PIN		[0..31]																															
B	RW	PORT		[0..1]																															
C	RW	CONNECT																																	
			Disconnected	1																															
			Connected	0																															

6.11.10.26 PSEL.SDIN

Address offset: 0x56C

Pin select for SDIN signal.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.11.10.27 PSEL.SDOUT

Address offset: 0x570

Pin select for SDOUT signal.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.11.11 Electrical specification

6.11.11.1 I2S timing specification

Symbol	Description	Min.	Typ.	Max.	Units
t_{S_SDIN}	SDIN setup time before SCK rising	20			ns
t_{H_SDIN}	SDIN hold time after SCK rising	15			ns
t_{S_SDOUT}	SDOUT setup time after SCK falling	40			ns
t_{H_SDOUT}	SDOUT hold time before SCK falling	6			ns
t_{SCK_LRCK}	SCLK falling to LRCK edge	-5	0	5	ns
f_{MCK}	MCK frequency			4000	kHz
f_{LRCK}	LRCK frequency			48	kHz
f_{SCK}	SCK frequency			2000	kHz
DC_{CK}	Clock duty cycle (MCK, LRCK, SCK)	45		55	%

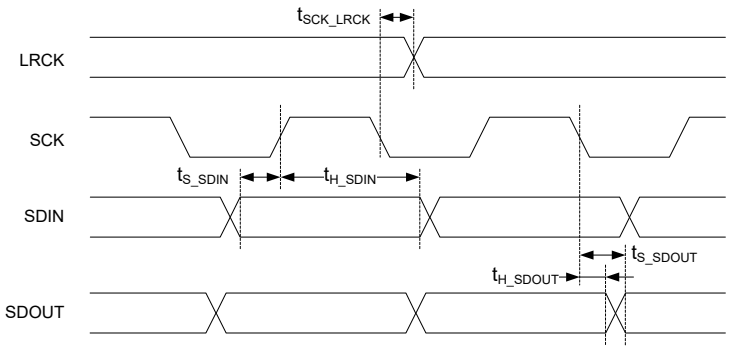


Figure 62: I2S timing diagram

6.12 LPCOMP — Low-power comparator

Low-power comparator (LPCOMP) compares an input voltage against a reference voltage.

Listed here are the main features of LPCOMP:

- 0 - VDD input range
- Ultra-low power
- Eight input options (**AIN0** to **AIN7**)
- Reference voltage options:
  - Two external analog reference inputs, or
  - 15-level internal reference ladder (VDD/16)
- Optional hysteresis enable on input
- Can be used as a wakeup source from System OFF mode

In System ON, the LPCOMP can generate separate events on rising and falling edges of a signal, or sample the current state of the pin as being above or below the selected reference. The block can be configured to use any of the analog inputs on the device. Additionally, the low-power comparator can be used as an analog wakeup source from System OFF or System ON. The comparator threshold can be programmed to a range of fractions of the supply voltage.

Note: LPCOMP cannot be used (STARTed) at the same time as COMP. Only one comparator can be used at a time.

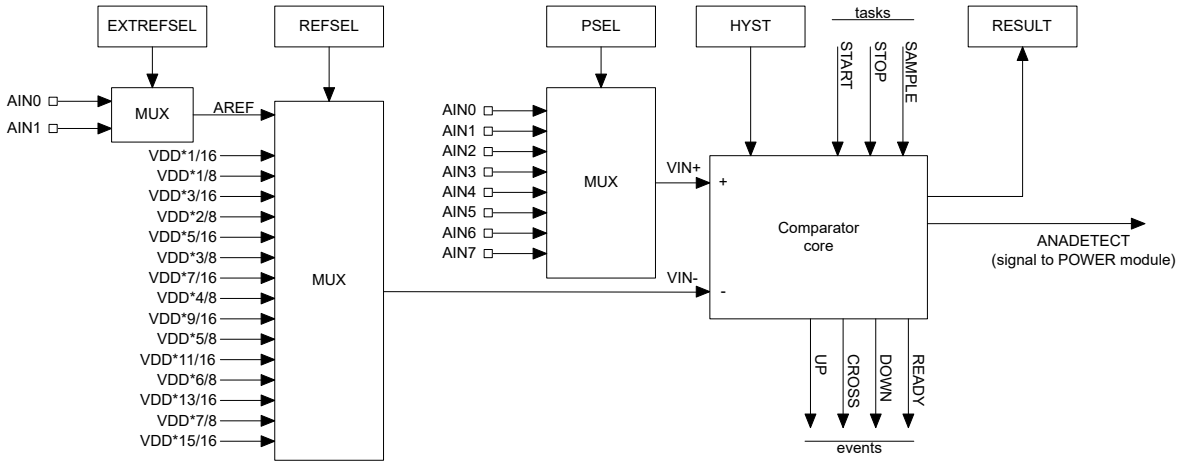


Figure 63: Low-power comparator

The wakeup comparator (LPCOMP) compares an input voltage (VIN+), which comes from an analog input pin selected via the PSEL register, against a reference voltage (VIN-) selected via registers REFSEL on page 416 and EXTREFSEL.

The PSEL, REFSEL, and EXTREFSEL registers must be configured before the LPCOMP is enabled through the ENABLE register.

The HYST register allows enabling an optional hysteresis in the comparator core. This hysteresis shall prevent noise on the signal to create unwanted events. Figure below illustrates the effect of an active hysteresis on a noisy input signal. It is disabled by default, and shall be configured before enabling LPCOMP as well.

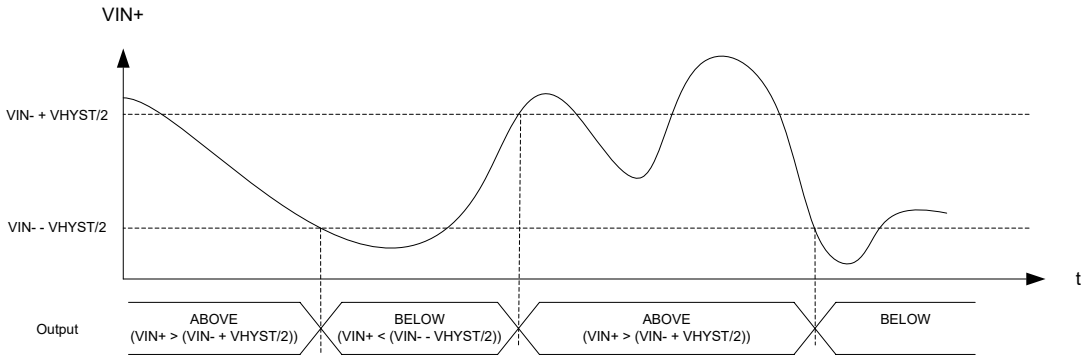


Figure 64: Effect of hysteresis on a noisy input signal

The LPCOMP is started by triggering the START task. After a startup time of $t_{LPCOMP,STARTUP}$, the LPCOMP will generate a READY event to indicate that the comparator is ready to use and the output of the LPCOMP is correct. The LPCOMP will generate events every time VIN+ crosses VIN-. More specifically, every time VIN+ rises above VIN- (upward crossing) an UP event is generated along with a CROSS event. Every time VIN+ falls below VIN- (downward crossing), a DOWN event is generated along with a CROSS event. When hysteresis is enabled, the upward crossing level becomes (VIN- + VHYST/2), and the downward crossing level becomes (VIN- - VHYST/2).

The LPCOMP is stopped by triggering the STOP task.

LPCOMP will be operational in both System ON and System OFF mode when it is enabled through the ENABLE register. See POWER — Power supply on page 81 for more information about power modes. Note that it is not allowed to go to System OFF when a READY event is pending to be generated.

All LPCOMP registers, including ENABLE, are classified as retained registers when the LPCOMP is enabled. However, when the device wakes up from System OFF, all LPCOMP registers will be reset.

The LPCOMP can wake up the system from System OFF by asserting the ANADETECT signal. The ANADETECT signal can be derived from any of the event sources that generate the UP, DOWN and CROSS events. In case of wakeup from System OFF, no events will be generated, only the ANADETECT signal. See the ANADETECT register (ANADETECT on page 417) for more information on how to configure the ANADETECT signal.

The immediate value of the LPCOMP can be sampled to RESULT on page 415 by triggering the SAMPLE task.

See RESETREAS on page 95 for more information on how to detect a wakeup from LPCOMP.

6.12.1 Shared resources

The LPCOMP shares analog resources with SAADC. While it is possible to use the SAADC at the same time as the LPCOMP, selecting the same analog input pin for both modules is not supported.

Additionally, LPCOMP shares registers and other resources with other peripherals that have the same ID as the LPCOMP. See Peripherals with shared ID on page 174 for more information.

The LPCOMP peripheral shall not be disabled (by writing to the ENABLE register) before the peripheral has been stopped. Failing to do so may result in unpredictable behavior.

6.12.2 Pin configuration

You can use the LPCOMP.PSEL register to select one of the analog input pins, **AIN0** through **AIN7**, as the analog input pin for the LPCOMP.

See GPIO — General purpose input/output on page 322 for more information about the pins. Similarly, you can use EXTREFSEL on page 417 to select one of the analog reference input pins, **AIN0** and **AIN1**,

as input for AREF in case AREF is selected in EXTREFSEL on page 417. The selected analog pins will be acquired by the LPCOMP when it is enabled through ENABLE on page 416.

6.12.3 Registers

Instances

Instance	Base address	Description
LPCOMP	0x40013000	Low power comparator

Register overview

Register	Offset	Description
TASKS_START	0x000	Start comparator
TASKS_STOP	0x004	Stop comparator
TASKS_SAMPLE	0x008	Sample comparator value
EVENTS_READY	0x100	LPCOMP is ready and output is valid
EVENTS_DOWN	0x104	Downward crossing
EVENTS_UP	0x108	Upward crossing
EVENTS_CROSS	0x10C	Downward or upward crossing
SHORTS	0x200	Shortcuts between local events and tasks
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
RESULT	0x400	Compare result
ENABLE	0x500	Enable LPCOMP
PSEL	0x504	Input pin select
REFSEL	0x508	Reference select
EXTREFSEL	0x50C	External reference select
ANADETECT	0x520	Analog detect configuration
HYST	0x538	Comparator hysteresis enable

6.12.3.1 TASKS_START

Address offset: 0x000

Start comparator

Bit number																															
ID																															A
Reset 0x00000000																															
																															0

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_START			Start comparator
			Trigger	1	Trigger task

6.12.3.2 TASKS_STOP

Address offset: 0x004

Stop comparator

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STOP			Stop comparator
			Trigger	1	Trigger task

6.12.3.3 TASKS_SAMPLE

Address offset: 0x008

Sample comparator value

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_SAMPLE			Sample comparator value
			Trigger	1	Trigger task

6.12.3.4 EVENTS_READY

Address offset: 0x100

LPCOMP is ready and output is valid

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_READY			LPCOMP is ready and output is valid
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.12.3.5 EVENTS_DOWN

Address offset: 0x104

Downward crossing

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_DOWN			Downward crossing
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.12.3.6 EVENTS_UP

Address offset: 0x108

Upward crossing

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_UP			Upward crossing
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.12.3.7 EVENTS_CROSS

Address offset: 0x10C

Downward or upward crossing

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_CROSS			Downward or upward crossing
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.12.3.8 SHORTS

Address offset: 0x200

Shortcuts between local events and tasks

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	READY_SAMPLE			Shortcut between event READY and task SAMPLE
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
B	RW	READY_STOP			Shortcut between event READY and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
C	RW	DOWN_STOP			Shortcut between event DOWN and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
D	RW	UP_STOP			Shortcut between event UP and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
E	RW	CROSS_STOP			Shortcut between event CROSS and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

6.12.3.9 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	READY			Write '1' to enable interrupt for event READY
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	DOWN			Write '1' to enable interrupt for event DOWN
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	UP			Write '1' to enable interrupt for event UP
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	CROSS			Write '1' to enable interrupt for event CROSS
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.12.3.10 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	READY			Write '1' to disable interrupt for event READY
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	DOWN			Write '1' to disable interrupt for event DOWN
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	UP			Write '1' to disable interrupt for event UP
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	CROSS			Write '1' to disable interrupt for event CROSS
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.12.3.11 RESULT

Address offset: 0x400

Compare result

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	RESULT			Result of last compare. Decision point SAMPLE task.
			Below	0	Input voltage is below the reference threshold (VIN+ < VIN-)
			Above	1	Input voltage is above the reference threshold (VIN+ > VIN-)

6.12.3.12 ENABLE

Address offset: 0x500

Enable LPCOMP

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ENABLE			Enable or disable LPCOMP
			Disabled	0	Disable
			Enabled	1	Enable

6.12.3.13 PSEL

Address offset: 0x504

Input pin select

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PSEL			Analog pin select
			AnalogInput0	0	AIN0 selected as analog input
			AnalogInput1	1	AIN1 selected as analog input
			AnalogInput2	2	AIN2 selected as analog input
			AnalogInput3	3	AIN3 selected as analog input
			AnalogInput4	4	AIN4 selected as analog input
			AnalogInput5	5	AIN5 selected as analog input
			AnalogInput6	6	AIN6 selected as analog input
			AnalogInput7	7	AIN7 selected as analog input

6.12.3.14 REFSEL

Address offset: 0x508

Reference select

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A
Reset 0x00000004				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	REFSEL			Reference select
			Ref1_8Vdd	0	VDD * 1/8 selected as reference
			Ref2_8Vdd	1	VDD * 2/8 selected as reference

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A	A	A	A
Reset 0x00000004				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
			Ref3_8Vdd	2																															
			Ref4_8Vdd	3																															
			Ref5_8Vdd	4																															
			Ref6_8Vdd	5																															
			Ref7_8Vdd	6																															
			ARef	7																															
			Ref1_16Vdd	8																															
			Ref3_16Vdd	9																															
			Ref5_16Vdd	10																															
			Ref7_16Vdd	11																															
			Ref9_16Vdd	12																															
			Ref11_16Vdd	13																															
			Ref13_16Vdd	14																															
			Ref15_16Vdd	15																															

6.12.3.15 EXTREFSEL

Address offset: 0x50C

External reference select

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A-	RW	EXTREFSEL																																	
			AnalogReference0	0																															
			AnalogReference1	1																															

6.12.3.16 ANADETECT

Address offset: 0x520

Analog detect configuration

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																		A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	ANADETECT																																	
			Cross	0																															
			Up	1																															
			Down	2																															

6.12.3.17 HYST

Address offset: 0x538

Comparator hysteresis enable

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	HYST			Comparator hysteresis enable
			Disabled	0	Comparator hysteresis disabled
			Enabled	1	Comparator hysteresis enabled

6.12.4 Electrical specification

6.12.4.1 LPCOMP Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
$t_{LPCANADET}$	Time from VIN crossing (>=50 mV above threshold) to ANADETECT signal generated		5		µs
$V_{INPOFFSET}$	Input offset including reference ladder error	-40		40	mV
V_{HYST}	Optional hysteresis		35		mV
$t_{STARTUP}$	Startup time for LPCOMP		140		µs

6.13 MWU — Memory watch unit

The Memory watch unit (MWU) can be used to generate events when a memory region is accessed by the CPU. The MWU can be configured to trigger events for access to Data RAM and Peripheral memory segments. The MWU allows an application developer to generate memory access events during development for debugging or during production execution for failure detection and recovery.

Listed here are the main features for MWU:

- Six memory regions, four user-configurable and two fixed regions in peripheral address space
- Flexible configuration of regions with START and END addresses
- Generate events on CPU read and/or write to a defined region of Data RAM or peripheral memory address space
- Programmable maskable or non-maskable (NMI) interrupt on events
- Peripheral interfaces can be watched for read and write access using subregions of the two fixed memory regions

Memory region	START address	END address
REGION[0..3]	Configurable	Configurable
PREGION[0]	0x40000000	0x4001FFFF
PREGION[1]	0x40020000	0x4003FFFF

Table 24: Memory regions

Each MWU region is defined by a start address and an end address, configured by the START and END registers respectively. These addresses are byte aligned and inclusive. The END register value has to be greater or equal to the START register value. Each region is associated with a pair of events that indicate that either a write access or a read access from the CPU has been detected inside the region.

For regions containing subregions (see below), a set of status registers PERREGION[0..1].SUBSTATWA and PERREGION[0..1].SUBSTATRA indicate which subregion(s) caused the EVENT_PREGION[0..1].WA and EVENT_PREGION[0..1].RA respectively.

The MWU is only able to detect memory accesses in the Data RAM and Peripheral memory segments from the CPU, see Memory on page 21 for more information about the different memory segments. EasyDMA

accesses are not monitored by the MWU. The MWU requires two HCLK cycles to detect and generate the event.

The peripheral regions, PREGION[0...1], are divided into 32 equally sized subregions, SR[0...31]. All subregions are excluded in the main region by default, and any can be included by specifying them in the SUBS register. When a subregion is excluded from the main region, the memory watch mechanism will not trigger any events when that subregion is accessed.

Subregions in PREGION[0..1] cannot be individually configured for read or write access watch. Watch configuration is only possible for a region as a whole. The PRGNiRA and PRGNiWA (i=0..1) fields in the REGIONEN register control watching read and write access.

REGION[0..3] can be individually enabled for read and/or write access watching through their respective RGNiRA and RGNiWA (i=0..3) fields in the REGIONEN register.

REGIONENSET and REGIONENCLR allow respectively enabling and disabling one or multiple REGIONs or PREGIONs watching in a single write access.

6.13.1 Registers

Instances

Instance	Base address	Description
MWU	0x40020000	Memory watch unit

Register overview

Register	Offset	Description
EVENTS_REGION[0].WA	0x100	Write access to region 0 detected
EVENTS_REGION[0].RA	0x104	Read access to region 0 detected
EVENTS_REGION[1].WA	0x108	Write access to region 1 detected
EVENTS_REGION[1].RA	0x10C	Read access to region 1 detected
EVENTS_REGION[2].WA	0x110	Write access to region 2 detected
EVENTS_REGION[2].RA	0x114	Read access to region 2 detected
EVENTS_REGION[3].WA	0x118	Write access to region 3 detected
EVENTS_REGION[3].RA	0x11C	Read access to region 3 detected
EVENTS_PREGION[0].WA	0x160	Write access to peripheral region 0 detected
EVENTS_PREGION[0].RA	0x164	Read access to peripheral region 0 detected
EVENTS_PREGION[1].WA	0x168	Write access to peripheral region 1 detected
EVENTS_PREGION[1].RA	0x16C	Read access to peripheral region 1 detected
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
NMIEN	0x320	Enable or disable interrupt
NMIENSET	0x324	Enable interrupt
NMIENCLR	0x328	Disable interrupt
PERREGION[0].SUBSTATWA	0x400	Source of event/interrupt in region 0, write access detected while corresponding subregion was enabled for watching
PERREGION[0].SUBSTATRA	0x404	Source of event/interrupt in region 0, read access detected while corresponding subregion was enabled for watching
PERREGION[1].SUBSTATWA	0x408	Source of event/interrupt in region 1, write access detected while corresponding subregion was enabled for watching
PERREGION[1].SUBSTATRA	0x40C	Source of event/interrupt in region 1, read access detected while corresponding subregion was enabled for watching

Register	Offset	Description
REGIONEN	0x510	Enable/disable regions watch
REGIONENSET	0x514	Enable regions watch
REGIONENCLR	0x518	Disable regions watch
REGION[0].START	0x600	Start address for region 0
REGION[0].END	0x604	End address of region 0
REGION[1].START	0x610	Start address for region 1
REGION[1].END	0x614	End address of region 1
REGION[2].START	0x620	Start address for region 2
REGION[2].END	0x624	End address of region 2
REGION[3].START	0x630	Start address for region 3
REGION[3].END	0x634	End address of region 3
PREGION[0].START	0x6C0	Reserved for future use
PREGION[0].END	0x6C4	Reserved for future use
PREGION[0].SUBS	0x6C8	Subregions of region 0
PREGION[1].START	0x6D0	Reserved for future use
PREGION[1].END	0x6D4	Reserved for future use
PREGION[1].SUBS	0x6D8	Subregions of region 1

6.13.1.1 EVENTS_REGION[0]

Peripheral events.

6.13.1.1.1 EVENTS_REGION[0].WA

Address offset: 0x100

Write access to region 0 detected

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	WA			Write access to region 0 detected
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.13.1.1.2 EVENTS_REGION[0].RA

Address offset: 0x104

Read access to region 0 detected

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	RA			Read access to region 0 detected
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.13.1.2 EVENTS_REGION[1]

Peripheral events.

6.13.1.2.1 EVENTS_REGION[1].WA

Address offset: 0x108

Write access to region 1 detected

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	WA			Write access to region 1 detected
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.13.1.2.2 EVENTS_REGION[1].RA

Address offset: 0x10C

Read access to region 1 detected

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RA			Read access to region 1 detected
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.13.1.3 EVENTS_REGION[2]

Peripheral events.

6.13.1.3.1 EVENTS_REGION[2].WA

Address offset: 0x110

Write access to region 2 detected

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	WA			Write access to region 2 detected
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.13.1.3.2 EVENTS_REGION[2].RA

Address offset: 0x114

Read access to region 2 detected

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RA			Read access to region 2 detected
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.13.1.4 EVENTS_REGION[3]

Peripheral events.

6.13.1.4.1 EVENTS_REGION[3].WA

Address offset: 0x118

Write access to region 3 detected

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	WA			Write access to region 3 detected
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.13.1.4.2 EVENTS_REGION[3].RA

Address offset: 0x11C

Read access to region 3 detected

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RA			Read access to region 3 detected
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.13.1.5 EVENTS_PREGION[0]

Peripheral events.

6.13.1.5.1 EVENTS_PREGION[0].WA

Address offset: 0x160

Write access to peripheral region 0 detected

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	WA			Write access to peripheral region 0 detected
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.13.1.5.2 EVENTS_PREGION[0].RA

Address offset: 0x164

Read access to peripheral region 0 detected

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RA			Read access to peripheral region 0 detected
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.13.1.6 EVENTS_PREGION[1]

Peripheral events.

6.13.1.6.1 EVENTS_PREGION[1].WA

Address offset: 0x168

Write access to peripheral region 1 detected

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	WA			Write access to peripheral region 1 detected
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.13.1.6.2 EVENTS_PREGION[1].RA

Address offset: 0x16C

Read access to peripheral region 1 detected

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RA			Read access to peripheral region 1 detected
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.13.1.7 INTEN

Address offset: 0x300

Enable or disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID								L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID			Value																						Description							
A	RW	REGION0WA																										Enable or disable interrupt for event REGION0WA							
			Disabled			0																						Disable							
			Enabled			1																						Enable							
B	RW	REGION0RA																										Enable or disable interrupt for event REGION0RA							
			Disabled			0																						Disable							
			Enabled			1																						Enable							
C	RW	REGION1WA																										Enable or disable interrupt for event REGION1WA							
			Disabled			0																						Disable							
			Enabled			1																						Enable							
D	RW	REGION1RA																										Enable or disable interrupt for event REGION1RA							
			Disabled			0																						Disable							
			Enabled			1																						Enable							
E	RW	REGION2WA																										Enable or disable interrupt for event REGION2WA							
			Disabled			0																						Disable							
			Enabled			1																						Enable							
F	RW	REGION2RA																										Enable or disable interrupt for event REGION2RA							
			Disabled			0																						Disable							
			Enabled			1																						Enable							
G	RW	REGION3WA																										Enable or disable interrupt for event REGION3WA							
			Disabled			0																						Disable							
			Enabled			1																						Enable							
H	RW	REGION3RA																										Enable or disable interrupt for event REGION3RA							
			Disabled			0																						Disable							
			Enabled			1																						Enable							
I	RW	PREGION0WA																										Enable or disable interrupt for event PREGION0WA							
			Disabled			0																						Disable							
			Enabled			1																						Enable							
J	RW	PREGION0RA																										Enable or disable interrupt for event PREGION0RA							
			Disabled			0																						Disable							
			Enabled			1																						Enable							
K	RW	PREGION1WA																										Enable or disable interrupt for event PREGION1WA							
			Disabled			0																						Disable							
			Enabled			1																						Enable							
L	RW	PREGION1RA																										Enable or disable interrupt for event PREGION1RA							
			Disabled			0																						Disable							
			Enabled			1																						Enable							

6.13.1.8 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	REGION0WA			Write '1' to enable interrupt for event REGION0WA
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	REGION0RA			Write '1' to enable interrupt for event REGION0RA
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	REGION1WA			Write '1' to enable interrupt for event REGION1WA
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	REGION1RA			Write '1' to enable interrupt for event REGION1RA
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	REGION2WA			Write '1' to enable interrupt for event REGION2WA
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	REGION2RA			Write '1' to enable interrupt for event REGION2RA
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
G	RW	REGION3WA			Write '1' to enable interrupt for event REGION3WA
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
H	RW	REGION3RA			Write '1' to enable interrupt for event REGION3RA
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
I	RW	PREGION0WA			Write '1' to enable interrupt for event PREGION0WA
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
J	RW	PREGION0RA			Write '1' to enable interrupt for event PREGION0RA
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
K	RW	PREGION1WA			Write '1' to enable interrupt for event PREGION1WA
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
L	RW	PREGION1RA			Write '1' to enable interrupt for event PREGION1RA
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.13.1.9 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID								L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	REGION0WA																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	REGION0RA																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	REGION1WA																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
D	RW	REGION1RA																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
E	RW	REGION2WA																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
F	RW	REGION2RA																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
G	RW	REGION3WA																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
H	RW	REGION3RA																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
I	RW	PREGION0WA																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
J	RW	PREGION0RA																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
K	RW	PREGION1WA																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
L	RW	PREGION1RA			Write '1' to disable interrupt for event PREGION1RA
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.13.1.10 NMIEN

Address offset: 0x320

Enable or disable interrupt

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	REGION0WA			Enable or disable interrupt for event REGION0WA
			Disabled	0	Disable
			Enabled	1	Enable
B	RW	REGION0RA			Enable or disable interrupt for event REGION0RA
			Disabled	0	Disable
			Enabled	1	Enable
C	RW	REGION1WA			Enable or disable interrupt for event REGION1WA
			Disabled	0	Disable
			Enabled	1	Enable
D	RW	REGION1RA			Enable or disable interrupt for event REGION1RA
			Disabled	0	Disable
			Enabled	1	Enable
E	RW	REGION2WA			Enable or disable interrupt for event REGION2WA
			Disabled	0	Disable
			Enabled	1	Enable
F	RW	REGION2RA			Enable or disable interrupt for event REGION2RA
			Disabled	0	Disable
			Enabled	1	Enable
G	RW	REGION3WA			Enable or disable interrupt for event REGION3WA
			Disabled	0	Disable
			Enabled	1	Enable
H	RW	REGION3RA			Enable or disable interrupt for event REGION3RA
			Disabled	0	Disable
			Enabled	1	Enable
I	RW	PREGION0WA			Enable or disable interrupt for event PREGION0WA
			Disabled	0	Disable
			Enabled	1	Enable
J	RW	PREGION0RA			Enable or disable interrupt for event PREGION0RA
			Disabled	0	Disable
			Enabled	1	Enable
K	RW	PREGION1WA			Enable or disable interrupt for event PREGION1WA
			Disabled	0	Disable
			Enabled	1	Enable
L	RW	PREGION1RA			Enable or disable interrupt for event PREGION1RA
			Disabled	0	Disable

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
			Enabled	1	Enable

6.13.1.11 NMIENSET

Address offset: 0x324

Enable interrupt

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	REGION0WA			Write '1' to enable interrupt for event REGION0WA
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	REGION0RA			Write '1' to enable interrupt for event REGION0RA
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	REGION1WA			Write '1' to enable interrupt for event REGION1WA
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	REGION1RA			Write '1' to enable interrupt for event REGION1RA
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	REGION2WA			Write '1' to enable interrupt for event REGION2WA
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	REGION2RA			Write '1' to enable interrupt for event REGION2RA
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
G	RW	REGION3WA			Write '1' to enable interrupt for event REGION3WA
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
H	RW	REGION3RA			Write '1' to enable interrupt for event REGION3RA
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
I	RW	PREGION0WA			Write '1' to enable interrupt for event PREGION0WA
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
J	RW	PREGION0RA			Write '1' to enable interrupt for event PREGION0RA
			Set	1	Enable

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
K	RW	PREGION1WA			Write '1' to enable interrupt for event PREGION1WA
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
L	RW	PREGION1RA			Write '1' to enable interrupt for event PREGION1RA
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.13.1.12 NMIENCLR

Address offset: 0x328

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	REGION0WA			Write '1' to disable interrupt for event REGION0WA
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	REGION0RA			Write '1' to disable interrupt for event REGION0RA
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	REGION1WA			Write '1' to disable interrupt for event REGION1WA
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	REGION1RA			Write '1' to disable interrupt for event REGION1RA
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	REGION2WA			Write '1' to disable interrupt for event REGION2WA
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	REGION2RA			Write '1' to disable interrupt for event REGION2RA
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
G	RW	REGION3WA			Write '1' to disable interrupt for event REGION3WA
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
H	RW	REGION3RA			Write '1' to disable interrupt for event REGION3RA

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
I	RW	PREGION0WA			Write '1' to disable interrupt for event PREGION0WA
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
J	RW	PREGION0RA			Write '1' to disable interrupt for event PREGION0RA
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
K	RW	PREGION1WA			Write '1' to disable interrupt for event PREGION1WA
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
L	RW	PREGION1RA			Write '1' to disable interrupt for event PREGION1RA
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.13.1.13 PERREGION[0].SUBSTATWA

Address offset: 0x400

Source of event/interrupt in region 0, write access detected while corresponding subregion was enabled for watching

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW W1C	SR0			Subregion 0 in region 0 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
B	RW W1C	SR1			Subregion 1 in region 0 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
C	RW W1C	SR2			Subregion 2 in region 0 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
D	RW W1C	SR3			Subregion 3 in region 0 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
E	RW W1C	SR4			Subregion 4 in region 0 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field		Value ID			Value																												
				Access			1																												
				Write access(es) occurred in this subregion																															
F	RW W1C	SR5																																	
				Subregion 5 in region 0 (write '1' to clear)																																
				NoAccess			0																												
				No write access occurred in this subregion																															
				Access			1																												
				Write access(es) occurred in this subregion																															
G	RW W1C	SR6																																	
				Subregion 6 in region 0 (write '1' to clear)																																
				NoAccess			0																												
				No write access occurred in this subregion																															
				Access			1																												
				Write access(es) occurred in this subregion																															
H	RW W1C	SR7																																	
				Subregion 7 in region 0 (write '1' to clear)																																
				NoAccess			0																												
				No write access occurred in this subregion																															
				Access			1																												
				Write access(es) occurred in this subregion																															
I	RW W1C	SR8																																	
				Subregion 8 in region 0 (write '1' to clear)																																
				NoAccess			0																												
				No write access occurred in this subregion																															
				Access			1																												
				Write access(es) occurred in this subregion																															
J	RW W1C	SR9																																	
				Subregion 9 in region 0 (write '1' to clear)																																
				NoAccess			0																												
				No write access occurred in this subregion																															
				Access			1																												
				Write access(es) occurred in this subregion																															
K	RW W1C	SR10																																	
				Subregion 10 in region 0 (write '1' to clear)																																
				NoAccess			0																												
				No write access occurred in this subregion																															
				Access			1																												
				Write access(es) occurred in this subregion																															
L	RW W1C	SR11																																	
				Subregion 11 in region 0 (write '1' to clear)																																
				NoAccess			0																												
				No write access occurred in this subregion																															
				Access			1																												
				Write access(es) occurred in this subregion																															
M	RW W1C	SR12																																	
				Subregion 12 in region 0 (write '1' to clear)																																
				NoAccess			0																												
				No write access occurred in this subregion																															
				Access			1																												
				Write access(es) occurred in this subregion																															
N	RW W1C	SR13																																	
				Subregion 13 in region 0 (write '1' to clear)																																
				NoAccess			0																												
				No write access occurred in this subregion																															
				Access			1																												
				Write access(es) occurred in this subregion																															
O	RW W1C	SR14																																	
				Subregion 14 in region 0 (write '1' to clear)																																
				NoAccess			0																												
				No write access occurred in this subregion																															
				Access			1																												
				Write access(es) occurred in this subregion																															
P	RW W1C	SR15																																	
				Subregion 15 in region 0 (write '1' to clear)																																
				NoAccess			0																												
				No write access occurred in this subregion																															
				Access			1																												
				Write access(es) occurred in this subregion																															
Q	RW W1C	SR16																																	
				Subregion 16 in region 0 (write '1' to clear)																																
				NoAccess			0																												
				No write access occurred in this subregion																															
				Access			1																												
				Write access(es) occurred in this subregion																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
R	RW W1C	SR17																																	Subregion 17 in region 0 (write '1' to clear)
			NoAccess	0																															No write access occurred in this subregion
			Access	1																															Write access(es) occurred in this subregion
S	RW W1C	SR18																																	Subregion 18 in region 0 (write '1' to clear)
			NoAccess	0																															No write access occurred in this subregion
			Access	1																															Write access(es) occurred in this subregion
T	RW W1C	SR19																																	Subregion 19 in region 0 (write '1' to clear)
			NoAccess	0																															No write access occurred in this subregion
			Access	1																															Write access(es) occurred in this subregion
U	RW W1C	SR20																																	Subregion 20 in region 0 (write '1' to clear)
			NoAccess	0																															No write access occurred in this subregion
			Access	1																															Write access(es) occurred in this subregion
V	RW W1C	SR21																																	Subregion 21 in region 0 (write '1' to clear)
			NoAccess	0																															No write access occurred in this subregion
			Access	1																															Write access(es) occurred in this subregion
W	RW W1C	SR22																																	Subregion 22 in region 0 (write '1' to clear)
			NoAccess	0																															No write access occurred in this subregion
			Access	1																															Write access(es) occurred in this subregion
X	RW W1C	SR23																																	Subregion 23 in region 0 (write '1' to clear)
			NoAccess	0																															No write access occurred in this subregion
			Access	1																															Write access(es) occurred in this subregion
Y	RW W1C	SR24																																	Subregion 24 in region 0 (write '1' to clear)
			NoAccess	0																															No write access occurred in this subregion
			Access	1																															Write access(es) occurred in this subregion
Z	RW W1C	SR25																																	Subregion 25 in region 0 (write '1' to clear)
			NoAccess	0																															No write access occurred in this subregion
			Access	1																															Write access(es) occurred in this subregion
a	RW W1C	SR26																																	Subregion 26 in region 0 (write '1' to clear)
			NoAccess	0																															No write access occurred in this subregion
			Access	1																															Write access(es) occurred in this subregion
b	RW W1C	SR27																																	Subregion 27 in region 0 (write '1' to clear)
			NoAccess	0																															No write access occurred in this subregion
			Access	1																															Write access(es) occurred in this subregion
c	RW W1C	SR28																																	Subregion 28 in region 0 (write '1' to clear)
			NoAccess	0																															No write access occurred in this subregion
			Access	1																															Write access(es) occurred in this subregion
d	RW W1C	SR29																																	Subregion 29 in region 0 (write '1' to clear)

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
e	RW W1C	SR30			Subregion 30 in region 0 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
f	RW W1C	SR31			Subregion 31 in region 0 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion

6.13.1.14 PERREGION[0].SUBSTATRA

Address offset: 0x404

Source of event/interrupt in region 0, read access detected while corresponding subregion was enabled for watching

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW W1C	SR0			Subregion 0 in region 0 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
B	RW W1C	SR1			Subregion 1 in region 0 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
C	RW W1C	SR2			Subregion 2 in region 0 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
D	RW W1C	SR3			Subregion 3 in region 0 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
E	RW W1C	SR4			Subregion 4 in region 0 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
F	RW W1C	SR5			Subregion 5 in region 0 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
G	RW W1C	SR6			Subregion 6 in region 0 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
H	RW W1C	SR7			Subregion 7 in region 0 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
I	RW W1C	SR8			Subregion 8 in region 0 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
J	RW W1C	SR9			Subregion 9 in region 0 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
K	RW W1C	SR10			Subregion 10 in region 0 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
L	RW W1C	SR11			Subregion 11 in region 0 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
M	RW W1C	SR12			Subregion 12 in region 0 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
N	RW W1C	SR13			Subregion 13 in region 0 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
O	RW W1C	SR14			Subregion 14 in region 0 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
P	RW W1C	SR15			Subregion 15 in region 0 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
Q	RW W1C	SR16			Subregion 16 in region 0 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
R	RW W1C	SR17			Subregion 17 in region 0 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
S	RW W1C	SR18			Subregion 18 in region 0 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
T	RW W1C	SR19			Subregion 19 in region 0 (write '1' to clear)

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
			NoAccess	0																															
			Access	1																															
U	RW W1C	SR20																																	
			NoAccess	0																															
			Access	1																															
V	RW W1C	SR21																																	
			NoAccess	0																															
			Access	1																															
W	RW W1C	SR22																																	
			NoAccess	0																															
			Access	1																															
X	RW W1C	SR23																																	
			NoAccess	0																															
			Access	1																															
Y	RW W1C	SR24																																	
			NoAccess	0																															
			Access	1																															
Z	RW W1C	SR25																																	
			NoAccess	0																															
			Access	1																															
a	RW W1C	SR26																																	
			NoAccess	0																															
			Access	1																															
b	RW W1C	SR27																																	
			NoAccess	0																															
			Access	1																															
c	RW W1C	SR28																																	
			NoAccess	0																															
			Access	1																															
d	RW W1C	SR29																																	
			NoAccess	0																															
			Access	1																															
e	RW W1C	SR30																																	
			NoAccess	0																															
			Access	1																															
f	RW W1C	SR31																																	
			NoAccess	0																															
			Access	1																															

6.13.1.15 PERREGION[1].SUBSTATWA

Address offset: 0x408

Source of event/interrupt in region 1, write access detected while corresponding subregion was enabled for watching

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW W1C	SR0																																	Subregion 0 in region 1 (write '1' to clear)
			NoAccess	0																															No write access occurred in this subregion
			Access	1																															Write access(es) occurred in this subregion
B	RW W1C	SR1																																	Subregion 1 in region 1 (write '1' to clear)
			NoAccess	0																															No write access occurred in this subregion
			Access	1																															Write access(es) occurred in this subregion
C	RW W1C	SR2																																	Subregion 2 in region 1 (write '1' to clear)
			NoAccess	0																															No write access occurred in this subregion
			Access	1																															Write access(es) occurred in this subregion
D	RW W1C	SR3																																	Subregion 3 in region 1 (write '1' to clear)
			NoAccess	0																															No write access occurred in this subregion
			Access	1																															Write access(es) occurred in this subregion
E	RW W1C	SR4																																	Subregion 4 in region 1 (write '1' to clear)
			NoAccess	0																															No write access occurred in this subregion
			Access	1																															Write access(es) occurred in this subregion
F	RW W1C	SR5																																	Subregion 5 in region 1 (write '1' to clear)
			NoAccess	0																															No write access occurred in this subregion
			Access	1																															Write access(es) occurred in this subregion
G	RW W1C	SR6																																	Subregion 6 in region 1 (write '1' to clear)
			NoAccess	0																															No write access occurred in this subregion
			Access	1																															Write access(es) occurred in this subregion
H	RW W1C	SR7																																	Subregion 7 in region 1 (write '1' to clear)
			NoAccess	0																															No write access occurred in this subregion
			Access	1																															Write access(es) occurred in this subregion
I	RW W1C	SR8																																	Subregion 8 in region 1 (write '1' to clear)
			NoAccess	0																															No write access occurred in this subregion
			Access	1																															Write access(es) occurred in this subregion
J	RW W1C	SR9																																	Subregion 9 in region 1 (write '1' to clear)
			NoAccess	0																															No write access occurred in this subregion
			Access	1																															Write access(es) occurred in this subregion
K	RW W1C	SR10																																	Subregion 10 in region 1 (write '1' to clear)
			NoAccess	0																															No write access occurred in this subregion

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
			Access	1																															
L	RW W1C	SR11																																	
			NoAccess	0																															
			Access	1																															
M	RW W1C	SR12																																	
			NoAccess	0																															
			Access	1																															
N	RW W1C	SR13																																	
			NoAccess	0																															
			Access	1																															
O	RW W1C	SR14																																	
			NoAccess	0																															
			Access	1																															
P	RW W1C	SR15																																	
			NoAccess	0																															
			Access	1																															
Q	RW W1C	SR16																																	
			NoAccess	0																															
			Access	1																															
R	RW W1C	SR17																																	
			NoAccess	0																															
			Access	1																															
S	RW W1C	SR18																																	
			NoAccess	0																															
			Access	1																															
T	RW W1C	SR19																																	
			NoAccess	0																															
			Access	1																															
U	RW W1C	SR20																																	
			NoAccess	0																															
			Access	1																															
V	RW W1C	SR21																																	
			NoAccess	0																															
			Access	1																															
W	RW W1C	SR22																																	
			NoAccess	0																															
			Access	1																															

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
X	RW W1C	SR23			Subregion 23 in region 1 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
Y	RW W1C	SR24			Subregion 24 in region 1 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
Z	RW W1C	SR25			Subregion 25 in region 1 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
a	RW W1C	SR26			Subregion 26 in region 1 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
b	RW W1C	SR27			Subregion 27 in region 1 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
c	RW W1C	SR28			Subregion 28 in region 1 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
d	RW W1C	SR29			Subregion 29 in region 1 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
e	RW W1C	SR30			Subregion 30 in region 1 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion
f	RW W1C	SR31			Subregion 31 in region 1 (write '1' to clear)
			NoAccess	0	No write access occurred in this subregion
			Access	1	Write access(es) occurred in this subregion

6.13.1.16 PERREGION[1].SUBSTATRA

Address offset: 0x40C

Source of event/interrupt in region 1, read access detected while corresponding subregion was enabled for watching

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW W1C	SR0			Subregion 0 in region 1 (write '1' to clear)

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
			NoAccess	0	No read access occurred in this subregion																														
			Access	1	Read access(es) occurred in this subregion																														
B	RW W1C	SR1			Subregion 1 in region 1 (write '1' to clear)																														
			NoAccess	0	No read access occurred in this subregion																														
			Access	1	Read access(es) occurred in this subregion																														
C	RW W1C	SR2			Subregion 2 in region 1 (write '1' to clear)																														
			NoAccess	0	No read access occurred in this subregion																														
			Access	1	Read access(es) occurred in this subregion																														
D	RW W1C	SR3			Subregion 3 in region 1 (write '1' to clear)																														
			NoAccess	0	No read access occurred in this subregion																														
			Access	1	Read access(es) occurred in this subregion																														
E	RW W1C	SR4			Subregion 4 in region 1 (write '1' to clear)																														
			NoAccess	0	No read access occurred in this subregion																														
			Access	1	Read access(es) occurred in this subregion																														
F	RW W1C	SR5			Subregion 5 in region 1 (write '1' to clear)																														
			NoAccess	0	No read access occurred in this subregion																														
			Access	1	Read access(es) occurred in this subregion																														
G	RW W1C	SR6			Subregion 6 in region 1 (write '1' to clear)																														
			NoAccess	0	No read access occurred in this subregion																														
			Access	1	Read access(es) occurred in this subregion																														
H	RW W1C	SR7			Subregion 7 in region 1 (write '1' to clear)																														
			NoAccess	0	No read access occurred in this subregion																														
			Access	1	Read access(es) occurred in this subregion																														
I	RW W1C	SR8			Subregion 8 in region 1 (write '1' to clear)																														
			NoAccess	0	No read access occurred in this subregion																														
			Access	1	Read access(es) occurred in this subregion																														
J	RW W1C	SR9			Subregion 9 in region 1 (write '1' to clear)																														
			NoAccess	0	No read access occurred in this subregion																														
			Access	1	Read access(es) occurred in this subregion																														
K	RW W1C	SR10			Subregion 10 in region 1 (write '1' to clear)																														
			NoAccess	0	No read access occurred in this subregion																														
			Access	1	Read access(es) occurred in this subregion																														
L	RW W1C	SR11			Subregion 11 in region 1 (write '1' to clear)																														
			NoAccess	0	No read access occurred in this subregion																														
			Access	1	Read access(es) occurred in this subregion																														
M	RW W1C	SR12			Subregion 12 in region 1 (write '1' to clear)																														
			NoAccess	0	No read access occurred in this subregion																														
			Access	1	Read access(es) occurred in this subregion																														

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
N	RW W1C	SR13			Subregion 13 in region 1 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
O	RW W1C	SR14			Subregion 14 in region 1 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
P	RW W1C	SR15			Subregion 15 in region 1 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
Q	RW W1C	SR16			Subregion 16 in region 1 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
R	RW W1C	SR17			Subregion 17 in region 1 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
S	RW W1C	SR18			Subregion 18 in region 1 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
T	RW W1C	SR19			Subregion 19 in region 1 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
U	RW W1C	SR20			Subregion 20 in region 1 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
V	RW W1C	SR21			Subregion 21 in region 1 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
W	RW W1C	SR22			Subregion 22 in region 1 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
X	RW W1C	SR23			Subregion 23 in region 1 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
Y	RW W1C	SR24			Subregion 24 in region 1 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
Z	RW W1C	SR25			Subregion 25 in region 1 (write '1' to clear)

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
a	RW W1C	SR26			Subregion 26 in region 1 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
b	RW W1C	SR27			Subregion 27 in region 1 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
c	RW W1C	SR28			Subregion 28 in region 1 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
d	RW W1C	SR29			Subregion 29 in region 1 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
e	RW W1C	SR30			Subregion 30 in region 1 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion
f	RW W1C	SR31			Subregion 31 in region 1 (write '1' to clear)
			NoAccess	0	No read access occurred in this subregion
			Access	1	Read access(es) occurred in this subregion

6.13.1.17 REGIONEN

Address offset: 0x510

Enable/disable regions watch

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RGN0WA			Enable/disable write access watch in region[0]
			Disable	0	Disable write access watch in this region
			Enable	1	Enable write access watch in this region
B	RW	RGN0RA			Enable/disable read access watch in region[0]
			Disable	0	Disable read access watch in this region
			Enable	1	Enable read access watch in this region
C	RW	RGN1WA			Enable/disable write access watch in region[1]
			Disable	0	Disable write access watch in this region
			Enable	1	Enable write access watch in this region
D	RW	RGN1RA			Enable/disable read access watch in region[1]
			Disable	0	Disable read access watch in this region
			Enable	1	Enable read access watch in this region
E	RW	RGN2WA			Enable/disable write access watch in region[2]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Disable	0	Disable write access watch in this region
			Enable	1	Enable write access watch in this region
F	RW	RGN2RA			Enable/disable read access watch in region[2]
			Disable	0	Disable read access watch in this region
			Enable	1	Enable read access watch in this region
G	RW	RGN3WA			Enable/disable write access watch in region[3]
			Disable	0	Disable write access watch in this region
			Enable	1	Enable write access watch in this region
H	RW	RGN3RA			Enable/disable read access watch in region[3]
			Disable	0	Disable read access watch in this region
			Enable	1	Enable read access watch in this region
I	RW	PRGN0WA			Enable/disable write access watch in PREGION[0]
			Disable	0	Disable write access watch in this PREGION
			Enable	1	Enable write access watch in this PREGION
J	RW	PRGN0RA			Enable/disable read access watch in PREGION[0]
			Disable	0	Disable read access watch in this PREGION
			Enable	1	Enable read access watch in this PREGION
K	RW	PRGN1WA			Enable/disable write access watch in PREGION[1]
			Disable	0	Disable write access watch in this PREGION
			Enable	1	Enable write access watch in this PREGION
L	RW	PRGN1RA			Enable/disable read access watch in PREGION[1]
			Disable	0	Disable read access watch in this PREGION
			Enable	1	Enable read access watch in this PREGION

6.13.1.18 REGIONENSET

Address offset: 0x514

Enable regions watch

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RGN0WA			Enable write access watch in region[0]
			Set	1	Enable write access watch in this region
			Disabled	0	Write access watch in this region is disabled
			Enabled	1	Write access watch in this region is enabled
B	RW	RGN0RA			Enable read access watch in region[0]
			Set	1	Enable read access watch in this region
			Disabled	0	Read access watch in this region is disabled
			Enabled	1	Read access watch in this region is enabled
C	RW	RGN1WA			Enable write access watch in region[1]
			Set	1	Enable write access watch in this region
			Disabled	0	Write access watch in this region is disabled
			Enabled	1	Write access watch in this region is enabled
D	RW	RGN1RA			Enable read access watch in region[1]
			Set	1	Enable read access watch in this region
			Disabled	0	Read access watch in this region is disabled
			Enabled	1	Read access watch in this region is enabled

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
E	RW	RGN2WA			Enable write access watch in region[2]
			Set	1	Enable write access watch in this region
			Disabled	0	Write access watch in this region is disabled
			Enabled	1	Write access watch in this region is enabled
F	RW	RGN2RA			Enable read access watch in region[2]
			Set	1	Enable read access watch in this region
			Disabled	0	Read access watch in this region is disabled
			Enabled	1	Read access watch in this region is enabled
G	RW	RGN3WA			Enable write access watch in region[3]
			Set	1	Enable write access watch in this region
			Disabled	0	Write access watch in this region is disabled
			Enabled	1	Write access watch in this region is enabled
H	RW	RGN3RA			Enable read access watch in region[3]
			Set	1	Enable read access watch in this region
			Disabled	0	Read access watch in this region is disabled
			Enabled	1	Read access watch in this region is enabled
I	RW	PRGN0WA			Enable write access watch in PREGION[0]
			Set	1	Enable write access watch in this PREGION
			Disabled	0	Write access watch in this PREGION is disabled
			Enabled	1	Write access watch in this PREGION is enabled
J	RW	PRGN0RA			Enable read access watch in PREGION[0]
			Set	1	Enable read access watch in this PREGION
			Disabled	0	Read access watch in this PREGION is disabled
			Enabled	1	Read access watch in this PREGION is enabled
K	RW	PRGN1WA			Enable write access watch in PREGION[1]
			Set	1	Enable write access watch in this PREGION
			Disabled	0	Write access watch in this PREGION is disabled
			Enabled	1	Write access watch in this PREGION is enabled
L	RW	PRGN1RA			Enable read access watch in PREGION[1]
			Set	1	Enable read access watch in this PREGION
			Disabled	0	Read access watch in this PREGION is disabled
			Enabled	1	Read access watch in this PREGION is enabled

6.13.1.19 REGIONENCLR

Address offset: 0x518

Disable regions watch

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RGN0WA			Disable write access watch in region[0]
			Clear	1	Disable write access watch in this region
			Disabled	0	Write access watch in this region is disabled
			Enabled	1	Write access watch in this region is enabled
B	RW	RGN0RA			Disable read access watch in region[0]
			Clear	1	Disable read access watch in this region
			Disabled	0	Read access watch in this region is disabled

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID								L	K	J	I																	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
			Enabled	1																															
C	RW	RGN1WA																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
D	RW	RGN1RA																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
E	RW	RGN2WA																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
F	RW	RGN2RA																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
G	RW	RGN3WA																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
H	RW	RGN3RA																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
I	RW	PRGN0WA																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
J	RW	PRGN0RA																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
K	RW	PRGN1WA																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
L	RW	PRGN1RA																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															

6.13.1.20 REGION[0].START

Address offset: 0x600

Start address for region 0

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	START			Start address for region

6.13.1.21 REGION[0].END

Address offset: 0x604

End address of region 0

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	END			End address of region.

6.13.1.22 REGION[1].START

Address offset: 0x610

Start address for region 1

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	START			Start address for region

6.13.1.23 REGION[1].END

Address offset: 0x614

End address of region 1

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	END			End address of region.

6.13.1.24 REGION[2].START

Address offset: 0x620

Start address for region 2

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	START			Start address for region

6.13.1.25 REGION[2].END

Address offset: 0x624

End address of region 2

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	END			End address of region.

6.13.1.26 REGION[3].START

Address offset: 0x630

Start address for region 3

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	START			Start address for region

6.13.1.27 REGION[3].END

Address offset: 0x634

End address of region 3

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	END			End address of region.

6.13.1.28 PREGION[0].START

Address offset: 0x6C0

Reserved for future use

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	START			Reserved for future use

6.13.1.29 PREGION[0].END

Address offset: 0x6C4

Reserved for future use

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	R	END																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
			Include	1	Include
N	RW	SR13			Include or exclude subregion 13 in region
			Exclude	0	Exclude
			Include	1	Include
O	RW	SR14			Include or exclude subregion 14 in region
			Exclude	0	Exclude
			Include	1	Include
P	RW	SR15			Include or exclude subregion 15 in region
			Exclude	0	Exclude
			Include	1	Include
Q	RW	SR16			Include or exclude subregion 16 in region
			Exclude	0	Exclude
			Include	1	Include
R	RW	SR17			Include or exclude subregion 17 in region
			Exclude	0	Exclude
			Include	1	Include
S	RW	SR18			Include or exclude subregion 18 in region
			Exclude	0	Exclude
			Include	1	Include
T	RW	SR19			Include or exclude subregion 19 in region
			Exclude	0	Exclude
			Include	1	Include
U	RW	SR20			Include or exclude subregion 20 in region
			Exclude	0	Exclude
			Include	1	Include
V	RW	SR21			Include or exclude subregion 21 in region
			Exclude	0	Exclude
			Include	1	Include
W	RW	SR22			Include or exclude subregion 22 in region
			Exclude	0	Exclude
			Include	1	Include
X	RW	SR23			Include or exclude subregion 23 in region
			Exclude	0	Exclude
			Include	1	Include
Y	RW	SR24			Include or exclude subregion 24 in region
			Exclude	0	Exclude
			Include	1	Include
Z	RW	SR25			Include or exclude subregion 25 in region
			Exclude	0	Exclude
			Include	1	Include
a	RW	SR26			Include or exclude subregion 26 in region
			Exclude	0	Exclude
			Include	1	Include
b	RW	SR27			Include or exclude subregion 27 in region
			Exclude	0	Exclude
			Include	1	Include
c	RW	SR28			Include or exclude subregion 28 in region
			Exclude	0	Exclude
			Include	1	Include
d	RW	SR29			Include or exclude subregion 29 in region

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Exclude	0	Exclude
			Include	1	Include
e	RW	SR30			Include or exclude subregion 30 in region
			Exclude	0	Exclude
			Include	1	Include
f	RW	SR31			Include or exclude subregion 31 in region
			Exclude	0	Exclude
			Include	1	Include

6.13.1.31 PREGION[1].START

Address offset: 0x6D0

Reserved for future use

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	START			Reserved for future use

6.13.1.32 PREGION[1].END

Address offset: 0x6D4

Reserved for future use

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	END			Reserved for future use

6.13.1.33 PREGION[1].SUBS

Address offset: 0x6D8

Subregions of region 1

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	SR0			Include or exclude subregion 0 in region
			Exclude	0	Exclude
			Include	1	Include
B	RW	SR1			Include or exclude subregion 1 in region
			Exclude	0	Exclude
			Include	1	Include
C	RW	SR2			Include or exclude subregion 2 in region
			Exclude	0	Exclude

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
			Include	1	Include
D	RW	SR3			Include or exclude subregion 3 in region
			Exclude	0	Exclude
			Include	1	Include
E	RW	SR4			Include or exclude subregion 4 in region
			Exclude	0	Exclude
			Include	1	Include
F	RW	SR5			Include or exclude subregion 5 in region
			Exclude	0	Exclude
			Include	1	Include
G	RW	SR6			Include or exclude subregion 6 in region
			Exclude	0	Exclude
			Include	1	Include
H	RW	SR7			Include or exclude subregion 7 in region
			Exclude	0	Exclude
			Include	1	Include
I	RW	SR8			Include or exclude subregion 8 in region
			Exclude	0	Exclude
			Include	1	Include
J	RW	SR9			Include or exclude subregion 9 in region
			Exclude	0	Exclude
			Include	1	Include
K	RW	SR10			Include or exclude subregion 10 in region
			Exclude	0	Exclude
			Include	1	Include
L	RW	SR11			Include or exclude subregion 11 in region
			Exclude	0	Exclude
			Include	1	Include
M	RW	SR12			Include or exclude subregion 12 in region
			Exclude	0	Exclude
			Include	1	Include
N	RW	SR13			Include or exclude subregion 13 in region
			Exclude	0	Exclude
			Include	1	Include
O	RW	SR14			Include or exclude subregion 14 in region
			Exclude	0	Exclude
			Include	1	Include
P	RW	SR15			Include or exclude subregion 15 in region
			Exclude	0	Exclude
			Include	1	Include
Q	RW	SR16			Include or exclude subregion 16 in region
			Exclude	0	Exclude
			Include	1	Include
R	RW	SR17			Include or exclude subregion 17 in region
			Exclude	0	Exclude
			Include	1	Include
S	RW	SR18			Include or exclude subregion 18 in region
			Exclude	0	Exclude
			Include	1	Include
T	RW	SR19			Include or exclude subregion 19 in region

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
			Exclude	0																															
			Include	1																															
U	RW	SR20		Include or exclude subregion 20 in region																															
			Exclude	0																															
			Include	1																															
V	RW	SR21		Include or exclude subregion 21 in region																															
			Exclude	0																															
			Include	1																															
W	RW	SR22		Include or exclude subregion 22 in region																															
			Exclude	0																															
			Include	1																															
X	RW	SR23		Include or exclude subregion 23 in region																															
			Exclude	0																															
			Include	1																															
Y	RW	SR24		Include or exclude subregion 24 in region																															
			Exclude	0																															
			Include	1																															
Z	RW	SR25		Include or exclude subregion 25 in region																															
			Exclude	0																															
			Include	1																															
a	RW	SR26		Include or exclude subregion 26 in region																															
			Exclude	0																															
			Include	1																															
b	RW	SR27		Include or exclude subregion 27 in region																															
			Exclude	0																															
			Include	1																															
c	RW	SR28		Include or exclude subregion 28 in region																															
			Exclude	0																															
			Include	1																															
d	RW	SR29		Include or exclude subregion 29 in region																															
			Exclude	0																															
			Include	1																															
e	RW	SR30		Include or exclude subregion 30 in region																															
			Exclude	0																															
			Include	1																															
f	RW	SR31		Include or exclude subregion 31 in region																															
			Exclude	0																															
			Include	1																															

6.14 NFCT — Near field communication tag

The NFCT peripheral is an implementation of an NFC Forum compliant listening device NFC-A.

With appropriate software, the NFCT peripheral can be used as the listening device NFC-A as specified by the NFC Forum.

Listed here are the main features for the NFCT peripheral:

- NFC-A listen mode operation
  - 13.56 MHz input frequency

- Bit rate 106 kbps
- Wake-on-field low power field detection (SENSE) mode
- Frame assemble and disassemble for the NFC-A frames specified by the NFC Forum
- Programmable frame timing controller
- Integrated automatic collision resolution, cyclic redundancy check (CRC), and parity functions

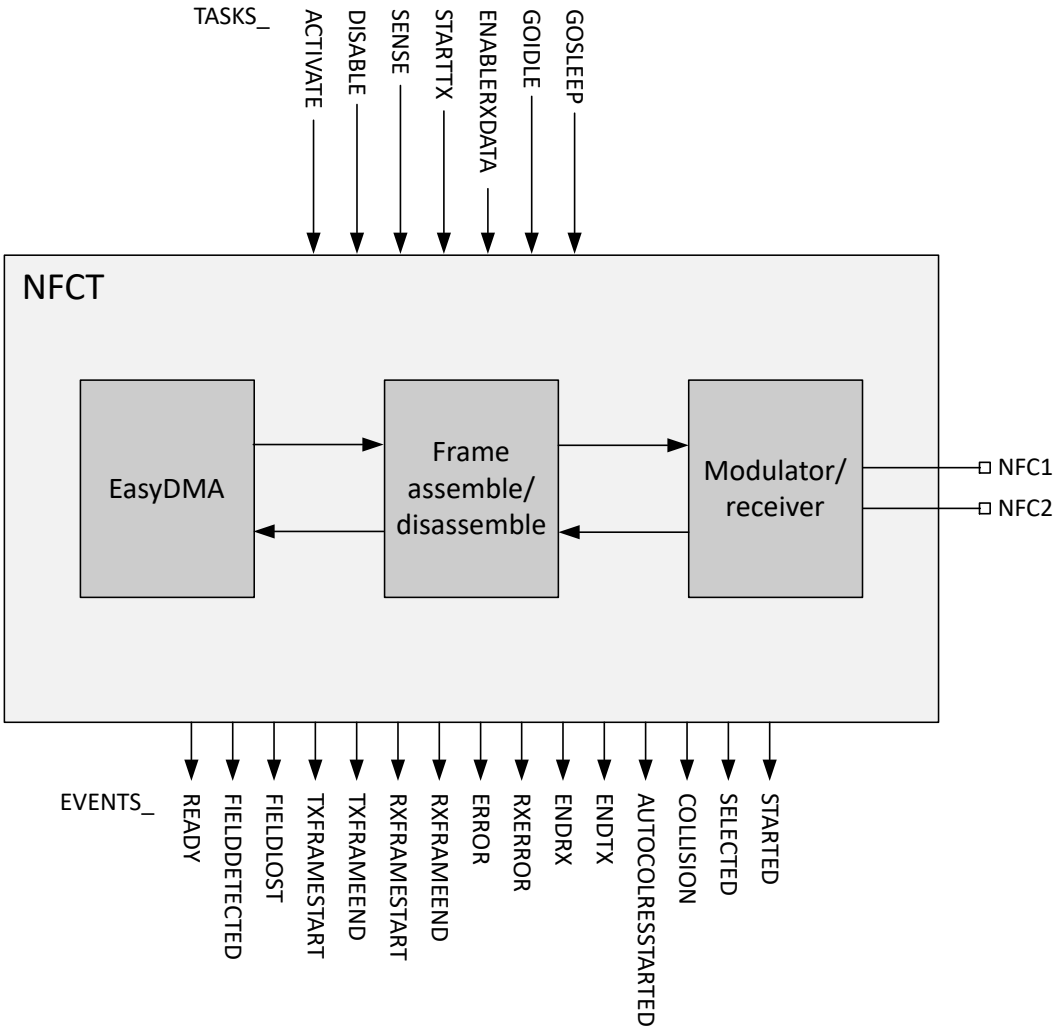


Figure 65: NFCT block diagram

6.14.1 Overview

The NFCT peripheral contains a 13.56 MHz AM receiver and a 13.56 MHz load modulator with 106 kbps data rate as defined by the NFC Forum.

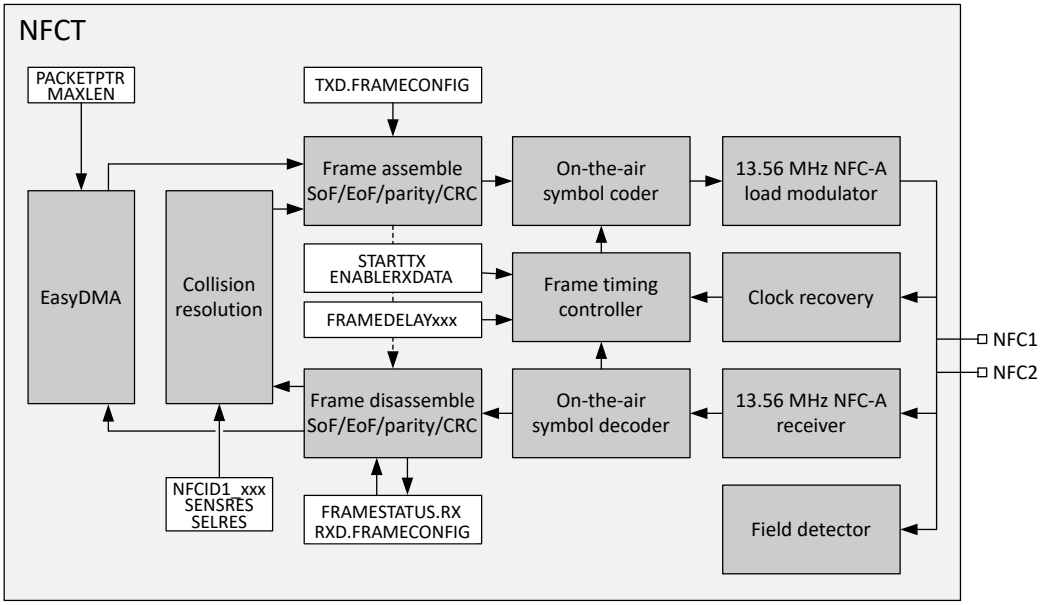


Figure 66: NFCT overview

When transmitting, the frame data will be transferred directly from RAM and transmitted with configurable frame type and delay timing. The system will be notified by an event whenever a complete frame is received or sent. The received frames will be automatically disassembled and the data part of the frame transferred to RAM.

The NFCT peripheral also supports the collision detection and resolution ("anticollision") as defined by the NFC Forum.

Wake-on-field is supported in SENSE mode while the device is either in System OFF or System ON mode. When the antenna enters an NFC field, an event will be triggered notifying the system to activate the NFCT functionality for incoming frames. In System ON, if the energy detected at the antenna increases beyond a threshold value, the module will generate a FIELDDETECTED event. When the strength of the field no longer supports NFC communication, the module will generate a FIELDLOST event. For the Low Power Field Detect threshold values, refer to NFCT Electrical Specification on page 479.

In System OFF, the NFCT Low Power Field Detect function can wake the system up through a reset. See RESETREAS on page 95 for more information on how to detect a wakeup from NFCT.

If the system is put into System OFF mode while a field is already present, the NFCT Low Power Field Detect function will wake the system up right away and generate a reset.

Important: As a consequence of a reset, NFCT is disabled, and therefore the reset handler will have to activate NFCT again and set it up properly.

The HFXO must be running before the NFCT peripheral goes into ACTIVATED state. Note that the NFCT peripheral calibration is automatically done on ACTIVATE task. The HFXO can be turned off when the NFCT peripheral goes into SENSE mode. The shortcut FIELDDETECTED_ACTIVATE can be used when the HFXO is already running while in SENSE mode.

Outgoing data will be collected from RAM with the EasyDMA function and assembled according to theTXD.FRAMECONFIG on page 476 register. Incoming data will be disassembled according to the RXD.FRAMECONFIG register and the data section in the frame will be written to RAM via the EasyDMA function.

The NFCT peripheral includes a frame timing controller that can be used to accurately control the inter-frame delay between the incoming frame and a corresponding outgoing frame. It also includes optional CRC functionality.

6.14.2 Operating states

Tasks and events are used to control the operating state of the peripheral. The module can change state by triggering a task, or when specific operations are finalized. Events and tasks allow software to keep track of and change the current state.

See NFCT block diagram on page 452 and NFCT state diagram, automatic collision resolution enabled on page 454 for more information. See *NFC Forum, NFC Activity Technical Specification* for description on NFCT operating states.

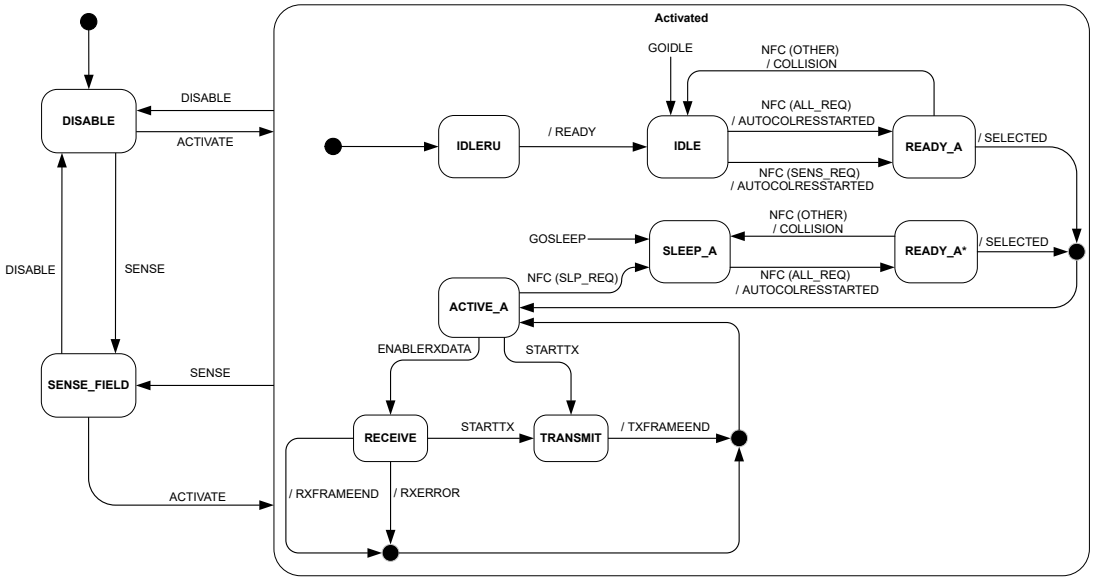


Figure 67: NFCT state diagram, automatic collision resolution enabled

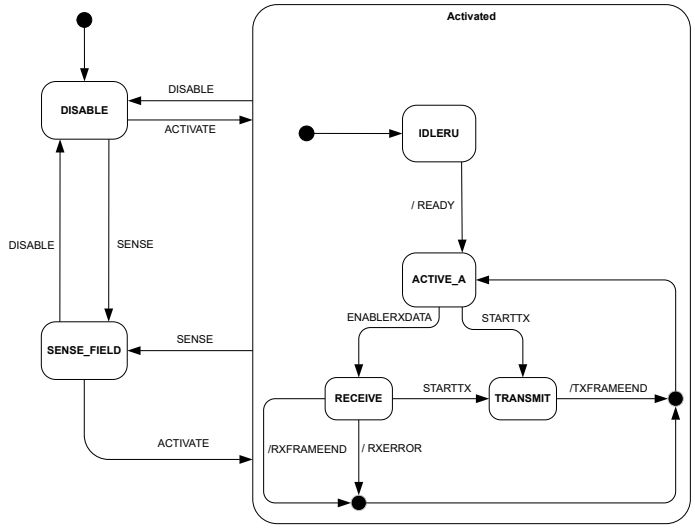


Figure 68: NFCT state diagram, automatic collision resolution disabled

Important:

- FIELDLOST event is not generated in SENSE mode.
- Sending SENSE task while field is still present does not generate FIELDDETECTED event.
- If the FIELDDETECTED event is cleared before sending the ACTIVATE task, then the FIELDDETECTED event shows up again after sending the ACTIVATE task. The shortcut FIELDDETECTED_ACTIVATE can be used to avoid this condition.

6.14.3 Pin configuration

NFCT uses two pins to connect the antenna and these pins are shared with GPIOs.

The PROTECT field in the NFCPINS register in UICR defines the usage of these pins and their protection level against excessive voltages. The content of the NFCPINS register is reloaded at every reset. See Pin assignments on page 926 for the pins used by the NFCT peripheral.

When NFCPINS.PROTECT=NFC, a protection circuit will be enabled on the dedicated pins, preventing the chip from being damaged in the presence of a strong NFC field. The protection circuit will short the two pins together if voltage difference exceeds approximately 2V. The GPIO function on those pins will also be disabled.

When NFCPINS.PROTECT=Disabled, the device will not be protected against strong NFC field damages caught by a connected NFCT antenna, and the NFCT peripheral will not operate as expected, as it will never leave the DISABLE state.

The pins dedicated to the NFCT antenna function will have some limitation when the pins are configured for normal GPIO operation. The pin capacitance will be higher on those (refer to C_{PAD_NFC} in the Electrical Specification of GPIO — General purpose input/output on page 322), and some increased leakage current between the two pins is to be expected if they are used in GPIO mode, and are driven to different logical values. To save power, the two pins should always be set to the same logical value whenever entering one of the device power saving modes. For details, refer to I_{NFC_LEAK} in the Electrical Specification of GPIO — General purpose input/output on page 322.

6.14.4 EasyDMA

The NFCT peripheral implements EasyDMA for reading and writing of data packets from and to the Data RAM.

The NFCT EasyDMA utilizes a pointer called PACKETPTR on page 475 for receiving and transmitting packets.

The NFCT peripheral uses EasyDMA to read or write RAM, but not both at the same time. The event RXFRAMESTART indicates that the EasyDMA has started writing to the RAM for a receive frame and the event RXFRAMEND indicates that the EasyDMA has completed writing to the RAM. Similarly, the event TXFRAMESTART indicates that the EasyDMA has started reading from the RAM for a transmit frame and the event TXFRAMEND indicates that the EasyDMA has completed reading from the RAM. If a transmit and a receive operation is issued at the same time, the transmit operation would be prioritized.

Starting a transmit operation while the EasyDMA is writing a receive frame to the RAM will result in unpredictable behavior. Starting an EasyDMA operation when there is an ongoing EasyDMA operation may result in unpredictable behavior. It is recommended to wait for the TXFRAMEEND or RXFRAMEEND event for the ongoing transmit or receive before starting a new receive or transmit operation.

The MAXLEN on page 475 register determines the maximum number of bytes that can be read from or written to the RAM. This feature can be used to ensure that the NFCT peripheral does not overwrite, or read beyond, the RAM assigned to a packet. Note that if the RXD.AMOUNT or TXD.AMOUNT register indicates longer data packets than set in MAXLEN, the frames sent to or received from the physical layer

will be incomplete. In that situation, in RX, the OVERRUN bit in the FRAMESTATUS.RX register will be set and an RXERROR event will be triggered.

Important: The RXD.AMOUNT and TXD.AMOUNT define a frame length in bytes and bits excluding start of frame (SoF), end of frame (EoF), and parity, but including CRC for RXD.AMOUNT only. Make sure to take potential additional bits into account when setting MAXLEN.

Only sending task ENABLERXDATA ensures that a new value in PACKETPTR pointing to the RX buffer in Data RAM is taken into account.

If PACKETPTR is not pointing to the Data RAM region, an EasyDMA transfer may result in a hard fault or RAM corruption. For more information about the different memory regions, see Chapter Memory on page 21.

The NFCT peripherals normally do alternative receive and transmit frames. Therefore, to prepare for the next frame, the PACKETPTR, MAXLEN, TXD.FRAMECONFIG and TXD.AMOUNT can be updated while the receive is in progress, and, similarly, the PACKETPTR, MAXLEN and RXD.FRAMECONFIG can be updated while the transmit is in progress. They can be updated and prepared for the next NFC frame immediately after the STARTED event of the current frame has been received. Updating the TXD.FRAMECONFIG and TXD.AMOUNT during the current transmit frame or updating RXD.FRAMECONFIG during current receive frame may cause unpredictable behaviour.

In accordance with *NFC Forum, NFC Digital Protocol Technical Specification*, the least significant bit (LSB) from the least significant byte (LSByte) is sent on air first. The bytes are stored in increasing order, starting at the lowest address in the EasyDMA buffer in RAM.

6.14.5 Frame assembler

The NFCT peripheral implements a frame assembler in hardware.

When the NFCT peripheral is in the ACTIVE_A state, the software can decide to enter RX or TX mode. For RX, see Frame disassembler on page 457. For TX, the software must indicate the address of the source buffer in Data RAM and its size through programming the PACKETPTR and MAXLEN registers respectively, then issuing a STARTTX task.

MAXLEN must be set so that it matches the size of the frame to be sent.

The STARTED event indicates that the PACKETPTR and MAXLEN registers have been captured by the frame assembler EasyDMA.

When asserting the STARTTX task, the frame assembler module will start reading TXD.AMOUNT.TXDATABYTES bytes (plus one additional byte if TXD.AMOUNT.TXDATABITS > 0) from the RAM position set by the PACKETPTR.

The NFCT peripheral transmits the data as read from RAM, adding framing and the CRC calculated on the fly if set in TXD.FRAMECONFIG. The NFCT peripheral will take (8*TXD.AMOUNT.TXDATABYTES + TXD.AMOUNT.TXDATABITS) bits and assemble a frame according to the settings in TXD.FRAMECONFIG. Both short frames, standard frames, and bit-oriented SDD frames as specified in the *NFC Forum, NFC Digital Protocol Technical Specification* can be assembled by the correct setting of the TXD.FRAMECONFIG register.

The bytes will be transmitted on air in the same order as they are read from RAM with a rising bit order within each byte, least significant bit (LSB) first. That is, b0 will be transmitted on air before b1, and so on. The bits read from RAM will be coded into symbols as defined in the *NFC Forum, NFC Digital Protocol Technical Specification*.

Important: Some NFC Forum documents, such as *NFC Forum, NFC Digital Protocol Technical Specification*, define bit numbering in a byte from b1 (LSB) to b8 (most significant bit (MSB)), while most other technical documents from the NFC Forum, and also the Nordic Semiconductor documentation, traditionally number them from b0 to b7. The present document uses the b0–b7 numbering scheme. Be aware of this when comparing the *NFC Forum, NFC Digital Protocol Technical Specification* to others.

The frame assembler can be configured in TXD.FRAMECONFIG to add SoF symbol, calculate and add parity bits, and calculate and add CRC to the data read from RAM when assembling the frame. The total frame will then be longer than what is defined by TXD.AMOUNT.TXDATABYTES. TXDATABITS. DISCARDMODE will select if the first bits in the first byte read from RAM or the last bits in the last byte read from RAM will be discarded if TXD.AMOUNT.TXDATABITS are not equal to zero. Note that if TXD.FRAMECONFIG.PARITY = Parity and TXD.FRAMECONFIG.DISCARDMODE=DiscardStart, a parity bit will be included after the non-complete first byte. No parity will be added after a non-complete last byte.

The frame assemble operation is illustrated in Frame assemble illustration on page 457 for different settings in TXD.FRAMECONFIG. All shaded bit fields are added by the frame assembler. Some of these bits are optional and appearances are configured in TXD.FRAMECONFIG. Note that the frames illustrated do not necessarily comply with the NFC specification. The figure is only to illustrate the behavior of the NFCT peripheral.

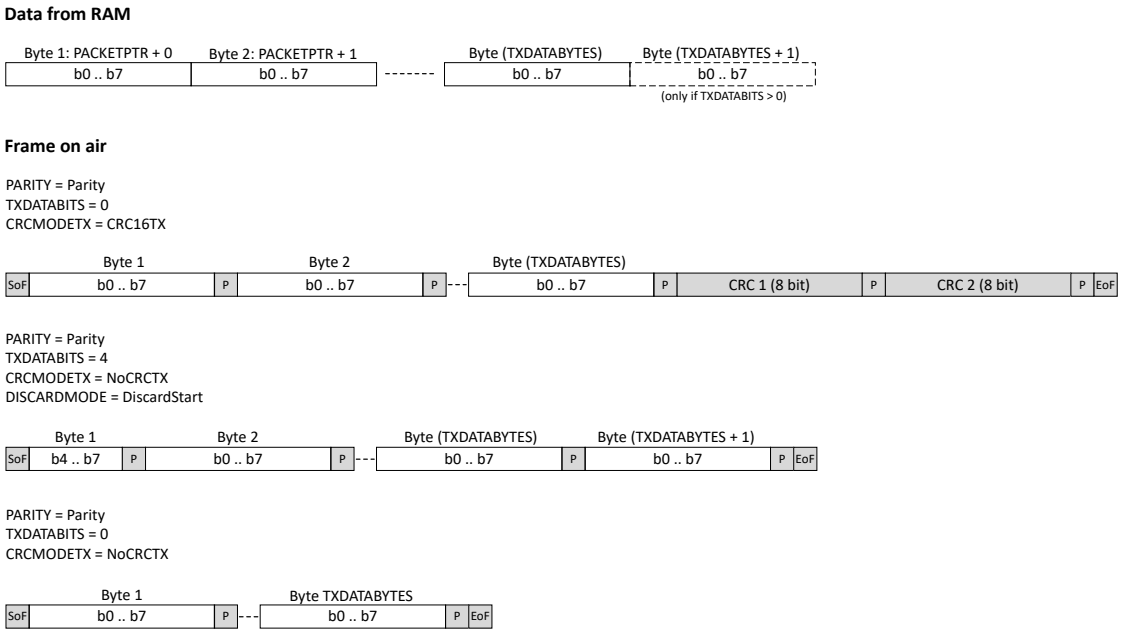


Figure 69: Frame assemble illustration

The accurate timing for transmitting the frame on air is set using the frame timing controller settings.

6.14.6 Frame disassembler

The NFCT peripheral implements a frame disassembler in hardware.

When the NFCT peripheral is in the ACTIVE_A state, the software can decide to enter RX or TX mode. For TX, see Frame assembler on page 456. For RX, the software must indicate the address and size of the destination buffer in Data RAM through programming the PACKETPTR and MAXLEN registers before issuing an ENABLERXDATA task.

The STARTED event indicates that the PACKETPTR and MAXLEN registers have been captured by the frame disassembler EasyDMA.

When an incoming frame starts, the RXFRAMESTART event will get issued and data will be written to the buffer in Data RAM. The frame disassembler will verify and remove any parity bits, start of frame (SoF) and

end of frame (EoF) symbols on the fly based on RXD.FRAMECONFIG register configuration. It will, however, verify and transfer the CRC bytes into RAM, if the CRC is enabled through RXD.FRAMECONFIG.

When an EoF symbol is detected, the NFCT peripheral will assert the RXFRAMEEND event and write the RXD.AMOUNT register to indicate numbers of received bytes and bits in the data packet. The module does not interpret the content of the data received from the remote NFC device, except for SoF, EoF, parity, and CRC checking, as described above. The frame disassemble operation is illustrated below.

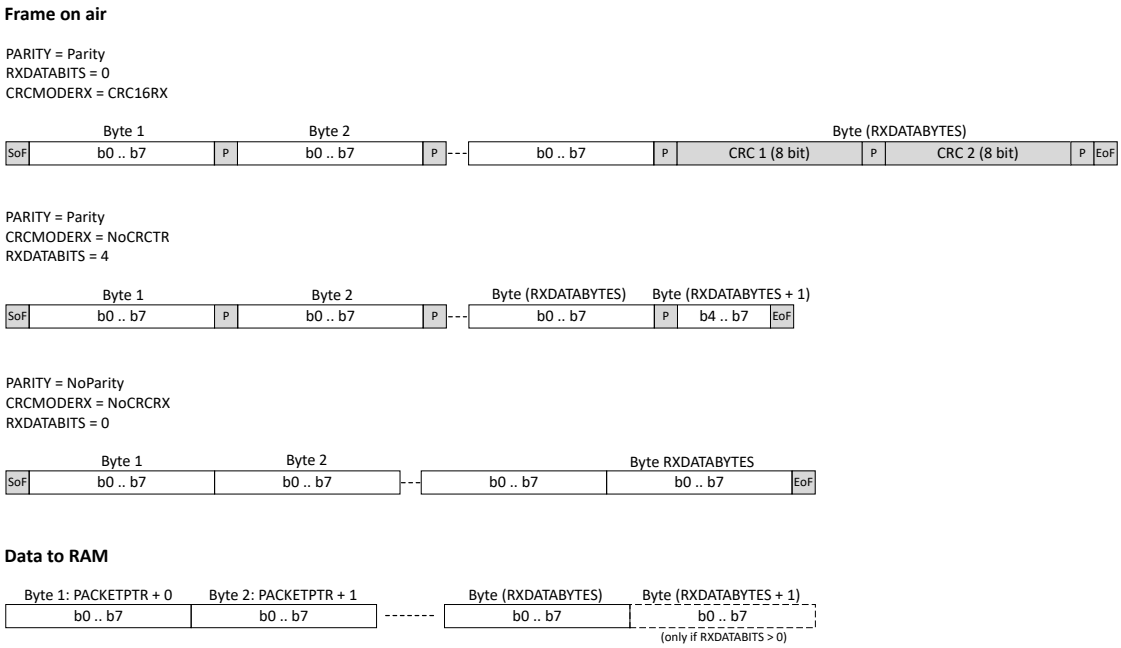


Figure 70: Frame disassemble illustration

Per NFC specification, the time between EoF to the next SoF can be as short as 86 µs, and thefore care must be taken that PACKETPTR and MAXLEN are ready and ENABLERXDATA is issued on time after the end of previous frame. The use of a PPI shortcut from TXFRAMEEND to ENABLERXDATA is recommended.

6.14.7 Frame timing controller

The NFCT peripheral includes a frame timing controller that continuously keeps track of the number of the 13.56 MHz RF carrier clock periods since the end of the EoF of the last received frame.

The NFCT peripheral can be programmed to send a responding frame within a time window or at an exact count of RF carrier periods. In case of FRAMEDELAYMODE = Window, a STARTTX task triggered before the frame timing controller counter is equal to FRAMEDELAYMIN will force the transmission to halt until the counter is equal to FRAMEDELAYMIN. If the counter is within FRAMEDELAYMIN and FRAMEDELAYMAX when the STARTTX task is triggered, the NFCT peripheral will start the transmission straight away. In case of FRAMEDELAYMODE = ExactVal, a STARTTX task triggered before the frame delay counter is equal to FRAMEDELAYMAX will halt the actual transmission start until the counter is equal to FRAMEDELAYMAX.

In case of FRAMEDELAYMODE = WindowGrid, the behaviour is similar to the FRAMEDELAYMODE = Window, but the actual transmission between FRAMEDELAYMIN and FRAMEDELAYMAX starts on a bit grid as defined for NFC-A Listen frames (slot duration of 128 RF carrier periods).

An ERROR event (with FRAMEDELAYTIMEOUT cause in ERRORSTATUS) will be asserted if the frame timing controller counter reaches FRAMEDELAYMAX without any STARTTX task triggered. This may happen even when the response is not required as per *NFC Forum, NFC Digital Protocol Technical Specification*. Any commands handled by the automatic collision resolution that don't involve a response being generated may also result in an ERROR event (with FRAMEDELAYTIMEOUT cause in ERRORSTATUS).
The FRAMEDELAYMIN and FRAMEDELAYMAX values shall only be updated before the STARTTX task is triggered. Failing to do so may cause unpredictable behaviour.

The frame timing controller operation is illustrated in Frame timing controller (FRAMEDELAYMODE=Window) on page 459. The frame timing controller automatically adjusts the frame timing counter based on the last received data bit according to NFC-A technology in the *NFC Forum, NFC Digital Protocol Technical Specification*.

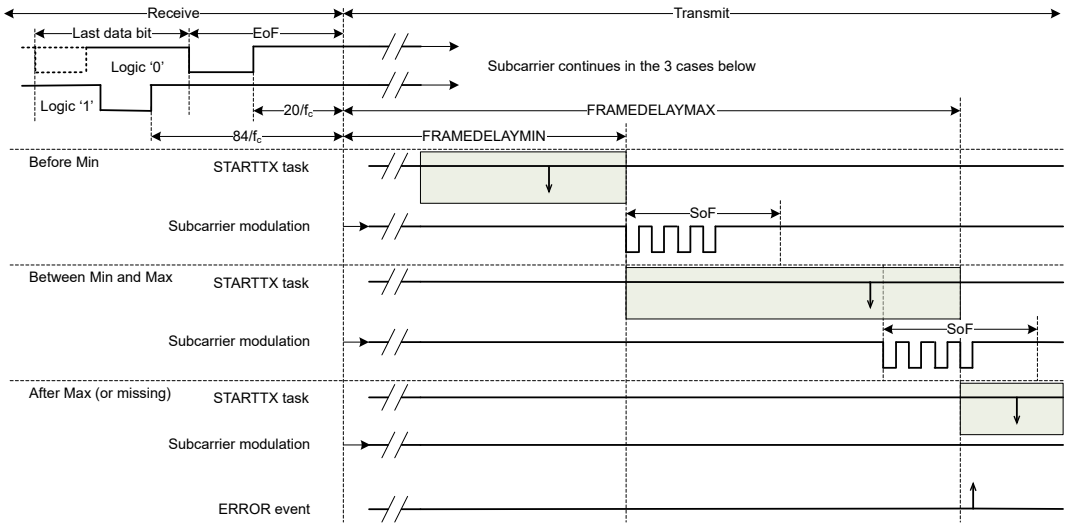


Figure 71: Frame timing controller (FRAMEDELAYMODE=Window)

6.14.8 Collision resolution

The NFCT peripheral implements an automatic collision resolution function as defined by the NFC Forum.

Automatic collision resolution is enabled by default, and it is recommended that the feature is used since it is power efficient and reduces the complexity of software handling the collision resolution sequence. This feature can be disabled through the MODE field in the AUTOCOLRESCONFIG register. When the automatic collision resolution is disabled, all commands will be sent over EasyDMA as defined in frame disassembler.

The SENSRES and SELRES registers need to be programmed upfront in order for the collision resolution to behave correctly. Depending on the NFCIDSIZE field in SENSRES, the following registers also need to be programmed upfront:

- NFCID1_LAST if NFCID1SIZE=NFCID1Single (ID = 4 bytes);
- NFCID1_2ND_LAST and NFCID1_LAST if NFCID1SIZE=NFCID1Double (ID = 7 bytes);
- NFCID1_3RD_LAST, NFCID1_2ND_LAST and NFCID1_LAST if NFCID1SIZE=NFCID1Triple (ID = 10 bytes);

A pre-defined set of registers, NFC.TAGHEADER0..3, containing a valid NFCID1 value, is available in FICR and can be used by software to populate the NFCID1_3RD_LAST, NFCID1_2ND_LAST, and NFCID1_LAST registers.

NFCID1 byte allocation (top sent first on air) on page 460 explains the position of the ID bytes in NFCID1_3RD_LAST, NFCID1_2ND_LAST, and NFCID1_LAST, depending on the ID size, and as compared to the definition used in the *NFC Forum, NFC Digital Protocol Technical Specification*.

	ID = 4 bytes	ID = 7 bytes	ID = 10 bytes
NFCID1_Q			$nfcid1_0$
NFCID1_R			$nfcid1_1$
NFCID1_S			$nfcid1_2$
NFCID1_T		$nfcid1_0$	$nfcid1_3$
NFCID1_U		$nfcid1_1$	$nfcid1_4$
NFCID1_V		$nfcid1_2$	$nfcid1_5$
NFCID1_W	$nfcid1_0$	$nfcid1_3$	$nfcid1_6$
NFCID1_X	$nfcid1_1$	$nfcid1_4$	$nfcid1_7$
NFCID1_Y	$nfcid1_2$	$nfcid1_5$	$nfcid1_8$
NFCID1_Z	$nfcid1_3$	$nfcid1_6$	$nfcid1_9$

Table 25: NFCID1 byte allocation (top sent first on air)

The hardware implementation can handle the states from IDLE to ACTIVE_A automatically as defined in the *NFC Forum, NFC Activity Technical Specification*, and the other states are to be handled by software. The software keeps track of the state through events. The collision resolution will trigger an AUTOCOLRESSTARTED event when it has started. Reaching the ACTIVE_A state is indicated by the SELECTED event.

If collision resolution fails, a COLLISION event is triggered. Note that errors occurring during automatic collision resolution may also cause ERROR and/or RXERROR events to be generated. Other events may also get generated. It is recommended that the software ignores any event except COLLISION, SELECTED and FIELDLOST during automatic collision resolution. Software shall also make sure that any unwanted SHORT or PPI shortcut is disabled during automatic collision resolution.

The automatic collision resolution will be restarted, if the packets are received with CRC or parity errors while in ACTIVE_A state. The automatic collision resolution feature can be disabled while in ACTIVE_A state to avoid this.

The SLP_REQ is automatically handled by the NFCT peripheral when the automatic collision resolution is enabled. However, this results in an ERROR event (with FRAMEDELAYTIMEOUT cause in ERRORSTATUS) since the SLP_REQ has no response. This error must be ignored until the SELECTED event is triggered and this error should be cleared by the software when the SELECTED event is triggered.

6.14.9 Antenna interface

In ACTIVATED state, an amplitude regulator will adjust the voltage swing on the antenna pins to a value that is within the V_{swing} limit.

Refer to NFCT Electrical Specification on page 479.

6.14.10 NFCT antenna recommendations

The NFCT antenna coil must be connected differential between **NFC1** and **NFC2** pins of the device.

Two external capacitors should be used to tune the resonance of the antenna circuit to 13.56 MHz.

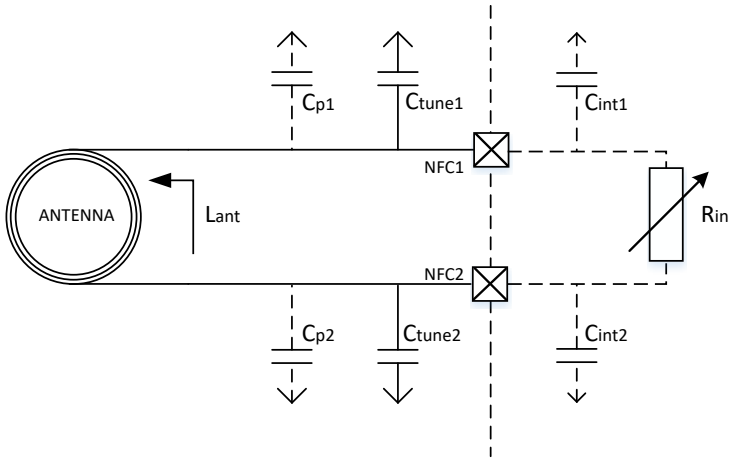


Figure 72: NFCT antenna recommendations

The required tuning capacitor value is given by the below equations:

$$C'_{tune} = \frac{1}{(2\pi \cdot 13.56\ MHz)^2 \cdot L_{ant}} \quad where\ C'_{tune} = \frac{1}{2} \cdot (C_p + C_{\text{int}} + C_{tune})$$

$$and\ C_{tune1} = C_{tune2} = C_{tune} \qquad C_{p1} = C_{p2} = C_p \qquad C_{\text{int1}} = C_{\text{int2}} = C_{\text{int}}$$

$$C_{tune} = \frac{2}{(2\pi \cdot 13.56\ MHz)^2 \cdot L_{ant}} - C_p - C_{\text{int}}$$

An antenna inductance of L_{ant} = 2 µH will give tuning capacitors in the range of 130 pF on each pin. The total capacitance on **NFC1** and **NFC2** must be matched.

6.14.11 Battery protection

If the antenna is exposed to a strong NFC field, current may flow in the opposite direction on the supply due to parasitic diodes and ESD structures.

If the battery used does not tolerate return current, a series diode must be placed between the battery and the device in order to protect the battery.

6.14.12 References

NFC Forum, NFC Analog Specification version 1.0, www.nfc-forum.org

NFC Forum, NFC Digital Protocol Technical Specification version 1.1, www.nfc-forum.org

NFC Forum, NFC Activity Technical Specification version 1.1, www.nfc-forum.org

6.14.13 Registers

Instances

Instance	Base address	Description
NFCT	0x40005000	Near field communication tag

Register overview

Register	Offset	Description
TASKS_ACTIVATE	0x000	Activate NFCT peripheral for incoming and outgoing frames, change state to activated
TASKS_DISABLE	0x004	Disable NFCT peripheral
TASKS_SENSE	0x008	Enable NFC sense field mode, change state to sense mode
TASKS_STARTTX	0x00C	Start transmission of an outgoing frame, change state to transmit
TASKS_ENABLERXDATA	0x01C	Initializes the EasyDMA for receive.
TASKS_GOIDLE	0x024	Force state machine to IDLE state
TASKS_GOSLEEP	0x028	Force state machine to SLEEP_A state
EVENTS_READY	0x100	The NFCT peripheral is ready to receive and send frames
EVENTS_FIELDDETECTED	0x104	Remote NFC field detected
EVENTS_FIELDLOST	0x108	Remote NFC field lost
EVENTS_TXFRAMESTART	0x10C	Marks the start of the first symbol of a transmitted frame
EVENTS_TXFRAMEEND	0x110	Marks the end of the last transmitted on-air symbol of a frame
EVENTS_RXFRAMESTART	0x114	Marks the end of the first symbol of a received frame
EVENTS_RXFRAMEEND	0x118	Received data has been checked (CRC, parity) and transferred to RAM, and EasyDMA has ended accessing the RX buffer
EVENTS_ERROR	0x11C	NFC error reported. The ERRORSTATUS register contains details on the source of the error.
EVENTS_RXERROR	0x128	NFC RX frame error reported. The FRAMESTATUS.RX register contains details on the source of the error.
EVENTS_ENDRX	0x12C	RX buffer (as defined by PACKETPTR and MAXLEN) in Data RAM full.
EVENTS_ENDTX	0x130	Transmission of data in RAM has ended, and EasyDMA has ended accessing the TX buffer
EVENTS_AUTOCOLRESSTARTED	0x138	Auto collision resolution process has started
EVENTS_COLLISION	0x148	NFC auto collision resolution error reported.
EVENTS_SELECTED	0x14C	NFC auto collision resolution successfully completed
EVENTS_STARTED	0x150	EasyDMA is ready to receive or send frames.
SHORTS	0x200	Shortcuts between local events and tasks
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
ERRORSTATUS	0x404	NFC Error Status register
FRAMESTATUS.RX	0x40C	Result of last incoming frame
NFCTAGSTATE	0x410	NfcTag state register
SLEEPSTATE	0x420	Sleep state during automatic collision resolution
FIELDPRESENT	0x43C	Indicates the presence or not of a valid field
FRAMEDELAYMIN	0x504	Minimum frame delay
FRAMEDELAYMAX	0x508	Maximum frame delay
FRAMEDELAYMODE	0x50C	Configuration register for the Frame Delay Timer
PACKETPTR	0x510	Packet pointer for TXD and RXD data storage in Data RAM
MAXLEN	0x514	Size of the RAM buffer allocated to TXD and RXD data storage each
TXD.FRAMECONFIG	0x518	Configuration of outgoing frames
TXD.AMOUNT	0x51C	Size of outgoing frame
RXD.FRAMECONFIG	0x520	Configuration of incoming frames
RXD.AMOUNT	0x524	Size of last incoming frame
NFCID1_LAST	0x590	Last NFCID1 part (4, 7 or 10 bytes ID)
NFCID1_2ND_LAST	0x594	Second last NFCID1 part (7 or 10 bytes ID)
NFCID1_3RD_LAST	0x598	Third last NFCID1 part (10 bytes ID)
AUTOCOLRESCONFIG	0x59C	Controls the auto collision resolution function. This setting must be done before the NFCT peripheral is activated.
SENSRES	0x5A0	NFC-A SENS_RES auto-response settings
SELRES	0x5A4	NFC-A SEL_RES auto-response settings

6.14.13.1 TASKS_ACTIVATE

Address offset: 0x000

Activate NFCT peripheral for incoming and outgoing frames, change state to activated

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_ACTIVATE			Activate NFCT peripheral for incoming and outgoing frames, change state to activated
			Trigger	1	Trigger task

6.14.13.2 TASKS_DISABLE

Address offset: 0x004

Disable NFCT peripheral

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_DISABLE			Disable NFCT peripheral
			Trigger	1	Trigger task

6.14.13.3 TASKS_SENSE

Address offset: 0x008

Enable NFC sense field mode, change state to sense mode

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_SENSE			Enable NFC sense field mode, change state to sense mode
			Trigger	1	Trigger task

6.14.13.4 TASKS_STARTTX

Address offset: 0x00C

Start transmission of an outgoing frame, change state to transmit

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTTX			Start transmission of an outgoing frame, change state to transmit
			Trigger	1	Trigger task

6.14.13.5 TASKS_ENABLERXDATA

Address offset: 0x01C

Initializes the EasyDMA for receive.

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_ENABLERXDATA			Initializes the EasyDMA for receive.
			Trigger	1	Trigger task

6.14.13.6 TASKS_GOIDLE

Address offset: 0x024

Force state machine to IDLE state

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_GOIDLE			Force state machine to IDLE state
			Trigger	1	Trigger task

6.14.13.7 TASKS_GOSLEEP

Address offset: 0x028

Force state machine to SLEEP_A state

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_GOSLEEP			Force state machine to SLEEP_A state
			Trigger	1	Trigger task

6.14.13.8 EVENTS_READY

Address offset: 0x100

The NFCT peripheral is ready to receive and send frames

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_READY			The NFCT peripheral is ready to receive and send frames
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.14.13.9 EVENTS_FIELDDETECTED

Address offset: 0x104

Remote NFC field detected

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value	Description																														
A	RW	EVENTS_FIELDDETECTED			Remote NFC field detected																														
			NotGenerated	0	Event not generated																														
			Generated	1	Event generated																														

6.14.13.10 EVENTS_FIELDLOST

Address offset: 0x108

Remote NFC field lost

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value	Description																														
A	RW	EVENTS_FIELDLOST			Remote NFC field lost																														
			NotGenerated	0	Event not generated																														
			Generated	1	Event generated																														

6.14.13.11 EVENTS_TXFRAMESTART

Address offset: 0x10C

Marks the start of the first symbol of a transmitted frame

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value	Description																														
A	RW	EVENTS_TXFRAMESTART			Marks the start of the first symbol of a transmitted frame																														
			NotGenerated	0	Event not generated																														
			Generated	1	Event generated																														

6.14.13.12 EVENTS_TXFRAMEEND

Address offset: 0x110

Marks the end of the last transmitted on-air symbol of a frame

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value	Description																														
A	RW	EVENTS_TXFRAMEEND			Marks the end of the last transmitted on-air symbol of a frame																														
			NotGenerated	0	Event not generated																														
			Generated	1	Event generated																														

6.14.13.13 EVENTS_RXFRAMESTART

Address offset: 0x114

Marks the end of the first symbol of a received frame

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_RXFRAMESTART			Marks the end of the first symbol of a received frame
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.14.13.14 EVENTS_RXFRAMEEND

Address offset: 0x118

Received data has been checked (CRC, parity) and transferred to RAM, and EasyDMA has ended accessing the RX buffer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_RXFRAMEEND			Received data has been checked (CRC, parity) and transferred to RAM, and EasyDMA has ended accessing the RX buffer
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.14.13.15 EVENTS_ERROR

Address offset: 0x11C

NFC error reported. The ERRORSTATUS register contains details on the source of the error.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ERROR			NFC error reported. The ERRORSTATUS register contains details on the source of the error.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.14.13.16 EVENTS_RXERROR

Address offset: 0x128

NFC RX frame error reported. The FRAMESTATUS.RX register contains details on the source of the error.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_RXERROR			NFC RX frame error reported. The FRAMESTATUS.RX register contains details on the source of the error.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.14.13.17 EVENTS_ENDRX

Address offset: 0x12C

RX buffer (as defined by PACKETPTR and MAXLEN) in Data RAM full.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDRX			RX buffer (as defined by PACKETPTR and MAXLEN) in Data RAM full.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.14.13.18 EVENTS_ENDTX

Address offset: 0x130

Transmission of data in RAM has ended, and EasyDMA has ended accessing the TX buffer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDTX			Transmission of data in RAM has ended, and EasyDMA has ended accessing the TX buffer
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.14.13.19 EVENTS_AUTOCOLRESSTARTED

Address offset: 0x138

Auto collision resolution process has started

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_AUTOCOLRESSTARTED			Auto collision resolution process has started
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.14.13.20 EVENTS_COLLISION

Address offset: 0x148

NFC auto collision resolution error reported.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_COLLISION			NFC auto collision resolution error reported.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.14.13.21 EVENTS_SELECTED

Address offset: 0x14C

NFC auto collision resolution successfully completed

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_SELECTED			NFC auto collision resolution successfully completed
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.14.13.22 EVENTS_STARTED

Address offset: 0x150

EasyDMA is ready to receive or send frames.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_STARTED			EasyDMA is ready to receive or send frames.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.14.13.23 SHORTS

Address offset: 0x200

Shortcuts between local events and tasks

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	FIELDDETECTED_ACTIVATE			Shortcut between event FIELDDETECTED and task ACTIVATE
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
B	RW	FIELDLOST_SENSE			Shortcut between event FIELDLOST and task SENSE
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																											F					B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
F	RW	TXFRAMEEND_ENABLERXDATA			Shortcut between event TXFRAMEEND and task ENABLERXDATA
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

6.14.13.24 INTEN

Address offset: 0x300

Enable or disable interrupt

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID												T	S	R				N		M	L	K			H	G	F	E	D	C	B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	READY			Enable or disable interrupt for event READY
			Disabled	0	Disable
			Enabled	1	Enable
B	RW	FIELDDETECTED			Enable or disable interrupt for event FIELDDETECTED
			Disabled	0	Disable
			Enabled	1	Enable
C	RW	FIELDLOST			Enable or disable interrupt for event FIELDLOST
			Disabled	0	Disable
			Enabled	1	Enable
D	RW	TXFRAMESTART			Enable or disable interrupt for event TXFRAMESTART
			Disabled	0	Disable
			Enabled	1	Enable
E	RW	TXFRAMEEND			Enable or disable interrupt for event TXFRAMEEND
			Disabled	0	Disable
			Enabled	1	Enable
F	RW	RXFRAMESTART			Enable or disable interrupt for event RXFRAMESTART
			Disabled	0	Disable
			Enabled	1	Enable
G	RW	RXFRAMEEND			Enable or disable interrupt for event RXFRAMEEND
			Disabled	0	Disable
			Enabled	1	Enable
H	RW	ERROR			Enable or disable interrupt for event ERROR
			Disabled	0	Disable
			Enabled	1	Enable
K	RW	RXERROR			Enable or disable interrupt for event RXERROR
			Disabled	0	Disable
			Enabled	1	Enable
L	RW	ENDRX			Enable or disable interrupt for event ENDRX
			Disabled	0	Disable
			Enabled	1	Enable
M	RW	ENDTX			Enable or disable interrupt for event ENDTX
			Disabled	0	Disable
			Enabled	1	Enable
N	RW	AUTOCOLRESSTARTED			Enable or disable interrupt for event AUTOCOLRESSTARTED
			Disabled	0	Disable
			Enabled	1	Enable

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID												T	S	R				N		M	L	K			H	G	F	E	D	C	B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
R	RW	COLLISION			Enable or disable interrupt for event COLLISION
			Disabled	0	Disable
			Enabled	1	Enable
S	RW	SELECTED			Enable or disable interrupt for event SELECTED
			Disabled	0	Disable
			Enabled	1	Enable
T	RW	STARTED			Enable or disable interrupt for event STARTED
			Disabled	0	Disable
			Enabled	1	Enable

6.14.13.25 INTENSET

Address offset: 0x304

Enable interrupt

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID												T	S	R				N		M	L	K			H	G	F	E	D	C	B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	READY			Write '1' to enable interrupt for event READY
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	FIELDDETECTED			Write '1' to enable interrupt for event FIELDDETECTED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	FIELDLOST			Write '1' to enable interrupt for event FIELDLOST
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	TXFRAMESTART			Write '1' to enable interrupt for event TXFRAMESTART
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	TXFRAMEEND			Write '1' to enable interrupt for event TXFRAMEEND
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	RXFRAMESTART			Write '1' to enable interrupt for event RXFRAMESTART
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
G	RW	RXFRAMEEND			Write '1' to enable interrupt for event RXFRAMEEND
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
H	RW	ERROR			Write '1' to enable interrupt for event ERROR
			Set	1	Enable

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				T S R N M L K H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
K	RW	RXERROR			Write '1' to enable interrupt for event RXERROR
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
L	RW	ENDRX			Write '1' to enable interrupt for event ENDRX
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
M	RW	ENDTX			Write '1' to enable interrupt for event ENDTX
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
N	RW	AUTOCOLRESSTARTED			Write '1' to enable interrupt for event AUTOCOLRESSTARTED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
R	RW	COLLISION			Write '1' to enable interrupt for event COLLISION
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
S	RW	SELECTED			Write '1' to enable interrupt for event SELECTED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
T	RW	STARTED			Write '1' to enable interrupt for event STARTED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.14.13.26 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				T S R N M L K H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	READY			Write '1' to disable interrupt for event READY
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	FIELDDETECTED			Write '1' to disable interrupt for event FIELDDETECTED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	FIELDLOST			Write '1' to disable interrupt for event FIELDLOST

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID												T	S	R				N		M	L	K			H	G	F	E	D	C	B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	TXFRAMESTART			Write '1' to disable interrupt for event TXFRAMESTART
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	TXFRAMEEND			Write '1' to disable interrupt for event TXFRAMEEND
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	RXFRAMESTART			Write '1' to disable interrupt for event RXFRAMESTART
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
G	RW	RXFRAMEEND			Write '1' to disable interrupt for event RXFRAMEEND
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
H	RW	ERROR			Write '1' to disable interrupt for event ERROR
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
K	RW	RXERROR			Write '1' to disable interrupt for event RXERROR
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
L	RW	ENDRX			Write '1' to disable interrupt for event ENDRX
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
M	RW	ENDTX			Write '1' to disable interrupt for event ENDTX
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
N	RW	AUTOCOLRESSTARTED			Write '1' to disable interrupt for event AUTOCOLRESSTARTED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
R	RW	COLLISION			Write '1' to disable interrupt for event COLLISION
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
S	RW	SELECTED			Write '1' to disable interrupt for event SELECTED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
T	RW	STARTED			Write '1' to disable interrupt for event STARTED
			Clear	1	Disable
			Disabled	0	Read: Disabled

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	T S R N M L K H G F E D C B A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
			Enabled	1	Read: Enabled

6.14.13.27 ERRORSTATUS

Address offset: 0x404

NFC Error Status register

Note: Write a bit to '1' to clear it. Writing '0' has no effect.

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW W1C	FRAMEDELAYTIMEOUT			No STARTTX task triggered before expiration of the time set in FRAMEDELAYMAX

6.14.13.28 FRAMESTATUS.RX

Address offset: 0x40C

Result of last incoming frame

Note: Write a bit to '1' to clear it. Writing '0' has no effect.

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	C B A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW W1C	CRCERROR			No valid end of frame (EoF) detected
			CRCCorrect	0	Valid CRC detected
			CRCError	1	CRC received does not match local check
B	RW W1C	PARITYSTATUS			Parity status of received frame
			ParityOK	0	Frame received with parity OK
			ParityError	1	Frame received with parity error
C	RW W1C	OVERRUN			Overrun detected
			NoOverrun	0	No overrun detected
			Overrun	1	Overrun error

6.14.13.29 NFCTAGSTATE

Address offset: 0x410

NfcTag state register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	NFCTAGSTATE			NfcTag state
			Disabled	0	Disabled or sense
			RampUp	2	RampUp
			Idle	3	Idle
			Receive	4	Receive
			FrameDelay	5	FrameDelay
			Transmit	6	Transmit

6.14.13.30 SLEEPSTATE

Address offset: 0x420

Sleep state during automatic collision resolution

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	SLEEPSTATE			Reflects the sleep state during automatic collision resolution. Set to IDLE by a GOIDLE task. Set to SLEEP_A when a valid SLEEP_REQ frame is received or by a GOSLEEP task.
			Idle	0	State is IDLE.
			SleepA	1	State is SLEEP_A.

6.14.13.31 FIELDPRESENT

Address offset: 0x43C

Indicates the presence or not of a valid field

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	FIELDPRESENT			Indicates if a valid field is present. Available only in the activated state.
			NoField	0	No valid field detected
			FieldPresent	1	Valid field detected
B	R	LOCKDETECT			Indicates if the low level has locked to the field
			NotLocked	0	Not locked to field
			Locked	1	Locked to field

6.14.13.32 FRAMEDELAYMIN

Address offset: 0x504

Minimum frame delay

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A
Reset 0x00000480				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 1 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	FRAMEDELAYMIN			Minimum frame delay in number of 13.56 MHz clocks

6.14.13.33 FRAMEDELAYMAX

Address offset: 0x508

Maximum frame delay

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A
Reset 0x00001000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	FRAMEDELAYMAX			Maximum frame delay in number of 13.56 MHz clocks

6.14.13.34 FRAMEDELAYMODE

Address offset: 0x50C

Configuration register for the Frame Delay Timer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	FRAMEDELAYMODE			Configuration register for the Frame Delay Timer
			FreeRun	0	Transmission is independent of frame timer and will start when the STARTTX task is triggered. No timeout.
			Window	1	Frame is transmitted between FRAMEDELAYMIN and FRAMEDELAYMAX
			ExactVal	2	Frame is transmitted exactly at FRAMEDELAYMAX
			WindowGrid	3	Frame is transmitted on a bit grid between FRAMEDELAYMIN and FRAMEDELAYMAX

6.14.13.35 PACKETPTR

Address offset: 0x510

Packet pointer for TXD and RXD data storage in Data RAM

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Packet pointer for TXD and RXD data storage in Data RAM. This address is a byte-aligned RAM address.

Note: See the memory chapter for details about which memories are available for EasyDMA.

6.14.13.36 MAXLEN

Address offset: 0x514

Size of the RAM buffer allocated to TXD and RXD data storage each

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	MAXLEN		[0..257]	Size of the RAM buffer allocated to TXD and RXD data storage each

6.14.13.37 TXD.FRAMECONFIG

Address offset: 0x518

Configuration of outgoing frames

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				D C B A
Reset 0x00000017				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PARITY			Indicates if parity is added to the frame
			NoParity	0	Parity is not added to TX frames
			Parity	1	Parity is added to TX frames
B	RW	DISCARDMODE			Discarding unused bits at start or end of a frame
			DiscardEnd	0	Unused bits are discarded at end of frame (EoF)
			DiscardStart	1	Unused bits are discarded at start of frame (SoF)
C	RW	SOF			Adding SoF or not in TX frames
			NoSoF	0	SoF symbol not added
			SoF	1	SoF symbol added
D	RW	CRCMODETX			CRC mode for outgoing frames
			NoCRCTX	0	CRC is not added to the frame
			CRC16TX	1	16 bit CRC added to the frame based on all the data read from RAM that is used in the frame

6.14.13.38 TXD.AMOUNT

Address offset: 0x51C

Size of outgoing frame

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B B B B B B B B A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	TXDATABITS		[0..7]	Number of bits in the last or first byte read from RAM that shall be included in the frame (excluding parity bit). The DISCARDMODE field in FRAMECONFIG.TX selects if unused bits is discarded at the start or at the end of a frame. A value of 0 data bytes and 0 data bits is invalid.
B	RW	TXDATABYTES		[0..257]	Number of complete bytes that shall be included in the frame, excluding CRC, parity and framing

6.14.13.39 RXD.FRAMECONFIG

Address offset: 0x520

Configuration of incoming frames

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A
Reset 0x00000015				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 1 0 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PARITY			Indicates if parity expected in RX frame
			NoParity	0	Parity is not expected in RX frames
			Parity	1	Parity is expected in RX frames
B	RW	SOF			SoF expected or not in RX frames
			NoSoF	0	SoF symbol is not expected in RX frames
			SoF	1	SoF symbol is expected in RX frames
C	RW	CRCMODERX			CRC mode for incoming frames
			NoCRCRX	0	CRC is not expected in RX frames
			CRC16RX	1	Last 16 bits in RX frame is CRC, CRC is checked and CRCSTATUS updated

6.14.13.40 RXD.AMOUNT

Address offset: 0x524

Size of last incoming frame

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B B B B B B B B A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	RXDATABITS			Number of bits in the last byte in the frame, if less than 8 (including CRC, but excluding parity and SoF/EoF framing). Frames with 0 data bytes and less than 7 data bits are invalid and are not received properly.
B	R	RXDATABYTES			Number of complete bytes received in the frame (including CRC, but excluding parity and SoF/EoF framing)

6.14.13.41 NFCID1_LAST

Address offset: 0x590

Last NFCID1 part (4, 7 or 10 bytes ID)

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				D D D D D D D D C C C C C C C C B B B B B B B B A A A A A A A A
Reset 0x00006363				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0 0 0 1 1 0 1 1 0 0 0 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	NFCID1_Z			NFCID1 byte Z (very last byte sent)
B	RW	NFCID1_Y			NFCID1 byte Y
C	RW	NFCID1_X			NFCID1 byte X
D	RW	NFCID1_W			NFCID1 byte W

6.14.13.42 NFCID1_2ND_LAST

Address offset: 0x594

Second last NFCID1 part (7 or 10 bytes ID)

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C C C C C C C C B B B B B B B B A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	NFCID1_V			NFCID1 byte V
B	RW	NFCID1_U			NFCID1 byte U
C	RW	NFCID1_T			NFCID1 byte T

6.14.13.43 NFCID1_3RD_LAST

Address offset: 0x598

Third last NFCID1 part (10 bytes ID)

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C C C C C C C C B B B B B B B B A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	NFCID1_S			NFCID1 byte S
B	RW	NFCID1_R			NFCID1 byte R
C	RW	NFCID1_Q			NFCID1 byte Q

6.14.13.44 AUTOCOLRESCONFIG

Address offset: 0x59C

Controls the auto collision resolution function. This setting must be done before the NFCT peripheral is activated.

Note: When modifiying this register bit 1 must be written to '1'.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	MODE			Enables/disables auto collision resolution
			Enabled	0	Auto collision resolution enabled
			Disabled	1	Auto collision resolution disabled

6.14.13.45 SENSRES

Address offset: 0x5A0

NFC-A SENS_RES auto-response settings

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E E E D D D D C C B A A A A A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	BITFRAMESDD			Bit frame SDD as defined by the b5:b1 of byte 1 in SENS_RES response in the NFC Forum, NFC Digital Protocol Technical Specification
			SDD00000	0	SDD pattern 00000
			SDD00001	1	SDD pattern 00001
			SDD00010	2	SDD pattern 00010

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E E E E D D D D C C B A A A A A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
			SDD00100	4	SDD pattern 00100
			SDD01000	8	SDD pattern 01000
			SDD10000	16	SDD pattern 10000
B	RW	RFU5			Reserved for future use. Shall be 0.
C	RW	NFCIDSIZE			NFCID1 size. This value is used by the auto collision resolution engine.
			NFCID1Single	0	NFCID1 size: single (4 bytes)
			NFCID1Double	1	NFCID1 size: double (7 bytes)
			NFCID1Triple	2	NFCID1 size: triple (10 bytes)
D	RW	PLATFCONFIG			Tag platform configuration as defined by the b4:b1 of byte 2 in SENS_RES response in the NFC Forum, NFC Digital Protocol Technical Specification
E	RW	RFU74			Reserved for future use. Shall be 0.

6.14.13.46 SELRES

Address offset: 0x5A4

NFC-A SEL_RES auto-response settings

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D D C C B A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RFU10			Reserved for future use. Shall be 0.
B	RW	CASCADE			Cascade as defined by the b3 of SEL_RES response in the NFC Forum, NFC Digital Protocol Technical Specification (controlled by hardware, shall be 0)
C	RW	RFU43			Reserved for future use. Shall be 0.
D	RW	PROTOCOL			Protocol as defined by the b7:b6 of SEL_RES response in the NFC Forum, NFC Digital Protocol Technical Specification
E	RW	RFU7			Reserved for future use. Shall be 0.

6.14.14 Electrical specification

6.14.14.1 NFCT Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
f_c	Frequency of operation		13.56		MHz
C_{MI}	Carrier modulation index	95			%
DR	Data Rate		106		kbps
V_{sense}	Peak differential Field detect threshold level on NFC1-NFC2[17]		1.2		Vp
I_{max}	Maximum input current on NFCT pins			80	mA

[17] Input is high impedance in sense mode

6.14.14.2 NFCT Timing Parameters

Symbol	Description	Min.	Typ.	Max.	Units
$t_{activate}$	Time from task_ACTIVATE in SENSE or DISABLE state to ACTIVATE_A or IDLE state[18]			500	µs
t_{sense}	Time from remote field is present in SENSE mode to FIELDDETECTED event is asserted			20	µs

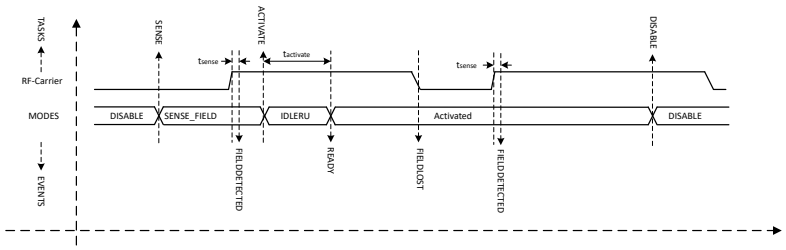


Figure 73: NFCT timing parameters (Shortcuts for FIELDDETECTED and FIELDLOST are disabled)

6.15 PDM — Pulse density modulation interface

The pulse density modulation (PDM) module enables input of pulse density modulated signals from external audio frontends, for example, digital microphones. The PDM module generates the PDM clock and supports single-channel or dual-channel (left and right) data input. Data is transferred directly to RAM buffers using EasyDMA.

Listed here are the main features for PDM:

- Up to two PDM microphones configured as a left/right pair using the same data input
- 16 kHz output sample rate, 16-bit samples
- EasyDMA support for sample buffering
- HW decimation filters
- Selectable ratio of 64 or 80 between PDM_CLK and output sample rate

The PDM module illustrated below is interfacing up to two digital microphones with the PDM interface. EasyDMA is implemented to relieve the real-time requirements associated with controlling of the PDM slave from a low priority CPU execution context. It also includes all the necessary digital filter elements to produce pulse code modulation (PCM) samples. The PDM module allows continuous audio streaming.

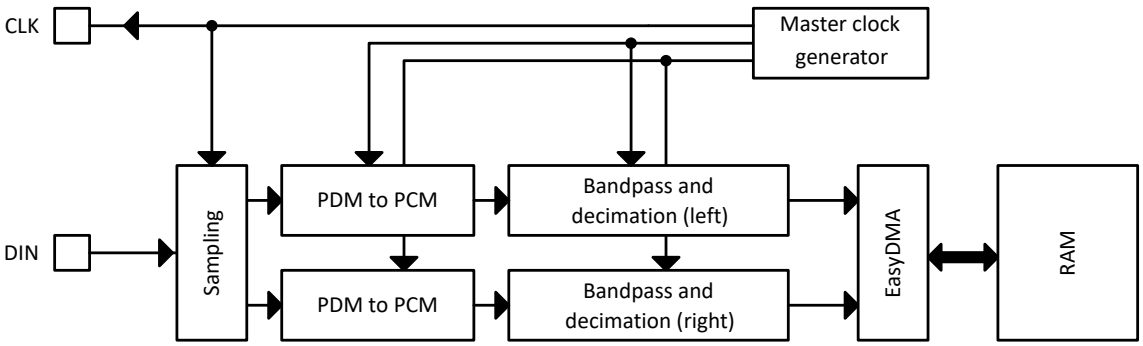


Figure 74: PDM module

[18] Does not account for voltage supply and oscillator startup times

6.15.1 Master clock generator

The master clock generator's PDMCLKCTRL register allows adjusting the PDM clock's frequency.

The master clock generator does not add any jitter to the HFCLK source chosen. It is recommended (but not mandatory) to use the Xtal as HFCLK source.

6.15.2 Module operation

By default, bits from the left PDM microphone are sampled on PDM_CLK falling edge, and bits for the right are sampled on the rising edge of PDM_CLK, resulting in two bitstreams. Each bitstream is fed into a digital filter which converts the PDM stream into 16-bit PCM samples, then filters and down-samples them to reach the appropriate sample rate.

The EDGE field in the MODE register allows swapping left and right, so that left will be sampled on rising edge, and right on falling.

The PDM module uses EasyDMA to store the samples coming out from the filters into one buffer in RAM. Depending on the mode chosen in the OPERATION field in the MODE register, memory either contains alternating left and right 16-bit samples (Stereo), or only left 16-bit samples (Mono). To ensure continuous PDM sampling, it is up to the application to update the EasyDMA destination address pointer as the previous buffer is filled.

The continuous transfer can be started or stopped by sending the START and STOP tasks. STOP becomes effective after the current frame has finished transferring, which will generate the STOPPED event. The STOPPED event indicates that all activity in the module is finished, and that the data is available in RAM (EasyDMA has finished transferring as well). Attempting to restart before receiving the STOPPED event may result in unpredictable behavior.

6.15.3 Decimation filter

In order to convert the incoming data stream into PCM audio samples, a decimation filter is included in the PDM interface module.

The input of the filter is the two-channel PDM serial stream (with left channel on clock high, right channel on clock low). Depending on the RATIO selected, its output is 2 × 16-bit PCM samples at a sample rate either 64 times or 80 times (depending on the RATIO register) lower than the PDM clock rate.

The filter stage of each channel is followed by a digital volume control, to attenuate or amplify the output samples in a range of -20 dB to +20 dB around the default (reset) setting, defined by $G_{PDM,default}$. The gain is controlled by the GAINL and GAINR registers.

As an example, if the goal is to achieve 2500 RMS output samples (16-bit) with a 1 kHz 90 dBA signal into a -26 dBFS sensitivity PDM microphone, do the following:

- Sum the PDM module's default gain ($G_{PDM,default}$) and the gain introduced by the microphone and acoustic path of his implementation (an attenuation would translate into a negative gain)
- Adjust GAINL and GAINR by the above summed amount. Assuming that only the PDM module influences the gain, GAINL and GAINR must be set to -$G_{PDM,default}$ dB to achieve the requirement.

With $G_{PDM,default}$=3.2 dB, and as GAINL and GAINR are expressed in 0.5 dB steps, the closest value to program would be 3.0 dB, which can be calculated as:

```
GAINL = GAINR = (DefaultGain - (2 * 3))
```

Remember to check that the resulting values programmed into GAINL and GAINR fall within MinGain and MaxGain.

6.15.4 EasyDMA

Samples will be written directly to RAM, and EasyDMA must be configured accordingly.

The address pointer for the EasyDMA channel is set in SAMPLE.PTR register. If the destination address set in SAMPLE.PTR is not pointing to the Data RAM region, an EasyDMA transfer may result in a HardFault or RAM corruption. See Memory on page 21 for more information about the different memory regions.

DMA supports Stereo (Left+Right 16-bit samples) and Mono (Left only) data transfer, depending on the setting in the OPERATION field in the MODE register. The samples are stored little endian.

MODE.OPERATION	Bits per sample	Result stored per RAM word	Physical RAM allocated (32-bit words)	Result boundary indexes in RAM	Note
Stereo	32 (2x16)	L+R	ceil(SAMPLE.MAXCNT/2)	R0=[31:16]; L0=[15:0]	Default
Mono	16	2xL	ceil(SAMPLE.MAXCNT/2)	L1=[31:16]; L0=[15:0]	

Table 26: DMA sample storage

The destination buffer in RAM consists of one block, the size of which is set in SAMPLE.MAXCNT register. Format is number of 16-bit samples. The physical RAM allocated is always:

```
(RAM allocation, in bytes) = SAMPLE.MAXCNT * 2;
```

(but the mapping of the samples depends on MODE.OPERATION.

If OPERATION=Stereo, RAM will contain a succession of left and right samples.

If OPERATION=Mono, RAM will contain a succession of left only samples.

For a given value of SAMPLE.MAXCNT, the buffer in RAM can contain half the stereo sampling time as compared to the mono sampling time.

The PDM acquisition can be started by the START task, after the SAMPLE.PTR and SAMPLE.MAXCNT registers have been written. When starting the module, it will take some time for the filters to start outputting valid data. Transients from the PDM microphone itself may also occur. The first few samples (typically around 50) might hence contain invalid values or transients. It is therefore advised to discard the first few samples after a PDM start.

As soon as the STARTED event is received, the firmware can write the next SAMPLE.PTR value (this register is double-buffered), to ensure continuous operation.

When the buffer in RAM is filled with samples, an END event is triggered. The firmware can start processing the data in the buffer. Meanwhile, the PDM module starts acquiring data into the new buffer pointed to by SAMPLE.PTR, and sends a new STARTED event, so that the firmware can update SAMPLE.PTR to the next buffer address.

6.15.5 Hardware example

PDM can be configured with a single microphone (mono), or with two microphones.

When a single microphone is used, connect the microphone clock to CLK, and data to DIN.

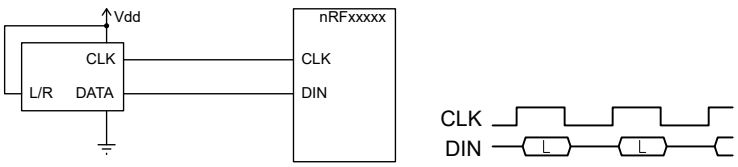


Figure 75: Example of a single PDM microphone, wired as left

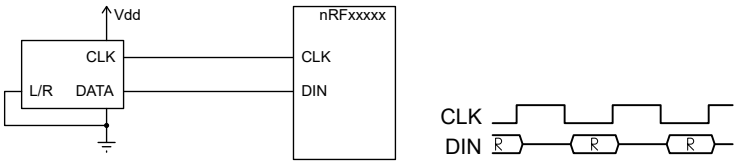


Figure 76: Example of a single PDM microphone, wired as right

Note that in a single-microphone (mono) configuration, depending on the microphone's implementation, either the left or the right channel (sampled at falling or rising CLK edge respectively) will contain reliable data.

If two microphones are used, one of them has to be set as left, the other as right (L/R pin tied high or to GND on the respective microphone). It is strongly recommended to use two microphones of exactly the same brand and type so that their timings in left and right operation match.

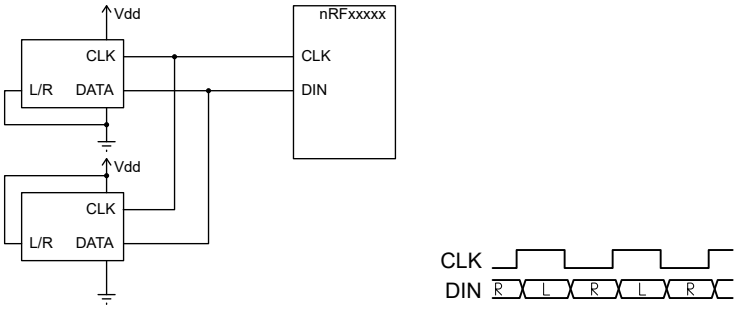


Figure 77: Example of two PDM microphones

6.15.6 Pin configuration

The CLK and DIN signals associated to the PDM module are mapped to physical pins according to the configuration specified in the PSEL.CLK and PSEL.DIN registers respectively. If the CONNECT field in any PSEL register is set to Disconnected, the associated PDM module signal will not be connected to the required physical pins, and will not operate properly.

The PSEL.CLK and PSEL.DIN registers and their configurations are only used as long as the PDM module is enabled, and retained only as long as the device is in System ON mode. See POWER — Power supply on page 81 for more information about power modes. When the peripheral is disabled, the pins will behave as regular GPIOs, and use the configuration in their respective OUT bit field and PIN_CNF[n] register.

To ensure correct behavior in the PDM module, the pins used by the PDM module must be configured in the GPIO peripheral as described in GPIO configuration before enabling peripheral on page 483 before enabling the PDM module. This is to ensure that the pins used by the PDM module are driven correctly if the PDM module itself is temporarily disabled or the device temporarily enters System OFF. This configuration must be retained in the GPIO for the selected I/Os as long as the PDM module is supposed to be connected to an external PDM circuit.

Only one peripheral can be assigned to drive a particular GPIO pin at a time. Failing to do so may result in unpredictable behavior.

PDM signal	PDM pin	Direction	Output value	Comment
CLK	As specified in PSEL.CLK	Output	0	
DIN	As specified in PSEL.DIN	Input	Not applicable	

Table 27: GPIO configuration before enabling peripheral

6.15.7 Registers

Instances

Instance	Base address	Description
PDM	0x4001D000	Pulse Density modulation (digital microphone) interface

Register overview

Register	Offset	Description
TASKS_START	0x000	Starts continuous PDM transfer
TASKS_STOP	0x004	Stops PDM transfer
EVENTS_STARTED	0x100	PDM transfer has started
EVENTS_STOPPED	0x104	PDM transfer has finished
EVENTS_END	0x108	The PDM has written the last sample specified by SAMPLE.MAXCNT (or the last sample after a STOP task has been received) to Data RAM
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
ENABLE	0x500	PDM module enable register
PDMCLKCTRL	0x504	PDM clock generator control
MODE	0x508	Defines the routing of the connected PDM microphones' signals
GAINL	0x518	Left output gain adjustment
GAINR	0x51C	Right output gain adjustment
RATIO	0x520	Selects the ratio between PDM_CLK and output sample rate. Change PDMCLKCTRL accordingly.
PSEL.CLK	0x540	Pin number configuration for PDM CLK signal
PSEL.DIN	0x544	Pin number configuration for PDM DIN signal
SAMPLE.PTR	0x560	RAM address pointer to write samples to with EasyDMA
SAMPLE.MAXCNT	0x564	Number of samples to allocate memory for in EasyDMA mode

6.15.7.1 TASKS_START

Address offset: 0x000

Starts continuous PDM transfer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_START			Starts continuous PDM transfer
			Trigger	1	Trigger task

6.15.7.2 TASKS_STOP

Address offset: 0x004

Stops PDM transfer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STOP			Stops PDM transfer
			Trigger	1	Trigger task

6.15.7.3 EVENTS_STARTED

Address offset: 0x100

PDM transfer has started

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_STARTED			PDM transfer has started
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.15.7.4 EVENTS_STOPPED

Address offset: 0x104

PDM transfer has finished

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_STOPPED			PDM transfer has finished
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.15.7.5 EVENTS_END

Address offset: 0x108

The PDM has written the last sample specified by SAMPLE.MAXCNT (or the last sample after a STOP task has been received) to Data RAM

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_END			The PDM has written the last sample specified by SAMPLE.MAXCNT (or the last sample after a STOP task has been received) to Data RAM
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.15.7.6 INTEN

Address offset: 0x300

Enable or disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	STARTED			Enable or disable interrupt for event STARTED
			Disabled	0	Disable
			Enabled	1	Enable
B	RW	STOPPED			Enable or disable interrupt for event STOPPED
			Disabled	0	Disable
			Enabled	1	Enable
C	RW	END			Enable or disable interrupt for event END
			Disabled	0	Disable
			Enabled	1	Enable

6.15.7.7 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	STARTED			Write '1' to enable interrupt for event STARTED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	STOPPED			Write '1' to enable interrupt for event STOPPED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	END			Write '1' to enable interrupt for event END
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.15.7.8 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	STARTED			Write '1' to disable interrupt for event STARTED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	STOPPED			Write '1' to disable interrupt for event STOPPED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
C	RW	END			Write '1' to disable interrupt for event END
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.15.7.9 ENABLE

Address offset: 0x500

PDM module enable register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ENABLE			Enable or disable PDM module
			Disabled	0	Disable
			Enabled	1	Enable

6.15.7.10 PDMCLKCTRL

Address offset: 0x504

PDM clock generator control

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x08400000				**0 0 0 0 1 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	FREQ			PDM_CLK frequency configuration
			1000K	0x08000000	PDM_CLK = 32 MHz / 32 = 1.000 MHz
			Default	0x08400000	PDM_CLK = 32 MHz / 31 = 1.032 MHz. Nominal clock for RATIO=Ratio64.
			1067K	0x08800000	PDM_CLK = 32 MHz / 30 = 1.067 MHz
			1231K	0x09800000	PDM_CLK = 32 MHz / 26 = 1.231 MHz
			1280K	0x0A000000	PDM_CLK = 32 MHz / 25 = 1.280 MHz. Nominal clock for RATIO=Ratio80.
			1333K	0x0A800000	PDM_CLK = 32 MHz / 24 = 1.333 MHz

6.15.7.11 MODE

Address offset: 0x508

Defines the routing of the connected PDM microphones' signals

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	OPERATION			Mono or stereo operation
			Stereo	0	Sample and store one pair (left + right) of 16-bit samples per RAM word R=[31:16]; L=[15:0]
			Mono	1	Sample and store two successive left samples (16 bits each) per RAM word L1=[31:16]; L0=[15:0]
B	RW	EDGE			Defines on which PDM_CLK edge left (or mono) is sampled
			LeftFalling	0	Left (or mono) is sampled on falling edge of PDM_CLK
			LeftRising	1	Left (or mono) is sampled on rising edge of PDM_CLK

6.15.7.12 GAINL

Address offset: 0x518

Left output gain adjustment

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A
Reset 0x00000028				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 1 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	GAINL			Left output gain adjustment, in 0.5 dB steps, around the default module gain (see electrical parameters) 0x00 -20 dB gain adjust 0x01 -19.5 dB gain adjust (...) 0x27 -0.5 dB gain adjust 0x28 0 dB gain adjust 0x29 +0.5 dB gain adjust (...) 0x4F +19.5 dB gain adjust 0x50 +20 dB gain adjust
			MinGain	0x00	-20 dB gain adjustment (minimum)
			DefaultGain	0x28	0 dB gain adjustment
			MaxGain	0x50	+20 dB gain adjustment (maximum)

6.15.7.13 GAINR

Address offset: 0x51C

Right output gain adjustment

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A
Reset 0x00000028				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 1 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	GAINR			Right output gain adjustment, in 0.5 dB steps, around the default module gain (see electrical parameters)
			MinGain	0x00	-20 dB gain adjustment (minimum)
			DefaultGain	0x28	0 dB gain adjustment
			MaxGain	0x50	+20 dB gain adjustment (maximum)

6.15.7.14 RATIO

Address offset: 0x520

Selects the ratio between PDM_CLK and output sample rate. Change PDMCLKCTRL accordingly.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RATIO			Selects the ratio between PDM_CLK and output sample rate
			Ratio64	0	Ratio of 64
			Ratio80	1	Ratio of 80

6.15.7.15 PSEL.CLK

Address offset: 0x540

Pin number configuration for PDM CLK signal

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.15.7.16 PSEL.DIN

Address offset: 0x544

Pin number configuration for PDM DIN signal

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.15.7.17 SAMPLE.PTR

Address offset: 0x560

RAM address pointer to write samples to with EasyDMA

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	SAMPLEPTR																																	
				Address to write PDM samples to over DMA																																	

Note: See the memory chapter for details about which memories are available for EasyDMA.

6.15.7.18 SAMPLE.MAXCNT

Address offset: 0x564

Number of samples to allocate memory for in EasyDMA mode

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																					A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	BUFFSIZE		[0..32767]																															
				Length of DMA RAM allocation in number of samples																																	

6.15.8 Electrical specification

6.15.8.1 PDM Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
$f_{PDM,CLK,64}$	PDM clock speed. PDMCLKCTRL = Default (Setting needed for 16 MHz sample frequency @ RATIO = Ratio64)		1.032		MHz
$f_{PDM,CLK,80}$	PDM clock speed. PDMCLKCTRL = 1280K (Setting needed for 16 MHz sample frequency @ RATIO = Ratio80)		1.280		MHz
$t_{PDM,JITTER}$	Jitter in PDM clock output			20	ns
$T_{dPDM,CLK}$	PDM clock duty cycle	40	50	60	%
$t_{PDM,DATA}$	Decimation filter delay			5	ms
$t_{PDM,cv}$	Allowed clock edge to data valid			125	ns
$t_{PDM,ci}$	Allowed (other) clock edge to data invalid	0			ns
$t_{PDM,s}$	Data setup time at $f_{PDM,CLK}$=1.024 MHz or 1.280 MHz	65			ns
$t_{PDM,h}$	Data hold time at $f_{PDM,CLK}$=1.024 MHz or 1.280 MHz	0			ns
$G_{PDM,default}$	Default (reset) absolute gain of the PDM module		3.2		dB

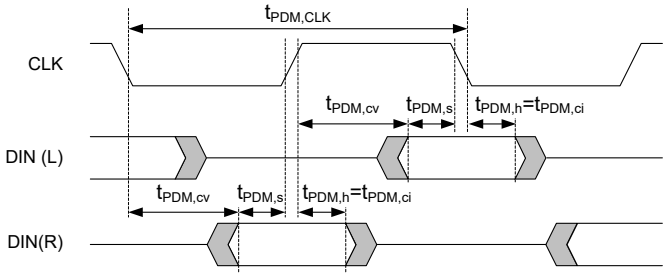


Figure 78: PDM timing diagram

6.16 PPI — Programmable peripheral interconnect

The programmable peripheral interconnect (PPI) enables peripherals to interact autonomously with each other using tasks and events independent of the CPU. The PPI allows precise synchronization between peripherals when real-time application constraints exist and eliminates the need for CPU activity to implement behavior which can be predefined using PPI.

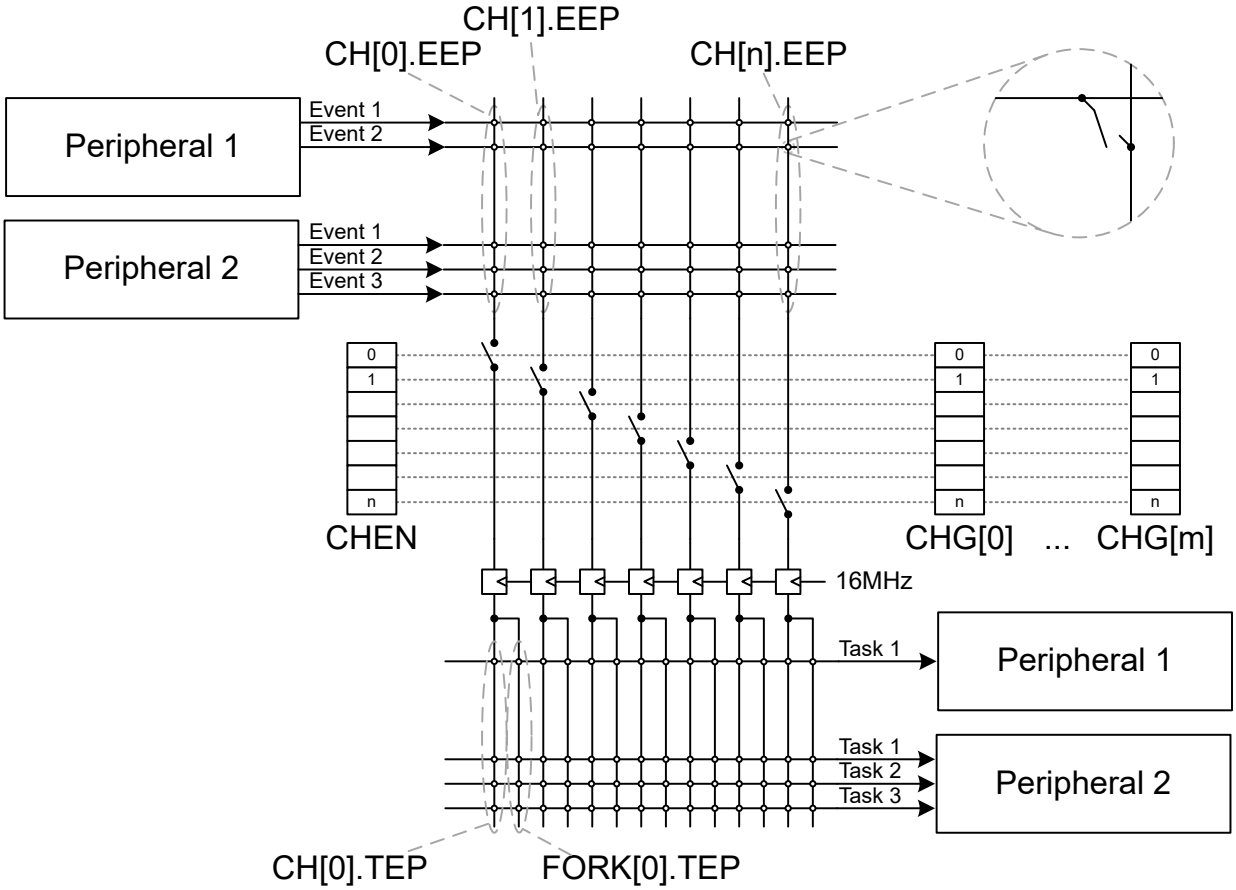


Figure 79: PPI block diagram

The PPI system has, in addition to the fully programmable peripheral interconnections, a set of channels where the event end point (EEP) and task end points (TEP) are fixed in hardware. These fixed channels can be individually enabled, disabled, or added to PPI channel groups (see CHG[n] registers), in the same way as ordinary PPI channels.

Instance	Channel	Number of channels
PPI	0-19	20
PPI (fixed)	20-31	12

Table 28: Configurable and fixed PPI channels

The PPI provides a mechanism to automatically trigger a task in one peripheral as a result of an event occurring in another peripheral. A task is connected to an event through a PPI channel. The PPI channel is composed of three end point registers, one EEP, and two TEPs. A peripheral task is connected to a TEP using the address of the task register associated with the task. Similarly, a peripheral event is connected to an EEP using the address of the event register associated with the event.

On each PPI channel, the signals are synchronized to the 16 MHz clock to avoid any internal violation of setup and hold timings. As a consequence, events that are synchronous to the 16 MHz clock will be delayed by one clock period, while other asynchronous events will be delayed by up to one 16 MHz clock period.

Note: Shortcuts (as defined in the SHORTS register in each peripheral) are not affected by this 16 MHz synchronization, and are therefore not delayed.

Each TEP implements a fork mechanism that enables a second task to be triggered at the same time as the task specified in the TEP is triggered. This second task is configured in the task end point register in the FORK registers groups, e.g. FORK.TEP[0] is associated with PPI channel CH[0].

There are two ways of enabling and disabling PPI channels:

- Enable or disable PPI channels individually using the CHEN, CHENSET, and CHENCLR registers.
- Enable or disable PPI channels in PPI channel groups through the groups' ENABLE and DISABLE tasks. Prior to these tasks being triggered, the PPI channel group must be configured to define which PPI channels belong to which groups.

Note: When a channel belongs to two groups m and n, and the tasks CHG[m].EN and CHG[n].DIS occur simultaneously (m and n can be equal or different), the CHG[m].EN on that channel has priority.

PPI tasks (for example, CHG[0].EN) can be triggered through the PPI like any other task, which means they can be hooked to a PPI channel as a TEP. One event can trigger multiple tasks by using multiple channels and one task can be triggered by multiple events in the same way.

6.16.1 Pre-programmed channels

Some of the PPI channels are pre-programmed. These channels cannot be configured by the CPU, but can be added to groups and enabled and disabled like the general purpose PPI channels. The FORK TEP for these channels are still programmable and can be used by the application.

For a list of pre-programmed PPI channels, see the following table.

Channel	EEP	TEP
20	TIMER0->EVENTS_COMPARE[0]	RADIO->TASKS_TXEN
21	TIMER0->EVENTS_COMPARE[0]	RADIO->TASKS_RXEN
22	TIMER0->EVENTS_COMPARE[1]	RADIO->TASKS_DISABLE
23	RADIO->EVENTS_BCMATCH	AAR->TASKS_START
24	RADIO->EVENTS_READY	CCM->TASKS_KSGEN
25	RADIO->EVENTS_ADDRESS	CCM->TASKS_CRYPT
26	RADIO->EVENTS_ADDRESS	TIMER0->TASKS_CAPTURE[1]
27	RADIO->EVENTS_END	TIMER0->TASKS_CAPTURE[2]
28	RTC0->EVENTS_COMPARE[0]	RADIO->TASKS_TXEN
29	RTC0->EVENTS_COMPARE[0]	RADIO->TASKS_RXEN
30	RTC0->EVENTS_COMPARE[0]	TIMER0->TASKS_CLEAR
31	RTC0->EVENTS_COMPARE[0]	TIMER0->TASKS_START

Table 29: Pre-programmed channels

6.16.2 Registers

Instances

Instance	Base address	Description
PPI	0x4001F000	Programmable peripheral interconnect

Configuration

Instance	Configuration
PPI	This PPI instance has 20 configurable channels (CH[0..19]) and 12 pre-programmed channels (CH[20..31])

Register overview

Register	Offset	Description
TASKS_CHG[0].EN	0x000	Enable channel group 0
TASKS_CHG[0].DIS	0x004	Disable channel group 0
TASKS_CHG[1].EN	0x008	Enable channel group 1
TASKS_CHG[1].DIS	0x00C	Disable channel group 1
TASKS_CHG[2].EN	0x010	Enable channel group 2
TASKS_CHG[2].DIS	0x014	Disable channel group 2
TASKS_CHG[3].EN	0x018	Enable channel group 3
TASKS_CHG[3].DIS	0x01C	Disable channel group 3
TASKS_CHG[4].EN	0x020	Enable channel group 4
TASKS_CHG[4].DIS	0x024	Disable channel group 4
TASKS_CHG[5].EN	0x028	Enable channel group 5
TASKS_CHG[5].DIS	0x02C	Disable channel group 5
CHEN	0x500	Channel enable register
CHENSET	0x504	Channel enable set register
CHENCLR	0x508	Channel enable clear register
CH[0].EEP	0x510	Channel 0 event endpoint
CH[0].TEP	0x514	Channel 0 task endpoint
CH[1].EEP	0x518	Channel 1 event endpoint
CH[1].TEP	0x51C	Channel 1 task endpoint
CH[2].EEP	0x520	Channel 2 event endpoint

Register	Offset	Description
CH[2].TEP	0x524	Channel 2 task endpoint
CH[3].EEP	0x528	Channel 3 event endpoint
CH[3].TEP	0x52C	Channel 3 task endpoint
CH[4].EEP	0x530	Channel 4 event endpoint
CH[4].TEP	0x534	Channel 4 task endpoint
CH[5].EEP	0x538	Channel 5 event endpoint
CH[5].TEP	0x53C	Channel 5 task endpoint
CH[6].EEP	0x540	Channel 6 event endpoint
CH[6].TEP	0x544	Channel 6 task endpoint
CH[7].EEP	0x548	Channel 7 event endpoint
CH[7].TEP	0x54C	Channel 7 task endpoint
CH[8].EEP	0x550	Channel 8 event endpoint
CH[8].TEP	0x554	Channel 8 task endpoint
CH[9].EEP	0x558	Channel 9 event endpoint
CH[9].TEP	0x55C	Channel 9 task endpoint
CH[10].EEP	0x560	Channel 10 event endpoint
CH[10].TEP	0x564	Channel 10 task endpoint
CH[11].EEP	0x568	Channel 11 event endpoint
CH[11].TEP	0x56C	Channel 11 task endpoint
CH[12].EEP	0x570	Channel 12 event endpoint
CH[12].TEP	0x574	Channel 12 task endpoint
CH[13].EEP	0x578	Channel 13 event endpoint
CH[13].TEP	0x57C	Channel 13 task endpoint
CH[14].EEP	0x580	Channel 14 event endpoint
CH[14].TEP	0x584	Channel 14 task endpoint
CH[15].EEP	0x588	Channel 15 event endpoint
CH[15].TEP	0x58C	Channel 15 task endpoint
CH[16].EEP	0x590	Channel 16 event endpoint
CH[16].TEP	0x594	Channel 16 task endpoint
CH[17].EEP	0x598	Channel 17 event endpoint
CH[17].TEP	0x59C	Channel 17 task endpoint
CH[18].EEP	0x5A0	Channel 18 event endpoint
CH[18].TEP	0x5A4	Channel 18 task endpoint
CH[19].EEP	0x5A8	Channel 19 event endpoint
CH[19].TEP	0x5AC	Channel 19 task endpoint
CHG[0]	0x800	Channel group 0
CHG[1]	0x804	Channel group 1
CHG[2]	0x808	Channel group 2
CHG[3]	0x80C	Channel group 3
CHG[4]	0x810	Channel group 4
CHG[5]	0x814	Channel group 5
FORK[0].TEP	0x910	Channel 0 task endpoint
FORK[1].TEP	0x914	Channel 1 task endpoint
FORK[2].TEP	0x918	Channel 2 task endpoint
FORK[3].TEP	0x91C	Channel 3 task endpoint
FORK[4].TEP	0x920	Channel 4 task endpoint
FORK[5].TEP	0x924	Channel 5 task endpoint
FORK[6].TEP	0x928	Channel 6 task endpoint
FORK[7].TEP	0x92C	Channel 7 task endpoint
FORK[8].TEP	0x930	Channel 8 task endpoint
FORK[9].TEP	0x934	Channel 9 task endpoint
FORK[10].TEP	0x938	Channel 10 task endpoint
FORK[11].TEP	0x93C	Channel 11 task endpoint

Register	Offset	Description
FORK[12].TEP	0x940	Channel 12 task endpoint
FORK[13].TEP	0x944	Channel 13 task endpoint
FORK[14].TEP	0x948	Channel 14 task endpoint
FORK[15].TEP	0x94C	Channel 15 task endpoint
FORK[16].TEP	0x950	Channel 16 task endpoint
FORK[17].TEP	0x954	Channel 17 task endpoint
FORK[18].TEP	0x958	Channel 18 task endpoint
FORK[19].TEP	0x95C	Channel 19 task endpoint
FORK[20].TEP	0x960	Channel 20 task endpoint
FORK[21].TEP	0x964	Channel 21 task endpoint
FORK[22].TEP	0x968	Channel 22 task endpoint
FORK[23].TEP	0x96C	Channel 23 task endpoint
FORK[24].TEP	0x970	Channel 24 task endpoint
FORK[25].TEP	0x974	Channel 25 task endpoint
FORK[26].TEP	0x978	Channel 26 task endpoint
FORK[27].TEP	0x97C	Channel 27 task endpoint
FORK[28].TEP	0x980	Channel 28 task endpoint
FORK[29].TEP	0x984	Channel 29 task endpoint
FORK[30].TEP	0x988	Channel 30 task endpoint
FORK[31].TEP	0x98C	Channel 31 task endpoint

6.16.2.1 TASKS_CHG[0]

Channel group tasks

6.16.2.1.1 TASKS_CHG[0].EN

Address offset: 0x000

Enable channel group 0

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	EN			Enable channel group 0
			Trigger	1	Trigger task

6.16.2.1.2 TASKS_CHG[0].DIS

Address offset: 0x004

Disable channel group 0

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	DIS			Disable channel group 0
			Trigger	1	Trigger task

6.16.2.2 TASKS_CHG[1]

Channel group tasks

6.16.2.2.1 TASKS_CHG[1].EN

Address offset: 0x008

Enable channel group 1

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field		Value ID				Value																		Description									
A	W	EN																								Enable channel group 1									
				Trigger				1																		Trigger task									

6.16.2.2.2 TASKS_CHG[1].DIS

Address offset: 0x00C

Disable channel group 1

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field		Value ID				Value																		Description									
A	W	DIS																								Disable channel group 1									
				Trigger				1																		Trigger task									

6.16.2.3 TASKS_CHG[2]

Channel group tasks

6.16.2.3.1 TASKS_CHG[2].EN

Address offset: 0x010

Enable channel group 2

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field		Value ID				Value																		Description									
A	W	EN																								Enable channel group 2									
				Trigger				1																		Trigger task									

6.16.2.3.2 TASKS_CHG[2].DIS

Address offset: 0x014

Disable channel group 2

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field		Value ID				Value																		Description									
A	W	DIS																								Disable channel group 2									
				Trigger				1																		Trigger task									

6.16.2.4 TASKS_CHG[3]

Channel group tasks

6.16.2.4.1 TASKS_CHG[3].EN

Address offset: 0x018

Enable channel group 3

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	EN			Enable channel group 3
			Trigger	1	Trigger task

6.16.2.4.2 TASKS_CHG[3].DIS

Address offset: 0x01C

Disable channel group 3

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	DIS			Disable channel group 3
			Trigger	1	Trigger task

6.16.2.5 TASKS_CHG[4]

Channel group tasks

6.16.2.5.1 TASKS_CHG[4].EN

Address offset: 0x020

Enable channel group 4

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	EN			Enable channel group 4
			Trigger	1	Trigger task

6.16.2.5.2 TASKS_CHG[4].DIS

Address offset: 0x024

Disable channel group 4

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	DIS			Disable channel group 4
			Trigger	1	Trigger task

6.16.2.6 TASKS_CHG[5]

Channel group tasks

6.16.2.6.1 TASKS_CHG[5].EN

Address offset: 0x028

Enable channel group 5

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	EN			Enable channel group 5
			Trigger	1	Trigger task

6.16.2.6.2 TASKS_CHG[5].DIS

Address offset: 0x02C

Disable channel group 5

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	DIS			Disable channel group 5
			Trigger	1	Trigger task

6.16.2.7 CHEN

Address offset: 0x500

Channel enable register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	CH0			Enable or disable channel 0
			Disabled	0	Disable channel
			Enabled	1	Enable channel
B	RW	CH1			Enable or disable channel 1
			Disabled	0	Disable channel
			Enabled	1	Enable channel
C	RW	CH2			Enable or disable channel 2
			Disabled	0	Disable channel

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
			Enabled	1	Enable channel
D	RW	CH3			Enable or disable channel 3
			Disabled	0	Disable channel
			Enabled	1	Enable channel
E	RW	CH4			Enable or disable channel 4
			Disabled	0	Disable channel
			Enabled	1	Enable channel
F	RW	CH5			Enable or disable channel 5
			Disabled	0	Disable channel
			Enabled	1	Enable channel
G	RW	CH6			Enable or disable channel 6
			Disabled	0	Disable channel
			Enabled	1	Enable channel
H	RW	CH7			Enable or disable channel 7
			Disabled	0	Disable channel
			Enabled	1	Enable channel
I	RW	CH8			Enable or disable channel 8
			Disabled	0	Disable channel
			Enabled	1	Enable channel
J	RW	CH9			Enable or disable channel 9
			Disabled	0	Disable channel
			Enabled	1	Enable channel
K	RW	CH10			Enable or disable channel 10
			Disabled	0	Disable channel
			Enabled	1	Enable channel
L	RW	CH11			Enable or disable channel 11
			Disabled	0	Disable channel
			Enabled	1	Enable channel
M	RW	CH12			Enable or disable channel 12
			Disabled	0	Disable channel
			Enabled	1	Enable channel
N	RW	CH13			Enable or disable channel 13
			Disabled	0	Disable channel
			Enabled	1	Enable channel
O	RW	CH14			Enable or disable channel 14
			Disabled	0	Disable channel
			Enabled	1	Enable channel
P	RW	CH15			Enable or disable channel 15
			Disabled	0	Disable channel
			Enabled	1	Enable channel
Q	RW	CH16			Enable or disable channel 16
			Disabled	0	Disable channel
			Enabled	1	Enable channel
R	RW	CH17			Enable or disable channel 17
			Disabled	0	Disable channel
			Enabled	1	Enable channel
S	RW	CH18			Enable or disable channel 18
			Disabled	0	Disable channel
			Enabled	1	Enable channel
T	RW	CH19			Enable or disable channel 19

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
			Disabled	0																															
			Enabled	1																															
U	RW	CH20																																	
			Disabled	0																															
			Enabled	1																															
V	RW	CH21																																	
			Disabled	0																															
			Enabled	1																															
W	RW	CH22																																	
			Disabled	0																															
			Enabled	1																															
X	RW	CH23																																	
			Disabled	0																															
			Enabled	1																															
Y	RW	CH24																																	
			Disabled	0																															
			Enabled	1																															
Z	RW	CH25																																	
			Disabled	0																															
			Enabled	1																															
a	RW	CH26																																	
			Disabled	0																															
			Enabled	1																															
b	RW	CH27																																	
			Disabled	0																															
			Enabled	1																															
c	RW	CH28																																	
			Disabled	0																															
			Enabled	1																															
d	RW	CH29																																	
			Disabled	0																															
			Enabled	1																															
e	RW	CH30																																	
			Disabled	0																															
			Enabled	1																															
f	RW	CH31																																	
			Disabled	0																															
			Enabled	1																															

6.16.2.8 CHENSET

Address offset: 0x504

Channel enable set register

Note: Read: reads value of CHi field in CHEN register.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW W1S	CH0			Channel 0 enable set register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Set	1	Write: Enable channel
B	RW W1S	CH1			Channel 1 enable set register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Set	1	Write: Enable channel
C	RW W1S	CH2			Channel 2 enable set register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Set	1	Write: Enable channel
D	RW W1S	CH3			Channel 3 enable set register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Set	1	Write: Enable channel
E	RW W1S	CH4			Channel 4 enable set register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Set	1	Write: Enable channel
F	RW W1S	CH5			Channel 5 enable set register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Set	1	Write: Enable channel
G	RW W1S	CH6			Channel 6 enable set register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Set	1	Write: Enable channel
H	RW W1S	CH7			Channel 7 enable set register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Set	1	Write: Enable channel
I	RW W1S	CH8			Channel 8 enable set register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Set	1	Write: Enable channel
J	RW W1S	CH9			Channel 9 enable set register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Set	1	Write: Enable channel

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
K	RW W1S	CH10																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
L	RW W1S	CH11																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
M	RW W1S	CH12																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
N	RW W1S	CH13																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
O	RW W1S	CH14																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
P	RW W1S	CH15																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
Q	RW W1S	CH16																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
R	RW W1S	CH17																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
S	RW W1S	CH18																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
T	RW W1S	CH19																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
U	RW W1S	CH20																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
V	RW W1S	CH21																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
W	RW W1S	CH22																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
X	RW W1S	CH23																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
Y	RW W1S	CH24																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
Z	RW W1S	CH25																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
a	RW W1S	CH26																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
b	RW W1S	CH27																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
c	RW W1S	CH28																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															
d	RW W1S	CH29																																	
			Disabled	0																															
			Enabled	1																															
			Set	1																															

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
e	RW W1S	CH30			Channel 30 enable set register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Set	1	Write: Enable channel
f	RW W1S	CH31			Channel 31 enable set register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Set	1	Write: Enable channel

6.16.2.9 CHENCLR

Address offset: 0x508

Channel enable clear register

Note: Read: reads value of CHi field in CHEN register.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW W1C	CH0			Channel 0 enable clear register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Clear	1	Write: disable channel
B	RW W1C	CH1			Channel 1 enable clear register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Clear	1	Write: disable channel
C	RW W1C	CH2			Channel 2 enable clear register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Clear	1	Write: disable channel
D	RW W1C	CH3			Channel 3 enable clear register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Clear	1	Write: disable channel
E	RW W1C	CH4			Channel 4 enable clear register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Clear	1	Write: disable channel
F	RW W1C	CH5			Channel 5 enable clear register. Writing '0' has no effect.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
G	RW W1C	CH6																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
H	RW W1C	CH7																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
I	RW W1C	CH8																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
J	RW W1C	CH9																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
K	RW W1C	CH10																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
L	RW W1C	CH11																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
M	RW W1C	CH12																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
N	RW W1C	CH13																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
O	RW W1C	CH14																																	
			Disabled	0																															
			Enabled	1																															
			Clear	1																															
P	RW W1C	CH15																																	

Description
Read: channel disabled
Read: channel enabled
Write: disable channel
Channel 6 enable clear register. Writing '0' has no effect.
Read: channel disabled
Read: channel enabled
Write: disable channel
Channel 7 enable clear register. Writing '0' has no effect.
Read: channel disabled
Read: channel enabled
Write: disable channel
Channel 8 enable clear register. Writing '0' has no effect.
Read: channel disabled
Read: channel enabled
Write: disable channel
Channel 9 enable clear register. Writing '0' has no effect.
Read: channel disabled
Read: channel enabled
Write: disable channel
Channel 10 enable clear register. Writing '0' has no effect.
Read: channel disabled
Read: channel enabled
Write: disable channel
Channel 11 enable clear register. Writing '0' has no effect.
Read: channel disabled
Read: channel enabled
Write: disable channel
Channel 12 enable clear register. Writing '0' has no effect.
Read: channel disabled
Read: channel enabled
Write: disable channel
Channel 13 enable clear register. Writing '0' has no effect.
Read: channel disabled
Read: channel enabled
Write: disable channel
Channel 14 enable clear register. Writing '0' has no effect.
Read: channel disabled
Read: channel enabled
Write: disable channel
Channel 15 enable clear register. Writing '0' has no effect.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Clear	1	Write: disable channel
Q	RW W1C	CH16			Channel 16 enable clear register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Clear	1	Write: disable channel
R	RW W1C	CH17			Channel 17 enable clear register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Clear	1	Write: disable channel
S	RW W1C	CH18			Channel 18 enable clear register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Clear	1	Write: disable channel
T	RW W1C	CH19			Channel 19 enable clear register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Clear	1	Write: disable channel
U	RW W1C	CH20			Channel 20 enable clear register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Clear	1	Write: disable channel
V	RW W1C	CH21			Channel 21 enable clear register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Clear	1	Write: disable channel
W	RW W1C	CH22			Channel 22 enable clear register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Clear	1	Write: disable channel
X	RW W1C	CH23			Channel 23 enable clear register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Clear	1	Write: disable channel
Y	RW W1C	CH24			Channel 24 enable clear register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Clear	1	Write: disable channel
Z	RW W1C	CH25			Channel 25 enable clear register. Writing '0' has no effect.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Clear	1	Write: disable channel
a	RW W1C	CH26			Channel 26 enable clear register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Clear	1	Write: disable channel
b	RW W1C	CH27			Channel 27 enable clear register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Clear	1	Write: disable channel
c	RW W1C	CH28			Channel 28 enable clear register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Clear	1	Write: disable channel
d	RW W1C	CH29			Channel 29 enable clear register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Clear	1	Write: disable channel
e	RW W1C	CH30			Channel 30 enable clear register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Clear	1	Write: disable channel
f	RW W1C	CH31			Channel 31 enable clear register. Writing '0' has no effect.
			Disabled	0	Read: channel disabled
			Enabled	1	Read: channel enabled
			Clear	1	Write: disable channel

6.16.2.10 CH[0]

PPI Channel

6.16.2.10.1 CH[0].EEP

Address offset: 0x510

Channel 0 event endpoint

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

6.16.2.10.2 CH[0].TEP

Address offset: 0x514

Channel 0 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

6.16.2.11 CH[1]

PPI Channel

6.16.2.11.1 CH[1].EEP

Address offset: 0x518

Channel 1 event endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

6.16.2.11.2 CH[1].TEP

Address offset: 0x51C

Channel 1 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

6.16.2.12 CH[2]

PPI Channel

6.16.2.12.1 CH[2].EEP

Address offset: 0x520

Channel 2 event endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

6.16.2.12.2 CH[2].TEP

Address offset: 0x524

Channel 2 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

6.16.2.13 CH[3]

PPI Channel

6.16.2.13.1 CH[3].EEP

Address offset: 0x528

Channel 3 event endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

6.16.2.13.2 CH[3].TEP

Address offset: 0x52C

Channel 3 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

6.16.2.14 CH[4]

PPI Channel

6.16.2.14.1 CH[4].EEP

Address offset: 0x530

Channel 4 event endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

6.16.2.14.2 CH[4].TEP

Address offset: 0x534

Channel 4 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

6.16.2.15 CH[5]

PPI Channel

6.16.2.15.1 CH[5].EEP

Address offset: 0x538

Channel 5 event endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

6.16.2.15.2 CH[5].TEP

Address offset: 0x53C

Channel 5 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

6.16.2.16 CH[6]

PPI Channel

6.16.2.16.1 CH[6].EEP

Address offset: 0x540

Channel 6 event endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

6.16.2.16.2 CH[6].TEP

Address offset: 0x544

Channel 6 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

6.16.2.17 CH[7]

PPI Channel

6.16.2.17.1 CH[7].EEP

Address offset: 0x548

Channel 7 event endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

6.16.2.17.2 CH[7].TEP

Address offset: 0x54C

Channel 7 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

6.16.2.18 CH[8]

PPI Channel

6.16.2.18.1 CH[8].EEP

Address offset: 0x550

Channel 8 event endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

6.16.2.18.2 CH[8].TEP

Address offset: 0x554

Channel 8 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

6.16.2.19 CH[9]

PPI Channel

6.16.2.19.1 CH[9].EEP

Address offset: 0x558

Channel 9 event endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

6.16.2.19.2 CH[9].TEP

Address offset: 0x55C

Channel 9 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

6.16.2.20 CH[10]

PPI Channel

6.16.2.20.1 CH[10].EEP

Address offset: 0x560

Channel 10 event endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

6.16.2.20.2 CH[10].TEP

Address offset: 0x564

Channel 10 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

6.16.2.21 CH[11]

PPI Channel

6.16.2.21.1 CH[11].EEP

Address offset: 0x568

Channel 11 event endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

6.16.2.21.2 CH[11].TEP

Address offset: 0x56C

Channel 11 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

6.16.2.22 CH[12]

PPI Channel

6.16.2.22.1 CH[12].EEP

Address offset: 0x570

Channel 12 event endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

6.16.2.22.2 CH[12].TEP

Address offset: 0x574

Channel 12 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

6.16.2.23 CH[13]

PPI Channel

6.16.2.23.1 CH[13].EEP

Address offset: 0x578

Channel 13 event endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

6.16.2.23.2 CH[13].TEP

Address offset: 0x57C

Channel 13 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

6.16.2.24 CH[14]

PPI Channel

6.16.2.24.1 CH[14].EEP

Address offset: 0x580

Channel 14 event endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

6.16.2.24.2 CH[14].TEP

Address offset: 0x584

Channel 14 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

6.16.2.25 CH[15]

PPI Channel

6.16.2.25.1 CH[15].EEP

Address offset: 0x588

Channel 15 event endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

6.16.2.25.2 CH[15].TEP

Address offset: 0x58C

Channel 15 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

6.16.2.26 CH[16]

PPI Channel

6.16.2.26.1 CH[16].EEP

Address offset: 0x590

Channel 16 event endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

6.16.2.26.2 CH[16].TEP

Address offset: 0x594

Channel 16 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

6.16.2.27 CH[17]

PPI Channel

6.16.2.27.1 CH[17].EEP

Address offset: 0x598

Channel 17 event endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

6.16.2.27.2 CH[17].TEP

Address offset: 0x59C

Channel 17 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

6.16.2.28 CH[18]

PPI Channel

6.16.2.28.1 CH[18].EEP

Address offset: 0x5A0

Channel 18 event endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EEP			Pointer to event register. Accepts only addresses to registers from the Event group.

6.16.2.28.2 CH[18].TEP

Address offset: 0x5A4

Channel 18 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register. Accepts only addresses to registers from the Task group.

6.16.2.29 CH[19]

PPI Channel

6.16.2.29.1 CH[19].EEP

Address offset: 0x5A8

Channel 19 event endpoint

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	EEP																																	

Pointer to event register. Accepts only addresses to registers from the Event group.

6.16.2.29.2 CH[19].TEP

Address offset: 0x5AC

Channel 19 task endpoint

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	TEP																																	

Pointer to task register. Accepts only addresses to registers from the Task group.

6.16.2.30 CHG[0]

Address offset: 0x800

Channel group 0

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	CH0																																	
			Excluded	0																															
			Included	1																															
B	RW	CH1																																	
			Excluded	0																															
			Included	1																															
C	RW	CH2																																	
			Excluded	0																															
			Included	1																															
D	RW	CH3																																	
			Excluded	0																															
			Included	1																															
E	RW	CH4																																	
			Excluded	0																															
			Included	1																															
F	RW	CH5																																	
			Excluded	0																															
			Included	1																															
G	RW	CH6																																	

Include or exclude channel 6

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
			Excluded	0	Exclude
			Included	1	Include
H	RW	CH7			Include or exclude channel 7
			Excluded	0	Exclude
			Included	1	Include
I	RW	CH8			Include or exclude channel 8
			Excluded	0	Exclude
			Included	1	Include
J	RW	CH9			Include or exclude channel 9
			Excluded	0	Exclude
			Included	1	Include
K	RW	CH10			Include or exclude channel 10
			Excluded	0	Exclude
			Included	1	Include
L	RW	CH11			Include or exclude channel 11
			Excluded	0	Exclude
			Included	1	Include
M	RW	CH12			Include or exclude channel 12
			Excluded	0	Exclude
			Included	1	Include
N	RW	CH13			Include or exclude channel 13
			Excluded	0	Exclude
			Included	1	Include
O	RW	CH14			Include or exclude channel 14
			Excluded	0	Exclude
			Included	1	Include
P	RW	CH15			Include or exclude channel 15
			Excluded	0	Exclude
			Included	1	Include
Q	RW	CH16			Include or exclude channel 16
			Excluded	0	Exclude
			Included	1	Include
R	RW	CH17			Include or exclude channel 17
			Excluded	0	Exclude
			Included	1	Include
S	RW	CH18			Include or exclude channel 18
			Excluded	0	Exclude
			Included	1	Include
T	RW	CH19			Include or exclude channel 19
			Excluded	0	Exclude
			Included	1	Include
U	RW	CH20			Include or exclude channel 20
			Excluded	0	Exclude
			Included	1	Include
V	RW	CH21			Include or exclude channel 21
			Excluded	0	Exclude
			Included	1	Include
W	RW	CH22			Include or exclude channel 22
			Excluded	0	Exclude
			Included	1	Include

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
X	RW	CH23																																	
			Excluded	0																															
			Included	1																															
Y	RW	CH24																																	
			Excluded	0																															
			Included	1																															
Z	RW	CH25																																	
			Excluded	0																															
			Included	1																															
a	RW	CH26																																	
			Excluded	0																															
			Included	1																															
b	RW	CH27																																	
			Excluded	0																															
			Included	1																															
c	RW	CH28																																	
			Excluded	0																															
			Included	1																															
d	RW	CH29																																	
			Excluded	0																															
			Included	1																															
e	RW	CH30																																	
			Excluded	0																															
			Included	1																															
f	RW	CH31																																	
			Excluded	0																															
			Included	1																															

6.16.2.31 CHG[1]

Address offset: 0x804

Channel group 1

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	CH0																																	
			Excluded	0																															
			Included	1																															
B	RW	CH1																																	
			Excluded	0																															
			Included	1																															
C	RW	CH2																																	
			Excluded	0																															
			Included	1																															
D	RW	CH3																																	
			Excluded	0																															
			Included	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
E	RW	CH4			Include or exclude channel 4
			Excluded	0	Exclude
			Included	1	Include
F	RW	CH5			Include or exclude channel 5
			Excluded	0	Exclude
			Included	1	Include
G	RW	CH6			Include or exclude channel 6
			Excluded	0	Exclude
			Included	1	Include
H	RW	CH7			Include or exclude channel 7
			Excluded	0	Exclude
			Included	1	Include
I	RW	CH8			Include or exclude channel 8
			Excluded	0	Exclude
			Included	1	Include
J	RW	CH9			Include or exclude channel 9
			Excluded	0	Exclude
			Included	1	Include
K	RW	CH10			Include or exclude channel 10
			Excluded	0	Exclude
			Included	1	Include
L	RW	CH11			Include or exclude channel 11
			Excluded	0	Exclude
			Included	1	Include
M	RW	CH12			Include or exclude channel 12
			Excluded	0	Exclude
			Included	1	Include
N	RW	CH13			Include or exclude channel 13
			Excluded	0	Exclude
			Included	1	Include
O	RW	CH14			Include or exclude channel 14
			Excluded	0	Exclude
			Included	1	Include
P	RW	CH15			Include or exclude channel 15
			Excluded	0	Exclude
			Included	1	Include
Q	RW	CH16			Include or exclude channel 16
			Excluded	0	Exclude
			Included	1	Include
R	RW	CH17			Include or exclude channel 17
			Excluded	0	Exclude
			Included	1	Include
S	RW	CH18			Include or exclude channel 18
			Excluded	0	Exclude
			Included	1	Include
T	RW	CH19			Include or exclude channel 19
			Excluded	0	Exclude
			Included	1	Include
U	RW	CH20			Include or exclude channel 20
			Excluded	0	Exclude

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Included	1	Include
V	RW	CH21			Include or exclude channel 21
			Excluded	0	Exclude
			Included	1	Include
W	RW	CH22			Include or exclude channel 22
			Excluded	0	Exclude
			Included	1	Include
X	RW	CH23			Include or exclude channel 23
			Excluded	0	Exclude
			Included	1	Include
Y	RW	CH24			Include or exclude channel 24
			Excluded	0	Exclude
			Included	1	Include
Z	RW	CH25			Include or exclude channel 25
			Excluded	0	Exclude
			Included	1	Include
a	RW	CH26			Include or exclude channel 26
			Excluded	0	Exclude
			Included	1	Include
b	RW	CH27			Include or exclude channel 27
			Excluded	0	Exclude
			Included	1	Include
c	RW	CH28			Include or exclude channel 28
			Excluded	0	Exclude
			Included	1	Include
d	RW	CH29			Include or exclude channel 29
			Excluded	0	Exclude
			Included	1	Include
e	RW	CH30			Include or exclude channel 30
			Excluded	0	Exclude
			Included	1	Include
f	RW	CH31			Include or exclude channel 31
			Excluded	0	Exclude
			Included	1	Include

6.16.2.32 CHG[2]

Address offset: 0x808

Channel group 2

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	CH0			Include or exclude channel 0
			Excluded	0	Exclude
			Included	1	Include
B	RW	CH1			Include or exclude channel 1
			Excluded	0	Exclude

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
			Included	1																															
C	RW	CH2																																	
			Excluded	0																															
			Included	1																															
D	RW	CH3																																	
			Excluded	0																															
			Included	1																															
E	RW	CH4																																	
			Excluded	0																															
			Included	1																															
F	RW	CH5																																	
			Excluded	0																															
			Included	1																															
G	RW	CH6																																	
			Excluded	0																															
			Included	1																															
H	RW	CH7																																	
			Excluded	0																															
			Included	1																															
I	RW	CH8																																	
			Excluded	0																															
			Included	1																															
J	RW	CH9																																	
			Excluded	0																															
			Included	1																															
K	RW	CH10																																	
			Excluded	0																															
			Included	1																															
L	RW	CH11																																	
			Excluded	0																															
			Included	1																															
M	RW	CH12																																	
			Excluded	0																															
			Included	1																															
N	RW	CH13																																	
			Excluded	0																															
			Included	1																															
O	RW	CH14																																	
			Excluded	0																															
			Included	1																															
P	RW	CH15																																	
			Excluded	0																															
			Included	1																															
Q	RW	CH16																																	
			Excluded	0																															
			Included	1																															
R	RW	CH17																																	
			Excluded	0																															
			Included	1																															
S	RW	CH18																																	

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
			Excluded	0																															
			Included	1																															
T	RW	CH19																																	
			Excluded	0																															
			Included	1																															
U	RW	CH20																																	
			Excluded	0																															
			Included	1																															
V	RW	CH21																																	
			Excluded	0																															
			Included	1																															
W	RW	CH22																																	
			Excluded	0																															
			Included	1																															
X	RW	CH23																																	
			Excluded	0																															
			Included	1																															
Y	RW	CH24																																	
			Excluded	0																															
			Included	1																															
Z	RW	CH25																																	
			Excluded	0																															
			Included	1																															
a	RW	CH26																																	
			Excluded	0																															
			Included	1																															
b	RW	CH27																																	
			Excluded	0																															
			Included	1																															
c	RW	CH28																																	
			Excluded	0																															
			Included	1																															
d	RW	CH29																																	
			Excluded	0																															
			Included	1																															
e	RW	CH30																																	
			Excluded	0																															
			Included	1																															
f	RW	CH31																																	
			Excluded	0																															
			Included	1																															

6.16.2.33 CHG[3]

Address offset: 0x80C

Channel group 3

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				f e d c b a Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	CH0			Include or exclude channel 0
			Excluded	0	Exclude
			Included	1	Include
B	RW	CH1			Include or exclude channel 1
			Excluded	0	Exclude
			Included	1	Include
C	RW	CH2			Include or exclude channel 2
			Excluded	0	Exclude
			Included	1	Include
D	RW	CH3			Include or exclude channel 3
			Excluded	0	Exclude
			Included	1	Include
E	RW	CH4			Include or exclude channel 4
			Excluded	0	Exclude
			Included	1	Include
F	RW	CH5			Include or exclude channel 5
			Excluded	0	Exclude
			Included	1	Include
G	RW	CH6			Include or exclude channel 6
			Excluded	0	Exclude
			Included	1	Include
H	RW	CH7			Include or exclude channel 7
			Excluded	0	Exclude
			Included	1	Include
I	RW	CH8			Include or exclude channel 8
			Excluded	0	Exclude
			Included	1	Include
J	RW	CH9			Include or exclude channel 9
			Excluded	0	Exclude
			Included	1	Include
K	RW	CH10			Include or exclude channel 10
			Excluded	0	Exclude
			Included	1	Include
L	RW	CH11			Include or exclude channel 11
			Excluded	0	Exclude
			Included	1	Include
M	RW	CH12			Include or exclude channel 12
			Excluded	0	Exclude
			Included	1	Include
N	RW	CH13			Include or exclude channel 13
			Excluded	0	Exclude
			Included	1	Include
O	RW	CH14			Include or exclude channel 14
			Excluded	0	Exclude
			Included	1	Include
P	RW	CH15			Include or exclude channel 15
			Excluded	0	Exclude
			Included	1	Include
Q	RW	CH16			Include or exclude channel 16
			Excluded	0	Exclude

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
			Included	1	Include
R	RW	CH17			Include or exclude channel 17
			Excluded	0	Exclude
			Included	1	Include
S	RW	CH18			Include or exclude channel 18
			Excluded	0	Exclude
			Included	1	Include
T	RW	CH19			Include or exclude channel 19
			Excluded	0	Exclude
			Included	1	Include
U	RW	CH20			Include or exclude channel 20
			Excluded	0	Exclude
			Included	1	Include
V	RW	CH21			Include or exclude channel 21
			Excluded	0	Exclude
			Included	1	Include
W	RW	CH22			Include or exclude channel 22
			Excluded	0	Exclude
			Included	1	Include
X	RW	CH23			Include or exclude channel 23
			Excluded	0	Exclude
			Included	1	Include
Y	RW	CH24			Include or exclude channel 24
			Excluded	0	Exclude
			Included	1	Include
Z	RW	CH25			Include or exclude channel 25
			Excluded	0	Exclude
			Included	1	Include
a	RW	CH26			Include or exclude channel 26
			Excluded	0	Exclude
			Included	1	Include
b	RW	CH27			Include or exclude channel 27
			Excluded	0	Exclude
			Included	1	Include
c	RW	CH28			Include or exclude channel 28
			Excluded	0	Exclude
			Included	1	Include
d	RW	CH29			Include or exclude channel 29
			Excluded	0	Exclude
			Included	1	Include
e	RW	CH30			Include or exclude channel 30
			Excluded	0	Exclude
			Included	1	Include
f	RW	CH31			Include or exclude channel 31
			Excluded	0	Exclude
			Included	1	Include

6.16.2.34 CHG[4]

Address offset: 0x810

Channel group 4

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	CH0																																	
			Excluded	0																															
			Included	1																															
B	RW	CH1																																	
			Excluded	0																															
			Included	1																															
C	RW	CH2																																	
			Excluded	0																															
			Included	1																															
D	RW	CH3																																	
			Excluded	0																															
			Included	1																															
E	RW	CH4																																	
			Excluded	0																															
			Included	1																															
F	RW	CH5																																	
			Excluded	0																															
			Included	1																															
G	RW	CH6																																	
			Excluded	0																															
			Included	1																															
H	RW	CH7																																	
			Excluded	0																															
			Included	1																															
I	RW	CH8																																	
			Excluded	0																															
			Included	1																															
J	RW	CH9																																	
			Excluded	0																															
			Included	1																															
K	RW	CH10																																	
			Excluded	0																															
			Included	1																															
L	RW	CH11																																	
			Excluded	0																															
			Included	1																															
M	RW	CH12																																	
			Excluded	0																															
			Included	1																															
N	RW	CH13																																	
			Excluded	0																															
			Included	1																															
O	RW	CH14																																	
			Excluded	0																															
			Included	1																															
P	RW	CH15																																	
			Excluded	0																															
			Included	1																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
Q	RW	CH16			Include or exclude channel 16
			Excluded	0	Exclude
			Included	1	Include
R	RW	CH17			Include or exclude channel 17
			Excluded	0	Exclude
			Included	1	Include
S	RW	CH18			Include or exclude channel 18
			Excluded	0	Exclude
			Included	1	Include
T	RW	CH19			Include or exclude channel 19
			Excluded	0	Exclude
			Included	1	Include
U	RW	CH20			Include or exclude channel 20
			Excluded	0	Exclude
			Included	1	Include
V	RW	CH21			Include or exclude channel 21
			Excluded	0	Exclude
			Included	1	Include
W	RW	CH22			Include or exclude channel 22
			Excluded	0	Exclude
			Included	1	Include
X	RW	CH23			Include or exclude channel 23
			Excluded	0	Exclude
			Included	1	Include
Y	RW	CH24			Include or exclude channel 24
			Excluded	0	Exclude
			Included	1	Include
Z	RW	CH25			Include or exclude channel 25
			Excluded	0	Exclude
			Included	1	Include
a	RW	CH26			Include or exclude channel 26
			Excluded	0	Exclude
			Included	1	Include
b	RW	CH27			Include or exclude channel 27
			Excluded	0	Exclude
			Included	1	Include
c	RW	CH28			Include or exclude channel 28
			Excluded	0	Exclude
			Included	1	Include
d	RW	CH29			Include or exclude channel 29
			Excluded	0	Exclude
			Included	1	Include
e	RW	CH30			Include or exclude channel 30
			Excluded	0	Exclude
			Included	1	Include
f	RW	CH31			Include or exclude channel 31
			Excluded	0	Exclude
			Included	1	Include

6.16.2.35 CHG[5]

Address offset: 0x814

Channel group 5

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	CH0																																	
			Excluded	0																															
			Included	1																															
B	RW	CH1																																	
			Excluded	0																															
			Included	1																															
C	RW	CH2																																	
			Excluded	0																															
			Included	1																															
D	RW	CH3																																	
			Excluded	0																															
			Included	1																															
E	RW	CH4																																	
			Excluded	0																															
			Included	1																															
F	RW	CH5																																	
			Excluded	0																															
			Included	1																															
G	RW	CH6																																	
			Excluded	0																															
			Included	1																															
H	RW	CH7																																	
			Excluded	0																															
			Included	1																															
I	RW	CH8																																	
			Excluded	0																															
			Included	1																															
J	RW	CH9																																	
			Excluded	0																															
			Included	1																															
K	RW	CH10																																	
			Excluded	0																															
			Included	1																															
L	RW	CH11																																	
			Excluded	0																															
			Included	1																															
M	RW	CH12																																	
			Excluded	0																															
			Included	1																															
N	RW	CH13																																	
			Excluded	0																															
			Included	1																															
O	RW	CH14																																	
			Excluded	0																															

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
			Included	1	Include
P	RW	CH15			Include or exclude channel 15
			Excluded	0	Exclude
			Included	1	Include
Q	RW	CH16			Include or exclude channel 16
			Excluded	0	Exclude
			Included	1	Include
R	RW	CH17			Include or exclude channel 17
			Excluded	0	Exclude
			Included	1	Include
S	RW	CH18			Include or exclude channel 18
			Excluded	0	Exclude
			Included	1	Include
T	RW	CH19			Include or exclude channel 19
			Excluded	0	Exclude
			Included	1	Include
U	RW	CH20			Include or exclude channel 20
			Excluded	0	Exclude
			Included	1	Include
V	RW	CH21			Include or exclude channel 21
			Excluded	0	Exclude
			Included	1	Include
W	RW	CH22			Include or exclude channel 22
			Excluded	0	Exclude
			Included	1	Include
X	RW	CH23			Include or exclude channel 23
			Excluded	0	Exclude
			Included	1	Include
Y	RW	CH24			Include or exclude channel 24
			Excluded	0	Exclude
			Included	1	Include
Z	RW	CH25			Include or exclude channel 25
			Excluded	0	Exclude
			Included	1	Include
a	RW	CH26			Include or exclude channel 26
			Excluded	0	Exclude
			Included	1	Include
b	RW	CH27			Include or exclude channel 27
			Excluded	0	Exclude
			Included	1	Include
c	RW	CH28			Include or exclude channel 28
			Excluded	0	Exclude
			Included	1	Include
d	RW	CH29			Include or exclude channel 29
			Excluded	0	Exclude
			Included	1	Include
e	RW	CH30			Include or exclude channel 30
			Excluded	0	Exclude
			Included	1	Include
f	RW	CH31			Include or exclude channel 31

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				f	e	d	c	b	a	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
			Excluded	0																															
			Included	1																															

6.16.2.36 FORK[0]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.36.1 FORK[0].TEP

Address offset: 0x910

Channel 0 task endpoint

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	TEP																																	
				Pointer to task register																															

6.16.2.37 FORK[1]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.37.1 FORK[1].TEP

Address offset: 0x914

Channel 1 task endpoint

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	TEP																																	
				Pointer to task register																															

6.16.2.38 FORK[2]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.38.1 FORK[2].TEP

Address offset: 0x918

Channel 2 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.39 FORK[3]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.39.1 FORK[3].TEP

Address offset: 0x91C

Channel 3 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.40 FORK[4]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.40.1 FORK[4].TEP

Address offset: 0x920

Channel 4 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.41 FORK[5]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.41.1 FORK[5].TEP

Address offset: 0x924

Channel 5 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.42 FORK[6]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.42.1 FORK[6].TEP

Address offset: 0x928

Channel 6 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.43 FORK[7]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.43.1 FORK[7].TEP

Address offset: 0x92C

Channel 7 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.44 FORK[8]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.44.1 FORK[8].TEP

Address offset: 0x930

Channel 8 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.45 FORK[9]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.45.1 FORK[9].TEP

Address offset: 0x934

Channel 9 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.46 FORK[10]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.46.1 FORK[10].TEP

Address offset: 0x938

Channel 10 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.47 FORK[11]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.47.1 FORK[11].TEP

Address offset: 0x93C

Channel 11 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.48 FORK[12]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.48.1 FORK[12].TEP

Address offset: 0x940

Channel 12 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.49 FORK[13]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.49.1 FORK[13].TEP

Address offset: 0x944

Channel 13 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.50 FORK[14]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.50.1 FORK[14].TEP

Address offset: 0x948

Channel 14 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.51 FORK[15]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.51.1 FORK[15].TEP

Address offset: 0x94C

Channel 15 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.52 FORK[16]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.52.1 FORK[16].TEP

Address offset: 0x950

Channel 16 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.53 FORK[17]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.53.1 FORK[17].TEP

Address offset: 0x954

Channel 17 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.54 FORK[18]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.54.1 FORK[18].TEP

Address offset: 0x958

Channel 18 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.55 FORK[19]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.55.1 FORK[19].TEP

Address offset: 0x95C

Channel 19 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.56 FORK[20]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.56.1 FORK[20].TEP

Address offset: 0x960

Channel 20 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.57 FORK[21]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.57.1 FORK[21].TEP

Address offset: 0x964

Channel 21 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.58 FORK[22]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.58.1 FORK[22].TEP

Address offset: 0x968

Channel 22 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.59 FORK[23]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.59.1 FORK[23].TEP

Address offset: 0x96C

Channel 23 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.60 FORK[24]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.60.1 FORK[24].TEP

Address offset: 0x970

Channel 24 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.61 FORK[25]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.61.1 FORK[25].TEP

Address offset: 0x974

Channel 25 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.62 FORK[26]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.62.1 FORK[26].TEP

Address offset: 0x978

Channel 26 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.63 FORK[27]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.63.1 FORK[27].TEP

Address offset: 0x97C

Channel 27 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.64 FORK[28]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.64.1 FORK[28].TEP

Address offset: 0x980

Channel 28 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.65 FORK[29]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.65.1 FORK[29].TEP

Address offset: 0x984

Channel 29 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.66 FORK[30]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.66.1 FORK[30].TEP

Address offset: 0x988

Channel 30 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.16.2.67 FORK[31]

Fork

This fork provides a second set of task endpoints for each of the channels in the PPI.

6.16.2.67.1 FORK[31].TEP

Address offset: 0x98C

Channel 31 task endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TEP			Pointer to task register

6.17 PWM — Pulse width modulation

The pulse with modulation (PWM) module enables the generation of pulse width modulated signals on GPIO. The module implements an up or up-and-down counter with four PWM channels that drive assigned GPIOs.

The following are the main features of a PWM module:

- Programmable PWM frequency
- Up to four PWM channels with individual polarity and duty cycle values
- Edge or center-aligned pulses across PWM channels
- Multiple duty cycle arrays (sequences) defined in RAM
- Autonomous and glitch-free update of duty cycle values directly from memory through EasyDMA (no CPU involvement)

- Change of polarity, duty cycle, and base frequency possibly on every PWM period
- RAM sequences can be repeated or connected into loops

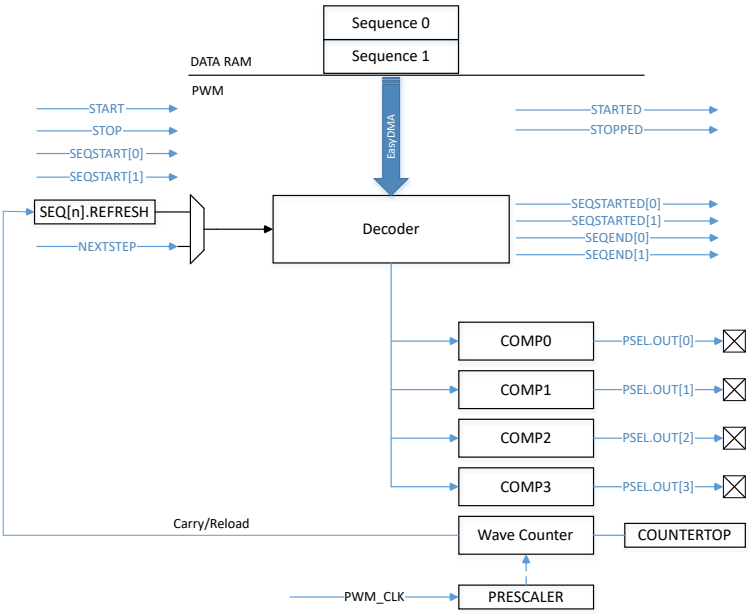


Figure 80: PWM module

6.17.1 Wave counter

The wave counter is responsible for generating the pulses at a duty cycle that depends on the compare values, and at a frequency that depends on COUNTERTOP.

There is one common 15-bit counter with four compare channels. Thus, all four channels will share the same period (PWM frequency), but can have individual duty cycle and polarity. The polarity is set by the most significant bit (MSb) of the 16-bit value read from RAM (see figure Decoder memory access modes on page 545). When the MSb is '1' the channel is configured as FallingEdge polarity, OUT[n] starts high to become low during the given PWM cycle, whereas the inverse occurs when configured for RisingEdge polarity. Whether the counter counts up, or up and down, is controlled by the MODE register.

The timer top value is controlled by the COUNTERTOP register. This register value, in conjunction with the selected PRESCALER of the PWM_CLK, will result in a given PWM period. A COUNTERTOP value smaller than the compare setting will result in a state where no PWM edges are generated. OUT[n] is held high, given that the polarity is set to FallingEdge. All compare registers are internal and can only be configured through decoder presented later. COUNTERTOP can be safely written at any time.

Sampling follows the START task. If DECODER.LOAD=WaveForm, the register value is ignored and taken from RAM instead (see section Decoder with EasyDMA on page 545 for more details). If DECODER.LOAD is anything else than the WaveForm, it is sampled following a STARTSEQ[n] task and when loading a new value from RAM during a sequence playback.

The following figure shows the counter operating in up mode (MODE=PWM_MODE_Up), with two PWM channels with the same frequency but different duty cycle:

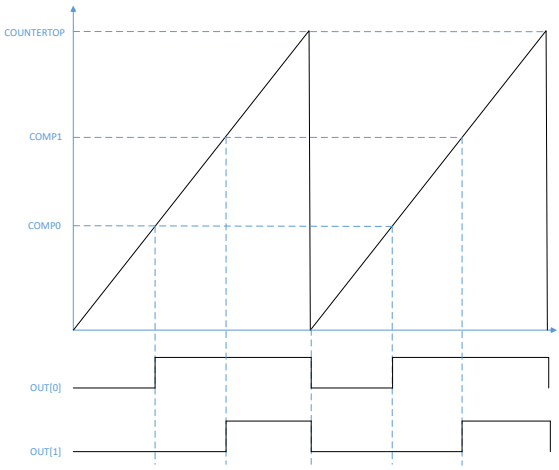


Figure 81: PWM counter in up mode example - RisingEdge polarity

The counter is automatically reset to zero when COUNTERTOP is reached and OUT[n] will invert. OUT[n] is held low if the compare value is 0 and held high if set to COUNTERTOP, given that the polarity is set to FallingEdge. Counter running in up mode results in pulse widths that are edge-aligned. The following is the code for the counter in up mode example:

```
uint16_t pwm_seq[4] = {PWM_CH0_DUTY, PWM_CH1_DUTY, PWM_CH2_DUTY, PWM_CH3_DUTY};
NRF_PWM0->PSEL.OUT[0] = (first_pin << PWM_PSEL_OUT_PIN_Pos) |
                        (PWM_PSEL_OUT_CONNECT_Connected <<
                                                PWM_PSEL_OUT_CONNECT_Pos);
NRF_PWM0->PSEL.OUT[1] = (second_pin << PWM_PSEL_OUT_PIN_Pos) |
                        (PWM_PSEL_OUT_CONNECT_Connected <<
                                                PWM_PSEL_OUT_CONNECT_Pos);
NRF_PWM0->ENABLE      = (PWM_ENABLE_ENABLE_Enabled << PWM_ENABLE_ENABLE_Pos);
NRF_PWM0->MODE        = (PWM_MODE_UPDOWN_Up << PWM_MODE_UPDOWN_Pos);
NRF_PWM0->PRESCALER   = (PWM_PRESCALER_PRESCALER_DIV_1 <<
                                                PWM_PRESCALER_PRESCALER_Pos);
NRF_PWM0->COUNTERTOP  = (16000 << PWM_COUNTERTOP_COUNTERTOP_Pos); //1 msec
NRF_PWM0->LOOP        = (PWM_LOOP_CNT_Disabled << PWM_LOOP_CNT_Pos);
NRF_PWM0->DECODER   = (PWM_DECODER_LOAD_Individual << PWM_DECODER_LOAD_Pos) |
                      (PWM_DECODER_MODE_RefreshCount << PWM_DECODER_MODE_Pos);
NRF_PWM0->SEQ[0].PTR  = ((uint32_t)(pwm_seq) << PWM_SEQ_PTR_PTR_Pos);
NRF_PWM0->SEQ[0].CNT  = ((sizeof(pwm_seq) / sizeof(uint16_t)) <<
                                                PWM_SEQ_CNT_CNT_Pos);
NRF_PWM0->SEQ[0].REFRESH  = 0;
NRF_PWM0->SEQ[0].ENDDELAY = 0;
NRF_PWM0->TASKS_SEQSTART[0] = 1;
```

When the counter is running in up mode, the following formula can be used to compute the PWM period and the step size:

PWM period: $T_{PWM(Up)} = T_{PWM_CLK} * COUNTERTOP$

Step width/Resolution: $T_{steps} = T_{PWM_CLK}$

The following figure shows the counter operating in up-and-down mode (MODE=PWM_MODE_UpAndDown), with two PWM channels with the same frequency but different duty cycle and output polarity:

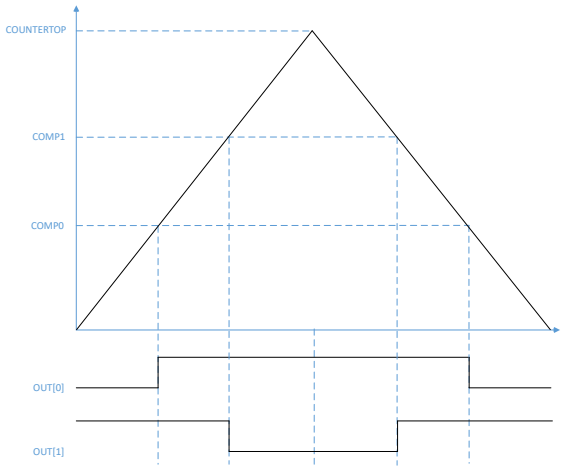


Figure 82: PWM counter in up-and-down mode example

The counter starts decrementing to zero when COUNTERTOP is reached and will invert the OUT[n] when compare value is hit for the second time. This results in a set of pulses that are center-aligned. The following is the code for the counter in up-and-down mode example:

```
uint16_t pwm_seq[4] = {PWM_CH0_DUTY, PWM_CH1_DUTY, PWM_CH2_DUTY, PWM_CH3_DUTY};
NRF_PWM0->PSEL.OUT[0] = (first_pin << PWM_PSEL_OUT_PIN_Pos) |
                        (PWM_PSEL_OUT_CONNECT_Connected <<
                                                PWM_PSEL_OUT_CONNECT_Pos);
NRF_PWM0->PSEL.OUT[1] = (second_pin << PWM_PSEL_OUT_PIN_Pos) |
                        (PWM_PSEL_OUT_CONNECT_Connected <<
                                                PWM_PSEL_OUT_CONNECT_Pos);
NRF_PWM0->ENABLE      = (PWM_ENABLE_ENABLE_Enabled << PWM_ENABLE_ENABLE_Pos);
NRF_PWM0->MODE        = (PWM_MODE_UPDOWN_UpAndDown << PWM_MODE_UPDOWN_Pos);
NRF_PWM0->PRESCALER   = (PWM_PRESCALER_PRESCALER_DIV_1 <<
                                                PWM_PRESCALER_PRESCALER_Pos);
NRF_PWM0->COUNTERTOP  = (16000 << PWM_COUNTERTOP_COUNTERTOP_Pos); //1 msec
NRF_PWM0->LOOP        = (PWM_LOOP_CNT_Disabled << PWM_LOOP_CNT_Pos);
NRF_PWM0->DECODER   = (PWM_DECODER_LOAD_Individual << PWM_DECODER_LOAD_Pos) |
                      (PWM_DECODER_MODE_RefreshCount << PWM_DECODER_MODE_Pos);
NRF_PWM0->SEQ[0].PTR  = ((uint32_t)(pwm_seq) << PWM_SEQ_PTR_PTR_Pos);
NRF_PWM0->SEQ[0].CNT  = ((sizeof(pwm_seq) / sizeof(uint16_t)) <<
                                                PWM_SEQ_CNT_CNT_Pos);
NRF_PWM0->SEQ[0].REFRESH  = 0;
NRF_PWM0->SEQ[0].ENDDELAY = 0;
NRF_PWM0->TASKS_SEQSTART[0] = 1;
```

When the counter is running in up-and-down mode, the following formula can be used to compute the PWM period and the step size:

$T_{PWM(Up\ And\ Down)} = T_{PWM_CLK} * 2 * COUNTERTOP$

Step width/Resolution: $T_{steps} = T_{PWM_CLK} * 2$

6.17.2 Decoder with EasyDMA

The decoder uses EasyDMA to take PWM parameters stored in RAM and update the internal compare registers of the wave counter, based on the mode of operation.

PWM parameters are organized into a sequence containing at least one half word (16 bit). Its most significant bit[15] denotes the polarity of the OUT[n] while bit[14:0] is the 15-bit compare value.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
Id				B A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
Id	RW	Field	Value Id	Value	Description
A	RW	COMPARE			Duty cycle setting - value loaded to internal compare register
B	RW	POLARITY			Edge polarity of GPIO.
			RisingEdge	0	First edge within the PWM period is rising
			FallingEdge	1	First edge within the PWM period is falling

The DECODER register controls how the RAM content is interpreted and loaded into the internal compare registers. The LOAD field controls if the RAM values are loaded to all compare channels, or to update a group or all channels with individual values. The following figure illustrates how parameters stored in RAM are organized and routed to various compare channels in different modes:

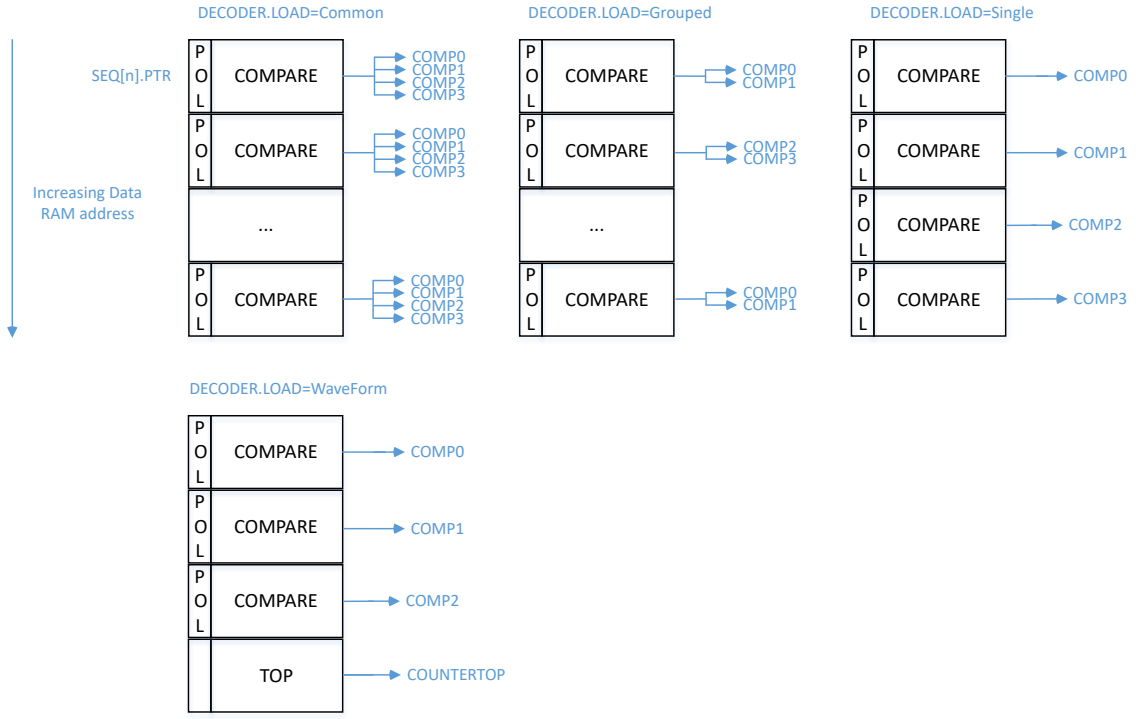


Figure 83: Decoder memory access modes

A special mode of operation is available when DECODER.LOAD is set to WaveForm. In this mode, up to three PWM channels can be enabled - OUT[0] to OUT[2]. In RAM, four values are loaded at a time: the first, second and third location are used to load the values, and the fourth RAM location is used to load the COUNTERTOP register. This way one can have up to three PWM channels with a frequency base that changes on a per PWM period basis. This mode of operation is useful for arbitrary wave form generation in applications, such as LED lighting.

The register SEQ[n].REFRESH=N (one per sequence n=0 or 1) will instruct a new RAM stored pulse width value on every (N+1)th PWM period. Setting the register to zero will result in a new duty cycle update every PWM period, as long as the minimum PWM period is observed.

Note that registers SEQ[n].REFRESH and SEQ[n].ENDDELAY are ignored when DECODER.MODE=NextStep. The next value is loaded upon every received NEXTSTEP task.

SEQ[n].PTR is the pointer used to fetch COMPARE values from RAM. If the SEQ[n].PTR is not pointing to a RAM region, an EasyDMA transfer may result in a HardFault or RAM corruption. See Memory on page 21 for more information about the different memory regions. After the SEQ[n].PTR is set to the desired RAM location, the SEQ[n].CNT register must be set to number of 16-bit half words in the sequence. It is important to observe that the Grouped mode requires one half word per group, while the Single mode requires one half word per channel, thus increasing the RAM size occupation. If PWM generation is not running when the SEQSTART[n] task is triggered, the task will load the first value from RAM and then start the PWM generation. A SEQSTARTED[n] event is generated as soon as the EasyDMA has read the first PWM parameter from RAM and the wave counter has started executing it. When LOOP.CNT=0, sequence n=0 or 1 is played back once. After the last value in the sequence has been loaded and started executing, a SEQEND[n] event is generated. The PWM generation will then continue with the last loaded value. The following figure illustrates an example of such simple playback:

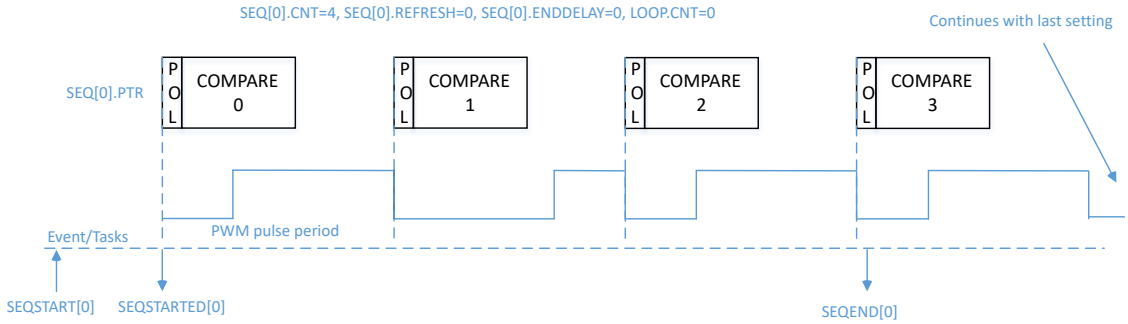


Figure 84: Simple sequence example

Figure depicts the source code used for configuration and timing details in a sequence where only sequence 0 is used and only run once with a new PWM duty cycle for each period.

```
NRF_PWM0->PSEL.OUT[0] = (first_pin << PWM_PSEL_OUT_PIN_Pos) |
                        (PWM_PSEL_OUT_CONNECT_Connected <<
                                                PWM_PSEL_OUT_CONNECT_Pos);
NRF_PWM0->ENABLE      = (PWM_ENABLE_ENABLE_Enabled << PWM_ENABLE_ENABLE_Pos);
NRF_PWM0->MODE        = (PWM_MODE_UPDOWN_Up << PWM_MODE_UPDOWN_Pos);
NRF_PWM0->PRESCALER   = (PWM_PRESCALER_PRESCALER_DIV_1 <<
                                                PWM_PRESCALER_PRESCALER_Pos);
NRF_PWM0->COUNTERTOP  = (16000 << PWM_COUNTERTOP_COUNTERTOP_Pos); //1 msec
NRF_PWM0->LOOP        = (PWM_LOOP_CNT_Disabled << PWM_LOOP_CNT_Pos);
NRF_PWM0->DECODER   = (PWM_DECODER_LOAD_Common << PWM_DECODER_LOAD_Pos) |
                      (PWM_DECODER_MODE_RefreshCount << PWM_DECODER_MODE_Pos);
NRF_PWM0->SEQ[0].PTR  = ((uint32_t)(seq0_ram) << PWM_SEQ_PTR_PTR_Pos);
NRF_PWM0->SEQ[0].CNT  = ((sizeof(seq0_ram) / sizeof(uint16_t)) <<
                                                PWM_SEQ_CNT_CNT_Pos);
NRF_PWM0->SEQ[0].REFRESH  = 0;
NRF_PWM0->SEQ[0].ENDDELAY = 0;
NRF_PWM0->TASKS_SEQSTART[0] = 1;
```

To completely stop the PWM generation and force the associated pins to a defined state, a STOP task can be triggered at any time. A STOPPED event is generated when the PWM generation has stopped at the end of currently running PWM period, and the pins go into their idle state as defined in GPIO OUT register. PWM generation can then only be restarted through a SEQSTART[n] task. SEQSTART[n] will resume PWM generation after having loaded the first value from the RAM buffer defined in the SEQ[n].PTR register.

The table below indicates when specific registers get sampled by the hardware. Care should be taken when updating these registers to avoid that values are applied earlier than expected.

Register	Taken into account by hardware	Recommended (safe) update
SEQ[n].PTR	When sending the SEQSTART[n] task	After having received the SEQSTARTED[n] event
SEQ[n].CNT	When sending the SEQSTART[n] task	After having received the SEQSTARTED[n] event
SEQ[0].ENDDELAY	When sending the SEQSTART[0] task	Before starting sequence [0] through a SEQSTART[0] task
	Every time a new value from sequence [0] has been loaded from RAM and gets applied to the Wave Counter (indicated by the PWMPERIODEND event)	When no more value from sequence [0] gets loaded from RAM (indicated by the SEQEND[0] event)
		At any time during sequence [1] (which starts when the SEQSTARTED[1] event is generated)
SEQ[1].ENDDELAY	When sending the SEQSTART[1] task	Before starting sequence [1] through a SEQSTART[1] task
	Every time a new value from sequence [1] has been loaded from RAM and gets applied to the Wave Counter (indicated by the PWMPERIODEND event)	When no more value from sequence [1] gets loaded from RAM (indicated by the SEQEND[1] event)
		At any time during sequence [0] (which starts when the SEQSTARTED[0] event is generated)
SEQ[0].REFRESH	When sending the SEQSTART[0] task	Before starting sequence [0] through a SEQSTART[0] task
	Every time a new value from sequence [0] has been loaded from RAM and gets applied to the Wave Counter (indicated by the PWMPERIODEND event)	At any time during sequence [1] (which starts when the SEQSTARTED[1] event is generated)
SEQ[1].REFRESH	When sending the SEQSTART[1] task	Before starting sequence [1] through a SEQSTART[1] task
	Every time a new value from sequence [1] has been loaded from RAM and gets applied to the Wave Counter (indicated by the PWMPERIODEND event)	At any time during sequence [0] (which starts when the SEQSTARTED[0] event is generated)
COUNTERTOP	In DECODER.LOAD=WaveForm: this register is ignored.	Before starting PWM generation through a SEQSTART[n] task
	In all other LOAD modes: at the end of current PWM period (indicated by the PWMPERIODEND event)	After a STOP task has been triggered, and the STOPPED event has been received.
MODE	Immediately	Before starting PWM generation through a SEQSTART[n] task
		After a STOP task has been triggered, and the STOPPED event has been received.
DECODER	Immediately	Before starting PWM generation through a SEQSTART[n] task
		After a STOP task has been triggered, and the STOPPED event has been received.
PRESCALER	Immediately	Before starting PWM generation through a SEQSTART[n] task
		After a STOP task has been triggered, and the STOPPED event has been received.
LOOP	Immediately	Before starting PWM generation through a SEQSTART[n] task
		After a STOP task has been triggered, and the STOPPED event has been received.
PSEL.OUT[n]	Immediately	Before enabling the PWM instance through the ENABLE register

Table 30: When to safely update PWM registers

Note: SEQ[n].REFRESH and SEQ[n].ENDDELAY are ignored at the end of a complex sequence, indicated by a LOOPSDONE event. The reason for this is that the last value loaded from RAM is maintained until further action from software (restarting a new sequence, or stopping PWM generation).

A more complex example, where LOOP.CNT>0, is shown in the following figure:

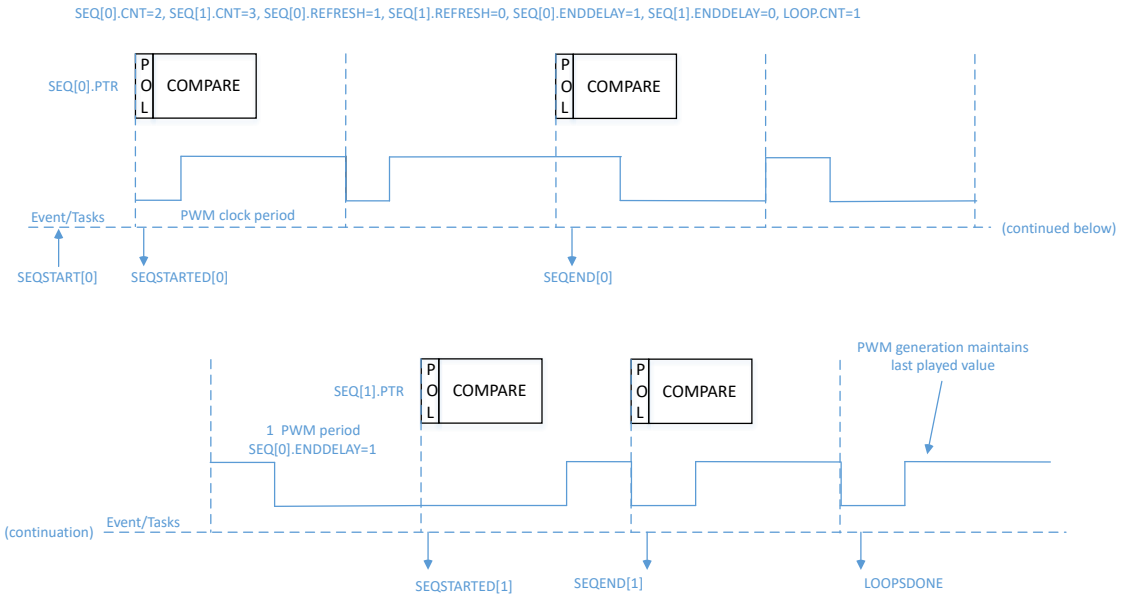


Figure 85: Example using two sequences

In this case, an automated playback takes place, consisting of SEQ[0], delay 0, SEQ[1], delay 1, then again SEQ[0], etc. The user can choose to start a complex playback with SEQ[0] or SEQ[1] through sending the SEQSTART[0] or SEQSTART[1] task. The complex playback always ends with delay 1.

The two sequences 0 and 1 are defined by the addresses of value tables in RAM (pointed to by SEQ[n].PTR) and the buffer size (SEQ[n].CNT). The rate at which a new value is loaded is defined individually for each sequence by SEQ[n].REFRESH. The chaining of sequence 1 following the sequence 0 is implicit, the LOOP.CNT register allows the chaining of sequence 1 to sequence 0 for a determined number of times. In other words, it allows to repeat a complex sequence a number of times in a fully automated way.

In the following code example, sequence 0 is defined with SEQ[0].REFRESH set to 1, meaning that a new PWM duty cycle is pushed every second PWM period. This complex sequence is started with the SEQSTART[0] task, so SEQ[0] is played first. Since SEQ[0].ENDDELAY=1 there will be one PWM period delay between last period on sequence 0 and the first period on sequence 1. Since SEQ[1].ENDDELAY=0 there is no delay 1, so SEQ[0] would be started immediately after the end of SEQ[1]. However, as LOOP.CNT is

1, the playback stops after having played SEQ[1] only once, and both SEQEND[1] and LOOPSDONE are generated (their order is not guaranteed in this case).

```
NRF_PWM0->PSEL.OUT[0] = (first_pin << PWM_PSEL_OUT_PIN_Pos) |
                        (PWM_PSEL_OUT_CONNECT_Connected <<
                                                  PWM_PSEL_OUT_CONNECT_Pos);
NRF_PWM0->ENABLE      = (PWM_ENABLE_ENABLE_Enabled << PWM_ENABLE_ENABLE_Pos);
NRF_PWM0->MODE        = (PWM_MODE_UPDOWN_Up << PWM_MODE_UPDOWN_Pos);
NRF_PWM0->PRESCALER   = (PWM_PRESCALER_PRESCALER_DIV_1 <<
                                                  PWM_PRESCALER_PRESCALER_Pos);
NRF_PWM0->COUNTERTOP  = (16000 << PWM_COUNTERTOP_COUNTERTOP_Pos); //1 msec
NRF_PWM0->LOOP        = (1 << PWM_LOOP_CNT_Pos);
NRF_PWM0->DECODER   = (PWM_DECODER_LOAD_Common << PWM_DECODER_LOAD_Pos) |
                      (PWM_DECODER_MODE_RefreshCount << PWM_DECODER_MODE_Pos);
NRF_PWM0->SEQ[0].PTR  = ((uint32_t)(seq0_ram) << PWM_SEQ_PTR_PTR_Pos);
NRF_PWM0->SEQ[0].CNT  = ((sizeof(seq0_ram) / sizeof(uint16_t)) <<
                                                  PWM_SEQ_CNT_CNT_Pos);
NRF_PWM0->SEQ[0].REFRESH  = 1;
NRF_PWM0->SEQ[0].ENDDELAY = 1;
NRF_PWM0->SEQ[1].PTR  = ((uint32_t)(seq1_ram) << PWM_SEQ_PTR_PTR_Pos);
NRF_PWM0->SEQ[1].CNT  = ((sizeof(seq1_ram) / sizeof(uint16_t)) <<
                                                  PWM_SEQ_CNT_CNT_Pos);
NRF_PWM0->SEQ[1].REFRESH  = 0;
NRF_PWM0->SEQ[1].ENDDELAY = 0;
NRF_PWM0->TASKS_SEQSTART[0] = 1;
```

The decoder can also be configured to asynchronously load new PWM duty cycle. If the DECODER.MODE register is set to NextStep, then the NEXTSTEP task will cause an update of internal compare registers on the next PWM period.

The following figures provide an overview of each part of an arbitrary sequence, in various modes (LOOP.CNT=0 and LOOP.CNT>0). In particular, the following are represented:

- Initial and final duty cycle on the PWM output(s)
- Chaining of SEQ[0] and SEQ[1] if LOOP.CNT>0
- Influence of registers on the sequence
- Events generated during a sequence
- DMA activity (loading of next value and applying it to the output(s))

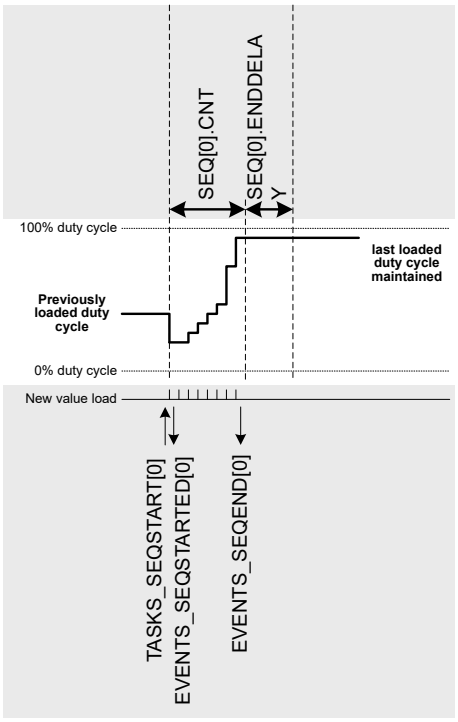


Figure 86: Single shot (LOOP.CNT=0)

Note: The single-shot example also applies to SEQ[1]. Only SEQ[0] is represented for simplicity.

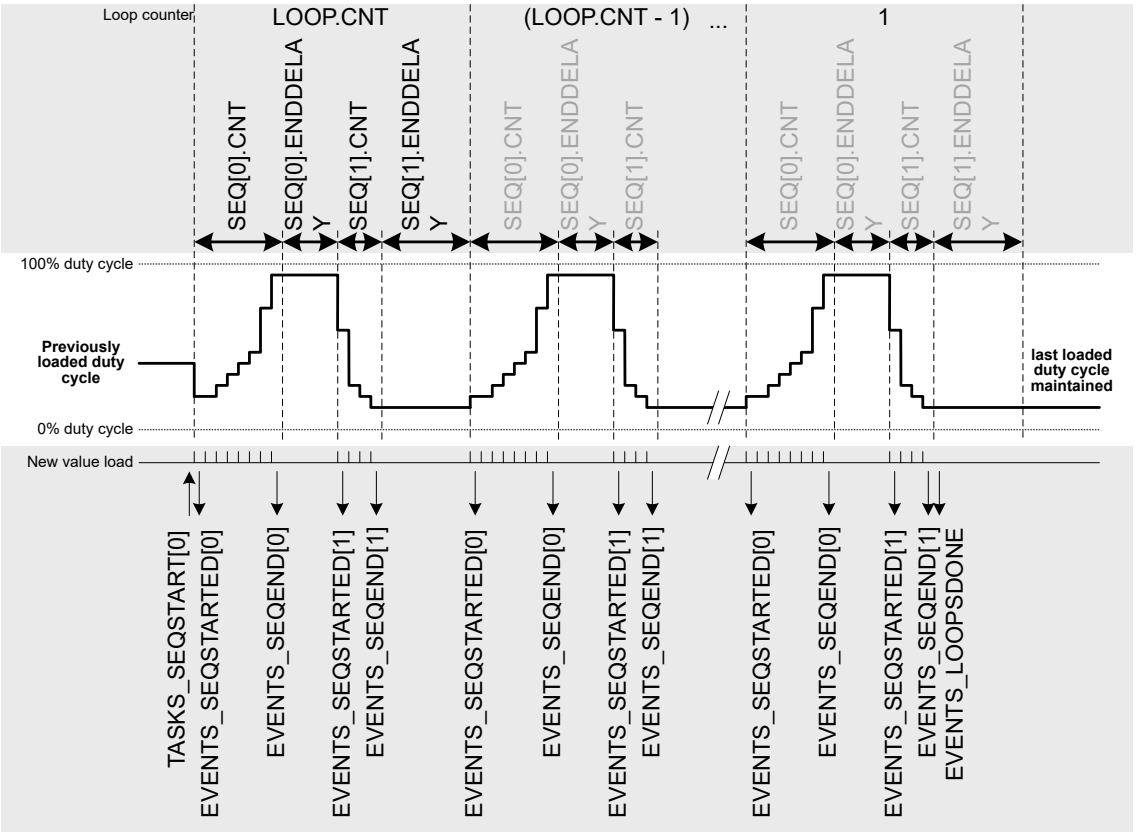


Figure 87: Complex sequence (LOOP.CNT>0) starting with SEQ[0]

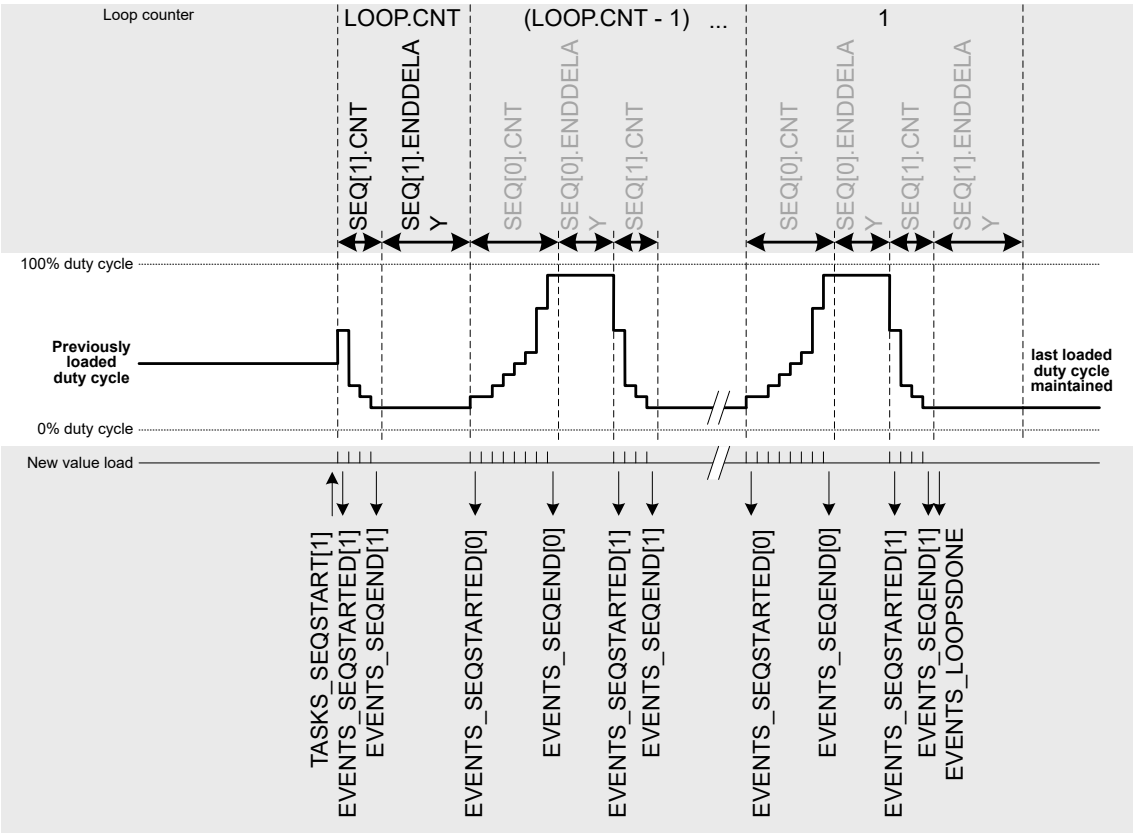


Figure 88: Complex sequence (LOOP.CNT>0) starting with SEQ[1]

Note: If a sequence is in use in a simple or complex sequence, it must have a length of SEQ[n].CNT > 0.

6.17.3 Limitations

Previous compare value is repeated if the PWM period is shorter than the time it takes for the EasyDMA to retrieve from RAM and update the internal compare registers. This is to ensure a glitch-free operation even for very short PWM periods.

6.17.4 Pin configuration

The OUT[n] (n=0..3) signals associated with each PWM channel are mapped to physical pins according to the configuration of PSEL.OUT[n] registers. If PSEL.OUT[n].CONNECT is set to Disconnected, the associated PWM module signal will not be connected to any physical pins.

The PSEL.OUT[n] registers and their configurations are used as long as the PWM module is enabled and the PWM generation active (wave counter started). They are retained only as long as the device is in System ON mode (see section POWER for more information about power modes).

To ensure correct behavior in the PWM module, the pins that are used must be configured in the GPIO peripheral in the following way before the PWM module is enabled:

PWM signal	PWM pin	Direction	Output value	Comment
OUT[n]	As specified in PSEL.OUT[n] (n=0..3)	Output	0	Idle state defined in GPIO OUT register

Table 31: Recommended GPIO configuration before starting PWM generation

The idle state of a pin is defined by the OUT register in the GPIO module, to ensure that the pins used by the PWM module are driven correctly. If PWM generation is stopped by triggering a STOP task, the PWM module itself is temporarily disabled or the device temporarily enters System OFF. This configuration must be retained in the GPIO for the selected pins (I/Os) for as long as the PWM module is supposed to be connected to an external PWM circuit.

Only one peripheral can be assigned to drive a particular GPIO pin at a time. Failing to do so may result in unpredictable behavior.

6.17.5 Registers

Instances

Instance	Base address	Description
PWM0	0x4001C000	Pulse width modulation unit 0
PWM1	0x40021000	Pulse width modulation unit 1
PWM2	0x40022000	Pulse width modulation unit 2
PWM3	0x4002D000	Pulse width modulation unit 3

Register overview

Register	Offset	Description
TASKS_STOP	0x004	Stops PWM pulse generation on all channels at the end of current PWM period, and stops sequence playback
TASKS_SEQSTART[0]	0x008	Loads the first PWM value on all enabled channels from sequence 0, and starts playing that sequence at the rate defined in SEQ[0]REFRESH and/or DECODER.MODE. Causes PWM generation to start if not running.
TASKS_SEQSTART[1]	0x00C	Loads the first PWM value on all enabled channels from sequence 1, and starts playing that sequence at the rate defined in SEQ[1]REFRESH and/or DECODER.MODE. Causes PWM generation to start if not running.
TASKS_NEXTSTEP	0x010	Steps by one value in the current sequence on all enabled channels if DECODER.MODE=NextStep. Does not cause PWM generation to start if not running.
EVENTS_STOPPED	0x104	Response to STOP task, emitted when PWM pulses are no longer generated
EVENTS_SEQSTARTED[0]	0x108	First PWM period started on sequence 0
EVENTS_SEQSTARTED[1]	0x10C	First PWM period started on sequence 1
EVENTS_SEQEND[0]	0x110	Emitted at end of every sequence 0, when last value from RAM has been applied to wave counter
EVENTS_SEQEND[1]	0x114	Emitted at end of every sequence 1, when last value from RAM has been applied to wave counter
EVENTS_PWMPERIODEND	0x118	Emitted at the end of each PWM period
EVENTS_LOOPSDONE	0x11C	Concatenated sequences have been played the amount of times defined in LOOP.CNT
SHORTS	0x200	Shortcuts between local events and tasks
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
ENABLE	0x500	PWM module enable register
MODE	0x504	Selects operating mode of the wave counter
COUNTERTOP	0x508	Value up to which the pulse generator counter counts
PRESCALER	0x50C	Configuration for PWM_CLK
DECODER	0x510	Configuration of the decoder
LOOP	0x514	Number of playbacks of a loop
SEQ[0].PTR	0x520	Beginning address in RAM of this sequence
SEQ[0].CNT	0x524	Number of values (duty cycles) in this sequence
SEQ[0].REFRESH	0x528	Number of additional PWM periods between samples loaded into compare register
SEQ[0].ENDDELAY	0x52C	Time added after the sequence

Register	Offset	Description
SEQ[1].PTR	0x540	Beginning address in RAM of this sequence
SEQ[1].CNT	0x544	Number of values (duty cycles) in this sequence
SEQ[1].REFRESH	0x548	Number of additional PWM periods between samples loaded into compare register
SEQ[1].ENDDELAY	0x54C	Time added after the sequence
PSEL.OUT[0]	0x560	Output pin select for PWM channel 0
PSEL.OUT[1]	0x564	Output pin select for PWM channel 1
PSEL.OUT[2]	0x568	Output pin select for PWM channel 2
PSEL.OUT[3]	0x56C	Output pin select for PWM channel 3

6.17.5.1 TASKS_STOP

Address offset: 0x004

Stops PWM pulse generation on all channels at the end of current PWM period, and stops sequence playback

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STOP			Stops PWM pulse generation on all channels at the end of current PWM period, and stops sequence playback
			Trigger	1	Trigger task

6.17.5.2 TASKS_SEQSTART[0]

Address offset: 0x008

Loads the first PWM value on all enabled channels from sequence 0, and starts playing that sequence at the rate defined in SEQ[0]REFRESH and/or DECODER.MODE. Causes PWM generation to start if not running.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_SEQSTART			Loads the first PWM value on all enabled channels from sequence 0, and starts playing that sequence at the rate defined in SEQ[0]REFRESH and/or DECODER.MODE. Causes PWM generation to start if not running.
			Trigger	1	Trigger task

6.17.5.3 TASKS_SEQSTART[1]

Address offset: 0x00C

Loads the first PWM value on all enabled channels from sequence 1, and starts playing that sequence at the rate defined in SEQ[1]REFRESH and/or DECODER.MODE. Causes PWM generation to start if not running.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_SEQSTART			Loads the first PWM value on all enabled channels from sequence 1, and starts playing that sequence at the rate defined in SEQ[1]REFRESH and/or DECODER.MODE. Causes PWM generation to start if not running.
			Trigger	1	Trigger task

6.17.5.4 TASKS_NEXTSTEP

Address offset: 0x010

Steps by one value in the current sequence on all enabled channels if DECODER.MODE=NextStep. Does not cause PWM generation to start if not running.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_NEXTSTEP			Steps by one value in the current sequence on all enabled channels if DECODER.MODE=NextStep. Does not cause PWM generation to start if not running.
			Trigger	1	Trigger task

6.17.5.5 EVENTS_STOPPED

Address offset: 0x104

Response to STOP task, emitted when PWM pulses are no longer generated

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_STOPPED			Response to STOP task, emitted when PWM pulses are no longer generated
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.17.5.6 EVENTS_SEQSTARTED[0]

Address offset: 0x108

First PWM period started on sequence 0

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_SEQSTARTED			First PWM period started on sequence 0
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.17.5.7 EVENTS_SEQSTARTED[1]

Address offset: 0x10C

First PWM period started on sequence 1

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_SEQSTARTED			First PWM period started on sequence 1
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.17.5.8 EVENTS_SEQEND[0]

Address offset: 0x110

Emitted at end of every sequence 0, when last value from RAM has been applied to wave counter

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_SEQEND			Emitted at end of every sequence 0, when last value from RAM has been applied to wave counter
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.17.5.9 EVENTS_SEQEND[1]

Address offset: 0x114

Emitted at end of every sequence 1, when last value from RAM has been applied to wave counter

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_SEQEND			Emitted at end of every sequence 1, when last value from RAM has been applied to wave counter
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.17.5.10 EVENTS_PWMPERIODEND

Address offset: 0x118

Emitted at the end of each PWM period

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_PWMPERIODEND			Emitted at the end of each PWM period
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.17.5.11 EVENTS_LOOPSDONE

Address offset: 0x11C

Concatenated sequences have been played the amount of times defined in LOOP.CNT

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_LOOPSDONE			Concatenated sequences have been played the amount of times defined in LOOP.CNT
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.17.5.12 SHORTS

Address offset: 0x200

Shortcuts between local events and tasks

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	SEQEND0_STOP			Shortcut between event SEQEND[0] and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
B	RW	SEQEND1_STOP			Shortcut between event SEQEND[1] and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
C	RW	LOOPSDONE_SEQSTART0			Shortcut between event LOOPSDONE and task SEQSTART[0]
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
D	RW	LOOPSDONE_SEQSTART1			Shortcut between event LOOPSDONE and task SEQSTART[1]
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
E	RW	LOOPSDONE_STOP			Shortcut between event LOOPSDONE and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

6.17.5.13 INTEN

Address offset: 0x300

Enable or disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				H G F E D C B
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
B	RW	STOPPED			Enable or disable interrupt for event STOPPED
			Disabled	0	Disable
			Enabled	1	Enable
C	RW	SEQSTARTED[0]			Enable or disable interrupt for event SEQSTARTED[0]
			Disabled	0	Disable
			Enabled	1	Enable
D	RW	SEQSTARTED[1]			Enable or disable interrupt for event SEQSTARTED[1]
			Disabled	0	Disable
			Enabled	1	Enable
E	RW	SEQEND[0]			Enable or disable interrupt for event SEQEND[0]
			Disabled	0	Disable
			Enabled	1	Enable
F	RW	SEQEND[1]			Enable or disable interrupt for event SEQEND[1]
			Disabled	0	Disable
			Enabled	1	Enable
G	RW	PWMPERIODEND			Enable or disable interrupt for event PWMPERIODEND
			Disabled	0	Disable
			Enabled	1	Enable
H	RW	LOOPSDONE			Enable or disable interrupt for event LOOPSDONE
			Disabled	0	Disable
			Enabled	1	Enable

6.17.5.14 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				H G F E D C B
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
B	RW	STOPPED			Write '1' to enable interrupt for event STOPPED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	SEQSTARTED[0]			Write '1' to enable interrupt for event SEQSTARTED[0]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	SEQSTARTED[1]			Write '1' to enable interrupt for event SEQSTARTED[1]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	SEQEND[0]			Write '1' to enable interrupt for event SEQEND[0]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	SEQEND[1]			Write '1' to enable interrupt for event SEQEND[1]
			Set	1	Enable

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				H G F E D C B
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
G	RW	PWMPERIODEND			Write '1' to enable interrupt for event PWMPERIODEND
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
H	RW	LOOPSDONE			Write '1' to enable interrupt for event LOOPSDONE
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.17.5.15 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				H G F E D C B
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
B	RW	STOPPED			Write '1' to disable interrupt for event STOPPED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	SEQSTARTED[0]			Write '1' to disable interrupt for event SEQSTARTED[0]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	SEQSTARTED[1]			Write '1' to disable interrupt for event SEQSTARTED[1]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	SEQEND[0]			Write '1' to disable interrupt for event SEQEND[0]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	SEQEND[1]			Write '1' to disable interrupt for event SEQEND[1]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
G	RW	PWMPERIODEND			Write '1' to disable interrupt for event PWMPERIODEND
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
H	RW	LOOPSDONE			Write '1' to disable interrupt for event LOOPSDONE
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.17.5.16 ENABLE

Address offset: 0x500

PWM module enable register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ENABLE			Enable or disable PWM module
			Disabled	0	Disabled
			Enabled	1	Enable

6.17.5.17 MODE

Address offset: 0x504

Selects operating mode of the wave counter

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	UPDOWN			Selects up mode or up-and-down mode for the counter
			Up	0	Up counter, edge-aligned PWM duty cycle
			UpAndDown	1	Up and down counter, center-aligned PWM duty cycle

6.17.5.18 COUNTERTOP

Address offset: 0x508

Value up to which the pulse generator counter counts

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A
Reset 0x000003FF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	COUNTERTOP		[3..32767]	Value up to which the pulse generator counter counts. This register is ignored when DECODER.MODE=WaveForm and only values from RAM are used.

6.17.5.19 PRESCALER

Address offset: 0x50C

Configuration for PWM_CLK

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PRESCALER			Prescaler of PWM_CLK
			DIV_1	0	Divide by 1 (16 MHz)
			DIV_2	1	Divide by 2 (8 MHz)
			DIV_4	2	Divide by 4 (4 MHz)
			DIV_8	3	Divide by 8 (2 MHz)
			DIV_16	4	Divide by 16 (1 MHz)
			DIV_32	5	Divide by 32 (500 kHz)
			DIV_64	6	Divide by 64 (250 kHz)
			DIV_128	7	Divide by 128 (125 kHz)

6.17.5.20 DECODER

Address offset: 0x510

Configuration of the decoder

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	LOAD			How a sequence is read from RAM and spread to the compare register
			Common	0	1st half word (16-bit) used in all PWM channels 0..3
			Grouped	1	1st half word (16-bit) used in channel 0..1; 2nd word in channel 2..3
			Individual	2	1st half word (16-bit) in ch.0; 2nd in ch.1; ...; 4th in ch.3
			WaveForm	3	1st half word (16-bit) in ch.0; 2nd in ch.1; ...; 4th in COUNTERTOP
B	RW	MODE			Selects source for advancing the active sequence
			RefreshCount	0	SEQ[n].REFRESH is used to determine loading internal compare registers
			NextStep	1	NEXTSTEP task causes a new value to be loaded to internal compare registers

6.17.5.21 LOOP

Address offset: 0x514

Number of playbacks of a loop

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	CNT			Number of playbacks of pattern cycles
			Disabled	0	Looping disabled (stop at the end of the sequence)

6.17.5.22 SEQ[0].PTR

Address offset: 0x520

Beginning address in RAM of this sequence

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Beginning address in RAM of this sequence

Note: See the memory chapter for details about which memories are available for EasyDMA.

6.17.5.23 SEQ[0].CNT

Address offset: 0x524

Number of values (duty cycles) in this sequence

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	CNT			Number of values (duty cycles) in this sequence
			Disabled	0	Sequence is disabled, and shall not be started as it is empty

6.17.5.24 SEQ[0].REFRESH

Address offset: 0x528

Number of additional PWM periods between samples loaded into compare register

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID									A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000001	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	CNT			Number of additional PWM periods between samples loaded into compare register (load every REFRESH.CNT+1 PWM periods)
			Continuous	0	Update every PWM period

6.17.5.25 SEQ[0].ENDDELAY

Address offset: 0x52C

Time added after the sequence

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID									A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	CNT			Time added after the sequence in PWM periods

6.17.5.26 SEQ[1].PTR

Address offset: 0x540

Beginning address in RAM of this sequence

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Beginning address in RAM of this sequence

Note: See the memory chapter for details about which memories are available for EasyDMA.

6.17.5.27 SEQ[1].CNT

Address offset: 0x544

Number of values (duty cycles) in this sequence

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	CNT			Number of values (duty cycles) in this sequence
			Disabled	0	Sequence is disabled, and shall not be started as it is empty

6.17.5.28 SEQ[1].REFRESH

Address offset: 0x548

Number of additional PWM periods between samples loaded into compare register

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID									A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000001	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	CNT			Number of additional PWM periods between samples loaded into compare register (load every REFRESH.CNT+1 PWM periods)
			Continuous	0	Update every PWM period

6.17.5.29 SEQ[1].ENDDELAY

Address offset: 0x54C

Time added after the sequence

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID									A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	CNT			Time added after the sequence in PWM periods

6.17.5.30 PSEL.OUT[0]

Address offset: 0x560

Output pin select for PWM channel 0

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.17.5.31 PSEL.OUT[1]

Address offset: 0x564

Output pin select for PWM channel 1

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.17.5.32 PSEL.OUT[2]

Address offset: 0x568

Output pin select for PWM channel 2

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.17.5.33 PSEL.OUT[3]

Address offset: 0x56C

Output pin select for PWM channel 3

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				C																										B	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.18 QDEC — Quadrature decoder

The Quadrature decoder (QDEC) provides buffered decoding of quadrature-encoded sensor signals. It is suitable for mechanical and optical sensors.

The sample period and accumulation are configurable to match application requirements. The QDEC provides the following:

- Digital waveform decoding from off-chip quadrature encoder
- Sample accumulation eliminating hard real-time requirements to be enforced on application
- Optional input de-bounce filters.
- Optional LED output signal for optical encoders

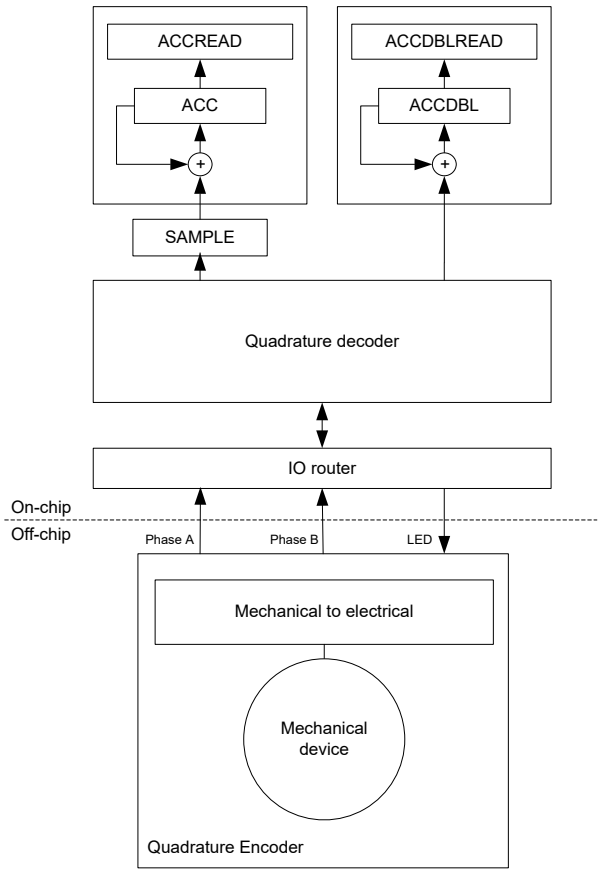


Figure 89: Quadrature decoder configuration

6.18.1 Sampling and decoding

The QDEC decodes the output from an incremental motion encoder by sampling the QDEC phase input pins (A and B).

The off-chip quadrature encoder is an incremental motion encoder outputting two waveforms, phase A and phase B. The two output waveforms are always 90 degrees out of phase, meaning that one always changes level before the other. The direction of movement is indicated by the waveform that changes level first. Invalid transitions may occur, meaning the two waveforms simultaneously switch. This may occur if the wheel rotates too fast relative to the sample rate set for the decoder.

The QDEC decodes the output from the off-chip encoder by sampling the QDEC phase input pins (A and B) at a fixed rate as specified in the SAMPLEPER register.

If the SAMPLEPER value needs to be changed, the QDEC shall be stopped using the STOP task. SAMPLEPER can be then changed upon receiving the STOPPED event, and QDEC can be restarted using the START task. Failing to do so may result in unpredictable behavior.

It is good practice to only change registers LEDPOL, REPORTPER, DBFEN, and LEDPRE when the QDEC is stopped.

When started, the decoder continuously samples the two input waveforms and decodes these by comparing the current sample pair (n) with the previous sample pair (n-1).

The decoding of the sample pairs is described in the table below.

Previous sample pair(n - 1)		Current samples pair(n)		SAMPLE register	ACC operation	ACCDBL operation	Description
A	B	A	B				
0	0	0	0	0	No change	No change	No movement
0	0	0	1	1	Increment	No change	Movement in positive direction
0	0	1	0	-1	Decrement	No change	Movement in negative direction
0	0	1	1	2	No change	Increment	Error: Double transition
0	1	0	0	-1	Decrement	No change	Movement in negative direction
0	1	0	1	0	No change	No change	No movement
0	1	1	0	2	No change	Increment	Error: Double transition
0	1	1	1	1	Increment	No change	Movement in positive direction
1	0	0	0	1	Increment	No change	Movement in positive direction
1	0	0	1	2	No change	Increment	Error: Double transition
1	0	1	0	0	No change	No change	No movement
1	0	1	1	-1	Decrement	No change	Movement in negative direction
1	1	0	0	2	No change	Increment	Error: Double transition
1	1	0	1	-1	Decrement	No change	Movement in negative direction
1	1	1	0	1	Increment	No change	Movement in positive direction
1	1	1	1	0	No change	No change	No movement

Table 32: Sampled value encoding

6.18.2 LED output

The LED output follows the sample period. The LED is switched on for a set period before sampling and then switched off immediately after. The period the LED is switched on before sampling is given in the LEDPRE register.

The LED output pin polarity is specified in the LEDPOL register.

When using off-chip mechanical encoders not requiring an LED, the LED output can be disabled by writing value 'Disconnected' to the CONNECT field of the PSEL.LED register. In this case, the QDEC will not acquire access to a pin for the LED output.

6.18.3 Debounce filters

Each of the two-phase inputs have digital debounce filters.

When enabled through the DBFEN register, the filter inputs are sampled at a fixed 1 MHz frequency during the entire sample period (which is specified in the SAMPLEPER register). The filters require all of the samples within this sample period to equal before the input signal is accepted and transferred to the output of the filter.

As a result, only input signal with a steady state longer than twice the period specified in SAMPLEPER are guaranteed to pass through the filter. Any signal with a steady state shorter than SAMPLEPER will always be suppressed by the filter. It is assumed that the frequency during the debounce period never exceeds 500 kHz (as required by the Nyquist theorem when using a 1 MHz sample frequency).

The LED will always be ON when the debounce filters are enabled, as the inputs in this case will be sampled continuously.

When the debounce filters are enabled, displacements reported by the QDEC peripheral are delayed by one SAMPLEPER period.

6.18.4 Accumulators

The quadrature decoder contains two accumulator registers, ACC and ACCDBL. These registers accumulate valid motion sample values and the number of detected invalid samples (double transitions), respectively.

The ACC register accumulates all valid values (1/-1) written to the SAMPLE register. This can be useful for preventing hard real-time requirements from being enforced on the application. When using the ACC register, the application can fetch data when necessary instead of reading all SAMPLE register output. The ACC register holds the relative movement of the external mechanical device from the previous clearing of the ACC register. Sample values indicating a double transition (2) will not be accumulated in the ACC register.

An ACCOF event is generated if the ACC receives a SAMPLE value that would cause the register to overflow or underflow. Any SAMPLE value that would cause an ACC overflow or underflow will be discarded, but any samples that do not cause the ACC to overflow or underflow will still be accepted.

The accumulator ACCDBL accumulates the number of detected double transitions since the previous clearing of the ACCDBL register.

The ACC and ACCDBL registers can be cleared by the READCLRACC and subsequently read using the ACCREAD and ACCDBLREAD registers.

The ACC register can be separately cleared by the RDCLRACC and subsequently read using the ACCREAD registers.

The ACCDBL register can be separately cleared by the RDCLRDBL and subsequently read using the ACCDBLREAD registers.

The REPORTPER register allows automated capture of multiple samples before sending an event. When a non-null displacement is captured and accumulated, a REPORTRDY event is sent. When one or more double-displacements are captured and accumulated, a DBLRDY event is sent. The REPORTPER field in this register determines how many samples must be accumulated before the contents are evaluated and a REPORTRDY or DBLRDY event is sent.

Using the RDCLRACC task (manually sent upon receiving the event, or using the DBLRDY_RDCLRACC shortcut), ACCREAD can then be read.

When a double transition has been captured and accumulated, a DBLRDY event is sent. Using the RDCLRDBL task (manually sent upon receiving the event, or using the DBLRDY_RDCLRDBL shortcut), ACCDBLREAD can then be read.

6.18.5 Output/input pins

The QDEC uses a three-pin interface to the off-chip quadrature encoder.

These pins are acquired when the QDEC is enabled in the ENABLE register. The pins acquired by the QDEC cannot be written by the CPU, but they can still be read by the CPU.

The pin numbers used for the QDEC are selected using the PSEL.n registers.

6.18.6 Pin configuration

The Phase A, Phase B, and LED signals are mapped to physical pins according to the configuration specified in the PSEL.A, PSEL.B, and PSEL.LED registers respectively.

If the CONNECT field value 'Disconnected' is specified in any of these registers, the associated signal will not be connected to any physical pin. The PSEL.A, PSEL.B, and PSEL.LED registers and their configurations are only used as long as the QDEC is enabled, and retained only as long as the device is in ON mode. When the peripheral is disabled, the pins will behave as regular GPIOs, and use the configuration in their respective OUT bit field and PIN_CNF[n] register.

To secure correct behavior in the QDEC, the pins used by the QDEC must be configured in the GPIO peripheral as described in GPIO configuration before enabling peripheral on page 568 before enabling the QDEC. This configuration must be retained in the GPIO for the selected I/Os as long as the QDEC is enabled.

Only one peripheral can be assigned to drive a particular GPIO pin at a time. Failing to do so may result in unpredictable behavior.

QDEC signal	QDEC pin	Direction	Output value	Comment
Phase A	As specified in PSEL.A	Input	Not applicable	
Phase B	As specified in PSEL.B	Input	Not applicable	
LED	As specified in PSEL.LED	Input	Not applicable	

Table 33: GPIO configuration before enabling peripheral

6.18.7 Registers

Instances

Instance	Base address	Description
QDEC	0x40012000	Quadrature decoder

Register overview

Register	Offset	Description
TASKS_START	0x000	Task starting the quadrature decoder
TASKS_STOP	0x004	Task stopping the quadrature decoder
TASKS_READCLRACC	0x008	Read and clear ACC and ACCDBL
TASKS_RDCLRACC	0x00C	Read and clear ACC
TASKS_RDCLRDBL	0x010	Read and clear ACCDBL
EVENTS_SAMPLERDY	0x100	Event being generated for every new sample value written to the SAMPLE register

Register	Offset	Description
EVENTS_REPORTRDY	0x104	Non-null report ready
EVENTS_ACCOF	0x108	ACC or ACCDBL register overflow
EVENTS_DBLRDY	0x10C	Double displacement(s) detected
EVENTS_STOPPED	0x110	QDEC has been stopped
SHORTS	0x200	Shortcuts between local events and tasks
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
ENABLE	0x500	Enable the quadrature decoder
LEDPOL	0x504	LED output pin polarity
SAMPLEPER	0x508	Sample period
SAMPLE	0x50C	Motion sample value
REPORTPER	0x510	Number of samples to be taken before REPORTRDY and DBLRDY events can be generated
ACC	0x514	Register accumulating the valid transitions
ACCREAD	0x518	Snapshot of the ACC register, updated by the READCLRACC or RDCLRACC task
PSEL.LED	0x51C	Pin select for LED signal
PSEL.A	0x520	Pin select for A signal
PSEL.B	0x524	Pin select for B signal
DBFEN	0x528	Enable input debounce filters
LEDPRE	0x540	Time period the LED is switched ON prior to sampling
ACCDBL	0x544	Register accumulating the number of detected double transitions
ACCDBLREAD	0x548	Snapshot of the ACCDBL, updated by the READCLRACC or RDCLRDBL task

6.18.7.1 TASKS_START

Address offset: 0x000

Task starting the quadrature decoder

When started, the SAMPLE register will be continuously updated at the rate given in the SAMPLEPER register.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_START			Task starting the quadrature decoder When started, the SAMPLE register will be continuously updated at the rate given in the SAMPLEPER register.
			Trigger	1	Trigger task

6.18.7.2 TASKS_STOP

Address offset: 0x004

Task stopping the quadrature decoder

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STOP			Task stopping the quadrature decoder
			Trigger	1	Trigger task

6.18.7.3 TASKS_READCLRACC

Address offset: 0x008

Read and clear ACC and ACCDBL

Task transferring the content of ACC to ACCREAD and the content of ACCDBL to ACCDBLREAD, and then clearing the ACC and ACCDBL registers. These read-and-clear operations will be done atomically.

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_READCLRACC			Read and clear ACC and ACCDBL
					Task transferring the content of ACC to ACCREAD and the content of ACCDBL to ACCDBLREAD, and then clearing the ACC and ACCDBL registers. These read-and-clear operations will be done atomically.
			Trigger	1	Trigger task

6.18.7.4 TASKS_RDCLRACC

Address offset: 0x00C

Read and clear ACC

Task transferring the content of ACC to ACCREAD, and then clearing the ACC register. This read-and-clear operation will be done atomically.

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_RDCLRACC			Read and clear ACC
					Task transferring the content of ACC to ACCREAD, and then clearing the ACC register. This read-and-clear operation will be done atomically.
			Trigger	1	Trigger task

6.18.7.5 TASKS_RDCLRDBL

Address offset: 0x010

Read and clear ACCDBL

Task transferring the content of ACCDBL to ACCDBLREAD, and then clearing the ACCDBL register. This read-and-clear operation will be done atomically.

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_RDCLRDBL			Read and clear ACCDBL
					Task transferring the content of ACCDBL to ACCDBLREAD, and then clearing the ACCDBL register. This read-and-clear operation will be done atomically.
			Trigger	1	Trigger task

6.18.7.6 EVENTS_SAMPLERDY

Address offset: 0x100

Event being generated for every new sample value written to the SAMPLE register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_SAMPLERDY			Event being generated for every new sample value written to the SAMPLE register
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.18.7.7 EVENTS_REPORTRDY

Address offset: 0x104

Non-null report ready

Event generated when REPORTPER number of samples has been accumulated in the ACC register and the content of the ACC register is not equal to 0. (Thus, this event is only generated if a motion is detected since the previous clearing of the ACC register).

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_REPORTRDY			Non-null report ready Event generated when REPORTPER number of samples has been accumulated in the ACC register and the content of the ACC register is not equal to 0. (Thus, this event is only generated if a motion is detected since the previous clearing of the ACC register).
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.18.7.8 EVENTS_ACCOF

Address offset: 0x108

ACC or ACCDBL register overflow

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ACCOF			ACC or ACCDBL register overflow
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.18.7.9 EVENTS_DBLRDY

Address offset: 0x10C

Double displacement(s) detected

Event generated when REPORTPER number of samples has been accumulated and the content of the ACCDBL register is not equal to 0. (Thus, this event is only generated if a double transition is detected since the previous clearing of the ACCDBL register).

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_DBLRDY			Double displacement(s) detected
					Event generated when REPORTPER number of samples has been accumulated and the content of the ACCDBL register is not equal to 0. (Thus, this event is only generated if a double transition is detected since the previous clearing of the ACCDBL register).
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.18.7.10 EVENTS_STOPPED

Address offset: 0x110

QDEC has been stopped

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_STOPPED			QDEC has been stopped
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.18.7.11 SHORTS

Address offset: 0x200

Shortcuts between local events and tasks

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					G F E D C B A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	REPORTRDY_READCLRACC			Shortcut between event REPORTRDY and task READCLRACC
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
B	RW	SAMPLERDY_STOP			Shortcut between event SAMPLERDY and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
C	RW	REPORTRDY_RDCLRACC			Shortcut between event REPORTRDY and task RDCLRACC
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
D	RW	REPORTRDY_STOP			Shortcut between event REPORTRDY and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
E	RW	DBLRDY_RDCLRDBL			Shortcut between event DBLRDY and task RDCLRDBL
			Disabled	0	Disable shortcut

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
			Enabled	1	Enable shortcut
F	RW	DBLRDY_STOP			Shortcut between event DBLRDY and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
G	RW	SAMPLERDY_READCLRACC			Shortcut between event SAMPLERDY and task READCLRACC
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

6.18.7.12 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	SAMPLERDY			Write '1' to enable interrupt for event SAMPLERDY
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	REPORTRDY			Write '1' to enable interrupt for event REPORTRDY Event generated when REPORTPER number of samples has been accumulated in the ACC register and the content of the ACC register is not equal to 0. (Thus, this event is only generated if a motion is detected since the previous clearing of the ACC register).
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	ACCOF			Write '1' to enable interrupt for event ACCOF
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	DBLRDY			Write '1' to enable interrupt for event DBLRDY Event generated when REPORTPER number of samples has been accumulated and the content of the ACCDBL register is not equal to 0. (Thus, this event is only generated if a double transition is detected since the previous clearing of the ACCDBL register).
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	STOPPED			Write '1' to enable interrupt for event STOPPED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.18.7.13 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	SAMPLERDY			Write '1' to disable interrupt for event SAMPLERDY
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	REPORTRDY			Write '1' to disable interrupt for event REPORTRDY Event generated when REPORTPER number of samples has been accumulated in the ACC register and the content of the ACC register is not equal to 0. (Thus, this event is only generated if a motion is detected since the previous clearing of the ACC register).
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	ACCOF			Write '1' to disable interrupt for event ACCOF
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	DBLRDY			Write '1' to disable interrupt for event DBLRDY Event generated when REPORTPER number of samples has been accumulated and the content of the ACCDBL register is not equal to 0. (Thus, this event is only generated if a double transition is detected since the previous clearing of the ACCDBL register).
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	STOPPED			Write '1' to disable interrupt for event STOPPED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.18.7.14 ENABLE

Address offset: 0x500

Enable the quadrature decoder

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A-	RW	ENABLE			Enable or disable the quadrature decoder When enabled the decoder pins will be active. When disabled the quadrature decoder pins are not active and can be used as GPIO .
			Disabled	0	Disable
			Enabled	1	Enable

6.18.7.15 LEDPOL

Address offset: 0x504

LED output pin polarity

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	LEDPOL			LED output pin polarity
			ActiveLow	0	Led active on output pin low
			ActiveHigh	1	Led active on output pin high

6.18.7.16 SAMPLEPER

Address offset: 0x508

Sample period

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	SAMPLEPER			Sample period. The SAMPLE register will be updated for every new sample
			128us	0	128 µs
			256us	1	256 µs
			512us	2	512 µs
			1024us	3	1024 µs
			2048us	4	2048 µs
			4096us	5	4096 µs
			8192us	6	8192 µs
			16384us	7	16384 µs
			32ms	8	32768 µs
			65ms	9	65536 µs
			131ms	10	131072 µs

6.18.7.17 SAMPLE

Address offset: 0x50C

Motion sample value

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	SAMPLE		[-1..2]	Last motion sample
					The value is a 2's complement value, and the sign gives the direction of the motion. The value '2' indicates a double transition.

6.18.7.18 REPORTPER

Address offset: 0x510

Number of samples to be taken before REPORTRDY and DBLRDY events can be generated

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	REPORTPER			Specifies the number of samples to be accumulated in the ACC register before the REPORTRDY and DBLRDY events can be generated. The report period in [µs] is given as: RPUS = SP * RP Where RPUS is the report period in [µs/report], SP is the sample period in [µs/sample] specified in SAMPLEPER, and RP is the report period in [samples/report] specified in REPORTPER .
			10Smpl	0	10 samples/report
			40Smpl	1	40 samples/report
			80Smpl	2	80 samples/report
			120Smpl	3	120 samples/report
			160Smpl	4	160 samples/report
			200Smpl	5	200 samples/report
			240Smpl	6	240 samples/report
			280Smpl	7	280 samples/report
			1Smpl	8	1 sample/report

6.18.7.19 ACC

Address offset: 0x514

Register accumulating the valid transitions

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	ACC		[-1024..1023]	Register accumulating all valid samples (not double transition) read from the SAMPLE register. Double transitions (SAMPLE = 2) will not be accumulated in this register. The value is a 32 bit 2's complement value. If a sample that would cause this register to overflow or underflow is received, the sample will be ignored and an overflow event (ACCOF) will be generated. The ACC register is cleared by triggering the READCLRACC or the RDCLRACC task.

6.18.7.20 ACCREAD

Address offset: 0x518

Snapshot of the ACC register, updated by the READCLRACC or RDCLRACC task

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	ACCREAD		[-1024..1023]	Snapshot of the ACC register. The ACCREAD register is updated when the READCLRACC or RDCLRACC task is triggered.

6.18.7.21 PSEL.LED

Address offset: 0x51C

Pin select for LED signal

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.18.7.22 PSEL.A

Address offset: 0x520

Pin select for A signal

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.18.7.23 PSEL.B

Address offset: 0x524

Pin select for B signal

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.18.7.24 DBFEN

Address offset: 0x528

Enable input debounce filters

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DBFEN			Enable input debounce filters
			Disabled	0	Debounce input filters disabled
			Enabled	1	Debounce input filters enabled

6.18.7.25 LEDPRE

Address offset: 0x540

Time period the LED is switched ON prior to sampling

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A
Reset 0x00000010				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	LEDPRE		[1..511]	Period in µs the LED is switched on prior to sampling

6.18.7.26 ACCDBL

Address offset: 0x544

Register accumulating the number of detected double transitions

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	ACCDBL		[0..15]	Register accumulating the number of detected double or illegal transitions. (SAMPLE = 2). When this register has reached its maximum value, the accumulation of double/illegal transitions will stop. An overflow event (ACCOF) will be generated if any double or illegal transitions are detected after the maximum value was reached. This field is cleared by triggering the READCLRACC or RDCLRDBL task.

6.18.7.27 ACCDBLREAD

Address offset: 0x548

Snapshot of the ACCDBL, updated by the READCLRACC or RDCLRDBL task

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	ACCDBLREAD		[0..15]	Snapshot of the ACCDBL register. This field is updated when the READCLRACC or RDCLRDBL task is triggered.

6.18.8 Electrical specification

6.18.8.1 QDEC Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
t_{SAMPLE}	Time between sampling signals from quadrature decoder	128		131072	µs
t_{LED}	Time from LED is turned on to signals are sampled	0		511	µs

6.19 QSPI — Quad serial peripheral interface

The QSPI peripheral provides support for communicating with an external flash memory device using SPI.

Listed here are the main features for the QSPI peripheral:

- Single/dual/quad SPI input/output
- 2–32 MHz configurable clock frequency
- Single-word read/write access from/to external flash
- EasyDMA for block read and write transfers
- Up to 16 MB/sec EasyDMA read rate
- Execute in place (XIP) for executing program directly from external flash

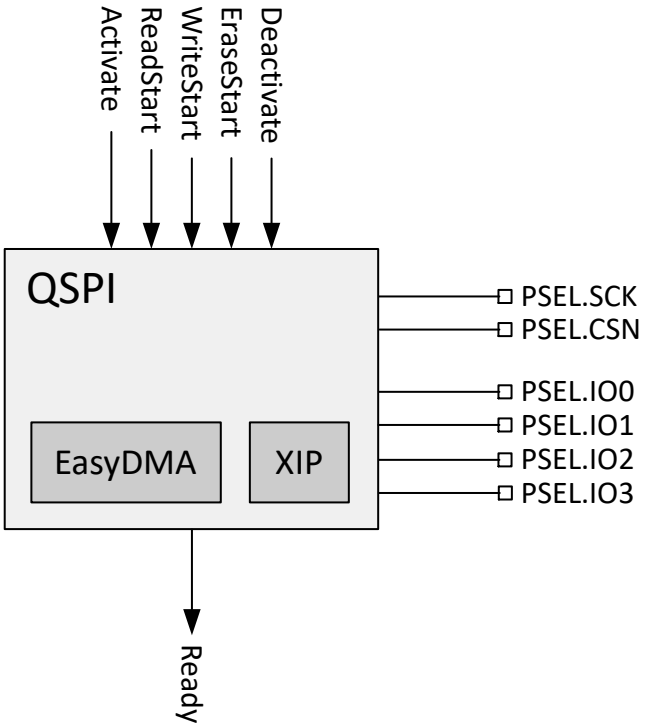


Figure 90: Block diagram

6.19.1 Configuring peripheral

Before any data can be transferred to or from the external flash memory, the peripheral needs to be configured.

1. Select input/output pins in PSEL.SCK on page 593, PSEL.CSN on page 593, PSEL.IO0 on page 594, PSEL.IO1 on page 594, PSEL.IO2 on page 594, and PSEL.IO3 on page 594. See Reference circuitry on page 937 for the recommended pins.

2. To ensure stable operation, set the GPIO drive strength to “high drive”. See the GPIO — General purpose input/output on page 322 chapter for details on how to configure GPIO drive strength.
3. Configure the interface towards the external flash memory using IFCONFIG0 on page 595, IFCONFIG1 on page 596, and ADDRCONF on page 597.
4. Enable the QSPI peripheral and acquire I/O pins using ENABLE on page 591.
5. Activate the external flash memory interface using the ACTIVATE task. The READY event will be generated when the interface has been activated and the external flash memory is ready for access.

Important:

If the IFCONFIG0 on page 595 register is configured to use the quad mode, the external flash device also needs to be set in the quad mode before any data transfers can take place.

This can be done by sending custom instructions to the external flash device, as described in Sending custom instructions on page 581.

6.19.2 Write operation

A write operation to the external flash is configured using the WRITE.DST on page 592, WRITE.SRC on page 592, and WRITE.CNT on page 592 registers and started using the WRITESTART task.

The READY event is generated when the transfer is complete.

The QSPI peripheral automatically takes care of splitting DMA transfers into page writes.

6.19.3 Read operation

A read operation from the external flash is configured using the READ.SRC on page 591, READ.DST on page 591, and READ.CNT on page 592 registers and started using the READSTART task.

The READY event is generated when the transfer is complete.

6.19.4 Erase operation

Erase of pages/blocks of the external flash is configured using the ERASE.PTR on page 592 and ERASE.LEN on page 593 registers and started using the ERASESTART task.

The READY event is generated when the erase operation has been started.

Note that in this case the READY event will not indicate that the erase operation of the flash has been completed, but it only signals that the erase operation has been started. The actual status of the erase operation can normally be read from the external flash using a custom instruction, see Sending custom instructions on page 581.

6.19.5 Execute in place

Execute in place (XIP) allows the CPU to execute program code directly from the external flash.

After the external flash has been configured, the CPU can execute code from the external flash by accessing the XIP memory region. See the figure below and Memory map on page 22 for details.

Note that the XIP memory region is read-only, writing to it will result in a bus error.

When accessing the XIP memory region, the start address of this XIP memory region will map to the address XIPOFFSET on page 595 of the external flash.

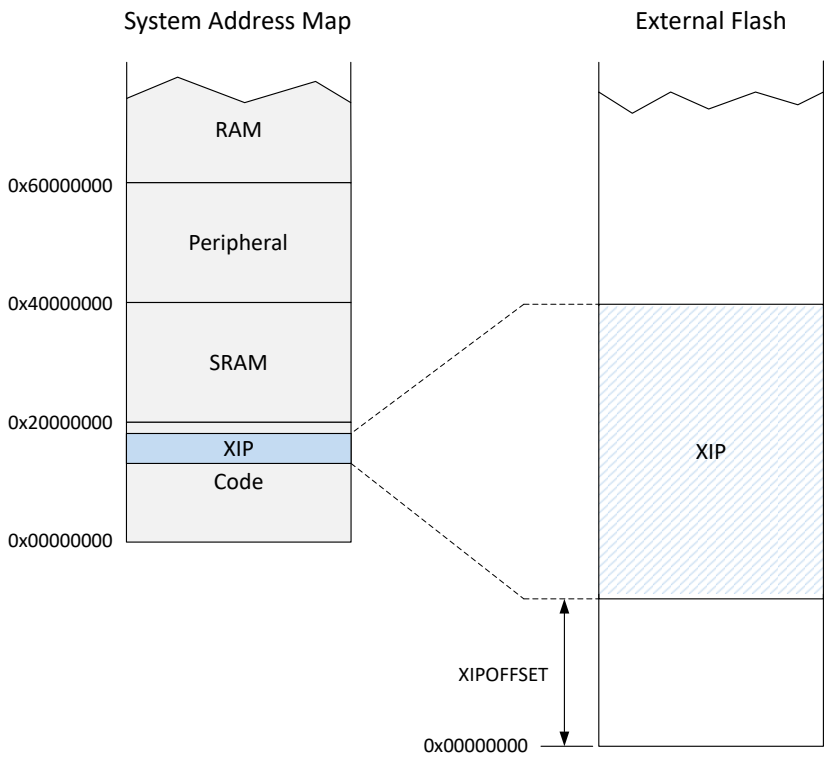


Figure 91: XIP memory map

6.19.6 Sending custom instructions

Custom instructions can be sent to the external flash using the CINSTRCONF on page 597, CINSTRDAT0 on page 598, and CINSTRDAT1 on page 598 registers. It is possible to send an instruction consisting of a one-byte opcode and up to 8 bytes of additional data and to read its response.

A custom instruction is prepared by first writing the data to be sent to CINSTRDAT0 on page 598 and CINSTRDAT1 on page 598 before writing the opcode and other configurations to the CINSTRCONF on page 597 register.

The custom instruction is sent when the CINSTRCONF on page 597 register is written and it is always sent on a single data line SPI interface.

The READY event will be generated when the custom instruction has been sent.

After a custom instruction has been sent, the CINSTRDAT0 on page 598 and CINSTRDAT1 on page 598 will contain the response bytes from the custom instruction.

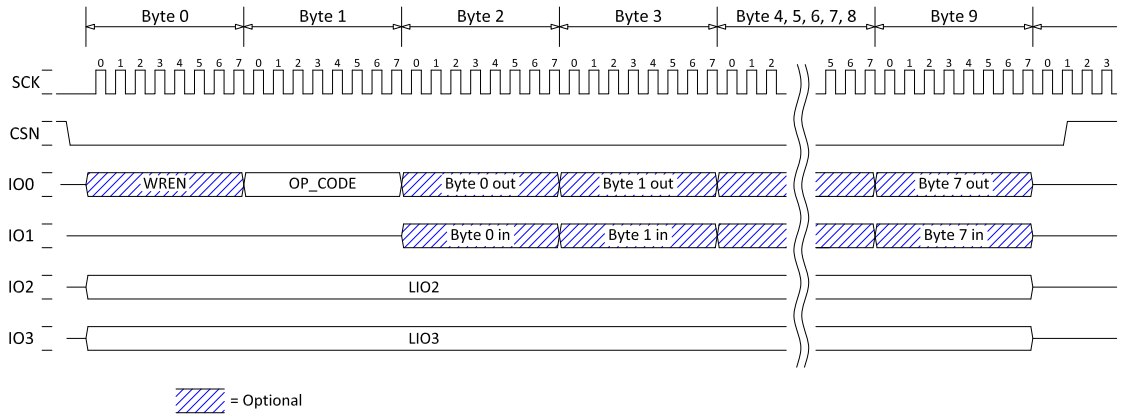


Figure 92: Sending custom instruction

6.19.6.1 Long frame mode

The LFEN and LFSTOP fields in the CINSTRCONF on page 597 control the operation of the custom instruction long frame mode. The long frame mode is a mechanism that permits arbitrary byte length custom instructions. While in long frame mode a long custom instruction sequence is split in multiple writes to the CINSTRDAT0 on page 598 and CINSTRDAT1 on page 598 registers.

To enable the long frame mode every write to the CINSTRCONF on page 597 register must have the LFEN field set to 1. The contents of the OPCODE field will be transmitted after the first write to CINSTRCONF on page 597 and will be omitted in every subsequent write to this register. For subsequent writes the number of data bytes as specified in the LENGTH field are transferred (that is the value of LENGTH - 1 data bytes). The values of the LIO2 and LIO3 fields are set in the first write to CINSTRCONF on page 597 and will apply for the entire custom instruction transmission until the long frame is finalized.

To finalize a long frame transmission, the LFSTOP field in CINSTRCONF on page 597 must be set to 1 in the last write to this register.

6.19.7 Deep power-down mode

The external flash memory can be put in deep power-down mode (DPM) to minimize its current consumption when there is no need to access the memory.

DPM is enabled in the IFCONFIG0 on page 595 register and configured in the DPMDUR on page 596 register. The DPM status of the external memory can be read in the STATUS on page 596 register. The DPMDUR register has to be configured according to the external flash specification to get the information in the STATUS register and the timing of the READY event correct.

Entering/exiting DPM is controlled using the IFCONFIG1 on page 596 register.

6.19.8 Instruction set

The table below shows the instruction set being used by the QSPI peripheral when communicating with an external flash device.

Instruction	Opcode	Description
WREN	0x06	Write enable
RDSR	0x05	Read status register
WRSR	0x01	Write status register
FASTREAD	0x0B	Read bytes at higher speed
READ2O	0x3B	Dual-read output
READ2IO	0xBB	Dual-read input/output
READ4O	0x6B	Quad-read output
READ4IO	0xEB	Quad-read input/output
PP	0x02	Page program
PP2O	0xA2	Dual-page program output
PP4O	0x32	Quad-page program output
PP4IO	0x38	Quad-page program input/output
SE	0x20	Sector erase
BE	0xD8	Block erase
CE	0xC7	Chip erase
DP	0xB9	Enter deep power-down mode
DPE	0xAB	Exit deep power-down mode
EN4B	Specified in the ADDRCONF on page 597 register	Enable 32 bit address mode

Table 34: Instruction set

6.19.9 Interface description

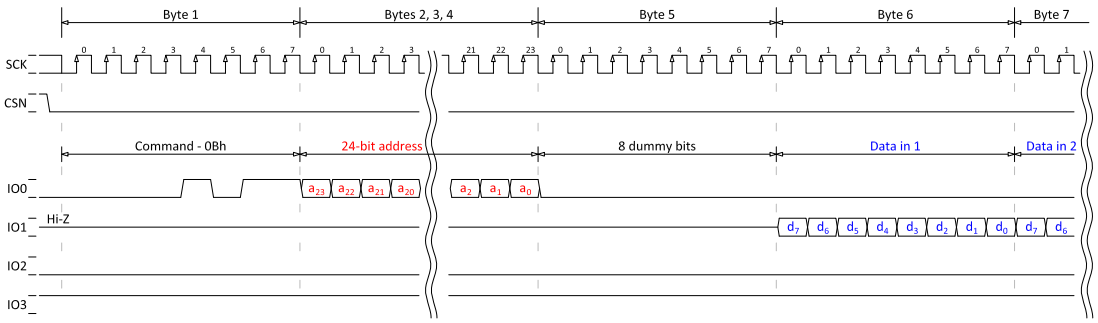


Figure 93: 24-bit FASTREAD, SPIMODE = MODE0

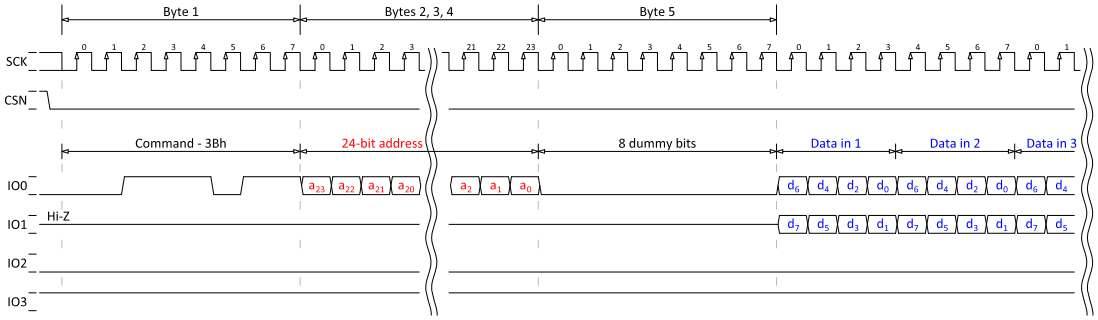


Figure 94: 24-bit READ2O (dual-read output), SPIMODE = MODE0

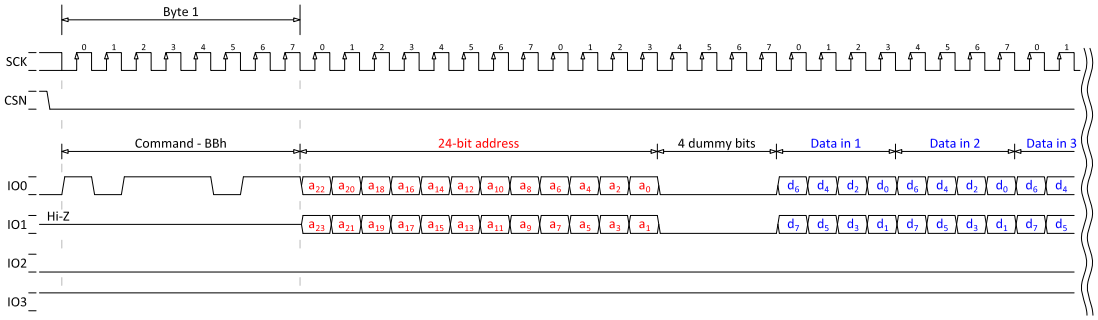


Figure 95: 24-bit READ2IO (dual read input/output), SPIMODE = MODE0

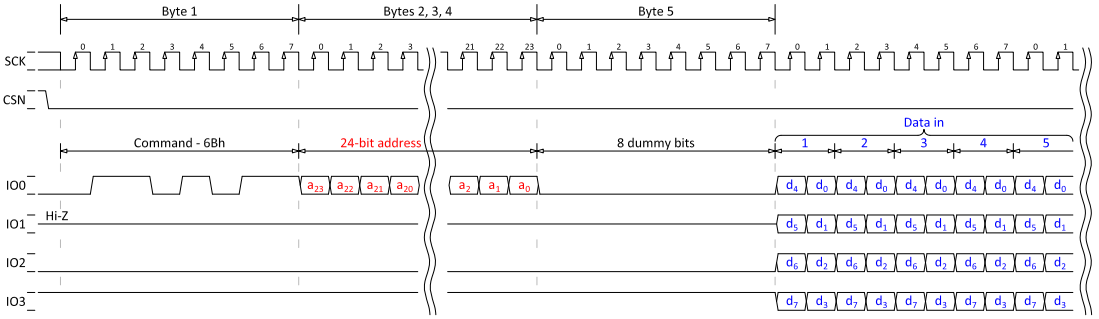


Figure 96: 24-bit READ4O (quad-read output), SPIMODE = MODE0

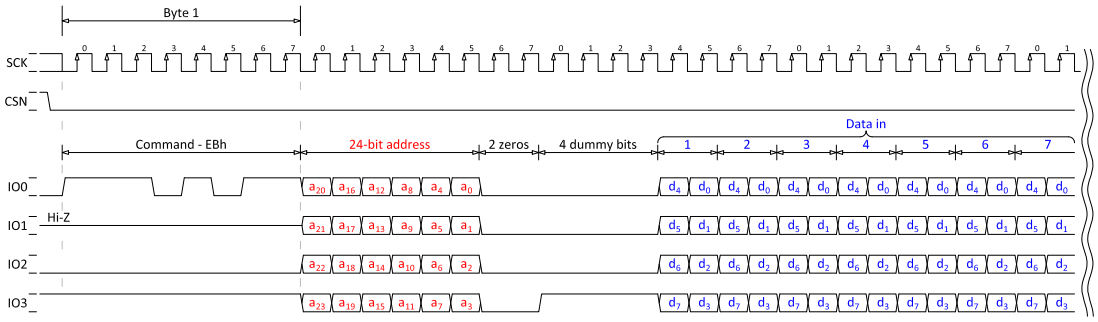


Figure 97: 24-bit READ4IO (quad-read input/output), SPIMODE = MODE0

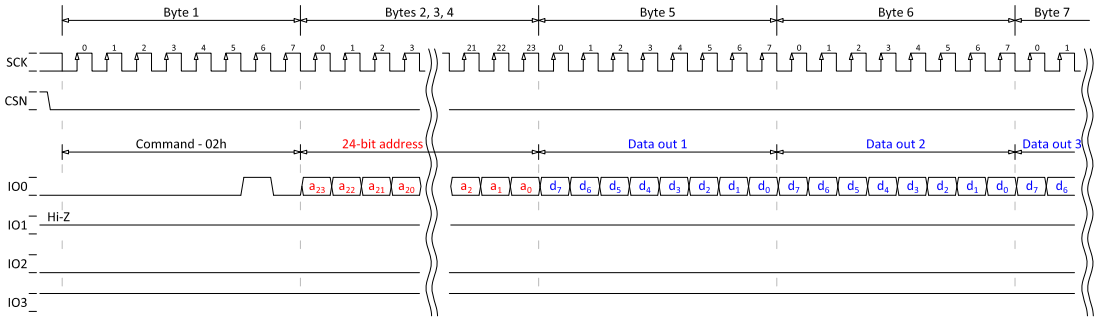


Figure 98: 24-bit PP (page program), SPIMODE = MODE0

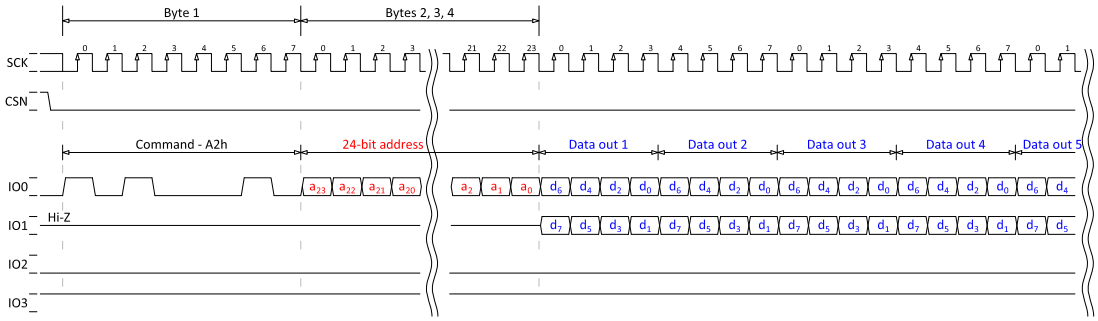


Figure 99: 24-bit PP2O (dual-page program output), SPIMODE = MODE0

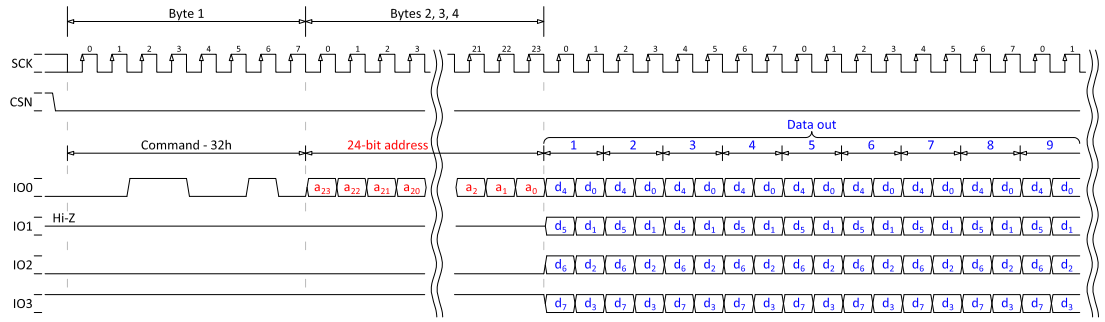


Figure 100: 24-bit PP4O (quad page program output), SPIMODE = MODE0

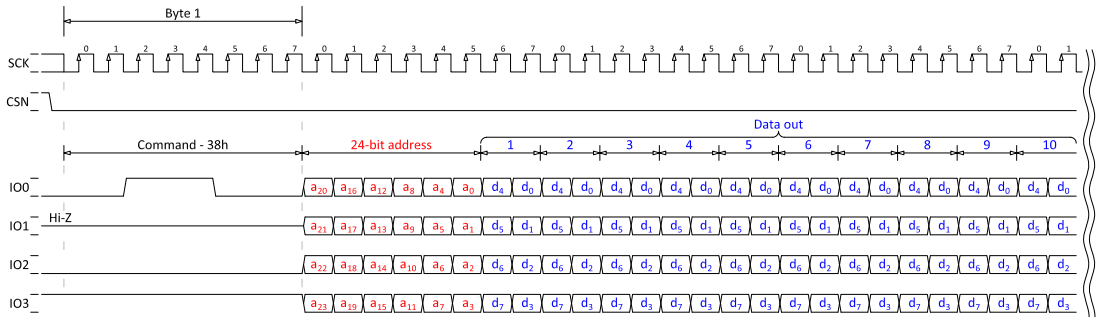


Figure 101: 24-bit PP4IO (quad page program input/output), SPIMODE = MODE0

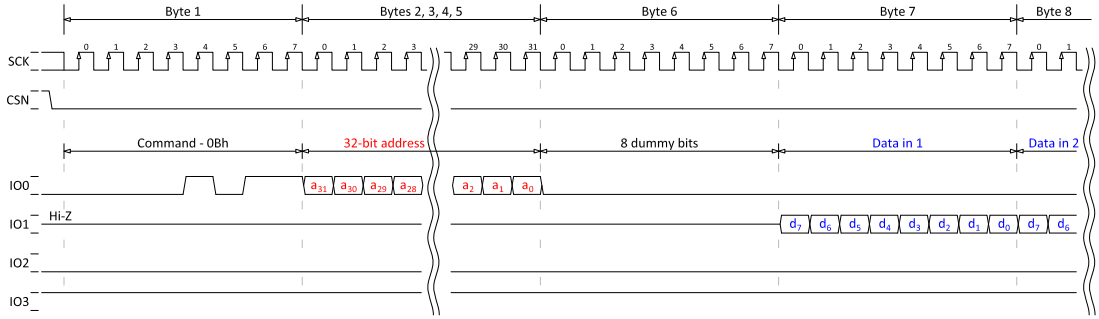


Figure 102: 32-bit FASTREAD, SPIMODE = MODE0

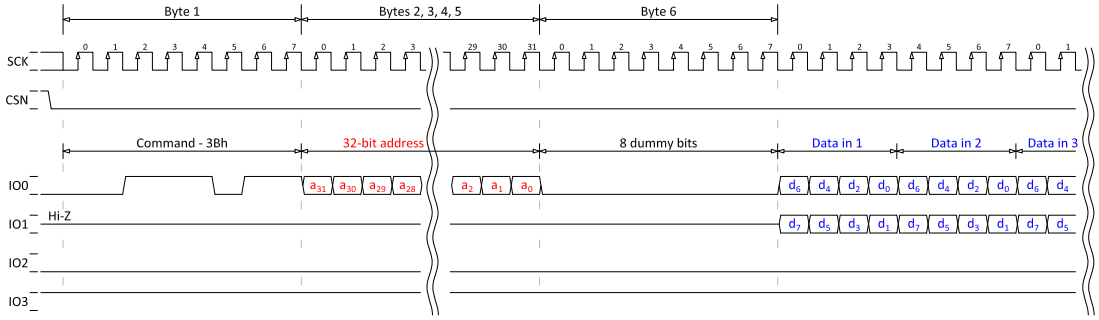


Figure 103: 32-bit READ2O (dual-read output), SPIMODE = MODE0

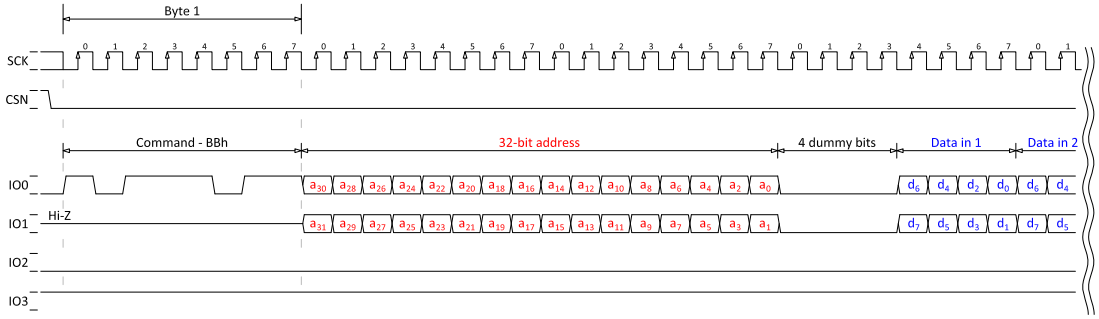


Figure 104: 32-bit READ2IO (dual read input/output), SPIMODE = MODE0

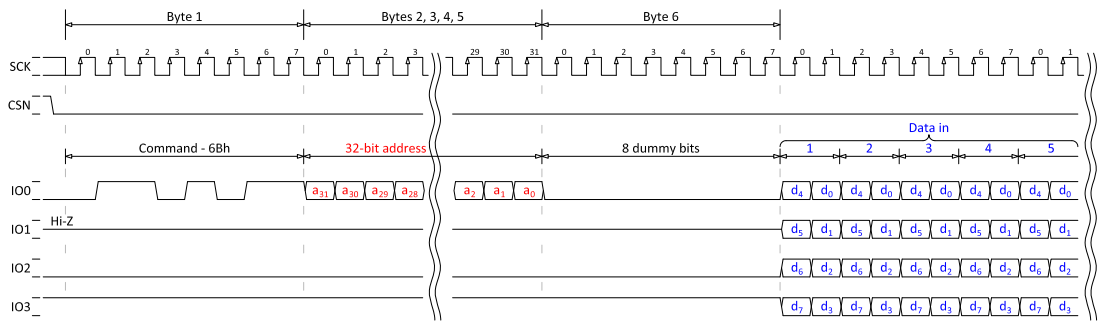


Figure 105: 32-bit READ4O (quad-read output), SPIMODE = MODE0

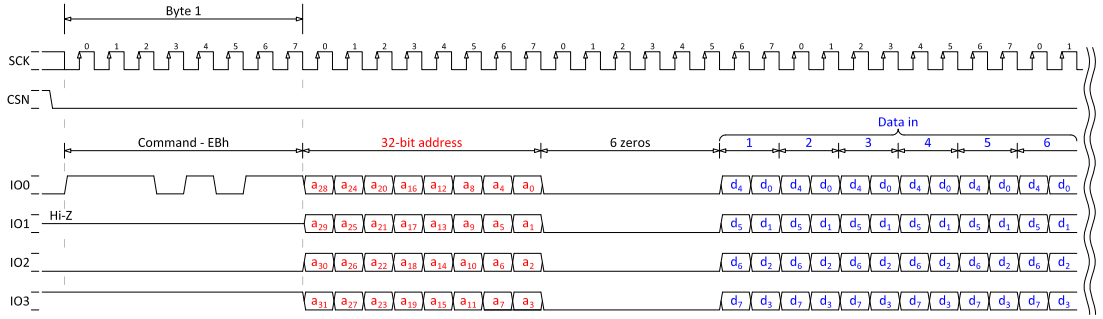


Figure 106: 32-bit READ4IO (quad-read input/output), SPIMODE = MODE0

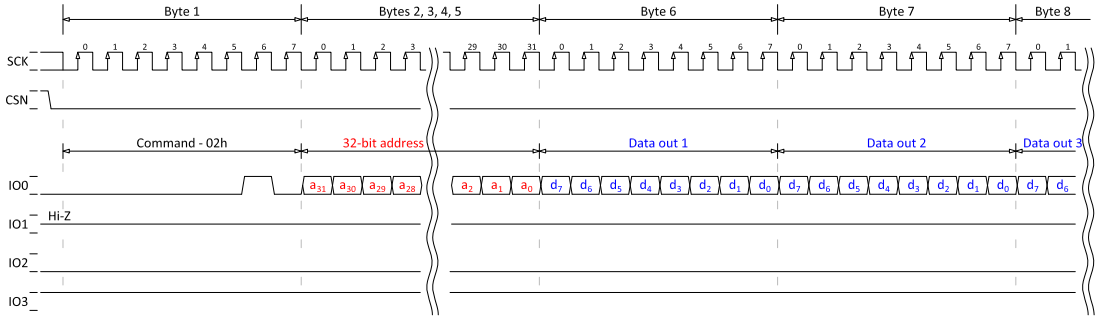


Figure 107: 32-bit PP (page program), SPIMODE = MODE0

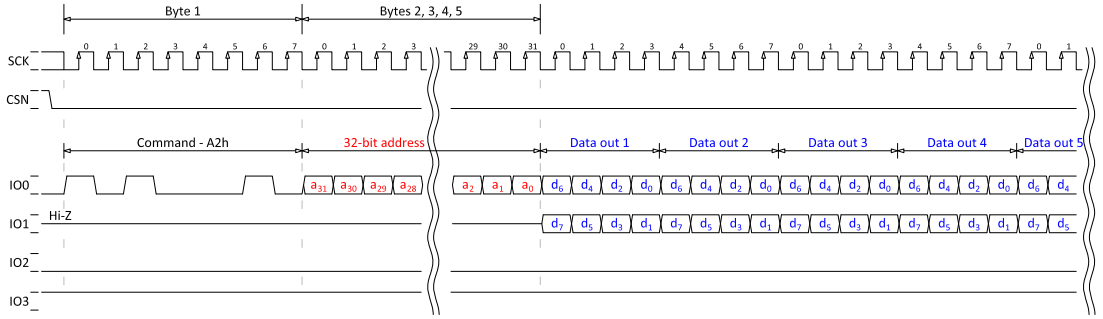


Figure 108: 32-bit PP2O (dual-page program output), SPIMODE = MODE0

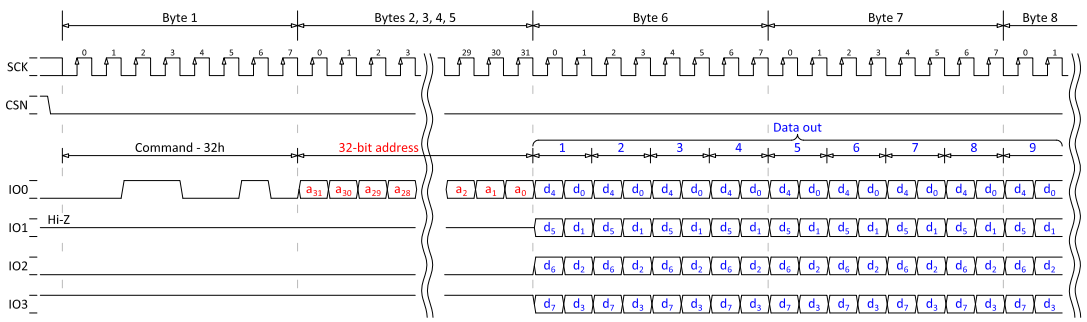


Figure 109: 32-bit PP4O (quad-page program output), SPIMODE = MODE0

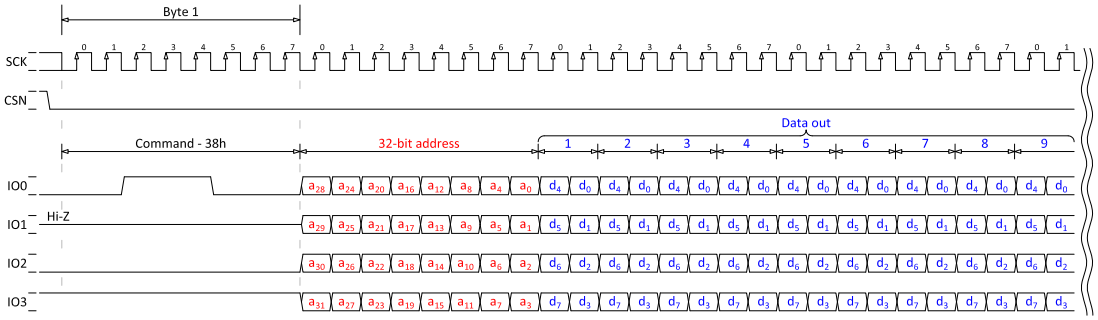


Figure 110: 32-bit PP4IO (quad page program input/output), SPIMODE = MODE0

6.19.10 Registers

Instances

Instance	Base address	Description
QSPI	0x40029000	External memory interface

Register overview

Register	Offset	Description
TASKS_ACTIVATE	0x000	Activate QSPI interface
TASKS_READSTART	0x004	Start transfer from external flash memory to internal RAM
TASKS_WRITESTART	0x008	Start transfer from internal RAM to external flash memory
TASKS_ERASESTART	0x00C	Start external flash memory erase operation
TASKS_DEACTIVATE	0x010	Deactivate QSPI interface
EVENTS_READY	0x100	QSPI peripheral is ready. This event will be generated as a response to any QSPI task.
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
ENABLE	0x500	Enable QSPI peripheral and acquire the pins selected in PSELn registers
READ.SRC	0x504	Flash memory source address
READ.DST	0x508	RAM destination address
READ.CNT	0x50C	Read transfer length
WRITE.DST	0x510	Flash destination address
WRITE.SRC	0x514	RAM source address
WRITE.CNT	0x518	Write transfer length
ERASE.PTR	0x51C	Start address of flash block to be erased

Register	Offset	Description
ERASE.LEN	0x520	Size of block to be erased.
PSEL.SCK	0x524	Pin select for serial clock SCK
PSEL.CSN	0x528	Pin select for chip select signal CSN.
PSEL.IO0	0x530	Pin select for serial data MOSI/IO0.
PSEL.IO1	0x534	Pin select for serial data MISO/IO1.
PSEL.IO2	0x538	Pin select for serial data IO2.
PSEL.IO3	0x53C	Pin select for serial data IO3.
XIPOFFSET	0x540	Address offset into the external memory for Execute in Place operation.
IFCONFIG0	0x544	Interface configuration.
IFCONFIG1	0x600	Interface configuration.
STATUS	0x604	Status register.
DPMDUR	0x614	Set the duration required to enter/exit deep power-down mode (DPM).
ADDRCONF	0x624	Extended address configuration.
CINSTRCONF	0x634	Custom instruction configuration register.
CINSTRDAT0	0x638	Custom instruction data register 0.
CINSTRDAT1	0x63C	Custom instruction data register 1.
IFTIMING	0x640	SPI interface timing.

6.19.10.1 TASKS_ACTIVATE

Address offset: 0x000

Activate QSPI interface

Triggering this task activates the external flash memory interface and initiates communication with the external memory. The READY event is generated when the activation has been completed.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	TASKS_ACTIVATE																																	
			Trigger	1																															

6.19.10.2 TASKS_READSTART

Address offset: 0x004

Start transfer from external flash memory to internal RAM

Start transfer from external flash memory to internal RAM. The READY event will be generated when transfer is complete.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	TASKS_READSTART																																	
			Trigger	1																															

6.19.10.3 TASKS_WRITESTART

Address offset: 0x008

Start transfer from internal RAM to external flash memory

Start transfer from internal RAM to external flash memory. The READY event will be generated when transfer is complete.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_WRITESTART			Start transfer from internal RAM to external flash memory
					Start transfer from internal RAM to external flash memory. The READY event will be generated when transfer is complete.
			Trigger	1	Trigger task

6.19.10.4 TASKS_ERASESTART

Address offset: 0x00C

Start external flash memory erase operation

Start external flash memory erase operation. The READY event will be generated when the erase operation has been started. Note, generation of the READY event does not imply that the erase operation is completed.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_ERASESTART			Start external flash memory erase operation
					Start external flash memory erase operation. The READY event will be generated when the erase operation has been started. Note, generation of the READY event does not imply that the erase operation is completed.
			Trigger	1	Trigger task

6.19.10.5 TASKS_DEACTIVATE

Address offset: 0x010

Deactivate QSPI interface

Deactivate QSPI interface. This task might be needed to optimize current consumption in case there are any added current consumption when QSPI interface is activated, but idle.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_DEACTIVATE			Deactivate QSPI interface
					Deactivate QSPI interface. This task might be needed to optimize current consumption in case there are any added current consumption when QSPI interface is activated, but idle.
			Trigger	1	Trigger task

6.19.10.6 EVENTS_READY

Address offset: 0x100

QSPI peripheral is ready. This event will be generated as a response to any QSPI task.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_READY			QSPI peripheral is ready. This event will be generated as a response to any QSPI task.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.19.10.7 INTEN

Address offset: 0x300

Enable or disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	READY			Enable or disable interrupt for event READY
			Disabled	0	Disable
			Enabled	1	Enable

6.19.10.8 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	READY			Write '1' to enable interrupt for event READY
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.19.10.9 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	READY																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															

6.19.10.10 ENABLE

Address offset: 0x500

Enable QSPI peripheral and acquire the pins selected in PSELn registers

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	ENABLE																																	
			Disabled	0																															
			Enabled	1																															

6.19.10.11 READ.SRC

Address offset: 0x504

Flash memory source address

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	SRC																																	

6.19.10.12 READ.DST

Address offset: 0x508

RAM destination address

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	DST																																	

6.19.10.13 READ.CNT

Address offset: 0x50C

Read transfer length

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID															A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	CNT		[1..0x3FFFF]	Read transfer length in number of bytes. The length must be a multiple of 4 bytes.

6.19.10.14 WRITE.DST

Address offset: 0x510

Flash destination address

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	DST			Word-aligned flash destination address.

6.19.10.15 WRITE.SRC

Address offset: 0x514

RAM source address

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	SRC			Word-aligned RAM source address.

6.19.10.16 WRITE.CNT

Address offset: 0x518

Write transfer length

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID															A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	CNT		[1..0x3FFFF]	Write transfer length in number of bytes. The length must be a multiple of 4 bytes.

6.19.10.17 ERASE.PTR

Address offset: 0x51C

Start address of flash block to be erased

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	PTR																																	
				Word-aligned start address of block to be erased.																															

6.19.10.18 ERASE.LEN

Address offset: 0x520

Size of block to be erased.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																		A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	LEN																																	
			4KB	0																															
			64KB	1																															
			All	2																															

6.19.10.19 PSEL.SCK

Address offset: 0x524

Pin select for serial clock SCK

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				C																										B	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW	PIN		[0..31]																															
B	RW	PORT		[0..1]																															
C	RW	CONNECT																																	
			Disconnected	1																															
			Connected	0																															

6.19.10.20 PSEL.CSN

Address offset: 0x528

Pin select for chip select signal CSN.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				C																										B	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW	PIN		[0..31]																															
B	RW	PORT		[0..1]																															
C	RW	CONNECT																																	
			Disconnected	1																															
			Connected	0																															

6.19.10.21 PSEL.IO0

Address offset: 0x530

Pin select for serial data MOSI/IO0.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.19.10.22 PSEL.IO1

Address offset: 0x534

Pin select for serial data MISO/IO1.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.19.10.23 PSEL.IO2

Address offset: 0x538

Pin select for serial data IO2.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.19.10.24 PSEL.IO3

Address offset: 0x53C

Pin select for serial data IO3.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.19.10.25 XIPOFFSET

Address offset: 0x540

Address offset into the external memory for Execute in Place operation.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	XIPOFFSET			Address offset into the external memory for Execute in Place operation. Value must be a multiple of 4.

6.19.10.26 IFCONFIG0

Address offset: 0x544

Interface configuration.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D C B B B A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	READOC			Configure number of data lines and opcode used for reading.
			FASTREAD	0	Single data line SPI. FAST_READ (opcode 0x0B).
			READ2O	1	Dual data line SPI. READ2O (opcode 0x3B).
			READ2IO	2	Dual data line SPI. READ2IO (opcode 0xBB).
			READ4O	3	Quad data line SPI. READ4O (opcode 0x6B).
			READ4IO	4	Quad data line SPI. READ4IO (opcode 0xEB).
B	RW	WRITEOC			Configure number of data lines and opcode used for writing.
			PP	0	Single data line SPI. PP (opcode 0x02).
			PP2O	1	Dual data line SPI. PP2O (opcode 0xA2).
			PP4O	2	Quad data line SPI. PP4O (opcode 0x32).
			PP4IO	3	Quad data line SPI. PP4IO (opcode 0x38).
C	RW	ADDRMODE			Addressing mode.
			24BIT	0	24-bit addressing.
			32BIT	1	32-bit addressing.
D	RW	DPMENABLE			Enable deep power-down mode (DPM) feature.
			Disable	0	Disable DPM feature.
			Enable	1	Enable DPM feature.
E	RW	PPSIZE			Page size for commands PP, PP2O, PP4O and PP4IO.
			256Bytes	0	256 bytes.
			512Bytes	1	512 bytes.

6.19.10.27 IFCONFIG1

Address offset: 0x600

Interface configuration.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				D	D	D	D			C	B																	A	A	A	A	A	A	A	A
Reset 0x00040480				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	SCKDELAY		[0..255]																															
				Minimum amount of time that the CSN pin must stay high before it can go low again. Value is specified in number of 16 MHz periods (62.5 ns).																																
B	RW	DPMEN																																	
				Enter/exit deep power-down mode (DPM) for external flash memory.																																
			Exit	0																															
				Exit DPM.																																
			Enter	1																															
				Enter DPM.																																
C	RW	SPIMODE																																	
				Select SPI mode.																																
			MODE0	0																															
				Mode 0: Data are captured on the clock rising edge and data is output on a falling edge. Base level of clock is 0 (CPOL=0, CPHA=0).																																
			MODE3	1																															
				Mode 3: Data are captured on the clock falling edge and data is output on a rising edge. Base level of clock is 1 (CPOL=1, CPHA=1).																																
D	RW	SCKFREQ		[0..15]																															
				SCK frequency is given as 32 MHz / (SCKFREQ + 1).																																

6.19.10.28 STATUS

Address offset: 0x604

Status register.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				C	C	C	C	C	C	C	C																					B	A		
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	R	DPM																																	
				Deep power-down mode (DPM) status of external flash.																																
			Disabled	0																															
				External flash is not in DPM.																																
			Enabled	1																															
				External flash is in DPM.																																
B	R	READY																																	
				Ready status.																																
			READY	1																															
				QSPI peripheral is ready. It is allowed to trigger new tasks, writing custom instructions or enter/exit DPM.																																
			BUSY	0																															
				QSPI peripheral is busy. It is not allowed to trigger any new tasks, writing custom instructions or enter/exit DPM.																																
C	R	SREG																																	
				Value of external flash device Status Register. When the external flash has two bytes status register this field includes the value of the low byte.																																

6.19.10.29 DPMDUR

Address offset: 0x614

Set the duration required to enter/exit deep power-down mode (DPM).

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B B B B B B B B B B B B B B B B A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	ENTER		[0..0xFFFF]	Duration needed by external flash to enter DPM. Duration is given as ENTER * 256 * 62.5 ns.
B	RW	EXIT		[0..0xFFFF]	Duration needed by external flash to exit DPM. Duration is given as EXIT * 256 * 62.5 ns.

6.19.10.30 ADDRCONF

Address offset: 0x624

Extended address configuration.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F E D D C C C C C C C C B B B B B B B B A A A A A A A A
Reset 0x000000B7				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 1 1 0 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	OPCODE		[0xFF..0]	Opcode that enters the 32-bit addressing mode.
B	RW	BYTE0		[0xFF..0]	Byte 0 following opcode.
C	RW	BYTE1		[0xFF..0]	Byte 1 following byte 0.
D	RW	MODE			Extended addressing mode.
			NoInstr	0	Do not send any instruction.
			Opcode	1	Send opcode.
			OpByte0	2	Send opcode, byte0.
			All	3	Send opcode, byte0, byte1.
E	RW	WIPWAIT			Wait for write complete before sending command.
			Disable	0	No wait.
			Enable	1	Wait.
F	RW	WREN			Send WREN (write enable opcode 0x06) before instruction.
			Disable	0	Do not send WREN.
			Enable	1	Send WREN.

6.19.10.31 CINSTRCONF

Address offset: 0x634

Custom instruction configuration register.

A new custom instruction is sent every time this register is written. The READY event will be generated when the custom instruction has been sent.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				H G F E D C B B B A A A A A A A A
Reset 0x00002000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	OPCODE		[0..255]	Opcode of Custom instruction.
B	RW	LENGTH			Length of custom instruction in number of bytes.
			1B	1	Send opcode only.
			2B	2	Send opcode, CINSTRDAT0.BYTE0.
			3B	3	Send opcode, CINSTRDAT0.BYTE0 -> CINSTRDAT0.BYTE1.
			4B	4	Send opcode, CINSTRDAT0.BYTE0 -> CINSTRDAT0.BYTE2.
			5B	5	Send opcode, CINSTRDAT0.BYTE0 -> CINSTRDAT0.BYTE3.
			6B	6	Send opcode, CINSTRDAT0.BYTE0 -> CINSTRDAT1.BYTE4.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				H G F E D C B B B B A A A A A A A A
Reset 0x00002000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			7B	7	Send opcode, CINSTRDAT0.BYTE0 -> CINSTRDAT1.BYTE5.
			8B	8	Send opcode, CINSTRDAT0.BYTE0 -> CINSTRDAT1.BYTE6.
			9B	9	Send opcode, CINSTRDAT0.BYTE0 -> CINSTRDAT1.BYTE7.
C	RW	LIO2		[0..1]	Level of the IO2 pin (if connected) during transmission of custom instruction.
D	RW	LIO3		[0..1]	Level of the IO3 pin (if connected) during transmission of custom instruction.
E	RW	WIPWAIT			Wait for write complete before sending command.
			Disable	0	No wait.
			Enable	1	Wait.
F	RW	WREN			Send WREN (write enable opcode 0x06) before instruction.
			Disable	0	Do not send WREN.
			Enable	1	Send WREN.
G	RW	LFEN			Enable long frame mode. When enabled, a custom instruction transaction has to be ended by writing the LFSTOP field.
			Disable	0	Long frame mode disabled
			Enable	1	Long frame mode enabled
H	RW	LFSTOP			Stop (finalize) long frame transaction
			Stop	1	Stop

6.19.10.32 CINSTRDAT0

Address offset: 0x638

Custom instruction data register 0.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				D D D D D D D D C C C C C C C C B B B B B B B B A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	BYTE0		[0..0xFF]	Data byte 0
B	RW	BYTE1		[0..0xFF]	Data byte 1
C	RW	BYTE2		[0..0xFF]	Data byte 2
D	RW	BYTE3		[0..0xFF]	Data byte 3

6.19.10.33 CINSTRDAT1

Address offset: 0x63C

Custom instruction data register 1.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				D D D D D D D D C C C C C C C C B B B B B B B B A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	BYTE4		[0..0xFF]	Data byte 4
B	RW	BYTE5		[0..0xFF]	Data byte 5
C	RW	BYTE6		[0..0xFF]	Data byte 6
D	RW	BYTE7		[0..0xFF]	Data byte 7

6.19.10.34 IFTIMING

Address offset: 0x640

SPI interface timing.

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A
Reset 0x00000200	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	RXDELAY		[7..0]	Timing related to sampling of the input serial data. The value of RXDELAY specifies the number of 64 MHz cycles (15.625 ns) delay from the the rising edge of the SPI Clock (SCK) until the input serial data is sampled. As en example, if set to 0 the input serial data is sampled on the rising edge of SCK.

6.19.11 Electrical specification

6.19.11.1 Timing specification

Symbol	Description	Min.	Typ.	Max.	Units
$F_{QSPI,CLK}$	SCK frequency			32	MHz
$DC_{QSPI,CLK}$	SCK duty cycle				%
$F_{QSPI,XIP,16}$	XIP fetch frequency for 16 bit instructions			8	MHz
$F_{QSPI,XIP,32}$	XIP fetch frequency for 32 bit instructions			4	MHz

6.20 RADIO — 2.4 GHz radio

The 2.4 GHz radio transceiver is compatible with multiple radio standards such as 1 Mbps, 2 Mbps and long range *Bluetooth* low energy. IEEE 802.15.4 250 kbps mode is fully supported as well as Nordic's proprietary 1 Mbps and 2 Mbps modes of operation.

Listed here are main features for the RADIO:

- Multidomain 2.4 GHz radio transceiver:
  - 1 Mbps, 2 Mbps and long range (125 kbps and 500 kbps mode) *Bluetooth* low energy modes
  - 250 kbps IEEE 802.15.4 mode
  - 1 Mbps and 2 Mbps Nordic proprietary modes
- Best in class link budget and low power operation
- Efficient data interface with EasyDMA support
- Automatic address filtering and pattern matching

EasyDMA in combination with an automated packet assembler and packet disassembler, and an automated CRC generator and CRC checker, make it very easy to configure and use the RADIO. See RADIO block diagram on page 600 for details.

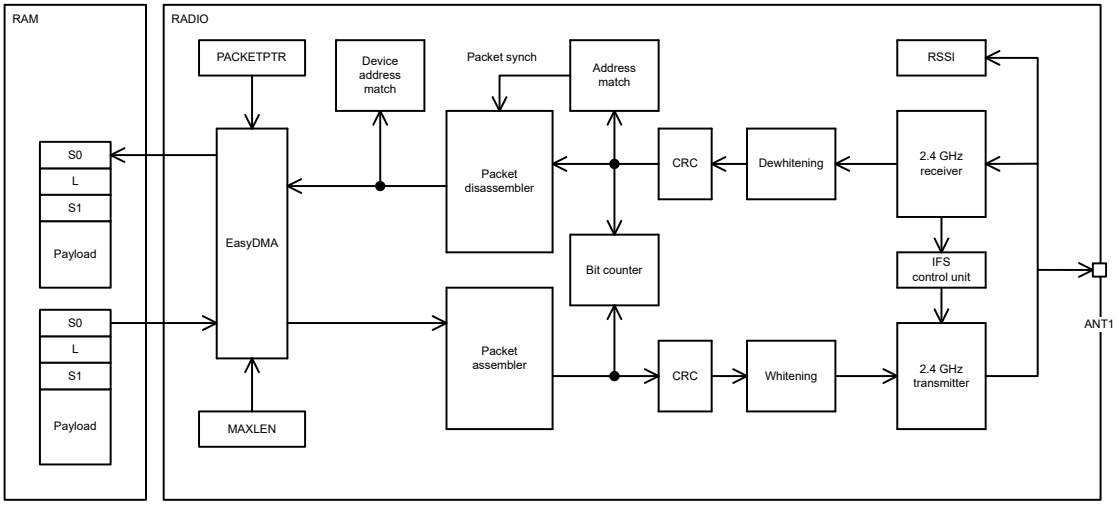


Figure 111: RADIO block diagram

The RADIO includes a device address match unit and an interframe spacing control unit that can be utilized to simplify address whitelisting and interframe spacing respectively in *Bluetooth* low energy and similar applications.

The RADIO also includes a received signal strength indicator (RSSI) and a bit counter. The bit counter generates events when a preconfigured number of bits have been sent or received by the RADIO.

6.20.1 Packet configuration

A radio packet contains the following fields: PREAMBLE, ADDRESS, S0, LENGTH, S1, PAYLOAD and CRC.

The content of a RADIO packet is illustrated in On air packet layout on page 600. The RADIO sends the different fields in the packet in the order they are illustrated below, from left to right:

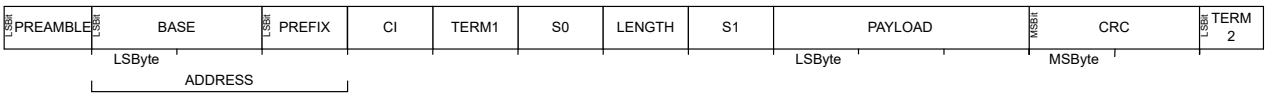


Figure 112: On air packet layout

Not shown in the figure is the static payload add-on (the length of which is defined in STATLEN, and which is 0 bytes long in a standard BLE packet). The static payload add-on is sent between PAYLOAD and CRC fields. The radio sends the different fields in the packet in the order they are illustrated above, from left to right. The preamble will be sent with least significant bit first on air.

Not shown in the figure above is the static payload add-on (the length of which is defined in PCNF1.STATLEN, and which is 0 bytes long in a standard BLE packet). The static payload add-on is sent between the PAYLOAD and CRC fields.

PREAMBLE is sent with least significant bit first on-air. The size of the PREAMBLE depends on the mode selected in the MODE register:

- The PREAMBLE is one byte for MODE = Ble_1Mbit as well as all Nordic proprietary operating modes (MODE = Nrf_1Mbit and MODE = Nrf_2Mbit), and the PLEN field in the PCNF0 register has to be set accordingly. If the first bit of the ADDRESS is 0 the preamble will be set to 0xAA otherwise the PREAMBLE will be set to 0x55.
- For MODE = Ble_2Mbit the PREAMBLE has to be set to 2 byte long through the PLEN field in the PCNF0 register. If the first bit of the ADDRESS is 0 the preamble will be set to 0xAAAA otherwise the PREAMBLE will be set to 0x5555.
- For MODE = Ble_LR125Kbit and MODE = Ble_LR500Kbit the PREAMBLE is 10 repetitions of 0x3C.

- For MODE = Ieee802154_250Kbit the PREAMBLE is 4 bytes long and set to all zeros.

Radio packets are stored in memory inside instances of a radio packet data structure as illustrated in In-RAM representation of radio packet - S0, LENGTH and S1 are optional on page 601. The PREAMBLE, ADDRESS, CI, TERM1, TERM2 and CRC fields are omitted in this data structure.

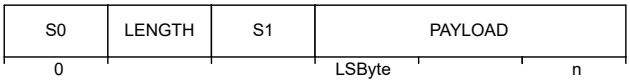


Figure 113: In-RAM representation of radio packet - S0, LENGTH and S1 are optional

The byte ordering on air is always least significant byte first for the ADDRESS and PAYLOAD fields and most significant byte first for the CRC field. The ADDRESS fields are always transmitted and received least significant bit first on air. The CRC field is always transmitted and received most significant bit first. The bit-endian, i.e. the order in which the bits are sent and received, of the S0, LENGTH, S1 and PAYLOAD fields can be configured via the ENDIAN in PCNF1.

The sizes of the S0, LENGTH and S1 fields can be individually configured via S0LEN, LFLEN and S1LEN in PCNF0 respectively. If any of these fields are configured to be less than 8 bits long, the least significant bits of the fields are used.

If S0, LENGTH or S1 are specified with zero length their fields will be omitted in memory, otherwise each field will be represented as a separate byte, regardless of the number of bits in their on air counterpart.

Independent of the configuration of MAXLEN, the combined length of S0, LENGTH, S1 and PAYLOAD cannot exceed 258 bytes.

6.20.2 Address configuration

The on air radio ADDRESS field is composed of two parts, the base address field and the address prefix field.

The size of the base address field is configurable via BALEN in PCNF1. The base address is truncated from the least significant byte if the BALEN is less than 4. See Definition of logical addresses on page 601.

Logical address	Base address	Prefix byte
0	BASE0	PREFIX0.AP0
1	BASE1	PREFIX0.AP1
2	BASE1	PREFIX0.AP2
3	BASE1	PREFIX0.AP3
4	BASE1	PREFIX1.AP4
5	BASE1	PREFIX1.AP5
6	BASE1	PREFIX1.AP6
7	BASE1	PREFIX1.AP7

Table 35: Definition of logical addresses

The on air addresses are defined in the BASEn and PREFIXn registers, and it is only when writing these registers the user will have to relate to actual on air addresses. For other radio address registers such as the TXADDRESS, RXADDRESSES and RXMATCH registers, logical radio addresses ranging from 0 to 7 are being used. The relationship between the on air radio addresses and the logical addresses is described in Definition of logical addresses on page 601.

6.20.3 Data whitening

The RADIO is able to do packet whitening and de-whitening.

See WHITEEN in PCNF1 register for how to enable whitening. When enabled, whitening and de-whitening will be handled by the RADIO automatically as packets are sent and received.

The whitening word is generated using polynomial $g(D) = D^7 + D^4 + 1$, which then is XORed with the data packet that is to be whitened, or de-whitened. See the figure below.

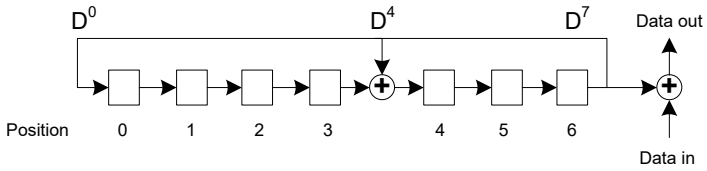


Figure 114: Data whitening and de-whitening

Whitening and de-whitening will be performed over the whole packet (except for the preamble and the address field).

The linear feedback shift register, illustrated in Data whitening and de-whitening on page 602 can be initialised via the DATAWHITEIV register.

6.20.4 CRC

The CRC generator in the RADIO calculates the CRC over the whole packet excluding the preamble. If desirable, the address field can be excluded from the CRC calculation as well

See CRCCNF register for more information.

The CRC polynomial is configurable as illustrated in CRC generation of an n bit CRC on page 602 where bit 0 in the CRCPOLY register corresponds to X^0 and bit 1 corresponds to X^1 etc. See CRCPOLY for more information.

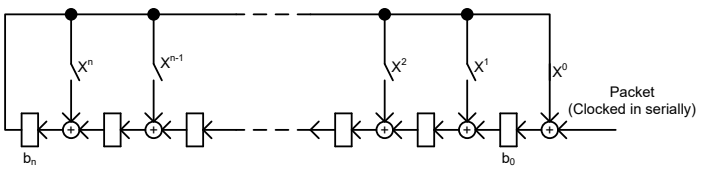


Figure 115: CRC generation of an n bit CRC

As illustrated in CRC generation of an n bit CRC on page 602, the CRC is calculated by feeding the packet serially through the CRC generator. Before the packet is clocked through the CRC generator, the CRC generator's latches b_0 through b_n will be initialized with a predefined value specified in the CRCINIT register. When the whole packet is clocked through the CRC generator, latches b_0 through b_n will hold the resulting CRC. This value will be used by the RADIO during both transmission and reception but it is not available to be read by the CPU at any time. A received CRC can however be read by the CPU via the RXCRC register independent of whether or not it has passed the CRC check.

The length (n) of the CRC is configurable, see CRCCNF for more information.

After the whole packet including the CRC has been received, the RADIO will generate a CRCOK event if no CRC errors were detected, or alternatively generate a CRCERROR event if CRC errors were detected.

The status of the CRC check can be read from the CRCSTATUS register after a packet has been received.

6.20.5 Radio states

Tasks and events are used to control the operating state of the RADIO.

The RADIO can enter the states described the table below.

State	Description
DISABLED	No operations are going on inside the radio and the power consumption is at a minimum
RXRU	The radio is ramping up and preparing for reception
RXIDLE	The radio is ready for reception to start
RX	Reception has been started and the addresses enabled in the RXADDRESSES register are being monitored
TXRU	The radio is ramping up and preparing for transmission
TXIDLE	The radio is ready for transmission to start
TX	The radio is transmitting a packet
RXDISABLE	The radio is disabling the receiver
TXDISABLE	The radio is disabling the transmitter

Table 36: RADIO state diagram

An overview state diagram for the RADIO is illustrated in Radio states on page 603.

Note: The END to START shortcut should not be used with Ble_LR125Kbit, Ble_LR500Kbit and Ieee802154_250Kbit modes. Rather the PHYEND to START shortcut.

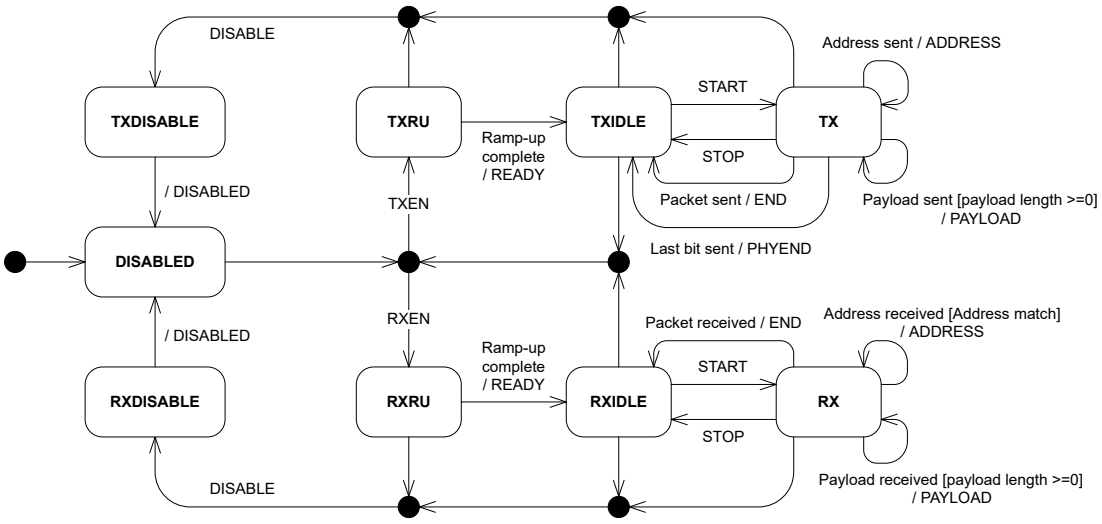


Figure 116: Radio states

This figure shows how the tasks and events relate to the RADIO's operation. The RADIO does not prevent a task from being triggered from the wrong state. If a task is triggered from the wrong state, for example if the RXEN task is triggered from the RXDISABLE state, this may lead to incorrect behaviour. As illustrated in Radio states on page 603, the PAYLOAD event is always generated even if the payload is zero.

6.20.6 Transmit sequence

Before the RADIO is able to transmit a packet, it must first ramp-up in TX mode.

See TXRU in Radio states on page 603 and Transmit sequence on page 604. A TXRU ramp-up sequence is initiated when the TXEN task is triggered. After the radio has successfully ramped up it will generate the READY event indicating that a packet transmission can be initiate. A packet transmission is initiated by triggering the START task. As illustrated in Radio states on page 603 the START task can first be triggered after the RADIO has entered into the TXIDLE state.

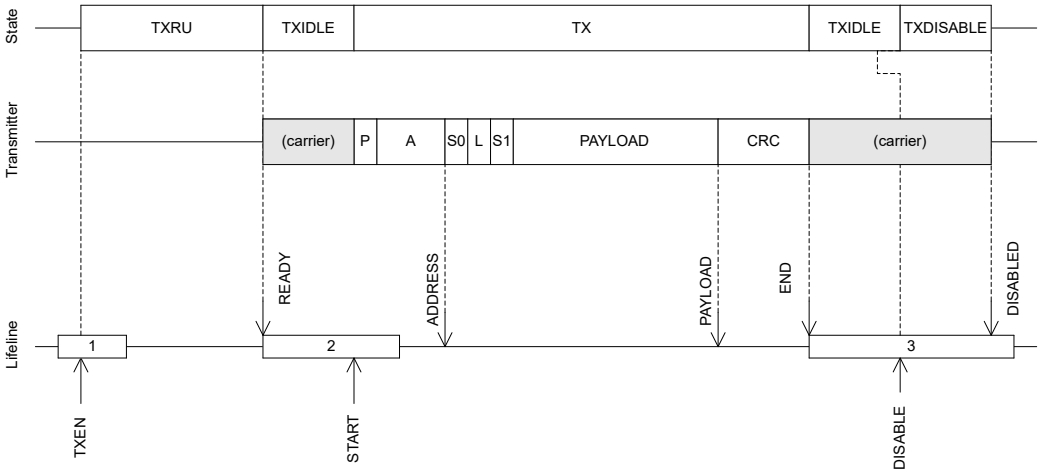


Figure 117: Transmit sequence

Transmit sequence on page 604 illustrates a single packet transmission where the CPU manually triggers the different tasks needed to control the flow of the RADIO, i.e. no shortcuts are used. If shortcuts are not used, a certain amount of delay caused by CPU execution is expected between READY and START, and between END and DISABLE. As illustrated in Transmit sequence on page 604 the RADIO will by default transmit '1's between READY and START, and between END and DISABLED. What is transmitted can be programmed through the DTX field in the MODECNF0 register.

A slightly modified version of the transmit sequence from Transmit sequence on page 604 is illustrated in Transmit sequence using shortcuts to avoid delays on page 604 where the RADIO is configured to use shortcuts between READY and START, and between END and DISABLE, which means that no delay is introduced.

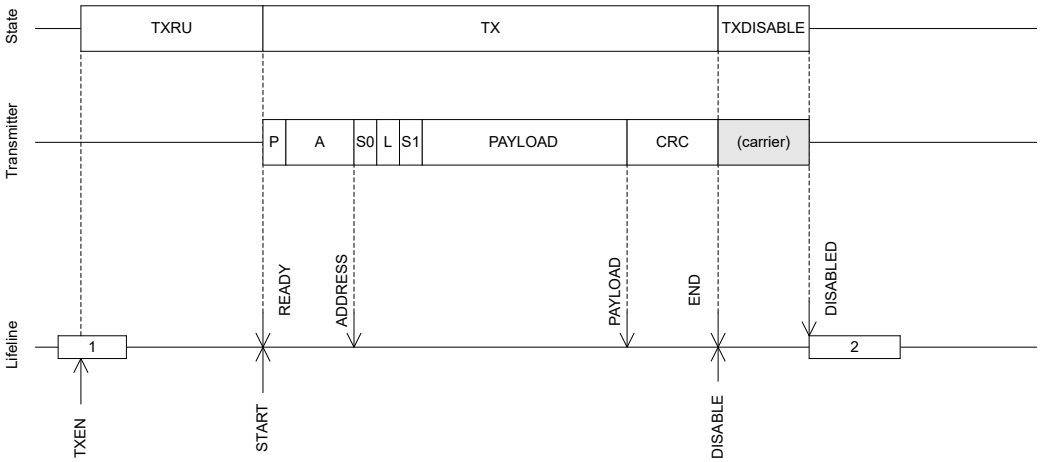


Figure 118: Transmit sequence using shortcuts to avoid delays

The RADIO is able to send multiple packets one after the other without having to disable and re-enable the RADIO between packets, this is illustrated in Transmission of multiple packets on page 605.

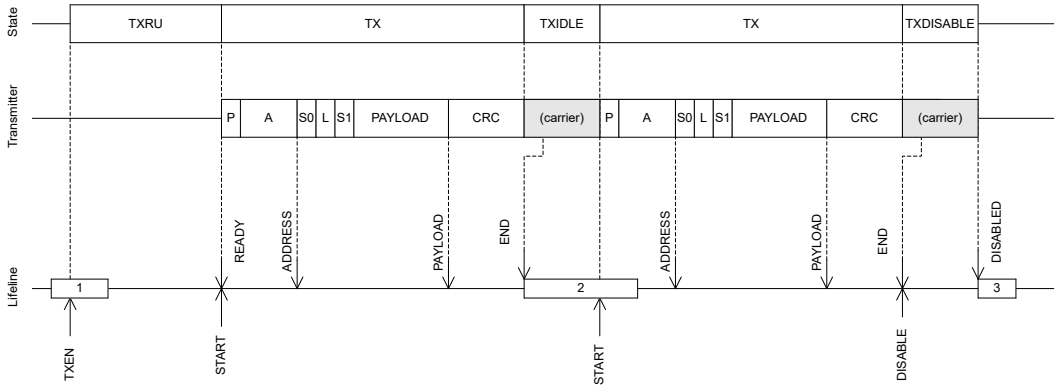


Figure 119: Transmission of multiple packets

6.20.7 Receive sequence

Before the RADIO is able to receive a packet, it must first ramp up in RX mode

See RXRU in Radio states on page 603 and Receive sequence on page 605.

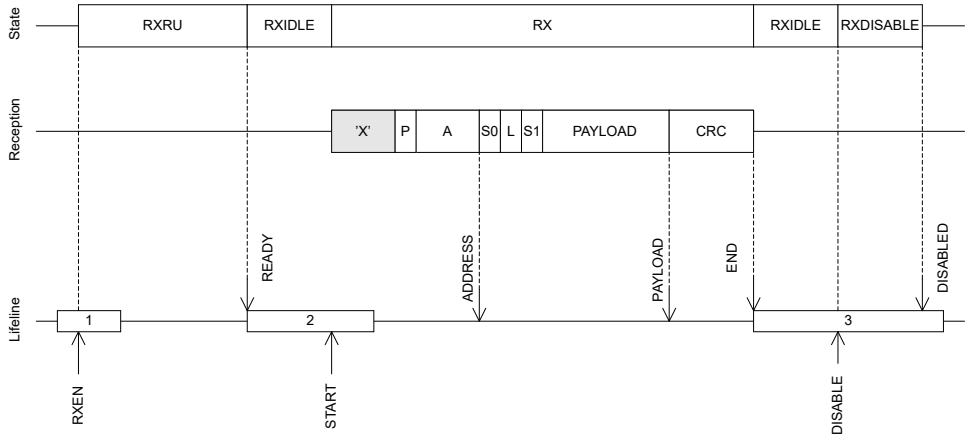


Figure 120: Receive sequence

An RXRU ramp up sequence is initiated when the RXEN task is triggered. After the radio has successfully ramped up it will generate the READY event indicating that a packet reception can be initiated. A packet reception is initiated by triggering the START task. As illustrated in Radio states on page 603 the START task can first be triggered after the RADIO has entered into the RXIDLE state.

Receive sequence on page 605 illustrates a single packet reception where the CPU manually triggers the different tasks needed to control the flow of the RADIO, i.e. no shortcuts are used. If shortcuts are not used, a certain amount of delay caused by CPU execution is expected between READY and START, and between END and DISABLE. As illustrated Receive sequence on page 605 the RADIO will be listening and possibly receiving undefined data, represented with an 'X', from START and until a packet with valid preamble (P) is received.

A slightly modified version of the receive sequence from Receive sequence on page 605 is illustrated in Receive sequence using shortcuts to avoid delays on page 606 where the RADIO is configured to use shortcuts between READY and START, and between END and DISABLE, which means that no delay is introduced.

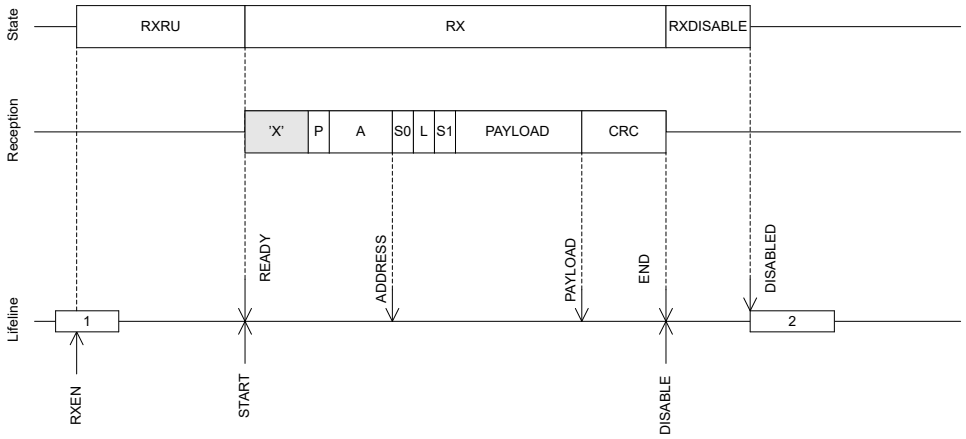


Figure 121: Receive sequence using shortcuts to avoid delays

The RADIO is able to receive multiple packets one after the other without having to disable and re-enable the RADIO between packets as illustrated in Reception of multiple packets on page 606.

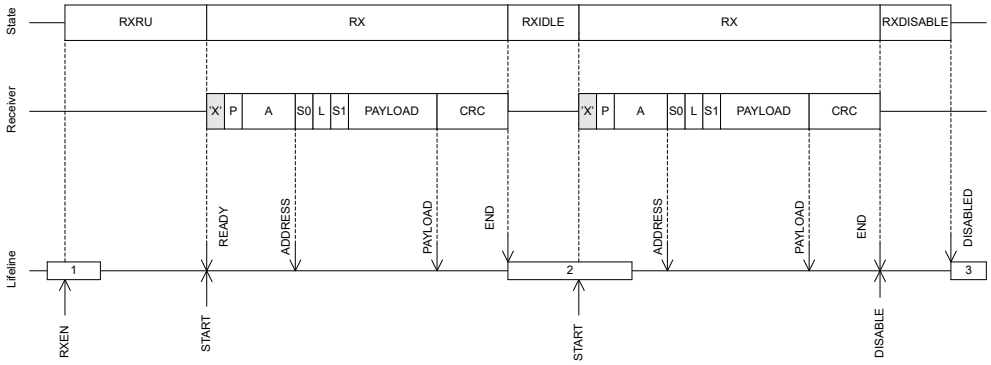


Figure 122: Reception of multiple packets

6.20.8 Received signal strength indicator (RSSI)

The RADIO implements a mechanism for measuring the power in the received signal. This feature is called received signal strength indicator (RSSI).

The RSSI is measured continuously and the value filtered using a single-pole IIR filter. After a signal level change, the RSSI will settle after approximately $RSSI_{SETTLE}$.

Sampling of the received signal strength is started by using the RSSISTART task. The sample can be read from the RSSISAMPLE register.

The sample period of the RSSI is defined by $RSSI_{PERIOD}$. The RSSISAMPLE will hold the filtered received signal strength after this sample period.

For the RSSI sample to be valid, the RADIO has to be enabled in receive mode (RXEN task) and the reception has to be started (READY event followed by START task).

6.20.9 Interframe spacing

Interframe spacing is the time interval between two consecutive packets.

It is defined as the time, in microseconds, from the end of the last bit of the previous packet received and to the start of the first bit of the subsequent packet that is transmitted. The RADIO is able to enforce this interval, as specified in the TIFS register, as long as the TIFS is not specified to be shorter than the RADIO's turnaround time, i.e. the time needed to switch off the receiver, and then switch the transmitter back on. The TIFS register can be written any time before the last bit on air is received.

This timing is illustrated in the figure below.

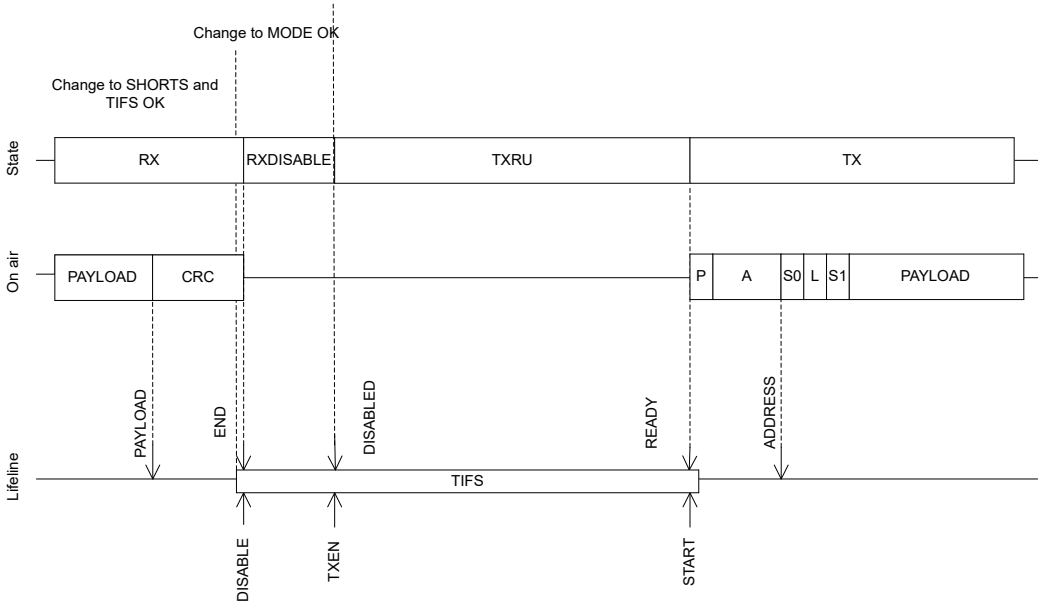


Figure 123: IFS timing detail

As illustrated, the TIFS duration starts after the last bit on air (just before the END event), and elapses with first bit being transmitted on air (just after READY event).

TIFS is only enforced if END_DISABLE and DISABLED_TXEN or END_DISABLE and DISABLED_RXEN shortcuts are enabled. TIFS is qualified for use in BLE_1MBIT, BLE_2MBIT, BLE_LR125KBIT, BLE_LR500KBIT and Ieee802154_250Kbit mode using the default ramp-up mode. SHORTS and TIFS are not double-buffered, and can be updated at any point in time before the last bit on air is received. The MODE register is double-buffered and sampled at the TXEN or RXEN task.

6.20.10 Device address match

The device address match feature is tailored for address whitelisting in a *Bluetooth* low energy and similar implementations.

This feature enables on-the-fly device address matching while receiving a packet on air. This feature only works in receive mode and as long as RADIO is configured for little endian, see PCNF1.ENDIAN.

The device address match unit assumes that the 48 first bits of the payload is the device address and that bit number 6 in S0 is the TxAdd bit. See the *Bluetooth* Core Specification for more information about device addresses, TxAdd and whitelisting.

The RADIO is able to listen for eight different device addresses at the same time. These addresses are specified in a DAB/DAP register pair, one pair per address, in addition to a TxAdd bit configured in the DACNF register. The DAB register specifies the 32 least significant bits of the device address, while the DAP register specifies the 16 most significant bits of the device address.

Each of the device addresses can be individually included or excluded from the matching mechanism. This is configured in the DACNF register.

6.20.11 Bit counter

The RADIO implements a simple counter that can be configured to generate an event after a specific number of bits have been transmitted or received.

By using shortcuts, this counter can be started from different events generated by the RADIO and hence count relative to these.

The bit counter is started by triggering the BCSTART task, and stopped by triggering the BCSTOP task. A BCMATCH event will be generated when the bit counter has counted the number of bits specified in the BCC register. The bit counter will continue to count bits until the DISABLED event is generated or until the BCSTOP task is triggered. The CPU can therefore, after a BCMATCH event, reconfigure the BCC value for new BCMATCH events within the same packet.

The bit counter can only be started after the RADIO has received the ADDRESS event.

The bit counter will stop and reset on BCSTOP, STOP, END and DISABLE tasks.

The figure below illustrates how the bit counter can be used to generate a BCMATCH event in the beginning of the packet payload, and again generate a second BCMATCH event after sending 2 bytes (16 bits) of the payload.

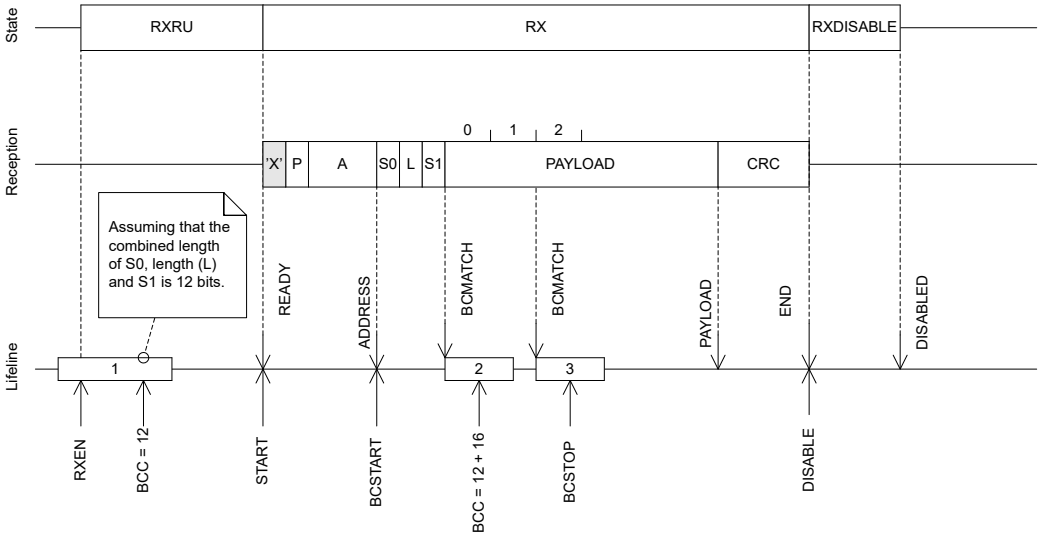


Figure 124: Bit counter example

6.20.12 IEEE 802.15.4 operation

With the MODE=Ieee802154_250kbit the radio module will comply with the IEEE 802.15.4-2006 standard implementing its 250 kbps 2450MHz O-QPSK PHY.

The IEEE 802.15.4 standard differs from Nordic's proprietary and *Bluetooth* low energy modes. Obvious differences are modulation scheme and channel structure, but also packet structure, security and medium access control.

The main features of the IEEE 802.15.4 mode are:

- Ultra low power 250 kbps 2450MHz IEEE 802.15.4-2006 compliant link
- Clear channel assessment
- Energy detection scan
- CRC generation

6.20.12.1 Packet structure

The IEEE 802.15.4 standard defines an on the air frame/packet that is different from what is used in BLE mode.

The following figure provides an overview of the physical frame structure and its timing:

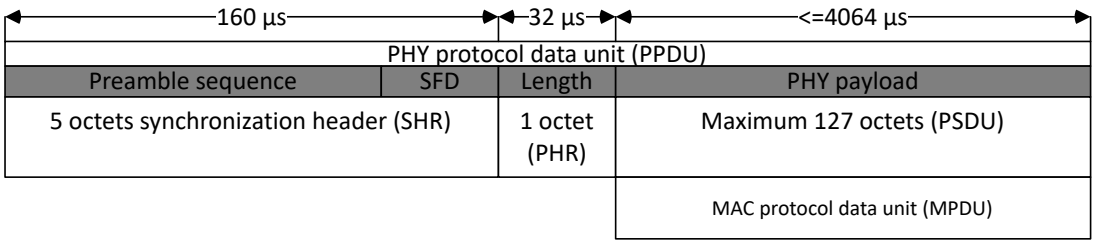


Figure 125: IEEE 802.15.4 frame format - PHY layer frame structure (PPDU)

The standard uses the term octet as storage unit for 8 bits within the PPDU. For timing, the value symbol is used, and it has the duration of 16 µs.

The total usable payload (PSDU) is 127 octets, but when CRC is being used, this is reduced to 125 octets of usable payload.

The preamble sequence consists of four octets that are all zero. These are used for the radio receiver to synchronize on. Following the four octets is a single octet named start of frame delimiter (SFD) with a fixed value of 0xA7. The user can program an alternative SFD through the SFD register. This feature is provided for an initial level of frame filtering for those who choose non-standard compliance. It is a valuable feature when operating in a congested or private network. The preamble sequence and the SFD are generated by the radio module, and are not programmed by the user into the frame buffer.

The PHY header (PHR) is a single octet following the synchronization header (SHR). The least significant seven bits denote the frame length of the following PSDU. The most significant bit is reserved and is set to zero for frames that are standard compliant. The radio module will report all eight bits and it can potentially be used to carry some information. The PHR is the first byte that will be written to the frame data memory pointed to by PACKETPTR. Frames with zero length will be discarded, and the FRAMESTART event will not be generated in this case.

The next N octets will carry the data of the PHY packet, where N equals the value of the PHR. For an implementation also using the IEEE 802.15.4 MAC layer, the PHY data will be a MAC frame of N-2 octets since two octets will occupy a CRC field.

An IEEE 802.15.4 MAC frame will always consist of a header (the frame control field (FCF), sequence number and addressing fields), a payload, and the 16-bit frame control sequence (FCS), as as illustrated in the figure below.

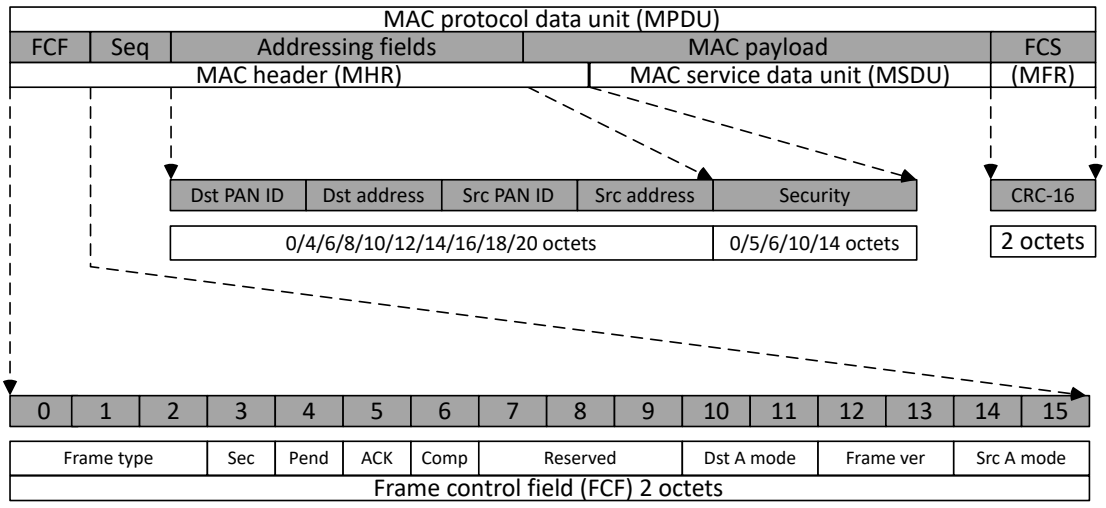


Figure 126: IEEE 802.15.4 frame format - MAC layer frame structure (MPDU)

The two FCF octets contain information about what type of frame this is, what addressing it uses, and other control flags. This field is decoded when using the assisted operating modes offered by the radio.

The sequence number is a single octet in size and is unique for a frame. It will be used in the associated acknowledgement frame sent upon successful frame reception.

The addressing field can be zero (acknowledgement frame) or up to 20 octets in size. The field is used to direct packets to the correct recipient as well as denoting its origin. IEEE 802.15.4 bases it's addressing on networks being organized in PANs with 16-bit identifier and nodes having a 16-bit or 64-bit address. In the assisted receive mode, these parameters are analyzed for address matching and acknowledgement.

The MAC payload carries the data of the next higher layer, or in the case of a MAC command frame information used by the MAC layer itself.

The two last octets contain the 16-bit ITU-T CRC. The FCS is calculated over the MAC header (MHR) and MAC payload (MSDU) parts of the frame. This field is calculated automatically when sending a frame, or indicated in the CRCSTATUS register when a frame is received. This feature is taken care of autonomously, by the CRC module (if configured).

6.20.12.2 Operating frequencies

The IEEE 802.15.4 standard defines 16 channels [11 - 26] of 5 MHz each in the 2450 MHz frequency band.

The FREQUENCY register of the radio module must be programmed according to table below for correct operation on the center frequency defined for each channel.

IEEE 802.15.4 channel	Center frequency (MHz)	FREQUENCY setting
Channel 11	2405	5
Channel 12	2410	10
Channel 13	2415	15
Channel 14	2420	20
Channel 15	2425	25
Channel 16	2430	30
Channel 17	2435	35
Channel 18	2440	40
Channel 19	2445	45
Channel 20	2450	50
Channel 21	2455	55
Channel 22	2460	60
Channel 23	2465	65
Channel 24	2470	70
Channel 25	2475	75
Channel 26	2480	80

Table 37: IEEE 802.15.4 center frequency definition

6.20.12.3 Energy detection (ED)

The IEEE 802.15.4 standard requires that it is possible to sample the received signal power within the bandwidth of a channel for the purpose of determining presence of activity.

There should be no attempt made to decode the signals on the channel, and this is done by disabling the shortcut between READY event and START task before putting the radio in receive mode. The energy detection (ED) measurement time where RSSI samples are averaged over is 8 symbol periods (128 µs). The standard further specifies the measurement to be a number between 0 and 0xFF - where 0 shall indicate received power less than 10 dB above the selected receiver sensitivity. The power range of the ED values must be at least 40 dB with a linear mapping with accuracy of ±6 dB. See section *6.9.7 Receiver ED* in the IEEE 802.15.4 standard for further details. An example of an ED scan is given below.

Below is a code snippet showing how to perform a single energy detection measurement and convert to IEEE 802.15.4 scale.

```
#define ED_RSSISCALE 4 // From electrical specifications
uint8_t sample_ed(void)
{
    int val;
    NRF_RADIO->TASKS_EDSTART = 1; // Start
    while (NRF_RADIO->EVENTS_EDEND != 1) {
        // CPU can sleep here or do something else
        // Use of interrupts are encouraged
        }
    val = NRF_RADIO->EDSAMPLE; // Read level
    return (uint8_t)(val>63 ? 255 : val*ED_RSSISCALE); // Convert to IEEE 802.15.4 scale
}
```

For scaling between hardware value and dBm, see Conversion between hardware value and dBm on page 612.

It is the mlme-scan.req primitive of the MAC layer that is using the ED measurement to detect channels where there might be wireless activity. To assist this primitive a taylored mode of operation is available where the ED measurement runs for a defined number of iterations where it keeps track of the maximum ED level. This is enganged by writing the EDCNT register to a value different from 0, it will then run the specified number of iterations reporting the maximum energy measurement in the EDSAMPLE register. The scan is started with EDSTART task and its end indicated with the EDEND event. This greatly reduces the interrupt frequency and hence power consumtion. The figure below shows how the ED measurement will operate depending on the EDCNT register.

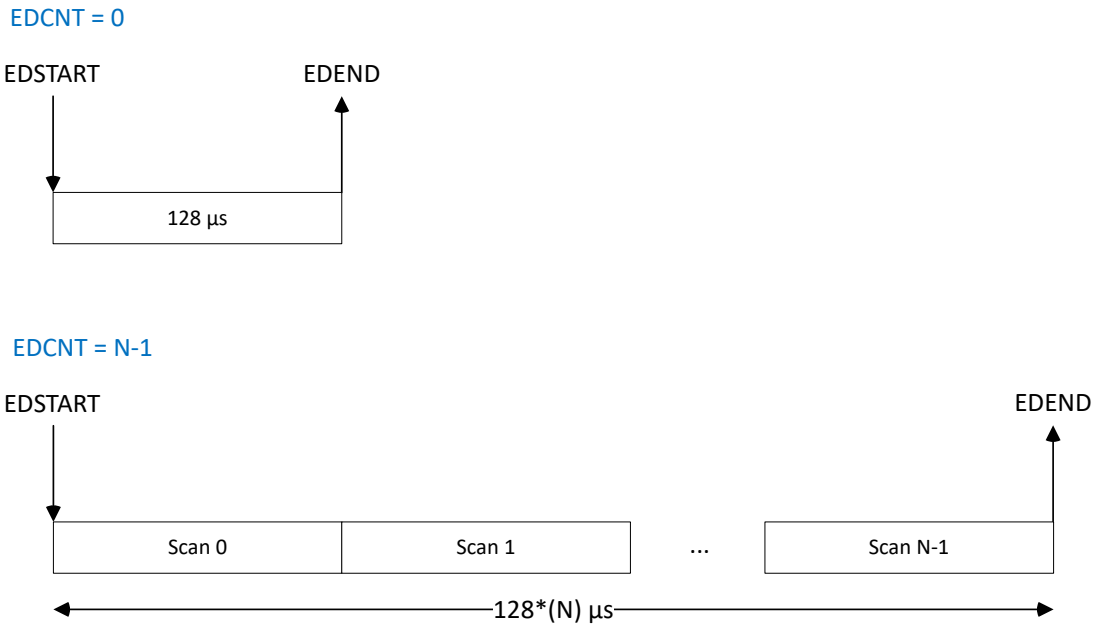


Figure 127: Energy detection measurement examples

An ongoing scan can always be stopped by writing the EDSTOP task. It will be followed by the EDSTOPPED event when the module has terminated.

6.20.12.4 Clear channel assessment (CCA)

IEEE 802.15.4 implements a listen-before-talk channel access method to avoid collisions when transmitting - namely carrier sense multiple access with collision avoidance (CSMA-CA). The key part of this is measuring if the wireless medium is busy or not.

At least three methods must be supported:

- Mode 1 (energy above threshold): The medium is reported busy upon detecting any energy above the ED threshold
- Mode 2 (carrier sense only): The medium is reported busy upon detection of a signal compliant with the IEEE 802.15.4 standard with the same modulation and spreading characteristics
- Mode 3 (carrier sense and threshold): The medium is reported busy by logically ANDing or ORing the results from mode 1 and mode 2.

It is furthermore specified that the clear channel assessment should survey a period equal to 8 symbols or 128 µs.

The radio module has to be in receive mode and be able to recived correct packets when performing the CCA. The shortcut between READY and START must be disabled if baseband processing is not to be performed while the measurement is running.

Mode 1 is enabled by first configuring the field CCAMODE=EdMode in CCACTRL and writing the CCAEDTHRES field to a chosen value. When the CCASTART task is written the radio module will perform a ED measurement for 8 symbols and compare the measured level with that found in the CCAEDTHRES field. If the measured value is higher than or equal to this threshold the CCABUSY event is generated - the CCAIDLE event is generated if the measured level is less than the threshold.

The conversion from CCAEDTHRES, CCA or EDLEVEL value to dBm can be done with the following equation, where $VAL_{HARDWARE}$ is the hardware-reported values, being either CCAEDTHRES, CCA or EDLEVEL, and constants ED_RSSISCALE and ED_RSSIOFFS are from electrical specifications:

$$P_{RF}[dBm] = ED_RSSIOFFS + ED_RSSISCALE \times VAL_{HARDWARE}$$

Figure 128: Conversion between hardware value and dBm

Mode 2 is enabled by configuring the CCAMODE=CarrierMode. In carrier mode the module will sample to see if a valid SFD is found during the 8 symbols. If a valid SFD is seen the CCABUSY event is generated and the node should not send any data. The CCABUSY event is also generated if the scan was performed during an ongoing frame reception. In the case where the measurement period completes with no SFD detection the CCAIDLE task is generated. With the CCA_CORR_COUNT unequal to zero the algorithm will look at the correlator output in addition to the SFD detection signal. If a SFD is reported during the scan period it will terminate immidiately indicating busy medium. Similarly, if the number of peaks above CCA_CORRTHRES crosses the CCA_CORR_COUNT the CCABUSY event is generated. If less than CCA_CORR_COUNT crossings are found and no SFD is reported the CCAIDLE signal will be generated and it is ok for the node to commence sending data.

With the CCA_MODE=CarrierAndEdMode or CCA_MODE=CarrierOrEdMode a logical combination of the result from running both mode 1 and mode 2 is performed. The CCABUSY or CCAIDLE signal will be generated based on an ANDing or ORing of the internal signals from performing both the energy detection and carrier detection scans.

An ongoing CCA can always be stopped by issuing the CCASTOP task. This will trigger the associated CCASTOPPED event.

For CCA mode automation there are three shortcuts available. One is between CCAIDLE and TXEN. This short must always be used in conjunction with the short between CCAIDLE and STOP. This automation is provided so that the radio can automatically switch between RX (when performing the CCA) and to TX where the packet is sent. The last shortcut associated with the CCA mode is between CCABUSY and DISABLE. This will cause the radio to be disabled whenever the CCA reports a busy medium.

Another handy shortcut is between RXREADY and CCASTART. When the radio has ramped up into RX mode it can immidiately start a CCA.

6.20.12.5 Cyclic redundancy check (CRC)

IEEE 802.15.4 uses a 16-bit ITU-T cyclic redundancy check (CRC) calculated over the MAC header (MHR) and MAC service data unit (MSDU).

The standard defines the following generator polynomial:

$G(x) = x^{16} + x^{12} + x^5 + 1$

In receive mode the radio will trigger the CRC module when the first octet after the frame length (PHR) is received. The CRC will then update on each consecutive octet received. When a complete frame is received the CRCSTATUS register will be updated accordingly and the EVENTS_CRCOK or EVENTS_CRCERROR generated. When the CRC module is enabled it will not write the two last octets (CRC) to the frame Data RAM. When transmitting the CRC will be computed on the fly, starting with the first octet after PHR, and inserted as the two last octets in the frame. The EasyDMA will fetch frame length - 2 octets from DataRAM and insert the CRC octets insitu.

Below is a code snippet for configuring the CRC module for correct operation when in IEEE 802.15.4 mode. The CRCCNF is written to 16-bit CRC and the CRCPOLY is written to 0x121. The start value used by IEEE 802.15.4 is zero and CRCINIT is configured to reflect this.

```
/* 16-bit CRC with ITU-T polynomial with 0 as start condition*/
write_reg(NRFRADIO_REG(CRCCNF), 0x202);
write_reg(NRFRADIO_REG(CRCPOLY), 0x11021);
write_reg(NRFRADIO_REG(CRCINIT), 0);
```

The ENDIANESS subregister must be set to little-endian since the FCS field is transmitted leftmost bit first.

6.20.12.6 Transmit sequence

The transmission is started by first putting the radio in receive mode sending the RXEN task.

An outline of the IEEE 802.15.4 transmission is illustrated in the figure below.

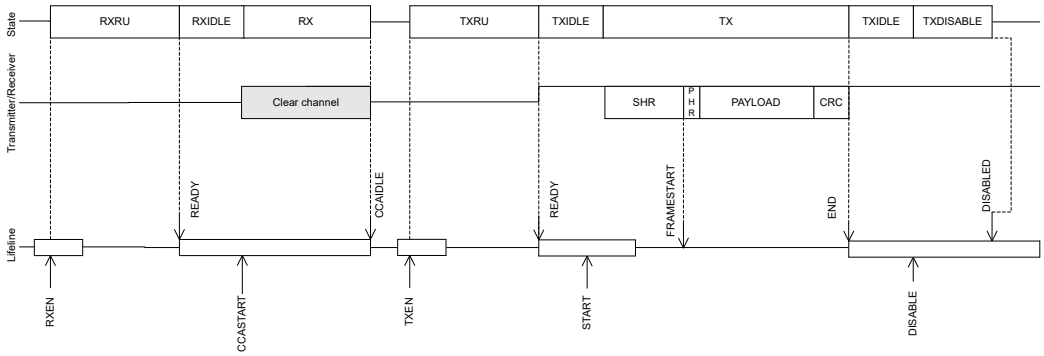


Figure 129: IEEE 802.15.4 transmit sequence

The receiver will ramp up and enter the RXIDLE state where the READY event is generated. Upon receiving the ready event the CCA is started by writing to the CCASTART task register. The chosen mode of assessment (CCA_MODE register) will be performed and signal the CCAIDLE or CCABUSY event 128 µs later. If the CCABUSY is received the radio will have to retry the CCA after a specific back off period as outlined in the IEEE 802.15.4 standard (see Figure 69 in section *7.5.1.4 The CSMA-CA algorithm* of the standard).

When the CCAIDLE event on the other hand is generated the user shall write to the TXEN task register to enter the TXRU state. The READY event will be generated when the radio is in TXIDLE state and ready

to transmit. With the PACKETPTR pointing to the length (PHR) field of the frame the START task can be written. The radio will send the four octet preamble sequence followed by the start of frame delimiter (SFD register). The first byte read from the Data RAM is the length field (PHR) followed by the transmission of the number of bytes indicated as the frame length. If the CRC module is configured it will run for PHR-2 octets. The last two octets will be substituted with the results from running the CRC. The necessary CRC parameters are sampled on the START task. The FCS field of the frame is little endian.

In addition to the already available shortcuts, one is provided between READY event and CCASTART task so that a CCA can automatically start when the receiver is ready. And a second shortcut has been added between CCAIDLE event and the TXEN task so that upon detecting a clear channel the radio can immediately enter transmit mode.

6.20.12.7 Receive sequence

The reception is started by first putting the radio in receive mode. Writing to the RXEN task the radio will start ramping up and enter the RXRU state.

When the READY event is generated the radio has entered the RXIDLE mode. For the baseband processing to be enabled the START task must be written. An outline of the IEEE 802.15.4 reception can be found in figure below.

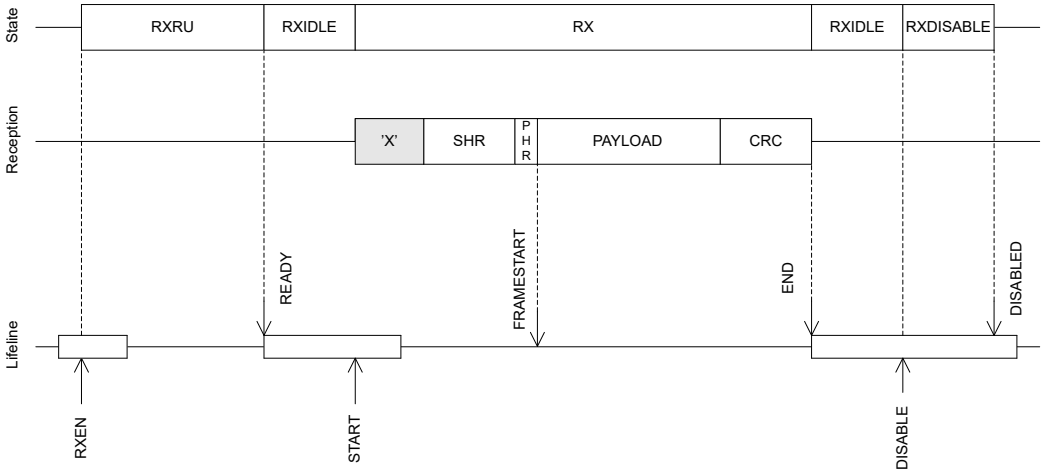


Figure 130: IEEE 802.15.4 receive sequence

When a valid SHR is received the radio will start storing future octets (starting with PHR) to the data memory pointed to by PACKETPTR. After the SFD octet is received the FRAMESTART event is generated. If the CRC module is enabled it will start updating with the second byte received (first byte in payload) and run for the full frame length. The two last bytes in the frame is not written to DataRAM when CRC is configured. However, if the result of the CRC after running the full frame is zero the CRCOK event will be generated. The END event is generated when the last octet has been received and is available in DataRAM.

When a packet is received a link quality indicator (LQI) is also generated and appended immediately after the last received octet. When using IEEE 802.15.4 compliant frame this will be just after the MSDU since the FCS is not reported. In the case of a non-complient frame it will be appended after the full frame. The LQI reported by hardware must be converted to IEEE 802.15.4 range by an 8-bit saturating multiplication by 4, as shown in the code example for ED sampling. The LQI is only valid for frames equal to or longer than three octets. When receiving a frame the RSSI (reported as negative dB) will be measured at three points during the reception. These three values will be sorted and the middle one selected (median 3) for then to be remapped within the LQI range. The following figure illustrates the LQI measurement and how the data is arranged in the DataRAM:

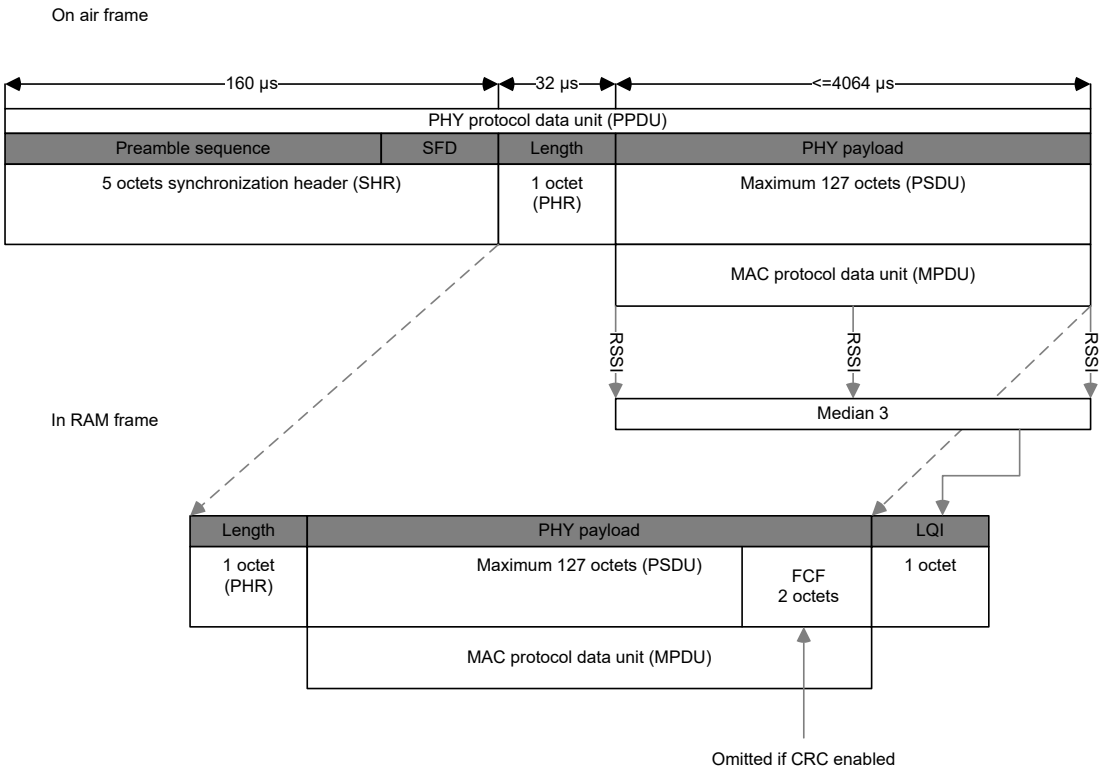


Figure 131: IEEE 802.15.4 frame in Data RAM

A shortcut has been added between FRAMESTART event and the BCSTART task. This can be used to trigger a BCMATCH event after N bits, such as when inspecting the MAC addressing fields.

6.20.12.8 Interframe spacing (IFS)

The IEEE 802.15.4 standard defines a specific time that is alotted for the MAC sublayer to process received data. Usage of this interframe spacing (IFS) comes into play to avoid that two frames are transmitted too close to eachother in time. If the a transmission is requesting an acknowledgement, the speration to the second frame shall be at least an IFS period.

The IFS is determined to be:

- IFS equals macMinSIFSPeriod (12 symbols) if the MPDU is less than or equal to aMaxSIFSFrameSize (18 octets) octets
- IFS equals macMinLIFSPeriod (40 symbols) if the MPDU is larger than aMaxSIFSFrameSize

Using the efficient assisted modes in the radio module the TIFS will be programmed with the correct value based on the frame being transmitted. If the assisted modes are not being used the user must update the TIFS register manually. The figure below provides details on what IFS period is valid in both acknowledged and unacknowledged transmissions.

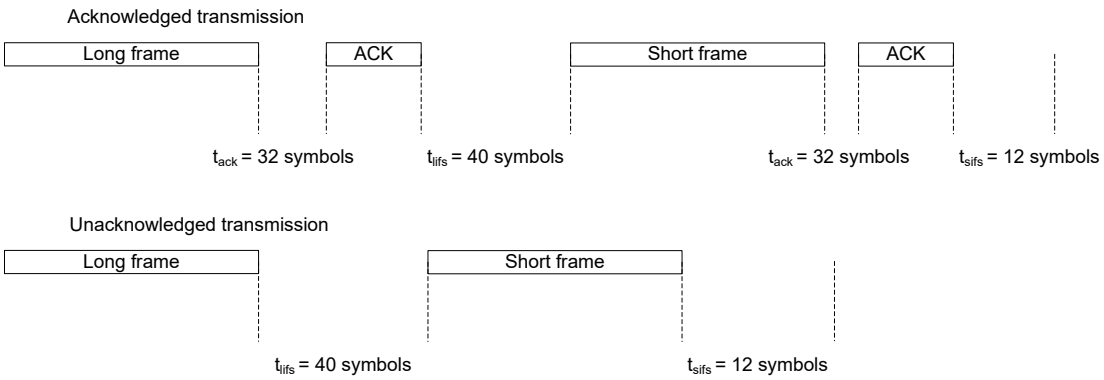


Figure 132: Interframe spacing examples

6.20.13 EasyDMA

The RADIO uses EasyDMA for reading of data packets from and writing to RAM, without CPU involvement.

As illustrated in RADIO block diagram on page 600, the RADIO's EasyDMA utilizes the same PACKETPTR for receiving and transmitting packets. This pointer should be reconfigured by the CPU each time before RADIO is started by the START task. The PACKETPTR registers is double-buffered, meaning that it can be updated and prepared for the next transmission.

Important: If the PACKETPTR is not pointing to the Data RAM region, an EasyDMA transfer may result in a HardFault or RAM corruption. See Memory on page 21 for more information about the different memory regions.

The END event indicates that the last bit has been processed by the radio. The DISABLED event is issued to acknowledge that a DISABLE task is done.

The structure of a radio packet is described in detail in Packet configuration on page 600. The data that is stored in Data RAM and transported by EasyDMA consists of the following fields:

- S0
- LENGTH
- S1
- PAYLOAD

In addition, a static add-on is sent immediately after the payload.

The size of each of the above fields in the frame is configurable (see Packet configuration on page 600), and the space occupied in RAM depends on these settings. A size of zero is possible for any of the fields, it is up to the user to make sure that the resulting frame complies with the RF protocol chosen.

All fields are extended in size to align with a byte boundary in RAM. For instance a 3 bit long field on air will occupy 1 byte in RAM while a 9 bit long field will be extended to 2 bytes.

The radio packets elements can be configured as follows:

- CI, TERM1 and TERM2 fields are only present in *Bluetooth* low energy long range mode
- S0 is configured through the S0LEN field in PCNF0
- LENGTH is configured through the LFLEN field in PCNF0
- S1 is configured through the S1LEN field in PCNF0
- Size of the payload is configured through the value in RAM corresponding to the LENGTH field
- Size of the static add-on to the payload is configured through the STATLEN field in PCNF1

The MAXLEN field in the PCNF1 register configures the maximum packet payload plus add-on size in number of bytes that can be transmitted or received by the RADIO. This feature can be used to ensure that the RADIO does not overwrite, or read beyond, the RAM assigned to the packet payload. This means

that if the packet payload length defined by PCNF1.STATLEN and the LENGTH field in the packet specifies a packet larger than MAXLEN, the payload will be truncated at MAXLEN.

> **Note:** The MAXLEN includes the payload and the add-on, but excludes the size occupied by the S0, LENGTH and S1 fields. This has to be taken into account when allocating RAM.

If the payload and add-on length is specified larger than MAXLEN, the RADIO will still transmit or receive in the same way as before, except the payload is now truncated to MAXLEN. The packet's LENGTH field will not be altered when the payload is truncated. The RADIO will calculate CRC as if the packet length is equal to MAXLEN.

> **Note:** If the PACKETPTR is not pointing to the Data RAM region, an EasyDMA transfer may result in a HardFault or RAM corruption. See Memory on page 21 for more information about the different memory regions.

The END event indicates that the last bit has been processed by the radio. The DISABLED event is issued to acknowledge that an DISABLE task is done.

6.20.14 Registers

Instances

Instance	Base address	Description
RADIO	0x40001000	2.4 GHz radio

Register overview

Register	Offset	Description
TASKS_TXEN	0x000	Enable RADIO in TX mode
TASKS_RXEN	0x004	Enable RADIO in RX mode
TASKS_START	0x008	Start RADIO
TASKS_STOP	0x00C	Stop RADIO
TASKS_DISABLE	0x010	Disable RADIO
TASKS_RSSISTART	0x014	Start the RSSI and take one single sample of the receive signal strength
TASKS_RSSISTOP	0x018	Stop the RSSI measurement
TASKS_BCSTART	0x01C	Start the bit counter
TASKS_BCSTOP	0x020	Stop the bit counter
TASKS_EDSTART	0x024	Start the energy detect measurement used in IEEE 802.15.4 mode
TASKS_EDSTOP	0x028	Stop the energy detect measurement
TASKS_CCASTART	0x02C	Start the clear channel assessment used in IEEE 802.15.4 mode
TASKS_CCASTOP	0x030	Stop the clear channel assessment
EVENTS_READY	0x100	RADIO has ramped up and is ready to be started
EVENTS_ADDRESS	0x104	Address sent or received
EVENTS_PAYLOAD	0x108	Packet payload sent or received
EVENTS_END	0x10C	Packet sent or received
EVENTS_DISABLED	0x110	RADIO has been disabled
EVENTS_DEVMATCH	0x114	A device address match occurred on the last received packet
EVENTS_DEVMISS	0x118	No device address match occurred on the last received packet
EVENTS_RSSIEND	0x11C	Sampling of receive signal strength complete
EVENTS_BCMATCH	0x128	Bit counter reached bit count value
EVENTS_CRCOK	0x130	Packet received with CRC ok
EVENTS_CRCERROR	0x134	Packet received with CRC error

Register	Offset	Description
EVENTS_FRAMESTART	0x138	IEEE 802.15.4 length field received
EVENTS_EDEND	0x13C	Sampling of energy detection complete. A new ED sample is ready for readout from the RADIO.EDSAMPLE register.
EVENTS_EDSTOPPED	0x140	The sampling of energy detection has stopped
EVENTS_CCAIDLE	0x144	Wireless medium in idle - clear to send
EVENTS_CCABUSY	0x148	Wireless medium busy - do not send
EVENTS_CCASTOPPED	0x14C	The CCA has stopped
EVENTS_RATEBOOST	0x150	Ble_LR CI field received, receive mode is changed from Ble_LR125Kbit to Ble_LR500Kbit.
EVENTS_TXREADY	0x154	RADIO has ramped up and is ready to be started TX path
EVENTS_RXREADY	0x158	RADIO has ramped up and is ready to be started RX path
EVENTS_MHRMATCH	0x15C	MAC header match found
EVENTS_SYNC	0x168	Preamble indicator.
EVENTS_PHYEND	0x16C	Generated in Ble_LR125Kbit, Ble_LR500Kbit and Ieee802154_250Kbit modes when last bit is sent on air.
SHORTS	0x200	Shortcuts between local events and tasks
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
CRCSTATUS	0x400	CRC status
RXMATCH	0x408	Received address
RXCRC	0x40C	CRC field of previously received packet
DAI	0x410	Device address match index
PDUSTAT	0x414	Payload status
PACKETPTR	0x504	Packet pointer
FREQUENCY	0x508	Frequency
TXPOWER	0x50C	Output power
MODE	0x510	Data rate and modulation
PCNF0	0x514	Packet configuration register 0
PCNF1	0x518	Packet configuration register 1
BASE0	0x51C	Base address 0
BASE1	0x520	Base address 1
PREFIX0	0x524	Prefixes bytes for logical addresses 0-3
PREFIX1	0x528	Prefixes bytes for logical addresses 4-7
TXADDRESS	0x52C	Transmit address select
RXADDRESSES	0x530	Receive address select
CRCCNF	0x534	CRC configuration
CRCPOLY	0x538	CRC polynomial
CRCINIT	0x53C	CRC initial value
TIFS	0x544	Interframe spacing in µs
RSSISAMPLE	0x548	RSSI sample
STATE	0x550	Current radio state
DATAWHITEIV	0x554	Data whitening initial value
BCC	0x560	Bit counter compare
DAB[0]	0x600	Device address base segment 0
DAB[1]	0x604	Device address base segment 1
DAB[2]	0x608	Device address base segment 2
DAB[3]	0x60C	Device address base segment 3
DAB[4]	0x610	Device address base segment 4
DAB[5]	0x614	Device address base segment 5
DAB[6]	0x618	Device address base segment 6
DAB[7]	0x61C	Device address base segment 7
DAP[0]	0x620	Device address prefix 0
DAP[1]	0x624	Device address prefix 1
DAP[2]	0x628	Device address prefix 2

Register	Offset	Description
DAP[3]	0x62C	Device address prefix 3
DAP[4]	0x630	Device address prefix 4
DAP[5]	0x634	Device address prefix 5
DAP[6]	0x638	Device address prefix 6
DAP[7]	0x63C	Device address prefix 7
DACNF	0x640	Device address match configuration
MHRMATCHCONF	0x644	Search pattern configuration
MHRMATCHMAS	0x648	Pattern mask
MODECNF0	0x650	Radio mode configuration register 0
SFD	0x660	IEEE 802.15.4 start of frame delimiter
EDCNT	0x664	IEEE 802.15.4 energy detect loop count
EDSAMPLE	0x668	IEEE 802.15.4 energy detect level
CCACTRL	0x66C	IEEE 802.15.4 clear channel assessment control
POWER	0xFFC	Peripheral power control

6.20.14.1 TASKS_TXEN

Address offset: 0x000

Enable RADIO in TX mode

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_TXEN			Enable RADIO in TX mode
			Trigger	1	Trigger task

6.20.14.2 TASKS_RXEN

Address offset: 0x004

Enable RADIO in RX mode

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_RXEN			Enable RADIO in RX mode
			Trigger	1	Trigger task

6.20.14.3 TASKS_START

Address offset: 0x008

Start RADIO

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_START			Start RADIO
			Trigger	1	Trigger task

6.20.14.4 TASKS_STOP

Address offset: 0x00C

Stop RADIO

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field		Value ID				Value																		Description									
A	W	TASKS_STOP																								Stop RADIO									
				Trigger				1																		Trigger task									

6.20.14.5 TASKS_DISABLE

Address offset: 0x010

Disable RADIO

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field		Value ID				Value																		Description									
A	W	TASKS_DISABLE																								Disable RADIO									
				Trigger				1																		Trigger task									

6.20.14.6 TASKS_RSSISTART

Address offset: 0x014

Start the RSSI and take one single sample of the receive signal strength

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field		Value ID				Value																		Description									
A	W	TASKS_RSSISTART																								Start the RSSI and take one single sample of the receive signal strength									
				Trigger				1																		Trigger task									

6.20.14.7 TASKS_RSSISTOP

Address offset: 0x018

Stop the RSSI measurement

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field		Value ID				Value																		Description									
A	W	TASKS_RSSISTOP																								Stop the RSSI measurement									
				Trigger				1																		Trigger task									

6.20.14.8 TASKS_BCSTART

Address offset: 0x01C

Start the bit counter

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_BCSTART			Start the bit counter
			Trigger	1	Trigger task

6.20.14.9 TASKS_BCSTOP

Address offset: 0x020

Stop the bit counter

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_BCSTOP			Stop the bit counter
			Trigger	1	Trigger task

6.20.14.10 TASKS_EDSTART

Address offset: 0x024

Start the energy detect measurement used in IEEE 802.15.4 mode

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_EDSTART			Start the energy detect measurement used in IEEE 802.15.4 mode
			Trigger	1	Trigger task

6.20.14.11 TASKS_EDSTOP

Address offset: 0x028

Stop the energy detect measurement

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_EDSTOP			Stop the energy detect measurement
			Trigger	1	Trigger task

6.20.14.12 TASKS_CCASTART

Address offset: 0x02C

Start the clear channel assessment used in IEEE 802.15.4 mode

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CCASTART			Start the clear channel assessment used in IEEE 802.15.4 mode
			Trigger	1	Trigger task

6.20.14.13 TASKS_CCASTOP

Address offset: 0x030

Stop the clear channel assessment

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CCASTOP			Stop the clear channel assessment
			Trigger	1	Trigger task

6.20.14.14 EVENTS_READY

Address offset: 0x100

RADIO has ramped up and is ready to be started

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_READY			RADIO has ramped up and is ready to be started
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.20.14.15 EVENTS_ADDRESS

Address offset: 0x104

Address sent or received

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ADDRESS			Address sent or received
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.20.14.16 EVENTS_PAYLOAD

Address offset: 0x108

Packet payload sent or received

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_PAYLOAD			Packet payload sent or received
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.20.14.17 EVENTS_END

Address offset: 0x10C

Packet sent or received

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_END			Packet sent or received
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.20.14.18 EVENTS_DISABLED

Address offset: 0x110

RADIO has been disabled

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_DISABLED			RADIO has been disabled
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.20.14.19 EVENTS_DEVMATCH

Address offset: 0x114

A device address match occurred on the last received packet

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_DEVMATCH			A device address match occurred on the last received packet
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.20.14.20 EVENTS_DEVMISS

Address offset: 0x118

No device address match occurred on the last received packet

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_DEVMISS			No device address match occurred on the last received packet
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.20.14.21 EVENTS_RSSIEND

Address offset: 0x11C

Sampling of receive signal strength complete

A new RSSI sample is ready for readout from the RADIO.RSSISAMPLE register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_RSSIEND			Sampling of receive signal strength complete A new RSSI sample is ready for readout from the RADIO.RSSISAMPLE register
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.20.14.22 EVENTS_BCMATCH

Address offset: 0x128

Bit counter reached bit count value

Bit counter value is specified in the RADIO.BCC register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_BCMATCH			Bit counter reached bit count value Bit counter value is specified in the RADIO.BCC register
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.20.14.23 EVENTS_CRCOK

Address offset: 0x130

Packet received with CRC ok

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_CRCOK			Packet received with CRC ok
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.20.14.24 EVENTS_CRCERROR

Address offset: 0x134

Packet received with CRC error

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_CRCERROR			Packet received with CRC error
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.20.14.25 EVENTS_FRAMESTART

Address offset: 0x138

IEEE 802.15.4 length field received

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_FRAMESTART			IEEE 802.15.4 length field received
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.20.14.26 EVENTS_EDEND

Address offset: 0x13C

Sampling of energy detection complete. A new ED sample is ready for readout from the RADIO.EDSAMPLE register.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_EDEND			Sampling of energy detection complete. A new ED sample is ready for readout from the RADIO.EDSAMPLE register.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.20.14.27 EVENTS_EDSTOPPED

Address offset: 0x140

The sampling of energy detection has stopped

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_EDSTOPPED			The sampling of energy detection has stopped
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.20.14.28 EVENTS_CCAIDLE

Address offset: 0x144

Wireless medium in idle - clear to send

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_CCAIDLE			Wireless medium in idle - clear to send
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.20.14.29 EVENTS_CCABUSY

Address offset: 0x148

Wireless medium busy - do not send

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_CCABUSY			Wireless medium busy - do not send
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.20.14.30 EVENTS_CCASTOPPED

Address offset: 0x14C

The CCA has stopped

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_CCASTOPPED			The CCA has stopped
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.20.14.31 EVENTS_RATEBOOST

Address offset: 0x150

Ble_LR CI field received, receive mode is changed from Ble_LR125Kbit to Ble_LR500Kbit.

Ble_LR CI field received, receive mode is changed from Ble_LR125Kbit to Ble_LR500Kbit.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_RATEBOOST			Ble_LR CI field received, receive mode is changed from Ble_LR125Kbit to Ble_LR500Kbit. Ble_LR CI field received, receive mode is changed from Ble_LR125Kbit to Ble_LR500Kbit.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.20.14.32 EVENTS_TXREADY

Address offset: 0x154

RADIO has ramped up and is ready to be started TX path

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TXREADY			RADIO has ramped up and is ready to be started TX path
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.20.14.33 EVENTS_RXREADY

Address offset: 0x158

RADIO has ramped up and is ready to be started RX path

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_RXREADY			RADIO has ramped up and is ready to be started RX path
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.20.14.34 EVENTS_MHRMATCH

Address offset: 0x15C

MAC header match found

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_MHRMATCH			MAC header match found
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.20.14.35 EVENTS_SYNC

Address offset: 0x168

Preamble indicator.

A possible preamble has been received in Ble_LR125Kbit, Ble_LR500Kbit or Ieee802154_250Kbit modes during an RX transaction. False triggering of the event is possible.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_SYNC			Preamble indicator. A possible preamble has been received in Ble_LR125Kbit, Ble_LR500Kbit or Ieee802154_250Kbit modes during an RX transaction. False triggering of the event is possible.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.20.14.36 EVENTS_PHYEND

Address offset: 0x16C

Generated in Ble_LR125Kbit, Ble_LR500Kbit and Ieee802154_250Kbit modes when last bit is sent on air.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_PHYEND			Generated in Ble_LR125Kbit, Ble_LR500Kbit and Ieee802154_250Kbit modes when last bit is sent on air.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.20.14.37 SHORTS

Address offset: 0x200

Shortcuts between local events and tasks

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				U T S R Q P O N M L K H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	READY_START			Shortcut between event READY and task START
			Disabled	0	Disable shortcut

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				U T S R Q P O N M L K H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Enabled	1	Enable shortcut
B	RW	END_DISABLE			Shortcut between event END and task DISABLE
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
C	RW	DISABLED_TXEN			Shortcut between event DISABLED and task TXEN
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
D	RW	DISABLED_RXEN			Shortcut between event DISABLED and task RXEN
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
E	RW	ADDRESS_RSSISTART			Shortcut between event ADDRESS and task RSSISTART
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
F	RW	END_START			Shortcut between event END and task START
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
G	RW	ADDRESS_BCSTART			Shortcut between event ADDRESS and task BCSTART
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
H	RW	DISABLED_RSSISTOP			Shortcut between event DISABLED and task RSSISTOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
K	RW	RXREADY_CCASTART			Shortcut between event RXREADY and task CCASTART
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
L	RW	CCAIDLE_TXEN			Shortcut between event CCAIDLE and task TXEN
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
M	RW	CCABUSY_DISABLE			Shortcut between event CCABUSY and task DISABLE
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
N	RW	FRAMESTART_BCSTART			Shortcut between event FRAMESTART and task BCSTART
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
O	RW	READY_EDSTART			Shortcut between event READY and task EDSTART
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
P	RW	EDEND_DISABLE			Shortcut between event EDEND and task DISABLE
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
Q	RW	CCAIDLE_STOP			Shortcut between event CCAIDLE and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
R	RW	TXREADY_START			Shortcut between event TXREADY and task START
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
S	RW	RXREADY_START			Shortcut between event RXREADY and task START
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
T	RW	PHYEND_DISABLE			Shortcut between event PHYEND and task DISABLE

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID											U	T	S	R	Q	P	O	N	M	L	K			H		G	F	E	D	C	B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
U	RW	PHYEND_START			Shortcut between event PHYEND and task START
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

6.20.14.38 INTENSET

Address offset: 0x304

Enable interrupt

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					Z	Y			V	U	T	S	R	Q	P	O	N	M	L	K		I		H	G	F	E	D	C	B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	READY			Write '1' to enable interrupt for event READY
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	ADDRESS			Write '1' to enable interrupt for event ADDRESS
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	PAYLOAD			Write '1' to enable interrupt for event PAYLOAD
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	END			Write '1' to enable interrupt for event END
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	DISABLED			Write '1' to enable interrupt for event DISABLED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	DEVMATCH			Write '1' to enable interrupt for event DEVMATCH
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
G	RW	DEVMISS			Write '1' to enable interrupt for event DEVMISS
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
H	RW	RSSIEND			Write '1' to enable interrupt for event RSSIEND A new RSSI sample is ready for readout from the RADIO.RSSISAMPLE register
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					Z	Y			V	U	T	S	R	Q	P	O	N	M	L	K		I			H	G	F	E	D	C	B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
I	RW	BCMATCH			Write '1' to enable interrupt for event BCMATCH Bit counter value is specified in the RADIO.BCC register
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
K	RW	CRCOK			Write '1' to enable interrupt for event CRCOK
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
L	RW	CRCERROR			Write '1' to enable interrupt for event CRCERROR
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
M	RW	FRAMESTART			Write '1' to enable interrupt for event FRAMESTART
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
N	RW	EDEND			Write '1' to enable interrupt for event EDEND
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
O	RW	EDSTOPPED			Write '1' to enable interrupt for event EDSTOPPED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
P	RW	CCAIDLE			Write '1' to enable interrupt for event CCAIDLE
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
Q	RW	CCABUSY			Write '1' to enable interrupt for event CCABUSY
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
R	RW	CCASTOPPED			Write '1' to enable interrupt for event CCASTOPPED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
S	RW	RATEBOOST			Write '1' to enable interrupt for event RATEBOOST Ble_LR CI field received, receive mode is changed from Ble_LR125Kbit to Ble_LR500Kbit.
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
T	RW	TXREADY			Write '1' to enable interrupt for event TXREADY
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
U	RW	RXREADY			Write '1' to enable interrupt for event RXREADY
			Set	1	Enable
			Disabled	0	Read: Disabled

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				Z Y V U T S R Q P O N M L K I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Enabled	1	Read: Enabled
V	RW	MHRMATCH			Write '1' to enable interrupt for event MHRMATCH
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
Y	RW	SYNC			Write '1' to enable interrupt for event SYNC A possible preamble has been received in Ble_LR125Kbit, Ble_LR500Kbit or Ieee802154_250Kbit modes during an RX transaction. False triggering of the event is possible.
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
Z	RW	PHYEND			Write '1' to enable interrupt for event PHYEND
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.20.14.39 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				Z Y V U T S R Q P O N M L K I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	READY			Write '1' to disable interrupt for event READY
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	ADDRESS			Write '1' to disable interrupt for event ADDRESS
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	PAYLOAD			Write '1' to disable interrupt for event PAYLOAD
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	END			Write '1' to disable interrupt for event END
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	DISABLED			Write '1' to disable interrupt for event DISABLED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	DEVMATCH			Write '1' to disable interrupt for event DEVMATCH
			Clear	1	Disable
			Disabled	0	Read: Disabled

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					Z	Y			V	U	T	S	R	Q	P	O	N	M	L	K		I			H	G	F	E	D	C	B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
			Enabled	1	Read: Enabled
G	RW	DEVMISS			Write '1' to disable interrupt for event DEVMISS
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
H	RW	RSSIEND			Write '1' to disable interrupt for event RSSIEND A new RSSI sample is ready for readout from the RADIO.RSSISAMPLE register
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
I	RW	BCMATCH			Write '1' to disable interrupt for event BCMATCH Bit counter value is specified in the RADIO.BCC register
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
K	RW	CRCOK			Write '1' to disable interrupt for event CRCOK
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
L	RW	CRCERROR			Write '1' to disable interrupt for event CRCERROR
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
M	RW	FRAMESTART			Write '1' to disable interrupt for event FRAMESTART
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
N	RW	EDEND			Write '1' to disable interrupt for event EDEND
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
O	RW	EDSTOPPED			Write '1' to disable interrupt for event EDSTOPPED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
P	RW	CCAIDLE			Write '1' to disable interrupt for event CCAIDLE
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
Q	RW	CCABUSY			Write '1' to disable interrupt for event CCABUSY
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
R	RW	CCASTOPPED			Write '1' to disable interrupt for event CCASTOPPED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID								Z	Y			V	U	T	S	R	Q	P	O	N	M	L	K		I			H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
S	RW	RATEBOOST																																	
				Write '1' to disable interrupt for event RATEBOOST Ble_LR CI field received, receive mode is changed from Ble_LR125Kbit to Ble_LR500Kbit.																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
T	RW	TXREADY																																	
				Write '1' to disable interrupt for event TXREADY																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
U	RW	RXREADY																																	
				Write '1' to disable interrupt for event RXREADY																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
V	RW	MHRMATCH																																	
				Write '1' to disable interrupt for event MHRMATCH																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
Y	RW	SYNC																																	
				Write '1' to disable interrupt for event SYNC A possible preamble has been received in Ble_LR125Kbit, Ble_LR500Kbit or Ieee802154_250Kbit modes during an RX transaction. False triggering of the event is possible.																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
Z	RW	PHYEND																																	
				Write '1' to disable interrupt for event PHYEND																																
			Clear	1																															
			Disabled	0																															
			Enabled	1																															

6.20.14.40 CRCSTATUS

Address offset: 0x400

CRC status

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	R	CRCSTATUS																																	
				CRC status of packet received																															
			CRCError	0																															
			CRCOk	1																															

6.20.14.41 RXMATCH

Address offset: 0x408

Received address

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	RXMATCH			Received address
					Logical address of which previous packet was received

6.20.14.42 RXCRC

Address offset: 0x40C

CRC field of previously received packet

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	RXCRC			CRC field of previously received packet
					CRC field of previously received packet

6.20.14.43 DAI

Address offset: 0x410

Device address match index

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	DAI			Device address match index
					Index (n) of device address, see DAB[n] and DAP[n], that got an address match

6.20.14.44 PDUSTAT

Address offset: 0x414

Payload status

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	B B A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	PDUSTAT			Status on payload length vs. PCNF1.MAXLEN
			LessThan	0	Payload less than PCNF1.MAXLEN
			GreaterThan	1	Payload greater than PCNF1.MAXLEN
B	R	CISTAT			Status on what rate packet is received with in Long Range
			LR125kbit	0	Frame is received at 125kbps
			LR500kbit	1	Frame is received at 500kbps

6.20.14.45 PACKETPTR

Address offset: 0x504

Packet pointer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	PACKETPTR																																	

Packet pointer

Packet address to be used for the next transmission or reception. When transmitting, the packet pointed to by this address will be transmitted and when receiving, the received packet will be written to this address. This address is a byte aligned RAM address.

Note: See the memory chapter for details about which memories are available for EasyDMA.

6.20.14.46 FREQUENCY

Address offset: 0x508

Frequency

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																											B		A	A	A	A	A	A	A
Reset 0x00000002				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	FREQUENCY		[0..100]																															
B	RW	MAP																																	
			Default	0																															
			Low	1																															

Frequency = 2400 + FREQUENCY (MHz).

Channel map selection.

Channel map between 2400 MHZ .. 2500 MHz

Frequency = 2400 + FREQUENCY (MHz)

Channel map between 2360 MHZ .. 2460 MHz

Frequency = 2360 + FREQUENCY (MHz)

6.20.14.47 TXPOWER

Address offset: 0x50C

Output power

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																												A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	TXPOWER																																	
			Pos8dBm	0x8																															
			Pos7dBm	0x7																															
			Pos6dBm	0x6																															
			Pos5dBm	0x5																															
			Pos4dBm	0x4																															
			Pos3dBm	0x3																															

Output power in number of dBm, i.e. if the value -20 is specified the output power will be set to -20dBm.

+8 dBm

+7 dBm

+6 dBm

+5 dBm

+4 dBm

+3 dBm

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
			Pos2dBm	0x2	+2 dBm
			0dBm	0x0	0 dBm
			Neg4dBm	0xFC	-4 dBm
			Neg8dBm	0xF8	-8 dBm
			Neg12dBm	0xF4	-12 dBm
			Neg16dBm	0xF0	-16 dBm
			Neg20dBm	0xEC	-20 dBm
			Neg30dBm	0xE2	-40 dBm
					This enumerator is deprecated.
			Neg40dBm	0xD8	-40 dBm

6.20.14.48 MODE

Address offset: 0x510

Data rate and modulation

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	MODE			Radio data rate and modulation setting. The radio supports frequency-shift keying (FSK) modulation.
			Nrf_1Mbit	0	1 Mbit/s Nordic proprietary radio mode
			Nrf_2Mbit	1	2 Mbit/s Nordic proprietary radio mode
			Ble_1Mbit	3	1 Mbit/s BLE
			Ble_2Mbit	4	2 Mbit/s BLE
			Ble_LR125Kbit	5	Long range 125 kbit/s TX, 125 kbit/s and 500 kbit/s RX
			Ble_LR500Kbit	6	Long range 500 kbit/s TX, 125 kbit/s and 500 kbit/s RX
			Ieee802154_250Kbit	15	IEEE 802.15.4-2006 250 kbit/s

6.20.14.49 PCNF0

Address offset: 0x514

Packet configuration register 0

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				H H G F F E E D C C C C B A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	LFLEN			Length on air of LENGTH field in number of bits.
B	RW	S0LEN			Length on air of S0 field in number of bytes.
C	RW	S1LEN			Length on air of S1 field in number of bits.
D	RW	S1INCL			Include or exclude S1 field in RAM
			Automatic	0	Include S1 field in RAM only if S1LEN > 0
			Include	1	Always include S1 field in RAM independent of S1LEN
E	RW	CILEN			Length of code indicator - long range
F	RW	PLEN			Length of preamble on air. Decision point: TASKS_START task
			8bit	0	8-bit preamble

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					H	H			G	F	F	E	E		D	C	C	C	C							B					A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
			16bit	1																															
			32bitZero	2																															
			LongRange	3																															
G	RW	CRCINC																																	
			Exclude	0																															
			Include	1																															
H	RW	TERMLEN																																	

Length of TERM field in Long Range operation

6.20.14.50 PCNF1

Address offset: 0x518

Packet configuration register 1

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID										E	D					C	C	C	B	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	MAXLEN		[0..255]																															

Maximum length of packet payload. If the packet payload is larger than MAXLEN, the radio will truncate the payload to MAXLEN.

B	RW	STATLEN		[0..255]																															

Static length in number of bytes

The static length parameter is added to the total length of the payload when sending and receiving packets, e.g. if the static length is set to N the radio will receive or send N bytes more than what is defined in the LENGTH field of the packet.

C	RW	BALEN		[2..4]																															

Base address length in number of bytes

The address field is composed of the base address and the one byte long address prefix, e.g. set BALEN=2 to get a total address of 3 bytes.

D	RW	ENDIAN																																	

On air endianness of packet, this applies to the S0, LENGTH, S1 and the PAYLOAD fields.

			Little	0																															

Least significant bit on air first

			Big	1																															

Most significant bit on air first

E	RW	WHITEEN																																	

Enable or disable packet whitening

			Disabled	0																															

Disable

			Enabled	1																															

Enable

6.20.14.51 BASE0

Address offset: 0x51C

Base address 0

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	BASE0																																	

Base address 0

Radio base address 0.

6.20.14.52 BASE1

Address offset: 0x520

Base address 1

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	BASE1																																	
				Base address 1 Radio base address 1.																																	

6.20.14.53 PREFIX0

Address offset: 0x524

Prefixes bytes for logical addresses 0-3

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				D	D	D	D	D	D	D	D	C	C	C	C	C	C	C	C	B	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	AP0																																	
				Address prefix 0.																																	
B	RW	AP1																																	
				Address prefix 1.																																	
C	RW	AP2																																	
				Address prefix 2.																																	
D	RW	AP3																																	
				Address prefix 3.																																	

6.20.14.54 PREFIX1

Address offset: 0x528

Prefixes bytes for logical addresses 4-7

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				D	D	D	D	D	D	D	D	C	C	C	C	C	C	C	C	B	B	B	B	B	B	B	B	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	AP4																																	
				Address prefix 4.																																	
B	RW	AP5																																	
				Address prefix 5.																																	
C	RW	AP6																																	
				Address prefix 6.																																	
D	RW	AP7																																	
				Address prefix 7.																																	

6.20.14.55 TXADDRESS

Address offset: 0x52C

Transmit address select

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	TXADDRESS			Transmit address select Logical address to be used when transmitting a packet.

6.20.14.56 RXADDRESSES

Address offset: 0x530

Receive address select

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ADDR0			Enable or disable reception on logical address 0.
			Disabled	0	Disable
			Enabled	1	Enable
B	RW	ADDR1			Enable or disable reception on logical address 1.
			Disabled	0	Disable
			Enabled	1	Enable
C	RW	ADDR2			Enable or disable reception on logical address 2.
			Disabled	0	Disable
			Enabled	1	Enable
D	RW	ADDR3			Enable or disable reception on logical address 3.
			Disabled	0	Disable
			Enabled	1	Enable
E	RW	ADDR4			Enable or disable reception on logical address 4.
			Disabled	0	Disable
			Enabled	1	Enable
F	RW	ADDR5			Enable or disable reception on logical address 5.
			Disabled	0	Disable
			Enabled	1	Enable
G	RW	ADDR6			Enable or disable reception on logical address 6.
			Disabled	0	Disable
			Enabled	1	Enable
H	RW	ADDR7			Enable or disable reception on logical address 7.
			Disabled	0	Disable
			Enabled	1	Enable

6.20.14.57 CRCCNF

Address offset: 0x534

CRC configuration

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B B A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	LEN		[1..3]	CRC length in number of bytes. **Note:** For MODE Ble_LR125Kbit and Ble_LR500Kbit, only LEN set to 3 is supported
			Disabled	0	CRC length is zero and CRC calculation is disabled
			One	1	CRC length is one byte and CRC calculation is enabled
			Two	2	CRC length is two bytes and CRC calculation is enabled
			Three	3	CRC length is three bytes and CRC calculation is enabled
B	RW	SKIPADDR			Include or exclude packet address field out of CRC calculation.
			Include	0	CRC calculation includes address field
			Skip	1	CRC calculation does not include address field. The CRC calculation will start at the first byte after the address.
			Ieee802154	2	CRC calculation as per 802.15.4 standard. Starting at first byte after length field.

6.20.14.58 CRCPOLY

Address offset: 0x538

CRC polynomial

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	CRCPOLY			CRC polynomial Each term in the CRC polynomial is mapped to a bit in this register which index corresponds to the term's exponent. The least significant term/ bit is hard-wired internally to 1, and bit number 0 of the register content is ignored by the hardware. The following example is for an 8 bit CRC polynomial: x8 + x7 + x3 + x2 + 1 = 1 1000 1101 .

6.20.14.59 CRCINIT

Address offset: 0x53C

CRC initial value

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	CRCINIT			CRC initial value Initial value for CRC calculation

6.20.14.60 TIFS

Address offset: 0x544

Interframe spacing in µs

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TIFS			Interframe spacing in µs
					Interframe space is the time interval between two consecutive packets. It is defined as the time, in microseconds, from the end of the last bit of the previous packet to the start of the first bit of the subsequent packet.

6.20.14.61 RSSISAMPLE

Address offset: 0x548

RSSI sample

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	RSSISAMPLE		[0..127]	RSSI sample
					RSSI sample result. The value of this register is read as a positive value while the actual received signal strength is a negative value. Actual received signal strength is therefore as follows: received signal strength = -A dBm

6.20.14.62 STATE

Address offset: 0x550

Current radio state

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	STATE			Current radio state
			Disabled	0	RADIO is in the Disabled state
			RxRu	1	RADIO is in the RXRU state
			RxIdle	2	RADIO is in the RXIDLE state
			Rx	3	RADIO is in the RX state
			RxDisable	4	RADIO is in the RXDISABLED state
			TxRu	9	RADIO is in the TXRU state
			TxIdle	10	RADIO is in the TXIDLE state
			Tx	11	RADIO is in the TX state
			TxDisable	12	RADIO is in the TXDISABLED state

6.20.14.63 DATAWHITEIV

Address offset: 0x554

Data whitening initial value

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A
Reset 0x00000040	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	DATAWHITEIV			Data whitening initial value. Bit 6 is hard-wired to '1', writing '0' to it has no effect, and it will always be read back and used by the device as '1'. Bit 0 corresponds to Position 6 of the LSFR, Bit 1 to Position 5, etc.

6.20.14.64 BCC

Address offset: 0x560

Bit counter compare

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	BCC			Bit counter compare Bit counter compare register

6.20.14.65 DAB[0]

Address offset: 0x600

Device address base segment 0

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	DAB			Device address base segment 0

6.20.14.66 DAB[1]

Address offset: 0x604

Device address base segment 1

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	DAB			Device address base segment 1

6.20.14.67 DAB[2]

Address offset: 0x608

Device address base segment 2

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	DAB			Device address base segment 2

6.20.14.68 DAB[3]

Address offset: 0x60C

Device address base segment 3

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	DAB			Device address base segment 3

6.20.14.69 DAB[4]

Address offset: 0x610

Device address base segment 4

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	DAB			Device address base segment 4

6.20.14.70 DAB[5]

Address offset: 0x614

Device address base segment 5

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	DAB			Device address base segment 5

6.20.14.71 DAB[6]

Address offset: 0x618

Device address base segment 6

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	DAB			Device address base segment 6

6.20.14.72 DAB[7]

Address offset: 0x61C

Device address base segment 7

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	DAB			Device address base segment 7

6.20.14.73 DAP[0]

Address offset: 0x620

Device address prefix 0

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	DAP			Device address prefix 0

6.20.14.74 DAP[1]

Address offset: 0x624

Device address prefix 1

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	DAP			Device address prefix 1

6.20.14.75 DAP[2]

Address offset: 0x628

Device address prefix 2

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	DAP			Device address prefix 2

6.20.14.76 DAP[3]

Address offset: 0x62C

Device address prefix 3

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	DAP			Device address prefix 3

6.20.14.77 DAP[4]

Address offset: 0x630

Device address prefix 4

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	DAP			Device address prefix 4

6.20.14.78 DAP[5]

Address offset: 0x634

Device address prefix 5

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	DAP			Device address prefix 5

6.20.14.79 DAP[6]

Address offset: 0x638

Device address prefix 6

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	DAP			Device address prefix 6

6.20.14.80 DAP[7]

Address offset: 0x63C

Device address prefix 7

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	DAP			Device address prefix 7

6.20.14.81 DACNF

Address offset: 0x640

Device address match configuration

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																				P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	ENA0																																	
			Disabled	0																															
			Enabled	1																															
B	RW	ENA1																																	
			Disabled	0																															
			Enabled	1																															
C	RW	ENA2																																	
			Disabled	0																															
			Enabled	1																															
D	RW	ENA3																																	
			Disabled	0																															
			Enabled	1																															
E	RW	ENA4																																	
			Disabled	0																															
			Enabled	1																															
F	RW	ENA5																																	
			Disabled	0																															
			Enabled	1																															
G	RW	ENA6																																	
			Disabled	0																															
			Enabled	1																															
H	RW	ENA7																																	
			Disabled	0																															
			Enabled	1																															
I	RW	TXADD0																																	
J	RW	TXADD1																																	
K	RW	TXADD2																																	
L	RW	TXADD3																																	
M	RW	TXADD4																																	
N	RW	TXADD5																																	
O	RW	TXADD6																																	
P	RW	TXADD7																																	

6.20.14.82 MHRMATCHCONF

Address offset: 0x644

Search pattern configuration

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	MHRMATCHCONF																																	

6.20.14.83 MHRMATCHMAS

Address offset: 0x648

Pattern mask

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	MHRMATCHMAS																																	

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																									A	A	A	A	A	A	A	A
Reset 0x000000A7	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**1**	**0**	**0**	**1**	**1**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	SFD			IEEE 802.15.4 start of frame delimiter

6.20.14.86 EDCNT

Address offset: 0x664

IEEE 802.15.4 energy detect loop count

Number of iterations to perform an ED scan. If set to 0 one scan is performed, otherwise the specified number + 1 of ED scans will be performed and the max ED value tracked in EDSAMPLE

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID												A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EDCNT			IEEE 802.15.4 energy detect loop count

6.20.14.87 EDSAMPLE

Address offset: 0x668

IEEE 802.15.4 energy detect level

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																									A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EDLVL		[0..127]	IEEE 802.15.4 energy detect level Register value must be converted to IEEE 802.15.4 range by an 8-bit saturating multiplication by factor ED_RSSISCALE, as shown in the code example for ED sampling

6.20.14.88 CCACTRL

Address offset: 0x66C

IEEE 802.15.4 clear channel assessment control

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				D	D	D	D	D	D	D	D	C	C	C	C	C	C	C	C	B	B	B	B	B	B	B	B						A	A	A
Reset 0x052D0000				**0**	**0**	**0**	**0**	**0**	**1**	**0**	**1**	**0**	**0**	**1**	**0**	**1**	**1**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	CCAMODE																																	
			EdMode	0																															
				Will report busy whenever energy is detected above CCAEDTHRES																															
			CarrierMode	1																															
				Will report busy whenever compliant IEEE 802.15.4 signal is seen																															
			CarrierAndEdMode	2																															
			CarrierOrEdMode	3																															
			EdModeTest1	4																															
				Energy above threshold test mode that will abort when first ED measurement over threshold is seen. No averaging.																															
B	RW	CCAEDTHRES		CCA energy busy threshold. Used in all the CCA modes except CarrierMode. Must be converted from IEEE 802.15.4 range by dividing by factor ED_RSSISCALE - similar to EDSAMPLE register																															
C	RW	CCACORRTHRES		CCA correlator busy threshold. Only relevant to CarrierMode, CarrierAndEdMode and CarrierOrEdMode.																															
D	RW	CCACORRCNT		Limit for occurances above CCACORRTHRES. When not equal to zero the corrolator based signal detect is enabled.																															

6.20.14.89 POWER

Address offset: 0xFFC

Peripheral power control

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000001				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW	POWER																																	
				Peripheral power control. The peripheral and its registers will be reset to its initial state by switching the peripheral off and then back on again.																															
			Disabled	0																															
			Enabled	1																															

6.20.15 Electrical specification

6.20.15.1 General radio characteristics

Symbol	Description	Min.	Typ.	Max.	Units
f_{OP}	Operating frequencies	2360		2500	MHz
$f_{PLL,CH,SP}$	PLL channel spacing		1		MHz
$f_{DELTA,1M}$	Frequency deviation @ 1 Mbps		±170		kHz
$f_{DELTA,BLE,1M}$	Frequency deviation @ BLE 1 Mbps		±250		kHz
$f_{DELTA,2M}$	Frequency deviation @ 2 Mbps		±320		kHz
$f_{DELTA,BLE,2M}$	Frequency deviation @ BLE 2 Mbps		±500		kHz
fsk_{BPS}	On-the-air data rate	125		2000	kbps
$f_{chip,\ IEEE\ 802.15.4}$	Chip rate in IEEE 802.15.4 mode		2000		kchip/s

6.20.15.2 Radio current consumption (transmitter)

Symbol	Description	Min.	Typ.	Max.	Units
$I_{TX,PLUS8dBM,DCDC}$	TX only run current (DC/DC, 3 V) P_{RF} = +8 dBm		14.8		mA
$I_{TX,PLUS8dBM}$	TX only run current P_{RF} = +8 dBm		32.7		mA
$I_{TX,PLUS4dBM,DCDC}$	TX only run current (DC/DC, 3 V) P_{RF} = +4 dBm		9.6		mA
$I_{TX,PLUS4dBM}$	TX only run current P_{RF} = +4 dBm		21.4		mA
$I_{TX,0dBM,DCDC,5V,REG0HIGH.}$	TX only run current (DC/DC, 5 V, REG0 out = 3.3 V)P_{RF} = 0 dBm		3.0		mA
$I_{TX,0dBM,DCDC,5V,REG0LOW}$	TX only run current (DC/DC, 5 V, REG0 out = 1.8 V)P_{RF} = 0 dBm		3.0		mA
$I_{TX,0dBM,DCDC}$	TX only run current (DC/DC, 3 V)P_{RF} = 0 dBm		4.8		mA
$I_{TX,0dBM}$	TX only run current P_{RF} = 0 dBm		10.6		mA
$I_{TX,MINUS4dBM,DCDC}$	TX only run current DC/DC, 3 V P_{RF} = -4 dBm		3.1		mA
$I_{TX,MINUS4dBM}$	TX only run current P_{RF} = -4 dBm		8.1		mA
$I_{TX,MINUS8dBM,DCDC}$	TX only run current DC/DC, 3 V P_{RF} = -8 dBm		3.3		mA
$I_{TX,MINUS8dBM}$	TX only run current P_{RF} = -8 dBm		7.2		mA
$I_{TX,MINUS12dBM,DCDC}$	TX only run current DC/DC, 3 V P_{RF} = -12 dBm		3.0		mA
$I_{TX,MINUS12dBM}$	TX only run current P_{RF} = -12 dBm		6.4		mA
$I_{TX,MINUS16dBM,DCDC}$	TX only run current DC/DC, 3 V P_{RF} = -16 dBm		2.8		mA
$I_{TX,MINUS16dBM}$	TX only run current P_{RF} = -16 dBm		6.0		mA
$I_{TX,MINUS20dBM,DCDC}$	TX only run current DC/DC, 3 V P_{RF} = -20 dBm		2.7		mA
$I_{TX,MINUS20dBM}$	TX only run current P_{RF} = -20 dBm		5.6		mA
$I_{TX,MINUS40dBM,DCDC}$	TX only run current DC/DC, 3 V P_{RF} = -40 dBm		2.3		mA
$I_{TX,MINUS40dBM}$	TX only run current P_{RF} = -40 dBm		4.6		mA
$I_{START,TX,DCDC}$	TX start-up current DC/DC, 3 V, P_{RF} = 4 dBm		5.2		mA
$I_{START,TX}$	TX start-up current, P_{RF} = 4 dBm		11.0		mA

6.20.15.3 Radio current consumption (Receiver)

Symbol	Description	Min.	Typ.	Max.	Units
$I_{RX,1M,DCDC}$	RX only run current (DC/DC, 3 V) 1 Mbps/1 Mbps BLE		4.6		mA
$I_{RX,1M}$	RX only run current (LDO, 3 V) 1 Mbps/1 Mbps BLE		9.9		mA
$I_{RX,2M,DCDC}$	RX only run current (DC/DC, 3 V) 2 Mbps/2 Mbps BLE		5.2		mA
$I_{RX,2M}$	RX only run current (LDO, 3 V) 2 Mbps/2 Mbps BLE		11.1		mA
$I_{START,RX,1M,DCDC}$	RX start-up current (DC/DC, 3 V) 1 Mbps/1 Mbps BLE		3.7		mA
$I_{START,RX,1M}$	RX start-up current 1 Mbps/1 Mbps BLE		6.7		mA

6.20.15.4 Transmitter specification

Symbol	Description	Min.	Typ.	Max.	Units
P_{RF}	Maximum output power		8.0		dBm
P_{RFC}	RF power control range		28.0		dB
P_{RFCR}	RF power accuracy			±4	dB
$P_{RF1,1}$	1st Adjacent Channel Transmit Power 1 MHz (1 Mbps)		-24.8		dBc
$P_{RF2,1}$	2nd Adjacent Channel Transmit Power 2 MHz (1 Mbps)		-54.0		dBc
$P_{RF1,2}$	1st Adjacent Channel Transmit Power 2 MHz (2 Mbps)		-25		dBc
$P_{RF2,2}$	2nd Adjacent Channel Transmit Power 4 MHz (2 Mbps)		-54.0		dBc
E_{vm}	Error vector magnitude IEEE 802.15.4		8		%rms
$P_{harm2nd,\ IEEE\ 802.15.4}$	2nd harmonics in IEEE 802.15.4 mode		-51.0		dBm
$P_{harm3rd,\ IEEE\ 802.15.4}$	3rd harmonics in IEEE 802.15.4		-48.0		dBm

Symbol	Description	Min.	Typ.	Max.	Units
$P_{ACP,R,\ IEEE\ 802.15.4}$	IEEE 802.15.4 Relative adjacent Channel Power, offset > 3.5 MHz [19]		-42		dBc
$P_{ACP,A,\ IEEE\ 802.15.4}$	IEEE 802 15.4 Absolute adjacent Channel Power, offset > 3.5 MHz [19]		-46		dBm

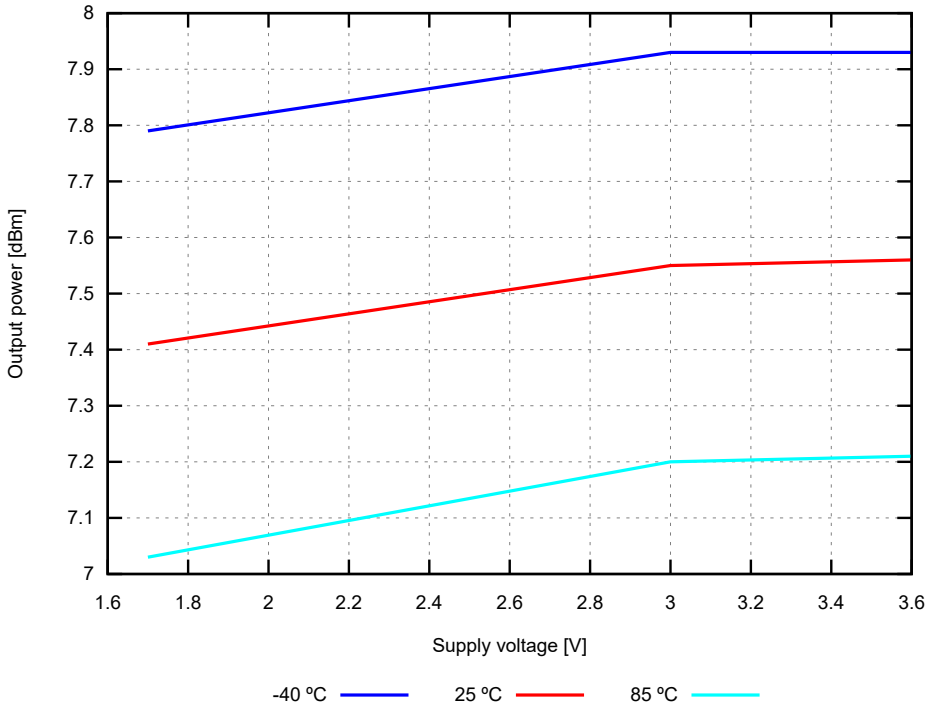


Figure 133: Output power, 1 Mbps Bluetooth low energy mode, at maximum TXPOWER setting (typical values)

[19] Output power set to maximum TXPOWER setting, resolution bandwidth (RBW) set to 100 kHz, and transmitter Duty-Cycle approximately 85%.

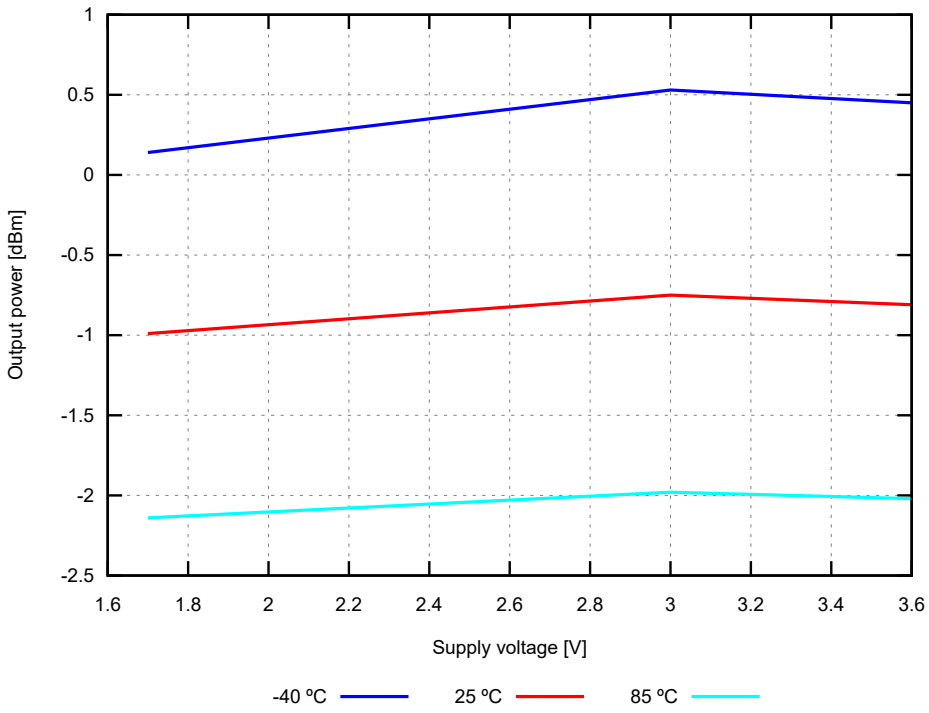


Figure 134: Output power, 1 Mbps Bluetooth low energy mode, at 0 dBm TXPOWER setting (typical values)

6.20.15.5 Receiver operation

Symbol	Description	Min.	Typ.	Max.	Units
$P_{RX,MAX}$	Maximum received signal strength at < 0.1% PER		0		dBm
$P_{SENS,IT,1M}$	Sensitivity, 1 Mbps nRF mode ideal transmitter[20]		-93		dBm
$P_{SENS,IT,2M}$	Sensitivity, 2 Mbps nRF mode ideal transmitter[21]		-89		dBm
$P_{SENS,IT,SP,1M,BLE}$	Sensitivity, 1 Mbps BLE ideal transmitter, packet length ≤ 37 bytes BER=1E-3[22]		-95		dBm
$P_{SENS,IT,LP,1M,BLE}$	Sensitivity, 1 Mbps BLE ideal transmitter, packet length ≥ 128 bytes BER=1E-4[23]		-94		dBm
$P_{SENS,IT,SP,2M,BLE}$	Sensitivity, 2 Mbps BLE ideal transmitter, packet length ≤ 37 bytes		-92		dBm
$P_{SENS,IT,BLE\ LE125k}$	Sensitivity, 125 kbps BLE mode		-103		dBm
$P_{SENS,IT,BLE\ LE500k}$	Sensitivity, 500 kbps BLE mode		-99		dBm
$P_{SENS,IEEE\ 802.15.4}$	Sensitivity in IEEE 802.15.4 mode		-100		dBm

[20] Typical sensitivity applies when ADDR0 is used for receiver address correlation. When ADDR[1...7] are used for receiver address correlation, the typical sensitivity for this mode is degraded by 3 dB.

[21] Typical sensitivity applies when ADDR0 is used for receiver address correlation. When ADDR[1..7] are used for receiver address correlation, the typical sensitivity for this mode is degraded by 3 dB.

[22] As defined in the *Bluetooth Core Specification v4.0 Volume 6: Core System Package (Low Energy Controller Volume)*

[23] Equivalent BER limit < 10E-04

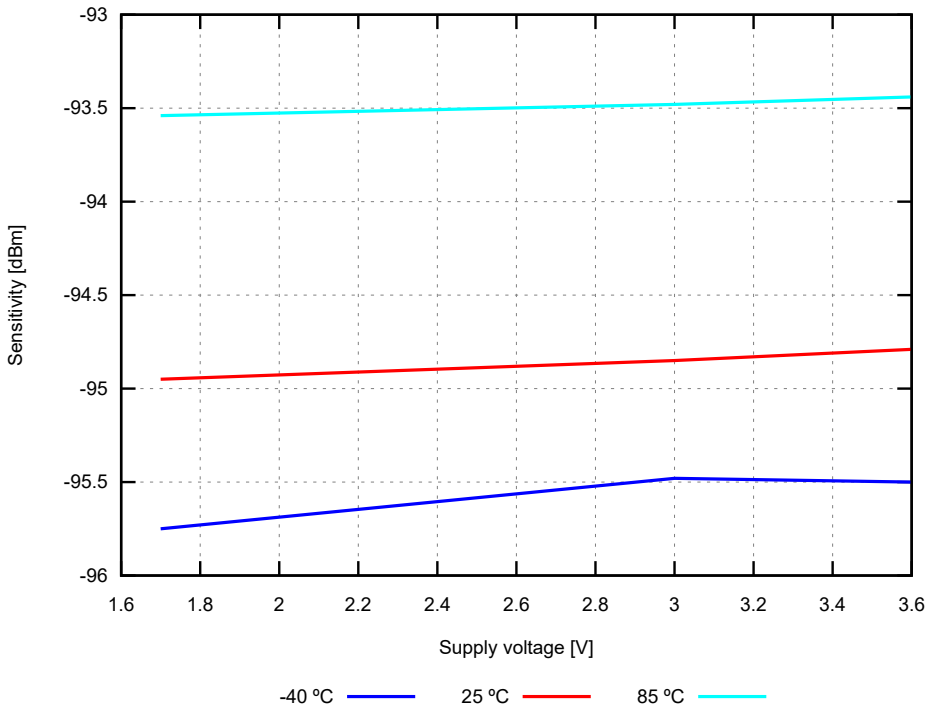


Figure 135: Sensitivity, 1 Mbps Bluetooth low energy mode, Regulator = LDO (typical values)

6.20.15.6 RX selectivity

RX selectivity with equal modulation on interfering signal[24]

Symbol	Description	Min.	Typ.	Max.	Units
$C/I_{1M,co\text{-}channel}$	1Mbps mode, Co-Channel interference		9		dB
$C/I_{1M,-1MHz}$	1 Mbps mode, Adjacent (-1 MHz) interference		-2		dB
$C/I_{1M,+1MHz}$	1 Mbps mode, Adjacent (+1 MHz) interference		-10		dB
$C/I_{1M,-2MHz}$	1 Mbps mode, Adjacent (-2 MHz) interference		-19		dB
$C/I_{1M,+2MHz}$	1 Mbps mode, Adjacent (+2 MHz) interference		-42		dB
$C/I_{1M,-3MHz}$	1 Mbps mode, Adjacent (-3 MHz) interference		-38		dB
$C/I_{1M,+3MHz}$	1 Mbps mode, Adjacent (+3 MHz) interference		-48		dB
$C/I_{1M,\pm6MHz}$	1 Mbps mode, Adjacent (≥6 MHz) interference		-50		dB
$C/I_{1MBLE,co\text{-}channel}$	1 Mbps BLE mode, Co-Channel interference		6		dB
$C/I_{1MBLE,-1MHz}$	1 Mbps BLE mode, Adjacent (-1 MHz) interference		-2		dB
$C/I_{1MBLE,+1MHz}$	1 Mbps BLE mode, Adjacent (+1 MHz) interference		-9		dB
$C/I_{1MBLE,-2MHz}$	1 Mbps BLE mode, Adjacent (-2 MHz) interference		-22		dB
$C/I_{1MBLE,+2MHz}$	1 Mbps BLE mode, Adjacent (+2 MHz) interference		-46		dB
$C/I_{1MBLE,>3MHz}$	1 Mbps BLE mode, Adjacent (≥3 MHz) interference		-50		dB
$C/I_{1MBLE,image}$	Image frequency interference		-22		dB
$C/I_{1MBLE,image,1MHz}$	Adjacent (1 MHz) interference to in-band image frequency		-35		dB
$C/I_{2M,co\text{-}channel}$	2 Mbps mode, Co-Channel interference		10		dB
$C/I_{2M,-2MHz}$	2 Mbps mode, Adjacent (-2 MHz) interference		6		dB
$C/I_{2M,+2MHz}$	2 Mbps mode, Adjacent (+2 MHz) interference		-19		dB
$C/I_{2M,-4MHz}$	2 Mbps mode, Adjacent (-4 MHz) interference		-20		dB
$C/I_{2M,+4MHz}$	2 Mbps mode, Adjacent (+4 MHz) interference		-44		dB

[24] Desired signal level at PIN = -67 dBm. One interferer is used, having equal modulation as the desired signal. The input power of the interferer where the sensitivity equals BER = 0.1% is presented

Symbol	Description	Min.	Typ.	Max.	Units
$C/I_{2M,-6MHz}$	2 Mbps mode, Adjacent (-6 MHz) interference		-42		dB
$C/I_{2M,+6MHz}$	2 Mbps mode, Adjacent (+6 MHz) interference		-42		dB
$C/I_{2M,\geq 12MHz}$	2 Mbps mode, Adjacent (≥12 MHz) interference		-52		dB
$C/I_{2MBLE,co\text{-}channel}$	2 Mbps BLE mode, Co-Channel interference		6.8		dB
$C/I_{2MBLE,\pm 2MHz}$	2 Mbps BLE mode, Adjacent (±2 MHz) interference		-10		dB
$C/I_{2MBLE,\pm 4MHz}$	2 Mbps BLE mode, Adjacent (±4 MHz) interference		-45		dB
$C/I_{2MBLE,\geq 6MHz}$	2 Mbps BLE mode, Adjacent (≥6 MHz) interference		-48		dB
$C/I_{2MBLE,image}$	Image frequency interference		-24		dB
$C/I_{2MBLE,image,\ 2MHz}$	Adjacent (2 MHz) interference to in-band image frequency		-35		dB
$C/I_{125k\ BLE\ LR,co\text{-}channel}$	125 kbps BLE LR mode, Co-Channel interference		4.4		dB
$C/I_{125k\ BLE\ LR,-1MHz}$	125 kbps BLE LR mode, Adjacent (-1 MHz) interference		-4.0		dB
$C/I_{125k\ BLE\ LR,+1MHz}$	125 kbps BLE LR mode, Adjacent (+1 MHz) interference		-12		dB
$C/I_{125k\ BLE\ LR,-2MHz}$	125 kbps BLE LR mode, Adjacent (-2 MHz) interference		-28		dB
$C/I_{125k\ BLE\ LR,+2MHz}$	125 kbps BLE LR mode, Adjacent (+2 MHz) interference		-50		dB
$C/I_{125k\ BLE\ LR,>3MHz}$	125 kbps BLE LR mode, Adjacent (≥3 MHz) interference		-55		dB
$C/I_{125k\ BLE\ LR,image}$	Image frequency interference		-29		dB

6.20.15.7 RX intermodulation

RX intermodulation[25]

Symbol	Description	Min.	Typ.	Max.	Units
$P_{IMD,5TH,1M}$	IMD performance, 1 Mbps, 5th offset channel, packet length ≤ 37 bytes		-33		dBm
$P_{IMD,5TH,1M,BLE}$	IMD performance, BLE 1 Mbps, 5th offset channel, packet length ≤ 37 bytes		-30		dBm
$P_{IMD,5TH,2M}$	IMD performance, 2 Mbps, 5th offset channel, packet length ≤ 37 bytes		-33		dBm
$P_{IMD,5TH,2M,BLE}$	IMD performance, BLE 2 Mbps, 5th offset channel, packet length ≤ 37 bytes		-31		dBm

6.20.15.8 Radio timing

Symbol	Description	Min.	Typ.	Max.	Units
$t_{TXEN,BLE,1M}$	Time between TXEN task and READY event after channel FREQUENCY configured (1 Mbps BLE and 150 µs TIFS)	140		140	µs
$t_{TXEN,FAST,BLE,1M}$	Time between TXEN task and READY event after channel FREQUENCY configured (1 Mbps BLE with fast ramp-up and 150 µs TIFS)	40		40	µs
$t_{TXDIS,BLE,1M}$	When in TX, delay between DISABLE task and DISABLED event for MODE = Nrf_1Mbit and MODE = Ble_1Mbit	6		6	µs
$t_{RXEN,BLE,1M}$	Time between the RXEN task and READY event after channel FREQUENCY configured (1 Mbps BLE)	140		140	µs
$t_{RXEN,FAST,BLE,1M}$	Time between the RXEN task and READY event after channel FREQUENCY configured (1 Mbps BLE with fast ramp-up)	40		40	µs
$t_{RXDIS,BLE,1M}$	When in RX, delay between DISABLE task and DISABLED event for MODE = Nrf_1Mbit and MODE = Ble_1Mbit	0		0	µs
$t_{TXDIS,BLE,2M}$	When in TX, delay between DISABLE task and DISABLED event for MODE = Nrf_2Mbit and MODE = Ble_2Mbit	4		4	µs
$t_{RXDIS,BLE,2M}$	When in RX, delay between DISABLE task and DISABLED event for MODE = Nrf_2Mbit and MODE = Ble_2Mbit	0		0	µs
$t_{TXEN,IEEE\ 802.15.4}$	Time between TXEN task and READY event after channel FREQUENCY configured (IEEE 802.15.4)	130		130	µs

[25] Desired signal level at PIN = -64 dBm. Two interferers with equal input power are used. The interferer closest in frequency is not modulated, the other interferer is modulated equal with the desired signal. The input power of the interferers where the sensitivity equals BER = 0.1% is presented.

Symbol	Description	Min.	Typ.	Max.	Units
$t_{TXEN,FAST,IEEE\ 802.15.4}$	Time between TXEN task and READY event after channel FREQUENCY configured (IEEE 802.15.4 with fast ramp-up)	40		40	µs
$t_{TXDIS,IEEE\ 802.15.4}$	When in TX, delay between DISABLE task and DISABLED event (IEEE 802.15.4)	21		21	µs
$t_{RXEN,IEEE\ 802.15.4}$	Time between the RXEN task and READY event after channel FREQUENCY configured (IEEE 802.15.4)	130		130	µs
$t_{RXEN,FAST,IEEE\ 802.15.4}$	Time between the RXEN task and READY event after channel FREQUENCY configured (IEEE 802.15.4 with fast ramp-up)	40		40	µs
$t_{RXDIS,IEEE\ 802.15.4}$	When in RX, delay between DISABLE task and DISABLED event (IEEE 802.15.4)	0.5		0.5	µs
$t_{RX\text{-}to\text{-}TX\ turnaround}$	Maximum TX-to-RX or RX-to-TX turnaround time in IEEE 802.15.4 mode		40		µs

6.20.15.9 Received signal strength indicator (RSSI) specifications

Symbol	Description	Min.	Typ.	Max.	Units
$RSSI_{ACC}$	RSSI accuracy valid range -90 to -20 dBm		±2		dB
$RSSI_{RESOLUTION}$	RSSI resolution		1		dB
$RSSI_{PERIOD}$	RSSI sampling time from RSSI_START task		0.25		µs
$RSSI_{SETTLE}$	RSSI settling time after signal level change		15		µs

6.20.15.10 Jitter

Symbol	Description	Min.	Typ.	Max.	Units
$t_{DISABLEDJITTER}$	Jitter on DISABLED event relative to END event when shortcut between END and DISABLE is enabled		0.25		µs
$t_{READYJITTER}$	Jitter on READY event relative to TXEN and RXEN task		0.25		µs

6.20.15.11 IEEE 802.15.4 energy detection constants

Symbol	Description	Min.	Typ.	Max.	Units
ED_RSSISCALE	Scaling value when converting between hardware-reported value and dBm	4	4	4	
ED_RSSIOFFS	Offset value when converting between hardware-reported value and dBm	-92	-92	-92	

6.21 RNG — Random number generator

The Random number generator (RNG) generates true non-deterministic random numbers based on internal thermal noise that are suitable for cryptographic purposes. The RNG does not require a seed value.

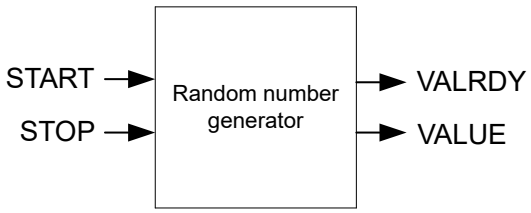


Figure 136: Random number generator

The RNG is started by triggering the START task and stopped by triggering the STOP task. When started, new random numbers are generated continuously and written to the VALUE register when ready. A VALRDY event is generated for every new random number that is written to the VALUE register. This means that after a VALRDY event is generated, the CPU has the time until the next VALRDY event to read out the random number from the VALUE register before it is overwritten by a new random number.

6.21.1 Bias correction

A bias correction algorithm is employed on the internal bit stream to remove any bias toward `1` or `0`. The bits are then queued into an eight-bit register for parallel readout from the VALUE register.

It is possible to enable bias correction in the CONFIG register. This will result in slower value generation, but will ensure a statistically uniform distribution of the random values.

6.21.2 Speed

The time needed to generate one random byte of data is unpredictable, and may vary from one byte to the next. This is especially true when bias correction is enabled.

6.21.3 Registers

Instances

Instance	Base address	Description
RNG	0x4000D000	Random number generator

Register overview

Register	Offset	Description
TASKS_START	0x000	Task starting the random number generator
TASKS_STOP	0x004	Task stopping the random number generator
EVENTS_VALRDY	0x100	Event being generated for every new random number written to the VALUE register
SHORTS	0x200	Shortcuts between local events and tasks
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
CONFIG	0x504	Configuration register
VALUE	0x508	Output random number

6.21.3.1 TASKS_START

Address offset: 0x000

Task starting the random number generator

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_START			Task starting the random number generator
			Trigger	1	Trigger task

6.21.3.2 TASKS_STOP

Address offset: 0x004

Task stopping the random number generator

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STOP			Task stopping the random number generator
			Trigger	1	Trigger task

6.21.3.3 EVENTS_VALRDY

Address offset: 0x100

Event being generated for every new random number written to the VALUE register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_VALRDY			Event being generated for every new random number written to the VALUE register
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.21.3.4 SHORTS

Address offset: 0x200

Shortcuts between local events and tasks

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	VALRDY_STOP			Shortcut between event VALRDY and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

6.21.3.5 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	VALRDY			Write '1' to enable interrupt for event VALRDY
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.21.3.6 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	VALRDY			Write '1' to disable interrupt for event VALRDY
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.21.3.7 CONFIG

Address offset: 0x504

Configuration register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DERCEN			Bias correction
			Disabled	0	Disabled
			Enabled	1	Enabled

6.21.3.8 VALUE

Address offset: 0x508

Output random number

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	VALUE		[0..255]	Generated random number

6.21.4 Electrical specification

6.21.4.1 RNG Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
$t_{RNG,START}$	Time from setting the START task to generation begins. This is a one-time delay on START signal and does not apply between samples.		128		µs
$t_{RNG,RAW}$	Run time per byte without bias correction. Uniform distribution of 0 and 1 is not guaranteed.		30		µs
$t_{RNG,BC}$	Run time per byte with bias correction. Uniform distribution of 0 and 1 is guaranteed. Time to generate a byte cannot be guaranteed.		120		µs

6.22 RTC — Real-time counter

The Real-time counter (RTC) module provides a generic, low power timer on the low-frequency clock source (LFCLK).

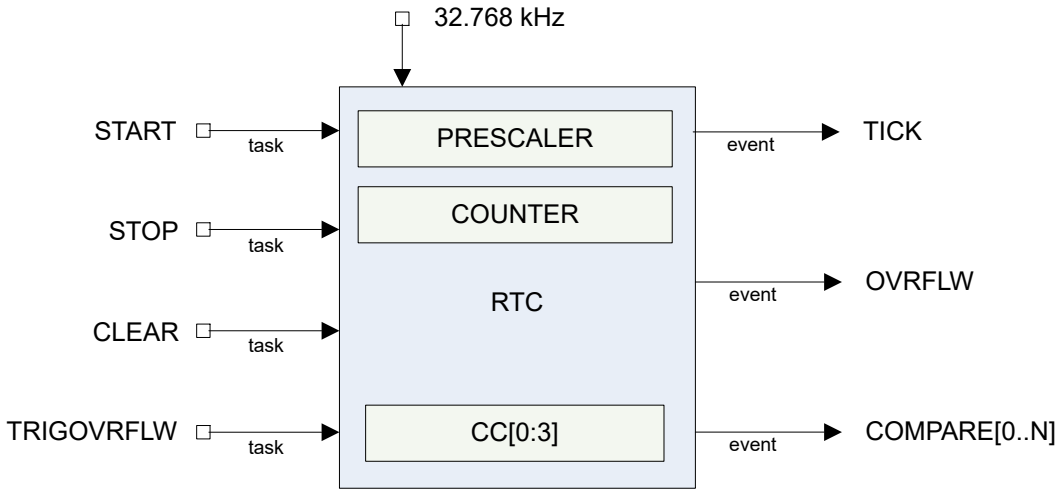


Figure 137: RTC block schematic

The RTC module features a 24-bit COUNTER, a 12-bit (1/X) prescaler, capture/compare registers, and a tick event generator for low power, tickless RTOS implementation.

6.22.1 Clock source

The RTC runs off the LFCLK.

The COUNTER resolution is 30.517 µs. Depending on the source, the RTC is able to run while the HFCLK is OFF and PCLK16M is not available.

The software has to explicitly start LFCLK before using the RTC.

See CLOCK — Clock control on page 157 for more information about clock sources.

6.22.2 Resolution versus overflow and the PRESCALER

Counter increment frequency:

$$f_{RTC}\ [kHz] = 32.768\ /\ (PRESCALER + 1\ )$$

The PRESCALER register is read/write when the RTC is stopped. The PRESCALER register is read-only once the RTC is STARTed. Writing to the PRESCALER register when the RTC is started has no effect.

The PRESCALER is restarted on START, CLEAR, and TRIGOVRFLW, meaning the prescaler value is latched to an internal register (<<PRESC>>) on these tasks.

Examples of different frequency configurations are as following:

- Desired COUNTER frequency 100 Hz (10 ms counter period)

  PRESCALER = round(32.768 kHz / 100 Hz) - 1 = 327

  f_{RTC} = 99.9 Hz

  10009.576 µs counter period
- Desired COUNTER frequency 8 Hz (125 ms counter period)

  PRESCALER = round(32.768 kHz / 8 Hz) – 1 = 4095

  f_{RTC} = 8 Hz

125 ms counter period

Prescaler	Counter resolution	Overflow
0	30.517 µs	512 seconds
2^8-1	7812.5 µs	131072 seconds
2^{12}-1	125 ms	582.542 hours

Table 38: RTC resolution versus overflow

6.22.3 COUNTER register

The COUNTER increments on LFCLK when the internal PRESCALER register (<<PRESC>>) is 0x00. <<PRESC>> is reloaded from the PRESCALER register. If enabled, the TICK event occurs on each increment of the COUNTER. The TICK event is disabled by default.

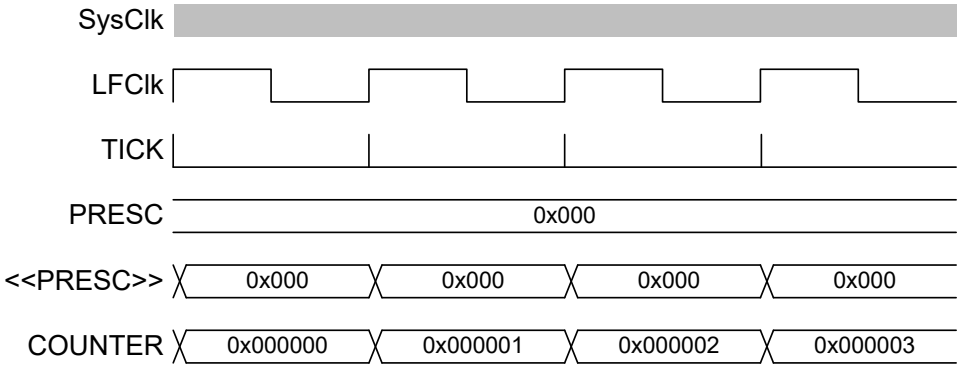


Figure 138: Timing diagram - COUNTER_PRESCALER_0

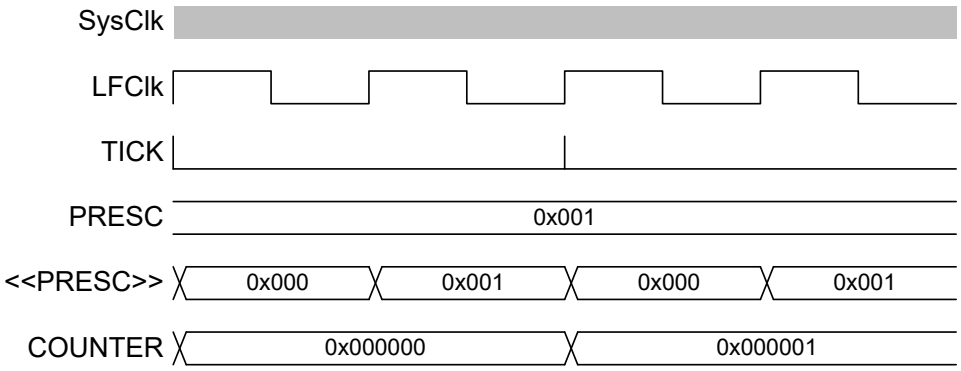


Figure 139: Timing diagram - COUNTER_PRESCALER_1

6.22.4 Overflow features

The TRIGOVRFLW task sets the COUNTER value to 0xFFFFF0 to allow SW test of the overflow condition.

OVRFLW occurs when COUNTER overflows from 0xFFFFFF to 0.

Note: The OVRFLW event is disabled by default.

6.22.5 TICK event

The TICK event enables low power tickless RTOS implementation as it optionally provides a regular interrupt source for a RTOS without the need to use the ARM SysTick feature.

Using the RTC TICK event rather than the SysTick allows the CPU to be powered down while still keeping RTOS scheduling active.

Note: The TICK event is disabled by default.

6.22.6 Event control feature

To optimize RTC power consumption, events in the RTC can be individually disabled to prevent PCLK16M and HFCLK being requested when those events are triggered. This is managed using the EVTEN register.

For example, if the TICK event is not required for an application, this event should be disabled as it is frequently occurring and may increase power consumption if HFCLK otherwise could be powered down for long durations.

This means that the RTC implements a slightly different task and event system compared to the standard system described in Peripheral interface on page 173. The RTC task and event system is illustrated in Tasks, events, and interrupts in the RTC on page 662.

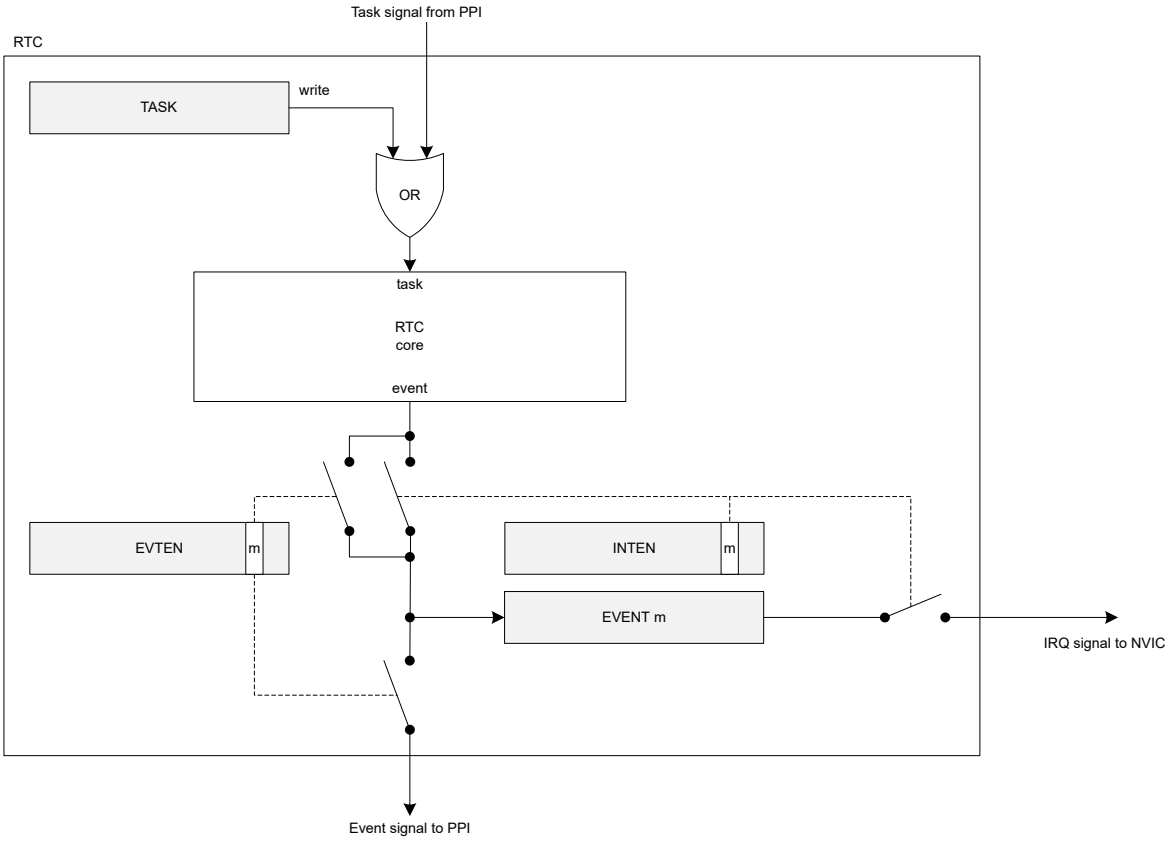


Figure 140: Tasks, events, and interrupts in the RTC

6.22.7 Compare feature

There are a number of Compare registers.

For more information, see Registers on page 667.

When setting a compare register, the following behavior of the RTC compare event should be noted:

- If a CC register value is 0 when a CLEAR task is set, this will not trigger a COMPARE event.

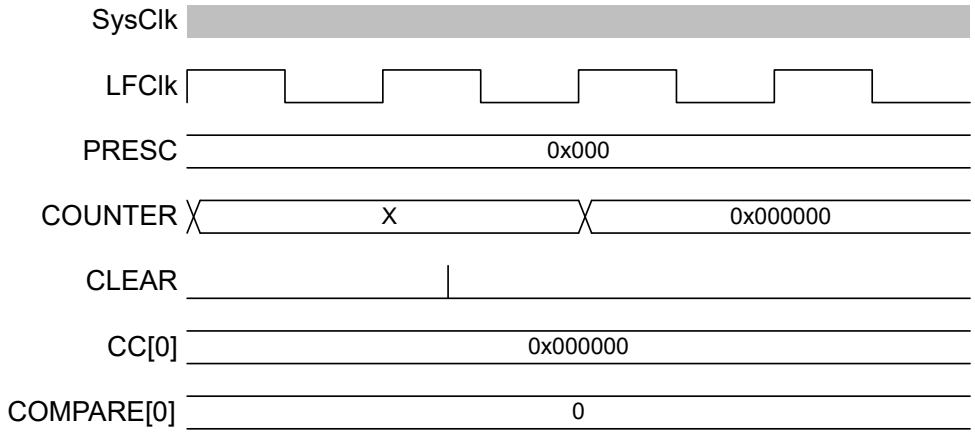


Figure 141: Timing diagram - COMPARE_CLEAR

- If a CC register is N and the COUNTER value is N when the START task is set, this will not trigger a COMPARE event.

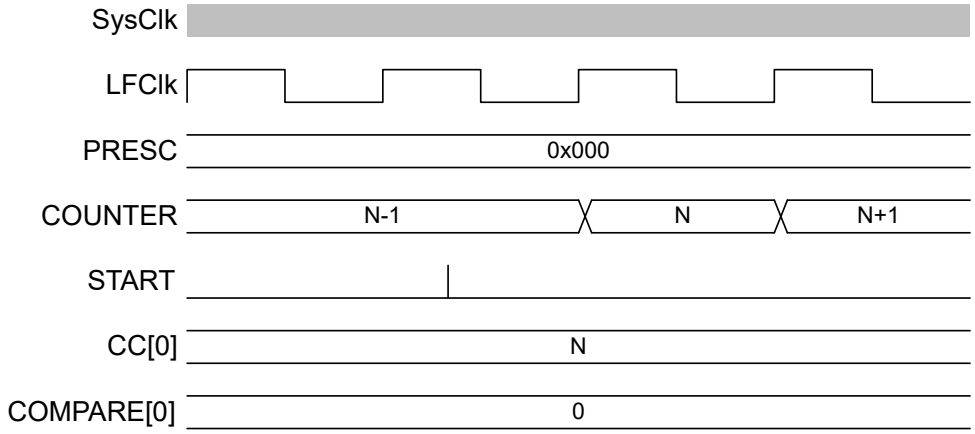


Figure 142: Timing diagram - COMPARE_START

- COMPARE occurs when a CC register is N and the COUNTER value transitions from N-1 to N.

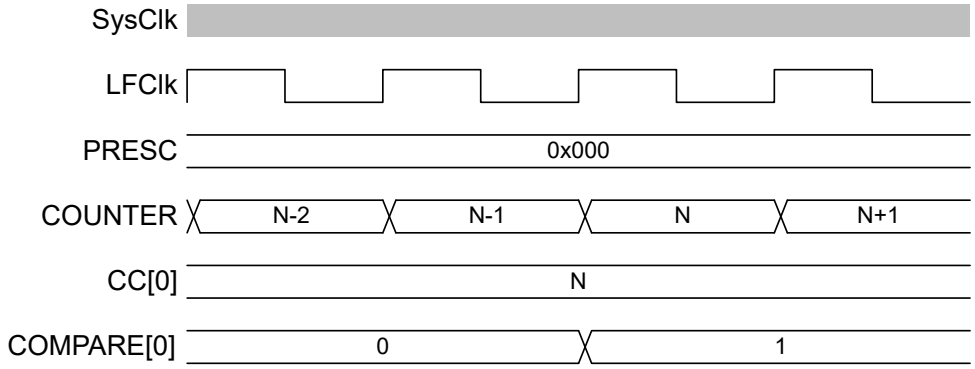


Figure 143: Timing diagram - COMPARE

- If the COUNTER is N, writing N+2 to a CC register is guaranteed to trigger a COMPARE event at N+2.

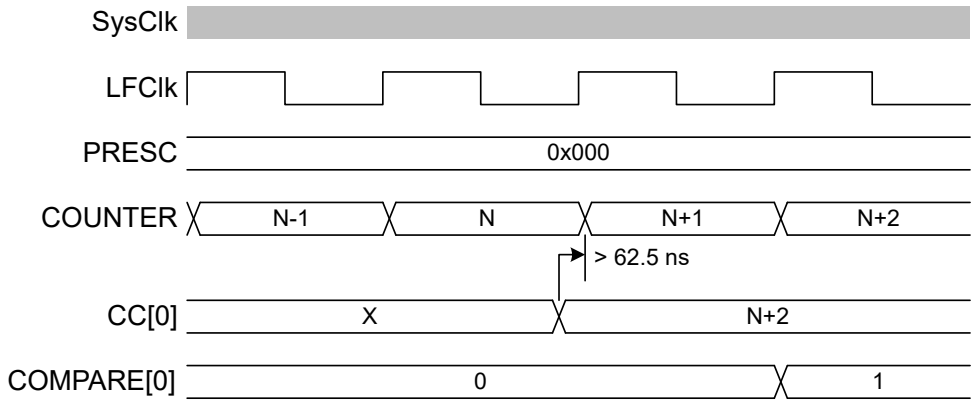


Figure 144: Timing diagram - COMPARE_N+2

- If the COUNTER is N, writing N or N+1 to a CC register may not trigger a COMPARE event.

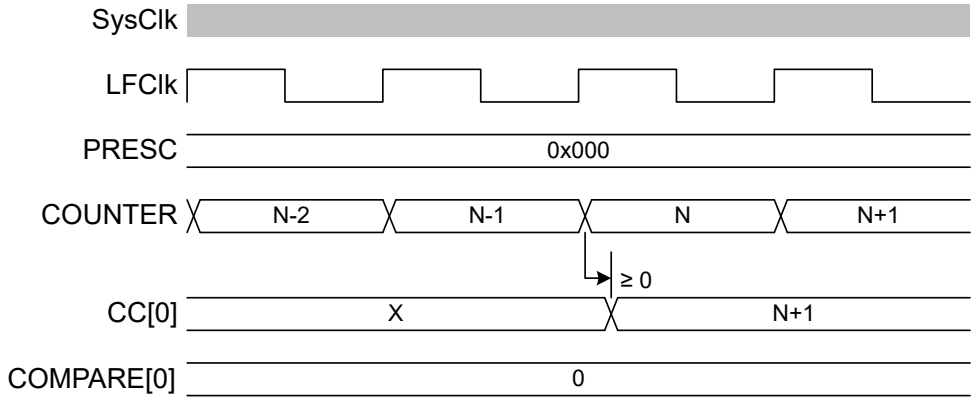


Figure 145: Timing diagram - COMPARE_N+1

- If the COUNTER is N and the current CC register value is N+1 or N+2 when a new CC value is written, a match may trigger on the previous CC value before the new value takes effect. If the current CC value is greater than N+2 when the new value is written, there will be no event due to the old value.

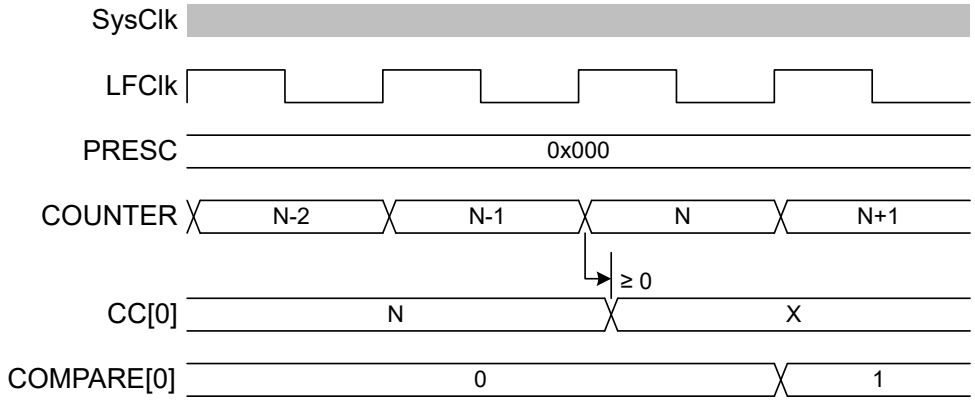


Figure 146: Timing diagram - COMPARE_N-1

6.22.8 TASK and EVENT jitter/delay

Jitter or delay in the RTC is due to the peripheral clock being a low frequency clock (LFCLK) which is not synchronous to the faster PCLK16M.

Registers in the peripheral interface, part of the PCLK16M domain, have a set of mirrored registers in the LFCLK domain. For example, the COUNTER value accessible from the CPU is in the PCLK16M domain and is latched on read from an internal register called COUNTER in the LFCLK domain. COUNTER is the register which is actually modified each time the RTC ticks. These registers must be synchronised between clock domains (PCLK16M and LFCLK).

The following is a summary of the jitter introduced on tasks and events.

Task	Delay
CLEAR, STOP, START, TRIGOVRFLOW	+15 to 46 µs

Table 39: RTC jitter magnitudes on tasks

Operation/Function	Jitter
START to COUNTER increment	+/- 15 µs
COMPARE to COMPARE [26]	+/- 62.5 ns

Table 40: RTC jitter magnitudes on events

Note: 32.768 kHz clock jitter is additional to the numbers provided above.

CLEAR and STOP (and TRIGOVRFLW; not shown) will be delayed as long as it takes for the peripheral to clock a falling edge and rising of the LFCLK. This is between 15.2585 µs and 45.7755 µs – rounded to 15 µs and 46 µs for the remainder of the section.

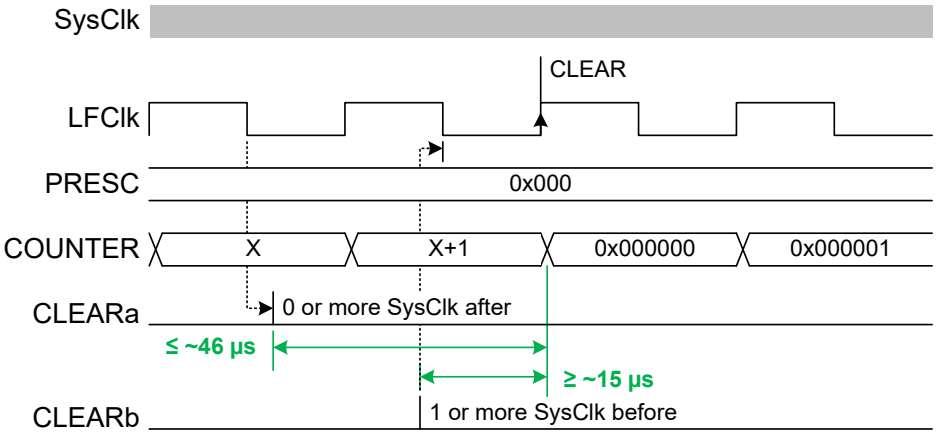


Figure 147: Timing diagram - DELAY_CLEAR

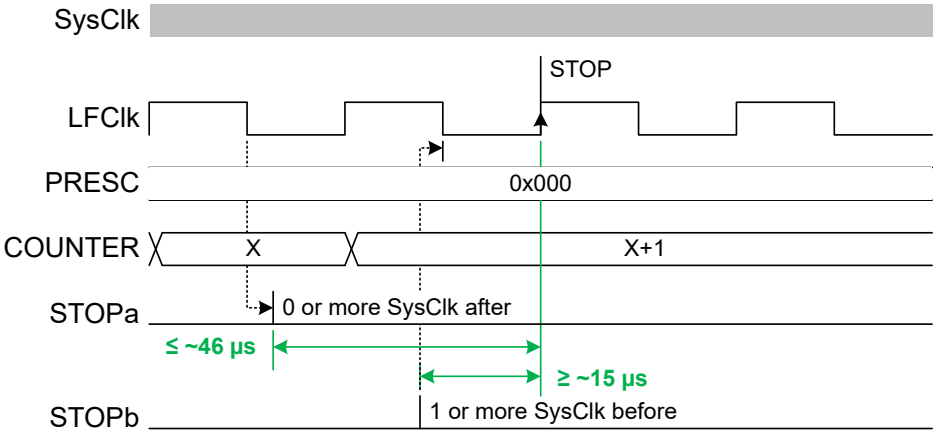


Figure 148: Timing diagram - DELAY_STOP

The START task will start the RTC. Assuming that the LFCLK was previously running and stable, the first increment of COUNTER (and instance of TICK event) will be typically after 30.5 µs +/-15 µs. In some cases, in particular if the RTC is STARTed before the LFCLK is running, that timing can be up to ~250 µs. The software should therefore wait for the first TICK if it has to make sure the RTC is running. Sending a

[26] Assumes RTC runs continuously between these events.

TRIGOVRFLW task sets the COUNTER to a value close to overflow. However, since the update of COUNTER relies on a stable LFCLK, sending this task while LFCLK is not running will start LFCLK, but the update will then be delayed by the same amount of time of up to ~250 µs. The figures show the shortest and longest delays on the START task which appears as a +/-15 µs jitter on the first COUNTER increment.

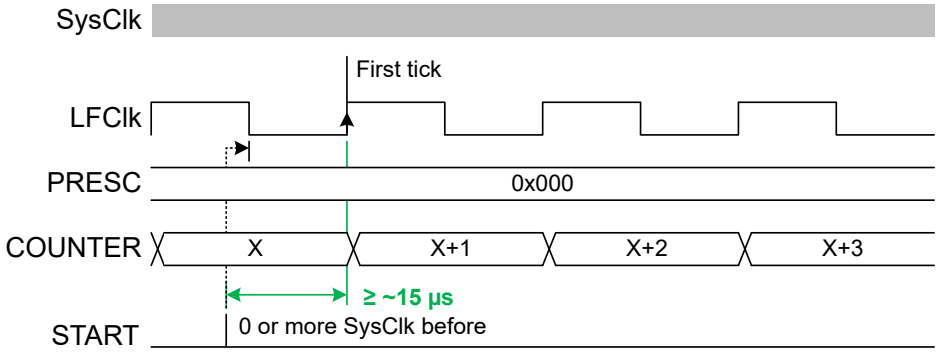


Figure 149: Timing diagram - JITTER_START-

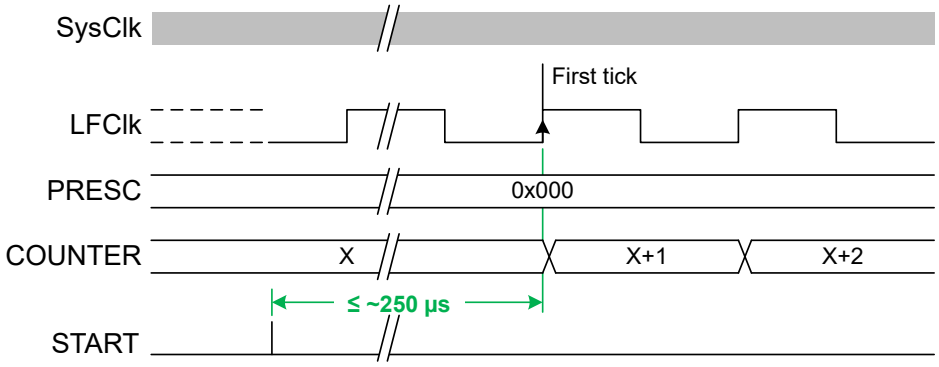


Figure 150: Timing diagram - JITTER_START+

6.22.9 Reading the COUNTER register

To read the COUNTER register, the internal <<COUNTER>> value is sampled.

To ensure that the <<COUNTER>> is safely sampled (considering an LFCLK transition may occur during a read), the CPU and core memory bus are halted for three cycles by lowering the core PREADY signal. The Read takes the CPU 2 cycles in addition resulting in the COUNTER register read taking a fixed five PCLK16M clock cycles.

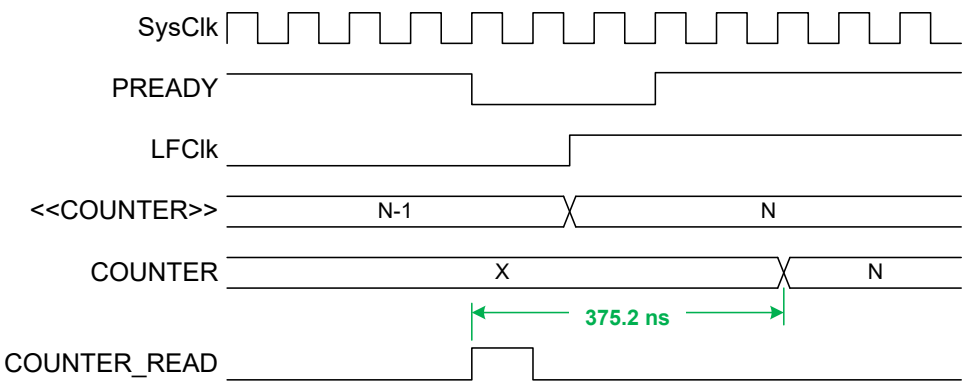


Figure 151: Timing diagram - COUNTER_READ

6.22.10 Registers

Instances

Instance	Base address	Description
RTC0	0x4000B000	Real-time counter 0
RTC1	0x40011000	Real-time counter 1
RTC2	0x40024000	Real-time counter 2

Configuration

Instance	Configuration
RTC0	CC[0..2] implemented, CC[3] not implemented
RTC1	CC[0..3] implemented
RTC2	CC[0..3] implemented

Register overview

Register	Offset	Description
TASKS_START	0x000	Start RTC COUNTER
TASKS_STOP	0x004	Stop RTC COUNTER
TASKS_CLEAR	0x008	Clear RTC COUNTER
TASKS_TRIGOVRFLW	0x00C	Set COUNTER to 0xFFFFF0
EVENTS_TICK	0x100	Event on COUNTER increment
EVENTS_OVRFLW	0x104	Event on COUNTER overflow
EVENTS_COMPARE[0]	0x140	Compare event on CC[0] match
EVENTS_COMPARE[1]	0x144	Compare event on CC[1] match
EVENTS_COMPARE[2]	0x148	Compare event on CC[2] match
EVENTS_COMPARE[3]	0x14C	Compare event on CC[3] match
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
EVTEN	0x340	Enable or disable event routing
EVTENSET	0x344	Enable event routing
EVTENCLR	0x348	Disable event routing
COUNTER	0x504	Current COUNTER value
PRESCALER	0x508	12 bit prescaler for COUNTER frequency (32768/(PRESCALER+1)). Must be written when RTC is stopped.
CC[0]	0x540	Compare register 0
CC[1]	0x544	Compare register 1
CC[2]	0x548	Compare register 2
CC[3]	0x54C	Compare register 3

6.22.10.1 TASKS_START

Address offset: 0x000

Start RTC COUNTER

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_START			Start RTC COUNTER
			Trigger	1	Trigger task

6.22.10.2 TASKS_STOP

Address offset: 0x004

Stop RTC COUNTER

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STOP			Stop RTC COUNTER
			Trigger	1	Trigger task

6.22.10.3 TASKS_CLEAR

Address offset: 0x008

Clear RTC COUNTER

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CLEAR			Clear RTC COUNTER
			Trigger	1	Trigger task

6.22.10.4 TASKS_TRIGOVRFLW

Address offset: 0x00C

Set COUNTER to 0xFFFFF0

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_TRIGOVRFLW			Set COUNTER to 0xFFFFF0
			Trigger	1	Trigger task

6.22.10.5 EVENTS_TICK

Address offset: 0x100

Event on COUNTER increment

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TICK			Event on COUNTER increment
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.22.10.6 EVENTS_OVRFLW

Address offset: 0x104

Event on COUNTER overflow

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_OVRFLW			Event on COUNTER overflow
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.22.10.7 EVENTS_COMPARE[0]

Address offset: 0x140

Compare event on CC[0] match

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_COMPARE			Compare event on CC[0] match
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.22.10.8 EVENTS_COMPARE[1]

Address offset: 0x144

Compare event on CC[1] match

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_COMPARE			Compare event on CC[1] match
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.22.10.9 EVENTS_COMPARE[2]

Address offset: 0x148

Compare event on CC[2] match

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_COMPARE			Compare event on CC[2] match
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.22.10.10 EVENTS_COMPARE[3]

Address offset: 0x14C

Compare event on CC[3] match

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_COMPARE			Compare event on CC[3] match
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.22.10.11 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TICK			Write '1' to enable interrupt for event TICK
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	OVRFLW			Write '1' to enable interrupt for event OVRFLW
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	COMPARE[0]			Write '1' to enable interrupt for event COMPARE[0]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	COMPARE[1]			Write '1' to enable interrupt for event COMPARE[1]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	COMPARE[2]			Write '1' to enable interrupt for event COMPARE[2]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	COMPARE[3]			Write '1' to enable interrupt for event COMPARE[3]
			Set	1	Enable
			Disabled	0	Read: Disabled

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID													F	E	D	C															B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
			Enabled	1	Read: Enabled

6.22.10.12 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID													F	E	D	C															B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TICK			Write '1' to disable interrupt for event TICK
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	OVRFLW			Write '1' to disable interrupt for event OVRFLW
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	COMPARE[0]			Write '1' to disable interrupt for event COMPARE[0]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	COMPARE[1]			Write '1' to disable interrupt for event COMPARE[1]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	COMPARE[2]			Write '1' to disable interrupt for event COMPARE[2]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	COMPARE[3]			Write '1' to disable interrupt for event COMPARE[3]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.22.10.13 EVTEN

Address offset: 0x340

Enable or disable event routing

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID													F	E	D	C															B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TICK			Enable or disable event routing for event TICK
			Disabled	0	Disable
			Enabled	1	Enable
B	RW	OVRFLW			Enable or disable event routing for event OVRFLW

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
			Disabled	0	Disable
			Enabled	1	Enable
C	RW	COMPARE[0]			Enable or disable event routing for event COMPARE[0]
			Disabled	0	Disable
			Enabled	1	Enable
D	RW	COMPARE[1]			Enable or disable event routing for event COMPARE[1]
			Disabled	0	Disable
			Enabled	1	Enable
E	RW	COMPARE[2]			Enable or disable event routing for event COMPARE[2]
			Disabled	0	Disable
			Enabled	1	Enable
F	RW	COMPARE[3]			Enable or disable event routing for event COMPARE[3]
			Disabled	0	Disable
			Enabled	1	Enable

6.22.10.14 EVTENSET

Address offset: 0x344

Enable event routing

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TICK			Write '1' to enable event routing for event TICK
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
			Set	1	Enable
B	RW	OVRFLW			Write '1' to enable event routing for event OVRFLW
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
			Set	1	Enable
C	RW	COMPARE[0]			Write '1' to enable event routing for event COMPARE[0]
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
			Set	1	Enable
D	RW	COMPARE[1]			Write '1' to enable event routing for event COMPARE[1]
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
			Set	1	Enable
E	RW	COMPARE[2]			Write '1' to enable event routing for event COMPARE[2]
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
			Set	1	Enable
F	RW	COMPARE[3]			Write '1' to enable event routing for event COMPARE[3]
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
			Set	1	Enable

6.22.10.15 EVTENCLR

Address offset: 0x348

Disable event routing

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																F	E	D	C																B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	TICK																									Write '1' to disable event routing for event TICK								
			Disabled	0																							Read: Disabled								
			Enabled	1																							Read: Enabled								
			Clear	1																							Disable								
B	RW	OVRFLW																									Write '1' to disable event routing for event OVRFLW								
			Disabled	0																							Read: Disabled								
			Enabled	1																							Read: Enabled								
			Clear	1																							Disable								
C	RW	COMPARE[0]																									Write '1' to disable event routing for event COMPARE[0]								
			Disabled	0																							Read: Disabled								
			Enabled	1																							Read: Enabled								
			Clear	1																							Disable								
D	RW	COMPARE[1]																									Write '1' to disable event routing for event COMPARE[1]								
			Disabled	0																							Read: Disabled								
			Enabled	1																							Read: Enabled								
			Clear	1																							Disable								
E	RW	COMPARE[2]																									Write '1' to disable event routing for event COMPARE[2]								
			Disabled	0																							Read: Disabled								
			Enabled	1																							Read: Enabled								
			Clear	1																							Disable								
F	RW	COMPARE[3]																									Write '1' to disable event routing for event COMPARE[3]								
			Disabled	0																							Read: Disabled								
			Enabled	1																							Read: Enabled								
			Clear	1																							Disable								

6.22.10.16 COUNTER

Address offset: 0x504

Current COUNTER value

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID												A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	R	COUNTER																									Counter value								

6.22.10.17 PRESCALER

Address offset: 0x508

12 bit prescaler for COUNTER frequency (32768/(PRESCALER+1)). Must be written when RTC is stopped.

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PRESCALER			Prescaler value

6.22.10.18 CC[0]

Address offset: 0x540

Compare register 0

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	COMPARE			Compare value

6.22.10.19 CC[1]

Address offset: 0x544

Compare register 1

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	COMPARE			Compare value

6.22.10.20 CC[2]

Address offset: 0x548

Compare register 2

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	COMPARE			Compare value

6.22.10.21 CC[3]

Address offset: 0x54C

Compare register 3

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	COMPARE			Compare value

6.23 SAADC — Successive approximation analog-to-digital converter

The SAADC peripheral is a differential successive approximation register (SAR) analog-to-digital converter.

The main features of the SAADC are the following:

- 8/10/12-bit resolution, 14-bit resolution with oversampling
- Up to eight channels for single-ended inputs and four channels for differential inputs, depending on the package variant
- Full scale input range (0 V to VDD)
- Multiple inputs
  - Input pins **AIN0** to **AIN7**
  - VDD input
  - VDDHDIV5 input
- Individual reference selection for each channel
  - VDD
  - Internal reference
- Continuous sampling
- Output samples automatically stored in RAM using EasyDMA as 16-bit two's complement values
- Internal resistor string

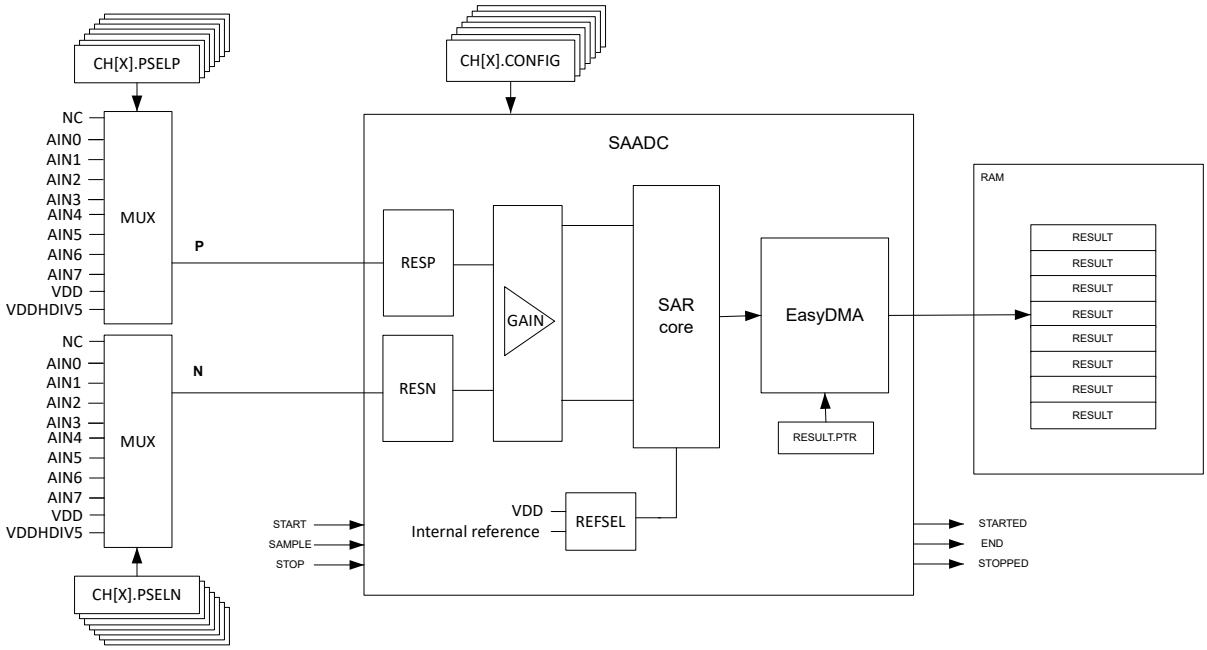


Figure 152: Block diagram

6.23.1 Channel and input configuration

Up to eight SAADC channels can be enabled and configured for SAADC.

A channel is connected to an analog input using the registers CH[n].PSELP and CH[n].PSELN. Each SAADC channel can be configured to use either single-ended mode or differential input mode. Setting register CH[n].PSELP enables the corresponding channel. Setting register CH[n].PSELN has no effect for single-ended channels.

Register CH[n].CONFIG configures SAADC channels. In Single-ended mode, the negative channel input is shorted to ground internally. In Single-ended mode, the assumption is that the internal ground of the ADC is the same as the external ground that the measured voltage is referred to. This makes SAADC sensitive to ground bounce on the PCB. To avoid this, use the differential input mode instead.

Before sampling is started, the length and location of the memory buffer in RAM must be configured. Use registers RESULT.MAXCNT on page 714 and RESULT.PTR on page 713 to configure the length and location where the output values are to be written. The START task must be triggered to apply the configuration. See EasyDMA on page 679 for details on memory configuration and how the results are stored in memory.

SAADC is stopped by triggering the STOP task. The STOP task also terminates the ongoing sampling. SAADC generates a STOPPED event when it has stopped. If SAADC is not started when the STOP task is triggered, the STOPPED event is still generated.

6.23.1.1 Shared resources

The SAADC peripheral shares analog resources with other analog peripherals.

While it is possible to use COMP and SAADC at the same time, selecting the same analog input pin for both peripherals is not supported.

6.23.1.2 Acquisition time

To sample input voltage, SAADC connects a capacitor to the input, as shown in the following figure.

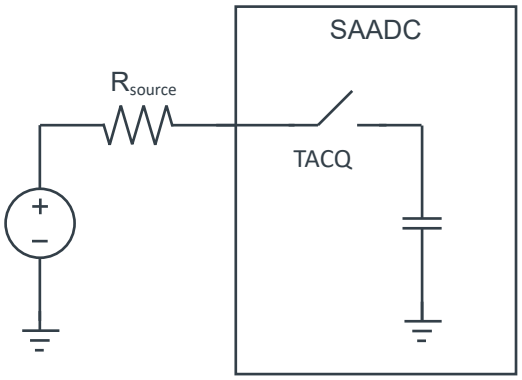


Figure 153: Simplified SAADC sample network

Acquisition time is the amount of time the capacitor is connected, see the TACQ field in the CH[n].CONFIG register. The required acquisition time depends on the source resistance R_{source}. For high source resistance, increase the acquisition time according to the following table:

TACQ [µs]	Maximum source resistance [kΩ]
3	10
5	40
10	100
15	200
20	400
40	800

Table 41: Acquisition time

When using `VDDHDIV5` as input, the acquisition time must be 10 µs or longer.

6.23.1.3 Internal resistor string (resistor ladder)

SAADC has an internal resistor string for positive and negative input. The resistors are controlled in register CH[n].CONFIG.

The following figure illustrates the resistor ladder for positive and negative input:

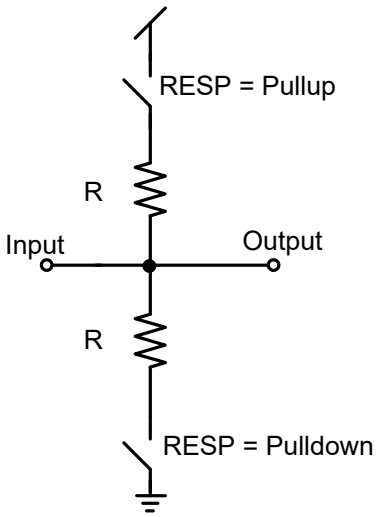


Figure 154: Resistor ladder for positive input (negative input is equivalent, using RESN instead of RESP)

6.23.1.4 Reference voltage and gain settings

Each SAADC channel can have individual reference and gain settings. These settings are configured in register CH[n].CONFIG.

The following configuration options are available:

- VDD/4 or internal 0.6 V reference
- Gain ranging from 1/6 to 4

The gain setting controls the effective input range of SAADC, as shown in the following equation:

```
Input range = (±0.6 V or ±VDD/4)/gain
```

For example, selecting VDD as reference, single-ended input (grounded negative input), and a gain of 1/4 gives the following input range:

```
Input range = (VDD/4)/(1/4) = VDD
```

With internal reference, single-ended input (grounded negative input) and a gain of 1/6, the input range is the following:

```
Input range = (0.6 V)/(1/6) = 3.6 V
```

Inputs **AIN0** to **AIN7** cannot exceed VDD or be lower than VSS.

6.23.2 Operation modes

The SAADC configuration supports Single-channel Single Conversion mode, Single-channel Continuous Conversion mode, and scan mode. Only one mode can be enabled at a time.

Single-channel sampling happens in either Single Conversion mode or Continuous Conversion mode. Scan mode is entered when more than one channel is enabled.

Oversampling can be used to improve the signal-to-noise ratio (SNR). It is not recommended to use oversampling for scan mode. For more information about oversampling, see Oversampling on page 678.

6.23.2.1 Single-channel Single Conversion mode

SAADC performs one conversion of a single channel and stops once complete.

This mode of operation is configured by enabling only one of the available channels defined by the registers CH[n].PSELP, CH[n].PSELN, and CH[n].CONFIG. When the SAMPLE task is triggered, SAADC starts sampling the input voltage.

A DONE event signals that the sample was taken. In this mode, the RESULTDONE and DONE events are equal when oversampling does not happen. Both events can occur before EasyDMA transfers the value to RAM. For more information, see EasyDMA on page 679. The END event is generated when RESULT.MAXCNT on page 714 values are transferred to RAM.

6.23.2.2 Single-channel Continuous Conversion mode

In Single-channel Continuous Conversion mode, a single channel is sampled continuously.

Continuous sampling is achieved with an internal timer in the SAADC peripheral. The register SAMPLERATE on page 713 configures the Single-channel Continuous Conversion mode and sample rate.

The sample rate must fulfill the following criteria:

$$f_{SAMPLE} < 1 / (t_{ACQ} + t_{conv})$$

When Single-channel Continuous Conversion mode is selected, SAADC is started by triggering the SAMPLE task once. Triggering the STOP task stops sampling. A DONE event signals that one sample was taken. In this mode, the RESULTDONE and DONE events are equal when oversampling does not happen. Both events may occur before EasyDMA transfers the value to RAM. For more information, see EasyDMA on page 679. The END event is generated when RESULT.MAXCNT on page 714 values are transferred to RAM.

6.23.2.3 Scan mode

If more than one channel is enabled, SAADC functions in scan mode.

In scan mode, one SAMPLE task triggers one conversion per enabled channel.

The time it takes to sample all channels is less than the sum of the conversion time of all enabled channels. The conversion time for a channel is defined as the sum of the acquisition time t_{ACQ} and the conversion time t_{CONV}.

The events DONE and RESULTDONE are generated when one sample is taken. Both events may occur before EasyDMA transfers the values into RAM, see EasyDMA on page 679 for more information.

Note: Continuous conversion mode is not supported in scan mode.

6.23.2.4 Oversampling

An accumulator in SAADC can be used to find the average of several analog input samples. In general, oversampling improves the signal-to-noise ratio (SNR). Oversampling does not improve the integral non-linearity (INL) or differential non-linearity (DNL).

Oversampling is configured in register OVERSAMPLE. When oversampling, $2^{OVERSAMPLE}$ samples are averaged before one result is transferred to memory. The mode used to sample the input determines when and how those samples are taken.

When oversampling is configured, DONE event is generated for every input sample taken. RESULTDONE event is generated for every averaged value ready to be transferred into RAM. END event is generated when RESULT.MAXCNT averaged values are transferred into RAM.

Note: Oversampling should only be used when a single input channel is enabled, as averaging is performed over all enabled channels.

6.23.3 Digital output

The digital output value is calculated using the following formula.

$$RESULT = [V(P) - V(N)] * (GAIN/REFERENCE) * 2^{(RESOLUTION - m)}$$

where `V(P)` is the voltage at input P, `V(N)` is the voltage at input N, `GAIN` is the selected gain, `REFERENCE` is the selected reference voltage, `RESOLUTION` is output resolution in bits, as configured in register RESOLUTION on page 712, and `m` being 0 for single-ended channels and 1 for differential channels.

Results are sign extended to 16 bits and stored in RAM in little-endian byte order.

Results generated by SAADC deviate due to DC errors like offset, gain, differential non-linearity (DNL), and integral non-linearity (INL). See Electrical specification for details on these parameters. The result can also vary due to AC errors like non-linearities in the gain block, settling errors due to high source impedance, and sampling jitter. DC errors affect the most for battery measurement.

6.23.4 EasyDMA

SAADC resources are started by triggering the START task. The SAADC uses EasyDMA to store results in a buffer in RAM.

Registers RESULT.PTR on page 713 and RESULT.MAXCNT on page 714 must be configured before SAADC is started.

The result buffer is located at the address specified in register RESULT.PTR on page 713. This register is double-buffered, and it can be updated and prepared for the next START task immediately after the STARTED event is generated. Register RESULT.MAXCNT on page 714 specifies the size of the result buffer. SAADC generates an END event when the result buffer is full, as shown in the following figure.

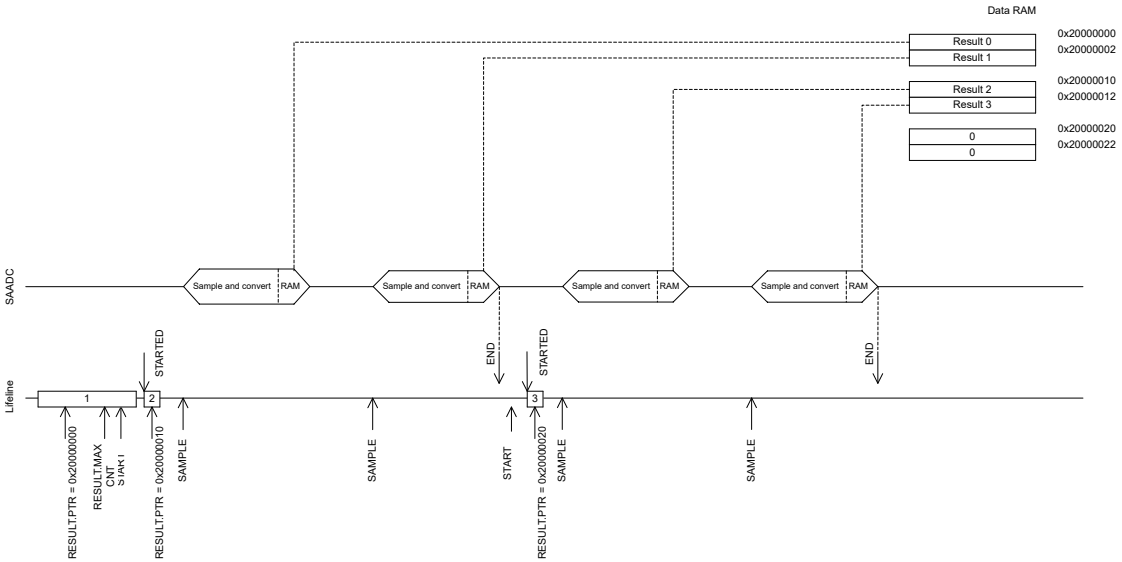


Figure 155: SAADC

The following figure provides an example of results in Data RAM with an even RESULT.MAXCNT on page 714 and channels 1, 2, and 5 enabled.

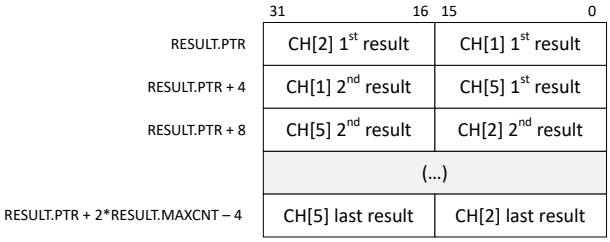


Figure 156: Example of RAM placement (even RESULT.MAXCNT), channels 1, 2, and 5 enabled

The following figure provides an example of results in Data RAM with an odd RESULT.MAXCNT on page 714 and channels 1, 2, and 5 enabled.

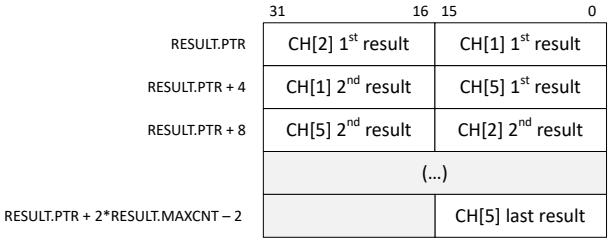


Figure 157: Example of RAM placement (odd RESULT.MAXCNT), channels 1, 2, and 5 enabled

The last 32-bit word is populated with only one 16-bit result.

See Memory on page 21 for more information about the different memory regions.

EasyDMA is finished accessing RAM when events END or STOPPED are generated. To see the number of results transferred to the RAM result buffer since the START task was triggered, read register RESULT.AMOUNT on page 714.

6.23.5 Event monitoring using limits

Using limits allows event monitoring on channels.

A high and low limit can be configured in the CH[n].LIMIT register. The high limit must be higher than or equal to the low limit.

Relevant events are generated when the conversion results (sampled input signals) are outside of the defined limits. It is not possible to generate an event when the input signal is inside a defined range by switching the high and low limit. An example of event monitoring using limits is illustrated in the following figure:

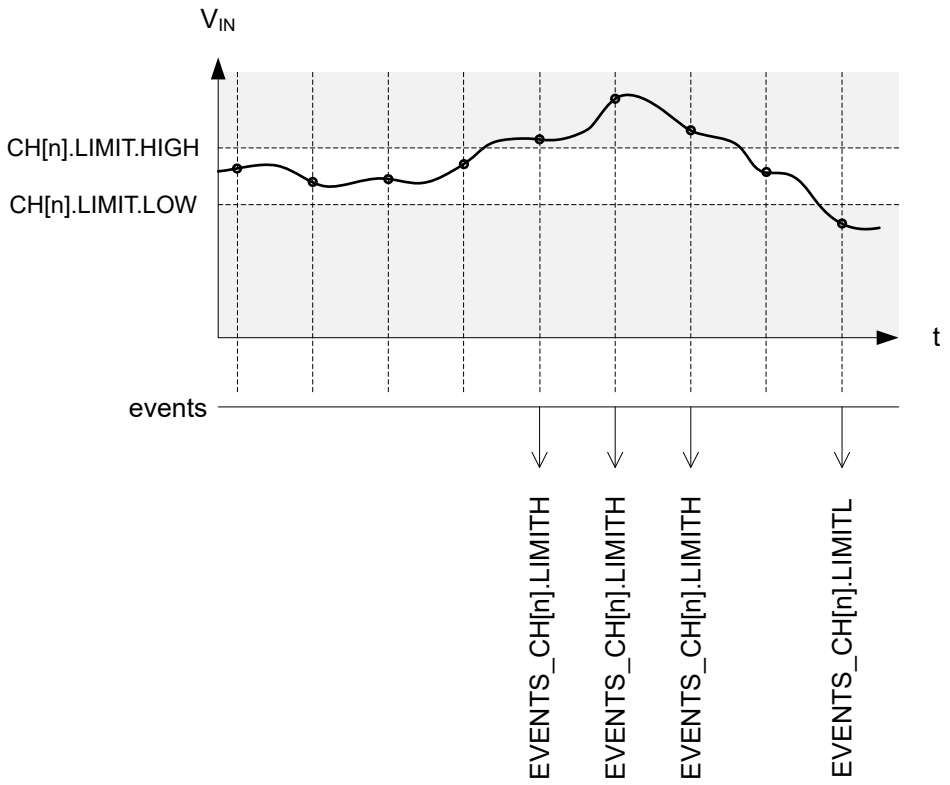


Figure 158: Event monitoring on channel[n] using limits

Limit comparison does not need to be enabled. If event monitoring is not required, related events should be ignored.

6.23.6 Calibration

The SAADC peripheral has a temperature dependent offset.

It is recommended to calibrate SAADC at least once before use, and recalibrate when the ambient temperature changes by more than 10°C.

Offset calibration is started by triggering the CALIBRATEOFFSET task. The CALIBRATEDONE event is generated when calibration is finished.

6.23.7 Registers

Instances

Instance	Base address	Description
SAADC	0x40007000	Analog to digital converter

Register overview

Register	Offset	Description
TASKS_START	0x000	Starts the SAADC and prepares the result buffer in RAM
TASKS_SAMPLE	0x004	Takes one SAADC sample
TASKS_STOP	0x008	Stops the SAADC and terminates all on-going conversions
TASKS_CALIBRATEOFFSET	0x00C	Starts offset auto-calibration
EVENTS_STARTED	0x100	The SAADC has started

Register	Offset	Description
EVENTS_END	0x104	The SAADC has filled up the result buffer
EVENTS_DONE	0x108	A conversion task has been completed. Depending on the configuration, multiple conversions might be needed for a result to be transferred to RAM.
EVENTS_RESULTDONE	0x10C	Result ready for transfer to RAM
EVENTS_CALIBRATEDONE	0x110	Calibration is complete
EVENTS_STOPPED	0x114	The SAADC has stopped
EVENTS_CH[0].LIMITH	0x118	Last result is equal or above CH[0].LIMIT.HIGH
EVENTS_CH[0].LIMITL	0x11C	Last result is equal or below CH[0].LIMIT.LOW
EVENTS_CH[1].LIMITH	0x120	Last result is equal or above CH[1].LIMIT.HIGH
EVENTS_CH[1].LIMITL	0x124	Last result is equal or below CH[1].LIMIT.LOW
EVENTS_CH[2].LIMITH	0x128	Last result is equal or above CH[2].LIMIT.HIGH
EVENTS_CH[2].LIMITL	0x12C	Last result is equal or below CH[2].LIMIT.LOW
EVENTS_CH[3].LIMITH	0x130	Last result is equal or above CH[3].LIMIT.HIGH
EVENTS_CH[3].LIMITL	0x134	Last result is equal or below CH[3].LIMIT.LOW
EVENTS_CH[4].LIMITH	0x138	Last result is equal or above CH[4].LIMIT.HIGH
EVENTS_CH[4].LIMITL	0x13C	Last result is equal or below CH[4].LIMIT.LOW
EVENTS_CH[5].LIMITH	0x140	Last result is equal or above CH[5].LIMIT.HIGH
EVENTS_CH[5].LIMITL	0x144	Last result is equal or below CH[5].LIMIT.LOW
EVENTS_CH[6].LIMITH	0x148	Last result is equal or above CH[6].LIMIT.HIGH
EVENTS_CH[6].LIMITL	0x14C	Last result is equal or below CH[6].LIMIT.LOW
EVENTS_CH[7].LIMITH	0x150	Last result is equal or above CH[7].LIMIT.HIGH
EVENTS_CH[7].LIMITL	0x154	Last result is equal or below CH[7].LIMIT.LOW
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
STATUS	0x400	Status
ENABLE	0x500	Enable or disable SAADC
CH[0].PSELP	0x510	Input positive pin selection for CH[0]
CH[0].PSELN	0x514	Input negative pin selection for CH[0]
CH[0].CONFIG	0x518	Input configuration for CH[0]
CH[0].LIMIT	0x51C	High/low limits for event monitoring of a channel
CH[1].PSELP	0x520	Input positive pin selection for CH[1]
CH[1].PSELN	0x524	Input negative pin selection for CH[1]
CH[1].CONFIG	0x528	Input configuration for CH[1]
CH[1].LIMIT	0x52C	High/low limits for event monitoring of a channel
CH[2].PSELP	0x530	Input positive pin selection for CH[2]
CH[2].PSELN	0x534	Input negative pin selection for CH[2]
CH[2].CONFIG	0x538	Input configuration for CH[2]
CH[2].LIMIT	0x53C	High/low limits for event monitoring of a channel
CH[3].PSELP	0x540	Input positive pin selection for CH[3]
CH[3].PSELN	0x544	Input negative pin selection for CH[3]
CH[3].CONFIG	0x548	Input configuration for CH[3]
CH[3].LIMIT	0x54C	High/low limits for event monitoring of a channel
CH[4].PSELP	0x550	Input positive pin selection for CH[4]
CH[4].PSELN	0x554	Input negative pin selection for CH[4]
CH[4].CONFIG	0x558	Input configuration for CH[4]
CH[4].LIMIT	0x55C	High/low limits for event monitoring of a channel
CH[5].PSELP	0x560	Input positive pin selection for CH[5]
CH[5].PSELN	0x564	Input negative pin selection for CH[5]
CH[5].CONFIG	0x568	Input configuration for CH[5]
CH[5].LIMIT	0x56C	High/low limits for event monitoring of a channel
CH[6].PSELP	0x570	Input positive pin selection for CH[6]
CH[6].PSELN	0x574	Input negative pin selection for CH[6]

Register	Offset	Description
CH[6].CONFIG	0x578	Input configuration for CH[6]
CH[6].LIMIT	0x57C	High/low limits for event monitoring of a channel
CH[7].PSELP	0x580	Input positive pin selection for CH[7]
CH[7].PSELN	0x584	Input negative pin selection for CH[7]
CH[7].CONFIG	0x588	Input configuration for CH[7]
CH[7].LIMIT	0x58C	High/low limits for event monitoring of a channel
RESOLUTION	0x5F0	Resolution configuration
OVERSAMPLE	0x5F4	Oversampling configuration. The RESOLUTION is applied before averaging, thus for high OVERSAMPLE a higher RESOLUTION should be used.
SAMPLERATE	0x5F8	Controls normal or continuous sample rate
RESULT.PTR	0x62C	Data pointer
RESULT.MAXCNT	0x630	Maximum number of 16-bit samples to be written to output RAM buffer
RESULT.AMOUNT	0x634	Number of 16-bit samples written to output RAM buffer since the previous START task

6.23.7.1 TASKS_START

Address offset: 0x000

Starts the SAADC and prepares the result buffer in RAM

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_START			Starts the SAADC and prepares the result buffer in RAM
			Trigger	1	Trigger task

6.23.7.2 TASKS_SAMPLE

Address offset: 0x004

Takes one SAADC sample

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_SAMPLE			Takes one SAADC sample
			Trigger	1	Trigger task

6.23.7.3 TASKS_STOP

Address offset: 0x008

Stops the SAADC and terminates all on-going conversions

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STOP			Stops the SAADC and terminates all on-going conversions
			Trigger	1	Trigger task

6.23.7.4 TASKS_CALIBRATEOFFSET

Address offset: 0x00C

Starts offset auto-calibration

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CALIBRATEOFFSET			Starts offset auto-calibration
			Trigger	1	Trigger task

6.23.7.5 EVENTS_STARTED

Address offset: 0x100

The SAADC has started

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_STARTED			The SAADC has started
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.23.7.6 EVENTS_END

Address offset: 0x104

The SAADC has filled up the result buffer

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_END			The SAADC has filled up the result buffer
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.23.7.7 EVENTS_DONE

Address offset: 0x108

A conversion task has been completed. Depending on the configuration, multiple conversions might be needed for a result to be transferred to RAM.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_DONE			A conversion task has been completed. Depending on the configuration, multiple conversions might be needed for a result to be transferred to RAM.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.23.7.8 EVENTS_RESULTDONE

Address offset: 0x10C

Result ready for transfer to RAM

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_RESULTDONE			Result ready for transfer to RAM
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.23.7.9 EVENTS_CALIBRATEDONE

Address offset: 0x110

Calibration is complete

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_CALIBRATEDONE			Calibration is complete
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.23.7.10 EVENTS_STOPPED

Address offset: 0x114

The SAADC has stopped

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_STOPPED			The SAADC has stopped
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.23.7.11 EVENTS_CH[0]

Peripheral events.

6.23.7.11.1 EVENTS_CH[0].LIMITH

Address offset: 0x118

Last result is equal or above CH[0].LIMIT.HIGH

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	LIMITH																																	
			NotGenerated	0																															
			Generated	1																															

6.23.7.11.2 EVENTS_CH[0].LIMITL

Address offset: 0x11C

Last result is equal or below CH[0].LIMIT.LOW

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	LIMITL																																	
			NotGenerated	0																															
			Generated	1																															

6.23.7.12 EVENTS_CH[1]

Peripheral events.

6.23.7.12.1 EVENTS_CH[1].LIMITH

Address offset: 0x120

Last result is equal or above CH[1].LIMIT.HIGH

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	LIMITH																																	
			NotGenerated	0																															
			Generated	1																															

6.23.7.12.2 EVENTS_CH[1].LIMITL

Address offset: 0x124

Last result is equal or below CH[1].LIMIT.LOW

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	LIMITL			Last result is equal or below CH[1].LIMIT.LOW
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.23.7.13 EVENTS_CH[2]

Peripheral events.

6.23.7.13.1 EVENTS_CH[2].LIMITH

Address offset: 0x128

Last result is equal or above CH[2].LIMIT.HIGH

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	LIMITH			Last result is equal or above CH[2].LIMIT.HIGH
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.23.7.13.2 EVENTS_CH[2].LIMITL

Address offset: 0x12C

Last result is equal or below CH[2].LIMIT.LOW

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	LIMITL			Last result is equal or below CH[2].LIMIT.LOW
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.23.7.14 EVENTS_CH[3]

Peripheral events.

6.23.7.14.1 EVENTS_CH[3].LIMITH

Address offset: 0x130

Last result is equal or above CH[3].LIMIT.HIGH

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	LIMITH			Last result is equal or above CH[3].LIMIT.HIGH
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.23.7.14.2 EVENTS_CH[3].LIMITL

Address offset: 0x134

Last result is equal or below CH[3].LIMIT.LOW

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	LIMITL			Last result is equal or below CH[3].LIMIT.LOW
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.23.7.15 EVENTS_CH[4]

Peripheral events.

6.23.7.15.1 EVENTS_CH[4].LIMITH

Address offset: 0x138

Last result is equal or above CH[4].LIMIT.HIGH

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	LIMITH			Last result is equal or above CH[4].LIMIT.HIGH
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.23.7.15.2 EVENTS_CH[4].LIMITL

Address offset: 0x13C

Last result is equal or below CH[4].LIMIT.LOW

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	LIMITL			Last result is equal or below CH[4].LIMIT.LOW
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.23.7.16 EVENTS_CH[5]

Peripheral events.

6.23.7.16.1 EVENTS_CH[5].LIMITH

Address offset: 0x140

Last result is equal or above CH[5].LIMIT.HIGH

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	LIMITH			Last result is equal or above CH[5].LIMIT.HIGH
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.23.7.16.2 EVENTS_CH[5].LIMITL

Address offset: 0x144

Last result is equal or below CH[5].LIMIT.LOW

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	LIMITL			Last result is equal or below CH[5].LIMIT.LOW
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.23.7.17 EVENTS_CH[6]

Peripheral events.

6.23.7.17.1 EVENTS_CH[6].LIMITH

Address offset: 0x148

Last result is equal or above CH[6].LIMIT.HIGH

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	LIMITH			Last result is equal or above CH[6].LIMIT.HIGH
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.23.7.17.2 EVENTS_CH[6].LIMITL

Address offset: 0x14C

Last result is equal or below CH[6].LIMIT.LOW

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	LIMITL			Last result is equal or below CH[6].LIMIT.LOW
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.23.7.18 EVENTS_CH[7]

Peripheral events.

6.23.7.18.1 EVENTS_CH[7].LIMITH

Address offset: 0x150

Last result is equal or above CH[7].LIMIT.HIGH

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	LIMITH			Last result is equal or above CH[7].LIMIT.HIGH
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.23.7.18.2 EVENTS_CH[7].LIMITL

Address offset: 0x154

Last result is equal or below CH[7].LIMIT.LOW

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	LIMITL			Last result is equal or below CH[7].LIMIT.LOW
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.23.7.19 INTEN

Address offset: 0x300

Enable or disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	STARTED			Enable or disable interrupt for event STARTED
			Disabled	0	Disable
			Enabled	1	Enable
B	RW	END			Enable or disable interrupt for event END
			Disabled	0	Disable

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Enabled	1	Enable
C	RW	DONE			Enable or disable interrupt for event DONE
			Disabled	0	Disable
			Enabled	1	Enable
D	RW	RESULTDONE			Enable or disable interrupt for event RESULTDONE
			Disabled	0	Disable
			Enabled	1	Enable
E	RW	CALIBRATEDONE			Enable or disable interrupt for event CALIBRATEDONE
			Disabled	0	Disable
			Enabled	1	Enable
F	RW	STOPPED			Enable or disable interrupt for event STOPPED
			Disabled	0	Disable
			Enabled	1	Enable
G	RW	CH0LIMITH			Enable or disable interrupt for event CH0LIMITH
			Disabled	0	Disable
			Enabled	1	Enable
H	RW	CH0LIMITL			Enable or disable interrupt for event CH0LIMITL
			Disabled	0	Disable
			Enabled	1	Enable
I	RW	CH1LIMITH			Enable or disable interrupt for event CH1LIMITH
			Disabled	0	Disable
			Enabled	1	Enable
J	RW	CH1LIMITL			Enable or disable interrupt for event CH1LIMITL
			Disabled	0	Disable
			Enabled	1	Enable
K	RW	CH2LIMITH			Enable or disable interrupt for event CH2LIMITH
			Disabled	0	Disable
			Enabled	1	Enable
L	RW	CH2LIMITL			Enable or disable interrupt for event CH2LIMITL
			Disabled	0	Disable
			Enabled	1	Enable
M	RW	CH3LIMITH			Enable or disable interrupt for event CH3LIMITH
			Disabled	0	Disable
			Enabled	1	Enable
N	RW	CH3LIMITL			Enable or disable interrupt for event CH3LIMITL
			Disabled	0	Disable
			Enabled	1	Enable
O	RW	CH4LIMITH			Enable or disable interrupt for event CH4LIMITH
			Disabled	0	Disable
			Enabled	1	Enable
P	RW	CH4LIMITL			Enable or disable interrupt for event CH4LIMITL
			Disabled	0	Disable
			Enabled	1	Enable
Q	RW	CH5LIMITH			Enable or disable interrupt for event CH5LIMITH
			Disabled	0	Disable
			Enabled	1	Enable
R	RW	CH5LIMITL			Enable or disable interrupt for event CH5LIMITL
			Disabled	0	Disable
			Enabled	1	Enable
S	RW	CH6LIMITH			Enable or disable interrupt for event CH6LIMITH

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Disabled	0	Disable
			Enabled	1	Enable
T	RW	CH6LIMITL			Enable or disable interrupt for event CH6LIMITL
			Disabled	0	Disable
			Enabled	1	Enable
U	RW	CH7LIMITH			Enable or disable interrupt for event CH7LIMITH
			Disabled	0	Disable
			Enabled	1	Enable
V	RW	CH7LIMITL			Enable or disable interrupt for event CH7LIMITL
			Disabled	0	Disable
			Enabled	1	Enable

6.23.7.20 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	STARTED			Write '1' to enable interrupt for event STARTED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	END			Write '1' to enable interrupt for event END
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	DONE			Write '1' to enable interrupt for event DONE
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	RESULTDONE			Write '1' to enable interrupt for event RESULTDONE
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	CALIBRATEDONE			Write '1' to enable interrupt for event CALIBRATEDONE
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	STOPPED			Write '1' to enable interrupt for event STOPPED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
G	RW	CH0LIMITH			Write '1' to enable interrupt for event CH0LIMITH
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
H	RW	CH0LIMITL			Write '1' to enable interrupt for event CH0LIMITL
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
I	RW	CH1LIMITH			Write '1' to enable interrupt for event CH1LIMITH
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
J	RW	CH1LIMITL			Write '1' to enable interrupt for event CH1LIMITL
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
K	RW	CH2LIMITH			Write '1' to enable interrupt for event CH2LIMITH
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
L	RW	CH2LIMITL			Write '1' to enable interrupt for event CH2LIMITL
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
M	RW	CH3LIMITH			Write '1' to enable interrupt for event CH3LIMITH
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
N	RW	CH3LIMITL			Write '1' to enable interrupt for event CH3LIMITL
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
O	RW	CH4LIMITH			Write '1' to enable interrupt for event CH4LIMITH
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
P	RW	CH4LIMITL			Write '1' to enable interrupt for event CH4LIMITL
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
Q	RW	CH5LIMITH			Write '1' to enable interrupt for event CH5LIMITH
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
R	RW	CH5LIMITL			Write '1' to enable interrupt for event CH5LIMITL
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
S	RW	CH6LIMITH			Write '1' to enable interrupt for event CH6LIMITH
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
T	RW	CH6LIMITL			Write '1' to enable interrupt for event CH6LIMITL
			Set	1	Enable

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
U	RW	CH7LIMITH			Write '1' to enable interrupt for event CH7LIMITH
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
V	RW	CH7LIMITL			Write '1' to enable interrupt for event CH7LIMITL
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.23.7.21 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	STARTED			Write '1' to disable interrupt for event STARTED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	END			Write '1' to disable interrupt for event END
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	DONE			Write '1' to disable interrupt for event DONE
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	RESULTDONE			Write '1' to disable interrupt for event RESULTDONE
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	CALIBRATEDONE			Write '1' to disable interrupt for event CALIBRATEDONE
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	STOPPED			Write '1' to disable interrupt for event STOPPED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
G	RW	CH0LIMITH			Write '1' to disable interrupt for event CH0LIMITH
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
H	RW	CH0LIMITL			Write '1' to disable interrupt for event CH0LIMITL

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
I	RW	CH1LIMITH			Write '1' to disable interrupt for event CH1LIMITH
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
J	RW	CH1LIMITL			Write '1' to disable interrupt for event CH1LIMITL
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
K	RW	CH2LIMITH			Write '1' to disable interrupt for event CH2LIMITH
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
L	RW	CH2LIMITL			Write '1' to disable interrupt for event CH2LIMITL
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
M	RW	CH3LIMITH			Write '1' to disable interrupt for event CH3LIMITH
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
N	RW	CH3LIMITL			Write '1' to disable interrupt for event CH3LIMITL
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
O	RW	CH4LIMITH			Write '1' to disable interrupt for event CH4LIMITH
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
P	RW	CH4LIMITL			Write '1' to disable interrupt for event CH4LIMITL
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
Q	RW	CH5LIMITH			Write '1' to disable interrupt for event CH5LIMITH
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
R	RW	CH5LIMITL			Write '1' to disable interrupt for event CH5LIMITL
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
S	RW	CH6LIMITH			Write '1' to disable interrupt for event CH6LIMITH
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
T	RW	CH6LIMITL			Write '1' to disable interrupt for event CH6LIMITL
			Clear	1	Disable
			Disabled	0	Read: Disabled

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
			Enabled	1	Read: Enabled
U	RW	CH7LIMITH			Write '1' to disable interrupt for event CH7LIMITH
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
V	RW	CH7LIMITL			Write '1' to disable interrupt for event CH7LIMITL
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.23.7.22 STATUS

Address offset: 0x400

Status

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	STATUS			Status
			Ready	0	SAADC is ready. No on-going conversions.
			Busy	1	SAADC is busy. Conversion in progress.

6.23.7.23 ENABLE

Address offset: 0x500

Enable or disable SAADC

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ENABLE			Enable or disable SAADC
			Disabled	0	Disable SAADC
			Enabled	1	Enable SAADC
					When enabled, the SAADC will acquire access to analog input pins specified in registers CH[n].PSELP and CH[n].PSELN

6.23.7.24 CH[0].PSELP

Address offset: 0x510

Input positive pin selection for CH[0]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PSELP			Analog positive input channel
			NC	0	Not connected
			AnalogInput0	1	AIN0
			AnalogInput1	2	AIN1
			AnalogInput2	3	AIN2
			AnalogInput3	4	AIN3
			AnalogInput4	5	AIN4
			AnalogInput5	6	AIN5
			AnalogInput6	7	AIN6
			AnalogInput7	8	AIN7
			VDD	9	VDD
			VDDHDIV5	0x0D	VDDH/5

6.23.7.25 CH[0].PSELN

Address offset: 0x514

Input negative pin selection for CH[0]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PSELN			Analog negative input, enables differential channel
			NC	0	Not connected
			AnalogInput0	1	AIN0
			AnalogInput1	2	AIN1
			AnalogInput2	3	AIN2
			AnalogInput3	4	AIN3
			AnalogInput4	5	AIN4
			AnalogInput5	6	AIN5
			AnalogInput6	7	AIN6
			AnalogInput7	8	AIN7
			VDD	9	VDD
			VDDHDIV5	0x0D	VDDH/5

6.23.7.26 CH[0].CONFIG

Address offset: 0x518

Input configuration for CH[0]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				G F E E E D C C C B B A A
Reset 0x00020000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RESP			Positive channel resistor control
			Bypass	0	Bypass resistor ladder
			Pulldown	1	Pull-down to GND
			Pullup	2	Pull-up to VDD
			VDD1_2	3	Set input at VDD/2

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				G F E E E D C C C B B A A
Reset 0x00020000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
B	RW	RESN			Negative channel resistor control
			Bypass	0	Bypass resistor ladder
			Pulldown	1	Pull-down to GND
			Pullup	2	Pull-up to VDD
			VDD1_2	3	Set input at VDD/2
C	RW	GAIN			Gain control
			Gain1_6	0	1/6
			Gain1_5	1	1/5
			Gain1_4	2	1/4
			Gain1_3	3	1/3
			Gain1_2	4	1/2
			Gain1	5	1
			Gain2	6	2
			Gain4	7	4
D-	RW	REFSEL			Reference control
			Internal	0	Internal reference (0.6 V)
			VDD1_4	1	VDD/4 as reference
E	RW	TACQ			Acquisition time, the time the SAADC uses to sample the input voltage
			3us	0	3 µs
			5us	1	5 µs
			10us	2	10 µs
			15us	3	15 µs
			20us	4	20 µs
			40us	5	40 µs
F	RW	MODE			Enable differential mode
			SE	0	Single-ended, PSELN will be ignored, negative input to SAADC shorted to GND
			Diff	1	Differential
G	RW	BURST			Enable burst mode
			Disabled	0	Burst mode is disabled (normal operation)
			Enabled	1	Burst mode is enabled. SAADC takes 2^OVERSAMPLE number of samples as fast as it can, and sends the average to Data RAM.

6.23.7.27 CH[0].LIMIT

Address offset: 0x51C

High/low limits for event monitoring of a channel

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B B B B B B B B B B B B B B B B A A A A A A A A A A A A A A A A
Reset 0x7FFF8000				**0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	LOW		[-32768 to +32767]	Low level limit
B	RW	HIGH		[-32768 to +32767]	High level limit

6.23.7.28 CH[1].PSELP

Address offset: 0x520

Input positive pin selection for CH[1]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PSELP			Analog positive input channel
			NC	0	Not connected
			AnalogInput0	1	AIN0
			AnalogInput1	2	AIN1
			AnalogInput2	3	AIN2
			AnalogInput3	4	AIN3
			AnalogInput4	5	AIN4
			AnalogInput5	6	AIN5
			AnalogInput6	7	AIN6
			AnalogInput7	8	AIN7
			VDD	9	VDD
			VDDHDIV5	0x0D	VDDH/5

6.23.7.29 CH[1].PSELN

Address offset: 0x524

Input negative pin selection for CH[1]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PSELN			Analog negative input, enables differential channel
			NC	0	Not connected
			AnalogInput0	1	AIN0
			AnalogInput1	2	AIN1
			AnalogInput2	3	AIN2
			AnalogInput3	4	AIN3
			AnalogInput4	5	AIN4
			AnalogInput5	6	AIN5
			AnalogInput6	7	AIN6
			AnalogInput7	8	AIN7
			VDD	9	VDD
			VDDHDIV5	0x0D	VDDH/5

6.23.7.30 CH[1].CONFIG

Address offset: 0x528

Input configuration for CH[1]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				G F E E E D C C C B B A A
Reset 0x00020000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RESP			Positive channel resistor control
			Bypass	0	Bypass resistor ladder
			Pulldown	1	Pull-down to GND
			Pullup	2	Pull-up to VDD
			VDD1_2	3	Set input at VDD/2

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				G F E E E D C C C B B A A
Reset 0x00020000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
B	RW	RESN			Negative channel resistor control
			Bypass	0	Bypass resistor ladder
			Pulldown	1	Pull-down to GND
			Pullup	2	Pull-up to VDD
			VDD1_2	3	Set input at VDD/2
C	RW	GAIN			Gain control
			Gain1_6	0	1/6
			Gain1_5	1	1/5
			Gain1_4	2	1/4
			Gain1_3	3	1/3
			Gain1_2	4	1/2
			Gain1	5	1
			Gain2	6	2
			Gain4	7	4
D-	RW	REFSEL			Reference control
			Internal	0	Internal reference (0.6 V)
			VDD1_4	1	VDD/4 as reference
E	RW	TACQ			Acquisition time, the time the SAADC uses to sample the input voltage
			3us	0	3 µs
			5us	1	5 µs
			10us	2	10 µs
			15us	3	15 µs
			20us	4	20 µs
			40us	5	40 µs
F	RW	MODE			Enable differential mode
			SE	0	Single-ended, PSELN will be ignored, negative input to SAADC shorted to GND
			Diff	1	Differential
G	RW	BURST			Enable burst mode
			Disabled	0	Burst mode is disabled (normal operation)
			Enabled	1	Burst mode is enabled. SAADC takes 2^OVERSAMPLE number of samples as fast as it can, and sends the average to Data RAM.

6.23.7.31 CH[1].LIMIT

Address offset: 0x52C

High/low limits for event monitoring of a channel

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B B B B B B B B B B B B B B B B A A A A A A A A A A A A A A
Reset 0x7FFF8000				**0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	LOW		[-32768 to +32767]	Low level limit
B	RW	HIGH		[-32768 to +32767]	High level limit

6.23.7.32 CH[2].PSELP

Address offset: 0x530

Input positive pin selection for CH[2]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PSELP			Analog positive input channel
			NC	0	Not connected
			AnalogInput0	1	AIN0
			AnalogInput1	2	AIN1
			AnalogInput2	3	AIN2
			AnalogInput3	4	AIN3
			AnalogInput4	5	AIN4
			AnalogInput5	6	AIN5
			AnalogInput6	7	AIN6
			AnalogInput7	8	AIN7
			VDD	9	VDD
			VDDHDIV5	0x0D	VDDH/5

6.23.7.33 CH[2].PSELN

Address offset: 0x534

Input negative pin selection for CH[2]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PSELN			Analog negative input, enables differential channel
			NC	0	Not connected
			AnalogInput0	1	AIN0
			AnalogInput1	2	AIN1
			AnalogInput2	3	AIN2
			AnalogInput3	4	AIN3
			AnalogInput4	5	AIN4
			AnalogInput5	6	AIN5
			AnalogInput6	7	AIN6
			AnalogInput7	8	AIN7
			VDD	9	VDD
			VDDHDIV5	0x0D	VDDH/5

6.23.7.34 CH[2].CONFIG

Address offset: 0x538

Input configuration for CH[2]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				G F E E E D C C C B B A A
Reset 0x00020000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RESP			Positive channel resistor control
			Bypass	0	Bypass resistor ladder
			Pulldown	1	Pull-down to GND
			Pullup	2	Pull-up to VDD
			VDD1_2	3	Set input at VDD/2

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				G F E E E D C C C B B A A
Reset 0x00020000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
B	RW	RESN			Negative channel resistor control
			Bypass	0	Bypass resistor ladder
			Pulldown	1	Pull-down to GND
			Pullup	2	Pull-up to VDD
			VDD1_2	3	Set input at VDD/2
C	RW	GAIN			Gain control
			Gain1_6	0	1/6
			Gain1_5	1	1/5
			Gain1_4	2	1/4
			Gain1_3	3	1/3
			Gain1_2	4	1/2
			Gain1	5	1
			Gain2	6	2
			Gain4	7	4
D-	RW	REFSEL			Reference control
			Internal	0	Internal reference (0.6 V)
			VDD1_4	1	VDD/4 as reference
E	RW	TACQ			Acquisition time, the time the SAADC uses to sample the input voltage
			3us	0	3 µs
			5us	1	5 µs
			10us	2	10 µs
			15us	3	15 µs
			20us	4	20 µs
			40us	5	40 µs
F	RW	MODE			Enable differential mode
			SE	0	Single-ended, PSELN will be ignored, negative input to SAADC shorted to GND
			Diff	1	Differential
G	RW	BURST			Enable burst mode
			Disabled	0	Burst mode is disabled (normal operation)
			Enabled	1	Burst mode is enabled. SAADC takes 2^OVERSAMPLE number of samples as fast as it can, and sends the average to Data RAM.

6.23.7.35 CH[2].LIMIT

Address offset: 0x53C

High/low limits for event monitoring of a channel

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B B B B B B B B B B B B B B B B A A A A A A A A A A A A A A
Reset 0x7FFF8000				**0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	LOW		[-32768 to +32767]	Low level limit
B	RW	HIGH		[-32768 to +32767]	High level limit

6.23.7.36 CH[3].PSELP

Address offset: 0x540

Input positive pin selection for CH[3]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PSELP			Analog positive input channel
			NC	0	Not connected
			AnalogInput0	1	AIN0
			AnalogInput1	2	AIN1
			AnalogInput2	3	AIN2
			AnalogInput3	4	AIN3
			AnalogInput4	5	AIN4
			AnalogInput5	6	AIN5
			AnalogInput6	7	AIN6
			AnalogInput7	8	AIN7
			VDD	9	VDD
			VDDHDIV5	0x0D	VDDH/5

6.23.7.37 CH[3].PSELN

Address offset: 0x544

Input negative pin selection for CH[3]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PSELN			Analog negative input, enables differential channel
			NC	0	Not connected
			AnalogInput0	1	AIN0
			AnalogInput1	2	AIN1
			AnalogInput2	3	AIN2
			AnalogInput3	4	AIN3
			AnalogInput4	5	AIN4
			AnalogInput5	6	AIN5
			AnalogInput6	7	AIN6
			AnalogInput7	8	AIN7
			VDD	9	VDD
			VDDHDIV5	0x0D	VDDH/5

6.23.7.38 CH[3].CONFIG

Address offset: 0x548

Input configuration for CH[3]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				G F E E E D C C C B B A A
Reset 0x00020000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RESP			Positive channel resistor control
			Bypass	0	Bypass resistor ladder
			Pulldown	1	Pull-down to GND
			Pullup	2	Pull-up to VDD
			VDD1_2	3	Set input at VDD/2

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				G F E E E D C C C B B A A
Reset 0x00020000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
B	RW	RESN			Negative channel resistor control
			Bypass	0	Bypass resistor ladder
			Pulldown	1	Pull-down to GND
			Pullup	2	Pull-up to VDD
			VDD1_2	3	Set input at VDD/2
C	RW	GAIN			Gain control
			Gain1_6	0	1/6
			Gain1_5	1	1/5
			Gain1_4	2	1/4
			Gain1_3	3	1/3
			Gain1_2	4	1/2
			Gain1	5	1
			Gain2	6	2
			Gain4	7	4
D-	RW	REFSEL			Reference control
			Internal	0	Internal reference (0.6 V)
			VDD1_4	1	VDD/4 as reference
E	RW	TACQ			Acquisition time, the time the SAADC uses to sample the input voltage
			3us	0	3 µs
			5us	1	5 µs
			10us	2	10 µs
			15us	3	15 µs
			20us	4	20 µs
			40us	5	40 µs
F	RW	MODE			Enable differential mode
			SE	0	Single-ended, PSELN will be ignored, negative input to SAADC shorted to GND
			Diff	1	Differential
G	RW	BURST			Enable burst mode
			Disabled	0	Burst mode is disabled (normal operation)
			Enabled	1	Burst mode is enabled. SAADC takes 2^OVERSAMPLE number of samples as fast as it can, and sends the average to Data RAM.

6.23.7.39 CH[3].LIMIT

Address offset: 0x54C

High/low limits for event monitoring of a channel

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B B B B B B B B B B B B B B B B A A A A A A A A A A A A A A A A
Reset 0x7FFF8000				**0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	LOW		[-32768 to +32767]	Low level limit
B	RW	HIGH		[-32768 to +32767]	High level limit

6.23.7.40 CH[4].PSELP

Address offset: 0x550

Input positive pin selection for CH[4]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PSELP			Analog positive input channel
			NC	0	Not connected
			AnalogInput0	1	AIN0
			AnalogInput1	2	AIN1
			AnalogInput2	3	AIN2
			AnalogInput3	4	AIN3
			AnalogInput4	5	AIN4
			AnalogInput5	6	AIN5
			AnalogInput6	7	AIN6
			AnalogInput7	8	AIN7
			VDD	9	VDD
			VDDHDIV5	0x0D	VDDH/5

6.23.7.41 CH[4].PSELN

Address offset: 0x554

Input negative pin selection for CH[4]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PSELN			Analog negative input, enables differential channel
			NC	0	Not connected
			AnalogInput0	1	AIN0
			AnalogInput1	2	AIN1
			AnalogInput2	3	AIN2
			AnalogInput3	4	AIN3
			AnalogInput4	5	AIN4
			AnalogInput5	6	AIN5
			AnalogInput6	7	AIN6
			AnalogInput7	8	AIN7
			VDD	9	VDD
			VDDHDIV5	0x0D	VDDH/5

6.23.7.42 CH[4].CONFIG

Address offset: 0x558

Input configuration for CH[4]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				G F E E E D C C C B B A A
Reset 0x00020000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RESP			Positive channel resistor control
			Bypass	0	Bypass resistor ladder
			Pulldown	1	Pull-down to GND
			Pullup	2	Pull-up to VDD
			VDD1_2	3	Set input at VDD/2

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				G F E E E D C C C B B A A
Reset 0x00020000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
B	RW	RESN			Negative channel resistor control
			Bypass	0	Bypass resistor ladder
			Pulldown	1	Pull-down to GND
			Pullup	2	Pull-up to VDD
			VDD1_2	3	Set input at VDD/2
C	RW	GAIN			Gain control
			Gain1_6	0	1/6
			Gain1_5	1	1/5
			Gain1_4	2	1/4
			Gain1_3	3	1/3
			Gain1_2	4	1/2
			Gain1	5	1
			Gain2	6	2
			Gain4	7	4
D-	RW	REFSEL			Reference control
			Internal	0	Internal reference (0.6 V)
			VDD1_4	1	VDD/4 as reference
E	RW	TACQ			Acquisition time, the time the SAADC uses to sample the input voltage
			3us	0	3 µs
			5us	1	5 µs
			10us	2	10 µs
			15us	3	15 µs
			20us	4	20 µs
			40us	5	40 µs
F	RW	MODE			Enable differential mode
			SE	0	Single-ended, PSELN will be ignored, negative input to SAADC shorted to GND
			Diff	1	Differential
G	RW	BURST			Enable burst mode
			Disabled	0	Burst mode is disabled (normal operation)
			Enabled	1	Burst mode is enabled. SAADC takes 2^OVERSAMPLE number of samples as fast as it can, and sends the average to Data RAM.

6.23.7.43 CH[4].LIMIT

Address offset: 0x55C

High/low limits for event monitoring of a channel

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B B B B B B B B B B B B B B B B A A A A A A A A A A A A A A A A
Reset 0x7FFF8000				**0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	LOW		[-32768 to +32767]	Low level limit
B	RW	HIGH		[-32768 to +32767]	High level limit

6.23.7.44 CH[5].PSELP

Address offset: 0x560

Input positive pin selection for CH[5]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PSELP			Analog positive input channel
			NC	0	Not connected
			AnalogInput0	1	AIN0
			AnalogInput1	2	AIN1
			AnalogInput2	3	AIN2
			AnalogInput3	4	AIN3
			AnalogInput4	5	AIN4
			AnalogInput5	6	AIN5
			AnalogInput6	7	AIN6
			AnalogInput7	8	AIN7
			VDD	9	VDD
			VDDHDIV5	0x0D	VDDH/5

6.23.7.45 CH[5].PSELN

Address offset: 0x564

Input negative pin selection for CH[5]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PSELN			Analog negative input, enables differential channel
			NC	0	Not connected
			AnalogInput0	1	AIN0
			AnalogInput1	2	AIN1
			AnalogInput2	3	AIN2
			AnalogInput3	4	AIN3
			AnalogInput4	5	AIN4
			AnalogInput5	6	AIN5
			AnalogInput6	7	AIN6
			AnalogInput7	8	AIN7
			VDD	9	VDD
			VDDHDIV5	0x0D	VDDH/5

6.23.7.46 CH[5].CONFIG

Address offset: 0x568

Input configuration for CH[5]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				G F E E E D C C C B B A A
Reset 0x00020000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RESP			Positive channel resistor control
			Bypass	0	Bypass resistor ladder
			Pulldown	1	Pull-down to GND
			Pullup	2	Pull-up to VDD
			VDD1_2	3	Set input at VDD/2

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				G F E E E D C C C B B A A
Reset 0x00020000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
B	RW	RESN			Negative channel resistor control
			Bypass	0	Bypass resistor ladder
			Pulldown	1	Pull-down to GND
			Pullup	2	Pull-up to VDD
			VDD1_2	3	Set input at VDD/2
C	RW	GAIN			Gain control
			Gain1_6	0	1/6
			Gain1_5	1	1/5
			Gain1_4	2	1/4
			Gain1_3	3	1/3
			Gain1_2	4	1/2
			Gain1	5	1
			Gain2	6	2
			Gain4	7	4
D-	RW	REFSEL			Reference control
			Internal	0	Internal reference (0.6 V)
			VDD1_4	1	VDD/4 as reference
E	RW	TACQ			Acquisition time, the time the SAADC uses to sample the input voltage
			3us	0	3 µs
			5us	1	5 µs
			10us	2	10 µs
			15us	3	15 µs
			20us	4	20 µs
			40us	5	40 µs
F	RW	MODE			Enable differential mode
			SE	0	Single-ended, PSELN will be ignored, negative input to SAADC shorted to GND
			Diff	1	Differential
G	RW	BURST			Enable burst mode
			Disabled	0	Burst mode is disabled (normal operation)
			Enabled	1	Burst mode is enabled. SAADC takes 2^OVERSAMPLE number of samples as fast as it can, and sends the average to Data RAM.

6.23.7.47 CH[5].LIMIT

Address offset: 0x56C

High/low limits for event monitoring of a channel

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B B B B B B B B B B B B B B B B A A A A A A A A A A A A A A A A
Reset 0x7FFF8000				**0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	LOW		[-32768 to +32767]	Low level limit
B	RW	HIGH		[-32768 to +32767]	High level limit

6.23.7.48 CH[6].PSELP

Address offset: 0x570

Input positive pin selection for CH[6]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PSELP			Analog positive input channel
			NC	0	Not connected
			AnalogInput0	1	AIN0
			AnalogInput1	2	AIN1
			AnalogInput2	3	AIN2
			AnalogInput3	4	AIN3
			AnalogInput4	5	AIN4
			AnalogInput5	6	AIN5
			AnalogInput6	7	AIN6
			AnalogInput7	8	AIN7
			VDD	9	VDD
			VDDHDIV5	0x0D	VDDH/5

6.23.7.49 CH[6].PSELN

Address offset: 0x574

Input negative pin selection for CH[6]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PSELN			Analog negative input, enables differential channel
			NC	0	Not connected
			AnalogInput0	1	AIN0
			AnalogInput1	2	AIN1
			AnalogInput2	3	AIN2
			AnalogInput3	4	AIN3
			AnalogInput4	5	AIN4
			AnalogInput5	6	AIN5
			AnalogInput6	7	AIN6
			AnalogInput7	8	AIN7
			VDD	9	VDD
			VDDHDIV5	0x0D	VDDH/5

6.23.7.50 CH[6].CONFIG

Address offset: 0x578

Input configuration for CH[6]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				G F E E E D C C C B B A A
Reset 0x00020000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RESP			Positive channel resistor control
			Bypass	0	Bypass resistor ladder
			Pulldown	1	Pull-down to GND
			Pullup	2	Pull-up to VDD
			VDD1_2	3	Set input at VDD/2

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				G F E E E D C C C B B A A
Reset 0x00020000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
B	RW	RESN			Negative channel resistor control
			Bypass	0	Bypass resistor ladder
			Pulldown	1	Pull-down to GND
			Pullup	2	Pull-up to VDD
			VDD1_2	3	Set input at VDD/2
C	RW	GAIN			Gain control
			Gain1_6	0	1/6
			Gain1_5	1	1/5
			Gain1_4	2	1/4
			Gain1_3	3	1/3
			Gain1_2	4	1/2
			Gain1	5	1
			Gain2	6	2
			Gain4	7	4
D-	RW	REFSEL			Reference control
			Internal	0	Internal reference (0.6 V)
			VDD1_4	1	VDD/4 as reference
E	RW	TACQ			Acquisition time, the time the SAADC uses to sample the input voltage
			3us	0	3 µs
			5us	1	5 µs
			10us	2	10 µs
			15us	3	15 µs
			20us	4	20 µs
			40us	5	40 µs
F	RW	MODE			Enable differential mode
			SE	0	Single-ended, PSELN will be ignored, negative input to SAADC shorted to GND
			Diff	1	Differential
G	RW	BURST			Enable burst mode
			Disabled	0	Burst mode is disabled (normal operation)
			Enabled	1	Burst mode is enabled. SAADC takes 2^OVERSAMPLE number of samples as fast as it can, and sends the average to Data RAM.

6.23.7.51 CH[6].LIMIT

Address offset: 0x57C

High/low limits for event monitoring of a channel

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B B B B B B B B B B B B B B B B A A A A A A A A A A A A A A
Reset 0x7FFF8000				**0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	LOW		[-32768 to +32767]	Low level limit
B	RW	HIGH		[-32768 to +32767]	High level limit

6.23.7.52 CH[7].PSELP

Address offset: 0x580

Input positive pin selection for CH[7]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PSELP			Analog positive input channel
			NC	0	Not connected
			AnalogInput0	1	AIN0
			AnalogInput1	2	AIN1
			AnalogInput2	3	AIN2
			AnalogInput3	4	AIN3
			AnalogInput4	5	AIN4
			AnalogInput5	6	AIN5
			AnalogInput6	7	AIN6
			AnalogInput7	8	AIN7
			VDD	9	VDD
			VDDHDIV5	0x0D	VDDH/5

6.23.7.53 CH[7].PSELN

Address offset: 0x584

Input negative pin selection for CH[7]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PSELN			Analog negative input, enables differential channel
			NC	0	Not connected
			AnalogInput0	1	AIN0
			AnalogInput1	2	AIN1
			AnalogInput2	3	AIN2
			AnalogInput3	4	AIN3
			AnalogInput4	5	AIN4
			AnalogInput5	6	AIN5
			AnalogInput6	7	AIN6
			AnalogInput7	8	AIN7
			VDD	9	VDD
			VDDHDIV5	0x0D	VDDH/5

6.23.7.54 CH[7].CONFIG

Address offset: 0x588

Input configuration for CH[7]

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				G F E E E D C C C B B A A
Reset 0x00020000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RESP			Positive channel resistor control
			Bypass	0	Bypass resistor ladder
			Pulldown	1	Pull-down to GND
			Pullup	2	Pull-up to VDD
			VDD1_2	3	Set input at VDD/2

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				G F E E E D C C C B B A A
Reset 0x00020000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
B	RW	RESN			Negative channel resistor control
			Bypass	0	Bypass resistor ladder
			Pulldown	1	Pull-down to GND
			Pullup	2	Pull-up to VDD
			VDD1_2	3	Set input at VDD/2
C	RW	GAIN			Gain control
			Gain1_6	0	1/6
			Gain1_5	1	1/5
			Gain1_4	2	1/4
			Gain1_3	3	1/3
			Gain1_2	4	1/2
			Gain1	5	1
			Gain2	6	2
			Gain4	7	4
D-	RW	REFSEL			Reference control
			Internal	0	Internal reference (0.6 V)
			VDD1_4	1	VDD/4 as reference
E	RW	TACQ			Acquisition time, the time the SAADC uses to sample the input voltage
			3us	0	3 µs
			5us	1	5 µs
			10us	2	10 µs
			15us	3	15 µs
			20us	4	20 µs
			40us	5	40 µs
F	RW	MODE			Enable differential mode
			SE	0	Single-ended, PSELN will be ignored, negative input to SAADC shorted to GND
			Diff	1	Differential
G	RW	BURST			Enable burst mode
			Disabled	0	Burst mode is disabled (normal operation)
			Enabled	1	Burst mode is enabled. SAADC takes 2^OVERSAMPLE number of samples as fast as it can, and sends the average to Data RAM.

6.23.7.55 CH[7].LIMIT

Address offset: 0x58C

High/low limits for event monitoring of a channel

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B B B B B B B B B B B B B B B B A A A A A A A A A A A A A A A A
Reset 0x7FFF8000				**0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	LOW		[-32768 to +32767]	Low level limit
B	RW	HIGH		[-32768 to +32767]	High level limit

6.23.7.56 RESOLUTION

Address offset: 0x5F0

Resolution configuration

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	VAL			Set the resolution
			8bit	0	8 bits
			10bit	1	10 bits
			12bit	2	12 bits
			14bit	3	14 bits

6.23.7.57 OVERSAMPLE

Address offset: 0x5F4

Oversampling configuration. The RESOLUTION is applied before averaging, thus for high OVERSAMPLE a higher RESOLUTION should be used.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	OVERSAMPLE			Oversample control
			Bypass	0	Bypass oversampling
			Over2x	1	Oversample 2x
			Over4x	2	Oversample 4x
			Over8x	3	Oversample 8x
			Over16x	4	Oversample 16x
			Over32x	5	Oversample 32x
			Over64x	6	Oversample 64x
			Over128x	7	Oversample 128x
			Over256x	8	Oversample 256x

6.23.7.58 SAMPLERATE

Address offset: 0x5F8

Controls normal or continuous sample rate

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	CC		[80..2047]	Capture and compare value. Sample rate is 16 MHz/CC
B	RW	MODE			Select mode for sample rate control
			Task	0	Rate is controlled from SAMPLE task
			Timers	1	Rate is controlled from local timer (use CC to control the rate)

6.23.7.59 RESULT

RESULT EasyDMA channel

6.23.7.59.1 RESULT.PTR

Address offset: 0x62C

Data pointer

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Data pointer

Note: See Memory on page 21 for details about memories available to EasyDMA.

6.23.7.59.2 RESULT.MAXCNT

Address offset: 0x630

Maximum number of 16-bit samples to be written to output RAM buffer

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	MAXCNT			Maximum number of 16-bit samples to be written to output RAM buffer

6.23.7.59.3 RESULT.AMOUNT

Address offset: 0x634

Number of 16-bit samples written to output RAM buffer since the previous START task

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	AMOUNT			Number of 16-bit samples written to output RAM buffer since the previous START task. This register can be read after an END or STOPPED event.

6.23.8 Electrical specification

6.23.8.1 SAADC electrical specification

Symbol	Description	Min.	Typ.	Max.	Units
DNL_{10}	Differential non-linearity, 10-bit resolution	-0.95	<1		LSB10b
INL_{10}	Integral non-linearity, 10-bit resolution		1		LSB10b
DNL_{12}	Differential non-linearity, 12-bit resolution	-0.95	1.3		LSB12b
INL_{12}	Integral non-linearity, 12-bit resolution		4.7		LSB12b
V_{OS}	Differential offset error (calibrated), 10-bit resolution [27]		±2		LSB10b
$E_{VDDHDIV5}$	Error on VDDHDIV5 input		±1		%
C_{EG}	Gain error temperature coefficient		0.02		%/°C
f_{SAMPLE}	Maximum sampling rate			200	kHz
t_{CONV}	Conversion time		<2		µs
$E_{G1/6}$	Error[28] for gain = 1/6	-3		3	%
$E_{G1/4}$	Error[28] for gain = 1/4	-3		3	%

[27] Digital output code at zero volt differential input.
[28] Does not include temperature drift

Symbol	Description	Min.	Typ.	Max.	Units
$E_{G1/2}$	Error[28] for gain = 1/2	-3		4	%
E_{G1}	Error[28] for gain = 1	-3		4	%
C_{SAMPLE}	Sample and hold capacitance at maximum gain[29]		2.5		pF
R_{INPUT}	Input resistance		>1		MΩ
E_{NOB}	Effective number of bits, differential mode, 12-bit resolution, 1/1 gain, 3 µs acquisition time, crystal HFCLK, 200 ksps		9		Bit
S_{NDR}	Peak signal to noise and distortion ratio, differential mode, 12-bit resolution, 1/1 gain, 3 µs acquisition time, crystal HFCLK, 200 ksps		56		dB
S_{FDR}	Spurious free dynamic range, differential mode, 12-bit resolution, 1/1 gain, 3 µs acquisition time, crystal HFCLK, 200 ksps		70		dBc
R_{LADDER}	Ladder resistance		160		kΩ

6.24 SPI — Serial peripheral interface master

The SPI master provides a simple CPU interface which includes a TXD register for sending data and an RXD register for receiving data. This section is added for legacy support for now.

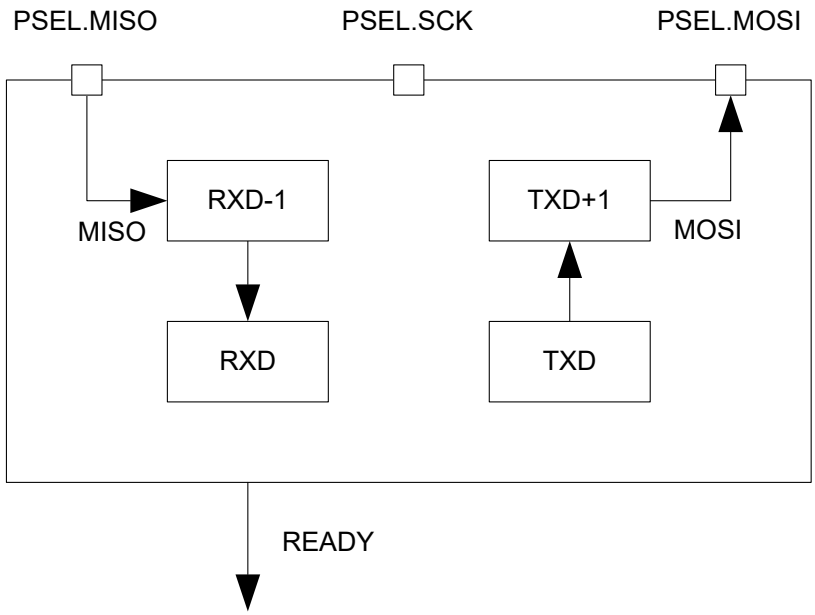


Figure 159: SPI master

RXD-1 and TXD+1 illustrate the double buffered version of RXD and TXD respectively.

6.24.1 Functional description

The TXD and RXD registers are double-buffered to enable some degree of uninterrupted data flow in and out of the SPI master.

The SPI master does not implement support for chip select directly. Therefore, the CPU must use available GPIOs to select the correct slave and control this independently of the SPI master. The SPI master supports SPI modes 0 through 3.

[29] Maximum gain corresponds to highest capacitance.

Mode	Clock polarity	Clock phase
	CPOL	CPHA
SPI_MODE0	0 (Leading)	0 (Active high)
SPI_MODE1	0 (Leading)	1 (Active low)
SPI_MODE2	1 (Trailing)	0 (Active high)
SPI_MODE3	1 (Trailing)	1 (Active low)

Table 42: SPI modes

6.24.1.1 SPI master mode pin configuration

The different signals SCK, MOSI, and MISO associated with the SPI master are mapped to physical pins.

This mapping is according to the configuration specified in the PSEL.SCK, PSEL.MOSI, and PSEL.MISO registers respectively. If the CONNECT field of a PSEL.xxx register is set to Disconnected, the associated SPI master signal is not connected to any physical pin. The PSEL.SCK, PSEL.MOSI, and PSEL.MISO registers and their configurations are only used as long as the SPI master is enabled, and retained only as long as the device is in ON mode. PSEL.SCK, PSEL.MOSI, and PSEL.MISO must only be configured when the SPI master is disabled.

To secure correct behavior in the SPI, the pins used by the SPI must be configured in the GPIO peripheral as described in GPIO configuration on page 716 prior to enabling the SPI. The SCK must always be connected to a pin, and that pin's input buffer must always be connected for the SPI to work. This configuration must be retained in the GPIO for the selected I/Os as long as the SPI is enabled.

Only one peripheral can be assigned to drive a particular GPIO pin at a time. Failing to do so may result in unpredictable behavior.

SPI master signal	SPI master pin	Direction	Output value
SCK	As specified in PSEL.SCK	Output	Same as CONFIG.CPOL
MOSI	As specified in PSEL.MOSI	Output	0
MISO	As specified in PSEL.MISO	Input	Not applicable

Table 43: GPIO configuration

6.24.1.2 Shared resources

The SPI shares registers and other resources with other peripherals that have the same ID as the SPI. Therefore, the user must disable all peripherals that have the same ID as the SPI before the SPI can be configured and used.

Disabling a peripheral that has the same ID as the SPI will not reset any of the registers that are shared with the SPI. It is therefore important to configure all relevant SPI registers explicitly to secure that it operates correctly.

See the Instantiation table in Instantiation on page 24 for details on peripherals and their IDs.

6.24.1.3 SPI master transaction sequence

An SPI master transaction is started by writing the first byte, which is to be transmitted by the SPI master, to the TXD register.

Since the transmitter is double buffered, the second byte can be written to the TXD register immediately after the first one. The SPI master will then send these bytes in the order they are written to the TXD register.

The SPI master is a synchronous interface, and for every byte that is sent, a different byte will be received at the same time. This is illustrated in SPI master transaction on page 717. Bytes that are received will be moved to the RXD register where the CPU can extract them by reading the register. The RXD register is double buffered in the same way as the TXD register, and a second byte can therefore be received at the

same time as the first byte is being extracted from RXD by the CPU. The SPI master will generate a READY event every time a new byte is moved to the RXD register. The double buffered byte will be moved from RXD-1 to RXD as soon as the first byte is extracted from RXD. The SPI master will stop when there are no more bytes to send in TXD and TXD+1.

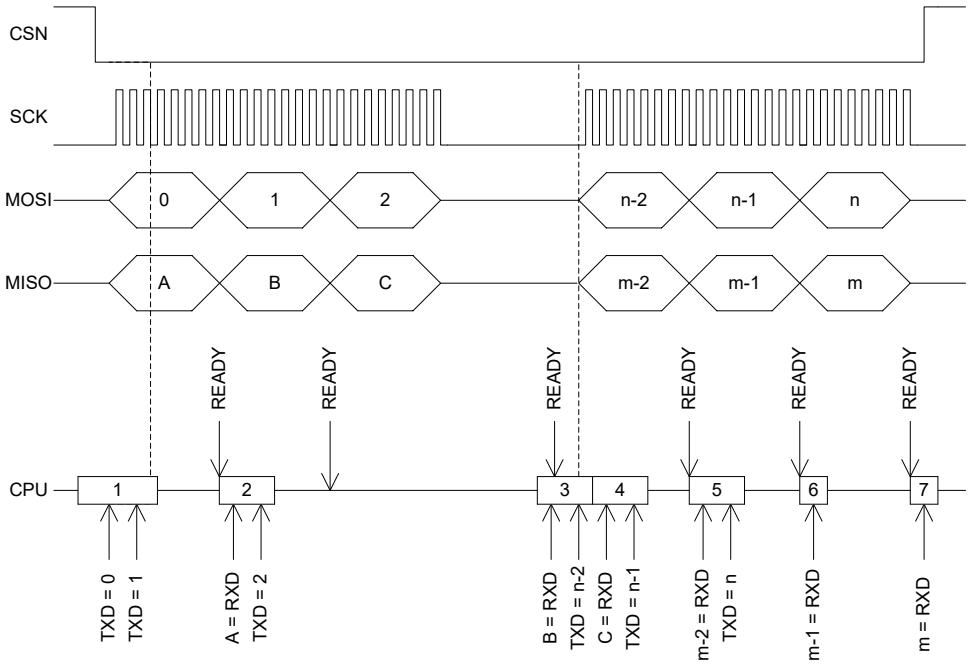


Figure 160: SPI master transaction

The READY event of the third byte transaction is delayed until B is extracted from RXD in occurrence number 3 on the horizontal lifeline. The reason for this is that the third event is generated first when C is moved from RXD-1 to RXD after B is read.

The SPI master will move the incoming byte to the RXD register after a short delay following the SCK clock period of the last bit in the byte. This also means that the READY event will be delayed accordingly, see SPI master transaction on page 718. Therefore, it is important that you always clear the READY event, even if the RXD register and the data that is being received is not used.

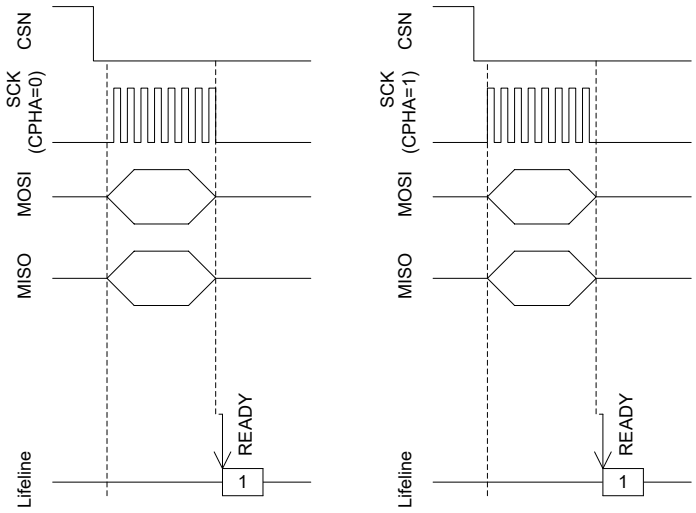


Figure 161: SPI master transaction

6.24.2 Registers

Instances

Instance	Base address	Description
SPI0	0x40003000	SPI master 0 This instance is deprecated.
SPI1	0x40004000	SPI master 1 This instance is deprecated.
SPI2	0x40023000	SPI master 2 This instance is deprecated.

Register overview

Register	Offset	Description
EVENTS_READY	0x108	TXD byte sent and RXD byte received
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
ENABLE	0x500	Enable SPI
PSEL.SCK	0x508	Pin select for SCK
PSEL.MOSI	0x50C	Pin select for MOSI signal
PSEL.MISO	0x510	Pin select for MISO signal
RXD	0x518	RXD register. Register is cleared on read and the buffer pointer will be modified if read.
TXD	0x51C	TXD register
FREQUENCY	0x524	SPI frequency. Accuracy depends on the HFCLK source selected.
CONFIG	0x554	Configuration register

6.24.2.1 EVENTS_READY

Address offset: 0x108

TXD byte sent and RXD byte received

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_READY			TXD byte sent and RXD byte received
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.24.2.2 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	READY			Write '1' to enable interrupt for event READY
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.24.2.3 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	READY			Write '1' to disable interrupt for event READY
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.24.2.4 ENABLE

Address offset: 0x500

Enable SPI

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ENABLE			Enable or disable SPI
			Disabled	0	Disable SPI
			Enabled	1	Enable SPI

6.24.2.5 PSEL.SCK

Address offset: 0x508

Pin select for SCK

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				C																										B	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW	PIN		[0..31]																															
B	RW	PORT		[0..1]																															
C	RW	CONNECT																																	
			Disconnected	1																															
			Connected	0																															

6.24.2.6 PSEL.MOSI

Address offset: 0x50C

Pin select for MOSI signal

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				C																										B	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW	PIN		[0..31]																															
B	RW	PORT		[0..1]																															
C	RW	CONNECT																																	
			Disconnected	1																															
			Connected	0																															

6.24.2.7 PSEL.MISO

Address offset: 0x510

Pin select for MISO signal

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				C																										B	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW	PIN		[0..31]																															
B	RW	PORT		[0..1]																															
C	RW	CONNECT																																	
			Disconnected	1																															
			Connected	0																															

6.24.2.8 RXD

Address offset: 0x518

RXD register. Register is cleared on read and the buffer pointer will be modified if read.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																												A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																								Description							
A	R	RXD																										RX data received. Double buffered							
	RME																																		

6.24.2.9 TXD

Address offset: 0x51C

TXD register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																												A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																								Description							
A	RW	TXD																										TX data to send. Double buffered.							

6.24.2.10 FREQUENCY

Address offset: 0x524

SPI frequency. Accuracy depends on the HFCLK source selected.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x04000000				**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																								Description							
A	RW	FREQUENCY																										SPI master data rate							
			K125	0x02000000																								125 kbps							
			K250	0x04000000																								250 kbps							
			K500	0x08000000																								500 kbps							
			M1	0x10000000																								1 Mbps							
			M2	0x20000000																								2 Mbps							
			M4	0x40000000																								4 Mbps							
			M8	0x80000000																								8 Mbps							

6.24.2.11 CONFIG

Address offset: 0x554

Configuration register

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	ORDER																																	
			MsbFirst	0																															
			LsbFirst	1																															
B	RW	CPHA																																	
			Leading	0																															
			Trailing	1																															
C	RW	CPOL																																	
			ActiveHigh	0																															
			ActiveLow	1																															

6.24.3 Electrical specification

6.24.3.1 SPI master interface electrical specifications

Symbol	Description	Min.	Typ.	Max.	Units
f_{SPI}	Bit rates for SPI[30]			8[31]	Mbps
$t_{SPI,START}$	Time from writing TXD register to transmission started		1		µs

6.24.3.2 Serial Peripheral Interface (SPI) Master timing specifications

Symbol	Description	Min.	Typ.	Max.	Units
$t_{SPI,CSCK}$	SCK period	125			ns
$t_{SPI,RSCK,LD}$	SCK rise time, standard drive [32]			$t_{RF,25pF}$	
$t_{SPI,RSCK,HD}$	SCK rise time, high drive[32]			$t_{HRF,25pF}$	
$t_{SPI,FSCK,LD}$	SCK fall time, standard drive[32]			$t_{RF,25pF}$	
$t_{SPI,FSCK,HD}$	SCK fall time, high drive[32]			$t_{HRF,25pF}$	
$t_{SPI,WHSCK}$	SCK high time[32]	$(t_{CSCK}/2) - t_{RSCK}$			
$t_{SPI,WLSCK}$	SCK low time[32]	$(t_{CSCK}/2) - t_{FSCK}$			
$t_{SPI,SUMI}$	MISO to CLK edge setup time	19			ns
$t_{SPI,HMI}$	CLK edge to MISO hold time	18			ns
$t_{SPI,VMO}$	CLK edge to MOSI valid			59	ns
$t_{SPI,HMO}$	MOSI hold time after CLK edge	20			ns

30 High bit rates may require GPIOs to be set as High Drive, see GPIO chapter for more details.

31 The actual maximum data rate depends on the slave's CLK to MISO and MOSI setup and hold timings.

32 At 25 pF load, including GPIO capacitance, see GPIO electrical specification.

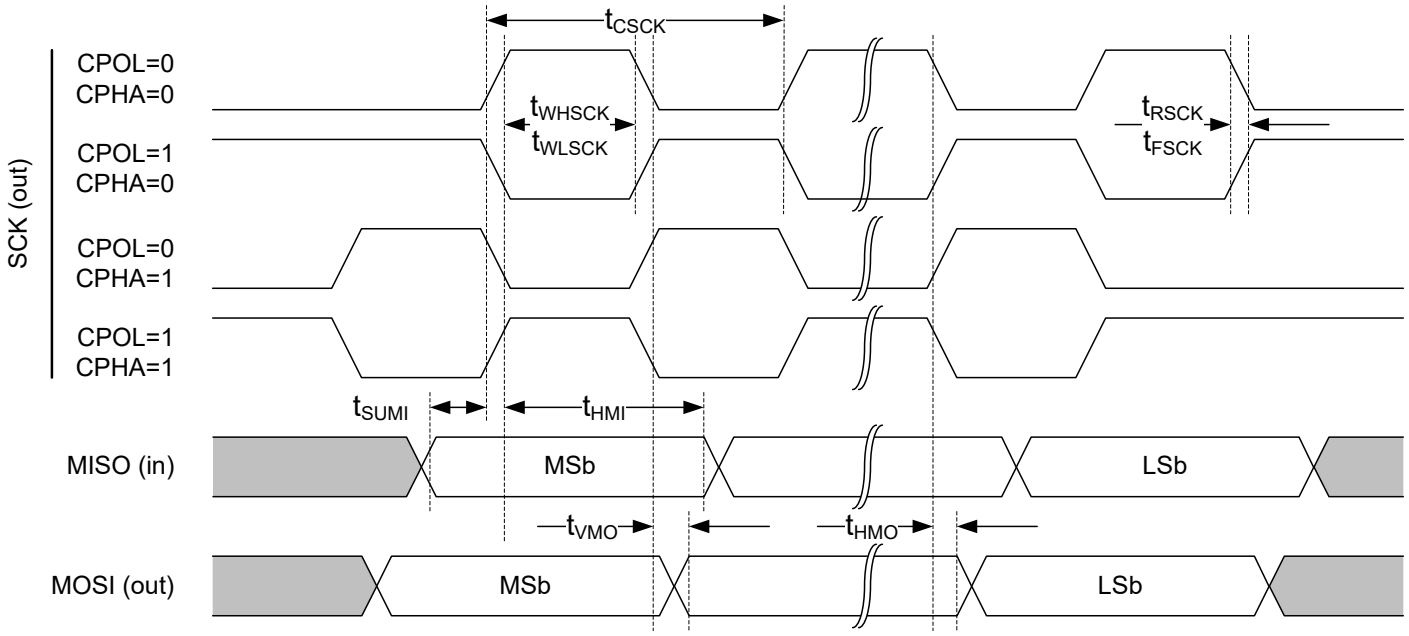


Figure 162: SPI master timing diagram

6.25 SPIM — Serial peripheral interface master with EasyDMA

The SPI master can communicate with multiple SPI slaves using individual chip select signals for each slave.

The the main features of SPIM are:

- EasyDMA direct transfer to/from RAM
- SPI mode 0-3
- Individual selection of I/O pins
- Optional D/CX output line for distinguishing between command and data bytes

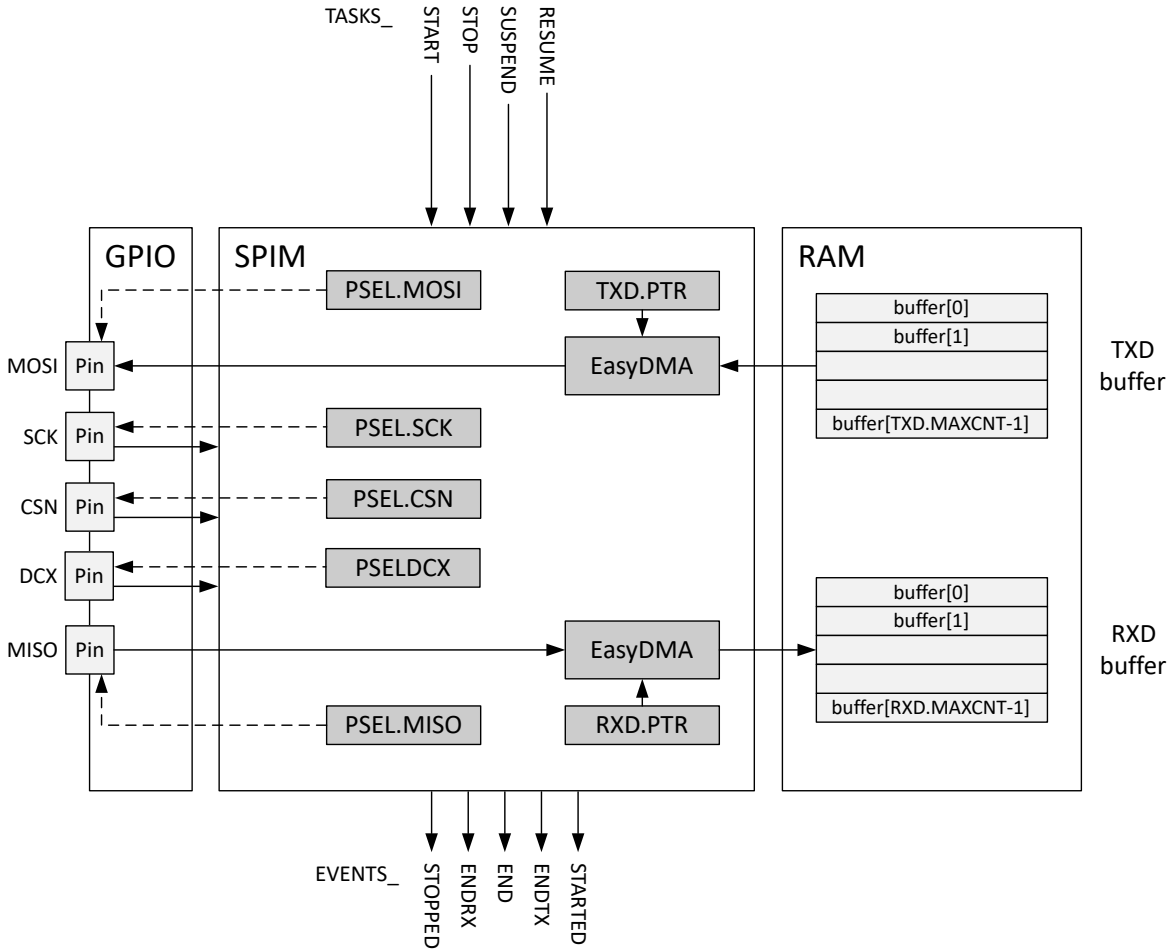


Figure 163: SPIM — SPI master with EasyDMA

6.25.1 SPI master transaction sequence

An SPI master transaction is started by triggering the START task. When started, a number of bytes will be transmitted/received on MOSI/MISO.

The following figure illustrates an SPI master transaction.

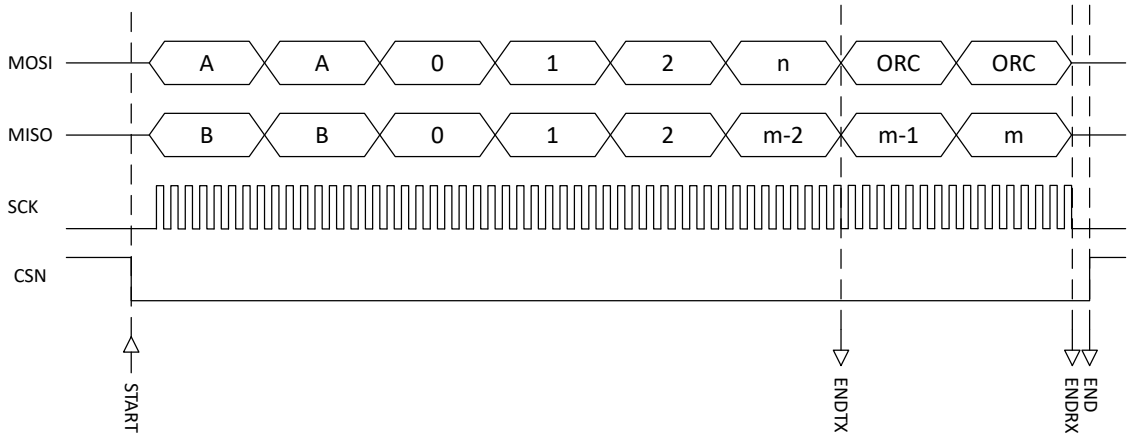


Figure 164: SPI master transaction

The ENDTX is generated when all bytes in buffer TXD.PTR on page 735 are transmitted. The number of bytes in the transmit buffer is specified in register TXD.MAXCNT on page 736. The ENDRX event will be generated when buffer RXD.PTR on page 734 is full, meaning the number of bytes specified in register RXD.MAXCNT on page 735 have been received. The transaction stops automatically after all bytes have been transmitted/received. When the maximum number of bytes in the receive buffer is larger than the number of bytes in the transmit buffer, the contents of register ORC on page 738 will be transmitted after the last byte in the transmit buffer has been transmitted.

The END event will be generated after both the ENDRX and ENDTX events have been generated.

The SPI master can be stopped by triggering the STOP task. A STOPPED event is generated when the SPI master has stopped. If the STOP task is triggered in the middle of a transaction, SPIM will complete the transmission/reception of the current byte before stopping. The STOPPED event is generated even if the STOP task is triggered while there is no ongoing transaction.

If the ENDTX event has not already been generated when the SPI master has come to a stop, the ENDTX event will be generated even if all bytes in the buffer TXD.PTR on page 735 have not been transmitted.

If the ENDRX event has not already been generated when the SPI master has come to a stop, the ENDRX event will be generated even if the buffer RXD.PTR on page 734 is not full.

A transaction can be suspended and resumed using the SUSPEND and RESUME tasks. When the SUSPEND task is triggered, the SPI master will complete transmitting and receiving the current ongoing byte before it is suspended.

6.25.2 D/CX functionality

Some SPI slaves, for example display drivers, require an additional signal from the SPI master to distinguish between command and data bytes. For display drivers this line is often called D/CX.

SPIM provides support for such a D/CX output line. The D/CX line is set low during transmission of command bytes and high during transmission of data bytes.

The D/CX pin number is selected using PSELDCX on page 738 and the number of command bytes preceding the data bytes is configured using DCXCNT on page 738.

It is not allowed to write to the DCXCNT on page 738 during an ongoing transmission.

The following figure shows the use of D/CX, using SPIM.DCXCNT=1.

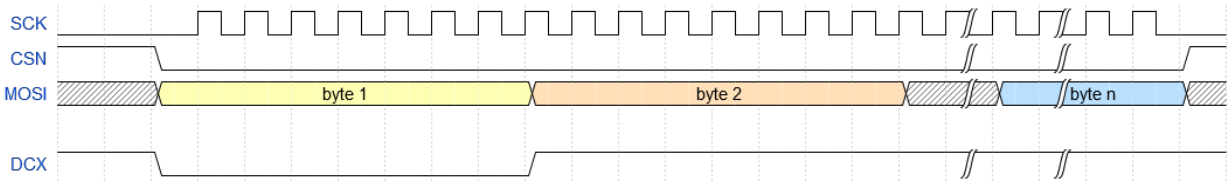


Figure 165: D/CX example. SPIM.DCXCNT = 1.

6.25.3 Pin configuration

The SCK, CSN, DCX, MOSI, and MISO signals associated with SPIM are mapped to physical pins according to the configuration specified in the PSEL.n registers.

The contents of registers PSEL.SCK on page 733, PSEL.CSN on page 734, PSELDCX on page 738, PSEL.MOSI on page 733, and PSEL.MISO on page 733 are only used when SPIM is enabled, and retained only as long as the device is in System ON mode. The PSEL.n registers can only be configured when SPIM is disabled. Enabling/disabling is done using register ENABLE on page 732.

To ensure correct behavior, the pins used by SPIM must be configured in the GPIO peripheral as described in GPIO configuration on page 726 before SPIM is enabled.

Only one peripheral can be assigned to drive a particular GPIO pin at a time. Failing to do so may result in unpredictable behavior.

SPI master signal	SPI master pin	Direction	Output value
SCK	As specified in PSEL.SCK on page 733	Output	Same as CONFIG.CPOL
CSN	As specified in PSEL.CSN on page 734	Output	Same as CONFIG.CPOL
DCX	As specified in PSELDCX on page 738	Output	1
MOSI	As specified in PSEL.MOSI on page 733	Output	0
MISO	As specified in PSEL.MISO on page 733	Input	Not applicable

Table 44: GPIO configuration

Some SPIM instances do not support automatic control of CSN, and for those the available GPIO pins need to be used to control CSN directly. See Instances on page 727 for information about what features are supported in the various SPIM instances.

SPIM supports SPI modes 0 through 3. The clock polarity (CPOL) and the clock phase (CPHA) are configured in register CONFIG on page 736.

Mode	Clock polarity	Clock phase
	CPOL	CPHA
SPI_MODE0	0 (Active High)	0 (Leading)
SPI_MODE1	0 (Active High)	1 (Trailing)
SPI_MODE2	1 (Active Low)	0 (Leading)
SPI_MODE3	1 (Active Low)	1 (Trailing)

Table 45: SPI modes

6.25.4 EasyDMA

SPIM implements EasyDMA for accessing RAM without CPU involvement.

SPIM peripheral implements the following EasyDMA channels.

Channel	Type	Register Cluster
TXD	READER	TXD
RXD	WRITER	RXD

Table 46: SPIM EasyDMA Channels

For detailed information regarding the use of EasyDMA, see EasyDMA on page 63.

The .PTR and .MAXCNT registers are double-buffered. They can be updated and prepared for the next transmission immediately after having received the STARTED event.

The SPI master will automatically stop transmitting after TXD.MAXCNT bytes have been transmitted and RXD.MAXCNT bytes have been received. If RXD.MAXCNT is larger than TXD.MAXCNT, the remaining transmitted bytes will contain the value defined in the ORC register. If TXD.MAXCNT is larger than RXD.MAXCNT, the additional received bytes will be discarded.

The ENDRX/ENDTX events indicate that EasyDMA has finished accessing respectively the RX/TX buffer in RAM. The END event gets generated when both RX and TX are finished accessing the buffers in RAM.

If several AHB bus masters try to access the same AHB slave at the same time, AHB bus congestion might occur, and the behavior of the EasyDMA channel will depend on the SPIM instance. Refer to Instances on page 727 for information about what behavior is supported in the various instances.

6.25.5 Low power

To ensure lowest possible power consumption when the peripheral is not needed stop and disable SPIM.

The STOP task may not be always needed (the peripheral might already be stopped), but if it is sent, software shall wait until the STOPPED event was received as a response before disabling the peripheral through the ENABLE register.

6.25.6 Registers

Instances

Instance	Base address	Description
SPIM0	0x40003000	SPI master 0
SPIM1	0x40004000	SPI master 1
SPIM2	0x40023000	SPI master 2
SPIM3	0x4002F000	SPI master 3

Configuration

Instance	Configuration
SPIM0	Not supported: > 8 Mbps data rate, CSNPOL register, DCX functionality, IFTIMING.x registers, hardware CSN control (PSEL.CSN), stalling mechanism during AHB bus contention.
SPIM1	Not supported: > 8 Mbps data rate, CSNPOL register, DCX functionality, IFTIMING.x registers, hardware CSN control (PSEL.CSN), stalling mechanism during AHB bus contention.
SPIM2	Not supported: > 8 Mbps data rate, CSNPOL register, DCX functionality, IFTIMING.x registers, hardware CSN control (PSEL.CSN), stalling mechanism during AHB bus contention.
SPIM3	

Register overview

Register	Offset	Description
TASKS_START	0x010	Start SPI transaction
TASKS_STOP	0x014	Stop SPI transaction
TASKS_SUSPEND	0x01C	Suspend SPI transaction
TASKS_RESUME	0x020	Resume SPI transaction
EVENTS_STOPPED	0x104	SPI transaction has stopped
EVENTS_ENDRX	0x110	End of RXD buffer reached
EVENTS_END	0x118	End of RXD buffer and TXD buffer reached
EVENTS_ENDTX	0x120	End of TXD buffer reached
EVENTS_STARTED	0x14C	Transaction started
SHORTS	0x200	Shortcuts between local events and tasks
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
STALLSTAT	0x400	Stall status for EasyDMA RAM accesses. The fields in this register are set to STALL by hardware whenever a stall occurs and can be cleared (set to NOSTALL) by the CPU.
ENABLE	0x500	Enable SPIM
PSEL.SCK	0x508	Pin select for SCK
PSEL.MOSI	0x50C	Pin select for MOSI signal
PSEL.MISO	0x510	Pin select for MISO signal
PSEL.CSN	0x514	Pin select for CSN
FREQUENCY	0x524	SPI frequency. Accuracy depends on the HFCLK source selected.
RXD.PTR	0x534	Data pointer
RXD.MAXCNT	0x538	Maximum number of bytes in receive buffer
RXD.AMOUNT	0x53C	Number of bytes transferred in the last transaction
RXD.LIST	0x540	EasyDMA list type
TXD.PTR	0x544	Data pointer
TXD.MAXCNT	0x548	Number of bytes in transmit buffer
TXD.AMOUNT	0x54C	Number of bytes transferred in the last transaction
TXD.LIST	0x550	EasyDMA list type
CONFIG	0x554	Configuration register
IFTIMING.RXDELAY	0x560	Sample delay for input serial data on MISO
IFTIMING.CSNDUR	0x564	Minimum duration between edge of CSN and edge of SCK at the start and the end of a transaction, and minimum duration CSN will stay high between transactions if END-START shortcut is used
CSNPOL	0x568	Polarity of CSN output
PSELDCX	0x56C	Pin select for DCX signal
DCXCNT	0x570	DCX configuration
ORC	0x5C0	Byte transmitted after TXD.MAXCNT bytes have been transmitted in the case when RXD.MAXCNT is greater than TXD.MAXCNT

6.25.6.1 TASKS_START

Address offset: 0x010

Start SPI transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_START			Start SPI transaction
			Trigger	1	Trigger task

6.25.6.2 TASKS_STOP

Address offset: 0x014

Stop SPI transaction

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STOP			Stop SPI transaction
			Trigger	1	Trigger task

6.25.6.3 TASKS_SUSPEND

Address offset: 0x01C

Suspend SPI transaction

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_SUSPEND			Suspend SPI transaction
			Trigger	1	Trigger task

6.25.6.4 TASKS_RESUME

Address offset: 0x020

Resume SPI transaction

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_RESUME			Resume SPI transaction
			Trigger	1	Trigger task

6.25.6.5 EVENTS_STOPPED

Address offset: 0x104

SPI transaction has stopped

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_STOPPED			SPI transaction has stopped
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.25.6.6 EVENTS_ENDRX

Address offset: 0x110

End of RXD buffer reached

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDRX			End of RXD buffer reached
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.25.6.7 EVENTS_END

Address offset: 0x118

End of RXD buffer and TXD buffer reached

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_END			End of RXD buffer and TXD buffer reached
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.25.6.8 EVENTS_ENDTX

Address offset: 0x120

End of TXD buffer reached

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDTX			End of TXD buffer reached
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.25.6.9 EVENTS_STARTED

Address offset: 0x14C

Transaction started

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_STARTED			Transaction started
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.25.6.10 SHORTS

Address offset: 0x200

Shortcuts between local events and tasks

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																			A																
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	END_START																																	
			Disabled	0																															
			Enabled	1																															

6.25.6.11 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																E										D		C		B			A		
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	STOPPED																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	ENDRX																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	END																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
D	RW	ENDTX																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
E	RW	STARTED																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															

6.25.6.12 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	STOPPED			Write '1' to disable interrupt for event STOPPED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	ENDRX			Write '1' to disable interrupt for event ENDRX
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	END			Write '1' to disable interrupt for event END
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	ENDTX			Write '1' to disable interrupt for event ENDTX
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	STARTED			Write '1' to disable interrupt for event STARTED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.25.6.13 STALLSTAT

Address offset: 0x400

Stall status for EasyDMA RAM accesses. The fields in this register are set to STALL by hardware whenever a stall occurs and can be cleared (set to NOSTALL) by the CPU.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TX		[1..0]	Stall status for EasyDMA RAM reads
			NOSTALL	0	No stall
			STALL	1	A stall has occurred
B	RW	RX		[1..0]	Stall status for EasyDMA RAM writes
			NOSTALL	0	No stall
			STALL	1	A stall has occurred

6.25.6.14 ENABLE

Address offset: 0x500

Enable SPIM

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ENABLE			Enable or disable SPIM
			Disabled	0	Disable SPIM
			Enabled	7	Enable SPIM

6.25.6.15 PSEL.SCK

Address offset: 0x508

Pin select for SCK

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**

ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.25.6.16 PSEL.MOSI

Address offset: 0x50C

Pin select for MOSI signal

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**

ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.25.6.17 PSEL.MISO

Address offset: 0x510

Pin select for MISO signal

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**

ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.25.6.18 PSEL.CSN

Address offset: 0x514

Pin select for CSN

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.25.6.19 FREQUENCY

Address offset: 0x524

SPI frequency. Accuracy depends on the HFCLK source selected.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x04000000				**0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	FREQUENCY			SPI master data rate
			K125	0x02000000	125 kbps
			K250	0x04000000	250 kbps
			K500	0x08000000	500 kbps
			M1	0x10000000	1 Mbps
			M2	0x20000000	2 Mbps
			M4	0x40000000	4 Mbps
			M8	0x80000000	8 Mbps
			M16	0x0A000000	16 Mbps
			M32	0x14000000	32 Mbps

6.25.6.20 RXD

RXD EasyDMA channel

6.25.6.20.1 RXD.PTR

Address offset: 0x534

Data pointer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Data pointer See Memory chapter for details about which memories are available for EasyDMA.

6.25.6.20.2 RXD.MAXCNT

Address offset: 0x538

Maximum number of bytes in receive buffer

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	MAXCNT		[0..0xFFFF]	Maximum number of bytes in receive buffer

6.25.6.20.3 RXD.AMOUNT

Address offset: 0x53C

Number of bytes transferred in the last transaction

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	AMOUNT		[0..0xFFFF]	Number of bytes transferred in the last transaction

6.25.6.20.4 RXD.LIST

Address offset: 0x540

EasyDMA list type

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	LIST			List type
			Disabled	0	Disable EasyDMA list
			ArrayList	1	Use array list

6.25.6.21 TXD

TXD EasyDMA channel

6.25.6.21.1 TXD.PTR

Address offset: 0x544

Data pointer

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Data pointer See Memory chapter for details about which memories are available for EasyDMA.

6.25.6.21.2 TXD.MAXCNT

Address offset: 0x548

Number of bytes in transmit buffer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	MAXCNT		[0..0xFFFF]	Maximum number of bytes in transmit buffer

6.25.6.21.3 TXD.AMOUNT

Address offset: 0x54C

Number of bytes transferred in the last transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	AMOUNT		[0..0xFFFF]	Number of bytes transferred in the last transaction

6.25.6.21.4 TXD.LIST

Address offset: 0x550

EasyDMA list type

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																															A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	LIST			List type
			Disabled	0	Disable EasyDMA list
			ArrayList	1	Use array list

6.25.6.22 CONFIG

Address offset: 0x554

Configuration register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ORDER			Bit order
			MsbFirst	0	Most significant bit shifted out first
			LsbFirst	1	Least significant bit shifted out first
B	RW	CPHA			Serial clock (SCK) phase
			Leading	0	Sample on leading edge of clock, shift serial data on trailing edge
			Trailing	1	Sample on trailing edge of clock, shift serial data on leading edge
C	RW	CPOL			Serial clock (SCK) polarity
			ActiveHigh	0	Active high
			ActiveLow	1	Active low

6.25.6.23 IFTIMING.RXDELAY

Address offset: 0x560

Sample delay for input serial data on MISO

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RXDELAY		[7..0]	Sample delay for input serial data on MISO. The value specifies the number of 64 MHz clock cycles (15.625 ns) delay from the the sampling edge of SCK (leading edge for CONFIG.CPHA = 0, trailing edge for CONFIG.CPHA = 1) until the input serial data is sampled. As en example, if RXDELAY = 0 and CONFIG.CPHA = 0, the input serial data is sampled on the rising edge of SCK.

6.25.6.24 IFTIMING.CSNDUR

Address offset: 0x564

Minimum duration between edge of CSN and edge of SCK at the start and the end of a transaction, and minimum duration CSN will stay high between transactions if END-START shortcut is used

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A
Reset 0x00000002				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	CSNDUR		[0xFF..0]	Minimum duration between edge of CSN and edge of SCK at the start and end of a transaction. If END-START shortcut is used, minimum duration CSN will stay high between transactions. The value is specified in number of 64 MHz clock cycles (15.625 ns). Note that for low values of CSNDUR, the system turnaround time will dominate the actual time between transactions.

6.25.6.25 CSNPOL

Address offset: 0x568

Polarity of CSN output

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	CSNPOL			Polarity of CSN output
			LOW	0	Active low (idle state high)
			HIGH	1	Active high (idle state low)

6.25.6.26 PSELDCX

Address offset: 0x56C

Pin select for DCX signal

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**

ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.25.6.27 DCXCNT

Address offset: 0x570

DCX configuration

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	DCXCNT		0x0..0xF	This register specifies the number of command bytes preceding the data bytes. The PSEL.DCX line will be low during transmission of command bytes and high during transmission of data bytes. Value 0xF indicates that all bytes are command bytes.

6.25.6.28 ORC

Address offset: 0x5C0

Byte transmitted after TXD.MAXCNT bytes have been transmitted in the case when RXD.MAXCNT is greater than TXD.MAXCNT

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ORC			Byte transmitted after TXD.MAXCNT bytes have been transmitted in the case when RXD.MAXCNT is greater than TXD.MAXCNT.

6.25.7 Electrical specification

6.25.7.1 Timing specifications

Symbol	Description	Min.	Typ.	Max.	Units
f_{SPIM}	Bit rates for SPIM[33]			32	Mbps
$t_{SPIM,START}$	Time from START task to transmission started		1		µs
$t_{SPIM,CSCK}$	SCK period	31.25			ns
$t_{SPIM,RSCK,LD}$	SCK rise time, standard drive[34]			$t_{RF,25pF}$	
$t_{SPIM,RSCK,HD}$	SCK rise time, high drive[34]			$t_{HRF,25pF}$	
$t_{SPIM,FSCK,LD}$	SCK fall time, standard drive[34]			$t_{RF,25pF}$	
$t_{SPIM,FSCK,HD}$	SCK fall time, high drive[34]			$t_{HRF,25pF}$	
$t_{SPIM,WHSCK}$	SCK high time[34]	$(t_{CSCK}/2) - t_{RSCK}$			
$t_{SPIM,WLSCK}$	SCK low time[34]	$(t_{CSCK}/2) - t_{FSCK}$			
$t_{SPIM,SUMI}$	MISO to CLK edge setup time	19			ns
$t_{SPIM,HMI}$	CLK edge to MISO hold time	18			ns
$t_{SPIM,VMO}$	CLK edge to MOSI valid, SCK frequency ≤ 8 MHz			59	ns
$t_{SPIM,VMO,HS}$	CLK edge to MOSI valid, SCK frequency > 8 MHz			8	ns
$t_{SPIM,HMO}$	MOSI hold time after CLK edge	20			ns

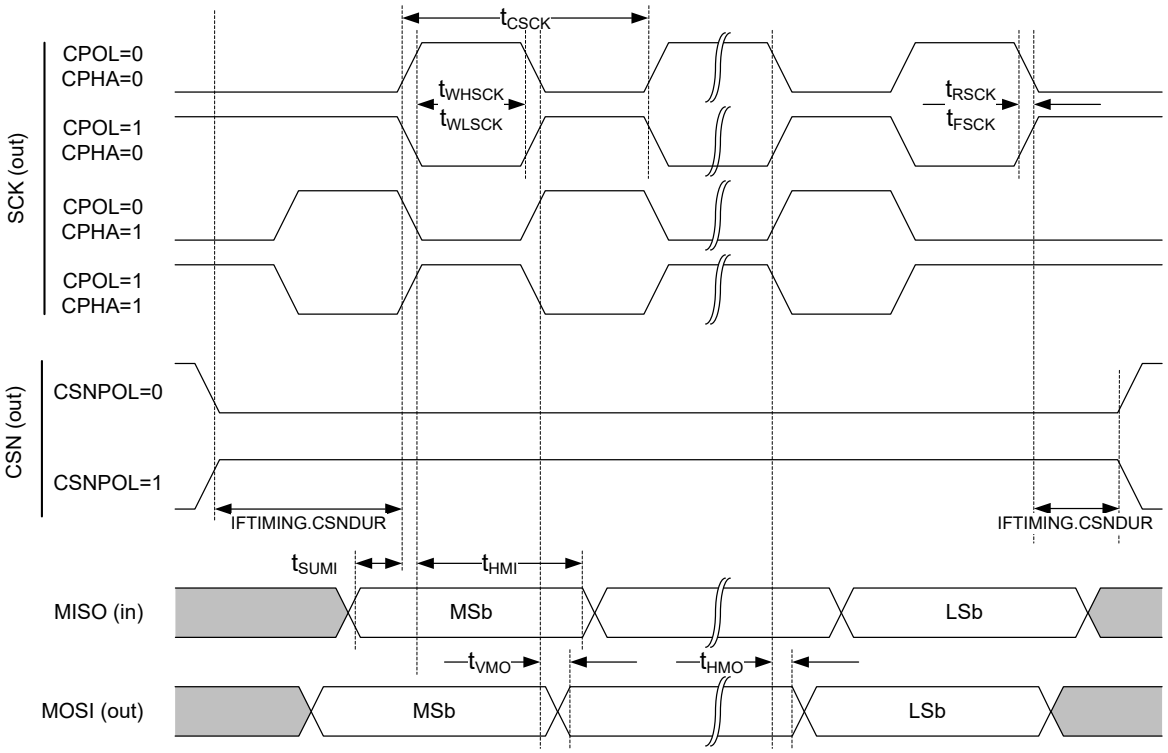


Figure 166: SPIM timing diagram

[33] High bit rates may require GPIOs to be set as High Drive, see GPIO chapter for more details.
[34] At 25 pF load, including GPIO pin capacitance, see GPIO electrical specification.

6.26 SPIS — Serial peripheral interface slave with EasyDMA

SPI slave (SPIS) is implemented with EasyDMA support for ultra-low power serial communication from an external SPI master. EasyDMA, in conjunction with hardware-based semaphore mechanisms, removes all real-time requirements associated with controlling the SPI slave from a low priority CPU execution context.

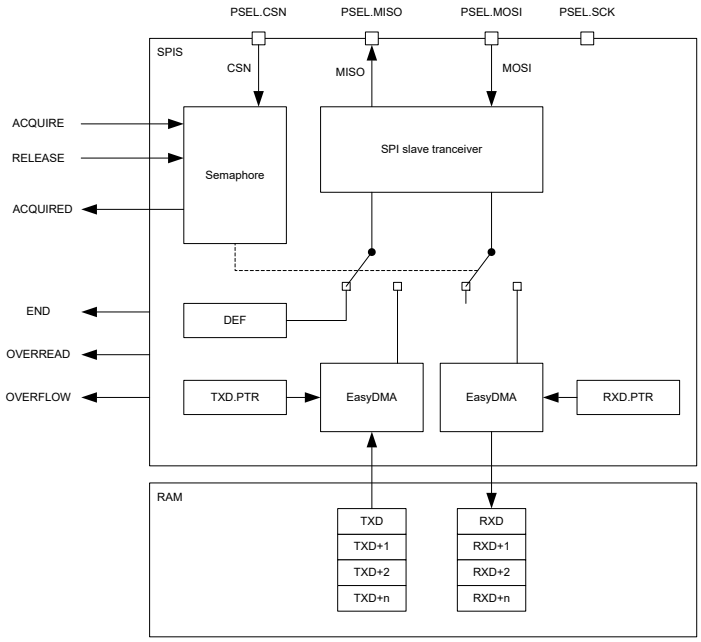


Figure 167: SPI slave

The SPIS supports SPI modes 0 through 3. The CONFIG register allows setting CPOL and CPHA appropriately.

Mode	Clock polarity	Clock phase
	CPOL	CPHA
SPI_MODE0	0 (Active High)	0 (Trailing Edge)
SPI_MODE1	0 (Active High)	1 (Leading Edge)
SPI_MODE2	1 (Active Low)	0 (Trailing Edge)
SPI_MODE3	1 (Active Low)	1 (Leading Edge)

Table 47: SPI modes

6.26.1 Shared resources

The SPI slave shares registers and other resources with other peripherals that have the same ID as the SPI slave. Therefore, you must disable all peripherals that have the same ID as the SPI slave before the SPI slave can be configured and used.

Disabling a peripheral that has the same ID as the SPI slave will not reset any of the registers that are shared with the SPI slave. It is important to configure all relevant SPI slave registers explicitly to secure that it operates correctly.

The Instantiation table in Instantiation on page 24 shows which peripherals have the same ID as the SPI slave.

6.26.2 EasyDMA

The SPIS implements EasyDMA for accessing RAM without CPU involvement.

The SPIS peripheral implements the following EasyDMA channels.

Channel	Type	Register Cluster
TXD	READER	TXD
RXD	WRITER	RXD

Table 48: SPIS EasyDMA Channels

For detailed information regarding the use of EasyDMA, see EasyDMA on page 63.

If RXD.MAXCNT is larger than TXD.MAXCNT, the remaining transmitted bytes will contain the value defined in the ORC register.

The END event indicates that EasyDMA has finished accessing the buffer in RAM.

6.26.3 SPI slave operation

SPI slave uses two memory pointers, RXD.PTR and TXD.PTR, that point to the RXD buffer (receive buffer) and TXD buffer (transmit buffer) respectively. Since these buffers are located in RAM, which can be accessed by both the SPI slave and the CPU, a hardware based semaphore mechanism is implemented to enable safe sharing.

See SPI transaction when shortcut between END and ACQUIRE is enabled on page 743.

Before the CPU can safely update the RXD.PTR and TXD.PTR pointers, it must first acquire the SPI semaphore. The CPU can acquire the semaphore by triggering the ACQUIRE task and then receiving the ACQUIRED event. When the CPU has updated the RXD.PTR and TXD.PTR pointers the CPU must release the semaphore before the SPI slave will be able to acquire it. The CPU releases the semaphore by triggering the RELEASE task. This is illustrated in SPI transaction when shortcut between END and ACQUIRE is enabled on page 743. Triggering the RELEASE task when the semaphore is not granted to the CPU will have no effect.

The semaphore mechanism does not, at any time, prevent the CPU from performing read or write access to the RXD.PTR register, the TXD.PTR registers, or the RAM that these pointers are pointing to. The semaphore is only telling when these can be updated by the CPU so that safe sharing is achieved.

The semaphore is by default assigned to the CPU after the SPI slave is enabled. No ACQUIRED event will be generated for this initial semaphore handover. An ACQUIRED event will be generated immediately if the ACQUIRE task is triggered while the semaphore is assigned to the CPU.

The SPI slave will try to acquire the semaphore when CSN goes low. If the SPI slave does not manage to acquire the semaphore at this point, the transaction will be ignored. This means that all incoming data on MOSI will be discarded, and the DEF (default) character will be clocked out on the MISO line throughout the whole transaction. This will also be the case even if the semaphore is released by the CPU during the transaction. In case of a race condition where the CPU and the SPI slave try to acquire the semaphore at the same time, as illustrated in lifeline item 2 in SPI transaction when shortcut between END and ACQUIRE is enabled on page 743, the semaphore will be granted to the CPU.

If the SPI slave acquires the semaphore, the transaction will be granted. The incoming data on MOSI will be stored in the RXD buffer and the data in the TXD buffer will be clocked out on MISO.

When a granted transaction is completed and CSN goes high, the SPI slave will automatically release the semaphore and generate the END event.

As long as the semaphore is available, the SPI slave can be granted multiple transactions one after the other. If the CPU is not able to reconfigure the TXD.PTR and RXD.PTR between granted transactions, the same TX data will be clocked out and the RX buffers will be overwritten. To prevent this from happening, the END_ACQUIRE shortcut can be used. With this shortcut enabled, the semaphore will be handed over to the CPU automatically after the granted transaction has completed. This enables the CPU to update the TXPTR and RXPTR between every granted transaction.

If the CPU tries to acquire the semaphore while it is assigned to the SPI slave, an immediate handover will not be granted. However, the semaphore will be handed over to the CPU as soon as the SPI slave has released the semaphore after the granted transaction is completed. If the END_ACQUIRE shortcut is enabled and the CPU has triggered the ACQUIRE task during a granted transaction, only one ACQUIRE request will be served following the END event.

The MAXRX register specifies the maximum number of bytes the SPI slave can receive in one granted transaction. If the SPI slave receives more than MAXRX number of bytes, an OVERFLOW will be indicated in the STATUS register and the incoming bytes will be discarded.

The MAXTX parameter specifies the maximum number of bytes the SPI slave can transmit in one granted transaction. If the SPI slave is forced to transmit more than MAXTX number of bytes, an OVERREAD will be indicated in the STATUS register and the ORC character will be clocked out.

The RXD.AMOUNT and TXD.AMOUNT registers are updated when a granted transaction is completed. The TXD.AMOUNT register indicates how many bytes were read from the TX buffer in the last transaction. This does not include the ORC (over-read) characters. Similarly, the RXD.AMOUNT register indicates how many bytes were written into the RX buffer in the last transaction.

The ENDRX event is generated when the RX buffer has been filled.

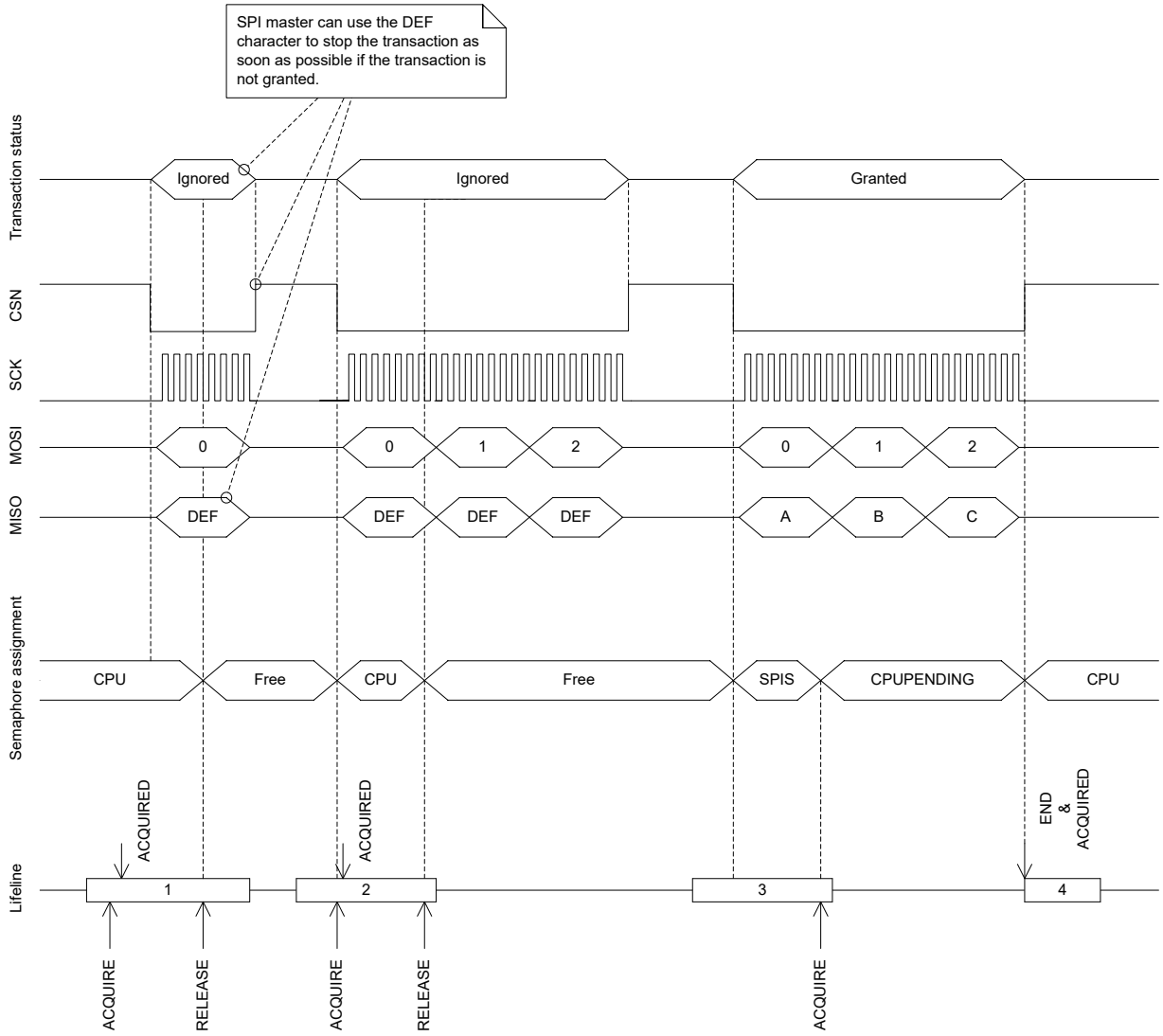


Figure 168: SPI transaction when shortcut between END and ACQUIRE is enabled

6.26.4 Pin configuration

The CSN, SCK, MOSI, and MISO signals associated with the SPI slave are mapped to physical pins according to the configuration specified in the PSEL.CSN, PSEL.SCK, PSEL.MOSI, and PSEL.MISO registers respectively. If the CONNECT field of any of these registers is set to Disconnected, the associated SPI slave signal will not be connected to any physical pins.

The PSEL.CSN, PSEL.SCK, PSEL.MOSI, and PSEL.MISO registers and their configurations are only used as long as the SPI slave is enabled, and retained only as long as the device is in System ON mode. See POWER — Power supply on page 81 chapter for more information about power modes. When the peripheral is disabled, the pins will behave as regular GPIOs and use the configuration in their respective OUT bit field and PIN_CNF[n] register. PSEL.CSN, PSEL.SCK, PSEL.MOSI, and PSEL.MISO must only be configured when the SPI slave is disabled.

To secure correct behavior in the SPI slave, the pins used by the SPI slave must be configured in the GPIO peripheral as described in GPIO configuration before enabling peripheral on page 744 before enabling the SPI slave. This is to secure that the pins used by the SPI slave are driven correctly if the SPI slave itself is temporarily disabled, or if the device temporarily enters System OFF. This configuration must be retained in the GPIO for the selected I/Os as long as the SPI slave is to be recognized by an external SPI master.

The MISO line is set in high impedance as long as the SPI slave is not selected with CSN.

Only one peripheral can be assigned to drive a particular GPIO pin at a time. Failing to do so may result in unpredictable behavior.

SPI signal	SPI pin	Direction	Output value	Comment
CSN	As specified in PSEL.CSN	Input	Not applicable	
SCK	As specified in PSEL.SCK	Input	Not applicable	
MOSI	As specified in PSEL.MOSI	Input	Not applicable	
MISO	As specified in PSEL.MISO	Input	Not applicable	Emulates that the SPI slave is not selected.

Table 49: GPIO configuration before enabling peripheral

6.26.5 Registers

Instances

Instance	Base address	Description
SPIS0	0x40003000	SPI slave 0
SPIS1	0x40004000	SPI slave 1
SPIS2	0x40023000	SPI slave 2

Register overview

Register	Offset	Description
TASKS_ACQUIRE	0x024	Acquire SPI semaphore
TASKS_RELEASE	0x028	Release SPI semaphore, enabling the SPI slave to acquire it
EVENTS_END	0x104	Granted transaction completed
EVENTS_ENDRX	0x110	End of RXD buffer reached
EVENTS_ACQUIRED	0x128	Semaphore acquired
SHORTS	0x200	Shortcuts between local events and tasks
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
SEMSTAT	0x400	Semaphore status register
STATUS	0x440	Status from last transaction
ENABLE	0x500	Enable SPI slave
PSEL.SCK	0x508	Pin select for SCK
PSEL.MISO	0x50C	Pin select for MISO signal
PSEL.MOSI	0x510	Pin select for MOSI signal
PSEL.CSN	0x514	Pin select for CSN signal
RXD.PTR	0x534	RXD data pointer
RXD.MAXCNT	0x538	Maximum number of bytes in receive buffer
RXD.AMOUNT	0x53C	Number of bytes received in last granted transaction
RXD.LIST	0x540	EasyDMA list type
TXD.PTR	0x544	TXD data pointer
TXD.MAXCNT	0x548	Maximum number of bytes in transmit buffer
TXD.AMOUNT	0x54C	Number of bytes transmitted in last granted transaction
TXD.LIST	0x550	EasyDMA list type
CONFIG	0x554	Configuration register
DEF	0x55C	Default character. Character clocked out in case of an ignored transaction.
ORC	0x5C0	Over-read character

6.26.5.1 TASKS_ACQUIRE

Address offset: 0x024

Acquire SPI semaphore

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_ACQUIRE			Acquire SPI semaphore
			Trigger	1	Trigger task

6.26.5.2 TASKS_RELEASE

Address offset: 0x028

Release SPI semaphore, enabling the SPI slave to acquire it

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_RELEASE			Release SPI semaphore, enabling the SPI slave to acquire it
			Trigger	1	Trigger task

6.26.5.3 EVENTS_END

Address offset: 0x104

Granted transaction completed

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_END			Granted transaction completed
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.26.5.4 EVENTS_ENDRX

Address offset: 0x110

End of RXD buffer reached

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDRX			End of RXD buffer reached
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.26.5.5 EVENTS_ACQUIRED

Address offset: 0x128

Semaphore acquired

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ACQUIRED			Semaphore acquired
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.26.5.6 SHORTS

Address offset: 0x200

Shortcuts between local events and tasks

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	END_ACQUIRE			Shortcut between event END and task ACQUIRE
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

6.26.5.7 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	END			Write '1' to enable interrupt for event END
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	ENDRX			Write '1' to enable interrupt for event ENDRX
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	ACQUIRED			Write '1' to enable interrupt for event ACQUIRED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.26.5.8 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	END			Write '1' to disable interrupt for event END
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	ENDRX			Write '1' to disable interrupt for event ENDRX
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	ACQUIRED			Write '1' to disable interrupt for event ACQUIRED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.26.5.9 SEMSTAT

Address offset: 0x400

Semaphore status register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**

ID	R/W	Field	Value ID	Value	Description
A	R	SEMSTAT			Semaphore status
			Free	0	Semaphore is free
			CPU	1	Semaphore is assigned to CPU
			SPIS	2	Semaphore is assigned to SPI slave
			CPUPending	3	Semaphore is assigned to SPI but a handover to the CPU is pending

6.26.5.10 STATUS

Address offset: 0x440

Status from last transaction

Individual bits are cleared by writing a `1` to the bits that shall be cleared

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	OVERREAD			TX buffer over-read detected, and prevented
			NotPresent	0	Read: error not present
			Present	1	Read: error present
			Clear	1	Write: clear error on writing '1'
B	RW	OVERFLOW			RX buffer overflow detected, and prevented
			NotPresent	0	Read: error not present
			Present	1	Read: error present
			Clear	1	Write: clear error on writing '1'

6.26.5.11 ENABLE

Address offset: 0x500

Enable SPI slave

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	ENABLE																																	
			Disabled	0																															
			Enabled	2																															

6.26.5.12 PSEL.SCK

Address offset: 0x508

Pin select for SCK

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				C																										B	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW	PIN		[0..31]																															
B	RW	PORT		[0..1]																															
C	RW	CONNECT																																	
			Disconnected	1																															
			Connected	0																															

6.26.5.13 PSEL.MISO

Address offset: 0x50C

Pin select for MISO signal

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				C																										B	A	A	A	A	A
Reset 0xFFFFFFFF				**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
ID	R/W	Field	Value ID	Value																															
A	RW	PIN		[0..31]																															
B	RW	PORT		[0..1]																															
C	RW	CONNECT																																	
			Disconnected	1																															
			Connected	0																															

6.26.5.14 PSEL.MOSI

Address offset: 0x510

Pin select for MOSI signal

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.26.5.15 PSEL.CSN

Address offset: 0x514

Pin select for CSN signal

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.26.5.16 RXD.PTR

Address offset: 0x534

RXD data pointer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			RXD data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.26.5.17 RXD.MAXCNT

Address offset: 0x538

Maximum number of bytes in receive buffer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	MAXCNT		[0..0xFFFF]	Maximum number of bytes in receive buffer

6.26.5.18 RXD.AMOUNT

Address offset: 0x53C

Number of bytes received in last granted transaction

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	AMOUNT		[0..0xFFFF]	Number of bytes received in the last granted transaction

6.26.5.19 RXD.LIST

Address offset: 0x540

EasyDMA list type

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	LIST			List type
			Disabled	0	Disable EasyDMA list
			ArrayList	1	Use array list

6.26.5.20 TXD.PTR

Address offset: 0x544

TXD data pointer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			TXD data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.26.5.21 TXD.MAXCNT

Address offset: 0x548

Maximum number of bytes in transmit buffer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	MAXCNT		[0..0xFFFF]	Maximum number of bytes in transmit buffer

6.26.5.22 TXD.AMOUNT

Address offset: 0x54C

Number of bytes transmitted in last granted transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	AMOUNT		[0..0xFFFF]	Number of bytes transmitted in last granted transaction

6.26.5.23 TXD.LIST

Address offset: 0x550

EasyDMA list type

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																															A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	LIST			List type
			Disabled	0	Disable EasyDMA list
			ArrayList	1	Use array list

6.26.5.24 CONFIG

Address offset: 0x554

Configuration register

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																														C	B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ORDER			Bit order
			MsbFirst	0	Most significant bit shifted out first
			LsbFirst	1	Least significant bit shifted out first
B	RW	CPHA			Serial clock (SCK) phase
			Leading	0	Sample on leading edge of clock, shift serial data on trailing edge
			Trailing	1	Sample on trailing edge of clock, shift serial data on leading edge
C	RW	CPOL			Serial clock (SCK) polarity
			ActiveHigh	0	Active high
			ActiveLow	1	Active low

6.26.5.25 DEF

Address offset: 0x55C

Default character. Character clocked out in case of an ignored transaction.

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																									A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	DEF			Default character. Character clocked out in case of an ignored transaction.

6.26.5.26 ORC

Address offset: 0x5C0

Over-read character

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																									A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ORC			Over-read character. Character clocked out after an over-read of the transmit buffer.

6.26.6 Electrical specification

6.26.6.1 SPIS slave interface electrical specifications

Symbol	Description	Min.	Typ.	Max.	Units
f_{SPIS}	Bit rates for SPIS[35]			8[36]	Mbps
$t_{SPIS,START}$	Time from RELEASE task to receive/transmit (CSN active)		0.125		µs

6.26.6.2 Serial Peripheral Interface Slave (SPIS) timing specifications

Symbol	Description	Min.	Typ.	Max.	Units
$t_{SPIS,CSCKIN}$	SCK input period	125			ns
$t_{SPIS,RFSCKIN}$	SCK input rise/fall time			30	ns
$t_{SPIS,WHSCKIN}$	SCK input high time	30			ns
$t_{SPIS,WLSCKIN}$	SCK input low time	30			ns
$t_{SPIS,SUCSN}$	CSN to CLK setup time	1000			ns
$t_{SPIS,HCSN}$	CLK to CSN hold time	1000			ns
$t_{SPIS,ASA}$	CSN to MISO driven	0			ns
$t_{SPIS,ASO}$	CSN to MISO valid [37]			1000	ns
$t_{SPIS,DISSO}$	CSN to MISO disabled[37]			68	ns
$t_{SPIS,CWH}$	CSN inactive time	300			ns
$t_{SPIS,VSO}$	CLK edge to MISO valid			59	ns
$t_{SPIS,HSO}$	MISO hold time after CLK edge	20[38]			ns
$t_{SPIS,SUSI}$	MOSI to CLK edge setup time	19			ns
$t_{SPIS,HSI}$	CLK edge to MOSI hold time	18			ns

35 High bit rates may require GPIOs to be set as High Drive, see GPIO chapter for more details.

36 The actual maximum data rate depends on the master's CLK to MISO and MOSI setup and hold timings.

37 At 25 pF load, including GPIO capacitance, see GPIO electrical specification.

38 This is to ensure compatibility to SPI masters sampling MISO on the same edge as MOSI is output.

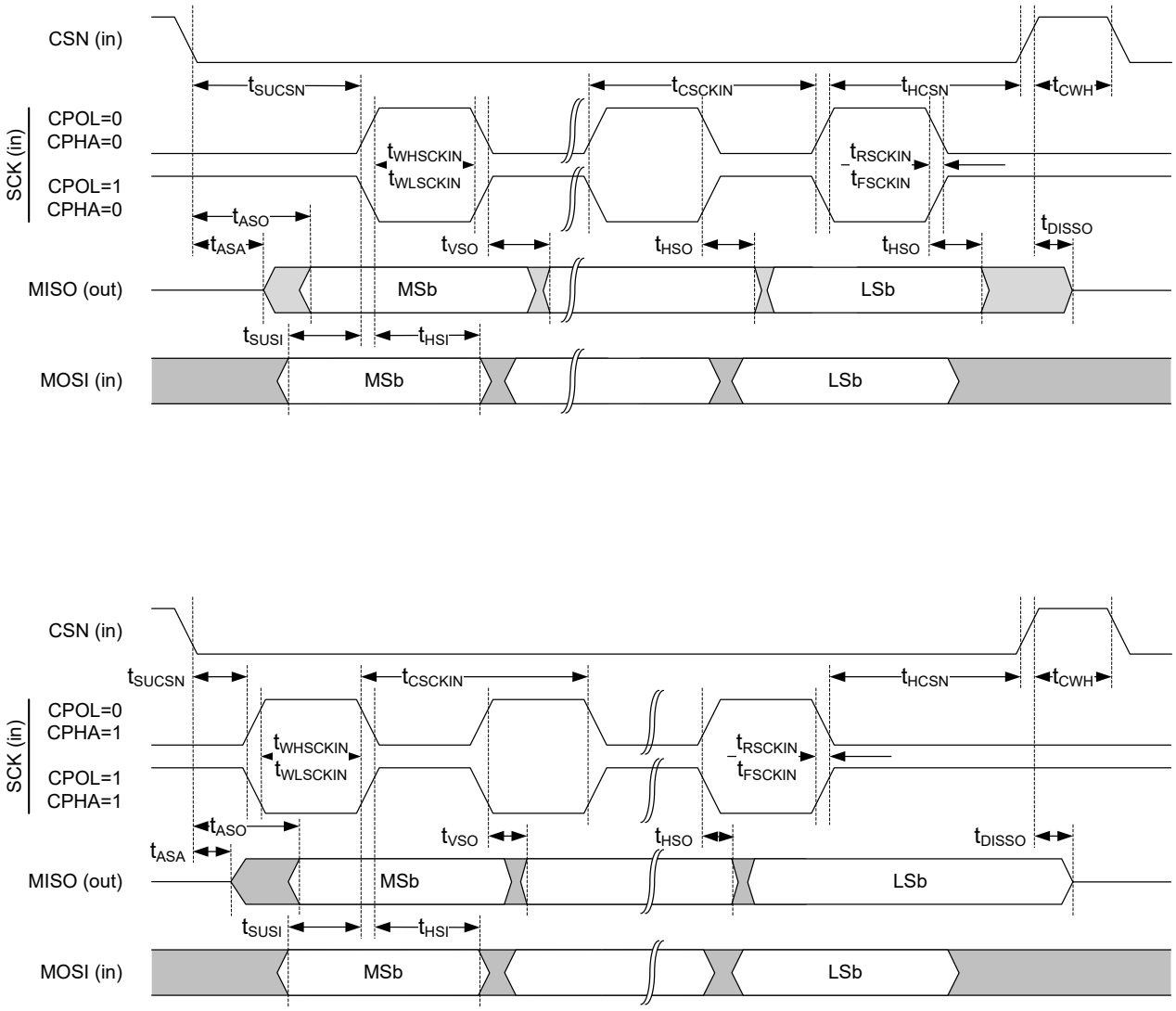


Figure 169: SPIS timing diagram

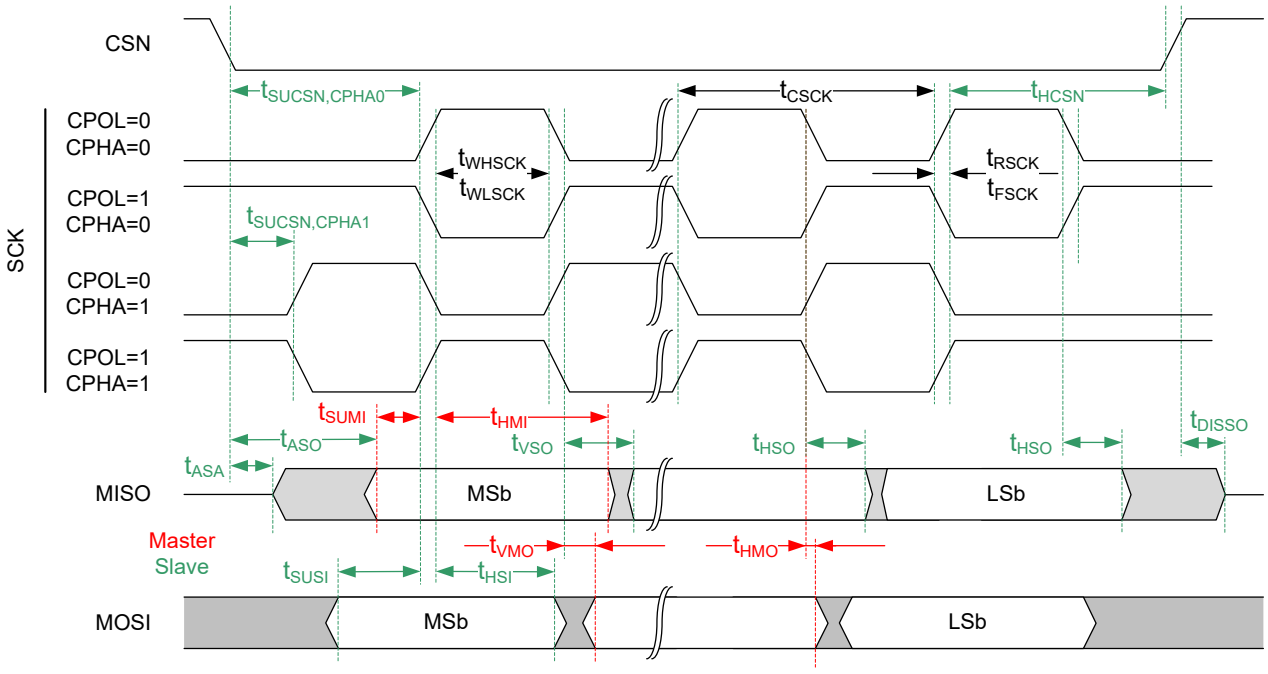


Figure 170: Common SPIM and SPIS timing diagram

6.27 SWI — Software interrupts

A set of interrupts have been reserved for use as software interrupts.

6.27.1 Registers

Instances

Instance	Base address	Description
SWI0	0x40014000	Software interrupt 0
SWI1	0x40015000	Software interrupt 1
SWI2	0x40016000	Software interrupt 2
SWI3	0x40017000	Software interrupt 3
SWI4	0x40018000	Software interrupt 4
SWI5	0x40019000	Software interrupt 5

6.28 TEMP — Temperature sensor

The temperature sensor measures die temperature over the temperature range of the device. Linearity compensation can be implemented if required by the application.

Listed here are the main features for TEMP:

- Temperature range is greater than or equal to operating temperature of the device
- Resolution is 0.25 degrees

TEMP is started by triggering the START task.

When the temperature measurement is completed, a DATARDY event will be generated and the result of the measurement can be read from the TEMP register.

To achieve the measurement accuracy stated in the electrical specification, the crystal oscillator must be selected as the HFCLK source, see CLOCK — Clock control on page 157 for more information.

When the temperature measurement is completed, TEMP analog electronics power down to save power.

TEMP only supports one-shot operation, meaning that every TEMP measurement has to be explicitly started using the START task.

6.28.1 Registers

Instances

Instance	Base address	Description
TEMP	0x4000C000	Temperature sensor

Register overview

Register	Offset	Description
TASKS_START	0x000	Start temperature measurement
TASKS_STOP	0x004	Stop temperature measurement
EVENTS_DATARDY	0x100	Temperature measurement complete, data ready

Register	Offset	Description
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
TEMP	0x508	Temperature in °C (0.25° steps)
A0	0x520	Slope of first piecewise linear function
A1	0x524	Slope of second piecewise linear function
A2	0x528	Slope of third piecewise linear function
A3	0x52C	Slope of fourth piecewise linear function
A4	0x530	Slope of fifth piecewise linear function
A5	0x534	Slope of sixth piecewise linear function
B0	0x540	y-intercept of first piecewise linear function
B1	0x544	y-intercept of second piecewise linear function
B2	0x548	y-intercept of third piecewise linear function
B3	0x54C	y-intercept of fourth piecewise linear function
B4	0x550	y-intercept of fifth piecewise linear function
B5	0x554	y-intercept of sixth piecewise linear function
T0	0x560	End point of first piecewise linear function
T1	0x564	End point of second piecewise linear function
T2	0x568	End point of third piecewise linear function
T3	0x56C	End point of fourth piecewise linear function
T4	0x570	End point of fifth piecewise linear function

6.28.1.1 TASKS_START

Address offset: 0x000

Start temperature measurement

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																				A
Reset 0x00000000					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field		Value ID					Value																											
A	W	TASKS_START																																		
				Trigger					1																											

6.28.1.2 TASKS_STOP

Address offset: 0x004

Stop temperature measurement

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																				A
Reset 0x00000000					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field		Value ID					Value																											
A	W	TASKS_STOP																																		
				Trigger					1																											

6.28.1.3 EVENTS_DATARDY

Address offset: 0x100

Temperature measurement complete, data ready

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_DATARDY			Temperature measurement complete, data ready
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.28.1.4 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DATARDY			Write '1' to enable interrupt for event DATARDY
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.28.1.5 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	DATARDY			Write '1' to disable interrupt for event DATARDY
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.28.1.6 TEMP

Address offset: 0x508

Temperature in °C (0.25° steps)

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	TEMP			Temperature in °C (0.25° steps) Result of temperature measurement. Die temperature in °C, 2's complement format, 0.25 °C steps. Decision point: DATARDY

6.28.1.7 A0

Address offset: 0x520

Slope of first piecewise linear function

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																					A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000326	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**0**	**0**	**1**	**0**	**0**	**1**	**1**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	A0			Slope of first piecewise linear function

6.28.1.8 A1

Address offset: 0x524

Slope of second piecewise linear function

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																					A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000348	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**0**	**1**	**0**	**0**	**1**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	A1			Slope of second piecewise linear function

6.28.1.9 A2

Address offset: 0x528

Slope of third piecewise linear function

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																					A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x000003AA	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**0**	**1**	**0**	**1**	**0**	**1**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	A2			Slope of third piecewise linear function

6.28.1.10 A3

Address offset: 0x52C

Slope of fourth piecewise linear function

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																					A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x0000040E	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	A3			Slope of fourth piecewise linear function

6.28.1.11 A4

Address offset: 0x530

Slope of fifth piecewise linear function

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A
Reset 0x000004BD				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 1 0 1 1 1 1 0 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	A4			Slope of fifth piecewise linear function

6.28.1.12 A5

Address offset: 0x534

Slope of sixth piecewise linear function

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A
Reset 0x000005A3				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 1 1 0 1 0 0 0 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	A5			Slope of sixth piecewise linear function

6.28.1.13 B0

Address offset: 0x540

y-intercept of first piecewise linear function

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A
Reset 0x00003FEF				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 0 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	B0			y-intercept of first piecewise linear function

6.28.1.14 B1

Address offset: 0x544

y-intercept of second piecewise linear function

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A
Reset 0x00003FBE				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 0 1 1 1 1 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	B1			y-intercept of second piecewise linear function

6.28.1.15 B2

Address offset: 0x548

y-intercept of third piecewise linear function

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A
Reset 0x00003FBE				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 0 1 1 1 1 1 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	B2			y-intercept of third piecewise linear function

6.28.1.16 B3

Address offset: 0x54C

y-intercept of fourth piecewise linear function

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																						A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000012				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**1**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	B3																																	
				y-intercept of fourth piecewise linear function																															

6.28.1.17 B4

Address offset: 0x550

y-intercept of fifth piecewise linear function

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																						A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000124				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**1**	**0**	**0**	**1**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	B4																																	
				y-intercept of fifth piecewise linear function																															

6.28.1.18 B5

Address offset: 0x554

y-intercept of sixth piecewise linear function

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																						A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x0000027C				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	B5																																	
				y-intercept of sixth piecewise linear function																															

6.28.1.19 T0

Address offset: 0x560

End point of first piecewise linear function

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																												A	A	A	A	A	A	A	A
Reset 0x000000E2				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**0**	**0**	**0**	**1**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	T0																																	
				End point of first piecewise linear function																															

6.28.1.20 T1

Address offset: 0x564

End point of second piecewise linear function

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																									A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	T1			End point of second piecewise linear function

6.28.1.21 T2

Address offset: 0x568

End point of third piecewise linear function

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																									A	A	A	A	A	A	A	A
Reset 0x00000019	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**0**	**0**	**1**

ID	R/W	Field	Value ID	Value	Description
A	RW	T2			End point of third piecewise linear function

6.28.1.22 T3

Address offset: 0x56C

End point of fourth piecewise linear function

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																									A	A	A	A	A	A	A	A
Reset 0x0000003C	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	T3			End point of fourth piecewise linear function

6.28.1.23 T4

Address offset: 0x570

End point of fifth piecewise linear function

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																									A	A	A	A	A	A	A	A
Reset 0x00000050	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**1**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	T4			End point of fifth piecewise linear function

6.28.2 Electrical specification

6.28.2.1 Temperature Sensor Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
t_{TEMP}	Time required for temperature measurement		36		µs
$T_{TEMP,RANGE}$	Temperature sensor range	-40		85	°C
$T_{TEMP,ACC}$	Temperature sensor accuracy	-5		5	°C
$T_{TEMP,RES}$	Temperature sensor resolution		0.25		°C
$T_{TEMP,STB}$	Sample to sample stability at constant device temperature	-0.25		0.25	°C
$T_{TEMP,OFFST}$	Sample offset at 25°C	-2.5		2.5	°C

6.29 TWI — I²C compatible two-wire interface

The TWI master is compatible with I²C operating at 100 kHz and 400 kHz.

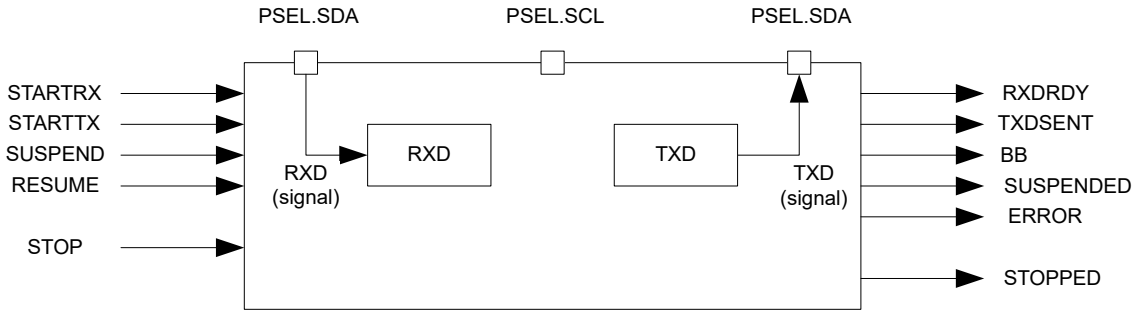


Figure 171: TWI master's main features

6.29.1 Functional description

This TWI master is not compatible with CBUS. The TWI transmitter and receiver are single buffered.

See TWI master's main features on page 761.

A TWI setup with one master and three slaves is shown in the following figure. This TWI master is only able to operate as the only master on the TWI bus.

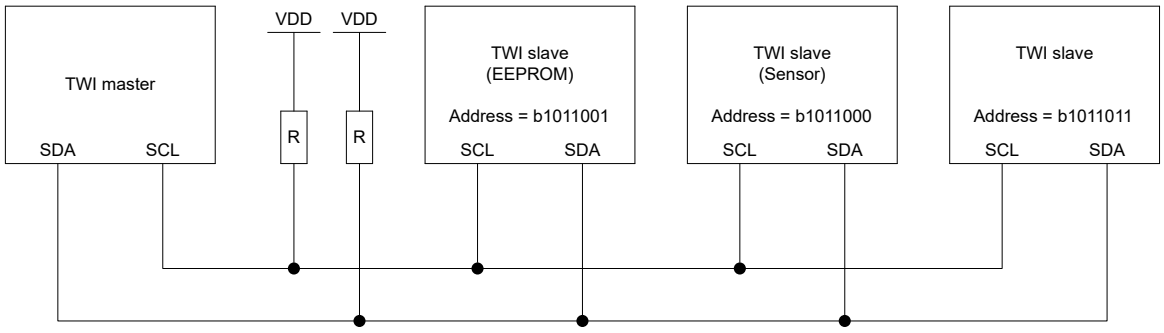


Figure 172: A typical TWI setup with one master and three slaves

This TWI master supports clock stretching performed by the slaves. The TWI master is started by triggering the STARTTX or STARTRX tasks, and stopped by triggering the STOP task.

If a NACK is clocked in from the slave, the TWI master will generate an ERROR event.

6.29.2 Master mode pin configuration

The different signals SCL and SDA associated with the TWI master are mapped to physical pins according to the configuration specified in the PSEL.SCL and PSEL.SDA registers respectively.

If the CONNECT field of a PSEL.xxx register is set to Disconnected, the associated TWI signal is not connected to any physical pin. The PSEL.SCL and PSEL.SDA registers and their configurations are only used as long as the TWI master is enabled, and retained only as long as the device is in ON mode. PSEL.SCL and PSEL.SDA must only be configured when TWI is disabled.

To secure correct signal levels on the pins used by the TWI master when the system is in OFF mode, and when the TWI master is disabled, these pins must be configured in the GPIO peripheral as described in GPIO configuration on page 762.

Only one peripheral can be assigned to drive a particular GPIO pin at a time, failing to do so may result in unpredictable behavior.

TWI master signal	TWI master pin	Direction	Drive strength	Output value
SCL	As specified in PSEL.SCL	Input	S0D1	Not applicable
SDA	As specified in PSEL.SDA	Input	S0D1	Not applicable

Table 50: GPIO configuration

6.29.3 Shared resources

TWI shares registers and other resources with other peripherals that have the same ID as TWI.

Therefore, you must disable all peripherals that have the same ID as TWI before TWI can be configured and used. Disabling a peripheral that has the same ID as TWI will not reset any of the registers that are shared with TWI. It is therefore important to configure all relevant TWI registers explicitly to secure that it operates correctly.

The Instantiation table in Instantiation on page 24 shows which peripherals have the same ID as TWI.

6.29.4 Master write sequence

A TWI master write sequence is started by triggering the STARTTX task. After the STARTTX task has been triggered, the TWI master will generate a start condition on the TWI bus, followed by clocking out the address and the READ/WRITE bit set to 0 (WRITE=0, READ=1).

The address must match the address of the slave device that the master wants to write to. The READ/WRITE bit is followed by an ACK/NACK bit (ACK=0 or NACK=1) generated by the slave.

After receiving the ACK bit, the TWI master will clock out the data bytes that are written to the TXD register. Each byte clocked out from the master will be followed by an ACK/NACK bit clocked in from the slave. A TXDSENT event will be generated each time the TWI master has clocked out a TXD byte, and the associated ACK/NACK bit has been clocked in from the slave.

The TWI master transmitter is single buffered. A second byte can only be written to the TXD register after the previous byte has been clocked out and the ACK/NACK bit clocked in, that is, after the TXDSENT event has been generated.

If the CPU is prevented from writing to TXD when the TWI master is ready to clock out a byte, the TWI master will stretch the clock until the CPU has written a byte to the TXD register.

A typical TWI master write sequence is illustrated in The TWI master writing data to a slave on page 763. Occurrence 3 in the figure illustrates delayed processing of the TXDSENT event associated with TXD byte 1. In this scenario the TWI master will stretch the clock to prevent writing erroneous data to the slave.

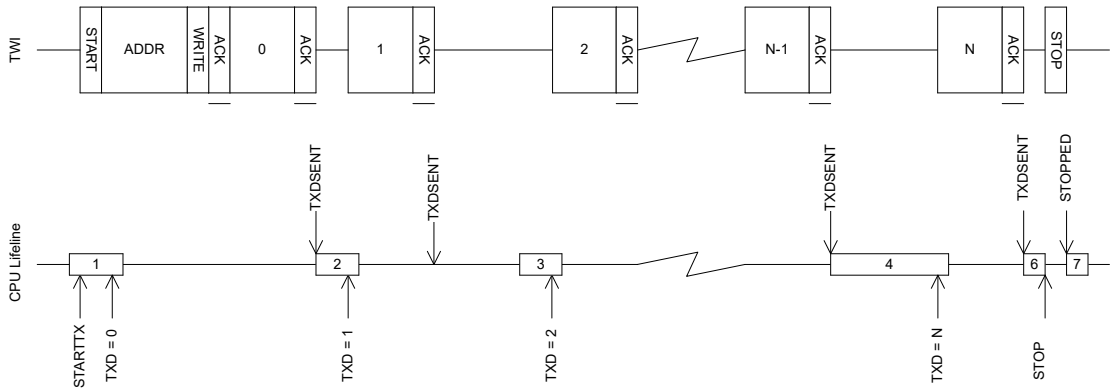


Figure 173: The TWI master writing data to a slave

The TWI master write sequence is stopped when the STOP task is triggered, causing the TWI master to generate a stop condition on the TWI bus.

6.29.5 Master read sequence

A TWI master read sequence is started by triggering the STARTRX task. After the STARTRX task has been triggered, the TWI master will generate a start condition on the TWI bus, followed by clocking out the address and the READ/WRITE bit set to 1 (WRITE = 0, READ = 1).

The address must match the address of the slave device that the master wants to read from. The READ/WRITE bit is followed by an ACK/NACK bit (ACK=0 or NACK = 1) generated by the slave.

After having sent the ACK bit, the TWI slave will send data to the master using the clock generated by the master.

The TWI master will generate a RXDRDY event every time a new byte is received in the RXD register.

After receiving a byte, the TWI master will delay sending the ACK/NACK bit by stretching the clock until the CPU has extracted the received byte, by reading the RXD register.

The TWI master read sequence is stopped by triggering the STOP task. This task must be triggered before the last byte is extracted from RXD to ensure that the TWI master sends a NACK back to the slave before generating the stop condition.

A typical TWI master read sequence is illustrated in The TWI master reading data from a slave on page 764. Occurrence 3 in this figure illustrates delayed processing of the RXDRDY event associated with RXD byte B. In this scenario the TWI master will stretch the clock to prevent the slave from overwriting the contents of the RXD register.

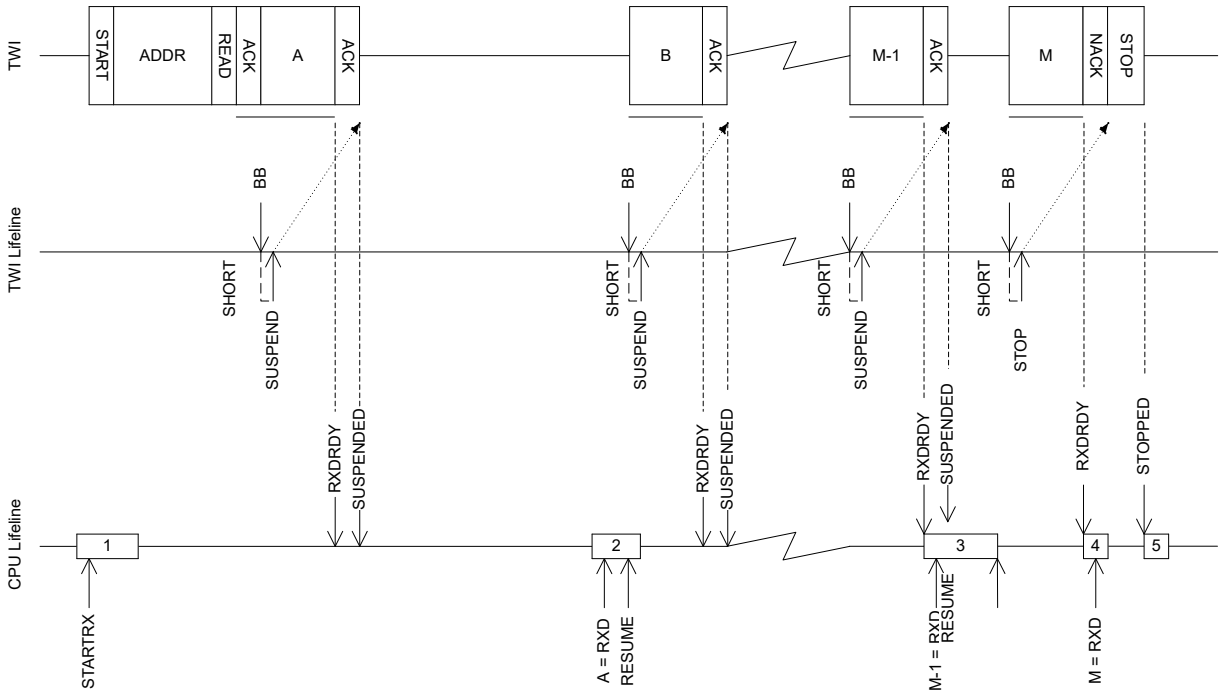


Figure 174: The TWI master reading data from a slave

6.29.6 Master repeated start sequence

A typical repeated start sequence is one in which the TWI master writes one byte to the slave followed by reading M bytes from the slave. Any combination and number of transmit and receive sequences can be combined in this fashion. Only one shortcut to STOP can be enabled at any given time.

The following figure shows a repeated start sequence where the TWI master writes one byte, followed by reading M bytes from the slave without performing a stop in-between.

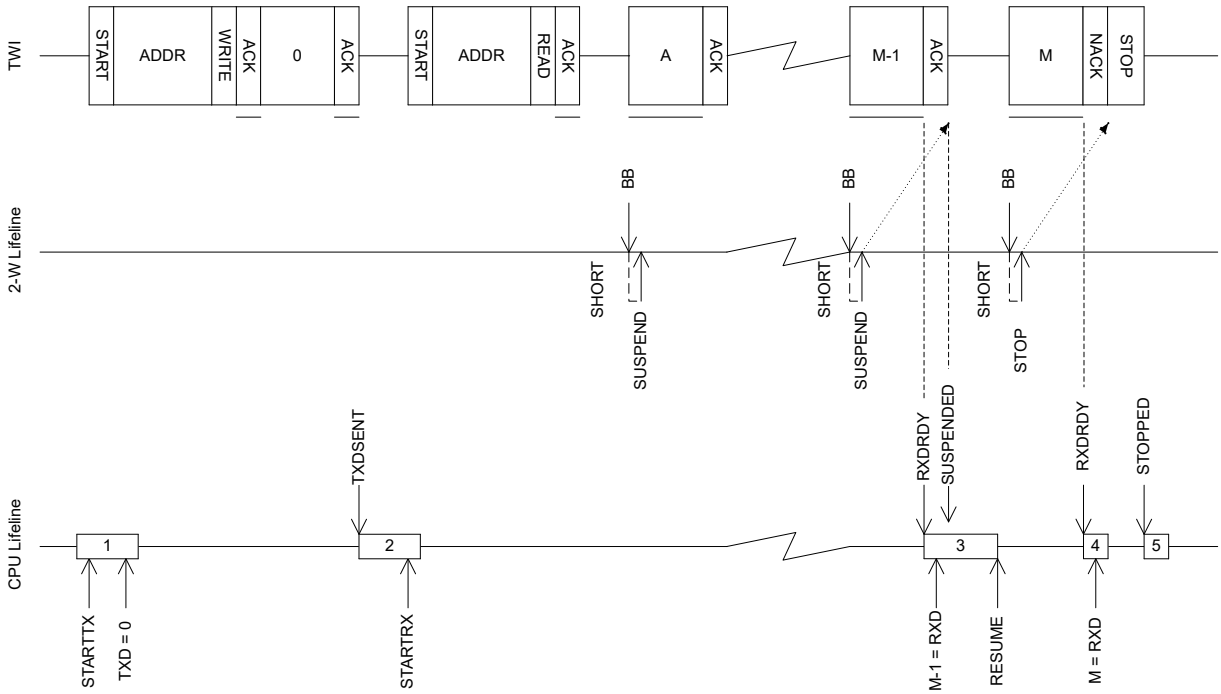


Figure 175: Repeated start sequence illustration

To generate a repeated start after a read sequence, a second start task, STARTRX or STARTTX, must be triggered instead of the STOP task. This start task must be triggered before the last byte is extracted from RXD to ensure that the TWI master sends a NACK back to the slave before generating the repeated start condition.

6.29.7 Low power

When putting the system in low power and the peripheral is not needed, lowest possible power consumption is achieved by stopping, and then disabling the peripheral.

The STOP task is not always needed, like when the peripheral is already stopped. If the STOP task is sent, the software shall wait until the STOPPED event was received as a response before disabling the peripheral through the ENABLE register.

6.29.8 Registers

Instances

Instance	Base address	Description
TWI0	0x40003000	Two-wire interface master 0 This instance is deprecated.
TWI1	0x40004000	Two-wire interface master 1 This instance is deprecated.

Register overview

Register	Offset	Description
TASKS_STARTRX	0x000	Start TWI receive sequence
TASKS_STARTTX	0x008	Start TWI transmit sequence
TASKS_STOP	0x014	Stop TWI transaction
TASKS_SUSPEND	0x01C	Suspend TWI transaction
TASKS_RESUME	0x020	Resume TWI transaction
EVENTS_STOPPED	0x104	TWI stopped
EVENTS_RXDREADY	0x108	TWI RXD byte received
EVENTS_TXDSENT	0x11C	TWI TXD byte sent
EVENTS_ERROR	0x124	TWI error
EVENTS_BB	0x138	TWI byte boundary, generated before each byte that is sent or received
EVENTS_SUSPENDED	0x148	TWI entered the suspended state
SHORTS	0x200	Shortcuts between local events and tasks
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
ERRORSRC	0x4C4	Error source
ENABLE	0x500	Enable TWI
PSEL.SCL	0x508	Pin select for SCL
PSEL.SDA	0x50C	Pin select for SDA
RXD	0x518	RXD register. Register is cleared on read and the buffer pointer will be modified if read.
TXD	0x51C	TXD register
FREQUENCY	0x524	TWI frequency. Accuracy depends on the HFCLK source selected.
ADDRESS	0x588	Address used in the TWI transfer

6.29.8.1 TASKS_STARTRX

Address offset: 0x000

Start TWI receive sequence

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTRX			Start TWI receive sequence
			Trigger	1	Trigger task

6.29.8.2 TASKS_STARTTX

Address offset: 0x008

Start TWI transmit sequence

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTTX			Start TWI transmit sequence
			Trigger	1	Trigger task

6.29.8.3 TASKS_STOP

Address offset: 0x014

Stop TWI transaction

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STOP			Stop TWI transaction
			Trigger	1	Trigger task

6.29.8.4 TASKS_SUSPEND

Address offset: 0x01C

Suspend TWI transaction

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_SUSPEND			Suspend TWI transaction
			Trigger	1	Trigger task

6.29.8.5 TASKS_RESUME

Address offset: 0x020

Resume TWI transaction

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_RESUME			Resume TWI transaction
			Trigger	1	Trigger task

6.29.8.6 EVENTS_STOPPED

Address offset: 0x104

TWI stopped

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_STOPPED			TWI stopped
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.29.8.7 EVENTS_RXDREADY

Address offset: 0x108

TWI RXD byte received

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_RXDREADY			TWI RXD byte received
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.29.8.8 EVENTS_TXDSENT

Address offset: 0x11C

TWI TXD byte sent

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TXDSENT			TWI TXD byte sent
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.29.8.9 EVENTS_ERROR

Address offset: 0x124

TWI error

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ERROR			TWI error
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.29.8.10 EVENTS_BB

Address offset: 0x138

TWI byte boundary, generated before each byte that is sent or received

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_BB			TWI byte boundary, generated before each byte that is sent or received
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.29.8.11 EVENTS_SUSPENDED

Address offset: 0x148

TWI entered the suspended state

Generated just after ACK bit has been transferred in a read transaction, and only if SUSPEND has been requested earlier.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_SUSPENDED			TWI entered the suspended state Generated just after ACK bit has been transferred in a read transaction, and only if SUSPEND has been requested earlier.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.29.8.12 SHORTS

Address offset: 0x200

Shortcuts between local events and tasks

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	BB_SUSPEND			Shortcut between event BB and task SUSPEND
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
B	RW	BB_STOP			Shortcut between event BB and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

6.29.8.13 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	STOPPED			Write '1' to enable interrupt for event STOPPED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	RXDREADY			Write '1' to enable interrupt for event RXDREADY
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	TXDSENT			Write '1' to enable interrupt for event TXDSENT
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	ERROR			Write '1' to enable interrupt for event ERROR
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	BB			Write '1' to enable interrupt for event BB
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	SUSPENDED			Write '1' to enable interrupt for event SUSPENDED Generated just after ACK bit has been transferred in a read transaction, and only if SUSPEND has been requested earlier.
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.29.8.14 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	STOPPED			Write '1' to disable interrupt for event STOPPED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	RXDREADY			Write '1' to disable interrupt for event RXDREADY
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	TXDSENT			Write '1' to disable interrupt for event TXDSENT
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	ERROR			Write '1' to disable interrupt for event ERROR
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	BB			Write '1' to disable interrupt for event BB
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	SUSPENDED			Write '1' to disable interrupt for event SUSPENDED Generated just after ACK bit has been transferred in a read transaction, and only if SUSPEND has been requested earlier.
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.29.8.15 ERRORSRC

Address offset: 0x4C4

Error source

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW W1C	OVERRUN			Overrun error A new byte was received before previous byte got read by software from the RXD register. (Previous data is lost)
			NotPresent	0	Read: no overrun occured
			Present	1	Read: overrun occured
B	RW W1C	ANACK			NACK received after sending the address (write '1' to clear)
			NotPresent	0	Read: error not present
			Present	1	Read: error present
C	RW W1C	DNACK			NACK received after sending a data byte (write '1' to clear)
			NotPresent	0	Read: error not present
			Present	1	Read: error present

6.29.8.16 ENABLE

Address offset: 0x500

Enable TWI

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ENABLE			Enable or disable TWI
			Disabled	0	Disable TWI
			Enabled	5	Enable TWI

6.29.8.17 PSEL.SCL

Address offset: 0x508

Pin select for SCL

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**

ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.29.8.18 PSEL.SDA

Address offset: 0x50C

Pin select for SDA

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**

ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.29.8.19 RXD

Address offset: 0x518

RXD register. Register is cleared on read and the buffer pointer will be modified if read.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	RXD			RXD register
	RME				

6.29.8.20 TXD

Address offset: 0x51C

TXD register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	TXD			TXD register

6.29.8.21 FREQUENCY

Address offset: 0x524

TWI frequency. Accuracy depends on the HFCLK source selected.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x04000000				**0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	FREQUENCY			TWI master clock frequency
			K100	0x01980000	100 kbps
			K250	0x04000000	250 kbps
			K400	0x06680000	400 kbps (actual rate 410.256 kbps)

6.29.8.22 ADDRESS

Address offset: 0x588

Address used in the TWI transfer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ADDRESS			Address used in the TWI transfer

6.29.9 Electrical specification

6.29.9.1 TWI interface electrical specifications

Symbol	Description	Min.	Typ.	Max.	Units
$f_{TWI,SCL}$	Bit rates for TWI[39]	100		400	kbps
$t_{TWI,START}$	Time from STARTRX/STARTTX task to transmission started		1.5		µs

6.29.9.2 Two Wire Interface (TWI) timing specifications

Symbol	Description	Min.	Typ.	Max.	Units
t_{TWI,SU_DAT}	Data setup time before positive edge on SCL – all modes	300			ns
t_{TWI,HD_DAT}	Data hold time after negative edge on SCL – all modes	500			ns
$t_{TWI,HD_STA,100kbps}$	TWI master hold time for START and repeated START condition, 100 kbps	10000			ns
$t_{TWI,HD_STA,250kbps}$	TWI master hold time for START and repeated START condition, 250kbps	4000			ns
$t_{TWI,HD_STA,400kbps}$	TWI master hold time for START and repeated START condition, 400 kbps	2500			ns
$t_{TWI,SU_STO,100kbps}$	TWI master setup time from SCL high to STOP condition, 100 kbps	5000			ns
$t_{TWI,SU_STO,250kbps}$	TWI master setup time from SCL high to STOP condition, 250 kbps	2000			ns
$t_{TWI,SU_STO,400kbps}$	TWI master setup time from SCL high to STOP condition, 400 kbps	1250			ns
$t_{TWI,BUF,100kbps}$	TWI master bus free time between STOP and START conditions, 100 kbps	5800			ns
$t_{TWI,BUF,250kbps}$	TWI master bus free time between STOP and START conditions, 250 kbps	2700			ns
$t_{TWI,BUF,400kbps}$	TWI master bus free time between STOP and START conditions, 400 kbps	2100			ns

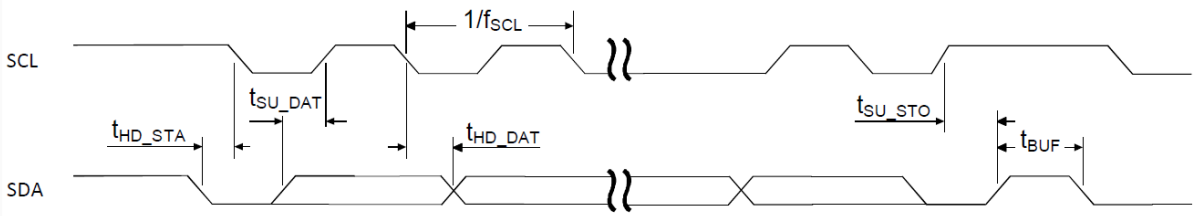


Figure 176: TWI timing diagram, 1 byte transaction

6.30 TIMER — Timer/counter

This peripheral is a general purpose timer designed to keep track of time in user-selective time intervals, it can operate in two modes: timer and counter.

[39] High bit rates or stronger pull-ups may require GPIOs to be set as High Drive, see GPIO chapter for more details.

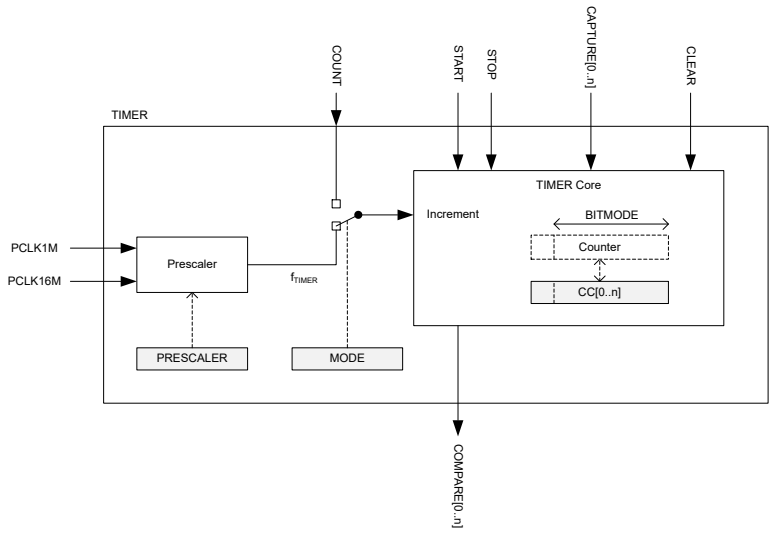


Figure 177: Block schematic for timer/counter

The timer/counter runs on the high-frequency clock source (HFCLK) and includes a four-bit (1/2X) prescaler that can divide the timer input clock from the HFCLK controller. Clock source selection between PCLK16M and PCLK1M is automatic according to TIMER base frequency set by the prescaler. The TIMER base frequency is always given as 16 MHz divided by the prescaler value.

The PPI system allows a TIMER event to trigger a task of any other system peripheral of the device. The PPI system also enables the TIMER task/event features to generate periodic output and PWM signals to any GPIO. The number of input/outputs used at the same time is limited by the number of GPIOTE channels.

TIMER can operate in two modes: Timer mode and Counter mode. In both modes, TIMER is started by triggering the START task, and stopped by triggering the STOP task. After the timer is stopped the timer can resume timing/counting by triggering the START task again. When timing/counting is resumed, the timer will continue from the value it had prior to being stopped.

In Timer mode, the TIMER's internal Counter register is incremented by one for every tick of the timer frequency f_{TIMER} as illustrated in Block schematic for timer/counter on page 774. The timer frequency is derived from PCLK16M as shown in the following example, using the values specified in the PRESCALER register.

$$f_{TIMER} = 16\ MHz\ /\ (2^{PRESCALER})$$

When $f_{TIMER} \leq 1$ MHz, TIMER will use PCLK1M instead of PCLK16M for reduced power consumption.

In counter mode, the TIMER's internal Counter register is incremented by one each time the COUNT task is triggered, meaning the timer frequency and the prescaler are not utilized in counter mode. Similarly, the COUNT task has no effect in Timer mode.

The TIMER's maximum value is configured by changing the bit-width of the timer in register BITMODE on page 783.

PRESCALER on page 783 and BITMODE on page 783 must only be updated when the timer is stopped. If these registers are updated while the timer is started, unpredictable behavior may occur.

When the timer is incremented beyond its maximum value, the Counter register will overflow and the timer will automatically start over from zero.

The Counter register can be cleared by triggering the CLEAR task. This will explicitly set the internal value to zero.

TIMER implements multiple capture/compare registers.

Independent of prescaler setting, the accuracy of TIMER is equivalent to one tick of the timer frequency f_{TIMER} as illustrated in Block schematic for timer/counter on page 774.

6.30.1 Capture

TIMER implements one capture task for every available capture/compare register.

Every time the CAPTURE[n] task is triggered, the Counter value is copied to the CC[n] register.

6.30.2 Compare

TIMER implements one COMPARE event for every available capture/compare register.

A COMPARE event is generated when the Counter is incremented and then becomes equal to the value specified in one of the capture compare registers. When the Counter value becomes equal to the value specified in a capture compare register CC[n], the corresponding compare event COMPARE[n] is generated.

BITMODE on page 783 specifies how many bits of the Counter register and the capture/compare register that are used when the comparison is performed. Other bits will be ignored.

6.30.3 Task delays

After TIMER is started, the CLEAR, COUNT, and STOP tasks are guaranteed to take effect within one clock cycle of the PCLK16M.

6.30.4 Task priority

If the START task and the STOP task are triggered at the same time, meaning within the same period of PCLK16M, the STOP task will be prioritized.

6.30.5 Registers

Instances

Instance	Base address	Description
TIMER0	0x40008000	Timer 0
TIMER1	0x40009000	Timer 1
TIMER2	0x4000A000	Timer 2
TIMER3	0x4001A000	Timer 3
TIMER4	0x4001B000	Timer 4

Configuration

Instance	Configuration
TIMER0	This timer instance has 4 CC registers (CC[0..3])
TIMER1	This timer instance has 4 CC registers (CC[0..3])
TIMER2	This timer instance has 4 CC registers (CC[0..3])
TIMER3	This timer instance has 6 CC registers (CC[0..5])
TIMER4	This timer instance has 6 CC registers (CC[0..5])

Register overview

Register	Offset	Description
TASKS_START	0x000	Start Timer
TASKS_STOP	0x004	Stop Timer
TASKS_COUNT	0x008	Increment Timer (Counter mode only)
TASKS_CLEAR	0x00C	Clear time
TASKS_SHUTDOWN	0x010	Shut down timer This register is deprecated.
TASKS_CAPTURE[0]	0x040	Capture Timer value to CC[0] register
TASKS_CAPTURE[1]	0x044	Capture Timer value to CC[1] register
TASKS_CAPTURE[2]	0x048	Capture Timer value to CC[2] register
TASKS_CAPTURE[3]	0x04C	Capture Timer value to CC[3] register
TASKS_CAPTURE[4]	0x050	Capture Timer value to CC[4] register
TASKS_CAPTURE[5]	0x054	Capture Timer value to CC[5] register
EVENTS_COMPARE[0]	0x140	Compare event on CC[0] match
EVENTS_COMPARE[1]	0x144	Compare event on CC[1] match
EVENTS_COMPARE[2]	0x148	Compare event on CC[2] match
EVENTS_COMPARE[3]	0x14C	Compare event on CC[3] match
EVENTS_COMPARE[4]	0x150	Compare event on CC[4] match
EVENTS_COMPARE[5]	0x154	Compare event on CC[5] match
SHORTS	0x200	Shortcuts between local events and tasks
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
MODE	0x504	Timer mode selection
BITMODE	0x508	Configure the number of bits used by the TIMER
PRESCALER	0x510	Timer prescaler register
CC[0]	0x540	Capture/Compare register 0
CC[1]	0x544	Capture/Compare register 1
CC[2]	0x548	Capture/Compare register 2
CC[3]	0x54C	Capture/Compare register 3
CC[4]	0x550	Capture/Compare register 4
CC[5]	0x554	Capture/Compare register 5

6.30.5.1 TASKS_START

Address offset: 0x000

Start Timer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_START			Start Timer
			Trigger	1	Trigger task

6.30.5.2 TASKS_STOP

Address offset: 0x004

Stop Timer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STOP			Stop Timer
			Trigger	1	Trigger task

6.30.5.3 TASKS_COUNT

Address offset: 0x008

Increment Timer (Counter mode only)

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_COUNT			Increment Timer (Counter mode only)
			Trigger	1	Trigger task

6.30.5.4 TASKS_CLEAR

Address offset: 0x00C

Clear time

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CLEAR			Clear time
			Trigger	1	Trigger task

6.30.5.5 TASKS_SHUTDOWN (Deprecated)

Address offset: 0x010

Shut down timer

This register is deprecated.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_SHUTDOWN			Shut down timer This field is deprecated.
			Trigger	1	Trigger task

6.30.5.6 TASKS_CAPTURE[0]

Address offset: 0x040

Capture Timer value to CC[0] register

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CAPTURE			Capture Timer value to CC[0] register
			Trigger	1	Trigger task

6.30.5.7 TASKS_CAPTURE[1]

Address offset: 0x044

Capture Timer value to CC[1] register

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CAPTURE			Capture Timer value to CC[1] register
			Trigger	1	Trigger task

6.30.5.8 TASKS_CAPTURE[2]

Address offset: 0x048

Capture Timer value to CC[2] register

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CAPTURE			Capture Timer value to CC[2] register
			Trigger	1	Trigger task

6.30.5.9 TASKS_CAPTURE[3]

Address offset: 0x04C

Capture Timer value to CC[3] register

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CAPTURE			Capture Timer value to CC[3] register
			Trigger	1	Trigger task

6.30.5.10 TASKS_CAPTURE[4]

Address offset: 0x050

Capture Timer value to CC[4] register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CAPTURE			Capture Timer value to CC[4] register
			Trigger	1	Trigger task

6.30.5.11 TASKS_CAPTURE[5]

Address offset: 0x054

Capture Timer value to CC[5] register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_CAPTURE			Capture Timer value to CC[5] register
			Trigger	1	Trigger task

6.30.5.12 EVENTS_COMPARE[0]

Address offset: 0x140

Compare event on CC[0] match

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_COMPARE			Compare event on CC[0] match
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.30.5.13 EVENTS_COMPARE[1]

Address offset: 0x144

Compare event on CC[1] match

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_COMPARE			Compare event on CC[1] match
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.30.5.14 EVENTS_COMPARE[2]

Address offset: 0x148

Compare event on CC[2] match

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_COMPARE			Compare event on CC[2] match
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.30.5.15 EVENTS_COMPARE[3]

Address offset: 0x14C

Compare event on CC[3] match

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_COMPARE			Compare event on CC[3] match
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.30.5.16 EVENTS_COMPARE[4]

Address offset: 0x150

Compare event on CC[4] match

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_COMPARE			Compare event on CC[4] match
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.30.5.17 EVENTS_COMPARE[5]

Address offset: 0x154

Compare event on CC[5] match

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_COMPARE			Compare event on CC[5] match
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.30.5.18 SHORTS

Address offset: 0x200

Shortcuts between local events and tasks

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	COMPARE0_CLEAR			Shortcut between event COMPARE[0] and task CLEAR
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
B	RW	COMPARE1_CLEAR			Shortcut between event COMPARE[1] and task CLEAR
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
C	RW	COMPARE2_CLEAR			Shortcut between event COMPARE[2] and task CLEAR
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
D	RW	COMPARE3_CLEAR			Shortcut between event COMPARE[3] and task CLEAR
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
E	RW	COMPARE4_CLEAR			Shortcut between event COMPARE[4] and task CLEAR
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
F	RW	COMPARE5_CLEAR			Shortcut between event COMPARE[5] and task CLEAR
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
G	RW	COMPARE0_STOP			Shortcut between event COMPARE[0] and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
H	RW	COMPARE1_STOP			Shortcut between event COMPARE[1] and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
I	RW	COMPARE2_STOP			Shortcut between event COMPARE[2] and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
J	RW	COMPARE3_STOP			Shortcut between event COMPARE[3] and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
K	RW	COMPARE4_STOP			Shortcut between event COMPARE[4] and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
L	RW	COMPARE5_STOP			Shortcut between event COMPARE[5] and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

6.30.5.19 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	COMPARE[0]			Write '1' to enable interrupt for event COMPARE[0]
			Set	1	Enable
			Disabled	0	Read: Disabled

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Enabled	1	Read: Enabled
B	RW	COMPARE[1]			Write '1' to enable interrupt for event COMPARE[1]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	COMPARE[2]			Write '1' to enable interrupt for event COMPARE[2]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	COMPARE[3]			Write '1' to enable interrupt for event COMPARE[3]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	COMPARE[4]			Write '1' to enable interrupt for event COMPARE[4]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	COMPARE[5]			Write '1' to enable interrupt for event COMPARE[5]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.30.5.20 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	COMPARE[0]			Write '1' to disable interrupt for event COMPARE[0]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	COMPARE[1]			Write '1' to disable interrupt for event COMPARE[1]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	COMPARE[2]			Write '1' to disable interrupt for event COMPARE[2]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	COMPARE[3]			Write '1' to disable interrupt for event COMPARE[3]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	COMPARE[4]			Write '1' to disable interrupt for event COMPARE[4]
			Clear	1	Disable

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	COMPARE[5]			Write '1' to disable interrupt for event COMPARE[5]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.30.5.21 MODE

Address offset: 0x504

Timer mode selection

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	MODE			Timer mode
			Timer	0	Select Timer mode
			Counter	1	Select Counter mode
					This enumerator is deprecated.
			LowPowerCounter	2	Select Low Power Counter mode

6.30.5.22 BITMODE

Address offset: 0x508

Configure the number of bits used by the TIMER

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	BITMODE			Timer bit width
			16Bit	0	16 bit timer bit width
			08Bit	1	8 bit timer bit width
			24Bit	2	24 bit timer bit width
			32Bit	3	32 bit timer bit width

6.30.5.23 PRESCALER

Address offset: 0x510

Timer prescaler register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A
Reset 0x00000004				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PRESCALER		[0..9]	Prescaler value

6.30.5.24 CC[0]

Address offset: 0x540

Capture/Compare register 0

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	CC			Capture/Compare value
					Only the number of bits indicated by BITMODE will be used by the TIMER.

6.30.5.25 CC[1]

Address offset: 0x544

Capture/Compare register 1

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	CC			Capture/Compare value
					Only the number of bits indicated by BITMODE will be used by the TIMER.

6.30.5.26 CC[2]

Address offset: 0x548

Capture/Compare register 2

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	CC			Capture/Compare value
					Only the number of bits indicated by BITMODE will be used by the TIMER.

6.30.5.27 CC[3]

Address offset: 0x54C

Capture/Compare register 3

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	CC			Capture/Compare value
					Only the number of bits indicated by BITMODE will be used by the TIMER.

6.30.5.28 CC[4]

Address offset: 0x550

Capture/Compare register 4

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value	Description																															
A	RW	CC			Capture/Compare value Only the number of bits indicated by BITMODE will be used by the TIMER.																																	

6.30.5.29 CC[5]

Address offset: 0x554

Capture/Compare register 5

Bit number					31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID					A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000					**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value	Description																															
A	RW	CC			Capture/Compare value Only the number of bits indicated by BITMODE will be used by the TIMER.																																	

6.31 TWIM — I²C compatible two-wire interface master with EasyDMA

TWI master with EasyDMA (TWIM) is a two-wire half-duplex master which can communicate with multiple slave devices connected to the same bus.

Listed here are the main features for TWIM:

- I²C compatible
- Supported baud rates: 100, 250, 400 kbps
- Support for clock stretching (non I²C compliant)
- EasyDMA

The two-wire interface can communicate with a bi-directional wired-AND bus with two lines (SCL, SDA). The protocol makes it possible to interconnect up to 127 individually addressable devices. TWIM is not compatible with CBUS.

The GPIOs used for each two-wire interface line can be chosen from any GPIO on the device and are independently configurable. This enables great flexibility in device pinout and efficient use of board space and signal routing.

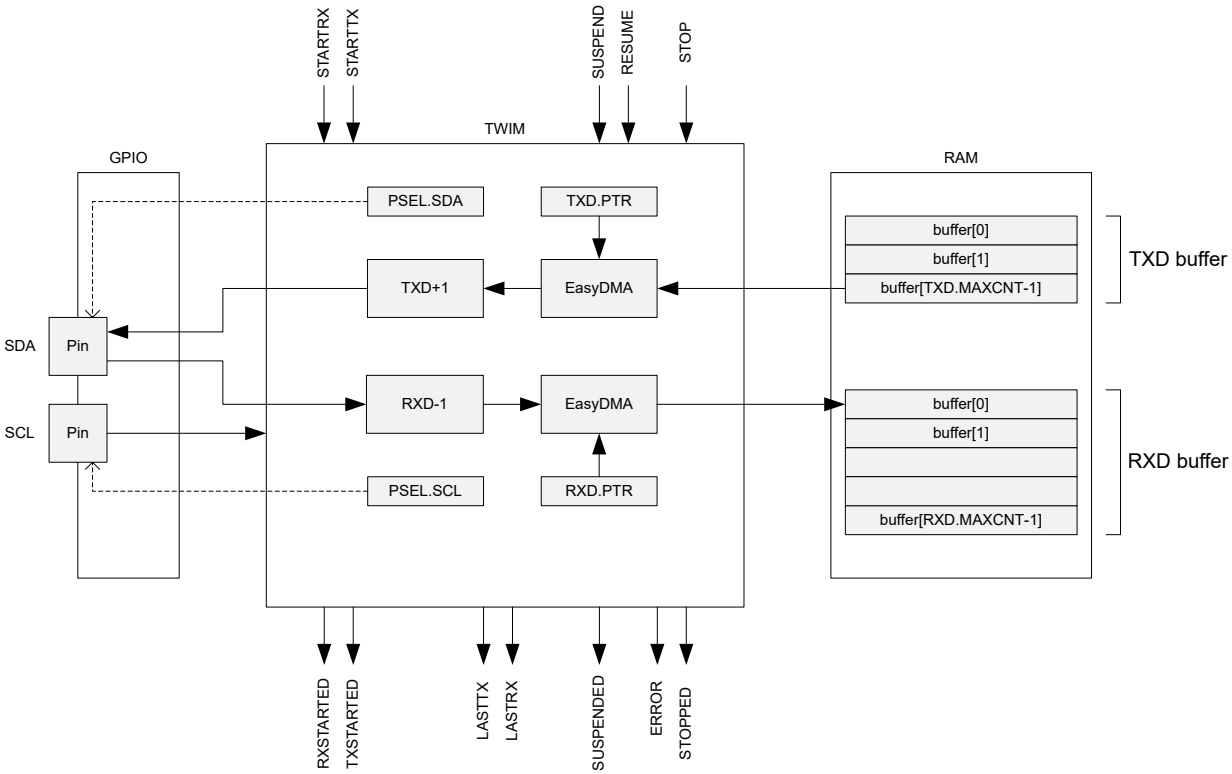


Figure 178: TWI master with EasyDMA

A typical TWI setup consists of one master and one or more slaves. For an example, see the following figure. This TWIM is only able to operate as a single master on the TWI bus. Multi-master bus configuration is not supported.

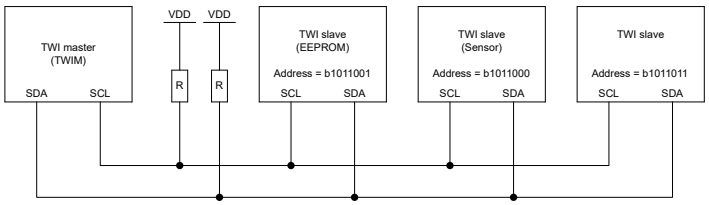


Figure 179: A typical TWI setup comprising one master and three slaves

This TWI master supports clock stretching performed by the slaves. The SCK pulse following a stretched clock cycle may be shorter than specified by the I2C specification.

The TWI master is started by triggering the STARTTX or STARTRX tasks, and stopped by triggering the STOP task. The TWI master will generate a STOPPED event when it has stopped following a STOP task.

After the TWI master is started, the STARTTX or STARTRX tasks should not be triggered again until the TWI master has issued a LASTRX, LASTTX, or STOPPED event.

The TWI master can be suspended using the SUSPEND task, this can be used when using the TWI master in a low priority interrupt context. When the TWIM enters suspend state, will automatically issue a SUSPENDED event while performing a continuous clock stretching until it is instructed to resume operation via a RESUME task. The TWI master cannot be stopped while it is suspended, thus the STOP task has to be issued after the TWI master has been resumed.

Note: Any ongoing byte transfer will be allowed to complete before the suspend is enforced. A SUSPEND task has no effect unless the TWI master is actively involved in a transfer.

If a NACK is clocked in from the slave, the TWI master will generate an ERROR event.

6.31.1 EasyDMA

The TWIM implements EasyDMA for accessing RAM without CPU involvement.

The TWIM peripheral implements the EasyDMA channels found in the following table.

Channel	Type	Register Cluster
TXD	READER	TXD
RXD	WRITER	RXD

Table 51: TWIM EasyDMA Channels

For detailed information regarding the use of EasyDMA, see EasyDMA on page 63.

The .PTR and .MAXCNT registers are double-buffered. They can be updated and prepared for the next RX/TX transmission immediately after having received the RXSTARTED/TXSTARTED event.

The STOPPED event indicates that EasyDMA has finished accessing the buffer in RAM.

6.31.2 Master write sequence

A TWI master write sequence is started by triggering the STARTTX task. After the STARTTX task has been triggered, the TWI master will generate a start condition on the TWI bus, followed by clocking out the address and the READ/WRITE bit set to 0 (WRITE=0, READ=1).

The address must match the address of the slave device that the master wants to write to. The READ/WRITE bit is followed by an ACK/NACK bit (ACK=0 or NACK=1) generated by the slave.

After receiving the ACK bit, the TWI master will clock out the data bytes found in the transmit buffer located in RAM at the address specified in the TXD.PTR register. Each byte clocked out from the master will be followed by an ACK/NACK bit clocked in from the slave.

A typical TWI master write sequence is shown in the following figure. Occurrence 2 in the figure illustrates clock stretching performed by the TWI master following a SUSPEND task.

A SUSPENDED event indicates that the SUSPEND task has taken effect. This event can be used to synchronize the software.

The TWI master will generate a LASTTX event when it starts to transmit the last byte, this is shown in the following figure.

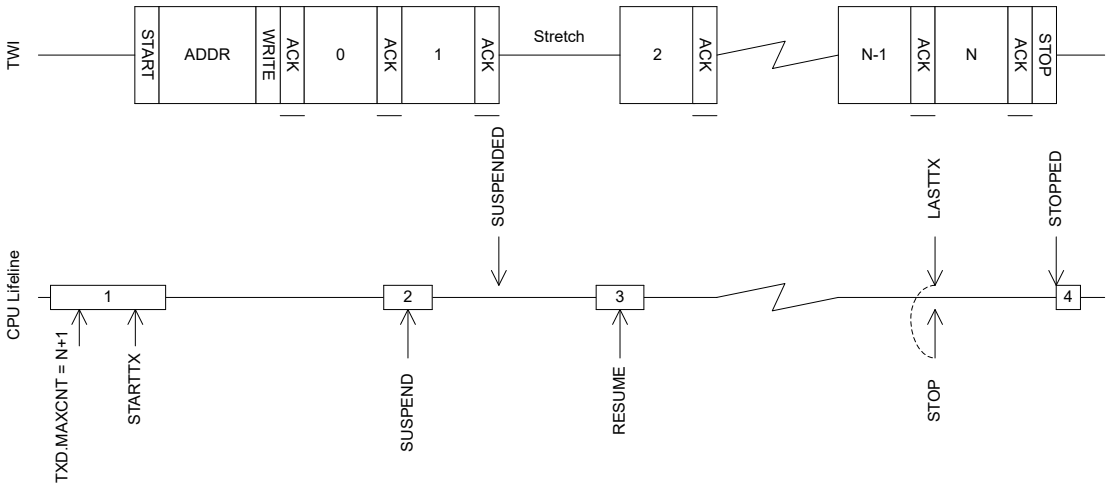


Figure 180: TWI master writing data to a slave

The TWI master is stopped by triggering the STOP task. This task should be triggered during the transmission of the last byte to secure that the TWI master will stop as fast as possible after sending the last byte. The shortcut between LASTTX and STOP can alternatively be used to accomplish this.

Note: The TWI master does not stop by itself when the entire RAM buffer has been sent, or when an error occurs. The STOP task must be issued, through the use of a local or PPI shortcut, or in software as part of the error handler.

6.31.3 Master read sequence

A TWI master read sequence is started by triggering the STARTRX task. After the STARTRX task has been triggered, the TWI master will generate a start condition on the TWI bus, followed by clocking out the address and the READ/WRITE bit set to 1 (WRITE = 0, READ = 1). The address must match the address of the slave device that the master wants to read from. The READ/WRITE bit is followed by an ACK/NACK bit (ACK=0 or NACK = 1) generated by the slave.

After sending the ACK bit, the TWI slave will send data to the master using the clock generated by the master.

Data received will be stored in RAM at the address specified in the RXD.PTR register. The TWI master will generate an ACK after all but the last byte have been received from the slave. The TWI master will generate a NACK after the last byte received to indicate that the read sequence shall stop.

A typical TWI master read sequence is illustrated in The TWI master reading data from a slave on page 789. Occurrence 2 in the figure illustrates clock stretching performed by the TWI master following a SUSPEND task.

A SUSPENDED event indicates that the SUSPEND task has taken effect. This event can be used to synchronize the software.

The TWI master will generate a LASTRX event when it is ready to receive the last byte, as shown in The TWI master reading data from a slave on page 789. If RXD.MAXCNT > 1, the LASTRX event is generated after sending the ACK of the previously received byte. If RXD.MAXCNT = 1, the LASTRX event is generated after receiving the ACK following the address and READ bit.

The TWI master is stopped by triggering the STOP task. This task must be triggered before the NACK bit is supposed to be transmitted. The STOP task can be triggered at any time during the reception of the last byte. It is recommended to use the shortcut between LASTRX and STOP to accomplish this.

The TWI master does not stop by itself when the RAM buffer is full, or when an error occurs. The STOP task must be issued, through the use of a local or PPI shortcut, or in software as part of the error handler.

The TWI master cannot be stopped while suspended, so the STOP task must be issued after the TWI master has been resumed.

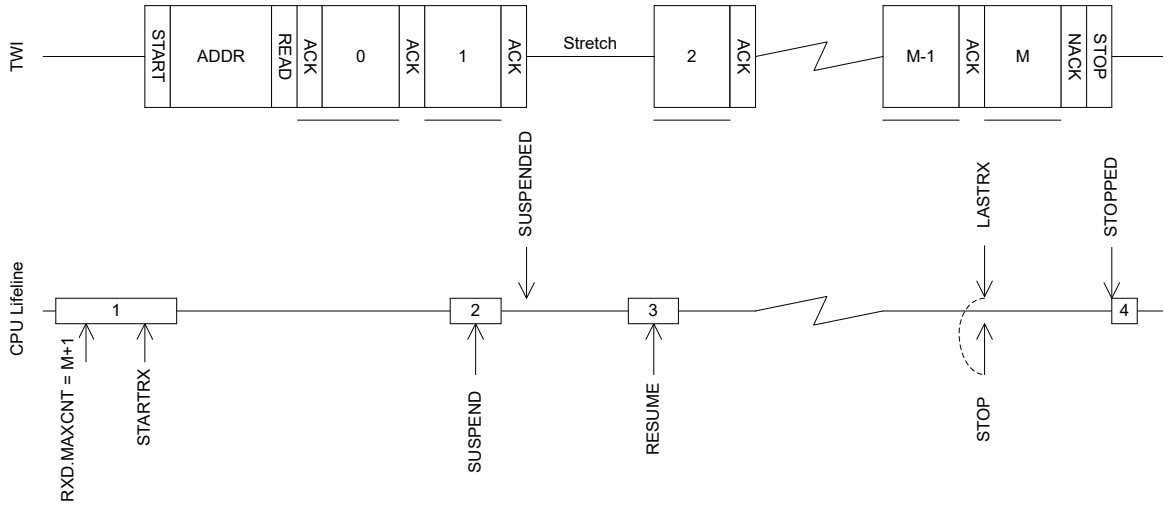


Figure 181: The TWI master reading data from a slave

6.31.4 Master repeated start sequence

A typical repeated start sequence is one in which the TWI master writes two bytes to the slave followed by reading four bytes from the slave. This example uses shortcuts to perform the simplest type of repeated start sequence, i.e. one write followed by one read. The same approach can be used to perform a repeated start sequence where the sequence is read followed by write.

The following figure shows an example of a repeated start sequence where the TWI master writes two bytes followed by reading four bytes from the slave.

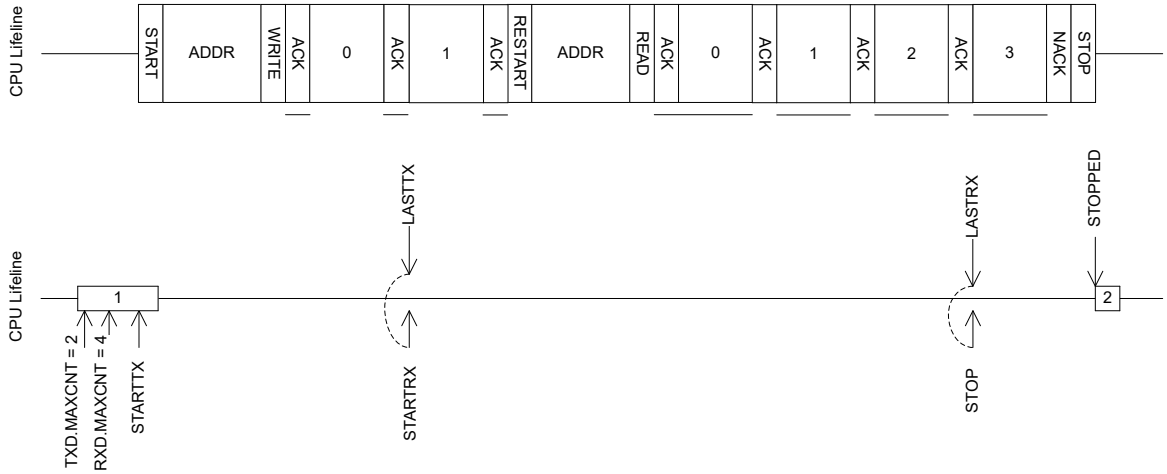


Figure 182: Master repeated start sequence

If a more complex repeated start sequence is needed, and the TWI firmware drive is serviced in a low priority interrupt, it may be necessary to use the SUSPEND task and SUSPENDED event to guarantee that the correct tasks are generated at the correct time. A double repeated start sequence using the SUSPEND task to secure safe operation in low priority interrupts is shown in the following figure.

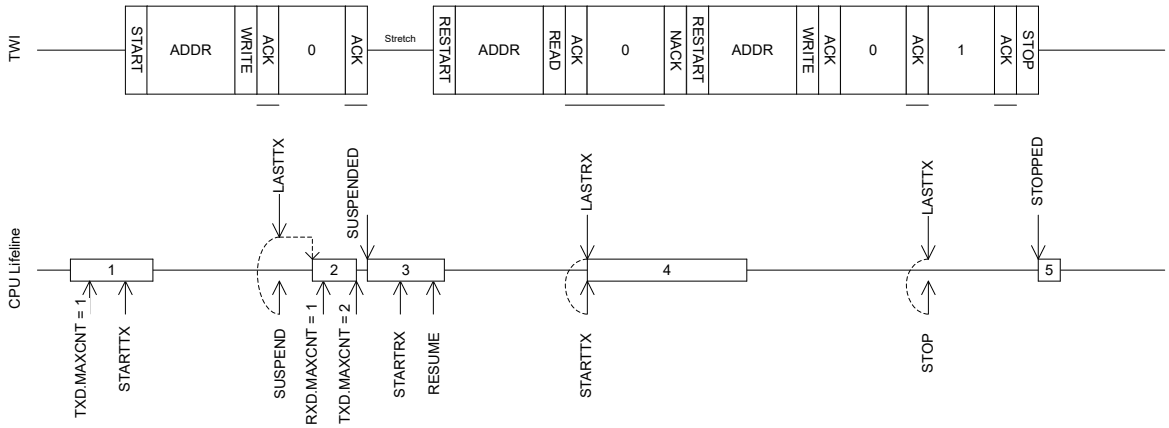


Figure 183: Double repeated start sequence

6.31.5 Low power

When putting the system in low power and the peripheral is not needed, lowest possible power consumption is achieved by stopping, and then disabling the peripheral.

When the STOP task is sent, the software shall wait until the STOPPED event is received as a response before disabling the peripheral through the ENABLE register. If the peripheral is already stopped, the STOP task is not required.

6.31.6 Master mode pin configuration

The SCL and SDA signals associated with the TWI master are mapped to physical pins according to the configuration specified in the PSEL.SCL and PSEL.SDA registers respectively.

The PSEL.SCL and PSEL.SDA registers and their configurations are only used as long as the TWI master is enabled, and retained only as long as the device is in ON mode. When the peripheral is disabled, the pins will behave as regular GPIOs, and use the configuration in their respective OUT bit field and PIN_CNF[n] register. PSEL.SCL, PSEL.SDA must only be configured when the TWI master is disabled.

To secure correct signal levels on the pins used by the TWI master when the system is in OFF mode, and when the TWI master is disabled, these pins must be configured in the GPIO peripheral as described in the following table.

Only one peripheral can be assigned to drive a particular GPIO pin at a time. Failing to do so may result in unpredictable behavior.

TWI master signal	TWI master pin	Direction	Output value	Drive strength
SCL	As specified in PSEL.SCL	Input	Not applicable	S0D1
SDA	As specified in PSEL.SDA	Input	Not applicable	S0D1

Table 52: GPIO configuration before enabling peripheral

6.31.7 Registers

Instances

Instance	Base address	Description
TWIM0	0x40003000	Two-wire interface master 0
TWIM1	0x40004000	Two-wire interface master 1

Register overview

Register	Offset	Description
TASKS_STARTRX	0x000	Start TWI receive sequence
TASKS_STARTTX	0x008	Start TWI transmit sequence
TASKS_STOP	0x014	Stop TWI transaction. Must be issued while the TWI master is not suspended.
TASKS_SUSPEND	0x01C	Suspend TWI transaction
TASKS_RESUME	0x020	Resume TWI transaction
EVENTS_STOPPED	0x104	TWI stopped
EVENTS_ERROR	0x124	TWI error
EVENTS_SUSPENDED	0x148	SUSPEND task has been issued, TWI traffic is now suspended.
EVENTS_RXSTARTED	0x14C	Receive sequence started
EVENTS_TXSTARTED	0x150	Transmit sequence started
EVENTS_LASTRX	0x15C	Byte boundary, starting to receive the last byte
EVENTS_LASTTX	0x160	Byte boundary, starting to transmit the last byte
SHORTS	0x200	Shortcuts between local events and tasks
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
ERRORSRC	0x4C4	Error source
ENABLE	0x500	Enable TWIM
PSEL.SCL	0x508	Pin select for SCL signal
PSEL.SDA	0x50C	Pin select for SDA signal
FREQUENCY	0x524	TWI frequency. Accuracy depends on the HFCLK source selected.
RXD.PTR	0x534	Data pointer
RXD.MAXCNT	0x538	Maximum number of bytes in receive buffer
RXD.AMOUNT	0x53C	Number of bytes transferred in the last transaction
RXD.LIST	0x540	EasyDMA list type
TXD.PTR	0x544	Data pointer
TXD.MAXCNT	0x548	Maximum number of bytes in transmit buffer
TXD.AMOUNT	0x54C	Number of bytes transferred in the last transaction
TXD.LIST	0x550	EasyDMA list type
ADDRESS	0x588	Address used in the TWI transfer

6.31.7.1 TASKS_STARTRX

Address offset: 0x000

Start TWI receive sequence

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTRX			Start TWI receive sequence
			Trigger	1	Trigger task

6.31.7.2 TASKS_STARTTX

Address offset: 0x008

Start TWI transmit sequence

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTTX			Start TWI transmit sequence
			Trigger	1	Trigger task

6.31.7.3 TASKS_STOP

Address offset: 0x014

Stop TWI transaction. Must be issued while the TWI master is not suspended.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STOP			Stop TWI transaction. Must be issued while the TWI master is not suspended.
			Trigger	1	Trigger task

6.31.7.4 TASKS_SUSPEND

Address offset: 0x01C

Suspend TWI transaction

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_SUSPEND			Suspend TWI transaction
			Trigger	1	Trigger task

6.31.7.5 TASKS_RESUME

Address offset: 0x020

Resume TWI transaction

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_RESUME			Resume TWI transaction
			Trigger	1	Trigger task

6.31.7.6 EVENTS_STOPPED

Address offset: 0x104

TWI stopped

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_STOPPED			TWI stopped
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.31.7.7 EVENTS_ERROR

Address offset: 0x124

TWI error

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ERROR			TWI error
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.31.7.8 EVENTS_SUSPENDED

Address offset: 0x148

SUSPEND task has been issued, TWI traffic is now suspended.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_SUSPENDED			SUSPEND task has been issued, TWI traffic is now suspended.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.31.7.9 EVENTS_RXSTARTED

Address offset: 0x14C

Receive sequence started

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_RXSTARTED			Receive sequence started
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.31.7.10 EVENTS_TXSTARTED

Address offset: 0x150

Transmit sequence started

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TXSTARTED			Transmit sequence started
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.31.7.11 EVENTS_LASTRX

Address offset: 0x15C

Byte boundary, starting to receive the last byte

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_LASTRX			Byte boundary, starting to receive the last byte
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.31.7.12 EVENTS_LASTTX

Address offset: 0x160

Byte boundary, starting to transmit the last byte

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_LASTTX			Byte boundary, starting to transmit the last byte
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.31.7.13 SHORTS

Address offset: 0x200

Shortcuts between local events and tasks

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	LASTTX_STARTRX			Shortcut between event LASTTX and task STARTRX
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
B	RW	LASTTX_SUSPEND			Shortcut between event LASTTX and task SUSPEND
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
C	RW	LASTTX_STOP			Shortcut between event LASTTX and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
D	RW	LASTRX_STARTTX			Shortcut between event LASTRX and task STARTTX
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
E	RW	LASTRX_SUSPEND			Shortcut between event LASTRX and task SUSPEND
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
F	RW	LASTRX_STOP			Shortcut between event LASTRX and task STOP
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

6.31.7.14 INTEN

Address offset: 0x300

Enable or disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				J I H G F D A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	STOPPED			Enable or disable interrupt for event STOPPED
			Disabled	0	Disable
			Enabled	1	Enable
D	RW	ERROR			Enable or disable interrupt for event ERROR
			Disabled	0	Disable
			Enabled	1	Enable
F	RW	SUSPENDED			Enable or disable interrupt for event SUSPENDED
			Disabled	0	Disable
			Enabled	1	Enable
G	RW	RXSTARTED			Enable or disable interrupt for event RXSTARTED
			Disabled	0	Disable
			Enabled	1	Enable
H	RW	TXSTARTED			Enable or disable interrupt for event TXSTARTED
			Disabled	0	Disable
			Enabled	1	Enable
I	RW	LASTRX			Enable or disable interrupt for event LASTRX
			Disabled	0	Disable
			Enabled	1	Enable
J	RW	LASTTX			Enable or disable interrupt for event LASTTX
			Disabled	0	Disable
			Enabled	1	Enable

6.31.7.15 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				J I H G F D A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	STOPPED			Write '1' to enable interrupt for event STOPPED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	ERROR			Write '1' to enable interrupt for event ERROR
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	SUSPENDED			Write '1' to enable interrupt for event SUSPENDED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
G	RW	RXSTARTED			Write '1' to enable interrupt for event RXSTARTED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
H	RW	TXSTARTED			Write '1' to enable interrupt for event TXSTARTED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
I	RW	LASTRX			Write '1' to enable interrupt for event LASTRX
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
J	RW	LASTTX			Write '1' to enable interrupt for event LASTTX
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.31.7.16 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				J I H G F D A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	STOPPED			Write '1' to disable interrupt for event STOPPED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	ERROR			Write '1' to disable interrupt for event ERROR
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	SUSPENDED			Write '1' to disable interrupt for event SUSPENDED
			Clear	1	Disable
			Disabled	0	Read: Disabled

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				J I H G F D A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Enabled	1	Read: Enabled
G	RW	RXSTARTED			Write '1' to disable interrupt for event RXSTARTED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
H	RW	TXSTARTED			Write '1' to disable interrupt for event TXSTARTED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
I	RW	LASTRX			Write '1' to disable interrupt for event LASTRX
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
J	RW	LASTTX			Write '1' to disable interrupt for event LASTTX
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.31.7.17 ERRORSRC

Address offset: 0x4C4

Error source

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW W1C	OVERRUN			Overrun error A new byte was received before previous byte got transferred into RXD buffer. (Previous data is lost)
			NotReceived	0	Error did not occur
			Received	1	Error occurred
B	RW W1C	ANACK			NACK received after sending the address (write '1' to clear)
			NotReceived	0	Error did not occur
			Received	1	Error occurred
C	RW W1C	DNACK			NACK received after sending a data byte (write '1' to clear)
			NotReceived	0	Error did not occur
			Received	1	Error occurred

6.31.7.18 ENABLE

Address offset: 0x500

Enable TWIM

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ENABLE			Enable or disable TWIM
			Disabled	0	Disable TWIM
			Enabled	6	Enable TWIM

6.31.7.19 PSEL.SCL

Address offset: 0x508

Pin select for SCL signal

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.31.7.20 PSEL.SDA

Address offset: 0x50C

Pin select for SDA signal

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.31.7.21 FREQUENCY

Address offset: 0x524

TWI frequency. Accuracy depends on the HFCLK source selected.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x04000000				**0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	FREQUENCY			TWI master clock frequency
			K100	0x01980000	100 kbps
			K250	0x04000000	250 kbps
			K400	0x06400000	400 kbps

6.31.7.22 RXD

RXD EasyDMA channel

6.31.7.22.1 RXD.PTR

Address offset: 0x534

Data pointer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.31.7.22.2 RXD.MAXCNT

Address offset: 0x538

Maximum number of bytes in receive buffer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	MAXCNT		[0..0xFFFF]	Maximum number of bytes in receive buffer

6.31.7.22.3 RXD.AMOUNT

Address offset: 0x53C

Number of bytes transferred in the last transaction

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	AMOUNT		[0..0xFFFF]	Number of bytes transferred in the last transaction. In case of NACK error, includes the NACK'ed byte.

6.31.7.22.4 RXD.LIST

Address offset: 0x540

EasyDMA list type

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	LIST			List type
			Disabled	0	Disable EasyDMA list
			ArrayList	1	Use array list

6.31.7.23 TXD

TXD EasyDMA channel

6.31.7.23.1 TXD.PTR

Address offset: 0x544

Data pointer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.31.7.23.2 TXD.MAXCNT

Address offset: 0x548

Maximum number of bytes in transmit buffer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	MAXCNT		[0..0xFFFF]	Maximum number of bytes in transmit buffer

6.31.7.23.3 TXD.AMOUNT

Address offset: 0x54C

Number of bytes transferred in the last transaction

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	AMOUNT		[0..0xFFFF]	Number of bytes transferred in the last transaction. In case of NACK error, includes the NACK'ed byte.

6.31.7.23.4 TXD.LIST

Address offset: 0x550

EasyDMA list type

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	LIST			List type
			Disabled	0	Disable EasyDMA list
			ArrayList	1	Use array list

6.31.7.24 ADDRESS

Address offset: 0x588

Address used in the TWI transfer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ADDRESS			Address used in the TWI transfer

6.31.8 Electrical specification

6.31.8.1 TWIM interface electrical specifications

Symbol	Description	Min.	Typ.	Max.	Units
$f_{TWIM,SCL}$	Bit rates for TWIM[40]	100		400	kbps
$t_{TWIM,START}$	Time from STARTRX/STARTTX task to transmission started		1.5		µs

6.31.8.2 Two Wire Interface Master (TWIM) timing specifications

Symbol	Description	Min.	Typ.	Max.	Units
t_{TWIM,SU_DAT}	Data setup time before positive edge on SCL – all modes	300			ns
t_{TWIM,HD_DAT}	Data hold time after negative edge on SCL – 100, 250 and 400 kbps	500			ns
$t_{TWIM,HD_STA,100kbps}$	TWIM master hold time for START and repeated START condition, 100 kbps	9937.5			ns
$t_{TWIM,HD_STA,250kbps}$	TWIM master hold time for START and repeated START condition, 250 kbps	3937.5			ns
$t_{TWIM,HD_STA,400kbps}$	TWIM master hold time for START and repeated START condition, 400 kbps	2437.5			ns
$t_{TWIM,SU_STO,100kbps}$	TWIM master setup time from SCL high to STOP condition, 100 kbps	5000			ns
$t_{TWIM,SU_STO,250kbps}$	TWIM master setup time from SCL high to STOP condition, 250 kbps	2000			ns
$t_{TWIM,SU_STO,400kbps}$	TWIM master setup time from SCL high to STOP condition, 400 kbps	1250			ns
$t_{TWIM,BUF,100kbps}$	TWIM master bus free time between STOP and START conditions, 100 kbps	5800			ns
$t_{TWIM,BUF,250kbps}$	TWIM master bus free time between STOP and START conditions, 250 kbps	2700			ns
$t_{TWIM,BUF,400kbps}$	TWIM master bus free time between STOP and START conditions, 400 kbps	2100			ns

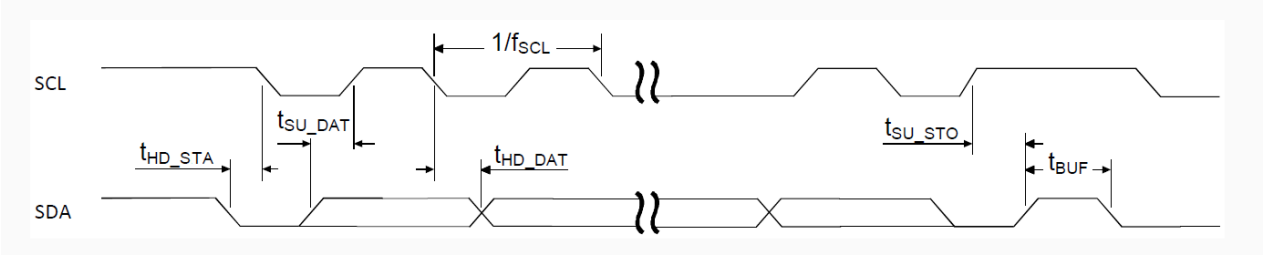


Figure 184: TWIM timing diagram, 1 byte transaction

[40] High bit rates or stronger pull-ups may require GPIOs to be set as High Drive, see GPIO — General purpose input/output on page 322 for more details.

6.31.9 Pullup resistor

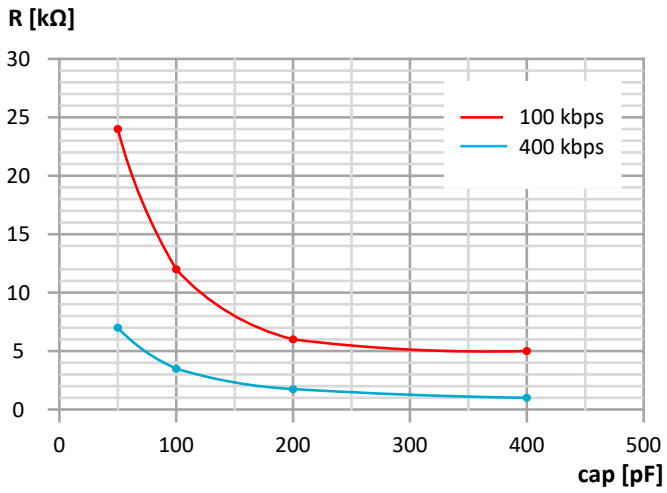


Figure 185: Recommended TWIM pullup value vs. line capacitance

- The I2C specification allows a line capacitance of 400 pF at most.
- The value of internal pullup resistor (R_{PU}) for nRF52840 can be found in GPIO — General purpose input/output on page 322.

6.32 TWIS — I²C compatible two-wire interface slave with EasyDMA

TWI slave with EasyDMA (TWIS) is compatible with I²C operating at 100 kHz and 400 kHz. The TWI transmitter and receiver implement EasyDMA.

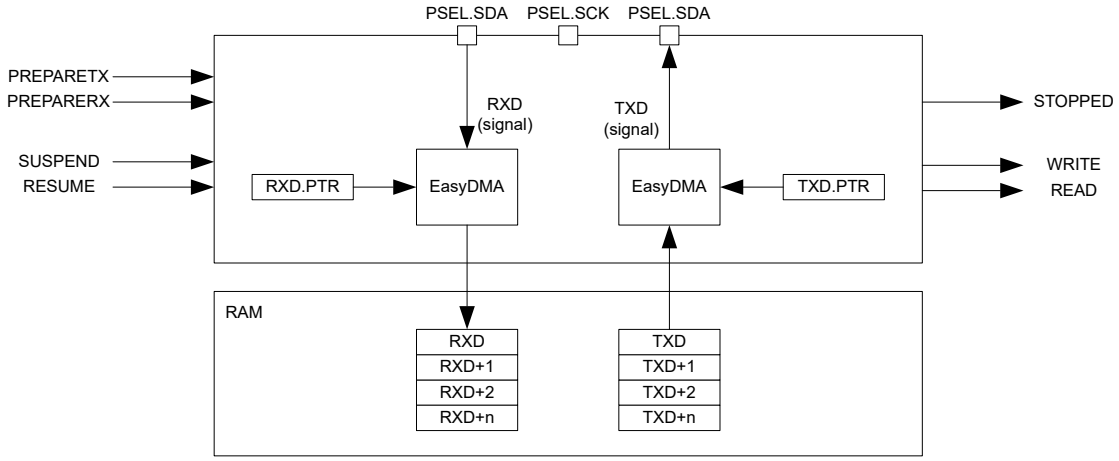


Figure 186: TWI slave with EasyDMA

A typical TWI setup consists of one master and one or more slaves. For an example, see the following figure. TWIS is only able to operate with a single master on the TWI bus.

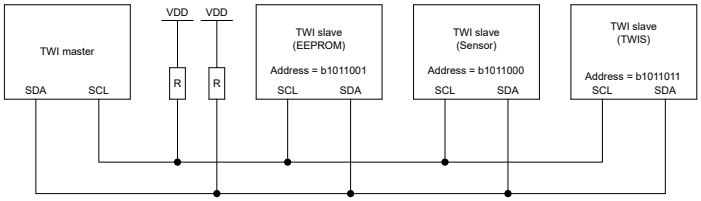


Figure 187: A typical TWI setup comprising one master and three slaves

The following figure shows the TWI slave state machine.

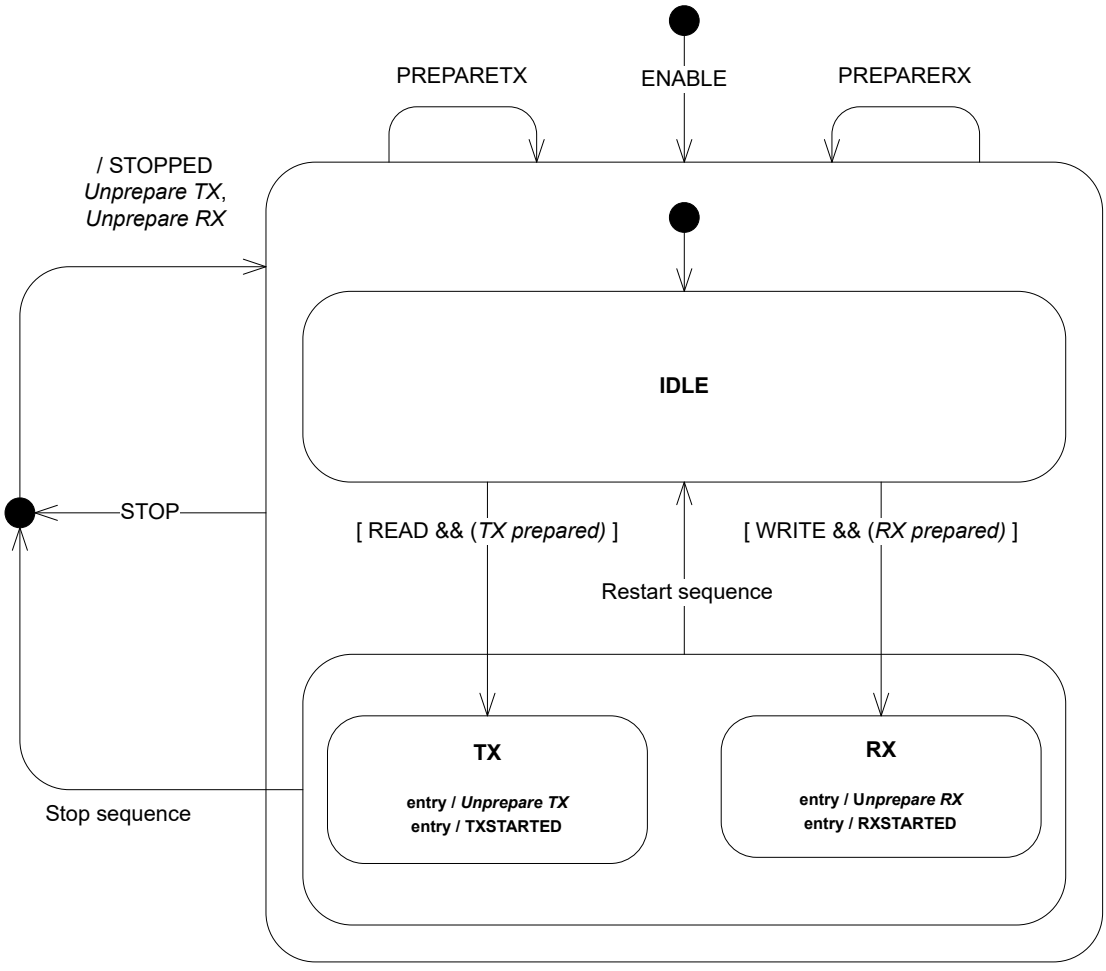


Figure 188: TWI slave state machine

The following table contains descriptions of the symbols used in the state machine.

Symbol	Type	Description
ENABLE	Register	The TWI slave has been enabled via the ENABLE register.
PREPARETX	Task	The TASKS_PREPARETX task has been triggered.
STOP	Task	The TASKS_STOP task has been triggered.
PREPARERX	Task	The TASKS_PREPARERX task has been triggered.
STOPPED	Event	The EVENTS_STOPPED event was generated.
RXSTARTED	Event	The EVENTS_RXSTARTED event was generated.
TXSTARTED	Event	The EVENTS_TXSTARTED event was generated.
TX prepared	Internal	Internal flag indicating that a TASKS_PREPARETX task has been triggered. This flag is not visible to the user.
RX prepared	Internal	Internal flag indicating that a TASKS_PREPARERX task has been triggered. This flag is not visible to the user.
Unprepare TX	Internal	Clears the internal 'TX prepared' flag until next TASKS_PREPARETX task.
Unprepare RX	Internal	Clears the internal 'RX prepared' flag until next TASKS_PREPARERX task.
Stop condition	TWI protocol	A TWI stop condition was detected.
Restart condition	TWI protocol	A TWI restart condition was detected.

Table 53: TWI slave state machine symbols

The TWI slave can perform clock stretching, with the premise that the master is able to support it.

The TWI slave operates in a low power mode while waiting for a TWI master to initiate a transfer. As long as the TWI slave is not addressed, it will remain in this low power mode.

To secure correct behavior of the TWI slave, PSEL.SCL, PSEL.SDA, CONFIG, and the ADDRESS[n] registers must be configured prior to enabling the TWI slave through the ENABLE register. Similarly, changing these settings must be performed while the TWI slave is disabled. Failing to do so may result in unpredictable behavior.

6.32.1 EasyDMA

The TWIS implements EasyDMA for accessing RAM without CPU involvement.

The following table shows the Easy DMA channels that the TWIS peripheral implements.

Channel	Type	Register Cluster
TXD	READER	TXD
RXD	WRITER	RXD

Table 54: TWIS EasyDMA Channels

For detailed information regarding the use of EasyDMA, see EasyDMA on page 63.

The STOPPED event indicates that EasyDMA has finished accessing the buffer in RAM.

6.32.2 TWI slave responding to a read command

Before the TWI slave can respond to a read command, the TWI slave must be configured correctly and enabled via the ENABLE register. When enabled, the TWI slave will be in its IDLE state. .

A read command is started when the TWI master generates a start condition on the TWI bus, followed by clocking out the address and the READ/WRITE bit set to 1 (WRITE=0, READ=1). The READ/WRITE bit is followed by an ACK/NACK bit (ACK=0 or NACK=1) response from the TWI slave.

The TWI slave is able to listen for up to two addresses at the same time. This is configured in the ADDRESS registers and the CONFIG register.

The TWI slave will only acknowledge (ACK) the read command if the address presented by the master matches one of the addresses the slave is configured to listen for. The TWI slave will generate a READ event when it acknowledges the read command.

The TWI slave is only able to detect a read command from the IDLE state.

The TWI slave will set an internal 'TX prepared' flag when the PREPARETX task is triggered.

When the read command is received, the TWI slave will enter the TX state if the internal 'TX prepared' flag is set.

If the internal 'TX prepared' flag is not set when the read command is received, the TWI slave will stretch the master's clock until the PREPARETX task is triggered and the internal 'TX prepared' flag is set.

The TWI slave will generate the TXSTARTED event and clear the 'TX prepared' flag ('unprepare TX') when it enters the TX state. In this state the TWI slave will send the data bytes found in the transmit buffer to the master using the master's clock.

The TWI slave will go back to the IDLE state if the TWI slave receives a restart command when it is in the TX state.

The TWI slave is stopped when it receives the stop condition from the TWI master. A STOPPED event will be generated when the transaction has stopped. The TWI slave will clear the 'TX prepared' flag ('unprepare TX') and go back to the IDLE state when it has stopped.

The transmit buffer is located in RAM at the address specified in the TXD.PTR register. The TWI slave will only be able to send TXD.MAXCNT bytes from the transmit buffer for each transaction. If the TWI master forces the slave to send more than TXD.MAXCNT bytes, the slave will send the byte specified in the ORC register to the master instead. If this happens, an ERROR event will be generated.

The EasyDMA configuration registers, see TXD.PTR etc., are latched when the TXSTARTED event is generated.

The TWI slave can be forced to stop by triggering the STOP task. A STOPPED event will be generated when the TWI slave has stopped. The TWI slave will clear the 'TX prepared' flag and go back to the IDLE state when it has stopped, see also Terminating an ongoing TWI transaction on page 807.

Each byte sent from the slave will be followed by an ACK/NACK bit sent from the master. The TWI master will generate a NACK following the last byte that it wants to receive to tell the slave to release the bus so that the TWI master can generate the stop condition. The TXD.AMOUNT register can be queried after a transaction to see how many bytes were sent.

A typical TWI slave read command response is shown in the following figure. Occurrence 2 in the figure illustrates clock stretching performed by the TWI slave following a SUSPEND task.

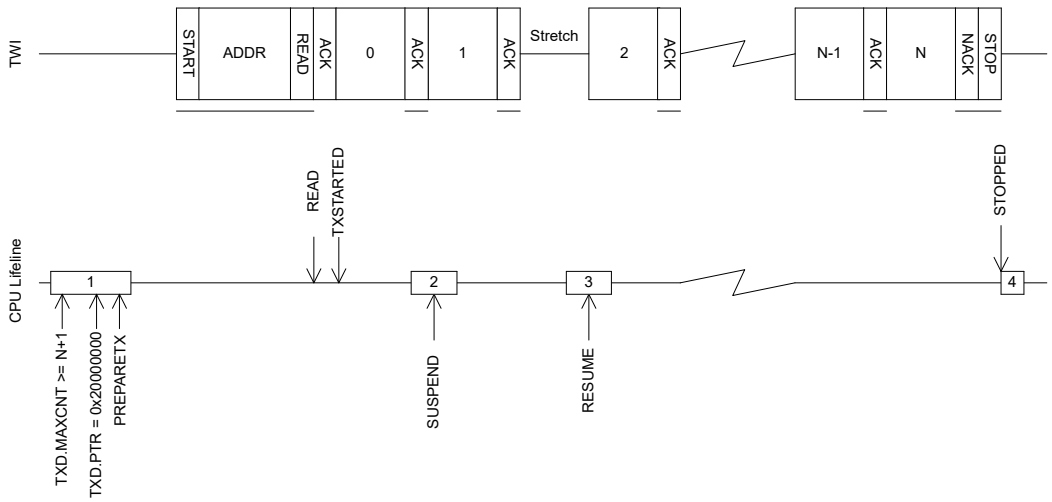


Figure 189: The TWI slave responding to a read command

6.32.3 TWI slave responding to a write command

Before the TWI slave can respond to a write command, the TWI slave must be configured correctly and enabled via the ENABLE register. When enabled, the TWI slave will be in its IDLE state.

A write command is started when the TWI master generates a start condition on the TWI bus, followed by clocking out the address and the READ/WRITE bit set to 0 (WRITE=0, READ=1). The READ/WRITE bit is followed by an ACK/NACK bit (ACK=0 or NACK=1) response from the slave.

The TWI slave is able to listen for up to two addresses at the same time. This is configured in the ADDRESS registers and the CONFIG register.

The TWI slave will only acknowledge (ACK) the write command if the address presented by the master matches one of the addresses the slave is configured to listen for. The TWI slave will generate a WRITE event if it acknowledges the write command.

The TWI slave is only able to detect a write command from the IDLE state.

The TWI slave will set an internal 'RX prepared' flag when the PREPARERX task is triggered.

When the write command is received, the TWI slave will enter the RX state if the internal 'RX prepared' flag is set.

If the internal 'RX prepared' flag is not set when the write command is received, the TWI slave will stretch the master's clock until the PREPARERX task is triggered and the internal 'RX prepared' flag is set.

The TWI slave will generate the RXSTARTED event and clear the internal 'RX prepared' flag ('unprepare RX') when it enters the RX state. In this state, the TWI slave will be able to receive the bytes sent by the TWI master.

The TWI slave will go back to the IDLE state if the TWI slave receives a restart command when it is in the RX state.

The TWI slave is stopped when it receives the stop condition from the TWI master. A STOPPED event will be generated when the transaction has stopped. The TWI slave will clear the internal 'RX prepared' flag ('unprepare RX') and go back to the IDLE state when it has stopped.

The receive buffer is located in RAM at the address specified in the RXD.PTR register. The TWI slave will only be able to receive as many bytes as specified in the RXD.MAXCNT register. If the TWI master tries to send more bytes to the slave than it can receive, the extra bytes are discarded and NACKed by the slave. If this happens, an ERROR event will be generated.

The EasyDMA configuration registers, see RXD.PTR etc., are latched when the RXSTARTED event is generated.

The TWI slave can be forced to stop by triggering the STOP task. A STOPPED event will be generated when the TWI slave has stopped. The TWI slave will clear the internal 'RX prepared' flag and go back to the IDLE state when it has stopped, see also Terminating an ongoing TWI transaction on page 807.

The TWI slave will generate an ACK after every byte received from the master. The RXD.AMOUNT register can be queried after a transaction to see how many bytes were received.

A typical TWI slave write command response is show in the following figure. Occurrence 2 in the figure illustrates clock stretching performed by the TWI slave following a SUSPEND task.

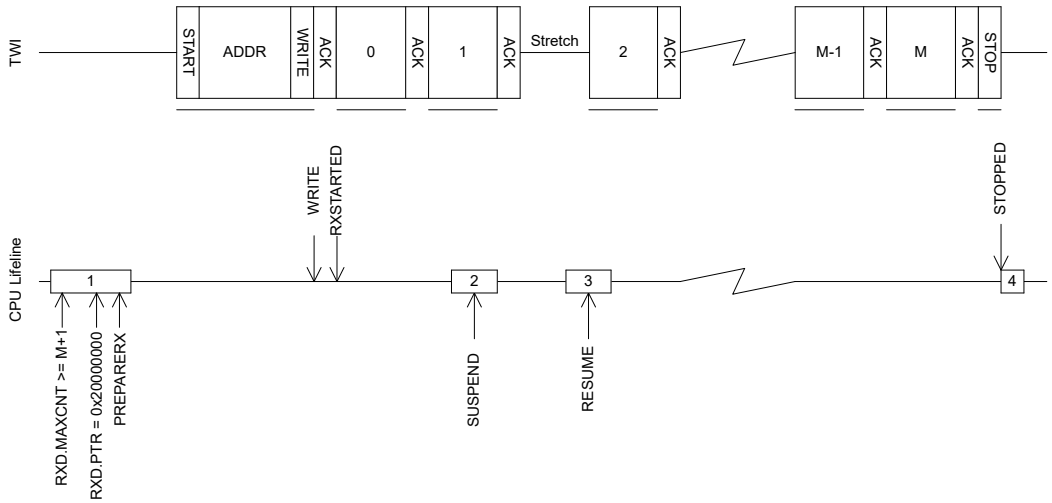


Figure 190: The TWI slave responding to a write command

6.32.4 Master repeated start sequence

An example of a repeated start sequence is one in which the TWI master writes two bytes to the slave followed by reading four bytes from the slave.

This is illustrated in the following figure.

In this example, the receiver does not know what the master wants to read in advance. This information is in the first two received bytes of the write in the repeated start sequence. To guarantee that the CPU is able to process the received data before the TWI slave starts to reply to the read command, the SUSPEND task is triggered via a shortcut from the READ event generated when the read command is received. When the CPU has processed the incoming data and prepared the correct data response, the CPU will resume the transaction by triggering the RESUME task.

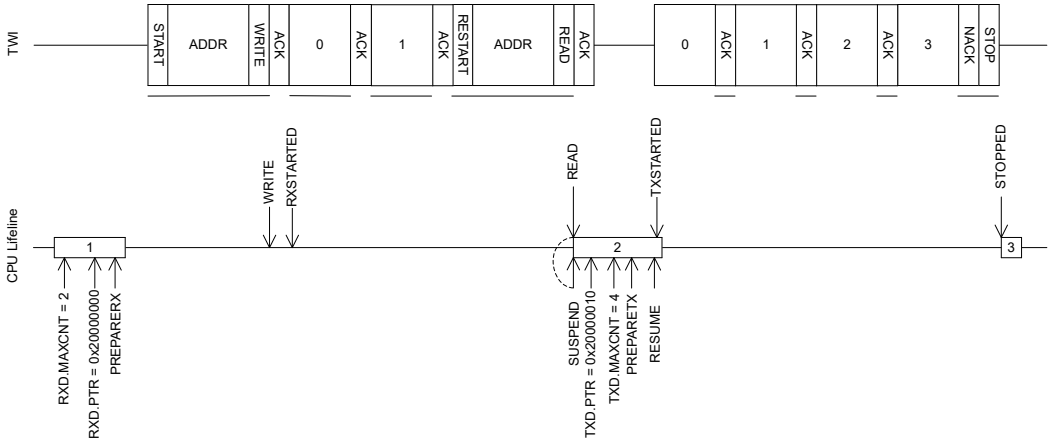


Figure 191: Repeated start sequence

6.32.5 Terminating an ongoing TWI transaction

In some situations, e.g. if the external TWI master is not responding correctly, it may be required to terminate an ongoing transaction.

This can be achieved by triggering the STOP task. In this situation, a STOPPED event will be generated when the TWI has stopped independent of whether or not a STOP condition has been generated on the TWI bus. The TWI slave will release the bus when it has stopped and go back to its IDLE state.

6.32.6 Low power

When putting the system in low power and the peripheral is not needed, lowest possible power consumption is achieved by stopping, and then disabling the peripheral.

The STOP task may not be always needed (the peripheral might already be stopped), but if it is sent, software shall wait until the STOPPED event was received as a response before disabling the peripheral through the ENABLE register.

6.32.7 Slave mode pin configuration

The SCL and SDA signals associated with the TWI slave are mapped to physical pins according to the configuration specified in the PSEL.SCL and PSEL.SDA registers respectively.

The PSEL.SCL and PSEL.SDA registers and their configurations are only used as long as the TWI slave is enabled, and retained only as long as the device is in ON mode. When the peripheral is disabled, the pins will behave as regular GPIOs, and use the configuration in their respective OUT bit field and PIN_CNF[n] register. PSEL.SCL and PSEL.SDA must only be configured when the TWI slave is disabled.

To secure correct signal levels on the pins used by the TWI slave when the system is in OFF mode, and when the TWI slave is disabled, these pins must be configured in the GPIO peripheral as described in the following table.

Only one peripheral can be assigned to drive a particular GPIO pin at a time. Failing to do so may result in unpredictable behavior.

TWI slave signal	TWI slave pin	Direction	Output value	Drive strength
SCL	As specified in PSEL.SCL	Input	Not applicable	S0D1
SDA	As specified in PSEL.SDA	Input	Not applicable	S0D1

Table 55: GPIO configuration before enabling peripheral

6.32.8 Registers

Instances

Instance	Base address	Description
TWIS0	0x40003000	Two-wire interface slave 0
TWIS1	0x40004000	Two-wire interface slave 1

Register overview

Register	Offset	Description
TASKS_STOP	0x014	Stop TWI transaction
TASKS_SUSPEND	0x01C	Suspend TWI transaction
TASKS_RESUME	0x020	Resume TWI transaction
TASKS_PREPARERX	0x030	Prepare the TWI slave to respond to a write command
TASKS_PREPARETX	0x034	Prepare the TWI slave to respond to a read command
EVENTS_STOPPED	0x104	TWI stopped
EVENTS_ERROR	0x124	TWI error
EVENTS_RXSTARTED	0x14C	Receive sequence started
EVENTS_TXSTARTED	0x150	Transmit sequence started
EVENTS_WRITE	0x164	Write command received
EVENTS_READ	0x168	Read command received
SHORTS	0x200	Shortcuts between local events and tasks

Register	Offset	Description
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
ERRORSRC	0x4D0	Error source
MATCH	0x4D4	Status register indicating which address had a match
ENABLE	0x500	Enable TWIS
PSEL.SCL	0x508	Pin select for SCL signal
PSEL.SDA	0x50C	Pin select for SDA signal
RXD.PTR	0x534	RXD Data pointer
RXD.MAXCNT	0x538	Maximum number of bytes in RXD buffer
RXD.AMOUNT	0x53C	Number of bytes transferred in the last RXD transaction
RXD.LIST	0x540	EasyDMA list type
TXD.PTR	0x544	TXD Data pointer
TXD.MAXCNT	0x548	Maximum number of bytes in TXD buffer
TXD.AMOUNT	0x54C	Number of bytes transferred in the last TXD transaction
TXD.LIST	0x550	EasyDMA list type
ADDRESS[0]	0x588	TWI slave address 0
ADDRESS[1]	0x58C	TWI slave address 1
CONFIG	0x594	Configuration register for the address match mechanism
ORC	0x5C0	Over-read character. Character sent out in case of an over-read of the transmit buffer.

6.32.8.1 TASKS_STOP

Address offset: 0x014

Stop TWI transaction

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field		Value ID				Value																Description											
A	W	TASKS_STOP																						Stop TWI transaction											
				Trigger				1																Trigger task											

6.32.8.2 TASKS_SUSPEND

Address offset: 0x01C

Suspend TWI transaction

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field		Value ID				Value																Description											
A	W	TASKS_SUSPEND																						Suspend TWI transaction											
				Trigger				1																Trigger task											

6.32.8.3 TASKS_RESUME

Address offset: 0x020

Resume TWI transaction

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	TASKS_RESUME																																	Resume TWI transaction
			Trigger	1																															Trigger task

6.32.8.4 TASKS_PREPARERX

Address offset: 0x030

Prepare the TWI slave to respond to a write command

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	TASKS_PREPARERX																																	Prepare the TWI slave to respond to a write command
			Trigger	1																															Trigger task

6.32.8.5 TASKS_PREPARETX

Address offset: 0x034

Prepare the TWI slave to respond to a read command

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	TASKS_PREPARETX																																	Prepare the TWI slave to respond to a read command
			Trigger	1																															Trigger task

6.32.8.6 EVENTS_STOPPED

Address offset: 0x104

TWI stopped

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	EVENTS_STOPPED																																	TWI stopped
			NotGenerated	0																															Event not generated
			Generated	1																															Event generated

6.32.8.7 EVENTS_ERROR

Address offset: 0x124

TWI error

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ERROR			TWI error
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.32.8.8 EVENTS_RXSTARTED

Address offset: 0x14C

Receive sequence started

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_RXSTARTED			Receive sequence started
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.32.8.9 EVENTS_TXSTARTED

Address offset: 0x150

Transmit sequence started

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TXSTARTED			Transmit sequence started
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.32.8.10 EVENTS_WRITE

Address offset: 0x164

Write command received

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_WRITE			Write command received
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.32.8.11 EVENTS_READ

Address offset: 0x168

Read command received

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_READ			Read command received
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.32.8.12 SHORTS

Address offset: 0x200

Shortcuts between local events and tasks

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	WRITE_SUSPEND			Shortcut between event WRITE and task SUSPEND
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
B	RW	READ_SUSPEND			Shortcut between event READ and task SUSPEND
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

6.32.8.13 INTEN

Address offset: 0x300

Enable or disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				H G F E B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	STOPPED			Enable or disable interrupt for event STOPPED
			Disabled	0	Disable
			Enabled	1	Enable
B	RW	ERROR			Enable or disable interrupt for event ERROR
			Disabled	0	Disable
			Enabled	1	Enable
E	RW	RXSTARTED			Enable or disable interrupt for event RXSTARTED
			Disabled	0	Disable
			Enabled	1	Enable
F	RW	TXSTARTED			Enable or disable interrupt for event TXSTARTED
			Disabled	0	Disable
			Enabled	1	Enable
G	RW	WRITE			Enable or disable interrupt for event WRITE
			Disabled	0	Disable
			Enabled	1	Enable
H	RW	READ			Enable or disable interrupt for event READ
			Disabled	0	Disable
			Enabled	1	Enable

6.32.8.14 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID									H	G					F	E									B									A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	STOPPED																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	ERROR																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
E	RW	RXSTARTED																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
F	RW	TXSTARTED																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
G	RW	WRITE																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
H	RW	READ																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															

6.32.8.15 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID									H	G					F	E									B									A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	STOPPED																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	ERROR																																	
			Clear	1																															
			Disabled	0																															
			Enabled	1																															
E	RW	RXSTARTED																																	
			Clear	1																															

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				H G F E B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	TXSTARTED			Write '1' to disable interrupt for event TXSTARTED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
G	RW	WRITE			Write '1' to disable interrupt for event WRITE
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
H	RW	READ			Write '1' to disable interrupt for event READ
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.32.8.16 ERRORSRC

Address offset: 0x4D0

Error source

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW W1C	OVERFLOW			RX buffer overflow detected, and prevented
			NotDetected	0	Error did not occur
			Detected	1	Error occurred
B	RW W1C	DNACK			NACK sent after receiving a data byte
			NotReceived	0	Error did not occur
			Received	1	Error occurred
C	RW W1C	OVERREAD			TX buffer over-read detected, and prevented
			NotDetected	0	Error did not occur
			Detected	1	Error occurred

6.32.8.17 MATCH

Address offset: 0x4D4

Status register indicating which address had a match

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	MATCH		[0..1]	Indication of which address in ADDRESS that matched the incoming address

6.32.8.18 ENABLE

Address offset: 0x500

Enable TWIS

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	ENABLE			Enable or disable TWIS
			Disabled	0	Disable TWIS
			Enabled	9	Enable TWIS

6.32.8.19 PSEL.SCL

Address offset: 0x508

Pin select for SCL signal

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**

ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.32.8.20 PSEL.SDA

Address offset: 0x50C

Pin select for SDA signal

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**

ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.32.8.21 RXD

RXD EasyDMA channel

6.32.8.21.1 RXD.PTR

Address offset: 0x534

RXD Data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			RXD Data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.32.8.21.2 RXD.MAXCNT

Address offset: 0x538

Maximum number of bytes in RXD buffer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	MAXCNT		[0..0xFFFF]	Maximum number of bytes in RXD buffer

6.32.8.21.3 RXD.AMOUNT

Address offset: 0x53C

Number of bytes transferred in the last RXD transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	AMOUNT		[0..0xFFFF]	Number of bytes transferred in the last RXD transaction

6.32.8.21.4 RXD.LIST

Address offset: 0x540

EasyDMA list type

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																															A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	LIST			List type
			Disabled	0	Disable EasyDMA list
			ArrayList	1	Use array list

6.32.8.22 TXD

TXD EasyDMA channel

6.32.8.22.1 TXD.PTR

Address offset: 0x544

TXD Data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			TXD Data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.32.8.22.2 TXD.MAXCNT

Address offset: 0x548

Maximum number of bytes in TXD buffer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	MAXCNT		[0..0xFFFF]	Maximum number of bytes in TXD buffer

6.32.8.22.3 TXD.AMOUNT

Address offset: 0x54C

Number of bytes transferred in the last TXD transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	AMOUNT		[0..0xFFFF]	Number of bytes transferred in the last TXD transaction

6.32.8.22.4 TXD.LIST

Address offset: 0x550

EasyDMA list type

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																															A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	LIST			List type
			Disabled	0	Disable EasyDMA list
			ArrayList	1	Use array list

6.32.8.23 ADDRESS[0]

Address offset: 0x588

TWI slave address 0

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ADDRESS			TWI slave address

6.32.8.24 ADDRESS[1]

Address offset: 0x58C

TWI slave address 1

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ADDRESS			TWI slave address

6.32.8.25 CONFIG

Address offset: 0x594

Configuration register for the address match mechanism

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	ADDRESS0			Enable or disable address matching on ADDRESS[0]
			Disabled	0	Disabled
			Enabled	1	Enabled
B	RW	ADDRESS1			Enable or disable address matching on ADDRESS[1]
			Disabled	0	Disabled
			Enabled	1	Enabled

6.32.8.26 ORC

Address offset: 0x5C0

Over-read character. Character sent out in case of an over-read of the transmit buffer.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ORC			Over-read character. Character sent out in case of an over-read of the transmit buffer.

6.32.9 Electrical specification

6.32.9.1 TWIS slave timing specifications

Symbol	Description	Min.	Typ.	Max.	Units
$f_{TWIS,SCL}$	Bit rates for TWIS[41]	100		400	kbps
$t_{TWIS,START}$	Time from PREPARERX/PREPARETX task to ready to receive/transmit		1.5		µs
t_{TWIS,SU_DAT}	Data setup time before positive edge on SCL – all modes	300			ns
t_{TWIS,HD_DAT}	Data hold time after negative edge on SCL – all modes	500			ns
$t_{TWIS,HD_STA,100kbps}$	TWI slave hold time from for START condition (SDA low to SCL low), 100 kbps	5200			ns
$t_{TWIS,HD_STA,400kbps}$	TWI slave hold time from for START condition (SDA low to SCL low), 400 kbps	1300			ns
$t_{TWIS,SU_STO,100kbps}$	TWI slave setup time from SCL high to STOP condition, 100 kbps	5200			ns
$t_{TWIS,SU_STO,400kbps}$	TWI slave setup time from SCL high to STOP condition, 400 kbps	1300			ns
$t_{TWIS,BUF,100kbps}$	TWI slave bus free time between STOP and START conditions, 100 kbps		4700		ns
$t_{TWIS,BUF,400kbps}$	TWI slave bus free time between STOP and START conditions, 400 kbps		1300		ns

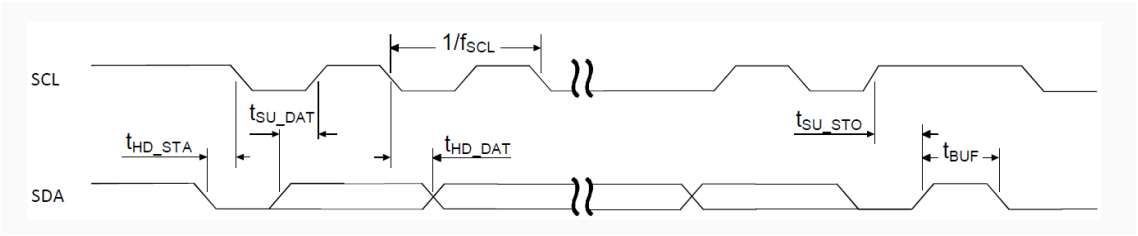


Figure 192: TWIS timing diagram, 1 byte transaction

6.33 UART — Universal asynchronous receiver/transmitter

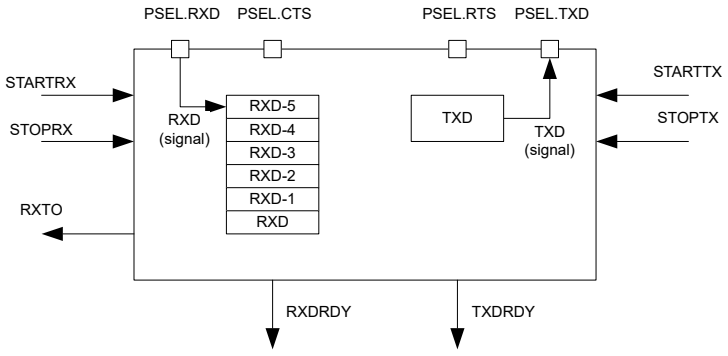


Figure 193: UART configuration

6.33.1 Functional description

Listed here are the main features of UART.

The UART implements support for the following features:

- Full-duplex operation

[41] High bit rates or stronger pull-ups may require GPIOs to be set as High Drive, see GPIO chapter for more details.

- Automatic flow control
- Parity checking and generation for the 9th data bit

As illustrated in UART configuration on page 819, the UART uses the TXD and RXD registers directly to transmit and receive data. The UART uses one stop bit.

Note: The external crystal oscillator must be enabled to obtain sufficient clock accuracy for stable communication. See CLOCK — Clock control on page 157 for more information.

6.33.2 Pin configuration

The different signals RXD, CTS (Clear To Send, active low), RTS (Request To Send, active low), and TXD associated with the UART are mapped to physical pins according to the configuration specified in the PSEL.RXD, PSEL.CTS, PSEL.RTS, and PSEL.TXD registers respectively.

If the CONNECT field of a PSEL.xxx register is set to Disconnected, the associated UART signal will not be connected to any physical pin. The PSEL.RXD, PSEL.CTS, PSEL.RTS, and PSEL.TXD registers and their configurations are only used as long as the UART is enabled, and retained only for the duration the device is in ON mode. PSEL.RXD, PSEL.CTS, PSEL.RTS, and PSEL.TXD must only be configured when the UART is disabled.

To secure correct signal levels on the pins by the UART when the system is in OFF mode, the pins must be configured in the GPIO peripheral as described in Pin configuration on page 820.

Only one peripheral can be assigned to drive a particular GPIO pin at a time. Failing to do so may result in unpredictable behavior.

UART pin	Direction	Output value
RXD	Input	Not applicable
CTS	Input	Not applicable
RTS	Output	1
TXD	Output	1

Table 56: GPIO configuration

6.33.3 Shared resources

The UART shares registers and resources with other peripherals that have the same ID as the UART.

All peripherals with the same ID as the UART must be disabled before configuring and using the UART. Disabling a peripheral that has the same ID as the UART will not reset any of the registers that are shared with the UART. It is therefore important to configure all relevant UART registers explicitly to ensure that it operates correctly.

See Instantiation on page 24 for details on peripherals and their IDs.

6.33.4 Transmission

A UART transmission sequence is started by triggering the STARTTX task.

Bytes are transmitted by writing to the TXD register. When a byte has been successfully transmitted, the UART will generate a TXDRDY event after which a new byte can be written to the TXD register. A UART transmission sequence is stopped immediately by triggering the STOPTX task.

If flow control is enabled, a transmission will be automatically suspended when CTS is deactivated, and resumed when CTS is activated again, as shown in the following figure. A byte that is in transmission when CTS is deactivated will be fully transmitted before the transmission is suspended. For more information, see Suspending the UART on page 822.

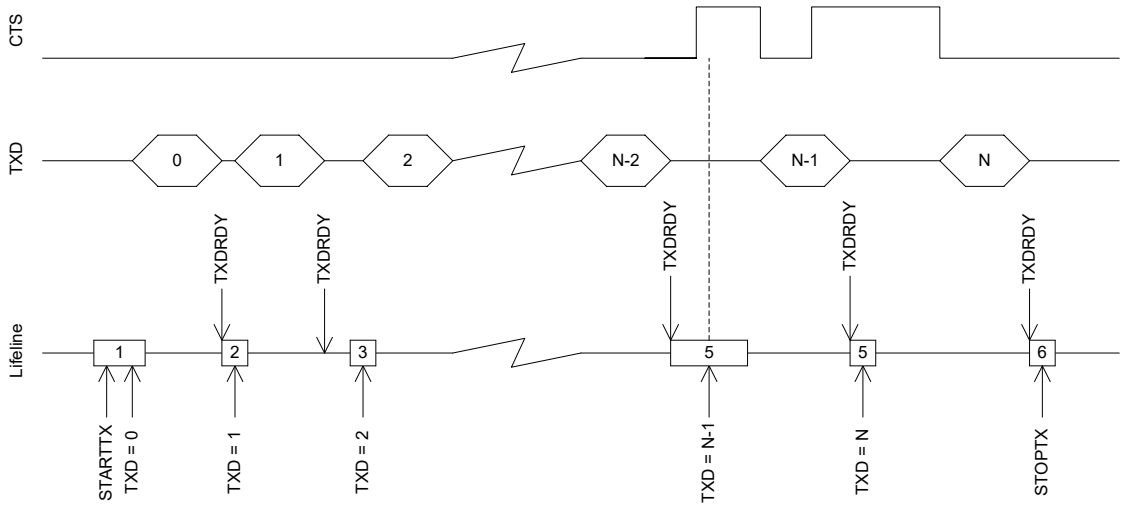


Figure 194: UART transmission

6.33.5 Reception

A UART reception sequence is started by triggering the STARTRX task.

The UART receiver chain implements a FIFO capable of storing six incoming RXD bytes before data is overwritten. Bytes are extracted from this FIFO by reading the RXD register. When a byte is extracted from the FIFO, a new byte pending in the FIFO will be moved to the RXD register. The UART will generate an RXDRDY event every time a new byte is moved to the RXD register.

When flow control is enabled, the UART will deactivate the RTS signal when there is only space for four more bytes in the receiver FIFO. The counterpart transmitter is therefore able to send up to four bytes after the RTS signal is deactivated before data is being overwritten. To prevent overwriting data in the FIFO, the counterpart UART transmitter must therefore make sure to stop transmitting data within four bytes after the RTS line is deactivated.

The RTS signal will first be activated again when the FIFO has been emptied, that is, when all bytes in the FIFO have been read by the CPU, see UART reception on page 822.

The RTS signal will also be deactivated when the receiver is stopped through the STOPRX task as illustrated in UART reception on page 822. The UART is able to receive four to five additional bytes if they are sent in succession immediately after the RTS signal has been deactivated. This is possible because the UART is, even after the STOPRX task is triggered, able to receive bytes for an extended period of time dependent on the configured baud rate. The UART will generate a receiver timeout event (RXTO) when this period has elapsed.

To prevent loss of incoming data, the RXD register must only be read one time following every RXDRDY event.

To secure that the CPU can detect all incoming RXDRDY events through the RXDRDY event register, the RXDRDY event register must be cleared before the RXD register is read. The reason for this is that the UART is allowed to write a new byte to the RXD register, and can generate a new event immediately after the RXD register is read (emptied) by the CPU.

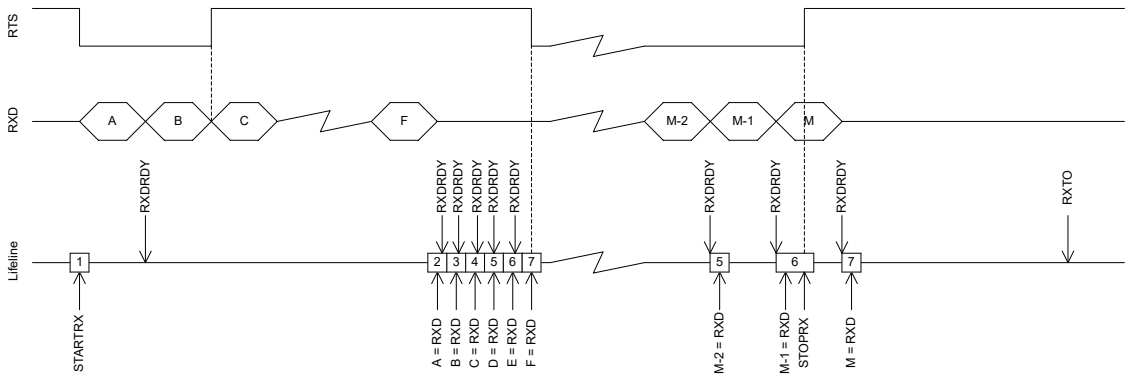


Figure 195: UART reception

As indicated in occurrence 2 in the figure, the RXDRDY event associated with byte B is generated first after byte A has been extracted from RXD.

6.33.6 Suspending the UART

The UART can be suspended by triggering the SUSPEND task.

SUSPEND will affect both the UART receiver and the UART transmitter, i.e. the transmitter will stop transmitting and the receiver will stop receiving. UART transmission and reception can be resumed, after being suspended, by triggering STARTTX and STARTRX respectively.

Following a SUSPEND task, an ongoing TXD byte transmission will be completed before the UART is suspended.

When the SUSPEND task is triggered, the UART receiver will behave in the same way as it does when the STOPRX task is triggered.

6.33.7 Error conditions

An ERROR event, in the form of a framing error, will be generated if a valid stop bit is not detected in a frame. Another ERROR event, in the form of a break condition, will be generated if the RXD line is held active low for longer than the length of a data frame. Effectively, a framing error is always generated before a break condition occurs.

6.33.8 Using the UART without flow control

If flow control is not enabled, the interface will behave as if the CTS and RTS lines are kept active all the time.

6.33.9 Parity and stop bit configuration

Automatic even parity generation for both transmission and reception can be configured using the register CONFIG on page 830. See the register description for details.

The amount of stop bits can also be configured through the register CONFIG on page 830.

6.33.10 Registers

Instances

Instance	Base address	Description
UART0	0x40002000	Universal asynchronous receiver/transmitter This instance is deprecated.

Register overview

Register	Offset	Description
TASKS_STARTRX	0x000	Start UART receiver
TASKS_STOPRX	0x004	Stop UART receiver
TASKS_STARTTX	0x008	Start UART transmitter
TASKS_STOPTX	0x00C	Stop UART transmitter
TASKS_SUSPEND	0x01C	Suspend UART
EVENTS_CTS	0x100	CTS is activated (set low). Clear To Send.
EVENTS_NCTS	0x104	CTS is deactivated (set high). Not Clear To Send.
EVENTS_RXDRDY	0x108	Data received in RXD
EVENTS_TXDRDY	0x11C	Data sent from TXD
EVENTS_ERROR	0x124	Error detected
EVENTS_RXTO	0x144	Receiver timeout
SHORTS	0x200	Shortcuts between local events and tasks
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
ERRORSRC	0x480	Error source
ENABLE	0x500	Enable UART
PSEL.RTS	0x508	Pin select for RTS
PSEL.TXD	0x50C	Pin select for TXD
PSEL.CTS	0x510	Pin select for CTS
PSEL.RXD	0x514	Pin select for RXD
RXD	0x518	RXD register. Register is cleared on read and the double buffered byte will be moved to RXD if it exists.
TXD	0x51C	TXD register
BAUDRATE	0x524	Baud rate. Accuracy depends on the HFCLK source selected.
CONFIG	0x56C	Configuration of parity and hardware flow control

6.33.10.1 TASKS_STARTRX

Address offset: 0x000

Start UART receiver

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	TASKS_STARTRX																																	Start UART receiver
			Trigger	1																															Trigger task

6.33.10.2 TASKS_STOPRX

Address offset: 0x004

Stop UART receiver

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	TASKS_STOPRX																																	Stop UART receiver
			Trigger	1																															Trigger task

6.33.10.3 TASKS_STARTTX

Address offset: 0x008

Start UART transmitter

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field		Value ID				Value																		Description									
A	W	TASKS_STARTTX																								Start UART transmitter									
				Trigger				1																		Trigger task									

6.33.10.4 TASKS_STOPTX

Address offset: 0x00C

Stop UART transmitter

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field		Value ID				Value																		Description									
A	W	TASKS_STOPTX																								Stop UART transmitter									
				Trigger				1																		Trigger task									

6.33.10.5 TASKS_SUSPEND

Address offset: 0x01C

Suspend UART

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field		Value ID				Value																		Description									
A	W	TASKS_SUSPEND																								Suspend UART									
				Trigger				1																		Trigger task									

6.33.10.6 EVENTS_CTS

Address offset: 0x100

CTS is activated (set low). Clear To Send.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field		Value ID				Value																		Description									
A	RW	EVENTS_CTS																								CTS is activated (set low). Clear To Send.									
				NotGenerated				0																		Event not generated									
				Generated				1																		Event generated									

6.33.10.7 EVENTS_NCTS

Address offset: 0x104

CTS is deactivated (set high). Not Clear To Send.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	EVENTS_NCTS																																	
			NotGenerated	0																															
			Generated	1																															

6.33.10.8 EVENTS_RXDRDY

Address offset: 0x108

Data received in RXD

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	EVENTS_RXDRDY																																	
			NotGenerated	0																															
			Generated	1																															

6.33.10.9 EVENTS_TXDRDY

Address offset: 0x11C

Data sent from TXD

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	EVENTS_TXDRDY																																	
			NotGenerated	0																															
			Generated	1																															

6.33.10.10 EVENTS_ERROR

Address offset: 0x124

Error detected

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	EVENTS_ERROR																																	
			NotGenerated	0																															
			Generated	1																															

6.33.10.11 EVENTS_RXTO

Address offset: 0x144

Receiver timeout

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	EVENTS_RXTO																																	
			NotGenerated	0																															
			Generated	1																															

6.33.10.12 SHORTS

Address offset: 0x200

Shortcuts between local events and tasks

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																															B	A			
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	CTS_STARTRX																																	
			Disabled	0																															
			Enabled	1																															
B	RW	NCTS_STOPRX																																	
			Disabled	0																															
			Enabled	1																															

6.33.10.13 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																		F							E		D					C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	CTS																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
B	RW	NCTS																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
C	RW	RXDRDY																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
D	RW	TXDRDY																																	
			Set	1																															

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	ERROR			Write '1' to enable interrupt for event ERROR
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	RXTO			Write '1' to enable interrupt for event RXTO
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.33.10.14 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	CTS			Write '1' to disable interrupt for event CTS
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	NCTS			Write '1' to disable interrupt for event NCTS
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	RXDRDY			Write '1' to disable interrupt for event RXDRDY
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	TXDRDY			Write '1' to disable interrupt for event TXDRDY
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	ERROR			Write '1' to disable interrupt for event ERROR
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	RXTO			Write '1' to disable interrupt for event RXTO
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.33.10.15 ERRORSRC

Address offset: 0x480

Error source

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	OVERRUN			Overrun error
	W1C				A start bit is received while the previous data still lies in RXD. (Previous data is lost.)
			NotPresent	0	Read: error not present
			Present	1	Read: error present
B	RW	PARITY			Parity error
	W1C				A character with bad parity is received, if HW parity check is enabled.
			NotPresent	0	Read: error not present
			Present	1	Read: error present
C	RW	FRAMING			Framing error occurred
	W1C				A valid stop bit is not detected on the serial data input after all bits in a character have been received.
			NotPresent	0	Read: error not present
			Present	1	Read: error present
D	RW	BREAK			Break condition
	W1C				The serial data input is '0' for longer than the length of a data frame. (The data frame length is 10 bits without parity bit, and 11 bits with parity bit.).
			NotPresent	0	Read: error not present
			Present	1	Read: error present

6.33.10.16 ENABLE

Address offset: 0x500

Enable UART

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ENABLE			Enable or disable UART
			Disabled	0	Disable UART
			Enabled	4	Enable UART

6.33.10.17 PSEL.RTS

Address offset: 0x508

Pin select for RTS

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.33.10.18 PSEL.TXD

Address offset: 0x50C

Pin select for TXD

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**

ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.33.10.19 PSEL.CTS

Address offset: 0x510

Pin select for CTS

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**

ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.33.10.20 PSEL.RXD

Address offset: 0x514

Pin select for RXD

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**

ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.33.10.21 RXD

Address offset: 0x518

RXD register. Register is cleared on read and the double buffered byte will be moved to RXD if it exists.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	RXD			RX data received in previous transfers, double buffered
	RME				

6.33.10.22 TXD

Address offset: 0x51C

TXD register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TXD			TX data to be transferred

6.33.10.23 BAUDRATE

Address offset: 0x524

Baud rate. Accuracy depends on the HFCLK source selected.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x04000000				**0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	BAUDRATE			Baud rate
			Baud1200	0x0004F000	1200 baud (actual rate: 1205)
			Baud2400	0x0009D000	2400 baud (actual rate: 2396)
			Baud4800	0x0013B000	4800 baud (actual rate: 4808)
			Baud9600	0x00275000	9600 baud (actual rate: 9598)
			Baud14400	0x003B0000	14400 baud (actual rate: 14414)
			Baud19200	0x004EA000	19200 baud (actual rate: 19208)
			Baud28800	0x0075F000	28800 baud (actual rate: 28829)
			Baud31250	0x00800000	31250 baud
			Baud38400	0x009D5000	38400 baud (actual rate: 38462)
			Baud56000	0x00E50000	56000 baud (actual rate: 55944)
			Baud57600	0x00EBF000	57600 baud (actual rate: 57762)
			Baud76800	0x013A9000	76800 baud (actual rate: 76923)
			Baud115200	0x01D7E000	115200 baud (actual rate: 115942)
			Baud230400	0x03AFB000	230400 baud (actual rate: 231884)
			Baud250000	0x04000000	250000 baud
			Baud460800	0x075F7000	460800 baud (actual rate: 470588)
			Baud921600	0x0EBED000	921600 baud (actual rate: 941176)
			Baud1M	0x10000000	1Mega baud

6.33.10.24 CONFIG

Address offset: 0x56C

Configuration of parity and hardware flow control

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																															C	B	B	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	HWFC			Hardware flow control																														
			Disabled	0	Disabled																														
			Enabled	1	Enabled																														
B	RW	PARITY			Parity																														
			Excluded	0x0	Exclude parity bit																														
			Included	0x7	Include parity bit																														
C	RW	STOP			Stop bits																														
			One	0	One stop bit																														
			Two	1	Two stop bits																														

6.33.11 Electrical specification

6.33.11.1 UART electrical specification

Symbol	Description	Min.	Typ.	Max.	Units
f_{UART}	Baud rate for UART[42].			1000	kbps
$t_{UART,CTSH}$	CTS high time	1			µs
$t_{UART,START}$	Time from STARTRX/STARTTX task to transmission started		1		µs

6.34 UARTE — Universal asynchronous receiver/transmitter with EasyDMA

The Universal asynchronous receiver/transmitter with EasyDMA (UARTE) offers fast, full-duplex, asynchronous serial communication with built-in flow control (CTS, RTS) support in hardware at a rate up to 1 Mbps, and EasyDMA data transfer from/to RAM.

Listed here are the main features for UARTE:

- Full-duplex operation
- Automatic hardware flow control
- Optional even parity bit checking and generation
- EasyDMA
- Up to 1 Mbps baudrate
- Return to IDLE between transactions supported (when using HW flow control)
- One or two stop bit
- Least significant bit (LSB) first

[42] High baud rates may require GPIOs to be set as High Drive, see GPIO for more details.

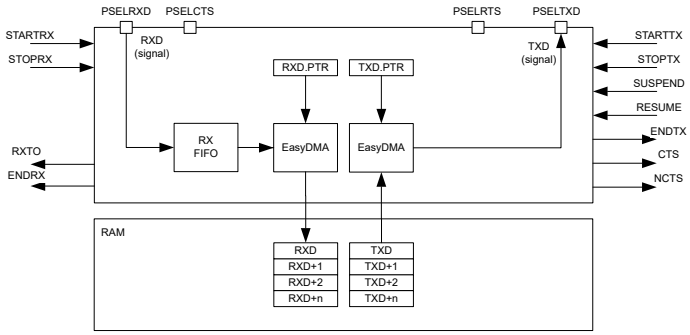


Figure 196: UARTE configuration

The GPIOs used for each UART interface can be chosen from any GPIO on the device and are independently configurable. This enables great flexibility in device pinout and efficient use of board space and signal routing.

Note: The external crystal oscillator must be enabled to obtain sufficient clock accuracy for stable communication. See CLOCK — Clock control on page 157 for more information.

6.34.1 EasyDMA

The UARTE implements EasyDMA for reading and writing to and from the RAM.

If the TXD.PTR and the RXD.PTR are not pointing to the Data RAM region, an EasyDMA transfer may result in a HardFault or RAM corruption. See Memory on page 21 for more information about the different memory regions.

The .PTR and .MAXCNT registers are double-buffered. They can be updated and prepared for the next RX/TX transmission immediately after having received the RXSTARTED/TXSTARTED event.

The ENDRX and ENDTX events indicate that the EasyDMA is finished accessing the RX or TX buffer in RAM.

6.34.2 Transmission

The first step of a DMA transmission is storing bytes in the transmit buffer and configuring EasyDMA. This is achieved by writing the initial address pointer to TXD.PTR, and the number of bytes in the RAM buffer to TXD.MAXCNT. The UARTE transmission is started by triggering the STARTTX task.

After each byte has been sent over the TXD line, a TXDRDY event will be generated.

When all bytes in the TXD buffer, as specified in the TXD.MAXCNT register, have been transmitted, the UARTE transmission will end automatically and an ENDTX event will be generated.

A UARTE transmission sequence is stopped by triggering the STOPTX task. A TXSTOPPED event will be generated when the UARTE transmitter has stopped.

If the ENDTX event has not already been generated when the UARTE transmitter has come to a stop, the UARTE will generate the ENDTX event explicitly even though all bytes in the TXD buffer, as specified in the TXD.MAXCNT register, have not been transmitted.

If flow control is enabled through the HWFC field in the CONFIG register, a transmission will be automatically suspended when CTS is deactivated and resumed when CTS is activated again, as shown in the following figure. A byte that is in transmission when CTS is deactivated will be fully transmitted before the transmission is suspended.

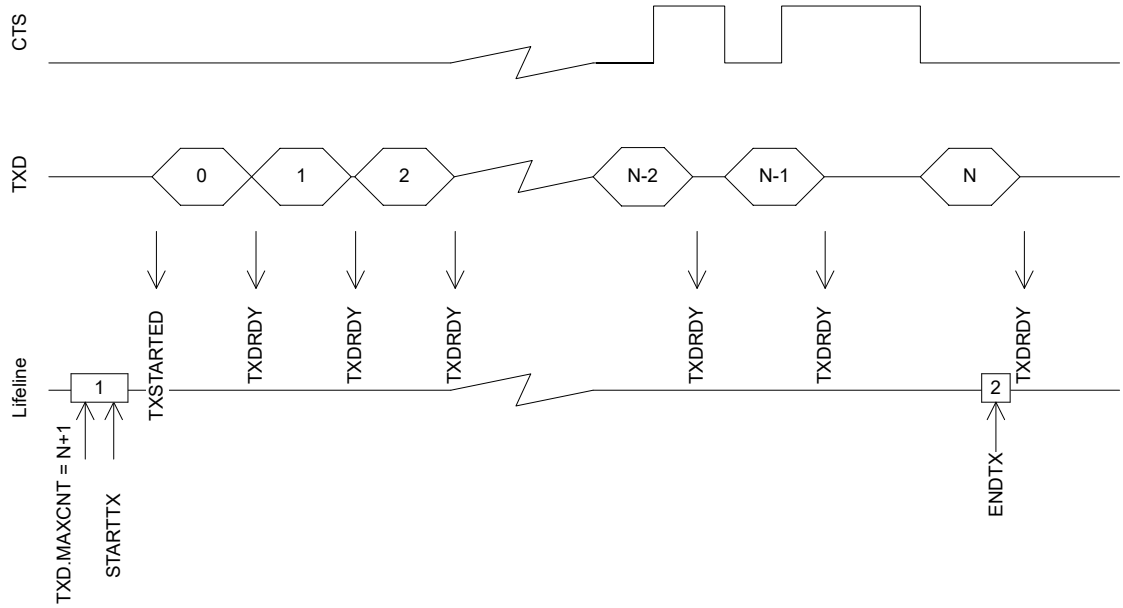


Figure 197: UARTE transmission

The UARTE transmitter will be in its lowest activity level, and consume the least amount of energy, when it is stopped, i.e. before it is started via STARTTX or after it has been stopped via STOPTX and the TXSTOPPED event has been generated. See POWER — Power supply on page 81 for more information about power modes.

6.34.3 Reception

The UARTE receiver is started by triggering the STARTRX task. The UARTE receiver is using EasyDMA to store incoming data in an RX buffer in RAM.

The RX buffer is located at the address specified in the RXD.PTR register. The RXD.PTR register is double-buffered and it can be updated and prepared for the next STARTRX task immediately after the RXSTARTED event is generated. The size of the RX buffer is specified in the RXD.MAXCNT register. The UARTE generates an ENDRX event when it has filled up the RX buffer, as seen in the following figure.

For each byte received over the RXD line, an RXDRDY event will be generated. This event is likely to occur before the corresponding data has been transferred to Data RAM.

The RXD.AMOUNT register can be queried following an ENDRX event to see how many new bytes have been transferred to the RX buffer in RAM since the previous ENDRX event.

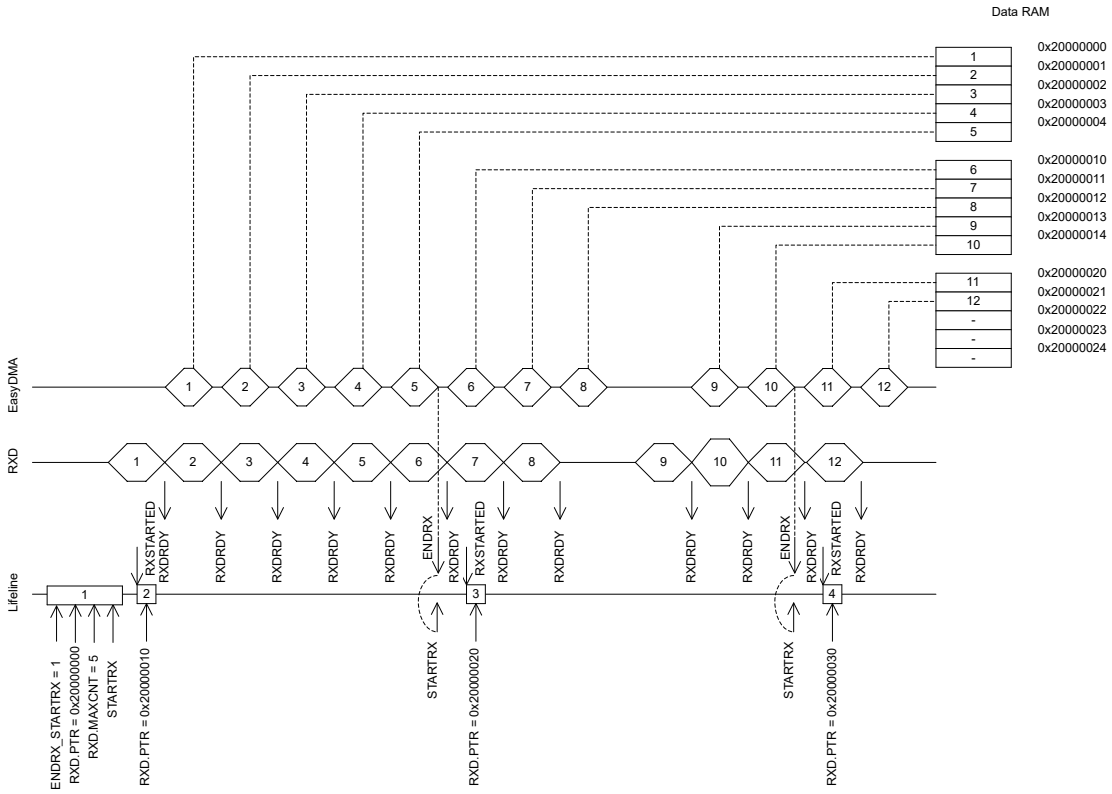


Figure 198: UARTE reception

The UARTE receiver is stopped by triggering the STOPRX task. An RXTO event is generated when the UARTE has stopped. The UARTE will make sure that an impending ENDRX event will be generated before the RXTO event is generated. This means that the UARTE will guarantee that no ENDRX event will be generated after RXTO, unless the UARTE is restarted or a FLUSHRX command is issued after the RXTO event is generated.

Note: If the ENDRX event has not been generated when the UARTE receiver stops, indicating that all pending content in the RX FIFO has been moved to the RX buffer, the UARTE will generate the ENDRX event explicitly even though the RX buffer is not full. In this scenario the ENDRX event will be generated before the RXTO event is generated.

To determine the amount of bytes the RX buffer has received, the CPU can read the RXD.AMOUNT register following the ENDRX event or the RXTO event.

The UARTE is able to receive up to four bytes after the STOPRX task has been triggered, as long as these are sent in succession immediately after the RTS signal is deactivated. After the RTS is deactivated, the UART is able to receive bytes for a period of time equal to the time needed to send four bytes on the configured baud rate.

After the RXTO event is generated the internal RX FIFO may still contain data, and to move this data to RAM the FLUSHRX task must be triggered. To make sure that this data does not overwrite data in the RX buffer, the RX buffer should be emptied or the RXD.PTR should be updated before the FLUSHRX task is triggered. To make sure that all data in the RX FIFO is moved to the RX buffer, the RXD.MAXCNT register must be set to RXD.MAXCNT > 4, as seen in the following figure. The UARTE will generate the ENDRX event after completing the FLUSHRX task even if the RX FIFO was empty or if the RX buffer does not get filled up. To be able to know how many bytes have actually been received into the RX buffer in this case, the CPU can read the RXD.AMOUNT register following the ENDRX event.

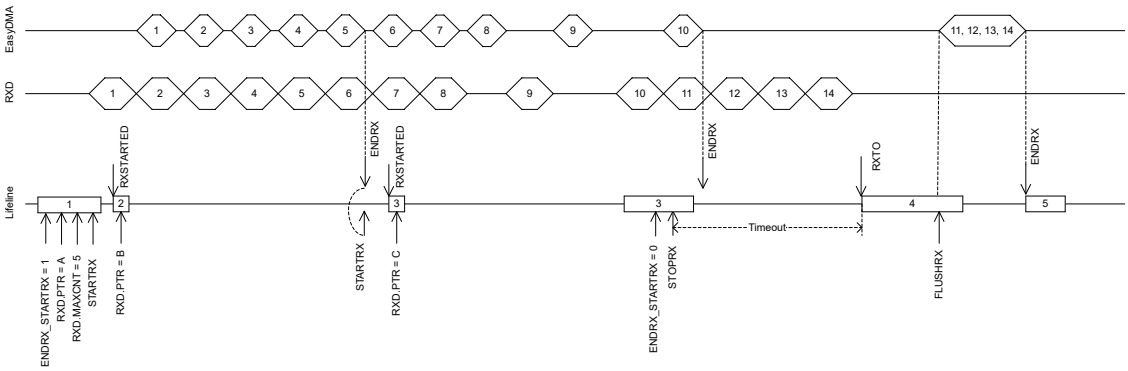


Figure 199: UARTE reception with forced stop via STOPRX

If HW flow control is enabled through the HWFC field in the CONFIG register, the RTS signal will be deactivated when the receiver is stopped via the STOPRX task or when the UARTE is only able to receive four more bytes in its internal RX FIFO.

With flow control disabled, the UARTE will function in the same way as when the flow control is enabled except that the RTS line will not be used. This means that no signal will be generated when the UARTE has reached the point where it is only able to receive four more bytes in its internal RX FIFO. Data received when the internal RX FIFO is filled up, will be lost.

The UARTE receiver will be in its lowest activity level, and consume the least amount of energy, when it is stopped, i.e. before it is started via STARTRX or after it has been stopped via STOPRX and the RXTO event has been generated. See POWER — Power supply on page 81 for more information about power modes.

6.34.4 Error conditions

An ERROR event, in the form of a framing error, will be generated if a valid stop bit is not detected in a frame. Another ERROR event, in the form of a break condition, will be generated if the RXD line is held active low for longer than the length of a data frame. Effectively, a framing error is always generated before a break condition occurs.

An ERROR event will not stop reception. If the error was a parity error, the received byte will still be transferred into Data RAM, and so will following incoming bytes. If there was a framing error (wrong stop bit), that specific byte will NOT be stored into Data RAM, but following incoming bytes will.

6.34.5 Using the UARTE without flow control

If flow control is not enabled, the interface will behave as if the CTS and RTS lines are kept active all the time.

6.34.6 Parity and stop bit configuration

Automatic even parity generation for both transmission and reception can be configured using the register CONFIG on page 849. See the register description for details.

The amount of stop bits can also be configured through the register CONFIG on page 849.

6.34.7 Low power

When putting the system in low power and the peripheral is not needed, lowest possible power consumption is achieved by stopping, and then disabling the peripheral.

The STOPTX and STOPRX tasks may not be always needed (the peripheral might already be stopped), but if STOPTX and/or STOPRX is sent, software shall wait until the TXSTOPPED and/or RXTO event is received in response, before disabling the peripheral through the ENABLE register.

6.34.8 Pin configuration

The different signals RXD, CTS (Clear To Send, active low), RTS (Request To Send, active low), and TXD associated with the UARTE are mapped to physical pins according to the configuration specified in the PSEL.RXD, PSEL.CTS, PSEL.RTS, and PSEL.TXD registers respectively.

The PSEL.RXD, PSEL.CTS, PSEL.RTS, and PSEL.TXD registers and their configurations are only used as long as the UARTE is enabled, and retained only for the duration the device is in ON mode. PSEL.RXD, PSEL.RTS, PSEL.RTS, and PSEL.TXD must only be configured when the UARTE is disabled.

To secure correct signal levels on the pins by the UARTE when the system is in OFF mode, the pins must be configured in the GPIO peripheral as described in the following table.

Only one peripheral can be assigned to drive a particular GPIO pin at a time. Failing to do so may result in unpredictable behavior.

UARTE signal	UARTE pin	Direction	Output value
RXD	As specified in PSEL.RXD	Input	Not applicable
CTS	As specified in PSEL.CTS	Input	Not applicable
RTS	As specified in PSEL.RTS	Output	1
TXD	As specified in PSEL.TXD	Output	1

Table 57: GPIO configuration before enabling peripheral

6.34.9 Registers

Instances

Instance	Base address	Description
UARTE0	0x40002000	Universal asynchronous receiver/transmitter with EasyDMA, unit 0
UARTE1	0x40028000	Universal asynchronous receiver/transmitter with EasyDMA, unit 1

Register overview

Register	Offset	Description
TASKS_STARTRX	0x000	Start UART receiver
TASKS_STOPRX	0x004	Stop UART receiver
TASKS_STARTTX	0x008	Start UART transmitter
TASKS_STOPTX	0x00C	Stop UART transmitter
TASKS_FLUSHRX	0x02C	Flush RX FIFO into RX buffer
EVENTS_CTS	0x100	CTS is activated (set low). Clear To Send.
EVENTS_NCTS	0x104	CTS is deactivated (set high). Not Clear To Send.
EVENTS_RXDRDY	0x108	Data received in RXD (but potentially not yet transferred to Data RAM)
EVENTS_ENDRX	0x110	Receive buffer is filled up
EVENTS_TXDRDY	0x11C	Data sent from TXD
EVENTS_ENDTX	0x120	Last TX byte transmitted
EVENTS_ERROR	0x124	Error detected
EVENTS_RXTO	0x144	Receiver timeout
EVENTS_RXSTARTED	0x14C	UART receiver has started
EVENTS_TXSTARTED	0x150	UART transmitter has started
EVENTS_TXSTOPPED	0x158	Transmitter stopped
SHORTS	0x200	Shortcuts between local events and tasks
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt

Register	Offset	Description
INTENCLR	0x308	Disable interrupt
ERRORSRC	0x480	Error source This register is read/write one to clear.
ENABLE	0x500	Enable UART
PSEL.RTS	0x508	Pin select for RTS signal
PSEL.TXD	0x50C	Pin select for TXD signal
PSEL.CTS	0x510	Pin select for CTS signal
PSEL.RXD	0x514	Pin select for RXD signal
BAUDRATE	0x524	Baud rate. Accuracy depends on the HFCLK source selected.
RXD.PTR	0x534	Data pointer
RXD.MAXCNT	0x538	Maximum number of bytes in receive buffer
RXD.AMOUNT	0x53C	Number of bytes transferred in the last transaction
TXD.PTR	0x544	Data pointer
TXD.MAXCNT	0x548	Maximum number of bytes in transmit buffer
TXD.AMOUNT	0x54C	Number of bytes transferred in the last transaction
CONFIG	0x56C	Configuration of parity and hardware flow control

6.34.9.1 TASKS_STARTRX

Address offset: 0x000

Start UART receiver

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field		Value ID				Value																		Description									
A	W	TASKS_STARTRX																								Start UART receiver									
				Trigger				1																		Trigger task									

6.34.9.2 TASKS_STOPRX

Address offset: 0x004

Stop UART receiver

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field		Value ID				Value																		Description									
A	W	TASKS_STOPRX																								Stop UART receiver									
				Trigger				1																		Trigger task									

6.34.9.3 TASKS_STARTTX

Address offset: 0x008

Start UART transmitter

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTTX			Start UART transmitter
			Trigger	1	Trigger task

6.34.9.4 TASKS_STOPTX

Address offset: 0x00C

Stop UART transmitter

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STOPTX			Stop UART transmitter
			Trigger	1	Trigger task

6.34.9.5 TASKS_FLUSHRX

Address offset: 0x02C

Flush RX FIFO into RX buffer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_FLUSHRX			Flush RX FIFO into RX buffer
			Trigger	1	Trigger task

6.34.9.6 EVENTS_CTS

Address offset: 0x100

CTS is activated (set low). Clear To Send.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_CTS			CTS is activated (set low). Clear To Send.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.34.9.7 EVENTS_NCTS

Address offset: 0x104

CTS is deactivated (set high). Not Clear To Send.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_NCTS			CTS is deactivated (set high). Not Clear To Send.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.34.9.8 EVENTS_RXDRDY

Address offset: 0x108

Data received in RXD (but potentially not yet transferred to Data RAM)

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_RXDRDY			Data received in RXD (but potentially not yet transferred to Data RAM)
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.34.9.9 EVENTS_ENDRX

Address offset: 0x110

Receive buffer is filled up

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDRX			Receive buffer is filled up
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.34.9.10 EVENTS_TXDRDY

Address offset: 0x11C

Data sent from TXD

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TXDRDY			Data sent from TXD
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.34.9.11 EVENTS_ENDTX

Address offset: 0x120

Last TX byte transmitted

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDTX			Last TX byte transmitted
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.34.9.12 EVENTS_ERROR

Address offset: 0x124

Error detected

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ERROR			Error detected
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.34.9.13 EVENTS_RXTO

Address offset: 0x144

Receiver timeout

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_RXTO			Receiver timeout
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.34.9.14 EVENTS_RXSTARTED

Address offset: 0x14C

UART receiver has started

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_RXSTARTED			UART receiver has started
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.34.9.15 EVENTS_TXSTARTED

Address offset: 0x150

UART transmitter has started

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TXSTARTED			UART transmitter has started
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.34.9.16 EVENTS_TXSTOPPED

Address offset: 0x158

Transmitter stopped

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TXSTOPPED			Transmitter stopped
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.34.9.17 SHORTS

Address offset: 0x200

Shortcuts between local events and tasks

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				D C
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
C	RW	ENDRX_STARTRX			Shortcut between event ENDRX and task STARTRX
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
D	RW	ENDRX_STOPRX			Shortcut between event ENDRX and task STOPRX
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

6.34.9.18 INTEN

Address offset: 0x300

Enable or disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				L J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	CTS			Enable or disable interrupt for event CTS
			Disabled	0	Disable
			Enabled	1	Enable
B	RW	NCTS			Enable or disable interrupt for event NCTS
			Disabled	0	Disable
			Enabled	1	Enable

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				L J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
C	RW	RXDRDY			Enable or disable interrupt for event RXDRDY
			Disabled	0	Disable
			Enabled	1	Enable
D	RW	ENDRX			Enable or disable interrupt for event ENDRX
			Disabled	0	Disable
			Enabled	1	Enable
E	RW	TXDRDY			Enable or disable interrupt for event TXDRDY
			Disabled	0	Disable
			Enabled	1	Enable
F	RW	ENDTX			Enable or disable interrupt for event ENDTX
			Disabled	0	Disable
			Enabled	1	Enable
G	RW	ERROR			Enable or disable interrupt for event ERROR
			Disabled	0	Disable
			Enabled	1	Enable
H	RW	RXTO			Enable or disable interrupt for event RXTO
			Disabled	0	Disable
			Enabled	1	Enable
I	RW	RXSTARTED			Enable or disable interrupt for event RXSTARTED
			Disabled	0	Disable
			Enabled	1	Enable
J	RW	TXSTARTED			Enable or disable interrupt for event TXSTARTED
			Disabled	0	Disable
			Enabled	1	Enable
L	RW	TXSTOPPED			Enable or disable interrupt for event TXSTOPPED
			Disabled	0	Disable
			Enabled	1	Enable

6.34.9.19 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				L J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	CTS			Write '1' to enable interrupt for event CTS
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	NCTS			Write '1' to enable interrupt for event NCTS
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	RXDRDY			Write '1' to enable interrupt for event RXDRDY
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				L J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
D	RW	ENDRX			Write '1' to enable interrupt for event ENDRX
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	TXDRDY			Write '1' to enable interrupt for event TXDRDY
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	ENDTX			Write '1' to enable interrupt for event ENDTX
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
G	RW	ERROR			Write '1' to enable interrupt for event ERROR
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
H	RW	RXTO			Write '1' to enable interrupt for event RXTO
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
I	RW	RXSTARTED			Write '1' to enable interrupt for event RXSTARTED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
J	RW	TXSTARTED			Write '1' to enable interrupt for event TXSTARTED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
L	RW	TXSTOPPED			Write '1' to enable interrupt for event TXSTOPPED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.34.9.20 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				L J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	CTS			Write '1' to disable interrupt for event CTS
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	NCTS			Write '1' to disable interrupt for event NCTS
			Clear	1	Disable
			Disabled	0	Read: Disabled

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID										L		J	I		H								G	F	E			D		C	B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
			Enabled	1	Read: Enabled
C	RW	RXDRDY			Write '1' to disable interrupt for event RXDRDY
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	ENDRX			Write '1' to disable interrupt for event ENDRX
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	TXDRDY			Write '1' to disable interrupt for event TXDRDY
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	ENDTX			Write '1' to disable interrupt for event ENDTX
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
G	RW	ERROR			Write '1' to disable interrupt for event ERROR
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
H	RW	RXTO			Write '1' to disable interrupt for event RXTO
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
I	RW	RXSTARTED			Write '1' to disable interrupt for event RXSTARTED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
J	RW	TXSTARTED			Write '1' to disable interrupt for event TXSTARTED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
L	RW	TXSTOPPED			Write '1' to disable interrupt for event TXSTOPPED
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.34.9.21 ERRORSRC

Address offset: 0x480

Error source

This register is read/write one to clear.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	OVERRUN			Overrun error
	W1C				A start bit is received while the previous data still lies in RXD. (Previous data is lost.)
			NotPresent	0	Read: error not present
			Present	1	Read: error present
B	RW	PARITY			Parity error
	W1C				A character with bad parity is received, if HW parity check is enabled.
			NotPresent	0	Read: error not present
			Present	1	Read: error present
C	RW	FRAMING			Framing error occurred
	W1C				A valid stop bit is not detected on the serial data input after all bits in a character have been received.
			NotPresent	0	Read: error not present
			Present	1	Read: error present
D	RW	BREAK			Break condition
	W1C				The serial data input is '0' for longer than the length of a data frame. (The data frame length is 10 bits without parity bit, and 11 bits with parity bit).
			NotPresent	0	Read: error not present
			Present	1	Read: error present

6.34.9.22 ENABLE

Address offset: 0x500

Enable UART

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	ENABLE			Enable or disable UARTE
			Disabled	0	Disable UARTE
			Enabled	8	Enable UARTE

6.34.9.23 PSEL.RTS

Address offset: 0x508

Pin select for RTS signal

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.34.9.24 PSEL.TXD

Address offset: 0x50C

Pin select for TXD signal

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**

ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.34.9.25 PSEL.CTS

Address offset: 0x510

Pin select for CTS signal

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**

ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.34.9.26 PSEL.RXD

Address offset: 0x514

Pin select for RXD signal

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**

ID	R/W	Field	Value ID	Value	Description
A	RW	PIN		[0..31]	Pin number
B	RW	PORT		[0..1]	Port number
C	RW	CONNECT			Connection
			Disconnected	1	Disconnect
			Connected	0	Connect

6.34.9.27 BAUDRATE

Address offset: 0x524

Baud rate. Accuracy depends on the HFCLK source selected.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x04000000				**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	BAUDRATE																																	
			Baud1200	0x0004F000																															
			Baud2400	0x0009D000																															
			Baud4800	0x0013B000																															
			Baud9600	0x00275000																															
			Baud14400	0x003AF000																															
			Baud19200	0x004EA000																															
			Baud28800	0x0075C000																															
			Baud31250	0x00800000																															
			Baud38400	0x009D0000																															
			Baud56000	0x00E50000																															
			Baud57600	0x00EB0000																															
			Baud76800	0x013A9000																															
			Baud115200	0x01D60000																															
			Baud230400	0x03B00000																															
			Baud250000	0x04000000																															
			Baud460800	0x07400000																															
			Baud921600	0x0F000000																															
			Baud1M	0x10000000																															

6.34.9.28 RXD

RXD EasyDMA channel

6.34.9.28.1 RXD.PTR

Address offset: 0x534

Data pointer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	PTR																																	
				See the memory chapter for details about which memories are available for EasyDMA.																																

6.34.9.28.2 RXD.MAXCNT

Address offset: 0x538

Maximum number of bytes in receive buffer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	MAXCNT		[0..0xFFFF]																															
				Maximum number of bytes in receive buffer																															

6.34.9.28.3 RXD.AMOUNT

Address offset: 0x53C

Number of bytes transferred in the last transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	AMOUNT		[0..0xFFFF]	Number of bytes transferred in the last transaction

6.34.9.29 TXD

TXD EasyDMA channel

6.34.9.29.1 TXD.PTR

Address offset: 0x544

Data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.34.9.29.2 TXD.MAXCNT

Address offset: 0x548

Maximum number of bytes in transmit buffer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	MAXCNT		[0..0xFFFF]	Maximum number of bytes in transmit buffer

6.34.9.29.3 TXD.AMOUNT

Address offset: 0x54C

Number of bytes transferred in the last transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	AMOUNT		[0..0xFFFF]	Number of bytes transferred in the last transaction

6.34.9.30 CONFIG

Address offset: 0x56C

Configuration of parity and hardware flow control

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					C B B B A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	HWFC			Hardware flow control
			Disabled	0	Disabled
			Enabled	1	Enabled
B	RW	PARITY			Parity
			Excluded	0x0	Exclude parity bit
			Included	0x7	Include even parity bit
C	RW	STOP			Stop bits
			One	0	One stop bit
			Two	1	Two stop bits

6.34.10 Electrical specification

6.34.10.1 UARTE electrical specification

Symbol	Description	Min.	Typ.	Max.	Units
f_{UARTE}	Baud rate for UARTE[43].			1000	kbps
$t_{UARTE,CTSH}$	CTS high time	1			µs
$t_{UARTE,START}$	Time from STARTRX/STARTTX task to transmission started		1		µs

6.35 USBD — Universal serial bus device

The USB device (USBD) controller implements a full speed USB device function that meets 2.0 revision of the USB specification.

[43] High baud rates may require GPIOs to be set as High Drive, see GPIO chapter for more details.

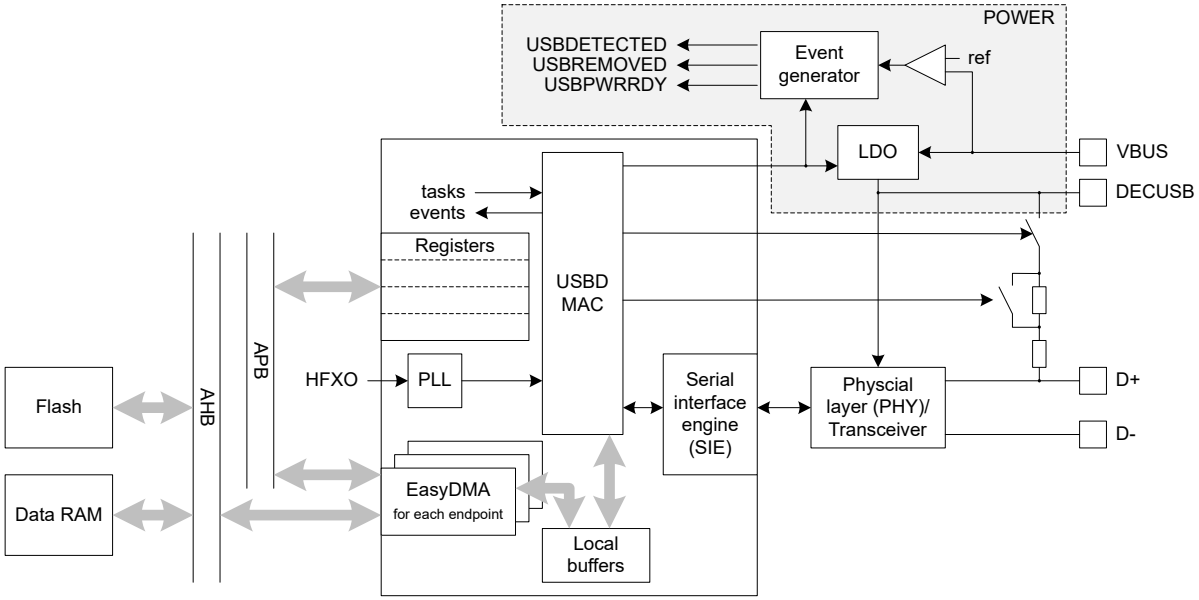


Figure 200: USB device block diagram

Listed here are the main features for USBD:

- Full-speed (12 Mbps) device fully compliant to Universal Serial Bus Specification Revision 2.0, including following engineering change notices (ECNs) issued by USB Implementers Forum:
  - *Pull-up/pull-down Resistors ECN*
  - *5V Short Circuit Withstand Requirement Change ECN*
- USB device stack available in the Nordic SDK
- Integrated (on-chip) USB transceiver (PHY)
- Software controlled on-chip pull-up on D+
- Endpoints:
  - Two control (1 IN, 1 OUT)
  - 14 bulk/interrupt (7 IN, 7 OUT)
  - Two isochronous (1 IN, 1 OUT)
- Double buffering for isochronous (ISO) endpoints (IN/OUT) support
- USB suspend, resume, and remote wake-up support
- 64 bytes buffer size for each bulk/interrupt endpoint
- Up to 1023 bytes buffer size for ISO endpoints
- EasyDMA for all data transfers

6.35.1 USB device states

The behavior of a USB device can be modelled through a state diagram.

The *USB 2.0 Specification* (see *Chapter 9 USB Device Framework*) defines a number of states for a USB device, as shown in the following figure.

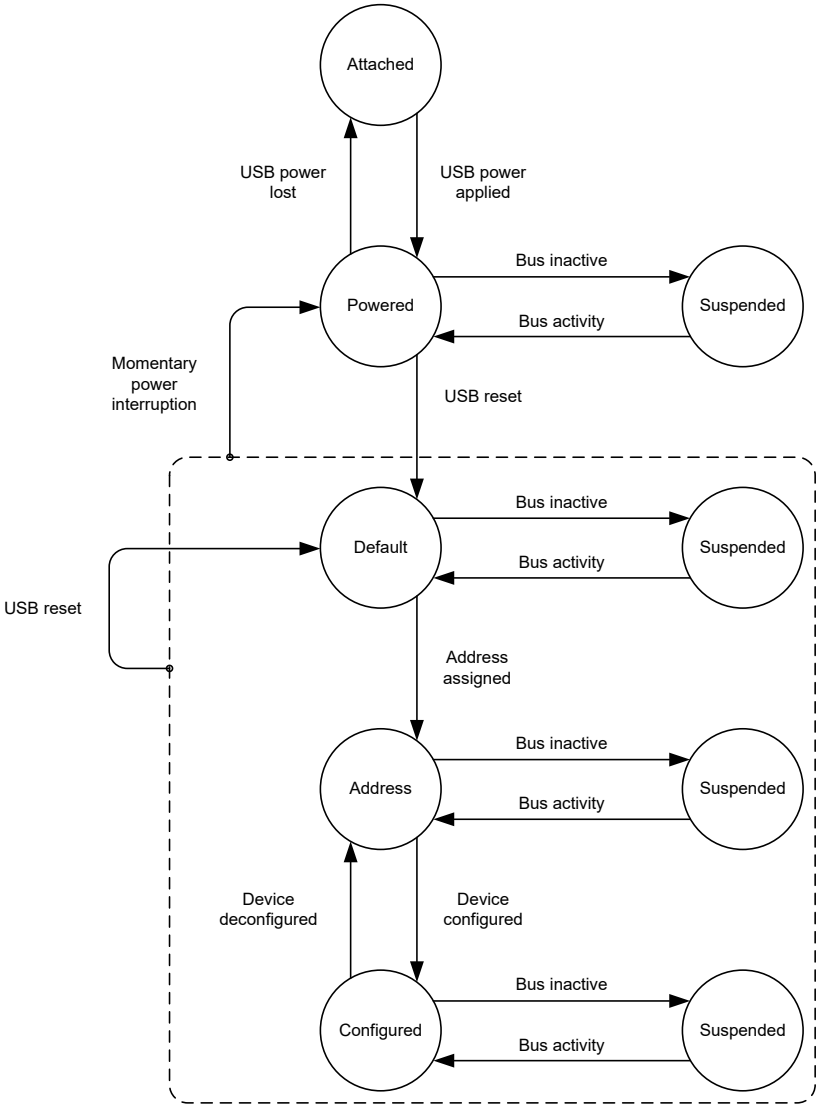


Figure 201: Device state diagram

The device must change state according to host-initiated traffic and USB bus states. It is up to the software to implement a state machine that matches the above definition. To detect the presence or absence of USB supply (VBUS), two events USBDETECTED and USBREMOVED can be used to implement the state machine. For more details on these events, see USB supply on page 86.

As a general rule when implementing the software, the host behavior shall never be assumed to be predictable. In particular the sequence of commands received during an enumeration. The software shall always react to the current bus conditions or commands sent by the host.

6.35.2 USB terminology

The USB specification defines bus states, rather than logic levels on the D+ and D- lines.

For a full speed device, the bus state where the D+ line is high and the D- line is low is defined as the J state. The bus state where D+ is low and D- high is called the K state.

An idle bus, where D+ and D- lines are only polarized through the pull-up on D+ and pull-downs on the host side, will be in J state.

Both lines low are called SE0 (single-ended 0), and both lines high SE1 (single-ended 1).

6.35.3 USB pins

The USBD peripheral features a number of dedicated pins.

The dedicated USB pins can be grouped in two categories, signal and power. The signal pins consist of the D+ and D- pins, which are to be connected to the USB host. They are dedicated pins, and not available as standard GPIOs. The USBD peripheral is implemented according to the USB specification revision 2.0, *5V Short Circuit Withstand ECN Requirement Change*, meaning these two pins are not 5 V tolerant.

The signal pins and the pull-up will operate only while VBUS is in its valid voltage range, and USBD is enabled through the ENABLE register. For details on the USB power supply and VBUS detection, see USB supply on page 86.

For more information about the pinout, see Pin assignments on page 926.

6.35.4 USBD power-up sequence

The physical layer interface (PHY)/USB transceiver is powered separately from the rest of the device (VBUS pin), which has some implications on the USBD power-up sequence.

The device is not able to properly signal its presence to the USB host and handle traffic from the host, unless the PHY's power supply is enabled and stable. Turning the PHY's power supply on/off is directly linked to register ENABLE. The device provides events that help synchronizing software to the various steps during the power-up sequence.

To make sure that all resources in USBD are available and the dedicated USB voltage regulator stabilized, the following is recommended:

- Enable USBD only after VBUS has been detected
- Turn the USB pull-up on after the following events have occurred:
  - USBPWRRDY
  - USBEVENT, with the READY condition flagged in EVENTCAUSE

The following sequence chart illustrates a typical handling of VBUS power-up:

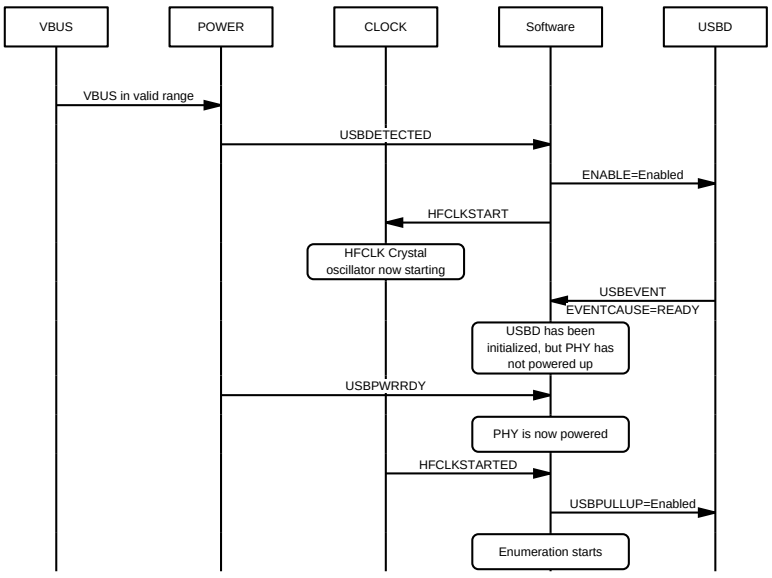


Figure 202: VBUS power-up sequence

Upon detecting VBUS removal, it is recommended to wait for ongoing EasyDMA transfers to finish before disabling USBD (relevant ENDEPIN[n], ENDISOIN, ENDEPOUT[n], or ENDISOOUT events, see EasyDMA on page 855). The USBREMOVED event, described in USB supply on page 86, signals when the VBUS is removed. Reading the ENABLE register will return Enabled until USBD is completely disabled.

6.35.5 USB pull-up

The USB pull-up serves two purposes: it indicates to the host that the device is connected to the USB bus, and it indicates the device's speed capability.

When no pull-up is connected to the USB bus, the host sees both D+ and D- lines low, as they are pulled down on the host side by 15 kΩ resistors. The device is not detected by the host, putting it in a detached state even if it is physically connected to the host. In this situation, the device is not allowed to draw current from VBUS, according to *USB 2.0 Specification*.

When a full-speed device connects its 1.5 kΩ pull-up to D+, the host sees the corresponding line high. The device is then in the attached state. During the enumeration process, the host attempts to determine if the full-speed device also supports higher speeds and initiates communication with the device to further identify it. The USBD peripheral implemented in this device supports only full-speed operation (12 Mbps), and thus ignores the negotiation for higher speeds in accordance with *USB 2.0 Specification*.

Register USBPULLUP enables software to connect or disconnect the pull-up on D+. This allows the software to control when USB enumeration takes place. It also allows to emulate a physical disconnect from the USB bus, for instance when re-enumeration is required. USBPULLUP has to be enabled to allow the USBD to handle USB traffic and generate appropriate events. This forbids the use of an external pull-up.

Note that disconnecting the pull-up through register USBPULLUP while connected to a host, will result in both D+ and D- lines to be pulled low by the host's pull-down resistors. However, as mentioned above, this will also inhibit the generation of the USBRESET event. The pull-up is disabled by default after a chip reset.

The pull-up shall only get connected after USBD has been enabled through register ENABLE. The USB pull-up value is automatically changed depending on the bus activity, as specified in *Resistor ECN* which amends the original *USB 2.0 Specification*. The user does not have access to this function as it is handled in hardware.

While they should never be used in normal traffic activity, lines D+ and D- may at any time be forced into state specified in register DPDMVALUE by the task DPDMDRIVE. The DPDMNODRIVE task stops driving them, and PHY returns to normal operation.

6.35.6 USB reset

The USB specification defines a USB reset, which is not be confused with a chip reset. The USB reset is a normal USB bus condition, and is used as part of the enumeration sequence, it does not reset the chip.

The USB reset results from a single-ended low state (SE0) on lines D+/D- for a $t_{USB,DETRST}$ amount of time. Only the host is allowed to drive a USB reset condition on the bus. The UBSD peripheral automatically interprets a SE0 longer than $t_{USB,DETRST}$ as a USB reset. When the device detects a USB reset and generates a USBRESET event, the device USB stack and related parts of the application shall re-initialize themselves, and go back to the default state.

Some of the registers in the USBD peripheral get automatically reset to a known state, in particular all data endpoints are disabled and the USBADDR reset to 0.

After the device has connected to the USB bus (i.e. after VBUS is applied), the device shall not respond to any traffic from the time the pull-up is enabled until it has seen a USB reset condition. This is automatically ensured by the USBD.

After a USB reset, the device shall be fully responsive after at most T_{RSTRCY} (according to chapter 7 in the USB specification). Software shall take into account this time that takes the hardware to recover from a USB reset condition.

6.35.7 USB suspend and resume

Normally, the host will maintain activity on the USB at least every millisecond according to USB specification. A USB device will enter suspend when there is no activity on the bus (idle) for a given time. The device will resume operation when it receives any non idle signalling.

To signal that the device shall go into low power mode (suspend), the host stops activity on the USB bus, which becomes idle. Only the device pull-up and host pull-downs act on D+ and D-, and the bus is thus kept at a constant J state. It is up to the device to detect this lack of activity, and enter the low power mode (suspend) within a specified time.

The USB host can decide to suspend or resume USB activity at any time. If remote wake-up is enabled, the device may signal to the host to resume from suspend.

6.35.7.1 Entering suspend

The USBD peripheral automatically detects lack of activity for more than a defined amount of time, and performs steps needed to enter suspend.

When no activity has been detected for longer than $t_{USB,SUSPEND}$, the USBD generates the USBEVENT event with SUSPEND bit set in register EVENTCAUSE. The software shall ensure that the current drawn from the USB supply line VBUS is within the specified limits before T_{2SUSP}, as defined in chapter 7 of the USB specification. In order to reduce idle current of USBD, the software must explicitly place the USBD in low power mode through writing `LowPower` to register LOWPOWER.

In order to save power, and provided that no other peripheral needs it, the crystal oscillator (HFXO) in CLOCK may be disabled by software during the USB suspend, while the USB pull-up is disconnected, or when VBUS is not present. Software must explicitly enable it at any other time. The USBD will not be able to respond to USB traffic unless HFXO is enabled and stable.

6.35.7.2 Host-initiated resume

Once the host resumes the bus activity, it has to be responsive to incoming requests on the USB bus within the time T_{RSMRCY} (as defined in chapter 7 of the USB specification) and revert to normal power consumption mode.

If the host resumes bus activity with or without a RESUME condition (in other words: bus activity is defined as any non-J state), the USBD peripheral will generate a USBEVENT event, with RESUME bit set in register EVENTCAUSE. If the host resumes bus activity simply by restarting sending frames, the USBD peripheral will generate SOF events.

6.35.7.3 Device-initiated remote wake-up

Assuming the remote wake-up is supported by the device and enabled by the host, the device can request the host to resume from suspend if wake-up condition is met.

To do so, the HFXO needs to be enabled first. After waking up the HFXO, the software must bring USBD out of the low power mode and into the normal power consumption mode through writing `ForceNormal` in register LOWPOWER. It can then instruct the USBD peripheral to drive a RESUME condition (K state) on the USB bus by triggering the DPDMDRIVE task, and hence attempt to wake up the host. By choosing `Resume` in DPDMVALUE, the duration of the RESUME state is under hardware control ($t_{USB,DRIVEK}$). By choosing `J` or `K`, the duration of that state is under software control (the J or K state is maintained until a DPDMNODRIVE task is triggered) and has to meet T_{DRSMUP} as specified in USB specification chapter 7.

Upon writing the `ForceNormal` in register LOWPOWER, a USBEVENT event is generated with the USBWUALLOWED bit set in register EVENTCAUSE.

The value in register DPDMVALUE on page 902 will only be captured and used when the DPDMDRIVE task is triggered. This value defines the state the bus will be forced into after the DPDMDRIVE task.

The device shall ensure that it does not initiate a remote wake-up request before T_{WTRSM} (according to USB specification chapter 7) after the bus has entered idle state. Using the recommended resume value in DPDMVALUE (rather than K) takes care of this, and postpones the RESUME state accordingly.

6.35.8 EasyDMA

The USBD peripheral implements EasyDMA for accessing memory without CPU involvement.

Each endpoint has an associated set of registers, tasks and events. EasyDMA and traffic on USB are tightly related. A number of events provide insight of what is happening on the USB bus with a number of tasks allowing an automated response to the traffic.

> **Note:** Endpoint 0 (IN and OUT) are implemented as control endpoint. For more information, see Control transfers on page 856.

Registers

Enabling endpoints is controlled through the EPINEN and EPOUTEN registers.

The following registers define the memory address of the buffer for a specific IN or OUT endpoint:

- EPIN[n].PTR, (n=0..7)
- EPOUT[n].PTR, (n=0..7)
- ISOIN.PTR
- ISOOUT.PTR

The following registers define the amount of bytes to be sent on USB for next transaction:

- EPIN[n].MAXCNT, (n=0..7)
- ISOIN.MAXCNT

The following registers define the length of the buffer (in bytes) for next transfer of incoming data:

- EPOUT[n].MAXCNT, (n=1..7)
- ISOOUT.MAXCNT

Since the host decides how many bytes are sent over USB, the MAXCNT value can be copied from register SIZE.EPOUT[n] (n=1..7) or register SIZE.ISOOUT.

Register EPOUT[0].MAXCNT defines the length of the OUT buffer (in bytes) for the control endpoint 0. Register SIZE.EPOUT[0] shall indicate the same value as `MaxPacketSize` from the device descriptor or `wLength` from the SETUP command, whichever is the least.

The .AMOUNT registers indicate how many bytes actually have been transferred over EasyDMA during the last transfer.

Stalling bulk/interrupt endpoints is controlled through the EPSTALL register.

> **Note:** Due to USB specification requirements, the effect of the stalling control endpoint 0 may be overridden by hardware, in particular when a new SETUP token is received.

EasyDMA will not copy the SETUP data to memory (it will only transfer data from the data stage). The following are separate registers in the USBD peripheral that have setup data.

- BMREQUESTTYPE
- BREQUEST
- WVALUEL
- WVALUEH
- WINDEXL
- WINDEXH

- WLENGTHL
- WLENGTHH

The EVENTCAUSE register provides details on what caused a given USBEVENT event, for instance if a CRC error is detected during a transaction, or if bus activity stops or resumes.

Tasks

Tasks STARTEPIN[n], STARTEPOUT[n] (n=0..7), STARTISOIN, and STARTISOOUT capture the values for .PTR and .MAXCNT registers. For IN endpoints, a transaction over USB gets automatically triggered when the EasyDMA transfer is complete. For OUT endpoints, it is up to software to allow the next transaction over USB. See the examples in Control transfers on page 856, Bulk and interrupt transactions on page 859, and Isochronous transactions on page 861.

For the control endpoint 0, OUT transactions are allowed through the EP0RCVOUT task. The EP0STATUS task allows a status stage to be initiated, and the EP0STALL task allows stalling further traffic (data or status stage) on the control endpoint.

Events

The STARTED event confirms that the values of the .PTR and .MAXCNT registers of the endpoints flagged in register EPSTATUS have been captured. Those can then be modified by software for the next transfer.

Events ENDEPIN[n], ENDEPOUT[n] (n=0..7), ENDISOIN, and ENDISOOUT events indicate that the entire buffer has been consumed. The buffer can be accessed safely by the software.

Only a single EasyDMA transfer can take place in USBD at any time. Software must ensure that tasks STARTEPIN[n] (n=0..7), STARTISOIN , STARTEPOUT[n] (n=0..7), or STARTISOOUT are not triggered before events ENDEPIN[n] (n=0..7), ENDISOIN, ENDEPOUT[n] (n=0..7), or ENDISOOUT are received from an on-going transfer.

The EPDATA event indicates that a successful (acknowledged) data transaction has occurred on the data endpoint(s) flagged in register EPDATASTATUS. A successful (acknowledged) data transaction on endpoint 0 is signalled by the EP0DATADONE event.

At any time a USBEVENT event may be sent, with details provided in EVENTCAUSE register.

The EP0SETUP event indicates that a SETUP token has been received on the control endpoint 0, and that the setup data is available in the setup data registers.

6.35.9 Control transfers

The USB specification mandates every USB device to implement endpoint 0 IN and OUT as control endpoints.

A control transfer consists of two or three stages:

- Setup stage
- Data stage (optional)
- Status stage

Each control transfer can be one of following types:

- Control read
- Control read no data
- Control write
- Control write no data

An EP0SETUP event indicates that the data in the setup stage (following the SETUP token) is available in registers.

The data in the data stage (following the IN or OUT token) is transferred from or to the desired location using EasyDMA.

The control endpoint buffer can be of any size.

After receiving the SETUP token, the USB controller will not accept (NAK) any incoming IN or OUT tokens until the software has finished decoding the command, determined the type of transfer, and prepared for the next stage (data or status) appropriately.

The software can stall a command when in the data and status stages, through the EP0STALL task, when the command is not supported or if its wValue, wIndex or wLength parameters are wrong. The following shows a stalled control read transfer, but the same mechanism (tasks) applies to stalling a control write transfer.

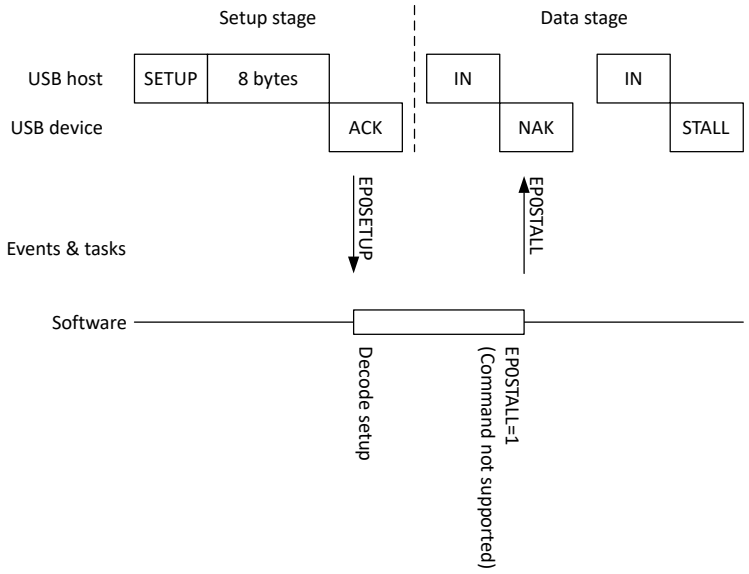


Figure 203: Control read gets stalled

See the *USB 2.0 Specification* and relevant class specifications for rules on stalling commands.

Note: The USBD peripheral handles the SetAddress transfer by itself. As a consequence, the software shall not process this command other than updating its state machine (see Device state diagram), nor initiate a status stage. If necessary, the address assigned by the host can be read out from the USBADDR register after the command has been processed.

6.35.9.1 Control read transfer

This section describes how the software behaves when responding to a control read transfer.

As mentioned earlier, the USB controller will not accept (NAK) any incoming IN tokens until software has finished decoding the command, determining the type of transfer, and preparing for the next stage (data or status) appropriately.

For a control read, transferring the data from memory into USBD will trigger a valid, acknowledged (ACK) IN transaction on USB.

The software has to prepare EasyDMA by pointing to the buffer containing the data to be transferred. If no other EasyDMA transfers are on-going with USBD, the software can send the STARTEPIN0 task, which will initiate the data transfer and transaction on USB.

A STARTED event (with EPIN0 bit set in the EPSTATUS register) will be generated as soon as the EPIN[0].PTR and .MAXCNT registers have been captured. Software may then prepare them for the next data transaction.

An ENDEPIN[0] event will be generated when the data has been transferred from memory to the USBD peripheral.

Finally, an EP0DATADONE event will be generated when the data has been transmitted over USB and acknowledged by the host.

The software can then either prepare and transmit the next data transaction by repeating the above sequence, or initiate the status stage through the EP0STATUS task.

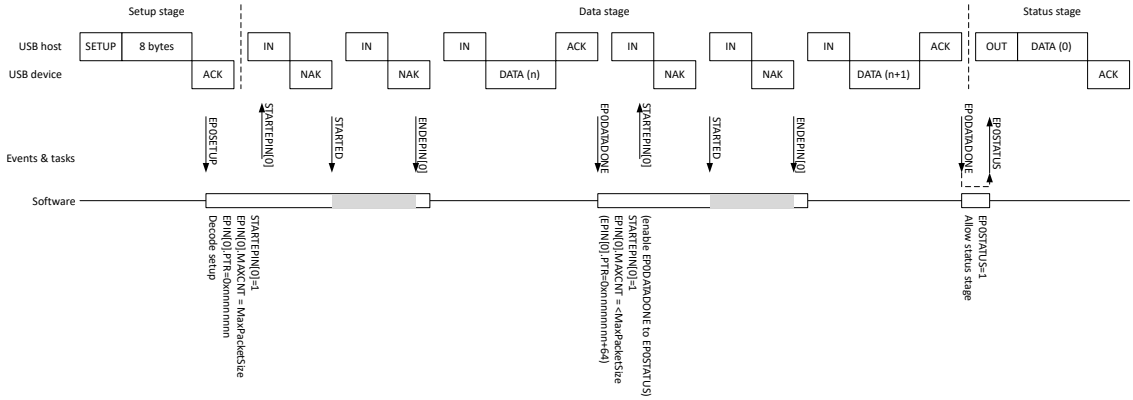


Figure 204: Control read transfer

It is possible to enable a shortcut from the EP0DATADONE event to the EP0STATUS task, typically if the data stage is expected to take a single transfer. If there is no data stage, the software can initiate the status stage through the EP0STATUS task right away, as as shown in the following figure.

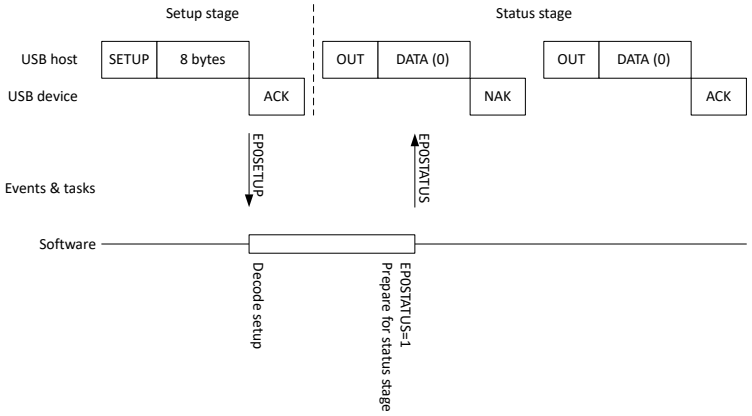


Figure 205: Control read no data transfer

6.35.9.2 Control write transfer

This section describes how the software responds to a control write transfer.

The software has to prepare EasyDMA by pointing to the buffer in memory that shall contain the incoming data. If no other EasyDMA transfers are ongoing with USBD, the software can then send the EP0RCVOUT task, which will make USBD acknowledge (ACK) the first OUT+DATA transaction from the host.

An EP0DATADONE event will be generated when a new OUT+DATA has been transmitted over USB, and is about to get acknowledged by the device.

After receiving the first transaction, a STARTED event (the EPOUT0 bit set in the EPSTATUS register) is generated when the EPOUT[0].PTR and .MAXCNT registers have been captured. Software may then prepare them for the next data transaction.

An ENDEPOUT[0] event will be generated when the data has been transferred from the USBD peripheral to memory. The software can then either prepare to receive the next data transaction by repeating the above sequence, or initiate the status stage through the EP0STATUS task. Until then, further incoming OUT +DATA transactions get a NAK response by the device.

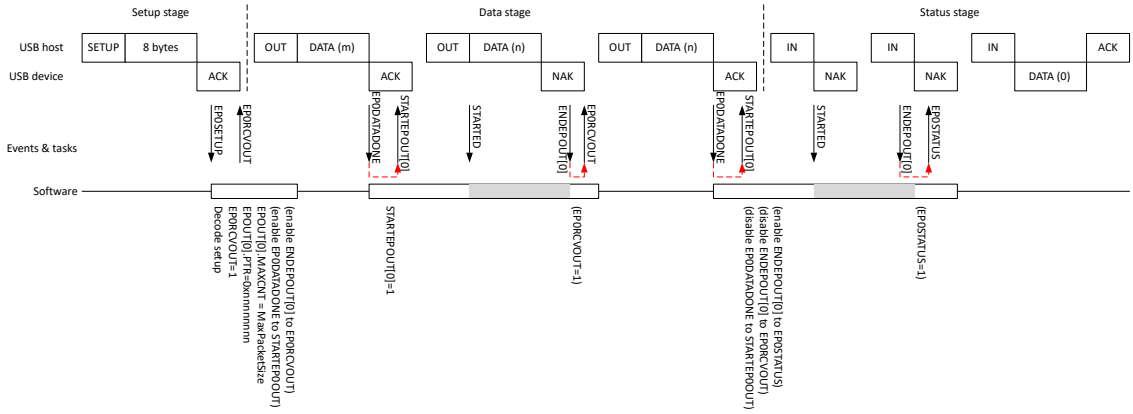


Figure 206: Control write transfer

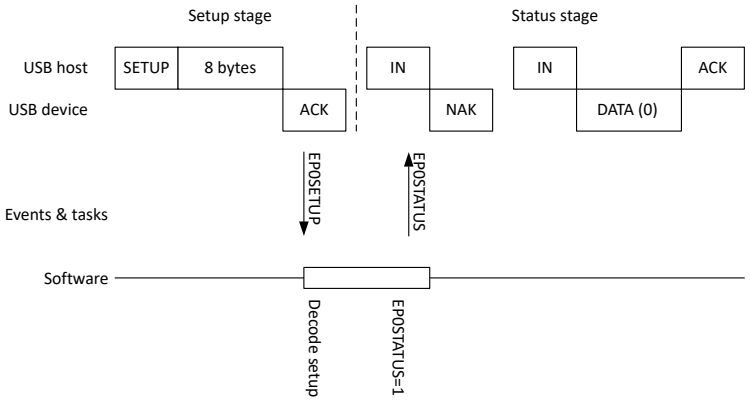


Figure 207: Control write no data transfer

6.35.10 Bulk and interrupt transactions

The USBD peripheral implements seven pairs of bulk/interrupt endpoints.

The bulk/interrupt endpoints have a fixed USB endpoint number, summarized in the following table.

Bulk endpoint #	USB IN endpoint	USB OUT endpoint
[1]	0x81	0x01
[2]	0x82	0x02
[3]	0x83	0x03
[4]	0x84	0x04
[5]	0x85	0x05
[6]	0x86	0x06
[7]	0x87	0x07

Table 58: Bulk/interrupt endpoint numbering

A bulk/interrupt transaction consists of a single data stage. Two consecutive, successful transactions are distinguished through alternating leading process ID (PID): DATA0 follows DATA1, DATA1 follows DATA0, etc. A repeated transaction is detected by re-using the same PID as previous transaction, i.e DATA0 follows DATA0, or DATA1 follows DATA1.

The USBD controller automatically toggles DATA0/DATA1 PIDs for every bulk/interrupt transaction.

If incoming data is corrupted (CRC does not match), the USBD controller automatically prevents DATA0/DATA1 from toggling, to request the host to resend the data.

In some specific cases, the software may want to force a data toggle (usually reset) on a specific IN endpoint, or force the expected toggle on an OUT endpoint, for instance as a consequence of the host issuing **`ClearFeature`**, **`SetInterface`**, or selecting an alternate setting. Controlling the data toggle of data IN or OUT endpoint n (n=1..7) is done through register DTOGGLE.

The bulk/interrupt transaction in USB full-speed can be of any size up to 64 bytes. It must be a multiple of four bytes and 32-bit aligned in memory.

When the USB transaction has completed, an EPDATA event is generated. Until new data has been transferred by EasyDMA from memory to the USBD peripheral (signalled by the ENDEPIN[n] event), the hardware will automatically respond with NAK to all incoming IN tokens. Software has to configure and start the EasyDMA transfer once it is ready to send more data.

Each IN or OUT data endpoint has to be explicitly enabled by software through register EPINEN or EPOUTEN, according to the configuration declared by the device and selected by the host through the **`SetConfig`** command.

A disabled data endpoint will not respond to any traffic from the host. An enabled data endpoint will normally respond NAK or ACK (depending on the readiness of the buffers), or STALL (if configured in register EPSTALL), in which case the endpoint is asked to halt. The halted (or not) state of a given endpoint can be read back from register HALTED.EPIN[n] or HALTED.EPOUT[n]. The format of the returned 16-bit value can be copied as is, as a response to a **`GetStatusEndpoint`** request from the host.

Enabling or disabling an endpoint will not change its halted state. However, a USB reset will disable and clear the halted state of all data endpoints.

The control endpoint 0 IN and OUT can also be enabled and/or halted using the same mechanisms, but due to USB specification, receiving a SETUP will override its state.

6.35.10.1 Bulk and interrupt IN transaction

The host issues IN tokens to receive bulk/interrupt data. In order to send data, the software has to enable the endpoint and prepare an EasyDMA transfer on the desired endpoint.

Bulk/interrupt IN endpoints are enabled or disabled through their respective INn bit (n=1..7) in EPINEN register.

It is also possible to stall or resume communication on an endpoint through the EPSTALL register.

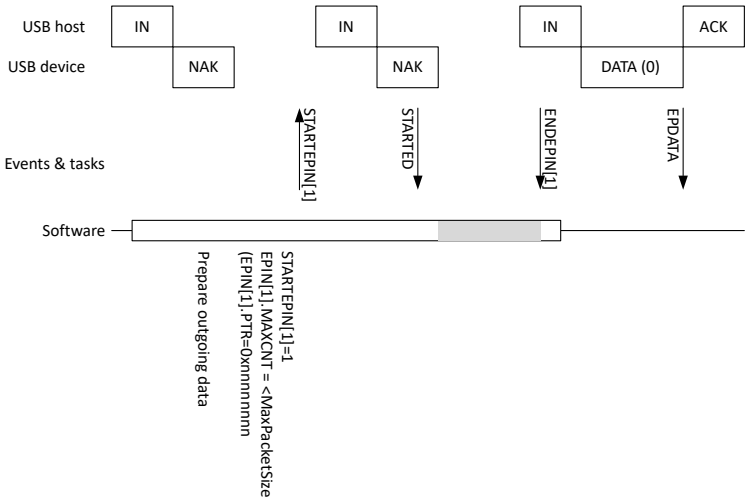


Figure 208: Bulk/interrupt IN transaction

It is possible (and in some situations it is required) to respond to an IN token with a zero-length data packet.

> **Note:** On many USB hosts, not responding (DATA+ACK or NAK) to three IN tokens on an interrupt endpoint would have the host disable that endpoint as a consequence. Re-enumerating the device (unplug-replug) may be required to restore functionality. Make sure that the relevant data endpoints are enabled for normal operation as soon as the device gets configured through a **`SetConfig`** request.

6.35.10.2 Bulk and interrupt OUT transaction

When the host wants to transmit bulk/interrupt data, it issues an OUT token (packet) followed by a DATA packet on a given endpoint n (n=1..7).

A NAK is returned until the software writes any value to register SIZE.EPOUT[n], indicating that the content of the local buffer can be overwritten. Upon receiving the next OUT+DATA transaction, an ACK is returned to the host while an EPDATA event is generated (and the EPDATASTATUS register flags are set to indicate on which endpoint this happened). Once the EasyDMA is prepared and enabled, by writing the EPOUT[n] registers and triggering the STARTEPOUT[n] task, the incoming data will be transferred to memory. Until that transfer is finished, the hardware will automatically NAK any other incoming OUT+DATA packets. Only when the EasyDMA transfer is done (signalled by the ENDEPOUT[n] event), or as soon as any values are written by the software in register SIZE.EPOUT[n], the endpoint n will accept incoming OUT+DATA again.

It is allowed for the host to send zero-length data packets.

Bulk/interrupt OUT endpoints are enabled or disabled through their respective OUTn bit (n=1..7) in the EPOUTEN register. It is also possible to stall or resume communication on an endpoint through the EPSTALL register.

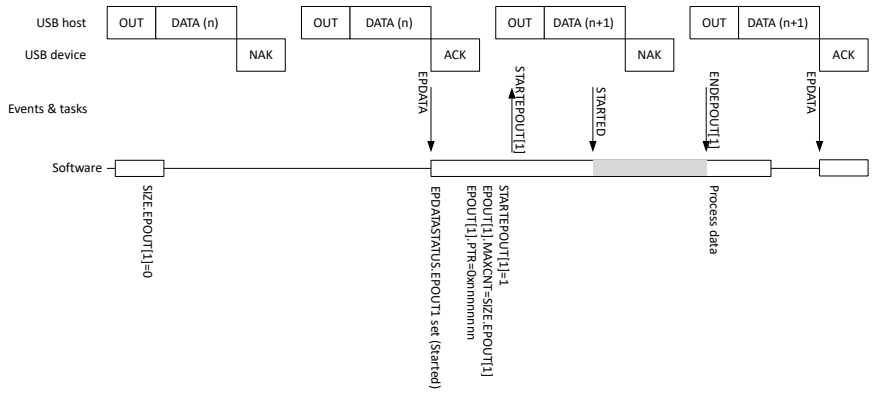


Figure 209: Bulk/interrupt OUT transaction

6.35.11 Isochronous transactions

The USBD peripheral implements isochronous (ISO) endpoints.

The ISO endpoints have a fixed USB endpoint number, summarized in the following table.

ISO endpoint #	USB IN endpoint	USB OUT endpoint
[0]	0x88	0x08

Table 59: Isochronous endpoint numbering

An isochronous transaction consists of a single, non-acknowledged data stage. The host sends out a start of frame at a regular interval (1 ms), and data follows IN or OUT tokens within each frame.

EasyDMA allows transferring ISO data directly from and to memory. EasyDMA transfers must be initiated by the software, which can synchronize with the SOF (start of frame) events.

Because the timing of the start of frame is very accurate, the SOF event can be used for jobs such as synchronizing a local timer through the SOF event and PPI. The SOF event gets synchronized to the 16 MHz clock prior to being made available to the PPI.

Every start of frame increments a free-running counter, which can be read by software through the FRAMECNTR register.

Each IN or OUT ISO data endpoint has to be explicitly enabled by software through register EPINEN or EPOUTEN, according to the configuration declared by the device and selected by the host through the **`SetConfig`** command. A disabled ISO IN data endpoint will not respond to any traffic from the host. A disabled ISO OUT data endpoint will ignore any incoming traffic from the host.

The USBD peripheral has an internal 1 kB buffer associated with ISO endpoints. The user can either allocate the full amount to the IN or the OUT endpoint, or split the buffer allocation between the two using register ISOSPLIT.

The internal buffer also sets the maximum size of the ISO OUT and ISO IN transfers: 1023 bytes when the full buffer is dedicated to either ISO OUT or ISO IN, and half when the buffer is split between the two.

6.35.11.1 Isochronous IN transaction

When the host wants to receive isochronous (ISO) data, it issues an IN token on the isochronous endpoint.

After the data has been transferred using the EasyDMA, the USB controller on the isochronous IN endpoint responds to the IN token with the transferred data using the ISOIN.MAXCNT for the size of the packet.

The ISO IN data endpoint has to be explicitly enabled by software through the ISOIN0 bit in register EPINEN.

When an ISO IN endpoint is enabled and no data transferred with EasyDMA, the response of the USBD depends on the setting of the RESPONSE field in register ISOINCONFIG. It can either provide no response to an IN token or respond with a zero-length data.

If the EasyDMA transfer on the isochronous endpoint is not completed before the next SOF event, the result of the transfer is undefined.

The maximum size of an ISO IN transfer in USB full-speed is 1023 bytes. The data buffer has to be a multiple of 4 bytes 32-bit aligned in memory. However, the amount of bytes transferred on the USB data endpoint can be of any size (up to 1023 bytes, if not shared with an OUT ISO endpoint).

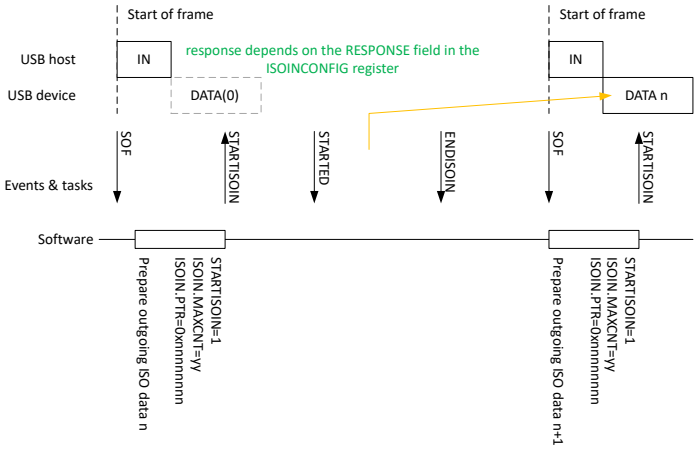


Figure 210: Isochronous IN transfer

6.35.11.2 Isochronous OUT transaction

When the host wants to send isochronous (ISO) data, it issues an OUT token on the isochronous endpoint, followed by data.

The ISO OUT data endpoint has to be explicitly enabled by software through the ISOOUT0 bit in register EPOUTEN.

The amount of last received ISO OUT data is provided in the SIZE.ISOOUT register. Software shall interpret the ZERO and SIZE fields as presented in the following table.

ZERO	SIZE	Last received data size
Normal	0	No data received at all
Normal	1..1023	1..1023 bytes of data received
ZeroData	(not of interest)	Zero-length data packet received

Table 60: ISO OUT incoming data size

When EasyDMA is prepared and started, triggering a STARTISOOUT task initiates an EasyDMA transfer to memory. Software shall synchronize ISO OUT transfers with the SOF events. EasyDMA uses the address in ISOOUT.PTR and size in ISOOUT.MAXCNT for every new transfer.

If the EasyDMA transfer on the isochronous endpoint is not completed before the next SOF event, the result of the transfer is undefined.

The maximum size of an isochronous OUT transfer in USB full-speed is 1023 bytes. The data buffer has to be a multiple of 4 bytes and 32-bit aligned in Data RAM. However, the amount of bytes transferred on the USB data endpoint can be of any size (up to 1023 bytes if not shared with an IN ISO endpoint).

If the last received ISO data packet is corrupted (wrong CRC), the USB controller generates an USBEVENT event (at the same time as SOF) and indicates a CRC error on ISOOUTCRC in register EVENTCAUSE. EasyDMA will transfer the data anyway if it has been set up properly.

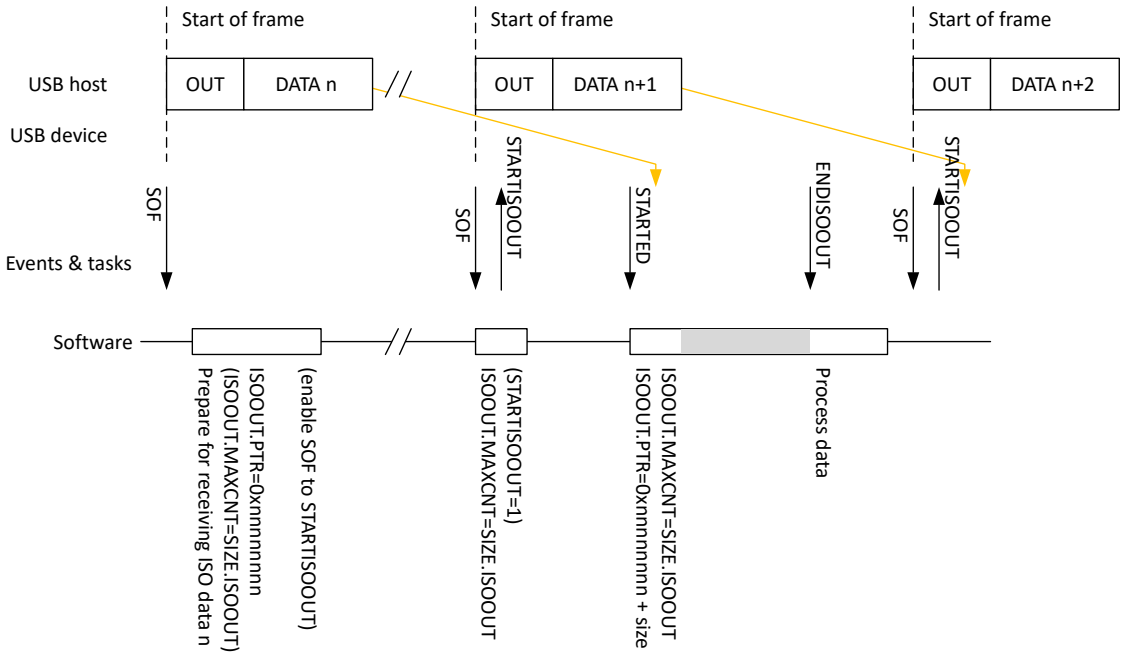


Figure 211: Isochronous OUT transfer

6.35.12 USB register access limitations

Some of the registers in USBD cannot be accessed in specific conditions.

This may be the case when USBD is not enabled (using the ENABLE register) and ready (signalled by the READY bit in EVENTCAUSE after a USBEVENT event), or when USBD is in low power mode while the USB bus is suspended.

Triggering any tasks, including the tasks triggered through the PPI, is affected by this behavior. In addition, the following registers are affected:

- HALTED.EPIN[0..7]
- HALTED.EPOUT[0..7]
- USBADDR
- BMREQUESTTYPE
- BREQUEST
- WVALUEL
- WVALUEH
- WINDEXL
- WINDEXH
- WLENGTHL
- WLENGTHH
- SIZE.EPOUT[0..7]
- SIZE.ISOOUT
- USBPULLUP
- DTOGGLE
- EPINEN
- EPOUTEN
- EPSTALL
- ISOSPLIT
- FRAMECNTR

6.35.13 Registers

Instances

Instance	Base address	Description
USBD	0x40027000	Universal serial bus device

Register overview

Register	Offset	Description
TASKS_STARTEPIN[0]	0x004	Captures the EPIN[0].PTR and EPIN[0].MAXCNT registers values, and enables endpoint IN 0 to respond to traffic from host
TASKS_STARTEPIN[1]	0x008	Captures the EPIN[1].PTR and EPIN[1].MAXCNT registers values, and enables endpoint IN 1 to respond to traffic from host
TASKS_STARTEPIN[2]	0x00C	Captures the EPIN[2].PTR and EPIN[2].MAXCNT registers values, and enables endpoint IN 2 to respond to traffic from host
TASKS_STARTEPIN[3]	0x010	Captures the EPIN[3].PTR and EPIN[3].MAXCNT registers values, and enables endpoint IN 3 to respond to traffic from host
TASKS_STARTEPIN[4]	0x014	Captures the EPIN[4].PTR and EPIN[4].MAXCNT registers values, and enables endpoint IN 4 to respond to traffic from host

Register	Offset	Description
TASKS_STARTEPIN[5]	0x018	Captures the EPIN[5].PTR and EPIN[5].MAXCNT registers values, and enables endpoint IN 5 to respond to traffic from host
TASKS_STARTEPIN[6]	0x01C	Captures the EPIN[6].PTR and EPIN[6].MAXCNT registers values, and enables endpoint IN 6 to respond to traffic from host
TASKS_STARTEPIN[7]	0x020	Captures the EPIN[7].PTR and EPIN[7].MAXCNT registers values, and enables endpoint IN 7 to respond to traffic from host
TASKS_STARTISOIN	0x024	Captures the ISOIN.PTR and ISOIN.MAXCNT registers values, and enables sending data on ISO endpoint
TASKS_STARTEPOUT[0]	0x028	Captures the EPOUT[0].PTR and EPOUT[0].MAXCNT registers values, and enables endpoint 0 to respond to traffic from host
TASKS_STARTEPOUT[1]	0x02C	Captures the EPOUT[1].PTR and EPOUT[1].MAXCNT registers values, and enables endpoint 1 to respond to traffic from host
TASKS_STARTEPOUT[2]	0x030	Captures the EPOUT[2].PTR and EPOUT[2].MAXCNT registers values, and enables endpoint 2 to respond to traffic from host
TASKS_STARTEPOUT[3]	0x034	Captures the EPOUT[3].PTR and EPOUT[3].MAXCNT registers values, and enables endpoint 3 to respond to traffic from host
TASKS_STARTEPOUT[4]	0x038	Captures the EPOUT[4].PTR and EPOUT[4].MAXCNT registers values, and enables endpoint 4 to respond to traffic from host
TASKS_STARTEPOUT[5]	0x03C	Captures the EPOUT[5].PTR and EPOUT[5].MAXCNT registers values, and enables endpoint 5 to respond to traffic from host
TASKS_STARTEPOUT[6]	0x040	Captures the EPOUT[6].PTR and EPOUT[6].MAXCNT registers values, and enables endpoint 6 to respond to traffic from host
TASKS_STARTEPOUT[7]	0x044	Captures the EPOUT[7].PTR and EPOUT[7].MAXCNT registers values, and enables endpoint 7 to respond to traffic from host
TASKS_STARTISOOUT	0x048	Captures the ISOOUT.PTR and ISOOUT.MAXCNT registers values, and enables receiving of data on ISO endpoint
TASKS_EP0RCVOUT	0x04C	Allows OUT data stage on control endpoint 0
TASKS_EP0STATUS	0x050	Allows status stage on control endpoint 0
TASKS_EP0STALL	0x054	Stalls data and status stage on control endpoint 0
TASKS_DPDMDRIVE	0x058	Forces D+ and D- lines into the state defined in the DPDMVALUE register
TASKS_DPDMNODRIVE	0x05C	Stops forcing D+ and D- lines into any state (USB engine takes control)
EVENTS_USBRESET	0x100	Signals that a USB reset condition has been detected on USB lines
EVENTS_STARTED	0x104	Confirms that the EPIN[n].PTR and EPIN[n].MAXCNT, or EPOUT[n].PTR and EPOUT[n].MAXCNT registers have been captured on all endpoints reported in the EPSTATUS register
EVENTS_ENDEPIN[0]	0x108	The whole EPIN[0] buffer has been consumed. The buffer can be accessed safely by software.
EVENTS_ENDEPIN[1]	0x10C	The whole EPIN[1] buffer has been consumed. The buffer can be accessed safely by software.
EVENTS_ENDEPIN[2]	0x110	The whole EPIN[2] buffer has been consumed. The buffer can be accessed safely by software.
EVENTS_ENDEPIN[3]	0x114	The whole EPIN[3] buffer has been consumed. The buffer can be accessed safely by software.
EVENTS_ENDEPIN[4]	0x118	The whole EPIN[4] buffer has been consumed. The buffer can be accessed safely by software.
EVENTS_ENDEPIN[5]	0x11C	The whole EPIN[5] buffer has been consumed. The buffer can be accessed safely by software.
EVENTS_ENDEPIN[6]	0x120	The whole EPIN[6] buffer has been consumed. The buffer can be accessed safely by software.
EVENTS_ENDEPIN[7]	0x124	The whole EPIN[7] buffer has been consumed. The buffer can be accessed safely by software.
EVENTS_EP0DATADONE	0x128	An acknowledged data transfer has taken place on the control endpoint
EVENTS_ENDISOIN	0x12C	The whole ISOIN buffer has been consumed. The buffer can be accessed safely by software.
EVENTS_ENDEPOUT[0]	0x130	The whole EPOUT[0] buffer has been consumed. The buffer can be accessed safely by software.
EVENTS_ENDEPOUT[1]	0x134	The whole EPOUT[1] buffer has been consumed. The buffer can be accessed safely by software.
EVENTS_ENDEPOUT[2]	0x138	The whole EPOUT[2] buffer has been consumed. The buffer can be accessed safely by software.
EVENTS_ENDEPOUT[3]	0x13C	The whole EPOUT[3] buffer has been consumed. The buffer can be accessed safely by software.
EVENTS_ENDEPOUT[4]	0x140	The whole EPOUT[4] buffer has been consumed. The buffer can be accessed safely by software.
EVENTS_ENDEPOUT[5]	0x144	The whole EPOUT[5] buffer has been consumed. The buffer can be accessed safely by software.
EVENTS_ENDEPOUT[6]	0x148	The whole EPOUT[6] buffer has been consumed. The buffer can be accessed safely by software.
EVENTS_ENDEPOUT[7]	0x14C	The whole EPOUT[7] buffer has been consumed. The buffer can be accessed safely by software.
EVENTS_ENDISOOUT	0x150	The whole ISOOUT buffer has been consumed. The buffer can be accessed safely by software.

Register	Offset	Description
EVENTS_SOF	0x154	Signals that a SOF (start of frame) condition has been detected on USB lines
EVENTS_USBEVENT	0x158	An event or an error not covered by specific events has occurred. Check EVENTCAUSE register to find the cause.
EVENTS_EP0SETUP	0x15C	A valid SETUP token has been received (and acknowledged) on the control endpoint
EVENTS_EPDATA	0x160	A data transfer has occurred on a data endpoint, indicated by the EPDATASTATUS register
SHORTS	0x200	Shortcuts between local events and tasks
INTEN	0x300	Enable or disable interrupt
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
EVENTCAUSE	0x400	Details on what caused the USBEVENT event
HALTED.EPIN[0]	0x420	IN endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
HALTED.EPIN[1]	0x424	IN endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
HALTED.EPIN[2]	0x428	IN endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
HALTED.EPIN[3]	0x42C	IN endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
HALTED.EPIN[4]	0x430	IN endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
HALTED.EPIN[5]	0x434	IN endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
HALTED.EPIN[6]	0x438	IN endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
HALTED.EPIN[7]	0x43C	IN endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
HALTED.EPOUT[0]	0x444	OUT endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
HALTED.EPOUT[1]	0x448	OUT endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
HALTED.EPOUT[2]	0x44C	OUT endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
HALTED.EPOUT[3]	0x450	OUT endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
HALTED.EPOUT[4]	0x454	OUT endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
HALTED.EPOUT[5]	0x458	OUT endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
HALTED.EPOUT[6]	0x45C	OUT endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
HALTED.EPOUT[7]	0x460	OUT endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
EPSTATUS	0x468	Provides information on which endpoint's EasyDMA registers have been captured
EPDATASTATUS	0x46C	Provides information on which endpoint(s) an acknowledged data transfer has occurred (EPDATA event)
USBADDR	0x470	Device USB address
BMREQUESTTYPE	0x480	SETUP data, byte 0, bmRequestType
BREQUEST	0x484	SETUP data, byte 1, bRequest
WVALUEL	0x488	SETUP data, byte 2, LSB of wValue
WVALUEH	0x48C	SETUP data, byte 3, MSB of wValue
WINDEXL	0x490	SETUP data, byte 4, LSB of wIndex
WINDEXH	0x494	SETUP data, byte 5, MSB of wIndex
WLENGTHL	0x498	SETUP data, byte 6, LSB of wLength
WLENGTHH	0x49C	SETUP data, byte 7, MSB of wLength
SIZE.EPOUT[0]	0x4A0	Number of bytes received last in the data stage of this OUT endpoint
SIZE.EPOUT[1]	0x4A4	Number of bytes received last in the data stage of this OUT endpoint
SIZE.EPOUT[2]	0x4A8	Number of bytes received last in the data stage of this OUT endpoint
SIZE.EPOUT[3]	0x4AC	Number of bytes received last in the data stage of this OUT endpoint
SIZE.EPOUT[4]	0x4B0	Number of bytes received last in the data stage of this OUT endpoint
SIZE.EPOUT[5]	0x4B4	Number of bytes received last in the data stage of this OUT endpoint
SIZE.EPOUT[6]	0x4B8	Number of bytes received last in the data stage of this OUT endpoint
SIZE.EPOUT[7]	0x4BC	Number of bytes received last in the data stage of this OUT endpoint
SIZE.ISOOUT	0x4C0	Number of bytes received last on this ISO OUT data endpoint
ENABLE	0x500	Enable USB
USBPULLUP	0x504	Control of the USB pull-up
DPDMVALUE	0x508	State D+ and D- lines will be forced into by the DPDMDRIVE task. The DPDMNODRIVE task reverts the control of the lines to MAC IP (no forcing).
DTOGGLE	0x50C	Data toggle control and status
EPINEN	0x510	Endpoint IN enable

Register	Offset	Description
EPOUTEN	0x514	Endpoint OUT enable
EPSTALL	0x518	STALL endpoints
ISOSPLIT	0x51C	Controls the split of ISO buffers
FRAMECNTR	0x520	Returns the current value of the start of frame counter
LOWPOWER	0x52C	Controls USBD peripheral low power mode during USB suspend
ISOINCONFIG	0x530	Controls the response of the ISO IN endpoint to an IN token when no data is ready to be sent
EPIN[0].PTR	0x600	Data pointer
EPIN[0].MAXCNT	0x604	Maximum number of bytes to transfer
EPIN[0].AMOUNT	0x608	Number of bytes transferred in the last transaction
EPIN[1].PTR	0x614	Data pointer
EPIN[1].MAXCNT	0x618	Maximum number of bytes to transfer
EPIN[1].AMOUNT	0x61C	Number of bytes transferred in the last transaction
EPIN[2].PTR	0x628	Data pointer
EPIN[2].MAXCNT	0x62C	Maximum number of bytes to transfer
EPIN[2].AMOUNT	0x630	Number of bytes transferred in the last transaction
EPIN[3].PTR	0x63C	Data pointer
EPIN[3].MAXCNT	0x640	Maximum number of bytes to transfer
EPIN[3].AMOUNT	0x644	Number of bytes transferred in the last transaction
EPIN[4].PTR	0x650	Data pointer
EPIN[4].MAXCNT	0x654	Maximum number of bytes to transfer
EPIN[4].AMOUNT	0x658	Number of bytes transferred in the last transaction
EPIN[5].PTR	0x664	Data pointer
EPIN[5].MAXCNT	0x668	Maximum number of bytes to transfer
EPIN[5].AMOUNT	0x66C	Number of bytes transferred in the last transaction
EPIN[6].PTR	0x678	Data pointer
EPIN[6].MAXCNT	0x67C	Maximum number of bytes to transfer
EPIN[6].AMOUNT	0x680	Number of bytes transferred in the last transaction
EPIN[7].PTR	0x68C	Data pointer
EPIN[7].MAXCNT	0x690	Maximum number of bytes to transfer
EPIN[7].AMOUNT	0x694	Number of bytes transferred in the last transaction
ISOIN.PTR	0x6A0	Data pointer
ISOIN.MAXCNT	0x6A4	Maximum number of bytes to transfer
ISOIN.AMOUNT	0x6A8	Number of bytes transferred in the last transaction
EPOUT[0].PTR	0x700	Data pointer
EPOUT[0].MAXCNT	0x704	Maximum number of bytes to transfer
EPOUT[0].AMOUNT	0x708	Number of bytes transferred in the last transaction
EPOUT[1].PTR	0x714	Data pointer
EPOUT[1].MAXCNT	0x718	Maximum number of bytes to transfer
EPOUT[1].AMOUNT	0x71C	Number of bytes transferred in the last transaction
EPOUT[2].PTR	0x728	Data pointer
EPOUT[2].MAXCNT	0x72C	Maximum number of bytes to transfer
EPOUT[2].AMOUNT	0x730	Number of bytes transferred in the last transaction
EPOUT[3].PTR	0x73C	Data pointer
EPOUT[3].MAXCNT	0x740	Maximum number of bytes to transfer
EPOUT[3].AMOUNT	0x744	Number of bytes transferred in the last transaction
EPOUT[4].PTR	0x750	Data pointer
EPOUT[4].MAXCNT	0x754	Maximum number of bytes to transfer
EPOUT[4].AMOUNT	0x758	Number of bytes transferred in the last transaction
EPOUT[5].PTR	0x764	Data pointer
EPOUT[5].MAXCNT	0x768	Maximum number of bytes to transfer
EPOUT[5].AMOUNT	0x76C	Number of bytes transferred in the last transaction
EPOUT[6].PTR	0x778	Data pointer
EPOUT[6].MAXCNT	0x77C	Maximum number of bytes to transfer

Register	Offset	Description
EPOUT[6].AMOUNT	0x780	Number of bytes transferred in the last transaction
EPOUT[7].PTR	0x78C	Data pointer
EPOUT[7].MAXCNT	0x790	Maximum number of bytes to transfer
EPOUT[7].AMOUNT	0x794	Number of bytes transferred in the last transaction
ISOOUT.PTR	0x7A0	Data pointer
ISOOUT.MAXCNT	0x7A4	Maximum number of bytes to transfer
ISOOUT.AMOUNT	0x7A8	Number of bytes transferred in the last transaction

6.35.13.1 TASKS_STARTEPIN[0]

Address offset: 0x004

Captures the EPIN[0].PTR and EPIN[0].MAXCNT registers values, and enables endpoint IN 0 to respond to traffic from host

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTEPIN			Captures the EPIN[0].PTR and EPIN[0].MAXCNT registers values, and enables endpoint IN 0 to respond to traffic from host
			Trigger	1	Trigger task

6.35.13.2 TASKS_STARTEPIN[1]

Address offset: 0x008

Captures the EPIN[1].PTR and EPIN[1].MAXCNT registers values, and enables endpoint IN 1 to respond to traffic from host

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTEPIN			Captures the EPIN[1].PTR and EPIN[1].MAXCNT registers values, and enables endpoint IN 1 to respond to traffic from host
			Trigger	1	Trigger task

6.35.13.3 TASKS_STARTEPIN[2]

Address offset: 0x00C

Captures the EPIN[2].PTR and EPIN[2].MAXCNT registers values, and enables endpoint IN 2 to respond to traffic from host

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTEPIN			Captures the EPIN[2].PTR and EPIN[2].MAXCNT registers values, and enables endpoint IN 2 to respond to traffic from host
			Trigger	1	Trigger task

6.35.13.4 TASKS_STARTEPIN[3]

Address offset: 0x010

Captures the EPIN[3].PTR and EPIN[3].MAXCNT registers values, and enables endpoint IN 3 to respond to traffic from host

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTEPIN			Captures the EPIN[3].PTR and EPIN[3].MAXCNT registers values, and enables endpoint IN 3 to respond to traffic from host
			Trigger	1	Trigger task

6.35.13.5 TASKS_STARTEPIN[4]

Address offset: 0x014

Captures the EPIN[4].PTR and EPIN[4].MAXCNT registers values, and enables endpoint IN 4 to respond to traffic from host

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTEPIN			Captures the EPIN[4].PTR and EPIN[4].MAXCNT registers values, and enables endpoint IN 4 to respond to traffic from host
			Trigger	1	Trigger task

6.35.13.6 TASKS_STARTEPIN[5]

Address offset: 0x018

Captures the EPIN[5].PTR and EPIN[5].MAXCNT registers values, and enables endpoint IN 5 to respond to traffic from host

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTEPIN			Captures the EPIN[5].PTR and EPIN[5].MAXCNT registers values, and enables endpoint IN 5 to respond to traffic from host
			Trigger	1	Trigger task

6.35.13.7 TASKS_STARTEPIN[6]

Address offset: 0x01C

Captures the EPIN[6].PTR and EPIN[6].MAXCNT registers values, and enables endpoint IN 6 to respond to traffic from host

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTEPIN			Captures the EPIN[6].PTR and EPIN[6].MAXCNT registers values, and enables endpoint IN 6 to respond to traffic from host
			Trigger	1	Trigger task

6.35.13.8 TASKS_STARTEPIN[7]

Address offset: 0x020

Captures the EPIN[7].PTR and EPIN[7].MAXCNT registers values, and enables endpoint IN 7 to respond to traffic from host

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTEPIN			Captures the EPIN[7].PTR and EPIN[7].MAXCNT registers values, and enables endpoint IN 7 to respond to traffic from host
			Trigger	1	Trigger task

6.35.13.9 TASKS_STARTISOIN

Address offset: 0x024

Captures the ISOIN.PTR and ISOIN.MAXCNT registers values, and enables sending data on ISO endpoint

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTISOIN			Captures the ISOIN.PTR and ISOIN.MAXCNT registers values, and enables sending data on ISO endpoint
			Trigger	1	Trigger task

6.35.13.10 TASKS_STARTEPOUT[0]

Address offset: 0x028

Captures the EPOUT[0].PTR and EPOUT[0].MAXCNT registers values, and enables endpoint 0 to respond to traffic from host

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTEPOUT			Captures the EPOUT[0].PTR and EPOUT[0].MAXCNT registers values, and enables endpoint 0 to respond to traffic from host
			Trigger	1	Trigger task

6.35.13.11 TASKS_STARTEPOUT[1]

Address offset: 0x02C

Captures the EPOUT[1].PTR and EPOUT[1].MAXCNT registers values, and enables endpoint 1 to respond to traffic from host

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTEPOUT			Captures the EPOUT[1].PTR and EPOUT[1].MAXCNT registers values, and enables endpoint 1 to respond to traffic from host
			Trigger	1	Trigger task

6.35.13.12 TASKS_STARTEPOUT[2]

Address offset: 0x030

Captures the EPOUT[2].PTR and EPOUT[2].MAXCNT registers values, and enables endpoint 2 to respond to traffic from host

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTEPOUT			Captures the EPOUT[2].PTR and EPOUT[2].MAXCNT registers values, and enables endpoint 2 to respond to traffic from host
			Trigger	1	Trigger task

6.35.13.13 TASKS_STARTEPOUT[3]

Address offset: 0x034

Captures the EPOUT[3].PTR and EPOUT[3].MAXCNT registers values, and enables endpoint 3 to respond to traffic from host

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTEPOUT			Captures the EPOUT[3].PTR and EPOUT[3].MAXCNT registers values, and enables endpoint 3 to respond to traffic from host
			Trigger	1	Trigger task

6.35.13.14 TASKS_STARTEPOUT[4]

Address offset: 0x038

Captures the EPOUT[4].PTR and EPOUT[4].MAXCNT registers values, and enables endpoint 4 to respond to traffic from host

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTEPOUT			Captures the EPOUT[4].PTR and EPOUT[4].MAXCNT registers values, and enables endpoint 4 to respond to traffic from host
			Trigger	1	Trigger task

6.35.13.15 TASKS_STARTEPOUT[5]

Address offset: 0x03C

Captures the EPOUT[5].PTR and EPOUT[5].MAXCNT registers values, and enables endpoint 5 to respond to traffic from host

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTEPOUT			Captures the EPOUT[5].PTR and EPOUT[5].MAXCNT registers values, and enables endpoint 5 to respond to traffic from host
			Trigger	1	Trigger task

6.35.13.16 TASKS_STARTEPOUT[6]

Address offset: 0x040

Captures the EPOUT[6].PTR and EPOUT[6].MAXCNT registers values, and enables endpoint 6 to respond to traffic from host

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTEPOUT			Captures the EPOUT[6].PTR and EPOUT[6].MAXCNT registers values, and enables endpoint 6 to respond to traffic from host
			Trigger	1	Trigger task

6.35.13.17 TASKS_STARTEPOUT[7]

Address offset: 0x044

Captures the EPOUT[7].PTR and EPOUT[7].MAXCNT registers values, and enables endpoint 7 to respond to traffic from host

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTEPOUT			Captures the EPOUT[7].PTR and EPOUT[7].MAXCNT registers values, and enables endpoint 7 to respond to traffic from host
			Trigger	1	Trigger task

6.35.13.18 TASKS_STARTISOOUT

Address offset: 0x048

Captures the ISOOUT.PTR and ISOOUT.MAXCNT registers values, and enables receiving of data on ISO endpoint

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_STARTISOOUT			Captures the ISOOUT.PTR and ISOOUT.MAXCNT registers values, and enables receiving of data on ISO endpoint
			Trigger	1	Trigger task

6.35.13.19 TASKS_EP0RCVOUT

Address offset: 0x04C

Allows OUT data stage on control endpoint 0

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_EP0RCVOUT			Allows OUT data stage on control endpoint 0
			Trigger	1	Trigger task

6.35.13.20 TASKS_EP0STATUS

Address offset: 0x050

Allows status stage on control endpoint 0

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_EP0STATUS			Allows status stage on control endpoint 0
			Trigger	1	Trigger task

6.35.13.21 TASKS_EP0STALL

Address offset: 0x054

Stalls data and status stage on control endpoint 0

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_EP0STALL			Stalls data and status stage on control endpoint 0
			Trigger	1	Trigger task

6.35.13.22 TASKS_DPDMDRIVE

Address offset: 0x058

Forces D+ and D- lines into the state defined in the DPDMVALUE register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_DPDMDRIVE			Forces D+ and D- lines into the state defined in the DPDMVALUE register
			Trigger	1	Trigger task

6.35.13.23 TASKS_DPDMNODRIVE

Address offset: 0x05C

Stops forcing D+ and D- lines into any state (USB engine takes control)

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_DPDMNODRIVE			Stops forcing D+ and D- lines into any state (USB engine takes control)
			Trigger	1	Trigger task

6.35.13.24 EVENTS_USBRESET

Address offset: 0x100

Signals that a USB reset condition has been detected on USB lines

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_USBRESET			Signals that a USB reset condition has been detected on USB lines
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.25 EVENTS_STARTED

Address offset: 0x104

Confirms that the EPIN[n].PTR and EPIN[n].MAXCNT, or EPOUT[n].PTR and EPOUT[n].MAXCNT registers have been captured on all endpoints reported in the EPSTATUS register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_STARTED			Confirms that the EPIN[n].PTR and EPIN[n].MAXCNT, or EPOUT[n].PTR and EPOUT[n].MAXCNT registers have been captured on all endpoints reported in the EPSTATUS register
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.26 EVENTS_ENDEPIN[0]

Address offset: 0x108

The whole EPIN[0] buffer has been consumed. The buffer can be accessed safely by software.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDEPIN			The whole EPIN[0] buffer has been consumed. The buffer can be accessed safely by software.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.27 EVENTS_ENDEPIN[1]

Address offset: 0x10C

The whole EPIN[1] buffer has been consumed. The buffer can be accessed safely by software.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDEPIN			The whole EPIN[1] buffer has been consumed. The buffer can be accessed safely by software.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.28 EVENTS_ENDEPIN[2]

Address offset: 0x110

The whole EPIN[2] buffer has been consumed. The buffer can be accessed safely by software.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDEPIN			The whole EPIN[2] buffer has been consumed. The buffer can be accessed safely by software.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.29 EVENTS_ENDEPIN[3]

Address offset: 0x114

The whole EPIN[3] buffer has been consumed. The buffer can be accessed safely by software.

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDEPIN			The whole EPIN[3] buffer has been consumed. The buffer can be accessed safely by software.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.30 EVENTS_ENDEPIN[4]

Address offset: 0x118

The whole EPIN[4] buffer has been consumed. The buffer can be accessed safely by software.

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDEPIN			The whole EPIN[4] buffer has been consumed. The buffer can be accessed safely by software.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.31 EVENTS_ENDEPIN[5]

Address offset: 0x11C

The whole EPIN[5] buffer has been consumed. The buffer can be accessed safely by software.

Bit number					31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID					A
Reset 0x00000000					**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDEPIN			The whole EPIN[5] buffer has been consumed. The buffer can be accessed safely by software.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.32 EVENTS_ENDEPIN[6]

Address offset: 0x120

The whole EPIN[6] buffer has been consumed. The buffer can be accessed safely by software.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDEPIN			The whole EPIN[6] buffer has been consumed. The buffer can be accessed safely by software.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.33 EVENTS_ENDEPIN[7]

Address offset: 0x124

The whole EPIN[7] buffer has been consumed. The buffer can be accessed safely by software.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDEPIN			The whole EPIN[7] buffer has been consumed. The buffer can be accessed safely by software.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.34 EVENTS_EP0DATADONE

Address offset: 0x128

An acknowledged data transfer has taken place on the control endpoint

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_EP0DATADONE			An acknowledged data transfer has taken place on the control endpoint
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.35 EVENTS_ENDISOIN

Address offset: 0x12C

The whole ISOIN buffer has been consumed. The buffer can be accessed safely by software.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDISOIN			The whole ISOIN buffer has been consumed. The buffer can be accessed safely by software.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.36 EVENTS_ENDEPOUT[0]

Address offset: 0x130

The whole EPOUT[0] buffer has been consumed. The buffer can be accessed safely by software.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDEPOUT			The whole EPOUT[0] buffer has been consumed. The buffer can be accessed safely by software.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.37 EVENTS_ENDEPOUT[1]

Address offset: 0x134

The whole EPOUT[1] buffer has been consumed. The buffer can be accessed safely by software.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDEPOUT			The whole EPOUT[1] buffer has been consumed. The buffer can be accessed safely by software.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.38 EVENTS_ENDEPOUT[2]

Address offset: 0x138

The whole EPOUT[2] buffer has been consumed. The buffer can be accessed safely by software.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDEPOUT			The whole EPOUT[2] buffer has been consumed. The buffer can be accessed safely by software.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.39 EVENTS_ENDEPOUT[3]

Address offset: 0x13C

The whole EPOUT[3] buffer has been consumed. The buffer can be accessed safely by software.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDEPOUT			The whole EPOUT[3] buffer has been consumed. The buffer can be accessed safely by software.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.40 EVENTS_ENDEPOUT[4]

Address offset: 0x140

The whole EPOUT[4] buffer has been consumed. The buffer can be accessed safely by software.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDEPOUT			The whole EPOUT[4] buffer has been consumed. The buffer can be accessed safely by software.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.41 EVENTS_ENDEPOUT[5]

Address offset: 0x144

The whole EPOUT[5] buffer has been consumed. The buffer can be accessed safely by software.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDEPOUT			The whole EPOUT[5] buffer has been consumed. The buffer can be accessed safely by software.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.42 EVENTS_ENDEPOUT[6]

Address offset: 0x148

The whole EPOUT[6] buffer has been consumed. The buffer can be accessed safely by software.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDEPOUT			The whole EPOUT[6] buffer has been consumed. The buffer can be accessed safely by software.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.43 EVENTS_ENDEPOUT[7]

Address offset: 0x14C

The whole EPOUT[7] buffer has been consumed. The buffer can be accessed safely by software.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDEPOUT			The whole EPOUT[7] buffer has been consumed. The buffer can be accessed safely by software.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.44 EVENTS_ENDISOOUT

Address offset: 0x150

The whole ISOOUT buffer has been consumed. The buffer can be accessed safely by software.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_ENDISOOUT			The whole ISOOUT buffer has been consumed. The buffer can be accessed safely by software.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.45 EVENTS_SOF

Address offset: 0x154

Signals that a SOF (start of frame) condition has been detected on USB lines

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_SOF			Signals that a SOF (start of frame) condition has been detected on USB lines
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.46 EVENTS_USBEVENT

Address offset: 0x158

An event or an error not covered by specific events has occurred. Check EVENTCAUSE register to find the cause.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_USBEVENT			An event or an error not covered by specific events has occurred. Check EVENTCAUSE register to find the cause.
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.47 EVENTS_EP0SETUP

Address offset: 0x15C

A valid SETUP token has been received (and acknowledged) on the control endpoint

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_EP0SETUP			A valid SETUP token has been received (and acknowledged) on the control endpoint
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.48 EVENTS_EPDATA

Address offset: 0x160

A data transfer has occurred on a data endpoint, indicated by the EPDATASTATUS register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_EPDATA			A data transfer has occurred on a data endpoint, indicated by the EPDATASTATUS register
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.35.13.49 SHORTS

Address offset: 0x200

Shortcuts between local events and tasks

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EP0DATADONE_STARTEPIN0			Shortcut between event EP0DATADONE and task STARTEPIN[0]
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
B	RW	EP0DATADONE_STARTEPOUT0			Shortcut between event EP0DATADONE and task STARTEPOUT[0]
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																												E	D	C	B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
C	RW	EP0DATADONE_EP0STATUS			Shortcut between event EP0DATADONE and task EP0STATUS
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
D	RW	ENDEPOUT0_EP0STATUS			Shortcut between event ENDEPOUT[0] and task EP0STATUS
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut
E	RW	ENDEPOUT0_EP0RCVOUT			Shortcut between event ENDEPOUT[0] and task EP0RCVOUT
			Disabled	0	Disable shortcut
			Enabled	1	Enable shortcut

6.35.13.50 INTEN

Address offset: 0x300

Enable or disable interrupt

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID									Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	USBRESET			Enable or disable interrupt for event USBRESET
			Disabled	0	Disable
			Enabled	1	Enable
B	RW	STARTED			Enable or disable interrupt for event STARTED
			Disabled	0	Disable
			Enabled	1	Enable
C	RW	ENDEPIN[0]			Enable or disable interrupt for event ENDEPIN[0]
			Disabled	0	Disable
			Enabled	1	Enable
D	RW	ENDEPIN[1]			Enable or disable interrupt for event ENDEPIN[1]
			Disabled	0	Disable
			Enabled	1	Enable
E	RW	ENDEPIN[2]			Enable or disable interrupt for event ENDEPIN[2]
			Disabled	0	Disable
			Enabled	1	Enable
F	RW	ENDEPIN[3]			Enable or disable interrupt for event ENDEPIN[3]
			Disabled	0	Disable
			Enabled	1	Enable
G	RW	ENDEPIN[4]			Enable or disable interrupt for event ENDEPIN[4]
			Disabled	0	Disable
			Enabled	1	Enable
H	RW	ENDEPIN[5]			Enable or disable interrupt for event ENDEPIN[5]
			Disabled	0	Disable
			Enabled	1	Enable
I	RW	ENDEPIN[6]			Enable or disable interrupt for event ENDEPIN[6]
			Disabled	0	Disable
			Enabled	1	Enable
J	RW	ENDEPIN[7]			Enable or disable interrupt for event ENDEPIN[7]
			Disabled	0	Disable
			Enabled	1	Enable

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
K	RW	EP0DATADONE			Enable or disable interrupt for event EP0DATADONE
			Disabled	0	Disable
			Enabled	1	Enable
L	RW	ENDISOIN			Enable or disable interrupt for event ENDISOIN
			Disabled	0	Disable
			Enabled	1	Enable
M	RW	ENDEPOUT[0]			Enable or disable interrupt for event ENDEPOUT[0]
			Disabled	0	Disable
			Enabled	1	Enable
N	RW	ENDEPOUT[1]			Enable or disable interrupt for event ENDEPOUT[1]
			Disabled	0	Disable
			Enabled	1	Enable
O	RW	ENDEPOUT[2]			Enable or disable interrupt for event ENDEPOUT[2]
			Disabled	0	Disable
			Enabled	1	Enable
P	RW	ENDEPOUT[3]			Enable or disable interrupt for event ENDEPOUT[3]
			Disabled	0	Disable
			Enabled	1	Enable
Q	RW	ENDEPOUT[4]			Enable or disable interrupt for event ENDEPOUT[4]
			Disabled	0	Disable
			Enabled	1	Enable
R	RW	ENDEPOUT[5]			Enable or disable interrupt for event ENDEPOUT[5]
			Disabled	0	Disable
			Enabled	1	Enable
S	RW	ENDEPOUT[6]			Enable or disable interrupt for event ENDEPOUT[6]
			Disabled	0	Disable
			Enabled	1	Enable
T	RW	ENDEPOUT[7]			Enable or disable interrupt for event ENDEPOUT[7]
			Disabled	0	Disable
			Enabled	1	Enable
U	RW	ENDISOOUT			Enable or disable interrupt for event ENDISOOUT
			Disabled	0	Disable
			Enabled	1	Enable
V	RW	SOF			Enable or disable interrupt for event SOF
			Disabled	0	Disable
			Enabled	1	Enable
W	RW	USBEVENT			Enable or disable interrupt for event USBEVENT
			Disabled	0	Disable
			Enabled	1	Enable
X	RW	EP0SETUP			Enable or disable interrupt for event EP0SETUP
			Disabled	0	Disable
			Enabled	1	Enable
Y	RW	EPDATA			Enable or disable interrupt for event EPDATA
			Disabled	0	Disable
			Enabled	1	Enable

6.35.13.51 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	USBRESET			Write '1' to enable interrupt for event USBRESET
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
B	RW	STARTED			Write '1' to enable interrupt for event STARTED
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
C	RW	ENDEPIN[0]			Write '1' to enable interrupt for event ENDEPIN[0]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
D	RW	ENDEPIN[1]			Write '1' to enable interrupt for event ENDEPIN[1]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
E	RW	ENDEPIN[2]			Write '1' to enable interrupt for event ENDEPIN[2]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
F	RW	ENDEPIN[3]			Write '1' to enable interrupt for event ENDEPIN[3]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
G	RW	ENDEPIN[4]			Write '1' to enable interrupt for event ENDEPIN[4]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
H	RW	ENDEPIN[5]			Write '1' to enable interrupt for event ENDEPIN[5]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
I	RW	ENDEPIN[6]			Write '1' to enable interrupt for event ENDEPIN[6]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
J	RW	ENDEPIN[7]			Write '1' to enable interrupt for event ENDEPIN[7]
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
K	RW	EP0DATADONE			Write '1' to enable interrupt for event EP0DATADONE
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
L	RW	ENDISOIN			Write '1' to enable interrupt for event ENDISOIN
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
M	RW	ENDEPOUT[0]			Write '1' to enable interrupt for event ENDEPOUT[0]
			Set	1	Enable

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID											Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
			Disabled	0																															
			Enabled	1																															
N	RW	ENDEPOUT[1]																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
O	RW	ENDEPOUT[2]																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
P	RW	ENDEPOUT[3]																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
Q	RW	ENDEPOUT[4]																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
R	RW	ENDEPOUT[5]																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
S	RW	ENDEPOUT[6]																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
T	RW	ENDEPOUT[7]																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
U	RW	ENDISOOUT																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
V	RW	SOF																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
W	RW	USBEVENT																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
X	RW	EP0SETUP																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															
Y	RW	EPDATA																																	
			Set	1																															
			Disabled	0																															
			Enabled	1																															

6.35.13.52 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID											Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value	Description																														
A	RW	USBRESET			Write '1' to disable interrupt for event USBRESET																														
			Clear	1	Disable																														
			Disabled	0	Read: Disabled																														
			Enabled	1	Read: Enabled																														
B	RW	STARTED			Write '1' to disable interrupt for event STARTED																														
			Clear	1	Disable																														
			Disabled	0	Read: Disabled																														
			Enabled	1	Read: Enabled																														
C	RW	ENDEPIN[0]			Write '1' to disable interrupt for event ENDEPIN[0]																														
			Clear	1	Disable																														
			Disabled	0	Read: Disabled																														
			Enabled	1	Read: Enabled																														
D	RW	ENDEPIN[1]			Write '1' to disable interrupt for event ENDEPIN[1]																														
			Clear	1	Disable																														
			Disabled	0	Read: Disabled																														
			Enabled	1	Read: Enabled																														
E	RW	ENDEPIN[2]			Write '1' to disable interrupt for event ENDEPIN[2]																														
			Clear	1	Disable																														
			Disabled	0	Read: Disabled																														
			Enabled	1	Read: Enabled																														
F	RW	ENDEPIN[3]			Write '1' to disable interrupt for event ENDEPIN[3]																														
			Clear	1	Disable																														
			Disabled	0	Read: Disabled																														
			Enabled	1	Read: Enabled																														
G	RW	ENDEPIN[4]			Write '1' to disable interrupt for event ENDEPIN[4]																														
			Clear	1	Disable																														
			Disabled	0	Read: Disabled																														
			Enabled	1	Read: Enabled																														
H	RW	ENDEPIN[5]			Write '1' to disable interrupt for event ENDEPIN[5]																														
			Clear	1	Disable																														
			Disabled	0	Read: Disabled																														
			Enabled	1	Read: Enabled																														
I	RW	ENDEPIN[6]			Write '1' to disable interrupt for event ENDEPIN[6]																														
			Clear	1	Disable																														
			Disabled	0	Read: Disabled																														
			Enabled	1	Read: Enabled																														
J	RW	ENDEPIN[7]			Write '1' to disable interrupt for event ENDEPIN[7]																														
			Clear	1	Disable																														
			Disabled	0	Read: Disabled																														
			Enabled	1	Read: Enabled																														
K	RW	EP0DATADONE			Write '1' to disable interrupt for event EP0DATADONE																														
			Clear	1	Disable																														
			Disabled	0	Read: Disabled																														
			Enabled	1	Read: Enabled																														

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
L	RW	ENDISOIN			Write '1' to disable interrupt for event ENDISOIN
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
M	RW	ENDEPOUT[0]			Write '1' to disable interrupt for event ENDEPOUT[0]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
N	RW	ENDEPOUT[1]			Write '1' to disable interrupt for event ENDEPOUT[1]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
O	RW	ENDEPOUT[2]			Write '1' to disable interrupt for event ENDEPOUT[2]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
P	RW	ENDEPOUT[3]			Write '1' to disable interrupt for event ENDEPOUT[3]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
Q	RW	ENDEPOUT[4]			Write '1' to disable interrupt for event ENDEPOUT[4]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
R	RW	ENDEPOUT[5]			Write '1' to disable interrupt for event ENDEPOUT[5]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
S	RW	ENDEPOUT[6]			Write '1' to disable interrupt for event ENDEPOUT[6]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
T	RW	ENDEPOUT[7]			Write '1' to disable interrupt for event ENDEPOUT[7]
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
U	RW	ENDISOOUT			Write '1' to disable interrupt for event ENDISOOUT
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
V	RW	SOF			Write '1' to disable interrupt for event SOF
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
W	RW	USBEVENT			Write '1' to disable interrupt for event USBEVENT
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
X	RW	EP0SETUP			Write '1' to disable interrupt for event EP0SETUP
			Clear	1	Disable

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				Y X W V U T S R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled
Y	RW	EPDATA			Write '1' to disable interrupt for event EPDATA
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.35.13.53 EVENTCAUSE

Address offset: 0x400

Details on what caused the USBEVENT event

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW W1C	ISOOUTCRC			CRC error was detected on isochronous OUT endpoint 8. Write '1' to clear.
			NotDetected	0	No error detected
			Detected	1	Error detected
B	RW W1C	SUSPEND			Signals that USB lines have been idle long enough for the device to enter suspend. Write '1' to clear.
			NotDetected	0	Suspend not detected
			Detected	1	Suspend detected
C	RW W1C	RESUME			Signals that a RESUME condition (K state or activity restart) has been detected on USB lines. Write '1' to clear.
			NotDetected	0	Resume not detected
			Detected	1	Resume detected
D	RW W1C	USBWUALLOWED			USB MAC has been woken up and operational. Write '1' to clear.
			NotAllowed	0	Wake up not allowed
			Allowed	1	Wake up allowed
E	RW W1C	READY			USB device is ready for normal operation. Write '1' to clear.
			NotDetected	0	USBEVENT was not issued due to USBD peripheral ready
			Ready	1	USBD peripheral is ready

6.35.13.54 HALTED.EPIN[0]

Address offset: 0x420

IN endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	GETSTATUS			IN endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
			NotHalted	0	Endpoint is not halted
			Halted	1	Endpoint is halted

6.35.13.55 HALTED.EPIN[1]

Address offset: 0x424

IN endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	GETSTATUS			IN endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
			NotHalted	0	Endpoint is not halted
			Halted	1	Endpoint is halted

6.35.13.56 HALTED.EPIN[2]

Address offset: 0x428

IN endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	GETSTATUS			IN endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
			NotHalted	0	Endpoint is not halted
			Halted	1	Endpoint is halted

6.35.13.57 HALTED.EPIN[3]

Address offset: 0x42C

IN endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	GETSTATUS			IN endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
			NotHalted	0	Endpoint is not halted
			Halted	1	Endpoint is halted

6.35.13.58 HALTED.EPIN[4]

Address offset: 0x430

IN endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	R	GETSTATUS																																	
			NotHalted	0																															
			Halted	1																															

6.35.13.59 HALTED.EPIN[5]

Address offset: 0x434

IN endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	R	GETSTATUS																																	
			NotHalted	0																															
			Halted	1																															

6.35.13.60 HALTED.EPIN[6]

Address offset: 0x438

IN endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	R	GETSTATUS																																	
			NotHalted	0																															
			Halted	1																															

6.35.13.61 HALTED.EPIN[7]

Address offset: 0x43C

IN endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	GETSTATUS			IN endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
			NotHalted	0	Endpoint is not halted
			Halted	1	Endpoint is halted

6.35.13.62 HALTED.EPOUT[0]

Address offset: 0x444

OUT endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	GETSTATUS			OUT endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
			NotHalted	0	Endpoint is not halted
			Halted	1	Endpoint is halted

6.35.13.63 HALTED.EPOUT[1]

Address offset: 0x448

OUT endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	GETSTATUS			OUT endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
			NotHalted	0	Endpoint is not halted
			Halted	1	Endpoint is halted

6.35.13.64 HALTED.EPOUT[2]

Address offset: 0x44C

OUT endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	GETSTATUS			OUT endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
			NotHalted	0	Endpoint is not halted
			Halted	1	Endpoint is halted

6.35.13.65 HALTED.EPOUT[3]

Address offset: 0x450

OUT endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	R	GETSTATUS																																	
			NotHalted	0																															
			Halted	1																															

6.35.13.66 HALTED.EPOUT[4]

Address offset: 0x454

OUT endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	R	GETSTATUS																																	
			NotHalted	0																															
			Halted	1																															

6.35.13.67 HALTED.EPOUT[5]

Address offset: 0x458

OUT endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	R	GETSTATUS																																	
			NotHalted	0																															
			Halted	1																															

6.35.13.68 HALTED.EPOUT[6]

Address offset: 0x45C

OUT endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	GETSTATUS			OUT endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
			NotHalted	0	Endpoint is not halted
			Halted	1	Endpoint is halted

6.35.13.69 HALTED.EPOUT[7]

Address offset: 0x460

OUT endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	GETSTATUS			OUT endpoint halted status. Can be used as is as response to a GetStatus() request to endpoint.
			NotHalted	0	Endpoint is not halted
			Halted	1	Endpoint is halted

6.35.13.70 EPSTATUS

Address offset: 0x468

Provides information on which endpoint's EasyDMA registers have been captured

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW W1C	EPIN0			Captured state of endpoint's EasyDMA registers. Write '1' to clear.
			NoData	0	EasyDMA registers have not been captured for this endpoint
			DataDone	1	EasyDMA registers have been captured for this endpoint
B	RW W1C	EPIN1			Captured state of endpoint's EasyDMA registers. Write '1' to clear.
			NoData	0	EasyDMA registers have not been captured for this endpoint
			DataDone	1	EasyDMA registers have been captured for this endpoint
C	RW W1C	EPIN2			Captured state of endpoint's EasyDMA registers. Write '1' to clear.
			NoData	0	EasyDMA registers have not been captured for this endpoint
			DataDone	1	EasyDMA registers have been captured for this endpoint
D	RW W1C	EPIN3			Captured state of endpoint's EasyDMA registers. Write '1' to clear.
			NoData	0	EasyDMA registers have not been captured for this endpoint
			DataDone	1	EasyDMA registers have been captured for this endpoint
E	RW W1C	EPIN4			Captured state of endpoint's EasyDMA registers. Write '1' to clear.
			NoData	0	EasyDMA registers have not been captured for this endpoint
			DataDone	1	EasyDMA registers have been captured for this endpoint

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID								R	Q	P	O	N	M	L	K	J									I	H	G	F	E	D	C	B	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
F	RW W1C	EPIN5			Captured state of endpoint's EasyDMA registers. Write '1' to clear.
			NoData	0	EasyDMA registers have not been captured for this endpoint
			DataDone	1	EasyDMA registers have been captured for this endpoint
G	RW W1C	EPIN6			Captured state of endpoint's EasyDMA registers. Write '1' to clear.
			NoData	0	EasyDMA registers have not been captured for this endpoint
			DataDone	1	EasyDMA registers have been captured for this endpoint
H	RW W1C	EPIN7			Captured state of endpoint's EasyDMA registers. Write '1' to clear.
			NoData	0	EasyDMA registers have not been captured for this endpoint
			DataDone	1	EasyDMA registers have been captured for this endpoint
I	RW W1C	EPIN8			Captured state of endpoint's EasyDMA registers. Write '1' to clear.
			NoData	0	EasyDMA registers have not been captured for this endpoint
			DataDone	1	EasyDMA registers have been captured for this endpoint
J	RW W1C	EPOUT0			Captured state of endpoint's EasyDMA registers. Write '1' to clear.
			NoData	0	EasyDMA registers have not been captured for this endpoint
			DataDone	1	EasyDMA registers have been captured for this endpoint
K	RW W1C	EPOUT1			Captured state of endpoint's EasyDMA registers. Write '1' to clear.
			NoData	0	EasyDMA registers have not been captured for this endpoint
			DataDone	1	EasyDMA registers have been captured for this endpoint
L	RW W1C	EPOUT2			Captured state of endpoint's EasyDMA registers. Write '1' to clear.
			NoData	0	EasyDMA registers have not been captured for this endpoint
			DataDone	1	EasyDMA registers have been captured for this endpoint
M	RW W1C	EPOUT3			Captured state of endpoint's EasyDMA registers. Write '1' to clear.
			NoData	0	EasyDMA registers have not been captured for this endpoint
			DataDone	1	EasyDMA registers have been captured for this endpoint
N	RW W1C	EPOUT4			Captured state of endpoint's EasyDMA registers. Write '1' to clear.
			NoData	0	EasyDMA registers have not been captured for this endpoint
			DataDone	1	EasyDMA registers have been captured for this endpoint
O	RW W1C	EPOUT5			Captured state of endpoint's EasyDMA registers. Write '1' to clear.
			NoData	0	EasyDMA registers have not been captured for this endpoint
			DataDone	1	EasyDMA registers have been captured for this endpoint
P	RW W1C	EPOUT6			Captured state of endpoint's EasyDMA registers. Write '1' to clear.
			NoData	0	EasyDMA registers have not been captured for this endpoint
			DataDone	1	EasyDMA registers have been captured for this endpoint
Q	RW W1C	EPOUT7			Captured state of endpoint's EasyDMA registers. Write '1' to clear.
			NoData	0	EasyDMA registers have not been captured for this endpoint
			DataDone	1	EasyDMA registers have been captured for this endpoint
R	RW W1C	EPOUT8			Captured state of endpoint's EasyDMA registers. Write '1' to clear.

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				R Q P O N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
			NoData	0	EasyDMA registers have not been captured for this endpoint
			DataDone	1	EasyDMA registers have been captured for this endpoint

6.35.13.71 EPDATASTATUS

Address offset: 0x46C

Provides information on which endpoint(s) an acknowledged data transfer has occurred (EPDATA event)

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW W1C	EPIN1			Acknowledged data transfer on this IN endpoint. Write '1' to clear.
			NotDone	0	No acknowledged data transfer on this endpoint
			DataDone	1	Acknowledged data transfer on this endpoint has occurred
B	RW W1C	EPIN2			Acknowledged data transfer on this IN endpoint. Write '1' to clear.
			NotDone	0	No acknowledged data transfer on this endpoint
			DataDone	1	Acknowledged data transfer on this endpoint has occurred
C	RW W1C	EPIN3			Acknowledged data transfer on this IN endpoint. Write '1' to clear.
			NotDone	0	No acknowledged data transfer on this endpoint
			DataDone	1	Acknowledged data transfer on this endpoint has occurred
D	RW W1C	EPIN4			Acknowledged data transfer on this IN endpoint. Write '1' to clear.
			NotDone	0	No acknowledged data transfer on this endpoint
			DataDone	1	Acknowledged data transfer on this endpoint has occurred
E	RW W1C	EPIN5			Acknowledged data transfer on this IN endpoint. Write '1' to clear.
			NotDone	0	No acknowledged data transfer on this endpoint
			DataDone	1	Acknowledged data transfer on this endpoint has occurred
F	RW W1C	EPIN6			Acknowledged data transfer on this IN endpoint. Write '1' to clear.
			NotDone	0	No acknowledged data transfer on this endpoint
			DataDone	1	Acknowledged data transfer on this endpoint has occurred
G	RW W1C	EPIN7			Acknowledged data transfer on this IN endpoint. Write '1' to clear.
			NotDone	0	No acknowledged data transfer on this endpoint
			DataDone	1	Acknowledged data transfer on this endpoint has occurred
H	RW W1C	EPOUT1			Acknowledged data transfer on this OUT endpoint. Write '1' to clear.
			NotStarted	0	No acknowledged data transfer on this endpoint
			Started	1	Acknowledged data transfer on this endpoint has occurred
I	RW W1C	EPOUT2			Acknowledged data transfer on this OUT endpoint. Write '1' to clear.
			NotStarted	0	No acknowledged data transfer on this endpoint
			Started	1	Acknowledged data transfer on this endpoint has occurred

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				N M L K J I H G F E D C B A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
J	RW W1C	EPOUT3			Acknowledged data transfer on this OUT endpoint. Write '1' to clear.
			NotStarted	0	No acknowledged data transfer on this endpoint
			Started	1	Acknowledged data transfer on this endpoint has occurred
K	RW W1C	EPOUT4			Acknowledged data transfer on this OUT endpoint. Write '1' to clear.
			NotStarted	0	No acknowledged data transfer on this endpoint
			Started	1	Acknowledged data transfer on this endpoint has occurred
L	RW W1C	EPOUT5			Acknowledged data transfer on this OUT endpoint. Write '1' to clear.
			NotStarted	0	No acknowledged data transfer on this endpoint
			Started	1	Acknowledged data transfer on this endpoint has occurred
M	RW W1C	EPOUT6			Acknowledged data transfer on this OUT endpoint. Write '1' to clear.
			NotStarted	0	No acknowledged data transfer on this endpoint
			Started	1	Acknowledged data transfer on this endpoint has occurred
N	RW W1C	EPOUT7			Acknowledged data transfer on this OUT endpoint. Write '1' to clear.
			NotStarted	0	No acknowledged data transfer on this endpoint
			Started	1	Acknowledged data transfer on this endpoint has occurred

6.35.13.72 USBADDR

Address offset: 0x470

Device USB address

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	ADDR			Device USB address

6.35.13.73 BMREQUESTTYPE

Address offset: 0x480

SETUP data, byte 0, bmRequestType

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B B A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	RECIPIENT			Data transfer type
			Device	0	Device
			Interface	1	Interface
			Endpoint	2	Endpoint
			Other	3	Other
B	R	TYPE			Data transfer type
			Standard	0	Standard
			Class	1	Class

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B B A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
			Vendor	2	Vendor
C	R	DIRECTION			Data transfer direction
			HostToDevice	0	Host-to-device
			DeviceToHost	1	Device-to-host

6.35.13.74 BREQUEST

Address offset: 0x484

SETUP data, byte 1, bRequest

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	BREQUEST			SETUP data, byte 1, bRequest. Values provided for standard requests only, user must implement class and vendor values.
			STD_GET_STATUS	0	Standard request GET_STATUS
			STD_CLEAR_FEATURE	1	Standard request CLEAR_FEATURE
			STD_SET_FEATURE	3	Standard request SET_FEATURE
			STD_SET_ADDRESS	5	Standard request SET_ADDRESS
			STD_GET_DESCRIPTOR	6	Standard request GET_DESCRIPTOR
			STD_SET_DESCRIPTOR	7	Standard request SET_DESCRIPTOR
			STD_GET_CONFIGURATION	8	Standard request GET_CONFIGURATION
			STD_SET_CONFIGURATION	9	Standard request SET_CONFIGURATION
			STD_GET_INTERFACE	10	Standard request GET_INTERFACE
			STD_SET_INTERFACE	11	Standard request SET_INTERFACE
			STD_SYNCH_FRAME	12	Standard request SYNCH_FRAME

6.35.13.75 WVALUEL

Address offset: 0x488

SETUP data, byte 2, LSB of wValue

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	WVALUEL			SETUP data, byte 2, LSB of wValue

6.35.13.76 WVALUEH

Address offset: 0x48C

SETUP data, byte 3, MSB of wValue

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																									A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	WVALUEH			SETUP data, byte 3, MSB of wValue

6.35.13.77 WINDEXL

Address offset: 0x490

SETUP data, byte 4, LSB of wIndex

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																									A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	WINDEXL			SETUP data, byte 4, LSB of wIndex

6.35.13.78 WINDEXH

Address offset: 0x494

SETUP data, byte 5, MSB of wIndex

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																									A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	WINDEXH			SETUP data, byte 5, MSB of wIndex

6.35.13.79 WLENGTHL

Address offset: 0x498

SETUP data, byte 6, LSB of wLength

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																									A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	WLENGTHL			SETUP data, byte 6, LSB of wLength

6.35.13.80 WLENGTHH

Address offset: 0x49C

SETUP data, byte 7, MSB of wLength

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																									A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	R	WLENGTHH			SETUP data, byte 7, MSB of wLength

6.35.13.81 SIZE.EPOUT[0]

Address offset: 0x4A0

Number of bytes received last in the data stage of this OUT endpoint

Write to any value to accept further OUT traffic on this endpoint, and overwrite the intermediate buffer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW W0C	SIZE			Number of bytes received last in the data stage of this OUT endpoint

6.35.13.82 SIZE.EPOUT[1]

Address offset: 0x4A4

Number of bytes received last in the data stage of this OUT endpoint

Write to any value to accept further OUT traffic on this endpoint, and overwrite the intermediate buffer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW W0C	SIZE			Number of bytes received last in the data stage of this OUT endpoint

6.35.13.83 SIZE.EPOUT[2]

Address offset: 0x4A8

Number of bytes received last in the data stage of this OUT endpoint

Write to any value to accept further OUT traffic on this endpoint, and overwrite the intermediate buffer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW W0C	SIZE			Number of bytes received last in the data stage of this OUT endpoint

6.35.13.84 SIZE.EPOUT[3]

Address offset: 0x4AC

Number of bytes received last in the data stage of this OUT endpoint

Write to any value to accept further OUT traffic on this endpoint, and overwrite the intermediate buffer

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW W0C	SIZE			Number of bytes received last in the data stage of this OUT endpoint

6.35.13.85 SIZE.EPOUT[4]

Address offset: 0x4B0

Number of bytes received last in the data stage of this OUT endpoint

Write to any value to accept further OUT traffic on this endpoint, and overwrite the intermediate buffer

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW W0C	SIZE			Number of bytes received last in the data stage of this OUT endpoint

6.35.13.86 SIZE.EPOUT[5]

Address offset: 0x4B4

Number of bytes received last in the data stage of this OUT endpoint

Write to any value to accept further OUT traffic on this endpoint, and overwrite the intermediate buffer

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW W0C	SIZE			Number of bytes received last in the data stage of this OUT endpoint

6.35.13.87 SIZE.EPOUT[6]

Address offset: 0x4B8

Number of bytes received last in the data stage of this OUT endpoint

Write to any value to accept further OUT traffic on this endpoint, and overwrite the intermediate buffer

Bit number	31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID	A A A A A A
Reset 0x00000000	**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW W0C	SIZE			Number of bytes received last in the data stage of this OUT endpoint

6.35.13.88 SIZE.EPOUT[7]

Address offset: 0x4BC

Number of bytes received last in the data stage of this OUT endpoint

Write to any value to accept further OUT traffic on this endpoint, and overwrite the intermediate buffer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																													A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	SIZE																																	
	W0C																																		

6.35.13.89 SIZE.ISOOUT

Address offset: 0x4C0

Number of bytes received last on this ISO OUT data endpoint

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																			B							A	A	A	A	A	A	A	A	A	A
Reset 0x00010000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	R	SIZE																																	
B	R	ZERO																																	
			Normal	0																															
			ZeroData	1																															

6.35.13.90 ENABLE

Address offset: 0x500

Enable USB

After writing Disabled to this register, reading the register will return Enabled until USBD is completely disabled.

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A-	RW	ENABLE																																	
			Disabled	0																															
			Enabled	1																															

6.35.13.91 USBPULLUP

Address offset: 0x504

Control of the USB pull-up

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																			A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	CONNECT																																	
			Disabled	0																															
			Enabled	1																															

6.35.13.92 DPDMVALUE

Address offset: 0x508

State D+ and D- lines will be forced into by the DPDMDRIVE task. The DPDMNODRIVE task reverts the control of the lines to MAC IP (no forcing).

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	STATE			State D+ and D- lines will be forced into by the DPDMDRIVE task
			Resume	1	D+ forced low, D- forced high (K state) for a timing preset in hardware (50 µs or 5 ms, depending on bus state)
			J	2	D+ forced high, D- forced low (J state)
			K	4	D+ forced low, D- forced high (K state)

6.35.13.93 DTOGGLE

Address offset: 0x50C

Data toggle control and status

First write this register with VALUE=Nop to select the endpoint, then either read it to get the status from VALUE, or write it again with VALUE=Data0 or Data1

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C C B A A A
Reset 0x00000100				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	EP			Select bulk endpoint number
B	RW	IO			Selects IN or OUT endpoint
			Out	0	Selects OUT endpoint
			In	1	Selects IN endpoint
C	RW	VALUE			Data toggle value
			Nop	0	No action on data toggle when writing the register with this value
			Data0	1	Data toggle is DATA0 on endpoint set by EP and IO
			Data1	2	Data toggle is DATA1 on endpoint set by EP and IO

6.35.13.94 EPINEN

Address offset: 0x510

Endpoint IN enable

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				I H G F E D C B A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	IN0			Enable IN endpoint 0
			Disable	0	Disable endpoint IN 0 (no response to IN tokens)
			Enable	1	Enable endpoint IN 0 (response to IN tokens)
B	RW	IN1			Enable IN endpoint 1
			Disable	0	Disable endpoint IN 1 (no response to IN tokens)
			Enable	1	Enable endpoint IN 1 (response to IN tokens)

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				I H G F E D C B A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
C	RW	IN2			Enable IN endpoint 2
			Disable	0	Disable endpoint IN 2 (no response to IN tokens)
			Enable	1	Enable endpoint IN 2 (response to IN tokens)
D	RW	IN3			Enable IN endpoint 3
			Disable	0	Disable endpoint IN 3 (no response to IN tokens)
			Enable	1	Enable endpoint IN 3 (response to IN tokens)
E	RW	IN4			Enable IN endpoint 4
			Disable	0	Disable endpoint IN 4 (no response to IN tokens)
			Enable	1	Enable endpoint IN 4 (response to IN tokens)
F	RW	IN5			Enable IN endpoint 5
			Disable	0	Disable endpoint IN 5 (no response to IN tokens)
			Enable	1	Enable endpoint IN 5 (response to IN tokens)
G	RW	IN6			Enable IN endpoint 6
			Disable	0	Disable endpoint IN 6 (no response to IN tokens)
			Enable	1	Enable endpoint IN 6 (response to IN tokens)
H	RW	IN7			Enable IN endpoint 7
			Disable	0	Disable endpoint IN 7 (no response to IN tokens)
			Enable	1	Enable endpoint IN 7 (response to IN tokens)
I	RW	ISOIN			Enable ISO IN endpoint
			Disable	0	Disable ISO IN endpoint 8
			Enable	1	Enable ISO IN endpoint 8

6.35.13.95 EPOUTEN

Address offset: 0x514

Endpoint OUT enable

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				I H G F E D C B A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	OUT0			Enable OUT endpoint 0
			Disable	0	Disable endpoint OUT 0 (no response to OUT tokens)
			Enable	1	Enable endpoint OUT 0 (response to OUT tokens)
B	RW	OUT1			Enable OUT endpoint 1
			Disable	0	Disable endpoint OUT 1 (no response to OUT tokens)
			Enable	1	Enable endpoint OUT 1 (response to OUT tokens)
C	RW	OUT2			Enable OUT endpoint 2
			Disable	0	Disable endpoint OUT 2 (no response to OUT tokens)
			Enable	1	Enable endpoint OUT 2 (response to OUT tokens)
D	RW	OUT3			Enable OUT endpoint 3
			Disable	0	Disable endpoint OUT 3 (no response to OUT tokens)
			Enable	1	Enable endpoint OUT 3 (response to OUT tokens)
E	RW	OUT4			Enable OUT endpoint 4
			Disable	0	Disable endpoint OUT 4 (no response to OUT tokens)
			Enable	1	Enable endpoint OUT 4 (response to OUT tokens)
F	RW	OUT5			Enable OUT endpoint 5
			Disable	0	Disable endpoint OUT 5 (no response to OUT tokens)
			Enable	1	Enable endpoint OUT 5 (response to OUT tokens)

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				I H G F E D C B A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
G	RW	OUT6			Enable OUT endpoint 6
			Disable	0	Disable endpoint OUT 6 (no response to OUT tokens)
			Enable	1	Enable endpoint OUT 6 (response to OUT tokens)
H	RW	OUT7			Enable OUT endpoint 7
			Disable	0	Disable endpoint OUT 7 (no response to OUT tokens)
			Enable	1	Enable endpoint OUT 7 (response to OUT tokens)
I	RW	ISOOUT			Enable ISO OUT endpoint 8
			Disable	0	Disable ISO OUT endpoint 8
			Enable	1	Enable ISO OUT endpoint 8

6.35.13.96 EPSTALL

Address offset: 0x518

STALL endpoints

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				C B A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W RME	EP			Select endpoint number
B	W RME	IO			Selects IN or OUT endpoint
			Out	0	Selects OUT endpoint
			In	1	Selects IN endpoint
C	W RME	STALL			Stall selected endpoint
			UnStall	0	Don't stall selected endpoint
			Stall	1	Stall selected endpoint

6.35.13.97 ISOSPLIT

Address offset: 0x51C

Controls the split of ISO buffers

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	SPLIT			Controls the split of ISO buffers
			OneDir	0x0000	Full buffer dedicated to either ISO IN or OUT
			HalfIN	0x0080	Lower half for IN, upper half for OUT

6.35.13.98 FRAMECNTR

Address offset: 0x520

Returns the current value of the start of frame counter

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	R	FRAMECNTR			Returns the current value of the start of frame counter

6.35.13.99 LOWPOWER

Address offset: 0x52C

Controls USBD peripheral low power mode during USB suspend

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	LOWPOWER			Controls USBD peripheral low-power mode during USB suspend
			ForceNormal	0	Software must write this value to exit low power mode and before performing a remote wake-up
			LowPower	1	Software must write this value to enter low power mode after DMA and software have finished interacting with the USB peripheral

6.35.13.100 ISOINCONFIG

Address offset: 0x530

Controls the response of the ISO IN endpoint to an IN token when no data is ready to be sent

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	RESPONSE			Controls the response of the ISO IN endpoint to an IN token when no data is ready to be sent
			NoResp	0	Endpoint does not respond in that case
			ZeroData	1	Endpoint responds with a zero-length data packet in that case

6.35.13.101 EPIN[0].PTR

Address offset: 0x600

Data pointer

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.35.13.102 EPIN[0].MAXCNT

Address offset: 0x604

Maximum number of bytes to transfer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	RW	MAXCNT		[64..0]	Maximum number of bytes to transfer

6.35.13.103 EPIN[0].AMOUNT

Address offset: 0x608

Number of bytes transferred in the last transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	R	AMOUNT			Number of bytes transferred in the last transaction

6.35.13.104 EPIN[1].PTR

Address offset: 0x614

Data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.35.13.105 EPIN[1].MAXCNT

Address offset: 0x618

Maximum number of bytes to transfer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	RW	MAXCNT		[64..0]	Maximum number of bytes to transfer

6.35.13.106 EPIN[1].AMOUNT

Address offset: 0x61C

Number of bytes transferred in the last transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	R	AMOUNT			Number of bytes transferred in the last transaction

6.35.13.107 EPIN[2].PTR

Address offset: 0x628

Data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.35.13.108 EPIN[2].MAXCNT

Address offset: 0x62C

Maximum number of bytes to transfer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	RW	MAXCNT		[64..0]	Maximum number of bytes to transfer

6.35.13.109 EPIN[2].AMOUNT

Address offset: 0x630

Number of bytes transferred in the last transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	R	AMOUNT			Number of bytes transferred in the last transaction

6.35.13.110 EPIN[3].PTR

Address offset: 0x63C

Data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.35.13.111 EPIN[3].MAXCNT

Address offset: 0x640

Maximum number of bytes to transfer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	RW	MAXCNT		[64..0]	Maximum number of bytes to transfer

6.35.13.112 EPIN[3].AMOUNT

Address offset: 0x644

Number of bytes transferred in the last transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	R	AMOUNT			Number of bytes transferred in the last transaction

6.35.13.113 EPIN[4].PTR

Address offset: 0x650

Data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.35.13.114 EPIN[4].MAXCNT

Address offset: 0x654

Maximum number of bytes to transfer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	RW	MAXCNT		[64..0]	Maximum number of bytes to transfer

6.35.13.115 EPIN[4].AMOUNT

Address offset: 0x658

Number of bytes transferred in the last transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	R	AMOUNT			Number of bytes transferred in the last transaction

6.35.13.116 EPIN[5].PTR

Address offset: 0x664

Data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.35.13.117 EPIN[5].MAXCNT

Address offset: 0x668

Maximum number of bytes to transfer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	RW	MAXCNT		[64..0]	Maximum number of bytes to transfer

6.35.13.118 EPIN[5].AMOUNT

Address offset: 0x66C

Number of bytes transferred in the last transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	R	AMOUNT			Number of bytes transferred in the last transaction

6.35.13.119 EPIN[6].PTR

Address offset: 0x678

Data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.35.13.120 EPIN[6].MAXCNT

Address offset: 0x67C

Maximum number of bytes to transfer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	RW	MAXCNT		[64..0]	Maximum number of bytes to transfer

6.35.13.121 EPIN[6].AMOUNT

Address offset: 0x680

Number of bytes transferred in the last transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	R	AMOUNT			Number of bytes transferred in the last transaction

6.35.13.122 EPIN[7].PTR

Address offset: 0x68C

Data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.35.13.123 EPIN[7].MAXCNT

Address offset: 0x690

Maximum number of bytes to transfer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	RW	MAXCNT		[64..0]	Maximum number of bytes to transfer

6.35.13.124 EPIN[7].AMOUNT

Address offset: 0x694

Number of bytes transferred in the last transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	R	AMOUNT			Number of bytes transferred in the last transaction

6.35.13.125 ISOIN.PTR

Address offset: 0x6A0

Data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.35.13.126 ISOIN.MAXCNT

Address offset: 0x6A4

Maximum number of bytes to transfer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																							A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	RW	MAXCNT		[1023..1]	Maximum number of bytes to transfer

6.35.13.127 ISOIN.AMOUNT

Address offset: 0x6A8

Number of bytes transferred in the last transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																							A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	R	AMOUNT			Number of bytes transferred in the last transaction

6.35.13.128 EPOUT[0].PTR

Address offset: 0x700

Data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.35.13.129 EPOUT[0].MAXCNT

Address offset: 0x704

Maximum number of bytes to transfer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	RW	MAXCNT		[64..0]	Maximum number of bytes to transfer

6.35.13.130 EPOUT[0].AMOUNT

Address offset: 0x708

Number of bytes transferred in the last transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	R	AMOUNT			Number of bytes transferred in the last transaction

6.35.13.131 EPOUT[1].PTR

Address offset: 0x714

Data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.35.13.132 EPOUT[1].MAXCNT

Address offset: 0x718

Maximum number of bytes to transfer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	RW	MAXCNT		[64..0]	Maximum number of bytes to transfer

6.35.13.133 EPOUT[1].AMOUNT

Address offset: 0x71C

Number of bytes transferred in the last transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	R	AMOUNT			Number of bytes transferred in the last transaction

6.35.13.134 EPOUT[2].PTR

Address offset: 0x728

Data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.35.13.135 EPOUT[2].MAXCNT

Address offset: 0x72C

Maximum number of bytes to transfer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	RW	MAXCNT		[64..0]	Maximum number of bytes to transfer

6.35.13.136 EPOUT[2].AMOUNT

Address offset: 0x730

Number of bytes transferred in the last transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	R	AMOUNT			Number of bytes transferred in the last transaction

6.35.13.137 EPOUT[3].PTR

Address offset: 0x73C

Data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.35.13.138 EPOUT[3].MAXCNT

Address offset: 0x740

Maximum number of bytes to transfer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	RW	MAXCNT		[64..0]	Maximum number of bytes to transfer

6.35.13.139 EPOUT[3].AMOUNT

Address offset: 0x744

Number of bytes transferred in the last transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	R	AMOUNT			Number of bytes transferred in the last transaction

6.35.13.140 EPOUT[4].PTR

Address offset: 0x750

Data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.35.13.141 EPOUT[4].MAXCNT

Address offset: 0x754

Maximum number of bytes to transfer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	RW	MAXCNT		[64..0]	Maximum number of bytes to transfer

6.35.13.142 EPOUT[4].AMOUNT

Address offset: 0x758

Number of bytes transferred in the last transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	R	AMOUNT			Number of bytes transferred in the last transaction

6.35.13.143 EPOUT[5].PTR

Address offset: 0x764

Data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.35.13.144 EPOUT[5].MAXCNT

Address offset: 0x768

Maximum number of bytes to transfer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	RW	MAXCNT		[64..0]	Maximum number of bytes to transfer

6.35.13.145 EPOUT[5].AMOUNT

Address offset: 0x76C

Number of bytes transferred in the last transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	R	AMOUNT			Number of bytes transferred in the last transaction

6.35.13.146 EPOUT[6].PTR

Address offset: 0x778

Data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.35.13.147 EPOUT[6].MAXCNT

Address offset: 0x77C

Maximum number of bytes to transfer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	RW	MAXCNT		[64..0]	Maximum number of bytes to transfer

6.35.13.148 EPOUT[6].AMOUNT

Address offset: 0x780

Number of bytes transferred in the last transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	R	AMOUNT			Number of bytes transferred in the last transaction

6.35.13.149 EPOUT[7].PTR

Address offset: 0x78C

Data pointer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	PTR			Data pointer See the memory chapter for details about which memories are available for EasyDMA.

6.35.13.150 EPOUT[7].MAXCNT

Address offset: 0x790

Maximum number of bytes to transfer

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	RW	MAXCNT		[64..0]	Maximum number of bytes to transfer

6.35.13.151 EPOUT[7].AMOUNT

Address offset: 0x794

Number of bytes transferred in the last transaction

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A-	R	AMOUNT			Number of bytes transferred in the last transaction

6.35.13.152 ISOOUT.PTR

Address offset: 0x7A0

Data pointer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	RW	PTR																																	

Data pointer

See the memory chapter for details about which memories are available for EasyDMA.

6.35.13.153 ISOOUT.MAXCNT

Address offset: 0x7A4

Maximum number of bytes to transfer

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A-	RW	MAXCNT																																	

Maximum number of bytes to transfer

6.35.13.154 ISOOUT.AMOUNT

Address offset: 0x7A8

Number of bytes transferred in the last transaction

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																										A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A-	R	AMOUNT																																	

Number of bytes transferred in the last transaction

6.35.14 Electrical specification

6.35.14.1 USB Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
$R_{USB,PU,ACTIVE}$	Value of pull-up on D+, bus active (upstream device transmitting)	1425	2300	3090	Ω
$R_{USB,PU,IDLE}$	Value of pull-up on D+, bus idle	900	1200	1575	Ω
$t_{USB,DETRST}$	Minimum duration of an SE0 state to be detected as a USB reset condition				µs
$f_{USB,CLK}$	Frequency of local clock, USB active		48		MHz
$f_{USB,TOL}$	Accuracy of local clock, USB active[44]			±1000	ppm
$T_{USB,JITTER}$	Jitter on USB local clock, USB active			±1	ns

6.36 WDT — Watchdog timer

A countdown watchdog timer using the low-frequency clock source (LFCLK) offers configurable and robust protection against application lock-up.

The watchdog timer is started by triggering the START task.

[44] The local clock can be stopped during USB suspend

The watchdog can be paused during long CPU sleep periods for low power applications and when the debugger has halted the CPU. The watchdog is implemented as a down-counter that generates a TIMEOUT event when it wraps over after counting down to 0. When the watchdog timer is started through the START task, the watchdog counter is loaded with the value specified in the CRV register. This counter is also reloaded with the value specified in the CRV register when a reload request is granted.

The watchdog's timeout period is given by the following equation:

```
timeout [s] = ( CRV + 1 ) / 32768
```

When started, the watchdog will automatically force the 32.768 kHz RC oscillator on as long as no other 32.768 kHz clock source is running and generating the 32.768 kHz system clock, see chapter CLOCK — Clock control on page 157.

6.36.1 Reload criteria

The watchdog has eight separate reload request registers, which shall be used to request the watchdog to reload its counter with the value specified in the CRV register. To reload the watchdog counter, the special value 0x6E524635 needs to be written to all enabled reload registers.

One or more RR registers can be individually enabled through the RREN register.

6.36.2 Temporarily pausing the watchdog

By default, the watchdog will be active counting down the down-counter while the CPU is sleeping and when it is halted by the debugger. It is possible to configure the watchdog to automatically pause while the CPU is sleeping as well as when it is halted by the debugger.

6.36.3 Watchdog reset

A TIMEOUT event will automatically lead to a watchdog reset.

See Reset on page 89 for more information about reset sources. If the watchdog is configured to generate an interrupt on the TIMEOUT event, the watchdog reset will be postponed with two 32.768 kHz clock cycles after the TIMEOUT event has been generated. Once the TIMEOUT event has been generated, the impending watchdog reset will always be effectuated.

The watchdog must be configured before it is started. After it is started, the watchdog's configuration registers, which comprise registers CRV, RREN, and CONFIG, will be blocked for further configuration.

The watchdog can be reset from several reset sources, see Reset behavior on page 90.

When the device starts running again, after a reset, or waking up from OFF mode, the watchdog configuration registers will be available for configuration again.

6.36.4 Registers

Instances

Instance	Base address	Description
WDT	0x40010000	Watchdog timer

Register overview

Register	Offset	Description
TASKS_START	0x000	Start the watchdog
EVENTS_TIMEOUT	0x100	Watchdog timeout
INTENSET	0x304	Enable interrupt
INTENCLR	0x308	Disable interrupt
RUNSTATUS	0x400	Run status
REQSTATUS	0x404	Request status
CRV	0x504	Counter reload value
RREN	0x508	Enable register for reload request registers
CONFIG	0x50C	Configuration register
RR[0]	0x600	Reload request 0
RR[1]	0x604	Reload request 1
RR[2]	0x608	Reload request 2
RR[3]	0x60C	Reload request 3
RR[4]	0x610	Reload request 4
RR[5]	0x614	Reload request 5
RR[6]	0x618	Reload request 6
RR[7]	0x61C	Reload request 7

6.36.4.1 TASKS_START

Address offset: 0x000

Start the watchdog

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	W	TASKS_START			Start the watchdog
			Trigger	1	Trigger task

6.36.4.2 EVENTS_TIMEOUT

Address offset: 0x100

Watchdog timeout

Bit number	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID																																A
Reset 0x00000000	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

ID	R/W	Field	Value ID	Value	Description
A	RW	EVENTS_TIMEOUT			Watchdog timeout
			NotGenerated	0	Event not generated
			Generated	1	Event generated

6.36.4.3 INTENSET

Address offset: 0x304

Enable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TIMEOUT			Write '1' to enable interrupt for event TIMEOUT
			Set	1	Enable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.36.4.4 INTENCLR

Address offset: 0x308

Disable interrupt

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	RW	TIMEOUT			Write '1' to disable interrupt for event TIMEOUT
			Clear	1	Disable
			Disabled	0	Read: Disabled
			Enabled	1	Read: Enabled

6.36.4.5 RUNSTATUS

Address offset: 0x400

Run status

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**

ID	R/W	Field	Value ID	Value	Description
A	R	RUNSTATUS			Indicates whether or not the watchdog is running
			NotRunning	0	Watchdog not running
			Running	1	Watchdog is running

6.36.4.6 REQSTATUS

Address offset: 0x404

Request status

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				H G F E D C B A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**

ID	R/W	Field	Value ID	Value	Description
A	R	RR0			Request status for RR[0] register
			DisabledOrRequested	0	RR[0] register is not enabled, or is already requesting reload
			EnabledAndUnrequested	1	RR[0] register is enabled, and is not yet requesting reload
B	R	RR1			Request status for RR[1] register
			DisabledOrRequeste	0	RR[1] register is not enabled, or is already requesting reload
			EnabledAndUnreque	1	RR[1] register is enabled, and is not yet requesting reload
C	R	RR2			Request status for RR[2] register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				H G F E D C B A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
			DisabledOrRequested	0	RR[2] register is not enabled, or is already requesting reload
			EnabledAndUnrequested	1	RR[2] register is enabled, and is not yet requesting reload
D	R	RR3			Request status for RR[3] register
			DisabledOrRequested	0	RR[3] register is not enabled, or is already requesting reload
			EnabledAndUnreque	1	RR[3] register is enabled, and is not yet requesting reload
E	R	RR4			Request status for RR[4] register
			DisabledOrRequested	0	RR[4] register is not enabled, or is already requesting reload
			EnabledAndUnrequested	1	RR[4] register is enabled, and is not yet requesting reload
F	R	RR5			Request status for RR[5] register
			DisabledOrRequested	0	RR[5] register is not enabled, or is already requesting reload
			EnabledAndUnreque	1	RR[5] register is enabled, and is not yet requesting reload
G	R	RR6			Request status for RR[6] register
			DisabledOrRequested	0	RR[6] register is not enabled, or is already requesting reload
			EnabledAndUnrequested	1	RR[6] register is enabled, and is not yet requesting reload
H	R	RR7			Request status for RR[7] register
			DisabledOrRequested	0	RR[7] register is not enabled, or is already requesting reload
			EnabledAndUnreque	1	RR[7] register is enabled, and is not yet requesting reload

6.36.4.7 CRV

Address offset: 0x504

Counter reload value

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0xFFFFFFFF				**1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	CRV		[0xF..0xFFFFFFFF]	Counter reload value in number of cycles of the 32.768 kHz clock

6.36.4.8 RREN

Address offset: 0x508

Enable register for reload request registers

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				H G F E D C B A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	RR0			Enable or disable RR[0] register
			Disabled	0	Disable RR[0] register
			Enabled	1	Enable RR[0] register
B	RW	RR1			Enable or disable RR[1] register
			Disabled	0	Disable RR[1] register
			Enabled	1	Enable RR[1] register
C	RW	RR2			Enable or disable RR[2] register
			Disabled	0	Disable RR[2] register
			Enabled	1	Enable RR[2] register
D	RW	RR3			Enable or disable RR[3] register
			Disabled	0	Disable RR[3] register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				H G F E D C B A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
			Enabled	1	Enable RR[3] register
E	RW	RR4			Enable or disable RR[4] register
			Disabled	0	Disable RR[4] register
			Enabled	1	Enable RR[4] register
F	RW	RR5			Enable or disable RR[5] register
			Disabled	0	Disable RR[5] register
			Enabled	1	Enable RR[5] register
G	RW	RR6			Enable or disable RR[6] register
			Disabled	0	Disable RR[6] register
			Enabled	1	Enable RR[6] register
H	RW	RR7			Enable or disable RR[7] register
			Disabled	0	Disable RR[7] register
			Enabled	1	Enable RR[7] register

6.36.4.9 CONFIG

Address offset: 0x50C

Configuration register

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				B A
Reset 0x00000001				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1**
ID	R/W	Field	Value ID	Value	Description
A	RW	SLEEP			Configure the watchdog to either be paused, or kept running, while the CPU is sleeping
			Pause	0	Pause watchdog while the CPU is sleeping
			Run	1	Keep the watchdog running while the CPU is sleeping
B	RW	HALT			Configure the watchdog to either be paused, or kept running, while the CPU is halted by the debugger
			Pause	0	Pause watchdog while the CPU is halted by the debugger
			Run	1	Keep the watchdog running while the CPU is halted by the debugger

6.36.4.10 RR[0]

Address offset: 0x600

Reload request 0

Bit number				31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
ID				A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
Reset 0x00000000				**0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0**
ID	R/W	Field	Value ID	Value	Description
A	W	RR			Reload request register
			Reload	0x6E524635	Value to request a reload of the watchdog timer

6.36.4.11 RR[1]

Address offset: 0x604

Reload request 1

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	RR																																	
			Reload	0x6E524635																															

6.36.4.12 RR[2]

Address offset: 0x608

Reload request 2

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	RR																																	
			Reload	0x6E524635																															

6.36.4.13 RR[3]

Address offset: 0x60C

Reload request 3

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	RR																																	
			Reload	0x6E524635																															

6.36.4.14 RR[4]

Address offset: 0x610

Reload request 4

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	RR																																	
			Reload	0x6E524635																															

6.36.4.15 RR[5]

Address offset: 0x614

Reload request 5

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	RR																																	
			Reload	0x6E524635																															

6.36.4.16 RR[6]

Address offset: 0x618

Reload request 6

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	RR																																	
			Reload	0x6E524635																															

6.36.4.17 RR[7]

Address offset: 0x61C

Reload request 7

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ID				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x00000000				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
ID	R/W	Field	Value ID	Value																															
A	W	RR																																	
			Reload	0x6E524635																															

6.36.5 Electrical specification

6.36.5.1 Watchdog Timer Electrical Specification

Symbol	Description	Min.	Typ.	Max.	Units
t_{WDT}	Time out interval	458 µs		36 h	

7 Hardware and layout

7.1 Pin assignments

The pin assignment figures and tables describe the pinouts for the product variants of the chip.

The nRF52840 device provides flexibility regarding GPIO pin routing and configuration. However, some pins have limitations or recommendations for pin configurations and uses.

7.1.1 aQFN73 ball assignments

The ball assignment figure and table in the following section describe the assignments for this variant of the chip.

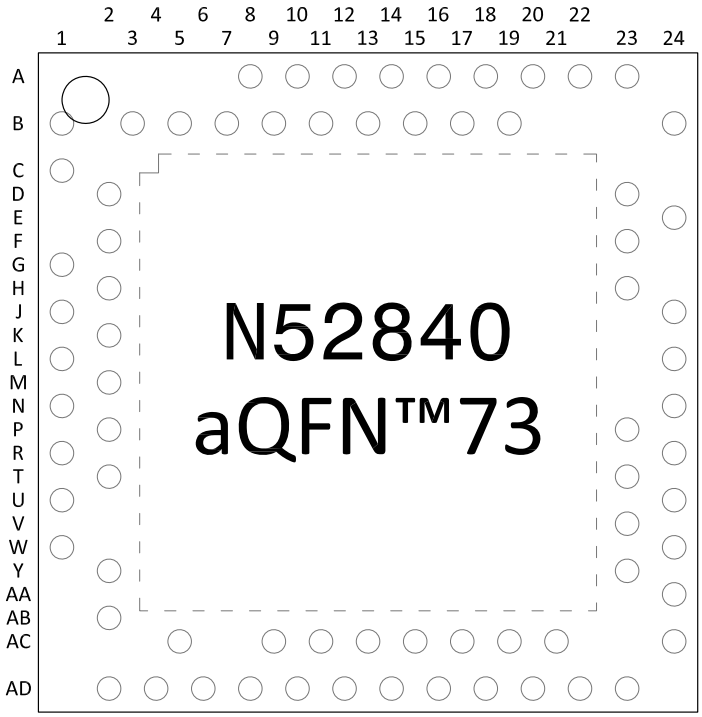


Figure 212: aQFN73 ball assignments, top view

Pin	Name	Function	Description	Recommended usage
A8	P0.31 AIN7	Digital I/O Analog input	General purpose I/O Analog input	Standard drive, low frequency I/O only
A10	P0.29 AIN5	Digital I/O Analog input	General purpose I/O Analog input	Standard drive, low frequency I/O only
A12	P0.02 AIN0	Digital I/O Analog input	General purpose I/O Analog input	Standard drive, low frequency I/O only
A14	P1.15	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
A16	P1.13	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
A18	DEC2	Power	1.3 V regulator supply decoupling	
A20	P1.10	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
A22	VDD	Power	Power supply	
A23	XC2	Analog input	Connection for 32 MHz crystal	
B1	VDD	Power	Power supply	
B3	DCC	Power	DC/DC converter output	
B5	DEC4	Power	1.3 V regulator supply decoupling	Must be connected to DEC6 (pin E24)
B7	VSS	Power	Ground	
B9	P0.30 AIN6	Digital I/O Analog input	General purpose I/O Analog input	Standard drive, low frequency I/O only
B11	P0.28 AIN4	Digital I/O Analog input	General purpose I/O Analog input	Standard drive, low frequency I/O only
B13	P0.03 AIN1	Digital I/O Analog input	General purpose I/O Analog input	Standard drive, low frequency I/O only
B15	P1.14	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
B17	P1.12	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
B19	P1.11	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
B24	XC1	Analog input	Connection for 32 MHz crystal	
C1	DEC1	Power	1.1 V regulator supply decoupling	
D2	P0.00 XL1	Digital I/O Analog input	General purpose I/O Connection for 32.768 kHz crystal	
D23	DEC3	Power	Power supply, decoupling	
E24	DEC6	Power	1.3 V regulator supply decoupling	Must be connected to DEC4 (pin B5)
F2	P0.01 XL2	Digital I/O Analog input	General purpose I/O Connection for 32.768 kHz crystal	
F23	VSS_PA	Power	Ground (radio supply)	
G1	P0.26	Digital I/O	General purpose I/O	
H2	P0.27	Digital I/O	General purpose I/O	
H23	ANT	RF	Single-ended radio antenna connection	See Reference circuitry on page 937 for guidelines on how to ensure good RF performance
J1	P0.04 AIN2	Digital I/O Analog input	General purpose I/O Analog input	
J24	P0.10	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only

Pin	Name	Function	Description	Recommended usage
	NFC2	NFC input	NFC antenna connection	
K2	P0.05	Digital I/O	General purpose I/O	
	AIN3	Analog input	Analog input	
L1	P0.06	Digital I/O	General purpose I/O	
L24	P0.09	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
	NFC1	NFC input	NFC antenna connection	
M2	P0.07	Digital I/O	General purpose I/O	
	TRACECLK	Trace clock	Trace buffer clock	
N1	P0.08	Digital I/O	General purpose I/O	
N24	DEC5	Power	1.3 V regulator supply decoupling for build codes Dxx and earlier.	
	Not connected		Not connected for build codes Fxx and later.	
P2	P1.08	Digital I/O	General purpose I/O	
P23	P1.07	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
R1	P1.09	Digital I/O	General purpose I/O	
	TRACEDATA3	Trace data	Trace buffer TRACEDATA[3]	
R24	P1.06	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
T2	P0.11	Digital I/O	General purpose I/O	
	TRACEDATA2	Trace data	Trace buffer TRACEDATA[2]	
T23	P1.05	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
U1	P0.12	Digital I/O	General purpose I/O	
	TRACEDATA1	Trace data	Trace buffer TRACEDATA[1]	
U24	P1.04	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
V23	P1.03	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
W1	VDD	Power	Power supply	
W24	P1.02	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
Y2	VDDH	Power	High voltage power supply	
Y23	P1.01	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
AA24	SWDCLK	Debug	Serial wire debug clock input for debug and programming	
AB2	DCCH	Power	DC/DC converter output	
AC5	DECUSB	Power	USB 3.3 V regulator supply decoupling	
AC9	P0.14	Digital I/O	General purpose I/O	
AC11	P0.16	Digital I/O	General purpose I/O	
AC13	P0.18	Digital I/O	General purpose I/O	QSPI/CSN
	nRESET		Configurable as pin RESET	
AC15	P0.19	Digital I/O	General purpose I/O	QSPI/SCK
AC17	P0.21	Digital I/O	General purpose I/O	QSPI
AC19	P0.23	Digital I/O	General purpose I/O	QSPI
AC21	P0.25	Digital I/O	General purpose I/O	
AC24	SWDIO	Debug	Serial wire debug I/O for debug and programming	
AD2	VBUS	Power	5 V input for USB 3.3 V regulator	
AD4	D-	USB	USB D-	
AD6	D+	USB	USB D+	

Pin	Name	Function	Description	Recommended usage
AD8	P0.13	Digital I/O	General purpose I/O	
AD10	P0.15	Digital I/O	General purpose I/O	
AD12	P0.17	Digital I/O	General purpose I/O	
AD14	VDD	Power	Power supply	
AD16	P0.20	Digital I/O	General purpose I/O	
AD18	P0.22	Digital I/O	General purpose I/O	QSPI
AD20	P0.24	Digital I/O	General purpose I/O	
AD22	P1.00 TRACEDATA0	Digital I/O Trace data	General purpose I/O Trace buffer TRACEDATA[0] Serial wire output (SWO)	QSPI
AD23	VDD	Power	Power supply	
Die pad	VSS	Power	Ground pad	Exposed die pad must be connected to ground (VSS) for proper device operation

Table 61: aQFN73 ball assignments

Note: For more information on standard drive, see GPIO — General purpose input/output on page 322. Low frequency I/O is a signal with a frequency up to 10 kHz.

7.1.2 QFN48 pin assignments

The pin assignment figure and table describe the assignments for this variant of the chip.

Note: VDD and VDDH are shortcircuited inside the package. Therefore the device is only usable in Normal Voltage supply mode, and not High Voltage supply mode.

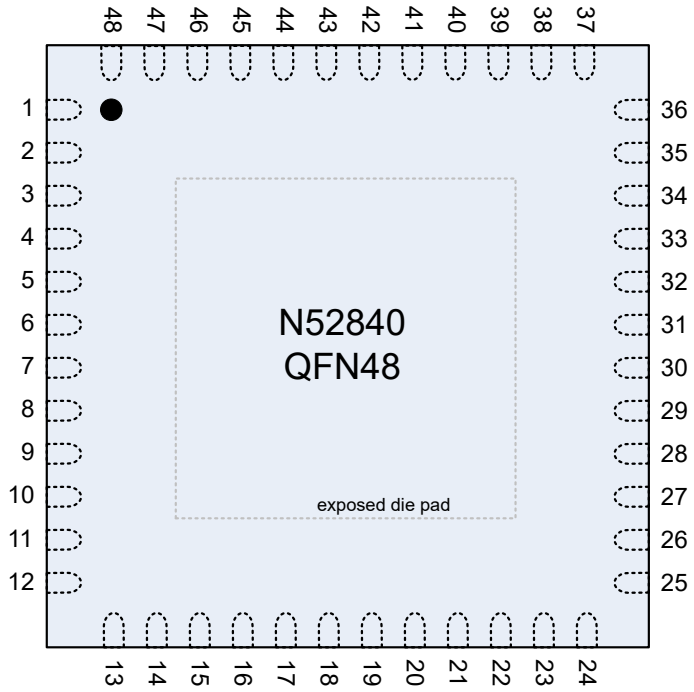


Figure 213: QFN48 pin assignments, top view

Pin	Name	Function	Description	Recommended usage
Left side of the chip				
1	DEC1	Power	1.1 V Digital supply decoupling	
2	P0.00	Digital I/O	General purpose I/O	
	XL1	Analog input	Connection for 32.768 kHz crystal	
3	P0.01	Digital I/O	General purpose I/O	
	XL2	Analog input	Connection for 32.768 kHz crystal	
4	P0.04	Digital I/O	General purpose I/O	
	AIN2	Analog input	Analog input	
5	P0.05	Digital I/O	General purpose I/O	
	AIN3	Analog input	Analog input	
6	P0.07	Digital I/O	General purpose I/O	
	TRACECLK	Trace clock	Trace buffer clock	
7	P0.08	Digital I/O	General purpose I/O	
8	P1.08	Digital I/O	General purpose I/O	
9	P1.09	Digital I/O	General purpose I/O	
	TRACEDATA3	Trace data	Trace buffer TRACEDATA[3]	
10	P0.11	Digital I/O	General purpose I/O	
	TRACEDATA2	Trace data	Trace buffer TRACEDATA[2]	
11	P0.12	Digital I/O	General purpose I/O	
	TRACEDATA1	Trace data	Trace buffer TRACEDATA[1]	
12	VDD	Power	Power supply	
Bottom side of chip				
13	P0.13	Digital I/O	General purpose I/O	
14	P0.14	Digital I/O	General purpose I/O	
15	P0.17	Digital I/O	General purpose I/O	
16	P0.18	Digital I/O	General purpose I/O	QSPI/CSN
	nRESET		Configurable as pin RESET	
17	VDD	Power	Power supply	
18	P0.19	Digital I/O	General purpose I/O	QSPI/SCK
19	P0.20	Digital I/O	General purpose I/O	
20	P0.21	Digital I/O	General purpose I/O	QSPI
21	P0.22	Digital I/O	General purpose I/O	QSPI
22	P0.23	Digital I/O	General purpose I/O	QSPI
23	P0.24	Digital I/O	General purpose I/O	
24	P1.00	Digital I/O	General purpose I/O	QSPI
	TRACEDATA0	Trace data	Trace buffer TRACEDATA[0]	
			Serial wire output (SWO)	
Right side of the chip				
25	VDD	Power	Power supply	
26	SWDIO	Debug	Serial wire debug I/O for debug and programming	
27	SWDCLK	Debug	Serial wire debug clock input for debug and programming	
28	NC		Not connected	
29	P0.09	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
	NFC1	NFC input	NFC antenna connection	
30	P0.10	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
	NFC2	NFC input	NFC antenna connection	

Pin	Name	Function	Description	Recommended usage
31	ANT	RF	Single-ended radio antenna connection	See Reference circuitry on page 937 for guidelines on how to ensure good RF performance
32	VSS_PA	Power	Ground (radio supply)	
33	DEC6	Power	1.3 V regulator supply decoupling	Must be connected to DEC4 (pin 46)
34	DEC3	Power	Power supply, decoupling	
35	XC1	Analog input	Connection for 32 MHz crystal	
36	XC2	Analog input	Connection for 32 MHz crystal	
Top side of the chip				
37	VDD	Power	Power supply	
38	P1.15	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
39	P0.03	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
	AIN1	Analog input	Analog input	
40	P0.02	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
	AIN0	Analog input	Analog input	
41	P0.28	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
	AIN4	Analog input	Analog input	
42	P0.29	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
	AIN5	Analog input	Analog input	
43	P0.30	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
	AIN6	Analog input	Analog input	
44	P0.31	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
	AIN7	Analog input	Analog input	
45	VSS	Power	Ground	
46	DEC4	Power	1.3 V regulator supply decoupling	Must be connected to DEC6 (pin 33)
47	DCC	Power	DC/DC converter output	
48	VDD	Power	Power supply	
Backside of the chip				
Die pad	VSS	Power	Ground pad	Exposed die pad must be connected to ground (VSS) for proper device operation

Table 62: QFN48 pin assignments

7.1.3 WLCSP ball assignments

The ball assignment figure and table describe the assignments for this variant of the chip.

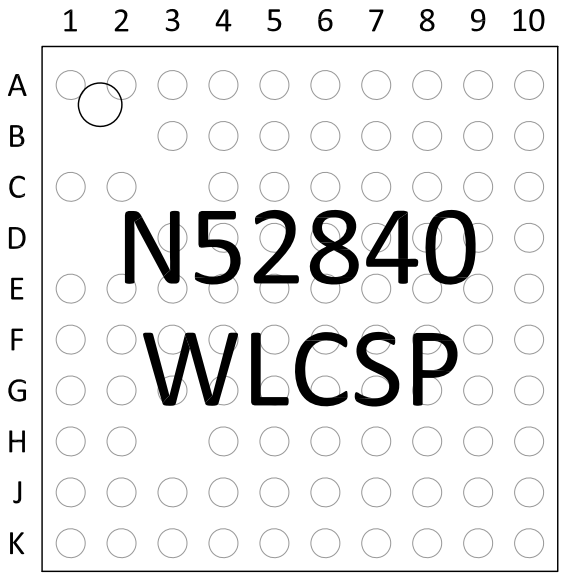


Figure 214: WLCSP ball assignments, top view

Pin	Name	Function	Description	Recommended usage
A1	XC1	Analog input	Connection for 32 MHz crystal	
A2	XC2	Analog input	Connection for 32 MHz crystal	
A3	P1.11	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
A4	P1.13	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
A5	P0.03	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
	AIN1	Analog input	Analog input	
A6	P0.28	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
	AIN4	Analog input	Analog input	
A7	P0.30	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
	AIN6	Analog input	Analog input	
A8	DEC4	Power	1.3 V regulator supply decoupling Must be connected to DEC6 (pin C2)	
A9	DCC	Power	DC/DC converter output	
A10	DEC1	Power	1.1 V regulator supply decoupling	
B3	VDD	Power	Power supply	
B4	P1.10	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
B5	P1.14	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
B6	P0.02	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
	AIN0	Analog input	Analog input	
B7	VSS	Power	Ground	
B8	VDD	Power	Power supply	
B9	P0.00	Digital I/O	General purpose I/O	
	XL1	Analog input	Connection for 32.768 kHz crystal	
B10	P0.01	Digital I/O	General purpose I/O	
	XL2	Analog input	Connection for 32.768 kHz crystal	
C1	VSS_PA	Power	Ground (radio supply)	
C2	DEC6	Power	1.3 V regulator supply decoupling Must be connected to DEC4 (pin A8)	
C4	VSS	Power	Ground	
C5	P1.12	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
C6	P1.15	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
C7	P0.29	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
	AIN5	Analog input	Analog input	
C8	P0.31	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
	AIN7	Analog input	Analog input	
C9	P0.26	Digital I/O	General purpose I/O	
C10	P0.04	Digital I/O	General purpose I/O	
	AIN2	Analog input	Analog input	
D3	VSS	Power	Ground	
D4	VSS	Power	Ground	
D5	VSS	Power	Ground	
D6	VSS	Power	Ground	
D7	VSS	Power	Ground	
D8	VSS	Power	Ground	

Pin	Name	Function	Description	Recommended usage
D9	P0.27	Digital I/O	General purpose I/O	
D10	P0.05	Digital I/O	General purpose I/O	
	AIN3	Analog input	Analog input	
E1	ANT	RF	Single-ended radio antenna connection	See Reference circuitry on page 937 for guidelines on how to ensure good RF performance
E2	P0.10	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
	NFC2	NFC input	NFC antenna connection	
E3	P1.06	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
E4	VSS	Power	Ground	
E5	VSS	Power	Ground	
E6	VSS	Power	Ground	
E7	VSS	Power	Ground	
E8	VSS	Power	Ground	
E9	P0.06	Digital I/O	General purpose I/O	
E10	P0.08	Digital I/O	General purpose I/O	
F1	P0.09	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
	NFC1	NFC input	NFC antenna connection	
F2	DEC5	Power	1.3 V regulator supply decoupling for build codes Dxx and earlier.	
	Not connected		Not connected for build codes Fxx and later.	
F3	P1.03	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
F4	VSS	Power	Ground	
F5	VSS	Power	Ground	
F6	VSS	Power	Ground	
F7	VSS	Power	Ground	
F8	VSS	Power	Ground	
F9	P0.07	Digital I/O	General purpose I/O	
	TRACECLK	Trace clock	Trace buffer clock	
F10	P1.09	Digital I/O	General purpose I/O	
	TRACEDATA3	Trace data	Trace buffer TRACEDATA[3]	
G1	P1.07	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
G2	P1.05	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
G3	P1.02	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
G4	VSS	Power	Ground	
G5	VSS	Power	Ground	
G6	VSS	Power	Ground	
G7	VSS	Power	Ground	
G8	VSS	Power	Ground	
G9	P1.08	Digital I/O	General purpose I/O	
G10	P0.12	Digital I/O	General purpose I/O	
	TRACEDATA1	Trace data	Trace buffer TRACEDATA[1]	
H1	P1.04	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
H2	SWDCLK	Digital input	Serial wire debug clock input for debug and programming	

Pin	Name	Function	Description	Recommended usage
H4	P0.24	Digital I/O	General purpose I/O	
H5	P0.23	Digital I/O	General purpose I/O	QSPI
H6	P0.16	Digital I/O	General purpose I/O	
H7	P0.13	Digital I/O	General purpose I/O	
H8	P0.11	Digital I/O	General purpose I/O	
	TRACEDATA2	Trace data	Trace buffer TRACEDATA[2]	
H9	DCCH	Power	DC/DC converter output	
H10	VDD	Power	Power supply	
J1	P1.01	Digital I/O	General purpose I/O	Standard drive, low frequency I/O only
J2	SWDIO	Digital I/O	Serial wire debug I/O for debug and programming	
J3	P1.00	Digital I/O	General purpose I/O	QSPI
	TRACEDATA0	Trace data	Trace buffer TRACEDATA[0]	
			Serial wire output (SWO)	
J4	P0.21	Digital I/O	General purpose I/O	QSPI
J5	P0.20	Digital I/O	General purpose I/O	
J6	P0.17	Digital I/O	General purpose I/O	
J7	P0.14	Digital I/O	General purpose I/O	
J8	D-	USB	USB D-	
J9	VBUS	Power	5 V input for USB 3.3 V regulator	
J10	VDDH	Power	High voltage power supply	
K1	VDD	Power	Power supply	
K2	P0.25	Digital I/O	General purpose I/O	
K3	P0.22	Digital I/O	General purpose I/O	QSPI
K4	P0.19	Digital I/O	General purpose I/O	QSPI/SCK
K5	VDD	Power	Power supply	
K6	P0.18	Digital I/O	General purpose I/O	QSPI/CSN
	nRESET		Configurable as pin reset	
K7	P0.15	Digital I/O	General purpose I/O	
K8	D+	USB	USB D+	
K9	DECUSB	Power	USB 3.3 V regulator supply decoupling	
K10	VSS	Power	Ground	

Table 63: WLCSP ball assignments

Note: For more information on standard drive, see GPIO — General purpose input/output on page 322. Low frequency I/O is a signal with a frequency up to 10 kHz.

7.2 Mechanical specifications

The mechanical specifications for the packages show the dimensions in millimeters.

7.2.1 aQFN73 7 x 7 mm package

Dimensions in millimeters for the aQFN73 7 x 7 mm package.

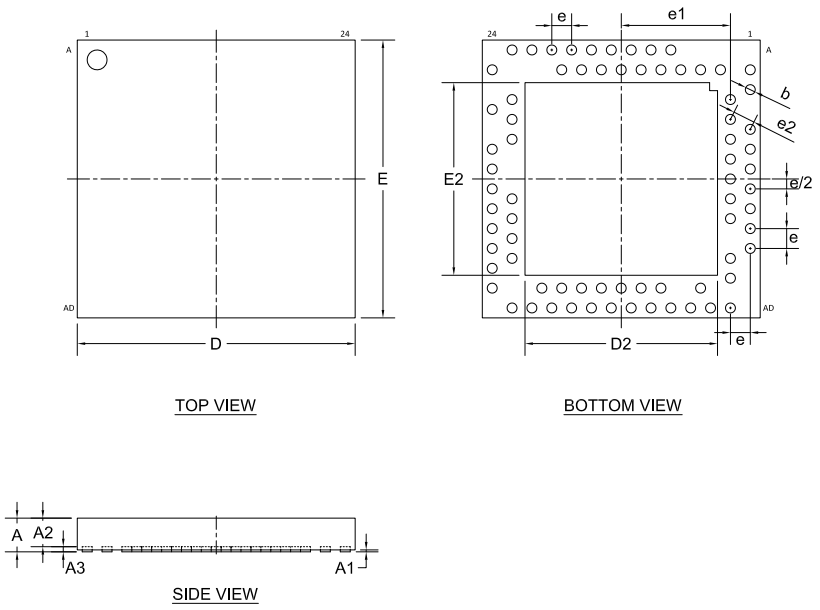


Figure 215: aQFN73 7 x 7 mm package

	A	A1	A2	A3	b	D, E	D2, E2	e	e1	e2
Min.		0.00			0.20	6.90	4.75			
Nom.			0.675	0.13	0.25	7.00	4.85	0.50	2.75	0.559
Max.	0.85	0.08			0.30	7.10	4.95			

Table 64: aQFN73 dimensions in millimeters

7.2.2 QFN48 6 x 6 mm package

Dimensions in millimeters for the QFN48 6 x 6 mm package.

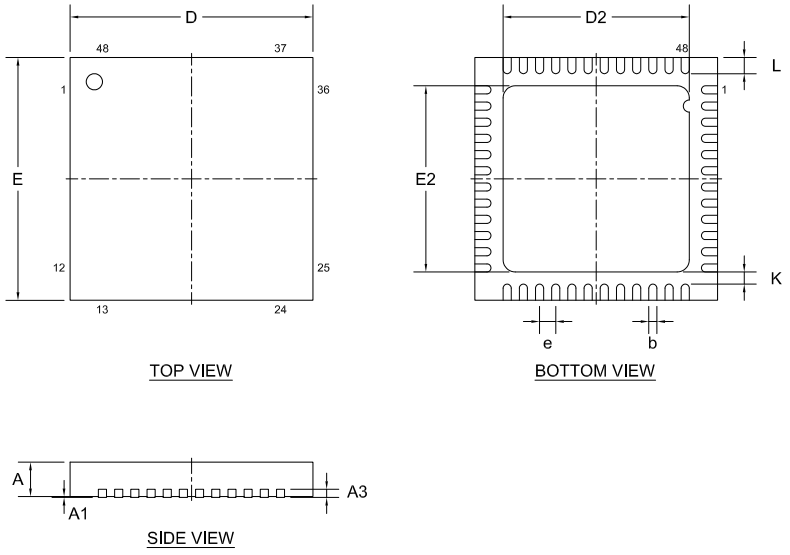


Figure 216: QFN48 6 x 6 mm package

	A	A1	A3	b	D, E	D2, E2	e	K	L
Min.	0.80	0.00		0.15	5.9	4.5		0.2	0.35
Nom.	0.85	0.035	0.203	0.20	6.0	4.6	0.4		0.40
Max.	0.90	0.05		0.25	6.1	4.7			0.45

Table 65: QFN48 dimensions in millimeters

7.2.3 WLCSP 3.544 x 3.607 mm package

Dimensions in millimeters for the WLCSP 3.544 x 3.607 mm package.

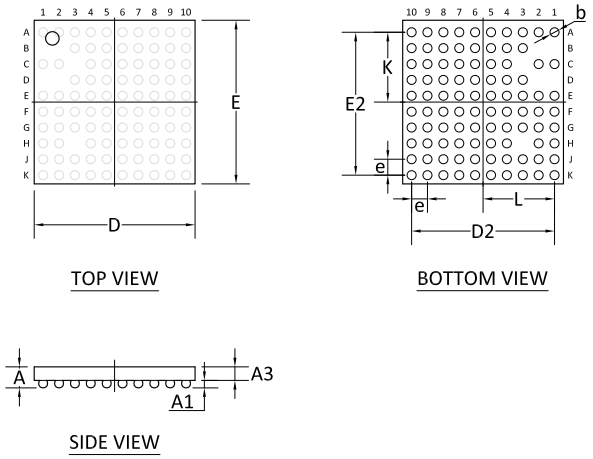


Figure 217: WLCSP 3.544 x 3.607 mm package

	A	A1	A3	b	D	E	D2	E2	e	K	L
Min.	0.464	0.148	0.303	0.184	3.514	3.577					
Nom.	0.489		0.325		3.544	3.607	3.15	3.15	0.35	1.575	1.575
Max.	0.514	0.18	0.347	0.244	3.574	3.637					

Table 66: WLCSP dimensions in millimeters

7.3 Reference circuitry

To ensure good RF performance when designing PCBs, it is highly recommended to use the PCB layouts and component values provided by Nordic Semiconductor.

Documentation for the different package reference circuits, including Altium Designer files, PCB layout files, and PCB production files can be downloaded from the product page for the nRF52840 on www.nordicsemi.com.

In this section there are reference circuits for QIAA aQFN73, QFAA QFN48, and CKAA WLCSP showing the components and component values to support on-chip features in a design.

Note: This is not a complete list of configurations, but all required circuitry is shown for further configurations.

Some general guidance is summarized here:

- External supply from VDD is only available when power is supplied to VDDH. External supply is annotated with the VEXT net name.
- When supplying power from a USB source only, VBUS must be connected to VDDH if USB is to be used.
- Components required for DC/DC function are only needed if DC/DC mode is enabled for that regulator.
- NFC can be used in any configuration.
- USB can be used in any configuration as long as VBUS is supplied by the USB host.
- The schematics include an optional series resistor on the USB supply for improved immunity to transient overvoltage during VBUS connection. Using the series resistor is recommended for new designs.
- Two component values for the RF-Match network for the QIAA aQFN73 package are given and referred to as v1.0 and v1.1 in the following tables. The reference schematics use v1.1 component values, which are recommended for new designs to improve the margin for spurious emissions during regulatory approval tests. However, both v1.0 and v1.1 are valid and can be used. All other RF parameters are unchanged.
- A new reference design with four-component RF match has been added for the QIAA aQFN73 package. The four-component RF match improves harmonic suppression when using Radio with TXPOWER equal to 5dBm or above. However, previous 3 component RF-match designs are valid and can be used. Using this four-component RF match is recommended for new designs.

Circuit configurations for QIAA aQFN73

Config no.	Supply configuration		Features that can be enabled for each configuration example				
	VDDH	VDD	EXTSUPPLY	DCDCEN0	DCDCEN1	USB	NFC
Config. 1	USB (VDDH = VBUS)	N/A	Yes	No	No	Yes	No
Config. 2	Battery/Ext. regulator	N/A	Yes	No	No	Yes	No
Config. 3	N/A	Battery/ Ext. regulator	No	No	No	Yes	No
Config. 4	Battery/Ext. regulator	N/A	Yes	Yes	Yes	Yes	No
Config. 5	N/A	Battery/ Ext. regulator	No	No	Yes	Yes	Yes
Config. 6	N/A	Battery/ Ext. regulator	No	No	No	No	No
Config. 7[1]	Battery/Ext. regulator	N/A	Yes	Yes	Yes	Yes	No

Table 67: Circuit configurations

[1]Reference design with four-component RF match.

Circuit configurations for QFAA QFN48

Config no.	Supply configuration		Features that can be enabled for each configuration example				
	VDDH[1]	VDD	EXTSUPPLY[2]	DCDCEN0[3]	DCDCEN1	USB[4]	NFC
Config. 1	N/A	Battery/Ext. regulator	N/A	N/A	Yes	N/A	No

Table 68: Circuit configurations

[1]High Voltage supply mode cannot be used because the VDDH pin is not routed in the QFN48 package.

[2]The external supply feature cannot be used because the VDDH pin is not routed in the QFN48 package.

[3]DCDC for REG0 stage cannot be used because the VDDH pin is not routed in the QFN48 package.

[4]USBD cannot be used because the USB pins are not routed in the QFN48 package.

Circuit configurations for CKAA WLCSP

Config no.	Supply configuration		Features that can be enabled for each configuration example				
	VDDH	VDD	EXTSUPPLY	DCDCEN0	DCDCEN1	USB	NFC
Config. 1	USB (VDDH = VBUS)	N/A	Yes	No	No	Yes	No
Config. 2	Battery/Ext. regulator	N/A	Yes	No	No	Yes	No
Config. 3	N/A	Battery/Ext. regulator	No	No	No	Yes	No
Config. 4	Battery/Ext. regulator	N/A	Yes	Yes	Yes	Yes	No
Config. 5	N/A	Battery/Ext. regulator	No	No	Yes	Yes	Yes
Config. 6	N/A	Battery/Ext. regulator	No	No	No	No	No

Table 69: Circuit configurations

7.3.1 Circuit configuration no. 1 for QIAA aQFN73

Circuit configuration number 1 for QIAA aQFN73, showing the schematic and the bill of materials.

Config no.	Supply configuration		Enabled features				
	VDDH	VDD	EXTSUPPLY	DCDCEN0	DCDCEN1	USB	NFC
Config. 1	USB (VDDH = VBUS)	N/A	Yes	No	No	Yes	No

Table 70: Configuration summary for circuit configuration no. 1

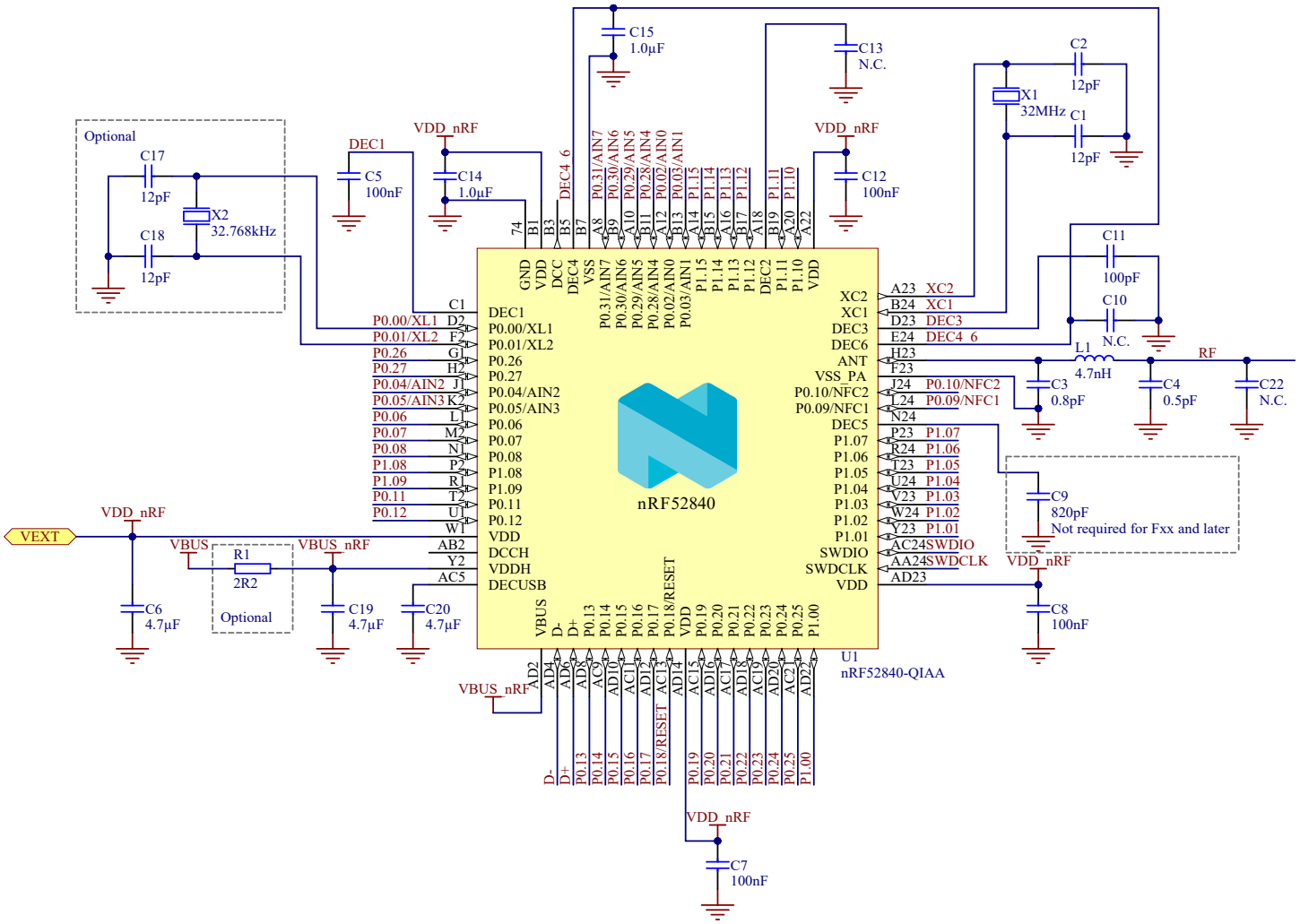


Figure 218: Circuit configuration no. 1 schematic

Note: For PCB reference layouts, see the product page for the nRF52840 on www.nordicsemi.com.

Designator	Value v1.0	Value v1.1	Description	Footprint
C1, C2, C17, C18	12 pF		Capacitor, NP0, ±2%	0402
C3	1 pF	0.8 pF	Capacitor, NP0, ±10%	0402
C4	1 pF	0.5 pF	Capacitor, NP0, ±10%	0402
C5, C7, C8, C12	100 nF		Capacitor, X7R, ±10%	0402
C6, C20	4.7 µF		Capacitor, X7R, ±10%	0603
C9	820 pF		Capacitor, NP0, ±5% Not required for Fxx and later	0402
C10, C13, C22	N.C.		Not mounted	0402
C11	100 pF		Capacitor, NP0, ±5%	0402
C14, C15	1.0 µF		Capacitor, X7R, ±10%	0603
C19	4.7 µF		Capacitor, X7S, ±10%	0603
L1	3.9 nH	4.7 nH	High frequency chip inductor ±5%	0402
R1	2R2		Resistor ±1%, 0.063W	0402
U1	nRF52840-QIAA		Multiprotocol Bluetooth low energy, IEEE 802.15.4, ANT, and 2.4 GHz proprietary System on Chip	AQFN-73
X1	32 MHz		XTAL SMD 2016, 32 MHz, Cl=8 pF, Total Tol: ±40 ppm	XTAL_2016
X2	32.768 kHz		XTAL SMD 3215, 32.768 kHz, Cl=9 pF, Total Tol: ±50 ppm	XTAL_3215

Table 71: Bill of material for circuit configuration no. 1

7.3.2 Circuit configuration no. 2 for QIAA aQFN73

Circuit configuration number 2 for QIAA aQFN73, showing the schematic and the bill of materials.

Config no.	Supply configuration		Enabled features				
	VDDH	VDD	EXTSUPPLY	DCDCEN0	DCDCEN1	USB	NFC
Config. 2	Battery/Ext. regulator	N/A	Yes	No	No	Yes	No

Table 72: Configuration summary for circuit configuration no. 2

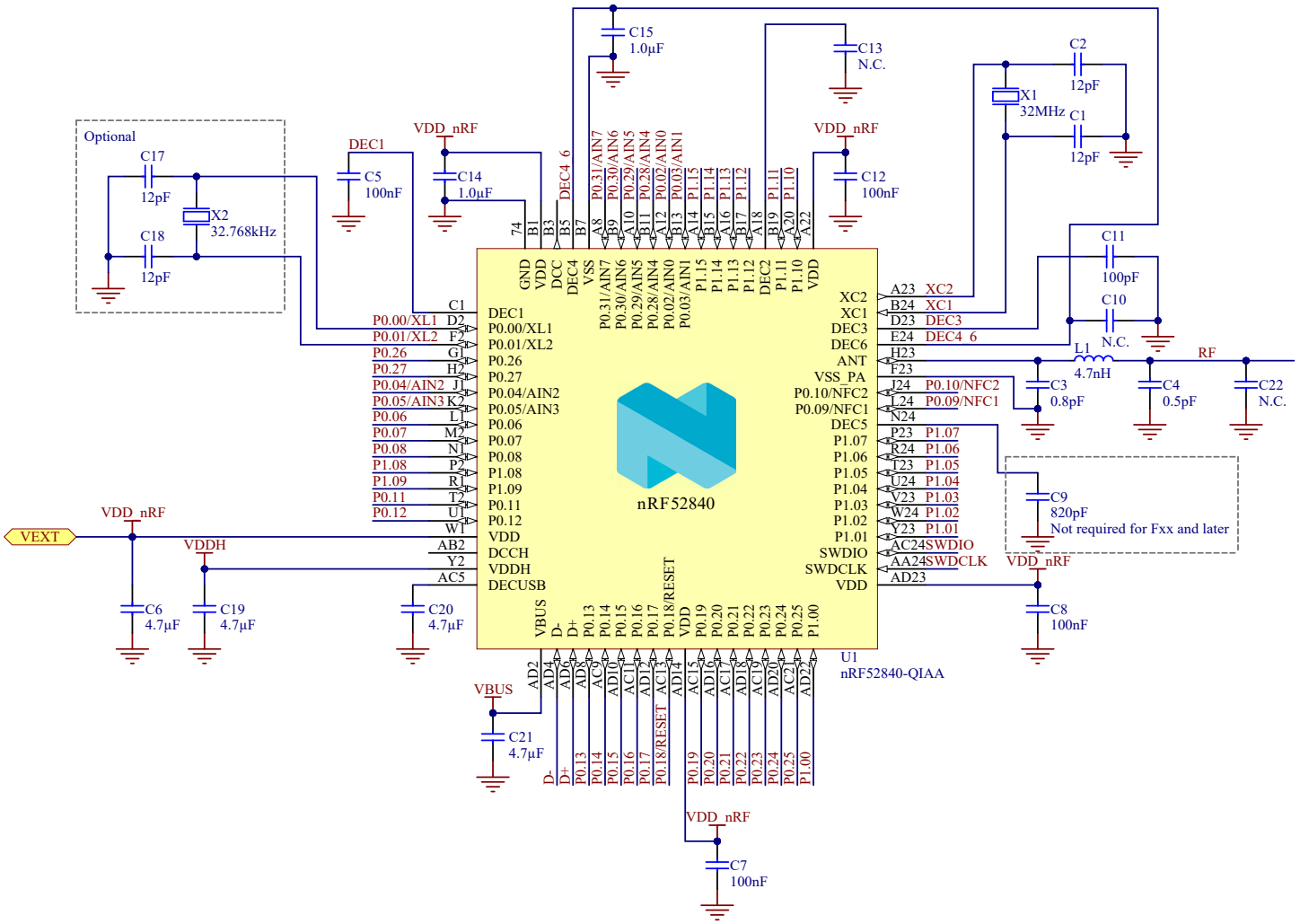


Figure 219: Circuit configuration no. 2 schematic

Note: For PCB reference layouts, see the product page for the nRF52840 on www.nordicsemi.com.

Designator	Value v1.0	Value v1.1	Description	Footprint
C1, C2, C17, C18	12 pF		Capacitor, NP0, ±2%	0402
C3	1 pF	0.8 pF	Capacitor, NP0, ±10%	0402
C4	1 pF	0.5 pF	Capacitor, NP0, ±10%	0402
C5, C7, C8, C12	100 nF		Capacitor, X7R, ±10%	0402
C6, C20	4.7 µF		Capacitor, X7R, ±10%	0603
C9	820 pF		Capacitor, NP0, ±5% Not required for Fxx and later	0402
C10, C13, C22	N.C.		Not mounted	0402
C11	100 pF		Capacitor, NP0, ±5%	0402
C14, C15	1.0 µF		Capacitor, X7R, ±10%	0603
C19, C21	4.7 µF		Capacitor, X7S, ±10%	0603
L1	3.9 nH	4.7 nH	High frequency chip inductor ±5%	0402
U1	nRF52840-QIAA		Multiprotocol Bluetooth low energy, IEEE 802.15.4, ANT, and 2.4 GHz proprietary System on Chip	AQFN-73
X1	32 MHz		XTAL SMD 2016, 32 MHz, Cl=8 pF, Total Tol: ±40 ppm	XTAL_2016
X2	32.768 kHz		XTAL SMD 3215, 32.768 kHz, Cl=9 pF, Total Tol: ±50 ppm	XTAL_3215

Table 73: Bill of material for circuit configuration no. 2

7.3.3 Circuit configuration no. 3 for QIAA aQFN73

Circuit configuration number 3 for QIAA aQFN73, showing the schematic and the bill of materials.

Config no.	Supply configuration		Enabled features				
	VDDH	VDD	EXTSUPPLY	DCDCEN0	DCDCEN1	USB	NFC
Config. 3	N/A	Battery/Ext. regulator	No	No	No	Yes	No

Table 74: Configuration summary for circuit configuration no. 3

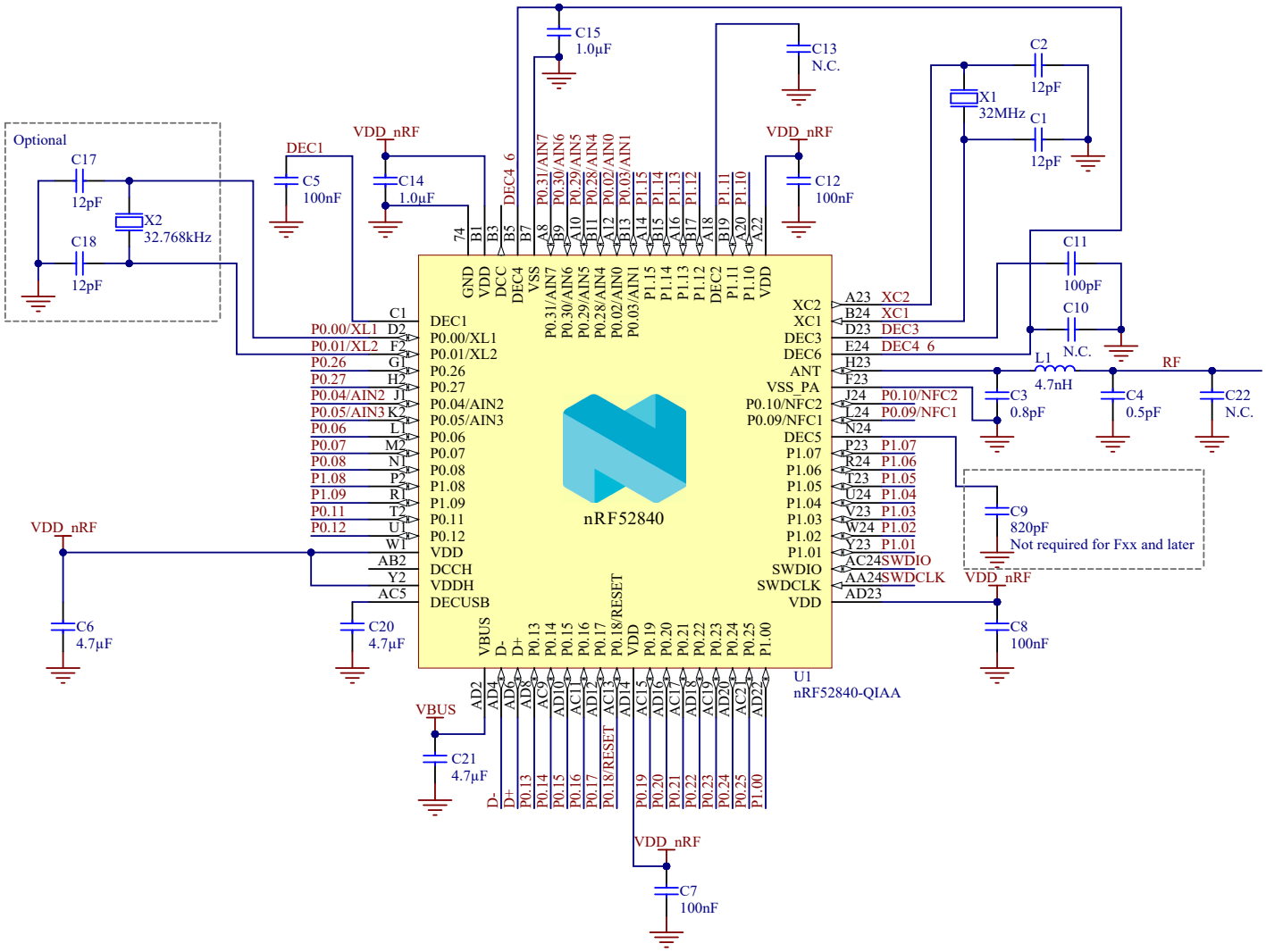


Figure 220: Circuit configuration no. 3 schematic

Note: For PCB reference layouts, see the product page for the nRF52840 on www.nordicsemi.com.

Designator	Value v1.0	Value v1.1	Description	Footprint
C1, C2, C17, C18	12 pF		Capacitor, NP0, ±2%	0402
C3	1 pF	0.8 pF	Capacitor, NP0, ±10%	0402
C4	1 pF	0.5 pF	Capacitor, NP0, ±10%	0402
C5, C7, C8, C12	100 nF		Capacitor, X7R, ±10%	0402
C6, C20	4.7 µF		Capacitor, X7R, ±10%	0603
C9	820 pF		Capacitor, NP0, ±5% Not required for Fxx and later	0402
C10, C13, C22	N.C.		Not mounted	0402
C11	100 pF		Capacitor, NP0, ±5%	0402
C14, C15	1.0 µF		Capacitor, X7R, ±10%	0603
C21	4.7 µF		Capacitor, X7S, ±10%	0603
L1	3.9 nH	4.7 nH	High frequency chip inductor ±5%	0402
U1	nRF52840-QIAA		Multiprotocol Bluetooth low energy, IEEE 802.15.4, ANT, and 2.4 GHz proprietary System on Chip	AQFN-73
X1	32 MHz		XTAL SMD 2016, 32 MHz, Cl=8 pF, Total Tol: ±40 ppm	XTAL_2016
X2	32.768 kHz		XTAL SMD 3215, 32.768 kHz, Cl=9 pF, Total Tol: ±50 ppm	XTAL_3215

Table 75: Bill of material for circuit configuration no. 3

7.3.4 Circuit configuration no. 4 for QIAA aQFN73

Circuit configuration number 4 for QIAA aQFN73, showing the schematic and the bill of materials.

Config no.	Supply configuration		Enabled features				
	VDDH	VDD	EXTSUPPLY	DCDCEN0	DCDCEN1	USB	NFC
Config. 4	Battery/Ext. regulator	N/A	Yes	Yes	Yes	Yes	No

Table 76: Configuration summary for circuit configuration no. 4

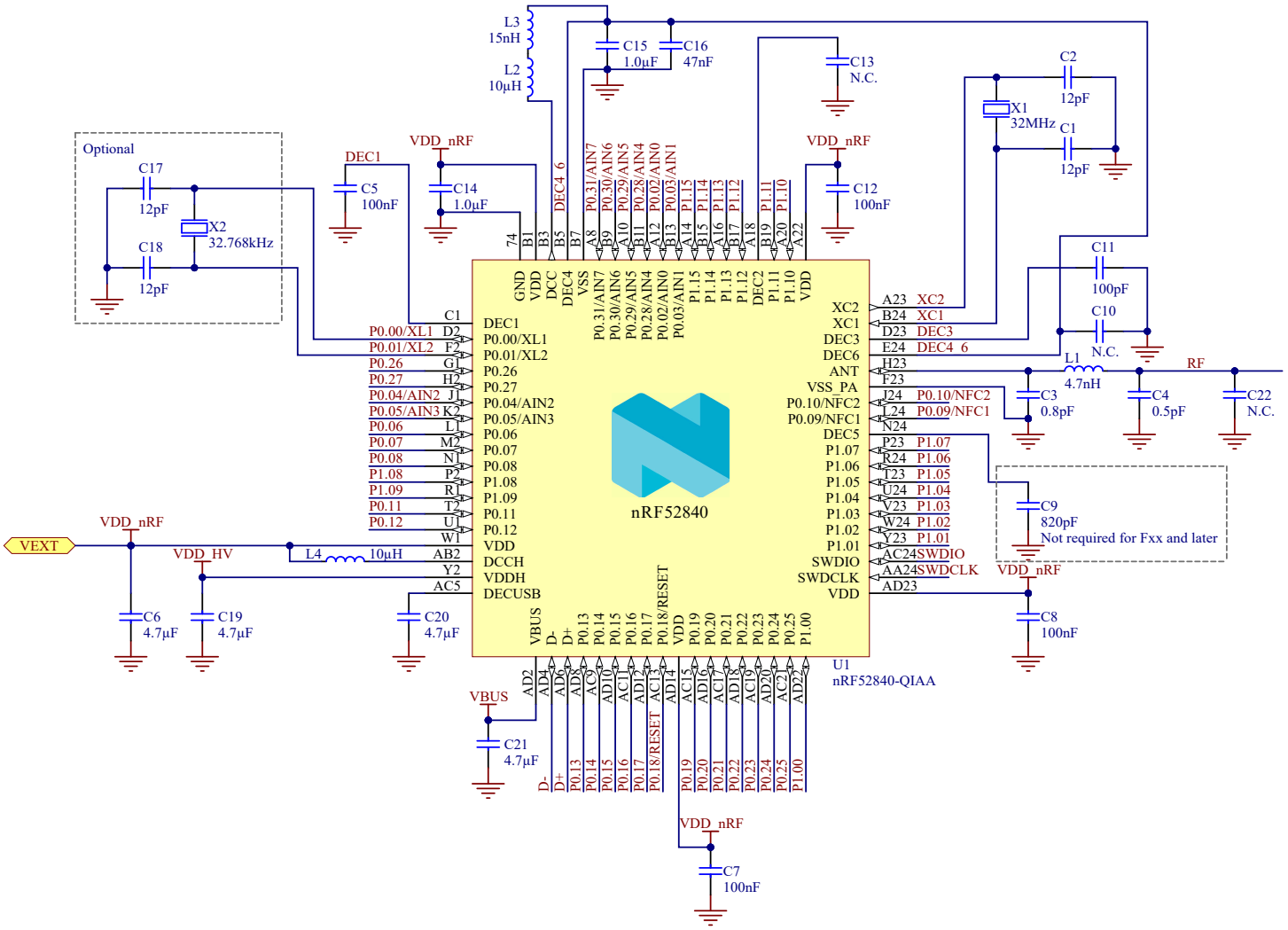


Figure 221: Circuit configuration no. 4 schematic for QIAA aQFN73

Note: For PCB reference layouts, see the product page for the nRF52840 on www.nordicsemi.com.

Designator	Value v1.0	Value v1.1	Description	Footprint
C1, C2, C17, C18	12 pF		Capacitor, NP0, ±2%	0402
C3	1 pF	0.8 pF	Capacitor, NP0, ±10%	0402
C4	1 pF	0.5 pF	Capacitor, NP0, ±10%	0402
C5, C7, C8, C12	100 nF		Capacitor, X7R, ±10%	0402
C6, C20	4.7 µF		Capacitor, X7R, ±10%	0603
C9	820 pF		Capacitor, NP0, ±5% Not required for Fxx and later	0402
C10, C13, C22	N.C.		Not mounted	0402
C11	100 pF		Capacitor, NP0, ±5%	0402
C14, C15	1.0 µF		Capacitor, X7R, ±10%	0603
C16	47 nF		Capacitor, X7R, ±10%	0402
C19, C21	4.7 µF		Capacitor, X7S, ±10%	0603
L1	3.9 nH	4.7 nH	High frequency chip inductor ±5%	0402
L2	10 µH		Chip inductor, IDC, min = 50 mA, ±20%	0603
L3	15 nH		High frequency chip inductor ±10%	0402
L4	10 µH		Chip inductor, IDC, min = 80 mA, ±20%	0603
U1	nRF52840-QIAA		Multiprotocol Bluetooth low energy, IEEE 802.15.4, ANT, and 2.4 GHz proprietary System on Chip	AQFN-73
X1	32 MHz		XTAL SMD 2016, 32 MHz, Cl=8 pF, Total Tol: ±40 ppm	XTAL_2016
X2	32.768 kHz		XTAL SMD 3215, 32.768 kHz, Cl=9 pF, Total Tol: ±50 ppm	XTAL_3215

Table 77: Bill of material for circuit configuration no. 4

7.3.5 Circuit configuration no. 5 for QIAA aQFN73

Circuit configuration number 5 for QIAA aQFN73, showing the schematic and the bill of materials.

Config no.	Supply configuration		Enabled features				
	VDDH	VDD	EXTSUPPLY	DCDCEN0	DCDCEN1	USB	NFC
Config. 5	N/A	Battery/Ext. regulator	No	No	Yes	Yes	Yes

Table 78: Configuration summary for circuit configuration no. 5

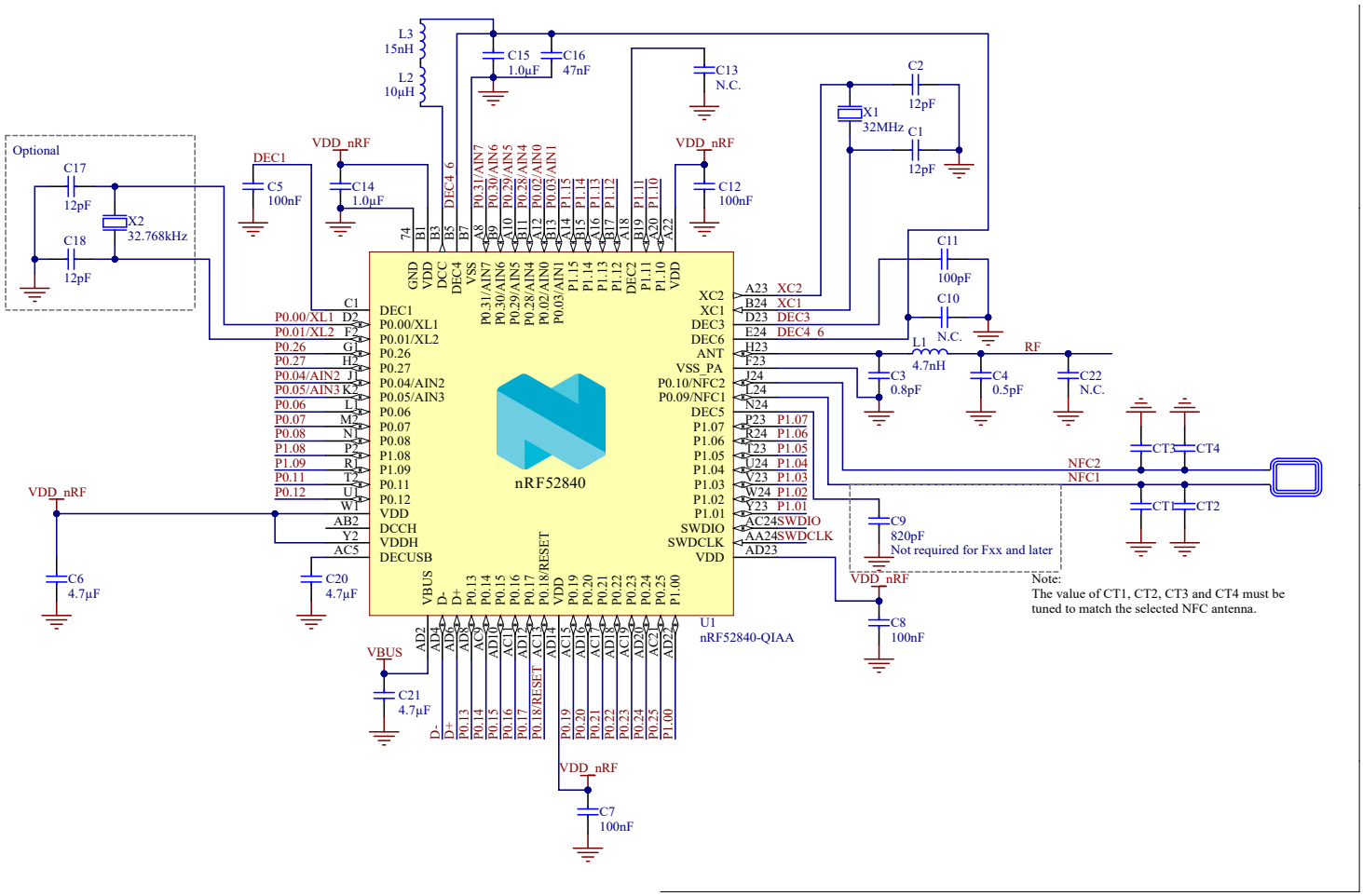


Figure 222: Circuit configuration no. 5 schematic

Note: For PCB reference layouts, see the product page for the nRF52840 on www.nordicsemi.com.

Designator	Value v1.0	Value v1.1	Description	Footprint
C1, C2, C17, C18	12 pF		Capacitor, NP0, ±2%	0402
C3	1 pF	0.8 pF	Capacitor, NP0, ±10%	0402
C4	1 pF	0.5 pF	Capacitor, NP0, ±10%	0402
C5, C7, C8, C12	100 nF		Capacitor, X7R, ±10%	0402
C6, C20	4.7 µF		Capacitor, X7R, ±10%	0603
C9	820 pF		Capacitor, NP0, ±5%	0402
C10, C13, C22	N.C.		Not mounted	0402
C11	100 pF		Capacitor, NP0, ±5%	0402
C14, C15	1.0 µF		Capacitor, X7R, ±10%	0603
C16	47 nF		Capacitor, X7R, ±10%	0402
C21	4.7 µF		Capacitor, X7S, ±10%	0603
CT1, CT2, CT3, CT4	Antenna dependent		Capacitor, NP0, ±5%	0402
L1	3.9 nH	4.7 nH	High frequency chip inductor ±5%	0402
L2	10 µH		Chip inductor, IDC, min = 50 mA, ±20%	0603
L3	15 nH		High frequency chip inductor ±10%	0402
U1	nRF52840-QIAA		Multiprotocol Bluetooth low energy, IEEE 802.15.4, ANT, and 2.4 GHz proprietary System on Chip	AQFN-73
X1	32 MHz		XTAL SMD 2016, 32 MHz, Cl=8 pF, Total Tol: ±40 ppm	XTAL_2016
X2	32.768 kHz		XTAL 3215, 32.768 kHz, Cl=9 pF, Total Tol: ±50 ppm	XTAL_3215

Table 79: Bill of material for circuit configuration no. 5

7.3.6 Circuit configuration no. 6 for QIAA aQFN73

Circuit configuration number 6 for QIAA aQFN73, showing the schematic and the bill of materials.

Config no.	Supply configuration		Enabled features				
	VDDH	VDD	EXTSUPPLY	DCDCEN0	DCDCEN1	USB	NFC
Config. 6	N/A	Battery/Ext. regulator	No	No	No	No	No

Table 80: Configuration summary for circuit configuration no. 6

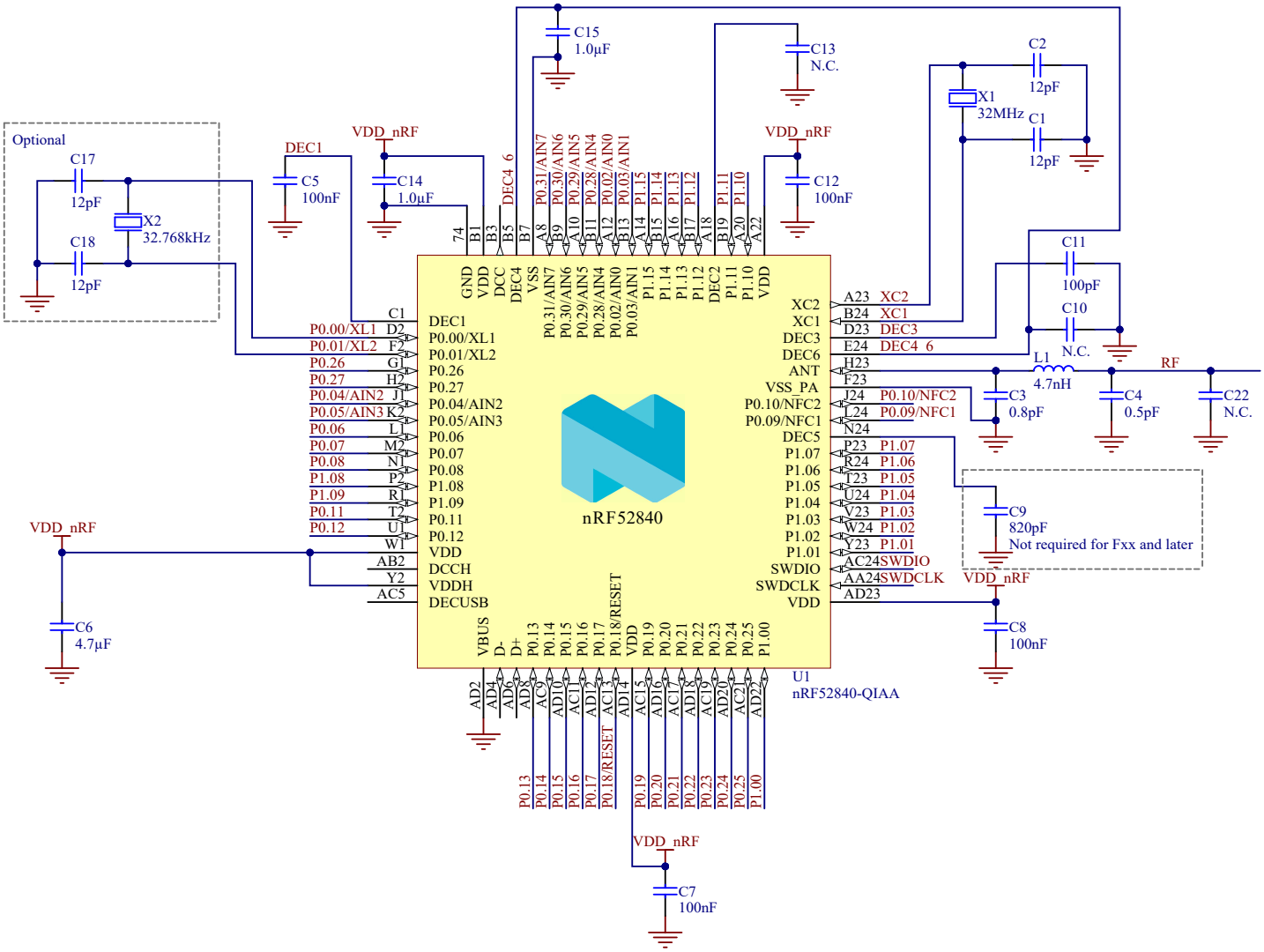


Figure 223: Circuit configuration no. 6 schematic

Note: For PCB reference layouts, see the product page for the nRF52840 on www.nordicsemi.com.

Designator	Value v1.0	Value v1.1	Description	Footprint
C1, C2, C17, C18	12 pF		Capacitor, NP0, ±2%	0402
C3	1 pF	0.8 pF	Capacitor, NP0, ±10%	0402
C4	1 pF	0.5 pF	Capacitor, NP0, ±10%	0402
C5, C7, C8, C12	100 nF		Capacitor, X7R, ±10%	0402
C6	4.7 µF		Capacitor, X7R, ±10%	0603
C9	820 pF		Capacitor, NP0, ±5% Not required for Fxx and later	0402
C10, C13, C22	N.C.		Not mounted	0402
C11	100 pF		Capacitor, NP0, ±5%	0402
C14, C15	1.0 µF		Capacitor, X7R, ±10%	0603
L1	3.9 nH	4.7 nH	High frequency chip inductor ±5%	0402
U1	nRF52840-QIAA		Multiprotocol Bluetooth low energy, IEEE 802.15.4, ANT, and 2.4 GHz proprietary System on Chip	AQFN-73
X1	32 MHz		XTAL SMD 2016, 32 MHz, Cl=8 pF, Total Tol: ±40 ppm	XTAL_2016
X2	32.768 kHz		XTAL SMD 3215, 32.768 kHz, Cl=9 pF, Total Tol: ±50 ppm	XTAL_3215

Table 81: Bill of material for circuit configuration no. 6

7.3.7 Circuit configuration no. 7 for QIAA aQFN73

Circuit configuration number 7 for QIAAaQFN73, showing the schematic and the bill of materials.

This circuit configuration includes a four-component RF match for improved harmonic suppression when using RADIO with TXPOWER ≥ 5 dBm.

Config no.	Supply configuration		Enabled features				
	VDDH	VDD	EXTSUPPLY	DCDCEN0	DCDCEN1	USB	NFC
Config. 7	Battery/Ext. regulator	N/A	Yes	Yes	Yes	Yes	No

Table 82: Configuration summary for circuit configuration no. 7

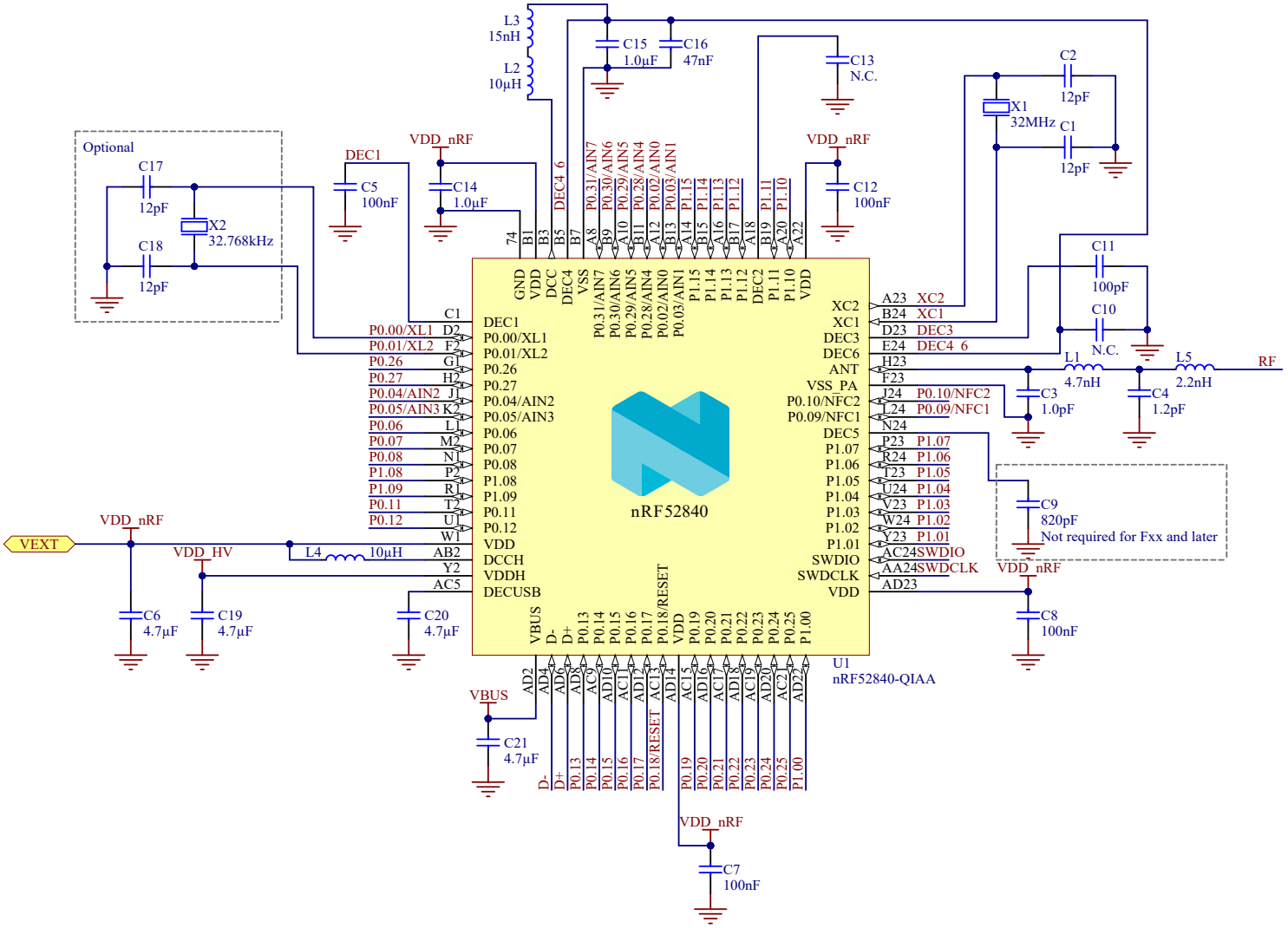


Figure 224: Circuit configuration no. 7 schematic for QIAA aQFN73

Note: For PCB reference layouts, see the product page for the nRF52840 on www.nordicsemi.com.

Designator	Value	Description	Footprint
C1, C2, C17, C18	12 pF	Capacitor, NP0, ±2%	0402
C3	1 pF	Capacitor, NP0, ±5%	0402
C4	1.2 pF	Capacitor, NP0, ±5%	0402
C5, C7, C8, C12	100 nF	Capacitor, X7R, ±10%	0402
C6, C20	4.7 µF	Capacitor, X7R, ±10%	0603
C9	820 pF	Capacitor, NP0, ±5% Not required for Fxx and later	0402
C10, C13	N.C.	Not mounted	0402
C11	100 pF	Capacitor, NP0, ±5%	0402
C14, C15	1.0 µF	Capacitor, X7R, ±10%	0603
C16	47 nF	Capacitor, X7R, ±10%	0402
C19, C21	4.7 µF	Capacitor, X7S, ±10%	0603
L1	4.7 nH	High frequency chip inductor ±5%	0402
L2	10 µH	Chip inductor, IDC, min = 50 mA, ±20%	0603
L3	15 nH	High frequency chip inductor ±10%	0402
L4	10 µH	Chip inductor, IDC, min = 80 mA, ±20%	0603
L5	2.2 nH	High frequency chip inductor ±5%	0402
U1	nRF52840-QIAA	Multiprotocol Bluetooth low energy, IEEE 802.15.4, ANT, and 2.4 GHz proprietary System on Chip	AQFN-73
X1	32 MHz	XTAL SMD 2016, 32 MHz, Cl=8 pF, Total Tol: ±40 ppm	XTAL_2016
X2	32.768 kHz	XTAL SMD 3215, 32.768 kHz, Cl=9 pF, Total Tol: ±50 ppm	XTAL_3215

Table 83: Bill of material for circuit configuration no. 7

7.3.8 Circuit configuration no. 1 for QFAA QFN48

Circuit configuration number 1 for QFAA QFN48, showing the schematic and the bill of materials.

Config no.	Supply configuration		Enabled features				
	VDDH	VDD	EXTSUPPLY	DCDCEN0	DCDCEN1	USB	NFC
Config. 1	N/A	Battery/Ext. regulator	N/A	N/A	Yes	N/A	No

Table 84: Configuration summary for circuit configuration no. 1

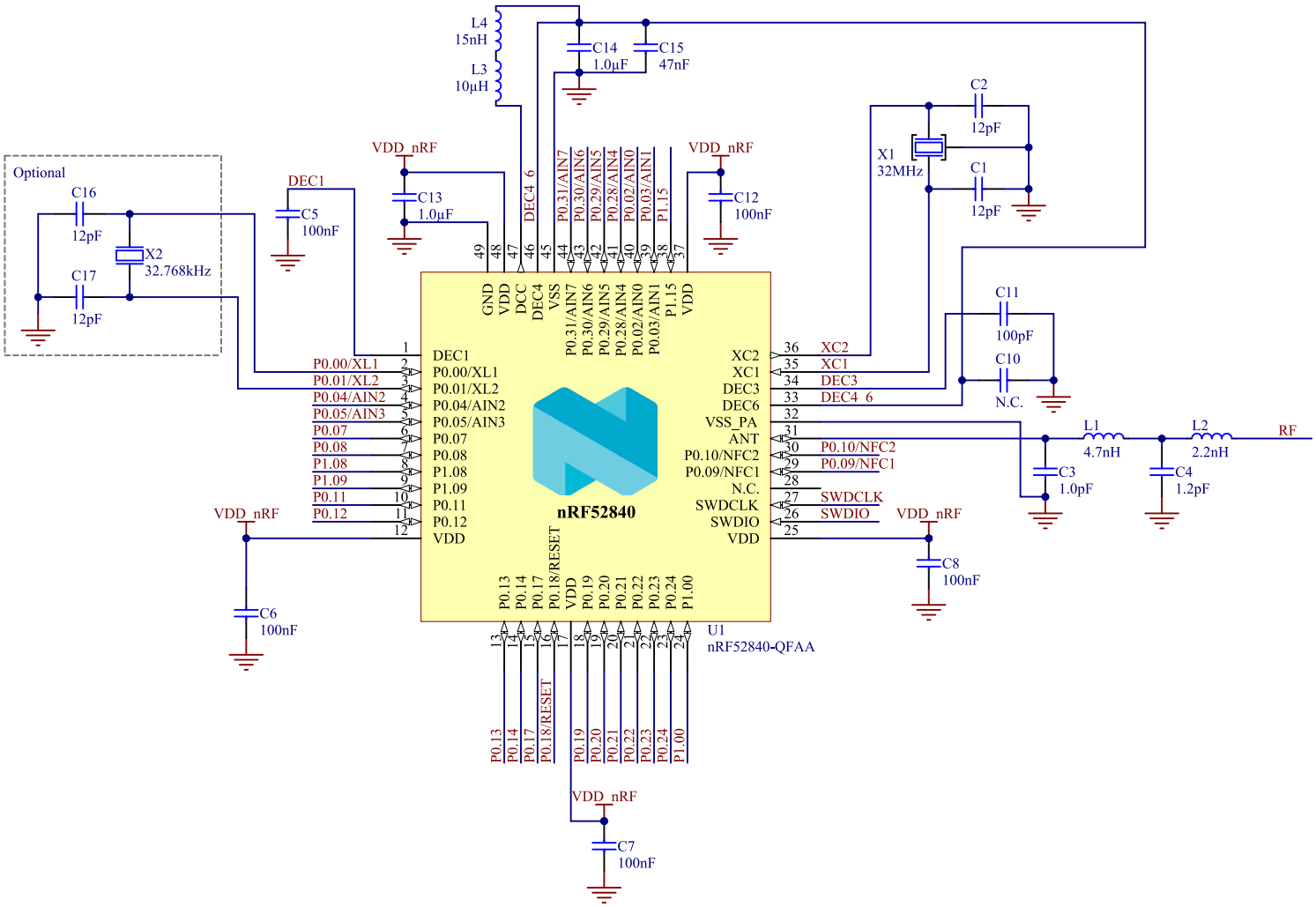


Figure 225: Circuit configuration no. 1 schematic

Note: For PCB reference layouts, see the product page for the nRF52840 on www.nordicsemi.com.

Designator	Value	Description	Footprint
C1, C2, C16, C17	12 pF	Capacitor, NP0, ±2%	0201
C3	1.0 pF	Capacitor, NP0, ±10%	0201
C4	1.2 pF	Capacitor, NP0, ±10%	0201
C5, C6, C7, C8, C12	100 nF	Capacitor, X5R, ±10%	0201
C10	N.C.	Not mounted	0201
C11	100 pF	Capacitor, NP0, ±5%	0201
C13, C14	1.0 µF	Capacitor, X5R, ±10%	0402
C15	47 nF	Capacitor, X5R, ±10%	0201
L1	4.7 nH	High frequency chip inductor ±5%	0201
L2	2.2 nH	High frequency chip inductor ±5%	0201
L3	10 µH	Chip inductor, IDC, min = 50 mA, ±20%	0603
L4	15 nH	High frequency chip inductor ±10%	0402
U1	nRF52840-QFAA	Multiprotocol Bluetooth low energy, IEEE 802.15.4, ANT, and 2.4 GHz proprietary System on Chip	QFN48
X1	32 MHz	XTAL SMD 2016, 32 MHz, Cl=8 pF, Total Tol: ±30 ppm	XTAL_2016
X2	32.768 kHz	XTAL SMD 2012, 32.768 kHz, Cl=9 pF, Total Tol: ±50 ppm	XTAL_2012

Table 85: Bill of material for circuit configuration no. 1

7.3.9 Circuit configuration no. 1 for CKAA WLCSP

Circuit configuration number 1 for CKAA WLCSP, showing the schematic and the bill of materials.

Config no.	Supply configuration		Enabled features				
	VDDH	VDD	EXTSUPPLY	DCDCEN0	DCDCEN1	USB	NFC
Config. 1	USB (VDDH = VBUS)	N/A	Yes	No	No	Yes	No

Table 86: Configuration summary for circuit configuration no. 1

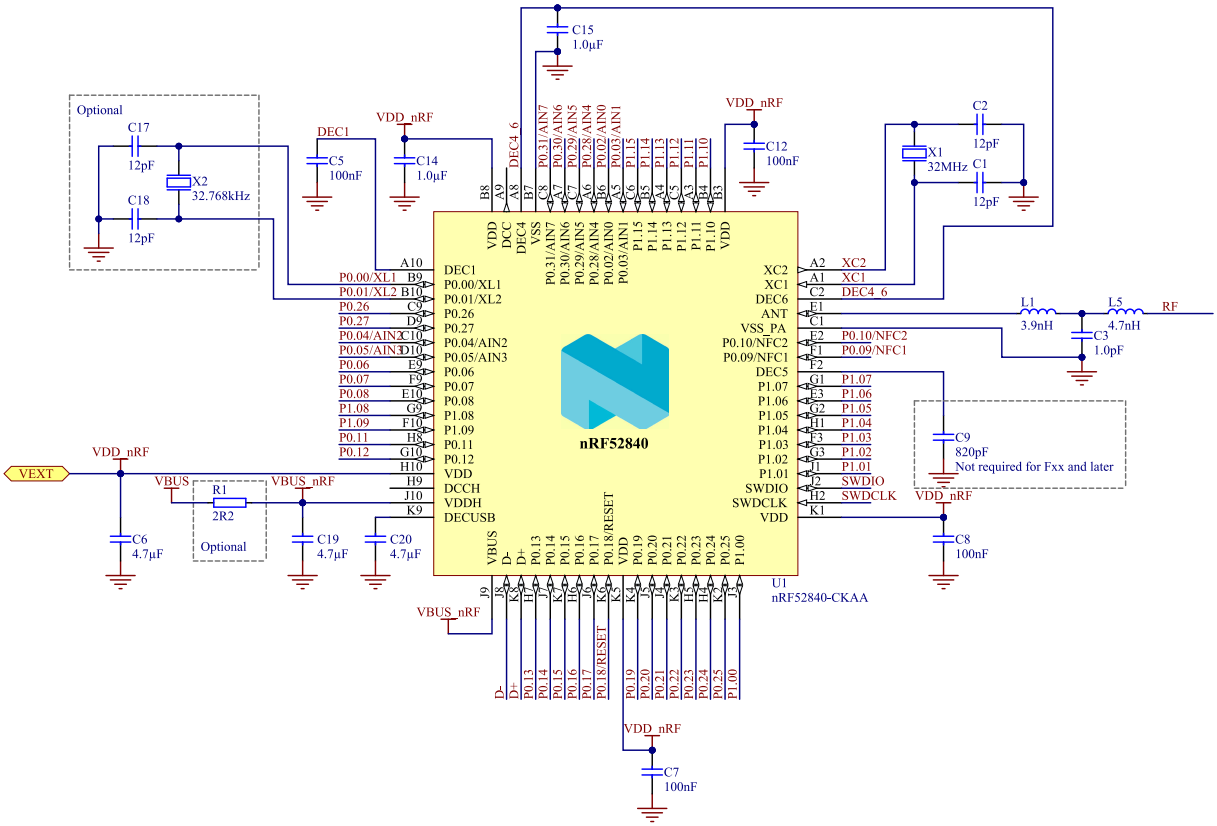


Figure 226: Circuit configuration no. 1 schematic for CKAA WLCSP

Note: For PCB reference layouts, see the product page for the nRF52840 on www.nordicsemi.com.

Designator	Value	Description	Footprint
C1, C2, C17, C18	12 pF	Capacitor, NP0, ±2%	0201
C3	1.0 pF	Capacitor, NP0, ±5%	0201
C5, C7, C8, C12	100 nF	Capacitor, X7R, ±10%	0201
C6, C20	4.7 µF	Capacitor, X7R, ±10%	0603
C9	820 pF	Capacitor, X7R, ±10% Not required for Fxx and later	0201
C14, C15	1.0 µF	Capacitor, X7R, ±10%	0603
C19	4.7 µF	Capacitor, X7S, ±10%	0603
L1	3.9 nH	High frequency chip inductor ±5%	0201
L5	4.7 nH	High frequency chip inductor ±5%	0201
R1	2R2	Resistor, ±1%, 0.05W	0201
U1	nRF52840-CKAA	Multiprotocol Bluetooth low energy, IEEE 802.15.4, ANT, and 2.4 GHz proprietary System on Chip	WLCSP-94
X1	32 MHz	XTAL SMD 2016, 32 MHz, Cl=8 pF, Total Tol: ±40 ppm	XTAL_2016
X2	32.768 kHz	XTAL SMD 2012, 32.768 kHz, Cl=9 pF, Total Tol: ±50 ppm	XTAL_2012

Table 87: Bill of material for circuit configuration no. 1

7.3.10 Circuit configuration no. 2 for CKAA WLCSP

Circuit configuration number 2 for CKAA WLCSP, showing the schematic and the bill of materials.

Config no.	Supply configuration		Enabled features				
	VDDH	VDD	EXTSUPPLY	DCDCEN0	DCDCEN1	USB	NFC
Config. 2	Battery/Ext. regulator	N/A	Yes	No	No	Yes	No

Table 88: Configuration summary for circuit configuration no. 2

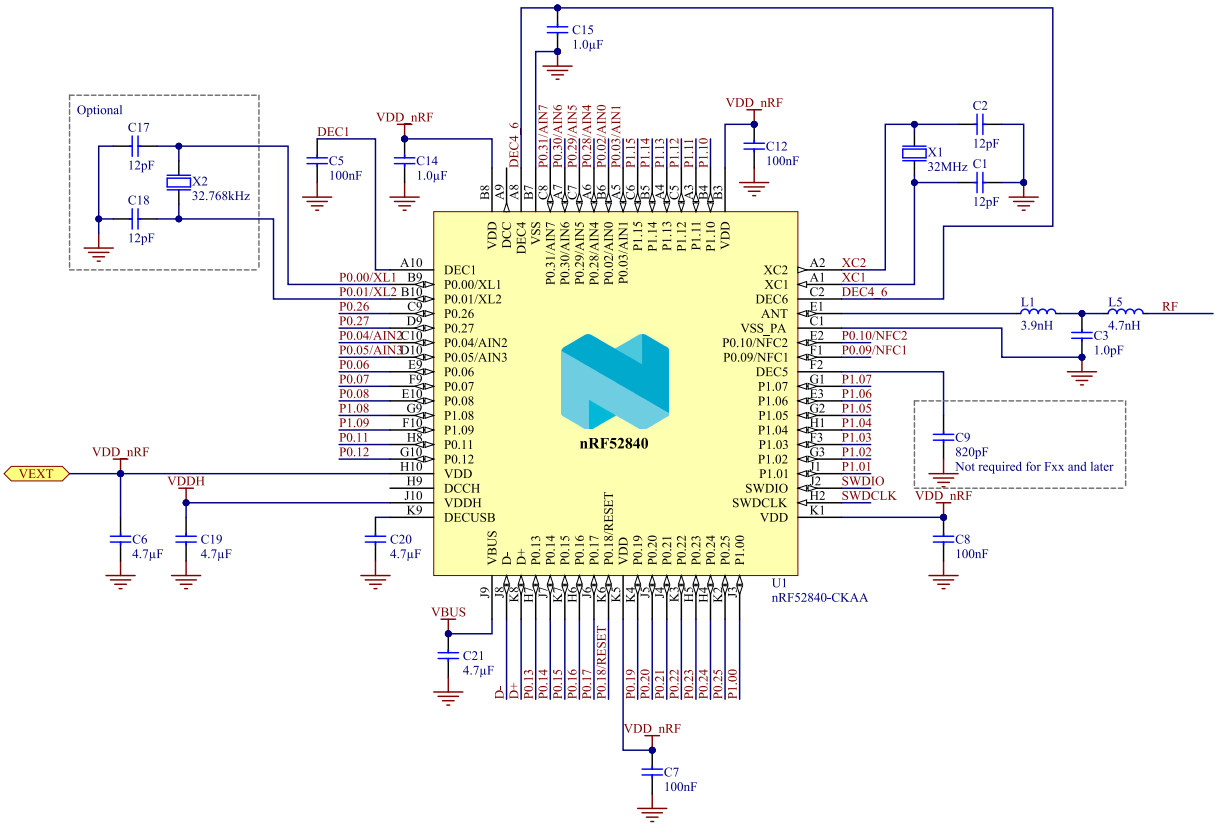


Figure 227: Circuit configuration no. 2 schematic for CKAA WLCSP

Note: For PCB reference layouts, see the product page for the nRF52840 on www.nordicsemi.com.

Designator	Value	Description	Footprint
C1, C2, C17, C18	12 pF	Capacitor, NP0, ±2%	0201
C3	1.0 pF	Capacitor, NP0, ±5%	0201
C5, C7, C8, C12	100 nF	Capacitor, X7R, ±10%	0201
C6, C20	4.7 µF	Capacitor, X7R, ±10%	0603
C9	820 pF	Capacitor, X7R, ±10% Not required for Fxx and later	0201
C14, C15	1.0 µF	Capacitor, X7R, ±10%	0603
C19, C21	4.7 µF	Capacitor, X7S, ±10%	0603
L1	3.9 nH	High frequency chip inductor ±5%	0201
L5	4.7 nH	High frequency chip inductor ±5%	0201
U1	nRF52840-CKAA	Multiprotocol Bluetooth low energy, IEEE 802.15.4, ANT, and 2.4 GHz proprietary System on Chip	WLCSP-94
X1	32 MHz	XTAL SMD 2016, 32 MHz, Cl=8 pF, Total Tol: ±40 ppm	XTAL_2016
X2	32.768 kHz	XTAL SMD 2012, 32.768 kHz, Cl=9 pF, Total Tol: ±50 ppm	XTAL_2012

Table 89: Bill of material for circuit configuration no. 2

7.3.11 Circuit configuration no. 3 for CKAA WLCSP

Circuit configuration number 3 for CKAA WLCSP, showing the schematic and the bill of materials.

Config no.	Supply configuration		Enabled features				
	VDDH	VDD	EXTSUPPLY	DCDCEN0	DCDCEN1	USB	NFC
Config. 3	N/A	Battery/Ext. regulator	No	No	No	Yes	No

Table 90: Configuration summary for circuit configuration no. 3

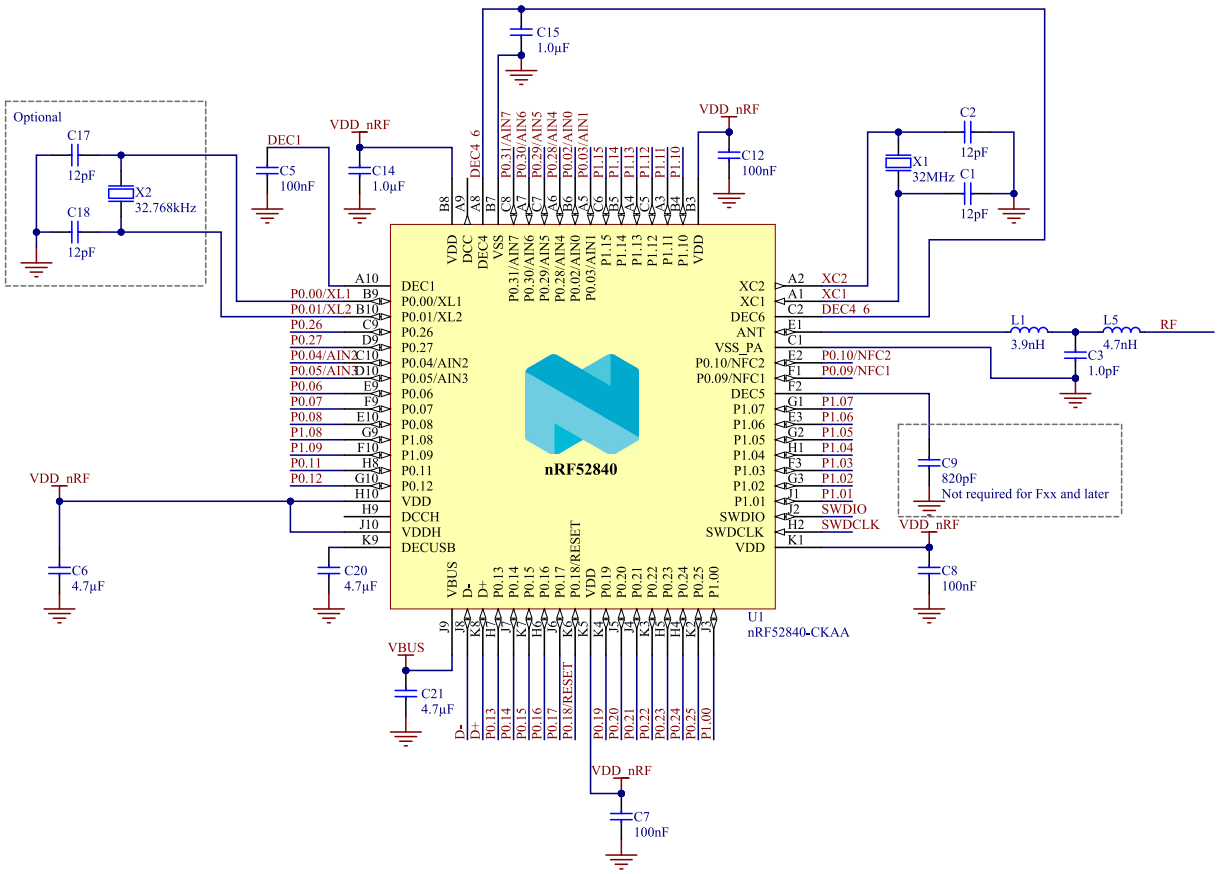


Figure 228: Circuit configuration no. 3 schematic for CKAA WLCSP

Note: For PCB reference layouts, see the product page for the nRF52840 on www.nordicsemi.com.

Designator	Value	Description	Footprint
C1, C2, C17, C18	12 pF	Capacitor, NP0, ±2%	0201
C3	1.0 pF	Capacitor, NP0, ±5%	0201
C5, C7, C8, C12	100 nF	Capacitor, X7R, ±10%	0201
C6, C20	4.7 µF	Capacitor, X7R, ±10%	0603
C9	820 pF	Capacitor, X7R, ±10% Not required for Fxx and later	0201
C14, C15	1.0 µF	Capacitor, X7R, ±10%	0603
C21	4.7 µF	Capacitor, X7S, ±10%	0603
L1	3.9 nH	High frequency chip inductor ±5%	0201
L5	4.7 nH	High frequency chip inductor ±5%	0201
U1	nRF52840-CKAA	Multiprotocol Bluetooth low energy, IEEE 802.15.4, ANT, and 2.4 GHz proprietary System on Chip	WLCSP-94
X1	32 MHz	XTAL SMD 2016, 32 MHz, Cl=8 pF, Total Tol: ±40 ppm	XTAL_2016
X2	32.768 kHz	XTAL SMD 2012, 32.768 kHz, Cl=9 pF, Total Tol: ±50 ppm	XTAL_2012

Table 91: Bill of material for circuit configuration no. 3

7.3.12 Circuit configuration no. 4 for CKAA WLCSP

Circuit configuration number 4 for CKAA WLCSP, showing the schematic and the bill of materials.

Config no.	Supply configuration		Enabled features				
	VDDH	VDD	EXTSUPPLY	DCDCEN0	DCDCEN1	USB	NFC
Config. 4	Battery/Ext. regulator	N/A	Yes	Yes	Yes	Yes	No

Table 92: Configuration summary for circuit configuration no. 4

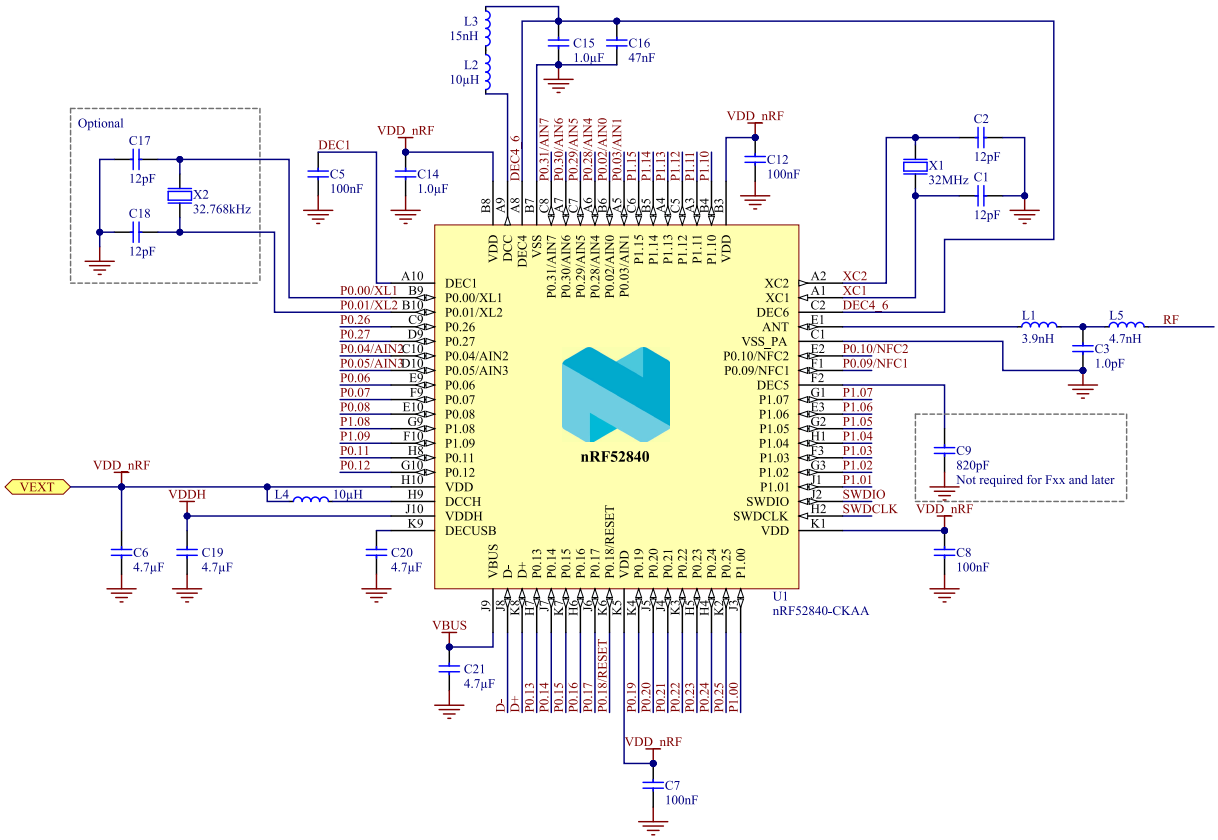


Figure 229: Circuit configuration no. 4 schematic for CKAA WLCSP

Note: For PCB reference layouts, see the product page for the nRF52840 on www.nordicsemi.com.

Designator	Value	Description	Footprint
C1, C2, C17, C18	12 pF	Capacitor, NP0, ±2%	0201
C3	1.0 pF	Capacitor, NP0, ±5%	0201
C5, C7, C8, C12	100 nF	Capacitor, X7R, ±10%	0201
C6, C20	4.7 µF	Capacitor, X7R, ±10%	0603
C9	820 pF	Capacitor, X7R, ±10% Not required for Fxx and later	0201
C14, C15	1.0 µF	Capacitor, X7R, ±10%	0603
C16	47 nF	Capacitor, X7R, ±10%	0201
C19, C21	4.7 µF	Capacitor, X7S, ±10%	0603
L1	3.9 nH	High frequency chip inductor ±5%	0201
L2	10 µH	Chip inductor, IDC, min = 50 mA, ±20%	0603
L3	15 nH	High frequency chip inductor ±10%	0402
L4	10 µH	Chip inductor, IDC, min = 80 mA, ±10%	0603
L5	4.7 nH	High frequency chip inductor ±5%	0201
U1	nRF52840-CKAA	Multiprotocol Bluetooth low energy, IEEE 802.15.4, ANT, and 2.4 GHz proprietary System on Chip	WLCSP-94
X1	32 MHz	XTAL SMD 2016, 32 MHz, Cl=8 pF, Total Tol: ±40 ppm	XTAL_2016
X2	32.768 kHz	XTAL SMD 2012, 32.768 kHz, Cl=9 pF, Total Tol: ±50 ppm	XTAL_2012

Table 93: Bill of material for circuit configuration no. 4

7.3.13 Circuit configuration no. 5 for CKAA WLCSP

Circuit configuration number 5 for CKAA WLCSP, showing the schematic and the bill of materials.

Config no.	Supply configuration		Enabled features				
	VDDH	VDD	EXTSUPPLY	DCDCEN0	DCDCEN1	USB	NFC
Config. 5	N/A	Battery/Ext. regulator	No	No	Yes	Yes	Yes

Table 94: Configuration summary for circuit configuration no. 5

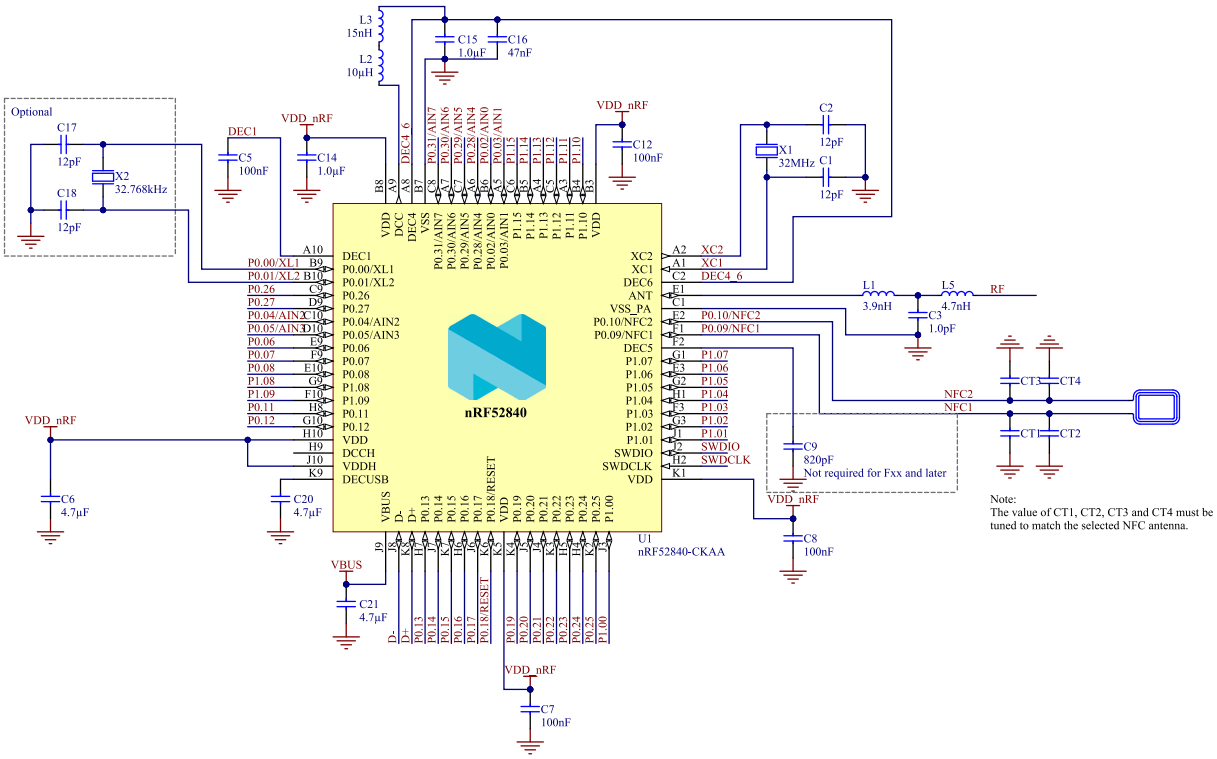


Figure 230: Circuit configuration no. 5 schematic for CKAA WLCSP

Note: For PCB reference layouts, see the product page for the nRF52840 on www.nordicsemi.com.

Designator	Value	Description	Footprint
C1, C2, C17, C18	12 pF	Capacitor, NP0, ±2%	0201
C3	1.0 pF	Capacitor, NP0, ±5%	0201
C5, C7, C8, C12	100 nF	Capacitor, X7R, ±10%	0201
C6, C20	4.7 µF	Capacitor, X7R, ±10%	0603
C9	820 pF	Capacitor, X7R, ±10% Not required for Fxx and later	0201
C14, C15	1.0 µF	Capacitor, X7R, ±10%	0603
C16	47 nF	Capacitor, X7R, ± 10%	0201
C21	4.7 µF	Capacitor, X7S, ± 10%	0603
CT1, CT2, CT3, CT4	Antenna dependent	Capacitor, X7R, ± 10%	0201
L1	3.9 nH	High frequency chip inductor ±5%	0201
L2	10 µH	Chip inductor, IDC, min = 50 mA, ±20%	0603
L3	15 nH	High frequency chip inductor ±10%	0402
L5	4.7 nH	High frequency chip inductor ±5%	0201
U1	nRF52840-CKAA	Multiprotocol Bluetooth low energy, IEEE 802.15.4, ANT, and 2.4 GHz proprietary System on Chip	WLCSP-94
X1	32 MHz	XTAL SMD 2016, 32 MHz, Cl=8 pF, Total Tol: ±40 ppm	XTAL_2016
X2	32.768 kHz	XTAL SMD 2012, 32.768 kHz, Cl=9 pF, Total Tol: ±50 ppm	XTAL_2012

Table 95: Bill of material for circuit configuration no. 5

7.3.14 Circuit configuration no. 6 for CKAA WLCSP

Circuit configuration number 6 for CKAA WLCSP, showing the schematic and the bill of materials.

Config no.	Supply configuration		Enabled features				
	VDDH	VDD	EXTSUPPLY	DCDCEN0	DCDCEN1	USB	NFC
Config. 6	N/A	Battery/Ext. regulator	No	No	No	No	No

Table 96: Configuration summary for circuit configuration no. 6

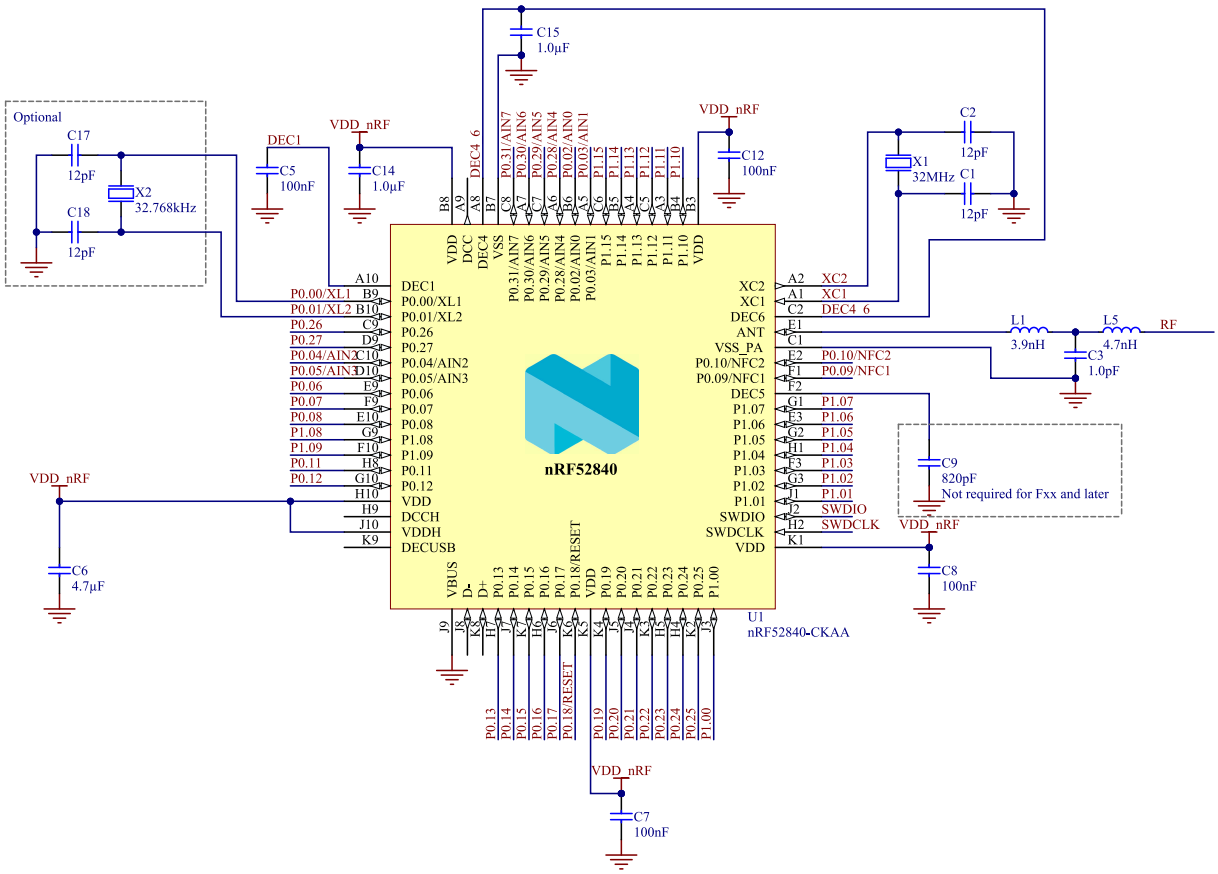


Figure 231: Circuit configuration no. 6 schematic for CKAA WLCSP

Note: For PCB reference layouts, see the product page for the nRF52840 on www.nordicsemi.com.

Designator	Value	Description	Footprint
C1, C2, C17, C18	12 pF	Capacitor, NP0, ±2%	0201
C3	1.0 pF	Capacitor, NP0, ±5%	0201
C5, C7, C8, C12	100 nF	Capacitor, X7R, ±10%	0201
C6	4.7 µF	Capacitor, X7R, ±10%	0603
C9	820 pF	Capacitor, X7R, ±10% Not required for Fxx and later	0201
C14, C15	1.0 µF	Capacitor, X7R, ±10%	0603
L1	3.9 nH	High frequency chip inductor ±5%	0201
L5	4.7 nH	High frequency chip inductor ±5%	0201
U1	nRF52840-CKAA	Multiprotocol Bluetooth low energy, IEEE 802.15.4, ANT, and 2.4 GHz proprietary System on Chip	WLCSP-94
X1	32 MHz	XTAL SMD 2016, 32 MHz, Cl=8 pF, Total Tol: ±40 ppm	XTAL_2016
X2	32.768 kHz	XTAL SMD 2012, 32.768 kHz, Cl=9 pF, Total Tol: ±50 ppm	XTAL_2012

Table 97: Bill of material for circuit configuration no. 6

7.3.15 PCB guidelines

A well designed PCB is necessary to achieve good RF performance. Poor layout can lead to loss in performance or functionality.

A qualified RF layout for the IC and its surrounding components, including matching networks, can be downloaded from www.nordicsemi.com.

To ensure optimal performance it is essential that you follow the schematics and layout references closely. Especially in the case of the antenna matching circuitry (components between device pin ANT and the antenna), any changes to the layout can change the behavior, resulting in degradation of RF performance or a need to change component values. All reference circuits are designed for use with a 50 Ω single-ended antenna.

A PCB with a minimum of two layers, including a ground plane, is recommended for optimal RF performance. On PCBs with more than two layers, put a keep-out area on the inner layers directly below the antenna matching circuitry (components between device pin ANT and the antenna) to reduce the stray capacitances that influence RF performance.

A matching network is needed between the RF pin ANT and the antenna, to match the antenna impedance (normally 50 Ω) to the optimum RF load impedance for the chip. For optimum performance, the impedance for the matching network should be set as described in the recommended aQFN73 package reference circuitry from Circuit configuration no. 1 for QIAA aQFN73 on page 939, the recommended QFN48 reference circuitry from Circuit configuration no. 1 for QFAA QFN48 on page 953, or the recommended WLCSP reference circuitry from Circuit configuration no. 1 for CKAA WLCSP on page 955 depending on the package variant used in your design.

The DC supply voltage should be decoupled as close as possible to the VDD pins with high performance RF capacitors. See the schematics for recommended decoupling capacitor values. The supply voltage for the chip should be filtered and routed separately from the supply voltages of any digital circuitry.

Long power supply lines on the PCB should be avoided. All device grounds, VDD connections, and VDD bypass capacitors must be connected as close as possible to the IC. For a PCB with a topside RF ground

plane, the VSS pins should be connected directly to the ground plane. For a PCB with a bottom ground plane, the best technique is to have via holes as close as possible to the VSS pads. A minimum of one via hole should be used for each VSS pin.

Fast switching digital signals should not be routed close to the crystal or the power supply lines. Capacitive loading of fast switching digital output lines should be minimized in order to avoid radio interference. Avoid routing the 32MHz crystal lines close to antenna line and antenna ground.

7.3.16 PCB layout example

The PCB layout shown in the following figures is a reference layout for the aQFN™ package with internal LDO setup and VBUS supply.

> **Note:** Pay attention to how the capacitor C3 is grounded. It is not directly connected to the ground plane, but grounded via VSS_PA pin F23. This is done to create additional filtering of harmonic components.

For all available reference layouts, see the product page for the nRF52840 on www.nordicsemi.com.

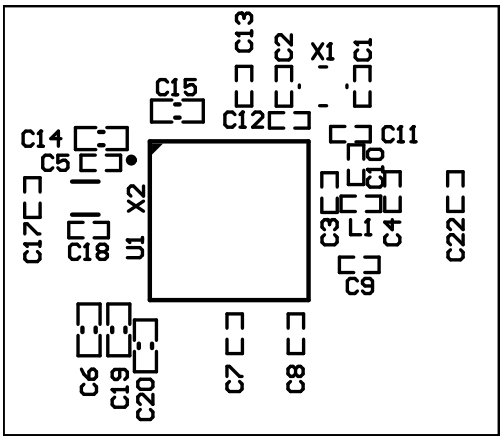


Figure 232: Top silk layer

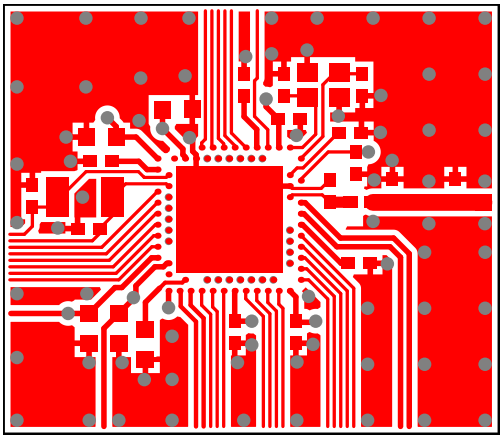


Figure 233: Top layer

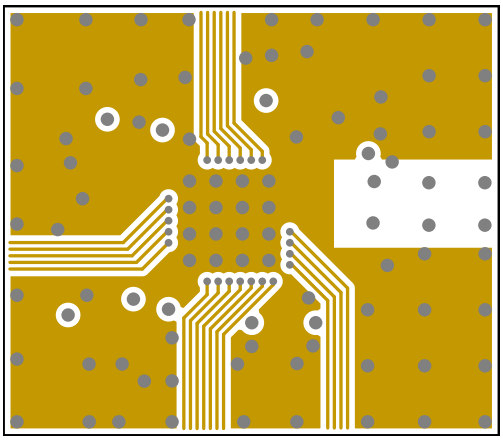


Figure 234: Mid layer 1

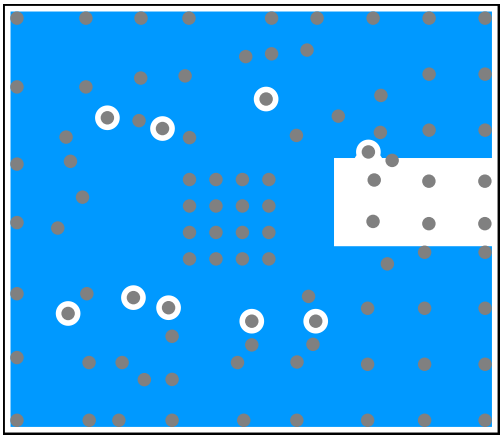


Figure 235: Mid layer 2

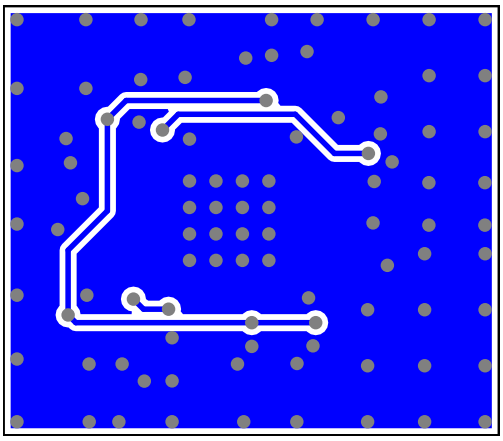


Figure 236: Bottom layer

Note: No components in bottom layer.

7.3.17 PMIC support

The nRF52 Series is comprehensively supported by Nordic Semiconductor's own range of PMICs (Power Management Integrated Circuits). These PMICs are meticulously designed to enhance the performance and efficiency of the nRF52 Series devices. This integration ensures the longest battery life and the highest reliability for the end application. The synergy between the nRF52 Series and the Nordic PMICs highlights

Nordic Semiconductor's commitment to providing a complete and cohesive solution for their customers' needs in wireless technology applications.

7.4 Package thermal characteristics

A summary of the thermal characteristics for the different packages available for the IC can be found below.

Symbol	Package	Typ.	Unit
$\theta_{JA,aQFN73}$	aQFN73	24.11	°C/W
$\theta_{JA,QFN48}$	QFN48	22.37	°C/W
$\theta_{JA,WLCSP}$	WLCSP	36.55	°C/W

Table 98: Package thermal characteristics

Values obtained by simulation following the EIA/JESD51-2 for still air condition using JEDEC PCB.

7.5 Package Variation

The following describes the variation between this specification and a specific device package.

7.5.1 QFN48

The following section describes the package variation of the QFN48 package.

- The QFN48 package does not support USBD.
- VDD and VDDH are short circuited inside the QFN48 package. Therefore the device is only usable in Normal Voltage supply mode, and not High Voltage supply mode. See POWER — Power supply on page 81 for more information.

The parameter variation when using the QFN48 package is as following:

Symbol	Min.	Typ.	Max.	Unit
$P_{SENS,IT,SP,1M,BLE}$		-95		dBm
P_{RF}		7.5[45]		dBm

Table 99: QFN48 package parameter variation

[45] Achieved using `Pos8dBm` setting in RADIO.TXPOWER

8 Recommended operating conditions

The operating conditions are the physical parameters that the chip can operate within.

Symbol	Parameter	Min.	Nom.	Max.	Units
VDD	VDD supply voltage, independent of DCDC enable	1.7	3.0	3.6	V
VDD_{POR}	VDD supply voltage needed during power-on reset	1.75			V
VDDH	VDDH supply voltage, independent of DCDC enable	2.5	3.7	5.5	V
VBUS	VBUS USB supply voltage	4.35	5.0	5.5	V
t_{R_VDD}	Supply rise time (0 V to 1.7 V)			60	ms
t_{R_VDDH}	Supply rise time (0 V to 3.7 V)			100	ms
TA	Operating temperature	-40	25	85	°C
T_J	Junction temperature			90	°C

Table 100: Recommended operating conditions

Note: The on-chip power-on reset circuitry may not function properly for rise times longer than the specified maximum.

8.1 WLCSP light sensitivity

All WLCSP package variants are sensitive to visible and close-range infrared light. This means that a final product design must shield the chip properly, either by final product encapsulation or by shielding/coating of the WLCSP device.

Some WLCSP package variants have a backside coating, where the marking side of the device is covered with a light absorbing film, while the side edges and the ball side of the device are still exposed and need to be protected. Other WLCSP package variants do not have any such protection.

The WLCSP package variant CKAA has a backside coating.

9 Absolute maximum ratings

Maximum ratings are the extreme limits to which the chip can be exposed for a limited amount of time without permanently damaging it. Exposure to absolute maximum ratings for prolonged periods of time may affect the reliability of the device.[46]

	Note	Min.	Max.	Unit
Supply voltages				
VDD		-0.3	+3.9	V
VDDH		-0.3	+5.8	V
VBUS		-0.3	+5.8	V
VSS			0	V
I/O pin voltage				
$V_{I/O}$, VDD ≤3.6 V		-0.3	VDD + 0.3	V
$V_{I/O}$, VDD >3.6 V		-0.3	3.9	V
NFC antenna pin current				
$I_{NFC1/2}$			80	mA
Radio				
RF input level			10	dBm
Environmental aQFN73 package				
Storage temperature		-40	+125	°C
MSL	Moisture Sensitivity Level		2	
ESD HBM	Human Body Model		2	kV
ESD HBM Class	Human Body Model Class		2	
ESD CDM	Charged Device Model		450	V
Environmental QFN48 package				
Storage temperature		-40	+125	°C
MSL	Moisture Sensitivity Level		2	
ESD HBM	Human Body Model		4	kV
ESD HBM Class	Human Body Model Class		3A	
ESD CDM	Charged Device Model		1	kV
Environmental WLCSP 3.544 x 3.607 mm package				
Storage temperature		-40	+125	°C
MSL	Moisture Sensitivity Level		1	
ESD HBM	Human Body Model		1	kV
ESD HBM Class	Human Body Model Class		1C	
ESD CDM	Charged Device Model		500	V
Flash memory				
Endurance		10 000		write/erase cycles
Retention at 85 °C		10		years

Table 101: Absolute maximum ratings

[46] For accelerated life time testing (HTOL, etc) supply voltage should not exceed the recommended operating conditions max value, see Recommended operating conditions on page 971.


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10 Ordering information

This chapter contains information on device marking, ordering codes, and container sizes.

10.1 Device marking

The nRF52840 package is marked as shown in the following figure. Only the first two characters of the function variant code are used in the <VV> entry.

N	5	2	8	4	0
<P	P>	<V	V>	<H>	<P>
<Y	Y>	<W	W>	<L	L>

Figure 237: Package marking

10.2 Box labels

The following figures show the box labels used for nRF52840.

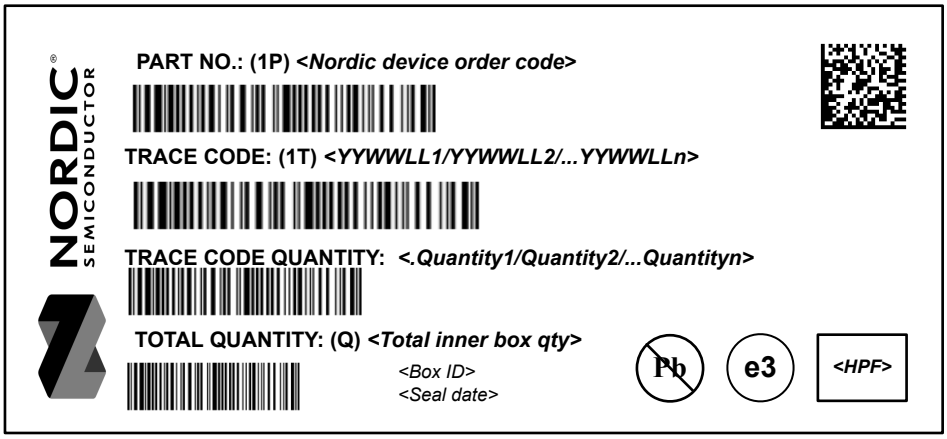


Figure 238: Inner box label

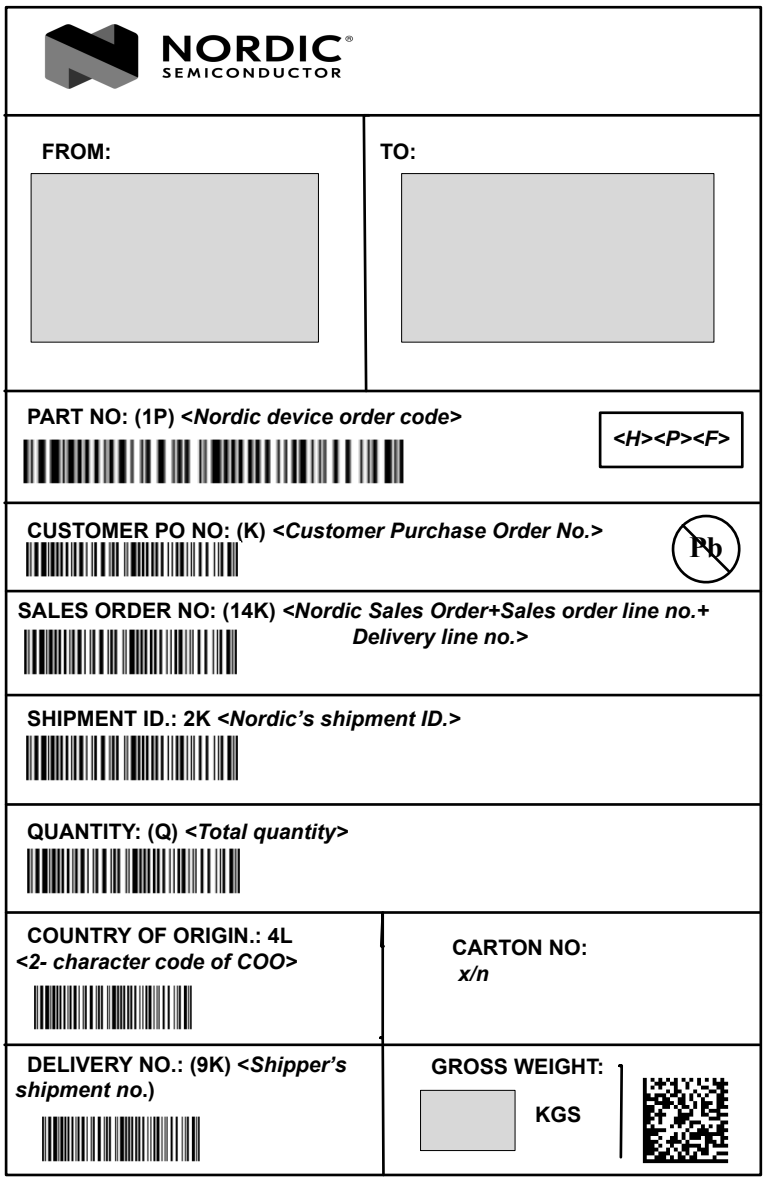


Figure 239: Outer box label

10.3 Order code

The following are the order codes and definitions for nRF52840.

n	R	F	5	2	8	4	0	-	<P	P>	<V	V>	-	<C	C>

Figure 240: Order code

Abbrevitation	Definition and implemented codes
N52/nRF52	nRF52 Series product
840	Part code
<PP>	Package variant code
<VV>	Function variant code
<H><P><F>	Build code H - Hardware version code P - Production configuration code (production site, etc.) F - Firmware version code (only visible on shipping container label)
<YY><WW><LL>	Tracking code YY - Year code WW - Assembly week number LL - Wafer lot code
<CC>	Container code

Table 102: Abbreviations

10.4 Code ranges and values

Defined here are the nRF52840 code ranges and values.

<PP>	Package	Size (mm)	Pin/Ball count	Pitch (mm)
QI	aQFN	7 x 7	73	0.5
QF	QFN	6 x 6	48	0.4
CK	WLCSP	3.544 x 3.607	94	0.35

Table 103: Package variant codes

<VV>	Flash (kB)	RAM (kB)	Access port protection
AA	1024	256	Controlled by hardware
AA-F	1024	256	Controlled by hardware and software

Table 104: Function variant codes

<H>	Description
[A . . Z]	Hardware version/revision identifier (incremental)

Table 105: Hardware version codes

<P>	Description
[0 . . 9]	Production device identifier (incremental)
[A . . Z]	Engineering device identifier (incremental)

Table 106: Production configuration codes

<F>	Description
[A . . N, P . . Z]	Version of preprogrammed firmware
[0]	Delivered without preprogrammed firmware

Table 107: Production version codes

<YY>	Description
[00 . . 99]	Production year: 2000 to 2099

Table 108: Year codes

<WW>	Description
[1 . . 52]	Week of production

Table 109: Week codes

<LL>	Description
[AA . . ZZ]	Wafer production lot identifier

Table 110: Lot codes

<CC>	Description
R7	7" Reel
R	13" Reel
T	Tray

Table 111: Container codes

10.5 Product options

Defined here are the nRF52840 product options.

Order code	MOQ[47]	Comment
nRF52840-QIAA-R7	800	Not recommended for new designs
nRF52840-QIAA-R	3000	Not recommended for new designs
nRF52840-QIAA-T	260	Not recommended for new designs
nRF52840-QIAA-F-R	3000	
nRF52840-QIAA-F-T	260	
nRF52840-QIAA-F-R7	800	
nRF52840-QFAA-F-R	3000	
nRF52840-QFAA-F-R7	1000	
nRF52840-CKAA-R	7000	Not recommended for new designs
nRF52840-CKAA-F-R	7000	

Table 112: nRF52840 order codes

Order code	Description
nRF52840-DK	nRF52840 Development Kit

Table 113: Development tools order code

[47] Minimum Ordering Quantity

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